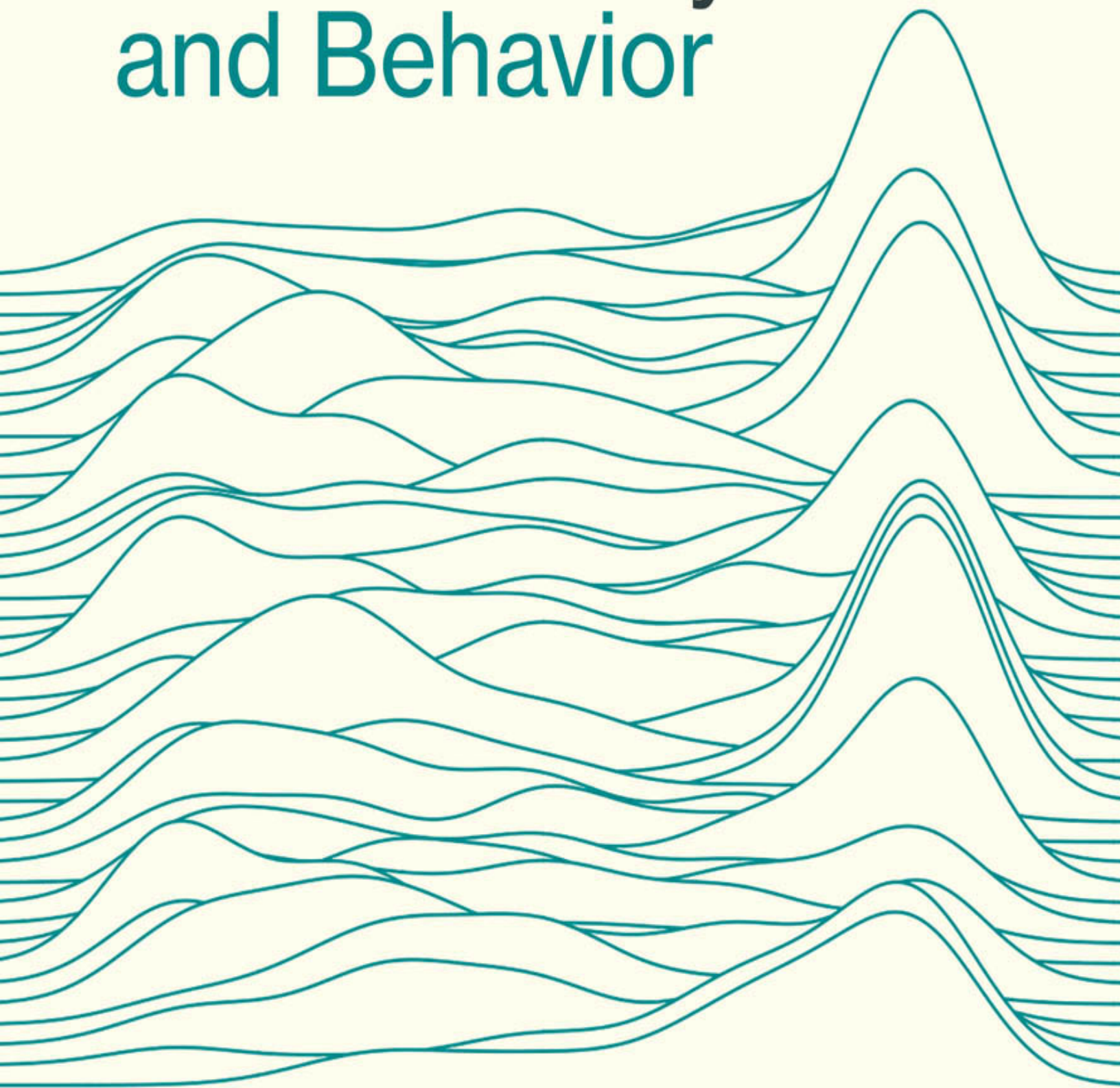
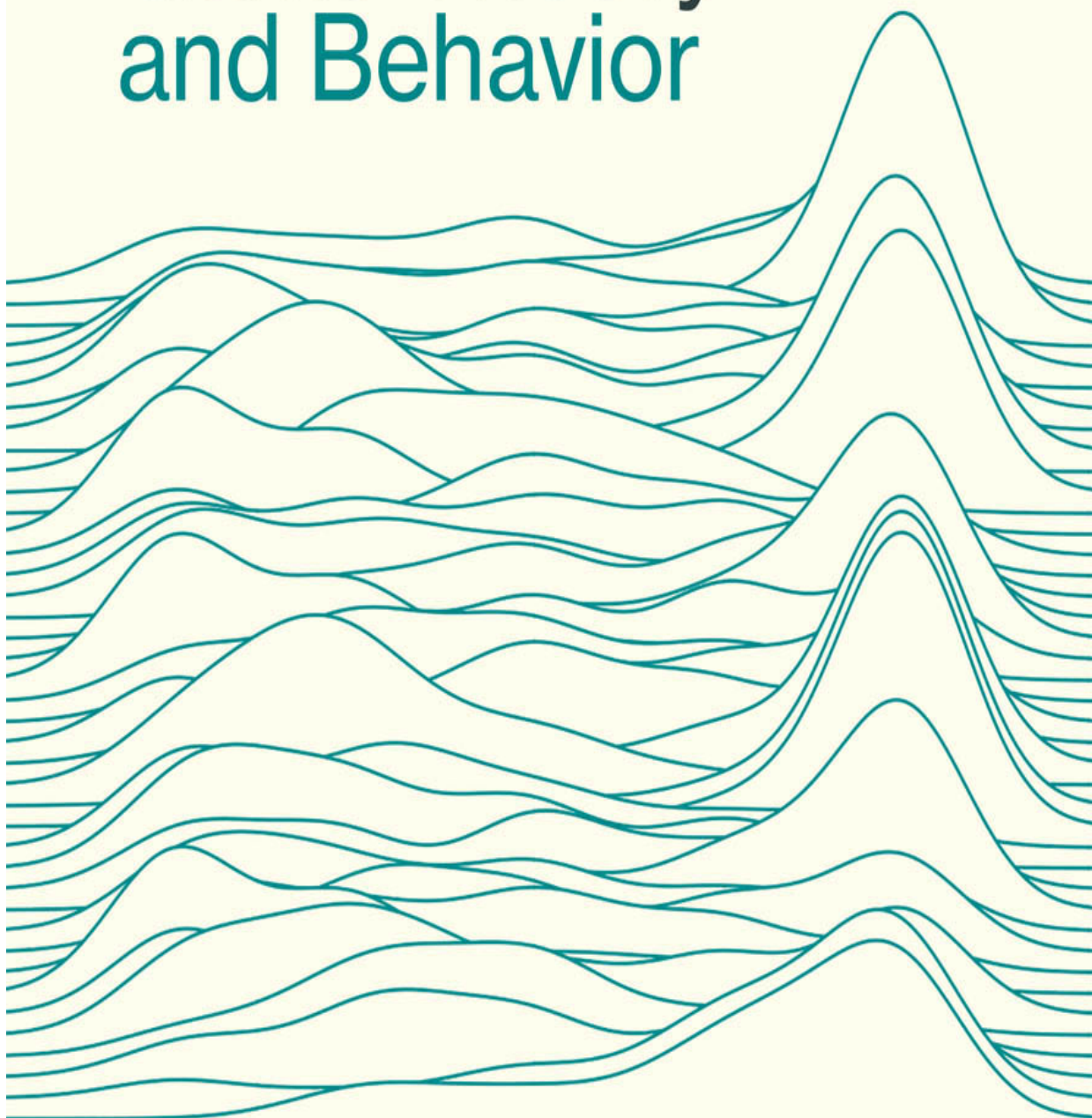


Game Theory and Behavior



Jeffrey Carpenter and Andrea Robbett

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Preface

The primary goal of this book is to provide a clear and accessible formal introduction to standard game theory. Where the book differs is that we are interested in both the normative and positive aspects of game theory. Not only do we ask how players *should* behave, given their preferences and the structure of the games they play, but we are equally interested in experimental evidence describing how they actually behave. Further, when carefully controlled studies suggest that there is a considerable gap between theory and behavior, we survey the recent developments in the game-theoretical literature seeking to narrow this gap. In this sense, we see game theory as a positive venture too—developing hypotheses based on the logic of game theory, testing them in the lab and field to see if they predict behavior, and then iterating the process to create versatile theoretical constructs that assume no more than is necessary about the players and yet predict what they will do. Our objective is to simultaneously provide students with both the theoretical tools to analyze situations through the logic of game theory and the intuition and behavioral insights to apply these tools to real-world situations.

The book was written to serve as the primary textbook in a first course in game theory at the undergraduate level and does not assume that students have any previous exposure to game theory or economics. However, it may also be a useful resource for graduate students or researchers who are looking for a synthesis of behavioral research or a concise introduction to some of the advanced topics covered here, such as auctions, cooperative game theory, matching market design, social dilemmas, voting games, evolutionary game theory, quantal response equilibrium (QRE), level- k reasoning, and psychological game theory. As a result, the book may also be used as a secondary text in graduate or undergraduate courses on behavioral

and experimental economics, behavioral game theory, and political game theory.

Most chapters follow the same basic format: We begin with a motivating game, which sets up the topic. Readers are then asked what they should (or would) do in this situation. We provide protocols and supplementary materials for instructors who want to run the motivating game in each chapter as an experiment during class, enabling students to learn experientially and build their intuition before diving in to the topic. From there, the chapter provides a traditional game-theoretic analysis of these types of games. We then present data from similar lab experiments, discuss when the predictions are likely to hold, and summarize the main behavioral insights. However, to the extent that behavior deviates from the predictions, we emphasize that this is not the end of the story and it doesn't imply that we should give up on game theory as a framework for understanding strategic interactions. Instead, we explain how these insights about human behavior have been woven back into the theory to better predict behavior of real people, such as using social preference models, QRE, level- k reasoning, cursed equilibrium, and so on. The goal is to demonstrate that a positive treatment of game theory—one that stresses the iterative process of advancing theory and gathering data—leads to more accurate predictions. These behavioral sections are optional; the instructor can choose whether to include these final sections in the lecture or course material.

We believe that there are three features that distinguish this book:

1. *A formal but accessible introduction.* Our first objective is to offer an introduction to game theory that is accessible to students with a wide range of academic backgrounds and mathematical preparation, but also precise and comprehensive enough to enable them to apply the tools that they learn more broadly and prepare them for advanced courses. Our goal throughout is to strike a balance between intuition and rigor that ensures that every student is capable of mastering this material. Whenever possible, we use a series of intuitive examples to slowly build up to more technical approaches. While we rely on notation and mathematical formulations, we also describe what we're doing (and why) in words and provide concrete examples. While we intend Calculus I to be a prerequisite for this course, the book does not assume any particular mathematical knowledge other

than the ability to take a derivative and a basic understanding of set notation and probability. All these tools are covered in the appendix to the level needed to solve the exercises in the book. It is also possible to navigate the book without calculus, as described at the end of the preface.

2. *Behavioral results and theoretical adaptations.* Alongside the formal theoretical treatment in each chapter, we present data from laboratory experiments testing the theoretical predictions in the games we analyze. We then demonstrate how the behavioral insights gleaned from these experiments can be incorporated within the existing framework of game theory to generate more accurate predictions. Part VIII of the book includes more in-depth treatments of behavioral models that either relax the rationality requirements of the players or incorporate belief-based motives. While behavioral research and models don't typically play a central role in game theory texts, we believe that this is essential for anyone who wishes to fully understand strategic behavior and predict the outcomes of game-theoretic interactions.

3. *Classroom games and experiential learning.* Experimental methods offer the opportunity to integrate carefully designed classroom demonstrations that give students firsthand experience with the situations they are learning to model. This supplemental content has thus far been missing in game theory texts, but it can be fundamental in helping students master the material. Each chapter opens with a motivating game, as noted previously and we provide the instructor with the materials necessary to seamlessly implement the corresponding experiment in the classroom. The game can be run using pencil and paper or online. The protocols, worksheets, and computer codes are provided in the online materials (mitpress.mit.edu/behavioral-game-theory).

The book contains 28 chapters, organized into eight parts, followed by a mathematical appendix. Each chapter is intended to correspond to one lecture period of material. Part I introduces the theory of games (chapter 1) and how to model strategic interactions as games of strategy (chapter 2). Part II analyzes how to solve relatively simple games, in which players each make a single decision simultaneously. This part begins with dominated strategies (chapter 3), and then analyzes how to solve for pure-strategy equilibria in two-player games (chapter 4), mixed-strategy equilibria

(chapter 5), and equilibria in continuous games and games with any arbitrary number of players (chapter 6), before addressing equilibrium selection in games with multiple Nash equilibria (chapter 7). In part III, we expand our scope to sequential-move games. We begin with subgame perfection (chapter 8) and then apply this solution concept to games that are finitely repeated (chapter 9) and infinitely repeated (chapter 10). Part IV introduces incomplete information in simultaneous-move games (chapter 11), signaling games (chapter 12), and auctions (chapter 13). We consider chapters 1 through 12 to comprise the nuts and bolts of any introductory game theory course (with chapter 13 as an optional application for advanced students). Parts V through VII apply the tools developed in the first half of the book to specific applications and extensions by presenting detailed treatments of prominent topics to which game theory has been applied: bargaining, cooperative game theory, and market design (part V), social dilemmas (part VI), and social choice and voting (part VII). Finally, part VIII provides a more in-depth discussion of behavioral game theory models. Each of the chapters in parts V through VIII is written as a stand-alone analysis of a specific topic and is not required for any other material. We therefore recommend that instructors first cover chapters 1 through 12 and then pick and choose among the remaining chapters based on the interests of the students, the focus of the course, and the remaining time in the term.

While the book assumes students know how to optimize functions and the definition of a derivative, it is possible to use the book in a course without calculus as a prerequisite because the majority of the chapters do not require calculus. Specifically, we rely on calculus in chapter 6, when we cover best responses in games with continuous strategy spaces; brief subsections of chapters 8 and 11, when finding subgame perfect Nash equilibria and Bayesian Nash equilibria, respectively, in games with continuous strategy spaces; chapter 13 (auctions); a subsection of chapter 18 (continuous social dilemmas); chapters 19 and 20 (public goods games and common pool resources); and chapter 25 (evolutionary game theory). These chapters or subsections could be skipped for courses that wish to cover the logic of game theory without using calculus.

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I

Introduction to the Theory of Games

1

Games of Strategy

Game theory provides a means of formally modeling strategic interactions among people and predicting their behavior. The tools of game theory can be used to analyze any situation in which individuals are affected by each other's actions and strategically attempt to anticipate each other's behavior. This chapter will provide a brief introduction to game theory and the overall approach of this textbook. We'll discuss several examples that illustrate the difference between game theoretic decision making and other types of choices, the importance of considering the strategic thinking of other players, and how strategic behavior can lead to outcomes that are bad for everyone. While traditional game theory is concerned with how rational, self-interested individuals *should* act in strategic interactions, one of the main objectives of this book is to consider how real people *actually* act in strategic situations and how standard game theoretic models can be modified to better reflect real-life behavior. This chapter provides a road map for the rest of the book and discusses how behavioral research is incorporated into the treatment of game theory.

Imagine that you are playing a guessing game with the rest of your class. Each person is asked to simultaneously write down a guess, which can be any real number from 0 to 100. The average of all the guesses in the class is then calculated, and the person whose guess is closest to two-thirds of the average of all the guesses will be declared the winner. What do you guess?

We will return to this guessing game later in this chapter. First, however, we will define what game theory is, consider the types of situations that it is used to study, and explain how game theoretic situations differ from other types of decision problems that individuals confront.

What Is Game Theory?

Game theory is the study of strategic interaction. It is sometimes referred to as “multiperson decision theory.” It is used to analyze situations in which two or more people act—independently and in their own self-interest—and then the outcomes depend on the decisions that each of them make. For example, perhaps you are bidding in an auction and whether you win

depends on whether your bid exceeds those of your competitors. Or perhaps you are shooting a soccer penalty kick and the likelihood that you score depends both on which side of the net you aim the ball and which direction the goalie dives. Or perhaps you are working on a group project and your success depends on how much effort each group member exerts. In these types of situations, the consequences of your own actions depend on the choices of others. Thus, when deciding what is best for yourself, you must put yourself in the shoes of the other individuals, reason through the situation from their perspective, and form beliefs about what they will choose—all the while knowing that they are simultaneously making the same calculations about you.

Game theoretic decision making differs from other types of economic decision making, in which each individual is assumed to have a negligible effect on the market as a whole. For instance, a consumer deciding whether to purchase a gallon of milk need only consider the price set by the grocer and how much she values milk compared to alternative uses of that same amount of money; she need not attempt to anticipate the actions of other shoppers and they need not attempt to anticipate her decision. Instead, game theory models the types of interactions you witness in your daily life—between you and your friends, between football teams, between Google, Apple, and Amazon, between presidential candidates, and so on. Almost any interaction can be modeled as a game, and we can predict behavior using the tools of game theory.

If you told your friends and family that you were planning to take a course on game theory, there is a decent chance that someone asked you: “Oh, you mean games like poker?” or “... games like soccer?” or “... video games?” While the answer to all these questions is “Yes,” game theory is not concerned purely with recreational games. Instead, the term “game” or “game of strategy” can be used to describe a vast array of social and economic interactions. In this book, a *game of strategy* is any situation in which the following conditions hold:

- There are two or more individuals, each of whom makes a choice about which course of action to take.
- The individuals’ actions jointly determine the outcomes.

- The individuals know about the available actions and incentives of others and use this information to strategically anticipate which actions they will choose.

Due to its generality, game theory is replacing supply and demand as the dominant framework in microeconomic theory; it has increasingly been adopted by many other social and behavioral sciences, including political science, psychology, sociology, and biology; and it provides a foundation for the design of algorithms in computer science.

Non-cooperative versus Cooperative Game Theory

There are two distinct branches of game theory: cooperative and non-cooperative. In practice, the term “game theory” is typically used as shorthand for *non-cooperative game theory*. Non-cooperative game theory involves, first, a careful specification of the rules of the interaction, including who is able to do what, when they’re able to do it, and the information that they have when they act. Then, we determine rational play, given the rules of the game. That is, we predict outcomes by assuming that each player acts independently with the goal of doing as well as they can within the structure of the game, and given the actions of others. Cooperative game theory provides a separate approach to modeling strategic interactions. Rather than focusing exclusively on what each individual can do independently, this approach considers what each subset or *coalition* of players could collectively achieve if they were to act as a single decision-making unit. This book mainly focuses on non-cooperative game theory, consistent with most introductory courses in game theory. However, we provide a treatment of cooperative game theory in chapter 16 and consider its application to bargaining (chapter 15) and market design (chapter 17).

Individual versus Game Theoretic Decision Making

An Individual Decision Problem

The standard paradigm for studying decision making in economics and political science is *rational choice theory*. Essentially, we assume that people know enough about all the alternatives available to them and about their own preferences that they are always able to calculate what is best for

themselves; then they perfectly execute this optimal course of action, without error. Under rational choice theory, an individual's decision consists of three elements:

- **A set of actions.** These are all the choices that the decision maker could select. For example, imagine a decision maker who must choose how to get to work and has two possible actions: drive or take the train.
- **A mapping of actions to outcomes.** Each action has different consequences, which the decision maker fully appreciates before acting. For example, the outcome associated with driving may be getting to work on time but feeling uptight and experiencing mild road rage, while the outcome associated with taking the train is being relaxed and prepared, but a bit tardy.
- **Preferences over outcomes.** A preference relation, $\geq$, specifies how the decision maker ranks any two outcomes. For outcomes a and b , $a \geq b$ indicates that the decision maker prefers a at least as much as b .

We don't impose many restrictions on what preferences rational decision makers are allowed to have. Faced with the same decision, some people may prefer driving, while others prefer the train. Neither is inherently irrational—what would be irrational is doing one thing when you prefer the other. However, there are two axioms or conditions that preferences must satisfy to be considered rational: *completeness* and *transitivity*.

- **Completeness.** Consider any two outcomes a and b . It must be the case that $a \geq b$, $b \geq a$, or both (i.e., the decision maker is indifferent between a and b).¹ The completeness axiom implies that the decision maker is never paralyzed by indecision; she can always compare any two outcomes and determine which is preferable or that she is indifferent.
- **Transitivity.** Consider any three outcomes, a , b , and c . If $a \geq b$ and $b \geq c$, then it must be the case that $a \geq c$.² The transitivity axiom implies that there are no weird cycles in the decision maker's preferences, such that a is preferred to b , which is preferred to c , which is preferred to a .

Combined, the completeness and transitivity axioms imply that when faced with a set of possible outcomes, the decision maker is always able to rank them from most preferred to least preferred (potentially including ties if the decision maker is indifferent between some outcomes). Given this, we can

represent the individual's preferences numerically by assigning a number to each outcome, such that more preferred outcomes are associated with higher numbers. In the parlance of microeconomic theory, these numbers are *utilities*; in game theory, we call them *payoffs*. Payoffs provide a convenient way of representing preferences since otherwise it would be unmanageable to keep track of an individual's full preference ranking when the set of outcomes is large or even infinite. Finally, rational choice theory assumes that individuals choose the action that leads to the outcome with the highest payoff.

Let's consider an example of a decision problem faced by an individual. Imagine that you are registering for courses and you must take one of three intermediate courses in the economics department this semester: Micro, Macro, or Stats. We assume that you fully understand all the knowable attributes of the courses that could affect your satisfaction, such as whether the course is a prerequisite for other classes, the time of day it meets, the style of the professor, and the workload, and are able to rank the courses from most to least preferred. Some features may not be knowable with certainty—perhaps the professor is new and her style is unfamiliar. In these cases, we assume that you accurately assess the probability of the professor being decent, as well as your own preference for risk, and incorporate this into your calculations.

For example, after reflecting on your preferences and the available information, perhaps you determine that $\text{Micro} \geq \text{Stats}$ and $\text{Stats} \geq \text{Macro}$. By transitivity, this implies that $\text{Micro} \geq \text{Macro}$, and so you go into registration with a complete ranking of your possible options, shown in [figure 1.1](#). We can then assign a payoff to each option such that higher payoffs reflect more preferred options, as shown in the second column of the table.³ Finally, we assume that you select your most preferred available option, which is signing up for Micro.

Preference Ranking	Payoff
1. Micro	35
2. Stats	25
3. Macro	22

Figure 1.1

Preference ranking and payoffs.

Other People Complicate Things

What happens when we add more people into the mix? Standard game theory is built on the same assumptions of rational choice: we assume that people can accurately calculate the payoff that they receive from each possible outcome and they always act to maximize their own payoff. The difference is that the outcomes aren't determined solely by their own actions, but also by the actions of others. In fact, some researchers think of decision theory as a special, simplified case of game theory that involves only a single player, although such scenarios would not be a game of strategy.

Let's reimagine this example as a game theoretic problem rather than an individual decision problem. Let's say that there is one other student whom you strongly prefer to be in the same class with, for whatever reason (e.g., you want to get to know him better, he would be an excellent study partner, or he is a complete knucklehead who will drag down the class average and improve your curved grade). For any given class, your payoff would be 30 units higher if this individual were to enroll as well, as shown in [figure 1.2](#).

Your Choice	Other Student's Choice	Your Payoff
Micro	Micro	65
Micro	Not Micro	35
Stats	Stats	55
Stats	Not Stats	25
Macro	Macro	52
Macro	Not Macro	22

Figure 1.2

Preference ranking and payoffs when another student's choice matters.

Now what do you do? You have just as much information as you did before about your own preferences over outcomes, but now Micro is no longer the obvious choice: if you were to select Micro and the other student signs up for something else, then you are worse off than you would have been if you had correctly anticipated this and managed to match. If, however, you knew what the other person was choosing, then your choice would once again be clear. The added challenge in game theoretic decision making is correctly anticipating the actions of other people. At the same time, it is essential to recognize that the other person's behavior is also not determined in isolation; it may be based on what he believes you will do. Do you think the other student wants to be in the same course as you or prefers to avoid you? Unless you think that he is oblivious to your existence, your belief about which class he will choose is at least partially driven by what class he believes you will choose ... which is, in turn, driven by which class he believes you believe he will choose, and so on. Thinking strategically thus requires you to attempt to anticipate the actions of another person, who is simultaneously attempting to anticipate your actions. Simply put, individual decision theory assumes that agents act to maximize their payoffs given their own preferences, while game theory assumes that agents act to maximize their payoffs given their own preferences, as well as the incentives of others and beliefs about how others will act.

As with individual decision making, standard game theory assumes that people are rational decision makers who do not make mistakes. They are always correct, both in their beliefs about how others will act and in their calculations of what is best for themselves given these beliefs. In game theory, however, we are not simply interested in how isolated decision makers behave, but rather in the outcome of an interaction among rational individuals who know their fates to be intertwined and strategically anticipate each other's moves. In each interaction, traditional game theory predicts that players' behavior will be in *equilibrium*—each person has correct beliefs about what other people are doing and maximizes their payoffs given these beliefs. In equilibrium, each player is doing as well as possible given the rules of the game and the actions of others. That is, we look for what outcomes could arise and persist if no one makes any mistakes.

Theory of Mind

While the process of placing oneself in another person's head and seeing that perspective is often complicated, it is something that most people become fairly adept at during childhood, at least in simple and familiar contexts. Psychologists call this ability "theory of mind." The Sally-Anne task was developed to test theory of mind in young children.

The test is shown in cartoon form in [figure 1.3](#), but is typically acted out with dolls. A blonde doll, named Sally, has a marble, which she stashes in her basket. Once Sally leaves, sneaky Anne comes along and moves the marble from the basket to the box. The question is: where will Sally look for her marble when she returns? Young children typically say that she'll look in the box, because that's where we all know it is. After the age of four or so, children learn to separate Sally's perspective from their own, and they say that she will look in her basket since that is the last place she saw it, even though they have private knowledge that it is elsewhere.

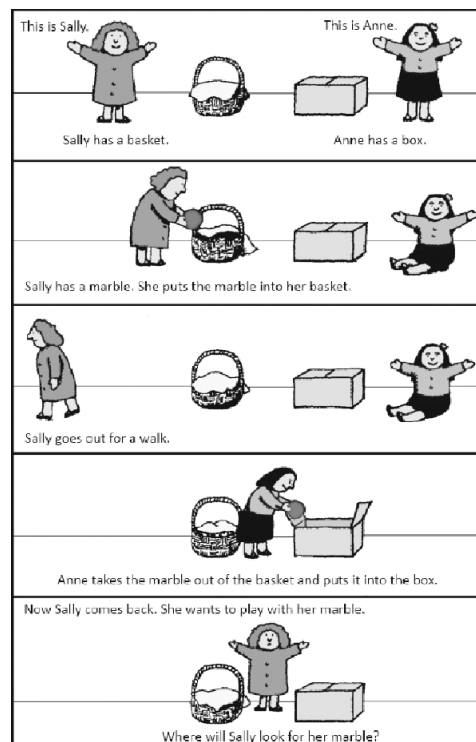


Figure 1.3
The Sally Anne task.

A good illustration of theory of mind can be found in a famous scene in the 1987 satirical adventure film *The Princess Bride*. The bad guy, Vizzini, has kidnapped the Princess Bride and our hero, Westley, is attempting to rescue her by challenging Vizzini to a battle of wits. Westley pours two glasses of wine and poisons one of them with iocaine powder out of Vizzini's line of sight. Then he places a glass in front of each of them. Vizzini's job is to decide who must drink from each glass.

This task propels Vizzini into a stream-of-consciousness soliloquy detailing his reasoning about which glass Westley poisoned, which you can find on YouTube by searching for "Princess Bride battle of wits." Each time Vizzini seems to settle on which glass Westley must have poisoned, he takes his reasoning about Westley's logic one step further and reverses his choice. Ultimately, Vizzini attempts to trick Westley by distracting him, switching glasses when his back is turned, and then suggesting that each drink from his own glass. Vizzini hesitates before taking a sip, but once he sees Westley promptly drink, he assumes that he has won and can drink freely himself. Vizzini continues to gloat until he keels over dead. Westley later reveals that in fact both glasses were poisoned but he has developed an immunity to iocaine powder.

While some of Vizzini's considerations are a bit unorthodox, this scene demonstrates the challenge of reasoning about the thought process of someone who is simultaneously reasoning about your thought process: Vizzini is attempting to think about what Westley thinks about what Vizzini thinks about what Westley thinks, and so on. It also illustrates how our ability to make predictions about choices and outcomes critically depends on a careful specification of the rules of the game, including the available actions (e.g., can Westley poison both cups?), how actions lead to outcomes (e.g., will Westley die if he drinks a poisoned cup?), and the information that players have (e.g., does Vizzini know that Westley could have an immunity to iocaine?).

Thinking Strategically: Reasoning about the Reasoning of Others

A Betting Game

Let's focus more closely on the importance of understanding the incentives of the people you are playing games with. Consider the following betting

game between two players, A and B, shown in [figure 1.4](#). There is a deck of three cards: two red and one black. Each player will draw a card and look at it privately, and then the two simultaneously decide whether to bet or pass. If one or both players pass, then the game ends and no one wins or loses any money. If both players bet, however, they compare cards. If both cards are red, then B wins, and A must pay him \$10. If the cards mismatch, then A wins, and the amount that B must pay her depends on who drew the red card. If A drew the red card and B the black, then B must pay \$20, and if A drew the black card and B the red, then B must pay \$5. The monetary payoffs are summarized in [figure 1.4](#).

	A Draws Red B Draws Black	Both players Draw Red	A Draws Black B Draws Red
A's payoff	\$20	−\$10	\$5
B's payoff	−\$20	\$10	−\$5

Figure 1.4
A betting game.

Imagine that you draw a red card in this game. Would you bet if you were in the position of A? What about if you were B?

If A draws a red card, then she knows that she is either in the situation in the first column of the table (i.e., the other player has the black card and she would win \$20 if they bet) or the situation in the second column (i.e., the other player also has a red card and she would lose \$10 if they bet). Since the other player is equally likely to have drawn the black card or the other red card, she reasons, she would gain \$5 on average. Thus, betting seems to be a good idea when she holds the red card. What about B? If he draws a red card, then he knows that he is either in the situation in the second column of the table (he would gain \$10) or in the last column (he would lose \$5). He thus reasons that he would gain \$2.50, on average, if he bets with a red card, so he also concludes that betting is in his interest. Notice that if a player draws a black card, then they know for sure which column of the table they are in and what the outcome would be if they bet. If A draws black, then she knows that she would gain \$5, so she prefers to bet.

In contrast, if B draws black, then he knows that he would lose \$20, so he prefers to pass.

On closer inspection, there's something a bit strange going on here: both players seem to believe that betting is an attractive gamble either all or most of the time. How can that be if one person's gain is always the other person's loss? The reasoning outlined here is problematic because it considers the situation only from the player's own perspective, failing to take into account the incentives and information of the other person. If both players reasoned through the situation from the other person's perspective as well as their own, then neither player would think that betting with a red card would be a good idea.

To see this, let's revisit A's decision to bet with a red card. She knows that B holds either a black card (in which case she wins \$20) or a red card (in which case she loses \$10). However, B would never bet if he holds the black card—in that case, he'd know for sure that he would lose. Once A takes B's information and thought process into account, she realizes that she'd never be able to win that \$20, and she can only lose by betting on the red card. (In other words, even though B is equally likely to hold the other red or the black card, A is not equally likely to win or lose if she bets because B will never bet in the situation in which A would win.) Thus, if A reasons through the situation from B's perspective, she would choose to bet only if she draws black, not if she draws red.

Returning to B, we previously saw that B prefers to bet with a red card if he believes that he is equally likely to win \$10 or lose \$5. If he thinks that A is not strategic and she bets even if she has the red card, then this is the case. However, if he anticipates that she will anticipate his reasoning, then he believes that she'll realize that it isn't in her interest to bet with red. And if she never bets when she sees a red card, then he can never win the \$10 and he can only lose if he bets on red. Thus, if B reasons through the situation from A's perspective—and assumes that she will reason from his—then B never finds it in his interest to bet.

The players therefore reach the opposite conclusion about whether it is their interest to bet with the red card once they consider the perspective of the other person.

The Beauty Contest Guessing Game

Let's return to the guessing game that was described at the beginning of the chapter: each person simultaneously guesses a number between 0 and 100, and whoever's guess is closest to two-thirds of the average wins a prize.

This game is known as the *beauty contest game*, because it is based on a passage by the famous economist John Maynard Keynes (1936, p. 156), in which he likens professional investment to contests that appeared in newspapers at the beginning of the twentieth century. The newspaper would run hundreds of photographs of faces, competitors would write in saying which six photos were the most attractive, and a prize was awarded to the individual whose choice was closest to the average response. As Keynes wrote, "each competitor has to pick not those faces which he himself finds prettiest, but those which he thinks likeliest to catch the fancy of the other competitors, all of whom are looking at the problem from the same point of view. It is not a case of choosing those which, to the best of one's judgment are really the prettiest, nor even those which average opinion genuinely thinks the prettiest. We have reached the third degree, where we devote our intelligences to anticipate what average opinion expects the average opinion to be. And there are some, I believe, who practise the fourth, fifth, and higher degrees."

The same type of logic applies in the guessing game. Let's assume that you put yourself in the shoes of the other players and form beliefs about what they will guess. In doing so, you determine that the average guess among your classmates will be x . You thus conclude that you will win by choosing roughly $\frac{2}{3}x$. Of course, you wouldn't precisely hit two-thirds of the average if you guessed exactly $\frac{2}{3}x$ because this doesn't take into account the inclusion of your own guess in the average. To be more precise, in a class of n people, you'd hit the winning number if you guessed $g = \frac{2}{3} \left(\frac{x(n-1)+g}{n} \right)$ or, after a bit of algebra, $g = \frac{2}{3}x \frac{n-1}{n-2/3}$. Since this quickly approaches $\frac{2}{3}x$ as n grows large, and it is within an integer even for fairly small classes, we'll save the headache and just assume that you want to guess $\frac{2}{3}x$. But wait—if you were able to reason your way to $\frac{2}{3}x$, shouldn't the other players be able to do the same? And if everyone manages to conclude that $\frac{2}{3}x$ is the best choice, then you need to stay one step ahead and choose $\frac{2}{3} \times \frac{2}{3}x$. But if you're able to anticipate that $\frac{2}{3} \times \frac{2}{3}x$ is the winning choice, surely everyone

else will as well, in which case you should actually choose $\frac{2}{3} \times \frac{2}{3} \times \frac{2}{3}x$. Where does this reasoning about the reasoning of others end? The only number that can hold up to this logic is to choose two-thirds of what you yourself would choose: that is, $x = \frac{2}{3}x$, or $x = 0$. Game theory thus predicts a unique outcome: everyone selects 0.

Another way to see this is to think about the range of guesses that could contain the winning number (rounding to the nearest integer for clarity). Since the highest anyone can guess is 100, the highest that two-thirds of the average could be is 67, and so no one should bother with guesses between 67 and 100. But if no one is guessing above 67, then the highest that two-thirds of the average could be is 44, and so no one should bother with guesses above 44. But if no one is guessing above 44, then the highest that two-thirds of the average could be is 29. Following this logic all day long, we again reason our way down to 0.

It may not surprise you to learn that very few people actually choose 0 in practice. In laboratory experiments with cash prizes, about 1 percent of people guess 0 and even professional game theorists guess 0 less than a quarter of the time. If you believe that other people will not reason their way down to 0, then it makes sense to guess a positive number as well. We have been running this game in our classes for years, and the winning number is typically in the high teens.

This game illustrates the process of thinking strategically by attempting to anticipate the actions of people who are simultaneously attempting to anticipate what you will do. It also illustrates some of the main themes of this book. When placed into strategic situations, real people do not always behave as rational agents who have limitless cognitive capabilities and assume that everyone else does as well. To be able to more accurately predict the strategic behavior of real people, it is important to test the game theoretic predictions by having people actually play the games for real stakes and use these findings to adapt the theory to better align with what people actually do. We'll revisit this game, along with the typical findings when it is played by actual people, in chapter 3, after we've developed additional tools to analyze the game.

The Prisoner's Dilemma

The most famous game in game theory is called the *prisoner's dilemma*. According to the most common origin story, it was developed in 1950 by Merrill Flood and Melvin Dresher, researchers at the RAND Corporation, which at the time was a think tank tasked with performing research for the US Armed Forces (Poundstone, 1992). The game was specifically created to be a challenge to non-cooperative game theory, which in this case predicts that rational play leads to an outcome that is seemingly irrational, in that it is undesirable for everyone. Another researcher, Albert Tucker, wanted to make the game more accessible in his presentations to general audiences, so he came up with a story of two prisoners deciding whether to cut a deal with the prosecution to reduce their own punishment at the other person's expense, which is where the dilemma gets its name.

Imagine two partners in crime, Bonnie and Clyde, who have been picked up by the police and are being held in separate cells, unable to communicate with each other. The detectives realize that they currently have only enough evidence to convict them of a lesser charge, carrying a prison sentence of 1 year for each crook. To convict on the more serious charge of bank robbery, they need a confession from at least one of them. The detectives therefore offer each of them the same deal: If you confess, then you will go free, while your partner will take all the heat and serve 10 years in prison. If both partners confess, then they will share the 10-year sentence, each getting 5 years. The situation is shown in [figure 1.5](#) for Bonnie and Clyde, with the outcomes for Bonnie bolded.

		Clyde	
		Silent	Confess
Bonnie	Silent	1 year, 1 year	10 years, Free
	Confess	Free, 10 years	5 years, 5 years

Figure 1.5

The prisoner's dilemma.

Question: If Bonnie is rational and always prefers outcomes in which she serves less time, what should she do?

If Bonnie believes that Clyde will stay silent, then she will get 1 year in prison if she remains silent as well but would go free if she confesses. She thus reasons that she is better off confessing if he remains silent. If she believes that Clyde will confess, then she would definitely want to confess too—she'd have to serve the full sentence of 10 years if she remains silent but only 5 years if she confesses. Thus, regardless of what Clyde chooses, Bonnie realizes that she minimizes her own prison sentence by confessing. Clyde faces a symmetric situation and also realizes that he is always better off confessing. Thus, two rational, purely self-interested players in the prisoner's dilemma would rat each other out, arm the prosecution with sufficient evidence to convict both of them, and wind up serving 5 years each. This outcome seems irrational, since both clearly would have been better off if they had kept their mouths shut and wound up with the more lenient sentence of 1 year.

Thus, in the prisoner's dilemma, each individual rationally pursuing their own self-interest leads to an outcome that is collectively irrational: it is bad for everyone. This paradox is why the prisoner's dilemma has been so extensively studied by game theorists and has become so widely known that it has been absorbed into popular culture. Although the players can agree that they would be better off by cooperating with each other, the problem arises because the players do not act as a single collaborative unit that can collectively decide either to both remain silent or both confess. Instead, each individual has the ability to act independently in accordance with their own interests. Players can never control what anyone else does; they can only do what is best for themselves, knowing that the other person's choice is out of their hands. Even if the prisoners were to make a pact to cooperate with each other and stay silent before being arrested, once they are offered the deal by the detectives, they will find it in their own self-interest to defect on their agreement and confess.

Furthermore, prisoner's dilemma scenarios are ubiquitous. They occur when two players face a choice between a cooperative action (which we'll call "Cooperate") and an uncooperative action (which we'll call "Defect"), as shown in [figure 1.6](#). (Notice that "Cooperate" refers to generally cooperating with the other player, not cooperating with the prosecution by confessing in the prisoner's dilemma story.) Each player is better off under mutual cooperation than under mutual defection, yet both always have an

individual incentive to defect. This general framework can represent a wide variety of situations. For instance, consider the following examples:

		Player 2	
		Cooperate	Defect
Player 1	Cooperate	Fine, Fine	Worst, Best
	Defect	Best, Worst	Bad, Bad

Figure 1.6

A generic prisoner's dilemma.

- Two firms each decide whether to set a high or low price. They are better off if they both set a high price rather than both set a low price, but if one firm sets a high price, the other firm prefers to undercut with a low price, and if one firm sets a low price, the other firm prefers to counter by offering a low price too.
- Two countries each decide whether to reduce their carbon dioxide emissions. They are better off when everyone has low emissions than when everyone has high emissions. However, both prefer not to have to reduce emissions themselves.
- Two parties in a legal dispute each decide whether to hire lawyers. Both are better off if they can resolve the dispute without any lawyers than they would be if both hired lawyers. However, regardless of what the other party does, the individuals find it in their own interest to retain lawyers.
- Two teammates each decide how much effort to exert on a team project. Both are better off if each works hard and the project succeeds than if neither works hard and the project fails. However, if the other person does the work, then each person would prefer to slack off, and if the other person doesn't work, why should they?

Back in the 1950s, when the prisoner's dilemma was developed by RAND researchers, it was viewed through the lens of the nuclear dilemma: There were two superpowers, the US and the Soviet Union, each capable of annihilating the other. Both would be better off in a world in which nuclear weapons were not amassed, and potentially deployed, by both sides. But

regardless of whether the other country had nuclear weapons, each side saw the advantage of accumulating weapons themselves. In fact, some people in the West, including a prominent game theorist, John von Neumann, came to see nuclear war as inevitable and advocated for a preemptive nuclear strike against the Soviet Union, before the other side had the chance to strike first. While Flood and Dresher originally hoped that game theory would eventually be able to solve the prisoner's dilemma, the logic of how self-interested, rational players would approach this game is unimpeachable. If the prisoner's dilemma is to be "solved," it must be by altering the incentives for defection and/or by allowing players to make binding agreements. We'll return to the prisoner's dilemma in later chapters, when we talk about social dilemmas more generally (chapters 18 through 20) and also discuss the effects of repetition (chapters 9 and 10) and the ability to make binding agreements (chapter 16).

This Book

As you've been reading about these games, we have asked you to consider how you would act in these scenarios. Perhaps you've been thinking that the way you would act in these situations doesn't quite align with the game theoretic predictions. Perhaps you think that you would not betray your partner in crime if you believed that he or she were remaining silent to protect you. Perhaps you guessed a positive number in the beauty contest game because either you didn't reason your way down to 0 or you (correctly) assumed that others would not. Perhaps you said you would have bet if you held a red card. In any of these cases, you wouldn't be alone. Standard game theory predicts the outcomes of strategic interactions among rational individuals who have perfect foresight and unbounded cognitive capabilities, such that they are always able to perfectly anticipate the strategies of others; calculate what is best for themselves, even in situations involving risk; and unwaveringly implement this course of action without error. You may be wondering how accurate the predictions of game theory can actually be if no one you have ever known in your life, no matter how nerdy, has ever navigated social interactions in this way. Is it time to drop this course and attempt to resell this textbook to some other sucker?

Not so fast. Predictions based on rational choice theory are a very useful benchmark, but the field is also continually evolving based on behavioral research that provides insights into how people actually behave when confronted with the types of situations that game theory models. Laboratory experiments that test the predictions of game theory have become increasingly common and allow us to answer the question of just how accurately game theory predicts interactions among real people. In a typical game theory experiment, human participants (usually undergraduate students) make decisions in the role of one of the players of the game, knowing that they will be paid in cash according to the payoffs that they accumulate. In other words, their decisions in the experiment are not hypothetical; they are actually incentivized in the same way as the players whose behavior we are attempting to model. In some situations, the standard game theoretic predictions are remarkably close to actual behavior. In other cases, they diverge widely. However, experiments also tend to reveal robust regularities in people's behavior: to the extent that people deviate from the game theoretic predictions, they do so in systematic, predictable ways.

Rather than implying that we should toss out game theory entirely and give up on attempting to model strategic behavior, the experimental results point toward adapting standard game theory to incorporate this behavioral evidence. Real people make mistakes in assessing which courses of action maximize their payoffs. Real people don't have the cognitive capacity to engage in unlimited steps of reasoning about the reasoning of others. Real people fail to fully appreciate that other people have private information that they are not privy to and their actions may reveal something about this information. And real people are not purely self-interested: they are willing to make sacrifices to repay those who have been kind to them and punish those who have behaved unfairly. These are all features of human behavior that can be incorporated within the existing framework of game theory to generate more accurate predictions.

The goal of this book is to provide a formal introduction to standard game theory, while at the same time addressing how people actually behave in these games and explaining how the standard theory can be expanded or updated to better predict the behavior of real people. While behavioral research and models don't typically play a central role in game theory texts,

we believe that this is essential for anyone who wishes to fully understand strategic behavior and predict the outcomes of game theoretic interactions.

Most chapters will follow the same basic format. We will begin with an illustrative example that is representative of the types of interactions that will be considered in the chapter, and you will be asked to consider how you would behave in this situation. From there, the chapter will provide a traditional game theoretic analysis, which predicts how rational people “should” behave in these types of games. After this analysis, we will present behavioral research that either confirms or challenges this traditional treatment. The chapter will end with examples of how the relevant theory has been updated to better align with the results of the experiments we have discussed. There are two exceptions to this format: chapter 2, in which we develop the basic structure for modeling games, and chapter 4, in which we develop the concept of equilibrium generally before solving for equilibria in more specific scenarios—and discussing when these predictions are likely to hold—in later chapters. The goal is to demonstrate that game theory is as much a positive endeavor (describing how people actually behave) as a normative one (predicting how people “should” behave) and that the field is continually advancing through an iterative process in which theory informs experiments and experimental results inform updated theory.

This book consists of eight parts. The remainder of part I develops the basic structure of modeling strategic interactions as games of strategy. Part II analyzes how to solve relatively simple games, in which players each make a single decision simultaneously. Part III analyzes games in which players act sequentially or repeatedly (e.g., the prisoners facing the same dilemma over and over), while part IV analyzes games in which players do not have complete information about the game that they are playing (e.g., Vizzini in the Princess Bride game not knowing that Westley is immune to iocane powder). We believe that the topics in parts I through IV comprise the nuts and bolts of any game theory course. Parts V through VII apply the tools developed in the first half of the book to specific applications and extensions, by presenting detailed treatments of three particularly prominent topics to which game theory has been applied: social dilemmas, such as the prisoner’s dilemma, in which what is best for the individual is harmful to the group as a whole; bargaining games and cooperative game theory; and

voting and political games. Each of the chapters in parts V through VII are written to stand alone and can be covered based on the interests of students and the focus of the course. Finally, part VIII provides a more in-depth discussion of the various behavioral game theory concepts that have been introduced in order to better describe how people play games, which are generally neglected in an introductory game theory course.

¹Stated in words, a is at least as preferred as b , b is at least as preferred as a , or a and b are equally preferred.

²Stated in words, if a is at least as preferred as b and b is at least as preferred as c , then a is at least as preferred as c . Transitivity allows a player to compare disparate outcomes by creating logical “chains” like this.

³For now, all that is important is that higher payoffs are assigned to more preferred outcomes. The absolute magnitudes of the payoffs do not matter, only the relative ordering. Later, we will assume that payoffs are numerically meaningful, as discussed in chapter 5.

2

Game Representations

The purpose of game theory is to make predictions about how people will respond in different strategic environments. Before making predictions, however, we must first settle on a framework to describe and represent the important elements of the situations we wish to understand. In this chapter, we will develop the basic structure for modeling games by defining the rules of the game, discussing several background assumptions, and introducing two common ways to depict games.

The philosopher Thomas Hobbes anticipated some of the central findings of game theory centuries before they were formalized. In what he calls the “natural condition,” for example, pairs of interacting people face what appears to be a simple choice: *cooperate* or act like a *knave*, and these choices largely determine whether life is “nasty, brutish, and short” (Hobbes [1668] 1994). Cooperating benefits both members of the pair, while being a knave is only good for the perpetrator. Although this description makes it seem like cooperators must be good, moral people while the parents of knaves must have failed to bring them up right, a full accounting of the incentives faced by the pair has revealed some of the nuances of a just society. How can we represent this choice as a game?

In this chapter, we explain the conventions that have evolved to formalize a strategic interaction as a game. We start by explaining the four basic elements of any game using Hobbes’s natural condition as an example. We then discuss what we usually assume about how players make choices and two ways to visually represent this information and structure.

The Elements of a Game

There are four foundational elements of any game: the *players* who interact, the *actions* that these players can choose among, the *payoffs* accruing to the players based on the actions taken, and the *information* that the players have

about the characteristics and actions taken by their counterparts. Each element is discussed next.

Players. We assume that there is a finite set of decision makers, or *players*. The number of players in a game can vary from just two to any number imaginable (say n). When there is just one player, the person is often playing against chance. This is the traditional purview of a large portion of microeconomic or decision theory. Game theory, in contrast, involves two or more decision makers who interact strategically, as discussed in chapter 1. In general, we will label players with indices from 1 to n and will refer to arbitrary players in the game by player i or player j . We will write the set of players as $N = \{1, \dots, n\}$. When there is a large number of players, predictions can become difficult unless those players are identical (i.e., they all have the same incentives and face a symmetric choice). In such a case, we can often focus on the symmetric solution to the game. After all, if players have identical incentives, why would they choose to pursue different actions? A good place to start, however, are situations in which there are just two players. For now, let's focus on the natural condition game with two players, called Hera and her brother Zeus.

Actions. Whenever players must make a choice, they weigh the consequences of taking different *actions*. Each player has a nonempty set of actions from which they can choose. The available actions can range from a finite set containing just a few elements to a continuum. For example, in the natural condition, Hera might have the binary choice between cooperating and being a knave, or her choice might be to pick the degree to which she wishes to be a knave (say on a continuous scale from 0 to 100). Whenever called upon to make a choice, both Hera and Zeus must each choose among a set of available actions, and the actions available may differ for each player.

Payoffs. The players' actions jointly determine their outcomes and, as described in the previous chapter, each outcome is associated with a *payoff* for each player, such that higher payoffs are associated with more preferred outcomes. Once all the choices have been made and the actions have been taken, players receive these payoffs. The payoffs may be equivalent to extrinsic rewards, such as amounts of money or other commodities, or they might reflect intrinsic psychological rewards like the warm glow of altruism or even the satisfaction that some might feel from revenge. A player's

payoffs thus reflect all his motives and concerns, even less tangible considerations, such as psychological costs and benefits. It is important to highlight that what differentiates game theory from other areas of decision or microeconomic theory is that in a game, the payoffs of players are linked—what Hera does in the natural condition affects the payoff received by Zeus and Zeus's choices have implications for the payoffs of Hera as well. This distinguishing feature of game theory is known as *strategic interdependence*. As discussed by Hobbes, in the natural condition, cooperating can benefit both Hera and Zeus, while acting like a knave also has implications for the payoffs of both players. For example, if Zeus acts like a knave, he will increase his payoff but reduce the payoff of his sister. Hence, there is strategic interdependence in the natural condition game.

Information. The final main element of any game is the *information* possessed by players as they make their choices. To make choices, players use information about the actions taken by the other players and the incentives of those players. There are two very broad classes of information that we'll think about in parts II and III of this book: perfect information and imperfect information. A game has *perfect information* if players move sequentially and, every time a player is called on to act, she knows everything that has happened up to that point. Some examples of well-known perfect information games are checkers, chess, and tic-tac-toe. In these games, all the moves unfold in front of the players on the board, so as long as they are awake and paying attention, everyone always knows the full history of moves. In contrast, a game has *imperfect information* when players are unsure about what the other players have done when making their own choices. Examples include the Princess Bride game between Westley and Vizzini (Vizzini doesn't know which glass Westley poisoned at the time he chooses which one to drink from), Battleship (you don't know where your opponent has placed her boats), and games where players move simultaneously or with no input from each other (such as the prisoner's dilemma). If the players in the natural condition game take turns, for example, Zeus knows whether Hera has chosen to cooperate (or be a knave) when he has to make his own choice. This would be a game of perfect information. However, if Hera and Zeus have to make their choices simultaneously, they have imperfect information because they are not informed about what the other has done when they make their choices. In

part IV, we will also consider games of *incomplete information*, in which one or more players are missing some critical piece of information about the game that they are playing, such as the other player's incentives or abilities. However, we will postpone the discussion of incomplete information until chapter 11 and focus on the perfect versus imperfect classification for now.

Background Assumptions

To accompany the game elements just described, game theorists typically invoke three assumptions about the motivations and cognitive ability of the players. We assume these things so that players are purposeful—they understand what they want and can figure out how to achieve it. That said, although we will invoke these assumptions in most instances, we will also explore what happens when we relax them in some of the final sections of each chapter, when we discuss behavioral results and alternative models, and then look at them in more detail in part VIII. These assumptions are explored next.

Assumption 1 (Players are rational) As discussed in the previous chapter, we use the framework of rational choice theory. We assume that players understand their own preferences or interests, which are captured by their payoffs, and they act to maximize their own payoffs. It is important to note, however, that rationality is not synonymous with being egoistic, or only caring about one's own outcomes. The altruistic preferences of Mother Teresa can be every bit as rational as the self-interested preferences demonstrated by Niccoló Machiavelli. In other words, rationality doesn't mean that a player must be self-interested in a narrow sense, in that she only cares about her own material outcomes. Instead, it means that the player's goal is to maximize her payoffs, which reflect everything that she cares about: all her concerns are already baked into these numbers. In short, rationality is not about what you prefer; it is merely about you logically following your preferences when you make choices.

Assumption 2 (The rules of the game and the rationality of the players are common knowledge) We assume that all the players understand and know that all the other players are rational. Further, we also assume that all the players know that all the players know that everyone is rational, that all

the players know that all the players know that all the players know that everyone is rational, and so on ad infinitum. This same logic applies to the rules of the game (e.g., the players' available actions and payoffs): each person knows the rules, knows that the others know the rules, knows that the others know that all the others know the rules, and so on. This is important because there can be instances in which information is available, but one player is not sure that the other observed or understood it, which affects their ability to reason about the reasoning of others. Common knowledge assumes that this problem doesn't happen.

What is the difference between everyone privately knowing a fact and that same fact being common knowledge? To see this, consider the following story, which goes back to at least colonial days and involved a group of cannibals, their unfaithful wives, and a missionary. However, we'll consider a more family-friendly version: Suppose that three children at a birthday party are wearing red hats. The children know that the birthday hats are either red or white and can observe the color of the hats worn by the other children, but not the color of their own hats. Imagine that we went around and asked each kid sequentially if they knew the color of their own hat. Of course, each would say no.

Now imagine that some blabbermouth at the party offhandedly announces that at least one child is wearing a red hat. This is not Earth-shattering information for any of the children—in fact, each can already clearly observe that the two others are wearing red. But now this piece of information is common knowledge, rather than just knowledge that was privately held by each person. And now what happens when we go around and ask the kids sequentially about the color of their hats? The first kid still doesn't know. Neither does the second kid. But the third kid now can definitively say, "Yes, I'm wearing a red hat."

How can this be? Let's look at the first kid's response. The blabbermouth had said that at least one person was wearing red, so if the first kid looked at the other two and saw them both wearing white, then she would be able to conclude that she must be the one wearing red. So, if the other two were wearing white, the first child would be able to answer the question. The fact that she couldn't answer tells the other kids that at least one of them is wearing a red hat. Again, this does not come as a surprise to the two remaining kids: they can each plainly see that the other kid is wearing red,

so they each knew previously that at least one of them was wearing red; but now it is common knowledge. Now let's move on to the second child's answer. If he were to look at the third child and see a white hat, then he would definitively conclude that he himself must be wearing red. The fact that he couldn't answer tells the third kid, "You are not wearing a white hat," and so she is now able to answer that she is wearing red. This conclusion was not possible when the fact "at least one child is wearing red" was privately held information; only when this fact became common knowledge could the children reason through this logic. Thus, the common knowledge distinction can be critical for solving for rational play in strategic interactions, which depends not just on what each person knows, but what they know the others to know, and what they know the others know about what they know, and so on.

Assumption 3 (Players have perfect recall) Players do not forget information that they may have acquired either before or during the course of play. If, for example, players make multiple choices in a game, we assume that they can remember what they did in the past, as well as what all the other players did.

Now that we have discussed the basic elements of a game of strategy, we can move on and describe how the game will be represented visually. We begin with a representation known as the *extensive form*, which includes all the elements mentioned here, including the structure of play (i.e., who moves when). We then discuss the *normal form*, which we can think of as a summary of any game—one that focuses on the strategies available to each player and the payoffs that result from the intersection of any set of strategies chosen.

Extensive Form Games

The extensive form is way of depicting a game using a directed graph called a *game tree*. The tree consists of a series of nodes and branches that connect the nodes and indicate how the players can progress through the game. An example is shown in [figure 2.1](#). Each white *decision node* is associated with a player, whose turn it is to act at that point, and that person's initial or number is printed on the node. The branches emerging from the white node show the actions available to the player at that point of decision. Each black

terminal node is associated with a final outcome of the game, which is uniquely determined by some path of actions through the tree. We will typically list the payoff of the first person to move in the game, then the second player, and so on, and clarify by using a legend whenever there is ambiguity.

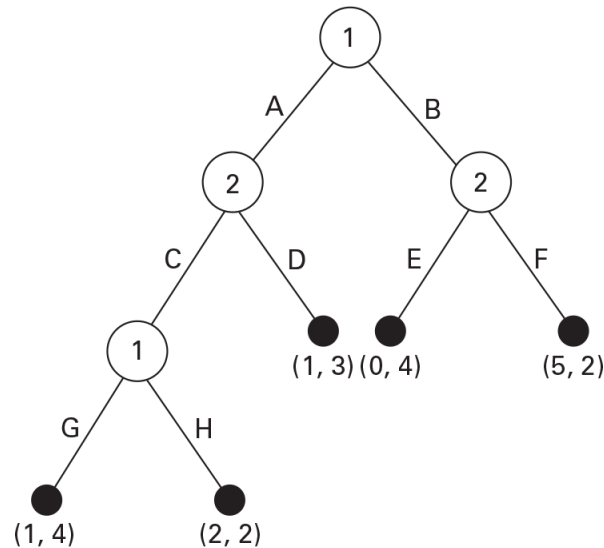


Figure 2.1

A game in extensive form.

Returning to the natural condition game, let's continue to assume that Hera chooses before Zeus and each player has two available actions: Cooperate and Knave. This extensive form representation of the natural condition game is shown in [figure 2.2\(a\)](#). The game starts with Hera's decision node, labeled H, and from this *initial* (or root) node, two branches extend out, representing the actions available to her, Cooperate or act as a Knave. Notice there are two decision nodes for Zeus, labeled Z, because he witnesses what Hera chooses before he has to make his own choice. In other words, he can condition his choice on what Hera chose. He can, for example, choose to Cooperate if Hera did and act as a Knave if she did. In total, there are four action plans that Zeus can follow. He can reciprocate—Cooperate if Hera did and be a Knave if she was. He can cooperate unconditionally—Cooperate after Hera does and Cooperate after she acts as a Knave. He can also be a knave unconditionally—choose Knave regardless of what Hera does. Or, lastly, Zeus can choose to do the opposite—pick

Knave after Hera chooses Cooperate and pick Cooperate after Hera chooses Knave.

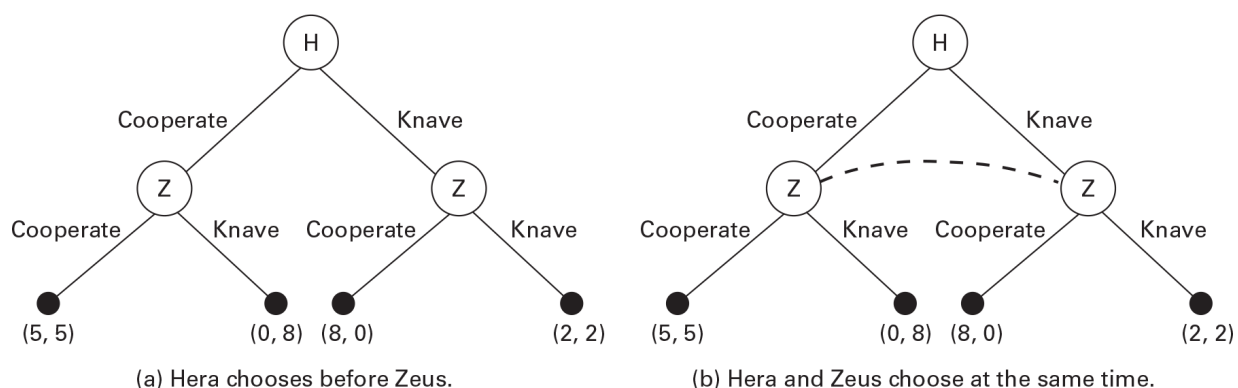


Figure 2.2

The natural condition game played (a) sequentially and (b) simultaneously.

What about the payoffs? We'll assume that each player most prefers the outcome in which she acts as a Knave at the cooperator's expense, then the outcome in which both Cooperate, and her least preferred outcome is when she Cooperates with a Knave. Staying on the left side of figure 2.2(a), if Hera Cooperates and Zeus does too, the game is over, and at this *terminal* node, the pair receive payoffs of, say, 5 each. Since Hera moves first, her payoff is listed first. If Zeus is instead a Knave after Hera Cooperates, then his payoff goes up to 8 and hers drops to 0 (and vice versa if she is Knavish while he Cooperates). Finally, if both siblings are Knaves, then each gets a payoff of 2. These payoffs reflect our initial description of the game, in which Cooperation benefits both players while being a Knave is good only for the individual. Given the players' preference orderings over the outcomes, we have a prisoner's dilemma, played sequentially.

Now suppose that the game is no longer played sequentially. Instead, Hera and Zeus must decide how to play simultaneously (as in the familiar game rock-paper-scissors, for example). It is clear that something fundamental has changed—Zeus can no longer condition his choice on what he has observed Hera to have done—but how can we account for this change to the structure of play in the extensive form? From the point of view of Zeus, what has changed is that in the original, sequential game, Zeus knew if he was at the left decision node or the right one in figure 2.2(a) when he was making a choice, but now he does not know which of

the two nodes he is at because he can't observe what his sister has done. In other words, this is now a game of imperfect information, as described previously.

To represent this information imperfection, game theorists use information sets. An *information set* is a collection of all the possible decision nodes that a player cannot distinguish among when called on to act. Based on what has already happened in the game and the information the player possesses, the player may not know for sure which of two or more nodes she is actually at. We indicate this by saying that the nodes are in the same information set. Because the game in [figure 2.2\(b\)](#) is not played sequentially, Zeus does not know if he is at the decision node on the left or the one on the right and, as a result, these two nodes are in the same information set. In [figure 2.2\(b\)](#), the information set is represented by a dashed line connecting these two decision nodes. In general, a dashed line or oval will connect any decision nodes that are part of the same information set.

What is important is that Zeus cannot condition his behavior based on whether he is at the left or right node, since he does not actually know which one he is at. Instead, he can choose only one action when he arrives at this information set, and this action will apply regardless of which node within the information set he is actually at. Thus, *each information set represents a single point of decision for a player*. Note that every decision node is always part of an information set. If the players always have perfect information, then each node will be the only element in the information set, indicating that the players have no confusion about where in the game tree they are. Hera has only one decision node, and so she has just a single information set containing this one node. Visually, if a node is the only element in the information set, then it will not be connected to any other with a dashed line.

Consider another example. The game rock-paper-scissors (aka Rochambeau) has evolved to be a familiar way to settle minor disputes and, more formally, is a two-person symmetric game of imperfect information in which participants choose among three actions: Rock, Paper, and Scissors. The rules of the game are as follows. The two players choose simultaneously and, because “paper covers rock,” a person playing Paper will beat a person playing Rock. Because “scissors cut paper,” Scissors

defeats Paper; and because “rock smashes scissors,” Rock beats Scissors. Hence, participants either win, lose, or the game is a draw. Let’s assume that the payoffs of these three outcomes are W , L , and D , respectively, where $W > D > L$. What does the extensive form of this game look like?

This game is depicted in extensive form in [figure 2.3](#). Each player (we call them “1” and “2” for simplicity) chooses either Rock, Paper, or Scissors without knowing what the other person has chosen. As a result, player 2’s information set includes all three of her decision nodes. The game is interesting because it is hard to know what to do since at any given node, there is the possibility of winning, losing, or drawing. As a matter of fact, the game would be very uninteresting if each of player 2’s decision nodes were in different information sets rather than all in the same information set. This would mean that player 1 would move first and player 2 would always observe what player 1 has done before making her own choice—in other words, player 2 could always assure herself of a win.

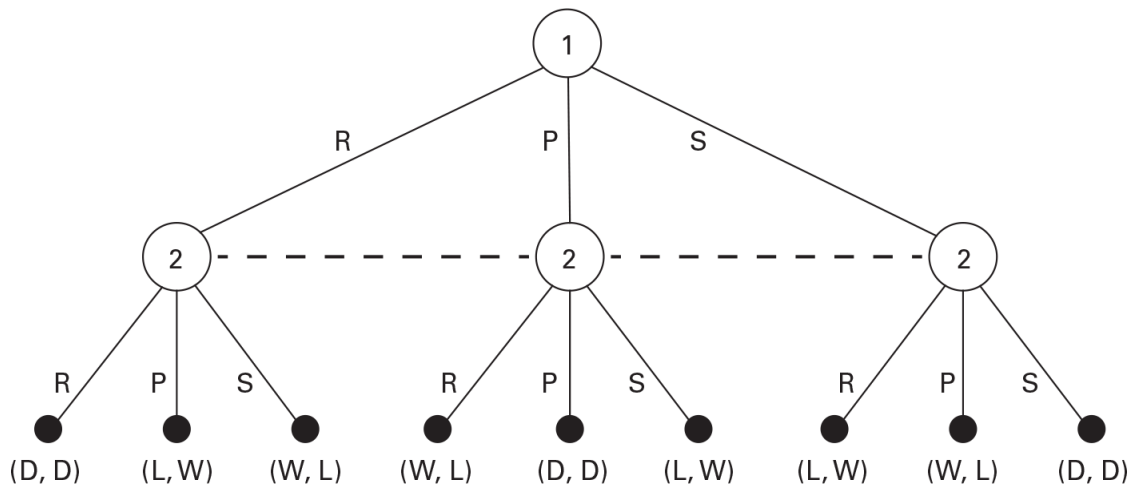


Figure 2.3

The extensive form of rock-paper-scissors. Players choose from three actions: (R)ock, (P)aper, and (S)cissors.

Now that we have the tools to represent a game, let’s circle back to the Princess Bride game discussed in chapter 1 and see what it looks like in extensive form. In the scene on which we focus, the hero, Westley, is matching wits against the villain, Vizzini. There are two goblets of wine, which Westley hides by putting his body between them and Vizzini. He then places poison in just one goblet, returns them, and instructs Vizzini to pick

the one from which he will drink. If he picks correctly, Vizzini lives and Westley dies. If he picks incorrectly, Vizzini dies. As we now know, the problem for Vizzini is that he has imperfect information—if he had seen which goblet was poisoned, he could live for sure. (We are playing a simplified version of the game in which Westley has not developed the immunity to iocaine that saves his life in the actual story.) Though a bit macabre, suppose the payoff to living can be normalized to 1 and the payoff to dying normalized to -1 , then the Princess Bride game can be represented as in [figure 2.4](#).

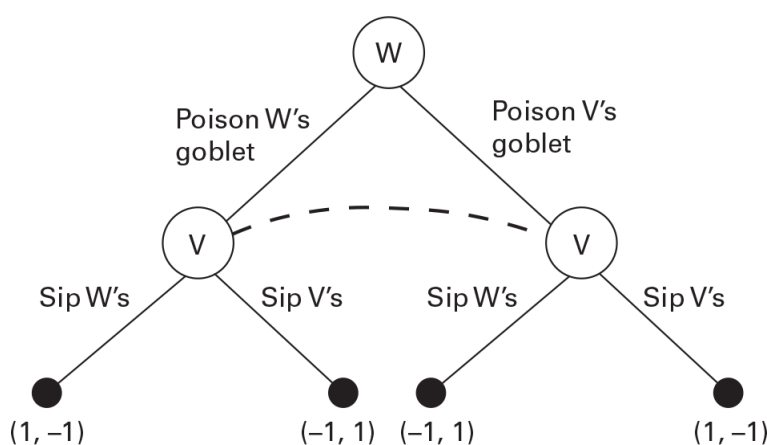


Figure 2.4

The extensive form of the Princess Bride game.

Normal Form Games

Contrary to the extensive form, which reveals the entire structure of a game, the actions, and the information available, the *normal form* of a game can be thought of as a summary of a game—one that focuses on just two aspects: the full strategies available to each of the players and the payoffs that result from the intersection of any set of strategies. We have already discussed payoffs, but we have yet to define exactly what we mean by a strategy.

Definition 1 (Strategy) *A strategy s_i for player i is a complete action plan for player i , stating which action is to be taken at every one of i 's decision nodes or information sets.*

In other words, players' strategies do not allow them to “wing it” and choose actions when and if they arrive at nodes. Instead, they are required

to state, for each possible information set, regardless of how likely the player thinks it is that she will arrive at that information set, what action she will take when she gets there. Another helpful way to think about a strategy is that it is a set of instructions that the player could leave for her agent to carry out in her absence. This means that a strategy is a collection of contingencies (if A happens, choose x; if B happens, choose y, etc.), and, of course, these instructions have to be complete and state for every possible contingency (i.e., information set) what action should be chosen.

It will be important to draw a distinction between two types of strategies: pure strategies and mixed strategies. A *pure strategy* is a plan that doesn't involve chance: for each information set, it specifies which action the player will take, with certainty, at that point of decision. In the sequential natural condition game, an example of a pure strategy for Zeus is to always Cooperate if Hera Cooperated and always act as a Knave if Hera was a Knave. In contrast, a *mixed strategy* involves randomizing according to a specific probability distribution. In the next two chapters, we will focus on predicting behavior when players can choose among their pure strategies. Beginning in chapter 5, we allow players to play mixed strategies by rolling a multisided die to choose among their possible pure strategies. For now, however, it is just important to be aware that this distinction exists.

Definition 2 (Strategy set) *Player i's strategy set lists all the available pure strategies for player i and is denoted as S_i .*

Notice that the strategy set can be finite, if the player chooses among several discrete actions at each of her information sets, or infinite, if the player can choose any real number within some interval for instance. We have a *finite game* if each player's strategy set is finite.

Given our definition of a strategy, the normal form of a two-person finite game can be represented as a matrix. Each row or column represents a possible strategy for one of the players. We will assign one player to be the "Row player," whose strategies are listed in the rows, and the other to be the "Column player," whose strategies are listed as columns. The payoffs determine the cell entries and are written as (Row player's payoff, Column player's payoff). If you have trouble remembering whose payoff is first, just think about reading the matrix from left to right. As your eyes move across the matrix, you would encounter the Row player's name and

strategies and then the Column player's name and strategies; so we write the Row player's payoff first.

If we return to the natural condition game, for example, what does the sequential game depicted in [figure 2.2\(a\)](#) look like in normal form? The first thing to determine are the dimensions of the matrix that represent this encounter. The number of strategies for Hera is straightforward because she faces just one decision node, and therefore her set of strategies overlaps exactly with the set of actions at this node. For Hera, the set of strategies is just $S_H = \{Cooperate, Knave\}$. Hence, there will be two rows in the normal form. The number of columns, which is determined by the number of strategies for Zeus, is a bit more complicated, however. Zeus has two information sets, and therefore his strategy must specify two actions. As described previously, one strategy available to Zeus is to Cooperate at both possible nodes (i.e., after Hera Cooperates and after she is a Knave). Instead, Zeus could be a Knave at both nodes. Zeus can also just do what Hera did (Cooperate if she did and be a Knave if she was). That makes three strategies. There is just one left—do the opposite of what Hera did. All told, there are four possible strategies for Zeus to follow: two actions at two possible nodes, or four permutations. If we abbreviate the actions, these strategies can be written as $S_Z = \{(C, C), (C, K), (K, C), (K, K)\}$, where the first letter dictates the action to be taken at the left node in the extensive form (i.e., following Cooperation) and the second letter dictates the action to be taken at the right node (following Hera being a Knave).

To find the payoffs in any given cell, we assume that Hera plays the strategy listed in the associated row and Zeus plays the strategy in the associated column, and then we trace the path through the tree that would result if they played according to those plans. Putting this all together results in the normal form of the natural condition game depicted in [figure 2.5](#). If, instead, the game is played simultaneously, notice that the sets of actions and strategies will overlap for both players. Because Zeus now does not know what Hera has done, he cannot condition his strategy on her action, and the game collapses to the two-by-two game shown in normal form in [figure 2.6](#):

		Zeus			
		(C,C)	(C,K)	(K,C)	(K,K)
Hera	C	(5, 5)	(5, 5)	(0, 8)	(0, 8)
	K	(8, 0)	(2, 2)	(8, 0)	(2, 2)

Figure 2.5

The natural condition game played sequentially.

		Zeus	
		C	K
Hera	C	(5, 5)	(0, 8)
	K	(8, 0)	(2, 2)

Figure 2.6

The natural condition game played simultaneously.

Can you anticipate what the normal form for rock-paper-scissors will look like? To figure this out, begin by noticing the structure of play and the information available to the players when making choices. This, in general, will determine the sets of strategies available for each player. What is also important for the normal form is that rock-paper-scissors is a symmetric game. We have used this term before, but let us now be more precise.

Definition 3 (Symmetric game) *A normal form game is symmetric when the players have identical strategy sets, earn the same payoff when they choose the same strategy, and achieve payoffs that are mirror images when they choose different strategies.*

Combined, the fact that the game of rock-paper-scissors is played simultaneously and is symmetric implies that the normal form matrix is three-by-three and the strategy sets for the two players are identical. Neither player 1 nor player 2 knows what the other player has done when choosing either Rock, Paper, or Scissors, so the shared strategy set is simply $S_i = \{Rock, Paper, Scissors\}$ for $i = 1, 2$. Further, along the main diagonal of the matrix, the players receive the same payoff because they pick the same strategy and the game is a draw, while off-diagonal, the payoffs are just mirror images because all that is happening is that the players are switching

roles in this symmetric encounter. The normal form is presented in [figure 2.7](#).

		Player 1		
		Rock	Paper	Scissors
Player 2	Rock	(D, D)	(L, W)	(W, L)
	Paper	(W, L)	(D, D)	(L, W)
	Scissors	(L, W)	(W, L)	(D, D)

Figure 2.7

Rock, paper, scissors played simultaneously.

As a final example of constructing the normal form of a game, let's consider the Princess Bride game again. Given what we just learned from rock, paper, scissors, the normal form of the simple Princess Bride game depicted in [figure 2.4](#) should be just a bit more complicated because, while the players pick strategies simultaneously, and therefore the matrix is just two-by-two, the game is not symmetric—the players do not achieve the same payoff when they choose the same strategy (i.e., the same goblet). Although one player is choosing which goblet to poison and the other is choosing which goblet to drink, we can think of the choice as a goblet, regardless of what they do with the chosen goblet. This means that the strategy sets are identical $S_i = \{Westley's, Vizzini's\}$ for $i = W, V$, even if the game is not symmetric. The normal form is shown in [figure 2.8](#).

		Vizzini	
		Westley's	Vizzini's
Westley	Westley's	(1, -1)	(-1, 1)
	Vizzini's	(-1, 1)	(1, -1)

Figure 2.8

The Princess Bride game.

As a last thought on game representations, consider the general relationship between the extensive and the normal forms of a game. Because the extensive form reveals more of the particulars of the

interaction, when we start with a game tree and endeavor to write the normal form, the resulting matrix will be unique. That is, there is only one normal form representation of any given extensive form. However, if you would like to go in the other direction and start with a normal form, in order to recover the extensive form, you are a bit stuck because the normal form is only a summary of the interaction, and therefore some information about the structure of play may be missing. In other words, given any normal form, one could potentially write multiple extensive form game trees that are consistent.

Perhaps an analogy will make the point. Recall that starting with a function, taking the derivative creates a unique result. However, if one starts with a derivative and tries to integrate to get back to the original function, one always needs to add an arbitrary constant. Going from extensive form to normal form is like taking a derivative, while going from normal form to extensive form is like integrating.

Figure 2.9 illustrates the relationship between game forms. If one were asked to write a normal form summary of either the game on the left of the figure or the game on the right, in both cases one would end up with the two-by-two matrix at the bottom. There would be no disagreement about this. If instead, one were asked to take the normal form at the bottom of the figure and write an extensive form game, people could disagree. Both the extensive form on the left and the one on the right are consistent with this normal form. The lost information is the structure of play in this instance. Is it the case that player 1 moves before player 2, as on the left, or do the players move at the same time, as on the right?

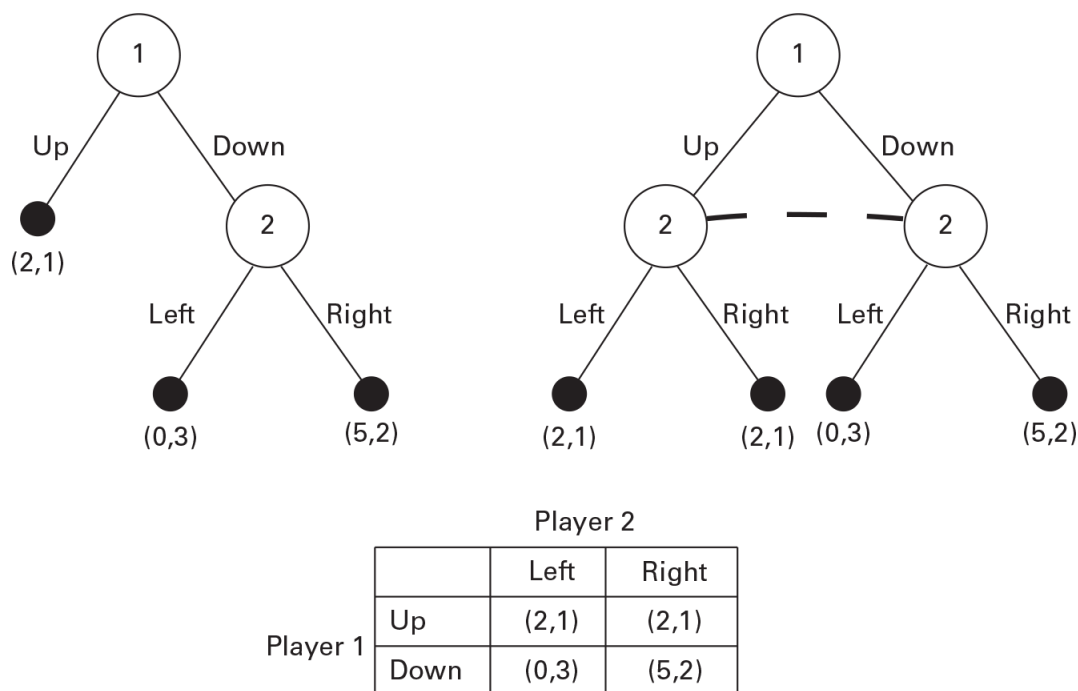


Figure 2.9

Two extensive forms consistent with the same normal form.

Worked Example

Let's return to the game shown in extensive form in [figure 2.1](#). Since no nodes are connected by dashed lines, each node is in its own information set, so we have a game of perfect information. Notice that each player has exactly two points of decision (i.e., two information sets) in the game and so each of their strategies must specify an action at each decision point. Player 1's strategy set is given by $S_1 = \{(A, G), (A, H), (B, G), (B, H)\}$, and player 2's strategy set is $S_2 = \{(C, E), (C, F), (D, E), (D, F)\}$. Player 2's strategy set is fairly intuitive: it specifies both what player 2 should do if she finds herself at the left node (as a result of player 1 choosing A) and at the right node (as a result of player 1 choosing B). However, it may seem unintuitive that player 1 has distinct strategies of (B, G) and (B, H) . After all, if player 1 chooses B at his first node, then he won't be called on to make another decision and will never actually have to choose between G and H. However, remember the definition of a strategy: it must specify the player's action at *every* one of his information sets. This is true even in examples like this, when the player's own action earlier in the game would preclude that information set from ever actually being reached. The reason

why this is necessary will become more clear when we start to solve for rational play, since player 1's optimal choice of A or B will depend on what he anticipates player 2 will choose, and player 2's optimal choice between C and D in turn depends on whether she anticipates that player 1 would choose G or H. To determine whether players are rationally anticipating and responding to each other's strategies, it will be important to have a full specification of the players' plans at every point in the game. Now that we know each player's strategy set, we can let these elements comprise the rows (for player 1) and columns (for player 2) and write the game in normal form.

		Player 2			
		(C, E)	(C, F)	(D, E)	(D, F)
Player 1	(A, G)	(1, 4)	(1, 4)	(1, 3)	(1, 3)
	(A, H)	(2, 2)	(2, 2)	(1, 3)	(1, 3)
	(B, G)	(0, 4)	(5, 2)	(0, 4)	(5, 2)
	(B, H)	(0, 4)	(5, 2)	(0, 4)	(5, 2)

Figure 2.10
Game from [figure 2.1](#) in normal form.

Strategy Profiles and Payoffs

In chapter 3, we will move from considering what players *can* do in games to predicting what they *will* do. Before we move on, we just need a little bit more notation. Thus far, we have let S_i denote the set of all of player i 's pure strategies, and let s_i denote a generic strategy for player i . However, we're not just interested in the individual strategies of isolated players. We want to predict the outcome of an interaction among strategic players whose actions affect each other. A *strategy profile*, s , tells us what each player's strategy is. It is an n -dimensional vector, where the first entry lists player 1's strategy, the second lists player 2's strategy, and so on, up to player n 's strategy.

Definition 4 (Strategy profile) *Consider a game with a set of players $N = \{1, \dots, n\}$. A strategy profile is an n -dimensional vector $s = (s_1, \dots, s_n)$ listing a strategy for each player.*

The strategy profile determines the ultimate outcome of the game, and thus each player's payoff. We will write the payoff for player i as $u_i(s)$. Notice that this function captures the notion of strategic interdependence in games: each player's payoff is a function of what each player is doing. While player i 's goal is to maximize $u_i(s)$, the only variable that is actually under i 's control is s_i . While he may predict what the other players are doing, from his perspective the other players' strategies are constants that he cannot control. Because of this, it is sometimes useful to separate the strategy profile into player i 's strategy and everyone else's strategies. To do so, let $-i$ refer to all players other than i . Then let s_{-i} refer to the strategy profile removing the entry of player i 's strategy. That is, s_{-i} is an $n - 1$ dimensional vector that contains all the players' strategies except for i 's. Given this, we can write the payoff of player i as $u_i(s_i, s_{-i})$. Stated in words, player i 's payoff is a function of the strategy he can control (s_i) and those he cannot (s_{-i}).

Now that we have established a language for describing games, in part II we begin a discussion of how rational players should behave in games. We start (in chapter 3) with situations in which players have strategies that always yield a lower payoff, regardless of what the other players are doing. We next expand our analysis to apply to any finite two-player game (in chapters 4 and 5), generalize to games with more players or continuous strategy sets (chapter 6), and specifically consider games with multiple predictions (chapter 7). Along the way, we will also consider the extent to which experimental participants follow these predictions.

Exercises

- (1) Consider the three games presented in [figures 2.11](#), [2.12](#), and [2.13](#). Which ones depict a game of perfect information?

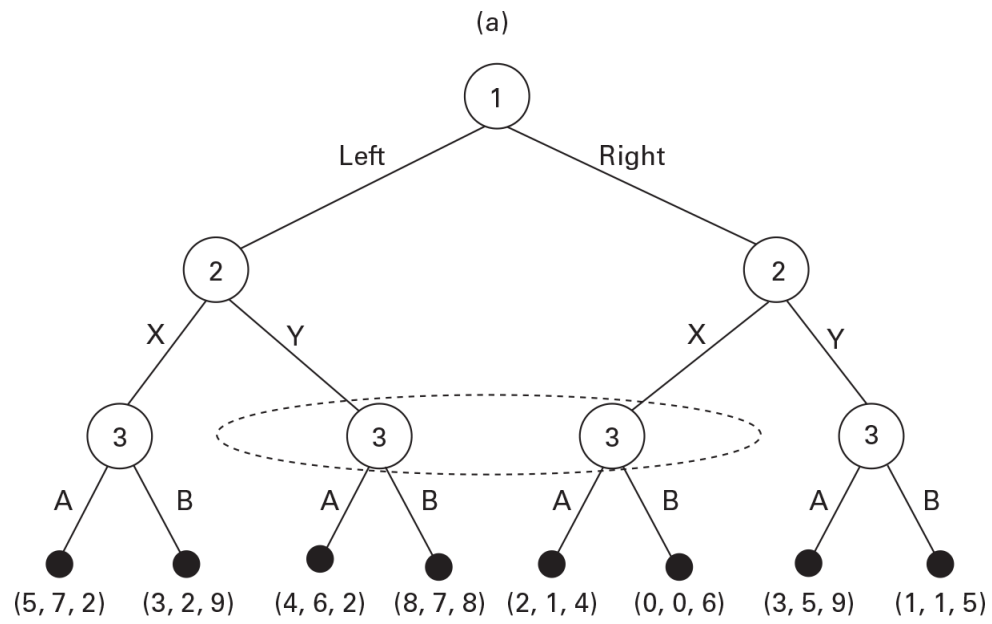


Figure 2.11

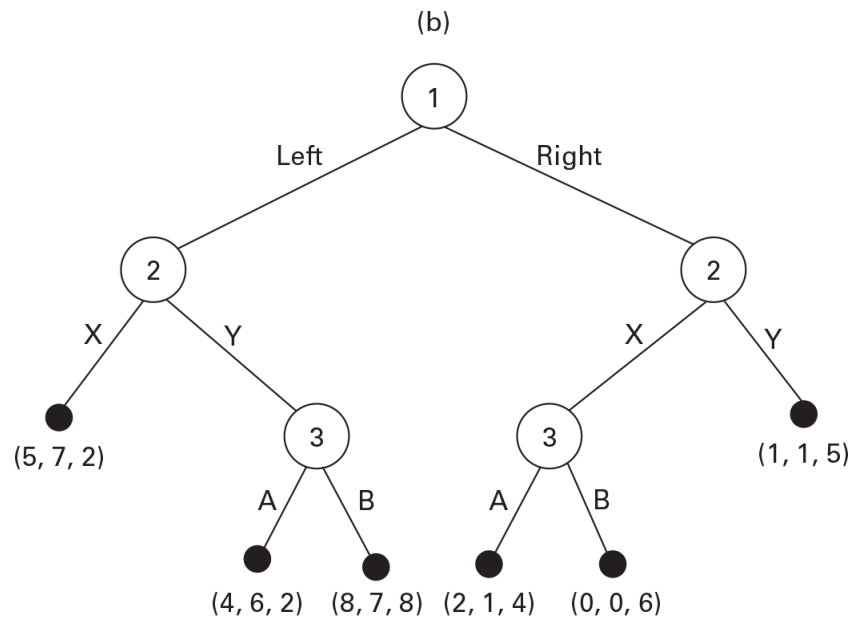


Figure 2.12

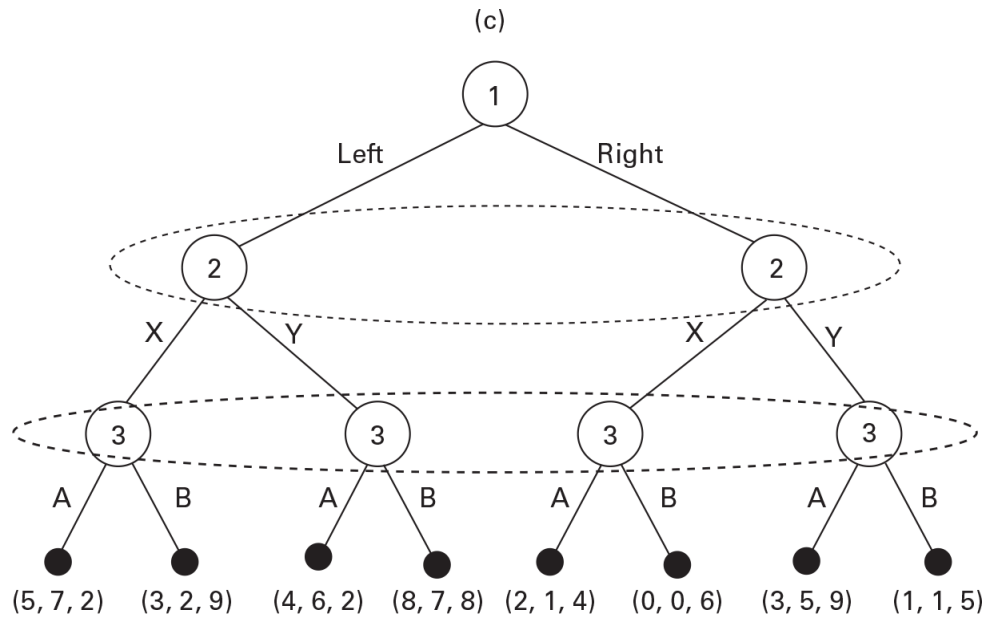


Figure 2.13

(2) Consider the three games presented in figures 2.14, 2.15, and 2.16. Which, if any, has an information set that is drawn in a way that is problematic?

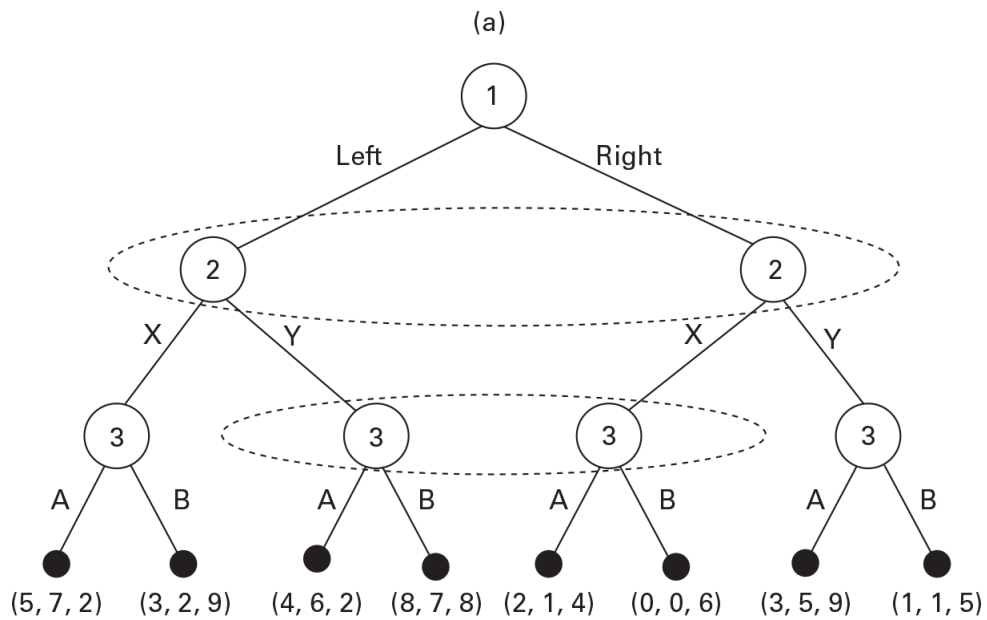


Figure 2.14

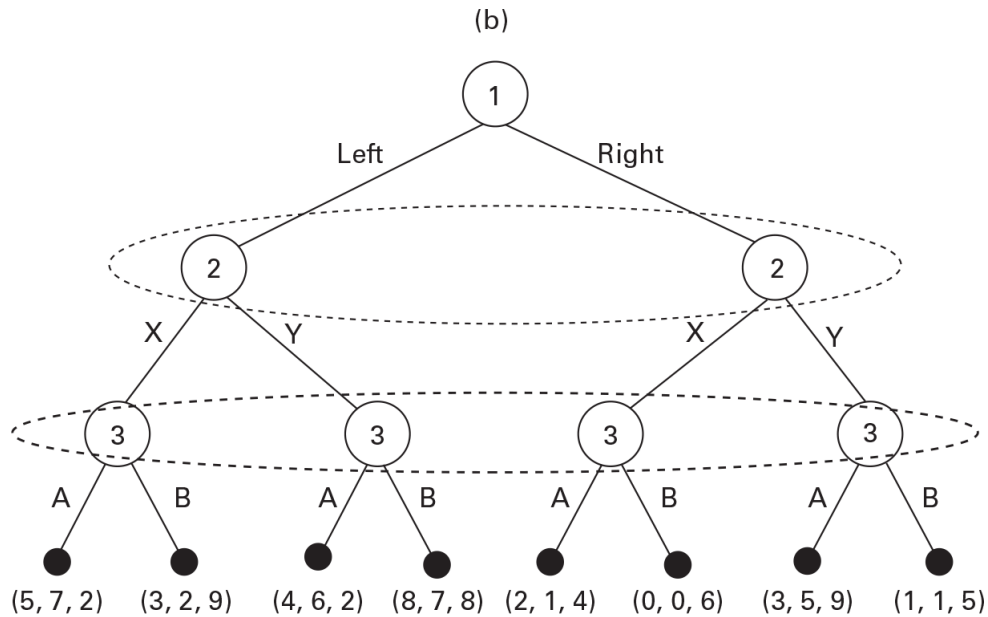


Figure 2.15

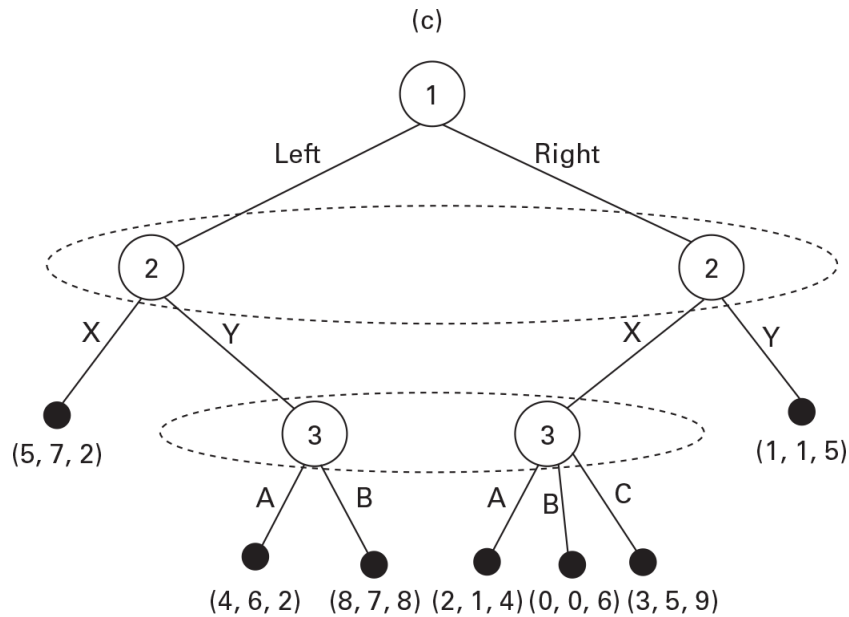


Figure 2.16

(3) Consider again the two-player betting game described in chapter 1. There is a deck of three cards: two red and one black. You and your opponent will each draw a card and look at it privately. Then you will simultaneously decide whether to bet or pass. At the time that you decide whether to bet, you thus know that you are at one of three nodes: (a) you

both have red cards, (b) you have a red card and the other player has a black card, (c) you have a black card and the other player has a red card.

Which nodes, if any, are in the same information set?

(4) Suppose that there are two firms, one of which (say, firm E) is considering entering the market already occupied by an incumbent second firm (call it firm I). Firm E can either *stay out* and earn a payoff (in terms of millions in profit) of 1 or *enter* and earn a payoff that depends on the choice of the second firm. If firm E enters and firm I tries to *smash* its competitor, firm E earns 0. If, however, firm I *accommodates* the entrant, firm E earns 3. Considering firm I, because it is profitable to be a monopolist, firm I earns 5 if it is alone in the market, but it is forced to share the market (earning 3, like firm E) if it accommodates the entrant. Finally, smashing an entrant is costly for the incumbent, so firm I's profits are only 2 if it pursues this action.

(a) Write down the extensive form of this game.

(b) Write down the normal form of this game.

(c) Is the game characterized by perfect or imperfect information?

(5) The following is known as *ultimatum bargaining*. Two players are endowed with a surplus that they can share if they can come to an agreement. A first-mover (player 1) decides on a split of the surplus, and a second-mover (player 2) *accepts* this ultimatum or *rejects* it. Suppose that the surplus is \$10 and the first-mover is restricted to making one of three offers: \$1, \$5, and \$8. If the second-mover accepts an offer, it is implemented and determines the allocation of the surplus. If the second-mover rejects the offer, both players receive \$0.

(a) Write down the extensive form of this game.

(b) Write down the normal form of this game.

(c) Is the game characterized by perfect or imperfect information?

(6) Auctions can be very complicated (as we will see in chapter 13), but let's think about a very simple one, in which there are just two bidders who vie for a prize by submitting sealed bids. The person who bids the most wins the prize and pays her bid. Suppose that bidder 1 values the prize at 5 and bidder 2's value is 3. Finally, suppose that the bidders can submit only

integer bids between \$0 and \$5, and if they submit the same bid, the winner is determined randomly.

- (a) Write down the normal form of this game.
- (b) Is the game characterized by perfect or imperfect information?

(7) While we might argue about the magnitude of the effect, workers typically increase their productivity by getting training. However, does it make sense for workers to invest in human capital? Imagine that a Worker first decides to *get training* or *not invest*, and then a Boss decides to offer the worker a *high wage* or a *low wage*. The Worker then decides whether to *accept* the offer or *reject* the wage offer presented.

As for payoffs, suppose that trained workers produce a value of 10 for the Boss, but untrained workers produce only 5. Further, suppose that the high wage is 5 and the low wage is just 1. Also, suppose that it is costly for the worker to invest in training. Let this cost be 2. Finally, if the worker rejects a wage offer, no surplus is produced, but any costs must still be paid.

- (a) Write down the extensive form of this game.
- (b) Now assume that the Boss only sees a résumé and cannot verify the skills of the worker when she applies for the job. What does this extensive form look like?
- (c) Is the new game described in part (b) perfect or imperfect information?
- (d) What are the dimensions of the normal form of the game described in part (b)?

(8) Much of the economics literature on contracts employs the *Principal-Agent* model in which a “Principal” tries to motivate his “Agent” to do something on his behalf. If the Principal can convince the Agent to *work hard*, a large surplus is generated, but if the Agent *shirks* instead, the surplus is small. Suppose that the Principal first decides to either offer a wage contract, w , or not. If the Principal does not offer a contract, the payoffs to both are just 0. Next, the Agent decides if she wants to *accept* the contract or *reject* it. If she rejects the contract, the game is again over, and the payoffs are 0. If, however, the Agent accepts the contract, she then has to decide how hard to work. If she works hard, the surplus is 10, $10 - w$ of which goes to the Principal and the remaining w goes to the Agent. If the

Agent shirks, though, the surplus is only 2. One last thing: working hard is costly. Suppose that it costs the Agent an amount equal to e_H when she works hard, but it costs nothing to shirk.

- (a) Write down the extensive form of this game.
- (b) Write down the normal form of this game.
- (c) Is the game characterized by perfect or imperfect information?
- (d) Can you anticipate how much the Principal should be willing to pay the Agent?

(9) Consider a *coordination game* in which the players do better when they coordinate on the same strategy. For example, producers in a supply chain can often do better if they establish an industry standard. Continuing with this example, imagine two keyboard makers. One standard, known as *qwerty*, is based on the first six letters in the top row. An alternative, known as *dvorak*, is actually preferred because its layout allows people to type faster. Suppose that the two producers have to independently pick one keyboard to produce or the other, and if they don't coordinate on the same design, their payoffs are both 0. If however, they both pick *qwerty*, their payoffs are 1 each, and if they both pick *dvorak* instead, they each get 2.

- (a) Write down the extensive form of this game.
- (b) Write down the normal form of this game.
- (c) Is the game characterized by perfect or imperfect information?
- (d) Now imagine there are three firms. Write down the extensive and normal forms of this new game.

II

Solving Games

3

Dominated Strategies

We now turn to considering how rational players will act. In chapter 2, we considered the set of possible strategies that a player *could* play. To address the question of which specific strategy a player *should* play, we start with a simpler question: Are there strategies that a rational player definitely wouldn't ever play? Put differently, are there some strategies that are clearly a bad plan, regardless of what the others are doing?

Imagine that you are participating in an auction in which you can bid on some item of value to you. You value this particular item at v dollars, meaning that you would be indifferent between purchasing it or not if the price were $\$v$, but happy to purchase it at any price lower than v . There are several other bidders in the auction, and each may value the item differently, but you don't know the values of the other participants. The auction rules are as follows: Each person will simultaneously submit a bid, and the person with the highest bid will win, therefore gaining her value, while the other players gain nothing. Of course, the winner still needs to pay a price for the item. In this particular auction, suppose that the price that the winner must pay is equal to the second highest bid submitted—this is known as a *second-price auction*.¹ If you value the item at v dollars, how much would you bid in this auction? Or, thinking about the question somewhat differently, are there certain bids that you definitely would not wish to submit?

The Logic of Dominance

Before we analyze the optimal bidding strategy in this auction, let's first consider a simpler scenario, with only two available strategies. In game 1, pictured in normal form in [figure 3.1](#), the Row player (Row) has strategy set $S_{row} = \{Up, Down\}$ and the Column player (Column) has strategy set S_{column}

= {Left, Right}. The Column player's payoffs are xx'd out for now so that we can focus on the Row player's incentives.

		Column	
		Left	Right
Row	Up	(5, xx)	(10, xx)
	Down	(2, xx)	(2, xx)

Figure 3.1
Game 1.

Question: What should Row do?

Even though we know nothing about Column's incentives and likely strategy, this is a very simple decision for Row. If Column were to choose Left, then Row would get 5 from Up and 2 from Down; if Column were to choose Right, then Row would get 10 from Up and 2 from Down. Thus, regardless of Column's actions, Row always receives a lower payoff from the strategy Down. Row therefore faces a very easy choice. She doesn't need to put herself in Column's shoes. She doesn't need to reason through the situation from his perspective and form beliefs about what he will do. She's simply always better off choosing Up instead of Down—*no matter what he does*. In this case, we say that the strategy Down *is strictly dominated* by the strategy Up.

Strictly Dominated Strategies

Definition 1 (Strictly dominated strategy) Consider strategies s_i and s'_i for player i . The strategies for the other players are denoted as s_{-i} . The strategy s_i is strictly dominated by s'_i if $u_i(s_i, s_{-i}) < u_i(s'_i, s_{-i})$ for all s_{-i} .²

The same logic can be applied to games with more than two strategies by comparing each pair. Consider game 2 pictured in [figure 3.2](#), in which Row now has three strategies to choose among and the payoffs for Column remain unseen.

		Column	
		Left	Right
Row	Up	(5, xx)	(10, xx)
	Middle	(10, xx)	(2, xx)
	Down	(4, xx)	(9, xx)

Figure 3.2
Game 2.

Question: Is one of Row's strategies strictly dominated by another in this game?

Comparing Up and Middle, we see that neither strategy is strictly dominated by the other: If Column plays Left, Middle yields a higher payoff, and if Column plays Right, Up yields a higher payoff: $u_{\text{row}}(\text{Up}, \text{Left}) < u_{\text{row}}(\text{Middle}, \text{Left})$ but $u_{\text{row}}(\text{Up}, \text{Right}) > u_{\text{row}}(\text{Middle}, \text{Right})$. Likewise, the comparison of Middle and Down reveals that neither strategy is dominated by the other. However, Down is dominated by Up in this game: Row receives a higher payoff from playing Up than from playing Down if Column plays Left and if Column plays Right.

Observation: A rational player will not play a strictly dominated strategy.

The logic of this statement is straightforward and uncontroversial. Recall that a rational player fully understands her own payoffs and aims to maximize them. A strictly dominated strategy, by definition, can never satisfy this aim since there always exists another strategy that provides a higher payoff in every scenario.

This statement tells us what a rational player will not do, but not necessarily what she *will* do. In game 2, for instance, we know that a rational Row will not choose the strictly dominated strategy Down, but we cannot yet say which of her two remaining strategies, Up or Middle, she will choose. In some cases, however, taking the dominated strategies off the table leaves us with only a single remaining option. This was the case in game 1: we know that Row will definitely not choose Down, and thus she must play Up, her only remaining strategy. If we eliminate all strictly

dominated strategies and are left with a single remaining strategy, we say that this strategy is *strictly dominant*.

Definition 2 (Strictly dominant strategy) *A strategy s_i for player i is a strictly dominant strategy if all other strategies s'_i in S_i are strictly dominated by s_i .*

Before we move on, note that, although we have described the thought process for Row in games 1 and 2, we still do not know exactly where the pair will end up because we have not yet considered the game from Column's perspective. Yes, rationality dictates that Row will play Up in game 1 and will not play Down in game 2, but without revealing the payoffs for Column, we do not know what he will do. Suppose that the fully revealed version of game 1 looks like game 1(a), shown in [figure 3.3](#).

		Column	
		Left	Right
Row	Up	(5, 5)	(10, 0)
	Down	(2, 1)	(2, 10)

Figure 3.3

Game 1(a).

In game 1(a), Column does not have a dominant strategy. Of course, as we'll see later in this chapter, it may be the case that Column can anticipate that Row will never play Down, and therefore there is no point in trying to achieve the 10 by playing Right. Instead, Column need only focus his attention to the payoffs he can receive from the undominated strategy Up of Row. Here, clearly the 5 from playing Left is better than earning 0 from playing Right—in this sense, conditional on Row not playing the strictly dominated strategy Down, Column does best by playing Left, and we expect the pair to end up at (Up, Left) and earn the payoffs (5, 5). We will explore this type of multistep reasoning in more depth later in this chapter.

Weakly Dominated Strategies

Let's now continue with the Column player and consider his choice in game 2 by examining his payoffs revealed in game 2(a). Column has two

strategies: Left and Right. Is Right dominated by Left? If Row chooses either Up or Middle, then Column gains a strictly higher payoff from playing Left (5 versus 0). If Row chooses Down, then Column gains the same payoff from playing either Left or Right (10). Therefore, Column is sometimes worse off playing Right, and never better off.

		Column	
		Left	Right
Row	Up	(5, 5)	(10, 0)
	Middle	(10, 5)	(2, 0)
	Down	(4, 10)	(9, 10)

Figure 3.4
Game 2(a).

Our definition of a strictly dominated strategy requires it to always yield strictly lower payoffs. Even though Right doesn't exactly adhere to this definition, the basic idea is the same: Right clearly seems like a worse choice than Left. In this case, we say that Right is *weakly dominated* by Left.

Definition 3 (Weakly dominated strategy) Consider strategies s_i and s'_i for player i . The strategies for the other players are denoted as s_{-i} . The strategy s_i is weakly dominated by s'_i if $u_i(s_i, s_{-i}) \leq u_i(s'_i, s_{-i})$ for all s_{-i} and $u_i(s_i, s_{-i}) < u_i(s'_i, s_{-i})$ for at least one s_{-i} .³

Definition 4 (Weakly dominant strategy) A strategy s_i for player i is a weakly dominant strategy if all other strategies s'_i in S_i are weakly dominated by s_i .

Question: Would a rational player ever play a weakly dominated strategy?

The answer to this question is not as straightforward as when the strategy in question is strictly dominated. On the one hand, playing a weakly dominated strategy is never the best plan for a player. By playing a weakly dominated strategy, the player is never better off and sometimes is worse off than he would have been if he instead had played the strategy that

dominates it. For this reason, game theorists often make the assumption that weakly dominated strategies will not be played. Unlike a strictly dominated strategy, however, there are some cases in which the player is completely indifferent to his choices, and therefore there is no wrong play.

What does this mean? Instead of the payoffs in game 1(a), suppose that Column has the payoffs in game 1(b) shown in [figure 3.5](#). If Column is absolutely certain that Row will play her dominant strategy Up, then it really doesn't matter to Column whether he plays Left or Right. Although Right weakly dominates Left according to definition 3, he will earn 5 in either case. In other words, if Column is sure that he will find himself in the "weak" part of his weakly dominated strategy, he has no incentive to move from Left to Right. Therefore, it is not unambiguously irrational to play a weakly dominated strategy. However, as long as there is any uncertainty about what the other players will do, players will be better off avoiding weakly dominated strategies.

		Column	
		Left	Right
Row	Up	(5, 5)	(10, 5)
	Down	(2, 1)	(2, 5)

Figure 3.5
Game 1(b).

Let's take what we've learned about dominance and apply it to two important examples, ones that have influenced the development of game theoretic reasoning.

Example: The Prisoner's Dilemma

Consider again the prisoner's dilemma game from chapter 1, depicted in game 3, shown in [figure 3.6](#). While we previously saw that, logically, the players have an incentive to choose Confess, we now have the terminology to explain the source of the dilemma. For both players, Silent is strictly dominated by Confess. From the perspective of cooperation, this is bad news: we now know that no rational player will choose the cooperative strategy Silent in this game.

		Prisoner 2	
		Silent	Confess
Prisoner 1	Silent	(5, 5)	(0, 10)
	Confess	(10, 0)	(1, 1)

Figure 3.6

Game 3 (the prisoner's dilemma).

Is the inferior outcome (Confess, Confess) therefore unavoidable? Or is there some solution to escape the dilemma? Often, people learning about this game wonder whether communication between the players could solve the dilemma. Unfortunately, communication alone does not change the incentives of the players. Even if the players could discuss the insight that both would be better off with the (Silent, Silent) outcome than the (Confess, Confess) outcome, and even if both agreed to remain Silent, each player would still be better off reneging on their agreement, so long as it is not binding and the payoffs are as depicted in game 3.

Another possibility is that the payoffs in the game don't truly reflect the players' preferences, or that with external interventions like communication or education, the payoffs might change. The standard prisoner's dilemma payoffs reflect the assumption that neither player is concerned about the fate of the other—or at least not sufficiently concerned that he would ever prefer to remain Silent. As we will see later in this chapter, the assumption that the players care only about the outcomes for themselves is often not realistic; many people would prefer to remain Silent if they knew for sure that their partner would reciprocate by staying Silent as well.

Another assumption of the game is that the players interact in a single encounter: the game ends after each decides whether to be Silent or to Confess. If, however, the players will encounter each other again—either with certainty or with some probability—then game 3 doesn't fully capture the situation. We will discuss the effect of repetition on this game in chapters 9 and 10.

Example: The Second-Price Auction

Let's now return to the second-price auction, the situation with which we began this chapter. In this game, each player i submits a bid, b_i , with the

highest bidder receiving her value of winning the item, v_i , but paying only the second-highest bid. The question was: what bid would you submit in this auction, given that your value is v ?

While this initially seems like an unfamiliar or unintuitive type of auction, it is very similar to, for instance, how participants submit bids on eBay and other online auctions. Each player submits a bid at which they would want to drop out of the auction. The auction begins at some initial low price of the seller's choosing and then ticks up. As long as multiple people are willing to pay the price, it continues to increase. As soon as the second-highest bidder drops out, the auction stops, and the one remaining bidder pays that price. That is, the auction does not continue with the lone remaining bidder bidding against herself. Therefore, the person who entered the highest bid wins, but she doesn't need to pay her own bid—she pays the price at which the second-highest bidder was no longer willing to continue.

It turns out that in this type of auction, each player has a weakly dominant strategy, which is to simply bid his own value. Any strategy in which he bids *more* than his value is weakly dominated by bidding v_i . Likewise, any strategy in which he bids *less* than his value is also weakly dominated by bidding v_i .

Proposition $b_i = v_i$ is a weakly dominant strategy in the second-price sealed-bid auction.

Proof. Imagine that player i is considering some alternative bid b_i that is not equal to v_i . We need to show that for any possible strategies (i.e., bids) played by the other players and any alternative bid b_i not equal to v_i , player i is always weakly worse off bidding b_i instead of v_i . To do this, we will first show that any bid $b_i < v_i$ is weakly dominated by v_i , and then show that any bid $b_i > v_i$ is also weakly dominated by v_i . The important point is to show that the bidder can't deviate away from bidding her value and expect to ever do better.

Step 1: Show that bidding $b_i < v_i$ is weakly dominated by bidding v_i .

Let $\overline{b_{-i}}$ equal the bid of the highest bidder other than player i (where $-i$ indicates the set of bidders, not including i , and the upper bar indicates the largest of these bids). Player i will therefore win the auction if and only his bid is greater than $\overline{b_{-i}}$. In other words, i 's payoff depends only on $\overline{b_{-i}}$, his

own bid, and his value. It does not depend on the bids of any of the other players. Notice that although player i doesn't know what $\overline{b}_{-i}$ is, there are only a few possible situations to consider when thinking about deviating below v_i , known as *shading* one's bid, or deviating above v_i , or *plumping* one's bid. This should prompt you to think about dominance. The various cases are depicted in [figure 3.7](#).

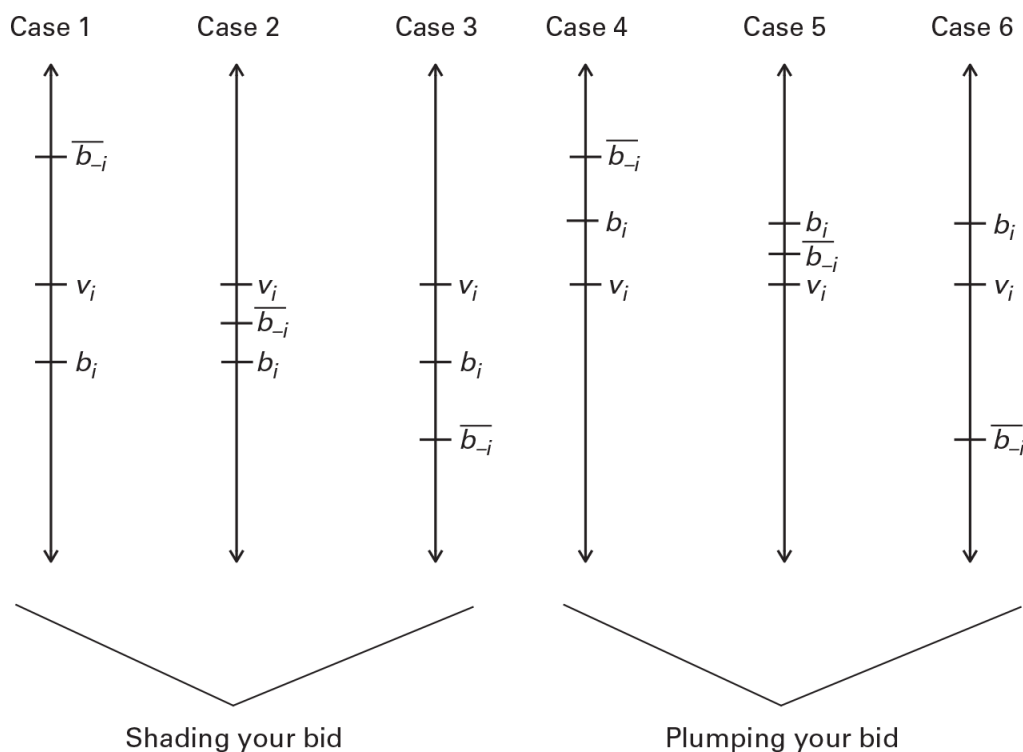


Figure 3.7

Shading your bid and plumping your bid are both weakly dominated strategies.

First, consider shading your bid by submitting $b_i < v_i$. Without knowing where $\overline{b}_{-i}$ is, you know that you'll end up in one of the three cases on the left side of [figure 3.7](#). In case 1, the realization of the auction is $b_i < v_i < \overline{b}_{-i}$: the highest bid is greater than your value, and thus you would not win by bidding v_i or by shading with b_i . Next consider case 2. Here, the realization of the auction is $b_i < \overline{b}_{-i} < v_i$, and because $b_i < \overline{b}_{-i}$, you lose the auction when you would have won it by bidding v_i . In this case, you are clearly worse off shading (and earning 0) than bidding your value (and earning $v_i - \overline{b}_{-i} > 0$). In case 3, you still shade, but not as aggressively, and the realization is $\overline{b}_{-i} < b_i < v_i$. The good news is that you win the auction. But notice that you

still would have won by bidding v_i and, most important, shading did not reduce the price you pay for the item because this is a second-price auction and you do not pay your own bid anyway. So regardless of whether you bid v_i or b_i , you still pay $\overline{b_{-i}}$ if you win. Putting these three results together, we can conclude that shading your bid in this auction can only hurt you (case 2) and never help you (cases 1 and 3).

Step 2: Show that bidding $b_i > v_i$ is weakly dominated by bidding v_i .

What about plumping your bid by submitting $b_i > v_i$? You could imagine that this makes sense because you only have to pay $\overline{b_{-i}}$, and bidding a bit more can only increase your chances of winning, right? It turns out that no, this is not correct. Return to [figure 3.7](#) to see why. In case 4, we have $v_i < b_i < \overline{b_{-i}}$, and you lose the auction, as you would have if you had bid v_i instead. Here, plumping your bid does not help you. In case 5, increasing your bid actually hurts you. Here $v_i < \overline{b_{-i}} < b_i$, and now you are stuck paying $\overline{b_{-i}}$ for something that you only value at v_i . Moreover, had you bid v_i , you would have lost, but your payoff would not be negative! The outcome of case 5 is known as the *winner's curse*. In the final case, case 6, your value is greater than the next highest bid, so you win the auction regardless of whether you bid your value or a bit more. Putting these three results together, we can conclude that plumping your bid in this auction can only hurt you (case 5) and never help you (cases 4 and 6).

The conclusion of these two steps is that $u_i(b_i, \overline{b_{-i}}) \leq u_i(v_i, \overline{b_{-i}})$ for all $\overline{b_{-i}}$. Bidding b_i generates lower (cases 2 and 5) or identical (cases 1, 3, 4, and 6) payoffs than bidding v_i , and therefore any b_i is thus weakly dominated by v_i .

It is important to understand the intuition for why you wouldn't want to gamble by bidding over your value. People often assume that there is little cost to doing so, since you don't pay your own bid anyway, and that it could increase your chances of winning. [Figure 3.7](#) shows the flaw in this reasoning: yes, it increases your chances of winning certain auctions—but only auctions that you wouldn't want to win anyway. If the other player is bidding less than your value, then you'd win even if you stuck with your value. But if the other bidder is bidding more than your value, then winning and paying her bid will make you worse off than losing.

How Far Can Dominance Take Us?

Thus far, we have seen that rational players will not play dominated strategies. If one strategy dominates all others, then this concept of dominance is sufficient to predict the behavior of rational players, as was the case in the prisoner's dilemma and the second-price auction. Often, however, a player will not have a strategy that dominates all others. Can the concept of dominance still enable us to predict rational behavior?

Consider game 4, depicted in [figure 3.8](#). Antonio and Beatrix are simultaneously deciding how much effort to put into a team project. Each player can either put in a lot of effort (Work), put in a moderate degree of effort (Coast), or do nothing (Shirk). If his or her teammate is either choosing Work or Coast, each player prefers to Coast. But if the teammate is choosing Shirk, he or she prefers to Work in order to compensate for the other person's lack of effort.

		Beatrix		
		Work	Coast	Shirk
Antonio	Work	(40, 40)	(20, 50)	(10, 30)
	Coast	(50, 20)	(25, 25)	(5, 10)
	Shirk	(30, 10)	(10, 5)	(0, 0)

Figure 3.8
Game 4.

Let's think about the game from Antonio's perspective. He does not have a dominant strategy: whether he prefers Work or Coast depends on what Beatrix does. Shirk, however, is strictly dominated by Coast, and therefore Antonio will not choose it if he is rational. Beatrix faces the same situation, so likewise she will not choose Shirk if she is rational. Furthermore, *if Antonio knows that Beatrix is rational*, then he knows that she will never play the dominated Shirk. Likewise, if Beatrix knows that Antonio is rational, then she knows that he, too, will never play Shirk.

So long as it is common knowledge among the players that Shirk will never be chosen, then they are essentially playing the abbreviated game 4(a), shown in [figure 3.9](#), in which Shirk is eliminated as a strategy for both

players. Within this abbreviated game, Coast dominates Work and is therefore a dominant strategy for both players. (Recall that Work was preferred only if the other player chose to Shirk—a scenario that we have eliminated.) We therefore conclude that both players will choose Coast. Notice that this process required not only the assumption that each player is rational (and therefore would not play the dominated Shirk), but that each player knows that the *other is rational*, such that both players knew that their teammate also would not choose Shirk. Recall from the previous chapter that we make the standard background assumption that rationality is common knowledge among the players, so it is not only the case that Antonio knows Beatrix is rational, but she knows that he knows she is rational, and he knows that she knows that he knows she is rational, and so on.

		B		
		Work	Coast	Shirk
A	Work	(40, 40)	(20, 50)	(10, 30)
	Coast	(50, 20)	(25, 25)	(5, 10)
	Shirk	(30, 10)	(10, 5)	(0, 0)

Figure 3.9

Game 4(a).

Iterated Elimination of Dominated Strategies

As long as rationality is common knowledge, we can continue indefinitely with this process of identifying dominated strategies, eliminating them, and then considering the resulting, truncated version of the game. We call this procedure the *iterated elimination of dominated strategies*. The process proceeds as described in [figure 3.10](#). We begin in step 1 by considering the full strategy set for each player. If neither player has a dominated strategy, we cannot advance. If any player has a dominated strategy or strategies, we eliminate them and continue to the next step. For all subsequent steps n , we consider the strategies remaining after step $n - 1$. If neither player has a dominated strategy, the process concludes. If any player has a dominated strategy or strategies, eliminate them and consider the truncated game in the next step.

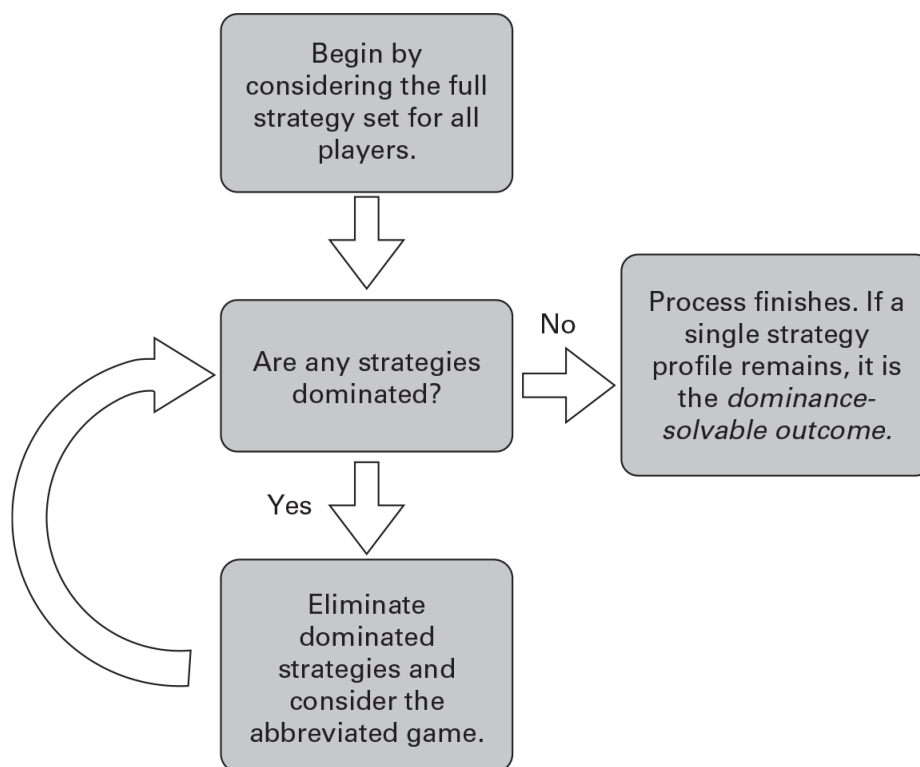


Figure 3.10

Iterated elimination of dominated strategies.

The process continues until no additional dominated strategies exist or only a single strategy remains for each player. If a unique strategy profile survives the iterated elimination of (strictly and weakly) dominated strategies, we say that the game is *dominance solvable* and call the remaining strategy profile the *dominance-solvable outcome*. In game 4, the unique dominance-solvable outcome is (Coast, Coast).⁴

Example

Let's consider game 5, shown in [figure 3.11](#), and work through the process of iterated elimination of dominated strategies.

		Column		
		Left	Center	Right
Row	Up	(3, 3)	(1, 6)	(4, 100)
	Middle	(4, 3)	(20, 5)	(5, 4)
	Down	(2, 2)	(2, 10)	(70, 0)

Figure 3.11
Game 5.

Step 1. The full strategy sets for the two players include three strategies each. Are any of these dominated? Yes. For Row, Up is dominated by Middle, and for Column, Left is dominated by Center. Therefore, we eliminate Up and Left from consideration. What assumptions were necessary to do so? Simply that both players are rational and so will not play these dominated strategies.

Step 2. We are now left with the truncated game 5(a), shown in [figure 3.12](#), in which each player has two undominated strategies remaining. For Row, neither strategy is dominated by the other, but for Column, Right is dominated by Center. Therefore, we eliminate Right. What rationality assumptions were necessary to do so? This elimination requires that Column is rational, but also that Row is rational and that Column knows that Row is rational (and therefore would not be playing Up). If Column didn't know that Up was eliminated for Row, then Column would not eliminate Right.

		Column		
		Left	Center	Right
Row	Up	(3, 3)	(1, 6)	(4, 100)
	Middle	(4, 3)	(20, 5)	(5, 4)
	Down	(2, 2)	(2, 10)	(70, 0)

Figure 3.12
Game 5(a).

		Column		
		Left	Center	Right
Row	Up	(3, 3)	(4, 6)	(4, 100)
	Middle	(4, 3)	(20, 5)	(5, 4)
	Down	(2, 2)	(2, 10)	(70, 0)

Figure 3.13
Game 5(b).

Step 3. This leaves us with the abbreviated game 5(b), in which Row has two remaining strategies and Column has only one. Given that Column can only play Center in this truncated game, Down is dominated by Middle and can be eliminated. What rationality assumptions were necessary to do so? Row must be rational in order for us to justify eliminating Down. But Row must also know that she is in the abbreviated game 5(b), which requires the following assumptions: Column is rational, Row knows Column is rational, Column knows that Row is rational, and Row knows that Column knows that Row is rational.

(Middle, Center) is therefore the dominance-solvable outcome in this game. It is important to note that all this reasoning is happening only in the players' minds. The game itself has only one stage, in which the players act simultaneously.

Does Order Matter?

The process of iterated elimination that we have been discussing involves, with each step, considering all the remaining strategies for each player and deleting every strategy that is dominated. You might wonder whether we would arrive at the same result if, rather than eliminating all dominated strategies possible in each step, we eliminated only one strategy at a time. For instance, what if we alternate between looking for dominated strategies for the Row player and the Column player? The answer depends on whether we only ever eliminate strictly dominated strategies or we also eliminate weakly dominated strategies. If the strategies to be eliminated are always strictly dominated, then the order in which you perform the elimination in any finite game doesn't matter—you will arrive at the same solution regardless of whether you eliminate all the dominated strategies in each step

or eliminate one at a time (in any order). If some of the strategies that would normally be eliminated are *weakly* dominated, however, then eliminating only one strategy at a time could lead to different results. Therefore, it is best to stick with the standard process of iterated elimination, described by [figure 3.10](#), in which we eliminate all dominated strategies for all players in each step, since it always leads to the same dominance-solvable outcomes.

		Column		
		<i>Left</i>	Center	<i>Right</i>
Row	<i>Up</i>	(3, 3)	(11, 6)	(4, 100)
	Middle	(4, 3)	(20, 5)	(8, 4)
	<i>Down</i>	(2, 2)	(2, 10)	(70, 0)

Figure 3.14
Game 5(c).

Now let's take what we've learned about iterated elimination of dominated strategies and apply it to two games that behavioral researchers often ask people to play.

The Beauty Contest

Recall the beauty contest game from chapter 1, in which a large group of players each guesses a number between 0 and 100. The winner is the person whose number is closest to *two-thirds* of the average guess. As we discussed in chapter 1, your guess should depend on what you think the other players are likely to guess, which may be based on your assessment of their rationality. If we assume that rationality is common knowledge, then we can arrive at a unique solution using the process of iterated deletion of dominated strategies. The first three steps of this process are shown in [figure 3.15](#).

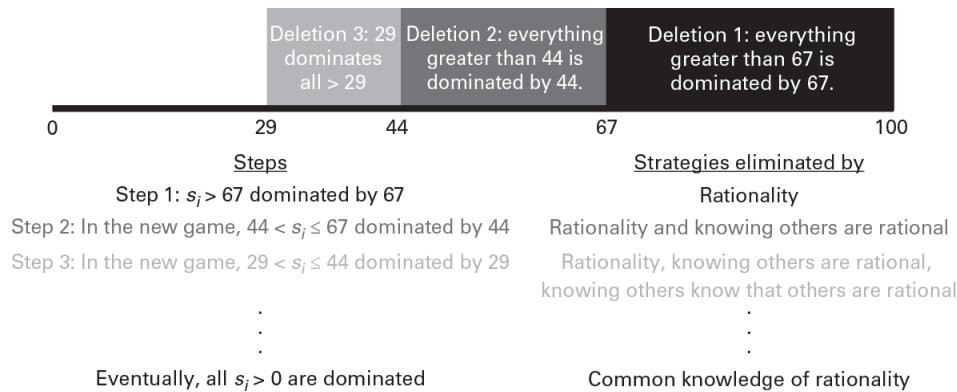


Figure 3.15

Iterated elimination of dominated strategies in the beauty contest.

In the first step, we consider the full strategy space, $[0,100]$, for all players. If the highest guess that anyone can choose is 100, then the largest that two-thirds of the average guess could possibly be is 67, which would occur if everyone else guesses 100. Since nothing greater than 67 could possibly be the winning guess, then guessing 67 will always get you closer to the winning number than anything above 67 (in the shaded black zone of the figure) will. Therefore, guesses greater than 67 are weakly dominated by 67 and so, for step 1, we eliminate everything greater than 67.

In step 2, we know that no one is choosing anything greater than 67. This means that the highest that two-thirds of the average could be is two-thirds of 67, or 44. Therefore, the winning number can never be greater than 44, and guessing any number greater than 44 is weakly dominated by 44. Notice that rational players should never play anything in the black zone of [figure 3.15](#), as it is numerically impossible for two-thirds of the average to fall in this range. However, it *could* make sense to play something in the dark gray zone, depending on what you believed the other players were doing. For instance, you would want to choose 50 if you believed that everyone else was choosing 75. But 75 falls in the black zone that rational players would not choose. So if you know that everyone else is rational and not choosing anything in the black zone, then you don't want to choose anything in the dark gray zone. Thus, the range from 44 to 67 is eliminated if all the players are rational *and* know that all the other players are rational, too.

In step 3, we know that no one is choosing anything greater than 44 (i.e., either in the dark gray or black zones). This means that the highest two-thirds of the average could be is two-thirds of 44, or 29. Then, the winning number can never be greater than 29, and we can eliminate any strategy that involves guessing more than 29. Again, notice that you want to write off the numbers between 29 and 44 (the light gray zone in [figure 3.15](#)) only if you are convinced that the other players have eliminated the dark gray zone. (If the average guess is in the dark gray zone, then you *would* want to pick a number in the light gray zone.) Like you, the others will have eliminated the dark gray zone if they are rational and know that all the other players are rational as well. Therefore, you will avoid the light gray zone if you are rational, know that the other players are rational, and know that the others know that the other players are rational.

Continuing this process of reasoning indefinitely, we converge toward the solution in which all players guess 0 and the prize is awarded randomly, a process that relies on common knowledge of rationality.

The Traveler's Dilemma

Imagine that you arrive home after a vacation and find that the airline has smashed a souvenir vase in your checked luggage. Another traveler on the same flight was transporting an identical vase, which is now identically smashed. Neither of you has a receipt to prove the replacement value of the vase. The airline offers the two of you the following arrangement: Each person will simultaneously write down a reimbursement amount from \$2 to \$100 in whole-dollar increments. If both amounts match, the airline will give you each this amount as compensation. If the amounts do not match, the airline will give you each the lesser amount. However, to incentivize honesty, the person who writes the smaller amount will get a bonus of \$2, while the other must pay a penalty of \$2. (Assume that the actual price paid was greater than \$100, so writing the true amount is not an option.) This game is known as the *traveler's dilemma* and was developed by Basu (1994).

How much do you claim the vase is worth?

Notice that you do not have a dominant strategy in this game. You have an incentive to undercut the other player so that you capture the honesty

bonus. However, you don't want to undercut by too much, since doing so drags down the compensation that the airline gives you both. Therefore, the amount that you want to write varies based on what the other player writes. Specifically, you're always best off by undercutting the other player's claim by \$1, which brings down the airline's compensation by only \$1 relative to matching the other person, but gives you the \$2 bonus.

With these incentives in mind, we can see that the players do have *dominated* strategies. In step 1, we consider the full strategy set for each player, which is the set of integers from 2 to 100. Claiming the maximum of \$100 is weakly dominated by coming in \$1 under and claiming \$99. To see this, we want to compare the payoffs that you receive from choosing \$100 and from choosing \$99 for each claim the other player could submit. This information is summarized in [figure 3.16](#). First, if the other traveler claims \$100, then you both receive \$100 if you make the same claim. If you claim \$99, however, the airline pays you both the lower amount (\$99), but you receive a \$2 bonus for submitting the smaller claim, bringing your total payoff to \$101. Likewise, if the other traveler claims \$99, claiming \$99 as well gives you a higher payoff (\$99) than claiming \$100 (in which case, the airline pays you \$99 and fines you \$2). Finally, if the other traveler claims \$98 or less, you get the same payoff regardless of whether you claim \$99 or \$100 (his claim minus the \$2 fine), since he undercuts you in both cases. Therefore claiming \$100 is weakly dominated, and we can eliminate it for rational players. Put a different way, claiming \$100 never satisfies your aim of undercutting the other player by \$1, since he can never claim more than \$100.

		Other Traveler		
		\$100	\$99	$x < \$99$
You	\$100	(100, 100)	(97, 101)	($x-2$, $x+2$)
	\$99	(101, 97)	(99, 99)	($x-2$, $x+2$)

Figure 3.16

Claiming \$100 is weakly dominated by claiming \$99.

In step 2, we consider the strategies remaining after step 1: the set of integers from 2 to 99. Notice that there was nothing special about the claim

of 100 in the previous step except that it was the maximum claim that either player could make. So long as both players are rational, we eliminate 100, and 99 becomes the maximum claim that either player makes. By the exact same reasoning as step 1, we can see that \$99 is now weakly dominated by \$98. The second step, therefore, is to eliminate \$99.

By following this same logic, in each step we eliminate the maximum remaining claim until, in the 98th step, we are left only with strategies \$2 and \$3 for each player and can eliminate the strategy of \$3. In this case, \$3 is dominated by \$2. If the other player claims \$3, then you get \$3 from matching and \$4 from claiming \$2. If the other player claims \$2, then you get \$2 from matching and \$0 from claiming \$3. Therefore, the dominance-solvable outcome in this game is (\$2, \$2). By following this continuous chain of logic and constantly working to undercut each other, the players drive their compensation down to the lowest possible amount. Notice, however, that the rationality assumptions required to derive this result are extreme: the chain of reasoning about the rationality of the other player must survive for 98 steps. In other words, if the first statement is that the players are rational, the second is that the players know that the other is rational, and the third statement is that the players know that the other knows that they are rational, etc., then 98 such statements must all hold. Do people actually behave this way?

Behavior in Games: Playing Dominated Strategies

The elimination of dominated strategies is a straightforward and intuitive concept that, as we've seen, can be used to solve for rational behavior in a variety of games. While logical, the question remains as to whether this process actually predicts behavior. In each of the games we've analyzed, participants regularly deviate from the predicted behavior in the lab, often for a variety of plausible reasons.

For the past several decades, economists have used experimental methods to test the predictions of game theory. In the typical game theory experiment, a group of participants—often undergraduate students—reports to a computer lab, where each person makes a series of anonymous decisions on a computer. Instead of receiving a flat monetary payment for their participation, the subjects are paid, in cash, according to the outcome

of the game they play. Thus, rather than being asked to imagine that they are playing a particular role (such as a prisoner who must decide whether to rat out his partner or a consumer who must decide how much to bid on a good), the participants are actually placed in the situation that we wish to study, with real payoffs on the line.

For instance, game 6, shown in [figure 3.17](#), depicts a version of the prisoner's dilemma that has been studied in the lab. Rather than choosing between actions with emotionally laden terms such as "Silent" and "Confess," the participants may be asked to choose between columns L(eft) and R(right) or rows U(p) and D(own). Before making their decisions, the participants are given detailed and accurate instructions about exactly how the choices they and the other participants make will map onto their ultimate cash payoff, thereby satisfying our assumption that the rules of the game are common knowledge.

		Blue player	
		L [Silent]	R [Confess]
Red Player	U [Silent]	(8, 8)	(0, 10)
	D [Confess]	(10, 0)	(3.5, 3.5)

Figure 3.17

Game 6 (prisoner's dilemma in the lab).

Playing Dominated Strategies

The prisoner's dilemma scenario was first conceived by researchers who wanted to test whether real people behave according to the predictions of game theory. In 1950, they recruited colleagues to play the game and conducted what many people consider the first formal game theory experiment. What happened? To the surprise of many, the economists playing the prisoner's dilemma game *regularly chose the strictly dominated strategy* of Silent. This finding has been replicated in many subsequent experiments, which generally find that Silent is chosen roughly one-quarter to one-half of the time by people playing the game for the first time.

Why is this? Perhaps the simplest explanation is that the participants are confused and don't fully understand the game. However, this is unlikely to be the primary motivation for playing the dominated Silent. For one reason,

people respond to changes in the relative payoffs, becoming even more cooperative as Confess becomes less lucrative. For another, many people still cooperate by choosing Silent after they have already played the game many times. A more common explanation is that many people care, not only about their own earnings, but about how fairly they are treated. In this sense, people are reciprocal—they are happy to help those who are helping them, but they also wish to hurt those who are hurting them (Rabin, 1993). Given this, the monetary payoffs presented in the game 6 matrix don't reflect the true preferences of the participants. In particular, a substantial proportion of people are “conditional cooperators”—they prefer to cooperate if the other person does. Therefore, if the other player chooses Silent, a conditional cooperator gets a higher payoff from choosing Silent as well. In other words, Silent is *not dominated* for these participants. Evidence of these types of preferences can be found in prisoner's dilemma experiments in which one player learns the other player's choice before deciding (Clark and Sefton, 2001). Very few participants are Silent after the other person has Confessed, but about half of people will choose Silent if the other person was Silent.

Evidence-Based Adaptations of the Standard Model

So how can the theory be updated to account for this? One common formulation allows players both to value material payoffs and to receive *psychological payoffs* that depend on the beliefs that players hold about the intentions of others. Specifically, suppose that Row knows that Column is playing Silent. Since Silent always pays less than Confess, Row recognizes that Column is sacrificing his own material payoff to help her, so she interprets his behavior as being kind. She may then want to reciprocate that help and be kind to him in return. Suppose that x is the additional payoff that players receive from being kind to those who are kind to them (and mean to those who are mean to them). If x is sufficiently large, then Silent is preferred to Confess when the other player is playing Silent, and confessing no longer dominates. This change to the players' payoffs is reflected in the updated game 6(a), shown in [figure 3.18](#).

		Blue player	
		L [Silent]	R [Confess]
Red Player	U [Silent]	$(8+x, 8+x)$	$(0, 10)$
	D [Confess]	$(10, 0)$	$(3.5+x, 3.5+x)$

Figure 3.18

Game 6(a) (prisoner's dilemma with "fairness" outcomes).

Notice that fairness cuts both ways, in that x is added to the payoffs for (Confess, Confess) too. In this case, each player knows that the other hurt them by confessing, so reciprocity requires them to feel good about being hostile toward the other player. That is, x is added here because not only does the player satisfy her material desire by playing Confess, she also satisfies her desire to hurt the other person, who she believes is treating her unfairly. We will consider updated payoffs that capture psychological motives, in addition to material benefits, in various games throughout the book, particularly those that involve bargaining (part V) and social dilemmas (part VI). In chapter 28, we will analyze *psychological games*, in which players are not just motivated by outcomes, but also beliefs, such as the belief that another player was intentionally unkind to them.

In the second-price auction, we have seen that bidding one's own value weakly dominates any other bid. In the lab, however, participants systematically *overbid* relative to their true value. Participants in these auction experiments bid on a fictitious good rather than an actual item. All of them are assigned private values by the experimenter, and they are told that the winner will be paid his private value for the item. Therefore, confusion or uncertainty about the prize's value cannot explain overbidding. What can? One explanation is that there is typically little consequence to overbidding, and thus little incentive for overbidders to change their behavior. Bidding one's value is only *weakly* dominant and, as we see from [figure 3.7](#), it is very often the case that submitting a different bid would generate an identical outcome. There is, therefore, little opportunity for learning that overbidding sometimes yields negative payoffs. In fact, even if the players were told the bids of all the other participants and then given the chance to change their own bid, they typically would have no real reason to adjust their behavior. We will revisit this question in chapter 24, when we

consider how games would eventually play out if, instead of being strategic, players could constantly adjust their strategies over time based on the previous choice of the other players. Experimental evidence for overbidding is discussed by Kagel and Levin (1993).

The beauty contest and traveler's dilemma games are even more interesting because their dominance-solvable outcomes rely on the common knowledge of rationality: The players must believe that no other player will play a dominated strategy, which we've just seen may be a precarious assumption. If they don't believe that the other players will be able to complete the full process of iterated deletion, then it would be foolish to choose 0 in the beauty contest or 2 in the traveler's dilemma. What is the rational way to play these games if the other players aren't rational?

Figure 3.19 shows the frequency with which people choose each possible number in the beauty contest game. The top panel presents data from a lab experiment, while the bottom panel presents data from a beauty contest challenge that appeared in three newspapers, with several thousand respondents.

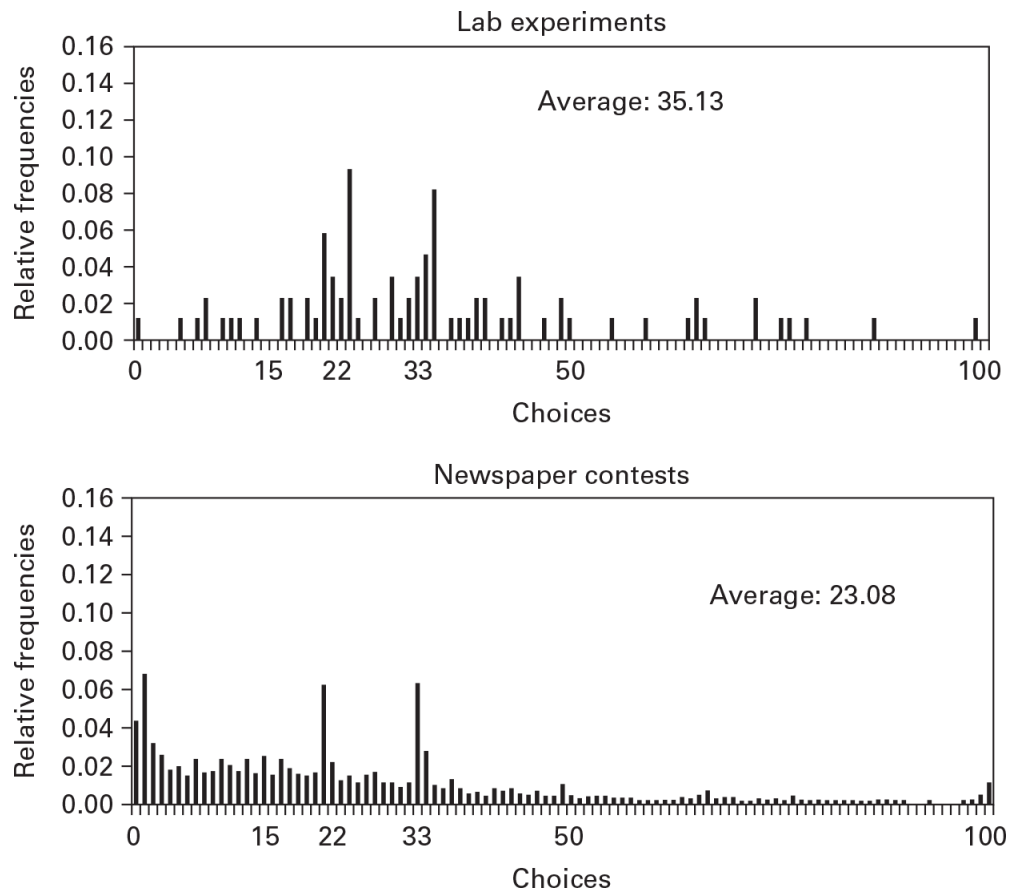


Figure 3.19

Beauty contest guesses reported in Bosch-Doménech et al. (2002).

The first thing to notice is that the actual average guess is 35 in the lab and 23 in the newspaper contests, so choosing 0 would not win you the contest. Very few people actually choose the dominance-solvable outcome of 0: Fewer than 2% of lab participants choose 0, and even the newspaper participants, who were able to spend at least a week considering and discussing their responses before submitting them, choose 0 less than 5% of the time. It is somewhat more common for people to choose a *dominated* strategy above 67. The bulk of the observations, however, are between 0 and 33. In fact, there seem to be large clusters around 33 and 22.

What is special about these numbers? Imagine that you think that none of your competitors are very bright and that no one but you will approach the question of which number to pick as a game of strategy. Instead of being strategic, you assume that the other players will choose a number essentially at random—picking a lucky number, or the number of their

favorite baseball player, or just going with their initial gut instinct. You thus conclude that the average chosen by these random players will be in the middle of the strategy set (50) and you should choose two-thirds of that (33). We can think of this conclusion as one step of strategic thinking, and this line of reasoning can explain the large spike around 33. Or perhaps you give your competitors a bit more credit. You assume that they will engage in one step of strategic thinking and reason their way to the response of 33. If everyone else is doing this single step of reasoning and choosing 33, then the winning number is two-thirds of 33, or 22. Participants engaging in two steps of strategic reasoning can explain the large spike around 22. The evidence that different people engage in different *levels* of strategic thinking has generated a new class of models called *level-k* models, which we will apply to several types of games throughout the book and discuss in more detail in chapter 27.

In the traveler's dilemma, the incentive to continually undercut each other leads players to choose the lowest possible claim of \$2—if each player completes the full process of iterated deletion and believes that the other player will as well. But what if you don't believe that the other player will work through the full chain of reasoning? If you think the other player might make a mistake and submit a high claim, then you would also want to choose a high claim. In fact, it would be extremely costly to play the “rational” strategy of 2 in this case. For instance, imagine that you believe it equally likely that the other player will choose \$2 or \$99. If you choose \$2, then your payoff will be \$2 if she also chooses \$2 and \$4 if she chooses \$99, for an average payoff of \$3. However, if you choose \$98 or \$99 then your payoff, on average, would be around \$50. Submitting a high claim can be perfectly reasonable if there is any significant chance that the other player will submit a high number, since the \$2 incentive to undercut the other player is small relative to the potential harm of choosing a low claim while the other player chooses a high claim.

How do people play this game? Unlike the beauty contest game, in which people engage in various levels of reasoning, most participants submit the maximum claim. In fact, as people gain more experience with the game, they become even more likely to choose the maximum claim (Capra et al., 1999). If players think that others are likely to make mistakes, then this behavior makes sense given the relatively small reward/penalty

scheme. But what if the incentives to undercut the other player are large relative to the range of possible claims (e.g., \$40 or \$60)? In this case, it may be much safer to choose \$2 than to risk being undercut and paying a huge penalty. In fact, this is exactly what we see in the lab: Once the reward and penalty terms become large, participants submit the minimal claim, especially if they are given a chance to learn.

How can we incorporate the possibility that players make mistakes into the theory? One approach is called *quantal response equilibrium* (QRE). The basic idea is that the players choose each of their strategies with some probability. Instead of always choosing the strategy that is most advantageous, they are simply *more likely* to choose more advantageous strategies. At the same time, the players believe that the others may also make mistakes in a similar fashion. Allowing some degree of randomness in players' abilities to implement their optimal strategy can explain deviations from predicted behavior in the traveler's dilemma, as well as the observed movement toward the minimal claim as the reward/penalty increases. We will discuss in greater detail QRE and the variety of games to which it has been applied in chapter 26.

As we learn more tools of game theory, these same concepts—caring about how you are treated, learning, level- k reasoning, and allowing players to make mistakes—will be common themes for explaining the behavior of real people in games of strategy. Crucially, behavioral evidence against the standard predictions does not mean that game theory itself is inadequate or should be disregarded. Instead, these results inform game theorists on how best to expand and refine the theory to incorporate these patterns of behavior. This process of testing and refining the theory is thus an essential component of behavioral game theory and will be a recurring theme through the rest of the book.

Exercises

(1) For each of the games in [figure 3.20](#), consider each of the strategies for the Row player and for the Column player. State which strategies are dominated and whether the dominance is strict or weak.

(a)			(b)				(c)						
Column			Column				Column						
Row		A	B	Row		L	C	R	Row		F	G	H
	C	(5, 5)	(4, 6)		U	(4, 4)	(3, 5)	(2, 3)		X	(9, 3)	(3, 5)	(6, 5)
	D	(2, 3)	(3, 4)		M	(1, 2)	(2, 2)	(1,11)		Y	(1, 2)	(5, 8)	(4, 8)
					D	(5, 1)	(1, 5)	(0, 0)		Z	(5, 1)	(1, 5)	(0, 0)

Figure 3.20

(2) Find the dominance-solvable outcome of the game in [figure 3.21](#) by eliminating all strictly and weakly dominated strategies in each step.

		Column		
		Left	Middle	Right
Row	Up	(1, 10)	(7, 6)	(10, 2)
	Center	(1, 7)	(5, 10)	(4, 3)
	Down	(5, 5)	(6, 1)	(9, 5)

Figure 3.21

(3) Consider the following game in [figure 3.22](#) between Jane and Rafael. Discuss the assumptions regarding the players' rationality and knowledge of rationality that are required to eliminate one of Jane's strategies.

		Rafael	
		Left	Right
Jane	Up	(1, 8)	(9, 1)
	Down	(9, 9)	(0, 1)

Figure 3.22

(4) Imagine that you are at dinner with an acquaintance and that you each independently choose what to order for dinner: a burger (which costs \$8), chicken (\$12), or steak (\$18). In terms of enjoyment, you each have the same preferences: you value the burger at \$13, chicken at \$16, and steak at

\$20. In other words, if you were dining alone and ordered a burger, your payoff would equal $\$13 - \$8 = \$5$. In this case, however, you will split the bill equally with your companion, and thus your payoff depends both on your own order and the order of your acquaintance. Specifically, each diner's payoff is equal to the value of their own meal minus half the cost of the total bill.

- (a) Write down the normal form representation of the game in which both players are independently choosing between the actions {Burger, Chicken, Steak}.
 - (b) Which dinner orders would maximize the sum of the two diners' payoffs?
 - (c) What is the dominance-solvable outcome?
- (5) Consider a modified version of the traveler's dilemma. Imagine that each traveler has only three possible claims they can submit: \$50, \$75, or \$100.
- (a) Write the game in normal form, assuming that there is a \$30 reward for submitting a lower claim and \$30 penalty for submitting a higher claim. What is the dominance-solvable outcome?
 - (b) If instead there were only a reward of \$30 for submitting a lower claim but no penalty for submitting a higher claim, is there dominance-solvable outcome? If yes, state the strategy profile; if not, explain why.
 - (c) If instead there were only a penalty of \$30 for submitting a higher claim but no reward for submitting a lower claim, is there a dominance-solvable outcome? If yes, state the strategy profile; if not, explain why.
 - (d) Finally, assume that there is still the same reward and penalty scheme as in part (a). However, similar to the situation described in [figure 3.18](#), the players care about fairness. In this case, they receive an additional psychological payoff from cooperating (submitting a high claim) with those who were cooperative with them and competing (submitting a low claim) with those who were competitive with them. Specifically, let each player receive an additional payoff of x if the two players' claims match. What is the lowest x such that there is no longer a dominance-solvable outcome?

(6) Notice that being asked to choose the number closest to two-thirds the average of the other guesses in the beauty contest is arbitrary. An obvious way to test the robustness of the standard experimental result is to see how people respond to different parameterizations of the underlying game. Before you run these follow-up experiments, however, you need to figure out the game theoretic predictions. With this in mind, solve the following problems.

- (a) What is the dominance-solvable outcome if people choose numbers between 0 and 100 and the winner is the person who chooses the number closest to one-half the average?
- (b) Now suppose that people pick numbers between 1 and 100 and the winner is the person who picks the number closest to four-thirds the average.
- (c) Why is it important to not allow a guess of 0 in part (b)?

(7) Suppose that Audrey works on her own and earns \$2 for each thingamajig that she produces. Her neighbor, Henry, who also produces thingamajigs, observes that if the two of them combine forces and work as a team, they can sell their output for \$3 per unit, perhaps because she is better at some aspect of the process and he is better at another aspect. If they each have 5 units of effort to divide between working for themselves or as part of the team and agree to split the proceeds made as part of the team equally, Audrey's payoff can be expressed as $\pi_a = 2(5 - x_a) + 3(x_a + x_h)/2$, where x_a is the amount of effort that Audrey expends working as part of the team. Henry's payoff is similar: $\pi_h = 2(5 - x_h) + 3(x_a + x_h)/2$.

- (a) Write the game in normal form for the case in which the two can chose integer values of x between 0 and 5.
- (b) Can you use dominance to figure out how much effort each worker will devote to the team?
- (c) If you use calculus, can you generalize the result you found in part (b) to any level of x between 0 and 5?

(8) Operation Deadstick: During the invasion of Normandy in World War II, British forces were tasked with capturing two bridges behind the German front line. The decision as to how many troops to send on this mission (and how many troops the Germans should leave behind to guard the bridges)

surely depended on the perceived benefits of keeping the bridges intact and the costs of getting troops behind enemy lines for the British (and taking troops away from the front for the Germans). Let's illustrate this in the following simplified strategic setting: There is just one bridge, and the commanders of each side choose to send zero, one, or two platoons to this bridge. We will assume that the commander devoting more troops to the fight always wins and the benefit from winning is captured in w . In addition, the cost of losing the bridge battle is l , the payoff for a draw (i.e., when both commanders devote the same number of platoons) is 0, and the cost of each platoon committed to the battle is 1.

- (a) Write the normal form of the resulting game in terms of w , l , and the troop cost.
- (b) What has to be true about the values of w and l in order for the commanders to devote all their available forces?
- (c) For what values of w and l will the commanders devote no resources to the fight?

¹*An example:* Suppose that five players submit bids of \$80, \$25, \$72, \$98, and \$3. Then the player who submitted the highest bid of \$98 would win the auction, receive his private value, and pay the price of \$80 (the second-highest bid). The other players' payoffs would equal zero, since they neither won nor paid for the item.

²*Definition 1 stated in words:* The definition compares two strategies for player i : s_i and s'_i . For each possible profile of strategies played by the other players (s_{-i}), player i receives some payoff from playing s_i (i.e., $u_i(s_i, s_{-i})$) and some payoff from playing s'_i (i.e., $u_i(s'_i, s_{-i})$). If, for each possible profile of strategies chosen by the other players, s_i always yields a lower payoff than s'_i for player i , then we say that s_i is *strictly dominated* by s'_i .

³*Definition 3 stated in words:* This definition looks nearly identical to definition 1, except that the less than sign is replaced by a less than or equal to sign. Stated in words, s_i is *weakly dominated* by s'_i if, for any possible profile of strategies chosen by the other players, s_i never yields a greater payoff than s'_i . Furthermore, we require one of the inequalities to be strict, implying that in at least one instance, s_i yields a lower payoff. In other words, if s_i and s'_i always yield exactly the same payoff, we wouldn't say that one strategy is dominated by the other.

⁴*Important!* Notice that we always refer to an outcome by the strategy profile (in this case, (Coast, Coast)), and not the payoffs that the players receive from the outcome. If we instead described the prediction by referring to the resulting payoffs, this wouldn't necessarily tell us what the players actually do, since other strategy profiles may be associated with the same payoffs.

4

Equilibrium

We now turn to solving for rational play in games without appealing to the concept of dominance. In chapter 3, we saw that rational players will not play dominated strategies, and thus eliminating these strategies from consideration can sometimes lead us to a unique solution. But in many games, players will not have dominated strategies, and the tools from the previous chapter will not allow us to predict the game's outcome. In this chapter, we will learn to solve two-player, finite games by looking for *equilibrium* outcomes. Essentially, an outcome is in equilibrium if neither player has incentive to change their strategy, even if they were able to perfectly foresee the other person's strategy.

Imagine that you are a politician about to announce your candidacy for your party's nomination for president. Your announcement is expected to give you a significant bump in the polls. Meanwhile, your more established primary opponent is scheduled to give a speech the same day. She has three choices: to ignore you and focus on the general election (Ignore), to treat your candidacy seriously and focus on making herself more attractive to the party base (Challenge), or to praise you and endear herself to your supporters (Praise).

In turn, you will use your announcement to establish your platform: you can set an extreme agenda (which will allow you to play to your strengths if she Challenges you but make you look like a nut if she Ignores you), set a moderate agenda (which will allow you to be most attractive to the party if she Ignores you but look uninteresting if she Challenges you), or keep your speech vague and avoid committing to anything, which will give you a moderate gain in the polls no matter what she does. This interaction is written in normal form in [figure 4.1](#), where you are the Row player (Bernie) and your opponent is the Column player (Hillary). The payoffs represent the number of points you will gain (or she will lose) in the polls in each case. What should you do?

		Hillary		
		Challenge	Ignore	Praise
Bernie	Extreme	(10, −10)	(−5, 5)	(0, 0)
	Moderate	(0, 0)	(7, −7)	(1, −1)
	Vague	(4, −4)	(3, −3)	(2, −2)

Figure 4.1

A zero-sum game.

Zero-Sum Games without Dominance

The first thing to note about this game is that the process of eliminating dominated strategies does not get us anywhere. Bernie does not have a dominated strategy that he can eliminate. Instead, any of Bernie's strategies could be a good plan, depending on what he believes Hillary will do: If she Challenges, he prefers Extreme; if she Ignores, he does best with Moderate, and if she Praises, his best response is Vague. Likewise, Hillary does not have a dominated strategy, so dominance does not provide Bernie with any clues about Hillary's likely behavior.

Next, notice that in every cell, the two players' payoffs sum to 0, indicating that every point Bernie gains is a point lost for Hillary, and vice versa. Two-player games of this type are known as *zero-sum games*, a term that has entered the broader popular lexicon to describe situations in which one person cannot gain without harming someone else. But note that this interaction could just as easily be captured by payoffs that always sum to a certain number other than zero. For instance, if we added 50 to each person's payoffs, then each cell would sum to 100, perhaps reflecting the percentage of the vote share that each candidate is expected to get. We will use the term *zero-sum game* (also known as a *constant game*) as shorthand for an interaction in which two players' interests are strictly conflicting.

Definition 1 (Zero-sum game) *A zero-sum game (also called a constant sum game) is a two-player game in which the two players' payoffs from any outcome always sum to the same constant, c . That is, for any strategy profile (s_i, s_j) , $u_i(s_i, s_j) + u_j(s_i, s_j) = c$.*

The important thing to remember with zero-sum games is that one player's gain is the other player's loss, and vice versa. So if your goal is to maximize

your own payoff, this is equivalent to minimizing your opponent's payoff. Zero-sum games yielded some of the initial results of game theory, as the diametrically opposing incentives create a very clear strategic tension between the two players. You can think of the interaction as a game of tug-of-war over the payoffs, with each player working to gain ground at the expense of the other.

The first person to demonstrate the rational way to play zero-sum games was John von Neumann, who was one of the most prestigious mathematicians of the 20th century, making contributions to almost every subfield in mathematics. He developed expected utility theory (still the standard framework for studying choice under risk), worked on the Manhattan Project, is assumed to be the inspiration for the character Dr. Strangelove, and is considered a father of the modern digital computer. He was also a remarkably bad poker player. Presumably, he was skilled at computing probabilities and expected values—he was, after all, the founder of expected utility theory. What he struggled with, though, was the human element of the game: when to bluff, how to guess if someone else was bluffing, and so on. As any ultimate math nerd would, von Neumann set about mathematically modeling this human element of poker. In doing so, he soon realized that the same framework could be applied, not just to recreational games, but to bargaining, politics, and military strategies. In 1928, von Neumann proved a general theorem demonstrating that there is a rational way to play any zero-sum game.

This approach assumes that players use a simple, very conservative rule of thumb. The basic idea is that each player assumes that, for each strategy she could play, she will end up with the lowest possible payoff associated with that strategy. She then chooses the best among these worst-case scenarios. Put differently, each player acts to *maximize* their own *minimum* payoff. We will call this strategy *maximin*. Since we are considering zero-sum games, this is the same as saying that the player acts to *minimize* their opponent's *maximum* payoff. As a result, von Neumann's solution is known as the *minimax theorem*.

Cake Cutting

Although it sounds technical, the strategy of maximin is likely familiar to anyone who has regularly shared food with a sibling or friend. For example, imagine that Jeff's kids, Henry and Audrey, have a single piece of cake to eat.

In an attempt to limit squabbling, Jeff tells Audrey that she can divide the cake into two slices—but that Henry will get to choose which slice goes to which kid. Assuming that each kid's payoff is equal to the amount of cake that he or she gets, then we have a zero (constant) sum game. No matter what happens, the cake is not going to get any larger—and the kids certainly aren't going to waste any—so the sum of the kids' payoffs is always equal to the total amount of cake.

How should Audrey divide the cake? She anticipates that no matter what she does, Henry will always give her the smaller slice—not simply because he is her older brother, but because this is equivalent to giving himself the larger slice. Knowing, then, that she will always get the smaller of the two slices, her goal is to ensure that the smaller slice is as large as possible, which is done by cutting the cake as equally as possible. In other words, Audrey's behavior is consistent with maximizing the minimum amount of cake she could receive. She is playing her maximin strategy, as are you whenever you dish out plates as equally as possible to make sure that you don't end up with too little or, more generally, whenever you make a decision based on the worst-case scenario associated with each option.

Solving Games Using the Maximin/Minimax Approach

Let's apply the maximin approach to our game between Bernie and Hillary. Since we have a zero-sum game, Hillary's payoff is always defined as the negative of Bernie's payoff. This means that we can think about the whole game just in terms of Bernie's payoffs, with Bernie aiming to maximize them and Hillary aiming to minimize them. The game is printed here with the polling bump that Bernie will receive in each case. When analyzing zero-sum games, we will generally consider the game in terms of the payoffs for the Row player, who we will refer to as player 1. However, nothing about the solution would change if we thought about the outcomes in terms of the other player's payoffs instead.

Maximin: Maximizing Your Minimum Possible Payoffs

Now, let's see what happens when Bernie follows the maximin rule of thumb. Imagine that Bernie's campaign staff are quite pessimistic (or a bit paranoid). They are convinced that no matter which strategy they choose, Hillary's team will anticipate it and respond with the strategy that minimizes Bernie's payoff. (Why would she do this? Because it is a zero-sum game, and this means

maximizing her own payoffs.) If Bernie chooses Extreme, Hillary will choose Ignore, and he will receive a payoff of -5 . If he chooses Moderate, she will minimize his payoffs by choosing Challenge, leaving him with 0 . And if he chooses Vague, she will choose Praise, giving him a payoff of 2 . These pessimistic, worst-case scenario payoffs for each of his three strategies are written to the right of the game shown in [figure 4.2](#). If these indeed are the payoffs associated with each of his strategies, what should Bernie do? He should choose the row in which the worst-outcome payoff is the highest. In other words, Bernie wants to maximize his minimum payoff. When given the choice between Extreme (payoff $= -5$), Moderate (payoff $= 0$), and Vague (payoff $= 2$), Bernie's best option is to choose Vague. Choosing Vague guarantees Bernie a payoff of at least 2 points, no matter what Hillary does. Put differently, Vague gives the best payoff (a 2-point bump) that Bernie can achieve against an omniscient opponent. We call Vague Bernie's maximin strategy and this guaranteed payoff of 2 the maximin value of the game.

		Hillary			
		Challenge	Ignore	Praise	<i>Bernie's worst outcome</i>
Bernie	Extreme	10	-5	0	-5
	Moderate	0	7	1	0
	Vague	4	3	2	2 $\leftarrow$

Figure 4.2

Bernie maximizes his minimum payoff. Cells show Bernie's payoff from each outcome.

More formally, we define these terms as follows.

Definition 2 (Maximin) Consider a two-player zero-sum game between player 1 and player 2, with the payoffs expressed in terms of player 1. Player 1 chooses the (pure or mixed) strategy s_1 , while player 2 chooses the (pure or mixed) strategy s_2 . The maximin value of the game is

$$v_1 = \max_{s_1} \min_{s_2} u_1(s_1, s_2)$$

We call the strategy for player 1 that achieves this the maximin strategy for player 1. By playing the maximin strategy, player 1 can guarantee a payoff of at least v_1 for any strategy that player 2 chooses.

The $u_1(s_1, s_2)$ piece tells us that we are considering player 1's payoff from player 1 playing strategy s_1 and player 2 playing strategy s_2 . In our example, we consider Bernie's payoffs, and he is player 1. Working outward, the min term tells us that s_2 is chosen to minimize player 1's payoff. This is exactly what we were doing when we assumed that Hillary always chooses the column that gives Bernie the lowest payoff. Finally, the max term tells us that player 1 chooses his strategy s_1 to maximize his payoff, conditional on player 2 minimizing player 1's payoff. This is what we did when we assumed that Bernie chose the row that would give himself the highest payoff, given Hillary always chooses a column to give him the lowest payoff.

Minimax: Minimizing Your Opponent's Maximum Possible Payoffs

Next, what about Hillary's decision? Remember that the payoffs we're considering are Bernie's, and Hillary's goal is to minimize his payoff. Let's assume that Hillary is equally paranoid and likewise assumes that Bernie will anticipate any move she makes and act to maximize his payoff at her expense. For each of her strategies, the maximum payoff that Bernie could get is the worst-case scenario for her and is written below each of her columns. Faced with the choice of Challenge (Bernie gets 10), Ignore (Bernie gets 7), and Praise (Bernie gets 2), she would prefer Praise, allowing her to hold Bernie's gain in the polls to 2 points. Praise gives the best outcome that Hillary can guarantee herself, no matter what Bernie does. As we go through this logic in [figure 4.3](#), it is important to remember that the payoffs displayed are Bernie's (not Hillary's), so Hillary's objective is to minimize this payoff as Bernie attempts to maximize it.

		Hillary		
		Challenge	Ignore	Praise
Bernie	Extreme	10	-5	0
	Moderate	0	7	1
	Vague	4	3	2
<i>Hillary's worst outcome</i>		10	7	2
				↑

Figure 4.3

Hillary minimizes Bernie's maximum payoff. Cells show *Bernie's* payoff from each outcome.

Definition 3 (Minimax) Consider a two-player zero-sum game between player 1 and player 2, with the payoffs expressed in terms of player 1 and the two players' payoffs summing to c . Player 1 chooses the (pure or mixed) strategy s_1 , while player 2 chooses the (pure or mixed) strategy s_2 . The minimax value of the game is

$$v_2 = \min_{s_2} \max_{s_1} u_1(s_1, s_2)$$

We call the strategy for player 2 that solves this problem the minimax strategy for player 2. By playing the minimax strategy, player 2 can guarantee that player 1 receives a payoff of no more than v_2 for any strategy that player 1 chooses. Note that v_2 is in terms of player 1's payoffs, and so this is equivalent to saying that player 2 is guaranteed a payoff of at least $c - v_2$.

The Solution to Zero-Sum Games: Maximin Playing against Minimax

To recap, we have seen that Bernie has a strategy (Vague) that guarantees him a payoff of at least 2, no matter what Hillary does. In turn, Hillary has a strategy (Praise) that guarantees she won't lose more than 2 to Bernie, no matter what he does. It is therefore logical that two rational players engaged in a tug-of-war battle over payoffs would necessarily arrive at the outcome (Vague, Praise), with Bernie gaining 2 points in the polls. The only way Bernie could do better is for Hillary to accept a payoff worse than what she could guarantee herself by simply playing her conservative minimax strategy of Praise. This would clearly be a mistake on Hillary's part and not consistent with rational play. Likewise, it is impossible for Hillary to do better than losing 2 points in the polls without Bernie making a mistake and moving away from his maximin strategy.

Definition 4 (Minimax solution) Consider a two-player zero-sum game, in which the two players' payoffs sum to c . The minimax solution to the game is a strategy profile, (s_1^*, s_2^*) , and a value V such that

- (1) s_1^* is player 1's maximin strategy
- (2) s_2^* is player 2's minimax strategy
- (3) The minimax and maximin values coincide, and we call this number the value of the game, $V = v_1 = v_2$. Under the minimax solution, player 1's payoff is V and player 2's payoff is $c - V$.

Note that a requirement of the minimax solution is $v_1 = v_2$. This is essentially telling us that there is no slack in our game of tug-of-war. One

player can't do any better than his guaranteed payoff without the other player doing worse than her own guaranteed payoff. If this equality doesn't hold, then we have not found a minimax solution. We call this number at which the maximin and minimax coincide the *value* of the game. It is the outcome that the players can guarantee themselves (in terms of player 1's payoffs) by playing the game rationally.

We've seen that one reasonable approach for players who do not know what the opponent will do is to act to guarantee themselves the value V . Another way to think about whether the minimax solution is rational is to ask whether both players would be happy with their own chosen strategy, given the behavior of the other player. If both play according to the minimax solution, would anyone have regrets? By looking at the payoff matrix, we can easily verify that if Bernie plays Vague and Hillary chooses Praise, neither player would regret the decision. If Hillary plays Praise, Bernie does best with Vague; if Bernie plays Vague, then Hillary does best with Praise. The minimax solution to zero-sum games is therefore *stable*, in the sense that neither player would want to revise their choice, even if they knew the other player's strategy.

Worked Examples

The worked examples here show the minimax solution in two slightly different games. In the first, imagine that the Offense in a baseball game is down by 1 run in their final opportunity at bat, and that they have a runner on third base. They have three options: the batter can swing away (Swing), bunt (Bunt), or bunt while the runner simultaneously runs home (Squeeze). At the same time, the Defense can either throw a pitch that is hittable (Hittable) or throw a pitch that the batter could not hit and potentially try to catch the runner heading home (Unhittable). Imagine that baseball analysts have estimated the likelihood that the Offense will manage to tie the game in each case and these probabilities represent the payoffs for Offense in the game shown in example 1, seen in [figure 4.4](#). In this case, the Defense's payoffs are equal to the probability that the Offense *does not* tie the game (i.e., 1 minus the Offense's payoff). Thus the game is zero-sum and we can analyze it using the minimax procedure. The Offense wants to maximize these payoffs (the probability they tie the game), and we see that the Offense's maximin strategy is Bunt (guaranteeing a likelihood of tying of at least 0.30). In contrast, the Defense wants to minimize the probability that Offense scores. The Defense's

minimax strategy is Hittable (guaranteeing that the Offense’s likelihood of tying is no more than 0.30). The minimax solution is therefore (Bunt, Hittable), and the value of the game is 0.30.

		Defense		
		Hittable	Unhittable	<i>Offense's worst outcome</i>
Offense	Swing	.25	.30	.25
	Bunt	.30	.32	.30 ←
	Squeeze	.15	.01	.01
<i>Defense's worst outcome</i>		.30	.32	
		↑		

Figure 4.4

Example 1: Minimax solution. Cells give Offense’s payoff from each outcome.

You may wonder whether the minimax solution is always unique or if there can be more than one outcome that satisfies our definition. This is addressed in example 2. When looking for player 1’s maximin strategy in this example, we find that there is a tie in player 1’s worst-case scenario payoffs. Both Up and Down are maximin strategies for player 1. There are thus two minimax solutions in this game: (Up, Right) and (Down, Right). However, both lead to the same payoffs for the players and are associated with the same value (− 2). This is the case more generally. More than one strategy profile may be a minimax solution of the game, but each will be associated with the same payoff outcome for the players. Therefore, the value of the game is always unique.

		Player 2		
		Left	Right	<i>1's worst outcome</i>
Player 1	Up	3	-2	-2 ←
	Middle	-4	-3	-4
	Down	5	-2	-2 ←
<i>2's worst outcome</i>		5	-2	
			↑	

Figure 4.5

Example 2: Minimax solutions.

Do the Maximin and Minimax Payoffs Always Correspond?

Is it just a coincidence that the maximin and minimax overlap in the three examples that we've seen so far? Consider again the Princess Bride game between Westley and Vizzini, in which Westley chooses whose cup to poison and Vizzini chooses whose cup to drink from, as shown in [figure 4.6](#). As in chapter 2, we'll consider the simplified game, in which Westley does not have an immunity to iocaine powder. Since exactly one player in the game survives, it is zero-sum, with Westley's payoff equal to 1 if he is the survivor and -1 if Vizzini is the survivor. To find the maximin payoff, we look for the worst-case scenario for Westley associated with each of his strategies, and find that the worst-case payoff is equal to -1 in each case. The best outcome that Westley can guarantee himself in this game is -1 , which he can do no matter which action he takes. Likewise, the best that Vizzini can guarantee is that Westley earns a payoff of 1 (i.e., Vizzini dies). Thus, $v_1 = -1 \neq v_2 = 1$.

		Vizzini		
		Sip W.'s	Sip V.'s	<i>Westley's worst outcome</i>
Westley	Poison W.'s	1	-1	-1 ←
	Poison V.'s	-1	1	-1 ←
<i>Vizzini's worst outcome</i>		1	1	
		↑	↑	

Figure 4.6

The Princess Bride game.

What is the solution? We will see in chapter 5 that the impasse can be broken by allowing players to use strategies in which they randomize over their possible actions. Once we open up the set of strategies that the players can consider to include mixed strategies, we will see that it is indeed the case that the maximin and minimax payoffs are aligned. Furthermore, this outcome is stable, in the sense that neither player could achieve a higher payoff given the other player's strategy. This is the theorem that John von Neumann proved in 1928.

General-Sum Games and the Collapse of the Minimax Approach

Thus far, we have seen a technique to solve for rational play in zero-sum games, where one player's gain is always the other's loss. However, this is a relatively narrow class of two-player games, and in many interactions, including the prisoner's dilemma, both players are better off under some outcomes relative to others. Let's now relax the assumption that the players' interests are in direct conflict. Returning to the Bernie-Hillary game, we will keep Bernie's payoffs the same but now allow Hillary's payoffs to vary beyond the negative of his payoffs, as shown in [figure 4.7](#). (For instance, imagine that Hillary has interests that go beyond minimizing Bernie's gain in the polls, such as supporting a specific platform, being more viable in the general election, or improving fundraising.) See the table shown here.

		Hillary		
		Challenge	Ignore	Praise
Bernie	Extreme	$(10, -2)$	$(-5, 0)$	$(0, 3)$
	Moderate	$(0, 4)$	$(7, 5)$	$(1, -1)$
	Vague	$(4, 3)$	$(3, 1)$	$(2, 2)$

Figure 4.7

A general-sum game.

Since Bernie's payoffs are the same, his maximin strategy is still Vague, which guarantees him a payoff of at least 2. Likewise, if Hillary acts to minimize Bernie's maximum payoff, she would still play Praise. But does she have any reason to do this? Even if she is able to anticipate his every move, she wouldn't necessarily want to choose the action that minimizes his payoff.

(For instance, if he chose Vague, she actually does the best by choosing Challenge, which gives him the best-case, rather than the worst-case, payoff associated with Vague.) When the players' payoffs are not diametrically opposed, it is no longer reasonable to assume that the other player will always act to minimize your payoffs and give you the worst-case outcome.

You may next wonder: What if Hillary adopted the conservative rule of thumb for *her own payoffs*, choosing the strategy for which her worst-case outcome was the best? If she Challenges, she could find herself with a payoff of -2 , and if she Praises, she could face a payoff of -1 , but if she Ignores, she is guaranteed to at least avoid a negative payoff, getting a payoff of at least 0 . Therefore, if both players play maximin over their own payoffs, we arrive at (Vague, Ignore). Would either player regret the chosen strategy if this were the outcome? Yes, we've already seen that Hillary would prefer Challenge if Bernie is choosing Vague. Furthermore, Bernie would prefer Moderate if Hillary is choosing Ignore. If players follow this conservative strategy in general-sum games, one or both may find they regret their choice. Maximizing the worst-case outcome may be best thought of as a *behavioral strategy* that players may adopt as a rule of thumb if they're unsure what to do, but it is not necessarily consistent with rational play beyond zero-sum games.

The logic of minimax thus breaks down when the players' interests are not diametrically opposed. The question of how to solve for rational play in more general games went unsolved until a first-year graduate student named John Nash walked into his advisor's office with a proof demonstrating the existence of a stable outcome in general-sum games. His paper was published in 1950, and ever since, the *Nash equilibrium* has been considered the primary solution concept in game theory.

Nash's Contribution

In the previous chapter, we saw solutions based on identifying which strategies players would *not* choose, and in this chapter, we saw solutions based on a simple rule of thumb that is reasonable in zero-sum games. We'll now take a broader approach, which works for any game regardless of the payoffs, and ask which strategy or strategies a player would prefer in response to each specific strategy that the other player could choose. We will continue to focus on two player, finite games in this chapter, and then expand this same

approach to games with continuous strategy sets and any arbitrary number of players in chapter 6.

The answer to the question of what Bernie should do clearly depends on what he believes Hillary will do. This is a fundamental aspect to most games of strategy. If he believes that Hillary will choose Challenge, it is clear that he does best with Extreme. If he believes that Hillary will choose Ignore, he should choose Moderate. And if he believes that Hillary will choose Praise, his best response is Vague. In fact, maybe he isn't sure what she'll do, and he believes each of her actions are equally likely. In this case, he is best off choosing Vague, as it gives the highest average payoff. We'll allow for probabilistic beliefs in the next chapter.

This reasoning illustrates the approach that we will take for the rest of this chapter—and going forward—to solve for rational play. How should a rational player approach this game? First, he should *form beliefs* about what the other player will do, and then *best respond* to those beliefs by choosing the strategy that would maximize his payoff in that case. Essentially, the Nash equilibrium is an outcome in which each player has correct beliefs about the other's strategy and, given what he knows the other will do, acts to maximize his own payoffs.

Best Responding

To determine the Nash equilibrium strategy profile, we first take stock of how each player would prefer to respond to each possible strategy played by her counterpart. We call the strategy that makes the player best off, conditional on the other player choosing x , the player's best response to x .

Definition 5 (Best response) *Player i 's strategy $s_i \in S_i$ is a best response to the other player's strategy, s_{-i} , if $u_i(s_i, s_{-i}) \geq u_i(s'_i, s_{-i})$ for all $s'_i \in S_i$. We write $s_i = BR_i(s_{-i})$.*

Stated in words, imagine that the other player is choosing s_{-i} . A strategy s_i is a best response if it gives at least as high a payoff as any alternative strategy s'_i .

Finding a player's best response in a normal-form game is simply a matter of inspection. To find player 1's best responses, we will go, one at a time, through each of player 2's columns. For each column, we will identify the row that gives player 1 the highest payoff. That is player 1's best response to player 2 choosing that particular column. Then we'll underline that payoff to mark it for later. In the game shown on the left in [figure 4.8](#), what is player 1's

best response to player 2 choosing Left? If player 1 knew that player 2 were choosing the Left column, then player 1's choice is simply whether he wants to have a payoff of 5 (which he'll get from Up), 4 (from Center), or 3 (from Down). Clearly, Up is the best response, yielding the highest-possible payoff conditional on player 2 choosing Left: $BR_1(Left) = Up$. We'll underline the payoff of 5 in this case to indicate that Up is a best response to Left. Similarly, we find that $BR_1(Middle) = Up$. What about the best response to Right? Here, we have a tie for the best choice if player 2 chooses Right, with both Center and Down yielding a payoff of 3. Therefore, there may be more than one best response if multiple strategies each lead to the same payoff conditional on the other player's behavior. In this case $BR_1(Right) = \{Center, Down\}$, and we underline both payoffs of 3.

		Player 2		
		Left	Middle	Right
Player 1	Up	(<u>5</u> , 5)	(<u>6</u> , 7)	(2, 7)
	Center	(4, 3)	(4, 5)	(<u>3</u> , 6)
	Down	(3, 10)	(4, 0)	(<u>3</u> , 1)

		Player 2		
		Left	Middle	Right
Player 1	Up	(5, 5)	(6, <u>7</u>)	(2, <u>7</u>)
	Center	(4, 3)	(4, 5)	(3, <u>6</u>)
	Down	(3, <u>10</u>)	(4, 0)	(3, 1)

Figure 4.8

Player 1's best responses to player 2's strategies (left) and player 2's best responses to player 1's strategies (right).

What about the best responses for player 2? In this case, we go through each row that player 1 could choose and identify the column that gives player 2 the highest payoff in the row. Then we underline player 2's payoff in that cell. This is done in the matrix to the right. Please confirm that $BR_2(Up) = \{Middle, Right\}$, $BR_2(Center) = Right$, and $BR_2(Down) = Left$.

Let's look at one more example, this time including the best responses for each player in the same matrix, as shown in [figure 4.9](#). Confirm that Sid's best responses to Nancy are indicated by his underlined payoffs, and Nancy's best responses to Sid are indicated by her underlined payoffs.

		Nancy		
		A	B	C
Sid	X	(1, 1)	(<u>3</u> , <u>3</u>)	(5, 0)
	Y	(5, <u>4</u>)	(2, 0)	(4, 3)
	Z	(<u>7</u> , 0)	(1, <u>8</u>)	(<u>6</u> , 1)

Figure 4.9

Both players' best responses.

Nash Equilibrium: Mutual Best Responses

Now that we know each player's best response to each of the other player's possible strategies, we look for overlap: a strategy profile that allows both players to best respond to each other at the same time. A strategy profile is a Nash equilibrium if player 1's strategy is a best response to player 2's strategy and player 2's strategy is a best response to player 1's strategy. In other words, neither player wants to *unilaterally deviate* by adjusting his or her own strategy *given what the other player is doing*. What the other player does is out of the player's control, but a rational player will always anticipate it and best respond.

Definition 6 (Nash equilibrium (two-player game)) A strategy profile (s_1^*, s_2^*) is a Nash equilibrium if $u_1(s_1^*, s_2^*) \geq u_1(s_1', s_2^*)$ for all $s_1' \in S_1$ and $u_2(s_1^*, s_2^*) \geq u_2(s_1^*, s_2')$ for all $s_2' \in S_2$. In other words, $s_1^* = BR_1(s_2^*)$ and $s_2^* = BR_2(s_1^*)$.

In a two-player, normal-form game, we can find the Nash equilibrium by looking at the payoffs that we underlined during our analysis of the players' best responses. In the game shown in [figure 4.9](#) between Sid and Nancy, for example, we see many cells with one number underlined, indicating that one player is best responding to the other. However, there is exactly one cell in which both players' payoffs are underlined: the cell associated with Sid playing X and Nancy playing B. Only in this cell are both players best responding to each other. (X, B) is therefore the unique Nash equilibrium of this game.

We can think about the approach that players take to games as a two-stage process: players form beliefs about what the other player will do, and then they decide on their own strategy. If the players' strategies form a Nash equilibrium, then two things must be true: (1) each player best responds to his or her beliefs and (2) those beliefs are always correct. Once someone has

formed beliefs regarding the other player's strategy, determining the best response is, as we've seen, relatively straightforward. It reduces the situation to a one-person decision problem, and determining the best path is simply a matter of calculation rather than strategy. The Nash equilibrium is equivalent to both players having perfect foresight: they are always able to correctly anticipate the other player's strategy and act to maximize their payoffs in response.

This means that if the players choose a Nash equilibrium strategy profile (s_1, s_2) , then we could go to player 1 and say "Pssst. Player 2 is playing s_2 . Are you sure that you still want to play s_1 ?" and player 1 would say "Yes!" Likewise, if we tipped off player 2 that player 1 was playing s_1 , player 2 would still be happy sticking with s_2 . *Perfect foresight* means that learning the other player's strategy provides no new information.

We've seen that (X, B) is the unique Nash equilibrium in the Sid and Nancy game. Now look at any other cell in the matrix. If we ended up there instead, would the players have regrets and want to change their choice if they were to learn the other player's strategy? For example, take a look at the cell associated with (Y, A). Both players do better than under the Nash equilibrium of (X, B). Why isn't this an equilibrium? Nancy is happy choosing A if Sid chooses Y. But if Nancy chooses A, then Sid would prefer to choose Z. The outcome (Y, A) is not stable because Sid would regret playing Y if he knew that Nancy chose A. This cannot be an equilibrium because Sid is not best responding to Nancy.

Is the Nash Equilibrium Unique?

Let's look again at the first game on which we conducted our best response analysis, but now include the best responses for each player on a single matrix, shown in [figure 4.10](#), so it is easier to look for mutual best responses. What is the Nash equilibrium in this game?

		Player 2		
		Left	Middle	Right
Player 1	Up	(<u>5</u> , 5)	(<u>6</u> , 7)	(2, <u>7</u>)
	Center	(4, 3)	(4, 5)	(<u>3</u> , <u>6</u>)
	Down	(3, <u>10</u>)	(4, 0)	(<u>3</u> , 1)

Figure 4.10

A game with multiple Nash equilibria.

Here, we see that there are two Nash equilibria: (Up, Middle) and (Center, Right). The Nash equilibrium is *not unique*, and there may be multiple strategy profiles that satisfy our definition. You may notice that both players are actually better off under the first outcome than the second. Even though both players would prefer (Up, Middle), neither would want to unilaterally deviate away from (Center, Right) in order to get there: If player 2 plays Right, player 1 doesn't want to deviate to Up, and if player 1 plays Center, player 2 doesn't want to deviate to Middle. In chapter 7, we will take up the question of whether some Nash equilibria are more likely than others to occur. You may also notice that both Nash equilibria in this example involve some degree of indifference on the part of one of the players. For instance, if player 2 believes that his counterpart will choose Up, he is equally happy with Middle and Right. If we told player 2 that player 1 is playing Up and asked if he wanted to stick with Middle or switch to Right, his answer would not be an unqualified "Yes!" but a more ambivalent "Eh, whatever." What is important is that player 2 doesn't have an incentive to switch away from Middle—he may be indifferent, but he wouldn't have any regrets about playing Middle if player 1 chose Up.

What Is the Relationship between the Nash Equilibrium and the Minimax Solution?

What is the relationship between the Nash equilibrium and the minimax solution for zero-sum games? These two solution concepts perfectly overlap whenever we have a zero-sum game. Let's return to the original Bernie-Hillary game, where Hillary's payoffs were the negative of Bernie's. The underlines below the payoffs in the game seen in [figure 4.11](#) show the best response for Bernie to each of Hillary's strategies and the best response for Hillary to each of Bernie's strategies. It also illustrates the connection to the

minimax approach. Let’s think about what we’re doing when we find Bernie’s best response. For each column that Hillary can choose, we find Bernie’s row that maximizes his payoffs. In doing so, we are finding Hillary’s worst-case scenario (i.e., Bernie’s best-case scenario) for that column. Note that Bernie’s underlined payoff in each column perfectly aligns with Hillary’s worst-case payoff listed below the column. This is only true because this is a zero-sum game, and his best response aligns with her worst-case scenario. Likewise, when we find Hillary’s best response, we look at each row that Bernie can choose and identify the column that maximizes her payoff, which is the worst-case scenario for Bernie. The cell with two underlines indicates the Nash equilibrium, but also where the minimax and maximin strategies coincide.

		Hillary			
		Challenge	Ignore	Praise	<i>Bernie’s worst outcome</i>
Bernie	Extreme	(<u>10</u> , −10)	(−5, <u>5</u>)	(0, 0)	−5
	Moderate	(0, <u>0</u>)	(<u>7</u> , −7)	(1, −1)	0
	Vague	(4, −4)	(3, −3)	(<u>2</u> , <u>−2</u>)	2 ←
<i>Hillary’s worst outcome</i>		10	7	2	
				↑	

Figure 4.11
Nash equilibria in the Bernie-Hillary game.

What Is the Relationship between the Nash Equilibrium and Dominance-Solvable Outcomes?

The Nash equilibrium allows us to make predictions about behavior in games that cannot be solved using the concept of dominance. But when a game *is* dominance solvable, what is the relationship between the outcome we reach through iterated elimination of dominated strategies and the Nash equilibrium?

First, notice that if a strategy is strictly dominated, then it cannot be chosen as part of a Nash equilibrium. This is because, by definition, a strictly dominated strategy always yields a worse outcome than the strategy that dominates it, and therefore can never be the best response. By the same token, when a player has a strictly *dominant* strategy, this is the only strategy that she could play in any Nash equilibrium. Why? If a player’s strategy *x* strictly

dominates all others, this means that x is the best response to every strategy the other player could choose. No matter what the other player does, x must always lead to the highest payoffs. Since the Nash equilibrium requires the player to best respond, this means that she must play x . As an example, let's find the Nash equilibrium in the prisoner's dilemma game from the previous chapter, seen in [figure 4.12](#). Our best response analysis indicates that Confess is always the best response for each player and (Confess, Confess) is the unique Nash equilibrium.

		Blue player	
		Silent	Confess
Red Player	Silent	(8, 8)	(0, <u>10</u>)
	Confess	(<u>10</u> , 0)	(<u>3.5</u> , <u>3.5</u>)

Figure 4.12

Nash equilibrium in the prisoner's dilemma.

In addition, if we arrive at a unique outcome using iterated elimination of *strictly* dominated strategies, then we know that this outcome is also the unique Nash equilibrium of the game. One way to think about this is to ask: as we go through the process of eliminating strictly dominated strategies, could we ever eliminate a cell that is a best response for both players? No. A cell could get eliminated only if, for at least one player, it provides a strictly worse outcome than another cell that the player could switch to (given the other person's strategy and the remaining outcomes). This could not be the case if the cell is a best response for both players. Therefore, a Nash equilibrium would never get deleted by eliminating strictly dominated strategies, and if we're left with a unique outcome, it must be the unique Nash equilibrium.

What if we also eliminate *weakly* dominated strategies? Let's look again at the game in [figure 4.10](#), which we saw earlier had two Nash equilibria. There are no strictly dominated strategies, but both players have a weakly dominated strategy. For player 1, Down is weakly dominated by Center and, for player 2, Middle is weakly dominated by Right. Once we eliminate these, we observe that Left is strictly dominated by Right. Once we eliminate Left, we observe that Up is strictly dominated by Center. Therefore, the dominance-solvable outcome is (Center, Right), which was one of the Nash equilibria that we identified. But what about the other one? (Up, Middle) was also a Nash

equilibrium, but we eliminated Middle in the very first step. The reason is based on the indifference of player 2. If player 1 chooses Up, player 2 is indifferent between Middle and Right, and both are best responses. Therefore, (Up, Middle) can be a Nash equilibrium. But if player 1 chooses anything other than Up, player 2 does worse with Middle compared to Right, indicating that it is weakly dominated, which is why we deleted it.

To summarize, if we find a dominance-solvable outcome, it must be a Nash equilibrium. If we find it by only ever eliminating strictly dominated strategies, then it is the unique Nash equilibrium. However, if we eliminate weakly dominated strategies as well, then there may be additional Nash equilibria that get eliminated in the process.

Does the Nash Equilibrium Always Exist?

The theorem that John Nash proved tells us that for any game in which players have a finite set of pure strategies, there exists at least one Nash equilibrium. However, there may be cases in which conducting our best response analysis on a normal-form game does not uncover any cells that are simultaneously best responses for both players. The Westley and Vizzini game shown in [figure 4.13](#) is an example. If Westley's only choices are to poison his own cup or poison Vizzini's, and Vizzini's only choices are to sip from his own cup or sip from Westley's, there can be no stable outcome. In the cells in which Vizzini guesses correctly (the top right and bottom left), Westley would want to change which cup he poisoned, and in cells in which Vizzini guesses wrong (the top left and bottom right), Vizzini would want to change which cup he drank from. However, Nash's theorem tells us that there is an equilibrium in this case, and finding it will be the topic of the next chapter.

		Vizzini	
		Sip W.'s	Sip V.'s
Westley	Poison W.'s	(<u>1</u> , -1)	(-1, <u>1</u>)
	Poison V.'s	(-1, <u>1</u>)	(<u>1</u> , -1)

Figure 4.13

Best responses in the Princess Bride game.

Exercises

(1) In the game seen in [figure 4.14](#), find the best responses for each player. That is, find $BR_{Row}(L)$, $BR_{Row}(C)$, $BR_{Row}(R)$, $BR_{Column}(U)$, $BR_{Column}(M)$, and $BR_{Column}(D)$.

		Column		
		L	C	R
Row	U	(4, 4)	(3, 5)	(2, 3)
	M	(1, 2)	(2, 2)	(1, 11)
	D	(5, 1)	(1, 5)	(0, 0)

Figure 4.14

(2) For each of the games seen in [figure 4.15](#), find the Nash equilibrium or equilibria. If you instead use the iterated elimination of weakly dominated strategies, do you arrive at a unique solution? If so, compare this solution to the equilibria you found.

(a)

		Column	
		A	B
Row	C	(7, 7)	(50, 50)
	D	(10, 10)	(0, 0)
	E	(30, 30)	(10, 10)

(b)

		Column			
		w	x	y	z
Row	a	(5, 3)	(4, 6)	(6, 3)	(8, 1)
	b	(5, 4)	(9, 3)	(9, 4)	(6, 1)
	c	(6, 4)	(8, 1)	(7, 4)	(10, 3)
	d	(7, 5)	(6, 2)	(8, 1)	(11, 6)

(c)

		Column				
		a	b	c	d	e
Row	A	(50, 2)	(25, 2)	(-5, 5)	(0, 30)	(-1, 20)
	B	(30, 3)	(5, 5)	(3, 10)	(30, 2)	(2, 7)
	C	(40, 1)	(75, 0)	(2, 2)	(5, 0)	(0, 5)
	D	(1, -20)	(-3, 50)	(0, 25)	(1, -15)	(-1, 10)
	E	(-16, 0)	(1, -10)	(0, 75)	(0, 40)	(-10, 50)

Figure 4.15

(3) For each of the games in [figure 4.16](#), find the minimax solution (including the game's value). Use best response analysis to confirm that this strategy profile is also a Nash equilibrium.

(a)

		Column	
		A	B
Row	C	$(5, -5)$	$(10, -10)$
	D	$(2, -2)$	$(3, -3)$

(b)

		Column		
		L	C	R
Row	U	$(4, -4)$	$(3, -3)$	$(2, -2)$
	M	$(-2, 2)$	$(0, 0)$	$(-4, 4)$
	D	$(5, -5)$	$(-2, 2)$	$(0, 0)$

(c)

		Column		
		F	G	H
Row	X	$(0.1, 0.9)$	$(0.4, 0.6)$	$(0, 1)$
	Y	$(0.3, 0.7)$	$(0.5, 0.5)$	$(0.8, 0.2)$
	Z	$(0.2, 0.8)$	$(0.7, 0.3)$	$(0.9, 0.1)$

Figure 4.16

(4) Consider the prisoner's dilemma game discussed in the previous chapter, in which players receive an additional payoff of x if they choose Confess when the other player chooses Confess or if they choose Silent when the other player chooses Silent (see figure 4.17).

		Blue Player	
		Silent	Confess
Red Player	Silent	$(8 + x, 8 + x)$	$(0, 10)$
	Confess	$(10, 0)$	$(3.5 + x, 3.5 + x)$

Figure 4.17

- (a) How high must x be for the strategy of Silent not to be weakly dominated by Confess?
- (b) In this case, what constitutes a Nash equilibrium of the game?
- (5) Consider again the modified version of the traveler's dilemma from exercise 5 in chapter 3. Imagine that each traveler has only three possible claims they can submit: \$50, \$75, or \$100.

- (a) Write the game in normal form, assuming that there is a \$40 reward for submitting a lower claim and \$40 penalty for submitting a higher claim. Use best response analysis to find the Nash equilibrium.
 - (b) Let's now assume that players are concerned about fairness, and each receives an additional payoff of 20 if their claims match. Find the Nash equilibrium in this game.
- (6) Imagine that Jeff comes up with a new procedure for Audrey and Henry to divide the final piece of cake. Each player simultaneously announces a "demand": either zero, one-quarter, one-half, three-quarters, or all the cake. If the demands are mutually feasible (i.e., they sum to 1 or less), then each kid gets exactly the proportion he or she demanded. If the demands sum to more than 1, then neither kid gets anything, and Jeff eats the cake. Assume that each kid's payoff is equal to the proportion of the cake that he or she gets.
- (a) Write the game in normal form. Is this version of the "divide the cake" game still zero sum (constant sum)?
 - (b) If we used iterated elimination of weakly dominated strategies, what strategies would remain for each child?
 - (c) Find the Nash equilibrium of the game.
 - (d) Now imagine that each kid gets an additional disutility of 2 if his or her sibling gets more cake than they do (i.e., you would have to give the kid two complete cakes to make up for this indignity). What is the Nash equilibrium?

5

Mixed Strategies

Thus far, we have solved for rational play in simultaneous games by assuming that players' strategies involve choosing one action from their available set. In this chapter, we will consider situations when players can play mixed strategies by randomizing over their available actions. We will consider the scenarios in which players want to do this (i.e., when is a mixed strategy a best response?) and learn how to solve for Nash equilibria with mixed strategies. The key insight is that a rational player will be willing to randomize over actions only when she is indifferent between those actions, given the strategy of the other player.

Imagine that you have the opportunity to play a card game against another player. You and your opponent are each given four playing cards—Joker, Ace, Two, and Three—and you must simultaneously select one of them. If both players select the Joker, you win \$5 from your opponent, while, if exactly one person plays Joker, then you lose and must pay \$5. If neither player chooses Joker, then you lose if the two cards match but win if the cards are mismatched. What strategy do you choose?

At first, you may think that Joker seems unlikely to be profitable—you only win if your opponent chooses Joker, and you lose in the other three cases. But if your opponent anticipated that you were avoiding the Joker card, then he could guarantee a win by playing Joker himself—in which case you would find Joker to be your best bet. But then if he anticipated you were using Joker, he would avoid it entirely, and you would always lose. How does this cycle end? In this chapter, we will learn how to solve for equilibrium play in two-player, finite games when there does not seem to be any stable outcome—being able to anticipate the other player's move means that at least one player would want to switch strategies. We will first consider a simpler situation, in which both players have only two actions, and then return to this card game later in the chapter.

The Trouble with Perfect Foresight

In a classic Sherlock Holmes story, Sherlock and his sidekick, Watson, are fleeing the evil Professor Moriarty. They have boarded a train from London to Dover, and from there they can take a ferry to France and escape through continental Europe. But just as the train is pulling away from the station, Moriarty spots them from the platform. Sherlock has described Moriarty as being “on the same intellectual plane as myself”—which is to say, the highest one—and so in determining Moriarty’s move, he explains that he simply needs to consider what he would do in the same situation. Sherlock reasons that Moriarty will hire a special, faster train to beat them to Dover and will be waiting to kill them when they arrive. Therefore, Sherlock and Watson get off at the only intermediary stop, Canterbury, and watch as Moriarty’s special train whizzes past them. Thus they survive another day.

However, if Moriarty truly were on the same intellectual plane, wouldn’t he have made the same calculations about his opponent and realized that Sherlock would stop in Canterbury instead? An economist named Oskar Morgenstern was fascinated by this problem. His interest was not in why Moriarty failed to foresee Holmes’s move in the story, but rather what it would mean if he had. In this case, Moriarty would have traveled to Canterbury, but then Sherlock should have foreseen that and stuck with his original plan to go to Dover, in which case Moriarty should have foreseen that and traveled to Dover, and so on. “Because of so much thinking,” Morgenstern wrote, “they might not have been able to act at all or the intellectually weaker of the two would have surrendered to the other in the [London] station, since the whole flight would have become unnecessary” (Morgenstern, 1935).

Recall our definition of equilibrium from chapter 4: neither player has an incentive to unilaterally change his own strategy, even if he were to perfectly foresee the other player’s strategy. In this scenario, however, stability and perfect foresight seem incompatible: if the players always know the station that the other is going to, one of them would want to shift his own plan. After posing this problem to a number of academics, Morgenstern eventually encountered John von Neumann (whom we met in the last chapter), who immediately provided a solution. His solution was to allow the players to randomize which station they are going to. Then, even if they know each other’s random strategies, they still don’t know which station the other

character is going to, and they don't necessarily have an incentive to change their own strategy.

This initial conversation between von Neumann and Morgenstern sparked a collaboration and they went on to write the very first game theory book, *Theory of Games and Economic Behavior* (1944). The book includes the zero-sum version of the Holmes-Moriarty game presented in [figure 5.1](#). Each player chooses whether to go to Canterbury or to Dover, and thus $S_S = S_M = \{\text{Canterbury}, \text{Dover}\}$. In the previous chapters, these two options would be the only two possible strategies for the players. In this chapter, we will distinguish between pure strategies and mixed strategies. A *pure strategy* involves playing a specific element of one's strategy set (such as Canterbury) with certainty, and this is equivalent to what we've been thinking of as a strategy thus far. The table in [figure 5.1](#), therefore, displays the possible outcomes associated with each player choosing a pure strategy. If they match, Moriarty kills Sherlock, but if they mismatch, Sherlock escapes. In addition, Sherlock prefers to escape in Dover, where he can easily flee England by boat. With the game in normal form, we can clearly see the problem that had baffled Morgenstern. If we conduct our best response analysis by underlining each player's best responses, we see that there is no cell in which the players are simultaneously best responding. However, this does not mean that a Nash equilibrium does not exist. We know from Nash's equilibrium theorem that an equilibrium always exists in finite games. It simply implies that there is not a Nash equilibrium in which both players are choosing pure strategies. To understand von Neumann's solution, we will therefore need to expand our best response analysis to allow players to play pure strategies probabilistically by using *mixed strategies*.

		Sherlock	
		Dover	Canterbury
Moriarty	Dover	(<u>100</u> , −100)	(0, <u>0</u>)
	Canterbury	(−50, <u>50</u>)	(<u>100</u> , −100)

Figure 5.1
The Holmes-Moriarty game.

A mixed strategy involves the player randomly selecting from among his pure strategies according to a specific probability assigned to each of the

player's pure strategies.¹ For example, one mixed strategy for Sherlock is to choose Dover and Canterbury each with probability 0.50, perhaps by flipping a coin and continuing to Dover if it comes up heads, but disembarking at Canterbury if it comes up tails. But Sherlock also has an infinite array of other mixed strategies available to him, such as Dover with probability 0.05 and Canterbury with probability 0.95, Dover with probability 0.651 and Canterbury with probability 0.349, and so on. In this game, Sherlock has two pure strategies, so a mixed strategy specifies a probability for each of them, with the only restriction being that the two probabilities are nonnegative and sum to 1. In the game rock-paper-scissors, each player has three pure strategies, and so a mixed strategy would specify the probability assigned to Rock, the probability assigned to Paper, and the probability assigned to Scissors, with the three numbers summing to 1. This is stated formally in definition 1.

Definition 1 (Mixed strategy) *A mixed strategy for player i is a probability distribution over the player's set of pure strategies, S_i . Imagine that player i has a set of m pure strategies, $S_i = (s_1, \dots, s_m)$. A mixed strategy is an m -dimensional vector, $(p_1, \dots, p_m)$, assigning a probability p_k to each pure strategy s_k , such that (1) $p_k \geq 0$ and (2) $\sum_{k=1}^m p_k = 1$.*

We will write i 's mixed strategy in the form $(p_1s_1, p_2s_2, \dots, p_ms_m)$. So, for instance, the mixed strategy in which Sherlock goes to Dover with probability 0.05 would be written as $(0.05Dover, 0.95Canterbury)$. Note that a mixed strategy can assign probability 0 to some elements of the player's strategy set. Therefore, a pure strategy can be thought of as a special case of a mixed strategy, in which the probability weights are equal to 0 for all pure strategies except for one, which is given weight 1.

When Best Responding Involves Risk

Before we can solve the Holmes-Moriarty game, we first need to answer two more general questions. First, how do we calculate a player's best response to a mixed strategy? Second, when is playing a mixed strategy itself a best response?

Expected Payoffs

If Moriarty is certain that Sherlock will choose Dover, then Moriarty's best response is clearly Dover, while Moriarty's best response to Canterbury is

Canterbury. But what if Moriarty believes that Sherlock will flip a coin and play Dover with probability 0.5 and Canterbury with probability 0.5? Even after forming these beliefs, Moriarty still does not know the payoffs associated with each of his pure strategies. If he chooses Dover, then he will earn either 100 or 0, each with probability 0.5. However, we can compute his payoff from choosing Dover *in expectation*. In other words, if the players were to have this interaction many times, with Sherlock always choosing Dover with probability 0.5, what would Moriarty's payoff from Dover be, on average, each time? We compute this by considering each of the possible payoffs from Dover (here, 100 or 0) and weighting them by the probability that they will occur (here, 0.5 and 0.5): $0.5(100) + 0.5(0) = 50$. We call this Moriarty's *expected payoff* of Dover and write $Eu_M(Dover) = 50$. Likewise, $Eu_M(Canterbury) = 0.5(-50) + 0.5(100) = 25$. In general, to find a player's expected payoff, we sum up the payoff from each possible outcome weighted by the likelihood that the outcome occurs. (Additional examples of computing expected payoffs are provided in the appendix.)

Definition 2 (Expected payoff) *Imagine that player i faces a risky decision or gamble X with possible outcomes $\{x_1, x_2, \dots, x_m\}$, each with respective probabilities $(p_1, p_2, \dots, p_m)$. Player i 's expected payoff from X is given by $Eu_i(X) = \sum_{k=1}^m p_k u_i(x_k) = p_1 u_i(x_1) + p_2 u_i(x_2) + \dots + p_m u_i(x_m)$.*

Comparing the expected payoffs across Moriarty's two pure strategies, we find that his expected payoff is larger when he chooses Dover. In general, we'll assume that agents who are facing a risky situation prefer the outcome associated with the highest expected payoff. Therefore, playing Dover with certainty is Moriarty's best response to Sherlock playing the mixed strategy $(0.5Dover, 0.5Canterbury)$.

This approach to evaluating agents' preferences under risk, called expected utility theory, was developed by von Neumann and Morgenstern in the first chapter of *Theory of Games and Economic Behavior* as a precursor to laying out the formal theory of games. Somewhat remarkably, it has been the dominant theory of choice under risk ever since. In general, when you're faced with the question of how a rational agent will act when facing a risky prospect, the steps are straightforward: identify the agents' possible strategies, compute the expected payoff associated with each, and select the strategy with the highest expected payoff.

To fully analyze Moriarty's best response to Sherlock playing a mixed strategy, we need to consider how he would respond to *any* mixed strategy Sherlock could choose, not just the specific $(0.5Dover, 0.5Canterbury)$. Since each player has only two pure strategies, we will simplify our notation a little by assuming that Sherlock plays Dover with some generic probability, p (implying that he must play Canterbury with probability $1 - p$), and Moriarty plays Dover with some generic probability q (implying that he plays Canterbury with probability $1 - q$). This notation is shown in the normal-form game seen in [figure 5.2](#). We then calculate Moriarty's expected payoff from Dover and Canterbury for any general p : $Eu_M(Dover) = (p)(100) + (1 - p)(0) = 100p$ and $Eu_M(Canterbury) = (p)(-50) + (1 - p)(100) = 100 - 150p$. Moriarty strictly prefers Dover when

		Sherlock	
		p	$1 - p$
Moriarty	q	Dover	Canterbury
	$1 - q$	Dover	Canterbury
		(100, -100)	(0, 0)
		(-50, 50)	(100, -100)

Figure 5.2

The Holmes-Moriarty game, assuming that Sherlock chooses Dover with probability p and Moriarty chooses Dover with probability q .

$$Eu_M(Dover) > Eu_M(Canterbury)$$

$$100p > 100 - 150p$$

$$p > 2/5$$

This tells us that whenever there is a relatively high likelihood that Sherlock goes to Dover ($p > 2/5$), Moriarty is best off traveling to Dover as well. When there is a relatively low likelihood that Sherlock goes to Dover ($p < 2/5$), meaning that he instead goes to Canterbury, Moriarty's best response is to travel to Canterbury. If p exactly equals $2/5$, then Moriarty gains the same expected payoff of 40 regardless of which station he goes to (since $Eu_M(Dover) = 100p = 100(2/5) = 40$ and $Eu_M(Canterbury) = 100 - 150p = 100 - 150(2/5) = 40$).

This can be seen visually if we graph Moriarty's expected payoff from Dover and from Canterbury as a function of the probability Sherlock plays

Dover, as shown in [figure 5.3](#). Notice that Moriarty's expected payoff from choosing Dover increases as the likelihood that Sherlock plays Dover goes from 0 to 1, which makes sense because Moriarty wishes to match Sherlock's location. Likewise, Moriarty's expected payoff from Canterbury decreases as Sherlock is more likely to choose Dover. The end points of these lines (when the probability equals 0 or 1) confirm what we already know from analyzing the game in normal form: When the probability is equal to 1, meaning Sherlock chooses Dover with certainty, we see that $Eu(Dover) = 100$, $Eu(Canterbury) = -50$, and Moriarty's best response is Dover; when the probability is 0, meaning Sherlock chooses Canterbury with certainty, the figure indicates that $Eu(Dover) = 0$, $Eu(Canterbury) = 100$, and Moriarty's best response is Canterbury. When Sherlock plays a mixed strategy, the graph confirms our arithmetic analysis, showing that Moriarty's best response is Canterbury for any $p < 2/5$ and Dover for any $p > 2/5$. When Sherlock chooses Dover with probability exactly equal to $2/5$, then Moriarty is indifferent between his two pure strategies, and both are best responses. Most important, it is only in this situation that Moriarty would be willing to play a mixed strategy himself.

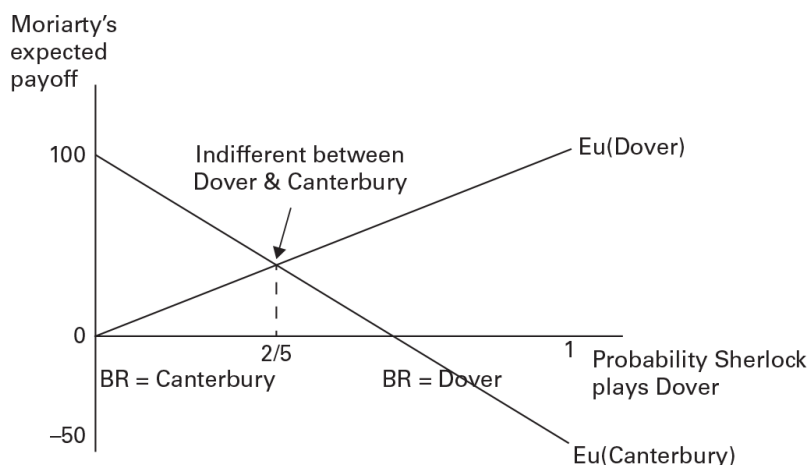


Figure 5.3

Moriarty's expected payoffs from Dover and Canterbury as functions of Sherlock's strategy.

Randomizing Requires Indifference

Why is Moriarty only willing to play a mixed strategy when he is perfectly indifferent between Dover and Canterbury? Remember that when we look for best responses, the other player's strategy is treated as known and fixed. For any known mixed strategy Sherlock is playing, we can compute Moriarty's

expected payoff from Dover and from Canterbury. If he plays Dover, his payoff will be $Eu_M(\text{Dover})$, if he plays Canterbury, his expected payoff will be $Eu_M(\text{Canterbury})$, and if he plays a mixed strategy, his payoff will be a weighted average of the two. No matter what mixed strategy Moriarty plays, his expected payoff will be within the range of $Eu_M(\text{Dover})$ and $Eu_M(\text{Canterbury})$, but no higher. By choosing a specific mixed strategy, he is deciding where within that range his expected payoff will fall—in other words, he is choosing how to weight $Eu_M(\text{Dover})$ and $Eu_M(\text{Canterbury})$ in his expected payoff.

As an analogy, imagine that you're taking a class with two tests, and you score 70 on the first and 90 on the second. No matter how your teacher weights the exams, your test score will be somewhere between 70 and 90, and the specific choice of weights will determine where within that range your grade falls. Now imagine you get to decide how to weight the tests. Your grade will clearly be highest if you put all of the weight on the second test, giving you a total score of 90. Mixing in the other test, no matter how little the weight, only drags down your grade. Or, if you scored 98 on the first test and 97 on the second, you would put all the weight on the first. Is there any situation in which you would be willing to give positive weights to both tests? Only if your scores on the two tests were identical. In this case, you are completely indifferent to how your tests are weighted—your score will be the same no matter how the tests are averaged. This is the exact situation we face when a player decides whether to play a mixed strategy. If one pure strategy offers a higher expected payoff than the others, it is rational to play that strategy with certainty. Mixing in another pure strategy, even with low probability, only drags down the player's expected payoff. It is only when the player gains the same expected payoff from two pure strategies that she would be willing to mix between them. Furthermore, she is willing to play any mixture since her expected payoff is the same no matter how the two pure strategies are weighted.

Observation: A mixed strategy is a best response only if each of the pure strategies receiving positive probability weight in the mix is also a best response. In this case, the player is indifferent between playing any mixed strategy that puts positive probability weight on these pure strategies and all are best responses.

In the Holmes-Moriarty game, this tells us that when Sherlock plays Dover with $p = 2/5$, *any* mixed strategy is a best response for Moriarty. Let q be the probability with which Moriarty chooses Dover, implying that he chooses Canterbury with probability $1 - q$. We then write Moriarty's best response q as a function of Sherlock's p . Moriarty's best response function is given by

$$BR_M(p) = \begin{cases} q=0 & \text{if } p < 2/5 \\ q \in [0, 1] & \text{if } p = 2/5 \\ q=1 & \text{if } p > 2/5 \end{cases}$$

Note the interpretation when $p = 2/5$. In this precise instance, Moriarty's best response is "Whatever." In this case, he is equally happy to play $q = 0$, $q = 1$, or any mixed strategy in between, which is why we say that his best response is any q in the $[0, 1]$ interval.

Turning to Sherlock's decision, we can now calculate his best response (probability p of going to Dover) to any strategy (probability q of going to Dover) that Moriarty chooses. First, we calculate Sherlock's expected payoff from Dover and from Canterbury, given that Moriarty plays Dover with probability q . $Eu_S(Dover) = (q)(-100) + (1 - q)(50) = -150q + 50$. $Eu_S(Canterbury) = (q)(0) + (1 - q)(-100) = 100q - 100$. Sherlock prefers Dover whenever

$$Eu_S(Dover) > Eu_S(Canterbury)$$

$$-150q + 50 > 100q - 100$$

$$q < 3/5$$

In this case, his best response is to play Dover with certainty by choosing $p = 1$. If $q > 3/5$, then Sherlock prefers Canterbury, and his best response is $p = 0$ (never go to Dover). If q exactly equals $3/5$, then Dover, Canterbury, and any mixed strategy all generate the same expected payoff (-40), and any q between 0 and 1 is a best response:

$$BR_S(q) = \begin{cases} p=1 & \text{if } q < 3/5 \\ p \in [0, 1] & \text{if } q = 3/5 \\ p=0 & \text{if } q > 3/5 \end{cases}$$

Mixed-Strategy Nash Equilibrium

Putting this together, Moriarty is willing to play a mixed strategy only when Sherlock plays Dover with probability $2/5$, and Sherlock is willing to play a mixed strategy only when Moriarty plays Dover with probability $3/5$. Therefore, the unique mixed-strategy Nash equilibrium occurs at $((\frac{2}{5}\text{Dover}, \frac{3}{5}\text{Canterbury}), (\frac{3}{5}\text{Dover}, \frac{2}{5}\text{Canterbury}))$. Furthermore, we already know that there is no Nash equilibrium in which the players choose pure strategies, so this is the only Nash equilibrium of the game.

One way to see this is to evaluate their best response functions graphically. Figure 5.4(a) plots the best response of Moriarty. For each p that Sherlock could choose (the horizontal axis), we graph the q that is Moriarty's best response. For all p up to $2/5$, the best response is $q = 0$, while for all p greater than $2/5$, the best response is $q = 1$. When $p = 2/5$, any q from 0 to 1 is a best response, and thus the best response function is a vertical line, capturing the "whatever" portion of his best response when he doesn't care how he weights Dover and Canterbury. Similarly, figure 5.4(b) plots Sherlock's best p as a function of each q Moriarty could choose.

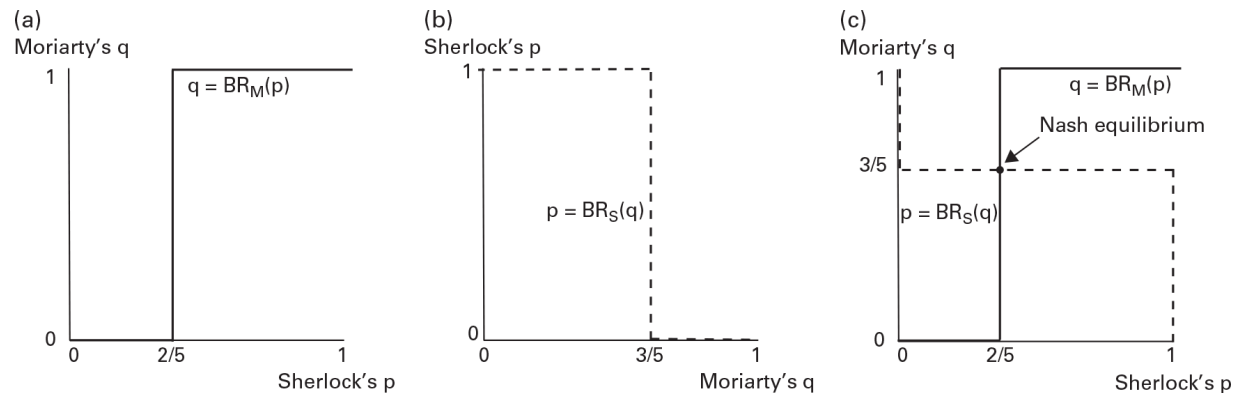


Figure 5.4

Moriarty's best response to Sherlock's mixed strategy p (a), Sherlock's best response to Moriarty's mixed strategy q (b), and both overlaid on the same graph with the Nash equilibrium at the intersection (c).

The Nash equilibrium requires the players to simultaneously best respond to each other, so we graph both best response functions in the same space. In figure 5.4(c), we arbitrarily assign p to the horizontal axis and q to the vertical. The space depicts all possible strategy profiles of the game, and our goal is to find a (p, q) pair that simultaneously lies on both players' best

response functions. Because we assigned p to the horizontal axis, Moriarty's best response is the same as in figure 5.4(a), but we must flip Sherlock's best response from figure 5.4(b) to correspond with these axes. There is exactly one point at which the best response functions intersect: $(2/5, 3/5)$, which indicates that $p = 2/5$ is a best response to $q = 3/5$ and vice versa, satisfying the definition of a Nash equilibrium.

This strategy profile is the unique equilibrium, since it is the only point (p, q) in this space that is stable. At any other point, one or both players would not be on their best response function and would therefore prefer to unilaterally change their strategy. For Sherlock, whose p is on the horizontal axis, this unilateral deviation would involve shifting the strategy profile horizontally until he reaches his best response function, while Moriarty would wish to move vertically until he is back on his own best response function. At $(2/5, 3/5)$, neither player has an incentive to move—but notice that neither has a strong incentive to stick with this strategy either. If we were to tip off Sherlock that Moriarty was choosing $3/5$ and asked if he wanted to stick with his choice of $2/5$, his answer would be less of a definitive “Yes!” and more of a “Yeah, sure, whatever.” He is completely indifferent between playing $p = 2/5$ or doing anything else. The Nash equilibrium falls on the “whatever” portion of both players' best response functions. But this is by design and is a general feature of mixed-strategy Nash equilibria. If Sherlock were not ambivalent, he would not be willing to play a mixed strategy and would instead switch to choosing one station with certainty—in which case Moriarty would also prefer a pure strategy to match Sherlock, in which case Sherlock would prefer to mismatch, and we would be back to the initial situation that baffled Morgenstern, in which the players chase each other around the normal-form matrix forever.

At the end of this chapter, we will consider whether it is reasonable to expect that players would choose these specific equilibrium probabilities when they are, in fact, completely indifferent between all their pure and mixed strategies.

Solving for Mixed-Strategy Nash Equilibria

The general procedure to solve for mixed-strategy Nash equilibria relies on our observation that a player will put positive weight only on pure strategies that are best responses to the other player's strategy. It involves four steps:

- (1) Eliminate any strictly dominated strategies, which are never a best response to any strategy of the other player.
- (2) Define the variables that you will use to refer to each player's mixed strategy over their remaining pure strategies. In our example, we used p and q , but these choices were arbitrary. All that is important is that you are consistent with your use of the notation and recall what it means at the end of the problem.
- (3) For each player i , write down the expected payoff i would receive from each of their remaining pure strategies as a function of the other player's mixed strategy (i.e., the variable defined in step 2).
- (4) For each player i , set i 's expected payoffs equal to each other and solve for the strategy that j must be choosing to make i indifferent (and willing to play a mixed strategy).

The most important thing to keep in mind when solving for mixed strategies is that, in equilibrium, one player's strategy is calculated using the *other* player's expected payoff.

The Princess Bride Game

Recall the game between Westley and Vizzini, which we know does not have a pure-strategy Nash equilibrium, as shown in [figure 5.5](#). To find the mixed-strategy Nash equilibrium, let's say that Vizzini sips Westley's cup with probability p and Westley poisons his own cup with probability w . Then Westley's expected payoffs are $Eu_w(PoisonW) = p(1) + (1 - p)(-1) = 2p - 1$ and $Eu_w(PoisonV) = p(-1) + (1 - p)(1) = 1 - 2p$, making him indifferent between his pure strategies when Vizzini chooses $p = 0.5$. Confirm that Vizzini is likewise indifferent between the two cups when Westley plays $w = 0.5$. Thus, the unique mixed-strategy Nash equilibrium in this game has each player choosing each pure strategy with equal probability.

		Vizzini	
		p	$1 - p$
Westley		Sip W.'s	Sip V.'s
	w	Poison W.'s	$(\underline{1}, -1)$
	$1 - w$	Poison V.'s	$(-1, \underline{1})$

Figure 5.5

The Princess Bride game.

Stag Hunt: A Game with Pure- and Mixed-Strategy Nash Equilibria

If a finite game does not have a pure-strategy Nash equilibrium, then we know that there exists an equilibrium in mixed strategies. However, many games have *both* pure- and mixed-strategy Nash equilibria. One classic game is the *stag hunt game*, which is based on a passage by Jean-Jacques Rousseau in which he describes the incentives of a group of hunters. Each of the hunters independently chooses whether to hunt for rabbit or for stag. Each hunter can catch a rabbit independently, regardless of what the other hunters do. But catching their preferred prize, a stag, requires mutual cooperation and will be successful only if the other members of the hunting party also commit to catching a stag. Stag is therefore a risky prospect: it has a high payoff if the others follow through on hunting stag, but it will fail if anyone pursues a rabbit instead. In contrast, Rabbit offers a guaranteed payoff.

The two-player version is shown here.

		Gale	
		p	$1 - p$
Katniss		Stag	Rabbit
	q	Stag	$(\underline{75}, \underline{75})$
	$1 - q$	Rabbit	$(60, 0)$

Figure 5.6

The stag hunt.

There are two pure-strategy Nash equilibria in the game: (Stag, Stag) and (Rabbit, Rabbit). Now we'll look for a mixed-strategy Nash equilibrium. Neither player has a dominated strategy, since each strictly prefers Stag when the other chooses Stag, but Rabbit when the other chooses Rabbit. Therefore

we will consider all the mixed strategies over the two pure strategies. Let's assume that Katniss chooses Stag with probability q (and therefore Rabbit with probability $1 - q$), while Gale chooses Stag with probability p . Katniss will be willing to play a mixed strategy only if Gale's p makes her indifferent between Stag and Rabbit, which happens when

$$\begin{aligned} Eu_K(Stag) &= Eu_K(Rabbit) \\ (75)(p) + (0)(1 - p) &= (60)(p) + (60)(1 - p) \\ p &= 4/5 \end{aligned}$$

If the probability that Gale chooses Stag is relatively high ($p > 4/5$), Katniss's best response is to take a chance on Stag. If it is relatively low ($p < 4/5$), Katniss's best response is to play it safe with Rabbit. But when Gale plays $p = 4/5$, Katniss is indifferent between choosing Stag and Rabbit and would therefore be willing to play a mixed strategy. Thus, Gale must be playing $p = 4/5$ in the mixed-strategy Nash equilibrium. To find the mixed strategy for Katniss, we perform the exact same calculation to determine the q that makes Gale indifferent between Stag and Rabbit: $Eu_G(Stag) = (75)(q) + (0)(1 - q) = 75q$ and $Eu_G(Rabbit) = (60)(q) + (60)(1 - q) = 60$. Setting his expected payoffs equal, we find $q = 4/5$. Therefore, our mixed-strategy Nash equilibrium in this stag hunt game is $((\frac{4}{5}Stag, \frac{1}{5}Hare), (\frac{4}{5}Stag, \frac{1}{5}Hare))$. We will return to the stag hunt in chapter 7, when we assess which types of equilibria are more likely to be played in games with multiple Nash equilibria.

While not necessary to solve for the mixed-strategy Nash equilibrium, we can also confirm our solution graphically by plotting the best response functions for the two players on one graph, as in [figure 5.7](#). Each point of intersection indicates one of the Nash equilibria that we have already identified. At the top right, the best response functions intersect when both players choose Stag with probability 1, corresponding to the (Stag, Stag) equilibrium. At the bottom left, the best response functions intersect when both choose Stag with probability 0, corresponding to the (Rabbit, Rabbit) equilibrium. Finally, we also have a mutual best response when each player chooses Stag with probability $4/5$ and Rabbit with probability $1/5$, corresponding to our mixed-strategy Nash equilibrium.

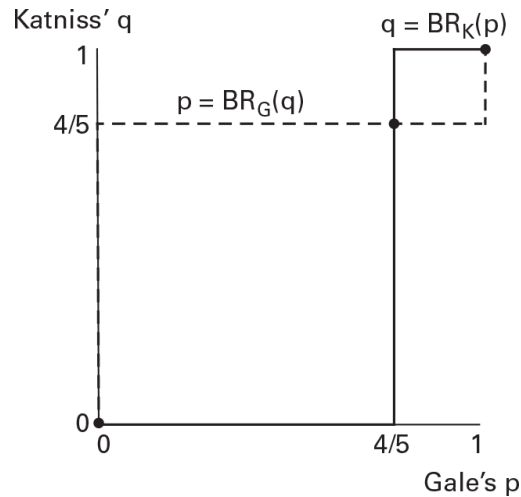


Figure 5.7

Best response functions in the stag hunt game with the three Nash equilibria at the points of intersection.

Mixing over Multiple Pure Strategies

In the two examples thus far, players randomize over two pure strategies. In other games, players may wish to put positive weight on more than two actions. One example is the classic game of rock-paper-scissors (as in exercise 4 at the end of this chapter). If you were playing this game, your best response would be to randomize your three actions only if the other player was putting equal probability on the three actions themselves. If your opponent put zero weight on Rock, then playing Paper would never be a good plan for you. Or if the other player chose Rock with a greater frequency than the other two actions, then you could gain an advantage by simply playing Paper all the time.

The card game described at the beginning of this chapter is a similar type of game, but the equilibrium probabilities are less transparent. This game was developed by Barry O'Neill (1987) for the purpose of testing whether people tend to play according to the mixed-strategy Nash equilibrium and is written in normal form in [figure 5.8](#).

		Column			
		j	a	t	$1 - j - a - t$
Row		Joker	Ace	Two	Three
	b	Joker	$(\underline{5}, -5)$	$(-5, \underline{5})$	$(-5, \underline{5})$
	c	Ace	$(-5, \underline{5})$	$(-5, \underline{5})$	$(\underline{5}, -5)$
	d	Two	$(-5, \underline{5})$	$(\underline{5}, -5)$	$(-5, \underline{5})$
	$1 - b - c - d$	Three	$(-5, \underline{5})$	$(\underline{5}, -5)$	$(\underline{5}, -5)$

Figure 5.8

Card game example from O'Neill (1987).

Since neither player has a strictly dominated strategy, we will allow all four of their pure strategies to receive positive weight in their mixed strategy, which requires us to use more parameters to describe each player's mix. Let's say that Column chooses Joker with probability j , Ace with probability a , and Two with probability t . Since the probabilities assigned all four pure strategies must sum to 1, this requires that Column chooses Three with probability $1 - j - a - t$. Now we can write Row's expected payoff from each of her pure strategies as a function of a, j , and t :

$$Eu_{Row}(Joker) = (5)(j) - (5)(a) - (5)(t) - 5(1 - j - a - t) = -5 + 10j$$

$$Eu_{Row}(Ace) = (-5)(j) - (5)(a) + (5)(t) + 5(1 - j - a - t) = 5 - 10j - 10a$$

$$Eu_{Row}(Two) = (-5)(j) + (5)(a) - (5)(t) + 5(1 - j - a - t) = 5 - 10j - 10t$$

$$Eu_{Row}(Three) = (-5)(j) + (5)(a) + (5)(t) - 5(1 - j - a - t) = -5 + 10a + 10t$$

In equilibrium, all four of these expressions must be equal, and so we can solve for j , a , and t by successively setting each expression equal to each other. Setting $Eu_{Row}(Ace) = Eu_{Row}(Two)$, we get $5 - 10j - 10a = 5 - 10j - 10t$ or $a = t$. Setting $Eu_{Row}(Joker) = Eu_{Row}(Three)$, we get $-5 + 10j = -5 + 10a + 10t$, and subbing in a for t , we find $j = 2a$. Finally, setting $Eu_{Row}(Joker) = Eu_{Row}(Ace)$, we get $-5 + 10j = 5 - 10j - 10a$, and subbing in for $t = a$ and $j = 2a$, we get $a = 0.2$. Since we already know $t = a$ and $j = 2a$, we have $t = 0.2$ and $j = 0.4$. Therefore, in equilibrium, Column plays Joker with probability 0.4, Ace with probability 0.2, Two with probability 0.2, and Three with probability 0.2.

You can confirm that $b = 0.4$, $c = 0.2$, and $d = 0.2$, meaning that Row's equilibrium mixed strategy also involves playing Joker with probability 0.4

and each of the other strategies with probability 0.2.

Best Responding, Dominated Strategies, and Rationalizability

We began our assessment of rational play in chapter 3 by asking whether there are strategies that rational players definitely would not want to play, regardless of their beliefs about the play of others. Now that we know how to assess players' best responses to any beliefs they may hold about the other player's strategy, we can turn this question on its head and ask: which strategies *would* be reasonable for rational players to play? That is, which strategies can be *rationalized* as a best response to some reasonable beliefs? As we've seen, the best response piece is a straightforward optimization problem, but the trick is putting oneself in the other player's shoes and forming beliefs about what they will do. The Nash equilibrium solution concept requires that these beliefs be correct. In this section, we'll consider what happens if we relax this assumption somewhat and only require that beliefs are consistent with common knowledge of rationality.

In this case, can players form beliefs about the other player's strategy purely through rational introspection? Let player i form some conjecture or belief about the strategy of player j , which is a probability distribution, p , over S_j . If player i is rational, then she will play only strategies that are a best response to p . Furthermore, if she knows that j is rational, then her belief p must be consistent with j in turn playing a best response to his belief about her strategy. And if she knows that he knows that she is rational, this belief of his must involve her playing a best response to some belief of hers, and so on. Strategies that survive this infinite chain of rationalizing are called *rationalizable*. The concept of rationalizability was originally developed in parallel by Pearce (1984) and Bernheim (1984).

In practice, how do we cut through this infinite chain of logic and actually determine which strategies are rationalizable? We can find the set of rationalizable strategies by successively deleting strategies that could never be a best response to any set of beliefs, in a procedure strongly reminiscent of the process of iterated elimination of dominated strategies. The strategies that remain are rationalizable. First, let's formally define strategies that are never a best response (NBR), as in definition 3.

Definition 3 (Never a best response) *Let p be a probability distribution over player j 's strategy set S_j . A strategy s_i for player i is never a best response*

(NBR) if there is no p such that $s_i \in BR_i(p)$. That is, there is no (pure or mixed) strategy that player j could play for which s_i is a best response.

If s_i is an NBR, then there are no beliefs that could rationalize playing it. What is the relationship between an NBR and a strictly dominated strategy? If a strategy is strictly dominated, then it is an NBR. But if a strategy is an NBR, does that necessarily mean that it is strictly dominated? Consider the Row player's three pure strategies in [figure 5.9](#). Can all of them be best responses? Let's say that Row believes that Column will play X with probability p . If we graph the expected payoff from each of Row's pure strategies over p (as shown in [figure 5.10](#)), we see that C is an NBR to any p : for $p < 1/2$, B is the best response; for $p > 1/2$, A is the best response; and for $p = 1/2$, both A and B , as well as any mix of the two, are best responses. Yet, if we were to inspect the matrix, we would not conclude that C is a strictly dominated strategy according to definition 1 from chapter 3. However, C is strictly dominated by a mixed strategy. Consider the mixed strategy $(\frac{1}{2}A, \frac{1}{2}B)$. This yields an expected payoff for Row of 5 over all p and thus strictly dominates C , for which the payoff is 4 for all p . In finite, two-player games, such as those we've been considering in the last couple of chapters, the set of strategies that are NBR are the same as the set of strategies that are strictly dominated *either by pure or mixed strategies*. However, in games with more players, there may be strategies that are NBR, and yet not strictly dominated, even by mixed strategies.

		Column	
		X	Y
Row	A	(10, 0)	(0, 5)
	B	(0, 5)	(10, 0)
	C	(4, 4)	(4, 4)

Figure 5.9

C is an NBR, but is undominated by A or B .

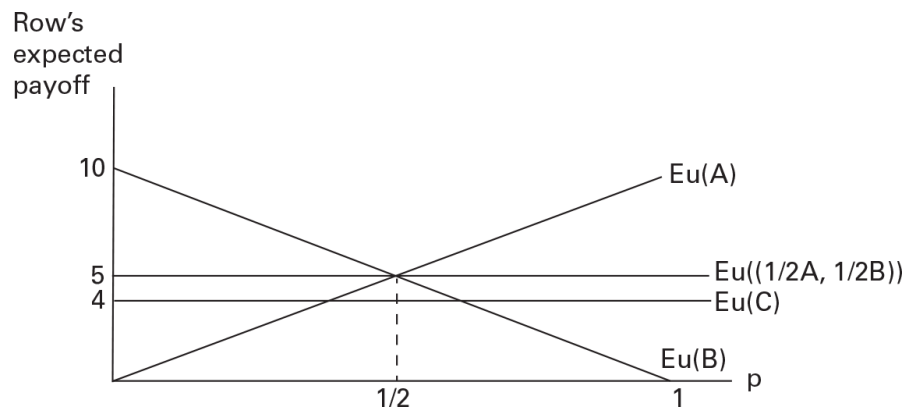


Figure 5.10

C is an NBR. While it is undominated by A or B, it is dominated by a mix of the two.

The set of rationalizable strategies can be found by iteratively eliminating strategies that are NBR. In a finite, two-player game, this process will lead to an identical set of strategies as iterated elimination of strictly dominated strategies *if* we also eliminated strategies that were dominated by a mixed strategy. In games with more than two players or a continuous strategy space, there may be strategies that aren't strictly dominated but are nevertheless NBR. As such, the set of rationalizable strategies is either equivalent to the set of strategies that remain after performing iterated elimination of dominated strategies or it is a subset of them.

What is the relationship between the Nash equilibrium solution concept and rationalizability? Rationalizability is a weaker solution concept than the Nash equilibrium because it requires only that players' strategies be logically consistent with common knowledge of rationality, whereas the Nash equilibrium requires each player to best respond to the other's actual strategy. A strategy is rationalizable if it is a best response to some beliefs of the other player's strategy, and those beliefs involve the other player best responding to some beliefs, and so on. Ultimately, however, none of these beliefs need be correct. For example, consider the stag hunt game. Both of the players' pure strategies are rationalizable. As such, the strategy profiles (Stag, Stag), (Rabbit, Rabbit), (Stag, Rabbit), and (Rabbit, Stag) are all consistent with rationalizability, although only the first two are Nash equilibria. Thus, in some games, rationalizability doesn't allow us to eliminate any strategies at all. Even in these cases, though, it is useful to provide an assessment of which strategies could be arrived at purely through rational introspection rather than relying on perfect foresight.

Behavior in Games

Playing Mixed-up Strategies

Is the mixed-strategy Nash equilibrium a good predictor of how people actually play these games? There is good reason to be skeptical. The equilibrium mix is challenging to compute, and there is no strong incentive to do so—in equilibrium, the players are indifferent between choosing their equilibrium strategy or literally any other mix.

The solution is most compelling in zero-sum games with repetition, as in O'Neill's experiment. If one player is not playing according to his equilibrium mixed strategy, his opponent can gain a payoff advantage by playing a single pure strategy. Since it is a zero-sum game, this advantage comes at his expense. Experimental evidence suggests that people playing against a computer that is programmed to implement nonequilibrium mixed strategies are actually quite adept at detecting and exploiting this advantage. What about when two humans play each other? The table shown in [figure 5.11](#) presents the Nash equilibrium predictions and the relative frequencies with which players chose each pure strategy in O'Neill's experiment, as well as a replication of the same game by Shachat (2002).

		Column				Nash	O'Neill	Shachat			
	Joker	Ace	Two	Three							
Row	Joker	$(\underline{5}, -5)$	$(-5, \underline{5})$	$(-5, \underline{5})$	$(-5, \underline{5})$				0.40	0.362	0.356
	Ace	$(-5, \underline{5})$	$(-5, \underline{5})$	$(\underline{5}, -5)$	$(\underline{5}, -5)$				0.20	0.221	0.212
	Two	$(-5, \underline{5})$	$(\underline{5}, -5)$	$(-5, \underline{5})$	$(\underline{5}, -5)$				0.20	0.215	0.222
	Three	$(-5, \underline{5})$	$(\underline{5}, -5)$	$(\underline{5}, -5)$	$(-5, \underline{5})$				0.20	0.203	0.210
Nash	0.40	0.20	0.20	0.20							
O'Neill	0.426	0.226	0.179	0.169							
Shachat	0.390	0.216	0.207	0.187							

Figure 5.11

Nash equilibrium predictions and experimental results from O'Neill (1987) and Shachat (2002).

At first glance, the relative frequencies (averaging across all participants) appear remarkably close to the Nash predictions, although they do differ statistically. At the individual level, however, the choices are less consistent

with the theory. In addition, there is significant correlation in the participants' choices from one round to the next, suggesting that they are not generating random sequences of pure strategies. Instead, people tend to switch back and forth between their pure strategies too frequently, and rarely play the same action multiple times in a row, as would be expected in a truly random sequence. To ascertain which strategies the subjects were trying to play—even if they were unable to successfully implement them—Shachat's study also allowed some pairs access to a mixed-strategy device. Instead of choosing a playing card each round, these players specified the mixed strategy that they wished to play, and the computer implemented an action according to the specified probabilities. While the mixed-strategy device eliminated the correlation in the participants' choices, it also revealed that very few deliberately chose the Nash mixed strategy. Participants played the Joker pure strategy in about 25% of the observations, placed equal probability on the four cards in 13.2% of the observations, and played the Nash mixed strategy in less than 3% of the observations.

These findings from the O'Neill game are fairly robust across a wide variety of games and experimental tests. In many experiments, the overall discrepancy between actual behavior and the Nash prediction is small but statistically significant, suggesting that the Nash equilibrium may be a reasonable prediction of the population as a whole. Behavior is most consistent with the theory when the Nash equilibrium involves *equiprobable play* (putting equal weight on each pure strategy), such as in rock-paper-scissors or the Princess Bride game. The further the Nash mixed strategy is from equiprobable play, the more behavior deviates from the prediction, with players typically overplaying low probability pure strategies and underplaying pure strategies that should be played with high frequency. As a result, the aggregate results generally fall between equiprobable play and the Nash equilibrium. At the individual level, in any case, it is clear that lab participants are not playing according to the equilibrium mixed strategies and their sequences of play are not truly random.

One notable exception comes from professional sports, where the participants have spent their lifetimes practicing these types of zero-sum games, such as baseball players mixing pitches, basketball players choosing between right- or left-handed layups, soccer players randomizing the direction they aim penalty kicks, and tennis players mixing the location of their serves.

Walker and Wooders (2001) studied tennis serves at major professional tournaments. In their model, the server chooses to direct his serve to the left or right, while the receiver attempts to anticipate the location. If the receiver is playing the equilibrium mixed strategy, then the server should be indifferent. In almost all the contests the researchers analyzed they found that the server is, in fact, equally likely to win by serving to the left or to the right, suggesting that the receiver is playing the equilibrium mix. Other studies have modeled soccer penalty kicks as a game, with the kicker choosing to kick to her natural or weaker side, as the goalie simultaneously decides in which direction to dive (Huertas-Palacios, 2003). Once kicked, the ball travels into the net in less than a third of a second, meaning the goalie must commit to a direction before observing the kicker's choice. Again, the professional athletes' behavior is remarkably close to the Nash predictions: the goalies are just as likely to prevent a goal when diving left or right (suggesting the kickers are playing the equilibrium mix), and the kickers are just as likely to score when aiming left or right (suggesting that the goalies are playing the equilibrium mix). In addition, the evidence suggests that professional athletes' choices are not correlated with their prior actions (Huertas-Palacios and Volij, 2008).

Thus far, we've been discussing behavioral evidence in zero-sum or constant-sum games. In variable-sum games, it is possible to alter one player's payoffs while holding the other player's payoffs constant. For instance, consider the three games (a)–(c) presented in [figure 5.12](#), in which Column's payoffs are consistent in all three games, but Row's payoff from (Top, Left) is lower in the second and higher in the third game. The mixed-strategy Nash equilibrium predicts that an increase in Row's payoff should not influence Row's behavior at all, but should instead decrease the probability that *Column* plays Left, in order to keep Row indifferent.

	Left	Right	Nash	Actual		Left	Right	Nash	Actual
Top	(80, 40)	(40, 80)	.50	.48	Top	(44, 40)	(40, 80)	.50	.08
Bottom	(40, 80)	(80, 40)	.50	.52	Bottom	(40, 80)	(80, 40)	.50	.92
Nash	.50	.50			Nash	.91	.09		
Actual	.48	.52			Actual	.80	.20		
Game (a)					Game (b)				

	Left	Right	Nash	Actual
Top	(320, 40)	(40, 80)	.50	.96
Bottom	(40, 80)	(80, 40)	.50	.04
Nash	.13	.87		
Actual	.16	.84		
Game (c)				

Figure 5.12

Nash equilibrium prediction and experimental data from three games in Goeree and Holt (2001).

Is behavior consistent with this prediction? Consider game (a) first, in which the players face a symmetric scenario, similar to the Princess Bride game, and the Nash equilibrium is for both players to choose their two pure strategies with equal likelihood. The overall findings are very close to this prediction. In the other two games, Row should also put equal probability on both actions. Instead, the Row players dramatically shift away from Top in game (b), when the payoff from (Top, Left) decreases to 44, and toward Top in game (c), when the payoff increases to 320. Therefore, behavior can differ substantially from the Nash prediction in variable-sum games, as players focus on their own payoffs rather than acting to make the other player indifferent.

Evidence-Based Adaptations of the Standard Model

Two models that have been used to predict and explain mixed-strategy play are quantal response equilibrium (QRE) and reinforcement learning (RL). Rather than requiring that players always perfectly best respond to correct beliefs, these models assume that players are simply more likely to play strategies that either have been particularly profitable in the past (RL) or that will be more profitable in the present, given that their opponent is also committing errors (QRE). While neither model perfectly predicts behavior,

both can be used to explain some of the systematic deviations from the Nash mixed strategy. In the remainder of the chapter, we'll broadly consider how these models relate to the standard mixed-strategy prediction (and refer readers to more detailed descriptions in part VIII of the book).

The QRE model (McKelvey and Palfrey, 1995) incorporates noise in the likelihood that both players choose their optimal strategies. Imagine that a player has correct beliefs about her opponent's strategy and, given these beliefs, computes the expected payoff from each of her strategies. However, she does so with some error. Instead of perfectly implementing the strategy associated with the highest expected payoff, the player is simply *more likely* to choose strategies that have higher payoffs. *How much* more likely is a parameter of the model, and rationality is therefore a matter of degree. At one extreme, the players' actions are unrelated to how profitable they are, and they simply randomize uniformly across their pure strategies. This looks like implementing a mixed strategy with each pure strategy receiving equal weight. At the other extreme, the players always best respond and play according to the Nash equilibrium. In between, the QRE estimates typically fall between equiprobable play and the Nash equilibrium, in line with the findings commonly observed in experimental tests. Because the player correctly anticipates that her opponent will also make mistakes, under the QRE she is not actually indifferent between her pure strategies and changes in the magnitudes of her own payoffs may influence her behavior, as we saw in games (a)–(c). For a game like game (c), QRE predicts that Row would play Top with probability 0.5 at the two extremes of rationality: that is, if the players were playing perfectly randomly or perfectly rationally. However, for intermediate levels of rationality, Row will choose Top more frequently, consistent with the experimental findings.

In the most basic RL model (Erev and Roth, 1995, 1998), each player starts with equal propensities for playing each of their pure strategies, but then learns over time that some strategies are more profitable than others and should be played more often. The player revises upward his propensity to play a strategy each time he chooses it and does well. In each round, the player plays each pure strategy with a probability equal to its relative propensity. Simulations of this process in two-player games with a unique mixed-strategy Nash equilibrium are often close to actual paths of behavior, especially if the model includes features such as forgetting earlier play or experimenting by

trying strategies that are similar to previously successful choices. For instance, when the Nash equilibrium requires that each player put equal weight on each pure strategy, the model predicts rapid convergence toward the Nash equilibrium, and these in fact tend to be the games in which people are most adept at playing Nash. In contrast, when the Nash equilibrium requires players to choose more extreme strategies, the model explains the tendency of players to initially overplay strategies that should be given little weight, but then to converge slowly in the direction of best responding over time.

In sum, there is no single foolproof model for describing behavior in games with unique mixed-strategy Nash equilibria, but models that relax the assumptions of superhuman rationality can be used to predict and explain many of the systematic deviations from the Nash prediction.

Exercises

(1) Consider the game shown here between Buffy and Willow. Imagine that Buffy believes that Willow chooses Left with probability w .

		Willow	
		Left	Right
Buffy	Up	(50, 50)	(30, 100)
	Middle	(10, 50)	(20, 100)
	Down	(30, 20)	(100, -100)

Figure 5.13

- Write Buffy's expected payoff from each of her three pure strategies.
- With w on the horizontal axis and expected payoff on the vertical, graph the expected payoffs associated with each of Buffy's pure strategies.
- Based on your graph from part (b), when would Buffy be willing to play a mixed strategy, and which pure strategies would receive positive weight?
- Find Buffy's best response function. Be sure to first define the probability with which Buffy plays her pure strategies (e.g., b is the probability that Buffy chooses Up).

(2) Johann and Igor are each simultaneously deciding what to do tonight: see a Bach concert or a Stravinsky concert. Each player prefers to match the other's decision, but the two disagree about which concert would be more enjoyable.

- (a) Find all the pure- and mixed-strategy Nash equilibria of the game shown in [figure 5.14](#).

		Igor	
		Bach	Stravinsky
Johann	Bach	(75, 25)	(0, 0)
	Stravinsky	(0, 0)	(25, 75)

Figure 5.14

- (b) On the same axes, graph both players' best response functions. Put Johann's probability of going to Bach on the vertical and Igor's probability of going to Bach on the horizontal. Explain how this graph confirms your answer from part (a).

(3) In the film *Footloose*, Kevin Bacon's character is challenged to a game of chicken by a fellow high school student. Both players drive tractors directly at each other and see who bails out first. If neither player bails, then both could be severely injured. If both bail, then the contest is a draw. But if one bails and the other does not, the person who bailed is branded a "chicken" and the other is victorious. The payoffs are shown in [figure 5.15](#). Find all the pure- and mixed-strategy Nash equilibria of this game.

		Not Kevin Bacon	
		Bail	Hang In
Kevin Bacon	Bail	(50, 50)	(0, 100)
	Hang In	(100, 0)	(-100, -100)

Figure 5.15

(4) Two players simultaneously choose between Rock, Paper, or Scissors. If both players play the same action, then the game is a tie and each earns a

payoff of 0. Otherwise, Rock beats Scissors, Scissors beats Paper, and Paper beats Rock. The loser must pay \$10 to the winner.

- (a) Show that the mixed-strategy Nash equilibrium has each player choosing each pure strategy with equal probability.
 - (b) Now imagine that the Column player has a hand injury that prevents him from playing Scissors (as in the plot of a *Bob's Burgers* episode). The Row player still can play Scissors. Write this game in normal form and find the Nash equilibrium.
- (5) One study of European professional soccer leagues analyzed the likelihood of scoring a penalty kick based on which direction the player kicks (to his natural or unnatural side) and which direction the goalie dives (to the kicker's unnatural or natural side). In the table seen in [figure 5.16](#), the kicker's payoff is the likelihood of scoring under each outcome, and the goalie's payoff is the likelihood that the goal will not be scored.

		Goalie	
		Natural	Unnatural
Kicker	Natural Side	(69.92, 30.08)	(92.91, 7.09)
	Unnatural	(94.97, 5.03)	(58.3, 41.7)

Figure 5.16

Find the mixed-strategy Nash equilibrium in this game.

- (6) An early replication of O'Neill's card game experiment (Budescu and Rapoport, 1994) also included the 3x3 game shown in [figure 5.17](#). Two players simultaneously announce 1, 2, or 3. If the announcements match, Column must pay Row \$2. Otherwise, Row pays Column \$1. Find the mixed-strategy Nash equilibrium.

		Column		
		1	2	3
Row	1	$(2, -2)$	$(-1, 1)$	$(-1, 1)$
	2	$(-1, 1)$	$(2, -2)$	$(-1, 1)$
	3	$(-1, 1)$	$(-1, 1)$	$(2, -2)$

Figure 5.17

(7) Hide-and-seek is another game that has been commonly used in experiments to test for mixed-strategy play. Two players simultaneously choose Black or Red. If the cards mismatch, Column wins. If the cards match, then a six-sided die is rolled. If they match on Black, Column wins if the die comes up 1 or 2 (and Row wins if a different number is rolled). If they match on Red, Row wins if the die comes up 1 or 2 (and Column wins if a different number is rolled). Assume that the winner gains 1 and the loser gets a payoff of 0. Find the mixed-strategy Nash equilibrium.

¹This chapter assumes that you have some familiarity with probabilities from your past statistics, math, or microeconomics courses, but a review of the background you will need is presented in the appendix.

6

Equilibrium in Nonmatrix Games

To this point, we have limited our attention to games that can be represented as normal-form matrices. This implies that we have considered games with just two players and a finite number of pure strategies. We now examine methods for solving games in which there may be an infinite number of possible strategies, an arbitrarily large number of players, or both.

Imagine that you are competing in a market with one other firm. You each sell identical products, and you will simultaneously choose how many units to produce this quarter. Each unit costs \$1 to produce. The price at which the firms can sell that number of units is then determined by market demand. If a total of Q units are produced, then the price at which the firms can sell each unit is $100 - Q$. Notice that, the more units that are produced in total, the lower the price will need to be to sell off all those units. We thus have a strategic game, where each firm's profits depend on its own production choice and the choice of its competitor. Specifically, if you produce q_i units and the other firm produces q_j units, your payoff is given by $q_i \times \text{price} - \text{cost}$, or $q_i(100 - q_i - q_j) - q_i$.

How many units will you produce? What if, instead, there were three other firms in the market?

Nonmatrix Games and Their Solutions

The game described here was first analyzed by Antoine Augustin Cournot in 1838 to model competition in a market with a small number of firms, known as an *oligopoly*. Firms in an oligopoly face a fundamentally different production decision than they would in a market with a different structure, since they are engaged in strategic interactions with their competitors. In a perfectly competitive market, with a large number of firms, each individual's production choice has a negligible effect on the market price, so firms

maximize their profits by producing up to the point that the price they get equals their marginal costs. In this example, this would imply a market price of $p = 1$ and total output of $Q = 99$. In contrast, if there were only a single firm in the market, then the monopolist's payoff would depend purely on its own decision, and it can simply act to maximize its own profits without needing to anticipate the strategies of its competitors. In our example, the monopolist's payoff of $Q(100 - Q) - Q$ is maximized when $Q = 49.5$ and $p = 50.5$. (The appendix of this book provides a refresher on optimization that may come in handy for this chapter.)

In an oligopoly, however, each firm's payoff depends on the independent choices of a small number of other players, each of whom attempts to anticipate the behavior of its competitors. We therefore have a strategic game, and we'll need game theory to predict the firms' optimal choices. In fact, Cournot proposed an equilibration process in which both players attempt to best respond to each other's production choice, which predated Nash's similar equilibrium concept by over a century. Since the firms have a continuum of possible strategies, we cannot write the game in matrix form and solve for the equilibrium using the method given in chapters 4 and 5, and so we'll need to learn additional tools to find the Cournot-Nash equilibrium.

The Cournot game is just one game that you cannot write in matrix form. Considering other possible *nonmatrix* games, there are two characteristics that will hinder us representing them in a normal-form matrix: the number of strategies, as described previously, and the number of players. If the strategy space is continuous or if the number of players is large, the dimensions of any normal-form matrix become unwieldy at best. This does not mean, however, that we cannot analyze the game; what it does mean is that we just have to be a bit more clever, taking advantage, when appropriate, of mathematical and logical tools. For example, if strategies are continuous, we can use calculus to ask which strategy maximizes a player's payoff, and if there are many players, all of whom have identical incentives, we can look for a symmetric outcome in which they all choose to do the same thing. To proceed, we first examine a few two-player games with continuous strategy spaces, for which we can use logic to determine players' best responses as a function of the other player's choice. We'll then see how to use calculus to solve for players' best response functions when payoff functions are continuous, as in the two-player Cournot oligopoly game (known as a *duopoly*) and a variant of the

traveler's dilemma. We then turn to games with many players but a finite number of pure strategies, starting with a classic volunteer's dilemma, where each person hopes that someone else will step up and complete an onerous task. Our analysis culminates with the case of continuous strategies and a large number of players, including examining the Cournot game for any general number of firms.

Demanding an Infinitely Divisible Cake

Consider again the game between two siblings deciding how to divide a cake, which was described in the exercises for chapter 4. The kids simultaneously announce a demand for the proportion of the cake that they personally should get. If the demands are mutually feasible (i.e., the proportions sum to 1 or less), then each kid gets the demand. Otherwise, both get nothing. We assume that the kids care only about the amount of cake they personally get, and each person's payoff simply is equal to his or her own share.¹

Let's now expand the kids' strategy sets to include any real number between 0 and 1—that is, $S_i = [0, 1]$. Assume that kid i demands d_i and the other kid demands d_j . Then kid i 's payoff is given by

$$u_i(d_i, d_j) = \begin{cases} d_i & \text{if } d_i + d_j \leq 1 \\ 0 & \text{if } d_i + d_j > 1 \end{cases}$$

Since each player now has an infinite number of strategies, we clearly cannot write this game in matrix form. However, the basic logic of our approach will remain the same as in the previous two chapters: we first consider each person's best response to each of the possible strategies played by the other person, and then we look for overlap. The difference is that instead of writing a player's best response to each element in the other player's strategy set, we will find the *best response function* over the range of the other player's possible strategies. (Notice that this is similar to our approach in chapter 5, when we found each player's best response as a function of the other player's mixed strategy.)

Let's look for the best response function for each kid in the continuous cake demand game. We need to consider player i 's best response to each $d_j \in [0, 1]$. Let's first consider the case where $d_j < 1$: that is, player i has some hope of getting some cake. In this case, player i maximizes her cake by demanding whatever remains, given player j 's demand. That is, player i 's

response is $1 - d_j$: if she were to demand any more, then they both would get nothing, and if she were to demand any less, then she would leave cake on the table. What if $d_j = 1$ (i.e., her sibling demanded all the cake for himself)? Well, then nothing she could say would get her any cake anyway, and so she's indifferent between all her strategies. If she were feeling particularly generous, she could say "0" and let him have the whole cake. If she were feeling bitter, she could demand a positive amount and ensure that he also gets nothing. But so long as we assume that she only cares about the cake that she gets, then any demand from 0 to 1 would be a best response. Thus, player i 's best response function is given by

$$BR_i(d_j) = \begin{cases} d_i = 1 - d_j & \text{if } 0 \leq d_j < 1 \\ d_i \in [0, 1] & \text{if } d_j = 1 \end{cases}$$

As we did in chapter 5, let's now graph both players' best response functions on the same axes and look for overlap—that is, for strategy profiles where both players are simultaneously best responding to each other (see [figure 6.1](#)). We'll put d_1 on the horizontal axis and d_2 on the vertical. The left panel shows player 2's best response (plotted on the vertical) to each d_1 from 0 to 1 (plotted on the horizontal). When $d_1 < 1$, the best response is $1 - d_1$, and when $d_1 = 1$, any demand is a best response, so the best response function is a vertical line at $d_1 = 1$. The right panel shows player 1's best response (plotted on the horizontal) to each d_2 from 0 to 1 (plotted on the vertical). Finally, the [figure 6.1\(c\)](#) overlays the two best response functions. Here, we see that any strategy profile $(d_1, 1 - d_1)$ (falling along the diagonal line in the graph) is on both players' best response functions, and thus is a Nash equilibrium. In addition, there is one other point of intersection: $(1, 1)$ falls on both players' best response functions and therefore is also a Nash equilibrium. Although neither player gets any cake under this outcome, neither player has an incentive to demand less, since doing so will not improve each person's share.

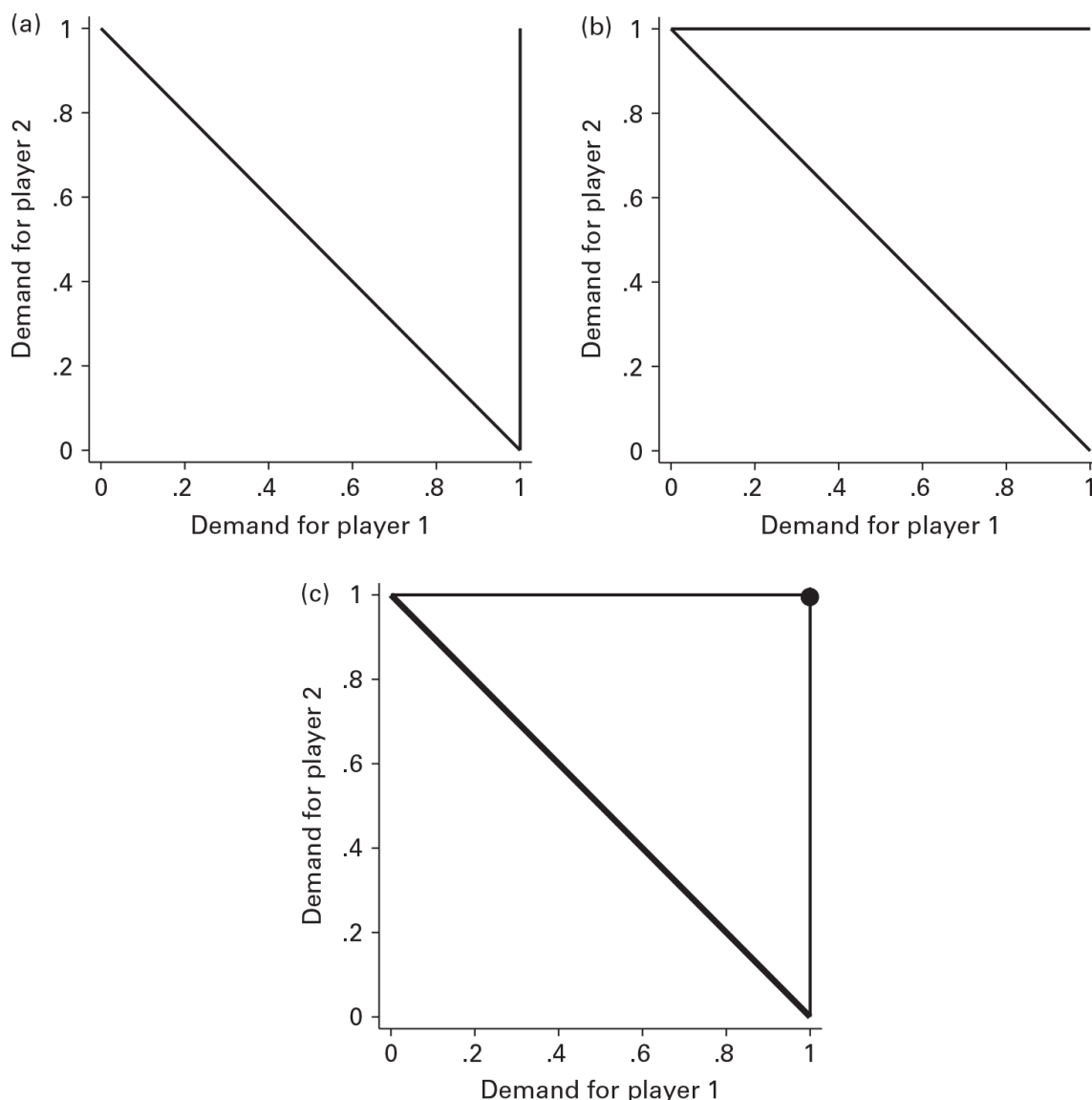


Figure 6.1

Player 2's best response (top left), player 1's best response (top right), and both best responses with the Nash equilibria along the thick line and at black dot (bottom).

The Bertrand Pricing Game

When thinking about the Cournot oligopoly game, it may have struck you as strange that the firms were choosing their quantities rather than their prices. After all, don't firms compete on price rather than quantity? While firms choose both their quantity and their price, they are constrained by the fixed relationship between price and quantity demanded in the market: in our example, $p = 100 - Q$. Therefore, it is equally plausible to think of the firms as choosing their price, with the market demand then determining how many

units they can sell at that price, or the firms choosing quantity, with the market demand determining the price that enables them to sell off their inventory. However, the two situations may lead to different strategic incentives. Joseph Louis François Bertrand made this point in his 1883 review of Cournot's book, sketching an alternative model in which the firms simultaneously choose their prices and consumers buy from the firm with the lowest price.

Let's analyze Bertrand's price competition model with two firms. If both firms set the same price, then they split the market demand at that price. Otherwise, the firm with the lower price captures the entire market demand and the other firm sells nothing. Let p_i be the price chosen by firm i and p_j be the price of its competitor. For simplicity, we'll assume that the prices do not exceed 100 (at which market demand hits 0), but this assumption does not matter for our analysis. Then firm i 's payoff is given by

$$u_i(p_i, p_j) = \begin{cases} 0 & \text{if } p_i > p_j \\ (p_i - 1)(100 - p_i) & \text{if } p_i < p_j \\ \frac{(p_i - 1)(100 - p_i)}{2} & \text{if } p_i = p_j \end{cases}$$

The Bertrand Game with a Large Strategy Set

We'll first assume that each player's price must be in 1-cent increments. That is, the players' strategy sets aren't the set of all real numbers, but rather the set $\{0.00, 0.01, 0.02, \dots, 99.99, 100\}$. While the strategy sets aren't continuous, we certainly wouldn't want to attempt to write this game in a matrix. As with the cake demand game, we'll think about the best responses of each player logically and then look for pairs of prices that are consistent with the players' mutual best responses.

Now we'll look for firm i 's best response to each possible p_j chosen by its competitor. What if $p_j = 100$? In this case, firm j has essentially priced itself out of the market: no one will buy its product no matter what firm i does. Firm i will want to come in at a lower price, but how much lower? Since firm j has made itself irrelevant, we can treat firm i as a monopolist and find the price that maximizes firm i 's payoff of $(p_i - 1)(100 - p_i)$. Taking the derivative with respect to p_i and setting it equal to zero, we find that firm i 's payoff is maximized when $100 - 2p_i + 1 = 0$, or $p_i = 50.5$. Thus, firm i 's best response to 100 is 50.50. Likewise, 50.50 would be firm i 's best response to 99.99, 99.98, or any price above 50.50.

Next, what if firm j chooses 50.50? If firm i chooses a higher price, it earns a payoff of 0. If the two firms match 50.50, then they split the market demand of $100 - 50.50$, and each gets a payoff of $\frac{(100-50.50)}{2}(50.50 - 1) = 1,225$. But if firm i undercuts firm j *just slightly*, by choosing 50.49, then it captures the entire market demand of $100 - 50.49$ and gets a payoff of $(100 - 50.49)(50.49 - 1) = 2,450.25$. Thus, firm i is better off undercutting firm j ever so slightly in order to capture the entire market demand at a price that is only a bit lower. Likewise, firm i 's best response to any lower prices chosen by firm j is to always undercut firm j by a penny—until we get down to a price of 1.01. In this case, firm i could match firm j 's price to capture half of the market demand and earn a penny on each sale (payoff: $\frac{(100-1.01)}{2}(1.01 - 1) = 0.495$), or it could undercut firm j to capture the entire market demand but earn zero on each sale for a payoff of zero. Thus, it prefers to match. In the event that firm j chooses 1, then firm i also does not wish to undercut—it would capture the market demand and earn a negative payoff on each sale, whereas if it chooses any price of 1 or above, then it earns a payoff of zero. Finally, if firm j chooses a price below 1, then firm i prefers any price that is greater than firm j 's price so it will earn zero rather than negative payoffs.

Thus, firm i 's best response is given by

$$BR_i(p_j) = \begin{cases} p_i = 50.5 & \text{if } p_j > 50.5 \\ p_i = p_j - 0.01 & \text{if } 1.01 < p_j \leq 50.5 \\ p_i = 1.01 & \text{if } p_j = 1.01 \\ p_i \in \{1, 1.01, \dots, 100\} & \text{if } p_j = 1 \\ p_i \in \{p_j + 0.01, p_j + 0.02, \dots, 100\} & \text{if } p_j < 1 \end{cases}$$

Putting the best response functions together, we find two Nash equilibria, (1, 1) and (1.01, 1.01): the perpetual incentive to undercut their competitor's price drives the firms (almost) all the way down to their marginal cost of 1. Under the first Nash equilibrium, both firms earn payoffs of 0, while under the second, they earn 0.495. In contrast, if they were able to collude on the optimal monopolist's price of 50.50, then they would each achieve payoffs of 1,225. Thus, Bertrand's price competition model suggests that adding a single competitor to the market is all that it takes to drive the market price down from the monopolist's price of 50.50 to the perfectly competitive price of 1.

The Bertrand Game with a Continuous Strategy Set

Now let's consider whether anything about our analysis would change if we allowed firms to choose any real number as their price rather than restricting them to increments of a penny. Does this allow the firms to cooperate and achieve higher than minimal profits? Or does it make them even more competitive?

The best response for firm i is very similar in this case, but it is undefined for the large range of p_j when their incentive is to narrowly undercut the other firm. We previously saw that firm i would wish to undercut firm j by a penny if its price were between 1.01 and 50.5. But now what is firm i 's best response in this range? It wants to choose the highest possible price that is still under firm j 's price—but since the price can be any real number, no such best response exists. No matter what it chose, there would always be some number between its choice and firm j 's choice that would be even better. Therefore, firm i 's best response is undefined whenever firm j 's choice is greater than 1 and less than or equal to 50.50.

Thus firm i 's best response function becomes

$$BR_i(p_j) = \begin{cases} p_i = 50.5 & \text{if } p_j > 50.5 \\ p_i = \emptyset & \text{if } 1 < p_j \leq 50.5 \\ p_i \in [1, 100] & \text{if } p_j = 1 \\ p_i \in (p_j, 100] & \text{if } p_j < 1 \end{cases}$$

Notice that this eliminates the possibility of a mutual best response at (1.01, 1.01), since now that the firms could potentially undercut by less than a penny, they would have an incentive to do so rather than matching the other firm's price of 1.01. The only Nash equilibrium in the continuous game is thus (1, 1): now the firms are unable to sustain even the minimal level of cooperation needed to achieve the positive payoff of around 50 cents each.

Expanding the Strategy Set in the Traveler's Dilemma

Recall the traveler's dilemma game, originally developed by Basu (1994) and described in chapter 3: An airline loses the luggage of two passengers whose bags happen to have identical contents. Say, for example, the passengers are youth soccer players returning home from a tournament abroad and have lost their kit bags. The claims agent offers the following deal: each passenger must write the value of his or her bag and contents without consulting the other passenger and submit the claim to the agent. If the claims match (i.e., the two

passengers write the same amount), the agent will consider this the true value and pay it to both passengers as compensation. If instead one passenger writes a figure that is larger than the other, the agent will assume that the higher claim is inflated and pay both passengers the lower amount. *Further*, the agent will give a bonus payment to the conservative, lower claimant and issue a penalty to the higher claimant. What claims should the two passengers submit?

In chapter 3, we restricted the players to integer claims between 2 and 100. Using iterated elimination of weakly dominated strategies, we saw that the maximum remaining claim was always weakly dominated by a claim that was 1 less, and eventually all claims except for the minimum of 2 would be eliminated.

This game bears a clear similarity to the Bertrand pricing game: the players would jointly be best off if they select a high price/claim, and yet there is a continual incentive to ever so slightly undercut the other person to achieve the reward or corner the market. As in the Bertrand game, we can solve for the Nash equilibrium by using logic to find the best response function of each player. Let's assume that the players can choose any claim from 2 to 100, but that the claims must be multiples of ϵ . For instance, suppose that they are restricted to submitting claims in dollars and cents, then $\epsilon = 0.01$. What is traveler i 's best response, c_i , to the possible claim made by the other passenger, c_j ? So long as $\epsilon < 2$, then you're best off undercutting the other player's claim by ϵ . Why? If you were to match the claim, then you earn c_j . If you were to undercut it by ϵ , then you each get a slightly lower compensation, but you also receive the 2 dollar reward, for a payoff of $c_j - \epsilon + 2$. Thus, your best response is

$$BR_i(c_j) = \begin{cases} c_i = c_j - \epsilon & \text{if } c_j > 2 \\ c_i = 2 & \text{if } c_j = 2 \end{cases}$$

The unique Nash equilibrium of this game has each player choosing the minimum claim of 2. As in the Bertrand game, if we were to allow the players to submit any real number between 2 and 100, we would again find that the best response function is undefined over the range in which the players have an incentive to undercut (in this case, when the other player chooses $c_j > 2$). However, 2 is the unique best response to 2, and thus (2, 2) is the Nash equilibrium.

Finding Best Response Functions Using Calculus

Clearly, the passengers would not like this outcome. What if the passengers proposed their own procedure for determining their compensation? For the proposal to be accepted, the passengers reckon that the agent will insist that greedy behavior continue to be punished, so they make the following offer: the passengers will submit claims, c_i for $i = 1, 2$ between 0 and 100, but this time the payoff u_i each will receive is

$$u_i = c_i - \frac{(c_i - c_j)}{100} c_i$$

Would the agent accept this method? The agent notices that as one passenger asks for more than the other, she is penalized. Specifically, she loses a $\frac{1}{100}$ share of her claim for each dollar that her claim exceeds the other's. Likewise, when her claim is lower than the other's, she receives a bonus of the same size. Shouldn't the agent be happy to accept this plan?

Before agreeing, it might be smart for the agent to solve for the Nash equilibrium claims of the two passengers under this new scheme. But how does he do this? What will help the agent do this is knowing that these payoffs are continuous in the claims of the two passengers. If the other passenger claims 50, for instance, [figure 6.2](#) illustrates the relationship between the first passenger's payoff and her claim.

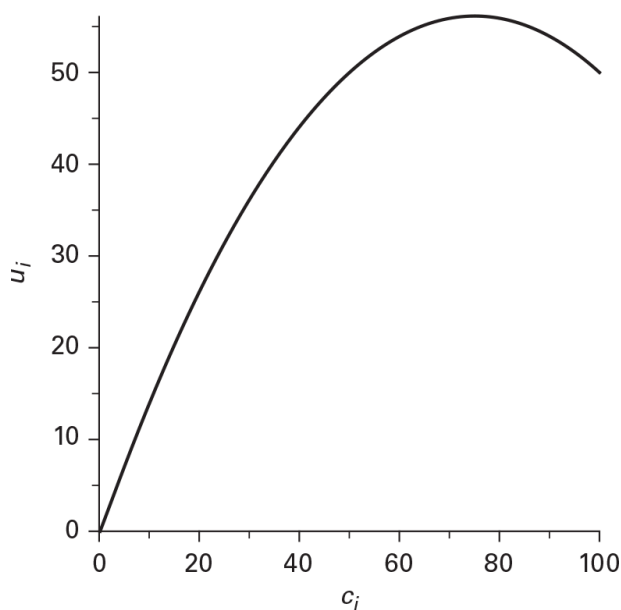


Figure 6.2

Utility is now continuous in the first passenger's claim.

Initially, as the passenger claims more, her payoff rises, but after some point, around a claim of 75, her payoff starts decreasing again. To see the intuition behind this, imagine that you start by claiming 0 and are thinking about how your payoff would change as you scale up your demand. So long as you're in the range below 50 (claiming less than the other passenger), it makes sense to keep demanding more because you will be awarded a bigger settlement and you continue to get the bonus for being the low claimant. Past 50, you will get more from scaling up your claim, but the bonus turns into a penalty. You're still better off increasing your claim, however, until you hit 75. In the end, for very high (relative) claims, the penalty outweighs the larger claim and your payoff starts to decrease.

Because the payoffs of the passengers are now continuous, we can use calculus to compute, for example, exactly what claim maximizes the first passenger's payoff when the other passenger submits a claim of 50. In this case, passenger 1's payoff is $u_1 = c_1 - \frac{(c_1 - 50)}{100} c_1 = c_1 \left(1 - \frac{(c_1 - 50)}{100}\right)$. If we use the product rule to find the derivative, $\frac{\partial u_1}{\partial c_1}$, (i.e., the slope of the payoff curve in [figure 6.2](#)), we can then set that derivative equal to zero to find the claim that places passenger 1 at the top of her payoff function. In other words, this c^* is the claim at which the slope of her payoff function becomes flat or equal to zero, indicating she is at the top or maximum payoff. In this case, we find

$$\frac{\partial u_1}{\partial c_1} = (c_1) \left(\frac{-1}{100}\right) + \left(1 - \frac{(c_1 - 50)}{100}\right) (1) = 0$$

or

$$\frac{-c_1}{100} + \frac{(150 - c_1)}{100} = 0$$

which quickly reduces to $c_1^* = 75$. This that matches our prediction based on [figure 6.2](#).

Notice, however, that the agent's calculations are a bit harder than this example because who is to say that the second passenger will submit a claim of 50, especially if she anticipates that the other passenger will ask for more? Indeed, the agent must account for this equilibrium of the two passengers responding to each other's claims. To do this, the agent must calculate how a passenger best responds to any possible claim made by the other passenger. This is actually easier that it sounds, however, because instead of substituting the claim of 50 for the other passenger, we simply leave the claim unspecified. Now, for the two passengers, we have

$$u_1 = c_1 - \frac{(c_1 - c_2)}{100}c_1 \quad \text{and} \quad u_2 = c_2 - \frac{(c_2 - c_1)}{100}c_2$$

To calculate the best response of each passenger to any general claim made by the other passenger, the agent will again maximize each passenger's payoff by taking the derivative.

First, consider passenger 1's payoff. While this is a function of both c_1 and c_2 , notice that only c_1 is actually under passenger 1's control. Since c_2 is what the other passenger claims, passenger 1 has no control over this, and so we treat it as a constant when calculating passenger 1's best response. In contrast, passenger 1 can control c_1 , and since we (and the claims agent) assume that she is rational, we also assume that she'll choose c_1 to maximize her payoff (see the appendix for a calculus refresher). To determine passenger 1's optimal response to c_2 , we take the derivative of u_1 with respect to the variable c_1 , treating c_2 just as we would any other constant:

$$\frac{\partial u_1}{\partial c_1} = (c_1) \left(\frac{-1}{100} \right) + \left(1 - \frac{(c_1 - c_2)}{100} \right) (1) = 0$$

Notice that we can confirm that the player's payoff is indeed maximized (rather than minimized) at this point, by checking the second derivative (in this case, $-2/100$) and ensuring that it is negative. Likewise, the agent finds that player 2's best response to any c_1 by taking the derivative of u_2 with respect to c_2 , while treating c_1 as a constant and setting it equal to zero:

$$\frac{\partial u_2}{\partial c_2} = (c_2) \left(\frac{-1}{100} \right) + \left(1 - \frac{(c_2 - c_1)}{100} \right) (1) = 0$$

Simplifying these expressions, the agent notices that each passenger's optimal claim is a function of the other passenger's claim, exactly what she was after. That is, she finds the players' best response functions to be

$$c_1 = \frac{100 + c_2}{2} \quad \text{and} \quad c_2 = \frac{100 + c_1}{2}$$

In each case, a passenger says to herself, with this function, if you tell me what the other passenger has claimed, I can tell you what is best for me to claim, and I know that it is the *best* option because I calculated the function by maximizing my payoff.

Graphing these two best response functions in [figure 6.3](#), we see that in both cases, the optimal claim of one passenger is increasing in the other passenger's demand. This realization should cause the agent to pause,

especially when she then realizes that there is only one claim at which this process can settle. That is, there is just one place where these functions intersect, and this intersection is key because it means that if the passengers stumble across this claim, they will simultaneously be best responding (in other words, they will have found a *fixed point*).

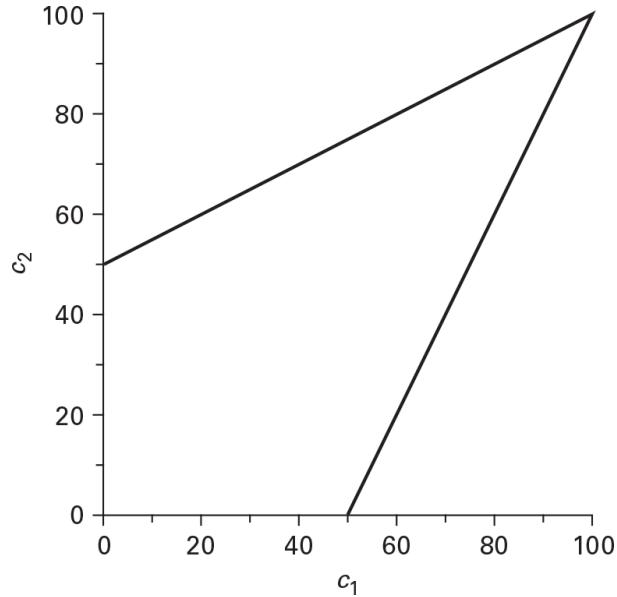


Figure 6.3

Best response functions for the two passengers.

To see how the process might work, imagine the passengers privately taking turns submitting provisional claims to each other before deciding on final ones to give the agent. What if they both start at 50 and then passenger 2 takes the next turn? She calculates $c_2 = \frac{100+50}{2} = 75$. Passenger 1 then takes passenger 2's updated claim and calculates $c_1 = \frac{100+75}{2} = 87.5$. Notice first that the claims are rising, but also notice what happens when one of the two (say, passenger 1) gets to 100. Now passenger 2 calculates her best response and finds $c_2 = \frac{100+100}{2} = 100$. Once they find their way to claims of 100, neither wants to change. By definition, this must be a Nash equilibrium.

Indeed, if we find the intersection of the best response functions algebraically by substituting passenger 2's into passenger 1's, we have

$$c_1 = \frac{100 + \left(\frac{100+c_1}{2}\right)}{2}$$

which reduces to $c_1^* = 100$. Further, we already know that if we substitute this result in the best response function for passenger 2, we will find $c_2^* = 100$ again.

Back to the question at hand—will the agent accept this counterproposal? Surely not! The agent goes from paying almost nothing to paying the maximum amount to both. So what is the intuition that explains the difference? The two claim schemes seem similar, in that they both punish claims that are higher than the other passenger's and reward those that are lower. The essence of the difference is that the original formulation given at the beginning of the chapter punishes high claims more severely. In the second scheme, for every dollar your claim is above the other passenger's, you lose just 1/100th of your claim, but in the first scenario, you lose, in essence, twice. First, you lose because you are assessed the fixed fine of \$2, which is always greater than the other fine; but you also lose a second time because your claim is reduced to that of the other passenger. In other words, starting from any situation in which you are claiming the same amount as the other passenger, in the first scenario, there is never a benefit from asking for another dollar—you will always lose money. By contrast, in the second scenario, the benefit of asking for another dollar is that dollar minus a penalty that rises to a dollar only when your claim is at its greatest.

The Cournot Duopoly

Let's apply what we've learned to the Cournot duopoly games that began the chapter. Two players, $i \in \{1, 2\}$, each simultaneously choose quantity q_i and earn payoff $u_i(q_i, q_j) = q_i(100 - q_i - q_j) - q_i$.

To find player i 's best response to the opponent's quantity, we take the derivative of the payoff function with respect to the variable under his control: q_i . Doing so, we get

$$\frac{\partial u_i(q_i, q_j)}{\partial q_i} = 100 - 2q_i - q_j - 1$$

Setting this equal to zero and solving for q_i , we get

$$q_i = \frac{99 - q_j}{2}$$

The best response functions for each of the two firms are shown graphically in [figure 6.4](#). Notice that, since firms cannot produce negative

quantities, the best response functions are bounded below by 0 and thus given by the following piecewise function:

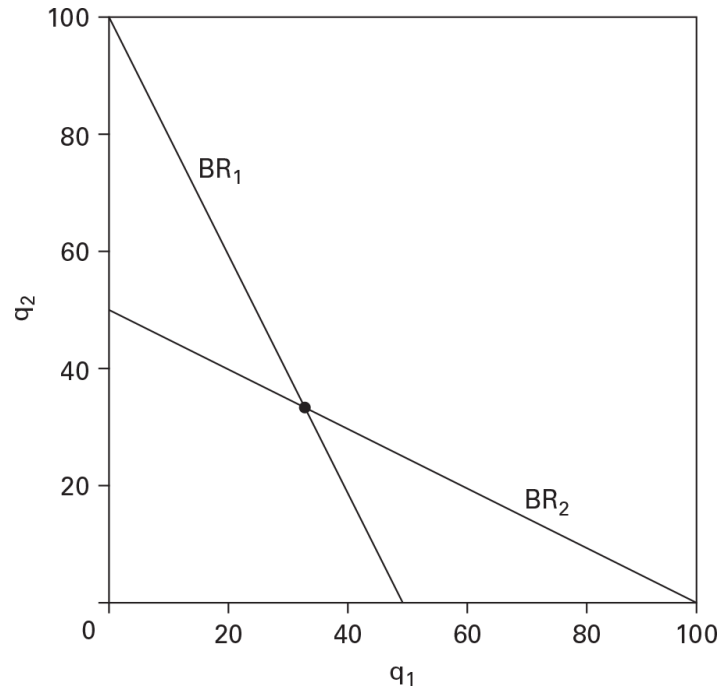


Figure 6.4

The players' best response functions in the Cournot duopoly game with a unique Nash equilibrium at (33, 33).

$$BR_i(q_j) = \begin{cases} q_i = \frac{99 - q_j}{2} & \text{if } q_j \leq 99 \\ q_i = 0 & \text{if } q_j > 99 \end{cases}$$

Now that we know each player's best response, we find the Nash equilibrium by looking for strategy profiles (q_1, q_2) , such that both players are best responding to each other—that is, they are maximizing their own profits given the output of their competitor. Algebraically, we're looking for the point such that $q_1 = \frac{99 - q_2}{2}$ and $q_2 = \frac{99 - q_1}{2}$. Substituting the second equation for q_2 in the first equation, we get

$$q_1 = \frac{99 - \left(\frac{99 - q_1}{2}\right)}{2}$$

Solving for q_1 , we have

$$\begin{aligned}
q_1 &= 99/2 - 99/4 + q_1/4 \\
3/4 q_1 &= 99/4 \\
q_1 &= 33
\end{aligned}$$

and

$$q_2 = \frac{99 - 33}{2} = 33$$

Thus, in equilibrium, a total of 66 units are produced and are sold for $100 - 66 = 34$ each. Each firm earns a profit of $33 \times 34 - 33 = 1,089$. Recall that when two firms competed on price, as in the Bertrand model, the strategic pressure to undercut their rival drove them to set their price equal to their per-unit cost, with both firms subsequently earning zero profit. The results of the Cournot quantity competition are less dramatic and firms in a duopoly are able to earn positive profits in equilibrium. However, they still *overproduce* (and thus achieve a lower market price) relative to the joint output level that would maximize the two firms' collective profits. To see this, imagine that a single corporation purchased both firms. Letting Q be the total output produced by the two firms ($Q = q_1 + q_2$), then the firms' joint profits are $u_1 + u_2 = q_1(100 - Q) - q_1 + q_2(100 - Q) - q_2 = Q(100 - Q) - Q$. Joint profits are maximized when $Q = 49.5$. In this case, the market price is $p = 50.5$, and so this confirms our previous conclusion that a monopolist maximizes profits by setting price equal to 50.5. Thus, we again have a social dilemma, as in the prisoner's dilemma, traveler's dilemma, and Bertrand pricing game: each player rationally following their own self-interest leads to both players receiving lower payoffs than they could achieve if they were able to cooperate.

Why do firms in a Cournot duopoly have an incentive to overproduce relative to the output that would maximize industry profits? Notice that there are pros and cons to a firm choosing to produce an extra unit: the firm sells more, but at a slightly lower price. This trade-off is worth it for firm i so long as $\partial u_i(q_i, q_j)/\partial q_i = 100 - 2q_i - q_j - 1 > 0$ (our first-order condition is positive), and so a firm that considers only how its choice affects its own profits would continue producing up to the point that $q_i = \frac{99 - q_j}{2}$. However, the choice to produce always has negative consequences for the other firm, since it drags down the price for the entire market. Thus, if the firm ignores the negative impact of its actions on its competitors' profits, then it will overproduce relative to the collective optimum.

***n*-Player Games with Discrete Strategies**

What happens if, instead of many strategies and just two players, we consider the opposite case, in which there are just two strategies but many players? It is straightforward to extend our definitions of best response and Nash equilibrium to games with any arbitrary number of players, n .

Definition 1 (Best response) *Player i 's strategy $s_i \in S_i$ is a best response to the other players' strategies s_{-i} if $u_i(s_i, s_{-i}) \geq u_i(s'_i, s_{-i})$ for all $s'_i \in S_i$. We write $s_i = \text{BR}_i(s_{-i})$.*

Definition 2 (Nash equilibrium [n-player game]) *A strategy profile $(s_1^*, \dots, s_n^*)$ is a Nash equilibrium if, for all players i , $u_i(s_i^*, s_{-i}^*) \geq u_i(s'_i, s_{-i}^*)$ $s'_i \in S_i$. In other words, $s_i^* = \text{BR}_i(s_{-i}^*)$ for all i .*

It would be clearly infeasible to analyze a game with an arbitrarily large number of players if all the players had their own actions and incentives. For this reason, we'll focus our attention on *symmetric games*, which are games that look the same from the perspective of each player: Each has the same available actions, payoffs, and information. Another way to think about a symmetric game is that we could change the names of all the players around (e.g., we switch who is assigned to the rows versus the columns of the payoff matrix) and no one would be the wiser—the game would remain exactly the same for each person. An example is shown in [figure 6.5](#).

		Column	
		A	B
Row	A	(20, 20)	(10, 40)
	B	(40, 10)	(0, 0)

Figure 6.5

A symmetric game.

Definition 3 (Symmetric Nash equilibrium) *A strategy profile s^* is a symmetric Nash equilibrium if $s_i^* = \text{BR}_i(s_{-i}^*)$ for all i (i.e., the strategy profile is a Nash equilibrium) and $s_1^* = s_2^* = s_3^* = \dots = s_i^* = \dots = s_n^*$ (i.e., the equilibrium strategies for each player are the same). In contrast, an asymmetric Nash equilibrium is a Nash equilibrium in which $s_i^* \neq s_j^*$ for some i and j .*

In general, there are two facts in symmetric games that will make our lives easier when looking for Nash equilibria in n -players games:

(1) If we find an asymmetric Nash equilibrium, then there are at least $n-1$ more asymmetric Nash equilibria, which we can find by swapping the players' strategies. For instance, in the game shown in [figure 6.5](#), we find that (A, B) is a Nash equilibrium. Since the game is symmetric, (B, A) must also be a Nash equilibrium. In a three-player game, if (X, X, Y) were a Nash equilibrium, then (X, Y, X) and (Y, X, X) must also be Nash equilibria, and so on.

(2) There always exists at least one symmetric Nash equilibrium, possibly in mixed strategies. For instance, in this game, there is no symmetric pure strategy Nash equilibria. However, there is a mixed-strategy Nash equilibrium in which each person chooses A with probability $1/3$.

The Volunteer's Dilemma

Consider the following scenario to introduce one example of this class of games. It is late—after 2 a.m. on March 13, 1964. You are awoken by screams in the courtyard of your apartment building in the borough of Queens, New York. As you rush to the window, you notice two things. First, a brutal assault on a young woman is happening in the courtyard below. Second, 37 other lights are coming on in the surrounding apartments, indicating these people are witnessing the crime too. All 38 of you face the same choice: stay silent or intervene (by shouting, calling the police, or heading outside to help). If you do nothing, someone is likely to be hurt badly, or even killed. If you intervene, you will surely help the victim, but at what cost? What if the perpetrator turns on you? Unfortunately, this was a real-life incident, and while some of the witnesses did call the police or shout out the window, most did not, and the victim, Catherine (Kitty) Genovese, died later that morning. An article published in the New York Times after the attack infamously—but erroneously—claimed that 38 neighbors witnessed the attack and all failed to intervene or call for help. Subsequently, this tragedy spawned an entire subfield of psychology that continues to study the “bystander effect,” while the basic decision has become known as the *volunteer's dilemma*.²

What is unique about the volunteer's dilemma is that it takes just one person to do the job and provide the public benefit (i.e., just one person has to call 911). In fact, any efforts by volunteers beyond the first are wasted. Hence,

not only does the difference between the benefit and the cost of volunteering play a role, so does coordination. To see this, let's first consider the game with two players. Suppose that volunteering results in a benefit, $b > 0$, for both of two players, but only volunteers incur the cost, $c > 0$, regardless of whether they are the first. Consequently, there are three possible payoffs. Each player earns 0 if nobody volunteers, earns b if the *other* person volunteers, or earns $b - c$ if he or she personally volunteers. Clearly, if $b - c < 0$, nobody ever volunteers—it just isn't worth it. However, if $b - c > 0$, there are two pure-strategy equilibria, wherein just one of the two players volunteers and the other doesn't. In both of these cases, the benefit is provided and therefore any nonvolunteers cannot do better by intervening—their payoff would go from b to only $b - c$. Further, the one volunteer does not want to step back in line because his payoff would fall from $b - c$ to 0. But the players now face the coordination aspect of the game—it is a waste for more than one person to volunteer, so who does it?

Since the game is symmetric, we know that there must be another, symmetric equilibrium. Since we just saw that the only pure-strategy Nash equilibria are asymmetric—different people take different actions—the symmetric equilibrium must involve mixing. Recall from chapter 5 that a mixed strategy is a best response only if the player is indifferent between the actions, and so, in equilibrium, the expected payoff of volunteering must equal the expected payoff of not volunteering. Volunteers get $b - c$ no matter what the other player does, while the expected payoff of not volunteering is given by $p(b) + (1 - p)(0)$, where p is the probability that the other player volunteers. Setting $Eu(\text{Volunteer}) = Eu(\text{Not Volunteer})$, we get $b - c = p(b) + (1 - p)(0)$. Solving this expression for the equilibrium rate of volunteering, we find $p^* = \frac{b-c}{b}$. This prediction of the rate of volunteering is intuitive; as the benefit increases, so does the rate of volunteering, but the opposite is true for increases in the cost.

Instead of just two players, now suppose there are a large number, n , of them who are considering an intervention. This time, there will be n pure-strategy equilibria, each with exactly one volunteer. There will also be a mixed-strategy equilibrium that highlights the notion of symmetry. Specifically, if all n individuals have exactly the same incentive, implying that their strategic situations are symmetric, why wouldn't they all be inclined to do exactly the same thing? Let's say that each person chooses to volunteer

with probability p . What does p need to be to ensure that no one wants to deviate from this mixed strategy to either volunteering with certainty or not volunteering with certainty?

As in the two-player version, volunteering is always associated with a payoff of $b - c$ regardless of the others' choices. At the same time, the payoff to not volunteering does depend on what the others do. If *none* of the other $n - 1$ people volunteer, then everyone gets a payoff of 0. Since each person's probability of not volunteering is $(1 - p)$, the probability that none of the other $n - 1$ people volunteer is $(1 - p)^{n-1}$, assuming that the players choose independently (the appendix goes through the basic laws of probability). This implies that in any other state of the world, at least one other person has volunteered (and it may be more than one person, but this doesn't matter for the payoff), which occurs with probability $1 - (1 - p)^{n-1}$. Therefore, the expected payoff of not volunteering is now a bit more complicated than in the two-player case:

$$Eu(\text{not volunteer}) = [1 - (1 - p)^{n-1}](b) + (1 - p)^{n-1}(0)$$

and after equating the two payoffs and a bit of algebra, we find that the equilibrium rate of volunteering in a population of n individuals is $p^* = 1 - \left(\frac{c}{b}\right)^{\frac{1}{n-1}}$.

In addition to the previous predictions about the effects of changes in the costs and benefits of volunteering, which still hold, our generalization to n players reveals an additional prediction that the coordination problem gets worse as the number of observers gets larger.³ Based on the logic given here, in equilibrium, the probability that nobody intervenes is just $(1 - p^*)^n$ or $\left[1 - \left(1 - \left(\frac{c}{b}\right)^{\frac{1}{n-1}}\right)\right]^n$. This expression simplifies to $\left(\frac{c}{b}\right)^{\frac{n}{n-1}}$, which is increasing in n and, in the limit, converges on c/b for a sufficiently large population, indicating that someone volunteering is far from certain. Given this logic, though horrible, it may no longer be surprising that so few people intervened on Kitty Genovese's behalf. There is no doubt that they all would have agreed that someone should volunteer, but because they were unable to coordinate, they mistakenly assumed that someone else would help.

n -Player Games with Continuous Strategies

In this chapter, we've learned how to solve for Nash equilibria in games with continuous strategy sets and those with an arbitrarily large number of players.

In this section, we'll see how to combine these two techniques. In doing so, we'll look at two important examples: a Cournot oligopoly with an arbitrarily large number of players and a contest in which players invest resources in the hope of winning a prize.

The n -Player Cournot Oligopoly

Let's now return to the Cournot oligopoly game, but instead of only two players, we'll assume that there are n players, each facing a symmetric situation. Now, each player's payoff is given by

$$u_i(q_1, \dots, q_n) = q_i \left(100 - \sum_{j=1}^n (q_j) \right) - q_i$$

We'll first find the best response for each player, and then we'll solve for the symmetric Nash equilibrium. Before we take the derivative, consider how we should deal with the sum of outputs, $\sum_{j=1}^n (q_j)$, which occurs in the payoff function. Notice that $\sum_{j=1}^n q_j$ sums up i 's output along with the other $n - 1$ players' outputs. Since we'll take the derivative with respect to i 's effort, let's break this summation into the part that has our variable of interest, which firm i can control (q_i), and the part that includes the production by other firms and is thus treated as a constant ($\sum_{j \neq i} q_j$). When we sub in for $\sum_{j=1}^n q_j = q_i + \sum_{j \neq i} q_j$, our payoff function becomes

$$u_i(q_1, \dots, q_n) = q_i \left(100 - q_i - \sum_{j \neq i} q_j \right) - q_i$$

Solving for the optimal level of q_i , we have

$$\partial u_i(q_1, \dots, q_n) / \partial q_i = 100 - 2q_i - \sum_{j \neq i} q_j - 1 = 0$$

and

$$q_i = \frac{99 - \sum_{j \neq i} q_j}{2}$$

Thus, each player best responds by choosing a quantity equal to half the amount by which 99 exceeds the total quantity produced by the other players. In equilibrium, this equation must be satisfied for all n players. This would give us n equations and n unknowns. However, our problem is made simpler by the fact that we're looking for a symmetric equilibrium in which all n members choose the same q . To find the symmetric Nash equilibrium, we look for a quantity q chosen by each player, such that q is a best response to

each of the other $n - 1$ players who choose q . That is, each player choosing q is a Nash equilibrium if $q = \frac{99 - (n-1)q}{2}$, which occurs when $q = \frac{99}{n+1}$. Thus, our symmetric Nash equilibrium strategy profile is the n -tuple $(\frac{99}{n+1}, \frac{99}{n+1}, \dots, \frac{99}{n+1})$.

If $n = 1$, then we see that the equilibrium quantity is 49.5: the optimum we found for a monopolist. If $n = 2$, then the equilibrium is $q = 33$ units produced by each firm—confirming our earlier result in the Cournot duopoly—and the total market output is 66. If $n = 4$, as in the game described at the beginning of the chapter, then the equilibrium quantity is 19.8 and a total of 79.2 is produced. Notice that as n increases, each firm produces less, but the total output in the market increases and approaches the perfectly competitive output of 99 and the price of the unit cost of 1. In other words, while the Bertrand model predicts that a market will reach the perfectly competitive outcome as soon as a second firm enters the market, the Cournot model predicts that the market price and quantity would fall between the perfectly competitive outcome of $p = 1$, $Q = 99$ and the monopoly outcome of $p = 50.5$, $Q = 49.5$, but the outcomes approach perfect competition as the number of firms grows larger.

Application: The Tullock Contest

In a Tullock contest, there are n players who each have an endowment of financial resources, E_i , that they can devote to a contest with a fixed prize, $P > 0$. You can think about this as the decision about how much to spend on research and development in a patent race, for instance. The chance of any individual winning the contest depends on how many resources are expended by all the players. Specifically, the chance of player i winning the prize is increasing in his investment, x_i , but decreasing in the investments made by others. If the sum of all the investments is $\sum_{j=1}^n x_j$ (the shorthand being $\sum x_j$), then we assume that the probability that contestant i wins is $\frac{x_i}{\sum x_j}$, which is his share of the total investment. Incorporating the endowment and the prize allows us to write player i 's expected payoff as

$$Eu_i = (E_i - x_i) + \frac{x_i}{\sum x_j} P$$

Notice that the n players are again symmetric, in that they face exactly the same investment incentives and they must decide among a continuum of strategies. As discussed previously, to make this choice, each of the players will wonder the same thing: what happens to my payoff when I increase my

investment a little bit? To answer this question, the i th contestant calculates the following derivative (treating the numerator of the second term above as $x_i P$ and using the quotient rule):

$$\frac{\partial E u_i}{\partial x_i} = -1 + \frac{\sum x_j (P) - x_i P(1)}{(\sum x_j)^2}$$

which he then sets equal to zero to find the x_i^* value that places him at the top of the utility hill where the slope is flat (i.e., equal to zero). Mathematically, the important thing to note here is the 1 in the numerator of this expression. This comes from taking the derivative of the denominator of the chance of winning the contest, $\sum x_j$. As we saw in the previous example, the derivative of this expression with respect to x_i is just 1, since we can rewrite this sum as $\sum x_j = x_i + \sum_{j \neq i} x_j$, and all the other x s in the second term are treated like constants, not to be determined by you. Doing the remaining algebra leads the contestant to an awkward intermediate step, where we find

$$P(\sum x_j - x_i) = (\sum x_j)^2$$

Here is where we rely on symmetry again to solve for the equilibrium investment. At the symmetric Nash equilibrium, all n contestants make exactly the same investment, which implies that $\sum x_j$ is effectively adding the same number (call it x because the subscripts are now meaningless) n times. Well, the sum of n x s is just nx . Making this substitution allows us to get to the equilibrium investment quickly because

$$P(nx - x) = (nx)^2$$

and therefore

$$x^* = \frac{P(n-1)}{n^2}$$

You can see that this expression predicts that investments increase as the prize value increases and decrease as the number of contestants rises because there is more competition.

Behavior in Games: Do Travelers Think of a Lost Bag as a Dilemma?

As is our convention, we now return to the scenarios that sparked off our conversation of nonmatrix games and ask how people actually play the Cournot, Bertrand, and traveler's dilemma games. We then discuss the impact

that this empirical research has begun to have on the behavioral theory of games.

The Cournot oligopoly game presented at the beginning of this chapter was investigated in the lab in a series of papers by Steffan Huck, Hans-Theo Normann, and Jörg Oechssler. One striking finding is that players' tendency to play according to the Nash equilibrium depends on the number of firms in the market. Huck, Normann, and Oechssler (2004) varied the number of firms from 2 to 5. The table in [figure 6.6](#) shows the Nash equilibrium total output in each case, as well as the actual average output in their experiments and this output as a proportion of the Nash prediction (such that 1 indicates average behavior consistent with the Nash equilibrium and numbers less than 1 indicate that players cooperate more than predicted). While collusion can sometimes emerge in markets with two firms, they find that this is rare when there are more firms in the market and the average output levels move closer to the Nash prediction. In fact, with four or more firms, the market output is often above the Nash equilibrium level. In a separate experiment (Huck, Normann, and Oechssler, 1999), the authors attempted to better understand behavior in the four-player game. They found that many subjects followed a process in which they best responded to the previous total output of the other three players (a process that we'll discuss in more detail in chapter 24, on best response dynamics). In addition, when players were given information about the individual output choices and payoffs of the other players, some participants followed an "imitate the best" rule of thumb, in which they matched the output of the player who earned the most in the previous period, a decision rule that lead to higher market output (and a learning process that we'll discuss in more detail in chapter 25, on evolutionary game theory).

Firms	Nash	All Data		Last Third of Game	
		Observed	Proportion of Nash	Observed	Proportion of Nash
2	66	59.36	0.89	60.44	0.91
3	74.25	73.47	0.99	72.59	0.98
4	79.20	77.26	0.98	80.67	1.02
5	82.50	86.21	1.05	88.43	1.07

Figure 6.6

Data from Huck et al. (2004).

In Bertrand pricing games, players are better able to sustain collusive outcomes (by choosing prices above the Nash equilibrium) than in the Cournot game (by choosing quantities below the Nash equilibrium). There are several differences between these games, including the frame of choosing prices versus quantities, whether cooperation requires choosing a higher or lower number than the equilibrium prediction, and the relative payoffs of the collusive and equilibrium outcomes. Potters and Suetens (2009) focus on whether players' choices are *strategic complements or substitutes*. Simply put, choices in the Bertrand game are strategic complements because the best response function is positively sloped. Similar to the traveler's dilemma, if you believe that the other person is attempting cooperation by choosing a higher price or claim, then you want to follow their lead and increase your claim as well (while still undercutting, naturally). In contrast, choices in the Cournot game are strategic substitutes, as can be seen from the downward-sloping best response function in [figure 6.4](#), earlier in this chapter: as the other player produces more, you prefer to produce less in response. As a result, if one player attempts cooperation by scaling down production, the other player is incentivized to fill this void by ramping up their own production.

In a carefully designed experiment, Potters and Suetens examined two games that differ only in whether the players' choices are strategic complements or substitutes, while keeping constant the framing, the strategy set, the Nash equilibrium and payoff-maximizing choices, the payoffs from these choices, and the absolute values of the best response functions' slopes. Consistent with their hypothesis, they found substantially more collusion in the strategic complements game.

While Bertrand competition is more amenable to cooperation, the level of collusion is affected by the number of firms in the market in these experiments as well. Experiments on Bertrand games have found that players are better able to sustain cooperative outcomes in two-player games, but that they tend to converge in the direction of the equilibrium prediction in games with three or four firms (Dufwenberg and Gneezy, 2000). One explanation for this finding is that players are not narrowly self-interested, but instead care about efficiency and the payoffs of others such that their best response is to cooperate in the event that the other players cooperate as well. Under these alternative payoffs, the monopolist outcome and the standard prediction may

both be Nash equilibria, and the challenge is for players to coordinate on one of these outcomes. As we will see in chapter 7, players are often more successful at coordinating on more efficient, but riskier, Nash equilibria when there are fewer players in the group.

The 1999 study by Monica Capra and her coauthors is both one of the first experimental examinations of the traveler's dilemma and one of the most comprehensive. In each of 10 rounds, randomly matched pairs of participants make simultaneous claims between \$0.80 and \$2.00. As before, the amount claimed is paid if the two claims are equal, but only the low claim is paid to both travelers when they are unequal. In addition, there is a reward, denoted as R , paid to the person making the lower of unequal claims and a punishment of equal value paid by the person making the higher claim. In other words, if c_1 is traveler 1's claim and c_2 is traveler 2's, when $c_1 > c_2$, for instance, traveler 2's payoff is $c_2 + R$ and traveler 1's is $c_2 - R$.

As [figure 6.7](#) indicates, there is a lot of heterogeneity in claims (pooled across rounds and treatments) when there shouldn't be. Given the parameterization of the experiment, the unique Nash equilibrium is to claim \$0.80. Despite this rather straightforward prediction, only a third or so of participants in the experiment claim \$0.80. Further, the distribution of midrange, nonequilibrium claims is roughly uniform, and more than 20% of the participants claim within 10 cents of the maximum, \$2.

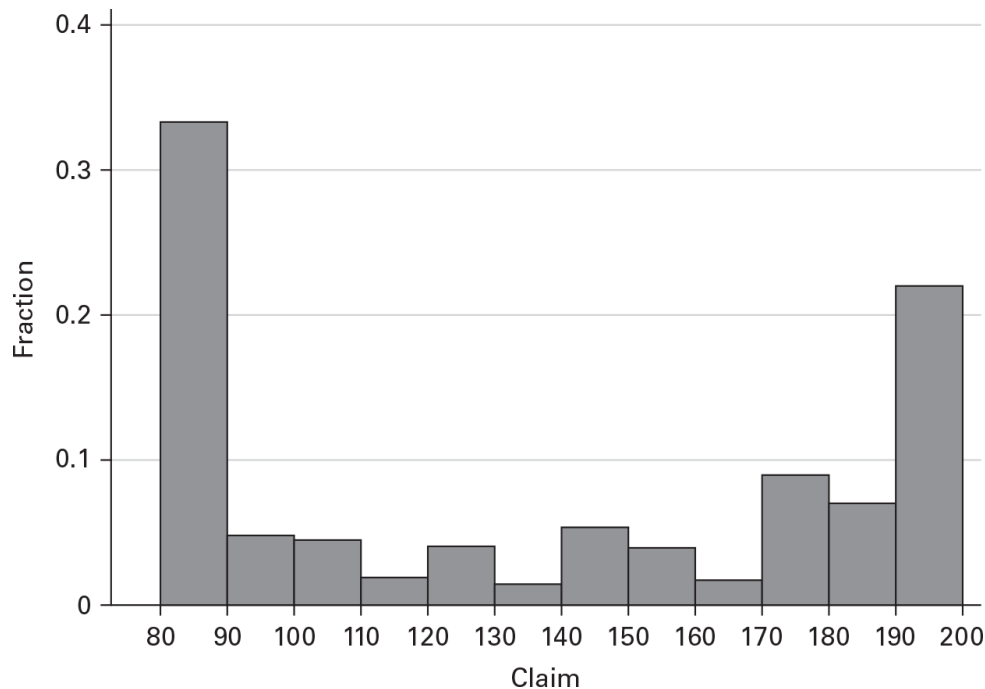


Figure 6.7
Claims in the traveler's dilemma.

While the heterogeneity of behavior seen in [figure 6.7](#) is interesting and unexpected given standard game theoretic reasoning, it is also important to remember that the equilibrium prediction is independent of the value of R . Put simply, so long as it is more than a penny, the exact value of R shouldn't matter. Indeed, for any R large enough to make travelers consider undercutting their fellow passengers, the Nash prediction is based on the *sign* of the payoff difference between maintaining a higher claim and undercutting, not the magnitude of the payoff differential between the two strategies. Put differently, it doesn't matter if undercutting the other traveler will increase your payoff by 5 cents or \$5; so long as the difference is positive, you are predicted to do it.

What is at odds with the concept of a Nash equilibrium is that R does appear to matter. As [figure 6.8](#) indicates, only when the rewards and penalties are substantial (e.g., equivalent to 80 cents, the value of the equilibrium claim) does behavior converge to the Nash prediction by the fifth of 10 periods. At the same time, when R is small (e.g., just 5 cents), claims converge to the opposite extreme—\$2, the social optimum.

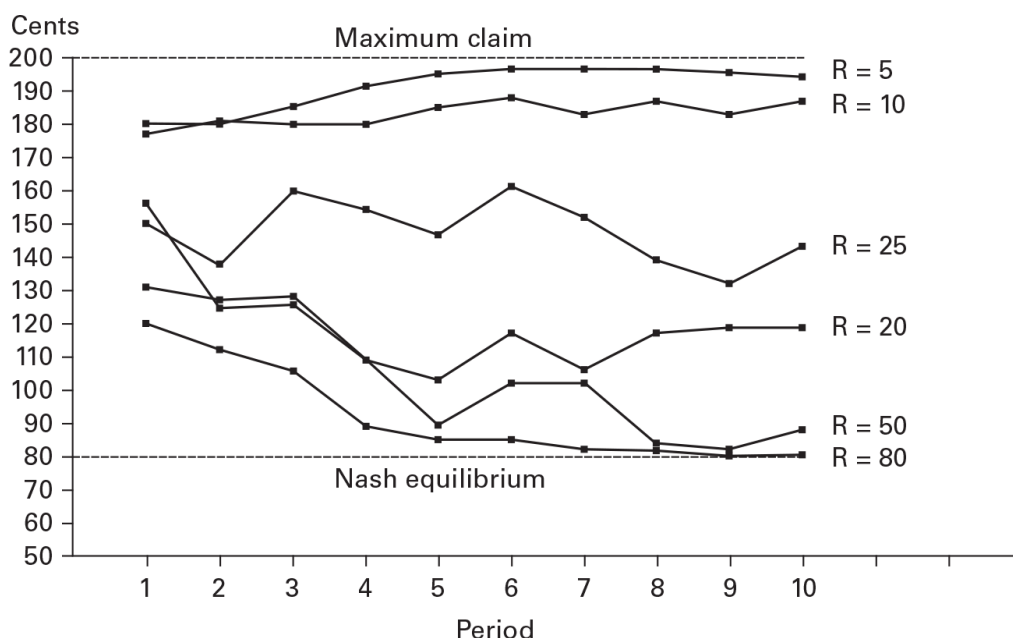


Figure 6.8
Treatment differences in the traveler's dilemma.

How can theory be altered to account for what the Nash prediction misses here? Given the pattern of responses, it seems clear that the decision to switch strategies depends on the magnitude of the difference in payoffs. When the benefit of undercutting your opponent is not very large, doing so is not common, but as the benefit increases, travelers are much more likely to do it. The quantal response equilibrium (QRE), mentioned in chapter 3 and discussed in detail in chapter 26, assumes that players don't always perfectly best respond with certainty, but instead are more likely to play better responses than worse ones, and the likelihood of making a mistake in implementing their best response decreases as the error becomes more costly. As a consequence, when R is low and the payoff consequences of making an error is low as well, players are more likely to make higher claims.

An alternative theory instead assumes that players avoid strategies that have the potential to inspire tremendous regret, such as choosing a low claim in the traveler's dilemma and then learning that your partner chose the maximum. In the model of *iterated regret minimization* offered by Halpern and Pass (2012), the various values of R (and therefore payoff magnitudes) affect the predicted claims in the traveler's dilemma. The authors ask how claims would be different if instead of picking strategies to best respond to your beliefs about what other players will do (even though in a one-shot

game, you can never observe or sample this behavior), players simply try to anticipate and minimize the regret that they may suffer from any choice. The concept of choosing actions to minimize regret has a long tradition in decision theory (e.g., Niehans, 1948; Savage, 1951) and is vastly easier to implement because there is no presumption that a player knows what the other players are likely to do.

In a nutshell, the regret minimizer considers each of her own possible strategies, c_i , and assesses how much she could wind up regretting this choice. To do so, she considers every possible claim that could be made by the other traveler, c_j , and calculates what her payoff would be (i.e., $u_i(c_i, c_j)$). She then compares it to the payoff she would get if she had managed to best respond to c_j instead of choosing c_i (i.e., $u_i(BR_i(c_j), c_j)$). This payoff difference tells us how much the player would regret playing c_i in the event that she later learns that her opponent played c_j . Of course, without knowing c_j , the decision maker doesn't know for sure how much she'll regret playing c_i , but she assumes the worst-case scenario and assesses c_i by the maximum possible regret associated with it. The regret minimizer, therefore, decides to submit the claim that is associated with the smallest maximum regret—that is, the claim for which the largest difference between the highest payoff she could have gotten and the payoff she did get is the smallest.

To illustrate the process of regret minimization, consider a discrete version of the traveler's dilemma in which there are just three possible claims: \$2, \$1.99 and \$0.8. In the top row of the table shown in [figure 6.9](#), we calculate the highest payoff that one could achieve for each of the claims the other traveler can make. These payoffs constitute our basis for feeling regret. If the other traveler claims \$2, the best you can do is claim \$1.99 and earn a payoff of $1.99 + R$. If she claims \$1.99, you do best by also claiming \$1.99 (conditional on R not being too large), in which case you both get 1.99. Finally, if the other traveler claims \$0.80, you get \$0.80 by matching this claim.

	Other's claim		
	2	1.99	0.8
$\max u$	$1.99 + R$	1.99	0.8
$u(2)$	2	$1.99 - R$	$0.8 - R$
$r(2)$	$R - 0.01$	R	R
$u(1.99)$	$1.99 + R$	1.99	$0.8 - R$
$r(1.99)$	0	0	R
$u(0.8)$	$0.8 + R$	$0.8 + R$	0.8
$r(0.8)$	1.19	$1.19 - R$	0

Figure 6.9

Regret minimization.

To calculate your regret from choosing to claim \$2, you compare the payoff that you get from claiming \$2 in each situation against the best you could possibly do in those situations. For instance, if you claim \$2 and the other traveler does too, you earn 2 but could have gotten $1.99 + R$ by claiming \$1.99 instead. Your regret is just the difference or $R - 0.01$. Using the same process, you see that your regret is R from claiming \$2 against claims of \$1.99 and \$0.8. The rest of the table lists these calculations for all three possible claims.

To make her final choice, the regret-minimizing traveler must account for the size of R , a feature of this decision rule that will make it perform better than Nash equilibrium. If, for example, R is low, then claiming \$0.80 doesn't seem like a good idea to the regret minimizer. Making this claim results in a potential regret of 1.19 when the worst she would regret claiming \$2 or \$1.99 is only R . On the other hand, when R is high enough (specifically, $R > 1.19$), the Nash equilibrium claim of \$0.80 minimizes regret. You can see that these predictions mimic the results seen in [figure 6.8](#).

Now that we've seen the intuition for regret minimization in the three-strategy version of the game, let's see what happens when we generalize to allowing participants to choose any number between 0.80 and 2.00 in 1 cent increments. While we wouldn't want to construct a table as we did in the three-strategy version for this entire strategy space, the problem isn't too bad because, for any claim you choose, there are at most two types of claims the other player can submit that could evoke the maximum regret for you: he

could submit 2, causing you to wish you submitted 1.99, or he could undercut you, causing you to wish you had undercut him instead. The possible regrets associated with each of your possible claims are summarized in [figure 6.10](#).

You Claim	Opponent Claims 2	Opponent Undercuts
0.80	1.19	–
0.81	1.18	R
0.82	1.17	$2R - 0.01$
$c_i \geq 0.82$	$1.99 - c_i$	$2R - 0.01$

Figure 6.10

Maximum regrets in the traveler's dilemma.

First, if you submit the minimum of 0.80, then you eliminate any regret associated with not undercutting, and your maximum regret would come from the other player submitting 2.00, in which case you get a payoff of $.80 + R$ but could have gotten $1.99 + R$ (regret: 1.19, as we already saw in the three-strategy version). If you submit a claim of 0.81, then your regret if you find that the other player submitted 2.00 is 1 cent lower (1.18), but now you have to worry about the other player undercutting you by choosing 0.80 (in which case you'd get a payoff of $0.8 - R$ but could have gotten 0.8, so your regret is just R). Your maximum regret associated with .81 is thus $\max\{1.18, R\}$. What if you submit 0.82? Again, your regret from learning that the other guy chose the maximum ticks down by 1 cent (1.17). But now the worst regret you could feel if he undercuts you is two-fold because not only do you have to pay the penalty, but you *could have* undercut him instead and gotten the reward. Specifically, if he chooses 0.81, then you get $0.81 - R$, but you could have gotten $0.80 + R$, so your regret is the difference, or $2R - 0.01$. Your maximum regret from 0.82 is thus $\max\{1.17, 2R - 0.01\}$. We could continue like this all the way up to your maximum claim of 2, but notice that the same pattern holds for any claims $c_i \geq 0.82$: the regret from learning the other player chose 2 is simply $1.99 - c_i$ and the maximum regret from being undercut is $2R - 0.01$, so your maximum regret from any claim is $\max\{1.99 - c_i, 2R - 0.01\}$.

By comparing the two columns, we can see that the size of R determines whether you worry more about undercutting too much and missing out on a high reimbursement versus being undercut and getting hit with the penalty

instead of extracting it from your partner. From studying the previous table, we can see that if R is very large (greater than 1.19), then the unique regret-minimizing claim is 0.80, and for R somewhat lower (between 0.595 and 1.19), regret is minimized by choosing 0.81. However, when R is less than 0.595, then any $c_i \in [2 - 2R, 2]$ minimizes regret, since all these choices are associated with a maximum regret of $2R - 0.01$. Notice when R is low, the lower bound on this range is close to the maximum claim, but as R increases, there is a broader range associated with regret minimization. Halpern and Pass (2012) take this reasoning one step further. If you believe that your opponent is a regret minimizer choosing $c_j \in [2 - 2R, 2]$, then what claim minimizes your regret? In this case, there is a unique regret-minimizing claim of $c_i = 2 - 2R + 0.01$. (If you feel inspired to work through the cases, you can show that this claim is associated with a maximum regret of $2R - 0.02$, while any other claim is associated with $2R - 0.01$.) The intuition is that situating your own claim 0.01 above the minimum figure that your opponent might choose avoids the double whammy of regret that you'd experience if you were undercut and could have undercut them.

What is important here is that a different decision rule comes closer than Nash equilibrium to describing the heterogeneity of claims that we see in the traveler's dilemma experiment. Not only does regret minimization account for the fact that the value of R substantially affects the choices people make, but the prediction made by the theory (i.e., $c_i = 2 - 2R + 0.01$ when $R < 0.595$) matches the data closely, and we are required to assume less about what players know when making a choice (i.e., regret minimizers need not form beliefs about what others will do). Halpern and Pass (2012) go on to show that regret minimization can better organize the data from other experiments in which the Nash equilibrium also often fails to predict behavior, including the Bertrand pricing game, Nash bargaining games like the cake demand game, and the centipede game that we'll discuss in exercise 10 and again in chapter 8.

Exercises

(1) Another traveler's dilemma: Imagine that our two travelers have lost another bag each. This time, the agent for the airline states that he is willing to compensate them for their losses but is sure that the bags are each worth between \$50 and \$100. Further, he will pay both travelers the amount of the

lower claim, and the reward he will grant the low claimant is \$2, which is the same as the penalty levied on the high claimant (as before, if the claims are equal, no rewards or penalties will be issued).

- (a) What is the Nash equilibrium claim for each traveler?
- (b) Now suppose that the airline agent changes the rule because punishing travelers seems like a bad public relations strategy. Specifically, only the lower claim will be paid and, the low claimant will still get a reward of \$2, but the high claimant will no longer face a penalty. What are the Nash equilibrium claims now?

(2) Price competition: Suppose that producers compete on price and that there are two firms producing amounts q_1 and q_2 in an industry for which market demand is $q_i = 10 - \min[p_1, p_2]$. Because the two firms produce a homogeneous product, all the buyers will flock to the low-price seller, in which case that seller provides the entire quantity demanded. If, however, the firms set the same price, they share the market demand. Finally, assume that each firm faces a constant marginal cost of production of \$2 per unit sold.

- (a) What are the prices charged by the two firms at the Nash equilibrium?
- (b) What if the firms both face the same quantity constraint—neither firm can produce more than 2 units? What are the prices now?

(3) Price competition with differentiated products: Often models of price competition assume that sellers offer identical products. In those simple settings, consumers care only about price and will buy from the seller whose price is lowest. In many markets, however, companies offer distinct—but similar—products, which enables them to charge different prices. For instance, some consumers will still buy Coke even if the price is slightly higher than the price of Pepsi.

Let's consider a price competition game between Dell and HP in the personal computer market. Assume that demand for Dell computers is given by $D_{Dell}(P_{Dell}, P_{HP}) = 100 - 2P_{Dell} + P_{HP}$, where P_{Dell} is the price Dell sets and P_{HP} is the price HP sets. In addition, Dell has a cost of manufacturing equal to 10 per unit. Dell's payoffs are therefore equal to the amount of computers it sells (given by the demand function) times the amount it makes on each sale ($P_{Dell} - 10$) or $u_{Dell}(P_{Dell}, P_{HP}) = (100 - 2P_{Dell} + P_{HP})(P_{Dell} - 10)$.

- (a) Find Dell's best response function. This should give you P_{Dell} as a function of P_{HP} . Explain the difference between what Dell's payoff function above tells you and what the best response function that you just found tells you.
 - (b) Assume that demand for HP computers is given by $D_{HP}(P_{Dell}, P_{HP}) = 100 - 2P_{HP} + P_{Dell}$ and that the manufacturing cost for HP is 30 per unit. Find HP's best response function.
 - (c) Draw both best response functions on the same graph. Put P_{HP} on the horizontal axis and P_{Dell} on the vertical axis.
 - (d) What is the Nash equilibrium price pair?
- (4) Market entry: There are n firms, each of which is trying to determine whether to enter a new market. Any firm that stays out continues to earn a payoff V , while the payoff to entering firms depends on how many decide to enter. Specifically, entrants earn $r(c - E)$ in addition to V , where E is the number of entering firms and c is the market capacity (i.e., the number of firms that the market can sustain). As you can see, entering is risky because when there are few entrants, the payoff to doing so is high, but if there are too many, you would be better off not entering.
- (a) Suppose that the market capacity is seven firms ($c = 7$). Describe the pure-strategy Nash equilibria of this game.
 - (b) Now suppose that $c = 7$, $V = 10$, and $r = 2$. Find the symmetric mixed-strategy equilibrium.
- (5) A volunteer's variant. Reconsider the volunteer's dilemma. As in the chapter, suppose there is a benefit, b , that accrues to a colony of individuals only if one of them steps forward to volunteer to stand guard against an attack. The cost of volunteering remains c .
- (a) Calculate the expected probability that *nobody* will volunteer for guard watch if $b = 4$ and $c = 2$.
 - (b) This probability seems relatively high, especially if the colony is filled with meerkats who are known to volunteer to stand guard at the perimeter of the colony. One reason offered as to why the observed probability of nobody volunteering is so low among this species is that they are guarding their families. If, on average, parents share half their genes with their children, and we account for this very roughly by augmenting the payoffs to volunteering (and not volunteering) by

adding half the benefits accruing to one other colony member, does the chance that nobody will volunteer respond as expected?

(6) Best responding: Suppose there are two countries, labeled $i = 1, 2$, whose expenditures on foreign aid are synergistic—the aid provided by one country enhances the aid of the other. You can see this in their payoff functions, $u_i = 100 + 40a_i - a_i^2 + a_i a_j$. Assume that a_1 and a_2 must be between 0 and 100 and that the countries make aid decisions independently.

- (a) Find the best response functions for the two countries.
- (b) Compute the Nash equilibrium aid levels.
- (c) Show that the countries would be better off if they made a binding agreement to increase aid past what they would do at the Nash equilibrium.

(7) A charity raffle: Sometimes contests are used to raise money for charity. A good example is the self-financing pari mutuel raffle. In this contest, participants decide how much of their wealth, W , they want to keep and how much they want to contribute to a public good. As an incentive to contribute, donors are entered into a raffle in which the prize is a share ($0 < s < 1$) of the funds raised. Naturally, one's chance of winning the raffle is equal to the ratio of one's donation to the sum of all the donations. Suppose each of n potential donors contributes x_i and the public good generates a payoff of $\frac{1}{n} < \alpha < 1$ times the sum of the contributions ($\sum x_j$) for each person, regardless of how much they contribute.

- (a) Write down the payoff function for the i th potential donor and figure out how much each will contribute at the symmetric Nash equilibrium. Is this result a surprise?
- (b) Now suppose that donors received a warm feeling inside when they contribute to the public good. Assume the functional form of this warm glow is $w_i = 2x_i^{\frac{1}{2}}$. Add warm glow to the payoff function from part (a) and recalculate one's optimal contribution.

(8) Tragedy of the commons: In his original article (Science, 1968), Garrett Hardin discussed the management problems concerning common pool resources—those that are rival in consumption (my consumption reduces your possible consumption) and nonexcludable (everyone has access). For example, suppose that the commons generates benefits according to the

following production function: $F(\sum e_j) = 1.6(\sum e_j) - 0.2(\sum e_j)^2$, where e_j is the individual level of extraction for the n members of the community. Suppose that we measure extraction in hours, individuals work at most 8 hours, and they have an outside wage, w , that they will be paid for the time they spend working outside the commons. Finally, if an individual's share of the commons benefits is determined by her share of the total extraction, then one's payoff can be summarized as $u_i = w(8 - e_i) + \frac{e_i}{\sum e_j} (1.6 \sum e_j - 0.2(\sum e_j)^2)$.

What is the symmetric Nash equilibrium level of extraction?

(9) Contest entry: Players decide both how much to devote to a contest (i.e., the intensive margin) and whether to participate at all (i.e., the extensive margin). Assume that there are n individuals who can enter this contest or take an outside option worth A . Upon entry, the contestants make simultaneous investments $x_i \in [0, E]$, where E is each contestant's endowment of resources. The contest success function is simple: the chance that each contestant wins the fixed prize of \$100 is proportional to how much they invest, such that $Pr(win) = \frac{x_i}{\sum x_j}$.

- (a) Conditional on entering, how much do contestants invest at the symmetric Nash equilibrium?
- (b) If the contestants that enter invest optimally, how many should we expect to participate?

(10) The extensive form game shown in [figure 6.11](#) is known as the centipede game. If player 1 passes an initial endowment of 1, it doubles (growing to 2), and if player 2 then passes, the surplus grows to 4. Along the way, however, either player can stop this process and claim the larger share of the existing surplus.

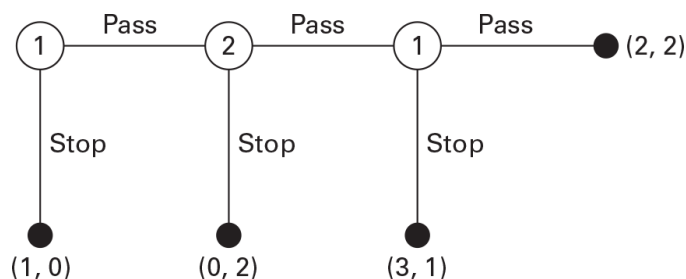


Figure 6.11
The Centipede Game.

- (a) Write the normal form of this game and identify any pure-strategy Nash equilibria. What is the predicted outcome?
 - (b) Now suppose that both players are *regret minimizers*. What is predicted to happen now (hint: consider building tables like the table on regret minimization earlier in the chapter)?
 - (c) Look online to find experimental studies of the centipede game; which prediction is closer to what these studies find (cite your sources)?
- (11) A Cournot example: Consider a market with two firms, each with a per-unit cost of 30, and market demand given by $p = 150 - Q$.
- (a) Assume that the firms each simultaneously choose their quantity, as in the Cournot model. According to the Nash equilibrium, how much will each firm produce? What is the market price?
 - (b) Now assume that the second firm's per-unit cost increases to 60. What is the Nash equilibrium?

¹This interaction is known as the Nash bargaining game and is discussed in more detail in chapter 16.

²In a representative social psychology experiment, you and a number of other students are brought to the lab, presumably to participate in a survey for pay. The graduate student in charge of the session comes into the room to greet you carrying a clipboard on which you are to sign in and a cigar box full of cash to pay you all. As you all sign in, the graduate student excuses himself to go into the next room. Shortly after the monitor leaves, one of the other participants (a confederate) stands up, walks over to the cigar box, shoves it into his backpack, and walks out of the lab. Does anyone choose to report this crime to the graduate student?

³In a review of the “bystander” literature, one of the authors of the original experiment notes that because of the “diffusion of responsibility” the likelihood of a volunteer can, indeed, fall as groups grow (Latane and Nida, 1981.)

7

Equilibrium Selection

Up to this point, we have focused on predicting behavior in games by finding equilibria. However, we've also seen that multiple equilibria can arise. The potential cost of not harmonizing at an equilibrium provides an incentive for players to coordinate. For these games, we'll consider whether it is possible to predict which one of the equilibria is most likely to obtain. In this chapter, we consider the deductive process of equilibrium selection based solely on general characteristics of the game.

A central tenet of New Keynesian economics is that the economy comprises a large game with multiple equilibria and, in terms of welfare, some equilibria are better than others. There are equilibria in which unemployment is high and few firms bother to post openings or offer attractive wages, and others in which most people are working and firms are eager to fill vacancies. One important factor thought to determine at which equilibrium the economy lands is expectations. There is no point in searching for a job if you don't expect there to be any openings. Likewise, there is no incentive to post openings if you don't think anyone will apply. In fact, in the spring of 2010, not long after the start of the so-called Great Recession, a Pew Research poll recorded that among the long-term unemployed, about as many people thought that they would eventually find a job as good as the one they lost (54%) as thought that when they were finally rehired, their job would be worse (46%). This is not a particularly confident expectation, especially considering that it was out of sync with reality. Indeed, many firms were posting good jobs. Among the previously long-term unemployed who were just rehired, a solid majority (71%) indicated that their new job was as good or better than their old one. It is this *strategic uncertainty* that creates the sort of coordination problems at the core of this chapter.

To capture the strategic uncertainty surrounding multiple equilibria in the labor market context, consider the *minimum effort game* (Van Huyck, Battalio, and Beil, 1990). There are n workers on a team, each of whom

chooses an integer effort level, e_i , and the payoff to any worker is determined by her effort and the *minimum* effort provided by the rest of the team, $\underline{e}$. In other words,

$$u_i = w[\min\{e_i, \underline{e}\}] - ce_i$$

where w and c determine the payoff from working and the cost of providing effort, respectively. It turns out that any effort level, if chosen by all the workers, is a Nash equilibrium, a fact that is easy to see in the following payoff matrix in [figure 7.1](#), which limits the effort choices, sets $w = 0.2$ and $c = 0.1$, and adds a constant of 0.3 to eliminate negative payoffs:

		$\underline{e}$				
		1	2	3	4	5
e_i	1	0.4	0.4	0.4	0.4	0.4
	2	0.3	0.5	0.5	0.5	0.5
	3	0.2	0.4	0.6	0.6	0.6
	4	0.1	0.3	0.5	0.7	0.7
	5	0	0.2	0.4	0.6	0.8

Figure 7.1

Player i 's payoffs in the minimum effort game for any level of own effort (e_i) and minimum effort of others ($\underline{e}$).

As you can see, you would not want to deviate and choose more than $e_i = 1$ in a team in which the minimum is also 1 because your extra effort would be wasted. Likewise, providing more than 2 units of effort when $\underline{e} = 2$ is also a waste, but providing less effort reduces both the team output and your payoff. The same logic holds for symmetric equilibria in which $e^* = 3, 4, 5$ and what drives this multiplicity of equilibria is the *weakest link* production technology.

What is also important about the minimum effort game is that the five equilibria can be ranked in terms of efficiency, similarly to the economies described previously. Specifically, one equilibrium *Pareto dominates* another if no player does worse there and some do better. The payoffs are highest when all workers choose the maximal effort level of 5 and lowest when they choose to provide only 1 unit of effort. Further, there are costs to not coordinating that are minimized in the low-effort equilibrium and maximized in the high-effort one. Regardless of what others do, if I play $e_i = 1$, I am sure

to get a payoff of 0.4. At the same time, if I play $e_i = 5$, instead, I earn the large payoff of 0.8 only if everyone else plays $e_i = 5$ too. In the worst case, I will earn 0 if anyone plays $e_i = 1$. This is much riskier and highlights the importance of coordination and expectations.

To get a better sense of what is at stake, suppose we consider a two-player version of this interaction first described by Jean-Jacques Rousseau in 1755 as the stag hunt (previously discussed in chapter 5). In Rousseau’s parable, stag hunt is a game played between two hunters who can either choose to provide low effort, hunting “hare,” which ensures a payoff of 4 or provide high effort in the hope of catching a “stag.” The problem is that they have to work together to bring down a stag, and therefore, this strategy is more risky, resulting in a payoff of 0 when the other hunter doesn’t help. The normal form of the game is shown in [figure 7.2](#).

		Hunter 2	
		Hare	Stag
Hunter 1	Hare	(4, 4)	(4, 0)
	Stag	(0, 4)	(6, 6)

Figure 7.2
The stag hunt game.

As in the more complicated minimum effort game, there are multiple symmetric equilibria in stag hunt: (Hare, Hare), (Stag, Stag), and a mixed strategy in which the hunters chase hare individually with probability $\frac{1}{3}$. These equilibria are also Pareto-rankable. Hunters do better when they coordinate to give high effort and worse when they both give low effort, but the situation is worse still when they miscoordinate. While the payoff at the mixed-strategy equilibrium is 4, as when they both hunt for hare, this is an expectation that is comprised partially of the pair choosing different strategies $\frac{4}{9}$ of the time and one of the hunters starving (i.e., earning 0). In other words, the hunters would be better off if they could resolve the strategic uncertainty that can lead to this outcome.

What if the hunters simply met for coffee before starting their day? Couldn’t they sort things out and agree to hunt stag? It seems obvious that this type of pre-play communication should remove the strategic uncertainty about

whether the two will play Stag, but does it? That is, wouldn't it be plausible for hunter 1, for example, to reason that him playing Stag is unambiguously better for hunter 2 because hunter 2 might then get 6 instead of 4? If this is the case, won't hunter 2 always agree to play Stag? Further, if hunter 2 will always agree to Stag, does this promise actually signal anything about his true intentions? If nonbinding pre-play communication like this does reveal the true intentions of a player, Aumann (1990) called such equilibria *self-enforcing*, but if the communication does not reveal the true intentions, the equilibrium is not self-enforcing. The key to an equilibrium being self-enforcing is one player's preferences over the other player's strategies. Suppose that the payoff each hunter got when choosing Hare when the other hunter chose Stag is 5 instead of 4. In this case, each hunter has a strict preference over the strategies of the other hunter—each would always prefer that the other choose Stag. In this case, agreeing to play Stag over coffee is not self-enforcing because the only reasonable signal to send the other hunter is Stag, regardless of what you will actually do. However, given that the players do not have a strict preference over the strategies of the other hunter in the stag hunt presented in [figure 7.2](#), an agreement to play Stag is self-enforcing.

To further clarify this point, consider the coordination game shown in [figure 7.3](#) in which, again, one player does not have strict preferences over the strategies played by the other. Without pre-play communication, it will be very hard for Alice and Bob to decide on a pure strategy (i.e., whether to go to the ballet or the prize fight). In this case, however, if they have coffee before choosing an event to attend, will they be motivated to keep whatever agreement they come up with? For instance, if they agree on going to the ballet, Alice, like the hunters, can reason that the agreement is a signal that Bob wants her to keep the agreement, and the fact that Bob wants Alice to keep the agreement implies that he plans to keep it too. For both of them, the agreement is strictly worth keeping, so (Ballet, Ballet) is a self-enforcing equilibrium. The normal form of this game is presented in [figure 7.3](#).

		Bob	
		Ballet	Fight
Alice	Ballet	(1, 2)	(0, 0)
	Fight	(0, 0)	(2, 1)

Figure 7.3

Coordination game between Alice and Bob.

In the remainder of this chapter, we consider four ways to overcome strategic uncertainty and increase player welfare by *selecting* among a set of equilibria. All these methods are based on deductive principles in which they rely on nothing more than a full description of the game. Methods to select equilibria based on *inductive* principles that depend on the history of play or that reduce the number of equilibria based on *refinements* that eliminate those that are not self-enforcing are discussed in part VIII of the book.

Payoff-Based Selection Criteria

The first selection criteria we will discuss is, perhaps, the most obvious. In the stag hunt, the hunters can strive to achieve 4s in the (Hare, Hare) equilibrium or 6s in the (Stag, Stag) equilibrium. Doesn't it seem clear that they should coordinate on (Stag, Stag)? In other words, shouldn't they be attracted to the most efficient or Pareto-dominant equilibrium? This concept, discussed in Schelling (1980), is known as *payoff dominance*.

Definition 1 (Payoff-dominant equilibrium) *A Nash equilibrium is payoff dominant if it is not strictly Pareto dominated by any other equilibrium. Specifically, if the strategy profile $(s_1^*, s_2^*, \dots, s_n^*)$ constitutes a Nash equilibrium and the profile $(s_1^{**}, s_2^{**}, \dots, s_n^{**})$ does as well, $(s_1^*, s_2^*, \dots, s_n^*)$ payoff dominates $(s_1^{**}, s_2^{**}, \dots, s_n^{**})$ if $u_i(s_1^*, s_2^*, \dots, s_n^*) \geq u_i(s_1^{**}, s_2^{**}, \dots, s_n^{**})$ for each player i , and for at least one player i , this inequality is strict: $u_i(s_1^*, s_2^*, \dots, s_n^*) > u_i(s_1^{**}, s_2^{**}, \dots, s_n^{**})$.*

Stated in words, definition 1 requires that, at the payoff-dominant equilibrium, players do better or no worse compared to the payoffs at any other equilibrium. Applying this definition, we see that, although there are three Nash equilibria in the stag hunt game, payoff dominance selects (Stag, Stag) as the most likely. Likewise, in the minimum effort game, payoff dominance selects the equilibrium in which all workers choose an effort level of 5.

The predictions of the payoff dominance principle are not always this clear, however. In symmetric games in which the equilibria are off-diagonal (i.e., where the players choose different strategies and thus achieve different payoffs), like the leave me alone game seen in [figure 7.4](#) (familiar to the parents of teenagers), there may or may not be a payoff dominant equilibrium. On the left, Ares and his mother, Hera, are better off picking rooms to be in so they can avoid each other. They both prefer to be alone in their cozy bedrooms, where they can read, to being alone in the kitchen. However, as all families know, most fights happen when there is a teenager and a parent in the kitchen, and because their rooms are both on the second floor, they can't truly enjoy peace when they are both upstairs.

		Hera				Hera	
		Kitchen	Bedroom			Kitchen	Bedroom
Ares	Kitchen	(−1, −1)	(1, 2)	Ares	Kitchen	(−1, −1)	(2, 3)
	Bedroom	(2, 1)	(0, 0)		Bedroom	(2, 1)	(0, 0)

Figure 7.4
The leave me alone game between Ares and Hera.

These preferences imply that there are two pure-strategy equilibria, but neither payoff dominates the other on the left (in addition, you can check that neither is payoff dominated by the expected payoff at the mixture). In this case, there is no payoff dominance prediction. On the right, however, when the Wi-Fi router is installed in the kitchen and the payoffs of both Ares, who loves to text his friends, and his mother, who is annoyed by the internet, are altered, we see that the equilibrium in which Ares is in the kitchen and Hera is upstairs now payoff dominates. Similar predictions based on payoff-dominance will also arise in asymmetric games where players have different strategy sets and potentially different payoffs.

Returning to the stag hunt game, someone might point out that, while payoffs are indeed highest at the Stag equilibrium, they also appear more risky. Failing to coordinate on Stag can cost hunters everything, but Hare-playing hunters get a payoff of 4 for sure. This is the basic logic of a second selection criteria—*risk dominance*—formalized for two-player games by Harsanyi and Selten (1988).

Definition 2 (Risk-dominant equilibrium) *Consider a game with two players, each of whom has two pure strategies. The risk-dominant Nash equilibrium obtains where the product of the deviation losses for both players is largest. Specifically, if the strategy profile s^* constitutes a Nash equilibrium and the deviation to profile (s'_i, s^*_{-i}) results in a payoff loss of $u_i(s^*_i, s^*_{-i}) - u_i(s'_i, s^*_{-i})$ for player i , then the risk-dominant equilibrium occurs where $\prod_{i=1}^n (u_i(s^*_i, s^*_{-i}) - u_i(s'_i, s^*_{-i}))$ is greatest.*

To understand the intuition for this definition of risk dominance, return to the stag hunt game and define a related, more intuitive term, the *risk factor* of an equilibrium: the smallest probability, p , such that if one's opponent is going to play her component of the equilibrium with probability at least p , then it is optimal for one to do so as well. For the (Hare, Hare) equilibrium, this probability solves a familiar equation: $4p + 4(1 - p) = 0p + 6(1 - p)$, the same one used to calculate the mixed strategy. With the slightest bit of algebra, the condition reduces to $(6 - 4) = 6p$, from which one gets the weight placed on Hare in the mixed strategy $\frac{1}{3}$. The alternative interpretation, however, is that if you believed that the other hunter was going to play Hare with at least one-third probability, your best response would be to play Hare too. In other words, the risk factor for {Hare, Hare} is $\frac{1}{3}$. The final and most important thing to notice here is the first part of the equation used to get the risk factor, $(6 - 4)$, is just the opportunity cost of not playing your part in the (Stag, Stag) equilibrium.

Now calculate the risk factor for the (Stag, Stag) equilibrium. Here, the perspective is just the opposite, so let q be the smallest belief that you need to have that the other hunter will play Stag for it to be optimal for you to do so as well. Now the equation is $4(1 - q) + 4q = 0(1 - q) + 6q$, from which we get $(4 - 0) = 6q$, with the first part this time being the opportunity cost of deviating from the (Hare, Hare) equilibrium. Solving the equation, we get the risk factor for the (Stag, Stag) equilibrium, $q = \frac{2}{3}$. It should be no surprise that in a symmetric game where each player has two strategies, $q = 1 - p$.

Now that we have all the parts, we can see how they fit together to get the formal definition of risk dominance. The first thing to notice is that if we take a short cut and just look at the risk factors, we see that the one for the (Hare, Hare) equilibrium is lower. By itself, this indicates that (Hare, Hare) is the risk-dominant equilibrium, but let's continue to see how this is consistent with definition 2. Applying the definition, if (Hare, Hare) is risk dominant, then the

product of the deviations from it for all players—in this symmetric case, $(4 - 0)(4 - 0) = 16$ —must be larger than the product of the deviations from the (Stag, Stag) equilibrium, $(6 - 4)(6 - 4) = 4$, and it is. To link the two concepts, notice that the opportunity costs of deviations are built into the risk factors as pointed out previously.

There is another way, however, to determine the risk-dominant equilibrium in the stag hunt game. To see it, consider the graph of the expected payoffs to playing Hare and Stag in [figure 7.5](#), in which p is still the likelihood of playing Hare. Because you always get 4 if you play Hare, regardless of what the other hunter does, this expected payoff, $Eu(Hare)$, is just a flat line that doesn't vary with the x -axis, p . By comparison, $Eu(Stag)$ is downward sloping in p because if you play Stag when $p = 0$, the other person you are playing against will be playing Stag for sure, and you will both get 6. However, when $p = 1$ and you play Stag, you will meet someone playing Hare for sure, and you will get 0. In between, the expected payoff to playing Stag falls as the other hunter becomes more likely to play Hare. Naturally, your eyes will drift to the intersection of the expected payoffs, where $p = \frac{1}{3}$, the mixed-strategy Nash equilibrium. To the left of this intersection, you can see that $Eu(Stag) > Eu(Hare)$, and to the right, we have the opposite, $Eu(Stag) < Eu(Hare)$. In other words, to the left, people are very unlikely to be playing Hare, so Stag is a best response; and to the right, people are relatively more likely to play Hare, so Hare becomes a best response.

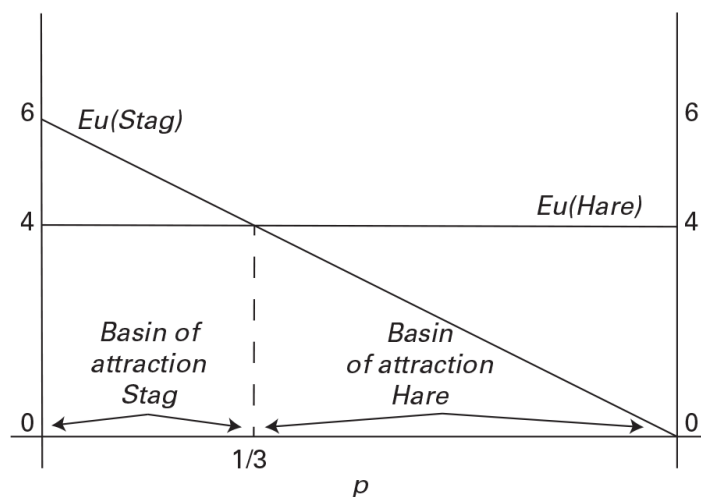


Figure 7.5

Risk dominance and basins of attraction.

What is important here is the asymmetry in how the mixed strategy bifurcates the space. Stag is a best response for only one-third of the axis, while Hare is a best response for two-thirds of it. Another way of putting this (which will become more significant in chapter 25), is that the *basin of attraction* of the Stag pure-strategy equilibrium is $0 \leq p < \frac{1}{3}$ and the basin of attraction of the Hare pure-strategy equilibrium is $\frac{1}{3} < p \leq 1$. As you can see, the Hare pure-strategy equilibrium has a larger basin of attraction than the Stag pure-strategy equilibrium. Importantly, for the current purpose, the equilibrium with the larger basin of attraction is also the risk-dominant equilibrium. The intuition is simple: the calculations to determine the mixed strategy are almost identical to those needed to find the risk factors.

It is important to note that, although the stag hunt game is generally characterized as one in which payoff dominance and risk dominance point to different equilibria, this is not true of all coordination games. As an example, consider the following asymmetric game played by Eye and Jay in [figure 7.6](#), in which there are two pure-strategy equilibria occurring off-diagonal at (Down, Left) and (Up, Right).

		Jay	
		Left	Right
Eye	Up	(1, 2)	(2, 3)
	Down	(5, 3)	(1, 0)

Figure 7.6

A game in which the payoff dominant and risk dominant equilibria overlap.

As we can see, Eye does better at (Down, Left) and Jay does no worse, so it is the payoff-dominant equilibrium. But now let's consider the deviations from each equilibrium. The product of the opportunity costs from the {Down, Left} equilibrium is $(5 - 1)(3 - 0) = 12$, while the product for the (Up, Right) equilibrium is $(2 - 1)(3 - 2) = 1$. This shows that (Down, Left) risk dominates too.

Focal Points

When the structure of a game gives no clues as to which equilibrium should be more likely (i.e., neither payoff nor risk dominance gains any traction),

Schelling (1960) posited that players may nonetheless spontaneously develop enough strategic sophistication to solve the coordination problem. They simply need to be provided with contextual clues. As a demonstration of this, Schelling quizzed people, asking them, for example, “You are to meet somebody in New York City. You have not been instructed where to meet; and you cannot communicate with each other. You are simply told the same thing and that you will just have to try to make your guesses coincide.” In terms of coordination, any place at which the two meet is as good as any other place, and therefore there are millions of possible equilibria, and yet when asked in the late 1950s, an absolute majority picked Grand Central Terminal, and nearly all of them settled on a noontime meeting. In Schelling’s terms, the train station is a *focal point*, a contextual clue that is both prominent and conspicuous enough to help players coordinate.

Here is another example to fix terms. Consider the game on the left in [figure 7.7](#). It is a pure coordination game in which neither payoff dominance nor risk dominance suggests an equilibrium on which to coordinate. Indeed, given the player and strategy names, it is not clear why one would favor one equilibrium over another. As a result, at the mixed-strategy equilibrium, we expect the players to miscoordinate half the time.

		Player 2				Player 2	
		1	A			Heads	Tails
Player 1	A	(1, 1)	(0, 0)	Player 1	Heads	(1, 1)	(0, 0)
	1	(0, 0)	(1, 1)		Tails	(0, 0)	(1, 1)

Figure 7.7
Pure coordination games.

Now consider the game on the right in [figure 7.7](#) in which only the strategy names have been changed. As a firm believer in the fact that “tails never fails,” Jeff would be tempted to play Tails. How about you? What if we told you that when Mehta, Starmer, and Sugden (1994) asked 88 people to pick one or the other, but there was nothing at stake, 76% picked Heads? Would that change your choice? It should considering that in his original experiment, 86% of Schelling participants picked Heads, as did 86% of the participants for whom the choice determined their payoff in the Mehta et al. study. In other

words, it seems that Heads is focal when paired with Tails in a way that A is not when paired with 1.

Forward Induction

For the last two selection criteria, we add something less innocuous to the game to make a prediction. In this first case, we add an outside option, but we will see that what is important is not the outside option as a construct, but what its presence signals to players. To be more concrete, let us return to the stag hunt game, but this time we will assume that hunter 1 has a slightly more complicated strategic choice. Suppose that, as before, both hunters choose Hare or Stag but only if hunter 1 decides to play the game. That is, hunter 1 must first decide whether to play the game at all or take an outside option and hunter 2 only chooses if the game is actually played.

Suppose the value of the outside option for the first hunter is 3. In other words, hunter 1 can either take a certain payment of 3 or opt into the game and get whatever she earns after the two hunters make their choices. Would this help? Well, no, it would not make sense for hunter 1 to ever take the outside option because she can guarantee a payoff of 4 by entering the game. However, what if the outside option rose to the value of 5? If hunter 2 now observes that hunter 1 enters the game, what should she infer? The principle of *forward induction* (Kohlberg and Mertens, 1986) requires the past actions of players to be rational. In other words, hunter 2 should not infer that hunter 1 made a mistake; forward induction implies that hunter 1 must have entered the game with the intention of playing Stag because entering to play Hare makes no sense (i.e., she could get more by staying out).

It is important to understand that forward induction is not necessarily linked with an outside option—the principle is more general. As an illustration, imagine that hunter 1 chooses between two games instead of between a game and an outside option. Suppose that one of the games continues to be stag hunt, and the other is a prisoner's dilemma in which the payoff to mutual defection is (5, 5). Again, using forward induction, hunter 2, observing the game to be played is stag hunt, infers that hunter 1 must be intending to coordinate on Stag. In essence, by adding another move for one of the players, forward induction helps them coordinate because it shrinks the set of plausible equilibria.

Correlated Equilibrium

We now consider adding something else to a coordination game—another way by which players can communicate, this time with a little bit of external intervention. To motivate this addition, let us consider a game context in which the relevant feature evolved naturally. Imagine an uncontrolled intersection of two country roads that are used intermittently. When two cars approach the intersection they can either “Stop” to make sure that they do not collide or “Go” and hope for the best. If we normalize the payoff to stopping at 0 (say, the benefit of not having to stop is 1 because of the time saved and the damage due to a collision is -2), then we get the following payoff matrix in [figure 7.8](#) (a version of the chicken game):

		Driver 2	
		Stop	Go
Driver 1	Stop	(0, 0)	(0, 1)
	Go	(1, 0)	(-2, -2)

Figure 7.8
Stop or go game.

It makes sense that there are two off-diagonal pure-strategy Nash equilibria, in which one of the two drivers stops and the other goes, and there is a mixed-strategy equilibrium in which both drivers stop with probability $\frac{2}{3}$. The problem is that without coordination, there is a $\frac{4}{9}$ chance that both drivers will stop and stare impatiently at each other. More important, there is also a $\frac{1}{9}$ chance that they will both go at the same time and crash. In the early twentieth century, this situation must have been common. How was it handled?

What if the town erected a simple, two-color traffic light that either signaled red, meaning Stop, or green, meaning Go. Critically, suppose that this *public signal*, as discussed by Aumann (1974), sends only one of two messages {driver 1 signal, driver 2 signal}: {red, green} or {green, red}. If these two messages are sent randomly, we can describe the information content of the traffic light in the following table. Each message is sent half the time and the chance that the other two unused messages are sent is zero, as shown in [figure 7.9](#).

	Red	Green
Red	0	$\frac{1}{2}$
Green	$\frac{1}{2}$	0

Figure 7.9

A public traffic signal.

What are the implications of the new public signal? First, should the drivers follow it? Can they do better by deviating and ignoring the signal sent? In general, the expected payoff of following a strategy now depends on both the payoffs and the signals received. The question is whether the players can use the public signal to coordinate play. In this symmetric case, the expected payoff of stopping, conditional on player i seeing a red signal is

$$Eu_i(\text{Stop}|\text{Red}) = \text{Prob}(\{\text{Red}, \text{Red}\})(0) + \text{Prob}(\{\text{Red}, \text{Green}\})(0) = 0$$

It is obvious that if you Stop, you will get a payoff of 0 regardless of what signal you saw, but the important things to focus on here are the drivers' expectations. Given that the signaling technology is public, both drivers should understand that $\text{Prob}(\{\text{Red}, \text{Red}\}) = 0$ because the traffic light never broadcasts red to both drivers. Likewise, both drivers should also realize that if they see a red signal, the other driver must be seeing green. Therefore, the conditional probability $\text{Prob}(\{\text{Red}, \text{Green}\}) = 1$, not $\frac{1}{2}$.

Also, what payoff should a driver expect if she ignores the red signal? In this case,

$$Eu_i(\text{Go}|\text{Red}) = \text{Prob}(\{\text{Red}, \text{Red}\})(1) + \text{Prob}(\{\text{Red}, \text{Green}\})(-2) = -2$$

This equation better illustrates the reason why one might want to deviate to Go when one sees red. If there is some chance that both drivers are seeing red, then they would each prefer to just go. Both drivers know, however, that this never happens, so if one driver is seeing red, the other driver is seeing green, and if that driver follows the signal and she doesn't, there will be an accident for sure. This implies that there is no incentive to deviate from Stop when the driver sees the red signal.

Checking to see that the same is true for a green signal, we see that, after a bit of algebra, $Eu_i(\text{Go}|\text{Green}) = 1$ and $Eu_i(\text{Stop}|\text{Green}) = 0$, indicating that there is also no incentive to deviate from Go when the driver sees green. Hence, following the signal is self-enforcing and constitutes a Nash

equilibrium. The implications of correlating play to a public signal are huge. Without the traffic light, the expected driver payoff is 0; after it is erected, that payoff increases to $\frac{1}{2}$ (because each driver has to wait half the time) and there are no accidents in equilibrium. This little bit of game theoretic magic is known as a *correlated equilibrium*, as described in definition 3.

Definition 3 (Correlated equilibrium) Suppose that players receive a signal, $s_i \in S_i$ according to the publicly known probability distribution, p . This probability distribution, conditioned on the signaling technology, is a correlated equilibrium if $\sum_{s_i \in S_i} p(s_i) u_i(s_i, s_{-i}) \geq \sum_{s_i \in S_i} p(s_i) u_i(s'_i, s_{-i})$ for all i and $s'_i \in S_i$.

Stated in words, a correlated equilibrium (first proposed by Aumann, 1974) occurs when players observe their component of a publicly available signal (but not the components sent to the other players) and have an incentive to follow this signal. In this way, the players' strategies are correlated with the public signal.

Sometimes the signaling technology is not perfect, however. Suppose, for instance, that instead of hitting each other, one driver hits the traffic light by accident one day and the signaling technology is damaged such that it now randomly sends three signals: {Red, Green}, {Green, Red} and {Red, Red}. In this case, the public signal can be summarized as in [figure 7.10](#):

	Red	Green
Red	$\frac{1}{3}$	$\frac{1}{3}$
Green	$\frac{1}{3}$	0

Figure 7.10

An imperfect public signal.

Should the drivers continue to follow their signals? You can anticipate the issue. If the signal now has both drivers waiting sometimes, it might be in their interests to ignore it. Let's see. First, the expected payoff of following a red signal continues to be 0 ($Eu_i(\text{Stop}|\text{red}) = (\frac{1}{2})(0) + (\frac{1}{2})(0) = 0$). However, the conditional probabilities $Prob(\{\text{Red}, \text{Red}\})$ and $Prob(\{\text{Red}, \text{Green}\})$ now both equal $\frac{1}{2}$. That is, when a driver sees a red signal, she knows that half the time the other driver sees green and half the time she sees red too. What can a driver expect from deviating? With the broken signal, $Eu_i(\text{Go}|\text{Red}) = (\frac{1}{2})(1) + (\frac{1}{2})(-2) = -\frac{1}{2}$. Because drivers know that when they see

green, the other driver still sees red, they continue to have an incentive to follow that signal too. Combined, these calculations suggest that, although imperfect, drivers should still heed the traffic signal, but there will be a welfare cost—the average payoff falls from $\frac{1}{2}$ to $\frac{1}{3}$.

Exactly how broken must the public signal be in this context before the drivers will start to ignore it? You can see that they will ignore it if they are both asked to wait too often. This happens when the relative frequency of the {Red, Red} signal is high compared to the {Red, Green} signal. Specifically, if p is the relative frequency with which the drivers see {Red, Red}, then they will deviate when $Eu_i(Go|Red) = p(1) + (1 - p)(-2) \geq 0 = Eu_i(Stop|Red)$. Solving for p , we see that if seeing red implies the message {Red, Red} at least $\frac{2}{3}$ of the time, the drivers do better, in expectation, by ignoring the signal. Working backward, this could happen if the {Red, Red} signal is sent half the time and the other two signals are each sent a quarter of the time, for example.

Before finishing our discussion of correlated equilibrium, let's return to the stag hunt to see if the concept generalizes to other types of coordination games. You can see that a public signal is an effective way to coordinate play when there is some conflict over which equilibrium to select. Nobody likes to wait in the driving game, so each player would prefer to be at the equilibrium in which the other player waits. However, this sort of conflict is not present in the games like minimum effort or stag hunt. In these games, all the players would prefer to be at the payoff-dominant equilibrium.

Can a public signal help the hunters in stag hunt, for instance? If hunters decide individually to run down deer when it is a cool, cloudy day, which happens a fraction $0 < x < 1$ of the time, and snare hare when it is sunny and hot (the other $1 - x$ of the days), so long as there is no ambiguity as to whether it is sunny or cloudy, then any value of $x > 0$ will suffice to correlate play at (Stag, Stag). With this in mind, and if you were a benevolent dictator, it seems like you could do better by the hunters. For instance, couldn't you tell them that unless the sun doesn't rise in the morning, they should plan to hunt stag?

Behavior in Games: Equilibrium Selection in Coordination Games

To what extent do the features of a game determine the equilibrium that arises? That is, do the deductive principles that we have studied in this chapter predict experimental play? We begin our review of the experimental literature

with the Harsanyi and Selten (1988) criteria: payoff and risk dominance, in coordination games. We then move on in turn to Schelling's (1960) notion of focal points, the Kohlberg and Mertens (1986) theory of forward induction, and Aumann's (1974) correlated equilibrium refinement. We finish the topic with a more recent contribution by Chen and Chen (2011), in which the idea of a focal point has been integrated directly into the equilibrium prediction, again demonstrating the interplay between theory and data.

One of the earliest and most influential equilibrium selection studies is the minimum effort experiment by Van Huyck et al. (1990), the experiment with which we began this chapter. Because it was among the first of its kind, the purpose of the study was to test whether players could solve the most basic coordination problem, one in which their interests are aligned. This game was generated using the payoff function, $u_i = 0.6 + 0.2[\min\{\underline{e}\}] - 0.1e_i$, in which, $\underline{e}$ refers to the minimum effort provided by anyone in the group of between 14 and 16 players, as before. Participants could choose integer effort levels between 1 and 7, and the game was presented in the following tabular form in [figure 7.11](#):

		Smallest Value of e Chosen						
		7	6	5	4	3	2	1
Your Choice of e	7	1.30	1.10	0.90	0.70	0.50	0.30	0.10
	6	—	1.20	1.00	0.80	0.60	0.40	0.20
	5	—	—	1.10	0.90	0.70	0.50	0.30
	4	—	—	—	1.00	0.80	0.60	0.40
	3	—	—	—	—	0.90	0.70	0.50
	2	—	—	—	—	—	0.80	0.60
	1	—	—	—	—	—	—	0.70

Figure 7.11

Payoffs in the minimum effort game from van Huyck et al. (1990).

As the reader can see, this presentation, while not completely intuitive, does indicate that the best response to the minimum effort offered is to provide the same amount. It also indicates that the payoffs achieved at the seven pure-strategy equilibria are increasing in the effort provided. In other

words, by looking up the diagonal, payoff dominance clearly predicts play at the $e_i = 7$ equilibrium.

As an alternative hypothesis, Van Huyck et al. (1990) suggest that the riskiness of the $e_i = 7$ equilibrium could lead to play at a more secure equilibrium. Though not equivalent, the notion of *security* based on the discussion in von Neumann and Morgenstern (1944) is similar to risk dominance; the problem is that risk dominance does not generalize well to games with more than two pure strategies. In this case, based on the maximin principle discussed in chapter 4, the secure equilibrium is clearly where $e_i = 1$. Here, players guarantee a payment of \$0.70, while anyone playing 7 runs the risk of only earning \$0.10.

In their main experiment, groups of 14 to 16 participants played the game for 10 periods. Focusing just on the distribution of choices in period 1, where the players could only rely on their prior expectations and deductive principles because there is no history of play, Van Huyck et al. report that at first blush, payoff dominance appears to predict better than security. A total of 31% of the participants chose $e_i = 7$, while only 2% chose $e_i = 1$. The problem, however, is that about half of the first-period play is at $e_i = 4$ or $e_i = 5$, indicating that the modal response was neither the payoff-dominant nor the secure effort level. Furthermore, the play over 10 periods tended to unravel to the secure ($e_i = 1$) equilibrium. In the final period, 72% of participants chose 1. Based on these results, the authors concluded that the first-best outcome, the payoff-dominant equilibrium, is unlikely to arise “either initially or in repeated play.” In a follow-up two-player game, however, they found that most pairs were able to coordinate on the efficient $e_i = 7$ over time.

In a simpler setting, Haruvy and Stahl (2004) asked participants to play a series of three-strategy normal-form games and in three of these games, the ones presented in [figure 7.12](#), payoff dominance and risk dominance predicted different strategies. As just mentioned, the complication with risk dominance is that even in games with just three strategies, the concept is not uniquely defined. The product of the deviation losses method may not be transitive when done pairwise, and so Haruvy and Stahl offered three predictions: Payoff dominance (*PD* in the figure), pairwise risk dominance (*PRD*), and risk dominance based more closely on the risk factor idea introduced earlier in this chapter (*RD*).

		P2					P2		
		A(95)	B(0)	C(5)			A(1)	B(97)	C(2)
P1	A(RD)	(7, 7)	(6, 6)	(9, 4)	P1	A(PD)	(7, 7)	(3, 6)	(2, 4.5)
	B(PRD)	(6, 6)	(8, 8)	(5, 2)		B(RD)	(6, 3)	(6, 6)	(3, 4.5)
	C(PD)	(4, 9)	(2, 5)	(10, 10)		C	(4.5, 2)	(4.5, 3)	(4, 4)

		P2		
		A(9)	B(87)	C(4)
P1	A(PRD)	(8, 8)	(6, 6)	(5, 0)
	B(RD)	(6, 6)	(7, 7)	(9, 0)
	C(PD)	(0, 5)	(0, 9)	(10, 10)

Figure 7.12

Three games from Haruvy and Stahl (2004).

The results are striking. The numbers in parentheses along the top row of the normal-form games indicate the percentage of play at that strategy. In the matrix on the left, to begin, 95% of participants chose strategy A, consistent with risk dominance, while only 5% chose the payoff-dominant strategy. As you can see, the results are very similar in the other two games. Overall, Haruvy and Stahl found that risk dominance is a much better predictor of play than payoff dominance and, to some extent, this is no surprise, considering that risk dominance takes into account the magnitude of the out-of-equilibrium payoffs, while payoff dominance ignores these consequences of deviating. Reinforcing this idea, Battalio, Samuelson, and Van Huyck (2001) found, in a related experiment that focused on the stag hunt game, that the larger the product of the deviation losses, the more likely was the play to converge on the risk-dominant equilibrium (a result that is also consistent with the discussion of the behavioral evidence on importance of seemingly irrelevant payoff differences in the traveler's dilemma from chapter 6).

Considering focal points, in addition to the Mehta et al. (1994) study discussed previously that extended Schelling's original results, there have also been a few studies using the minimum-effort experimental paradigm. In one strand of this literature, Bornstein, Gneezy, and Nagel (2002) and Riechmann and Weimann (2008) studied whether groups can coordinate on the payoff-dominant equilibrium when they are in competition with another group. The

basic idea is that intergroup competition makes the payoff-dominant equilibrium (i.e., $e_i = 7$) focal, an idea that has roots in Adam Smith's notion of the "invisible hand." In the same seven-strategy design as in van Huyck et al. (1990), Bornstein et al. (2002) found that intergroup competition raises the mean strategy choice from 3.6 to 5.3, a difference that is significant at better than the 1% level. In a follow-up study to Bornstein et al. (2002), Riechmann and Weimann (2008) found a similar coordinating effect of intergroup competition, which they attributed to the reduction of strategic uncertainty. On top of this, they also found that small groups perform better than large ones and too many groups can diminish the focal effect, both intuitive results.

In a variant of the minimum effort game, Manzini, Sadrieh, and Vriend (2009) employ simpler, more personal tactics to make the payoff-dominant equilibrium focal. Specifically, they ask whether a simple social cue such as a smile can subtly indicate trustworthiness and help players coordinate. While the investigators do find that electronic "smiles" are used, and that they help coordinate play closer to the payoff-dominant equilibrium, average payoffs do not improve because there is still considerable miscoordination. In a related study in which nationality instead of a smile is used to help coordination in a stag hunt experiment, Stoddard and Leibbrandt (2014) found that when players know just that they are of the same nationality, they are more likely to land at the payoff-dominant equilibrium. In summary, coordination is possible through the use of salient features like competition, smiles, and nationality, even though these features do not come into play when describing the equilibria of the core game.

In a clever study to test whether forward induction can lead to the selection of the payoff-dominant equilibrium, Cachon and Camerer (1996) asked participants to play a game in which one of seven integers is picked and the payoffs are determined by how close one's pick is to the median choice. Like the minimum effort game, there are seven pure-strategy Nash equilibria in this median effort game, wherein players choose the same number and choosing the number seven leads to the one that is payoff dominant. When the participants play the baseline game, choices quickly converge to 4 or 5 (with 71% of people choosing the median) by round 3. When the authors imposed a small entry cost, designed to not be large enough to move behavior above a median of 4 or 5, not much changes. However, when the entry cost was raised so that it made sense to enter the game only if the median choice were 6 or 7,

choices followed upward as predicted. These results are consistent with forward induction, but Cachon and Camerer pointed out that simple loss avoidance can also explain much of what they found.

In the simpler, stag hunt game setting, Blume, Kriss, and Weber (2017) compared a condition in which pre-play communication is unlimited and free to one in which a cost to communicate is imposed. The hypothesis here is that while “cheap talk” in the costless setting may trigger salient focal cues, being willing to incur the cost of communicating should signal that one is targeting the payoff-dominant equilibrium. The authors first discover that the cost reduces communication, as you might expect. Nonetheless, efficiency is not sacrificed because the rate of coordination does not fall overall. The few groups that use communication are at least as likely to coordinate at the payoff-dominant equilibrium, a result consistent with forward induction. Overall, there is some evidence of forward induction helping players to coordinate, but it is not overwhelming, and other factors may be at work.

The last selection principle to consider is correlated equilibrium. Here, Duffy and Feltovich (2010) found significant support for correlated equilibrium (as did Cason and Sharma, 2007). Participants play a two-by-two game of chicken, in which the worst the players can do is to get nothing and the best they can do is to capture 75% of a 12-unit surplus. Without a public signal, the rate of coordination is similar to what one would expect at the mixed-strategy equilibrium. With a public signal, however, coordination increases significantly, but only when the signals that are sent constitute a correlated equilibrium (i.e., sending noise through the signaling mechanism does not help, and participants correctly learn to ignore the information).

To see how theory has evolved to account for some of these experimental results, consider the example offered by Chen and Chen (2011), in which social identity (e.g., in-group status) is used to make equilibria focal, similar to the hypothesis tested by Stoddard and Leibbrandt (2014). To illustrate this, consider the two-player version of this interaction. Imagine that players care, not only about their own material payoff in the game, but also about the outcome that their partner gets. In this case, we say that players have *social preferences*. One way to model this is to adapt the player’s payoff such that it is a weighted average of her own material earnings and the earnings of her partner (as in Charness and Rabin, 2002, which we discuss in more depth in

chapters 14 and 18). Specifically, let's augment the player's payoff in the minimum effort game to be the following:

$$u_i = (1 - \alpha_j)(w[\min\{e_i, e_j\}] - ce_i) + \alpha_j(w[\min\{e_i, e_j\}] - ce_j)$$

where the first term is just player i 's material earnings in the original game (multiplied by $(1 - \alpha_j)$) and the second term is player j 's earnings (multiplied by α_j). The parameter α_j reflects how player i weights player j 's payoffs in her decision. If $\alpha_j = 0$, then the player doesn't care about the earnings of player j and simply seeks to maximize her own material earnings. When $\alpha_j > 0$, then player i is happier when player j does well and would even be willing to give up some of her own earnings to increase j 's earnings. (Specifically, she'd be willing to give up $\frac{\alpha_j}{1-\alpha_j}$ units of her own earnings to increase j 's earnings by 1.) When $\alpha_j < 0$, player i prefers that player j do poorly and would even be willing to sacrifice her own earnings to hurt j . Let α_j reflect the other player's group identity. Suppose that $\alpha_j \in [-1, 1]$ is negative when the other player is from an out-group and positive when she is from an in-group. In the following payoff matrices shown in [figure 7.13](#), we use the same parameters as before (i.e., $w = 0.2$ and $c = 0.1$) and compare two fairly extreme cases: where $\alpha_j = 0.9$ and $\alpha_j = -0.9$:

		e_j							e_j				
		1	2	3	4	5			1	2	3	4	5
e_i	1	0.1	0.01	-0.08	-0.17	-0.26	e_i	1	0.1	0.19	0.28	0.37	0.46
	2	0.09	0.2	0.11	0.02	-0.07		2	-0.09	0.2	0.29	0.39	0.48
	3	0.08	0.19	0.3	0.21	0.12		3	-0.28	0.01	0.3	0.39	0.48
	4	0.07	0.18	0.29	0.4	0.31		4	-0.47	-0.18	0.11	0.4	0.49
	5	0.06	0.17	0.28	0.39	0.5		5	-0.66	-0.37	-0.08	0.21	0.5
In-group, $\alpha_j = 0.9$							Out-group, $\alpha_j = -0.9$						

Figure 7.13

The minimum effort game with identity-contingent social preferences.

The stark difference between the two matrices indicates the impact of endogenizing the notion of a focal point. In both cases, we still have five efficiency-ranked equilibria. However, on the left, when identity is focal and the other player is a member of your own group, overshooting the other

player's strategy decreases your utility less than undershooting it. This is because you care about the other person and over-shooting only hurts you while undershooting hurts both of you. As a result, the choice of $e_i = 5$ actually seems like a safer bet. However, if identity is focal but the other player is not a member of your group and you don't want him to do well, as on the right, then the opposite result obtains: you don't mind guessing less than the other player as much because, although it harms you materially, you also gain utility from hurting the other player as well.

In the experiments that Chen and Chen (2011) ran to test this theory, participants are randomly sorted into groups and group identity is induced using a problem-solving task. Participants then play the minimum effort game with another person who is either in the same group or from the other group. As predicted, in-group pairs are much more likely to end up at the payoff-dominant equilibrium while out-group pairings drift toward the secure equilibrium, a result that is consistent with what Stoddard and Leibbrandt (2014) later found using nationality as a form of identity.

Exercises

(1) Consider the two-player game in normal form shown in [figure 7.14](#):

		Player 2	
		X	Y
Player 1	X	(45, 45)	(0, 35)
	Y	(35, 0)	(40, 40)

Figure 7.14

- Find all the Nash equilibria.
 - Does payoff dominance make a unique prediction?
 - Does risk dominance make a unique prediction?
 - Based on the behavioral evidence, which prediction do you think is more likely at the end of 10 rounds of play?
- (2) Consider the two-player game in normal form shown in [figure 7.15](#), this time with three strategies:

		Player 2		
		A	B	C
Player 1	A	(60, 60)	(10, 30)	(20, 70)
	B	(30, 10)	(70, 70)	(30, 20)
	C	(70, 20)	(20, 30)	(50, 50)

Figure 7.15

- Find all the Nash equilibria.
 - Does payoff dominance make a unique prediction?
 - Does risk dominance make a unique prediction?
 - Based on the behavioral evidence, which prediction do you think is more likely at the end of 10 rounds of play?
- (3) Figure 7.16 shows a three-strategy pure coordination game in which two players, who live on the same block, have to agree on a place to meet without explicitly communicating with each other:

		Player 2		
		Tree	House	Bridge
Player 1	Tree	(1, 1)	(0, 0)	(0, 0)
	House	(0, 0)	(1, 1)	(0, 0)
	Bridge	(0, 0)	(0, 0)	(1, 1)

Figure 7.16

- Find all the Nash equilibria.
 - Assume that the two neighbors have the following information about their block: there are 10 trees on the block, 2 of which are dead; there are 5 houses on the block, of which 2 are white and 3 are blue; and there are 2 bridges, where one is named after the founder of the town, while the other is unnamed. Which equilibrium do you predict to arise, and why?
- (4) Suppose that two players first play what is known as the Nash demand game depicted on the left in figure 7.17 in which they must make a demand over a share of a unit surplus. In this version, players can demand $\frac{1}{3}$, $\frac{1}{2}$, or $\frac{2}{3}$ of

the surplus, and then they play the coordination game on the right in [figure 7.17](#).

		Player 2					Player 2		
		$\frac{1}{3}$	$\frac{1}{2}$	$\frac{2}{3}$			$\frac{1}{3}$	$\frac{1}{2}$	$\frac{2}{3}$
Player 1	$\frac{1}{3}$	$(\frac{1}{3}, \frac{1}{3})$	$(\frac{1}{3}, \frac{1}{2})$	$(\frac{1}{3}, \frac{2}{3})$	Player 1	$\frac{1}{3}$	(1, 1)	(0, 0)	(0, 0)
	$\frac{1}{2}$	$(\frac{1}{2}, \frac{1}{3})$	$(\frac{1}{2}, \frac{1}{2})$	(0, 0)		$\frac{1}{2}$	(0, 0)	(1, 1)	(0, 0)
	$\frac{2}{3}$	$(\frac{2}{3}, \frac{1}{3})$	(0, 0)	(0, 0)		$\frac{2}{3}$	(0, 0)	(0, 0)	(1, 1)

Figure 7.17

- (a) Predict the outcome of the coordination game.
- (5) In the median effort game, a variant of the minimum effort game shown in [figure 7.13](#), payoffs are determined by how close a player is to the median effort provided by a group. In this case, suppose that x_i is the effort provided by player i and M is the median effort provided by all members of the group, then the payoff to any group member is $u_i = 0.1M - 0.05(M - x_i)^2 + 0.6$. As you can see, players are penalized for being above or below the median.
- (a) Suppose that players are restricted to the integer values of x_i between 1 and 7. Construct a matrix representing the payoff to any player, in which the rows represent the player's choice and the columns represent the median effort provided.
- (b) Describe the pure-strategy equilibria of the game.
- (c) Now suppose that players choose whether to enter the median effort game or stay out. To enter, assume that players must pay an entry fee of \$1.25, and players receive a payoff of \$0 if they opt out. Predict the outcome of the game.
- (6) Two firms are deciding simultaneously whether to enter a market. Firms that stay out save the fixed costs associated with entry but make no money. The market can accommodate just one entrant, so if both firms decide to enter, they both lose the fixed cost, F . When just one firm enters, that firm earns net profits of 1. The normal form of the game can be seen in [figure 7.18](#):

		Firm 2	
		<i>Stay Out</i>	<i>Enter</i>
Firm 1	<i>Stay Out</i>	(0, 0)	(0, 1)
	<i>Enter</i>	(1, 0)	(− F , − F)

Figure 7.18

- (a) What are all the Nash equilibria of this game?
- (b) Now suppose that firm 1 can preempt the process by making a sunk investment of a fraction $0 < x < 1$ of the fixed costs. What is likely to happen now?
- (7) In figure 7.19 showing a version of the Battle of the Sexes, player 1 prefers to go to the baseball game (B), but player 2 prefers to go to the soccer game (S). Above all, however, they prefer to go to an event together:

		Player 2	
		<i>B</i>	<i>S</i>
Player 1	<i>B</i>	(2, 1)	(0, 0)
	<i>S</i>	(0, 0)	(1, 2)

Figure 7.19

- (a) If the players fail to coordinate on a pure-strategy Nash equilibrium, what payoff are they likely to receive?
- (b) Can they ask someone to use a fair coin to help them?
- (8) Consider the normal-form game shown in figure 7.20:

		Player 2	
		<i>a</i>	<i>b</i>
Player 1	<i>A</i>	(8, 8)	(6, 10)
	<i>B</i>	(10, 6)	(0, 0)

Figure 7.20

- (a) Find *all* the Nash equilibria.

- (b) Suppose that a third person, named Fer, steps forward and offers to toss a coin for the two players. Find a correlated equilibrium, if one exists.
- (c) Now imagine that instead of a coin, Fer has a six-sided die in his pocket. Fer states that when he rolls the die, he will signal to player 1 B if the roll is 1 or 2, and A if the roll is 3, 4, 5, or 6. In addition, he will signal to player 2 a if the roll is 1, 2, 3, or 4, and b if the roll is 5 or 6. Is this a correlated equilibrium?

III

Analyzing Sequential-Move Games

8

Subgame Perfection

While part II considered play in simultaneous-move games, in this part we will shift our focus to sequential games, in which the timing of the players' moves is of critical importance. In sequential games, at least one player has some information about what other players have previously done at the time she is making her own choice. In this chapter, we will see how to solve for Nash equilibria in sequential games. However, we'll also see that the Nash equilibrium can generate nonsensical, seemingly irrational predictions and we need a stronger equilibrium concept to rule out these outcomes. We'll therefore introduce a stricter kind of Nash equilibrium called *subgame perfect*, and see how to solve for subgame perfect Nash equilibria using the process of backward induction.

In the 2015 space adventure film *The Martian*, Matt Damon's character is presumed dead and left on Mars by his crew, who are heading back to Earth. Once NASA discovers that he is alive, a rogue astrodynamacist develops an ingenious plan to slingshot the crew around Earth and back to Mars to fetch him. The plan requires that as they pass by Earth, the crew will link up with a shuttle carrying the supplies they'll need to survive the longer trip. NASA deems the plan too risky, but when the crew learns of it, they consider whether to return to Mars anyway and hope that NASA will send the necessary supply shuttle as they pass by Earth. The scenario is modeled as a game in extensive form in [figure 8.1](#).

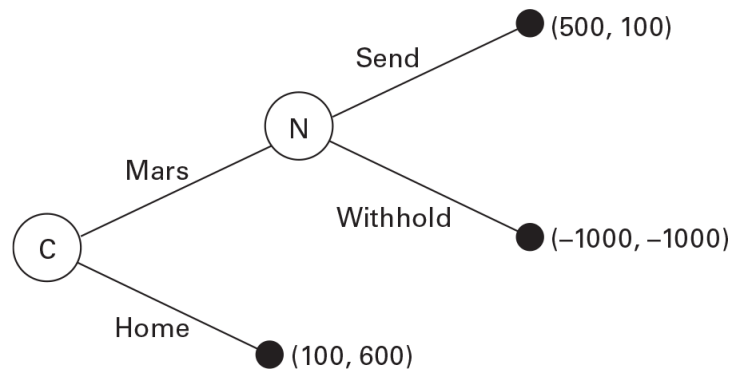


Figure 8.1

The Martian game between the Crew (C) and NASA (N).

In this game, the Crew gets to move first, by either continuing on their course back to Earth or overriding NASA and committing to the slingshot maneuver sending them back to Mars. If they choose to return to Mars, then NASA is forced to respond to their action by either sending the supply shuttle or withholding the needed supplies. Clearly, NASA withholding the supplies when the Crew has already committed to return to Mars is the worst outcome for everyone, since all of the astronauts—including Matt Damon—will die. Otherwise, NASA prefers the outcome in which the Crew returns home immediately, even if it means losing Matt Damon, and the Crew prefers the outcome in which they attempt the rescue mission. If you were the Crew, what would you do?

Nonsensical Nash Equilibria in Sequential Games

Solving for Nash equilibria in a two-player sequential-move game requires two steps. Luckily, we already know how to do both of them. First, we write the game in normal form, as we covered in chapter 2: we identify the strategy set for each player, use these strategies to form the rows and columns of the matrix, and then fill in the payoffs. Second, we solve for the Nash equilibrium or equilibria using our standard best response analysis.

The Martian Game

Let's use our procedure to find the Nash equilibria of the Martian game. In this game, each player has only one information set, and so each player's strategy requires only one action to be specified. The strategy sets are $S_{Crew} = \{Mars, Home\}$ and $S_{NASA} = \{Send, Withhold\}$. We arbitrarily assign the

Crew to be the Row player, with each row corresponding to one of its two strategies, and NASA to be the Column player, with the columns representing its two strategies. Finally, we fill in the payoffs. Note that if the Crew returns home, the outcome is the same regardless of NASA's choice, since the game ends before NASA is called on to act. The normal form representation of this interaction is shown in [figure 8.2](#). Finally, we use our standard best response analysis to find that there are two pure-strategy Nash equilibria in the game: (Mars, Send) and (Home, Withhold).

		NASA	
		Send	Withhold
Crew	Mars	(<u>500</u> , <u>100</u>)	(−1000, −1000)
	Home	(100, <u>600</u>)	(100, <u>600</u>)

Figure 8.2

The Martian game in normal form.

In following this procedure, we are essentially collapsing the sequential game into a simultaneous game, which we analyze in the usual way. We think of a player's strategy as her full contingency plan of action, which she can write down before the game even begins, specifying what she will do at each of her information sets. At the start of the game, the players form beliefs about the others' full contingency plans and then best respond to those beliefs. This means that players will be indifferent between their actions at information sets that they don't believe will be reached—in this example, NASA is indifferent between Withhold and Send if it believes that the Crew is going Home. However, by thinking of the strategy profile as a simultaneous choice before the game begins, we may be neglecting strategic considerations that are specific to the sequential nature of the game.

Let's return to the extensive form of the game and consider our two equilibria in the sequential context, shown in [figure 8.3](#). In sequential games, it is often useful to make a distinction between the parts of the strategy profile that are *on* or *off the equilibrium path*. The *equilibrium path* is the unique path of play that would be traced out through the tree so long as everyone stuck with the equilibrium strategy profile. This is what we

would see as an outside observer watching the game play out, and it is portrayed as thicker black edges in the diagram. In the first equilibrium, the equilibrium path is Mars followed by Send. In the second equilibrium, it would simply be Home, as we'd never actually observe NASA's strategy of Withhold. We would say that Withhold is *off the equilibrium path* and is highlighted in thick gray. This refers to the portion of the equilibrium strategy profile that never actually transpires in equilibrium: so long as the Crew sticks to the strategy of Home, NASA's strategy of Withhold never occurs. However, it is still important—the Crew's belief that NASA is playing Withhold is what incentivizes them to choose Home.

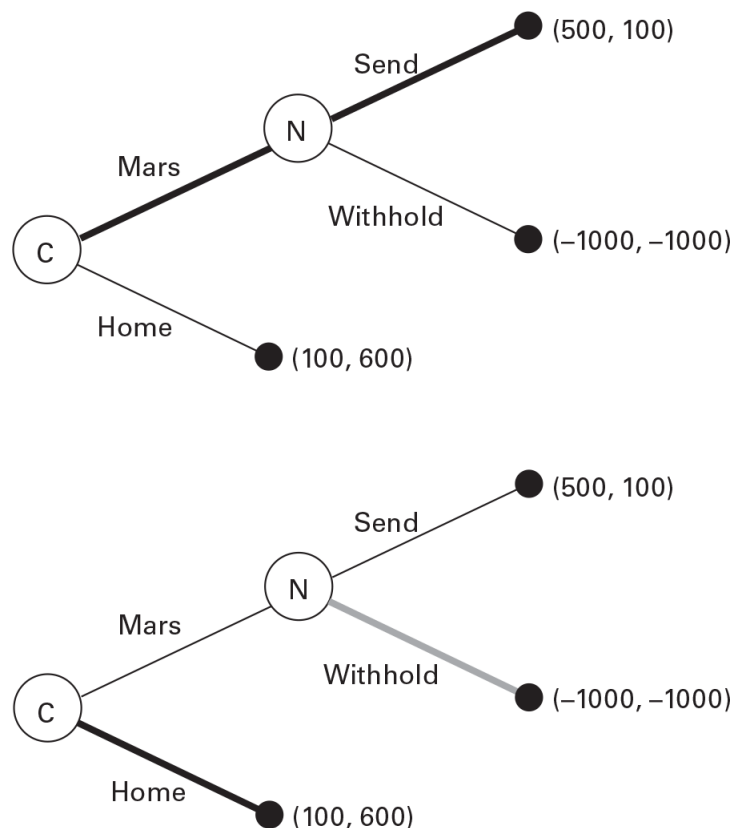


Figure 8.3

The Martian game equilibria. The thicker black lines show the portion of the strategy profile on the equilibrium path, and the gray lines show the portion that is off the equilibrium path.

Do both of these equilibria seem plausible? You may think that the (Home, Withhold) equilibrium is a bit strange. In the moment that NASA is called on to act, choosing Withhold means that they would be intentionally killing off their whole crew of astronauts. Yet this is a Nash equilibrium

because when they write down their strategies at the start of the game, Withhold can be a best response to Home because NASA will never actually have to follow through with it. NASA has accurate beliefs that the Crew is going Home, so NASA will never actually have to act anyway, and they are therefore indifferent between Send and Withhold. But what if NASA did wind up having to act, knowing that the Crew has already committed to Mars? If they were to actually reach this point in the game, it would be completely irrational for NASA to do anything but Send—by sticking with Withhold, they would get a payoff of -1000 instead of 100 . Furthermore, if the Crew recognizes (as they do in the movie) that NASA would always send the supplies if the situation arose, then they may also realize that they can get their best-case outcome by choosing Mars and “forcing NASA’s hand,” as they put it in the film. The sequential nature of the game could allow the Crew to get their way. However, these insights are neglected by the Nash equilibrium.

Clean Your Room or Die

Here is a more commonplace, but equally absurd, example. Imagine that a mom says to her kid, “If you don’t clean your room, I’m going to kill you.” This interaction is written in extensive form in [figure 8.4](#). Kid moves first by cleaning his room or not, then Mom comes home, observes whether the room is clean, and then decides whether to kill him. Both parties always prefer the outcome in which everyone survives and then, conditional on that, Kid prefers the room to be dirty while Mom prefers it to be clean.

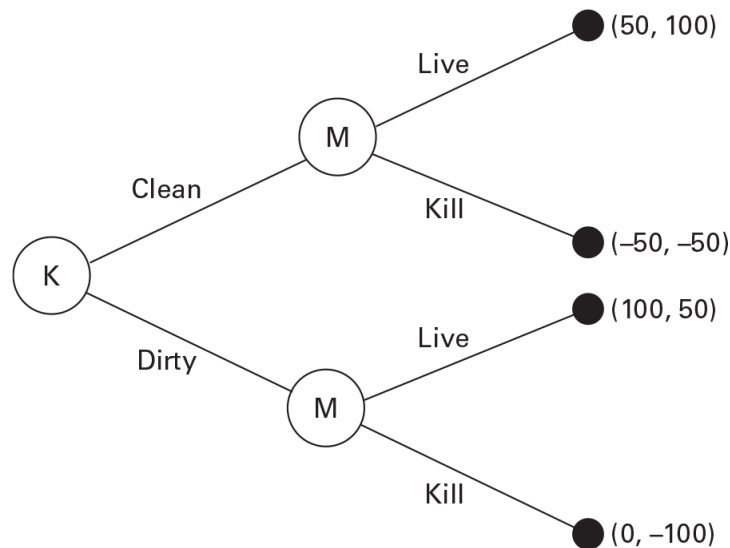


Figure 8.4

The clean your room game.

Let's find the Nash equilibria in this game. Kid's strategy set is {Clean, Dirty}, and these strategies will form the rows. But Mom has two information sets, so her strategy must specify an action at each of them: both what to do if she finds the room clean and what to do if she finds it dirty. When a player's strategy requires actions at multiple information sets, we will list the actions in the order that they come in the tree from left to right and from top to bottom. The actions specified at each information set are separated by commas, and we put parentheses around the player's entire strategy for clarity. For instance, (Live, Kill) is one strategy for Mom, indicating that she chooses Live if she finds the room clean and Kill if she finds it dirty. Her strategy set is thus: {(Kill, Kill), (Kill, Live), (Live, Kill), (Live, Live)}, and each of these strategies will be a column. The game is written in normal form in [figure 8.5](#):

		Mom			
		(Kill, Kill)	(Kill, Live)	(Live, Kill)	(Live, Live)
Kid	Clean	(-50, -50)	(-50, -50)	(<u>50</u> , <u>100</u>)	(50, <u>100</u>)
	Dirty	(<u>0</u> , -100)	(<u>100</u> , <u>50</u>)	(0, -100)	(<u>100</u> , <u>50</u>)

Figure 8.5

The clean your room game written in normal form.

We therefore identify three Nash equilibria. Two of these equilibria have Mom playing a strategy in which she would have to kill her kid in one of the scenarios. Just like the (Home, Withhold) example, they can be part of a Nash equilibrium only because she would not actually have to follow through with the threat to kill the kid, given the strategy the kid is playing. However, if she *were* to reach an information set in which her strategy specified “kill,” do you think she would actually go through with it? It would be completely irrational to do so, yielding a payoff that is lower by 150. In addition, Kid should anticipate that his mother will never follow through on her threat, he will survive the day no matter what he does, and therefore there is no point in cleaning his room. The threat to kill him is *not credible* because it requires Kid to believe that Mom would act contrary to her interests when the moment arose.

Subgame Perfect Nash Equilibria

We’ve now seen two examples of Nash equilibria that have the second-mover in a sequential game choosing actions that we know no rational actor would ever select if put in that position. Yet these strategies are consistent with our definition of rationality in simultaneous-move games, in that they are best responses to accurate beliefs about the other players’ actions. It therefore seems that we need a stricter definition of rationality in sequential games to rule out this type of behavior. Such a definition is given in definition 1.

Definition 1 (Sequential rationality) *A player’s strategy is sequentially rational if, at every information set, it maximizes her expected payoff, given her beliefs about how the remainder of the game will play out. That is, the strategy must be a best response at every information set, even those the player believes will not be reached.*

Sequential rationality requires that the player is actually willing to follow through on all parts of her strategy, given how she believes the rest of the game will proceed from that point. In other words, sequential rationality eliminates strategies that allow irrational behavior off the equilibrium path.

In the clean your room game, sequential rationality requires that Mom’s strategy maximize her expected payoff at both of her information sets. Once she reaches her top information set (finding the room clean), her best

response is clearly to let him live, providing her a payoff of 100 instead of -50 . Thus, the equilibrium (Dirty, (Kill, Live)) cannot be consistent with sequential rationality, as it has her choosing Kill at this information set. Likewise, once she reaches her bottom information set (finding the room dirty), her best response is also to let him live, eliminating the equilibrium (Clean, (Live, Kill)). What about Kid's best response? If he knows Mom to be sequentially rational, then he believes that she will always let him live, and therefore his payoff is highest from Dirty. Our third Nash equilibrium, (Dirty, (Live, Live)) is therefore consistent with common knowledge of sequential rationality. To indicate that this Nash equilibrium is special, we call it a *subgame perfect Nash equilibrium (SPNE)*. We'll provide a more formal definition (see definition 2) later in this chapter, including a discussion of where the name *subgame* comes from. For now, all you need to know is that an SPNE is an outcome that is consistent with (common knowledge of) sequential rationality. An SPNE is always a Nash equilibrium, but as we've seen, not every Nash equilibrium is subgame perfect.

What is the SPNE in the Martian game? If NASA reaches its information set, we said that the only reasonable move was to send the supplies, ruling out the (Home, Withhold) equilibrium. The Crew, anticipating this, best responds by choosing Mars. Of our two Nash equilibria, only (Mars, Send) is subgame perfect. Notice that, similar to chapter 7, subgame perfection can be used to select among Nash equilibria.

Solving for Subgame Perfect Nash Equilibria Using Backward Induction

Notice that in both of these games, we found the SPNE by starting with the second-mover's information sets and then working backward from there to see how the first-mover should behave if he anticipates that his counterpart is sequentially rational. This process is called *backward induction*, and this is the same procedure that we will use more generally to solve for SPNE.

To backward induct, we look ahead to the very last move or moves of the game. In the Martian game, this is NASA's choice; in the clean your room game, there are two: Mom's choice when finding the room clean and her choice when finding it dirty. At these nodes, the other player has already

acted, and the second player is simply choosing which payoff she wants. We assume that the player will take the action that maximizes her payoffs at this point, and we fix that as a given. Then we back up a step, to the move just prior, and determine the action that maximizes the player's payoff, given what we now know the subsequent move will be. In games with more than two steps, we continue working backward to the beginning of the game.

Backward Inducting

The example in [figure 8.6](#) works through the process of backward induction in a larger game tree. There are three final moves of the game—the last action taken before the game ends and the payoffs are given. Once Matt reaches the choice between A and B, he is simply choosing between a payoff of 100 and 20, and he maximizes his payoff by choosing A. Likewise, Jamie maximizes her payoff by choosing D over C, and Amelia maximizes her payoff by choosing F over E. We highlight those branches to remember the players' best responses at each of these decision nodes and then back up to the steps just prior. Amelia will choose In instead of Out because she correctly anticipates Matt's choice of A, such that In is associated with a payoff of 20 while Out provides 10. Note that we can never determine a player's action at a given node until we've established what would happen at all the nodes that follow it. So, even though Amelia's choice of Out has known payoffs attached to it, we can't determine what she will do until we establish whether Matt will choose A or B, and therefore the payoff she can expect from In. Similarly, Matt chooses Down instead of Up. Finally, we back all the way up to the first move. If Jamie chooses Over, we now know that she will ultimately end up with a payoff of 5, while Under leads to a better payoff of 10.

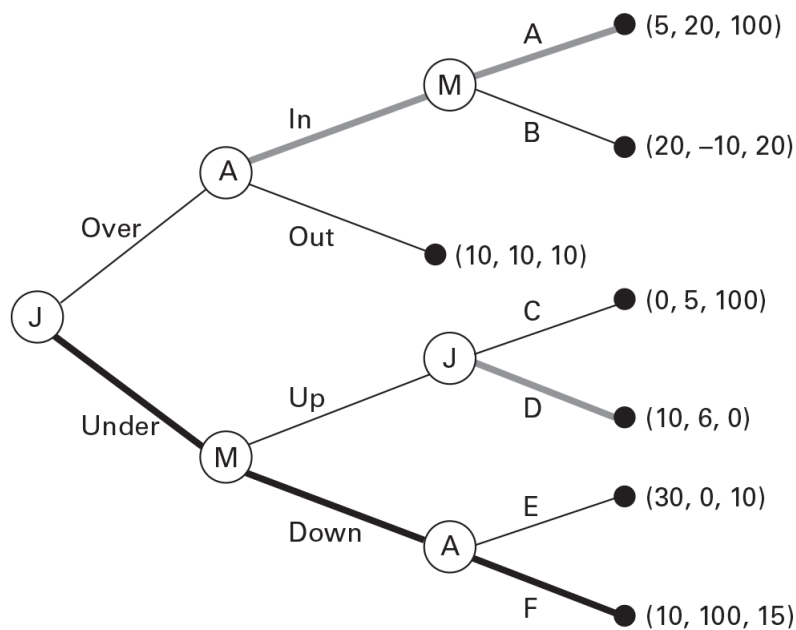


Figure 8.6

Multistep example where payoffs are written for player J(ameie), then player A(melia), then player M(att).

Our SPNE is thus ((Under, D), (In, F), (Down, A)). The equilibrium path, or what would transpire if the game were acted out, is highlighted in black. The other branches that we highlighted (shown in gray) are off the equilibrium path—but they are still an important part of the equilibrium strategy profile, since the players’ beliefs about what would happen going forward if they made a different choice are what keep them on the equilibrium path. For instance, even though Jamie never actually chooses D, Matt’s belief that she would is what causes him to choose Down. It is important to remember that the SPNE is the full strategy profile—specifying an action for each person at every information set—not just the equilibrium path.

You may wonder if the SPNE is unique or if there can be more than one. In a game of perfect information, the SPNE will be unique so long as there are no ties in payoffs. For example, imagine that Jamie’s payoff in the top outcome (after Matt chooses A) is 10 instead of 5. Then Over and Under are both best responses to the other players’ strategies, each yielding a payoff of 10. Thus, ((Under, D), (In, F), (Down, A)) and ((Over, D), (In, F), (Down, A)) are both subgame perfect Nash equilibria. As we’ll see later in the

chapter, games of imperfect information may have multiple SPNE even if every outcome is associated with different payoffs.

Centipede Game

Consider the game in [figure 8.7](#). Imagine that there is a pile of coins on the table and two players alternate deciding whether to Take the money, which ends the game, or Pass and allow the game to continue. Each time a player Passes, the amount of money on the table is doubled. If a player Takes, the game ends and he gets 80% of the pot while the other player receives only 20%. If the players Pass at all six nodes, then player 1 gets the larger share. This game was introduced by Rosenthal (1981) and later became known as the *centipede game*, after the critter that its extensive form resembles.

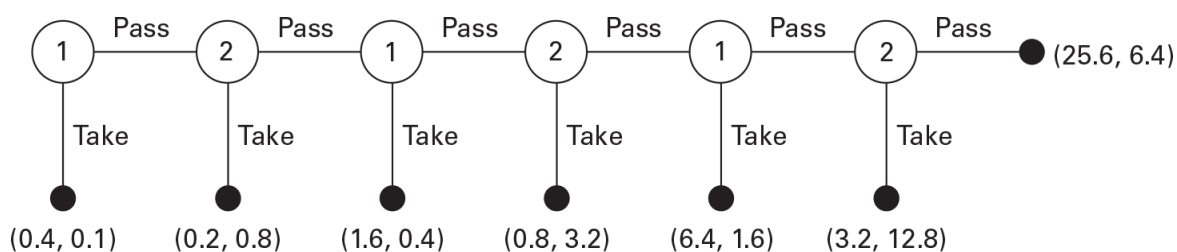


Figure 8.7

The centipede game.

What is the SPNE of this game? We start with the final decision node, in which player 2 decides between Pass, for a payoff of 6.4, or Take, for a payoff of 12.8. Player 2 will therefore prefer to Take at the final node. If player 1 knows that player 2 will Take in the final step, then player 1 would wish to Take first in the penultimate step (to get 6.4 instead of 3.2). If we follow this logic all the way backward, we find that the unique SPNE is for each player to Take at every opportunity. The incentive to grab the larger share before the other player gets the chance to do so leads to an inefficient outcome in which the players collectively earn only \$0.50, rather than the \$32 that could be achieved if they Pass on every turn.

Hit 15

In the hit 15 game, two players take turns announcing a number, either 1 or 2, while keeping a running tally of the cumulative sum of their numbers (Dufwenberg, Sundaram, and Butler, 2010; Gneezy, Rustichini, and

Vostroknutov 2010). The player whose announcement brings the total sum to 15 (i.e., “hits 15”) wins the game. In the centipede game, what one player prefers early in the game depends on what he anticipates the other will do later. In contrast, the second-mover in the hit 15 game has a dominant strategy, which guarantees her victory regardless of what her competitor does in later turns. We can see this by backward inducting from the end of the game.

If you hit 15, you win, but if you hit 14, then your opponent can choose 1 and win. Likewise, if you hit 13, then you’ll lose so long as your opponent rationally chooses 2 in response. However, if you hit 12, then you have a clear path to victory: if your opponent chooses 2, you can announce 1, and if your opponent chooses 1, you can win by announcing 2. By the same reasoning, once you hit 9, you can always ensure that you can hit 12, and then go on to hit 15. If we follow this logic all the way backward, we find that the second-mover always has a strategy that guarantees she’ll win. On each turn, if the first player just announced 1, the second player should announce 2, and vice versa. This way, the two announcements always sum to 3 and, after five turns of this, the second-mover hits 15.

Continuous Games

Let’s now consider games in which players have continuous strategy sets rather than finite sets of actions at each decision node. Imagine that Starbucks and Peet’s are simultaneously setting the price of their blended iced coffee drinks. Some loyal customers will always buy the drink from their favorite coffee house even if it is more expensive, but in general each company will lose customers as its price exceeds that of the competition, so each company’s profit will depend on the prices that they both choose. Let’s say that Starbucks chooses price s , Peet’s chooses price p , and their payoffs are given by $u_s = 4s - s^2 + ps$ and $u_p = 9p - 3/2p^2 + ps$.

In the simultaneous game, we would solve for the Nash equilibrium prices using the procedure discussed in chapter 6: we take the derivative of each player’s payoff with respect to their own price (since this is the variable under their control), set it equal to zero, and solve for their own price to get their best response as a function of their opponent’s price. For Starbucks, we find its best response price is $s = 2 + p/2$, and for Peet’s the

best response is $p = 3 + s/3$. Then we solve the two equations simultaneously to get $s = \$4.20$ and $p = \$4.40$.

Now, imagine that Starbucks makes its choice first—perhaps they are launching a national advertising campaign committing to a specific price—and Peet’s will price its drink only after learning what Starbucks chose. This game is depicted in extensive form in [figure 8.8](#). Since each player’s strategy set is infinite, we cannot draw a separate branch for each possible action, and instead depict each choice as being drawn from a somewhat pie-shaped wedge that represents a continuum of possible prices. How does this sequence of moves change our solution? First, notice that $(4.20, 4.40)$ is still a Nash equilibrium. If Peet’s knows that Starbucks has chosen 4.20, they will still prefer 4.40 since it is their best response to 4.20. And, if Starbucks is convinced that Peet’s will always choose 4.40 *no matter what Starbucks announces*, then Starbucks will still prefer 4.20. However, it would not be sequentially rational for Peet’s to stubbornly choose 4.40 regardless of the actual advertised price of Starbucks.

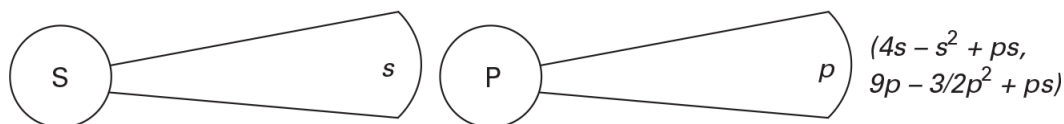


Figure 8.8

A sequential game with continuous strategies: Starbucks first chooses a price, s , and then Peet’s chooses a price, p .

To find the SPNE, let’s backward induct from the final decision of the game, in which Peet’s sets its price after observing what Starbucks does. Once Peet’s learns s , the only rational response is to maximize its payoff by plugging this number into its best response function and selecting $p = 3 + s/3$. Now we back up a step to the initial choice of Starbucks. It anticipates that Peet’s will always respond with $p = 3 + s/3$, so that p actually becomes a known function of its own choice s . Notice the difference from the simultaneous game, when p is a fixed variable outside of Starbucks’ control. In the sequential game, Starbucks effectively determines p by first choosing s . So how should Starbucks maximize its profit? To calculate this, we take the payoff for Starbucks, $u_s = 4s - s^2 + ps$, and, since p is a known function of s , plug in for Peet’s price $p = 3 + s/3$. Now, we have a function only of s :

$u_s = 4s - s^2 + (3 + s/3)s = 7s - (2/3)s^2$. To find the s that maximizes the payoff for Starbucks, we take the derivative, set it equal to 0, and solve for s , yielding: $s = 5.25$. The SPNE is therefore $(5.25, 3+s/3)$.¹ The equilibrium path—what actually transpires if the players choose prices according to the SPNE—is Starbucks advertises 5.25 and Peet’s responds with $3 + (5.25)/3 = 4.75$. Note, though, that when we write the SPNE, we need to specify Peet’s full strategy for any price that Starbucks could choose, and not simply write 4.75, which would incorrectly indicate that Peet’s would choose this price no matter what Starbucks did.

Example: Stackelberg Competition

In the Cournot duopoly model (described in chapter 6), two firms simultaneously choose their output quantity and then can sell their units at a market price determined by the total output of both firms. In equilibrium, the firms produce more (and experience a lower price) than if they acted to maximize their joint profits. In the example in chapter 6, two firms $i \in \{1, 2\}$ each simultaneously choose quantity q_i , each has marginal costs of 1, and the market price is given by $100 - q_i - q_j$. Each firm’s payoff is given by its quantity times price minus their costs: $u_i(q_i, q_j) = q_i(100 - q_i - q_j) - q_i$. We found that the equilibrium output for each firm was 33, resulting in an equilibrium market price of $100 - 33 - 33 = 34$.

How does this prediction change if one firm (say firm A) commits to an output level first, and firm B chooses its output only after observing what firm A has done? This model of a duopoly in which one firm is the “leader” and the other is the “follower” is known as *Stackelberg competition*. As usual, we begin with the final move of the game: firm B’s choice after learning that firm A produced q_A . At this point, the strategic uncertainty has been resolved and firm B’s decision uniquely determines its payoff. To maximize its payoff, firm B simply plugs q_A into its best response function, which we found in chapter 6 to be: $q_B = \frac{99 - q_A}{2}$. Now we back up to firm A’s decision. Firm A acts to maximize its payoff, $u_A(q_A, q_B) = q_A(100 - q_A - q_B) - q_A$. Since it correctly anticipates that $q_B = \frac{99 - q_A}{2}$, we plug this in for q_B in firm A’s payoff: $u_A(q_A, q_B) = q_A \left(100 - q_A - \frac{99 - q_A}{2} \right) - q_A$. We find that this function achieves its maximum when: $\frac{\partial u_A}{\partial q_A} = 100 - 2q_A - \frac{99}{2} + q_A - 1 = 0$, or $q_A = \frac{99}{2}$. On the equilibrium path, firm B chooses $q_B = \frac{99 - 50.5}{2} = \frac{99}{4}$ and the

equilibrium price is 25.75. Notice that, relative to the simultaneous Cournot outcome, more is produced in equilibrium (74.25 instead of 66) and the price is lower (25.75 instead of 34). The leader does better under this outcome, earning a payoff of 1, 225 instead of 1, 089. The follower, however, does worse in the sequential game on both counts: it produces less *and* sells that output at a lower price. Thus, a firm can gain an advantage if it is able to establish itself as the “leader” by credibly committing to an output level before its competitor makes its own production decision.

Games of Imperfect Information

Thus far, we have considered only sequential games of perfect information, in which each node is in its own information set and the players always know where in the game tree they are. In contrast, consider the game presented in [figure 8.9](#) between Rogelio and Xiomara. If we wanted to find the SPNE using backward induction, we might attempt to start with Xiomara’s final decision node at the top right. However, we have a problem. Although we know what point in the tree she is at, she does not know whether she is at this node or the one below it. It is impossible for her to consider this choice without backing up a step to Rogelio’s choice of X or Y and generating beliefs about what he is likely to do. Therefore, we’ll need to proceed in the same way. We can’t start the backward induction process with Xiomara’s final information set; we will need to back up to Rogelio’s simultaneous choice in order to understand her decision as described in more detail below.

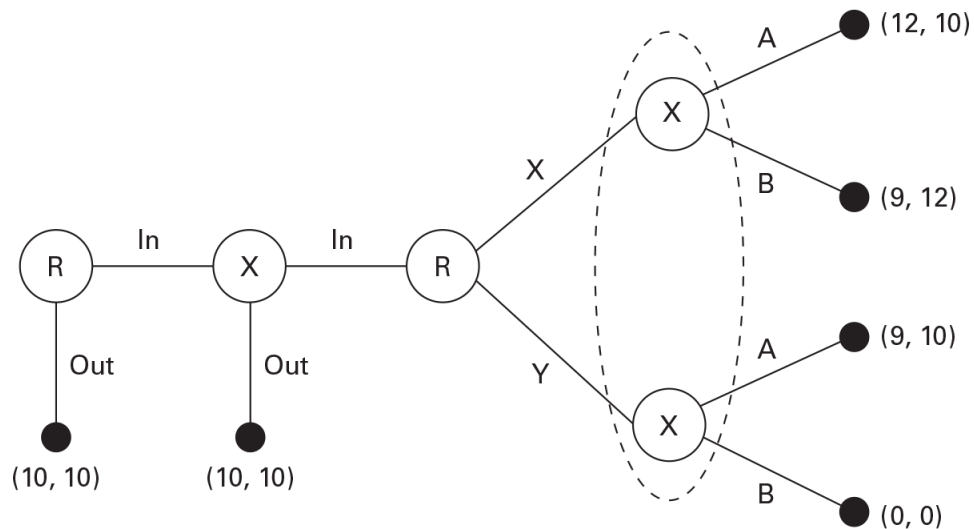


Figure 8.9

A sequential game of imperfect information between players R(ogelio) and X(iomara).

Subgames

In games of perfect information, we can start at any decision node in the tree, and there is always a well-defined set of sequential decisions for the players going forward from that point. This allows us to backward induct one node (or decision) at a time. In games of imperfect information, not every decision node is associated with a coherent decision or game from that point forward, as we saw with Xiomara's final nodes. Thus we can only backward induct over subgames. A *subgame* is a part of a game that can stand on its own as a self-contained game or decision. It always begins with a single node, which is the only node in this information set, and then contains all the decision nodes that follow it. A subgame can never cut through information sets, by including some nodes in the set and not others.

In a game of perfect information, every decision node starts a subgame. In the Martian game, we therefore have two subgames, each starting from one node and containing the remainder of the game from that point. Notice that the entire game itself is considered a subgame. In the clean your room game, there are three subgames: the one beginning when Mom finds the room clean, the one beginning when Mom finds the room dirty, and then one beginning with Kid's choice and continuing through Mom's two information sets (i.e., the entire game itself). How many subgames are shown in [figure 8.6](#)? There are six decision nodes in this game of perfect

information, so there are six subgames, each starting with one node and continuing for the rest of the game.

Now, what about the imperfect information game with Rogelio and Xiomara? We cannot start a subgame with one of Xiomara's final nodes, since neither is the unique node in the information set. However, we can consider the interaction beginning with Rogelio's choice of X or Y, and this node begins a subgame. There are also two others: one starting with Xiomara's choice of In or Out (and continuing through the rest of the game) and one starting with Rogelio's initial choice (the whole game itself). Now that we've defined the term *subgame*, we can offer a more formal definition of the SPNE solution concept.

Definition 2 (Subgame perfect Nash equilibrium [SPNE]) *A strategy profile is an SPNE if the players' strategies are a best response in every subgame.*

Note that, since the entire game itself is a subgame, definition 2 implies that SPNE are Nash equilibria (i.e., each player best responds in the game). In games of perfect information, every node begins a subgame, so the definition implies that, at each node, players best respond to what will come next, consistent with our process of backward induction.

To solve for the SPNE in games of imperfect information, we backward induct one subgame—rather than one decision node—at a time. In the Rogelio-Xiomara game, the final subgame is the interaction beginning with Rogelio choosing X or Y. At this point, we have a simultaneous-move game, which is written in normal form in [figure 8.10](#). There is a unique Nash equilibrium in this interaction: (X, B). In an SPNE, both players must best respond in this subgame, and so Rogelio must play X and Xiomara B. If the game gets to this point, Xiomara will get a payoff of 12 and Rogelio 9. Now, we back up a step, to the subgame beginning with Xiomara's choice of In or Out. Out yields a payoff of 10, but moving on to the final subgame will yield a payoff of 12, and so she will choose In. Finally, we back up to the first move of the game and conclude that Rogelio's best response is Out. Thus the SPNE of this game is ((Out, X), (In, B)).

		Xiomara	
		A	B
Rogelio	X	(<u>12</u> , 10)	(9, <u>12</u>)
	Y	(9, <u>10</u>)	(0, 0)

Figure 8.10
The final subgame.

Example with Multiple Subgame Perfect Nash Equilibria

Consider the sequential game with imperfect information given in [figure 8.11](#). Elmer first decides whether to call up his friend Sylvester and invite him hunting, or to stay home and both get a payoff of 70. Sylvester then decides whether to accept or decline. If he declines, he gets a payoff of 70, but Elmer's feelings are hurt and so Elmer only gets a payoff of 10. If he accepts, then the two hunters play the stag hunt game introduced in chapter 5, in which they simultaneously choose between Rabbit and Stag.

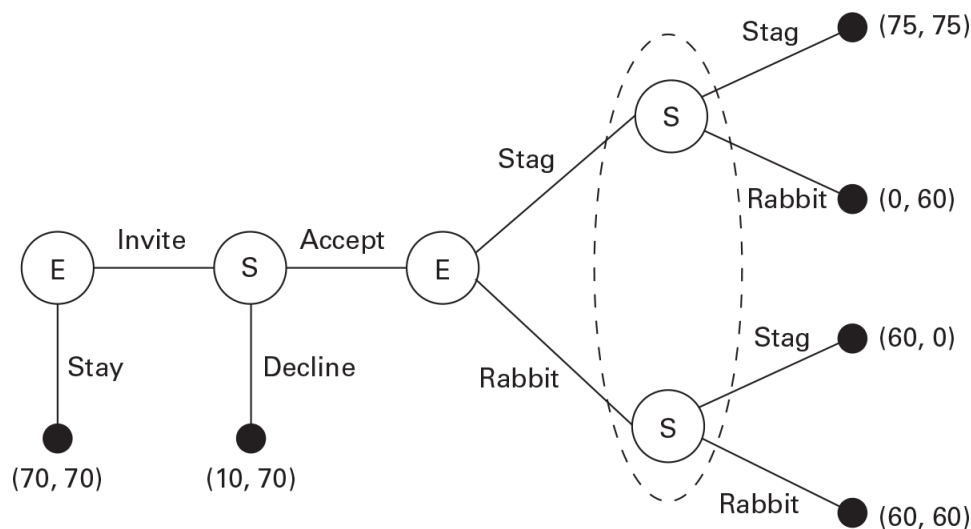


Figure 8.11
Stag hunt with a sensitive hunter.

Let's find the subgame perfect Nash equilibria in this game by backward inducting over each subgame. The final subgame is the stag hunt game interaction that begins when Sylvester accepts the invitation. As we saw in chapter 5, there are three Nash equilibria in this subgame: (Stag, Stag), (Rabbit, Rabbit), and a mixed-strategy Nash equilibrium in which each

player chooses Stag with probability $4/5$. Since each of these outcomes involve mutual best responses, each could be part of a different SPNE in this game.

To find each SPNE, we will backward induct from each of these final outcomes, one at a time. If (Stag, Stag) is played in the final subgame, then each hunter earns a final payoff of 75. Working backward, Sylvester would best respond by accepting the invitation, and so Elmer prefers to invite him. ((Invite, Stag), (Accept, Stag)) is therefore one SPNE. What if (Rabbit, Rabbit) is played in the final subgame? Then players would receive only 60 if they reach the final subgame, so Sylvester would prefer to Decline and Elmer wants to Stay Home. So ((Stay, Rabbit), (Decline, Rabbit)) is also an SPNE. What if the mixed strategy is played in the final subgame? Although the players anticipating the mixed strategy do not know what their payoff will be with certainty, they can compute their expected payoff, which is 60 for each player, and is worse than not going hunting. Therefore, ((Stay, $(\frac{4}{5}\text{Stag}, \frac{1}{5}\text{Rabbit})$), (Decline, $(\frac{4}{5}\text{Stag}, \frac{1}{5}\text{Rabbit})$)) is also an SPNE.

Behavior in Games

Playing Incredible Strategies

Do players who move early in the game act as though they anticipate sequentially rational play from later movers? Or do they act as if they believe that the other player might choose an action that is not credible? Consider the game in [figure 8.12](#), which has a similar structure to the Martian game. Just as in that interaction, the first player can choose a “Safe” option and end the game with a moderate payoff, or they can take the risk of allowing the other player to act, trusting that they will choose the rational, credible action instead of the “Punish” option, which is worse for both players. As in the Martian game, both (Safe, Punish) and (Risky, Credible) are Nash equilibria, but only (Risky, Credible) is subgame perfect.

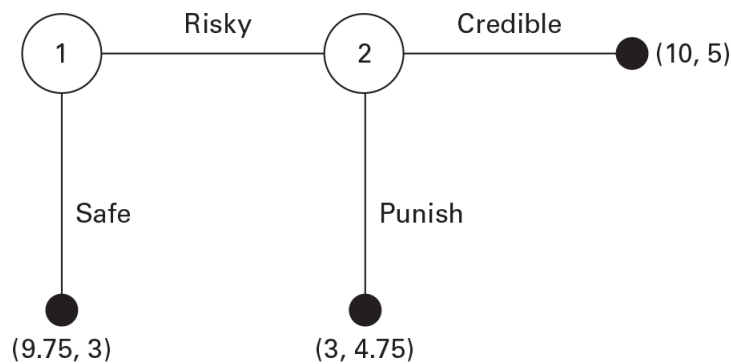


Figure 8.12

Game with noncredible punishment.

An early experimental test was conducted by Beard and Beil (1994), using the payoffs listed at the decision nodes in [figure 8.12](#). In this first game, in which player 1 has little to gain (0.25) from choosing Risky and player 2 has little to lose from choosing Punish (0.25), fully two-thirds of player 1s choose the Safe option, contrary to the SPNE. However, Beard and Beil found that the likelihood of reaching the SPNE depends on the relative payoffs associated with each outcome. They conducted modified versions of the game with slightly different payoffs (as listed in [figure 8.13](#)), but (Risky, Credible) was the unique SPNE in each case. In the second game, everything stays the same, except that the Safe payoff for player 1 becomes less attractive (dropping from 9.75 to 7). In that scenario, only one-fifth of players choose the Safe option and 80% of the pairs reached the SPNE. Player 1 may also be more willing to trust player 2 to play the Credible strategy if it's fairly costly for player 2 to mess up: game 3 is identical to game 1 except player 2 loses 2 by choosing Punish, and in this case, player 1 becomes more likely to play Risky.

Game	Payoffs from Outcomes			% Choosing Risky
	Safe	Punish	Credible	
Beard and Beil game 1	(9.75, 3)	(3, 4.75)	(10, 5)	34
Beard and Beil game 2	(7, 3)	(3, 4.75)	(10, 5)	80
Beard and Beil game 3	(9.75, 3)	(3, 3)	(10, 5)	53
Beard and Beil game 4	(9.75, 6)	(3, 4.75)	(10, 5)	14
Goeree and Holt game 1	(70, 60)	(60, 10)	(90, 50)	88
Goeree and Holt game 2	(70, 60)	(60, 48)	(90, 50)	68

Figure 8.13

Data from noncredible threat games. Notes: Payoffs are listed for (player 1, player 2) for each outcome. % choosing Risky indicates player 1 choices consistent with SPNE.

You may notice that the Martian game differs from these three games in that NASA did best when the Crew chose the safe Home option and therefore may be resentful after Crew chooses the more risky Return. Game 4 captures this resentment by increasing player 2's Safe payoff—and finds that 86% of player 1s are unwilling to choose Risky and face possible punishment by a resentful player 2.

Goeree and Holt (2001) find consistent results with a different payoff structure: when it would be very costly for player 2 to punish, the SPNE is played 88% of the time. But when the cost of punishment is small, only 68% trust that player 2 won't choose a strategy that is not credible, and nearly half of player 2s follow through on the punishment at a cost to themselves, such that only a little over one-third of pairs reach the SPNE.

In the centipede game, players need to backward induct over multiple steps. In addition, there is a clear conflict in the incentives: the SPNE requires the players to Take at their first opportunity, implying that the game ends with payoffs of (0.40, 0.10), but if they are able to make it to the end of the tree, they can gain payoffs *64 times* higher. So what happens? Pretty much everything—except, perhaps, the two extremes you might expect: McKelvey and Palfrey (1992) found that essentially no one plays the SPNE of Taking right off the bat (0.7% of observations) or the most socially efficient outcome of Passing all the way to the end of the game (1.4% of observations). Instead, most games end one step (25.3%), two steps (38.4%), or three steps (20%) before the last decision node, indicating that

the players are trying to capture the larger share, but either they don't fully backward induct until they are closer to the end of the game, they have some preference for cooperation and social efficiency, or they anticipate that their opponent is not always going to Take, and so they can let the pot grow a bit before Taking themselves.

There is some indication that a failure of backward induction is to blame for the infrequent SPNE play. When the game is repeated, players learn to Take earlier in the interaction. In addition, players are more likely to take earlier in games with fewer steps (centipede legs) than in longer ones. But there is also evidence that some of the behavioral deviation may be due to players having a preference for cooperation and social efficiency, or anticipating that their opponent might have such preferences. Fey, McKelvey, and Palfrey (1996) study a constant-sum centipede game which has the same SPNE, but Passing does not raise collective payoffs, which always sum to 3.2. Instead, the portion of the pot that the Taker gets increases as the game proceeds, such that Taking on the first step leads to equal payoffs while, if everyone Passes, player 1 gets 91% of the pot. Not only is there no social efficiency benefit to letting the game proceed, but a player who cares about fairness may find the later outcomes less appealing, as the payoffs become increasingly lopsided. In this experiment, almost 60% of participants Take on their first opportunity—far higher than in the original game—but a substantial portion still allow the game to continue.

Evidence-Based Adaptations of the Standard Model

How can we better predict outcomes in sequential-move games where large majorities of players do not play according to the SPNE? Although there are many potential factors at play, it is clear that early movers often anticipate that later players will not always act to maximize their own monetary gain, either due to decision errors or social preferences.

In simultaneous games, we've seen that the quantal response equilibrium (QRE) typically provides a more accurate prediction than the Nash equilibrium by assuming that players don't always best respond, but rather play better responses more frequently than worse ones and anticipate that others similarly make mistakes. Extensive-form games have an analogous concept called *agent quantal response equilibrium (AQRE)*. Rather than assuming that each player has a comprehensive strategy, imagine that each

player is represented at each information set by a different agent, each independently acting on the player's behalf. The agent doesn't necessarily implement the player's best response at her information set. Instead, she puts more probability weight on actions that are associated with higher expected payoffs, given her beliefs about the mixed strategies played by the other agents. The agents have accurate beliefs about the mixed strategies of the others, but never know for sure which actions the later agents will choose, even those representing the same player. This model fits the centipede game data best when it also incorporates the possibility of social preferences by assuming that all the agents believe that there is a certain probability that the others are altruists, who will always Pass at every opportunity. We will apply this solution concept to the centipede game more rigorously in chapter 26, but the intuition is straightforward. If you are making the first choice in the centipede game and you believe that there is a decent chance that the other player will Pass on the later moves, it would be very costly to end the game immediately. Instead, you would be better off gambling on the possibility that the game will last for multiple turns.

Level- k models, which we introduced in chapter 3 to explain beauty contest game behavior and will discuss in depth in chapter 27, have a similar flavor to QRE in that they assume that players anticipate the bounded rationality of others. Imagine that there are different types of players, each capable of different levels of reasoning in the centipede game (as described by Ho and Su, 2013). Each player assumes that her opponent uses one level of reasoning below her own and she best responds to that strategy. The strategy that each level of player would choose is listed in [figure 8.14](#). A level-0 player simply randomizes uniformly over the possible strategies and flips a coin at each node reached. A level-1 player best responds to his belief that his opponent is a level-0 randomizer. This involves always passing, unless it is the very last move of the game, in which case player 2 Takes. Why would a player 1 not wish to Take on his last turn if he believes that player 2 is randomizing? Taking is associated with a payoff of 6.40, while Passing provides a 50-50 chance at 3.20 or 25.60, with an expected payoff of 14.40. In turn, a level-2 player best responds to her belief that her opponent is acting as a level-1 player: since he always Passes except in the final move, she will wish to Pass until the second-to-last node of the game, at which point she Takes before he can.

We can follow this reasoning to find the strategies for each level of player in the game. Notice that, if all players are level-6 or higher, then we arrive at the SPNE. However, if we assume a distribution of types, then the predictions would be consistent with most games ending two or three steps from the end. Furthermore, the model is consistent with the findings that games with fewer steps are more likely to end at the first node and play is more consistent with the SPNE over time, as players may update their beliefs about which level-type they are dealing with.

Level	Player 1's Strategy	Player 2's Strategy
Level-0	(R, R, R)	(R, R, R)
Level-1	(P, P, P)	(P, P, T)
Level-2	(P, P, T)	(P, P, T)
Level-3	(P, P, T)	(P, T, T)
Level-4	(P, T, T)	(P, T, T)
Level-5	(P, T, T)	(T, T, T)
Level-6	(T, T, T)	(T, T, T)

Figure 8.14

Level- k strategies in the centipede game. Note: Each player's strategy lists their action in order of the turn to move. P is Pass; T is Take; R is Randomize.

Exercises

(1) Consider the three games depicted in [figure 8.15](#). The payoffs are listed for player 1, then player 2, and then player 3 in each case. For each game:

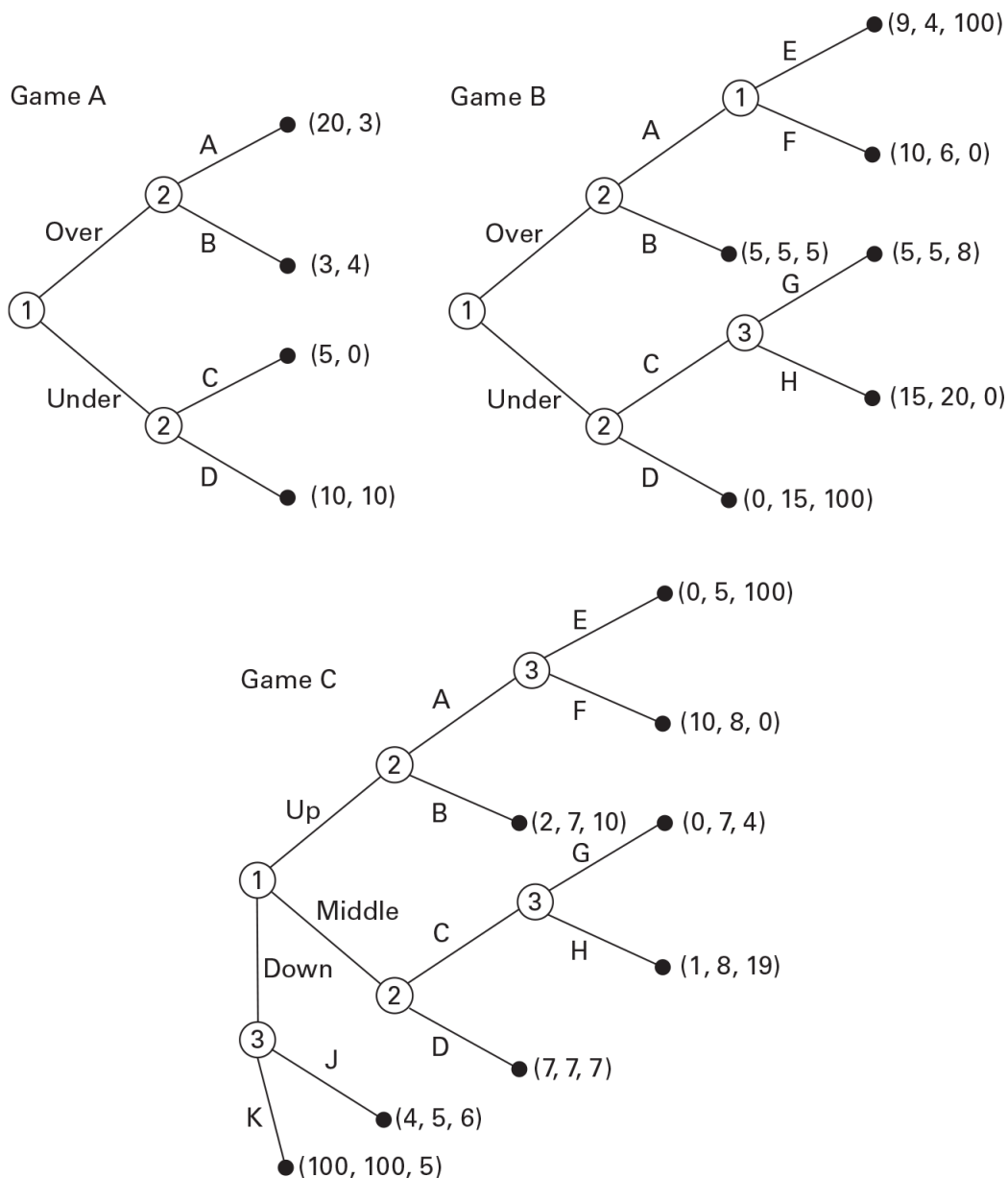


Figure 8.15

- Write the complete strategy set for each player.
 - Say how many subgames are associated with the game.
 - Find the subgame perfect Nash equilibrium or equilibria (note that any of these should consist of one element from each of the players' strategy sets that you identified in part [a]).
 - Which parts of the SPNE are on and off the equilibrium path?
- (2) Consider a game between two firms, BiffCo and Cyberdyne Systems. BiffCo first decides whether to enter the industry. If it stays out, BiffCo

earns 8 and Cyberdyne earns 20. If BiffCo enters, then the firms play the simultaneous pricing game listed in [figure 8.16](#).

		Cyberdyne		
		\$1	\$2	\$3
BiffCo	\$1	(6, 6)	(15, 0)	(15, 0)
	\$2	(0, 15)	(9, 9)	(18, 0)
	\$3	(0, 15)	(0, 18)	(12, 12)

Figure 8.16

- Write the game in extensive form.
 - How many subgames are associated with this game?
 - Find the subgame perfect Nash equilibrium or equilibria.
- (3) Consider the game depicted in [figure 8.17](#).

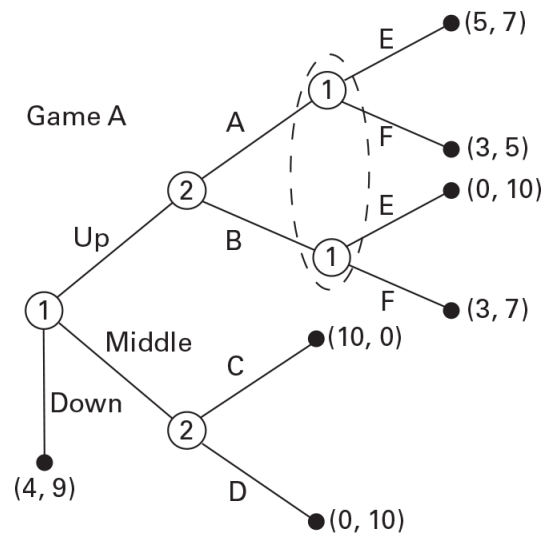


Figure 8.17

- Write the complete strategy set for each player.
- How many subgames are associated with the game?
- Find the subgame perfect Nash equilibrium or equilibria (note that any of these should consist of one element from each of the players' strategy sets that you identified in part [a]).

(4) Let's consider a sequential version of the Bach or Stravinsky game we looked at in the chapter 5 exercises. Each player is deciding what to do tonight: to see a Bach concert or a Stravinsky concert. Each player prefers to match the other's decision, but they disagree about which concert would be more enjoyable. Assume that Johann moves first, and Igor moves only after observing Johann's action.

		Igor	
		Bach	Stravinsky
Johann	Bach	(75, 25)	(0, 0)
	Stravinsky	(0, 0)	(25, 75)

Figure 8.18

- Write this sequential game in normal form.
 - Find all pure-strategy Nash equilibria of the sequential game.
 - What is the subgame perfect Nash equilibrium?
- (5) Consider a game between Sony and Microsoft in which both players are choosing the quantity of video game consoles to release between Thanksgiving and Christmas. Sony's payoff function is given by $u_s(q_s, q_m) = 1000q_s - q_s^2 - q_sq_m$, where q_s is the quantity of Playstations released by Sony and q_m is the quantity of Xboxes released by Microsoft. Microsoft's payoff function is: $u_m(q_s, q_m) = 800q_m - q_m^2 - q_sq_m$. Imagine that Sony will choose how many Playstations to release first. Then Microsoft will observe this decision and choose how many Xboxes to release. Find the subgame perfect Nash equilibrium of this game.
- (6) Consider the following game of Nim. There are two piles of matchsticks, each containing three matchsticks. Two players take turns removing matchsticks from one of the piles, starting with player 1. On each turn, the player must remove *at least one* matchstick and can remove matches from only one of the piles (but can choose which one). The player who removes the final match loses. Which player has a strategy that guarantees she will win, regardless of what the other player does (similar to the hit 15 game)? What is it?

(7) Consider an extended version of the hit 15 game described earlier in the chapter, in which two players alternate announcing numbers (Levitt, List, and Sadoff 2011).

(a) On each turn, the players can announce a number in the set $\{1, 2, 3, 4, 5, 6, 7, 8, 9\}$, and a running tally of the sum of the numbers is kept. The player who brings the tally to 100 wins the game. Which player has a winning strategy, and what is it?

(b) Now imagine that the strategy set for the players is expanded to also include 10. Which player has a winning strategy, and what is it?

(8) Consider the game in [figure 8.19](#).

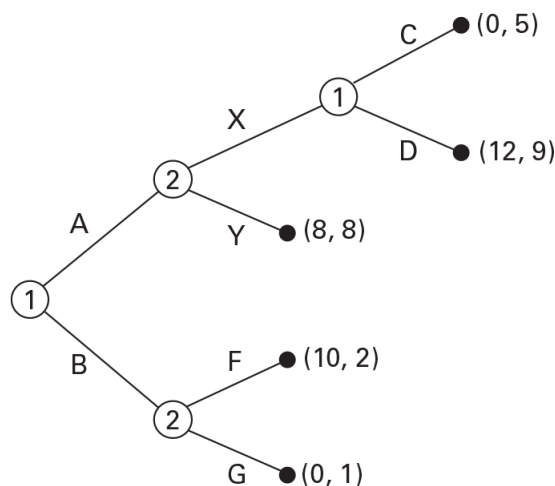


Figure 8.19

(a) Find the SPNE.

(b) Now let's look at this from the perspective of the level- k model. List the strategies that would be played by each player using level-0, level-1, level-2, and level-3 reasoning.

¹Notice that we do not use the Starbucks best response function for this process. This is because the best response function tells us how Starbucks will maximize its payoff given that Peet's choice is fixed (which is why we treat it as a constant when solving for the best response function). But in the case of a sequential game, Starbucks can influence Peet's price and it wants to determine the price that maximizes its profit conditional on this influence, which is what we do when we take the derivative of the Starbucks payoff after plugging in Peet's response as a function of what Starbucks chooses.

9

Finitely Repeated Games

Our discussion hitherto has assumed that agents interact just once. But what happens when the players meet repeatedly? In this case, we have a specific type of sequential game, called a *repeated game*. In a repeated game, the same players have an identical interaction multiple times. We'll see that the subgame perfect Nash equilibrium (SPNE) in repeated games depends on whether the game is repeated a finite or infinite number of times. In a finitely repeated game (the focus of this chapter), the SPNE depends on how many equilibria there are in the underlying game. In an infinitely repeated game (the subject of chapter 10), the equilibrium outcomes depend on how patient the players are.

The behavioral economist Colin Camerer once likened repeated games to the movie *Groundhog Day*, in which a jinxed protagonist is forced to relive the same day, interacting with exactly the same people, over and over. The only thing that changes from one day to the next is the knowledge he gathers from experimenting on the previous days. Because the movie targeted cineplexes and not art houses, this learning process ultimately leads to a cosmic equilibrium in which he lives happily ever after. Unfortunately, as we will see, this is not always how things work out in repeated games.

Consider, for example, a situation in which you are in a group with three other players, and you each have an endowment of 20 tokens. These tokens (redeemable for cash at the end of the game) can be invested in an “individual project” or a “group project,” which is a public good. Investments in the individual project just accumulate, one-for-one, but investments in the group project are summed up, doubled, and then shared equally among all four group members. If you play the game just once, it is not hard to figure out what to do, given the intuition provided in the previous chapters. Each token invested in the group project turns into 2 tokens. That is clearly good. However, those 2 tokens are shared equally among all four of you. That's not so good. In fact, this means that for every token you invest in the group project, you just get 0.5 tokens back. If you can put them in the individual project instead, you double your return and also get a share of whatever the

other players put in the group project. It is a dominant strategy to contribute nothing to the group project because it doesn't matter what the other players choose—it is always best for you to put everything you have into the individual project.

But what if you repeat this game 10 times? If each person invests everything in the individual project, you will end up with $10 \times 20 = 200$ tokens. However, if everyone invests in the group project instead, you'll each earn a total of $10[\frac{2(80)}{4}] = 400$ tokens. In other words, the group can leave a lot of tokens on the table by relying on the individual project. In this chapter (and the next), we consider games like this one, which are repeated, and ask whether players can take advantage of the repetitions to improve their outcomes.

Definitions

Let's begin by discussing how the structure and terminology of a game changes as we allow players to engage in *stationary replication*, meaning they play the exact same game over and over, accumulating utility along the way. The basic game, determining the structure of play and outcomes on each repetition, which we will assume does not change from one round to the next, is known as the *stage game*. This stage game can be repeated either a *finite* number of times or *infinitely*. Considering all the repetitions together is known as the *supergame*. In this chapter, we will focus on analyzing games played a finite number of times. In the next chapter, we examine infinitely repeated games. Finally, the payoffs that players achieve in the game are just the (perhaps weighted) sum of the utilities they earn in each period of the supergame. The table in [figure 9.1](#) illustrates the three sets of circumstances that we will discuss in these two chapters:

		Number of Stage Equilibria	
		Single	Multiple
Repetition	Finite	Selten's theorem	Mechanism design
	Infinite	Folk theorem	

Figure 9.1
A taxonomy of repeated games.

In this chapter on finitely-repeated games, we will analyze what can be classified as the easiest and the hardest of the three cases. As we will see, finitely repeated supergames with just one equilibrium in the stage game (the upper-left cell of the table) are relatively straightforward thanks to the work of Reinhard Selten (1978) on *induction theory*, what we now call *backward induction*. Much harder to analyze are supergames based on stage games that have more than one equilibrium (the upper-right cell). We call this category of games an *application of mechanism design* because if strategies are cleverly constructed, one can achieve what seem like equilibria that only an engineer would think up. Finally, in the next chapter, we will see that just about any strategy profile can be supported as an equilibrium when the stage game is played indefinitely, provided that the players are sufficiently patient. This result has come to be known as the *Folk theorem* because many game theorists knew this “folk wisdom” before it was finally written down by James Friedman in 1971.

Finitely Repeated Games with a Single Stage Equilibrium

It makes pedagogical sense to start with the simplest case—finitely repeated games with a unique stage game equilibrium. The classic case is a two-strategy, symmetric game that can be solved by dominance. Consider the example in [figure 9.2](#), which takes the form of a prisoner’s dilemma. In the stage game, players can either contribute to the public good (C) or free ride on the contributions of others (FR).

		Player 2	
		FR	C
Player 1	FR	(0, 0)	(6, −1)
	C	(−1, 6)	(3, 3)

Figure 9.2

A stage game with one equilibrium.

If they both free ride, the public good is not provided at all, whereas if they both contribute, they share the net benefits of the public good, 3 for each player. Finally, if one person contributes and the other doesn’t, the contributor pays the cost of provision while the other player reaps all the benefits without

incurring any of the costs. As was the case in the 20-token version, it is dominant for each player to try to free ride on the other and, correspondingly, there is just one Nash equilibrium, (FR, FR) , in this stage game.

Although we predict that rational players will not provide the public good if the stage game is played once, what if they play it twice? In a simple game like this, one way to analyze what should happen is to write down the payoffs at the end of the second round, conditional on any possible outcome of the first round. In essence, we are asking what can happen at the end of the four possible subgames—in this case, those that occur after every possible first-round outcome. These payoffs are recorded in [figure 9.3](#). To see how this works, consider what the payoffs at the end of round 2 could be if both players free rode in round 1. These payoffs are listed in the matrix at the end of the arrow from the (FR, FR) cell of the round 1 game at the center. If both players chose FR in round 1, they both got 0, and therefore, their payoffs at the end of round 2 will just be whatever they get out of that round. If instead, they both contributed in the first round, they each received 3, and the arrow to the southeast from the round 1 matrix indicates they will each receive 6 if they both cooperate again, for example. More generally, to get the payoffs listed in the cells of the four round 2 matrices, just add the payoffs from the appropriate first-round outcome to the stage game at the center.

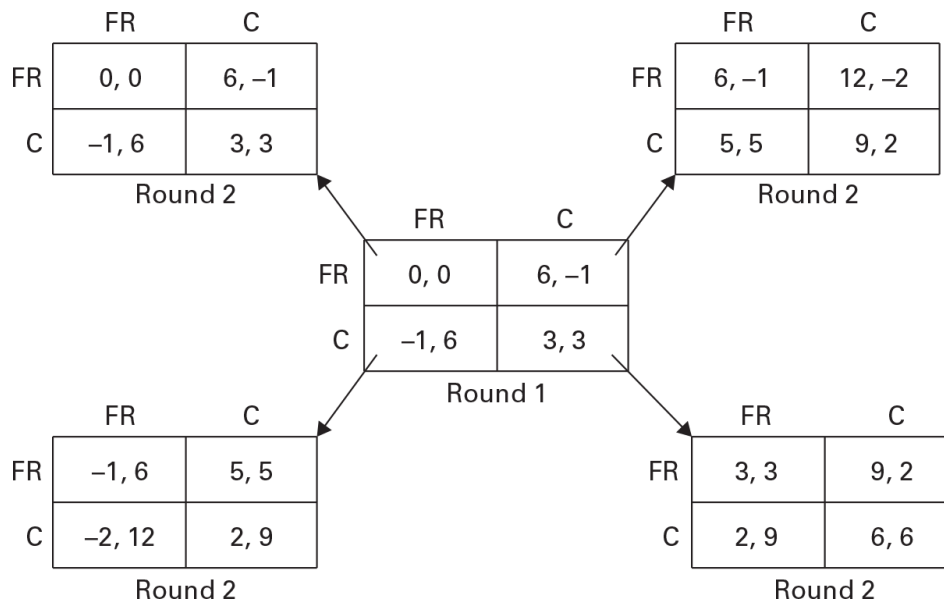


Figure 9.3

The provision game played twice.

Now look at each of the four potential round 2 games based on an assumption about what happened in round 1. Starting with the second-round game to the northwest that would result from both players free riding initially, because the game is identical to the stage game, our prediction is obvious—free riding dominates here too. What about if the first-round payoffs are not 0? Moving clockwise, if player 1 free rode on round 1 and player 2 contributed, the possible payoffs at the end of round 2 appear in the upper right side. Based on these payoff sums, what should the players do in round 2? As you can see, it is still the case that free riding dominates. OK, well, what about the strategy that, no doubt, has come to your mind—why not contribute in the first round and see what happens? As you can see from the bottom right, even after both players cooperate in the first round, they have an incentive to free ride on the second round. In fact, you can confirm that regardless of what the outcome of the first round is, both players in this game find it dominant to free ride in the second round. This is no accident, and the result can be extended to any finite number of rounds. Regardless of the how many rounds, t , have already occurred, the incremental utility gained on the $t + 1$ th round over the base accumulated during the previous t rounds will always be maximized by playing a best response. More practically, this means that rational players will always best respond in the last round, regardless of the accumulated pay from which they are working.

Back to our two-round setting, given we have established that players will always best respond and free ride in the second round, what should we expect them to do in the first round? Is there any chance, for example, that the players could support contributing in the first round as an equilibrium? Well, if they both contribute in round 1 and free ride in round 2, they get $3 + 0 = 3$ for the two rounds. As before, we can assess whether this constitutes an equilibrium by seeing if there is another strategy that would do better. The only other option is to free ride in both rounds. From [figure 9.3](#), we can see that if one of the two players deviated to this strategy, she would get $6 + 0 = 6$ instead, which is of course much better. Further, there is no strategy that does better than free riding in both rounds. Unfortunately, if both players use this strategy, the resulting equilibrium $((FR, FR), (FR, FR))$ leads both players to earn nothing.

We can go through one more example based on the original game used by Selten (1978) to refine this point. In the chain store paradox, there is a large

incumbent firm (I) established in L locations, and in each location, this incumbent faces a rival (R) that chooses sequentially whether to open up a store in that location. The game is played sequentially, in that the rival first decides whether to Enter the market or Stay Out in location 1, the incumbent decides whether to Fight this competitor or Accommodate her entry in location 1, then the rival firm decides whether to open at location 2, and so on.

The extensive and normal forms of the stage game are represented in figure 9.4. In the normal form on the right side, we see that there are actually two Nash equilibria, $(Enter, Accommodate)$ and $(Stay Out, Fight)$, but only one is subgame perfect, as can be seen in the extensive form on the left. When facing an entrant, the incumbent finds it in her interest to accommodate. Understanding this, the rival should enter. In other words, fighting entrants is not a credible threat for the incumbent, and therefore, the $(Stay Out, Fight)$ equilibrium that is based on this threat is not subgame perfect.

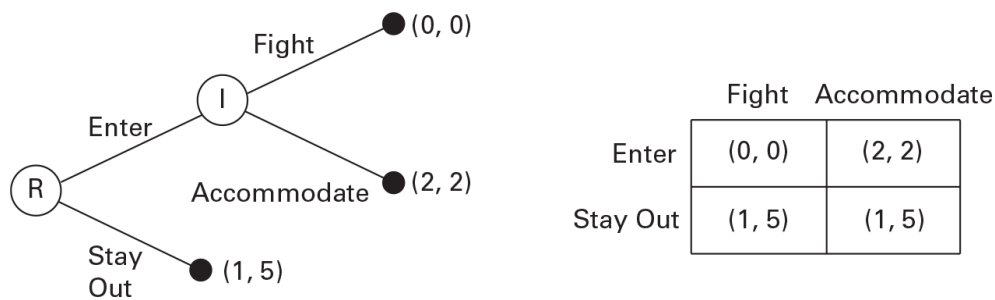


Figure 9.4
The chain store paradox.

What happens when this game is repeated for all L locations? The question is whether or not the incumbent will follow through on the threat to fight the rival in any location. Using backward induction, and considering the last location first, regardless of what the incumbent has done with the other $L - 1$ locations, she can earn nothing by fighting this one last battle (i.e., compare the stage payoff of fighting, 0, to that of accommodating, 2), so she won't do it. Well, if there is nothing to be gained from fighting in the L th location, there can be nothing to gain from fighting in the $L - 1$ th location either. Notice that, extending this logic according to what Selten called the "induction hypothesis," the game unravels in the sense that, no matter what she has done before or how many local rivals she has already faced, the incumbent's threat

to fight is always empty. It is a best response to accommodate regardless of where the incumbent finds herself in the supergame. This rationale can be summarized in theorem 1.

Theorem 1 (Selten's induction) *Suppose that the supergame involves $T < \infty$ repetitions of a stage game in which $(s_1^*, \dots, s_n^*)$ is the unique subgame perfect equilibrium. Then the only subgame perfect Nash equilibrium of the supergame is $(s_1^*, \dots, s_n^*)$ repeated in each of the T periods.*

To sum up this section, if the stage game has just one subgame perfect Nash equilibrium, that equilibrium played over and over is the only subgame perfect Nash equilibrium of the finitely repeated version of the game.

Finitely Repeated Games with Multiple Stage Equilibria

When there are multiple equilibria in the stage game, lots of different things can happen in equilibrium if the game is finitely repeated. This is what Benoit and Krishna (1985) discovered. Some of these outcomes will be straightforward and follow directly from Selten's induction argument, but others will be both more complicated and more interesting. Consider first the game of chicken in the normal form, as shown in [figure 9.5](#) and verify that there are three Nash equilibria, two in pure strategies and one mixture in which $p_A = p_B = 0.5$:

		Player 2	
		A	B
Player 1	A	(3, 3)	(1, 4)
	B	(4, 1)	(0, 0)

Figure 9.5

A stage game with multiple equilibria.

We know (from chapter 8) that for an equilibrium to be subgame perfect, all the players must be best responding in every subgame. In the two-round version of chicken, there are (as in the provision game in [figure 9.3](#)) four proper subgames other than the one comprising the entire game. Each of these subgames starts after the conclusion of the first round of play. The implication of subgame perfection in this case is that players must be playing their components of a Nash equilibrium in the last round, regardless of the history

of play to that point—they must end the supergame at an equilibrium. Why? Ending anywhere but at an equilibrium would be based on an incredible threat or promise. Once the first period is over and done for the players, the only rational behavior in the second period is to maximize one's second-period earnings by best responding to the other player's strategy, which means that they must play a stage Nash equilibrium in all the final subgames.

What about this remaining subgame, the one comprising the entire game? The agents need to be playing equilibrium strategies here too. To figure out what might be a best response in this subgame, let's assume a *fixed* outcome in the second round (which does not vary with first-period play) and backward induct, as before. In [figure 9.6](#) we write the matrices that result from assuming the payoffs at each of the three stage equilibria in the final round:

	A	B		A	B		A	B
A	(5, 5)	(3, 6)	A	(4, 7)	(2, 8)	A	(7, 4)	(5, 5)
B	(6, 3)	(2, 2)	B	(5, 5)	(1, 4)	B	(8, 2)	(4, 1)
Round 2 ends on:	(Mix, Mix)			(A, B)			(B, A)	

Figure 9.6

Accounting for the three possible outcomes of round 2.

If the players mix in round 2, we add the expected value, 2, to the possible payoffs from round 1, resulting in the payoff matrix on the left. If the game ends on the (A, B) equilibrium, we add 1 to the payoffs for player 1 and 4 to player 2's payoffs. This is seen in the middle matrix of [figure 9.6](#). Finally, if they end at (B, A) instead, we end up with the matrix on the right.

Examining each of these matrices closely, we see the same three stage equilibria arising again, regardless of what we assume about the end of the game (check the mixtures to confirm this). In other words, it seems that we can extend Selten's induction theorem to finitely repeated games with multiple stage equilibria. Although the subgame perfect equilibrium of the twice-played game of chicken is not unique as it was in the previous section, there are now more equilibria of the same flavor. That is, if agents simply play a stage equilibrium on each round, the result is subgame perfect. This means there are lots of equilibria now. For example, playing (A, B) twice is one, and

playing (A, B) then (B, A) is another. In fact, any combination of two stage equilibria will be subgame perfect.

The extension of Selten's induction theorem is not complete, however, because these will not be the only subgame perfect equilibria that can arise in this class of games. In fact, what is particularly interesting with these games is that, if designed properly, we can think of strategies as mechanisms used to improve the welfare (e.g., average payoff) of our players. For example, think about the worst that the players can do in equilibrium in the twice-repeated chicken game (see the illustration that follows). This happens when they decide to mix in both rounds because, in their expectation, they will spend a quarter of their time earning $(0, 0)$, and therefore their expected payoff is just 2. Can we help them do better? In particular, because the sum of the payoffs is highest at (A, A) , can we construct an equilibrium in which they spend more of their time there?

The problem is obvious: (A, A) is not a stage equilibrium, so how do we prevent players from deviating to B on the first round to try to get 4 instead of 3? Is there a way to punish them if they deviate, and if there is, will this punishment be credible (remember—we love subgame perfection)? Well, we know that the punishment will be credible if it results in a stage equilibrium because the punisher will have no incentive to deviate, and the punishee won't either. In fact, the punishee's best response will be to "take it."

Chicken is an interesting environment to implement this sort of mechanism because it has two very asymmetric pure-strategy equilibria. At (B, A) , player 1 does well but player 2 does pretty poorly, while at (A, B) , their fortunes are reversed. Mindful of this feature of the game, could a clever player propose the following: *Let's agree to play A in round 1 and mix in round 2. In addition, I commit to switching to B in round 2 if I see that you have played B in round 1. Yet further, if somehow, I screw up and play B instead of A in round 1, I also commit to playing A in round 2.* How would the other player react to this proposition?

Even though we will assume that a court cannot enforce it, perhaps the second player would like to see this agreement in writing. Filling in the blanks for the second player, the agreement would look something like the set of strategies hereunder (in [figure 9.7](#)). Summarizing, the two players agree to play the socially efficient A strategy in the first round and then punish anyone who deviates. If the deviation is one-sided, then the deviant is punished (and

the victim is rewarded), while if both players deviate, the two wink at each other in understanding and agree to mix on the second round. Will these strategies support an equilibrium?

Player 1	Player 2
Round 1: Play A	Round 1: Play A
Round 2: If (A, A), then Mix	Round 2: if (A, A), then Mix
If (A, B), then play B	if (A, B), then play A
If (B, A), then play A	if (B, A), then play B
If (B, B), then Mix	if (B, B), then Mix

Figure 9.7

Can clever players improve payoffs in the finitely repeated game of chicken?

Along the equilibrium path, the players both achieve a payoff of 3 in round 1 and then, in expectation, 2 in round 2. The sum is 5. Can someone do better by unilaterally deviating? Notice that there are two points at which someone can deviate, right away in round 1 and during round 2. Let’s start by looking at deviations during round 2. Given that the other player is mixing, a deviation from the mixture to a pure strategy would not improve your second-round payoff, by definition. In expectation, you are stuck with an expected payoff of 2 in the second round if the other player continues to mix. Hence, a second-round deviation will not make you better off. It seems, however, that if you are committed to cheating on the agreement, you would do it right away on round 1, switching to *B* from *A*. At first blush, this seems like a good idea because, given that the other player is committed to *A*, you will receive 4 instead of 3. But, what will now happen in round 2? You have triggered the other player’s commitment to playing *B*, so now you will actually do best by acquiescing and playing *A*. This means that although you get 4 from round 1, you will only get 1 from round 2 and the sum is still 5—you haven’t done any better.

We can also see this if we use a version of the trick of positing a final round-ending equilibrium and backward inducting. Based on what we have said that players will do in round 2 conditional on the round 1 outcomes, we can create the following matrix in figure 9.8, which is a bit different from the one on the left in figure 9.6. What is different is that the payoff to deviating

from A in round 1 is now 5 and not 6 because of the credible punishment being imposed. As a result, playing A in round 1 (as part of the more complicated mechanism) is a best response, given how we know the rest of the game will unfold.

		Player 2	
		A	B
Player 1	A	(5, 5)	(5, 5)
	B	(5, 5)	(2, 2)

Figure 9.8

Mechanism design in the game of chicken.

Since you cannot deviate along the equilibrium path and do better, these two strategies must result in a Nash equilibrium. But is it subgame perfect? To check this, we need to consider whether either player could do better deviating off the equilibrium path—that is, in one of the subgames that would be reached only if someone failed to adhere to (A, A) in the first round. From the full strategies listed out in the table shown here, we can see that the play always ends at a stage equilibrium. Thus, the players are always mutually best responding even off the equilibrium path, and this strategy profile is subgame perfect. What is particularly clever about this equilibrium is that, because the punishment is credible, the perpetrator best responds by accepting it. As a result, you would not need the courts to enforce the agreement or intervene at all.

To see the social value of these sorts of engineered equilibria, recall that if the players flip from (A, B) to (B, A) or flip in the reverse order, their average payoffs across the two rounds will be 2.5, while our mechanism does just as well even though we were only able to enforce (A, A) in a single round (the average under our strategy profile is also 2.5). In fact, if the interaction lasts any longer than two rounds, then the players do strictly better under similar mechanisms that incentivize (A, A) in the earlier rounds. As the number of rounds in the game increases, the average payoff under this type of strategy profile will converge on 3, indicating that mechanisms like this one can also raise the welfare of the players substantially.

To fix ideas, let's go through one more example of this kind of mechanism design in finitely repeated games. Consider the following asymmetric stage

game repeated twice, in which player 1 chooses from $S_1 = (A, B)$ and player 2 chooses from $S_2 = (X, Y, Z)$.

As always, we should begin by considering the Nash equilibria in the stage game. We see that there are two pure-strategy stage Nash equilibria: (A, Z) and (B, Y) . Unfortunately, (A, X) is not an equilibrium of the stage game (because player 2 would deviate to Z), so to achieve average payoffs close to the same efficiency, the players again will have to be clever.

		Player 2		
		X	Y	Z
Player 1	A	(4, 3)	(0, 0)	(1, 4)
	B	(0, 0)	(2, 1)	(0, 0)

Figure 9.9
Mechanism design in an asymmetric game.

As before, using any combination of stage equilibria during the two rounds will achieve a subgame perfect equilibrium of this supergame, so (A, Z) then (B, Y) , for example, works, but this results in average payoffs of just 1.5 and 2.5. Can the players increase payoffs by supporting play at (A, X) on round 1 using the equilibria of the stage game to reward or punish deviations? If you look closely, you will see that this sort of cooperation may be easier to establish here than in chicken because only one of the two players has an incentive to deviate from (A, X) . Player 1 is quite content to play A if player 2 is using X . It is just player 2 who would like to deviate to Z . Luckily for player 1, the stage equilibrium (B, Y) is particularly punishing for player 2.

Consider the following strategy profile. In the first round, player 1 plays A and player 2 plays X . In the second round, they play (A, Z) if (A, X) was played in round 1, and (B, Y) if anything other than (A, X) was the outcome of round 1. Since we prescribe a stage Nash equilibrium in the second round (either (A, Z) or (B, Y)), we know, by definition, that no one has an incentive to unilaterally deviate at that point in the game. What about in the first round? Along the proposed equilibrium path, player 1 earns $4 + 1 = 5$, and if she deviates to B in round 1, she will earn $0 + 2 = 2$ —not a smart play. For player 2, the payoff sum for the two rounds will be $3 + 4 = 7$ if he follows the plan, but only $4 + 1 = 5$ if he deviates to Z in the first round. Again, the two players

have constructed strategies from which they have no incentive to deviate in either round 1 or round 2, and we thus have an SPNE. Finally, in equilibrium, the players earn average payoffs of 2.5 and 3.5, which are both more efficient and more equitable than simply playing a combination of stage equilibria.

We can end our discussion of the theory of finitely repeated games by summarizing some design principles for the construction of efficiency-enhancing equilibria when there are multiple stage equilibria. A simple list like the one in [figure 9.10](#) should be helpful.

Strategies must:

- 1) Constitute a Nash equilibrium—deviations can't pay.
- 2) Be subgame perfect—no incredible promises or threats.

In practice, this means:

- 1) Cheaters must be punished and/or compliers must be rewarded.
 - 2) The interaction must end at a stage game equilibrium.
-

Figure 9.10

Design principles for finitely repeated games.

First and foremost, the proposed mechanisms have to constitute an equilibrium of the supergame. The first thing to check, as you have seen here, is whether players have an incentive to cheat on the mechanisms proposed. Further, payoff-enhancing deviations may occur in any round of the game, so your audit must be complete. In practical terms, this generally will mean that you have to anticipate the kinds of deviations that are likely to occur when trying to support a certain payoff vector and construct ways to reward and punish players so they don't stray from the equilibrium path. Subgame perfection is just as important. Not only do you need to devise rewards and punishments, these parts of the mechanism need to be credible. To a great extent, this means that you must tie the players' hands and make it in their interests not to pull back on these incentives. The way to do this is to limit your choices to things that constitute an equilibrium of the stage game, since the mechanism will never work unless it results in a stage equilibrium on the final round.

Behavior in Games—Playing (Un?)ravelled Strategies

The main theoretical prediction in many of the games that are repeated a finite number of times is that the game should unravel, in the sense that it will be hard to sustain efficiency-enhancing play unless that play coincides with an equilibrium of the stage game. In these cases, we should observe that stage equilibrium being played over and over again. Is this what we see in the experimental data? We will look first to the results of experiments in which the stage game has a single equilibrium, and then to other experiments in which the stage equilibrium is not unique.

In finitely repeated prisoner's dilemmas conducted in the lab with real incentives, people do regularly cooperate, and these rates of cooperation depend on the parameters of the game. Specifically, the rates of cooperation vary depending on (1) the temptation to defect and (2) the number of times the game is repeated. To analyze the impact of these features on cooperation in various experiments, Embrey, Fréchette, and Yuksel (2018) considered the generalized prisoner's dilemma game shown in [figure 9.11](#), in which the payoffs have been normalized such that the payoff from mutual cooperation is 1, the payoff from mutual defection is 0, g is the added gain that a player gets by defecting on the cooperating partner (instead of cooperating), and l is the loss that someone incurs if she cooperates and her partner defects on her. Notice that we have a prisoner's dilemma so long as g and l are positive and any prisoner's dilemma can be rewritten in this way. Reviewing past experiments, the researchers find that people are less likely to cooperate as g grows: that is, when there is greater temptation to defect on a cooperative partner.

	Cooperate	Defect
Cooperate	$(1, 1)$	$(-l, 1 + g)$
Defect	$(1 + g, -l)$	$(0, 0)$

Figure 9.11

The prisoner's dilemma in Embrey et al. (2018).

Lab participants are also significantly more cooperative the longer the time horizon is. Notice that there are two plausible explanations for this. First, as we saw in the last chapter, people may have trouble backward inducting and

this process is made more difficult when the game is longer so that people have to reason through more steps. Second, there is more to be gained from cooperating with someone you're going to be stuck with for a while—in the event that the players can achieve cooperation, then there are simply more periods to reap the rewards in the longer games. Embrey et al. attempted to tease these explanations apart.

To see this, imagine a simplified example in which there are only two possible strategies that the players in the finitely repeated game can choose between: “always defect” (AD), in which the player chooses Defect in every period; and “grim trigger” (GT), in which the player cooperates in the first period and as long their partner is cooperative, but if the other person ever defects, then the player switches to defecting forever. Notice that AD and GT each satisfies our definition of a strategy: both specify a full-contingency plan of action. If these were the only two possible strategies that players could choose between, then our game matrix looks like [figure 9.12](#) (where H is the number of periods in the game):

	GT	AD
GT	(H, H)	$(-l, 1 + g)$
AD	$(1 + g, -l)$	$(0, 0)$

Figure 9.12

Grim trigger in Embrey et al. (2018).

If GT meets GT, the players both cooperate in every period. They get 1 each period and play for H periods, for a total payoff of (H, H) . If AD meets AD, then each person gets 0 in each period, for a total payoff of $(0, 0)$. If GT meets AD, then the person playing GT cooperates in the first period while the other person defects (so the first person gets $-l$ in that period while the other gets $1 + g$). Then they both defect forever after that, and so each gets 0 for the remainder of the game. So the person playing AD gets $-l$ while the other gets $1 + g$.

In this very simplified game where these are the only two possible strategies, (GT, GT) and (AD, AD) are both Nash equilibria (since a strategy like only defecting in the final period is not an option). Furthermore, as H grows, the grim trigger strategy becomes more attractive, and the less assurance that the other person is cooperative you would need to give GT a

try. Specifically, let p be the probability that you think that the other person is playing GT (so with probability $1 - p$, they are playing AD). Then your expected payoff from each strategy is shown in [figure 9.13](#) (just focus on the left panel for now).

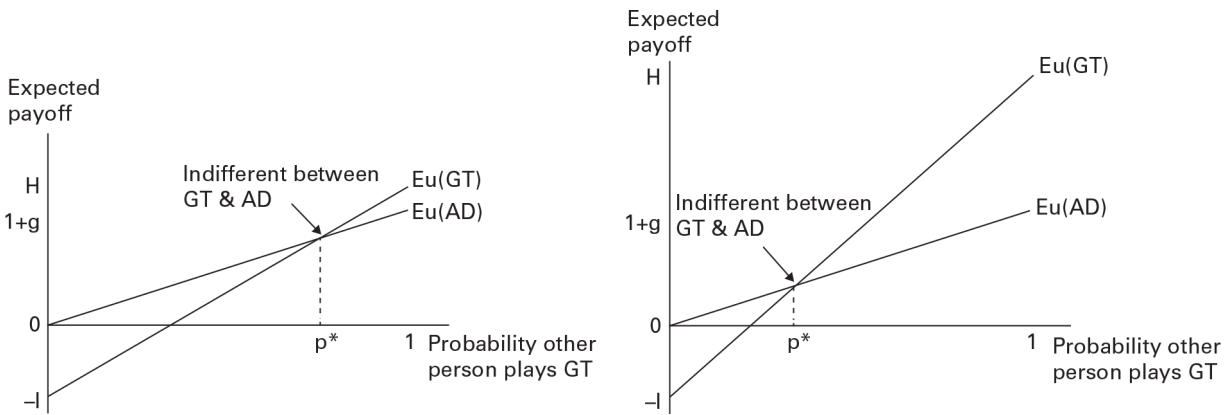


Figure 9.13

The expected payoff from AD and GT in games with a shorter time horizon (left) and a longer time horizon (right). The assurance that the other player will play GT (p^*) required for GT to be a best response is lower when the time horizon is longer.

Let p^* be the probability that the other player chooses GT that makes you indifferent between choosing GT and AD (i.e., where the expected payoff lines intersect, which is calculated as $\frac{l}{(H-1)+l-g}$). In the top left of [figure 9.13](#), notice that you need a reasonably high assurance that the other person is going to choose GT for you to be willing to try cooperating. For any probability lower than p^* , you prefer AD. Importantly, notice that this depends on H . What if H were higher, as in the graph at the right? Then the line $Eu(GT)$ swings up and the intersection point, p^* , decreases: the assurance you need that the other person is going to be cooperative decreases as the time horizon grows longer.

The authors computed this p^* for various past experiments and found that p^* has a significant effect on cooperation. This indicates that the time horizon affects cooperation not just because people are bad at backward inducting, but because they are responding to the value of establishing a long-run cooperative relationship. They then conduct an experiment in which they vary both the time horizon (either 4 or 8 periods) and g and l such that cooperation is either easy or difficult to achieve. They further make sure that p^* is the

same between the difficult-8 and easy-4 treatments. The idea is that if there is more cooperation in difficult-8 than in easy-4, then it must be because of backward induction issues, not because of the “long-run benefit of achieving cooperation” argument described previously. As predicted, they found that cooperation is higher in the easy treatments than in the difficult treatments, and that cooperation is higher in the 8-period game than the 4-period game. However, they did not find that there is more cooperation in difficult-8 than in easy-4: this provides evidence that the horizon primarily influences cooperation through its impact on the value of cooperation and p^* . Thus, it seems that the higher cooperation rates in games with longer time horizons are not the result of the more difficult backward induction problem but they are due to the increased value of cooperating in a longer-term relationships instead.

Let’s now return to the game from the start of this chapter. The stage game is known as the *voluntary contribution mechanism (VCM)*. Instead of two players and binary strategy choices as in the prisoner’s dilemma, in the VCM, four players decide how much (in integer amounts) to contribute to a public good, keeping the rest of their 20-token endowment for themselves. To determine the benefits provided by the public good, all the contributions are summed, multiplied by a productivity factor (in this case, 2), and then divided equally among the four players. The payoff for each player of this symmetric stage game can be expressed as follows:

$$u_i = (20 - c_i) + \frac{2 \sum c_j}{4} = (20 - c_i) + 0.5 \sum c_j$$

As in chapter 6, we can take the derivative of this function with respect to the player’s choice variable, c_i . Then setting the resulting slope equal to 0 (so we are at the top of the utility function) will tell us the level of individual contribution that maximizes the player’s payoff. Further, because the game is symmetric, if we find this payoff-maximizing contribution for one player, we have found it for them all. The derivative (recalling that $\sum c_j = c_1 + c_2 + c_3 + c_4$ and we are looking from the perspective of just one of these four players) is expressed as

$$\frac{du_i}{dc_i} = -1 + 0.5 = 0$$

This result should seem a bit strange for two reasons. First, what should we make of the fact that c falls out of the derivative here? Second, we can all

agree that $0.5 - 1$ does not equal 0, right? On further inspection, the fact that c falls out of the calculations should signal to you that there might be some dominance here. In other words, the payoff-maximizing contribution is likely to be everything or nothing. Can you already see which way this will go? If not, consider the second oddity—the fact that the first-order condition requires that $0.5 - 1 = 0$. The second number here, -1 , is just the opportunity cost of making a contribution. It is the token that you do not keep for yourself when you increase your contribution by 1 unit. The 0.5 is the marginal benefit of that contribution. To see this, imagine increasing your contribution by 1. That turns into 2 for the group, which is a very nice public good, but your share is only 0.5, which is not enough to outweigh the cost. Putting the two results together, we see that the only equilibrium of this stage game is $c_i^* = 0$, and it is based on dominance—regardless of how much your groupmates are contributing, you always do better by contributing less. Given there is just one equilibrium of this stage game, we can use Selten's induction theorem to predict the outcome of a supergame that is finitely repeated—nobody should ever contribute to the public good.

While we might not expect to see contributions fall to zero in the first round because novice players have to learn the strategic environment, the trend should be toward zero if the theory is valid. To get a sense of the related empirical literature, consider the experiment conducted by James Andreoni (1988), which used a similar setup with a 50-token endowment and five players per group. The graph in [figure 9.14](#) shows the evolution of the mean individual contribution over the course of the experiment.

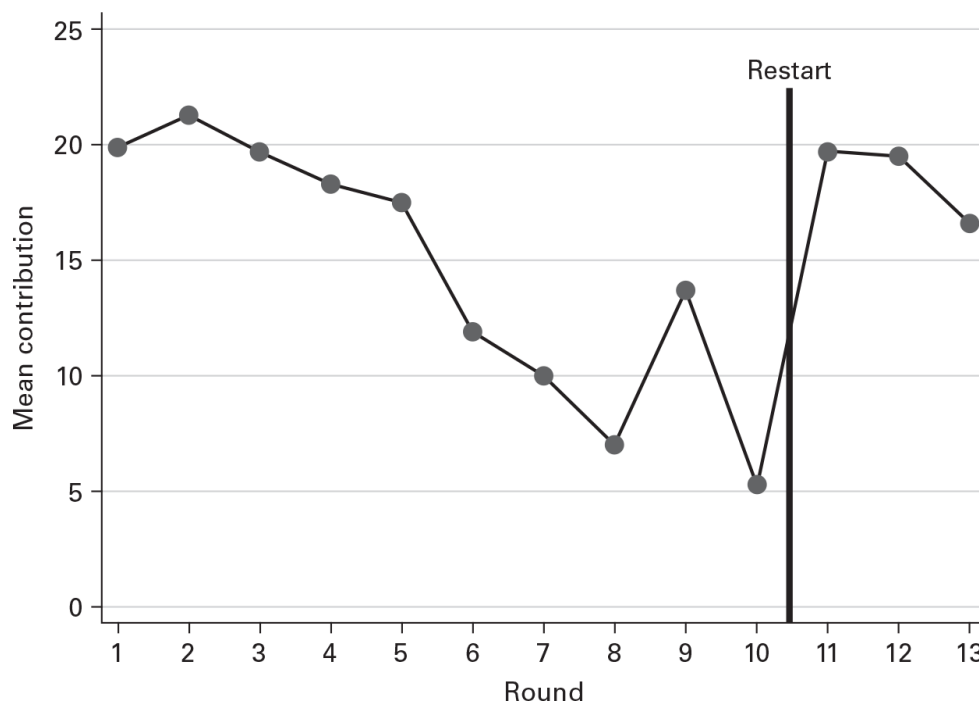


Figure 9.14

Behavior in the finitely repeated VCM (Andreoni, 1988).

Remember that the experiment ran for 10 repetitions, so for now, ignore the data to the right of the vertical line. What happens to contributions as the experiment proceeds? Interestingly, the average contribution on the first round is close to 25. This might indicate that the players had a lot to learn because it is consistent with them deciding to dump half their endowment in the public good, the other half in their private accounts, and seeing what happens. By the end of 10 rounds, the mean contribution has fallen to just 5 tokens, however, so it appears that players are learning the equilibrium as predicted by Selten's induction theorem—or are they?

Unbeknownst to the participants, at the end of the 10th round, the experimenters announced that there was still some more time before the session was scheduled to end, so they would restart the experiment and play more rounds. If learning explains what is happening in [figure 9.14](#), what should the data look like in rounds 11–13? Players should pick up where they left off in round 10 and continue the decline in average contributions to the equilibrium, $c_i^* = 0$, right? Well, as you can see, this is not at all what happened. In round 11, contributions actually jumped from 5 to almost 20 tokens before the decline set in again. Not only is this inconsistent with

learning the equilibrium, it is exactly contrary to it. Can you think of what might have been going on? Why else might contributions have declined?

An alternative hypothesis is that there is not that much to learn in the VCM, and players get it pretty quickly. In fact, the players are not only clever enough to get the gist of the game, they are smart enough to understand that the only way to get back at other players who don't behave *prosocially* by contributing (after all, payoffs are much higher if everyone contributes) is to withhold their own future contributions. This alternative can be thought of as the *bad apple theorem*—it takes only one bad, noncontributing apple to spoil the group. As the restart indicates, players are willing to wipe the slate clean at the beginning of a new experiment and give the bad apples the benefit of the doubt. It seems that these free-riding players don't learn, however, and, as a result, the game continues to unravel.

This alternative has since been bolstered by other VCM experiments in which players reveal *conditionally cooperative* strategies, suggesting they are perfectly happy to contribute so long as others do too (e.g., Fischbacher, Gächter, and Fehr, 2001) or group members are given another method to express their dissatisfaction with free riding, such as paying to reduce the earnings of others. In this second class of experiments, contributions tend to be high and stable, despite the seemingly incredible threats to explicitly fine/punish free riding (Fehr and Gächter, 2000). In other words, while Selten's induction theorem seems somewhat predictive, the tendency toward the stage equilibrium in the VCM is likely due to a number of factors, and learning might not be the strongest of them.

Let us now switch to experiments in which the stage game has multiple equilibria. What happens here? Do we see the simple repetition of stage equilibria, or do more complicated, efficiency-enhancing, strategies emerge? It turns out that there are just a few experimental studies of games similar to the game of chicken from before. Specifically, for mechanism design to be an interesting exercise, there needs to be an efficiency reason to try to support play away from the stage equilibria. Hence, in addition to focusing on supergame experiments in which pairings remain fixed, as in our discussion of the VCM, we consider only experiments where play at a strategy like *A* in [figure 9.5](#) can increase average payoffs in the supergame.

Shortly after theorists conceived of solution concepts for normal-form games, there was great interest in examining what people actually chose. In

two early studies by Alvin Scodel and his colleagues (Scodel et al., 1959, Minas et al., 1960), the experimenters studied chicken games, including the one presented in [figure 9.15](#). In these experiments, fixed pairs of participants played up to 50 repetitions of the stage game. While the authors rarely got into the details of individual strategies, there are some interesting session-level trends. First, despite the mixed-strategy equilibrium prediction of $p_{Black} = \frac{5}{6}$ in the game, only a quarter of the rounds end at (*Black*, *Black*). In fact, another quarter of the rounds end at (*Red*, *Red*), and the remaining half of the rounds end with one player choosing *Red* and the other choosing *Black*. Further, this distribution of outcomes did not change much when the payoffs were altered slightly. What might this indicate? The obvious conclusion is that players were confused and chose at random (resulting in a quarter of the rounds ending in each cell of the matrix). Perhaps the most important trend seen in these early experiments, however, is that the frequency of *Red* choices tended to grow over time. This could indicate the evolution of frustration or punishment for not settling on (*Black*, *Black*).

		Player 2	
		Black	Red
Player 1	Black	(6, 6)	(4, 7)
	Red	(7, 4)	(−1, −1)

Figure 9.15

Game H_1 from Minas et al. (1960).

More recently, and perhaps using more sophisticated methods, Feltovich (2011) found evidence that was even more consistent with the evolution of mechanisms in finitely repeated games with multiple stage equilibria. In his experiment, the way that participants were matched varied, which affected the results. When players were randomly re-paired after each of the 20 rounds, behavior quickly settled near the mixed-strategy equilibrium of his chicken game. By contrast, when the pairings were fixed, there was considerably more play at the socially efficient strategy pair. Feltovich concluded that the “subjects under fixed pairs learn to cooperate more.”

As in the VCM, conditional cooperation and reciprocity, more generally, could be the ultimate motivation on which players rely to develop trigger

strategies in games like chicken that coincide with subgame perfect equilibria. Let us now examine this idea further, showing how game theory has developed to account for these results. Our discussion will rely on the ideas developed more completely by Bowles (2004, chapter 7). To generalize our discussion to both types of games (VCM and chicken), consider the following two-strategy, symmetric normal form in [figure 9.16](#):

		Player 2	
		Cooperate	Defect
Player 1	Cooperate	(b, b)	(d, a)
	Defect	(a, d)	(c, c)

Figure 9.16

Generalized symmetric game from Bowles (2004).

If we assume $a > b > c > d$, we have a prisoner's dilemma, the two-strategy version of the VCM, but if we assume $a > b > d > c$, we have the game of chicken, instead.

Suppose that this game is played by conditional cooperators who pay an information cost, i , to look up the past interactions of the player they will interact with next. If the other player has built a strong cooperative reputation in her previous interactions, the conditional cooperator will take a chance and cooperate. If the other player's reputation is not "up to snuff," the conditional cooperator will play *Defect*. What does this do to the payoff matrix? As you can see in [figure 9.17](#), the first thing that happens is that when conditional cooperators choose to *Inspect* their partner's past, they pay the information cost, regardless of what they learn. You can see this in the top row of the matrix. Second, conditional cooperation effectively does away with the off-diagonal outcomes in the game. If players *Inspect* each other, there can be no surprises, and therefore, they are never taken advantage of.

		Player 2	
		Inspect	Defect
Player 1	Inspect	$(b - i, b - i)$	$(c - i, c)$
	Defect	$(c, c - i)$	(c, c)

Figure 9.17

Conditional cooperator game from Bowles (2004).

Analyzing the new game with conditionally cooperative players, we see there are now potentially two pure-strategy equilibria that exist in both VCM-like and chicken-like games. Further, one equilibrium is potentially more efficient than the other. As in the VCM, there will always be a less efficient equilibrium at *(Defect, Defect)*, but now if the information cost is not too large (i.e., $b - i > c$), then there is a second equilibrium in which players Inspect and condition their play on the information they receive. Considering the supergame, we know that inspecting on each round will be a subgame perfect Nash equilibrium of the finitely repeated interaction among conditionally cooperative players.

Not only can a model as simple as this one explain the cooperative behavior witnessed in the lab studies described in this chapter, it is the implied model on which lots of online trading now operates. Whether one is bidding in online auctions or taking a ride from a stranger, the ratings of past trading partners are increasingly checked and heeded.

Exercises

(1) Consider the two-player stage game in [figure 9.18](#), repeated 10 times:

		Player 2	
		X	Y
Player 1	X	$(3, 3)$	$(0, 4)$
	Y	$(4, 0)$	$(2, 2)$

Figure 9.18

Find all the subgame perfect Nash equilibria of this supergame. Also, briefly explain why each one is subgame perfect.

(2) Consider the two-player stage game in [figure 9.19](#), repeated 10 times:

		Player 2	
		L	R
Player 1	U	(5, 0)	(0, 3)
	D	(2, 7)	(6, 1)

Figure 9.19

Find all the subgame perfect Nash equilibria of this supergame. Briefly explain why each one is subgame perfect.

(3) Consider the two-player stage game in [figure 9.20](#), repeated five times:

		Player 2		
		1	2	3
Player 1	A	(2, -2)	(6, -6)	(14, -14)
	B	(8, -8)	(7, -7)	(10, -10)

Figure 9.20

(a) Find all the subgame perfect Nash equilibria of this supergame. Also briefly explain why each one is subgame perfect.

(4) Consider the following two-player stage game in [figure 9.21](#), repeated twice:

		Player 2		
		X	Y	Z
Player 1	X	(3, 3)	(0, 0)	(0, 0)
	Y	(0, 0)	(5, 5)	(9, 0)
	Z	(0, 0)	(0, 9)	(8, 8)

Figure 9.21

(a) Find all the pure-strategy Nash equilibria of the stage game.

- (b) Based on your answer to part (a), and restricting play to pure strategies, list the four most obvious subgame perfect Nash equilibria of the repeated game.
- (c) Can you propose a set of subgame perfect strategies that supports play at (Z, Z) in the first round?
- (5) Consider the following two-player stage game in [figure 9.22](#), repeated twice:

		Player 2		
		X	Y	Z
Player 1	X	(3, 3)	(0, 4)	(-2, 0)
	Y	(4, 0)	(1, 1)	(-2, 0)
	Z	(0, -2)	(0, -2)	(-1, -1)

Figure 9.22

- (a) Find all the pure-strategy Nash equilibria of the stage game.
- (b) Based on your answer to part (a), and restricting play to pure strategies, list the four most obvious subgame perfect Nash equilibria of the repeated game.
- (c) Can you propose a subgame perfect strategy profile that supports play at (X, X) in the first round?
- (6) Consider the following two-player, Hawk-Dove, stage game in [figure 9.23](#), repeated twice:

		Player 2	
		H	D
Player 1	H	(-1, -1)	(5, 0)
	D	(0, 5)	(2, 2)

Figure 9.23

- (a) Find all the Nash equilibria of the stage game.
- (b) Using only pure strategies, list two subgame perfect Nash equilibria of the two-round supergame that achieve average payoffs of 2.5 for each

player.

- (c) Can you propose a set of subgame perfect strategies that supports play at (D, D) in the first round?

(7) Imagine carpooling with a friend to Middlebury on a snowy morning in January. Coming down Seminary Street, your car gets stuck in a snowdrift. There are two shovels in the back of your truck, and if the two of you get out and dig, you can free the truck more easily than if only one of you digs. More specifically, let's say that the benefit to each of you of getting out of the snowdrift is 10, the cost of digging alone is 5, the cost of digging together is 2 (for each digger), and the cost resulting from nobody getting out to dig is -3 (amputated frostbitten toes).

- (a) Write down the normal form of the snowdrift game and identify all the equilibria.
- (b) Suppose that your trip to Middlebury lasts 3 days, and you find yourselves in the snowdrift each morning. Specify a subgame perfect Nash equilibrium of the supergame that maximizes your average payoff.
- (8) Consider the following prisoners' dilemma shown in [figure 9.24](#), played 5 times:

		Player 2	
		C	D
Player 1	C	(3, 3)	(1, 4)
	D	(4, 1)	(2, 2)

Figure 9.24

- (a) Suppose that conditional cooperators can read faces, and this allows them to discriminate among their opponents such that if they identify an unconditional cooperator, they play C too, when they identify another conditional cooperator, they also play C , but when they identify a defector, they refuse to play the game at all, resulting in both players earning 0. Add a third row and column to the normal form along with the payoff for conditional cooperation (CC), and construct a mechanism that allows the players to achieve a total payoff of 15 each.

- (b) Suppose that conditional cooperators can't read faces; rather, they have to pay a fee, $i > 0$, to get a sense of their opponent in this game. Add a third row and column to the normal form along with the payoff for conditional cooperation (CC). If the fee is not large, construct a mechanism that allows the players to achieve a total payoff that is close to 15 each.
- (9) Consider the following two-player stage game in [figure 9.25](#), repeated two times:

		Player 2	
		L	R
Player 1	U	(5, 10)	(9, 9)
	D	(5, 4)	(4, 3)

Figure 9.25

- (a) Find the pure-strategy stage equilibria.
- (b) Is there a subgame perfect Nash equilibrium in which (U, R) is played in the first period? If so, write the strategy profile that supports this outcome (i.e., what does each player do in the second period after each possible history). If not, explain why not.
- (10) Consider the following two-player stage game in [figure 9.26](#), repeated two times:

		Player 2		
		L	M	R
Player 1	U	(20, 20)	(0, 0)	(0, 0)
	C	(25, 10)	(3, 3)	(1, 1)
	D	(5, 5)	(15, 15)	(2, 2)

Figure 9.26

- (a) Find the pure-strategy stage equilibria.
- (b) Is there a subgame perfect Nash equilibrium in which (U, L) is played in the first period? If so, write the strategy profile that supports this

outcome (i.e., what does each player do in the second period after each possible history). If not, explain why not.

10

Infinitely Repeated Games

In chapter 9, we examined how rational agents play games that are repeated a finite number of times—typically just a few times in quick succession. We now extend our analysis by asking how the subgame perfect Nash equilibria change, if at all, as we increase the number of repetitions to the extreme: playing the game an infinite number of times.

An astonished British staff officer touring the trenches along the front lines dividing France and Belgium during World War I was dumbfounded that just across no-man’s-land, German soldiers walked about, upright, without any discernible fear despite being well within rifle range. To him, it did not seem like anyone knew there was a war on, and both sides seemed to have evolved to a state of “live and let live.”

Indeed, any reasonable (though simple) model of the interaction between opposing soldiers across the front lines would resemble a prisoner’s dilemma, like the game in [figure 10.1](#), in that it must be dominant to fire when the enemy is in range, even though both sides would jointly be better off if a truce, even an informal one, arose. As we know from the previous chapter, things should not change if the war drags on for more than one day because any proposed truce has got to be an incredible promise—one that you surely would not bet your life on. So how can we understand the “live and let live” understanding that arose over and over, up and down the lines, during the war?

		Soldier 2	
		Truce	Fire
Soldier 1	Truce	32, 32	12, 50
	Fire	50, 12	25, 25

Figure 10.1

Rifle fire across no-man's-land.

Along with retelling this story in his 1984 book, Robert Axelrod reported on a unique type of tournament that he had just conducted. Instead of an experiment with students playing the prisoner's dilemma over and over in the experimental lab, like the ones we have described in detail so far in this book, Axelrod solicited full forward- and backward-looking strategies from prominent game theorists in the fields of economics, psychology, mathematics, political science, and sociology, and had these strategies compete in a round-robin computer tournament lasting a large number of rounds (200, to be precise). What was interesting is that a simple strategy emerged as the winner, though it is not the one that you (or standard theory) might predict. Instead of defecting on every round, *tit-for-tat* (submitted by the mathematical psychologist Anatol Rapoport) always begins by cooperating and then simply does whatever the other agent did on the last round. In the tournament, *tit-for-tat* did best, beating "always defect," as well as a number of more sophisticated strategies, by a noticeable margin.

The reason why *tit-for-tat* did well in the tournament is because it was nice, forgiving, and does well against itself, and yet it is not easily exploited. *Tit-for-tat* is nice, in the sense that on the first round it cooperates, while many of the other entrants spoiled the relationship right away with an opening defection. *Tit-for-tat* is also forgiving, in the sense that if the interaction has somehow reached a truce and the other agent screws up by defecting, it just has to show a good-faith effort (by returning to cooperate) to get the cooperative relationship back on track. Further, without any errors, *tit-for-tat* starts off cooperating with itself and will remain there forever. As a result, it is impossible for another strategy to do better against itself. Finally, you fool *tit-for-tat* only once. The implication here is that any so-called mean strategy can never accumulate a big lead over *tit-for-tat* by defecting on the first round.

It is not hard to believe that the soldiers mentioned in the scenario beginning this chapter understood some of the rationale of *tit-for-tat*. They

must have understood the long-run benefits of cooperation, while also understanding that forgiveness for an erroneous defection (i.e., a shot across enemy lines) would be hard to come by. Is there a game-theoretic logic that can support both this observation and the tournament results?

Discounted Utility

There is an important difference in how game theory treats games played an infinite versus a finite number of times—namely, the way that benefits received in the future are accounted for. In finitely repeated games, as we saw in the previous chapter, a payoff gained in round 2 is treated identically to the same payoff gained in round 1, or in any other period. The payoffs accumulated at their face values in these games. Consider, instead, how much you should value the same payoff accumulating right now, in round 1, versus in round 101. Specifically, should you still treat these payoffs identically?

There are two rationales that might convince you to discount payoffs accruing in the distant future. (Please see the appendix for a further discussion of time discounting and intertemporal choice.) First, from the finance perspective, suppose that you are comparing two payments: receiving a dollar right now or receiving the same dollar one year (i.e., one period) from now. It is hard to imagine being indifferent between these two payments because if you took the dollar right now and put it in a savings account paying an interest rate, r , you would have $1 + r$ dollars in the next period. Surely $1 + r$ is better than 1, right? However, by the same logic, if you put $\frac{1}{1+r}$ dollars in the bank right now, that amount would grow to be $(1 + r)\frac{1}{(1+r)} = 1$ dollar next period. In other words, shouldn't you be indifferent between having a smaller amount, $\frac{1}{1+r}$ dollars, right now and 1 dollar, next period?

Continuing the logic one more step, shouldn't you also be indifferent between getting $\frac{1}{(1+r)^2}$ dollar right now and 1 dollar two periods from now? After all, if you put $\frac{1}{(1+r)^2}$ in the bank today, after one period, you would have $(1 + r)\frac{1}{(1+r)^2} = \frac{1}{1+r}$, and then after two periods, you would have $(1 + r)\frac{1}{1+r} = 1$. That is, the miracle of compound interest suggests that the further in the future a payoff is to be received, the smaller is its *present value*.

Definition 1 (Present value [PV]) *The present value of a payoff x , received t periods in the future, is $\frac{x}{(1+r)^t}$, when the relevant interest rate is r .*

Definition 2 (Discount factor) Refer to r as the discount rate and δ as the discount factor. We set $\delta = \frac{1}{1+r}$ so that the present value of a payoff, x , received t periods in the future is just $\delta^t x$.

Alternatively, if the recent international financial crisis has made you leery of finance, we can also motivate discounting psychologically with the notions of *delay of gratification* and patience. How hard do you find it to resist choosing a sooner, but smaller, reward over something that is larger, but will be received later? As an example, suppose that we put you in a room by yourself with a marshmallow and tell you that if, when we come back in 10 minutes, the marshmallow is still there, we will give you a second one. Oh, and imagine that (1) you like sugar and (2) you are four years old. Could you resist the temptation to gobble up the one treat when you know you could have a second one if you can hold out for just 10 minutes? Put differently, can you delay your gratification?

Notice that one way to think about this is identical to the finance logic. If you are impatient and have a small δ (or a high r), you will discount that second marshmallow a lot and give in. If instead, you can delay the gratification of tasting sugar, this is synonymous with having a larger δ and being able to hold out. Amazingly, when Walter Mischel conducted exactly this experiment in the 1970s (by the way, when you have a minute, look up some of the video footage on the internet—it is very interesting), he found considerable variation in the ability of children to delay gratification. Some kids ate the marshmallow before he even left the room, while others held out for a while and then crumbled. However, there was another group that lasted the 10 minutes. The amazing thing about this study, however, was that when Mischel followed up with the children and their families, he could see that the group of kids who managed to hold out got more than two marshmallows. These children, as they grew older, also tended to do better more generally. As teenagers, they achieved higher SAT scores and were perceived as more mature. Later yet, as adults, these same children were less likely to have drug problems, get divorced, or become overweight.

Let's end this section with an example to reinforce what we have discussed. Suppose that Peter loves marshmallows and faces the following decision. He can either have what remains of the current bag of marshmallows, 2 of them, right now, or he can have a stream of treats, one marshmallow each period, for 3 periods. Suppose that his $\delta = 0.5$, and he gets

his first marshmallow in the stream right now. What does he choose? Doing the discounting, we see that Peter compares 2 to the sum of the discounted utility he will receive from the stream of payoffs:

$$1 + \delta(1) + \delta^2(1) = 1 + 0.5 + 0.25 = 1.75$$

and he grabs the current bag because $2 > 1.75$. What if, however, this stream of treats does not end after 3 periods but rather goes on forever? Would he change his mind? We need to consider the PV of this payoff stream—that is, how much is it worth to Peter *right now* to get one marshmallow per period forever? Now the discounted stream of marshmallow utility becomes

$$PV = 1 + \delta(1) + \delta^2(1) + \delta^3(1) + \delta^4(1) + \dots$$

But how can we assess this sum, since it goes on forever? The thing to notice is that although the sum is of infinite length, each additional component is smaller than the previous one, so in the limit, the additional components are converging on zero. Let's do a little algebra and see what happens. What if we first subtract 1 from each side and then divide by δ . We get

$$\frac{PV - 1}{\delta} = 1 + \delta(1) + \delta^2(1) + \delta^3(1) + \dots$$

Because payoffs received in the distant future are effectively zero, dividing them by δ will have no effect on the sum. With this in mind, look at the right side of this equation. Isn't this exactly what we started with? If we referred to that sum as PV , then substituting, we get

$$\frac{PV - 1}{\delta} = PV$$

Using just a little more algebra, we see that the equation reduces to $PV = \frac{1}{1-\delta}$, and because Peter's $\delta = 0.5$, he must value the infinite stream of marshmallows at 2, so he will be indifferent between having 2 now and 1 marshmallow per period, in perpetuity.

Similarly, we can calculate the PV of getting a marshmallows in each period as $PV = \frac{a}{1-\delta}$. Finally, what if Peter doesn't get any marshmallows in the current period, but gets a marshmallows every period forever, starting next period? We just found that the PV of a marshmallows forever is $\frac{a}{1-\delta}$. But now he must wait a period to get this payoff stream, and so we must discount it by δ , giving us $\frac{a\delta}{1-\delta}$.

Now that we know how to calculate the present value of various streams of future payoffs, we can turn toward solving for equilibrium behavior in

infinitely repeated games.

Equilibria in Infinitely Repeated Games

Armed with the notions of present value and discounted utility, we can see how equilibria may change when games are repeated infinitely. As you may anticipate, if the sum grows forever instead of accumulating payoffs in just a few periods, players might reconsider cooperating with each other. In fact, given the dominance that characterizes the prisoner's dilemma stage game, there can be no setting in which it can be harder to support cooperation. For just this reason, Axelrod chose it as the setting for his tournament, and it now becomes the basis for our analysis.

Consider the same stage prisoner's dilemma game used in Axelrod's contest, reproduced in [figure 10.2](#). If this game is repeated infinitely, what would you expect, as player 1, to earn from cooperating? Well, this obviously depends on what player 2 does. If she also cooperates, the two of you will get 3 forever, and the discounted present utility of that pairing is, by the logic of the last section, $\frac{3}{1-\delta}$. But if she defects forever and you doggedly stick with cooperation, it is easy to see that you get 0 (formally, $\frac{0}{1-\delta}$).

		Player 2	
		C	D
Player 1	C	(3, 3)	(0, 5)
	D	(5, 0)	(1, 1)

Figure 10.2
Axelrod's prisoner's dilemma.

Likewise, if you choose to defect and player 2 cooperates, you can expect a discounted utility of $\frac{5}{1-\delta}$, while if she defects too, you will both get $\frac{1}{1-\delta}$. It still seems that defection dominates because regardless of δ , $\frac{5}{1-\delta} > \frac{3}{1-\delta}$ and $\frac{1}{1-\delta} > 0$. In this sense, we have a similar result to the case in which this game is repeated a finite number of times. Doggedly sticking with cooperation does worse than defecting.

Notice, however, that in finitely repeated games, we could sustain cooperation only with more sophisticated strategies. Let's try that here. What if instead of doggedly cooperating, your strategy was more nuanced, in that

you said to player 2, “I’m going to start off by cooperating and I will continue to cooperate until you defect on me. If you defect on me, then I will defect ever after.” We can call this a *trigger strategy*—the other player’s defection “triggers” you to defect in every subsequent period. Are these kinds of promises and threats now credible in ways they were not in the finitely repeated game?

Definition 3 (Trigger strategy) *In the repeated prisoner’s dilemma, a trigger strategy cooperates on every round up to T , the round in which the other player defects at least $\tau > 0$ times. On rounds $T + 1$ through ∞ , the trigger strategy defects.*

A simple example of a trigger strategy occurs when $\tau = 1$, in which case a single defection triggers a punishment stage that lasts forever. This extreme trigger strategy is called *grim trigger (GT)*, as mentioned in the previous chapter. To see the intuition for how this strategy could be played in equilibrium, we need to compare payoffs—but which ones? What if we start, similarly to the previous scenario, with two players both using the same strategy (in this case, GT). Notice that if both players use the grim trigger strategy, they both start off by cooperating and without any trembles, they continue to cooperate forever. As before, the payoff to this is just $\frac{3}{1-\delta}$. What if one of the players deviates to *always defect (AD)*. Does this increase her payoff? If she deviates right away, in the first round, she will get

$$PV(AD) = 5 + \delta(1) + \delta^2(1) + \delta^3(1) + \delta^4(1) + \dots$$

which is also not difficult to reduce because we know that the discounted utility from defecting always is $\frac{1}{1-\delta}$. This flow of utility is delayed one period, and you account for that by discounting it, so it is valued at $\frac{\delta}{1-\delta}$. Hence, $PV(AD) = 5 + \frac{\delta}{1-\delta}$. Comparing payoffs, it will not pay to deviate to “always defect,” so long as

$$PV(GT) = \frac{3}{1-\delta} > 5 + \frac{\delta}{1-\delta} = PV(AD)$$

Simplifying this inequality, we find that it does not pay to deviate (to AD) from a situation in which both players are using the grim trigger strategy if $\delta > \frac{1}{2}$. The implications of this result are huge. Before, there seemed to be no way to escape the prisoner's dilemma; players were doomed to defect on each other over and over again. However, now we see that when the game is repeated infinitely and players are sufficiently patient, the threat of retaliating against a defection becomes credible, and thus cooperation can be sustained.

To get a better sense of this result, consider the graphs in [figure 10.1](#), in which we plot the discounted utility involved in this choice for two levels of δ : one above the threshold and one below it. In each graph, the dark line traces the present value of receiving the mutual cooperation payoff of 3 some number of rounds in the future (i.e., 3 in round zero, 3δ in round 1, $3\delta^2$ in round 2, etc.). We can compare this to the dot at 5 (associated with the immediate payoff of defecting) plus the dotted line showing the present value of receiving the mutual defection payoff of 1 some number of rounds in the future. The continuation payoff over any number of rounds can thus be seen from the graph as the accumulation of the area under the curve up to that number of rounds. Starting with the graph on the left, when $\delta = 0.75$, we can see that the decision is between sticking with the trigger strategy and accumulating the area under the solid dark line, which represents the discounted utility from each of 15 rounds, and taking the 5 util payoff right now (the single dot) and the integral under the dotted line starting at 0.75, the discounted utility of 1 received one period from now. Essentially, you are giving up the gap in the two integrals, which in this case is sizable because you are relatively patient and future payoffs do not decay too rapidly.

The graph on the right of [figure 10.3](#) shows the case where players are relatively impatient ($\delta = 0.25$). Here, the value of payoffs decay quickly, and by round 5 they seem indistinguishable from zero (and each other). In contrast to the graph on the left, here the difference in the integrals under these two plots is much smaller. In fact, it is too small to outweigh the immediate payoff of 5, looming large in the mind of potential defectors.

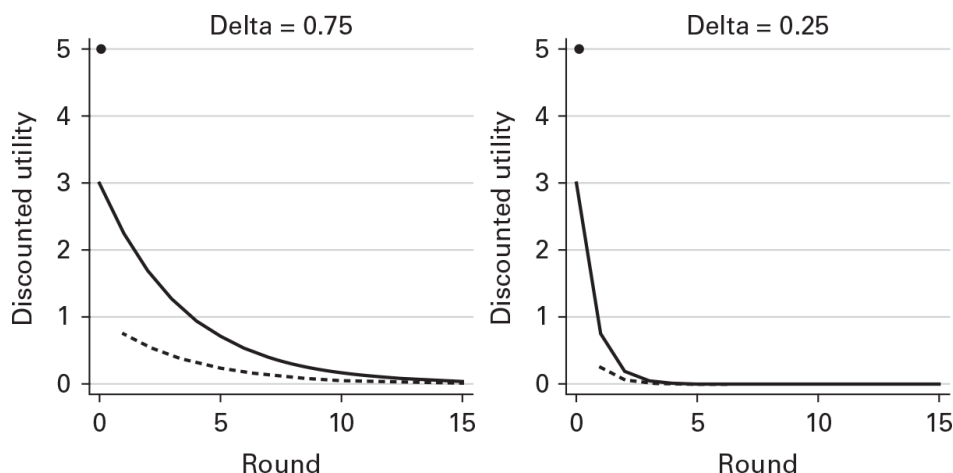


Figure 10.3

Discounted utility in the infinitely repeated prisoner's dilemma (for high and low δ). The dark line shows the PV of receiving the mutual cooperation payoff of 3 in a future round, while the dotted line shows the PV of receiving the mutual defection payoff of 1.

At this point, you might be thinking, “Well, yes, I get it, but doesn't this just mean that it is not obvious that you should deviate immediately? What if you wait a bit before you trigger the other player?” Let's see how this affects our results. If you wait one round before defecting, for example, you will get

$$PV(\text{Defect on round 2}) = 3 + \delta(5) + \delta^2(1) + \delta^3(1) + \delta^4(1) + \dots$$

which reduces to $3 + 5\delta + \frac{\delta^2}{1-\delta}$ because you've delayed the flow of 1s for another period. If we set this equal to the discounted present utility of GT played against itself, $\frac{3}{1-\delta}$, and solve for the new critical value, we find that the result is not so new because we find $\delta = \frac{1}{2}$ again. Further, if we delay defecting until the third round, the discounted utility is $3 + 3\delta + 5\delta^2 + \frac{\delta^3}{1-\delta}$, which, when set equal to the payoff from playing GT against itself, also yields $\delta = \frac{1}{2}$. In other words, it does not matter in which round you deviate and trigger the other player, the critical level of patience remains the same. This seems like magic, but upon reflection, the key is to understand that because the game is played infinitely, it doesn't matter when a player decides to defect, the critical trade-off is always the same, regardless of whether the players are interacting for the first time or years into their relationship. Specifically, at any moment in the game, you are essentially asking whether it makes sense to delay the net benefit of defecting one more period or accrue another 3. You will be indifferent between the two if

$$\left(5 + \frac{\delta}{1-\delta}\right) - \delta \left(5 + \frac{\delta}{1-\delta}\right) = 3$$

where the left side is the payoff difference, at any point in the game, between defecting now or delaying the defection one round, and the right side is the opportunity cost of doing so (3). Now notice that if you factor out $(1 - \delta)$ and divide both sides by it, you get the same condition with which we started the discussion (i.e., the condition that resulted in the critical level of patience, $\delta = \frac{1}{2}$).

At the same time, notice that the critical value we found, $\delta = \frac{1}{2}$, does depend on the payoffs of the game itself. For example, if the temptation to defect on a cooperator was larger (say 10), then the critical δ would adjust (to $\frac{7}{9}$), and as you would expect, it will be harder to sustain cooperation in this new environment—players will have to be very patient.

Definition 4 (Tit-for-tat) *In the repeated prisoner's dilemma, the tit-for-tat strategy cooperates on the initial round and then cooperates on every subsequent round with a one-round history of play (C,C) or (D,C) and defects on every subsequent round in which the immediate history of play is (C,D) or (D,D).*

Considering how well tit-for-tat did in Axelrod's tournament, let's now examine whether the theory of infinitely repeated games can justify its performance. As before, let's first calculate the payoff to playing tit-for-tat, defined formally in definition 4, against itself. We know two tit-for-tatters will start by cooperating and, without any trembles, continue to cooperate on every subsequent round. As with the grim trigger strategy, the discounted present utility of this is $\frac{3}{1-\delta}$. To see whether tit-for-tat is a best response to tit-for-tat, we need to consider the players' alternatives. That is, if they were to deviate from tit-for-tat, what would they deviate to? One possibility is to switch to always defect. If one player switches to always defect, then the realization of the game will be identical to the trigger analysis given previously.

Suppose that instead, the clever deviant decides to play a strategy of Defect in just the first round and then cooperate thereafter. She will receive $5 + \delta(0) + \delta^2(\frac{3}{1-\delta})$ which she compares against the payoff of not deviating, which can be written as $3 + \delta(3) + \delta^2(\frac{3}{1-\delta})$. Because she notices that the difference boils down to deviating if $5 > 3 + 3\delta$, she can calculate a different

critical value, $\delta \geq \frac{2}{3}$. In this case, we see that, because tit-for-tat players are more forgiving than “grim trigger” players, they need to be more patient to support cooperation.

Subgame Perfection and the One-Stage Deviation Principle

To this point, we have only demonstrated that strategies like grim trigger and tit-for-tat can be preferable to always defect when played against each other by sufficiently patient players in the infinitely repeated prisoner’s dilemma. However, in order for both players choosing grim trigger to be an equilibrium, the players must be better off playing these strategies than *any* other strategy profile, given that the other player is choosing grim trigger as well. In an infinitely repeated game, there are an infinite combination of alternative strategies that the player could choose. How can we compare the payoffs under grim trigger to all of them? In addition, as stressed in chapters 8 and 9, we need to also check to see if the strategies are based on incredible promises or threats—are they subgame perfect? Recall that in the case of repeated games, subgame perfection means that strategies must prescribe best responses after every possible history of play, regardless of how likely those histories are. For example, it must be in a player’s interest to come through on the threat to punish, even if the mere threat is enough to support an equilibrium (i.e., it is very unlikely that the player will be called on to punish anyone).

To better understand this concept, let’s continue with tit-for-tat and consider what would happen if one player accidentally clicked the “Defect” button on a round instead of “Cooperate” and then, noticing his error, immediately returned to tit-for-tat. While this seems unlikely, for a strategy to be subgame perfect, it has to be robust to this sort of perturbation. How would play proceed from that mistake onward? Well, the other player, not knowing it was a mistake, would immediately retaliate with Defect—resulting in (C,D)—and then, because the first player returned to tit-for-tat after his mistake, he would defect on the next round—resulting in (D,C)—and play would alternate between (C,D) and (D,C) forever after. Then the discounted present utility for the other player starting in round 2 would be

$$PV(\text{Play tit-for-tat after a mistake}) = 5 + \delta(0) + \delta^2(5) + \delta^3(0) + \delta^4(5) + \dots$$

Given the pattern we have seen so far, you can see that this will first reduce to $5 + 5(\frac{\delta^2}{1-\delta^2})$ and then, after a bit more algebra, $\frac{5}{1-\delta^2}$.

What if the second player deviated from tit-for-tat on the second round, and therefore did not retaliate after the mistake? Put differently, what if after the history of play (D,C), the second player deviated from the prescriptions of tit-for-tat just once? In this case, she cooperates on round 2, and play evolves as (D,C), (C,C), (C,C), ... Here, the second player gets 3s starting in round 2, which collapses, again, to $\frac{3}{1-\delta}$. Should the second player stick with tit-for-tat and retaliate, or would it be rational to deviate, even if just for one round? To figure this out, the second player reckons that she should stick with tit-for-tat so long as $\frac{5}{1-\delta^2} > \frac{3}{1-\delta}$. This condition reduces to $3\delta^2 - 5\delta + 2 > 0$, which is only true when $\delta < \frac{2}{3}$, the opposite of what we found at the end of the previous section. This means that the second player has an incentive to deviate away from tit-for-tat after a mistake is made (i.e., not punish the offender) to “save” the interaction. Thus, two players using tit-for-tat is *not* an SPNE.

The rationale that we have just described is a demonstration of Robert Aumann and Lloyd Shapley’s (1994) *one-stage deviation principle* for testing whether equilibrium strategies are subgame perfect (originally conceived in a 1976 mimeo). The more general idea is based on the notion of the *continuation value* associated with any strategy (say s_i), which is just the payoff at the terminal node that will be reached using s_i against the fixed strategies of the other players, s_{-i} , continuing from any specific node in the extensive form. For infinitely repeated games, the continuation values of specific strategies are just the discounted present utilities calculated previously.

Recall that what is important for subgame perfection is that there can be no payoff-enhancing deviations at any subgame in the extensive form. To assess whether deviations are profitable in repeated games, we can look at these continuation values. While there are an infinite variety of possible deviations in an infinite game, we can rely on a very handy result called the *one-stage deviation principle* (as described in definition 5), which basically says that if it is not profitable to deviate from s_i exactly once, but otherwise adhere to s_i in every other period, then it also is not profitable to deviate from s_i in more than one period. This implies that to assess whether a strategy is subgame perfect, we only have to check to see whether these one-time departures are rational.

Definition 5 (One-stage deviation principle) *Fixing other players' strategies, s_{-i} , s_i^* constitutes a subgame perfect Nash equilibrium of an extensive-form game if and only if, for every player and every subgame, no one-stage deviation from s_i^* will increase the player's subgame payoff.*

As another example, let's see if grim trigger is subgame perfect. In this case, there are two types of deviations that we need to check: the type that initiates punishment and the type that could occur after punishment is already occurring. We begin with the latter case in which the immediate history of play is (C,D), (D,C), or (D,D). In any of these subgames, would it make sense to deviate to Cooperate, just once, and then go back to adhering to grim trigger (which, at that point, would mean switching to Defect forever)? After doing so, a player's immediate payoff would fall to 0 from 1, so that doesn't seem rational. Further, after switching back to Defect, the continuation value of the game is just the same as it was before $\frac{1}{1-\delta}$. Hence, a one-stage deviation of this sort is not rational.

Now let's consider the implications of deviating on a cooperating grim trigger player just once. In the round of the deviation, the player would increase her payoff from 3 to 5, a gain of 2, but the continuation value would fall from $\frac{3}{1-\delta}$ to $\frac{1}{1-\delta}$, a loss of $\frac{2}{1-\delta}$. Considering her potential gains and losses, and accounting for the fact that the losses start in the next round (so they are discounted one period), the player checks to see when a deviation will not be rational—that is, when $2 < \delta(\frac{2}{1-\delta})$, which is true if $\delta > \frac{1}{2}$, the condition necessary for grim trigger to be an equilibrium. In other words, no one-stage deviation from grim trigger will be rational, so it is subgame perfect.

Let's recap what is happening here and why both players playing GT can be subgame perfect, but tit-for-tat is not. When GT players deviate, they immediately move to a stage equilibrium, and as we saw in chapter 9, this must be credible. By contrast, tit-for-tat asks players to do something they would regret if they were sufficiently patient. For the payoffs given in Axelrod's prisoner's dilemma, if players have δ s greater than $\frac{2}{3}$, they both should be patient enough to use tit-for-tat as an equilibrium strategy and reckon that resetting cooperation after a screw-up is better than tumbling off-diagonally in the normal form, as tit-for-tat requires.

The Folk Theorem

While we have discussed a few strategies that constitute equilibria in the infinitely repeated prisoner's dilemma (e.g., grim trigger and always defect), it turns out that there are many more, conditional on players being patient enough. In fact, almost anything goes, a result that was conventional wisdom among game theorists before it was ever published (by James Friedman in 1971). As a consequence, this logic has become known as the *folk theorem*. It is described in definition 6.

Definition 6 (Folk theorem) *Consider the stage game with a Nash equilibrium strategy profile $(s_1^*, \dots, s_n^*)$ that results in average (per-period) payoffs $(u_1^*, \dots, u_n^*)$. Any average payoff vector $(v_1, \dots, v_n)$, such that $v_i > u_i^*$ for all i , can be supported by a subgame perfect Nash equilibrium of the infinitely repeated game if δ is close enough to 1.*

Figure 10.4 summarizes this concept for the prisoner's dilemma we have been working with. The vertices of the larger diamond are created by plotting the average per-period payoff vectors that would result when each of the four outcomes of the stage game is played on each round. For example, if the players always played (D,C), the average payoff vector would just be (5, 0), or if both players always defected, the average payoffs would be (1, 1). The edges and interior of this diamond constitute all the possible average payoff vectors that could ever arise in the infinitely repeated game, regardless of whether the strategies form an equilibrium. The darker, smaller shape indicates all the payoff vectors that can be achieved as the result of subgame perfect strategies.

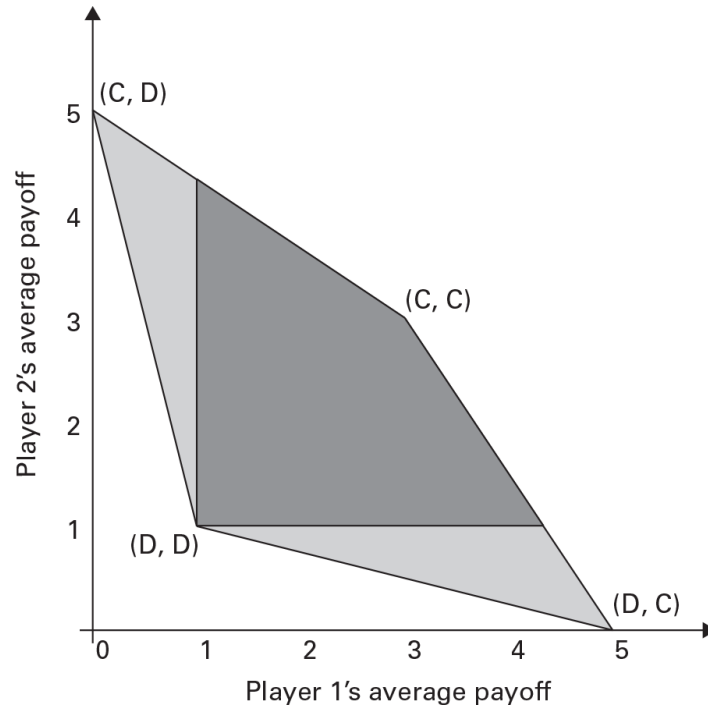


Figure 10.4

The feasible (average) payoffs in the infinitely repeated prisoner's dilemma.

The single stage equilibrium at (D,D) constitutes the baseline average payoff, above which higher payoffs can be achieved using the credible threat of falling back to (D,D) if there is a violation. For instance, imagine that a different strategy than the ones already considered—alternate between (C,C) and (D,D) and revert to (D,D) forever after a deviation—were played against itself. The payoffs would be

$$3 + \delta(1) + \delta^2(3) + \delta^3(1) + \dots = \frac{3 + \delta}{1 - \delta^2}$$

and, as we now know, comparing this payoff to the one that the player would receive from deviating, $5 + \frac{\delta}{1-\delta}$, serves two purposes. First, the comparison tells us if this strategy, played against itself, will form an equilibrium if the players are sufficiently patient. Second, because the future is always infinite, the comparison tells us if players ever have an incentive to deviate after the (C,C) history of play in odd rounds. While slightly more complicated, this comparison can be reduced to this: players don't deviate if $\delta > \left(\frac{1}{2}\right)^{\frac{1}{2}} = 0.71$.

Intuition suggests that there will also be no incentive to deviate on the even rounds because the players are scheduled to be at the stage equilibrium (D,D), and then, by definition, they cannot deviate and do better. Specifically, the

immediate payoff from deviating from the stage equilibrium must be worse, and the longer-term payoff will also be worse because, although the flow will be the same, it will begin only after your mistake. Hence, this alternating strategy forms a subgame perfect Nash equilibrium.

To end our discussion of the folk theorem, notice that the discounted utility accruing to players as a result of the subgame perfect strategy just analyzed is $\frac{3+\delta}{1-\delta^2}$, but to find the average per-period payoff achieved at this equilibrium, we can just multiply this by $(1 - \delta)$. While this multiplication may not be intuitive, consider a simpler illustration of the principle. The discounted utility of cooperating forever was $\frac{3}{1-\delta}$, and if we multiply this by $(1 - \delta)$, we get back to the average per-period payoff of 3. Now, because the denominator of the discounted payoff is the difference of perfect squares, the average equilibrium payoff of our third strategy reduces to $\frac{3+\delta}{1+\delta}$. As δ approaches 1, this payoff converges on 2, smack in the middle of the darker area highlighted in [figure 10.4](#) and exactly halfway between the stage equilibrium (D,D) and the socially efficient outcome (C,C).

Behavior in Games—Playing “Anything Goes” Strategies

A number of economists have been unsatisfied with the theory of infinitely repeated games because if players are sufficiently patient, it appears that almost anything goes. As we have shown, equilibria can be supported in which players are both cooperative and egoistic. As a result, experiments have served two important functions in this context. In addition to the more traditional testing of theory, experimental results on infinitely repeated games have helped sharpen our predictions about when cooperation is likely to evolve.

Considering design, the obvious problem for the experimenter is that players do not have infinite lives, so how can you run an experiment that lasts an infinite number of rounds? On top of this, player discount factors are unobserved (i.e., they are private information), so how could you ever know whether cooperation is theoretically possible or not? Luckily, Roth and Murnighan (1979) were the first to come up with a clever design feature that solves both these issues simultaneously.

Instead of playing an infinitely repeated game and discounting the payoffs you expect to receive in the future by some amount, δ , what if the game only proceeded to another round with probability δ ? That is, you play round 1 for

sure, but after that, the game proceeds to the next (and any future) rounds based on a random event, such as a coin toss. In this context, Roth and Murnighan showed that expected utility is isomorphic with discounted utility. For example, if one plays a grim trigger strategy as described earlier in this chapter, when the cooperative payoff is 3 per round, one would weigh the payoff of getting 3 in a third round, for example, by the odds that one gets to the third round, which are δ^2 because the coin tosses at the end of each round are independent of each other. Expanding this logic, the expected utility of two grim trigger strategies played against each other is just $Eu(GT) = 3 + 3\delta + 3\delta^2 + 3\delta^3 + \dots = \frac{3}{1-\delta}$. In other words, games with continuation probabilities of less than 1 will be of random length and should have strategies that are valued identically to those in infinitely repeated games. Instead of being *infinitely* repeated, these games are *indefinitely* repeated. Further, unobserved discount factors are unimportant because by picking δ , the experiment has effectively induced the important parameter.

Indefinitely repeated games of this nature have been run in the lab since Roth and Murnighan (1979), but a recent survey by Dal Bó and Fréchette (2018) summarizes the main results. To organize the prisoner's dilemma data from many experiments, the authors normalized the underlying stage games so they were completely determined by just two parameters, g , the gain from deviating on a cooperator, and l , the loss from being deviated on, also known as the *sucker's* payoff. All the games and experiments they considered can therefore be summarized as shown in [figure 10.5](#):

		Player 2	
		C	D
Player 1	C	(1, 1)	(-l, 1 + g)
	D	(1 + g, -l)	(0, 0)

Figure 10.5

The normalized prisoner's dilemma.

The obvious first thing to test in the lab is whether the likelihood of cooperation is linked to the continuation probability, δ . As the expected length of the supergame increases with δ because players are more likely to make it to the next round, do players place a higher value on the potential cooperative gains?

As we can see in the left most panel of [figure 10.6](#), the rate of cooperation observed in the experiments increases in the continuation probability, as expected (or hoped). The difference between the three upward-sloping lines is the amount of experience that the players had. Subgame 1 labels the effect of increasing the continuation probability in the first supergame that players experienced. The data labeled “Supergame 7” comes from the 7th supergame played, and “Supergame 15” indicates the results from the 15th. As you can see, experience matters a lot in this setting. With experience, players better appreciate the accrual of cooperative payoffs in games that are expected to last longer, and they respond more to the difference in the continuation probabilities.

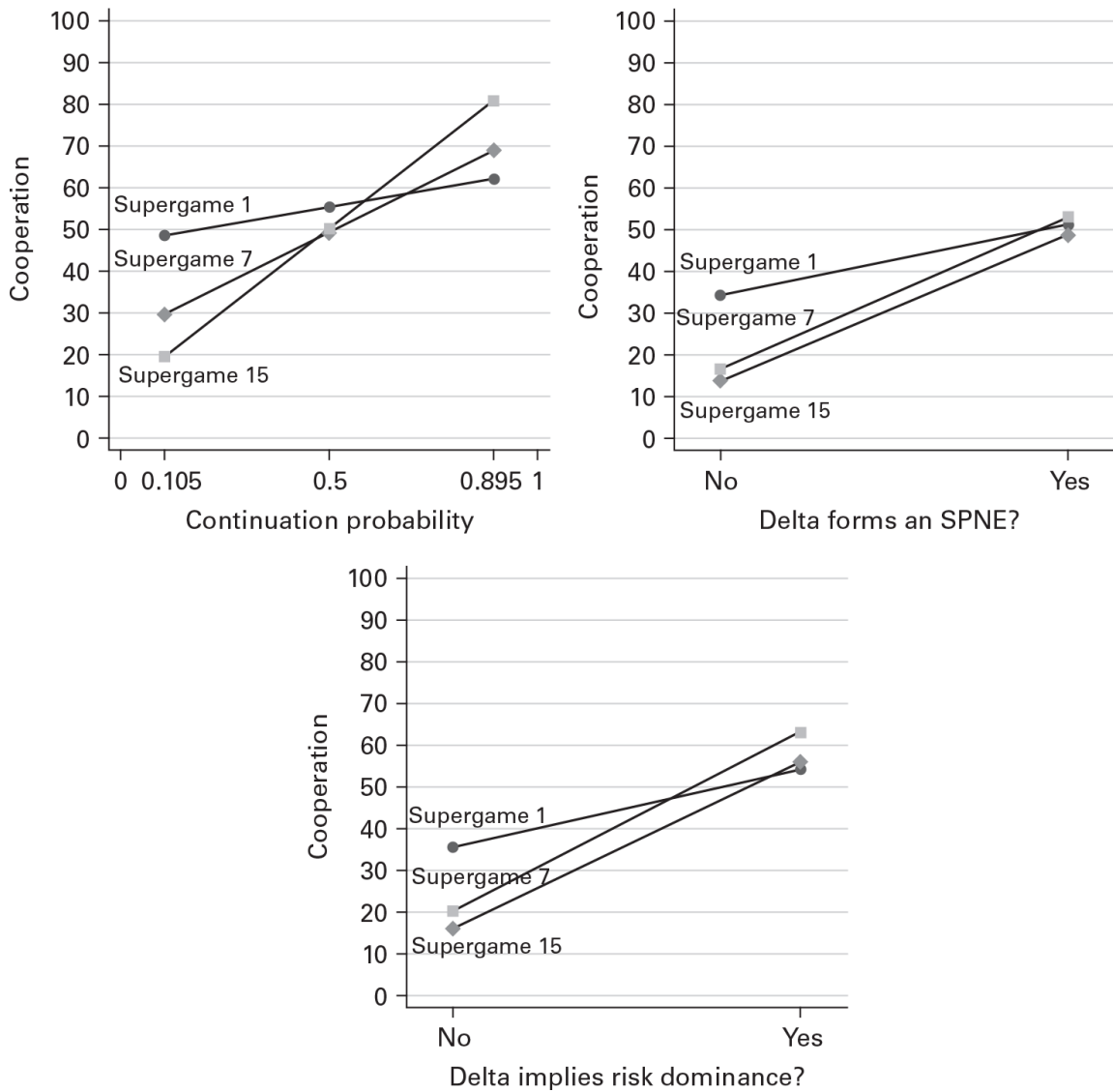


Figure 10.6

Cooperation in the indefinitely repeated prisoner's dilemma.

The problem with the result seen in the left panel of [figure 10.6](#) is that cooperation should not evolve in every setting. If the continuation probability, and therefore δ , is low enough, the only subgame perfect Nash equilibrium is to always defect. Using the normalized payoffs from the matrix discussed at the beginning of this section, we can calculate the cutoff continuation probability required to support cooperation as the consequence of a subgame perfect Nash equilibrium. Specifically, we need the expected utility of following a trigger strategy to outweigh the immediate benefit of defecting. This occurs when $\frac{1}{1-\delta} > 1+g$ or $\delta^{SPNE} > \frac{g}{1+g}$.

In the upper right panel of [figure 10.6](#), we list the rates of cooperation for parameter values placing δ above or below the subgame perfect threshold. Overall, we see that cooperation is more likely to evolve when it is supported by theory. When we differentiate the results by experience, we see that being exposed to more supergames mostly helps players to figure out when they should *not* be cooperating.

An interesting observation made by Blonski et al. (2011) in their indefinitely repeated experiment was the difference in observed behavior in the two different versions of the stage prisoner's dilemma shown in [figure 10.7](#):

		Player 2				Player 2	
		C	D			C	D
Player 1	C	(90, 90)	(70, 100)	Player 1	C	(90, 90)	(0, 100)
	D	(100, 70)	(80, 80)		D	(100, 0)	(80, 80)

Figure 10.7

The prisoner's dilemma with strategic uncertainty.

Specifically, even though the critical value of δ needed to support cooperation is exactly the same, $\frac{1}{2}$ (you can verify this calculation as an exercise), and, therefore, theory predicts that the incidence of cooperation should be the same when these two stage games are repeated, the observed rates of cooperation were very different. In the stage game on the left, the average rate of cooperation was 36%, while it was just 4% in the game on the right.

Intuitively, we all see the problem with the theory that we have been developing in this chapter (highlighting the second benefit of experimentation mentioned here). The two games differ just in the sucker's payoff, and this is not part of the theory. In particular, the sucker's payoff is much lower in the game on the right, and as a result, cooperation seems dramatically more risky, as it clearly did to the participants in the experiment. Digging a bit deeper reveals that indeed, the sucker's payoff is ignored when we calculate whether a trigger strategy can support cooperation. We calculated the discount factors for which grim trigger is an SPNE to be $\delta^{SPNE} > \frac{g}{1+g}$, but notice that the

sucker's payoff, $-l$, is missing, even though it clearly must matter because it is the only difference between the games.

To account for the strategic uncertainty that arises when the sucker's payoff gets low, Blonski et al. (2011) revisited Harsanyi and Selten's 1988 notion of risk dominance (defined in chapter 7) and adapted it to infinitely repeated games. The basic idea of the theory they developed is that players should stop cooperating when the strategic risk of doing so, as determined by the size of the sucker's payoff, becomes overwhelming. Calculating the boundary conditions on this uncertainty and translating them into a critical value of δ that can easily be compared with the other theoretical results in this literature, they found $\delta^{RD} > \frac{g+l}{1+g+l}$. If we normalize the payoffs of the two games described here, we find that $\delta^{RD} = 0.67$ for the game on the left and 0.90 for the game on the right, indicating that, indeed, players have to be much more likely to accrue the rewards from cooperating repeatedly for them to risk doing so when the sucker's payoff is so low. More generally, if you return to the bottom panel of [figure 10.6](#), you can see the same result, derived from a number of experiments. Cooperation is much more common in games wherein the grim trigger strategy is risk dominant (i.e., the sucker's payoff is not too low).

Exercises

(1) Consider the stage game shown in [figure 10.8](#), repeated infinitely with discount factor δ :

		Player 2	
		X	Y
Player 1	X	(3, 3)	(0, 8)
	Y	(8, 0)	(1, 1)

Figure 10.8

- Using a standard grim trigger strategy, how patient must the players be to support play at (X,X)?
- Use the one-stage deviation principle to test whether your strategy is subgame perfect.

(2) Consider the following stage game shown in [figure 10.9](#), repeated infinitely with discount factors (δ_1, δ_2) for the two players:

		Player 2	
		L	R
Player 1	U	(1, 1)	(-2, 4)
	D	(3, 0)	(0, 0)

Figure 10.9

- (a) Using a standard grim trigger strategy, how patient must the players be to support play at (U,L)?
- (b) Use the one-stage deviation principle to test whether your strategies are subgame perfect.
- (3) Consider the stage game shown in [figure 10.10](#), repeated infinitely with discount factors (δ_1, δ_2) for the two players:

		Player 2	
		L	R
Player 1	U	(1, 3)	(10, 0)
	D	(0, 10)	(4, 6)

Figure 10.10

- (a) Write a grim trigger strategy profile that has the players choosing (D, R) in equilibrium (on the equilibrium path).
- (b) Show that this strategy profile is subgame perfect if the players are sufficiently patient. What must (δ_1, δ_2) be for this strategy profile to be subgame perfect?
- (4) Consider the following stage game in [figure 10.11](#), repeated infinitely with discount factor δ :

		Player 2	
		A	B
Player 1	A	(0, 0)	(2, 1)
	B	(1, 2)	(0, 0)

Figure 10.11

Devise a subgame perfect Nash equilibrium in which players earn 1.5 per period, on average.

(5) Consider the following stage game in figure 10.12, repeated infinitely with discount factor δ :

		Player 2		
		A	B	C
Player 1	A	(2, 2)	(0, 3)	(-4, 4)
	B	(3, 0)	(1, 1)	(-5, 0)
	C	(4, -4)	(0, -5)	(-6, -6)

Figure 10.12

Using a grim trigger strategy, what is the minimum level of patience that can support play at (A,A) as a subgame perfect Nash equilibrium?

(6) In an n -player prisoner's dilemma, suppose that if everyone cooperates, they all get a payoff of n , if any *one* player defects, that player gets n^2 and all other players get -1 , and if more than one player defects, all the defectors get 0 while all the cooperators get -1 .

- What is the subgame perfect Nash equilibrium of this stage game?
- If this stage game is infinitely repeated and the discount factor is δ , can cooperation be sustained using a trigger strategy?
- What happens to the chances that the trigger strategy will work as the number of players increases?

(7) In the volunteers' dilemma (introduced in chapter 6), anyone who volunteers gets a payoff of $b - c$ and if at least one player volunteers, those who don't volunteer get b , and if nobody volunteers, all players get 0.

- (a) Write the two-player normal form for this game.
- (b) If this stage game is infinitely repeated and the discount factor is δ , can volunteering be sustained using a trigger strategy?

(8) A number of studies have found that many people are more likely to discount *quasi-hyperbolically* than exponentially, as we have been assuming. In a nutshell, quasi-hyperbolic discounters act like exponential discounters when they consider any two future outcomes, but they discount amounts that they do not receive immediately more deeply. A relatively easy quasi-hyperbolic discounting function to use is $PV = x + \beta\delta x + \beta\delta^2x + \dots$, where x is the per-period payoff, δ is the discount factor as discussed previously, and $0 < \beta < 1$ is the additional discounting that players attach to any payoff that they do not receive immediately:

		Player 2	
		C	D
Player 1	C	(3, 3)	(-1, 5)
	D	(5, -1)	(0, 0)

Figure 10.13

Considering the stage game shown in [figure 10.13](#), do quasi-hyperbolic discounters have to be more or less patient than exponential ones to sustain cooperation (and does this make sense)?

IV

Incomplete Information

11

Simultaneous Games of Incomplete Information

Thus far, we have assumed that the only uncertainty that players face is strategic: they don't necessarily know the strategies that the others are playing, but they do always know who the other players are and the available actions and payoffs associated with each person. In this section, we will consider situations in which players are missing some critical piece of information about the rules of the game they are playing. This typically takes the form of players not knowing the payoffs or capabilities of the others or how actions map to outcomes. For example, a bidder in an auction may not know how much her competitors value the item, a firm may not know the cost structure of its rivals, a hunter in the stag hunt game may not know whether his partner is actually a competent hunter or a poser, a poker player may (and indeed should) not know the cards the other players have been dealt, and so on. We model these scenarios by saying that players may be one of multiple possible types. In these situations, players form beliefs about what each type of player would do and act to maximize their own expected payoffs, based on the likelihood of dealing with different types. In this chapter, we will consider simultaneous interactions. In chapter 12, we will expand our consideration to games in which one player moves first, in which case her action may reveal some information about her type.

Imagine that you have the opportunity to purchase a company and must decide how much to offer the current owner. The problem is that you don't know what kind of shape the company is in, and thus how much it will be worth to you. Suppose that the current value of the company, v , is known to be an integer between 0 and 100, with each value equally likely. The current owner knows the value, and the company is worth v to her. However, if you were to take over it would, due to your superior leadership, be worth 50% more to you—that is, $1.5v$. Thus, regardless of the underlying quality, it is most efficient for the firm to change hands. The owner's job is to determine how much she is willing to accept for the company. Simultaneously, you will submit a bid, b . If your bid is less than the owner's willingness to accept, you earn nothing (payoff = 0) and the owner retains the company (payoff = v). If your bid is greater than or equal to the owner's willingness to accept, you gain the company and pay your bid (payoff = $1.5v - b$), while the owner receives the price (payoff = b). How much do you bid?

Up until this point, we've considered two types of information in games: perfect and imperfect. In both cases, however, we always assumed that rules of the game were common knowledge. This includes who the players are, the available actions for each player, how actions map to outcomes, the payoffs associated with each outcome, and what information each person possesses. This assumption of common knowledge allows us to use such solution concepts as iterated elimination of dominated strategies, backward induction, and the Nash equilibrium. But in this “acquiring a company” game and the other games we'll consider in this section, at least one player is missing important information about the rules. We call this a situation of *incomplete information*. How do we solve for rational play when some players don't even know what game they are playing?

The Conditional Cooperator's Dilemma

Before investigating your optimal bid in the acquiring a company game, let's examine incomplete information in a familiar normal-form game: the one-shot prisoner's dilemma. In our discussion of behavioral evidence, we have seen that many participants in these types of games (roughly half of experimental subjects) can be classified as *conditional cooperators*. Rather than following their narrow self-interest, these players prefer to cooperate if the other player cooperates, but defect if the other player defects. Let's assume that Rey is a conditional cooperator playing this game against Kylo. In addition to the standard payoffs, assume that Rey gains 3 additional units of utility if she either cooperates when Kylo cooperates or defects when Kylo defects. Her optimal strategy depends on her beliefs about the *type* of player Kylo is. Is he a Mensch who always prefers to cooperate, an egoistic Jerk who cares only about his own outcome, or is he a conditional cooperator, too?¹

Let's assume that there is a $p = 3/5$ chance that Kylo is a Jerk type with a dominant strategy to defect (as in the game shown in [figure 11.1](#) on the left, with standard prisoner's dilemma payoffs for Kylo). There is a $1 - p = 2/5$ chance that he is a Mensch type with a dominant strategy to cooperate (as in the game on the right, which simply reverses his payoffs from cooperating and defecting). If she knew which of these two games she was playing, Rey's decision would be simple, requiring only one step of reasoning in the process of iterated elimination of dominated strategies. If she is in the game on the left, she anticipates that Kylo will defect, and she wishes to match him. If she

is in the game on the right, her decision is equally straightforward: since a rational Kylo will cooperate, she wants to cooperate as well. The trouble is that she doesn't actually know which of these two games she is playing. This is like no scenario we have seen before. How can we proceed?

		(Jerk) Kylo				(Mensch) Kylo	
		Cooperate	Defect			Cooperate	Defect
Rey	Cooperate	(6, 3)	(0, 5)	Rey	Cooperate	(6, 5)	(0, 3)
	Defect	(5, 0)	(4, 1)		Defect	(5, 1)	(4, 0)
Game when Kylo is a Jerk				Game when Kylo is a Mensch			

Figure 11.1
A game of incomplete information.

Harsanyi Transformation

The question of how to deal with this type of scenario went unresolved until the late 1960s, when John Harsanyi proposed a solution.² Harsanyi's method was to take the situation of incomplete information, which we don't know how to solve, and transform it into a new game, called a *Bayesian game*. The Bayesian game will be one of imperfect—but complete—information, and we can then solve for the Nash equilibrium of this game.

Recall that a game of imperfect information is one in which players know the rules, but not all the actions that have been taken by others. In other words, we want to include the determination of the player's type *as just another move in the game*, which some players might know and others might not. Who will make this move? To address this, we introduce a new player, called "Nature." Nature is responsible for the resolution of all random forces in games. For instance, if a coin flip determines who kicks off in a football game, we say that Nature decides who kicks off. If players in a poker game are randomly dealt cards, we say that Nature determines their hands. And if Rey is facing a prisoner's dilemma game against a guy that could be either a Jerk or a Mensch, we say that Nature determines her partner. Nature differs from the other players in that it is not strategic (although if you're superstitious, you may disagree with this). Nature doesn't get payoffs or form beliefs about the other players' strategies. Instead, it always acts with a fixed, known probability. In our case, it acts to make Kylo a Jerk with probability $p = 3/5$ and a Mensch with probability $1 - p = 2/5$.

Consider the game depicted in [figure 11.2](#). This is our transformed *Bayesian game*, which we'll define in more detail in the next subsection. The first move is taken by Nature, to determine Kylo's type according to a set probability p . Then, Rey and Kylo simultaneously choose to Cooperate or Defect. Notice the information sets here. Rey's decision nodes are in the same information set, indicating that she does not know Kylo's type at the time she moves. Kylo does not know Rey's choice when he acts (since it is simultaneous-move game), but he does know his own type with certainty. Therefore, he can condition his behavior on his type.

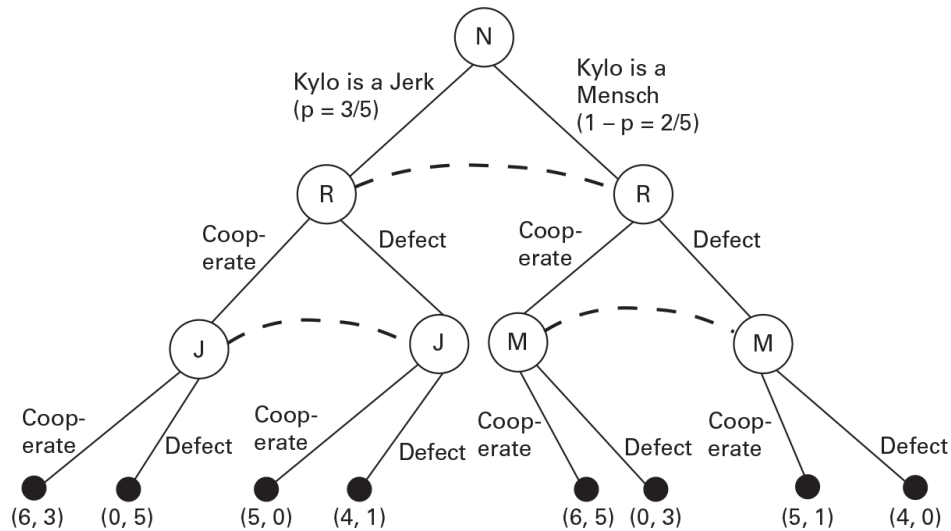


Figure 11.2

The Bayesian game between Rey (**R**), Jerk Kylo (**J**), and Mensch Kylo (**M**) after the initial move by Nature (**N**).

Note also that the probability p with which Nature chooses Jerk Kylo as Rey's partner is part of the game, and thus common knowledge for both players rather than just Rey's privately held belief about who Kylo is. There are two ways to interpret the determination of Kylo's type. First, it may be that there are both *Mensches* and *Jerks* in the population, and Nature is randomly selecting one member of the general population to be Rey's partner. In this case, we can think of the game as a three-player interaction between Rey, Jerk Kylo, and Mensch Kylo, each of whom simultaneously chooses between Cooperate and Defect. Alternatively, we can imagine that over the course of Kylo's life before he stumbles into this interaction with Rey, Nature has conspired to make him either a Mensch or a Jerk. This interpretation is a

bit trippy from Kylo's perspective. Since Kylo has two information sets, any strategy of his must specify both what to do if he is a Jerk *and* what to do if he is a Mensch, which is kind of an odd situation to be in.

Most important, our situation is now modeled as a fully specified, coherent game, the rules of which are common knowledge.

Bayesian Games

In general, Bayesian games are directly analogous to the standard games we've been analyzing thus far. The difference is that we allow each player to be associated with multiple possible *types*. For instance, in the prisoner's dilemma game with incomplete information about Kylo's preferences, we say that Kylo may be one of two types. If you were a goalie defending a penalty kick and were unsure about the competence of the kicker, we could represent this by allowing the kicker to be one of, say, four types: highly skilled, moderately skilled, mediocre, or a complete klutz. In the example we started with, there is a continuum of possible values for the owner's company, and we represent this by allowing the owner to take on a continuum of possible types. Players always know their own type, but not necessarily the types of those they interact with. Rather, they know the probability distribution over possible types. For instance, Rey knows that Kylo is a Jerk with probability $3/5$ and a Mensch with probability $2/5$, the prospective buyer of the company knows that any company value between 0 and 100 is equally likely, and so on.

The analogy between a standard game and a Bayesian game is depicted in [figure 11.3](#). As in our standard definition of a game of strategy, there exists a set of players, each with a set of possible actions. However, each player in a Bayesian game is also associated with a set of possible types and a probability distribution that specifies the likelihood that each of the types is realized.

Standard Game	Bayesian Game
• There is a set of players $\{1, \dots, n\}$. • Each player has a set of actions.	• There is a set of players $\{1, \dots, n\}$. • Each player has a set of actions, a set of types, and a probability distribution over types.
• A strategy s_i for player i specifies an action at each point of decision. • Each player i has a payoff function: $u_i(s_1, \dots, s_n)$.	• A strategy s_i for player i specifies an action at each point of decision for each of i's possible types. • Each player i has a payoff function: $u_i(s_1, \dots, s_n, \text{type}_1, \dots, \text{type}_n)$.

Figure 11.3

Standard games versus Bayesian games.

The main solution concept in games of incomplete information is called the *Bayesian Nash equilibrium* (as given in definition 1) and is equivalent to the Nash equilibrium of the Bayesian game. Recall that the Nash equilibrium is a strategy profile, specifying one strategy for each player, such that no player can receive a higher (expected) payoff from unilaterally deviating. Likewise, the Bayesian Nash equilibrium is a strategy profile, specifying one strategy for each player *and type of player*, such that no player-type can receive a higher expected payoff from unilaterally deviating. For example, in the Rey-Kylo game, any Bayesian Nash equilibrium would need to specify a strategy for Rey (who has only one type) and a strategy for Kylo that includes what he does for each of his possible types. In this way, players in Bayesian games can play *type-contingent strategies*, which specify different actions for different types. Furthermore, it must be the case that Rey would not wish to change her strategy given the strategy employed by Jerk Kylo and Mensch Kylo, and that neither type of Kylo would want to change his strategy given the strategy played by Rey.

Definition 1 (Bayesian Nash equilibrium) *A Bayesian Nash Equilibrium is a Nash equilibrium of a Bayesian game.*

Solving for Bayesian Nash Equilibria

To solve for Bayesian Nash equilibria, we must first identify the source of the incomplete information (i.e., the possible types) and transform the situation into a Bayesian game. Then the process of solving for Bayesian Nash equilibria is fundamentally the same as for Nash equilibria in standard games.

We determine the best response of each player's *types* and look for strategy profiles in which all types are simultaneously best responding. The best response for each type of Kylo is straightforward. Since each has a dominant strategy, Jerk Kylo's best response is always to Defect and Mensch Kylo's best response is always to Cooperate, regardless of Rey's strategy. In equilibrium, therefore, Kylo's strategy must be $s_{Kylo}(Jerk, Mensch) = (Defect, Cooperate)$. What is Rey's best response to Kylo playing (Defect, Cooperate)? She has two possible pure strategies, Cooperate and Defect (remember that she can't condition her behavior on Kylo's type, which she does not know), and to determine which is her best response, we calculate her expected payoff from each. If Rey plays Cooperate, there is a 3/5 chance that she is dealing with Jerk Kylo, who defects, and her ultimate payoff is 0 (the upper-left quadrant in the Rey–Jerk Kylo game), and a 2/5 chance that she is dealing with Mensch Kylo, who Cooperates, giving her a payoff of 6. In expectation, $Eu_{Rey}(Cooperate, (Defect, Cooperate)) = (3/5)(0) + (2/5)(6) = 2.4$. Similarly, her expected payoff from Defecting when Kylo plays (Defect, Cooperate) is $Eu_{Rey}(Defect, (Defect, Cooperate)) = (3/5)(4) + (2/5)(5) = 4.4$. Therefore, Defect is Rey's best response to the Kylos playing their dominant strategy of (Defect, Cooperate). The unique Bayesian Nash equilibrium of this game is (Defect, (Defect, Cooperate)). Likewise, we can find the best responses for Rey to the other possible strategies of Kylo, although we know none of these will be a Bayesian Nash equilibrium. Specifically, $BR_{Rey}(Cooperate, Cooperate) = Cooperate$, $BR_{Rey}(Defect, Defect) = Defect$, and $BR_{Rey}(Cooperate, Defect) = Defect$.

Another method of identifying the Bayesian Nash equilibria is to first write the Bayesian game in normal form and then conduct our standard best response analysis by underlining each player's best responses. This procedure is more systematic and clear but may require more computation. Writing the Bayesian game in normal form also works only if there are a limited number of types and strategies. First, we must convert the extensive-form Bayesian game to normal form, as shown in the matrix in [figure 11.4](#). Rey has two pure strategies, Cooperate and Defect, and these form the rows. Kylo has four pure strategies, which form the columns. Unlike other games we've seen, even when we know the pure strategy of each player, we still do not know their payoffs with certainty since they still depend on Kylo's type. The payoffs that we fill into the cells are therefore *expected payoffs*. Consider the top left cell,

corresponding to Rey and both types of Kylos Cooperating. In this case, Rey earns a payoff of 6 if Kylo is a Jerk and 6 if Kylo is a Mensch, so her expected payoff is 6. Kylo earns a payoff of 3 if he's a Jerk and they both Cooperate and 5 if he's a Mensch and they both Cooperate. His expected payoff is thus $3(3/5) + 5(2/5) = 3.8$. Similarly, consider the next cell to the right, corresponding to Rey Cooperating and Kylo Cooperating if he is a Jerk and Defecting if he is a Mensch. In this case, Rey earns a payoff of 6 if he's a Jerk (since his strategy has him Cooperating in this case) and 0 if he's a Mensch (since his strategy has him Defecting in this case), for an expected payoff of $6(3/5) + 0(2/5) = 3.6$. Proceeding in this way, we can fill in the expected payoffs of each player for every cell. Finally, we conduct our standard best response analysis and find that there is one cell in which the players are both best responding: (Defect, (Defect, Cooperate)), consistent with the Bayesian Nash equilibrium we found previously.

		Kylo			
		(Cooperate, Cooperate)	(Cooperate, Defect)	(Defect, Cooperate)	(Defect, Defect)
Rey	Cooperate	(<u>6</u> , 3.8)	(3.6, 3)	(2.4, <u>5</u>)	(0, 4.2)
	Defect	(5, .4)	(<u>4.6</u> , 0)	(<u>4.4</u> , <u>1</u>)	(<u>4</u> , .6)

Figure 11.4

The Bayesian game in normal form (Kylo is Jerk type with probability $p = 3/5$); (Defect, (Defect, Cooperate)) is the unique Bayesian Nash equilibrium.

You may wonder how the Bayesian Nash equilibrium depends on the assumption that Kylo is more likely to be a Jerk. After all, if the conditionally cooperative Rey were sufficiently certain that she was dealing with a cooperative type, wouldn't she wish to Cooperate as well? Rather than assuming that Kylo is the Jerk type with probability $3/5$, let's substitute a generic p into this calculation, such that p is strictly greater than 0 and strictly less than 1. In this case, we find that Rey's expected utilities are $Eu_{Rey}(Cooperate, (Defect, Cooperate)) = (p)(0) + (1 - p)(6) = 6 - 6p$ and $Eu_{Rey}(Defect, (Defect, Cooperate)) = (p)(4) + (1 - p)(5) = 5 - p$. Cooperate is therefore her best response when $6 - 6p \geq 5 - p$, or $p \leq 1/5$.³ In other words, if it is at all likely that Kylo is a Jerk ($p > 1/5$), she prefers to Defect, if it is unlikely that he is a Jerk ($p < 1/5$), she prefers to Cooperate, and when $p = 1/5$, she is indifferent. Therefore, the Bayesian Nash equilibrium we found

earlier holds only if $p \geq 1/5$, while (Cooperate, (Defect, Cooperate)) is a Bayesian Nash equilibrium if $p \leq 1/5$.

The game with a generic p is written in normal form in [figure 11.5](#) and the best response analysis confirms this conclusion. To calculate the expected payoff in each cell, we simply use p as the probability that Rey is dealing with Jerk Kylo and $1 - p$ as the probability that she is dealing with Mensch Kylo. The best response analysis is somewhat more opaque now that payoffs can vary in p , but the procedure is the same. For Kylo, (Defect, Cooperate) yields the highest payoff both when Rey Cooperates and when Rey Defects. (Since p is strictly greater than 0 and strictly less than 1, his payoff of 5 is the highest available in the top row and his payoff of 1 is the highest available in the bottom row. You can also confirm this by graphing his four possible payoffs in each row.) For Rey, the best responses to (Cooperate, Cooperate) and (Defect, Defect) are clear and do not depend on p . However, her best responses to Kylo's other possible strategies do depend on p . If Kylo plays the unintuitive (Cooperate, Defect), her expected payoff from Cooperating is $6p$ and from Defecting is $p + 4$. Thus, Cooperate is a best response if $6p \geq p + 4$, or $p \geq 4/5$, while Defect is a best response if $p \leq 4/5$. Since Kylo is being cooperative only if he is a Jerk, she finds it in her interest to Defect unless it is very likely that he is a Jerk ($p > 4/5$). Thus, both Cooperate and Defect can be best responses for Rey over different ranges of p , as indicated in [figure 11.5](#). The same is true for her best response to (Defect, Cooperate): since Kylo only cooperates if he is a Mensch, Rey prefers to Defect unless there is a very low probability that he is a Jerk ($p \leq 1/5$). We find that there are two cells with double underlines, but for different ranges of p . If $p \leq 1/5$, then (Cooperate, (Defect, Cooperate)) is the unique Bayesian Nash equilibrium, and if $p \geq 1/5$, then (Defect, (Defect, Cooperate)) is the unique Bayesian Nash equilibrium.

		Kylo			
		(Cooperate, Cooperate)	(Cooperate, Defect)	(Defect, Cooperate)	(Defect, Defect)
Rey	Cooperate	<u>(6, 5 - 2p)</u>	if $p \geq 4/5$ (<u>6p</u> , 3)	if $p \leq 1/5$ (<u>6 - 6p</u> , <u>5</u>)	(0, 2p + 3)
	Defect	(5, 1 - p)	if $p \leq 4/5$ (p + 4, 0)	if $p \geq 1/5$ (<u>5 - p</u> , <u>1</u>)	(<u>4</u> , p)

Figure 11.5

The Bayesian game in normal form (Kylo is Jerk type with any probability $0 < p < 1$); (Defect, (Defect, Cooperate)) is a Bayesian Nash equilibrium if $p \geq 1/5$; (Cooperate, (Defect, Cooperate)) is a Bayesian Nash equilibrium if $p \leq 1/5$.

Going Hunting with a Flatlander

Imagine that a hunter in Vermont is playing the stag hunt game with a *Flatlander*, the somewhat derogatory term that Vermonters use to refer to anyone not originally from the Green Mountain State. If the Flatlander is, in fact, a competent Hunter, then they are playing the stag hunt game previously seen in chapter 8 and depicted in the left matrix in figure 11.6. Alternatively, the other player may be a Poser who has no idea what they’re doing. In this case, there is absolutely no way that the pair would successfully manage a stag, and Rabbit is a dominant strategy for them both, as modeled in the game on the right.

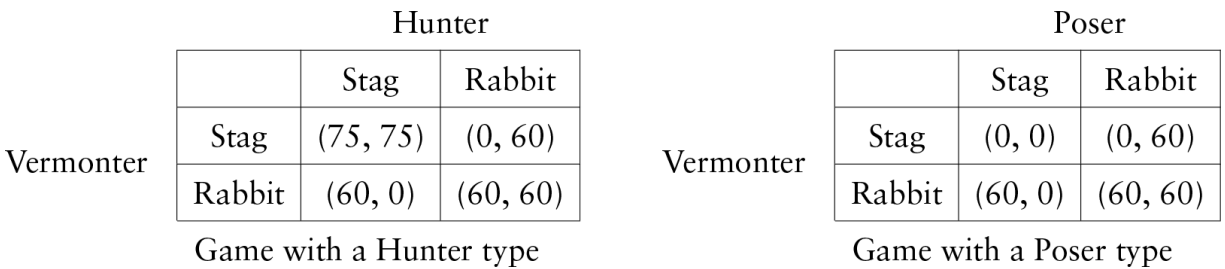


Figure 11.6
A stag hunt game of incomplete information.

To transform this situation of incomplete information into a Bayesian game, we first identify the source of the missing information. The Vermonter does not know whether the Flatlander is the *Hunter* type or the *Poser* type. We thus model the determination of the Flatlander’s type as the first move in the game, taken by Nature, as depicted in figure 11.7. Nature makes the Flatlander a Hunter with some probability p , which is a fixed parameter of the game known to both players. Note the information sets here.

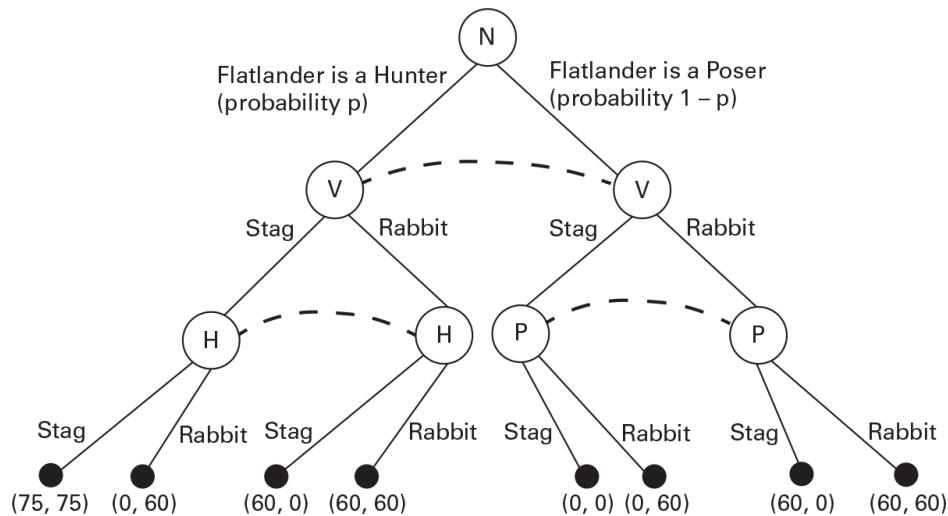


Figure 11.7

The Bayesian game between the Vermonter (V), Hunter (H), and Poser (P) after the initial move by Nature (N).

To find the Bayesian Nash equilibria, we can write the Bayesian game in normal form, as shown in [figure 11.8](#). The Vermonter cannot condition on the Flatlander's type, and thus has only two possible pure strategies (Stag and Rabbit). Each of the Flatlander's strategies must specify both what to do as a Hunter and what to do as a Poser, and thus there are four possible pure strategies ((Stag, Stag), (Stag, Rabbit), (Rabbit, Stag), and (Rabbit, Rabbit)). Finally, we fill in the expected payoffs for each of the players, and conduct a best response analysis. The Flatlander's best response to Stag is (Stag, Rabbit), since $60 + 15p$ is the highest payoff in this row, given that p must be strictly greater than zero and strictly less than 1. This makes intuitive sense: a competent Hunter prefers to match on Stag, while a Poser would rather go after a Rabbit. Likewise, the Flatlander's best response to Rabbit is (Rabbit, Rabbit), since Stag is no longer a good strategy even for a competent Hunter if the Vermonter is going after a Rabbit. Turning to the Vermonter's best responses, the first two columns involve the Flatlander choosing Stag when competent. In these cases, the Vermonter should prefer Stag only if it is highly probable that the Flatlander is actually a Hunter. Specifically, Stag is a best response only if its payoff ($75p$) is greater than the payoff from Rabbit (60), which is true so long as $p \geq 4/5$; otherwise, Rabbit is the best response in these two columns. In the last two columns, which involve the Hunter selecting the Rabbit strategy, we see that the Vermonter's best response is Rabbit, regardless of p . We therefore find two cells with mutual responses.

(Stag, (Stag, Rabbit)) is a Bayesian Nash equilibrium only if $p \geq 4/5$. If the likelihood that the Flatlander is a Hunter is any lower than 4 out of 5, then the Vermonter would prefer Rabbit instead. Additionally, (Rabbit, (Rabbit, Rabbit)) is a Bayesian Nash equilibrium regardless of the value of p .

		Flatlander			
		(Stag, Stag)	(Stag, Rabbit)	(Rabbit, Stag)	(Rabbit, Rabbit)
Vermont	Stag	if $p \geq 4/5$ (<u>75</u> p, 75p)	if $p \geq 4/5$ (<u>75</u> p, <u>15</u> p + 60)	(0, 60p)	(0, 60)
	Rabbit	if $p \leq 4/5$ (<u>60</u> , 0)	if $p \leq 4/5$ (<u>60</u> , 60-60p)	(<u>60</u> , 60p)	(<u>60</u> , <u>60</u>)

Figure 11.8

The Bayesian stag hunt game in normal form (Flatlander is a Hunter type with any probability $0 < p < 1$); (Stag, (Stag, Rabbit)) is a Bayesian Nash equilibrium if $p \geq 4/5$; (Rabbit, (Rabbit, Rabbit)) is a Bayesian Nash equilibrium for all p .

Could we have determined the Bayesian Nash equilibria without going to the trouble of writing the game in normal form and calculating all 16 expected payoffs for the players? A slightly less systematic and clear—but equally viable—method is to search for equilibria by identifying a strategy that one player could choose and then asking whether it is possible for that strategy to be part of an equilibrium strategy profile. (We will rely on this technique in chapter 12, when we consider sequential-move Bayesian games.) In the stag hunt game, there are only two possible pure strategies for the Vermonter: Stag and Rabbit. We can therefore ask: Is there a Bayesian Nash equilibrium in which the Vermonter plays Stag, and is there a Bayesian Nash equilibrium in which the Vermonter plays Rabbit?

If there is an equilibrium involving $s_{\text{Vermont}} = \text{Stag}$, then the Flatlander's strategy must be a best response to Stag and, in turn, Stag must be the Vermonter's best response to that strategy. We thus start by determining $BR_{\text{Flatlander}}(\text{Stag})$, which we can do by examining each of the two matrices in [figure 11.6](#), associated with the different types of Flatlanders. If the Vermonter plays Stag, then the Hunter type prefers to match on Stag (earning a payoff of 75 versus 60), while the Poser type prefers to play Rabbit (earning a payoff of 60 versus 0). Thus $BR_{\text{Flatlander}}(\text{Stag}) = (\text{Stag}, \text{Rabbit})$. To check if Stag is, in fact, the Vermonter's best response to this strategy, we calculate the Vermonter's expected utility from Stag and from Rabbit, given that the Flatlander is playing (Stag, Rabbit). $Eu_{\text{Vermont}}(\text{Stag}, (\text{Stag}, \text{Rabbit})) = 75p$ and

$Eu_{\text{Vermont}}(\text{Rabbit}, (\text{Stag}, \text{Rabbit})) = 60$. Stag is thus a best response if $75p \geq 60$, or $p \geq 4/5$, and so $(\text{Stag}, (\text{Stag}, \text{Rabbit}))$ is a Bayesian Nash equilibrium if $p \geq 4/5$. Note that this is the exact same calculation that we did in the second column of the Bayesian normal-form game—but we needed to compute only two expected payoffs to figure out this equilibrium. Likewise, we can ask if there is a Bayesian Nash equilibrium in which the Vermonter plays Rabbit. From the left payoff matrix in [figure 11.6](#) (the game between the Vermonter and Hunter), we see that the Hunter’s best response to Rabbit is Rabbit and, from the right matrix (the game between the Vermonter and Poser), we see that the Poser’s best response is also Rabbit. Thus, the Flatlander’s best response to Rabbit is $(\text{Rabbit}, \text{Rabbit})$. Is Rabbit the Vermonter’s best response to $(\text{Rabbit}, \text{Rabbit})$? The payoff from Rabbit is 60, regardless of the other player’s strategy or type, and the payoff from Stag is 0 if the other player is choosing Rabbit, regardless of type, so Rabbit is indeed the best response. Thus $(\text{Rabbit}, (\text{Rabbit}, \text{Rabbit}))$ is a Bayesian Nash equilibrium, and no restrictions on p are necessary for this to hold.

Continuous Strategies

When the players’ strategy sets are continuous, we cannot write the Bayesian game in normal form. Instead, we can approach the situation by calculating the best response function for each *type* and then finding the intersection of the best response functions. In other words, we’re essentially treating each type as a distinct player and determining how that person maximizes his payoffs (given the preferences of his type and the distribution of types that the other players could be).

Recall the pricing game between Starbucks and Peet’s, which we looked at as both a simultaneous-move game and sequential-move game in chapter 8. In the simultaneous version, Starbucks sets a price, s , for its coffee at the same time that Peet’s sets its price, p , and the two firms’ subsequent profits are given by $u_s = 4s - s^2 + ps$ and $u_p = 9p - 3/2p^2 + ps$, respectively. Now imagine that Starbucks is uncertain about the profits that Peet’s will actually achieve once the coffee shops have set their prices. Specifically, it believes that there is a 50% chance that this expression is in fact Peet’s payoff function, and a 50% chance that Peet’s profits are actually lower, so its payoff function is instead $u_p = 3p - 3/2p^2 + ps$.

Figure 11.9 depicts this situation as a Bayesian game. Nature first moves to determine whether Peet's is the High type (with the original payoff function) or the Low type (with the lower payoff function). Because Peet's can condition its strategy on its own type, it can set one price, h , if it is the High type and a different price, l , if it is the Low type. Starbucks can set only a single price, s , without knowing the type of Peet's it is dealing with. Finally, the two firms' payoffs are determined by h and s if Peet's is High and l and s if Peet's is Low. It is important to note that p no longer appears in any payoff functions.

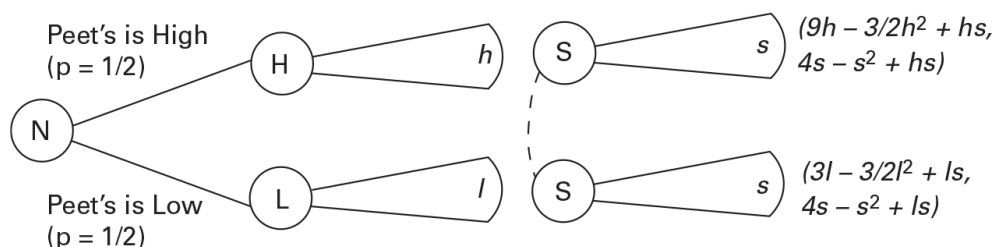


Figure 11.9

The Bayesian game between High Peet's (H), Low Peet's (L), and Starbucks (S) after the initial move by Nature (N).

Let's begin by deriving the best response functions for both types of Peet's. For High Peet's, we take the derivative of its payoff function, $u_h = 9h - 3/2h^2 + hs$, with respect to the variable under its control, h , and set it equal to zero. We find that its best response to Starbucks' s is $h = 3 + s/3$. Doing the same exercise for Low Peet's using the payoff function $u_l = 3l - 3/2l^2 + ls$, we find that its best response to Starbucks is $l = 1 + s/3$.

While Starbucks cannot condition on Peet's type, it can form beliefs about the price that each type of Peet's will choose. It then acts to maximize its expected payoff given these beliefs about h and l and the knowledge that it will face each type with equal probability. We therefore need to first figure out Starbucks' expected payoff, which is a function of h , l , and s . There is a 0.5 chance that Starbucks' payoff is given by the expression in the top node of figure 11.9, which depends on h and s , and a 0.5 chance that it is given by the bottom expression, which depends on l and s . Thus, Starbucks' expected payoff is given by $Eu_s = 0.5(4s - s^2 + hs) + 0.5(4s - s^2 + ls)$ or, combining terms, $Eu_s = 4s - s^2 + 0.5hs + 0.5ls$. As usual, we take the derivative with respect to the variable under Starbucks' control, s , set it equal to zero, and

solve for s . Doing so, we find that Starbucks' best response to High Peet's h and Low Peet's l is $s = 2 + h/4 + l/4$. This tells us the s that maximizes Starbucks' expected payoff if Peet's chooses h when it is the High type and l when it is the Low type.

We now have three equations (the best response functions associated with Starbucks, High Peet's, and Low Peet's) and three unknowns (s , h , and l), and can find the prices that solve these three equations simultaneously. To do so, we can plug Peet's best responses $h = 3 + s/3$ and $l = 1 + s/3$ into Starbucks' best response $s = 2 + h/4 + l/4$, which gives us $s = 2 + (3 + s/3)/4 + (1 + s/3)/4$ or $s = 3.60$. Plugging $s = 3.60$ back into Peet's best response functions, we arrive at $h = 4.20$ and $l = 2.20$. Our Bayesian Nash equilibrium is therefore $(3.60, (4.20, 2.20))$.

The Acquiring a Company Game and the Market for Lemons

Let's return now to the acquiring a company game from the beginning of this chapter, in which a buyer has the opportunity to purchase a company of unknown value v (an integer between 0 and 100, each with equal likelihood) from an owner who knows its quality with certainty. In this case, we can think of the company's value as the owner's type, which is determined by Nature at the beginning of the game. The buyer offers a bid, b , while at the same time the owner decides the *reserve price*, or minimum bid she is willing to accept. If the bid is greater than or equal to the reserve price, the buyer acquires the company at price b , for a final payoff of $1.5v - b$, while the owner's payoff is b . If the bid is lower than the owner's reserve price, the buyer's payoff is 0 and the seller retains the company, which she values at v . Since the owner knows her own type, she can condition her reserve price on the actual firm's quality. What is the minimum willingness to accept for an owner of quality type v_i ? Rejecting the bid gives a payoff of v_i , while accepting yields a payoff of b . Therefore, the owner prefers to accept if and only if $b \geq v_i$, and an owner of type v_i chooses a reserve price of v_i .

Now, how much should the buyer bid? While the buyer does not know the true value of the firm, he knows that each integer from 0 to 100 is equally likely and so, on average, the firm's quality will fall in the middle of this interval, at $v = 50$. In other words, on average, the firm is worth 50 to the seller and 75 to the buyer. Perhaps the buyer reasons that if he offers to pay between 50 and 75, the owner is likely to accept it, and both will make a

profit. But wait—this reasoning ignores the fact that the owner will only actually accept bids that are at least as high as her value. Conditional on the owner accepting the bid of 50, we know that the firm can't be worth more than 50 to the owner and may be worth far less. Once we observe the owner accepting the bid of 50, we need to update our assessment of the company's expected value. We now know that the value must fall between 0 and 50 and, on average, is in the middle of this range, at $v = 25$. This means that the expected value to the buyer is $1.5(25) = 37.5$ and, since he just agreed to pay 50 for it, he will earn a negative expected payoff of 12.5. This phenomenon is known as the *winner's curse*: discovering that your bid won is actually an unfortunate turn of events in this case, since it means that you likely overbid and lost money. We will revisit the winner's curse in various auction environments in which bidders do not know how much the item is worth to them in chapter 13. In general, the trick to avoiding the curse is to think through exactly what it means to win with a specific bid before actually submitting that bid as an offer, so that learning you won doesn't come as an unfortunate surprise.

The trick for bidders in the acquiring a company game is to recognize that the owner's strategy is conditional on her type, so the fact that she is willing to accept a bid provides additional information about what the true value actually is. In general, if the buyer's bid of b is accepted, then the value must be between 0 and b . Therefore, the expected value is not 50, but rather $b/2$, and it is thus worth $3b/4$ to the buyer in expectation. The buyer's expected payoff, conditional on the owner accepting the bid, is $3b/4 - b = -b/4$. The buyer thus maximizes his expected payoff by bidding 0, since any positive bid will lead to an expected loss. The Bayesian Nash equilibrium in this game thus has the owner setting a reserve price equal to v_i and the buyer bidding 0.

This game is an example of the classic "lemons problem," first described by the economist George Akerlof in 1970. Imagine that there are used cars of various qualities on the market. If buyers and sellers were always able to detect the quality level of any specific car, then, at each quality level, there would be buyers willing to pay more than the sellers' willingness to accept, and efficiency-enhancing trades would take place. Or, if neither the buyers nor the sellers knew the underlying quality of any particular car, then trades could still occur at a price based on the market participants' assessment of the average quality in the market. The problem is that information about the car's

quality is asymmetric: the sellers have better knowledge of the car's quality than the buyers do. Buyers who would be willing to pay top dollar for a car they knew to be of high quality will no longer be willing to do so, and will instead make offers assuming that the car is of average value. But sellers of the highest-quality cars would not be willing to accept this, since they know that their cars are worth more. But if sellers of high-quality cars are not willing to bring their cars to market, then this drives down the value of the average car that *is* for sale on the market. Buyers will then assume that the average car being offered for sale is inferior and would prefer to adjust their offer downward accordingly. But now sellers of average-quality cars may not find it in their interest to sell their car to a buyer who assumes that it is inferior. As a result, only low-quality cars are sold, for low prices, and mutually beneficial trades of high- and medium-quality cars will go unrealized. This phenomenon is also known as *adverse selection*: the information asymmetry leads to disproportionate selection into the market by types who know they can profit most off those with less information, and thus a withdrawal from the market of players with incomplete information. Other applications include the least healthy individuals choosing to purchase health insurance or the riskiest drivers purchasing more inclusive car insurance.

Behavior in Games

Failures of Contingent Thinking

A variety of studies in economics and psychology have demonstrated that people struggle to engage in contingent reasoning, or hypothetical thinking, on the basis of future events, such as the realization that one's bid has been accepted in the acquiring a company game. This game (also known as the takeover game) was first run in the lab by Samuelson and Bazerman (1985). While the unique Bayesian Nash equilibrium involves the bidder offering 0 for the company, only 7% of participants actually did so. Instead, the vast majority bid between 50 and 75. Transcripts of the buyers' explanations reveal that their logic is consistent naive reasoning about the value of the firm to the two players, without taking into account the private information that the seller possesses. The bidder reasons that, on average, the company will be worth 50 to the seller, and thus 75 to themselves, and concludes that bids in this range will therefore be mutually beneficial, completely ignoring the seller's ability to condition her decision on the true value of the company.

This finding has been replicated in several other experiments, and participants don't seem to learn the flaw in their logic even when they are given feedback on the outcomes over the course of 100 trials (Selten, Abbink, and Cox, 2005). You may wonder whether learning is impeded because people just don't like bidding 0, since they are foregoing the opportunity to acquire the company and make a profit. However, Selten et al. also found that participants are just as susceptible to overbidding when the game's parameters are altered such that the optimal bid is greater than 0. The one intervention that does seem to enable buyers to avoid overbidding is having them switch roles over the course of the experiment, so that the buyers choose their bids after already having the experience of playing the role of the owner, although the percentage of participants who eventually learned to bid 0 is still only 37% (Ball, Bazerman, and Carroll, 1991). This suggests that some subjects don't appreciate the role of the seller's private information until they have actually been in the seller's shoes and used this information to guide their decisions.

Evidence-Based Adaptations of the Standard Model

In 2005, Eyster and Rabin developed an alternative approach to modeling Bayesian games called the *cursed equilibrium* model (where *cursed* refers to the winner's curse). This model takes into account the finding that people seem to underappreciate that other players can condition their behavior on their own private information about their own type. Imagine that players always correctly anticipate the overall probability with which the other players will choose certain actions. In the acquiring a company game, this means that they correctly anticipate that a bid b will be accepted with probability $b/100$ (corresponding to the likelihood that the true value is less than b). However, they may underestimate the correlation between the players' actions and their type. In the most extreme case, a "fully cursed" player assumes that every type plays the same mixed strategy, ignoring their private information. In the example, this means that the buyer believes that every type accepts b with probability $b/100$, regardless of what the player's actual value is. Thus, fully cursed players act as if finding out that one's bid was accepted provides no new information about the value of the firm.

The cursed equilibrium model also allows for intermediate levels of rationality, in which players recognize that types may play strategies that are contingent on their private information but underestimate the extent to which

they do so. The parameter χ indicates the extent to which the player is cursed, with $\chi = 0$ indicating that the player is perfectly rational. The player believes that the others play their type-contingent optimal strategies with probability $1 - \chi$, and that all types behave the same way with probability χ (with each playing the average optimal strategy across all types). This can be illustrated with a simple lemons problem. Imagine that half of used cars are “lemons,” worth 0 to both buyers and sellers, and half are “peaches,” worth 2,000 to sellers and 3,000 to buyers. The buyers and sellers each decide whether to transact at a fixed market price. Assume that the market price is 1,000, and so sellers of peaches will never be willing to sell, while sellers of lemons always are. A buyer who is not cursed ($\chi = 0$) will recognize that the sellers’ choice of whether to sell their car is determined by their private information about the cars’ quality, and thus that the probability of getting a peach if they buy is 0, causing them to refrain from purchasing. A fully cursed buyer ($\chi = 1$) correctly anticipates that half of the sellers will sell, but assumes that this likelihood is the same regardless of whether the seller has a peach or a lemon. He therefore believes that the probability of getting a peach if he buys is $1/2$, and so the expected value of a car is $1/2(3000) = 1500$ and he should buy. In general, a buyer with parameter value χ believes the likelihood that he gets a peach is $(\chi)(1/2) + (1 - \chi)(0) = \chi/2$. His expected value of a car is thus $\chi/2(3000) = 1500\chi$ and he should buy whenever the expected value is greater than the market price: $1500\chi \geq 1000$, or $\chi \geq 2/3$. In other words, the buyer will choose to purchase a car in this market if he believes that there is a high likelihood that all types play the same strategy (sell with probability $1/2$), ignoring the actual quality of their car.

The cursed equilibrium model has been shown to explain behavior in a variety of Bayesian games, such as auctions with multiple bidders, jury voting, and scenarios in which players must decide how much private information to reveal. However, studies in simpler environments suggest that the problem may be even broader than players in Bayesian games failing to anticipate the cognition and private information of others. More generally, people seem to have difficulty engaging in hypothetical reasoning about future events. For example, Charness and Levin (2009) conducted a simplified version of the acquiring a company game, in which the role of the owner was removed. Instead, buyers confront 100 cards, labeled 0 through 99, which are displayed face down and correspond to the possible firm values.

The buyer submits a bid and then flips a card. If the value shown on the card is less than or equal to the bid, the subject earns 1.5 times the value of the card minus her bid; otherwise, she earns nothing. Note that this scenario is identical to an acquiring a company game in which the owner always accepts bids that are at least as high as her value and rejects the others, except now there is no other player whose logic the bidder needs to anticipate. It is purely a game of chance. However, the authors found that overbidding is still rampant in this environment, indicating that people have trouble recognizing that a future event (the outcome of the card flip) can be informative for their current decision making (choosing a bid).

Charness and Levin also included an even simpler treatment, in which there were only two card values, 0 and 99, although subjects could still bid any amount. First, notice that submitting a bid that is greater than 0 and less than 99 is a bad plan, as it leads to zero earnings if the 99 card is selected (since the bid doesn't exceed the value and the company doesn't change hands) and negative earnings if the 0 card is selected (since you must pay your positive bid). Therefore, the only reasonable options are to bid 0, which leads to a payoff of 0, or 99, in which case you have a 50% chance of earning $1.5(99) - 99 = 49.5$ and a 50% chance of earning $1.5 * 0 - 99 = -99$. The latter is clearly an unattractive gamble (and fewer than 9% choose to play such a gamble when it is presented as a lottery outside the context of this game). The authors found that, as expected, the vast majority of participants bid either 0 or 99. However, they choose 0 and 99 with roughly equal frequency, again demonstrating that a substantial portion of people have difficulty with contingent reasoning on future events.

Will Information Be Revealed?

We've seen that having private information about one's own type does not necessarily make a player better off: in some cases, it may cause the other player to avoid cooperating or transacting in situations that she otherwise would if her information were complete. Now imagine that players had some way of credibly, verifiably revealing their type to those they are interacting with. For instance, Kylo can allow Rey to see his true intentions, Peet's can allow a third party to conduct a review of its production and market research to share with Starbucks, the owner of a used car can share the car's complete history and allow it to be inspected by a mechanic of the buyer's choosing, or

the owner of the company can allow the prospective buyer to complete a full audit prior to submitting his bid. Would they do so?

In the game in which Kylo is a Jerk with probability $3/5$, Mensch Kylo would absolutely prefer Rey to know his type. In the face of incomplete information, Rey chooses to Defect. However, if she knew that Kylo were the Mensch type, she would switch to Cooperating as well, and both would be better off. Jerk Kylo is indifferent between Rey knowing his true type or not, since she will Defect either way. But if a means of credibly sharing types exists, then notice that Jerk Kylo's type will be revealed whether he shares it or not. Since Mensch Kylo always reveals his type, the act of not revealing tells Rey that she must be dealing with the Jerk type.

In the acquiring a company game, any seller who knows her value to be positive would choose to share that information with the buyer, since the buyer would bid 0 without that revelation. What about in the Peet's-Starbucks game? When Starbucks has incomplete information about Peet's profits, we found that Starbucks would choose a price of \$3.60, while High Peet's would choose a price of \$4.20. This yields a payoff of \$26.46 for High Peet's in equilibrium. What if Starbucks knew Peet's type for sure? In chapter 8, we looked at the complete information game between Starbucks and High Peet's and found that Starbucks would set a price of \$4.20 in equilibrium, while Peet's set a price of \$4.40. Peet's profit would be \$29.04 in the situation in which Starbucks knows Peet's is the High type. Therefore, High Peet's would prefer Starbucks to know its type. What about Low Peet's? In the Bayesian Nash equilibrium, we found that Low Peet's would choose a price of \$2.20, which yields a payoff of \$7.26 for Low Peet's in the situation where Starbucks does not know that Peet's is the Low type. In a situation of complete information, however, Low Peet's would set a price of \$2 while Starbucks chooses \$3, which is associated with a payoff of only \$6 for Peet's. Therefore, Low Peet's would *not* want to reveal its type. Once again, though, it would be hard for Low Peet's to keep this information private if it is clear to Starbucks that Peet's *could* reveal it. Since High Peet's would always prefer to share information with Starbucks, the fact that Low Peet's does not disclose tells Starbucks everything it needs to know about Peet's true profit function.

A classic example of this type of unraveling of information comes from the restaurant industry. In many cities, you may have noticed that restaurants post their health inspection rating (e.g., A, B, or C) in the window. When the health

inspectors began giving out these cards in Los Angeles, it was not mandatory to post them. Rather, restaurateurs were simply expected to have them on hand if anyone requested it. But who do you imagine would want to proudly display their ratings in the window? The A-rated restaurants would not wish to be mistaken for B or C restaurants, and so they began visibly displaying their cards. But if A restaurants typically display their cards, then B restaurants would want to display theirs as well: if they didn't post their rating, they would not be mistaken for an A restaurant (which would have its A card in the window), but they could be mistaken for a C restaurant. Eventually, even C restaurants were displaying their signs, perhaps to indicate, "Well, hey, at least we passed."

Therefore, when there is a verifiable, credible means of sharing information, private information tends to unravel and all parties become fully informed. The problem is that there typically is not such a means of credibly revealing private information. (Unfortunately, the health inspector of Los Angeles does not generally provide players with cards verifying their types that they could present to others prior to interacting.) In this case, how can players convince the others that they truly are of a certain type? One possibility is to take a costly action that only players of a certain type would be willing to do. For instance, Mensch Kylo could make a large donation to charity. If Jerk Kylo would never do such a thing, even if it meant tricking Rey into Cooperating, then Mensch Kylo may find it in his interest to make the donation in order to credibly signal his type to Rey. Likewise, the owner of a high-quality used car may be willing to offer a warranty or guarantee to the buyer. Doing so would be potentially costly to them, since even a high-quality car could experience problems. But it would be far more costly to the seller of a lemon, who would never wish to offer such a guarantee, and so the willingness to stand behind the car's performance may be an effective means of signaling its high quality. In the next chapter, we will consider sequential games of incomplete information, in which the initial move that a player makes may signal something to the other players about her true type and enable the others to update their beliefs about whom they are dealing with.

Exercises

(1) Consider the situation of incomplete information shown in [figure 11.10](#). There is a 0.5 probability that the Column player is type 1, and that Row and

Column are playing the game on the left, and a 0.5 probability that the Column player is type 2, and they are playing the game on the right.

		Column Type 1				Column Type 2	
		Left	Right			Left	Right
Row	Up	(10, 10)	(0, 4)	Row	Up	(0, 20)	(0, 2)
	Down	(4, 4)	(6, 6)		Down	(4, 14)	(6, 6)

Figure 11.10

- (a) Write this interaction as a Bayesian game in extensive form.
 (b) Write this interaction as a Bayesian game in normal form.
 (c) Find the pure-strategy Bayesian Nash equilibrium or equilibria.
- (2) In figure 11.11, Midge and Joel are simultaneously deciding what to do tonight. Midge prefers to go to a comedy club, while Joel prefers a movie. However, Joel's primary concern is that they arrive at the same activity, and he will get a payoff of zero if they mismatch and are not together tonight. With probability 1/2, Midge is interested in meeting Joel as well, and they are playing the game on the left. However, with probability 1/2, Midge would rather avoid Joel and will earn a payoff of zero if they wind up at the same activity, since she will be too annoyed with him to enjoy it. In this case, they are playing the game on the right. (Joel's payoffs are the same in both games, indicating that he doesn't care if she actually wants to meet him or not, which, frankly, explains why she might want to avoid him.) Midge is aware of her own type.

		Interested Midge				Annoyed Midge	
		Club	Movie			Club	Movie
Joel	Club	(10, 20)	(0, 0)	Joel	Club	(10, 0)	(0, 10)
	Movie	(0, 0)	(20, 10)		Movie	(0, 20)	(20, 0)

Figure 11.11

- (a) Write the Bayesian game in normal form.
 (b) Find the pure-strategy Bayesian Nash equilibrium.

- (c) Instead of Midge being interested with probability $1/2$, assume that she is interested with some generic probability p . Write this game in normal form and find the pure-strategy Bayesian Nash equilibrium or equilibria.
- (3) Consider the type of prisoner's dilemma game discussed in chapter 3 and shown in [figure 11.12](#), in which players receive an additional payoff of x if they choose Confess when the other player chooses Confess or if they choose Silent when the other player chooses Silent.

		Blue player	
		Silent	Confess
Red Player	Silent	$(8+x, 8+x)$	$(0, 10)$
	Confess	$(10, 0)$	$(3+x, 3+x)$

Figure 11.12

Imagine that it is known that $x = 4$ for the Red player. However, there is a p chance that $x = 1$ for the Blue player and a $1 - p$ chance that $x = 6$ for the Blue player. The Blue player knows their own type.

- (a) Write this interaction as a Bayesian game in extensive form.
- (b) Write this interaction as a Bayesian game in normal form.
- (c) Find all the Bayesian Nash equilibria.
- (4) Imagine that the president is negotiating with congressional leadership over the federal budget to avoid a government shutdown. Each party simultaneously decides whether to Stand Firm or Cave. With probability p , all parties see a shutdown as the worst case outcome, and they are playing the game of chicken illustrated in [figure 11.13](#) in the matrix on the left. However, with probability $1 - p$, the president actually prefers a shutdown to looking like s/he gave in, in which case, they are playing the game on the right.

		Pragmatic President		Headstrong President	
		Stand Firm	Cave	Stand Firm	Cave
Congress	Stand Firm	$(0, 0)$	$(100, 20)$	$(0, 20)$	$(100, 0)$
	Cave	$(20, 100)$	$(50, 50)$	$(20, 100)$	$(50, 50)$

Figure 11.13

- (a) Is there a Bayesian Nash equilibrium in which the Congress Caves? If so, what is it, and are there any restrictions on p that must hold for this to be an equilibrium?
- (b) Is there a Bayesian Nash equilibrium in which the Congress Stands Firm? If so, what is it, and are there any restrictions on p that must hold for this to be an equilibrium?

(5) Consider a pricing game between Samsung and Dell. The two companies produce similar products and choose their prices simultaneously. Let Dell's price be given by d and Samsung's price be given by s . Let's assume that Dell's marginal cost is 10, and therefore their payoff is given by $u_D(d, s) = (100 - 2d + s)(d - 10)$. There is a 0.25 chance that Samsung's costs will also be 10 this year, so that their payoff is $u_S(d, s) = (100 - 2s + d)(s - 10)$. There is a 0.75 chance that Samsung will have higher costs of 30, in which case their payoff is given by $u_S(d, s) = (100 - 2s + d)(s - 30)$.

- (a) Write the Bayesian game between Low Cost Samsung (who chooses price l), High Cost Samsung (who chooses price h), and Dell (who chooses price d) in extensive form.
- (b) Find the best response functions for Low Cost Samsung, High Cost Samsung, and Dell.
- (c) Find the Bayesian Nash equilibrium of this game.

(6) Elizabeth and Philip are Soviet spies living in the United States, who must simultaneously decide how much effort to put into their work. Elizabeth's payoff depends on how much effort, e , she puts into spying and how much effort, p , Philip puts in, such that

$$u_{Elizabeth} = 48e - e^2 + ep$$

With probability 0.5, Philip is committed to the cause and has a similar payoff:

$$u_{CommittedPhilip} = 48c - c^2 + ec, \text{ where } c \text{ is Committed Philip's strategy.}$$

However, with probability 0.5, Philip is becoming an American and is less devoted:

$$u_{AmericanPhilip} = 24a - a^2 + ea, \text{ where } a \text{ is American Philip's strategy.}$$

How much effort will Elizabeth, Committed Philip, and American Philip choose under the Bayesian Nash equilibrium?

(7) Consider a market for used cars that may be of three different qualities, each with equal likelihood. High-quality cars are worth 5, 000 to the seller and 7, 000 to the buyer; medium-quality cars are worth 3, 000 to the seller and 5, 000 to the buyer; low-quality cars are worth 0 to the seller and 1, 000 to the buyer. Assume that the market price is fixed at 4, 000. The seller always knows the value of his own car.

- (a) What is the expected payoff for the buyer of purchasing a car if: (i) all three types of sellers participate in the market; (ii) only low- and medium-quality sellers participate; (iii) only low-quality sellers participate.

For the remainder of the problem, assume that buyers and sellers simultaneously choose whether to participate in the market at the market price of 4, 000.

- (b) What is the Bayesian Nash equilibrium of this game?
(c) Imagine that buyers are fully cursed (i.e., $\chi = 1$ in the cursed equilibrium model). Will they purchase a car?
(d) For what values of χ would buyers prefer to purchase a car?

¹Note that there is nothing inherently irrational about Kylo being a Mensch or Rey being a conditional cooperator. Different people have different preferences, and we are simply making assumptions about the players' preferences for cooperating with each other. What *would* be irrational is if the players acted contrary to their preferences.

²Harsanyi received the Nobel Prize in Economics in 1994, alongside John Nash and Reinhard Selten, who developed the finitely repeated games theorem that we learned about in chapter 9. See Harsanyi (1967).

³Notice that Kylo's best response does not change if we vary the probability distribution of his type, since he already knows his type with certainty before acting and can condition on it.

12

Signaling: Sequential Games of Incomplete Information

In this chapter, we consider games of incomplete information in which the players act sequentially. We focus on a specific type of two-player sequential game, called a *signaling game*, in which the informed player moves first. The uninformed player can then condition her action on the first player's move, but not his type. If she believes that he is playing a type-contingent strategy, she may be able to infer some information about his type from the action he took. In equilibrium, the second-mover correctly anticipates the first-mover's strategy, rationally updates her beliefs about his type after observing his action, and best responds to these updated beliefs. In turn, the first-mover correctly anticipates how the second-mover will respond to each of his actions and chooses the action that will get him the best outcome, given her response. We consider two broad classes of games. In costly signaling games, the first-mover's action affects his payoff differently depending on his type, such that some types are more willing to take certain actions than others. In cheap talk games, the first-mover's action does not directly affect his ultimate payoff and is thus "all talk."

Imagine that you are making an offer to a job applicant. You know that the applicant is one of two types: a Go-getter, who will be very productive (probability 0.4), or a Slacker, who will not be productive (probability 0.6). You could offer the applicant a Managerial position, which would be a great fit for a Go-getter (your payoff = 100) but a disaster if the applicant turns out to be a Slacker (your payoff = 0). Alternatively, you could offer the applicant a Clerical position, which provides you a payoff of 60 regardless of the person's type. Before making the offer, you observe an action, or signal, taken by the applicant: whether he took Difficult or Easy courses in college. Assume that the extra investment in education has absolutely no productive value to you, the employer, and no intrinsic value to the job applicant. Instead, it is purely costly to the applicant, who will need to work harder and potentially spend longer in school. However, the cost of taking the harder courses varies by type. If the worker is a Go-getter type, the cost of taking Difficult courses is 20. If the applicant is a Slacker, the cost of Difficult coursework is 50. The cost of taking Easy courses is 0 for both types. The

payoff to the applicant is a salary of 100 if you offer the Managerial job and 60 if you offer the Clerical job, minus the cost of his coursework (either 0, 20, or 50 depending on his choice and type). Which job do you offer if you observe that the applicant has taken Difficult courses? What about if he has taken Easy courses?

Now imagine that the applicant does not have the possibility of taking a variety of courses prior to being hired. Instead, the applicant simply writes a cover letter, in which he states that he is a Go-getter or a Slacker. What would you do after seeing each of these possible messages?

Sequential Games of Incomplete Information

These two scenarios are illustrative of signaling games, which will be the focus of this chapter. In both cases, the first-mover's action sends a signal or a message to the second-mover. We therefore call the first player to move the *sender* and the second player the *receiver*. Both games consist of the following stages:

- (1) Nature determines the type of the sender, which is known to the sender but not the receiver.
- (2) The sender acts by sending a signal to the receiver.
- (3) The receiver observes the signal of the sender and can revise her beliefs about the sender's type.
- (4) The receiver acts.

The sender's strategy thus consists of actions (or signals) contingent on each of his possible types, and the receiver's strategy consists of an action contingent on each of the possible signals she could receive. An important feature of these types of games, compared to those discussed in chapter 11, is the ability of the receiver to adjust her beliefs about the sender's type as the game progresses. The receiver begins the game with some initial beliefs about the other player's type, which are based on the probability distribution with which Nature assigns each type. However, she may be able to revise these beliefs after witnessing the sender's action, but before acting herself. In particular, if the sender is playing a type-contingent strategy, then his action contains information about his type, and a rational player will need to update her beliefs about his type accordingly. In contrast, if the player believes that all types of senders play the same action or mix of actions, then witnessing the action provides no new information.

The first game, in which education is costly to acquire, is an example of a *costly signaling game*. The sender's actions affect his payoff, and the cost of taking different actions depend on his type. If certain types are more willing to take an action than others, it may be possible to use this initial action to signal one's type to the receiver. As we'll see, what is important in these games is the extent to which the costs differ. If it is much more costly for the Slacker type to take hard courses—so much so that it wouldn't be worth it, even if it convinced the employer to grant the better job—then there can exist an equilibrium in which Slackers take Easy courses, Go-getters take Difficult courses, and the employer grants the higher-paying job only to those who took hard courses. Even though Difficult courses are also costly for the Go-getter type, he finds it in his interest to take them in order to distinguish himself from a Slacker type, who would never be willing to do so. In contrast, if the hard courses aren't sufficiently costly for the Slacker types, they would choose to take them in order to mimic the Go-getter types, and the employer would, in turn, no longer wish to offer the high-salaried job. If the hard courses are prohibitively costly even for the Go-getter type, they likewise will not be an effective signal. The second game, in which the sender can send an unverifiable message that has no effect on either player's payoff, is an example of a *cheap talk game*. It is costless for either type to claim to be a Go-getter, and he never has to put his money where his mouth is. The message is, therefore, just talk, which is cheap and should be ignored by the boss.

We've already seen how to solve for equilibria in games of incomplete information (in chapter 11, by converting the situation to a Bayesian game and finding the Bayesian Nash equilibrium) and in sequential games (in chapter 8, by using backward induction to find the subgame perfect Nash equilibrium (SPNE)). Unfortunately, these tools are not quite enough to solve for rational play in sequential games of incomplete information. To see why, let's consider again the game from *The Martian* between NASA and the Crew (from chapter 8), this time adding incomplete information. With probability $1/2$, the Crew is Loyal to Matt Damon, and they are playing the same game as we saw before, in which the Crew's most preferred outcome is returning to Mars to get him and NASA sending them the supplies to do so. However, with probability $1/2$, the Crew is Disloyal and unwilling to risk their lives by returning to Mars, even if NASA were to send the supplies. This Bayesian

game is shown in [figure 12.1](#). (This game isn't all that interesting, since NASA simply always prefers to send the supplies if it observes the Crew returning to Mars, regardless of the Crew's type, and so the incomplete information has little relevance. However, it illustrates the trouble with attempting to use our existing tools to solve for rational play.)

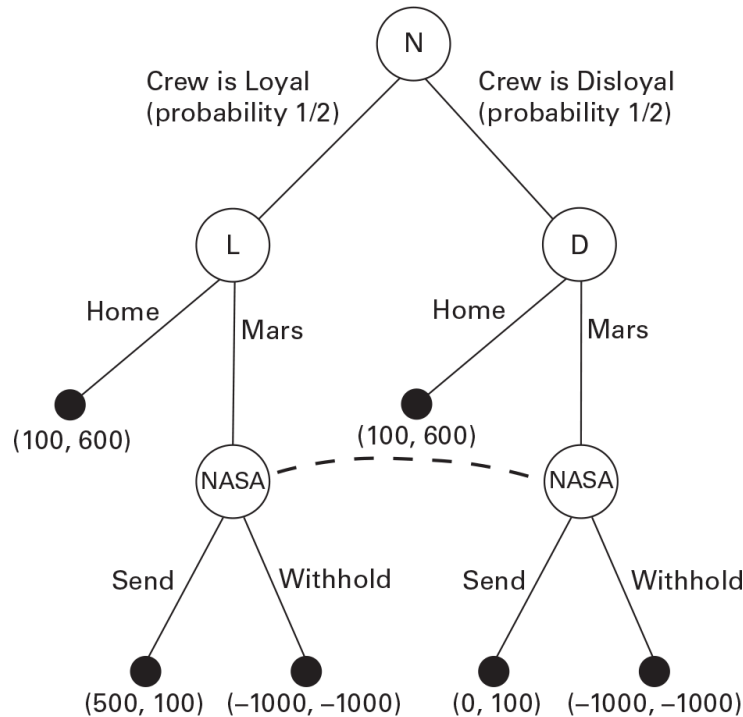


Figure 12.1

The Martian game as a Bayesian game between a Loyal Crew (L), a Disloyal Crew (D), and NASA after the initial move by Nature (N).

In chapter 11, we solved for the Bayesian Nash equilibrium in a simultaneous game by writing the game in normal form and looking for mutual best responses. In the Martian scenario, we find that there are two pure-strategy Bayesian Nash equilibria: ((Home, Home), Withhold) and ((Mars, Home), Send). You can confirm this in the normal-form version of the game shown in [figure 12.2](#). We thus run into the same issue we encountered in chapter 8: the first equilibrium does not seem plausible in the sequential game, since it has NASA playing a strategy in which it would Withhold supplies from its astronauts if ever called on to act. Since NASA would always prefer to send supplies if put into this situation—regardless of their

beliefs about NASA's type—this strategy is not credible and does not satisfy our condition of sequential rationality.

		NASA	
		Send	Withhold
Crew	(Home, Home)	(100, <u>600</u>)	(<u>100</u> , <u>600</u>)
	(Home, Mars)	(50, <u>350</u>)	(−450, −200)
	(Mars, Home)	(<u>300</u> , <u>350</u>)	(−450, −200)
	(Mars, Mars)	(250, <u>100</u>)	(−1000, −1000)

Figure 12.2

The Bayesian Martian game in normal form. Crew's strategy specifies (action if Loyal, action if Disloyal).

In chapter 8, we solved for equilibria that satisfied sequential rationality: that is, at every information set, the player best responded given his beliefs about how the rest of the game would play out. To do so, we identified subgames, or decisions and interactions that could be treated independently from the rest of the game, and backward inducted over subgames. However, this process gets us nowhere in the interaction in [figure 12.1](#), since the first move is Nature's, which the second player does not know. Even if NASA observes the Crew's action, it is still missing information about the Crew's type, and so we can't simply consider NASA's decision conditional on the Crew choosing Mars as a decision that is independent from the rest of the game. Put differently, the only subgame in [figure 12.1](#) is the entire game itself. This means that both of the equilibria we found satisfy the definition of subgame perfect Nash equilibria and we will need an equilibrium concept that takes into account the specifics of sequential games of *incomplete information* in order to rule out strategies that are not sequentially rational.

Rather than focusing on subgames, we will make an assessment of the player's best response at each of her information sets. However, this assessment cannot be considered independently from the game that came before. Before we can calculate a player's best response at an information set, we will first need to consider what that player's belief is about the likelihood of being at each node in the information set. In the [figure 12.1](#) game, this means asking: given that NASA has reached its information set (i.e., the Crew

has chosen Mars), what is the probability that we are at the left node (the Crew is Loyal) versus the right node (the Crew is Disloyal). We can then calculate the best response given those beliefs (and how they anticipate the rest of the game will play out). What is important is that both (1) the players are best responding to their beliefs and (2) those beliefs are rational (they are based on correct beliefs about the other player's strategies and the rational updating of those beliefs according to Bayes' rule, as discussed next).

Updating Beliefs

In the previous chapter, the beliefs that a player had about the other players' types were simply given by the probability distribution played by Nature. In this chapter, we will refer to these beliefs that are based solely on Nature's move as *prior beliefs*. This reflects the belief that a player has about another player's type prior to observing any actions by strategic players in the game. In sequential games, what matters for a player's behavior is not their prior belief, but their belief when they are called on to move—that is, at the information set they reach. At each information set, we will assess the belief that the player associated with that information set has about the relative likelihood of being at each node. We will refer to this belief as their *updated belief* at that information set. For example, consider again the game in [figure 12.1](#) between the Crew and NASA. NASA's prior belief about the Crew's type is simply given by the probability distribution played by Nature, in which each type is equally likely. In this game, there is one information set associated with NASA, which contains two nodes. What we want to know is: conditional on arriving at that information set, what is the probability that NASA believes that it is at the left node as opposed to the right node? This updated belief may or may not be different from its prior belief, depending on whether it believes the Crew is playing a type-contingent strategy. Thus, the updated belief depends on both the prior belief and the strategy that the other player is assumed to be playing. We will assume that a rational player formulates her updated beliefs at each of her information sets using Bayes' rule. The concept is given in definition 1.

Definition 1 (Updated belief at an information set) *Consider an information set and the player who is assigned to act at that information set. The updated belief is a probability distribution over the nodes in the information set. The probability assigned to each node reflects the player's*

belief that she is at that node, conditional on the information set being reached.

Bayes' Rule

Bayes' rule describes the probability of an event, conditional on another event occurring. For instance, we may use this to calculate the probability that the worker is the Slacker type, given that the worker chose hard courses. The statement of Bayes' rule is given in theorem 1, and it is likely very familiar to anyone who has taken a course in probability or statistics. For our purposes, what is most important is not that you memorize this formula, but that you understand the intuition behind it (this subsection) and how to apply it to sequential games (the next subsection).

Theorem 1 (Bayes' rule) *Consider two events, A and B . Let $p(A)$ refer to the probability of event A occurring, $p(B)$ refer to the probability of event B occurring, $p(B | A)$ refer to the probability of event B occurring, given that event A has occurred, and $p(A \cap B)$ refer to the probability that both events A and B occur. Assume that $p(B) > 0$. The probability that event A occurs given that event B has occurred is given by: $p(A | B) = \frac{p(A \cap B)}{p(B)} = \frac{p(B|A)p(A)}{p(B)}$.*

Imagine that you are a detective investigating a hit-and-run involving a taxi. You're interested in the event that the culprit was a blue taxi, which we'll call event A . In your city, 5% of cabs are blue and 95% are green, so $p(A) = 0.05$. Without knowing anything else, you therefore initially believe that the probability that the culprit drove a blue cab is 0.05. Now imagine that a witness comes forward and makes a sworn statement that the cab was blue. Let's call this event B . Based on your experience, you know that witnesses tend to be accurate 80% of the time. Thus $p(B | A) = 0.80$. Stated in words, the probability that the witness says the cab is blue, given that it is actually blue, is 0.8. Now what do you believe is the probability that you should be seeking a blue cab? You don't want to ignore the witness and stubbornly cling to your initial belief of 0.05. Nor do you want to fully discard your initial assessment and rely only on the witness's testimony to conclude that the probability is 0.80. Instead, you want to take your initial belief of 0.05 and *update* it with the new information, such that your resulting belief takes into account both the underlying distribution of cab types and what you know about the probability that the witness is correct. That is, we wish to compute $p(A | B)$.

Consider [figure 12.3](#), which diagrams the situation. First, the taxi causing the accident is blue with probability 0.05 and green with probability 0.95, simply based on the distribution of cabs in the city. (Nature randomly selects a taxi from the population to commit the hit-and-run.) If the taxi is truly blue, the witness correctly identifies it as blue with probability 0.80 and misidentifies it as green with probability 0.20. Likewise, if the taxi were green, the witness would state that it is green with probability 0.80 and that it is blue with probability 0.20. Finally, we can use the multiplication rule (which is reviewed in the appendix) to calculate the probability of each of the four possible outcomes listed at the end nodes. The probability that the cab was actually blue *and* the witness identifies it as such is $0.05 \times 0.80 = 0.04$. We can make the same calculations for the other three possible outcomes. However, we know that the witness has stated that the cab is blue, which happens with probability $0.04 + 0.19 = 0.23$. In other words, out of 100 such incidents, only 23 would involve the witness claiming to have seen a blue cab. In how many of those cases would the cab actually be blue? Only in 4 of the 23, or 17% of the time:

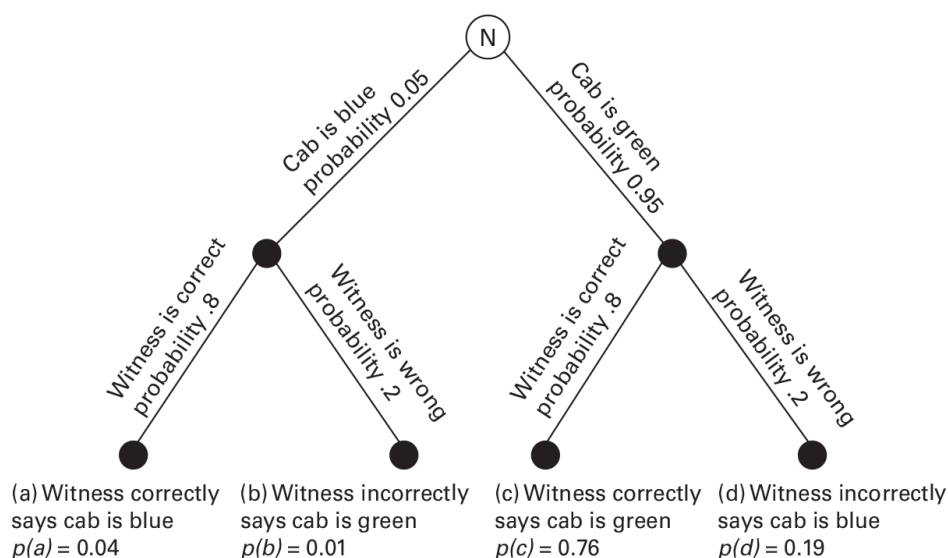


Figure 12.3

The prior probability of a blue cab is 0.05. Once the witness gives the statement, there are four possible situations, labeled (a) through (d) along with the probabilities of each. Conditional on the witness saying blue, know that we are in situation (a) or (d). The relative likelihood that we are at (a) vs. (d) is $0.04/(0.04 + 0.19) = 0.17$. Thus the probability that the cab was actually blue conditional on the statement of blue is 0.17.

Probability that the cab is blue, given the witness's statement of blue =

$$\frac{\text{probability witness correctly says cab is blue}}{\text{probability witness says cab is blue}} = \frac{0.04}{0.04 + 0.19} = 0.17$$

Learning the witness's statement allows you to update the probability of a blue cab being the culprit from 0.05 to 0.17.

Belief Updating in Sequential Games

The same procedure applies to updating beliefs in sequential games. Instead of the detective taking her initial belief about the distribution of cab types and updating it based on the witness's statement and the probability that the witness is correct, the receiver takes her belief about the initial distribution of sender types and updates it based on the sender's action and the probability that the sender's strategy assigns to that action. Notice that this requires the receiver to first form beliefs about the strategy the sender is playing for each of his possible types. This is analogous to the detective's belief that witnesses provide accurate information 80% of the time. Without this link, the sender's action wouldn't provide well-defined information about his type (in the terminology of Bayes' rule, we wouldn't know the $p(B | A)$ piece, which is the probability that the sender takes a specific action conditional on his type).

In sequential games, we want to separately consider each information set for the receiver. Within that information set, we ask: conditional on reaching this information set, what is the probability of being at each node within the set? Bayes' rule is rewritten in theorem 2 for the specific context we will be considering.

Theorem 2 (Bayes' rule applied to sequential games) *Consider the event that the sender is of type A and the event that information set B has been reached. Let $p(A)$ refer to the probability that Nature assigns the sender to type A and $p(B)$ refer to the probability of information set B being reached. $p(B | A)$ refers to the probability that the information set B is reached conditional on the sender being type A —in other words, the probability that a sender of type A takes the action necessary to lead to information set B according to the strategy that the sender is playing. Assume that $p(B) > 0$; that is, that the information set is reached with nonzero probability given the strategy played by the sender. The probability that the sender is type A given that information set B is reached is given by $p(A | B) = \frac{p(A \cap B)}{p(B)} = \frac{p(B|A)p(A)}{p(B)}$.*

Consider again the Martian game with incomplete information, which has only one information set for NASA. Initially, NASA believes that the Crew is Loyal with probability $p = 0.5$. However, NASA acts only after witnessing the Crew return to Mars. What we need to know is: what is NASA's updated belief about the Crew's type once it reaches its information set? In other words, *conditional* on having reached its information set (i.e., witnessing the Crew choose Mars), what is the probability that it is at the left node (dealing with Loyal Crew) versus the right node (dealing with Disloyal Crew)?

This depends on the strategy that NASA believes the Crew to be playing. There are three examples depicted in [figure 12.4](#). First, let's assume that NASA believes the Crew is playing (Mars, Home). This means that conditional on being Loyal, the Crew plays Home with probability 0 and Mars with probability 1; conditional on being Disloyal, the Crew plays Home with probability 1 and Mars with probability 0. This scenario is depicted in panel (1), which gives the probability with which the Crew plays each of its branches, conditional on its type. After the Crew acts, there are four possible points in the tree where we could be, labeled a, b, c, and d. We can determine the probability with which each point is reached using the multiplication rule, by multiplying the probabilities on the two branches leading to the point. Point a is reached if the Crew is Loyal (probability 0.5) and, conditional on being Loyal, chooses Home (probability 0, since we're assuming a Loyal Crew always chooses Mars). Thus, the probability of arriving at point a is $0.50 \times 0 = 0$. We can similarly find that the probability of arriving at point b is 0.5 (since they are Loyal with probability 0.5 and then always choose Mars), arriving at point c is 0.5 (since they are Disloyal with probability 0.5 and then always choose Home), and arriving at point d is 0 (since they never choose Mars if they are Disloyal). Now, conditional on arriving at the information set, what is the probability that the Crew is Loyal? In other words, what is the relative likelihood of being at point b (associated with NASA's left decision node) instead of point d (associated with NASA's right decision node)? The answer is $\frac{.5}{.5+0} = 1$. Since the Crew returns to Mars only if they are Loyal, witnessing the choice of Mars tells NASA that it is dealing with the Loyal Crew with certainty. In this case, NASA is able to update its beliefs about the Crew's type fully.

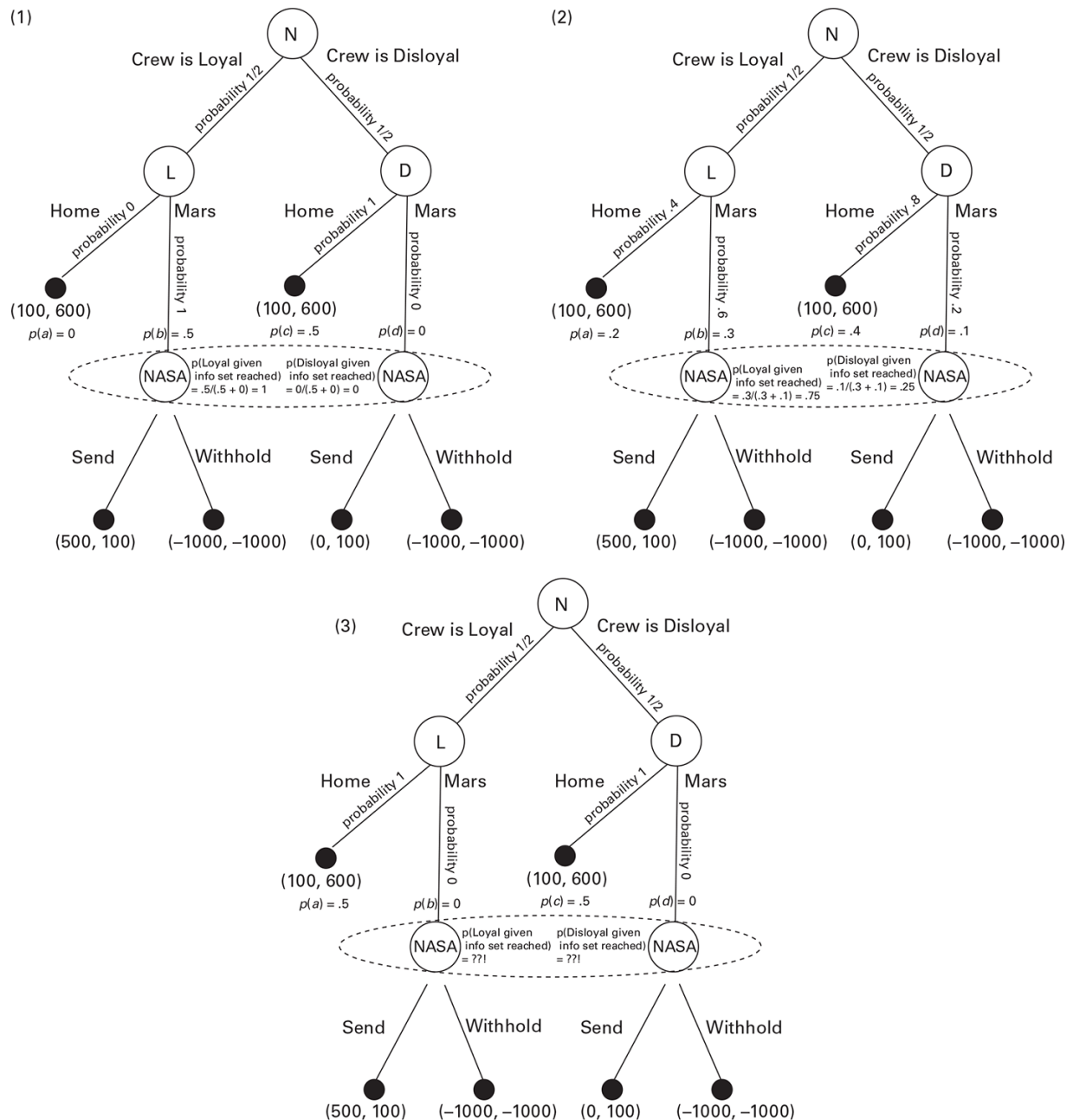


Figure 12.4

The Martian game as a Bayesian game between a Loyal Crew (L), a Disloyal Crew (D), and NASA after the initial move by Nature (N). In each panel, the Crew plays a different strategy, and NASA updates its beliefs about the Crew's type after observing that its information set is reached.

Now, assume that instead NASA believes the Crew is playing a mixed strategy, in which they choose Home with probability 0.4 if Loyal and Home with probability 0.8 if Disloyal. This scenario is depicted in [figure 12.4](#) panel (2), and we can once again compute the probability of arriving at points a, b,

c, and d by multiplying the probabilities on the branches leading to them. Once we arrive at NASA's information set, we know that we must be at either point b (probability 0.3) or d (probability 0.1). Therefore, the relative frequency with which we're at point b is $\frac{0.3}{0.3+0.1} = 0.75$. Conditional on being at this information set, NASA believes that there is a 0.75 chance that the Crew is Loyal.

Finally, let's assume that NASA believes that the Crew is playing (Home, Home). What can NASA then infer about the Crew's type if it reaches its information set—that is, they see the Crew go to Mars? Pretty much anything could be a reasonable conclusion in this case. It is unclear what inference NASA should draw about the Crew's type if it sees the Crew taking an action that neither type ever should. As we see in [figure 12.4](#) panel (3), the probability of being at point b is 0 and of being at point d is also 0, such that the relative frequency of being at point b is $\frac{0}{0+0} = \text{undefined}$. Note the requirement in Bayes' rule that $p(B)$ must be nonzero in order for the formula to hold. Since the Crew plays a strategy in which the probability of reaching the information set is zero, this assumption is violated. Therefore, Bayes' rule does not speak to this situation, and there are no specific beliefs that we would require the receiver to have in order to be consistent with “rational” Bayesian updating.

Perfect Bayesian Equilibrium

Over the course of the book thus far, we have seen games that can vary in whether they are played simultaneously or sequentially and whether the players have complete or incomplete information. [Figure 12.5](#) depicts the main equilibrium concept in each of these four classifications of games. In simultaneous games of incomplete information, we saw that the Bayesian Nash equilibrium was directly analogous to the Nash equilibrium, while allowing players to take on multiple possible types and condition their behavior on their own type. In sequential games of complete information, we use the subgame perfect Nash equilibrium. This is a somewhat more stringent than the Nash equilibrium, in that it requires sequential rationality: each player must play a best response at every information set, even those that are not reached in equilibrium.

	Complete Info	Incomplete Info
Simultaneous	Nash equilibrium	Bayesian Nash equilibrium
Sequential	Subgame perfect Nash equilibrium	Perfect Bayesian equilibrium

Figure 12.5

Solution concepts in different types of games.

The primary equilibrium concept in sequential games of incomplete information is the *perfect Bayesian equilibrium*. It is analogous to the subgame perfect Nash equilibrium, in that it too requires players (and types) to be sequentially rational and best respond at every information set. However, the beliefs that players possess about the types of others play a central role in the perfect Bayesian equilibrium. In fact, the updated beliefs at each information set are considered part of the equilibrium itself and must also be “rational.” We refer to this set of beliefs as the *system of beliefs*—it is similar to a strategy profile, but rather than specifying a strategy at each information set, it specifies a belief. In addition to requiring that players play a best response at each of their information sets, the perfect Bayesian equilibrium requires that players also formulate updated beliefs in accordance with Bayes’ rule whenever possible, given the strategies played by others, and act on these updated beliefs. The system of beliefs and perfect Bayesian equilibrium are formally defined in definitions 2 and 3.

Definition 2 (System of beliefs) *A system of beliefs is a collection of updated beliefs, one at each information set in the game. In other words, at each information set, the system of beliefs indicates a probability distribution over nodes. The probability assigned to each node reflects the belief of the player associated with the information set that he is at that node, conditional on the information set being reached.*

Definition 3 (Perfect Bayesian equilibrium) *A perfect Bayesian equilibrium is a system of beliefs and a strategy profile satisfying two conditions:*

Sequential rationality: Each player best responds at each information set.

Consistency of beliefs: At each information set that is reached with positive probability, the player’s updated belief is consistent with the strategy profile and Bayes’ rule.

The consistency of beliefs condition imposes requirements on beliefs only at information sets that are reached with positive probability, given the strategy profile. Some information sets may be *off the equilibrium path*: they are reached with zero probability, provided that everyone sticks to the strategy profile. This was the case in panel (3) of [figure 12.4](#), where the Crew was playing the (Home, Home) strategy, and thus the probability of NASA's information set being reached was zero. In this case, there are no specific requirements regarding the belief that a player must hold under the perfect Bayesian equilibrium in order for the consistency of beliefs assumption to be satisfied. (We note, however, that some beliefs cannot occur in specific perfect Bayesian equilibria because they would cause players to want to deviate—however, this is not a result of the beliefs being “irrational.”) Game theorists have considered refinements that impose restrictions on what the player should believe in this situation. For instance, perhaps NASA should stick with its prior belief, since it has no basis on which to update it, or perhaps it should assume that the Crew is more likely to be Loyal, since a Loyal Crew would be more willing to play the Mars strategy. However, these requirements are not part of the perfect Bayesian equilibria.

Finding Perfect Bayesian Equilibria in Costly Signaling Games

Recall that in costly signaling games, the informed sender first takes an action, which affects his own payoff, and the uninformed receiver acts after observing this action. There are three broad classifications of equilibria in signaling games, which refer to the extent to which different types of senders play different strategies, such that their actions are effective signals of their types.

- *Separating equilibria* The sender plays a strategy in which each type chooses a different action. Thus, the action taken by the sender fully reveals his type to the receiver.
- *Pooling equilibria* The sender plays a strategy in which all types choose the same action or mix of actions. Thus, the action taken by the sender does not reveal his type to the receiver. The receiver's updated beliefs at information sets reached with nonzero probability are the same as her prior belief. The consistency of beliefs assumption is silent about updated beliefs at information sets that are reached with zero probability.

- *Semi-separating equilibria* The sender plays a strategy in which some, but not all, types choose the same action, or in which types choose different mixed strategies. The receiver's beliefs are updated, but not with perfect certainty.

We will focus on finding pooling and separating perfect Bayesian equilibria. Our approach to identifying perfect Bayesian equilibria involves four steps:

1. Set a strategy for the sender.
2. Update the receiver's beliefs at each of her information sets according to the sender's strategy that is set in step 1 and Bayes' rule.
3. Determine the receiver's best response at each information set, given her updated beliefs from step 2. Set this as the receiver's strategy.
4. Check whether the sender's strategy that is set in step 1 is a best response to the receiver's strategy that is set in step 3. If not, there can be no perfect Bayesian equilibrium in which the sender plays the strategy from step 1. If yes, then there exists a perfect Bayesian equilibrium in which the sender plays the strategy from step 1, the receiver plays the strategy from step 3, and the belief system is the updated beliefs that you calculated in step 2.

Job Market Signaling

Let's use this procedure to find the perfect Bayesian equilibria in the job market signaling game introduced at the beginning of the chapter and depicted in [figure 12.6](#). The game is based on a scenario first described by Michael Spence (1973), who pointed out that investing in educational credentials could prove worthwhile for high-ability workers, even if it is common knowledge that the additional education does not improve the workers' skills or provide any other value. Provided that educational investment is much less onerous for smart, high-productivity workers than it is for lower-ability workers, high-ability workers can find it an effective means of signaling their type to employers, who will in turn grant them better jobs.

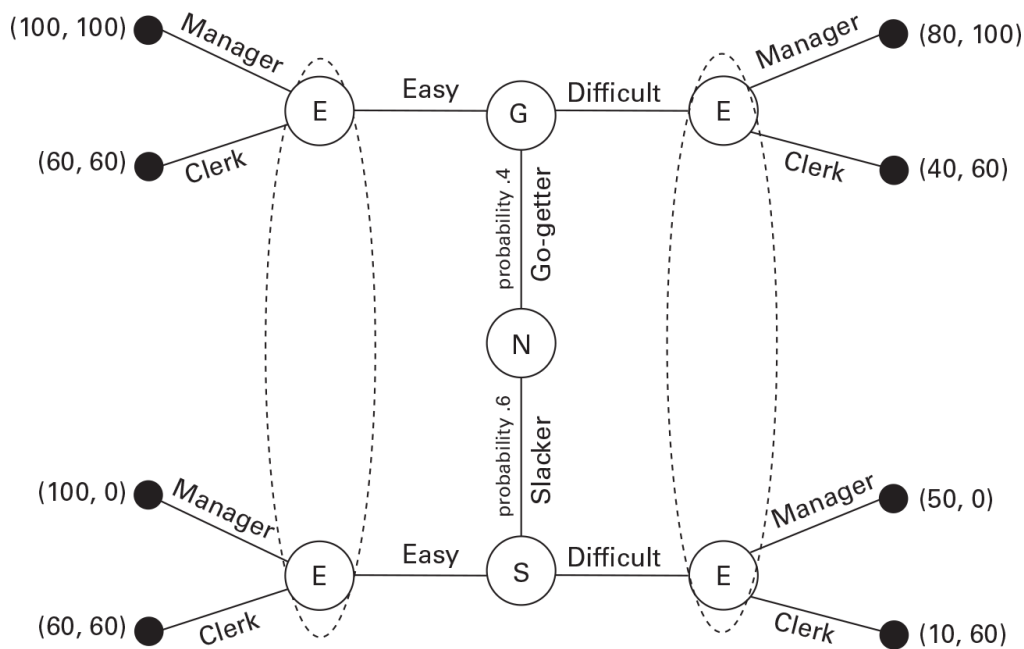


Figure 12.6

The job market game between a Go-getter (G), Slacker (S), and Employer (E) after the initial move by Nature (N).

First, we set a strategy for the sender. Let's begin by investigating whether there exists the most intuitive separating equilibrium, in which Slackers take Easy courses and Go-getters take Difficult courses. This strategy for senders is bolded in [figure 12.7](#). The second step is to determine the receiver's updated beliefs at her two information sets. After observing the signal of Easy courses, the employer knows for sure that she is dealing with a Slacker, since the sender's strategy specifies that only Slackers choose Easy courses. Thus, once she reaches the information set on the left, she believes that she is at the bottom node with probability 1 and at the top node with probability 0. Likewise, if she reaches the right information set, she believes that she is at the top node (dealing with a Go-getter who took Difficult courses) with probability 1 and at the bottom node (dealing with a Slacker who took Difficult courses) with probability 0. The third step is to determine the receiver's best response at each information set, given these updated beliefs. If she reaches the left information set (i.e., witnessed a signal of Easy), she believes herself to be at the bottom node with certainty and is thus choosing between a payoff of 0 (hiring a Slacker to a Managerial job) or 60 (hiring a Slacker to a Clerical job). Her best response is to offer the Clerical job after observing the signal of Easy. Likewise, her best response is to offer the

Managerial job after observing the signal of Difficult, since she believes that she's definitely dealing with the Go-getter applicant. The receiver's best response of (Clerical, Managerial) is highlighted in the game tree. Finally, we must check whether the strategy initially set for the sender, (Easy, Difficult), is in fact the best response to (Clerical, Managerial). Would either type of worker wish to deviate? Under the strategy profile we found, the Slacker earns a payoff of 60. If he were to deviate by taking Difficult courses, the employer would grant him the better job, but he earns a payoff of only 50, since the Difficult coursework carries a higher cost. It isn't worth it to him to pursue Difficult coursework, even if it meant receiving a higher-paying job, and so he would not wish to deviate. The Go-getter would also not wish to deviate, since he currently earns a payoff of 80, but pursuing Easy coursework would mean that he would be offered the lower-paying job and earn a payoff of 60. Therefore, we have found a perfect Bayesian equilibrium. Taking Difficult courses is an effective means for the Go-getter to signal his type, since the Slacker would not wish to mimic him.

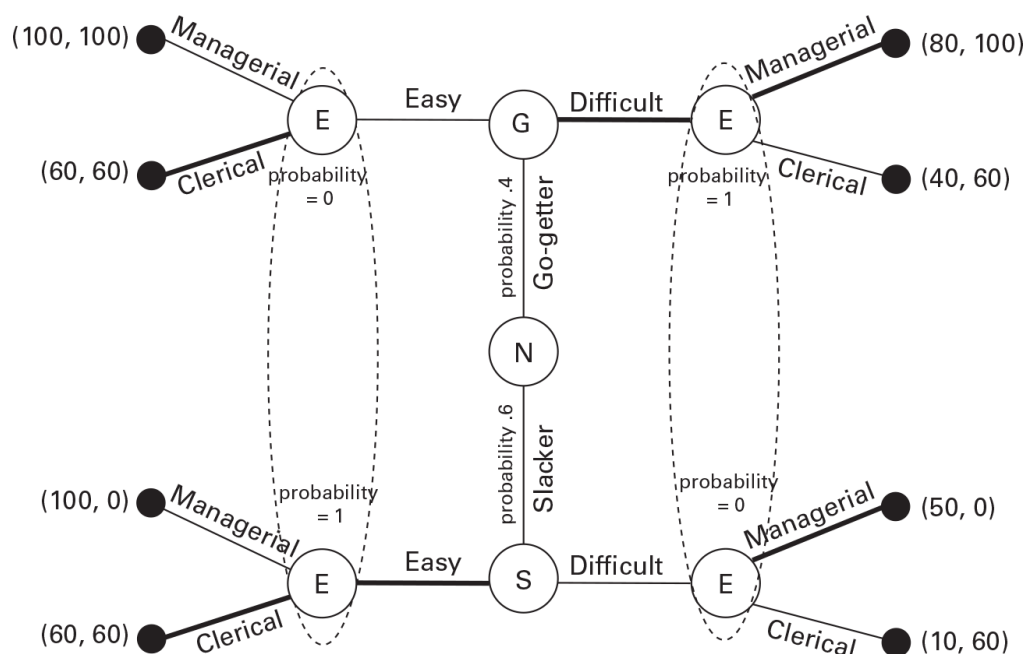


Figure 12.7

A separating equilibrium in which Go-getters take Difficult courses and Slackers take Easy courses. The receiver perfectly updates her beliefs and offers the Managerial job only to those taking Difficult courses.

What if we set the other, less intuitive separating strategy for the sender to start, namely, that the Go-getters choose Easy courses but the Slackers choose Difficult courses? This is shown in [figure 12.8](#). The receiver again perfectly updates her beliefs and best responds to the signal of Easy coursework by offering the Managerial job (knowing that she is hiring a Go-getter) and best responds to the signal of Difficult courses by offering the Clerical job (knowing that she is hiring a Slacker). The Go-getter has no incentive to unilaterally deviate in this situation: he is getting the best of all worlds, taking Easy courses and getting the better job. The Slacker, however, is getting the worst of all worlds: he takes hard courses and gets only the Clerical job (payoff = 10), but if he were to unilaterally deviate, he would take Easy courses, and in so doing, mimic the Go-getter and be given a superior job (payoff = 100). Thus we cannot have a separating equilibrium in which the Slackers are the ones choosing the Difficult courses, since the Slackers would never go along with this situation.

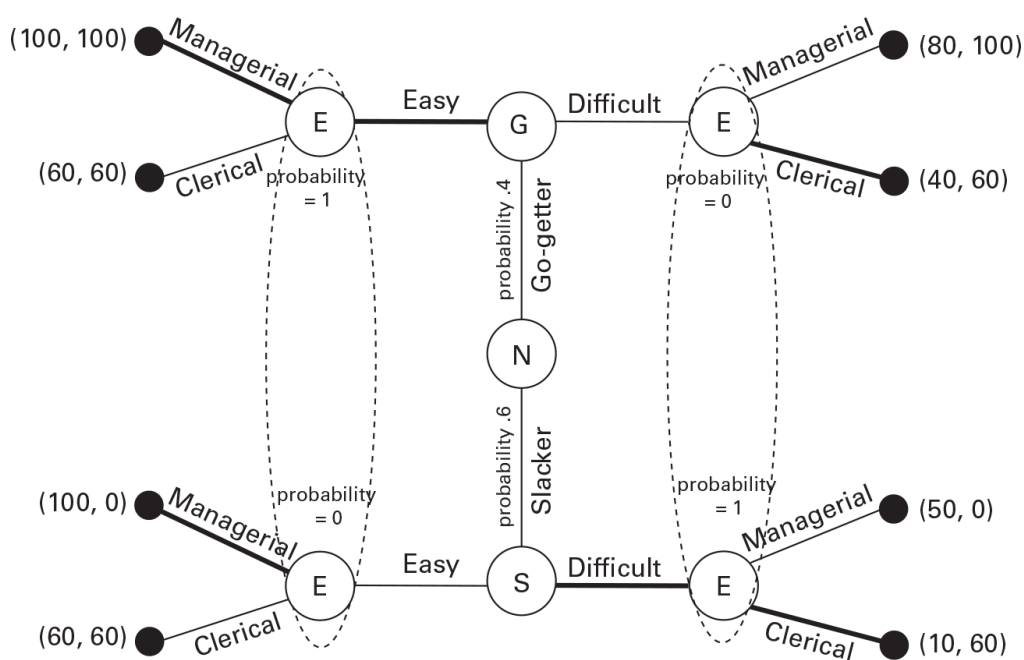


Figure 12.8

There is no separating equilibrium in which Go-getters take Easy courses and Slackers take Difficult courses. The receiver perfectly updates her beliefs and offers the Managerial job only to those taking Easy courses. The Slacker type would wish to deviate in this case.

Now, let's look for pooling equilibria. [Figure 12.9](#) shows the situation in which everyone takes Easy courses. How does the employer then update her

beliefs? You'll notice that [figure 12.9](#) includes two separate cases, which differ in the employer's updated beliefs after seeing the signal of Difficult. For now, just focus on the left side of the figure, which is the same in both cases. If the employer sees the signal of Easy, then her updated beliefs at this information set should be the same as her prior beliefs. Initially, she believed that 40% of applicants were Go-getters and that all applicants take Easy courses, so she should believe that 40% of those taking Easy courses are Go-getters. To calculate her best response at this information set, we must calculate her expected payoff from offering the Managerial job and from offering the Clerical job, given these beliefs. $Eu(Managerial) = 0.4(100) + 0.6(0) = 40$ and $Eu(Clerical) = 60$. Thus, her best response is to offer the Clerical job after seeing the signal of Easy courses.

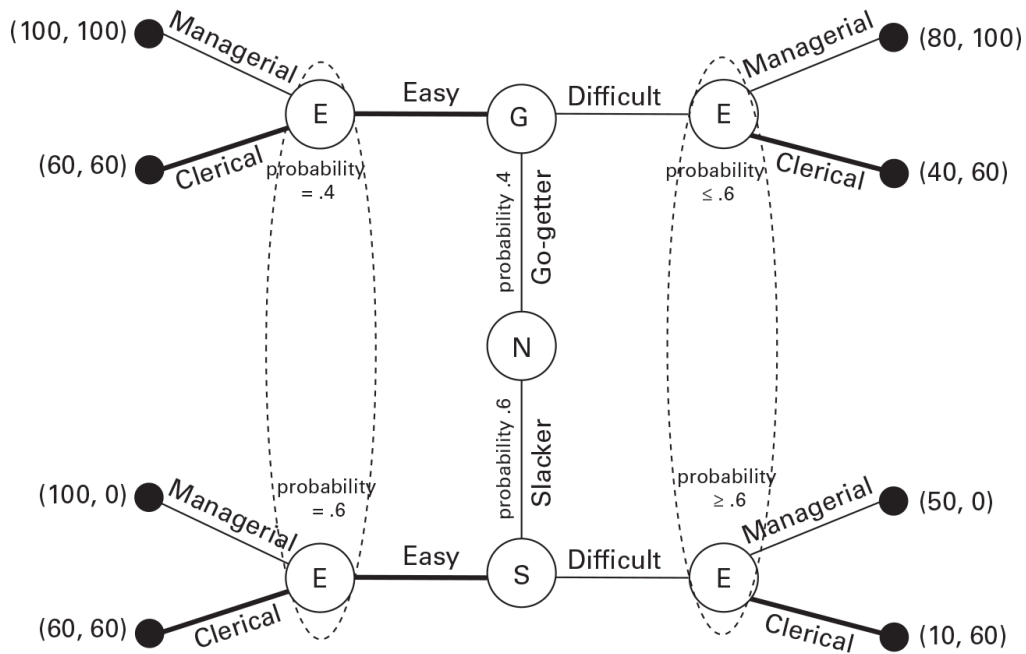
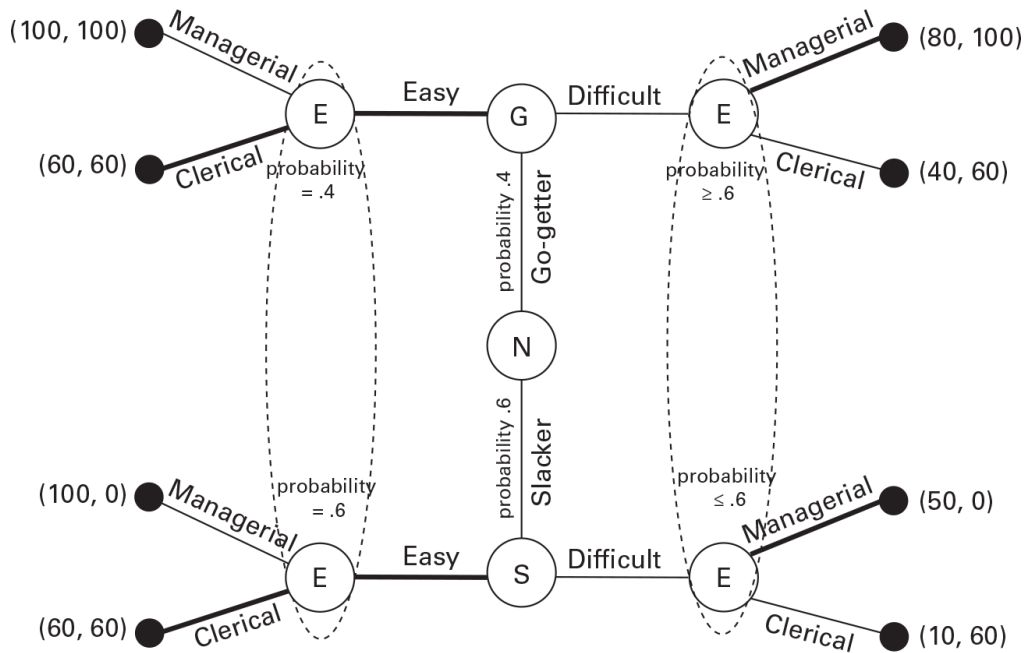


Figure 12.9

There is a pooling equilibrium in which both types take Easy courses. The receiver's belief after seeing the Easy signal is the same as her prior beliefs, and she offers the Clerical job. Two possible cases are presented for her off-equilibrium beliefs after seeing the signal of Difficult. Only case 2 is a perfect Bayesian equilibrium, since the Go-getter prefers to deviate in case 1.

What if she sees the signal of Difficult? Since she believes that all applicants choose Easy courses, this should never happen, and it is unclear what she should conclude about the applicants' productivity. This is the "off the equilibrium path" situation, in which Bayes' rule doesn't apply and there are no restrictions on the beliefs that she can hold. However, it is still important to consider what she would do in this case, since (1) her strategy must specify what she does at this information set and sequential rationality requires it to be a best response; and (2) the applicant's beliefs about how the manager would respond in this situation determine whether he would want to deviate to choosing Difficult courses. Let's say that the employer believes the applicant is a Go-getter with probability p and then calculate her best response. If she offers the Managerial job, her expected payoff is $Eu(\text{Managerial}) = p(100) + (1 - p)(0) = 100p$, and if she offers to the Clerical job, her payoff is 60. Therefore, her best response is to offer the Managerial job if $p \geq 0.6$ and to offer the Clerical job if $p \leq 0.6$.

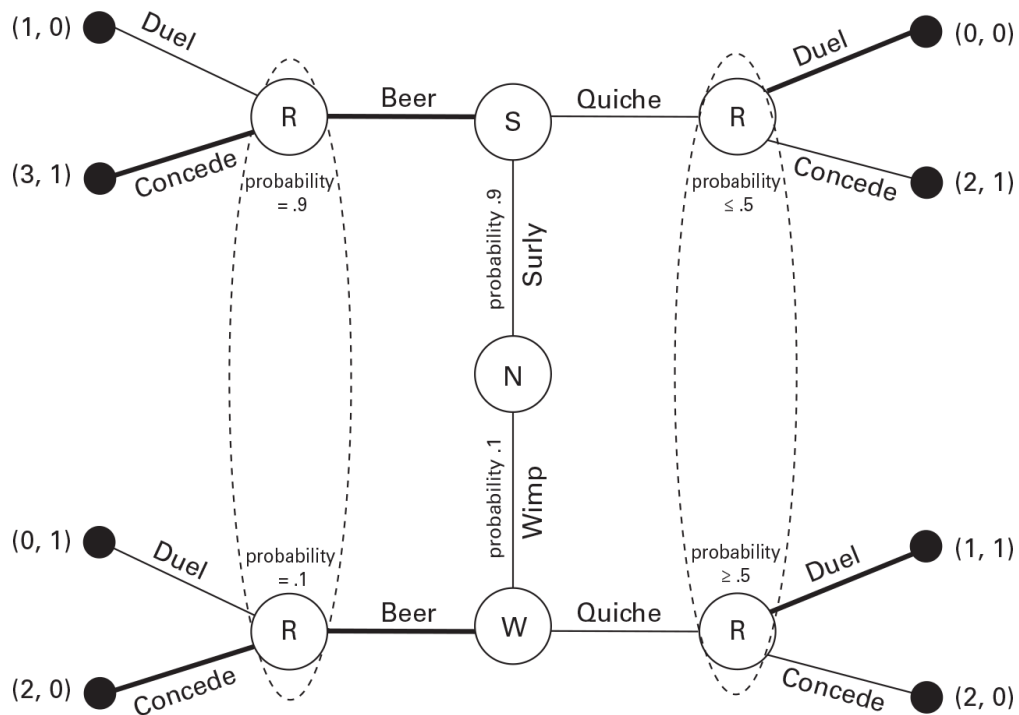
These two situations are depicted as case 1 and case 2 in [figure 12.9](#). If she believes that the likelihood of dealing with a Go-getter, after seeing the signal of Difficult, is at least 0.6, then her best response is to offer the Managerial job, as depicted in case 1. In this case, does either type of applicant wish to deviate? The Go-getter type would want to deviate in this situation—rather than taking Easy courses and getting the Clerical job (payoff of 60), he could take Difficult courses, in which case he would be offered the Managerial job (payoff of 80). Therefore, the situation in case 1 is not a perfect Bayesian equilibrium. There cannot be a perfect Bayesian equilibrium in which all types choose Easy, if the applicant believes there is a high likelihood the applicant is a Go-getter conditional on seeing someone take Difficult courses. In case 2, the employer believes that the applicant is a Go-getter with probability less than or equal to 0.6 conditional on observing the Difficult signal, and thus her best response is to offer those taking Difficult coursework the Clerical job as well. In this scenario, neither type of applicant wishes to deviate: doing so means he must put in the effort to take hard courses, but he still winds up in the same position. Therefore, we can have a perfect Bayesian equilibrium of ((Easy, Easy), (Clerical, Clerical)), in which the receiver's belief that she is dealing with the Go-getter after seeing the signal of Easy is 0.4 and after seeing the signal of Difficult is less than 0.6.

Note that we're not saying that the applicant *should* believe that the probability of the Go-getter type is less than 0.6 conditional on receiving the Difficult signal. Rather, we're saying that there can exist an equilibrium in which all types choose Easy courses *only if* the applicant believes this.

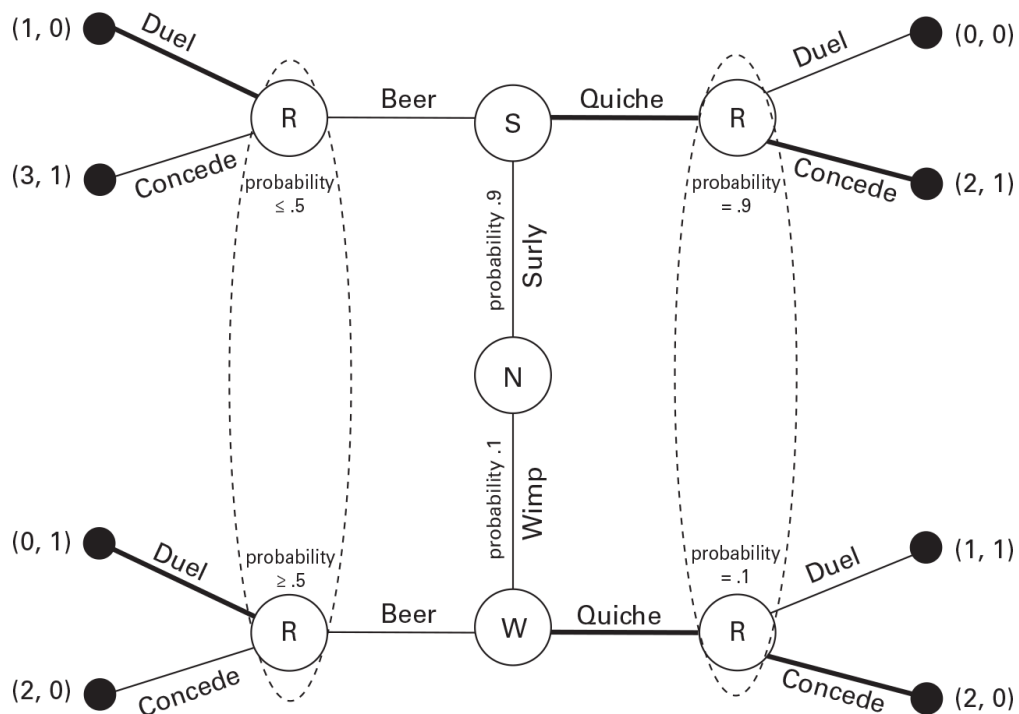
There is one other possible type of pooling equilibrium, involving all types of applicants choosing Difficult courses. In the exercises at the end of this chapter, you are asked to show that there does not exist a perfect Bayesian equilibrium of this type.

The Beer-Quiche Game

Imagine that you are in a diner and considering challenging another patron to a Duel (perhaps he stole your parking spot or favorite table). How well this will go for you depends on the person's type—you prefer to Duel if he is a Wimp (probability 0.1), but Concede if he is Surly (probability 0.9). Before issuing your challenge, you observe a signal: his breakfast order. He could order Quiche (the known breakfast of Wimps) or Beer (the known breakfast of Surly guys). We're not actually making this game up—it is a classic signaling game from a paper by Cho and Kreps (1987). (The choice of breakfast orders is based on a tongue-in-cheek book from the early 1980s called *Real Men Don't Eat Quiche*, a send-up of masculinity stereotypes.) The payoff to you (the receiver) is 0 if you Duel someone who is Surly or Concede to a Wimp, and 1 if you Duel a Wimp or back down from dueling a Surly type. The other patron (the sender) gets an extra payoff of 1 for having his preferred breakfast and 2 if he can eat in peace without being challenged to a Duel. The game is shown in [figure 12.10](#), along with the two pooling equilibria that exist in this game.



(1) Pooling equilibrium with both types ordering Beer



(2) Pooling equilibrium with both types ordering Quiche

Figure 12.10

The beer-quiche game between a Surly Sender (S), a Wimp Sender (W), and a Receiver (R) after the

initial move by Nature (N). Panel (1) presents a pooling equilibrium in which both types of senders choose Beer; panel (2) presents a pooling equilibrium in which both types of senders choose Quiche.

First, note that there does not exist a separating equilibrium in this game. This is due to the assumption that the benefit of getting one's preferred breakfast is low relative to the benefit of avoiding a Duel. As such, the Wimp type would always prefer to mimic the Surly type. This is obvious if we go through the steps for identifying a perfect Bayesian equilibrium that we outlined previously.

1. Let's set the sender's strategy to be (Beer, Quiche) and see why this cannot be an equilibrium.
2. We update the receiver's beliefs and note that she believes she's dealing with Surly with certainty after seeing the Beer signal and Wimp with certainty after seeing the Quiche signal.
3. Given these beliefs, the receiver best responds with (Concede, Duel).
4. We check whether either type of sender wishes to deviate. The Wimp sender is currently getting Quiche but ends up in a Duel (payoff of 1), but he could switch his signal to Beer and avoid the Duel (payoff of 2).

Therefore, at least one player wishes to unilaterally deviate, and this strategy profile cannot be a perfect Bayesian equilibrium.

If we instead search for pooling equilibria, we find that perfect Bayesian equilibria exist in which both types choose Beer (panel 1) and both types choose Quiche (panel 2). In the (Beer, Beer) scenario, the receiver's beliefs after observing Beer are equal to her prior beliefs: that is, there is a 0.9 chance that the sender is Surly. Her expected payoffs are thus: $Eu(Duel) = 0.9(0) + 0.1(1) = 0.1$ and $Eu(Concede) = 0.9(1) + 0.1(0) = 0.9$, and she best responds at this information set by conceding. After observing the off-equilibrium-path signal of Quiche, she may have any updated beliefs p about the likelihood that the sender is Surly. Her expected payoffs are $Eu(Duel) = p(0) + (1 - p)(1) = 1 - p$ and $Eu(Concede) = p(1) + (1 - p)(0) = p$, and so Duel is a best response for $p \leq 0.5$ and Concede is a best response for $p \geq 0.5$.

Would either type of sender wish to unilaterally deviate? The Surly sender certainly would not—as things stand, he's getting Beer and avoiding a Duel. What about the Wimp sender? This depends on how he anticipates the receiver will respond to a message of Quiche—if she will not Duel him, he wishes to switch breakfasts; if she will, he'll stick with Beer. We just said that she'll respond to an off-equilibrium message of Quiche with Duel if $p \leq 0.5$.

Therefore, if $p \leq 0.5$, then we can have a pooling equilibrium of ((Beer, Beer), (Concede, Duel)). If $p > 0.5$, this would fall apart, because the receiver would respond to a signal of Quiche by Conceding and the Wimp sender would correctly anticipate this and switch his breakfast order. You can also confirm that there is a pooling equilibrium in which both types of senders order Quiche, provided that the sender's updated beliefs after seeing the off-equilibrium signal of Beer assign a high probability to Wimp, such that she would best respond to the signal of Beer by initiating a Duel (shown in panel 2).

The Meaning of Beer and Quiche

While this example is undoubtedly goofy, the signals of Beer and Quiche are stand-ins for actions that players can take that have different payoffs associated with different types. In addition to the job market signaling game, there are many other scenarios in which one type of player would want to distinguish themselves, while another type would wish to mimic them. The extent to which the former type of players can (job market signaling game) or cannot (Beer and Quiche game) distinguish themselves depends on the relative costs and benefits of the game. In order to be effective, the signal must be (1) affordable for one type, such that they would be willing to send the signal if that meant that they could distinguish themselves from other types and thus get a better outcome from the receiver; and (2) too expensive for the other type, such that they would not want to mimic the costly signal even if it meant achieving a better outcome from the receiver. Examples include a firm with a high-quality product who is willing to spend money on stadium naming rights or other expensive advertising to signal the quality of its product, knowing that lower-quality firms could not afford to do so since they will not be able to retain any customers that they attract through advertising (see exercise 7); humans and animals engaging in courtship rituals in which they make some elaborate display or grand gesture in order to signal their fitness or commitment, knowing that weaker or lower-quality mates would not be willing to expend such resources (see exercise 6); or a monopolist reducing the price of its product to signal its low production costs to potential entrants, knowing that firms with higher costs would not be able to afford to do so (see the example in the Behavior in Games section later in this chapter).

Perfect Bayesian Equilibria in Cheap Talk Games

Thus far, we've considered costly signaling games, in which the sender's action had an impact on his payoff (e.g., investing in education, his breakfast order). Now we'll consider a special type of signaling game, in which the sender's signal has no bearing on his payoff. You can think of this as a message that he sends for free, rather than a consequential action that he takes. These types of games are referred to as *cheap talk* games, or information transmission games. The model of cheap talk games is from Crawford and Sobel (1982).

What do you think would happen if the job market applicants could only send a costless message about their type, such as "I am a Go-getter" or "I am a Slacker"? As you might anticipate, no useful information can be transmitted to the receiver in equilibrium, and the receiver makes her decision based on her prior beliefs. Since the messages are costless, the Slacker type always has an incentive to mimic the Go-getter type's message. There can be no equilibrium in which the Go-getter is able to transmit his type with sufficient effectiveness that the employer would be willing to offer him the Managerial job. If the Go-getter were able to successfully communicate his type to get the better job, then the Slacker type would prefer to deviate to imitate the Go-getter's message, and the manager would, in turn, prefer to deviate to offering only the Clerical job.

In cheap talk games, these types of pooling equilibria are called *babbling equilibria*. An example of a babbling equilibrium in this game is for both types to announce each message with equal probability, the receiver to believe the probability of dealing with the Slacker type is the same as her prior beliefs (0.6) at each information set, and to always offer the Clerical job. Since the receiver always offers the Clerical job, no matter which message she receives, neither type of sender has an incentive to switch strategies.

In *every* cheap talk game, there exists a babbling equilibrium. However, depending on the nature of the game, there may also be equilibria in which useful information is transmitted and the receiver can infer the sender's type (or at least narrow it down). The extent to which this is possible in equilibrium depends on how much the sender and receiver's incentives are aligned.

First, imagine that the sender and receiver's incentives are closely aligned, such that for each type of sender, the players would always agree on the

receiver's best course of action. In this case, there can exist a separating equilibrium in which senders' messages perfectly reveal the true state of the world, and receivers always follow the message that they were sent. For example, imagine that the Slacker doesn't actually want the Managerial job—just like taking Difficult courses, he assumes that it will be too much work. In this case, there exists a perfect Bayesian equilibrium in which Slackers state that they are low productivity, Go-getter applicants state that they are high productivity, and the employer is able to fully update her beliefs after seeing the message and best responds by granting Managerial jobs if and only if the worker sends a message that he is a high-productivity type. Neither type of worker would wish to deviate and send a different signal in this case. There also can exist an unintuitive separating equilibrium, in which Slackers state that they are high productivity and Go-getters state that they are low productivity, allowing the Employer to again perfectly identify them based on message and best responds by granting Managerial jobs if and only if the worker sends a message that he is a Slacker type. While somewhat odd, no players have an incentive to shift their behavior or updated beliefs in this scenario. Note, too, that there still exist babbling equilibria, even though both types would prefer that the Employer learn the true state of the world. Consider again the strategy profile and system of beliefs in which both types choose each message with equal probability, the applicant continues to believe that the probability of the Go-getter type is 0.4 regardless of the message she receives, and offers the Clerical job in both cases. The Go-getter is unhappy with this scenario, but given that the receiver is playing (Clerical, Clerical), he has no incentive to unilaterally deviate from the babbling message.

Perhaps the more interesting cases are those in between, when the sender wants some information to be transmitted. Consider the situation in which the sender's and receiver's incentives are somewhat aligned, such that for each type of sender, the players disagree only slightly on the receiver's best course of action. If the sender thinks that the receiver will always believe his message, he prefers to exaggerate somewhat.

Let's say that the applicant may be one of five possible productivity types, $t \in \{1, 3, 5, 7, 9\}$, each with equal probability, and can send a message $m \in \{1, 3, 5, 7, 9\}$. In turn, the Employer can take an action, a , which is to offer one of nine possible jobs $a \in \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$.

The receiver's payoff is given by $110 - 10|t - a|^{1.4}$, meaning that she always wants to choose $a = t$ and does increasingly worse as the job she offers differs from the worker's true type. The sender's payoff is $110 - 10|t + 1.2 - a|^{1.4}$, meaning that he does best when the receiver chooses $a = t + 1.2$. In other words, he always wants to get a job that is slightly superior to what the Employer thinks someone of his type should have, but he doesn't want the job to be too incompatible with his true type.

As always, there is a babbling equilibrium in which no information is transmitted. However, the following strategy profile and system of beliefs is also a perfect Bayesian equilibrium: workers of types 1 and 3 send messages 1 and 3 with equal probability, workers of higher types randomize over messages 5, 7, and 9 with equal probability; receivers who see message 1 or 3 believe that they are dealing with types 1 and 3 with equal probability and best respond by offering job 2; and receivers who see messages 5, 7, or 9 believe that they are dealing with types 5, 7, and 9 with equal probability and best respond by offering job 7. We can check that no type of sender would want to change his message. Someone of type 1 or 3 would receive job 2 under this strategy profile (payoff of $110 - 10|1 + 1.2 - 2|^{1.4} = 109$ or $110 - 10|3 + 1.2 - 2|^{1.4} = 79.8$, respectively), but switching to a higher message would get job 7, associated with lower payoffs (20.1 or 67.7, respectively). Likewise, someone of type 5, 7, or 9 would receive job 7 under this strategy profile (payoff of 102.7, 97.1, or 59, respectively), but switching to a lower message would get job 2, associated with lower payoffs (35.4, -18.6, or -80.3, respectively). Thus we have a semi-separating equilibrium, in which the two lower types pool together and play a mixed strategy over one set of messages, while the three higher types pool together and play a mix over a different set of messages. As a result, the receiver can distinguish only between the higher and lower types, but not between the specific types within each pool, and offers two distinct jobs based on the message she receives. In contrast, imagine that there is an incentive gap of 2, rather than 1.2. In this case, the most informative equilibrium involves the type 1 worker sending message 1 and the other four types pooling together to randomize over messages 3, 5, 7, and 9. As the incentive gap expands, the less the receiver can distinguish between types, until there is only a babbling equilibrium.

Other examples of this type of setting include financial analysts exaggerating the outlook of stocks in the hope that clients will make more

purchases, congressional committees whose views differ from the Congress as a whole exaggerating the effects of policy in the hope of obtaining more extreme outcomes, or college professors exaggerating about their students' abilities in recommendation letters in the hope of getting them better jobs.

We can thus summarize the existence of perfect Bayesian equilibria in cheap talk games as follows:

1. There always exists a babbling equilibrium in which no information is transmitted in the senders' messages and receivers maximize their expected payoff based on their prior beliefs.
2. If the senders' and receivers' incentives are closely aligned, then there exists a separating equilibrium in which the senders' messages fully reveal their types and the receivers take these messages at face value, choosing an action contingent on the message.
3. If the senders' and receivers' incentives are somewhat aligned, then there exists a semi-separating equilibrium, in which similar types pool together to send the same message and the receiver updates her beliefs so as to narrow down—but not perfectly identify—the type she is dealing with, choosing an action that differs for some messages and not for others.

Behavior in Games

Overcommunicating in Costly Signaling and Cheap Talk Games

Costly signaling games are both cognitively demanding for players and often have a multiplicity of perfect Bayesian equilibria. Theorists have considered placing additional assumptions on how senders update their beliefs off the equilibrium path in order to refine the set of possible outcomes and rule out less intuitive perfect Bayesian equilibria. Early experimental studies tested whether people play according to the equilibrium predictions and, if so, whether equilibria that are consistent with these additional assumptions are more likely to occur. Experiments typically find that, despite the complexity of signaling games, equilibrium behavior does tend to eventually emerge. However, it can take many repetitions for participants' behavior to converge toward anything resembling equilibrium play. In early periods, senders frequently choose the action that would be best for them in a complete information environment, ignoring the fact that they have private information and their signal may influence the receiver's subsequent beliefs. Over time, players learn and adapt their behavior by responding to these early patterns.

As a result, it seems that which equilibrium ultimately occurs depends on the initial behavior that arises in the game, rather than any theoretical assumptions about what types of equilibria are more intuitive or reasonable.

For example, consider the beer-quiche game, which had two pooling equilibria: one in which everyone chose Beer and one in which everyone chose Quiche, with the receiver conceding in both cases. First, consider the equilibrium in which both senders choose Beer. The only thing keeping the Wimp sender from switching to his more preferred breakfast is the fact that the receiver will choose Duel if she sees the Quiche signal, believing that the sender is likely to be a Wimp in that case. The receiver's off-the-equilibrium-path beliefs are fairly intuitive in this case: under the equilibrium strategy profile, the Surly sender is getting his preferred Beer and avoiding the Duel, so it seems reasonable for her to conclude that someone who unexpectedly deviates from this strategy and picks Quiche is likely the Wimp type. In contrast, the pooling equilibrium in which everyone orders Quiche requires the receiver to believe that someone who unexpectedly deviates to order Beer is likely a Wimp. Why would a Wimp type do such a thing? No matter how the receiver responded, the Wimp type would do worse than under the equilibrium outcome. Thus, it would be pretty strange for the receiver to believe that everyone always plays Quiche, suddenly see a signal of Beer, and conclude: "Oh, this is probably a Wimp." Put differently, if the Wimp sender is attempting to mimic the Surly sender, shouldn't it be on the Surly sender's (culinary) terms? When introducing this game, Cho and Kreps (1987) proposed the "intuitive criterion" as a means of ruling out this type of unintuitive pooling equilibrium.

Brandts and Holt (1992) conducted an early experiment that had a similar setup to the beer-quiche game. They found that, initially, Surly senders always chose Beer, while Wimp senders almost always chose Quiche. The receiver best responded by dueling those ordering Quiche and conceding when someone ordered Beer. Over time, the Wimp senders learned to mimic the Beer order, while the receivers continued to Concede whenever they saw the Beer signal, consistent with the intuitive pooling equilibrium. However, they noted that this equilibrium behavior didn't spontaneously arise from the beginning as some intuitive outcome—rather, it was arrived at by players initially playing naively and then slowly learning to best respond to previous outcomes. To test this, they made some tweaks to the payoffs, such that in the

event of a Duel, the Surly senders actually preferred Quiche while the Wimp senders preferred Beer. Perhaps they shouldn't be called Surly versus Wimpy, but rather, those preferring the optimal breakfast for the task at hand versus those who prefer to get drunk when facing a Duel. The equilibria remained the same, but Wimp senders may be more willing to send the Beer signal, while Surly senders could be more willing to send the Quiche signal. In this case, they found that reverse signals were sent earlier in the game and, after a process of adjustment, 78% of signals were Quiche, consistent with the unintuitive equilibrium.

A pair of studies by Cooper, Garvin, and Kagel (1997a, 1997b) explored a more complicated signaling game but reached similar conclusions. Consider a game between a monopolist and a potential entrant (based on the model of entry limit pricing by Milgrom and Roberts, 1982). The entrant wants to enter the market only if the monopolist has high costs, but otherwise wishes to stay out. The monopolist can signal its type by choosing an output quantity prior to the entrant making its decision. In the experimental setup, the monopolist can choose a quantity between 1 and 7. All else being equal, the high-cost monopolist always prefers an output of 2, while the low-cost monopolist prefers 4. Importantly, the high-cost monopolist would be willing to choose output levels up to 5 if it meant that it could mimic the low-cost type and avoid the entrant; it would not be willing to choose output levels above 5, and so outputs of 6 or 7 could serve as effective signals of the low-cost type. Cooper et al. (1997a, 1997b) conducted experimental treatments in which there were both pooling and separating equilibria versus only separating equilibria. They found that senders initially choose naively, with the high-cost player selecting 2 and low-cost selecting 4. Receivers can thus identify the types and enter when they see the lower-quantity choice. High-cost monopolists eventually learn that they can mimic the low-cost player's choice of 4 to avoid entry. When pooling at 4 is an equilibrium, this is where the process of adjustment stops. When there is no pooling equilibrium, the low-cost monopolists eventually further adjust upward to the separating equilibrium, in which they effectively signal their type by choosing a high quantity that high-cost types would find prohibitively costly to mimic. As further evidence that the equilibrium that is ultimately selected is primarily shaped by earlier experience, they found that subjects in the game with pooling equilibria who had previously played the game with only separating

equilibria continued to play according to the separating equilibrium, while those who had not had this experience settled into the pooling equilibrium.

Many experiments have also investigated cheap talk games. They generally have found that as the conflict of interest between the sender and the receiver increases, less information is transmitted by the sender and the receiver relies less on the message in choosing her action. However, senders tend to overcommunicate their private information (relative to the most informative possible equilibrium) and receivers are overly trusting of the messages they receive. This is even true when their interests are so opposed that only babbling equilibria exist. Recall the example from earlier in the chapter, in which there are five possible types, five possible messages for the sender, and nine possible actions for the receiver. In our example, we assumed that the incentive gap between the sender and receiver about the best course of action was 1.2, meaning that the receiver always preferred her action to match the true state of the world, while the sender preferred that the receiver choose an action slightly higher. Cai and Wang (2006) conducted an experiment using this setup, along with three other incentive gaps: (1) 0.5, in which case the players agree sufficiently that there exists an equilibrium with complete truth-telling; (2) 2, in which case senders of type 1 send one message and those with higher types pool together to send a different message; and (3) 4, in which the gap is so large that there is only a babbling equilibrium. [Figure 12.11](#) shows the correlations between states and messages, and between messages and actions for each of these values. In each case that the players' interests are not aligned, we see that there is significantly higher correlation between the state and the message than is predicted (suggesting overcommunication) and between message and action (suggesting that the receivers are more trusting of the message than is predicted).

Incentive Gap	Correlation Between State and Message		Correlation Between Message and Action	
	Predicted	Actual	Predicted	Actual
0.5	1.000	0.916	1.000	0.965
1.2	0.750	0.896	0.866	0.924
2	0.500	0.734	0.707	0.794
4	0.000	0.391	0.000	0.542

Figure 12.11

Predictions and observed behavior in a cheap talk game experiment (Cai and Wang, 2006).

Evidence-Based Adaptations of the Standard Model

As we've already discussed, the convergence in costly signaling games toward a perfect Bayesian equilibrium based on the earlier history of play can be well described by a learning model in which players adapt their behavior based on earlier outcomes; Cooper et al. (1997a) and Anderson and Camerer (2000) present such models of signaling games. How can the persistent over-communication of (and reliance on) information in cheap talk games be explained?

One explanation is that there actually are type-dependent costs of sending certain messages—they just aren't financial costs. Imagine that players experience a psychological or emotional cost of telling a lie. In other words, it is costless to send a message that is true, but there is a cost to other types mimicking this message. In this case, the situation is no longer a cheap talk game, but a costly signaling game. Sanchez-Pages and Vorsatz (2007) consider a simple game with diametrically opposed (constant-sum) incentives in which there are two states of the world and two messages that could be sent by the sender (truth or lie), and two actions that can be taken by the receiver (trust or distrust). In this case, there is only a babbling equilibrium in which the receiver tells the truth exactly half the time and the receiver trusts exactly half the time. They then add another stage in which the receiver can punish the sender after learning the ultimate payoffs (and thus whether the sender lied) by reducing both of their payoffs to zero. They find the subjects who are regularly willing to punish (thus reflecting a personal distaste for lying) also tell the truth the vast majority of the time when they are senders; the other participants tell the truth with the expected frequency of 50%. In addition,

Wang, Spezio, and Camerer (2010) find that participants' pupils tend to dilate more when they are sending a false message, further indicating that there may be some emotional or cognitive costs to lying.

Adaptations to the standard analysis of cheap talk games take into account the fact that there are different types of people, such as those who always tell the truth (either due to naivety or because they find lying to be psychologically costly) and those who anticipate that some portion of the population tells the truth. We've previously discussed level- k models, in which the level-0 type randomizes over his possible actions, the level-1 type's best response to level-0 play, and so on. Instead, imagine that the level-0 sender always tells the truth, while level-0 receivers always believe the message. A level-1 sender assumes that he is playing against a level-0 receiver, who will always believe him and will therefore send the message that makes him best off if the sender follows it. A level-1 receiver assumes that she's dealing with a level-1 sender, and therefore corrects the message for the bias that level-1 senders introduce, and so on. (Notice that the application of the level- k model in sender-receiver games is slightly different than in simultaneous-move games, in which the level-1 players always best respond to level-0 play. Instead, researchers typically assume that a level-1 receiver best responds to level-1 play.)

Figure 12.12 shows the level-0, level-1, and level-2 behavior in the cheap talk game with incentive gap 2. A level-0 sender always tells the truth, and a level-0 believer takes the message at face value. A level-1 sender, however, believes that the receiver will adhere to his message, so he inflates it by 2 units (with the maximum message being 9). A level-1 sender anticipates that the message will be inflated by 2 units and works to correct it. This means that if she sees a message of 3, 5, or 7, she knows to subtract 2 units for her action. If she sees the message of 9, she thinks it is equally likely that the state is 7 or 9 (since the sender sends the message of 9 in each case), and she chooses action 8. The message of 1 is off the equilibrium path, and she may have any beliefs after viewing this message, but researchers who have used this model to predict experimental behavior generally ascribe a belief of 1 in this case. A level-2 sender, meanwhile, must inflate his message by 4 in order to correct for the sender's correction, causing the level-2 receiver to deflate the message she receives by 4. We only list the behavior for these three levels,

since most subjects fall under one of these categories (or reason all the way to equilibrium).

Level	Sender's					Receiver's				
	Message by State					Action by Message				
	1	3	5	7	9	1	3	5	7	8
L-0	1	3	5	7	9	1	3	5	7	9
L-1	3	5	7	9	9	1	1	3	5	8
L-2	5	7	9	9	9	1	1	1	3	7

Figure 12.12

Behavior of level-0, level-1, and level-2 types in a cheap talk game with incentive gap 2.

What is important to notice is that, across these three levels, more information is being transmitted than is predicted by equilibrium, and the receiver is able to incorporate the information into her action more than is predicted. As the level of strategic sophistication increases, however, the signal becomes more muddled. This model can therefore explain the persistent overcommunication of types in cheap talk games.

As further support, Wang, Spezio, and Camerer (2010) conducted a nearly identical game using an eye-tracker: a small camera that sits below the subjects' monitor and records their gaze as they play the game. They found that the eye-tracking data was consistent with level- k behavior, with senders spending most of their time looking at the true state of the world and their own payoffs (suggesting that they weren't considering the asymmetry of the information and the receiver's strategic reasoning) and different senders displaying looking patterns that were consistent with different levels of reasoning.

Finally, it is worth noting that the standard costly signaling model has itself been used to explain behavioral evidence in situations to which you might not have expected it to be applicable. For instance, one reason that people may be generous to others is that they are playing a signaling game with themselves—that is, they want to be able to think of themselves as generous people. In this case, the “sender” is an individual who is making a choice about whether to give money to someone or take some other kind action at a cost to

themselves. The “receiver” is the person’s future self, who will recall their action (i.e., their signal), but not their motivation (i.e., their true type).

At first, it may seem somewhat wacky to think that people might undertake actions in an attempt to deceive their future selves about their true underlying motivations. However, this type of self-deception is consistent with behavioral evidence in several cleverly designed experiments that seek to determine whether people exploit “moral wiggle room.” For example, imagine that you are randomly matched with another person and can choose between two options: option A gives you \$6 and your partner \$1, while option B gives you each \$5 (as in Dana, Weber, and Kuang, 2007). In this case, the vast majority (three-quarters) of people choose the more fair option, B. Now imagine that you don’t know the impact of your decision on your partner—you only know that A gives you \$6 and B gives you \$5 (and that the other person will get either \$5 or \$1, but you don’t know which payoff is associated with which option). However, you can costlessly click a button and easily learn your partner’s payoff under both options. Now, people avoid the information and the majority choose A. This suggests that many of the people who chose option B when they knew the consequences didn’t really do so because they wanted to be fair—rather, they wanted to maintain a certain image of themselves as fair people. When presented with the possibility of getting the better outcome for themselves without knowing that they were being unfair, they took advantage of it. Grossman and van der Weele (2017) contributes a costly signaling model that explains this type of self-signaling in this and other similar experiments.

Exercises

(1) Consider a very simple two-player poker-style game, in which player 1 draws a card from a deck of half aces and half kings. Player 1 observes his card and then decides whether to bet or fold. If he folds, player 1 must pay \$1 to player 2. If he bets, then player 2 decides whether to fold or call, without knowing the card. If she folds, player 2 must pay \$1 to player 1. If she calls, they look at the card. If it is an ace, player 2 pays \$2 to player 1; if it is a king, player 1 pays \$2 to player 2. (This game is based on Reiley, Urbancic, and Walker, 2008.)

- (a) Write this sequential game of incomplete information in extensive form.

(b) Assume that player 1 is playing a strategy in which he always bets if he holds an ace and bets with probability $1/3$ if he holds a king. Use Bayes' rule to update player 2's belief at her information set. That is, conditional on player 1 betting, what is the probability that he has an ace?

(2) Consider the game presented in figure 12.13. Nature moves first to determine the Sender's type: either High (probability 0.8) or Low (probability 0.2). The Sender then chooses In or Out. In the event that the Sender chooses In, the Receiver can choose Left or Right. (The payoffs aren't given, since they aren't relevant for this question.) For each of the following strategies that the Sender could choose, give the Receiver's updated belief (consistent with Bayes' rule) about the likelihood that the Sender is the High type, conditional on the Sender choosing In. In other words, given that the Receiver reaches her information set, what is the likelihood that she believes that she is at the left node rather than the right one? In the event that Bayes' rule does not apply (and thus any possible beliefs could be consistent with Bayes' rule), state that.

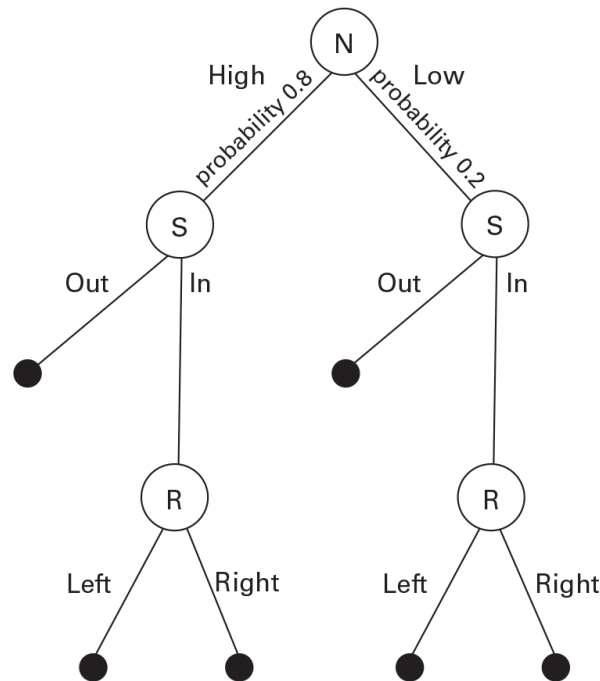


Figure 12.13

(a) Sender chooses In if High and Out if Low.

- (b) Sender chooses In with probability 0.4 (and Out with probability 0.6) if High and In with probability 0.4 (and Out with probability 0.6) if Low.
- (c) Sender chooses Out if High and Out if Low.
- (d) Sender chooses In with probability 0.4 (and Out with probability 0.6) if High and In with probability 0.9 (and Out with probability 0.1) if Low.
- (3) Consider the job market signaling game described in this chapter. Show that there cannot exist a perfect Bayesian equilibrium in which all types choose difficult courses.
- (4) The prominent game theorist Ariel Rubinstein has argued that PhD students looking for their first academic job should dress in jeans and t-shirts rather than in suits, since star researchers should not have trouble finding a job no matter how sloppily they dress and average researchers should want to attract attention and mimic the top students.

Consider the following signaling game shown in [figure 12.14](#), in which Nature determines whether the student is a Star or Average, the student decides whether to be Sloppy or Professional, and the department decides whether to Offer a job or to Pass. Departments prefer to hire better students and those that are more professionally dressed; Students prefer to dress sloppily and prefer to get the job offer, but the Star students do fine regardless of whether they are offered this particular job.

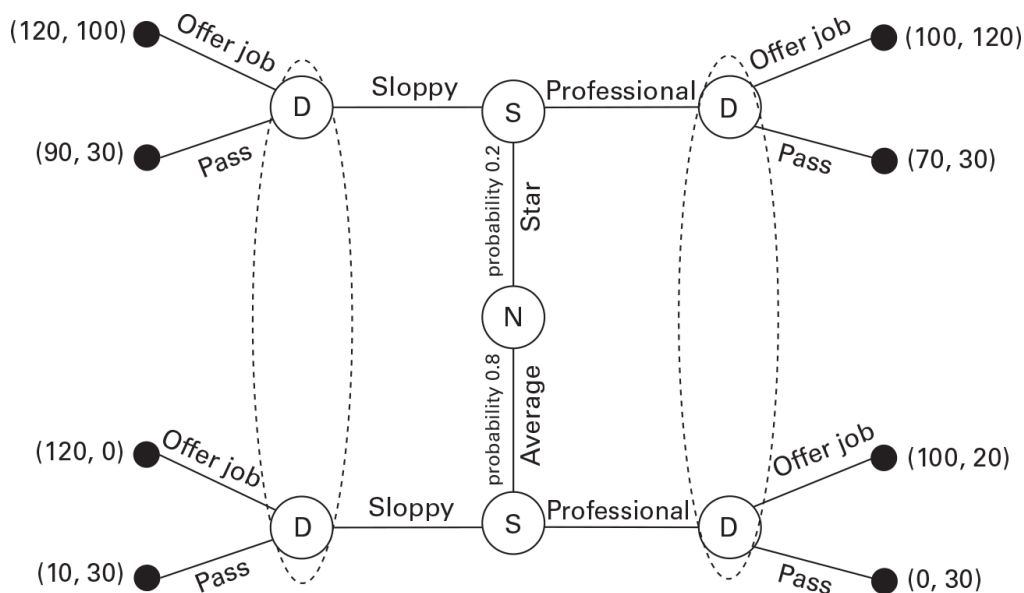


Figure 12.14

Is there a pooling equilibrium in which all students dress sloppily? If no, explain why not. If yes, say what it is (the strategy for each player and any restrictions on beliefs).

(5) Consider again the beer-quiche game described in this chapter. Assume that the receiver's payoffs remain the same, but let's change the sender's payoffs, such that he gets an additional payoff of 3 (instead of 1) if he orders his preferred breakfast (and earns 2 for avoiding a Duel). Find all the pooling and separating perfect Bayesian equilibria in this game.

(6) Why might animals develop traits that seem to handicap their ability to survive in nature, such as peacocks developing elaborate tails? Biologist Amotz Zahavi hypothesized that these traits may have evolved to signal an animal's genetic fitness to potential mates, since weaker types could not afford to create such an extravagant display and survive. While this idea has not been widely accepted, biologists have pointed out that it is consistent with a costly signaling model. Examples of costly signaling in the course of mate selection in the human world include buying expensive gifts, marriage proposals, or getting a tattoo of someone's name (all of which would be significantly more costly to someone not committed to the relationship). Consider a game in which the sender may be strong or weak, each with equal probability, and makes a proposal to a prospective mate. The receiver earns a payoff of 5 for selecting a strong mate, 0 for selecting a weak mate, and 1 for passing on the proposal. The sender earns a payoff of 10 if the prospective mate accepts and 2 otherwise. However, the sender may first mount an elaborate display, which costs 5 for high-quality types and 9 for low-quality types.

(a) Is there a separating perfect Bayesian equilibria in this game? If yes, state what it is (the strategy profile for each player and the receiver's beliefs at each information set). If no, explain why not.

(b) Is there a pooling perfect Bayesian equilibria in this game? If yes, state what it is (the strategy profile for each player and the receiver's beliefs at each information set). If no, explain why not.

(7) Consider a firm offering a product that may be either high quality (probability 0.3) or low quality (probability 0.7). The firm knows its own type, but the consumer does not. The consumer earns a payoff of 100 from purchasing the high-quality product but 0 from purchasing a low-quality

product, and she earns 50 if she doesn't make a purchase. Furthermore, the firm earns a profit of 100 from selling a high-quality product (since it will have gained a loyal customer), 30 from selling a low-quality product (since it will be a onetime sale and the customer avoids purchasing from the firm ever again), and 0 from not making a sale. Prior to the customer making the purchase, the firm can take a costly action: it can pay 80 for a 30-second Super Bowl ad.

- (a) Write the game in extensive form.
 - (b) Find any pooling perfect Bayesian equilibria.
 - (c) Is there a separating perfect Bayesian equilibrium in which the high-quality firm advertises and the low-quality firm does not?
 - (d) For what range of Super Bowl ad prices can there be a separating perfect Bayesian equilibrium in which the high-quality firm advertises and the low-quality firm does not? Find the perfect Bayesian equilibria if the advertising cost is greater than this range, and then if it is less than this range.
- (8) Consider a cheap talk game in which the sender is either type A or B, each with equal probability (see [figure 12.15](#)). The sender sends a message $m \in \{A, B\}$ and the receiver takes action $a \in \{X, Y, Z\}$. Finally, their payoffs are as shown in [figure 12.15](#) (the sender's payoffs are listed first). (Based on an experimental study by Kawagoe and Takizawa, 2009.)

		Action		
		X	Y	Z
Type	A	(3, 4)	(2, 1)	(4, 3)
	B	(2, 1)	(3, 4)	(4, 3)

Figure 12.15

- (a) Write the game in extensive form.
- (b) Is there a perfect Bayesian equilibrium in which the sender truthfully reveals his type—that is, chooses message A when he is type A and message B with type B? If yes, find the equilibrium (i.e., find the receiver's updated beliefs and best responses); if no, explain why not.
- (c) Is there a babbling perfect Bayesian equilibrium in which the sender chooses each message with equal probability, regardless of his type? If

yes, state what it is (i.e., find the receiver's updated beliefs and best responses); if no, explain why not.

(9) Consider a cheap talk game between a securities analyst and his client. The analyst observes the true state of the world $s \in \{1, 2, 3, 4, 5\}$ and sends a message $m \in \{1, 2, 3, 4, 5\}$. The client receives the message and chooses an action $a \in \{1, 2, 3, 4, 5\}$. Finally, the client receives payoff $u_c = 110 - 20|s-a|^{1.4}$, and the analyst receives payoff $u_a = 110 - 20|s + 0.9 - a|^{1.4}$ (Wang et al. 2010).

- (a) Show that there is a babbling perfect Bayesian equilibrium in which the analyst sends each message with equal probability, regardless of the state.
- (b) Show that there is a perfect Bayesian equilibrium in which the analyst sends message 1 if $s = 1$ and randomizes over messages $\{2, 3, 4, 5\}$ if $s > 1$. What is the sender's best response to each message in this equilibrium?
- (c) Now let's consider the predictions under the level- k model. Find the strategy for analysts and clients who use level-0, level-1, and level-2 reasoning (similar to [figure 12.12](#)).

13

Auctions

We end our discussion of games with incomplete information by considering an important application: auctions. The information that bidders have is incomplete because, while they know how to value what is being sold for themselves, they do not know how other bidders value the item. We first consider auctions in which bidders hold independent private values, and then discuss auctions in which the item for sale has a common value. As we will see at the end of the chapter, auction participants seem to account for their own values when bidding, but they struggle to anticipate both the values of the other bidders and how those values translate into bids.

Didius Julianus was emperor of the Roman Empire for only nine weeks. It turns out that he purchased it in one of the most historically significant auctions ever conducted, in this case by the Praetorian Guard who had just killed Pertinax, the previous emperor. Although this auction sparked a civil war and ultimately resulted in Didius's beheading (surely the most extreme example of the winner's curse), not all auctions are quite this fierce, nor do they regularly end in homicide. By encouraging competition among buyers, however, auctions typically help solve the fundamental problem faced by sellers—namely, they do not know the values of their buyers, but they do best by trading with the one whose value is highest.

We don't know for sure how the Praetorian Guard set the rules for the auction, but it is likely that offers were accepted from all the active bidders and the highest bid won. In this case, Didius's winnings would be the difference between how highly he valued the empire and the price he had to pay. As an illustration, suppose that four Roman senators each has randomly drawn a value between 0 and 100. This means that they all have incomplete information. Each senator knows his own value but can only guess at the values of the other three. How much should any senator bid? Bidding more increases one's chances of winning but reduces the amount won. If you could accurately predict the value held by your closest rival, you would have a sense of which bid would balance these two forces.

However, bidding in an auction can be more complicated because of the adverse selection problem: you have to make sure that winning does not imply that you have overbid and will suffer the winner's curse. Suppose, for example, that instead of independent valuations for the empire, there is one true value, but each senator only receives his own private signal, which could be more or less than the true value. If we expect the naive senator with the highest signal to bid the most and win the auction, shouldn't we also expect that having the highest signal is likely to imply that the true value of the empire is lower and this senator has bid too much?

Since classical antiquity, auctions have become quite common and they are used regularly to sell items both in stuffy auction houses and over the internet. Many different auction mechanisms have also evolved. There are dynamic live auctions in which auctioneers bark out prices for famous pieces of art, homes, or tractors, and there are sealed-bid auctions for contracts or government procurement that are conducted by bidders simultaneously submitting private tenders. Finally, there is also variation in who has to pay and how the price is determined. In most cases, just the winner has to pay her own bid, but in important deviations, winners pay a price that is different from their bid, and in some cases, everyone has to pay, regardless of winning or losing. We proceed here by first describing the auction formats we will consider. We then set the stage with a conceptual framework that will be the backbone of the rest of the chapter, characterize the Bayesian Nash equilibria in the auction formats considered, and review the experimental literature evaluating auction performance.

A Typography of Auction Formats

Although there are many auction formats, we will consider four classics from the theoretical and experimental literature, all of which are used frequently outside the lab. Before we get into the details of any of the formats, however, we begin by defining the terminology used to describe auctions. First, some auction mechanisms are *open*, meaning that the bidding process and dynamics can be viewed by all bidders, while other mechanisms are *closed*, indicating that bidders know nothing for sure other than their own values when they make a bid. Live or *open outcry* auctions are open (e.g., art auctions at the famous auction house of your choice, or the stock market), and sealed-bid auctions (e.g., construction contracts or fire truck replacements) are closed.

Auction formats can also vary by who has to pay. In most of the auction mechanisms that are used regularly, just the winner has to pay. In stark contrast, a few auction mechanisms ask everyone to pay their own bid, regardless of winning or losing. These are known as *all-pay* auctions. Another way of thinking about these auctions is as contests in which there is a prize, competition for the prize is costly, and the prize winner is often the person who invests the most in the competition. Common examples include patent races and political campaigns.

With this terminology, we can characterize the four auction formats that we will consider as follows:

- *English auction*: The English auction is an open, winner-pay format. Here, an auctioneer announces ascending prices and bidders acknowledge their willingness to pay the outstanding price. The bidder who acknowledges the highest outstanding price wins and pays this price.
- *Second-price auction*: The second-price auction is a closed, winner-pay format. In this format, bidders submit sealed tenders and the bidder offering the most wins, but she only has to pay the amount of the second-highest bid (see chapter 3).
- *First-price auction*: The first-price auction is a closed, winner-pay format. Bidders submit sealed tenders, and the bidder offering the most wins and has to pay her bid.
- *All-pay auction*: The (first-price) all-pay auction is a closed, all-pay format. Bidders submit sealed tenders and the bidder offering the most wins, but everyone has to pay his bid, regardless of winning or losing.

There is one final way that auctions can differ, but it has more to do with the prize than the rules of the game. In *private value* auctions, there will always be differences in how much players bid because they will have different values. For example (and borrowing from a famous minimal group paradigm manipulation), though both Paul Klee and Wassily Kandinsky are seen as founders of classical modern painting, people tend to prefer the work of one over the other. Hence, when a Klee fan goes to an art auction at which Kandinsky's *Der Blaue Reiter* is being auctioned, he is likely to have a lower valuation and bid less in equilibrium than many Kandinsky fans. In *common value* auctions, all bidders will instead always agree on the value of the item being auctioned, but bid heterogeneity comes from a different source. For instance, in auctions for drilling or mining rights, all bidders agree on the

current market price for the mineral they seek and value the site by simply multiplying the market price by the amount of the mineral at the site, but they don't know this last number for sure. Bidders can hire experts to estimate how much is under the ground, but these estimates are likely to vary, and this will be the source of differing bids. We proceed now by talking mostly about private value auctions, but we will consider optimal bidding in common value auctions later in the chapter.

Private Value Auction Preliminaries

Suppose that there is a seller of a unique item for which $n > 1$ bidders have values v_i indexed $i \in \{1, \dots, n\}$. Further, let's assume that the values of these bidders are independent draws from a uniform distribution bound from below by 0 and from above by 1. In other words, each bidder's value for the item is uncorrelated with the value drawn by anyone else, and it is equally likely to be any amount between 0 and 1. (Please see the appendix for a brief overview of the uniform distribution.) Player i submits bid b_i for the item, and therefore, the payoff (aka surplus) for any bidder is just the difference between how much bidder i values the item and how much she has to pay ($v_i - b_i$). As in any Bayesian game, our bidders do not know the payoffs of the other players. All that bidder i knows when submitting her bid is her own value, v_i , and that the value of any other bidder $j \neq i$ is equally likely to be any amount between 0 and 1. Hence, bidder i knows what her own potential surplus, $v_i - b_i$, is for any possible bid, but she can only guess what $v_j - b_j$ is.

Reducing auctions to their most basic elements, players make choices concerning their bids. In the context of incomplete information, one's auction type is just one's value, but there is nothing forcing bidder i to bid based on v_i . Instead of using a type-contingent bidding strategy, bidders are free to ignore their types when bidding. As a result, part of the mechanism design problem for auctions is to induce bidders to truthfully reveal their types. Put differently, bidder i must have an incentive to bid according to v_i , using her *bid function*, $b_i(v_i)$. While the bid function will not always be as simple as $b_i = v_i$, one way to ensure what is known as *incentive compatibility* is to take advantage of dominance. Any auction format in which it is a dominant strategy to bid one's value guarantees incentive compatibility. When dominance can't be ensured, however, we have to impose an incentive

compatibility constraint. Suppose that $Eu_i(v_i)$ is the expected utility bidder i receives from bidding according to her value, v_i , and $Eu_i(w_i, v_i)$ is the expected payoff to bidder i when she bids according to a different value, w_i , even though her true value is still v_i . In this case, we must assume that bidder i will bid truthfully whenever $Eu_i(v_i) \geq Eu_i(w_i, v_i)$ for any possible true value, v_i , and alternative value, w_i .

Once the incentive compatibility problem is sorted, bids are more straightforward, but there are still two assumptions that are particularly important and useful when it comes to characterizing equilibrium bidding in auctions. First, we assume that the higher one's value is, the more one bids. In other words, $b_i(v_i)$ is monotonically increasing. Second, as discussed in chapter 6, we will limit our attention to symmetric equilibria: in this case, those in which all bidders share a common bid function, $b^*(v_i)$, and $b^*(v_i)$ is a best response to the other players choosing $b^*(v_i)$, given the distribution of v in the population. For example, they might all bid the same fraction of their values because doing so is a best response to other bidders doing the same.

What are the implications of these assumptions? In addition to ignoring the possibility of asymmetric equilibria, the most important consequence of these assumptions has to do with the implied probability of winning the auction. Suppose, for instance, that there are just two bidders, and both decide to bid half their value. Further, let's also suppose that $v_i > v_j$. In this case, surely $\frac{1}{2}v_i > \frac{1}{2}v_j$, so bidder i should win. More generally, in the symmetric equilibrium when all bidders use the same bidding rule, $b(\cdot)$, the bidder with the highest value should always win and the chance that any bidder wins the auction is just v_i because values are drawn independently and randomly from the $[0, 1]$ interval. The intuition for this has a better chance of soaking in after viewing [figure 13.1](#). Here, if $v_i = \frac{3}{5}$ and if both bidders bid any common fraction of their values, then bidder i expects to top j 's bid $\frac{3}{5}$ of the time and lose $\frac{2}{5}$ of the time.

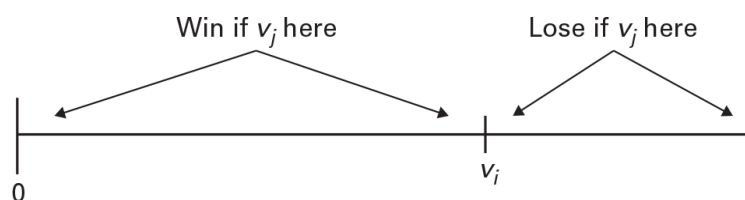


Figure 13.1

Assessing the probability of winning an auction.

It is important to note, however, that the example illustrated in [figure 13.1](#) is a two-bidder scenario using uniformly distributed values. In general, while the probability that a bidder has the highest value will always be key to the analysis of auction bidding, when more bidders are added (as we will see next) and the distribution of values is more complex, this probability (also known as the n th order statistic) is more difficult to assess. Getting into those details, however, distracts us from our main purpose—developing the intuition for equilibrium bidding in the standard auction formats.

Putting this all together, auction bidding is characterized by players seeking to maximize their expected payoffs in a Bayesian game wherein they don't know the values of the other players. Perhaps you can already see that this is going to be a balancing act. Conditional on bidding according to the value that I draw and committing to use a common bid function, as my bid increases, my chances of winning the auction increase, but the surplus that I will receive if I win falls.

Private Value Auctions with Two Bidders

To build some intuition, let's begin our discussion of equilibrium bidding in private value auctions by considering just two active bidders and how these bidders should respond to the four standard auction formats described previously.

To see how this balancing act plays out in each format first consider the English auction, in which an auctioneer continues to raise the item's price until only one bidder remains and is willing to pay this amount. To put a little more structure on this format, consider its *Japanese button* cousin, in which bidders press a button while the price continues to increase and release it only when they want to drop out of the auction. How should the bidders behave in this format? Imagine that the current price agreed to by bidder 2, p , is less than bidder 1's value. That is, $p < v_1$. If bidder 1 releases the button and drops

out of the auction, bidder 2 wins and bidder 1 earns nothing. If, instead, bidder 1 agrees to a small increment, say k , and $p + k < v_1$, then bidder 1 still has a chance to win and earn the difference, a positive surplus. This means that whenever $p < v_i$, then bidder i has an incentive to stay in the auction—even though the price will rise, a small potential surplus is still better than nothing. However, once the price has risen such that $p + k = v_i$, bidder i is now indifferent between potentially winning the item and having to pay her exact value for it (in which case the surplus is zero) and dropping out of the auction. Considering the only remaining case, staying in the auction when $p > v_i$, bidding above one's value in the English/Japanese auction never makes sense (i.e., it is always dominated by dropping out) because once you've bid above your value, your surplus can be negative only if you win. Hence, we have proposition 1.

Proposition 1 (Bidding in the English/Japanese auction) *In the Japanese button auction, it is weakly dominant to bid up to one's value and then stop. This results in a Bayesian Nash equilibrium.*

Notice that equilibrium bidding in the English/Japanese auction is efficient in the sense that the highest-value bidder inevitably receives the item. Also, notice that this winner earns some surplus because she ends up paying only an amount equal to the value of the second-highest bidder, the price at which the second bidder drops out of the auction. More specifically, if bidders use their equilibrium strategies and $v_1 > v_2$, then bidder 1 should win, she should pay v_2 and earn surplus $v_1 - v_2$, and bidder 2 should earn nothing. Finally, optimal bidding in the English/Japanese auction is incentive compatible because the format, via dominance, forces bidders to truthfully reveal their values.

Although we went through it in chapter 3, reconsider the second-price sealed-bid auction. As you may recall, in this format, also known as a *Vickrey auction*, the two bidders submit sealed tenders in a closed format and the winner is the bidder who submits the highest bid, but this person pays only the amount of the second-highest bid (Vickrey, 1961). Optimal bidding in this auction format is also governed by the principle of weak dominance. Hence, it too is incentive compatible.

Specifically, like the English/Japanese auction, in the second-price sealed-bid auction, players should bid their values. Put differently, this means that bidders can't deviate from submitting their values and do better. Here is the

rationale from the perspective of bidder 1. Suppose that bidder 1 submits a bid, b_1 , that is below her value. At face value, this seems smart because reducing her bid seems naturally like it should increase her surplus, but in a second-price auction this is not true. Remember, here the winner doesn't pay her bid, she pays the second highest bid, submitted in this case, by bidder 2. Hence, bidder 1 can't reduce her bid to increase her surplus. However, what can happen if bidder 1 reduces her bid? There are various possibilities, but only two really matter. If bidder 1 reduces her bid and we have $v_1 > b_1 > b_2$, then bidder 1 still wins, but she would have won by bidding her value, and as just mentioned, she continues to pay b_2 , so her surplus is unchanged. This is the weak part of the weakly dominant strategy—bidder 1 reduced her bid but it had no effect on her payoff. However, there is also some chance that when bidder 1 reduces her bid she will lose the auction, although she would have won if she had just bid her value. That is, by reducing her bid, bidder 1 could find herself in the following situation: $v_1 > b_2 > b_1$, in which she does strictly worse than bidding her value. In sum, reducing your bid below your value in the second-price auction only reduces your chances of winning the contest.

If decreasing one's bid doesn't help, how about increasing it? This is the inclination of many actual bidders, as we will see when we discuss the related experiments. It turns out, however, that these bidders erroneously posit that if they only have to pay the second-highest bid, they can increase their bid to improve their chances of winning the auction. As we will see, *plumping* your bid does increase your chance of winning, but only in situations in which you don't want to win! To understand this rationale, consider the first case that could befall bidder 1: $b_1 > v_1 > b_2$. Here, bidder 1 wins and still receives surplus $v_1 - b_2$, but she would have won and received exactly the same surplus by bidding her value.

The second relevant case is more of a problem. Here, bidder 1 increases her bid, only to discover that $b_1 > b_2 > v_1$. Now she wins the auction but earns a negative surplus because $b_2 > v_1$. This is known as the *winner's curse* (see chapter 3). As you can see, increasing your bid above your value in the second-price auction either doesn't raise your chance of winning or leads to you losing money. Proposition 2 summarizes this logic.

Proposition 2 (Bidding in the second-price sealed-bid auction) *In the second-price sealed-bid auction, it is weakly dominant to bid one's value. This results in a Bayesian Nash equilibrium.*

As with the English/Japanese auction, we can anticipate the result of any two-person second-price auction in which bidders use their equilibrium strategies. Because the high-value bidder will win again, this auction mechanism also maximizes allocative efficiency. In fact, for the same two bidders, the realization of the second-price auction should be identical to that of the English/Japanese auction, as was intended by William Vickrey.

Working out the Bayesian Nash equilibrium in the first-price sealed-bid auction is a bit more complicated, partially because dominance no longer plays a role. In the first-price sealed-bid format, bidders submit tenders simultaneously, the highest bidder wins and pays her bid. More than in the other two auction formats, in the first-price sealed-bid format, the balancing act described previously plays a role. For example, despite maximizing your chance of winning the auction, bidding one's value can't continue to be an equilibrium because when you do so your surplus is surely zero, making you indifferent between bidding and just opting out of the auction. This implies that the optimum bid in this format must be something less than one's value so that there is some surplus to win. The question, however, is how much lower than your value should you bid? You can't submit some arbitrarily low bid because this can result in you losing an auction that you would have won with a slightly larger bid. What this all implies is that the optimal bid must be some fraction of your value. That is, the optimal bidding strategy must be of the form $b_i^* = \alpha v_i$, where $0 < \alpha < 1$ is the amount that bidder i shades her bid.

Now we just have to determine the optimal amount of bid shading in the first-price sealed-bid auction. Exactly how much less than her value should bidder i bid to maximize her expected payoff from the auction? Or, what is α^* ? To begin, we need to figure out the payoff to bidding in this auction for any bidder. The expected payoff for bidder i is just the probability that she wins, $P(v_i)$, when having value v_i times the surplus she achieves if she wins, $v_i - b(v_i)$, where her bid is a function of the value she draws, as discussed previously. Putting these together, we have

$$Eu_i(v_i) = P(v_i)(v_i - b(v_i)) \tag{13.}$$

The first thing to notice is that everything is a function of one's value here. One's value determines how much one bids, the size of the surplus, and the likelihood of winning. The problem, however, is that bidders can't choose their value; they have to choose their bids. Luckily, in the case of the first-price winner-pay auction, we can use the assumptions we made before to

invert the expected payoff so it is a function of one's bid. Recall that we assumed that all the bidders use the common linear bid function, $b^* = \alpha v_i$. Inverting this, we get $v_i = \frac{b}{\alpha}$, which expresses the chance that bidder i wins the auction when there is just one other competitor, as discussed at the end of the previous section. Plugging this in and noticing that $b(v_i) = b$ when we are talking about bids gives us

$$Eu_i(b_i) = \left(\frac{b}{\alpha}\right)(v_i - b)$$

From here, we can solve for the optimal bid function directly by maximizing $Eu_i(b_i)$ with respect to b . Taking the derivative and setting it equal to zero at the maximum yields $\left(\frac{b}{\alpha}\right)(-1) + (v_i - b)\left(\frac{1}{\alpha}\right) = 0$, and if we solve this for b , we get $b^* = \frac{v_i}{2}$. This result is summarized in proposition 3.

Proposition 3 (Bidding in the two-bidder, first-price, sealed-bid auction)

In the first-price sealed-bid auction with two bidders (and values drawn from the uniform distribution over the $[0, 1]$ interval), each bidder should bid $\alpha = \frac{1}{2}$ of her value. This results in a Bayesian Nash equilibrium.

If both bidders behave according to the Bayesian Nash equilibrium for the first-price sealed bid format, what is the realization of the auction? First, at this symmetric equilibrium in which everyone shades the same amount, the high-value bidder will bid more than anyone else, and this person wins the auction. That is, the first-price sealed bid auction is allocatively efficient too. In addition, because this high bidder bids half her value, she earns a positive surplus in expectation, just like the winning bidders in the other two formats. To see this, consider a bidder whose value is $\frac{2}{3}$. This player should bid half of her value, or $\frac{1}{3}$, and earn in expectation $P(v_i)(v_i - b_i)$, or $\frac{2}{3}\left(\frac{2}{3} - \frac{1}{3}\right) = \frac{2}{9}$ based on equation (13.1).

A More General Analysis Is Needed

So far, we have gotten away with making assumptions that dramatically simplify the analysis of auctions, but neither dominance nor limiting the search to linear bid functions can work for all the common auction formats. In this section, we generalize our analysis of symmetric equilibria in two-bidder auctions. In particular, we no longer make assumptions about the specific form of the bid function (other than to continue to assume it is increasing and symmetric) and we forgo the use of the inverse bid function. The primitive of

the following analysis is simply one's value, the ultimate determiner of an auction.

Returning to equation (13.1), we know that in equilibrium, bidders will best respond by bidding according to the optimal symmetric rule, $b^*(v_i)$, so the first thing we can do is substitute this rule into the expected payoff, yielding the maximum value function: the expected payoff from equilibrium play conditional on knowing one's own value but not the values of the other bidders—namely,

$$Eu_i(v_i) = P(v_i)(v_i - b^*(v_i)) \quad (13.)$$

Now, to get a better sense of what happens when bidders bid according to their values in equilibrium, consider how i 's expected utility changes when her value increases a bit. Put differently, we ask: What happens to a bidder's expected surplus while playing the equilibrium bidding strategy when her value increases? In what way is she better off? To see this, let's take the derivative of i 's expected utility with respect to any possible value and evaluate it at the bidder's specific value:

$$Eu'_i(v_i) = \frac{dEu_i}{dv} \Big|_{v=v_i} = \left(\frac{\partial Eu_i}{\partial P} \times \frac{\partial P}{\partial v_i} \right) + \left(\frac{\partial Eu_i}{\partial b} \times \frac{\partial b^*}{\partial v_i} \right) + \frac{\partial Eu_i}{\partial v_i} \quad (13.)$$

What equation (13.3) says is that the result of increasing bidder i 's value a little in equilibrium can be parsed into three effects: the indirect effect of increasing v_i on the probability of winning, $\frac{\partial Eu_i}{\partial P} \times \frac{\partial P}{\partial v_i}$, the indirect effect on i 's bid, $\frac{\partial Eu_i}{\partial b} \times \frac{\partial b^*}{\partial v_i}$, and the direct effect, $\frac{\partial Eu_i}{\partial v_i}$, because one's value is higher and the surplus will be larger. The first thing to note about this equation is that according to the envelope theorem, the indirect effects will be zero. The intuition is that in equilibrium, for example, $\frac{\partial Eu_i}{\partial b}$ has to be zero because $b^*(v_i)$ is an optimum value that places i at the top of some expected utility hill where the function is flat. Hence, this derivative reduces to just the direct effect, $\frac{\partial Eu_i}{\partial v_i}$. Returning to equation (13.2), if you expand the equation and take the derivative, you can see that this direct effect is just

$$Eu'_i(v_i) = \frac{dEu_i}{dv_i} \Big|_{v=v_i} = \frac{\partial Eu_i}{\partial v_i} = P(v_i) \quad (13.)$$

Considering the intuition, the direct effect of increasing one's value a little is just the probability that one wins the auction, $P(v_i)$, which is determined by one's value. More generally, when one's value increases by a little (say by ϵ),

the increase in one's expected payoff is just ϵ times the probability that one wins the auction, $P(v_i)\epsilon$. Another way of thinking about this is to imagine that you mail in a bid that you formulated based on having value v_i . You later realize that your value is actually $v_i + \epsilon$. What happens to your expected payoff after this realization? Well, your bid has already been submitted so that doesn't change and, because you bid according to the equilibrium strategy (and your original value), your chances of winning don't change either. However, because your true value is higher, if you win, your potential surplus increases by ϵ and the expectation is that you get this increment with probability $P(v_i)$, again based on your original value. Summing up, the change in i 's expected utility when her value increases by ϵ is $dEu_i(v_i) = P(v_i)\epsilon$. Dividing both sides by ϵ gives us $\frac{dEu_i(v_i)}{\epsilon} = P(v_i)$, and if we take the limit of the left side as ϵ approaches zero, it becomes the marginal utility of an ϵ bump in value.

Can we extend this intuition to figure out what any bidder should expect to get out of the auction, given her value? The fact that expected utility is just a function of one's value is a reflection of all that we have discussed to this point—one's value determines both the chances of winning and the surplus that one gets when everyone uses a common bid function in equilibrium. Further, because equation (13.4) can be reduced to $Eu'_i(v_i) = P(v_i)$, we know that $P(v_i)$ must be denominated in utils. If we can add up all the marginal utils that accumulate to bidder i as her value rises, we can estimate her total expected payoff in equilibrium. Luckily, this is rather straightforward to accomplish here if we notice that the left side of equation (13.4) is a derivative, which makes it a very simple differential equation. To solve it, we just have to integrate both sides (i.e., sum up all those marginal utils). This gives us

$$Eu_i(v_i) = Eu_i(0) + \int_0^{v_i} P(v_i)dv \quad (13.5)$$

where $Eu_i(0)$ accounts for the constant of integration. We can simplify this a bit more, though, if we realize that this constant must be zero if we assume that the expected payoff of a bidder with a value of zero must also be zero. This leaves us with $Eu_i(v_i) = \int_0^{v_i} P(v_i)dv$, our expectation of what any bidder should earn from the auction. It turns out that equation (13.5) is the central

result of the theoretical auction literature, and we use it to infer the optimal bidding strategy in most auction formats.

As an example of the importance of equation (13.5), let's return to the first-price winner-pay auction. Here, the expected utility is $Eu_i(v_i) = P(v_i)(v_i - b(v_i))$. From [figure 13.1](#), we know that if both bidders adopt a common bidding strategy, then the probability of winning just reduces to one's value when there is just one other bidder. Hence, equation (13.1) becomes

$$Eu_i(v_i) = v_i(v_i - b(v_i)) \tag{13.}$$

Now remember that equation (13.5) also defines the expected utility that any bidder can achieve, this time when using the optimal bidding strategy, $b^*(v_i)$. After realizing a zero-value bidder must expect to get nothing out of this auction too, this equation reduces to $Eu_i(v_i) = \int_0^{v_i} P(v_i) dv$. If we equate these two equations (i.e., equations 13.5 and 13.6) because they both equal $Eu_i(v_i)$ we get

$$v_i(v_i - b(v_i)) = \int_0^{v_i} P(v_i) dv$$

Notice both that $P(v_i)$ appears on the right side of this equality and that $b(v_i)$ appears on the left side. If we substitute v_i for $P(v_i)$ again and rearrange the equation, we get (after some algebra)

$$b(v_i) = \frac{v_i^2 - \int_0^{v_i} v_i dv}{v_i}$$

Taking the anti-derivative on the right side (i.e., $\int v_i dv = \frac{v_i^2}{2}$) results in

$$b(v_i) = \frac{v_i^2 - \int_0^{v_i} v_i dv}{v_i}$$

From here, the optimal bidding strategy, $b^*(v_i) = \frac{v_i}{2}$, follows. This should build some confidence in the more general analysis because this is exactly the same result we got before.

Also, remember that the winning bidder's expected payoff in the first-price winner-pay auction was $\frac{2}{9}$. We should be able to replicate this result with the new analysis too. By equation (13.5), this bidder's expected payoff should be $\int_0^{\frac{2}{3}} \frac{1}{2} \times \frac{2}{3}^2$ which is just $\frac{2}{9}$, as anticipated. To provide a bit of intuition, a plot of $Eu_i(v_i)$ for this example is presented on the left side of [figure 13.2](#), where you can verify this result. In addition, on the right side of [figure 13.2](#), we plot

$Eu'_i(v_i)$, which in this case is just v_i . As you can also see on the right side of [figure 13.2](#), the integral under the curve $Eu'_i(v_i) = v_i$ from $v_i = 0$ to $v_i = \frac{2}{3}$ is approximately the triangle with area $\frac{1}{2} \times (\frac{2}{3} \times \frac{2}{3})$, again $\frac{2}{9}$. As this example illustrates, integrating over all the marginal expected utilities evaluated at the equilibrium captures the trade-off between increasing one's chances of winning and increasing the surplus if one wins.

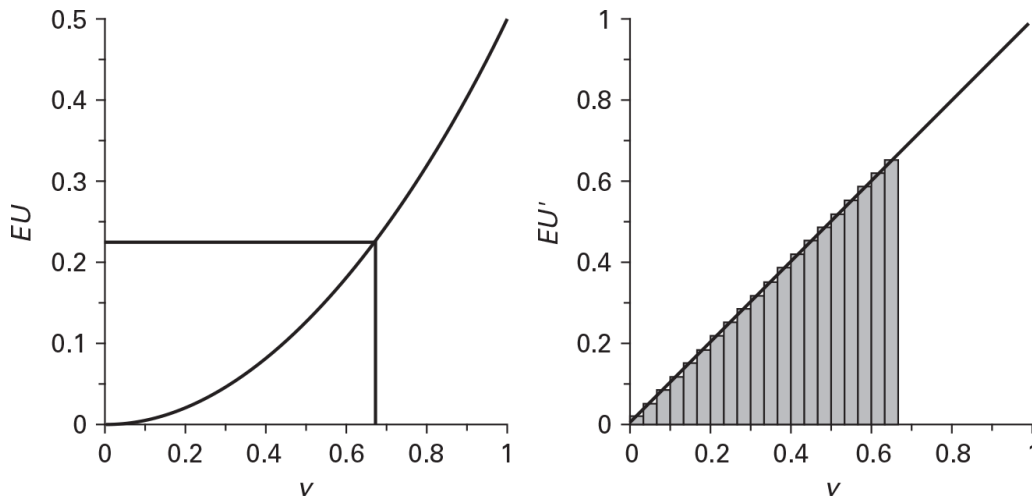


Figure 13.2

The expected payoff of a bidder with $v_i = \frac{2}{3}$ in the two-person first-price auction.

Before we move on, there is one final thing to point out about optimal bidding in the first-price sealed-bid auction, something that, if you are paying attention, might be stuck in your head. Think again of the realization of the auction with just two bidders. If $v_1 > v_2$, we expect bidder 1 to bid $\frac{v_1}{2}$ and win the auction. Given our discussion of [figure 13.1](#), you can see that this bid coincides with bidder 1's best guess as to what bidder 2's value must be, conditional on her winning. That is, if bidder 1 wins, bidder 2's value must be lower or to her left in [figure 13.1](#) and, given that the values are assigned randomly, her best guess is that bidder 2's value must just be halfway, or at $\frac{v_1}{2}$. But if bidder 2 follows the same rule, she bids $\frac{v_2}{2}$, not v_2 , so why can't bidder 1 shade even more and increase her surplus? The simple answer is that by shading more than $\frac{1}{2}$, bidder 1 is increasing the chance of losing the auction proportionately more than she is increasing her surplus. Remember that bidder 1's best guess is that bidder 2's value is at $\frac{v_1}{2}$, but this is still just a

guess. It is always the case that as bidders shade more, they are reducing their chances of winning.

The All-Pay Auction

Given our generalization, we can now tackle the symmetric Bayesian Nash equilibrium in the first-price all-pay auction. An important difference illustrated by the contrast between the all-pay and winner-pay formats is that we shouldn't assume that the symmetric bid function is linear in every auction format (reinforcing the need for a more general analysis). In the all-pay, for instance, you can imagine that if you draw a very low value, you might be hesitant to bid aggressively because you are pretty sure that there will be someone else with a larger value and you will forfeit your bid whether you win or lose. At the same time, if you are lucky enough to draw a high value, you might bid extra aggressively because you pay a heavy price if you lose when you could have won by not shading as much. In other words, there is good reason to believe that the bid function for the first-price all-pay is nonlinear.

Ironically, the all-pay auction is a bit easier to deal with precisely because bidders pay their bids regardless of the outcome. Although the expected profit of bidder i seems more complicated because you have to account for the probability of both winning and losing the auction as follows:

$$Eu_i(v_i) = P(v_i)(v_i - b(v_i)) - (1 - P(v_i))b(v_i) \quad (13.5)$$

a little algebra indicates that this just reduces to

$$Eu_i(v_i) = P(v_i)(v_i) - b(v_i) \quad (13.6)$$

Despite our hunch that the bid function in the all-pay auction isn't linear, we can still proceed as we just did with the first-price winner-pay format. In particular, equation (13.5) still applies in equilibrium (i.e., given your value, the expected benefit from bidding in the auction remains the same), and when we combine it with equation (13.8), we get

$$P(v_i)(v_i) - b(v_i) = \int_0^{v_i} P(v_i) dv \quad (13.7)$$

Because the game remains two-player, as in the winner-pay auction, it is still the case that $P(v_i) = v_i$ in the symmetric equilibrium. If we make this

substitution and rearrange (13.9) to isolate $b(v_i)$ we can find the equilibrium bid function for the all-pay auction. As a first step, we get

$$b(v_i) = v_i^2 - \int_0^{v_i} v_i dv$$

Integrating and subtracting results in $b^*(v) = \frac{v^2}{2}$, a nonlinear function of one's value, as expected. This result is summarized in proposition 4.

Proposition 4 (Bidding in the two-bidder, first-price all-pay auction) *In the first-price all-pay auction with two bidders (and values drawn from the uniform distribution over the $[0, 1]$ interval), each bidder should bid $b^*(v) = \frac{v^2}{2}$. This results in a Bayesian Nash equilibrium.*

To conclude this section, we present a summary of the equilibrium bid functions for the four auction formats in [figure 13.3](#). The 45° line represents bidding one's value in the English/Japanese and second-price auctions, the half as steep, but still linear, bid function reaching a maximum bid of $\frac{1}{2}$ when $v = 1$ is the optimal response of bidders in the first-price auction and the remaining bid function is the equilibrium response in the all-pay auction. As predicted, the all-pay bid function is increasing and nonlinear. Hence, when their values increase, all-pay bidders bid more aggressively.

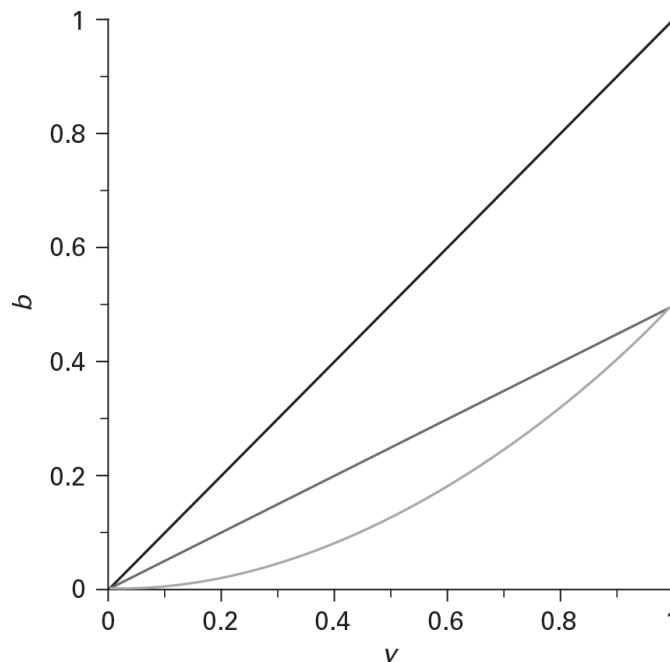


Figure 13.3

Comparing equilibrium bid functions across auction formats.

Revenue Equivalence

Thus, far we've talked a lot about bidding strategies, but only a little about the realization of any of these auction formats. What exactly can bidders, and for that matter sellers, expect to get out of an auction? After all, is there any point in discussing auction formats that are expected to perform poorly for sellers? If one format should always raise more revenue, why would sellers consider a different one? Let's calculate the predicted benefit for the winning bidder and the revenue accruing to the seller for each auction format to see how the formats compare.

Before we get into the details, we first need to discuss our expectations about the distribution of buyer values. If, as we have assumed all along, values are distributed independently and randomly on the $[0, 1]$ interval, and there are just two bidders, what do we expect the values of the two bidders to be? Put differently, on average, if we take two draws from this uniform distribution, what do we expect those two values to be? Intuitively, we expect the draws to split the interval into three equal pieces, with the high value being $\frac{2}{3}$ and the low value being $\frac{1}{3}$ (as assumed in [figure 13.2](#)). Let's continue to assume that bidder 1 is lucky and draws the high value, so that $v_1 = \frac{2}{3}$ and $v_2 = \frac{1}{3}$.

Now consider the English/Japanese auction, the first auction format we discussed in this chapter, in which buyers should bid up to their values. Here, in expectation, the auctioneer will raise the price to $\frac{1}{3}$ and bidder 2 will drop out (we assume that she drops out when indifferent). As a result, bidder 1 pays $\frac{1}{3}$ for something she values at $\frac{2}{3}$ and earns a surplus of $\frac{1}{3}$. However, before the auction takes place, bidder 1 can't know for sure that she has the high value (otherwise this wouldn't be a Bayesian game). What she does know is that she has a $v_1 = \frac{2}{3}$ chance of being the winner (recall [figure 13.1](#)). Hence, the winner's expected profit is $P(v_1)(v_1 - b_1)$. Substituting the value and bid into this equation, we get $(\frac{2}{3})(\frac{1}{3})$ or $\frac{2}{9}$. The seller, by comparison, earns revenue equal to the sales price, $\frac{1}{3}$ in this winner-pay format.

What about sealed-bid auctions? In the second-price format, bidders simply bid their values and bidder 1 wins again because her value is the highest. She only pays the second-highest bid, however. This means that the seller's revenue, in expectation, is $E[\text{revenue}] = E[\min\{v_1, v_2\}]$, or $\frac{1}{3}$. Speaking of expectations, the buyer, bidder 1, earns surplus $v_1 - b_2$ or $\frac{1}{3}$, but we have to

remember that bidder 1 could not be sure that she would win when she submitted her sealed tender because, as in the English/Japanese auction, she could only guess the value of bidder 2. Her expectation, when submitting her bid, was that she had a $v_1 = \frac{2}{3}$ chance of winning. Hence, bidder 1's expected profit is $(\frac{2}{3})(\frac{1}{3})$, or $\frac{2}{9}$ again. Further, in the first-price auction, bidder 1 wins by submitting $\frac{v_1}{2}$. The winner's expected profit when submitting this bid is, in general, $P(v_1)(v_1 - \frac{v_1}{2})$, which equals $v_1(v_1 - \frac{v_1}{2})$ and simplifies to $\frac{v_1^2}{2}$. Plugging $\frac{2}{3}$ in for bidder 1's value yields $\frac{2}{9}$ yet again, as illustrated in [figure 13.2](#). Considering the seller, $E[\text{revenue}] = E[\max\{\frac{v_1}{2}, \frac{v_2}{2}\}]$. This expression can be simplified to $\frac{1}{2}E[\max\{v_1, v_2\}] = (\frac{1}{2})(\frac{2}{3})$, or $\frac{1}{3}$.

We need to work through the last auction format, the first-price all-pay, to see if there truly is a pattern here, but doing that is a little more complicated. Bidder 1's optimal bid is $\frac{v_1^2}{2}$, which in equilibrium is $\frac{1}{2}(\frac{2}{3})^2$, or $\frac{2}{9}$, but this isn't all the revenue because here the loser, bidder 2, has to pay too. Bidder 2's payment is $\frac{1}{2}(\frac{1}{3})^2$, or $\frac{1}{9}$. The problem is that we can't simply add these two up because we are only accounting for one of the possible configurations of bidder values. We need to average over all the possibilities. Considering all the possible realizations of each bidder's value (and the fact there are two potential buyers), we have

$$E[\text{revenue}] = 2 \int_0^1 \frac{v_i^2}{2} dv_i = \int_0^1 v_i^2 dv_i = \frac{1}{3}$$

The fact that the seller always earns $\frac{1}{3}$ must seem like a coincidence. After all, while the English/Japanese and second-price auctions can be solved by dominance and may, therefore, yield similar behavior, the first-price auction format seems very different strategically. Further, though it is intuitive that if all bidders have to pay regardless of whether they win or lose, they will reduce their bids, it is not at all obvious that the sum of their reduced expected bids will be identical to the revenue gathered in all the other formats. Despite this, as we've seen, it now appears that it doesn't matter which format the seller chooses—they all can be expected to generate the same revenue!

Although the four auction formats that we have considered have different rules about what the bidders have to pay and how the bids are collected, in every case the highest bidder wins, so how different are all these auction mechanisms really? It turns out that all these formats are standard in the following sense: A *standard* auction is one in which: (1) the high-bidder wins;

(2) bidder i is asked to pay an amount $p(b_i, b_j)$ that may depend on the bids submitted by the other bidders; (3) at the Bayesian Nash equilibrium participants bid according to a rule, $b(v)$, that is increasing in v , assuring that that the highest value bidder wins; and (4) bidders with a value of zero expect to earn nothing in equilibrium (i.e., $Eu_i(0) = 0$), an assumption that we have already invoked. As you can see, the English/Japanese auction, the second- and first-price sealed-bid auctions, and the first-price all-pay auction are all standard. If we limit our attention to the large class of standard auctions, there is one foundational theorem in auction theory, which appears in theorem 1.

Theorem 1 (Revenue equivalence) *In two-person auctions with values drawn independently from the uniform distribution over the $[0, 1]$ interval, every standard auction yields the same revenue. In expectation, that revenue is $\frac{1}{3}$.*

For an intuitive proof of theorem 1, we just have to consider the total surplus at stake in an auction. The total surplus (TS) is simply the sum of the consumer surplus and the producer surplus. In an auction, this is the same as the sum of the bidder surplus (BS) and the seller's surplus (SS), so $TS = BS + SS$. Further, given that there is just one seller and if we assume the marginal cost to the seller of supplying the item for auction is zero, then the seller's surplus is just the revenue garnered in the auction (i.e., $SS = \text{revenue}$). This means that to prove that the revenue is the same in any standard auction, we just have to demonstrate that both the buyer surplus and the total surplus are constant.

Let's start with the buyer surplus. Here, we can return to equation (13.5) one more time. Rewriting it without subscripts, we have

$$Eu(v) = Eu(0) + \int_0^v P(v) dv$$

and for the uniform value, two-bidder, standard auction case, this just collapses to

$$Eu(v) = Eu(0) + \int_0^v v dv = \frac{v^2}{2}$$

which tells us the expected surplus of the bidder whose value is v . Considering that our values span the $[0, 1]$ interval, we can get the average or expected profit of any bidder in a standard auction by integrating again (i.e., recall that you integrate to get the expectation). In this case, we have

$$E[Eu(v)] = \int_0^1 Eu(v) dv = \int_0^1 \frac{v^2}{2} dv = \frac{1}{6}$$

Notice that this is not the $\frac{2}{9}$ we got at the beginning of this section because there we only considered one of the two bidders, the winner, but it is exactly half of what we found when we considered both bidders in the all-pay.

Speaking of the all-pay auction, recall that the total consumer surplus there was equal to $\frac{1}{3}$, the combined expected profit of the two bidders. As you can now see, in any standard auction with two bidders, the total bidder surplus is going to be $2 \times E[Eu(v)]$, which is just $2 \times \frac{1}{6} = \frac{1}{3}$. Hence, $BS = \frac{1}{3}$ for any standard auction.

What about the total expected surplus in a standard auction (TS)? In general, the total surplus when 1 unit is exchanged in the market is just the difference between the buyer's value and the seller's marginal or opportunity cost of providing the unit, something that we assumed to be zero in this setting. This means that the total surplus available is just the value of the auction winner, which will always be the highest value (in this case $E[\max\{v_1, v_2\}]$), and from the beginning of this section, we know this is just $\frac{2}{3}$.

Let's finish the proof by combining what we now know about standard auctions when there are two bidders and their values are uniform on the $[0, 1]$ interval. First, the total expected surplus available in the auction will always be $TS = \frac{2}{3}$, just the expected value of the winner. Second, the total expected bidder surplus in the auction will always be $BS = \frac{1}{3}$ by equation (13.5). As you can already see, both of these amounts are constant; they don't vary by auction format, so neither will the expected revenue. Here, the expected revenue is just $TS - BS = \frac{1}{3}$. The important insight here is that, if we can show that the potential surplus will always be the same in any standard auction, which makes sense because the auction format does not determine the underlying values of the bidders, and show that the expected consumer surplus will also be the same across formats, which is a little more difficult, then the revenue has to be the same.

Private Value Auctions with n Bidders

To get the intuition for how bidding works in auctions, we began our discussion by considering the competition between just two active bidders. However, most auctions tend to be much "thicker" (i.e., they have many more bidders) than this. In this section, we analyze equilibrium bidding in auctions

with any number of bidders, n . As before, we start the discussion with the English/Japanese auction and then consider the sealed-bid formats: second-price, first-price, and all-pay.

As the number of bidders increases, how should you adjust your bidding in the English/Japanese auction? Actually, should you change your strategy at all? Given that bidding one's value is weakly dominant in the two-person English/Japanese auction, there is no reason why adding bidders should weaken this dominance. So long as any bidder can raise her bid to stay in the auction (without exceeding her value, of course), she should do so, no matter how many other bidders there are. Hence, proposition 1 describes bidding in the English/Japanese auction for any number of bidders.

How about the second-price sealed-bid auction? Well, because the equilibrium bidding strategy in the second-price auction is also based on dominance, bidding your value in this format must also remain an equilibrium strategy as more active bidders are added. In fact, because there are more draws from the underlying distribution of values when we add bidders, you can see that not bidding more than your value to avoid the winner's curse is even more important in thick second-price auctions. If we continue to assume that values are drawn independently from the uniform distribution over the $[0, 1]$ interval, we now know that if there are just two bidders, we expect them to have values of $\frac{1}{3}$ and $\frac{2}{3}$, but if there are three bidders we expect them at $\frac{1}{4}$, $\frac{1}{2}$ and $\frac{3}{4}$. Importantly, you can see that as the number of bidders grows, the expected gap between your value and the value of the bidder directly below you shrinks, increasing the chance of ending up in a situation in which someone sneaks a bid in between your bid and your value when you bid above your value. The bottom line, however, is that in auctions that are dominance-solvable, the equilibrium bidding strategy is invariant to the number of active bidders.

What about auction formats in which dominance doesn't play a role? What is the role of adding bidders in the first-price auction, for instance? Perhaps we can build on the previous discussion to gain some intuition before we derive the result. As we saw in the two-person case, when the expectation was that the high bidder would have value $\frac{2}{3}$, she ended up shading her bid such that it equaled her expectation of where the second-highest bidder was, $\frac{1}{3}$. With three bidders, the one who, in expectation, receives a value of $\frac{3}{4}$ should expect the second-highest bidder to have a value of $\frac{1}{2}$ and if the basic logic

persists, she should then bid this amount. As you can see, when the number of bidders increases, we should probably see that each bidder shades less. With two bidders, the winner shades their bid by $\frac{1}{3}$ or half their value, but with three bidders, the winner shades only $\frac{1}{4}$, in this case one-third of her value. Perhaps you can see the pattern here. With a total of n bidders, each should shade by $\frac{1}{n}$ of his value. If they each do this, in equilibrium, they bid

$$b^*(v_i) = v_i - \frac{v_i}{n} = \frac{n-1}{n} v_i$$

Let's see if we can confirm this intuitive result using the framework adopted at the beginning of the chapter. The big difference between the n -bidder first-price auction and the 2-bidder case is the probability of winning the auction. As we saw in [figure 13.1](#), if everyone adopts the symmetric bidding strategy, your chance of beating one other bidder was just v_i . Now, if there are three total bidders, because the values are drawn independently, your chance of beating a second competitor is also just v_i , by the same logic. However, this implies that the likelihood of you beating them both is v_i^2 . More generally, your chance of beating any $n - 1$ competitors is just v_i^{n-1} . Equating equations (13.4) and (13.5) again, this time substituting $P(v_i) = v_i^{n-1}$, gives us (using the maintained assumption that the auction is standard so $Eu_i(0) = 0$)

$$v_i^{n-1}(v_i - b(v_i)) = \int_0^{v_i} v_i^{n-1} dv$$

That can be rearranged as

$$b(v_i) = v_i - \frac{\int_0^{v_i} v_i^{n-1} dv}{v_i^{n-1}}$$

Integrating gets us

$$b(v_i) = v_i - \frac{\frac{v_i^n}{n}}{v_i^{n-1}}$$

and some algebra confirms the result, summarized in proposition 5:

$$b(v_i) = \left(\frac{n-1}{n} \right) v_i$$

Proposition 5 (Bidding in the n -bidder, first-price sealed-bid auction) *In the first-price sealed-bid auction with n bidders (and values drawn from the*

uniform distribution over the $[0, 1]$ interval), each bidder should bid v_i^2 of her value. This results in a Bayesian Nash equilibrium.

As before, the same rationale for the chances of winning at the symmetric equilibrium extends to the all-pay auction. In this case, we can utilize equations (13.5) and (13.7) to derive the optimal bidding strategy in the n -person, first-price all-pay auction. Substituting again for $P(v_i) = v_i^{n-1}$ yields

$$v_i^{n-1} v_i - b(v_i) = \int_0^{v_i} v_i^{n-1} dv$$

or

$$b(v_i) = v_i^n - \int_0^{v_i} v_i^{n-1} dv$$

Integrating and a bit of algebra results in the following, which is summarized in proposition 6

$$b(v_i) = \left(\frac{n-1}{n} \right) v_i^n$$

Proposition 6 (Bidding in the n -bidder, first-price all-pay auction) *In the first-price all-pay auction with n bidders (and values drawn from the uniform distribution over the $[0, 1]$ interval), each bidder should bid $b^*(v) = \left(\frac{n-1}{n} \right) v^n$. This results in a Bayesian Nash equilibrium.*

Common Value Auctions

Recall that in many auctions, all the bidders would value the item being sold the same, if they had all the information they needed to assess this value. Instead, in *common value auctions*, the n bidders each have private information, a signal s_i , about the true value of the item. For instance, the bidders may be bidding on the rights to harvest trees in Maine and each one of them might hire a forester to inventory all the trees in one random acre of the forest. In a common value auction, the bidders must formulate their bids based on just this signal, but if they had access to the signals received by all the bidders, they would have a much better sense of the forest's value. As a result, compared to *private value auctions*, in which the payoff to bidders depends just on their own type/value and a common bid function, in common value auctions, one's payoff depends on the private information (i.e., the signals) held by the other bidders too. Further, you can anticipate the

consequences of your signal being very optimistic in this type of auction: if your forester paints too rosy a picture of the number and quality of the trees in the forest, almost by definition, you will overpay and suffer the winner's curse.

As is the pattern of this chapter, let's start with a two-bidder example to build the intuition. We will also approach the search for a symmetric equilibrium bid function slightly differently, an alternative method that you can return to in the exercises at the end of this chapter. To be consistent, let's continue to use the uniform distribution between 0 and 1, this time for the signal that each bidder receives. Hence, $s_i \in [0, 1]$ and the probability that $s_j < s_i$ is just s_i , similar to [figure 13.1](#). Let's also suppose that there is some benefit to knowing all the signals. Specifically, we can assume that collectively, the foresters accurately assess the true value, v , of the timber they see. In other words, $v = \sum_{i=1}^n s_i$. Finally, we can assume that the first-price winner-pay format will be used, along with the symmetric linear bid function $b_i = \alpha E[v]$, where $E[v]$ is any bidder's expectation about the true value of the item.

How much should the two bidders bid in this setting? As a place to start, what if the bidders are conservative and don't shade (i.e., $\alpha = 1$)? The first problem is that each bidder has to make an assessment of the value by estimating the signal received by the other bidder. That is, she calculates $E_i[v] = s_i + E_i[s_j]$. Given that each bidder receives her own signal, this boils down to figuring out $E_i[s_j]$, but isn't this just $\frac{1}{2}$, the expected signal, given signals are uniformly distributed between 0 and 1? Let's see where this expectation leads us. Suppose that bidder 1 receives $s_1 = \frac{2}{3}$ and, therefore, bids $b_1 = \frac{2}{3} + \frac{1}{2}$ or $\frac{7}{6}$. Here is the problem. Bidder 1 winning the auction has to mean that $s_2 < \frac{2}{3}$ if the second bidder is using the same rule. Further, this implies that the conditional expectation of what bidder 2's signal is, given 1 wins, must be $E_1[s_2 | s_1 > s_2] = \frac{1}{3}$. That is, if I win and the signals are uniformly distributed, my best guess is that the signal of my competitor has to be half of my own signal, not half of the maximum possible signal. This all implies that the true expected value of the timber is $E_i[v | s_1 > s_2] = \frac{2}{3} + \frac{1}{3}$, and bidder 1 is likely to suffer the winner's curse—she will pay $\frac{7}{6}$ for something only worth 1. This is the predicted result for unsophisticated bidders in common value auctions. The key insight, which we develop further next, is that bidders in common value auctions have to shade *twice* to avoid the winner's curse. They have to shade as is normal if, for example, the first-price auction is used, but they also

have to shade in the sense that when they form expectations about the signals received by the other bidders, they have to use the conditional expectation, assuming that they win, not the unconditional one.

To generalize our two-person result, consider an equilibrium in which people continue to use the common linear bid function, $b_i = \alpha E[v]$, but this time we leave α unspecified, as in the derivation of proposition 3. In this case, however, bidders only see private signals, so we assume that $b_i = \alpha s_i$. If we invert this bid function, we get $s_i = \frac{b_i}{\alpha}$ and we can use this to reformulate equation (1) in terms of bidder i 's bid instead of her value. Specifically, the bidder with signal s_i has a $\frac{b_i}{\alpha}$ chance of winning the first-price winner-pay auction in the symmetric equilibrium and, therefore, her expected utility, as a function of her bid, is

$$Eu_i(b_i) = \frac{b_i}{\alpha} (s_i + E[s_j | s_i > s_j] - b_i) \quad (13.1)$$

In addition, the conditional expectation of what bidder j 's signal must be, given that bidder i wins or $E[s_j | s_i > s_j]$, is just equal to $\frac{b_i}{2\alpha}$ in this setting. Substituting, we get

$$Eu_i(b_i) = \frac{b_i}{\alpha} \left(s_i + \frac{b_i}{2\alpha} - b_i \right)$$

and notice that if we take the derivative of this equation with respect to the bidder's choice variable, b_i , we can directly solve for the equilibrium bid function. That derivative, set equal to zero, is

$$\frac{dEu_i}{db_i} = \frac{b_i}{\alpha} \left(\frac{1}{2\alpha} - 1 \right) + \left(s_i + \frac{b_i}{2\alpha} - b_i \right) \frac{1}{\alpha} = 0$$

Solving this for b_i yields

$$b_i = \left(\frac{\alpha}{2\alpha - 1} \right) s_i$$

Finally, remember that we assumed that the bid function was linear such that $b_i = \alpha s_i$, so in equilibrium, it must be the case that

$$\alpha = \left(\frac{\alpha}{2\alpha - 1} \right) \quad (13.1)$$

and if we solve this we find that $\alpha = 1$ or $b_i = s_i$.

Proposition 7 (Bidding in the two-bidder, first-price common value auction) *In the first-price common value auction with two bidders (and*

values drawn from the uniform distribution over the $[0, 1]$ interval), each bidder should bid her signal, $b_i = s_i$. This results in a Bayesian Nash equilibrium.

The interpretation of the result in proposition 7 is interesting. Here, the bidders effectively assume that, conditional on winning, the value of the item must be approximately one and a half times their signal. This assumption, based on the conditional expectation discussed previously bakes in a first round of shading. A second round of shading comes from the fact that in equilibrium, bidders don't bid $1.5s_i$; they only submit a bid equal to s_i —the result, again, of this being a first-price auction.

As a final illustration, one that will also help with the exercises at the end of the chapter, let's see how this result generalizes to n -bidder auctions. As in the private value case, the big difference when we add bidders is the effect on the chances of winning. However, now there will be a similar effect driven by the private information held by the other bidders on any bidder's expectation about the true value of the item. Reformulating equation (13.10) to account for $n - 1$ other bidders, we have

$$Eu_i(b_i) = \left(\frac{b_i}{\alpha}\right)^{n-1} \left(s_i + (n-1)\frac{b_i}{2\alpha} - b_i\right) \quad (13.1)$$

In this equation, if $\frac{b_i}{\alpha}$ is the chance of beating any other bidder, as in equation (13.10), and the signals are assigned independently, then the chance of beating all $n - 1$ of them is $\left(\frac{b_i}{\alpha}\right)^{n-1}$. Also, if bidder i 's expectation, conditional on winning, is that the signal of some other bidder must be $\frac{b_i}{2\alpha}$, then her expectation about the sum of all the unseen signals must be $(n-1)\frac{b_i}{2\alpha}$. Here, after combining the two terms containing b_i in the parentheses in equation (13.12), the first-order condition for the equilibrium bidding strategy is

$$\frac{dEu_i}{db_i} = \left(\frac{b_i}{\alpha}\right)^{n-1} \left(\frac{n-1-2\alpha}{2\alpha}\right) + \left(s_i + \frac{b_i(n-1-2\alpha)}{2\alpha}\right) (n-1) \left(\frac{1}{\alpha}\right) \left(\frac{b_i}{\alpha}\right)^{n-2} = 0$$

Solving this condition for b_i yields something that looks familiar:

$$b_i = \left(\frac{2\alpha}{1+2\alpha-n}\right) \left(\frac{n-1}{n}\right) s_i$$

However, as before, the product before s_i must equal α in equilibrium, so

$$\alpha = \left(\frac{2\alpha}{1+2\alpha-n}\right) \left(\frac{n-1}{n}\right)$$

which can be reduced to

$$\alpha = \left(\frac{2+n}{2}\right) \left(\frac{n-1}{n}\right)$$

and is the basis for proposition 8.

Proposition 8 (Bidding in the n -bidder, first-price common value auction) *In the first-price common value auction with n bidders (and values drawn from the uniform distribution over the $[0, 1]$ interval), each bidder should bid $(\frac{2+n}{2})(\frac{n-1}{n})$ of her signal. This results in a Bayesian Nash equilibrium.*

Checking our results, you can see that for $b_i = (\frac{2+n}{2})(\frac{n-1}{n})s_i$, when there are just two bidders, this collapses to $b_i = s_i$, as in proposition 7. Again, we see bid shading resulting from bidder i 's expectation about the true value of the item auctioned, as well as the fact that this is a first-price auction. However, notice here that in the general case, when n increases, so does $\frac{2+n}{2}$, implying that in thicker auctions, there is less of both types of shading.

Behavior in Games—Revenue Nonequivalence?

Based on the areas of auction theory that we have covered to this point, we will focus on three topics of experimentation: the extent to which the revenue equivalence theorem holds in the lab, bidding behavior in all-pay auctions, and the ubiquity of the winner's curse in common value auctions. Beginning with revenue equivalence, in addition to comparing across all formats, there are pairwise comparisons that seem particularly telling. For instance, when you think about the English/Japanese and second-price sealed-bid auctions, it almost seems as if the second-price auction could have been concocted as a way to replicate the outcome of the English/Japanese auction in a sealed-bid format. In other words, shouldn't you use the same bidding logic in both the English/Japanese and second-price sealed-bid auctions? Indeed, regardless of how bids are collected, if two formats incentivized bidders to truthfully reveal their types (i.e., bid their values), in this case via dominance, then the formats are *strategically equivalent*.

As another example, consider what is known as the *Dutch* auction, wherein prices decrease instead of increase in an open format like the English/Japanese auction and the bidder who speaks first (indicating that she is willing to pay the current price) stops and wins the auction. As the name

implies, this format is used to auction flowers in the Netherlands (as well as fish in both Israel and Japan). Now think about the realization of this auction. In principle, the high-value bidder should start to get a little nervous when the price descends to her value because she does not know for sure how far to let it continue to decline before stopping the auction. The more the price drops, the greater is the high-value bidder's surplus, but if she lets it go for too long, the second-highest-valued bidder will surely stop it first. Notice that the essential conundrum of the high-value bidder is the same as it is in a first-price sealed-bid auction—where is that second-highest bidder? This just means that the Dutch and first-price sealed-bid auctions are also strategically equivalent.

As a rather low first hurdle for revenue equivalence, it should certainly be the case that strategically equivalent formats will produce the same pattern of bidding, and therefore the same revenue. It turns out, however, that this is not the case. Comparing the difference in experimental auction bids between English/Japanese and second-price sessions, Kagel, Harstad, and Levin (1987) found that while bidders in the English/Japanese format quickly adopt the equilibrium bidding strategy ($b_i = v_i$), participants in the second-price auction routinely submitted bids above their values. On average, second-price bidders exceed their values just a bit (11%, on average), but they do it very frequently, almost two-thirds of the time. This behavior is suspiciously consistent with the most difficult bit of equilibrium logic to understand with the second-price auction: bidding over your value increases your chance of winning only in auctions that you don't want to win. Further, one reason why overbidding persists is that it is punished infrequently. Only the winner of any given auction could ever suffer the winner's curse, so most overbidders never realize their losses. In fact, John Kagel and his coauthors found that only 6% of players lose money. But, what happens if the punishment is more severe? Do bidders learn not to overbid more quickly? When Georganas, Levin, and McGee (2017) took any losses and multiplied them by 20, bids did settle closer to the equilibrium predictions. In short, bidding in the second-price auction is harder to comprehend than in the English/Japanese auction, and where people get stuck is understandable; however, when the cost of screwing up is increased, they learn to bid more carefully.

Moving to an assessment of the theoretical equivalence of the Dutch and first-price auctions, Coppinger et al. (1980) and Cox et al. (1982) both found

that bidders tend to be more aggressive, again, in the sealed-bid format. On average, participants in Dutch auctions submitted bids that hovered around the equilibrium price but were 5% lower than bids coming from first-price participants. One hypothesis, consistent with the experiments that followed, is that bidders falsely update their priors about where their second-highest rival is as the price drops. That is, participants should not change their assessment of where the other bidder values are as the price starts to drop, but they seem to do just that.

Overall, we see that closed, sealed-bid formats tend to result in bids that are higher than both the equilibrium benchmark and in a strategically equivalent open format. Although the procedures and subject pools differ, in a crude attempt to aggregate across studies, we tend to see the most persistent overbidding in the first-price auction, followed by the second-price auction. Bids in the English/Japanese auction settle relatively quickly at the equilibrium, and Dutch bids hover at or below the equilibrium. As you can infer, if most of the bids deviate from their equilibrium values, revenue equivalence cannot hold, something shown a bit more recently in the field experiment described in Lucking-Reiley (1999).

Despite the increased risk of suffering a loss, in first-price all-pay auctions conducted in the lab, we also see a lot of overbidding. Returning to [figure 13.3](#), it is important to remember that the equilibrium bid function is nonlinear for the all-pay auction, and this implies very different behavior for people who have objectively low values (and should perceive a low chance of winning) than for other participants with high values. Noussair and Silver (2006) conducted experiments with very similar procedures as Cox et al. (1982), changing just the payment rule and found that bidding does fit this pattern, though only roughly, as you can see in [figure 13.4](#). In fact, bidding zero is common for participants with low values, even though these players should bid small positive amounts (you never know—while your value is small, there is still some chance that everyone else's is lower). Further, those bidders with higher values bid considerably more than their values on average. Combined, Noussair and Silver considered this behavior to be a perverse type of risk aversion, wherein bidders do not wish to commit resources unless they have a very high chance of winning. You can anticipate that the implications of these bidding errors are likely to be good for the seller. Not only did Noussair and Silver (2006) find that the all-pay auction

garners more revenue than predicted, they also compared it to the first-price winner-pay results of Cox et al. (1982), where we already know that revenues are higher than predicted, and found that the all-pay does even better than this benchmark. In sum, the all-pay experiments are another strike against the empirical validity of revenue equivalence.

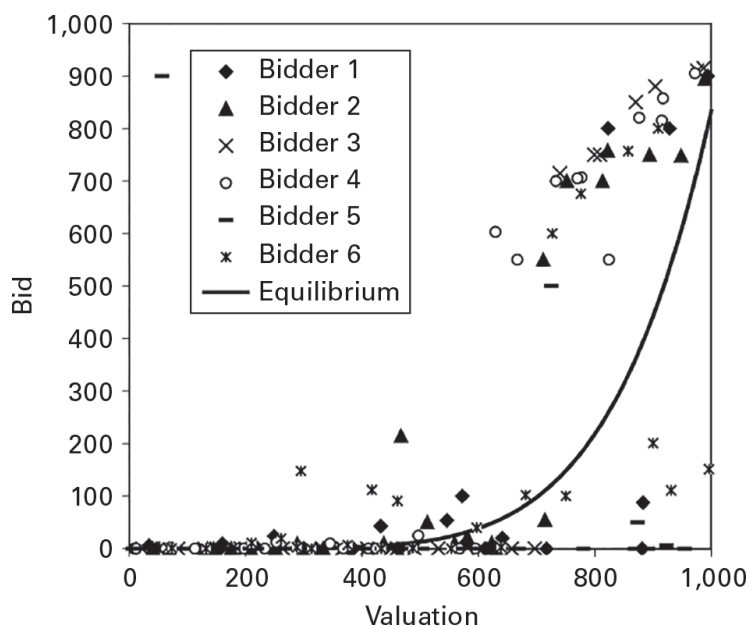


Figure 13.4

Actual versus predicted bids in an all-pay auction (Noussair and Silver, 2006, figure 2c).

Speaking of bidding more than you should, experiments have also discovered a lot of overbidding in common value auctions. In this case, the overbidding almost always results in the winner's curse. Considering an experiment using a similar setup to the models discussed in the previous section, Avery and Kagel (1997) conducted two-bidder auctions in which the common value of the item sold is just the sum of the signals received. The only difference is that Avery and Kagel used the second-price format instead of the first-price; however, this makes the equilibrium prediction a little bit easier for bidders to discover because they only have to shade once. Specifically, suppose that s_1 and s_2 are the signals received by the two bidders. How much should bidder 1 bid, accounting for the fact that if she wins, bidder 2 must have a lower signal?

It turns out that in the symmetric equilibrium, $b_i^* = 2s_i$. Here is the rationale: suppose that $s_1 > s_2$; then bidding $2s_1$ will always be larger than any sum $s_1 +$

s_2 , so bidder 1 wins. Further, $s_1 + s_2$ will always be larger than $2s_2$, the price that bidder 1 pays in equilibrium. Hence, by bidding twice her signal, bidder 1 guarantees both a win and a positive surplus. Notice that bidder 2 can't do better because she would have to raise her bid to $2s_1 + \epsilon$ to win, and by definition, this would result in the winner's curse.

As anticipated, when participants bid in the lab, many of the winners lose money. For the experiment, Avery and Kagel chose signals drawn randomly from the $[1, 4]$ interval, so on average, the high signal will be 3 and the low signal will be 2. This means that in equilibrium, their expectation is that the high-signal bidder bids 6, the low-signal bidder bids 4, and the value is $2 + 3 = 5$. This means that the high-signal bidder should win, pay 4 for something worth 5, and earn a surplus of 1. Instead of this prediction, the average price is closer to 4.60, and 40% of inexperienced winners lost money. When Avery and Kagel estimated the bid function actually used by inexperienced bidders, they found $b_{actual} = 1.13s_i + 2.64$, not $2s_i$.

What could be going on here? Recall the first example of the winner's curse in a common value auction we discussed earlier in this chapter, wherein the naive bidder reckons that she should bid the sum of her value and the *unconditional* expected signal as an estimate of the other bidder's signal. In this setting, that would be $b_{naive} = s_i + 2.5$ because the unconditional expectation of the uniform distribution over the $[1, 4]$ interval is 2.5. Notice how similar this is to the actual behavior. On average, auction participants behave according to this naive bidding rule, which we know will surely lead to the winner's curse. In fact, even with experience, bidders creep only slightly toward the equilibrium strategy. After playing many rounds, the estimated bid function for experienced players is $b_{actual} = 1.34s_i + 1.99$. As you can see, the weight put on one's own signal has increased and the intercept has decreased, but this behavior is still far from bidding twice one's signal. Even here, almost 30% of winners lose money.

The essential problem faced by bidders in common value auctions is understanding the implied adverse selection problem—the difference in relying on the conditional expectation of the item's true value instead of the unconditional expectation. Put differently, do bidders understand that whenever a bid is high enough to win, it must also be higher than the item's true value? At this point, the preponderance of evidence suggests that bidders fail to understand this logic, a fact that has been demonstrated in even more

simplistic decision environments (e.g., Charness and Levin, 2009). The question is: how can evidence-based advances in game theory deal with this issue to improve predictions?

Evidence-Based Adaptations of the Standard Model

There have been at least two promising attempts to alter the basic common value auction model to account for the biased decision making seen regularly among bidders. Viewed from 10, 000 feet, these two competing models aim to provide the same thing: predictions that better match the bidding behavior we observe using a mixture of impulses, part traditional rational actor and part biased decision maker. Where the models differ is the ultimate source of the equilibrium balance of these impulses. Crawford and Iriberri (2007) use the *level-k* approach, described in detail in chapter 27, to model the equilibrium mixture of different types of bidders. According to this explanation, there are some unsophisticated level-0 bidders, who just choose a bid randomly, some level-1 bidders, who best respond to the level-0 behavior, some level-2s, who best respond to the level-1s and so on. It turns out that in the common value auction, bidders do not have to be very unsophisticated to generate a significant incidence of the winner's curse. In fact, a combination of mostly level-1s and a few level-2s is all that is needed. To see this, notice that if the level-1s best respond to the level-0s who are bidding at random, level 1s must believe that winning conveys no new information about the true value of the object. Hence, level-1s do not use the conditional expectation; rather, they use the unconditional one, which, as we have seen, is the root of the winner's curse.

For another perspective on the winner's curse, consider the *cursed equilibrium model* offered by Eyster and Rabin (2005) and discussed previously in chapter 11. Here, bidders best respond to the distribution of other player bids but aren't necessarily clever enough to understand how the bids of others relate to their signals. To implement this adaptation, suppose that $0 \leq \chi \leq 1$ is the extent that any bidder is "cursed," meaning that the player uses the naive unconditional expectation of the signal received by the other players instead of the conditional expectation. As a result, for the second-price auction, in equilibrium, bidders show mixed impulses, bidding $b_i(s_i) = (1 - \chi)v(s_i, \bar{s}) + \chi r(s_i)$, where $v(s_i, \bar{s})$ is the conditional expectation of the item's common value, given i 's own signal and the highest of the other bidders'

signals (given i wins), $\bar{s}$, and $r(s_i)$ is the unconditional expectation of the item's value. To put a finer point on the prediction, using Avery and Kagel's experimental parameters (described previously), we have

$$b_i(s_i) = (1 - \chi)2s_i + \chi \left(s_i + \frac{5}{2} \right)$$

which reduces to $b_i(s_i) = (2 - \chi)s_i + (\frac{5}{2})\chi$. Observe that when $\chi = 0$ and the bidder is completely rational, $b_i = 2s_i$ (the Bayesian Nash prediction), and when the bidder is fully cursed and $\chi = 1$, she bids $b_i = s_i + \frac{5}{2}$, exactly the naive bidding strategy described previously. Finally, can we back out the implied level of the curse using the bidding data collected by Avery and Kagel? Using nonlinear least squares, Eyster and Rabin estimated that $\chi = 0.63$ fits the bid data best. In sum, there is evidence that both the concept of cursed equilibrium, in which the impulse to use the unconditional expectation is more likely, and the level- k approach dominated by level-1 agents best responding to the anticipation of the unconditional expectation being correct can explain the persistence of the winner's curse.

Exercises

(1) *The English/Japanese auction.* Suppose that values for five bidders in an English/Japanese auction are drawn at random from a uniform distribution bounded by 0 and 1.

- (a) If everyone bids according to the Bayesian Nash equilibrium, predict the resulting auction price and the surplus for the winner.
- (b) Now suppose that the values for four bidders are drawn independently from a uniform distribution bound from below by 25 and from above by 100. Predict the auction price and surplus to the winner.

(2) *The second-price auction.* Predict the winning bid, price paid, and surplus in the following second-price auctions:

- (a) There are three bidders, with values drawn independently and randomly from the interval bound from below 0 and from above by 8.
- (b) There are five bidders, with values drawn independently and randomly from the interval bound from below 10 and from above by 64.
- (c) There are eight bidders, with values drawn independently and randomly from the interval bound from below 10 and from above by 64.

(3) *Using the inverse function $b^{-1}(b)$ to find $b^*(v)$ directly.* Reconsider the two-bidder, first-price winner-pay auction. In the original derivation of equilibrium bidding, notice that expected utility was formed in terms of a bidder's value, v . What if we reformulate expected utility in terms of the bid instead, dropping the subscripts? If we continue to assume that in the symmetric equilibrium the bid function is $b = \alpha v$, we can invert this function, solving it for v instead, and get $v = \frac{b}{\alpha}$. Whereas v was the probability of having a larger value than a second bidder in the uniform $[0, 1]$ case before, now the probability is just $\frac{b}{\alpha}$.

- (a) Write the expected utility for any bidder using the inverted bid function and then maximize the expected utility with respect to the bidder's choice variable, b_i . What is the Bayesian Nash equilibrium bid function?
- (b) Here is one to think about: what does the fact that α fell out of the first-order condition say about bidding in equilibrium?

(4) *The inverse function $b^{-1}(b)$ for the n -bidder case.* Use the inverse bid function again to directly calculate the equilibrium bid in the first-price winner-pay auction when there are n bidders and values are still drawn randomly in the $[0, 1]$ interval.

(5) *The inverse function $b^{-1}(b)$ and the all-pay auction.* One more time: Use the inverse bid function again to directly calculate the equilibrium bid in the first-price all-pay auction when there are two bidders and values are still drawn randomly in the $[0, 1]$ interval. Here, notice that the symmetric equilibrium you are trying to prove is where $b = \kappa v^2$, and κ won't fall out this time.

(6) *All-pay auction versus Tullock contest.* The all-pay auction is also used to model wars of attrition and other kinds of contests. The Tullock contest is another form of contest in which the amount invested directly determines the chance that a contestant will win. Here, the probability of winning is $\frac{b_i}{\sum b_j}$, where b_i is i 's bid or investment in the contest. If the prize of this contest is one's value, v_i , and one has a common endowment, E , used to finance bids, contestant i 's expected utility is $Eu_i = (E - b_i) + \frac{b_i}{\sum b_j} v_i$.

- (a) Suppose that there are n contestants. Derive the symmetric equilibrium bid in the Tullock contest, b^* , as a function of n and v .

- (b) Compare the equilibrium bids in the Tullock contest and the n -person all-pay auction derived in this chapter. If there are just two bidders, when do you bid more aggressively in the all-pay auction?

(7) *First-price bidding with a known common component.* Suppose that there are two bidders who compete in a first-price winner-pay auction, and each bidder's value is partially idiosyncratic and partially common; however, all the bidders know that the common factor is 0.5. That is, each bidder's value equals $0.5 + v_i$, where v_i is an independent and random draw from the unit interval, $[0, 1]$, as before. What is the symmetric bid function in this case?

(8) *Additional common value auctions.* As in this chapter, suppose that there are two-bidders who each independently receives a uniformly distributed signal, s_i , this time on the $[0, 10]$ interval. The true, common value of the item for sale is $s_1 + s_2$.

- (a) Suppose that the auction is second price. What is the equilibrium bid function?
- (b) Suppose that the auction is first price. Can you prove that the conservative strategy, $b_i = s_i$ is a Bayesian Nash equilibrium?

V

Bargaining and Cooperative Game Theory

14

Non-cooperative Bargaining

Over the course of the next few chapters, we will consider bargaining situations, in which two or more players can be made better off by cooperating, joining forces, or transacting. There are some gains to cooperation, but only if they can agree on how to share those gains. For instance, consider a buyer and seller who could make a mutually beneficial transaction—but only if they can actually agree on a price. In this chapter, we will focus on two players bargaining over how to divide a monetary surplus. We will use the standard tools of non-cooperative game theory by formally modeling the negotiation process as a game in which players alternate making proposals until one is accepted, and solving for the subgame perfect Nash equilibrium (SPNE) agreement. We will consider the prediction in games where there is a fixed number of proposals that can be made before negotiations fail, as well as games where the negotiations can last indefinitely, and then extend the model to situations with more than two players. In chapter 15, we will abstract away from specific rules about who gets to offer what and when, and instead consider which agreements would plausibly arise in the course of any free-form negotiation.

Imagine that you and a stranger are bargaining over how to share \$100. You will make a take-it-or-leave-it offer to the other person. That is, you will make a proposal that specifies how much of the \$100 the other person receives, with the understanding that you will receive the remainder of the money. You can offer the other person any amount from \$0 to \$100, in 1-cent increments. After receiving your offer, the other person will either *accept it*, in which case she receives your offer (and you get \$100 minus your offer), or *reject it*, in which case the interaction ends with both players getting nothing. How much do you offer?

Now imagine that the game is slightly different. As before, you will make an offer regarding how to split \$100 and the other player can choose whether to accept your offer. If she accepts it, the interaction ends and both players are paid according to the accepted proposal. But in this version, the game doesn't automatically end if the other person rejects your offer. Instead, she gets to make a counteroffer. In this case, though, the two of you will have only \$25 to divide between you. That is, the other player's counteroffer specifies how much of the \$25 you will get (and she will keep the remainder of the \$25). If

you reject the counteroffer, the game ends and you both get nothing. Now, how much of the \$100 do you offer?

The Bargaining Problem

Before discussing the rules and predictions of specific bargaining games, such as those discussed earlier, let's consider more broadly the types of scenarios that we will be discussing in this chapter and the next. Imagine, for example, that you would like to sell your used car to an acquaintance. You know that the local dealership will pay you \$300 for your car and that, to buy a comparable car there, your acquaintance would need to pay \$700. In other words, you are better off selling the car to your acquaintance, so long as he pays you at least \$300, and he is better off buying the car from you, so long as he does not pay more than \$700. Both parties can therefore gain from trading at a price between \$300 and \$700. Coming to an agreement in this range means that you are creating a joint surplus for yourselves of \$400. And by choosing a specific price in this range, you are choosing how to split this surplus between the two of you. For instance, if you agree on a price of \$700, then you gain the entire \$400 surplus (since you are making \$400 more than you could get from the dealership, while the buyer is no better off than buying from the dealer); if you agree on a price of \$300, then he gains the entire surplus; if you agree on a price of \$500, then you split the surplus equally, since you each gain \$200 compared to what you would have if you don't transact. The question is: what price will be agreed upon?

The question of how two agents will share a surplus that they collectively generate is the focus of the *bargaining problem* (Nash, 1950; and for an excellent overview of various bargaining models, also see Osborne and Rubinstein, 1990). The bargaining problem occurs when there are gains to be had from two parties trading or otherwise cooperating with each other, but in order to actually realize these gains, they need to agree on how to share them. Examples include a homeowner negotiating with a prospective buyer over the sale price of a house, a baseball free agent negotiating a contract extension with his team, or two countries negotiating the terms of a peace treaty. The bargaining problem was first presented by Nash (1950), who defined a *bargaining* interaction to be a situation in which the following is the case:

- There is room for two individuals to reach a mutually beneficial agreement (e.g., agreeing to trade the car at a price between \$300 and \$700 makes

- both parties better off),
- but there is a conflict of interest between the parties (e.g., the buyer wants the price to be as low as possible while the seller wants it to be as high as possible),
- and no agreement can be imposed on someone against their will (e.g., you cannot, say, force the buyer to agree to a price of \$1,000).

Agreeing, Disagreeing, and Individually Rational Outcomes

Any bargaining problem must specify two features: the feasible set of agreements and the disagreement point. The *feasible set* (described in definition 1) consists of all possible agreements or bargaining outcomes that could be reached. For the games we consider in this chapter, the feasible set is all possible allocations of a monetary surplus that the agents can achieve by reaching an agreement. In the next chapter, we allow players to negotiate any type of agreement, not simply cash payments between them. For instance, the buyer and seller of a house may reach an agreement that concerns not only the price, but the closing date and terms of inspection; a firm and a job seeker may draw up an employment contract that specifies salary, benefits, how often the employee can work from home, or whether a parking spot is provided; a parent and child may negotiate an agreement in which the child does chores in exchange for getting a ride to a show from the parent; and so on. In these cases, the feasible set is the utility that each player receives from each possible negotiated outcome.

Definition 1 (Feasible set for two players bargaining over a monetary surplus) *Let S be the monetary surplus that the players can jointly create by reaching an agreement and let x_i be the amount of the surplus allocated to player i . Then the feasible set for players 1 and 2 bargaining over how to divide S is $F = \{(x_1, x_2) \in \mathbb{R}^2: x_1 + x_2 = S \text{ and } x_i \geq 0 \text{ for } i = 1, 2\}$.*

We assume that players' payoffs are simply equal to the share of the surplus that they receive. Note that two players bargaining over a monetary surplus is equivalent to a buyer and seller bargaining over a price (or an employer and job candidate bargaining over a salary), so long as players have complete information each other's willingness to pay and willingness to accept. Consider again the used car example, where agents bargain over a surplus of \$400. Let the buyer be player 1 and the seller be player 2. The feasible set is all (x_1, x_2) such that both players' allocations are nonzero and x_1

$+ x_2 = 400$. It is graphed in the left panel of [figure 14.1](#). Note that agreeing to the outcome (x_1, x_2) is equivalent to agreeing on a price of $300 + x_2$ (i.e., the seller's allocation specifies how much the agreed-upon price exceeds the dealer's offer of \$300) or, equivalently, $700 - x_1$ (i.e., the buyer's allocation specifies how much lower the agreed-upon price is compared to the dealer's price of \$700).

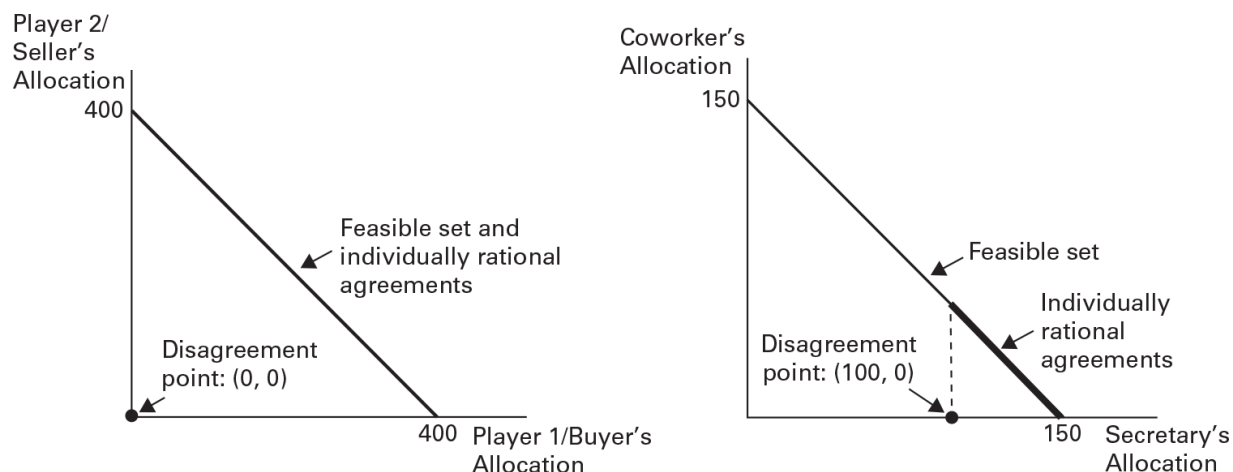


Figure 14.1

The feasible set, disagreement point, and individually rational agreements in the used car example (left) and RAND workplace example (right).

Alternatively, the players may fail to reach an agreement. The disagreement point (described in definition 2) specifies the outcome that each party would receive if negotiations fail. In the used car example (and most of the games we will consider in this chapter), the disagreement point is $(0, 0)$: if negotiations break down, the surplus is fully lost.

Definition 2 (Disagreement point) *The disagreement point is a vector $d = (d_1, d_2) \in \mathbb{R}^2$, such that $d_i \geq 0$ for $i = 1, 2$. If the players do not agree on an outcome in F , then player 1 receives d_1 and player 2 receives d_2 .*

Since agreements cannot be imposed on players without their consent, it is rational for individuals to agree only to outcomes that make them at least as well off as walking away and letting negotiations fail. The set of agreements in the feasible set that make each player at least as well off as the disagreement point are called *individually rational*, as described in definition 3.

Definition 3 (Individually rational agreements) *The set of individually rational agreements are all $(x_1, x_2) \in F$ such that $x_1 \geq d_1$ and $x_2 \geq d_2$.*

Another example is graphed in the right panel of [figure 14.1](#). In an early bargaining “experiment” (or perhaps just an exercise in annoying one’s colleagues), a game theorist at the RAND Corporation offered to give his secretary \$100... or to give her and a coworker \$150 if they could agree how to share it.

This bargaining problem is graphed in the right panel of [figure 14.1](#), with the secretary’s share on the horizontal axis. The feasible set of bargaining outcomes is all allocations summing to \$150 and the disagreement point is $(100, 0)$, since the secretary would still receive \$100 in the event that negotiations failed, while the coworker would not receive anything. The set of individually rational agreements is thus divisions of the \$150 surplus in which the secretary gets at least \$100 (highlighted in the figure).¹

While it is generally reasonable to assume that players won’t agree to outcomes that make them worse off than walking away from the bargaining table, this assumption alone does not necessarily allow us to predict what will actually happen, as indicated by the wide range of individually rational agreements shown in [figure 14.1](#). In this chapter, we will put a specific structure on the bargaining process, by modeling it as a non-cooperative game and specifying who gets to make offers when. This will allow us to predict which outcome (among those in the union of the feasible set and the disagreement point) will occur in equilibrium under different sets of rules. In the next chapter, we will take an *axiomatic* approach—that is, specify a set of axioms that we would like the bargaining solution to satisfy, and see which solutions are consistent with these axioms.

The Ultimatum Game

The simplest bargaining game consists of just a single stage, in which one player (the proposer) makes an offer to the other (the responder), who then decides to accept or reject it. If the responder rejects the offer, the players receive their disagreement payoff and there is no further opportunity for negotiation. This is called the *ultimatum game*, since one player issues a take-it-or-leave-it ultimatum to the other. (In the games considered in this chapter, we will use the pronoun *she* when referring to the first-mover and *he* when referring to the second-mover.) The ultimatum game described at the

beginning of the chapter is depicted in [figure 14.2](#). The players have \$100 to split between them. Player 1 makes a proposal offering a specific share, s , to the other player. Player 2 can then accept the proposal, in which case player 1 gets $100 - s$ and player 2 gets s , or he can reject the proposal and they both get 0. The left panel shows the game in extensive form, while the right panel shows the feasible set of allocation agreements that could be reached (all (x_1, x_2) in the positive quadrant such that $x_1 + x_2 = 100$) and the disagreement point $((0, 0))$.

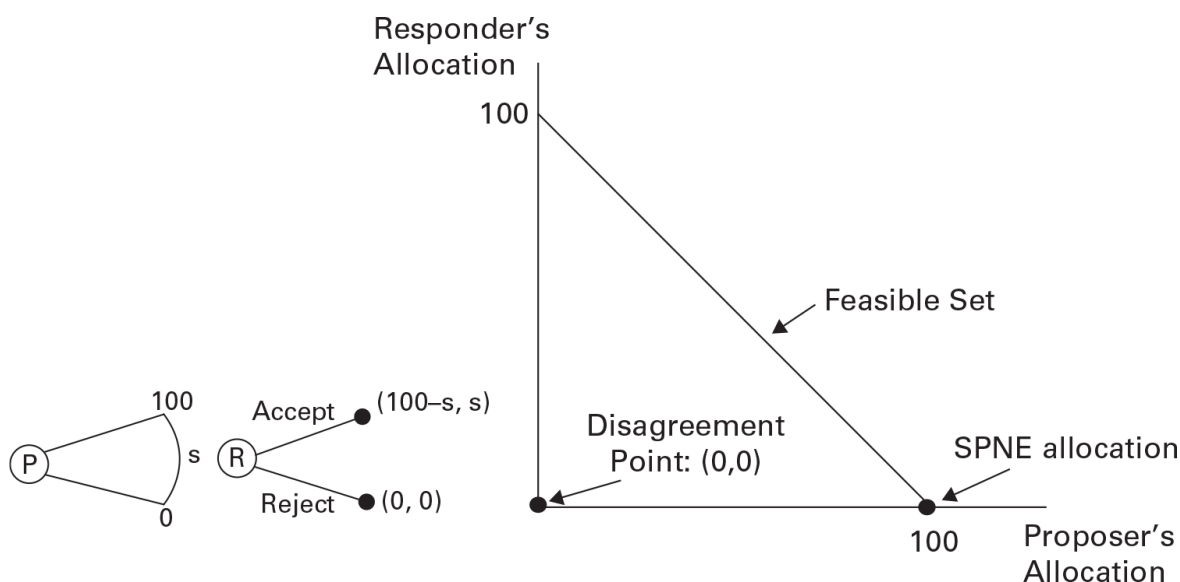


Figure 14.2
The ultimatum game.

What agreement do we expect players would arrive at in the ultimatum game? First, notice that *any offer* can be part of a Nash equilibrium strategy profile, so long as it is compatible with the receiver's threshold for rejection. For instance, a strategy profile in which the proposer offers \$75 and the receiver rejects any offer less than \$75 is a mutual best response. But would we really expect the receiver to reject \$74 in favor of nothing? Recall from chapter 8 that the standard solution concept in sequential-move games is the subgame perfect Nash equilibrium, which rules out these types of noncredible threats.

Let's use backward induction to find the subgame perfect Nash equilibrium of the ultimatum game. We begin with the final move of the game, at which point the responder has received some offer, s , from player 1. He can either

accept and receive s or reject and receive 0. Therefore, he is strictly better off accepting so long as $s > 0$, and so a rational, self-interested responder will always accept any positive amount offered to him. If $s = 0$, then he is indifferent between accepting and rejecting, and both are best responses. Now, we back up to the proposer. She maximizes her payoff by offering the smallest possible amount that will be accepted by the responder. If she believes that he will accept anything positive, she maximizes her payoff by offering him the smallest possible increment, \$0.01. The SPNE is thus ($s = 0.01$, (Accept if $s > 0$, Reject if $s = 0$)). If she believes that he will accept any $s \geq 0$, then she will offer nothing. Thus ($s = 0.00$, Accept all s) is also an SPNE, since if the proposer believes that he will accept all offers, she will offer him nothing. The first-mover has all of the bargaining power in this game, since the responder is forced to choose between whatever he is offered and getting nothing at all, and thus the proposer captures all, or essentially all, of the \$100 surplus. The SPNE allocation outcome is shown in the right panel of [figure 14.2](#). The SPNE outcome is in the feasible set since players reach an agreement in equilibrium, and it is at or just slightly above the responder's disagreement payoff, reflecting the fact that the proposer chooses her offer to just barely beat the outcome that the responder would get if he walked away.

The ultimatum game is one of the most widely studied games in the lab—and, as we will see later in this chapter, one of the games in which the behavior of real people varies most dramatically from the theoretical prediction.

Alternating Offers

Most negotiations are not limited to a single ultimatum offer. If the opening proposal is deemed unacceptable, the other party typically makes a counteroffer, which could in turn be countered by the first player, and so on. Ariel Rubinstein (1982)'s *alternating offers bargaining model* captures this natural back-and-forth inherent in most negotiations.

The alternating offers bargaining game lasts for T stages, with $T \in \{1, \dots, \infty\}$. In each stage, t , one player makes an offer about how to share the surplus. The other player can accept it, at which point the game ends with each player getting what was proposed. Or he can reject it, in which case the game moves to the next stage, $t + 1$, if $t + 1 \leq T$, or ends in disagreement if $t + 1 > T$. When an offer is rejected, it is off the table and has no relevance for which offers

can be made or rejected in later stages. The players alternate making proposals, so player 1 makes the offer (is the proposer) in odd-numbered stages and player 2 makes the offer in even-numbered stages. If a proposal is not accepted within T stages, the game ends with the players getting their disagreement payoffs. Note that the ultimatum game is a special case of this model with $T = 1$: there is one stage of proposals and if the initial proposal is not accepted, then the game ends in disagreement. We will look at a couple of examples of games with a finite number of stages ($T = 2$ and $T = 3$) and then look at an infinite-stage game ($T = \infty$), where there is no limit to the number of proposals that can be considered.

In games with many—or even infinite—stages, what stops players from bargaining indefinitely? A critical feature of Rubinstein's bargaining model is that the surplus shrinks each time an offer is rejected and the players move to a new stage of negotiations. The shrinking surplus represents the fact that time has value. Imagine that the players ultimately reach an agreement in which player 1 gets some proportion of the surplus and player 2 gets the remainder. All else equal, the players would prefer to reach this agreement right off the bat, rather than reach the same agreement after a prolonged, contentious negotiation. For instance, if a buyer and a seller take weeks to agree on the price of a house, then this is time that the seller does not have the money from the sale and that the buyer may be paying rent on another place to live; if an employee and a firm are in an extended negotiation over salary, then this is time that the employee is not earning that salary and the firm is losing revenue by not having the position filled; and so on. In other words, some of the surplus that the players could generate by transacting is being lost as long as time ticks by without an agreement.

This is represented in the model by having the pie shrink at the start of each new stage. Specifically, let $\delta \in (0, 1)$ be the players' discount factor. As in chapter 10 on infinitely repeated games, δ represents the players' patience. When δ is close to 1, the players are perfectly patient and do not discount the future; the closer δ is to 0, the more the players dislike waiting to receive a payoff. The surplus that the players bargain over in stage t is given by $\delta^{t-1}S$. For example, the surplus is S in the first stage, δS in the second stage, and so on. Given the changing pie size, it is simpler to think of the players as bargaining over the *proportion* of the remaining surplus that each gets. Consider a pair (p_1, p_2) such that $p_1 + p_2 = 1$ and $p_i \geq 0$ for $i = 1, 2$, where p_i is

the proportional share of the surplus that player i receives. Then if the players agree on a proportional allocation (p_1, p_2) in stage t , the payoff to player i is $p_i \delta^{t-1} S$.

How should rational players approach this game? The longer the bargaining drags on, the smaller the surplus is, and thus the smaller the expected payoffs. It is therefore in both players' interests to reach an agreement as soon as possible—on the very first offer. So what should player 1 propose in the first stage? She wants to make an opening offer that is *just* good enough to be accepted. She doesn't want to overshoot and offer any more than she has to, but she also doesn't want to have her offer rejected and allow the pie to shrink. To determine what this opening offer should be, she can use backward induction to calculate the best outcome that the other player can expect if he rejects her offer, given how the rest of the game is expected to play out. Then she should offer exactly that (or a hair higher) so that he will be willing to accept.

Specifically, consider a player making a proposal in stage t . Using backward induction, she forms beliefs about the proportion, p , of the remaining surplus that he will offer her in the subsequent stage if he were to reject her offer. This implies that he will receive the remaining proportion, $1 - p$. Thus, if she offers him $\delta(1 - p)$, then he would be willing to accept her offer in the current stage rather than wait until the next stage to receive $1 - p$ (at which point the surplus would have shrunk by the discount factor δ).² We will now look at examples of games with two stages, games with three stages, and games with infinite stages, solving for the SPNE using backward induction in each.

Finite Number of Stages

Two Stages

Let's begin with the two-stage bargaining game discussed at the beginning of the chapter, before moving on to the general two-stage alternating offers game. In that example, player 1 initially makes a proposal about how to share \$100, and if her proposal is rejected, we go on to stage 2, where player 2 makes a proposal about how to share \$25. If the second-stage proposal is rejected, both players get nothing. The game is written in extensive form in [figure 14.3](#), and we can solve for the SPNE using backward induction. Once we reach the second stage, we essentially have an ultimatum game over a pie of size \$25, with player 2 making the ultimatum. Therefore, player 2 can

essentially capture the entire \$25 if the game goes to stage 2. Now, we back up to stage 1. Player 1 determines that, if player 2 were to reject her offer, he expects to gain \$25 and will therefore accept any offer of \$25 or more. She thus offers \$25, keeping \$75 herself.

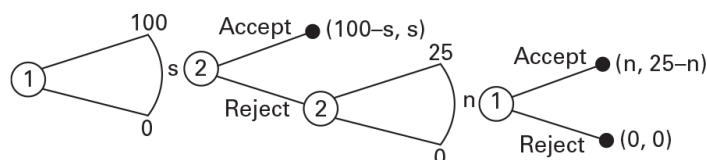


Figure 14.3

The two-stage example, in which player 1 makes an offer from a \$100 pie and if, it is rejected, player 2 makes a counteroffer from a \$25 pie.

Now let's consider any general two-stage game. The surplus is S in the first stage and shrinks to δS in the second stage. Following Rubinstein's model, the proposals that players make are the proportional share of the remaining surplus that they are offering to the other person. Let s be the proportion of the surplus that player 1 offers in the first stage and n be the proportion of the remaining surplus that player 2 offers in the second stage. The game is shown in extensive form in [figure 14.4](#). In the second stage, the players are playing an ultimatum game over the smaller surplus and so player 2 offers $n = 0$ and gains the entire remaining surplus if the game goes to stage 2.

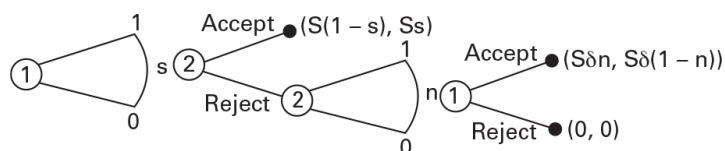


Figure 14.4

The generic two-stage alternating offers bargaining game.

What should player 1 offer in stage 1 to make player 2 indifferent between accepting immediately or rejecting? If he rejects, his proportion of the surplus will equal 1, but he needs to wait a stage to get it, and so it will be discounted by δ . In stage 1, therefore, player 1 offers proportion $s = \delta$ to player 2 (keeping $1 - \delta$ for herself), and player 2 accepts. The SPNE is thus ((Offer proportion $s = \delta$ in stage 1, Accept all $n \geq 0$ in stage 2), (Accept all $s \geq \delta$ in stage 1, Offer $n = 0$ in stage 2)). This is shown in [figure 14.5](#), which shows the feasible set of allocations in stage 1 to be all pairs summing to S , and the

feasible set of allocations in stage 2 to be all pairs summing to δS . In stage 2, player 1 must get at least her disagreement outcome (0), and so player 2 gets the δS surplus; in stage 1, player 2 must get at least his stage 2 outcome (δS), with player 1 getting the remainder of the surplus ($S - \delta S$).

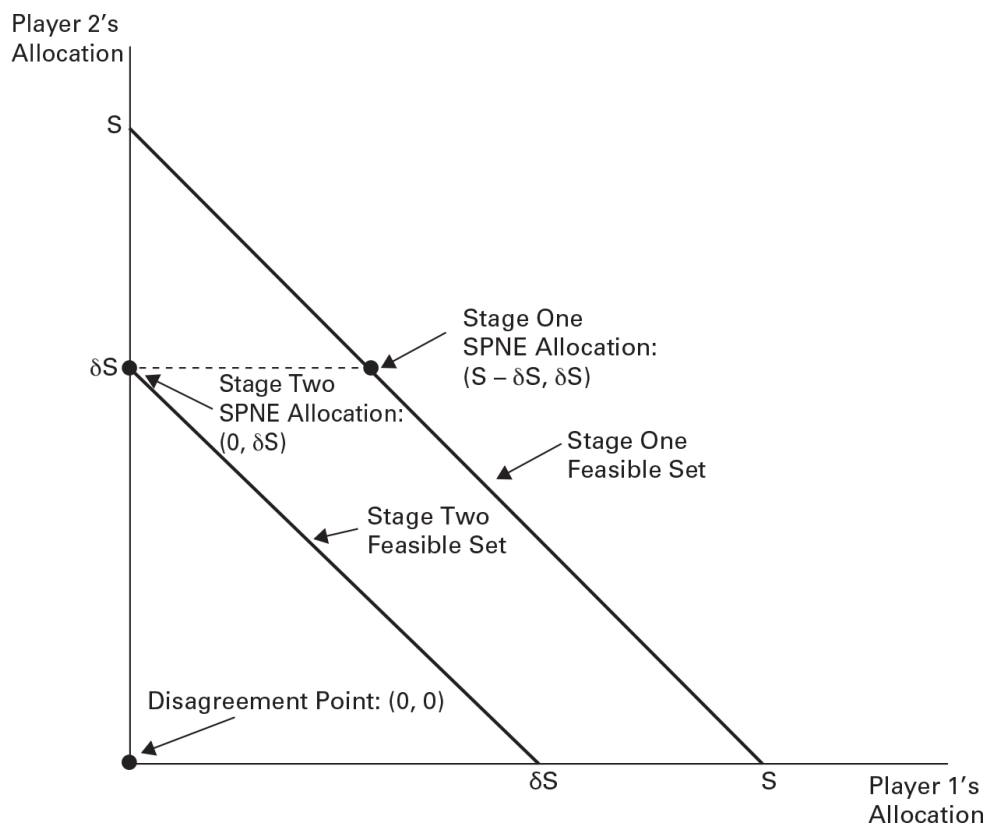


Figure 14.5

The generic two-stage alternating offers bargaining game.

If you were playing this game, would you prefer to be player 1 or player 2? When there are two stages, the SPNE outcome has player 1 earning a proportion $1 - \delta$ of the surplus and player 2 earning δ . This implies that the players split the surplus equally when $\delta = 0.5$. When δ is close to 0, the surplus essentially disappears in the second period, and so the game resembles an ultimatum game with player 1 making the offer and gaining most of the surplus. When δ is close to 1, the surplus essentially remains intact in the second stage, at which point it becomes an ultimatum game with player 2 making the offer and gaining most of the surplus, so player 1 must offer essentially all of the surplus initially to ensure her offer is accepted. Thus, when players are relatively impatient ($\delta < 0.5$), the person making the initial

offer is better off, since it would be very costly for the second-mover to reject an offer; when players are relatively patient ($\delta > 0.5$), the person making the counteroffer is better off, since there isn't much cost to rejecting a disadvantageous initial offer.

Three Stages

In the three-stage game, we backward induct from stage 3, which has player 1 making a take-it-or-leave-it offer to player 2. Player 1 thus winds up with proportion 1 of the remaining surplus if they get to stage 3. In stage 2, player 2 therefore must offer player 1 proportion δ in order for player 1 to accept, in which case he keeps proportion $1 - \delta$. Finally, in stage 1, player 1 must calculate the offer that makes player 2 indifferent between accepting and rejecting. If he rejects, he will get $1 - \delta$ in the next stage, but since he must wait a stage to get it, it is discounted by δ . Thus he is indifferent between accepting an offer of $\delta(1 - \delta) = \delta - \delta^2$ in the first stage and rejecting and getting $1 - \delta$ in the second stage. Player 1 thus offers proportion $\delta - \delta^2$ and keeps the remaining $1 - \delta + \delta^2$.

This can also be seen by looking at the feasible sets for the three-stage game, shown in [figure 14.6](#). The surplus is S in the first stage, δS in the second stage, and $\delta^2 S$ in the third stage. In stage 3, player 1's offer must simply match player 2's disagreement payoff of 0, and so the allocation if they get to stage 3 is $(\delta^2 S, 0)$. In stage 2, player 2 must match player 1's stage-3 payoff of $\delta^2 S$. Thus, the allocation if they get to stage 2 is $(\delta^2 S, \delta S - \delta^2 S)$. Finally, backing up to stage 1, player 1 must match player 2's stage-2 payoff of $(\delta S - \delta^2 S)$. Thus the stage-1 allocation is $(S - \delta S + \delta^2 S, \delta S - \delta^2 S)$.

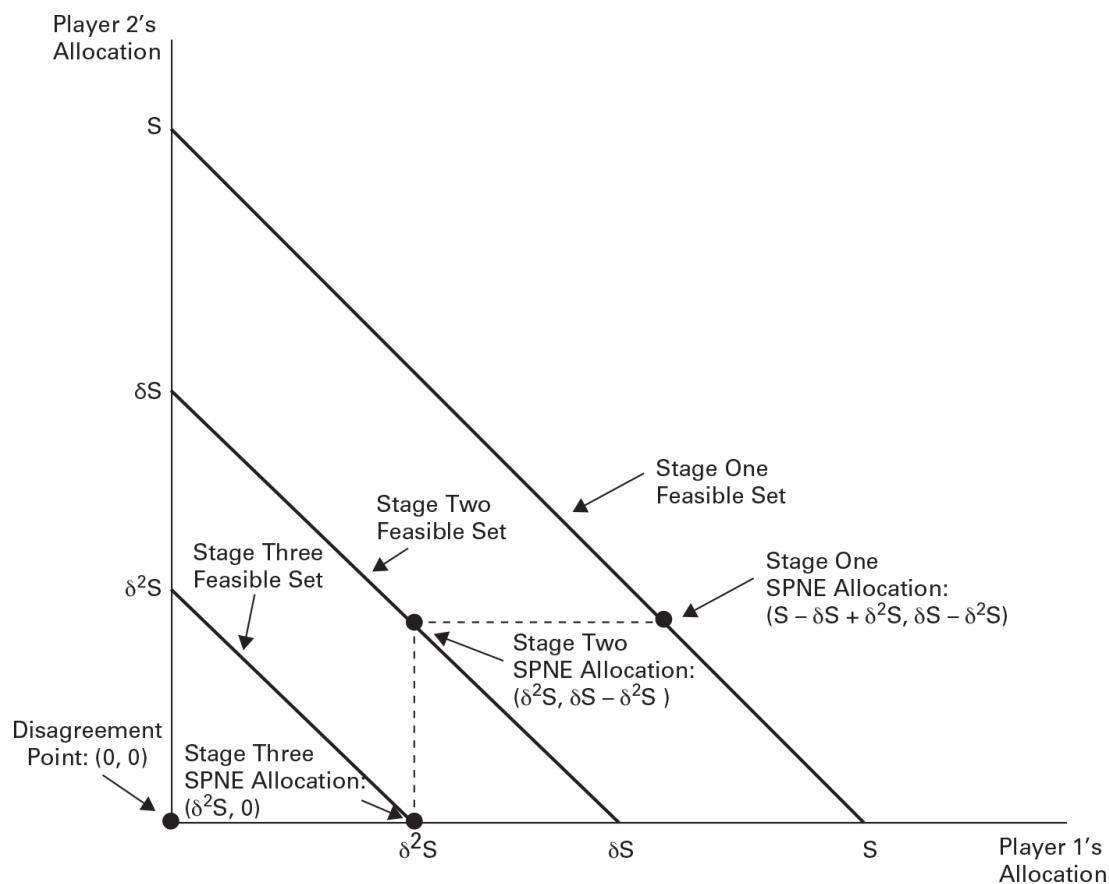


Figure 14.6

The generic three-stage alternating offers bargaining game.

Infinite Stages

In games with infinite stages, we cannot backward induct from a final stage, where the player making the ultimate offer holds all the bargaining power and captures the entire remaining pie. However, the basic approach remains the same: players determine their proposals by looking ahead and figuring out exactly how much they need to offer in order to make the other person indifferent between accepting immediately or rejecting and allowing the rest of the game to unfold. In the infinite game, notice that every odd-numbered stage looks the same as every other odd-numbered stage: player 1 makes an offer to player 2, and player 2 can either accept and end the game, or reject and allow the game to move to an even-numbered stage, where they bargain over a pie that has shrunk by δ , with an infinite number of stages still ahead of them. Likewise, every even-numbered stage looks the same as every other even-numbered stage. Therefore, it seems reasonable that player 1 would

propose the same offer in every odd-numbered period and player 2 would propose the same offer in every even-numbered period.

To help keep track of which player is offering what to whom, let's call player 1 Nancy and player 2 Sid. In every odd-numbered stage, Nancy makes the same equilibrium offer, s , to Sid. This specifies the proportion of the existing surplus that Sid would get if the offer were to be accepted, with Nancy receiving the remaining $1 - s$. In every even-numbered stage, Sid makes the same equilibrium offer, n , to Nancy, implying that he would receive proportion $1 - n$ of the remaining pie if the offer were accepted. So, how much should Nancy offer? If Sid accepts her offer, he receives s . If he rejects her offer, then we proceed to the next even-numbered stage, in which he offers her n and expects to keep $1 - n$. However, since we are now one stage later, the pie has shrunk, and so we need to discount his $1 - n$ payoff by δ . Thus, the offer s that makes Sid indifferent between accepting immediately and waiting one stage to get $1 - n$ is $s = \delta(1 - n)$. Likewise, in every even-numbered period, Sid chooses n to make Nancy indifferent between accepting n in the current stage or rejecting the offer and getting $1 - s$ in the next stage. Since she would need to wait one stage to gain the $1 - s$ outcome, she is indifferent between the two outcomes when $n = \delta(1 - s)$. This is summarized in figure 14.7.

Odd-numbered Stages	Even-numbered Stages
• Nancy offers s and keeps $1 - s$. • Sid accepts s if it is at least as good as proceeding to the next stage. • Nancy chooses s to make Sid indifferent between accepting s and proceeding to the next even-numbered period, where he expects $1 - n$ discounted by δ: $s = \delta(1 - n)$	• Sid offers n and keeps $1 - n$. • Nancy accepts n if it is at least as good as proceeding to the next stage. • Sid chooses n to make Nancy indifferent between accepting n and proceeding to the next odd-numbered period, where she expects $1 - s$ discounted by δ: $n = \delta(1 - s)$

Figure 14.7

Offers made by Nancy in odd-numbered stages (left) and by Sid in even-numbered stages (right) in an infinite-stage alternating offers game.

Thus, in the SPNE, it must be the case that both $s = \delta(1 - n)$ and $n = \delta(1 - s)$. We have two equations and two unknowns (s and n —remember that δ is a fixed parameter representing the players' patience). Now we need to solve the equations simultaneously. Plugging in for n , we have $s = \delta(1 - \delta(1 - s))$.

Distributing the right side, we get $s = \delta - \delta^2 + \delta^2 s$. Bringing the terms with s to the left side and factoring out s on the left side and δ on the right side, we get $s(1 - \delta^2) = \delta(1 - \delta)$. Dividing both sides by $(1 - \delta^2)$, we get $s = \frac{\delta(1-\delta)}{1-\delta^2}$. Finally, note that $1 - \delta^2$ can also be written $(1 - \delta)(1 + \delta)$. Replacing the denominator with $(1 - \delta)(1 + \delta)$, we get $s = \frac{\delta(1-\delta)}{(1-\delta)(1+\delta)}$, and canceling $(1 - \delta)$ from the numerator and denominator, we get $s = \frac{\delta}{(1+\delta)}$. Similarly, $n = \frac{\delta}{(1+\delta)}$ as well. Therefore, the SPNE is *((offer $\frac{\delta}{(1+\delta)}$ in odd-numbered stages, accept any offer $\geq \frac{\delta}{(1+\delta)}$ in even-numbered stages), (offer $\frac{\delta}{(1+\delta)}$ in odd-numbered stages, accept any offer $\geq \frac{\delta}{(1+\delta)}$ in even-numbered stages))*.

Note the interpretation of n . This is not telling us what first-mover Nancy ends up with in equilibrium. Instead, it is telling us how much Sid would offer Nancy in even-numbered stages—which are off-the-equilibrium path, since, in equilibrium, Sid accepts the initial proposal and the game does not progress far enough for Sid to offer n . Instead, the first-mover receives a share $1 - s$, or $\frac{1}{(1+\delta)}$, in equilibrium. Note also that, as δ approaches 1, the equilibrium division of the surplus approaches the equal split. So long as $\delta < 1$, player 1 earns more than half of the surplus, with her share increasing as δ decreases. That is, player 1 has an advantage by being able to make the first offer, and this advantage is greater the less patient the players are.

For example, let's imagine a buyer and seller with $\delta = 0.7$ bargaining over the price of a house. The buyer's willingness to pay is \$400, 000, while the seller's willingness to accept is \$370, 000. Therefore, they are bargaining over a surplus of \$30, 000. The buyer makes the initial offer and is thus player 1. Given our delta value of $\delta = 0.7$, we have $s = \frac{0.7}{1.7} = 0.41$ and $n = 0.41$. Notice that $s + n$ does not sum to 1 because these values are not telling us what each player gets in equilibrium; rather, they are telling us what each player offers when it is her or his turn to make a proposal. In equilibrium, the very first proposal is accepted, so the seller winds up with proportion 0.41 of the surplus, while the buyer gets the remaining 0.59. Given that the surplus is equal to \$30, 000 and they reach an agreement without delay, this means that the buyer gains \$17, 700 while the seller gains \$12, 300 by transacting. In practical terms, this is equivalent to the buyer and seller settling on a price of \$382, 300.

Bargaining in Committees: The Baron-Ferejohn Legislative Bargaining

Model

Thus far, we have considered bargaining interactions between two people. The *Baron-Ferejohn legislative bargaining model* extends the alternating offers model to groups of more than two players. For example, imagine a committee attempting to allocate a budget across their divisions or districts.

Consider a committee of n members. In each stage, one member is selected at random to make an offer, with each member equally likely to be chosen. Unlike the alternating offers game, there is no pre-established order in which players will be called to make an offer. Instead, they simply know that the likelihood of being selected to make the proposal in any given stage is $1/n$ and selection is independent across stages (i.e., the same person could be asked to make the proposal for multiple stages in a row). In each stage, t , the committee members learn which player has been selected, and that person proposes an allocation of the surplus, $p = (p_1, \dots, p_n)$, where $p_1 + \dots + p_n = 1$, $p_i \geq 0$ for all i , and p_i refers to the proportion of the surplus allocated to player i . All n players then vote on the proposal. If more than half of the committee members approve the proposal, the game ends and each player i earns payoff $p_i \delta^{t-1} S$. If half or fewer of the committee members approve the proposal, then it is rejected, and they move to stage $t + 1$. Note that, unlike in the two-player bargaining problem, agreements can be imposed on players without their approval if the majority votes in favor.

Let's consider an example with $n = 5$, $\delta = 0.8$, and $S = 25$, which are the parameters used in an experiment by Fréchet, Kagel, and Lehrer (2003). The game is symmetric, since all players have the same probability of being the proposer in each stage. We will therefore look for a symmetric SPNE in which each player always proposes the same allocation p whenever selected. As in the alternating offers model, the proposer wants to select an allocation that is *just* good enough to be accepted. To be passed, a proposal requires the approval of only two other members (in addition to the proposer's own approval). Thus the proposer can completely cut out two of the other four members, offering them nothing, and offer positive shares to the other two. The question is: how much do these players need to be offered such that they are willing to vote in favor of the proposal rather than take their chances that they will get a better outcome in the next stage? Let s be the amount that the proposer offers to the two players whose vote she is courting. If a player rejects s , then he faces three possibilities in the next stage:

- (1) He will be selected to be proposer, which occurs with probability $1/5$. In this case, he offers s to two other members and keeps $1 - 2s$.
- (2) He will be one of the two committee members offered a positive share by the proposer, which occurs with probability $2/5$. In this case, he receives s .
- (3) He will be one of the two committee members offered nothing, which occurs with probability $2/5$. In this case, he receives 0.

Therefore, his expected share in the next stage is $(1/5)(1 - 2s) + (2/5)(s) + (2/5)(0) = 1/5$. Since this expected payoff comes in the next stage, it must be discounted by $\delta = 0.8$. Thus, rejecting the proposal is associated with a discounted expected payoff of $1/5 \times 0.8 = 0.16$. Players therefore find it in their interest to approve proposals that give them an immediate share of at least 0.16 and reject proposals that give them less. The proposer should then propose an allocation that offers 0.16 to two players, 0 to the two other players, and allocates the remaining 0.68 to herself. In this example, with a surplus of \$25, this means that the first person chosen as proposer offers \$4 to each of two other players.

Behavior in Games

The Ultimatum Game

The first experimental test of the ultimatum game was conducted by Güth, Schmittberger, and Schwarze (1982). Despite the relative strategic simplicity of the ultimatum game, the results differed dramatically from the theoretical predictions. Most offers were in the 40%–50% range, with very few proposers offering the theoretical prediction. Also it turns out that offering a fair share was wise because lowball offers were routinely rejected. These results have been replicated around the world—from major cities to small-scale societies with little contact with the outside world, to places where researchers could afford to offer stake sizes that are triple the participants' monthly expenditures. Initially, however, economists found these findings to be shocking. They had viewed the assumption that players would aim to maximize their payoffs as trivial. Now here was a study showing that, when offered the choice between something and nothing, a substantial proportion of people said, "Give me nothing."

So why do responders reject free money? What are these people thinking? Thanks to the tools of neuroscience, economists are increasingly able to glimpse what is happening in players' brains during a game. Sanfey et al.

(2003) had responders view their offer and decide whether to accept or reject while in a functional magnetic resonance imaging (fMRI) scanner, which measures neural activity in the brain. They found that players who received unfair offers had significantly greater activation in several areas of their brain compared to those who received fair offers: namely, the right dorsal lateral prefrontal cortex (associated with planning), the anterior insula (associated with pain and disgust), and the anterior cingulate cortex (associated with resolving conflicts among different regions of the brain). In other words, if someone receives a lowball proposal and claims to be disgusted by the offer, this isn't just a figure of speech; they are literally disgusted—we can see it in their brain activity. Even more notably, this is predictive of their decision: those people who rejected the offer had significantly heightened activity in the “disgust region” of the brain. Responders thus seem willing to sacrifice their own earnings in order to express their disgust and punish stingy proposers for behaving unfairly.

So why do the proposers regularly offer 40%–50% of the surplus? Are they rationally best responding to people's tendency to reject unfair proposals? Or do they have a genuine preference for making a fair offer? One way to disentangle these two motives is to eliminate the responder's ability to reject the offer—then, if fair offers persist, it must be because the proposer preferred the fair split. To this end, we can compare offers in the ultimatum game with offers in a game called the *dictator game* (Kahneman, Knetsch, and Thaler, 1986; Forsythe et al., 1994): the proposer offers a split to the other player and ...that's that. The partner is not given the option to accept or reject the offer; that person is stuck with whatever the proposer dictates. Note that this game does not satisfy our definition of a game of strategy since only one player makes a decision; neither does it satisfy the definition of a bargaining interaction since one player can impose an outcome on the other. The offers of how to split a \$5 surplus in each of the two games in an experiment by Forsythe et al. (1994) are shown in [figure 14.8](#). In the ultimatum game, roughly three-quarters of people offered 40%–50%. In contrast, participants in the dictator game most commonly offered nothing. But still nearly two-thirds of people offered a positive amount, even though there was no possibility of their offer being rejected. This suggests that some proposers in the ultimatum game made fair proposals only because they didn't want the

offer to be rejected, while others had an intrinsic preference for acting (or appearing to act) fairly.

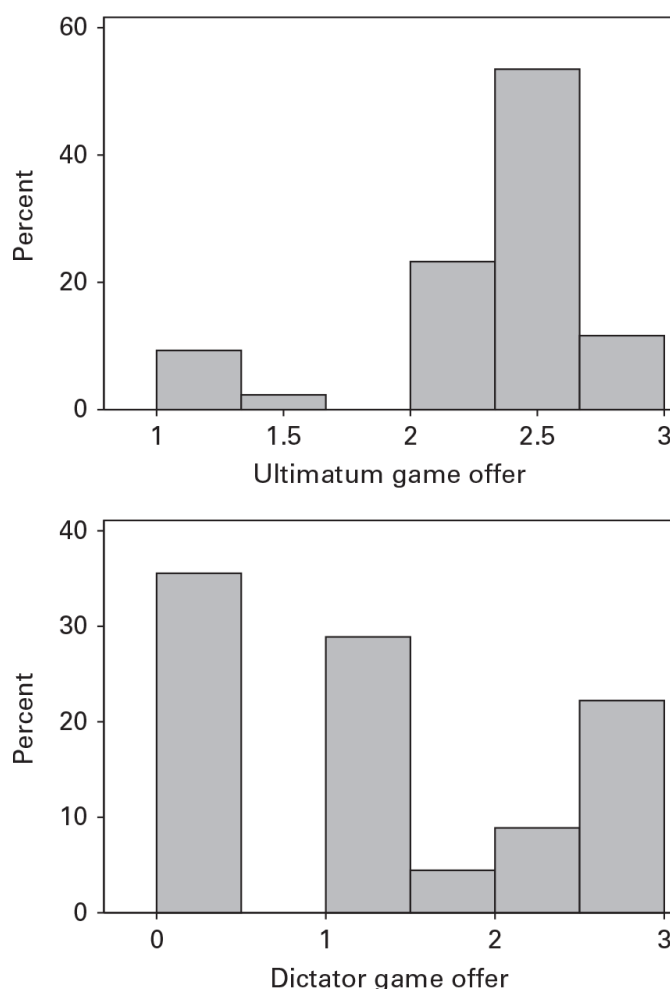


Figure 14.8

Ultimatum game offers (top) and dictator game offers (bottom) in Forsythe et al. (1994).

Bargaining with Multiple Stages

In the ultimatum game, there is a clear pull toward the fair allocation. What about games with multiple stages, when there is room for extended negotiations? Consider the two-, three-, and five-stage games described in the table in [figure 14.9](#), which were used in an experiment by Neelin et al. (1988). The table lists the surplus size in each stage for each of the three games. The stage-1 surplus is 5 in all three games and then decreases at different rates ($\delta = 0.25, 0.5$, and 0.34 , respectively). The surplus values were chosen so that all three games have the same SPNE initial offer: player 1 offers 1.25 (out of the

initial pie of 5), so stage 1 of each game is the same except for the number of remaining stages before the game ends in disagreement. The bottom panel of the table shows the relative frequency of initial offers in Neelin et al. Although players could make offers in 1-cent increments, for the purposes of displaying the data in the table, the offers are grouped together in 25-cent increments. (For example, the 1.25 row includes all offers from 1.13 to 1.38.) Looking at the two-stage game, we see that the majority of offers (65%) are in line with the equilibrium prediction of 1.25. So while play in the one-stage ultimatum game is characterized by proposers offering a fair and equal split, the SPNE actually seems to be a good prediction in the two-stage game. This is consistent with other experiments investigating two-stage games. Turning to the three-stage game, however, it seems we are back to the equal split again: 80% of the players offer \$2.50. So what happens in the five-stage game? The equal split? The SPNE? There again seems to be a clear consensus regarding what proposers offer, but the amount that the majority of proposers offer is between the equal split and SPNE: 60% of offers are in the neighborhood of \$1.75.

Surplus in Each Round			
	Two Stages	Three Stages	Five Stages
Stage 1	\$5.00	\$5.00	\$5.00
Stage 2	\$1.25	\$2.50	\$1.70
Stage 3	—	\$1.25	\$0.58
Stage 4	—	—	\$0.20
Stage 5	—	—	\$0.07

Relative Frequency of Initial Offers			
	Two Stages	Three Stages	Five Stages
\$1.00	0.05	0.00	0.05
\$1.25	0.65	0.075	0.025
\$1.50	0.225	0.00	0.15
\$1.75	0.00	0.00	0.60
\$2.00	0.025	0.075	0.125
\$2.25	0.00	0.05	0.00
\$2.50	0.05	0.80	0.05

Figure 14.9

Experiment and data from Neelin et al. (1988). Note: Initial offer of \$1.25 is SPNE in all three games.

What do you think is going on here? Look again at the top panel, showing the pie sizes in each stage, and compare these with the offers that players tend to make in the first stage of each game. In the two-stage game, the second-stage pie is \$1.25, and this is what most proposers offer. Because of the strategic incentives in the two-stage game, offering the second-stage pie is also the SPNE. In the three-stage game, the second-stage pie is \$2.50, which is what the vast majority of proposers offer. And in the five-stage game, the second-stage pie is \$1.70, which is consistent with what most proposers offer.

Thus, it seems that initial offers in multistage bargaining games are largely driven by limited strategic thinking. Many players are not willing or able to backward induct over multiple stages—or assume that their partner won't be—and so they simply offer the second-stage pie. After all, offering the other

player the entirety of what they could share if he rejects the offer ought to be sufficient to get him to accept, right?

The interpretation that proposers' initial offers are driven by limited backward induction is further confirmed by Johnson et al. (2002). The researchers assessed proposers' reasoning processes by using software that measures which information they are viewing on their computer screens. They had participants play a three-stage bargaining game on the computer, but the pie sizes in each stage were initially covered up. The proposer could see all three pie sizes before making her initial proposal, but she could only look at one at a time. To view the pie for a given stage, the player must click on this area of her screen to uncover it, and then cover it back up again before she can uncover the pie for a different stage. In this way, the researchers can observe which information the proposer looks at and for how long. While the SPNE offer requires proposers to click on the final-stage pie and then work backward over the three stages, they find that proposers spend the most time looking at the first-stage pie, and nearly half never even look at the third-stage pie at all.

Evidence-Based Adaptations of the Standard Model

While limited backward induction seems to influence initial offers in multistage bargaining games, proposals in the simpler ultimatum game seem to deviate from the SPNE because players are driven by motives other than maximizing their own financial gain. Instead, generous ultimatum offers appear to be the result of two factors: the responders' willingness to reject stingy offers (which is correctly anticipated by the proposers), and the proposers' willingness to share the surplus (as evidenced by the abundance of positive offers in the dictator game). These patterns can be explained by models that incorporate *social preferences*: players' payoffs depend not only on their own monetary allocation, but on the earnings of others.

One of the most widely used and straightforward social preference models is Fehr and Schmidt (1999)'s *inequality aversion* model. In their specification, a player's payoff is equal to her monetary payment minus the psychological disutility that she experiences when her share differs from her partner's. There are two types of inequality that could affect her payoff. *Advantageous inequality* occurs when her monetary payment exceeds her partner's, while *disadvantageous inequality* occurs when her partner gets more money than she does. Players may differ in how much they dislike each type of inequality,

but it is generally assumed that disadvantageous inequality is more distressing.

Formally, consider a two-player game in which the allocation of monetary payments is (x_i, x_j) . Under the Fehr-Schmidt model, player i 's payoff is given by the following expression, where α_i and β_i are individual-specific parameters reflecting player i 's distaste for disadvantageous and advantageous inequality, respectively:

$$u_i(x_i, x_j) = x_i - \alpha_i \max\{x_j - x_i, 0\} - \beta_i \max\{x_i - x_j, 0\}$$

where $\alpha_i \geq 0$, $\beta_i \geq 0$, $\beta_i \leq \alpha_i$.

The first term is simply the player's monetary payment, while the second and third terms incorporate the player's disutility from disadvantageous and advantageous inequality, respectively. The second term comes into play only when player j earns more. When player i earns at least as much, then $x_i \geq x_j$, so $\max\{x_j - x_i, 0\} = 0$, and this term disappears. When player i earns less, then the term reduces to $\alpha_i(x_j - x_i)$. That is, for each dollar that player j 's payment exceeds player i 's, player i 's payoff decreases by α_i . The third term comes into play only if player i receives more than her partner (and is otherwise zero). In this case, player i 's utility decreases by β_i for each extra dollar she gets compared to her partner. If the players care only about their overall payment (as in the standard model), not relative payments, then α_i and β_i are zero, and the utility function collapses to $u_i(x_i, x_j) = x_i$.

How does this explain behavior in the ultimatum and dictator games? The distaste for advantageous inequality can explain why a player would offer to share the surplus even when there is little or no chance of having a less fair offer rejected, while the distaste for disadvantageous inequality can explain why responders would reject a positive but unequal offer in favor of both players receiving nothing. First, consider a player in the dictator game who is deciding whether to split a \$10 surplus equally or take all the money for himself. If he splits the money equally, his Fehr-Schmidt payoff is simply 5. (There is no inequality in payoffs, so the last two terms of the utility function don't come into play.) If he takes all the money for himself, then his payoff is equal to his material payment of \$10 minus the disutility of $10\beta_i$ that he experiences from earning \$10 more than his counterpart, so his Fehr-Schmidt payoff is $10 - 10\beta_i$. This player would prefer the equal split so long as $5 \geq 10 - 10\beta_i$ or $\beta_i \geq 1/2$. Thus, the Fehr-Schmidt model predicts that some players

would prefer to share money rather than earn too much more than their partner; that is, players will share money if their distaste for advantageous inequality (β_i) is strong enough. Next, consider a responder in the ultimatum game who has been offered \$1 out of \$10. If he rejects the offer, then his Fehr-Schmidt payoff is 0 (he earns nothing in the game and there is no inequality in payments). If he accepts the offer, then he gets \$1, but experiences a disutility of $9\alpha_i$ because the proposer earns \$9 more than him. He will reject the offer whenever $0 \geq 1 - 9\alpha_i$, or $\alpha_i \geq 1/9$. Thus, the Fehr-Schmidt model predicts that some players would prefer to reject offers and earn nothing rather than allow the other player to earn too much more than them.

But do you think that responders reject stingy ultimatum game offers purely because they dislike unequal payments? Or because they want to punish their proposers for intentionally trying to stiff them, and potentially teach them a lesson about how to behave in the future? The Fehr-Schmidt model assumes that players care only about the ultimate outcomes, and not how those outcomes arose or the other player's intentions. Falk, Fehr, and Fischbacher (2003) designed a clever ultimatum game experiment to try to tease apart these motives.

In that experiment, proposers made offers about how to split a pie of 10. However, they had only two options for how much to offer. One of their options was always to offer 2 to the other player; the other option varied across treatments. The researchers then looked at whether willingness to reject the proposal of 2 depends on the other option that was available to the proposer. Although the proposal of 2 was the same, the responder may perceive the fairness intentions of the other player differently depending on what the proposer *could have* offered instead. Depending on the treatment, the alternative offer was either 5 (in which case the offer of 2 is likely to be viewed as unkind, since the proposer could have offered the fair split); 0 (in which case the offer of 2 is likely to be viewed as kind, since the proposer could have been more selfish); 2 (in which case the proposer had no choice at all, and so the proposal communicates nothing about the proposer's fairness intentions); or 8 (in which case the responder may not judge the proposer too harshly, since her only other option was to make an offer that was unfair to herself).

Note that the Fehr-Schmidt model predicts that the choice to reject the offer of 2 does not depend on the other offers that were available to the proposer. According to this model, the responder should reject the 2 offer if his payoff from both players getting 0 is greater than his payoff of getting 2 while the other person gets 8: that is, whenever $0 \geq 2 - 6\alpha_i$, or $\alpha_i \geq 1/3$. Since α_i is a fixed preference for player i , on average there should be no difference in rejection rates across treatments. Instead, the researchers find that the rate at which the 2 offer is rejected varied significantly depending on what the proposer could have offered instead. When the proposer could have offered 5, nearly half of responders rejected the offer of 2, but when the proposer's only other option was to offer 0, fewer than 10% rejected it. (This is also consistent with the fMRI study by Sanfey et al., which found that anterior insula activation was higher when an unfair offer came from a human partner than when it was generated by a computer.) This variation in rejection rates of the offer of 2 across the treatments suggests that players care about the intentions of the proposer to treat them fairly, not just the ultimate outcomes. However, the results also indicate that some players aren't willing to accept the short end of the stick, even when the proposer didn't intentionally shortchange them. In the treatment with the alternative offer of 2, the proposer had no choice, and the offer thus signals nothing about her fairness intentions—yet, 18% still reject the offer of 2.

Similar conclusions can be drawn from games other than bargaining interactions. Falk, Fehr, and Fischbacher (2008) considered a game in which the first-mover can either transfer some of her money to the responder or steal some money away, and the responder can then pay to either reward the first-mover by increasing her payment or punish her by reducing her payment. They found that responders were more likely to reward or punish the first-movers when the first-movers intentionally chose whether to give or take money than they were if the same transfer were decided by a roll of the dice.

This desire to reciprocate other players' intentionally kind or unkind behavior can be modeled using the theoretical framework of *psychological game theory* (discussed in detail in chapter 28), which allows players to have preferences over beliefs as well as outcomes (Rabin, 1993; Dufwenberg and Kirchsteiger, 2004; Falk and Fischbacher, 2006). These models assume that players assess each other's kindness based on their perception of the other

player's actions, beliefs, and intentions, and then they are motivated to reciprocate that (un)kindness.

Another approach, by Charness and Rabin (2002), is similar to the Fehr-Schmidt model in that it specifies a payoff function that depends on each player's allocation; however, it also allows the players' preferences over allocations to change if they believe their partner has treated them unfairly. The Charness-Rabin payoff function is given by

$$u_i(x_i, x_j) = (\rho r + \sigma s + \theta q)x_j + (1 - \rho r - \sigma s - \theta q)x_i$$

where $r = 1$ if $x_i > x_j$ and 0 otherwise; $s = 1$ if $x_i < x_j$ and 0 otherwise; $q = -1$ if player j has misbehaved and 0 otherwise; and ρ , σ , θ are preference parameters.

While it seems that there is a lot going on here, the player's payoff is simply a weighted sum of the two players' monetary allocations, where the relative weight put on each player's allocation can differ depending on whether three conditions are turned on or off in the payoff function: whether i earns more, whether j earns more, and whether j has previously behaved unfairly. To see this more clearly, let's first consider the situation in which j has not yet behaved unfairly, in which case $q = 0$; for instance, player i may be the proposer in a dictator game who has not previously interacted with the recipient. If both players earn the same amount, then r and s are also equal to zero, and the whole expression reduces to $u_i = x_i$. If player i earns more, then $r = 1$ and $s = 0$, so $u_i = \rho x_j + (1 - \rho)x_i$. If $\rho = 0$, then i doesn't care about j 's payment, and we have a situation equivalent to the standard model of self-interested preferences. Otherwise, ρ could be positive or negative. A negative value would indicate that i puts negative weight on j 's payment; that is, holding constant i 's own share, i is less happy as j earns more. In this case, i would be willing to give up some of her own money in order to reduce j 's allocation. (Specifically, i would be willing to pay up to $-\rho/(1 - \rho)$ to reduce j 's payment by 1.) A positive value of ρ would indicate that i puts positive weight on j 's payment: holding her own payment constant, she is happier the higher j 's share is. If, instead, player j earns more than player i , we have $u_i = \sigma x_j + (1 - \sigma)x_i$. Thus, the extent to which player i values player j 's payment depends on whether she is doing better or worse than him. The inequality aversion model is captured by the situation in which $\sigma < 0 < \rho < 1$: she prefers the other player's payoff to be increased when he is earning less than

she is and decreased when he is earning more. When player j has previously behaved unfairly, $q = -1$, which means that the weighting on his payment is reduced by θ . In other words, when $\theta > 0$, player i values player j 's payment (even) more negatively than if he hadn't behaved poorly.

For example, let's look at how a responder in Falk et al. (2003)'s ultimatum game considers an offer of 2 out of 10 according to the Charness-Rabin model. First, let's assume that the alternative proposal was also 2, so that the offer signals nothing about the proposer's fairness intentions and the responder doesn't think that the proposer has treated him poorly (i.e., $q = 0$ in the payoff function). If the offer is accepted, the proposer earns more than the responder, and so $r = 0$ and $s = 1$. The responder's payoff from accepting the offer is calculated as $u_i(2, 8) = (\sigma)(8) + (1 - \sigma)(2) = 6\sigma + 2$. The responder's payoff from rejecting the offer is $u_i(0, 0) = (0\rho + 0\sigma + 0\theta)(0) + (1 - 0\rho + 0\sigma + 0\theta)(0) = 0$. Thus, the responder would reject the offer when $\sigma < -1/3$ (i.e., when he's sufficiently willing to pay to reduce the proposer's payoff when she earns more than he does).

Now, let's consider the situation when the alternative proposal was 5, such that offering 2 instead is perceived as the proposer acting unkindly (i.e., $q = -1$ in the payoff function). Now there is a negative θ in the weight put on the proposer's share (and thus an additional θ in the weight put on the responder's share). The responder's payoff from accepting this offer is thus $u_i(2, 8) = (\sigma - \theta)(8) + (1 - \sigma + \theta)(2) = 6\sigma - 6\theta + 2$. As before, the payoff from rejecting the offer is 0, so the responder prefers to reject whenever $\sigma - \theta < -1/3$. Since θ is generally assumed to be positive, this condition is more likely to be satisfied than the previous condition of $\sigma < -1/3$. This thus explains (1) why the offer of 2 is sometimes rejected in Falk et al.'s experiment even when the proposer had no choice; and (2) why the offer of 2 is rejected more frequently when the proposer could have offered 5 instead. Specifically, players for whom $-1/3 < \sigma < -1/3 + \theta$ would accept the offer when the alternative was also 2, but reject it when the alternative was 5.

Exercises

(1) Consider the games in the Neelin et al. (1988) experiment described earlier in [figure 14.9](#).

- (a) For each of the three versions, show that \$1.25 is the SPNE opening offer in all three games.

- (b) Consider a five-stage bargaining game, similar to that in Neelin et al. (1988). The pie sizes for each stage are shown in [figure 14.10](#). Under the SPNE, how much does player 1 offer in the initial stage?

Surplus in Each Round	
Stage 1	\$100
Stage 2	\$68
Stage 3	\$30
Stage 4	\$12
Stage 5	\$5

Figure 14.10

(2) Consider a four-stage alternating offers bargaining game over a surplus of size S , in which the surplus shrinks by δ in each round. According to the SPNE, how much does player 1 offer in stages 1 and 3, and how much does player 2 offer in stages 2 and 4?

(3) Consider two players, a buyer and a seller, who are bargaining over the price of a house. The seller is not willing to accept any less than \$500, 000. The buyer is willing to pay up to \$512, 000. Therefore, there is a \$12, 000 surplus to be divided. Each player has the discount factor $\delta = 0.8$.

- (a) Consider the *two-stage* bargaining game: The buyer moves first by making an offer for how to split the surplus. The seller can then accept it or reject it and make a counteroffer. Finally, the buyer can accept or reject the counteroffer, and the game ends. According to the subgame perfect Nash equilibrium, what proportion of the surplus does each party receive and what price does the buyer initially offer?
- (b) Now assume that there is no limit to the number of bargaining stages. Continue to assume that the buyer moves first. According to the subgame perfect Nash equilibrium, what proportion of the surplus does each party receive and what price does the buyer initially offer?
- (4) Let's consider the same bargaining problem as in problem 3, in which the buyer and seller bargain over the price of a house. As before, the buyer's willingness to pay is \$512, 000, the seller's willingness to accept is \$500, 000,

and they are bargaining over a \$12,000 surplus. Also, as in problem 3, the buyer makes the initial offer and the buyer's discount factor is $\delta_b = 0.8$. However, now assume that the seller's discount factor differs, and is $\delta_s = 0.2$.

- (a) Consider the *two-stage* bargaining game: The buyer moves first by making an offer for how to split the surplus. The seller can then accept it or reject it and make a counteroffer. Finally, the buyer can accept or reject the counteroffer, and the game ends. According to the SPNE, what proportion of the surplus does each party receive, and what price does the buyer initially offer?
 - (b) Now assume that there is no limit to the number of bargaining stages. Continue to assume that the buyer moves first. According to the SPNE, what proportion of the surplus does each party receive, and what price does the buyer initially offer?
- (5) Consider a committee of 12 people, each with $\delta = 0.75$, who are bargaining over how to divide a surplus of \$8,000. The bargaining proceeds according to the Baron-Ferejohn legislative bargaining model: in each stage, one committee member is randomly chosen to make a proposal about how to divide the surplus, and then the committee votes on this proposal. More than half of the committee must approve of the proposal for it to be implemented (in this case, there must be at least 7 votes in support). If it is rejected, the pie shrinks by $\delta = 0.75$ and a committee member is randomly selected to make the next offer. As in the chapter, consider the SPNE in which each player always makes the same proposal regardless of the stage. What is the initial offer according to the SPNE?
- (6) There are five pirates, numbered pirate 1 through pirate 5, who must decide how to split 10 pieces of gold. (The gold pieces cannot be further split into smaller pieces.) The game starts with the highest-numbered pirate (pirate 5) making a proposal regarding how many gold pieces each pirate gets. All the pirates, including the proposer, then vote on the proposal. If one-half or more of the voters approve the split, then the proposal is implemented and the game ends. If not, the pirate who made the proposal is thrown overboard, and the game continues with the next-highest pirate making the proposal. The discarded pirates do not vote. Assume that the primary goal of any pirate is maximizing his own gold pieces. However, if two outcomes give a pirate the

same amount of gold pieces, then he prefers the outcome in which more pirates get tossed overboard.

Using backward induction, figure out the split that pirate 5 would propose under the SPNE. Hint: remember that only 50% of the vote is needed to pass a proposal (rather than more than 50%). So, if the game gets to pirate 2, only his own vote (and not pirate 1's) is needed to pass his proposal and end the game.

(7) Consider a proposer who has a Fehr-Schmidt utility function with $\alpha_p = 0.4$ and $\beta_p = 0.4$. There is a \$6 pie, and the proposer can offer the other person \$1 or \$3.

- (a) If the proposer is playing a dictator game, how much does she prefer to offer?
- (b) Now imagine the proposer is playing an ultimatum game. With probability 0.5, the responder has the same preference parameters as her ("type 1"), and with probability 0.5, the responder has preference parameters $\alpha_r = 0.2$ and $\beta_r = 0.5$ ("type 2"). Find the SPNE of this Bayesian game.

(8) Consider a decision maker who is choosing between two possible allocations, A and B, of money between himself and his partner. His partner does not have the option of rejecting, and so his choice determines the final outcome. For each of the situations given here, consider the situation under the Fehr-Schmidt model and under the Charness-Rabin model. State which conditions on the preference parameters (α and β in the Fehr-Schmidt model, ρ , σ , and θ in the Charness-Rabin model) must hold for the decision maker to prefer option A.

- (a) Option A: Each player gets \$500. Option B: The decision maker gets \$600 and his partner gets \$1, 000. The decision maker and his partner have not previously interacted.
- (b) Option A: Each player gets \$500. Option B: The decision maker gets \$600 and his partner gets \$1, 000. The decision maker believes his partner has previously acted unfairly.
- (c) Option A: The decision maker gets \$400 and his partner gets \$500. Option B: The decision maker gets \$500 and his partner gets \$0. The decision maker and his partner have not previously interacted.

- (d) Option A: The decision maker gets \$400 and his partner gets \$500.
Option B: The decision maker gets \$500 and his partner gets \$0. The decision maker believes that his partner has previously acted unfairly.

¹In fact, the two RAND employees agreed to split the \$150 equally, which is not individually rational for the secretary. As we will see in the experimental results discussed at the end of the chapter, there appears to be a strong pull toward fair splits in bargaining games. In addition, it is probably a safe assumption that this game was part of a larger, infinitely repeated interaction between the players that may make it in the secretary's interest to be appear generous, even if it means sacrificing \$25.

²Note that the player is indifferent between accepting the offer of $\delta(1 - p)$ immediately or proceeding to the next stage and getting $1 - p$ at that time, and so accepting and rejecting are both best responses. To ensure that the player would definitely accept the offer, she would need to add a tiny inducement, ϵ , to her proposal—that is, offer $\delta(1 - p) + \epsilon$. If offers can be made in arbitrarily small increments, however, this is essentially the same as just offering $\delta(1 - p)$. In the standard alternating offers model, players make the offer that would make the other player exactly as well off as they would be rejecting, and players accept this offer rather than force the game to continue.

15

Cooperative Bargaining

In chapter 14, we focused on a positive model of bargaining—how the structure of a negotiation leads to a non-cooperative prediction of the outcome. By contrast, in this chapter we take a normative view of the two-person bargaining problem. Here, we study the cooperative outcome that should arise if it is to embody a number of appealing principles. As always, we will also examine whether these principles actually steer experimental participants toward agreements.

In many cases, there just isn't enough structure to make a non-cooperative prediction of how a negotiation will resolve. It is often not obvious who will make the first offer, or whether, for that matter, the bargainers will take turns making offers. In addition, there is no guarantee that the surplus will shrink (in any sense) if an agreement is not made soon or that one of the bargainers can make an ultimatum just before the surplus vanishes. How, then, can bargainers come to an agreement? One idea, formulated by John Nash (1950), is that the bargainers could agree instead on a set of principles—axioms, if you will—that any bargaining outcome should embody, and then turn to an impartial arbitrator to simply calculate the solution based on the agreed axioms.

Here is an example founded on the influential experiment by Roth and Malouf (1979). Instead of bargaining over the division of, say, \$100, what if you were told that you and your opponent will bargain over the distribution of 100 lottery tickets? For each of you, every lottery ticket you gather is another chance of winning a \$100 prize. However, you are not explicitly told the size of the monetary prize that your opponent will receive if she wins. How many tickets would you demand? The 50-50 allocation, in which you have equal chances of winning your prize, seems intuitive, but why? First, though it seems obvious, it needs to be stated that all 100 tickets should be allocated between you and your opponent—there should be none left over. The 50-50 allocation does that. Second, because the situation looks completely

symmetric, there doesn't seem like any compelling reason to give one of you more than the other. In other words, for every outcome in which you get t_1 tickets and your opponent gets t_2 tickets, there is another one in which your opponent gets t_1 and you get t_2 . Third, imagine that I told you that your opponent could not receive more than 90 tickets. Do you think that would change the outcome? Probably not. Given the symmetry of the situation, there was no way your opponent was going to get 90+ tickets anyway. In other words, the 10-90 alternative is irrelevant, and all similarly irrelevant outcomes should be disregarded to ease the choice by shrinking the set of viable alternatives. Fourth, notice that at this point, you don't know what your opponent's monetary prize is and she doesn't know what yours is. Should that information matter? Though less obvious than the other reasons favoring the 50-50 split, one could argue that all that matters is that we both have a prize we could win. In fact, if you hired a good impartial arbitrator, she would tell you that the 50-50 split of the tickets is the *only* solution to this bargaining problem that satisfies all four principles.

Cooperative Versus Non-cooperative Bargaining

To better illustrate the difference between this chapter and the previous one, consider the following setup. This time, you and your opponent have a surplus to split by simultaneously making demands. If the sum of your demands can be satisfied by the surplus, you both get what you asked for, but if the sum of your demands is greater than the surplus, you both get a "disagreement payoff," d_1 for you and d_2 for your opponent. If the allowable demands for a share of the surplus are common to both of you and limited to, say, $\frac{1}{3}$, $\frac{1}{2}$, and $\frac{2}{3}$, we can represent the negotiation as the normal-form game shown in [figure 15.1](#), known as the Nash demand game:

		Opponent		
		$\frac{1}{3}$	$\frac{1}{2}$	$\frac{2}{3}$
You	$\frac{1}{3}$	$\frac{1}{3}, \frac{1}{3}$	$\frac{1}{3}, \frac{1}{2}$	$\frac{1}{3}, \frac{2}{3}$
	$\frac{1}{2}$	$\frac{1}{2}, \frac{1}{3}$	$\frac{1}{2}, \frac{1}{2}$	d_1, d_2
	$\frac{2}{3}$	$\frac{2}{3}, \frac{1}{3}$	d_1, d_2	d_1, d_2

Figure 15.1

The Nash demand game.

Consider the case where you and your opponent actually want to come to an agreement (i.e., where d_1 and d_2 are both less than $\frac{1}{3}$). Based on non-cooperative game theory, we have three predictions along the diagonal, all the Nash equilibria of the normal-form game. Here, no agreements or contracts are necessary, and the only axiom used by the bargainers is self-interest. That said, it is not clear which of the $(\frac{1}{3}, \frac{2}{3})$, the $(\frac{1}{2}, \frac{1}{2})$ or the $(\frac{2}{3}, \frac{1}{3})$ equilibria will arise. Without any context or guidance, they all seem plausible.

The non-cooperative prediction becomes even muddier when we consider the continuous Nash demand game depicted in [figure 15.2](#). Here, bargainers can submit any demand between 0 and 1 and, therefore, any allocation along the Pareto frontier connecting the points (1, 0) and (0, 1) is potentially a Nash equilibrium, so long as both bargainers get more than their disagreement payoff (i.e., along the frontier above the dashed triangle). Further, we call the set consisting of all the possible outcomes, the larger triangle below the entire Pareto frontier, the *feasible set*; and we see that the disagreement point (d_1, d_2) largely determines the set of viable (equilibrium) alternatives. As before, without any structure, which of these equilibria will the bargainers settle on?

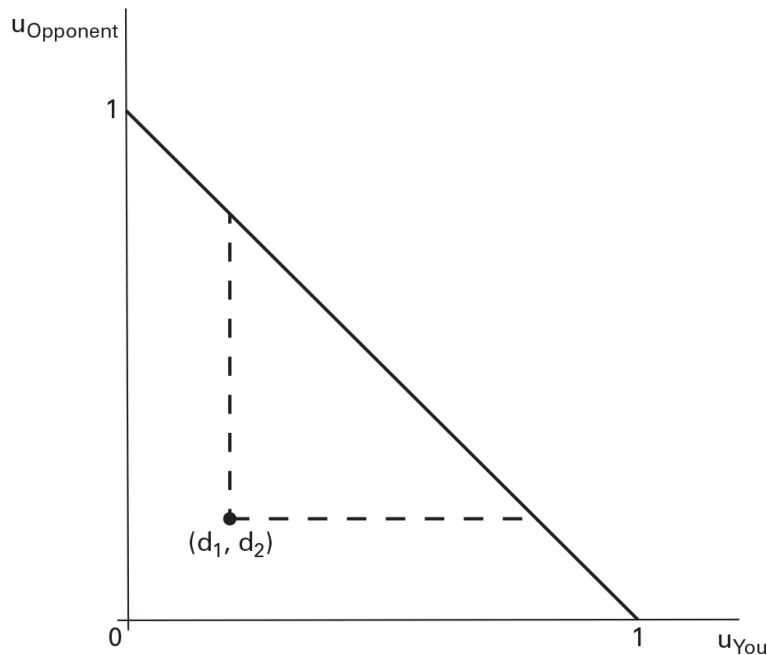


Figure 15.2
The bargaining problem.

As you might recall from chapter 7's discussion of multiple equilibria, Nash's bargaining problem now seems, at its essence, to be a problem of equilibrium selection, and therefore the question is, how does the cooperative approach to game theory help resolve this problem? The first important difference with the non-cooperative approach is that in a cooperative game, players can write irrevocable contracts on how the game should be played. In this case, the bargainers can agree, for instance, on two important aspects of the negotiation: the axioms that any bargaining solution should exemplify (i.e., the guidance or structure they need) and that an impartial arbitrator will have the final word (i.e., a way to implement the guidance). With these details sorted, the job of the arbitrator is to, as Thomas Schelling (1960) might put it, make one of the Nash equilibria "focal" (see chapter 7 for more on this term). This means that the cooperative approach to bargaining is better understood as a process of costless arbitration than a system of proposals and counterproposals.

Given the axioms have been agreed on, making an equilibrium focal boils down to finding one that satisfies as many axioms as possible. Before we examine how this is done, let's discuss the details of the axioms that Nash invoked, starting with Pareto optimality.

- **Pareto Optimality.** Any cooperative bargaining solution will represent an allocation, $U = (u_1, u_2)$, upon which no gains to either or both bargainers are possible.

In other words, Pareto optimality assures that the solution will be on the frontier illustrated in [figure 15.1](#), where all the possible gains are captured. This makes obvious sense because if the negotiations stopped before the bargainers took full advantage of the surplus (i.e., they were not at the Pareto frontier), then there would always exist the possibility to renegotiate another outcome that would make both parties better off. If you are bargaining over splitting a pie, make sure that the two of you eat it all!

- **Invariance.** Suppose that the bargaining solution is U . Now say that the utility of the bargainers is subjected to a linear rescaling, L . Then the new bargaining solution, U^* will also be rescaled such that $U^* = LU$.

The purpose of the invariance axiom is to account for the fact that utility representations are somewhat arbitrary and, therefore, no bargainer should be able to affect the solution by inflating or deflating their assessment of the surplus. That is, the solution that the bargainers end up agreeing to will remain the same in real terms, regardless of whether they rescale their utilities. The important caveat, of course, is that this is not too difficult to keep track of if the rescaling is linear (i.e., $U^* = mU + b$), but things quickly become intractable when the transformations are not linear. An example of the importance of this axiom is presented as an exercise at the end of the chapter.

- **Independence of irrelevant alternatives.** Suppose the bargaining solution is U when the feasible set of outcomes is F and the disagreement point is d . Now restrict the feasible set to $F' \subset F$ such that F' still contains U and d . Then the bargaining solution for F' and d is also U .

In almost every situation in which the independence of irrelevant alternatives axiom is invoked, you will hear an intuition similar to the following: You are out to eat at a restaurant, and the waiter tells you that the menu consists of chicken, beef, or fish. Imagine that you choose fish. When the waiter returns to fill your water glass and take the order of the next table over, he tells the other customers that there is no more beef. It would be very odd for you, overhearing this, to then insist that he changes your order to chicken. When you originally chose the fish, both chicken and beef were available. If you then remove an option that you did not prefer, it should not

affect your choice. In the bargaining context, if a solution survives all the possible objections in the original problem, it should still survive when the players have fewer objections to make. To put it succinctly, if bargainers typically agree on U when U' is available, they should never choose U' when U is available.

- **Symmetry.** Consider the feasible set F and the disagreement point d . The bargaining problem is symmetric if $d_1 = d_2$ and whenever the allocation $(u_1, u_2) \in F$ so is $(u_2, u_1) \in F$.

The feasible set, in other words, is symmetric whenever it is identical on either side of the 45 degree line through the disagreement point. That is, the problem is symmetric if it looks the same to both bargainers. If this is the case, then all the asymmetries between the bargainers have already been accounted for in the construction of the problem, and therefore, there should be no reason to favor one bargainer over the other.

The Nash Bargaining Solution

It turns out that there is a unique solution to the bargaining problem that embodies all four axioms. To characterize this solution, we first need to expand on the notation used earlier.

Suppose that there are two negotiators who bargain over the division of a surplus. For simplicity, think of the surplus as a pie of size $\pi > 0$ and the negotiators' demands as the pair (x_1, x_2) , where $0 \leq x_1 \leq \pi$ and $x_2 = \pi - x_1$. Each negotiator's utility from a given share of the pie, $u_i(x_i)$, is based on a function that is strictly increasing and concave. When the players fail to reach an agreement, they receive utility amounts (d_1, d_2) , the disagreement point. To ensure that there is a point to the negotiations, we assume that there exists some allocation of the pie such that $u_i(x_i) > d_i$ for the two. Finally, the feasible set of agreements is defined in terms of utilities: $F = \{(u_1, u_2), \text{ such that } u_1 = u_1(x_1) \text{ and } u_2 = u_2(x_2) \text{ for all pairs } (x_1, x_2)\}$. Hence, the bargaining problem can be characterized by F , the feasible set, and d , the disagreement point.

Definition 1 (Nash bargaining solution) *Nash's (1950) solution to the two-person bargaining problem is a utility pair, $U^N = (u_1^N, u_2^N)$, that solves the following constrained optimization problem: $\max(u_1 - d_1)(u_2 - d_2)$ subject to $(u_1, u_2) \in F$.*

Put differently, the Nash bargaining solution picks the point on the frontier of the feasible set that maximizes the product of the utility gains beyond the disagreement point. You can see that the Nash bargaining solution will be unique because maximizing this product is equivalent to maximizing the logarithm of the product, which is a strictly concave function. The general form of the solution is depicted on the left of [figure 15.3](#). Here the feasible set is concave, given our assumptions about the utility functions of the two negotiators, and the disagreement point is marked, as in [figure 15.2](#). What we have added is a series of dashed *iso-Nash product* contours, lines along which the Nash product, $(u_1 - d_1)(u_2 - d_2)$, equals a constant, k . As we move to northeast, the constant associated with the Nash product increases and the negotiators become better off. The Nash bargaining solution is found at the tangency between the frontier of the feasible set and the highest achievable iso-Nash product line.

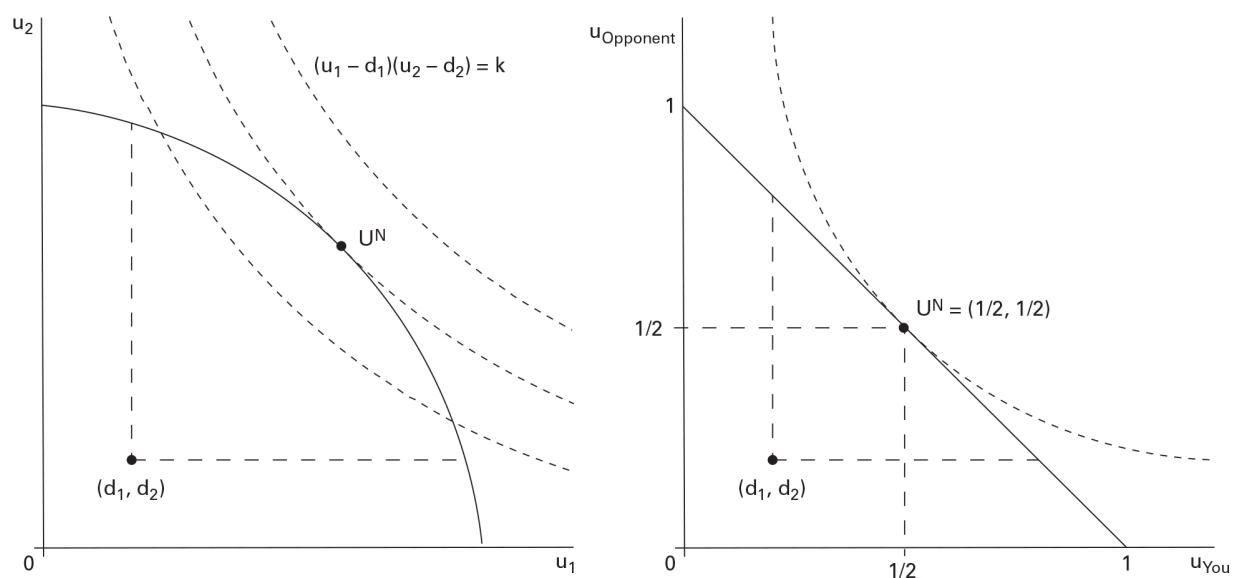


Figure 15.3

The Nash bargaining solution.

Proving the result requires two steps. The details are elegant but a bit cumbersome and adds little to the exposition, so we will only go through the intuition. The first step is to prove that the Nash bargaining solution satisfies the four axioms. The fact that the Nash product is increasing in both u_1 and u_2 guarantees that the solution will be on the frontier of the feasible set, and therefore is Pareto optimal. Next, suppose that we started with the Nash

product, $(u_1 - md_1 - b)(u_2 - md_2 - b)$, which has the solution U^* . If we transform utilities linearly such that $u'_i = mu_i + b$ for the two negotiators, we will find that $U^* = mU^N + b$. That is, the bargaining solution is invariant to linear transformations of utility. Now suppose that $F' \subset F$. The value, k from [figure 15.3](#), that the Nash product obtains at the optimal when considering F , by definition, must be larger than or equal to the optimal when the feasible set is F' . Further, if the optimum of the original problem is in F' too, then the value of k that it obtains must be equal to the value obtained at the optimal of F' . That is, it must be optimal in both feasible sets, and irrelevant alternatives can be ignored. Finally, assume that $d_1 = d_2$ and let (u_1^N, u_2^N) be the Nash solution. Then (u_2^N, u_1^N) must also be optimal. However, because the solution must be unique, these two conditions can hold only if the solution is symmetric (i.e., $u_1^N = u_2^N$).

The second step is to show that if a bargaining solution satisfies the four axioms, it must be the Nash solution. If we rescale any bargaining problem according to the invariance axiom so that $(d_1, d_2) = (0, 0)$ and the solution is $(\frac{1}{2}, \frac{1}{2})$, we can also show that there can be no rescaled u_1 and u_2 such that $u_1 + u_2 > 1$ because this contradicts the optimality of the solution. Hence, $(\frac{1}{2}, \frac{1}{2})$ must be on the frontier of the feasible set, F . Because the transformed bargaining problem is bounded (i.e., $u_1 + u_2 \leq 1$), we can fit a rectangle (call it F'), that is symmetric with respect to the line $u_1 = u_2$, such that $F \subset F'$ and the solution, $(\frac{1}{2}, \frac{1}{2})$ is on the boundary of F' . Then by Pareto optimality and symmetry, the solution for the set F' must also be $(\frac{1}{2}, \frac{1}{2})$, and by the independence of irrelevant alternatives, because $F \subset F'$, the solution for F' must be $(\frac{1}{2}, \frac{1}{2})$ too. Hence, any solution that satisfies the axioms must be $(\frac{1}{2}, \frac{1}{2})$, the Nash bargaining solution to the original problem.

As a simple first illustration of the Nash bargaining solution in action, let's reconsider the situation that resulted in [figure 15.1](#). Here, we assume $u_i(x_i) = x_i$. That is, for both you and your opponent, utilities are synonymous with the share of the surplus you receive. Let's say that you are negotiator 1 and your opponent is negotiator 2. If we substitute in the utility functions, we can put the entire problem in terms of the shares x_1 and x_2 . Now the maximand becomes $(x_1 - d_1)(x_2 - d_2)$ and the frontier of the feasible set is just $x_1 + x_2 = 1$. As is typical when solving a constrained optimization problem, we can

substitute the rearranged constraint, $x_2 = 1 - x_1$, into the maximand yielding the unconstrained problem:

$$\max(x_1 - d_1)(1 - x_1 - d_2)$$

Next, we take the derivative to maximize this version of the Nash product and get $\frac{d}{dx_1} = (x_1 - d_1)(-1) + (1 - x_1 - d_2)(1)$, which we then set equal to zero at the maximum. Solving the resulting equation for negotiator 1's (i.e., your) share returns $x_1^N = \frac{1+d_1-d_2}{2}$. From the equation for the feasible set frontier, we can also calculate that $x_2^N = \frac{1-d_1+d_2}{2}$. In the special case where a disagreement leaves the negotiators with nothing (i.e., $d_1 = d_2 = 0$), you can see that the Nash bargaining solution points to the equal split $(\frac{1}{2}, \frac{1}{2})$, a very intuitively appealing result. Alternatively, you can also see that your share of the surplus increases when your disagreement outcome improves and decreases when the disagreement outcome of your opponent increases. To see this, calculate that $\frac{dx_1^N}{dd_1} = \frac{1}{2} > 0$ and $\frac{dx_1^N}{dd_2} = -\frac{1}{2} < 0$. This should make sense. For example, it seems reasonable that employers will have to pay more to employees when the unemployment benefits are more generous. All of this is also illustrated on the right side of [figure 15.2](#).

Speaking of unemployment benefits, we can now consider another example, based on McDonald and Solow (1981), that examines the case of labor-management negotiations more closely. Suppose that the union and management negotiate over the wage rate, w , and the level of employment, L . Starting with management, suppose that the firm, for simplicity, has revenue, $R(L) = 10L$, and profit, $\pi(L, w) = 10L - wL^2$. What makes this example a bit different from the first is that no matter what the wage is, we can count on management to maximize the firm's profit. This will provide a needed link to the frontier of the feasible set of outcomes. Management's calculations will find that $\frac{d\pi}{dL} = 10 - 2wL$, and they will discover that this is zero where $L = \frac{5}{w}$. It should be no surprise that the higher the wage, the fewer workers management will employ. The interests of the union, however, are contrary to those of management. The union wants to raise wages for its members, but it understands that as wages rise, the firm will cut back on its workforce. This means the payoff for the union is just the total wages it negotiates for its members, wL .

To formulate the Nash product for this negotiation, we see the first part, representing management's outcome, is just the firm's profit. In other words,

we assume that if negotiations fail, the workers will strike and the firm will be forced to shut down. This implies that the disagreement payoff for management is zero. The second part of the product, corresponding to the union's outcome, will be the difference between the total wage package achieved and the disagreement point for the union. If we assume that when the plant is shut down, the workers receive unemployment benefits of β per worker, then the union's net gain will be $wL - L\beta$. Putting this all together, the Nash product is

$$(10L - wL^2)(wL - L\beta)$$

Accounting for management's commitment to financial prudence allows us to put the Nash product in terms of just the wage because the firm maximizes profit when $L = \frac{5}{w}$, as discussed previously. Making this substitution yields

$$\left(\frac{50}{w} - \frac{25}{w}\right)\left(5 - \frac{5\beta}{w}\right)$$

after a little algebra. We can now maximize the product to find the wage that conforms to the Nash bargaining solution. The derivative, $\frac{d}{dw}$, can be written as $\frac{125}{w^2}\left(\frac{2\beta}{w} - 1\right)$. Setting the second part equal to zero at the optimum (and assuming that $\frac{125}{w^2}$ is not equal to zero), we find that $w^N = 2\beta$. As anticipated, we see that the negotiated wage increases as the unemployment benefits become more generous. In the end, if the negotiators commit to the axioms at the heart of the Nash solution, the union should receive benefits equal to $5 - \frac{5\beta}{2\beta}$, or 2.5, and management should leave the bargaining table with $\frac{25}{2\beta}$.

Alternative Solutions to the Cooperative Bargaining Problem

The cooperative approach to bargaining received a lot of early attention from theorists and it was not long before a number of other bargaining solutions were developed. In many cases, the new solutions were created because people felt Nash's independence of irrelevant alternatives axiom was either too strong or a bit flawed. For example, consider [figure 15.4](#). On the left, when the feasible set is truncated from F to F' , it seems a bit odd that negotiator 2 should get just as much at the Nash bargaining solution as she did when the feasible set was larger, and in particular when she had a considerably better best outcome (u_2^{Fmax} versus $u_2^{F' max}$).

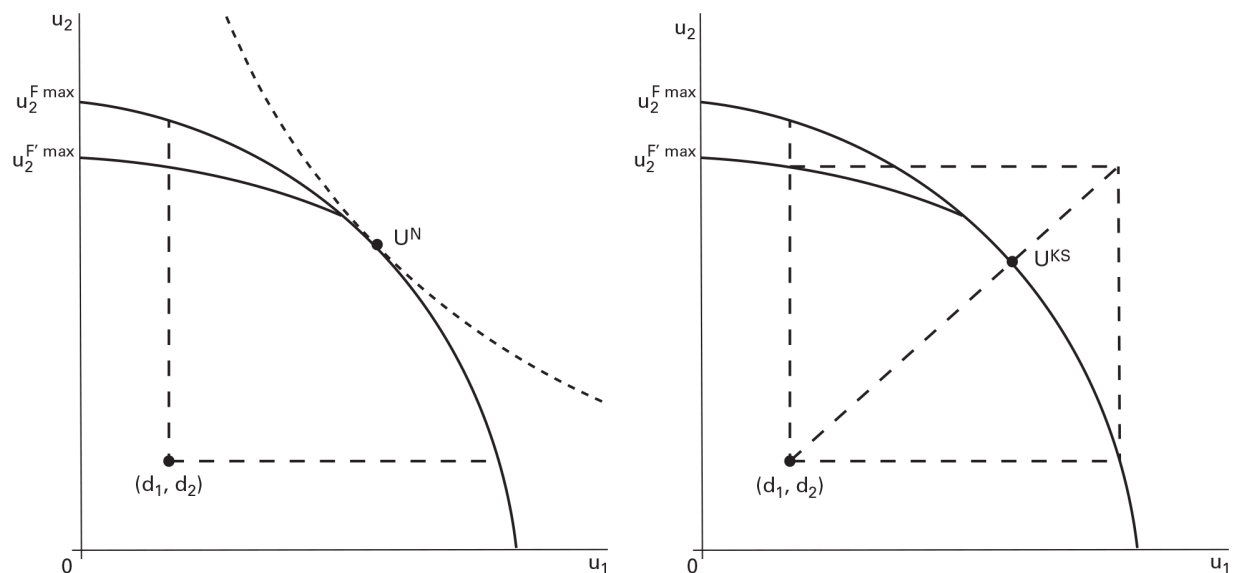


Figure 15.4

The Kalai-Smorodinsky bargaining solution.

Ehud Kalai and Meir Smorodinsky (1975) were the first to create an alternative solution to Nash's bargaining problem with this criticism in mind. Kalai and Smorodinsky introduced an axiom called *resource monotonicity*, which simply says that if the surplus over which the negotiators bargain increases, so there are more resources to share, no player should be worse off. When this axiom is substituted for the independence of irrelevant alternatives, Kalai and Smorodinsky showed that the result is an equally justifiable solution, based on preserving the ratio of maximal gains. Let's consider a social welfare function based on the most a bargainer could ever get, given her opponent was getting his disagreement utility, and call it $W(u_1, u_2)$. The Kalai-Smorodinsky solution can then be given as in definition 6.

Definition 2 (Kalai-Smorodinsky bargaining solution) *Kalai and Smorodinsky's solution to the two-person bargaining problem is a utility pair; $U^{KS} = (u_1^{KS}, u_2^{KS})$, that solves the following constrained optimization problem: $\max W(u_1, u_2) = \min \left\{ \frac{u_1 - d_1}{u_1^{\max} - d_1}, \frac{u_2 - d_2}{u_2^{\max} - d_2} \right\}$, subject to $u_i > d_i$.*

While the Kalai-Smorodinsky solution seems complicated, it is actually quite simple if you consider its geometric interpretation, which is best seen on the right of figure 15.4. Call $(u_1^{\max}, u_2^{\max})$ the *utopian point*, the point at which both negotiators get the most they could ever possibly receive, conditional on the other negotiator getting his or her disagreement utility. If we run a line

from the disagreement point to the utopian point, all the points along this line, including the one at its intersection with the frontier of the feasible set, will preserve the ratio of maximal possible gains. This intersection is the Kalai-Smorodinsky solution. Further, comparing the two panels of [figure 15.4](#), you can now see how the Kalai-Smorodinsky solution varies when the feasible set expands or contracts to favor or disadvantage the negotiators, and how this might not happen under the Nash solution.

Returning again to our example that generated the feasible set in [figure 15.1](#), suppose now that negotiator 2 can never receive more than three-quarters of the pie. If we lop off the small triangle above negotiator 2 getting $\frac{3}{4}$ from the feasible set, we can see that the independence of irrelevant alternatives axiom will assure that the Nash bargaining solution will continue to be $(\frac{1}{2}, \frac{1}{2})$ when the disagreement point is normalized to $(0, 0)$. However, what does the Kalai-Smorodinsky solution require? Again, the way to go here is to use the geometric interpretation of the Kalai-Smorodinsky solution. Using [figures 15.1](#) and [15.3](#) as references, you can see that the Kalai-Smorodinsky solution will lie at the intersection of the frontier of the feasible set and the line from the origin (because the disagreement point has been normalized) through the utopian point $(1, \frac{3}{4})$. If you draw the new bargaining problem, you will see that this intersection will occur along the original segment of the frontier described by $x_1 + x_2 = 1$, where it meets the line from the origin with slope $\frac{3}{4}$, to be precise. Substituting the line $x_2 = \frac{3}{4}x_1$ into the frontier returns the point $(\frac{4}{7}, \frac{3}{7})$, the Kalai-Smorodinsky solution. As expected, this alternative favors negotiator 1, but if the feasible set grows so that negotiator 2 can receive a maximum of 1 too, you can see that the Kalai-Smorodinsky solution will overlap with Nash's solution.

Ehud Kalai also offered what is known as the *egalitarian* bargaining solution (Kalai, 1977). Contrary to his first contribution, the egalitarian solution maintains Nash's independence of irrelevant alternatives and resource monotonicity but drops invariance. This solution, when the utility functions of the negotiators are the same, simply seeks to grant equal gains above the disagreement point. However, when the utilities are not the same, the egalitarian solution is generalized as in definition 7.

Definition 3 (Egalitarian bargaining solution) *Kalai's solution to the two-person bargaining problem is a utility pair, $U^E = (u_1^E, u_2^E)$, that solves the*

following constrained optimization problem: $\max W(u_1, u_2) = \min\{(u_1 - d_1), (u_2 - d_2)\}$, subject to $u_i > d_i$.

On the left side of [figure 15.5](#), you can see that the egalitarian solution just equalizes the utility gains, regardless of what the utility functions are for the two negotiators. Perhaps you can now understand the importance of the invariance axiom. Clearly, some gamesmanship is possible when invariance is not invoked. As the graphs are drawn, the negotiators seem to have similar utility functions and therefore share the surplus. But couldn't the negotiators strategically misrepresent their preferences? If, for instance, negotiator 2 downplayed the utility she was getting from the surplus, the intersection of the feasible set and the u_2 axis will fall. Now, to continue to equalize the gains, the slope of the ray from the disagreement point would have to increase, yielding more of the underlying resources to negotiator 2.

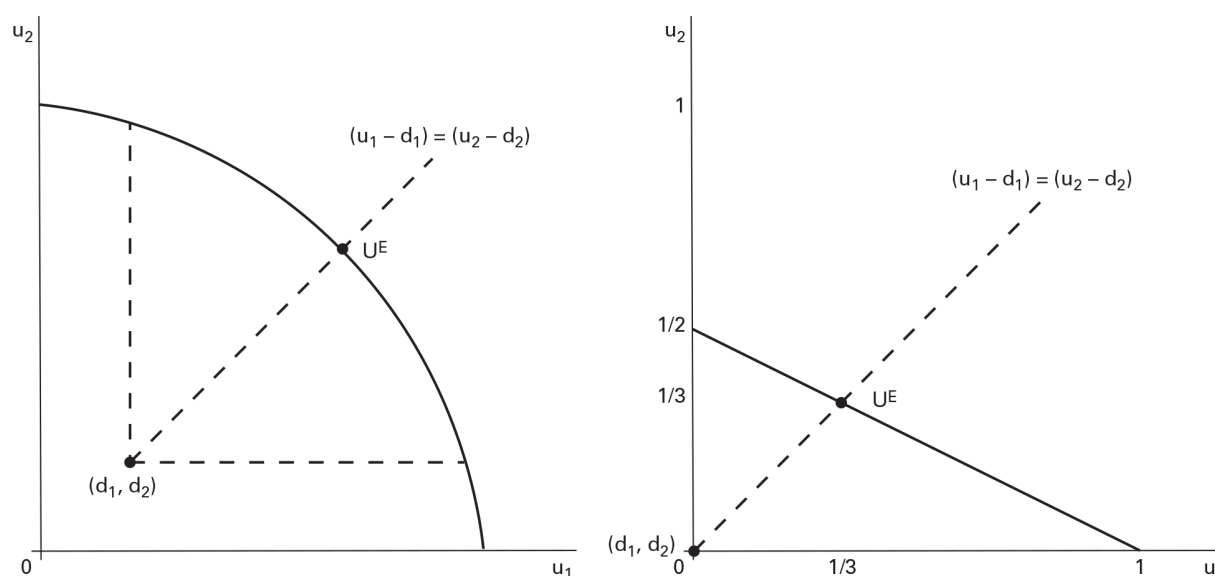


Figure 15.5

The egalitarian bargaining solution.

The other thing to notice about Kalai's egalitarian solution is its relationship to the work of John Rawls. In his *A Theory of Justice*, Rawls develops a thought experiment in which people must decide on a just set of rules when they do not know the position they will occupy in society. You don't know if you will be born with a silver spoon in your mouth or into poverty. That is, you have to make the choice behind the *veil of ignorance*

(Rawls, 1971). As Rawls points out, in this setting, it is reasonable to strive to improve the position of the worst off in society because that might be the position into which you will be born. More formally, according to Rawls's logic, in a two-person world, people should agree to maximize their minimum possible utility, or $\max \min(u_1, u_2)$. Notice that in a bargaining context, this is exactly the same rule as the one implied by the egalitarian bargaining solution.

Continuing our example, suppose that for negotiator 1, it is still the case that $u_1(x_1) = x_1$, but negotiator 2 now claims that $u_2(x_2) = \frac{1}{2}x_2$, consistent with downplaying one's utility. This scenario is depicted on the right side of [figure 15.5](#). Applying the egalitarian bargaining solution to this setting, again assuming that the disagreement point is $(0, 0)$, we see that we need to maximize $\min(x_1, \frac{1}{2}x_2)$, which will happen when $x_1 = \frac{1}{2}x_2$, when both quantities are simultaneously the minimum and, therefore, equal. Substituting this into the resource constraint, $x_1 + x_2 = 1$, we find that $U^E = (u_1^E, u_2^E) = (\frac{1}{3}, \frac{1}{3})$ but $(x_1^E, x_2^E) = (\frac{1}{3}, \frac{2}{3})$. To make negotiator 2 as happy as negotiator 1, we need to give her twice as much of the pie.

Let's consider one final solution to Nash's bargaining problem—one that emphasizes the resolution of situations in which player incentives lead them to suboptimal outcomes, the topic explored more completely in chapter 18. Specifically, what if we think of the bargaining problem as just another social dilemma? Without the help of an unbiased arbitrator, the negotiators may end up at the disagreement point or some other point inside the frontier of the feasible set in expectation because they play mixed strategies. Instead, suppose that the arbitrator used the same criteria as a social planner. In this case, the social planner maximizes the simple additive social welfare function. With two players, this is just $W(u_1, u_2) = u_1 + u_2$, and the *utilitarian* bargaining solution is the allocation of the pie that maximizes this function. This is described in definition 8.

Definition 4 (Utilitarian bargaining solution) *The social planner's solution to the two-person bargaining problem is a utility pair, $U^U = (u_1^U, u_2^U)$, that solves the following constrained optimization problem: $\max W(u_1, u_2) = u_1 + u_2$, subject to $u_i > d_i$.*

Notice, on the left side of [figure 15.5](#), that because the bargaining problem is formulated in terms of utility, this social welfare function is just a straight

line and the total utility, k , grows larger as we move to lines farther to the northeast. If the feasible set is strictly convex, you can see that the social planner will again be able to pick a unique solution to the bargaining problem. The problem, however, is that the utilitarian solution doesn't help solve the example that we have been using all along. Remember that at the beginning of the chapter, we commented that solving the bargaining problem was a bit like solving an equilibrium selection problem. All the points along the frontier of the feasible set are potential Nash equilibria, and it is up to the impartial arbitrator to make one equilibrium more focal than the others. Well, if we revert to the original example where $u_i(x_i) = x_i$, then the feasible set frontier reverts to $x_1 + x_2 = 1$. Perhaps you can see the problem: we now have two straight lines, and both of them have a slope of -1 . As you can see on the right side of [figure 15.5](#), in this example, the utilitarian solution is not unique because any allocation of the pie such that both parties get more than their disagreement utility maximizes the sum of the utilities.

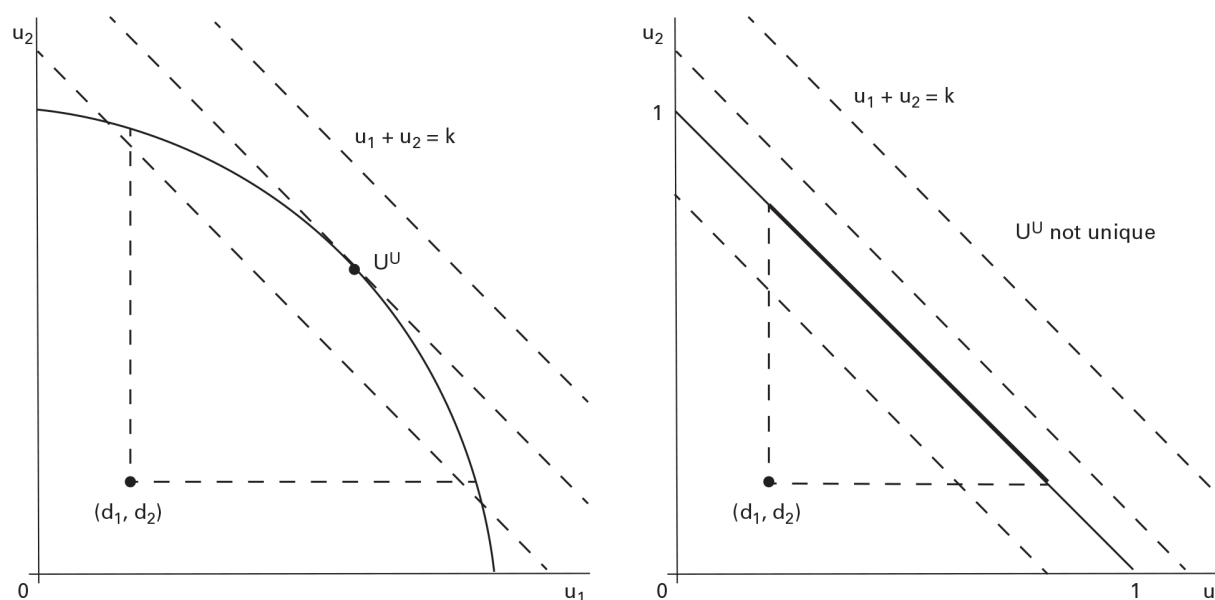


Figure 15.6
The utilitarian bargaining solution.

Behavior in Games—Which of Nash's Axioms Seem to Resonate with Bargainers?

Coterminously with the theoretical attention given Nash's bargaining problem, an interest arose in whether the Nash axioms could predict the

outcome of actual negotiations. As Nydegger and Owen (1974, p. 239) point out, “the logic of a mathematically derived axiom is not undermined by the irrational performance of a subject in an experiment, but in order to generalize from abstract models to real life behavior, care must be taken to fully appreciate the variety of variables that may enter into and affect bargaining behavior.” In their experiment, the authors, like Ehud Kalai and Meir Smorodinsky before them, were primarily concerned with the relevance of the independence of irrelevant alternatives axiom to experimental participants, but they also had a hunch that the invariance axiom might be unrealistic because bargainers would find interpersonal comparisons of utility, if available, irresistible.

To examine the relevance of Pareto optimality and symmetry, Nydegger and Owen tested whether two experimental participants, when given a dollar, would divide it equally. Indeed, all 10 pairs in this treatment did exactly that. In a second treatment, two participants were again given a dollar to divide, but one of them was not allowed to earn more than 60 cents. Here, the idea was that the Nash solution, 50-50, was still available and no player would have ever gotten more than 60 cents anyway, so the outcomes above that point were irrelevant. By contrast, in the Kalai-Smorodinsky solution, this alteration should matter, and it should have hurt the player whose best outcome was reduced. As in the first treatment, all 10 pairs agreed on the 50-50 split in this treatment too. This was taken as evidence in favor of Nash’s independence axiom. In a third treatment, the two participants were given 60 poker chips to divide, with an exchange rate of 2 cents a chip for the first participant and 1 cent per chip for the second one. Invariance suggests that the two participants should continue to split the chips evenly, even though the ultimate payoffs would now be different. This, however, is not how the experimental participants saw it. In this third treatment, instead of dividing the chips equally in number, all the pairs of participants decided to divide the money equally, with the first participant getting 20 chips and the second getting 40 of them. This was seen as a serious blow to the invariance axiom and a win for the egalitarian solution. It seemed that people were more interested in interpersonal comparisons of utility than first thought.

In real life, it is hard to compare utilities. As Philip K. Dick wrote more than once, “*de gustibus non est disputandum*,” or there is no point in arguing about differences in tastes and preferences. If this is the case, what happens if

we take Nydegger and Owen's experiment and make it harder to compare utilities—does this get us experimental results that are closer to what Nash had in mind? Following up on his previous work, Rudy Nydegger (1977) did exactly this. Instead of bargaining over money (as a proxy for utility), which is easy and natural to compare, in his replication of Nydegger and Owen (1974), Nydegger (1977) first asked participants to rank-order different lists of items from the college store and then assign arbitrary values to 10 of the items between 0 and 100. Once negotiations started, the participants were instructed to divide 100 tokens and told that the number of tokens they accrued would determine the store item that they would receive. What is innovative here is that it was clear that each person had a utility ordering over the items presented, but that it was impossible for one person to know the utility function of the other.

In this setting, and with the same three treatments used in the earlier study, Nydegger found very similar results in the first treatment, again supporting the relevance of Pareto optimality and symmetry. In the second condition, wherein one participant could get no more than 60 tokens, the results were divided. Instead of continuing to share the tokens 50-50, four of the eight groups did this, and the other four were much closer to the Kalai-Smorodinsky solution, providing some support for resource monotonicity over the independence of irrelevant alternatives. Finally, when invariance was tested in the third treatment, the results, on average, moved closer to the Nash solution. Here, the split was also four close to the egalitarian solution and four close to the Nash solution. The takeaway message is that when it is harder to compare utilities, both invariance and resource monotonicity seem more salient to bargainers.

Evidence-Based Adaptations of the Standard Model

The next major contribution to this area of work was the study by Alvin Roth and Michael Malouf (1979), discussed earlier in this chapter, which also dealt with the invariance axiom issues confronted in the previous studies. In response, the researchers decided to conform even more closely to the information conditions implicit in the theoretical bargaining problem. Specifically, they contended that much of the previous work provided too little of one kind of information (about utilities) and too much of another (about monetary payoffs) to be consistent with the theory, with the basic

problem being that knowing a player's monetary reward does not reveal her utility function. To deal with this problem, Roth and Malouf came up with an extremely clever design. Instead of bargaining over monetary rewards or proxy currency, negotiators bargained over how to divide 100 lottery tickets.

Here's the idea. Suppose that negotiator 1 has monetary prize $\$x_1$ and negotiator 2 has prize $\$x_2$. Let's assume that we normalize $u_i(0) = 0$ and $u_i(x_i) = 1$. This means that if negotiator i loses the lottery and gets nothing, her utility is zero, but if she wins the lottery, her utility is 1. Assuming that this sort of transformation leaves the essential preference ordering intact is consistent with any standard utility theory. Given all this, what is negotiator i 's expected utility from bargaining when she receives t_i lottery tickets? This is calculated as $Eu_i = t_i u_i(x_i) + (1 - t_i) u_i(0)$, but after all the substitutions, this collapses to $Eu_i = t_i$. In other words, by having participants negotiate over lottery tickets, Roth and Malouf effectively linearized participant utility functions and forced each participant's utility to just be the number of tickets gathered. This means that participants now know exactly what their and the other negotiator's utilities are for any outcome in the feasible set, without knowing anything about the monetary prizes they will receive.

To test the predictions of the Nash and Kalai-Smorodinsky solutions in the lab, Roth and Malouf (1979) considered treatments similar to those employed by Nydegger and Owen (1974). In game 1, there were 100 tickets to divide and the monetary prize was \$1 for each negotiator. In game 2, the prizes were the same as in game 1, but negotiator 2 could accrue no more than 60 lottery tickets. In game 3, both players could gather all 100 tickets, but negotiator 1's prize was \$1.25, while the prize for negotiator 2 was \$3.75. Finally, in game 4 the monetary prizes were the same as in game 3, but negotiator 2 could again gain no more than 60 tickets.

As in previous studies, when the maximum achievable ticket count is reduced for one of the negotiators, we can test for the independence of irrelevant alternatives, and when the monetary prizes differ, we can test for invariance (or resource monotonicity). Considering predictions in this environment, it should be clear by now that the Nash solution requires an even 50-50 split of the lottery tickets in all four games, while the Kalai-Smorodinsky solution requires the 50-50 split only in games 1 and 3. In games 2 and 4, the Kalai-Smorodinsky solution is where negotiator 1 gets 62.5% of the tickets and negotiator 2 gets the rest, 37.5%.

So what happens when experimental participants negotiate over lottery tickets? If the participants know their own monetary prize and the monetary prize of their opponents, the Nash solution, 50-50, arises in all but two instances in games 1 and 2, supporting Parteo optimality, symmetry, and the independence of irrelevant alternatives. In games 3 and 4, however, the equal split is almost nonexistent, but only in a few instances do bargainers divert to the Kalai-Smorodinsky solution. Instead, as in Nydegger and Owen (1974), most pairs end up splitting the utility gains as in the egalitarian solution. This means that neither invariance nor resource monotonicity find much support.

Remarkably, Roth and Malouf (1979) anticipated this outcome and also ran a partial information treatment in which participants know just their own monetary prizes, a setting much closer to the conditions on which Nash's theory is formulated. Here, negotiators know utilities, but not monetary payoffs, and Nash's solution does well in all four games as predicted. The upshot is that when the conditions of the model are satisfied, Nash's solution seems to apply. However, it will only be in rare instances that negotiators will bargain while knowing each other's utility functions but not the money stakes involved. In fact, in most instances, deals are stated in terms of some currency, and this means that any predictive bargaining solution has to account for the comparisons that negotiators will surely make. These are exactly the same theoretical advancements discussed in chapter 14.

Exercises

(1) *Finding the Nash bargaining solution.* Graph the bargaining problem and compute the Nash bargaining solution in the following situations:

- (a) Suppose that F is the convex hull enclosing the points $A = (0, 10)$, $B = (4, 8)$, $C = (10, 0)$ and $D = (0, 0)$ and the disagreement point is $(0, 2)$.
- (b) Suppose that F is the convex hull enclosing the points $A = (0, 7)$, $B = (6, 4)$, $C = (7, 0)$ and $D = (0, 0)$ and the disagreement point is $(1, 0)$.
- (c) Suppose that F is the convex hull enclosing the points $A = (0, 4)$, $B = (4, 2.8)$, $C = (4, 0)$ and $D = (0, 0)$ and the disagreement point is $(0, 0)$.

(2) *Finding Kalai-Smorodinsky.* Go back over the three scenarios listed in exercise 1 and identify the Kalai-Smorodinsky bargaining solutions.

(3) *Split the difference.* Now suppose that two negotiators, $i = \{1, 2\}$, have the utility function $u_i = x_i$ and the disagreement point $(d_1, d_2) = (\frac{1}{2}, 0)$ and bargain

over the division of a unit pie (i.e., $x_1 + x_2 = 1$).

- (a) What will each negotiator get at the Nash bargaining solution?
- (b) What will each negotiator get at the Kalai-Smorodinsky solution?
- (c) What will each negotiator get at the egalitarian solution?
- (d) What will each negotiator get at the utilitarian solution?

(4) *Split the difference—hard version (you might need a graphing calculator for this one)*. Now suppose that two negotiators, $i = \{1, 2\}$, have the utility function $u_i = x_i^{\frac{1}{2}}$ and the disagreement point $(d_1, d_2) = (\frac{1}{2}, 0)$ and bargain over the division of a unit pie (i.e., $x_1 + x_2 = 1$).

- (a) What will each negotiator get at the Nash bargaining solution?
- (b) Suppose that we can estimate the frontier of the feasible utility set to be $u_2 = 1 - u_1^4$. What will each negotiator get at the Kalai-Smorodinsky solution?
- (c) Continue with the frontier of the feasible utility set being $u_2 = 1 - u_1^4$. What will each negotiator get at the egalitarian solution?
- (d) Continue with the frontier of the feasible utility set being $u_2 = 1 - u_1^4$. What will each negotiator get at the utilitarian solution?

(5) *Invariance*. Reconsider the negotiators in exercise 3. This time the disagreement point is just $(0, 0)$. Show that negotiator 2 cannot restate her preferences such that $u_2 = 2x_2$ and change her share of the pie in the Nash bargaining solution.

(6) *Risk aversion and bargaining outcomes*. What is your intuition about how risk aversion should affect the result of bargaining? Suppose that two negotiators are trying to divide \$10 (i.e., $x_1 + x_2 = 10$), negotiator 1 has utility linear utility, $u_1(x_1) = x_1$, and is, therefore, risk neutral, and negotiator 2 has concave utility, $u_2(x_2) = 2x_2^{\frac{1}{2}}$, and is risk averse. Also, assume that both get nothing if they can't come to an agreement. Find the Nash bargaining solution. Does being risk averse matter, as you expected?

(7) *Bargaining power*. Now assume that some negotiators are more talented than others. Let $0 < \rho < 1$ be the bargaining power of negotiator 1 and $1 - \rho$ be negotiator 2's bargaining power. In this case, the Nash product is $(u_1 - d_1)^\rho (u_2 - d_2)^{1-\rho}$. Assume, again, that the negotiators bargain over the division

of a unit surplus and have linear utility functions, $u_i = x_i$, and a disagreement results in them getting nothing. Does it pay to be the better bargainer?

16

Cooperative Game Theory

The focus of this book thus far has been on non-cooperative games, in which players act independently in pursuit of their own self-interests. In a cooperative game, sets of players can join together to form coalitions and then make binding agreements regarding which strategy each person in the coalition will play and how they will share the resulting payoffs. The goal of cooperative game theory is to predict the outcomes of cooperative games: which coalitions form, the actions that each member of the coalition takes, and how the coalitions' total payoffs are distributed. Rather than modeling the structure of the bargaining process that leads to these agreements, cooperative game theory focuses on the possible outcomes that rational players might agree on. In this chapter, we will consider two prominent solution concepts in cooperative game theory and their applications to political power and cost sharing. In chapter 17, we will look at the application to matching markets.

Imagine that you are considering a merger between your company and one or two other firms. Merging will generate additional profits for the firms that participate, beyond what those firms would earn on their own. The total profits of the merger depend on which firms participate. The additional profits associated with each of the possible mergers are shown in [figure 16.1](#):

Merging Firms	Total Profits
	Associated with Merger
A and B	90
A and C	70
B and C	40
A, B, and C	100

Figure 16.1

The merger game.

Imagine that you own firm A and you are able to propose a merger. Which firm or firms would you approach, and how would you offer to split the associated profits?

Cooperative Games

In a cooperative game, players can team up to form a coalition that plays the game as a coordinated decision-making unit. Rather than a strategic interaction among individuals, a cooperative game is a strategic interaction among coalitions. These coalitions of players have the ability to write legal contracts, which bind all the coalition members and are exogenously enforced. Once a player signs on to join a coalition under specific terms, it is not possible to renege. We will focus on *transferable utility games*, wherein the members' collective payoffs are paid to the coalition as a whole, which can allocate them according to the agreement the members have reached. Cooperative game theory abstracts away from the specific structure of the bargaining process that leads to these agreements and instead considers which agreements are possible under various reasonable requirements. In this chapter, we will not be concerned with finding rational best responses to specific situations; rather, we will conduct a higher-level assessment of which agreements rational players would be willing to sign.

Consider the game shown in [figure 16.2](#), which was the very first version of the prisoner's dilemma game. After designing the payoffs, Merrill Flood and Melvin Dresher drafted two of their colleagues to play the game for 100 rounds: Armen Alchian was the Row player and John D. Williams was the Column player. While the unique subgame perfect Nash equilibrium (SPNE) in this game is for each player to defect in every period, there was a substantial amount of cooperation (73% of observations). Cooperation would likely have been even more frequent if not for the asymmetry of the payoffs, which seems to have caused a discrepancy in how the players believed that they should share the gains from cooperation, as can be seen in the players' notes during the game. While the players could not communicate, they could write down private comments about their thought process each period. Williams's commentary demonstrates that he believed that the players should always mutually cooperate to maximize their joint earnings. He attempted to enforce this by responding to any defection on Alchian's part by punishing him with a defection of his own, comparing this process of teaching Alchian to cooperate to "toilet training a child." Alchian, however, objected to the unequal payoffs associated with the (Cooperate, Cooperate) outcome and believed that he should be able to mix in an unanswered defection on

occasion to increase his own payoff, repeatedly complaining that Williams “will not share.”

		John D. Williams	
		Defect	Cooperate
Armen Alchian	Defect	(0, 0.5)	(1, -1)
	Cooperate	(-1, 2)	(0.5, 1)

Figure 16.2

The first prisoner’s dilemma.

Now imagine that, instead, this was a cooperative game played just once. Rather than independently choosing their actions while jotting down snide remarks about one another, the players can talk about the game and draw up a binding agreement describing which action each person will take and how they will share the resulting payoffs. For example, if the agreement specifies that they will play (Cooperate, Defect), then the players’ combined payoffs equal 1 and the agreement specifies how they will allocate it across the two of them. Since the players cannot renege on an agreement, the strategic interaction ends as soon as an agreement is reached. However, if they do not reach an agreement, then they simply play the game as independent agents, without an agreement in place. That is, the disagreement point is the standard, non-cooperative game.

Question: If you were playing this game as Williams, what agreement would you propose? That is: which actions do you propose that each player take, and how would you allocate the payoffs?

Since the players can negotiate how to share their combined payoffs, it is reasonable to expect that they would agree on the strategy profile that is associated with the highest collective payoffs rather than leave money on the table. Agreements in which players share the maximum possible collective payoffs are thus called *collectively rational*. In this case, (Cooperate, Cooperate) yields the highest collective payoffs, and so a collectively rational agreement would require each player to Cooperate and the players to split a joint payoff of 1.5 per period.

How might they agree to share this payoff? As with the other bargaining interactions we’ve previously studied, agreements cannot be imposed on

players without their consent, and so it is rational for individuals to agree only to outcomes that make them at least as well off as walking away and letting negotiations fail. Recall from the previous chapters that an agreement is *individually rational* if it makes each player as well off as the disagreement point. If the players in this example fail to reach an agreement, they play the prisoner's dilemma game without a cooperative arrangement. The predicted outcome of this interaction is the unique equilibrium of (Defect, Defect), with Alchian earning a payoff of 0 and Williams earning a payoff of 0.5. It would thus not be individually rational for Williams to consent to a payoff distribution in which he receives less than 0.5 (or for Alchian to consent to receiving less than 0), since he could always guarantee himself this amount by simply opting out of an agreement and playing Defect. An agreement is thus both individually and collectively rational if the players each Cooperate and share the resulting 1.5 in a way that provides Alchian with at least 0 and Williams with at least 0.5.

Let's look at another example. Consider the game shown in [figure 16.3](#). What agreements would be both individually and collectively rational?

		Column	
		Left	Right
Row	Up	(6, 20)	(2, 10)
	Down	(15, 2)	(3, 0)

Figure 16.3

Another two-person cooperative game.

The players' joint payoffs are maximized when Row plays Up and Column plays Left, yielding collective payoffs of 26. Thus, any collectively rational agreement must involve the players collectively earning 26. How much of this 26 must each player be offered in order for it to be individually rational to accept? If an agreement is not reached, we expect the outcome of the game to be (Down, Left). Thus, it is only individually rational for Row to accept an agreement granting him a payoff of at least 15 and for Column to accept an agreement that gives her at least 2. Any allocation of 26 in which Row receives at least 15 and Column receives at least 2 is both individually and collectively rational.

We will now apply this intuition to a wider range of scenarios, with more players.

Coalitions and Values

Rather than requiring that each individual player act as an independent decision maker, in a cooperative game *any subset* of players can join together and play the game as a unit. We call these subsets of players *coalitions*. We'll first consider the possible coalitions of players that can form and the joint payoff that they can share if they join forces. In the subsequent sections, we'll consider how they will divide those payoffs.

Consider a set of n players. A coalition of players, S , is a subset $S \subseteq \{1, \dots, n\}$. Note that the full set of all n players may comprise a coalition, which we call the *grand coalition*, and that a coalition may consist of a single player.

In non-cooperative games, we consider the players' strategy sets and the payoffs the players receive from each possible strategy profile. In contrast, a cooperative game is defined by its *characteristic function*. The characteristic function (as described in definition 1) assigns a value, $v(S)$, to each possible coalition S , including the empty set and the grand coalition. These values tell us the total payoffs that coalition could achieve on its own, without the cooperation of any other players outside the coalition.

Definition 1 (Characteristic function) *Consider a cooperative game with a set of players $N = \{1, \dots, n\}$. The game's characteristic function is a function $v: 2^n \rightarrow \mathbb{R}$ that maps each coalition of players to a payoff.*

In all cases, we'll assume the value of the empty set is zero ($v(\emptyset) = 0$) and the value of the grand coalition is equal to the highest achievable joint payoffs in the game. In cooperative games, we thus abstract away from considering which strategies or choices lead to various outcomes and instead consider only the ultimate payoffs that would be feasible for various coalitions.

Let's look at a few examples.

Prisoner's Dilemma

Let's first consider the asymmetric prisoner's dilemma game described earlier. The first column of the table in [figure 16.4](#) lists out all the possible coalitions. To find the characteristic function, we consider the payoffs that each coalition can achieve. The value of the empty set, $v(\emptyset)$, is always 0. Notice that this is just always true, by definition, and has nothing to do with the value to the

players of playing independently. The value of the grand coalition is $v(\{Alchian, Williams\}) = 1.5$, the joint payoff that the two players could achieve if they were to team up and operate as a single decision-making unit. Finally, to find the value of the single-person coalitions, we consider what the player's payoff would be in the event that he goes it alone. This is what we found earlier when we thought about which agreements would be individually rational: $v(\{Alchian\}) = 0$ and $v(\{Williams\}) = 0.5$.

S	$v(S)$
$\emptyset$	0
$\{Alchian, Williams\}$	1.5
$\{Alchian\}$	0
$\{Williams\}$	0.5

Figure 16.4

Characteristic function for the prisoner's dilemma.

The Merger Game

Let's consider the characteristic function in the game from the beginning of the chapter, with the three firms negotiating a merger. The first column in the table [figure 16.5](#) lists all the possible coalitions, while the second column lists the characteristic function for the coalition if those firms were to merge. If all three firms merge, they attain collective profits of 100, while if two of the three firms merge, they attain lower profits, which depend on which firms are involved in the agreement. The value of any single-member coalition is equal to zero, since a firm does not profit from the merger if it doesn't participate in it.

S	$v(S)$
$\emptyset$	0
{A, B, C}	100
{A, B}	90
{A, C}	70
{B, C}	40
{A}	0
{B}	0
{C}	0

Figure 16.5

Characteristic function for the merger game.

Buyers and Sellers

Let's consider again the bargaining problem game that we looked at in chapter 14, in which a buyer and seller are considering trading a used car. The buyer values the car at \$700, while the seller values it at \$300. Now let's push this example ever so slightly in the direction of a market by adding a second buyer to the mix. Let's say that buyer 2 values the car at \$550. All possible coalitions are listed in the first column of the table seen in [figure 16.6](#) and the characteristic function tells us the greatest surplus that coalition would be able to share if they were to team up and transact. For singleton coalitions, the value is 0, since they do not transact with the other actors in the market. If the coalition consists of buyer 1 and the seller, they can achieve a collective surplus of $\$700 - \$300 = \$400$, as we saw in chapter 14. If the coalition consists of buyer 2 and the seller, then their potential surplus is only \$250, since buyer 2 has a lower willingness to pay. If the coalition consists only of the two buyers, they do not achieve a surplus, since they are unable to buy the car if the seller does not participate. What is the value of the grand coalition? Although both buyers are in the coalition, there is only one seller, and thus only a single car available for the buyers to purchase. Thus, the value of this coalition is the maximum surplus that these three players can achieve if the car were to be traded among them, which is \$400.

S	$v(S)$
$\emptyset$	0
{Buyer 1}	0
{Buyer 2}	0
{Seller}	0
{Buyer 1, Buyer 2}	0
{Buyer 1, Seller}	400
{Buyer 2, Seller}	250
{Buyer 1, Buyer 2, Seller}	400

Figure 16.6

Characteristic function for the used car game.

Ride Sharing

In some cases, the characteristic function may represent the costs associated with each possible coalition rather than the value. Imagine that the players wish to achieve some costly goal, and by forming a coalition, they may share their collective costs. For example, say that A is taking a taxi home after a night out with two friends, B and C. It costs A \$10 to get home, while it costs B \$5 and C \$12. A and B could share a cab for \$12, A and C could share a cab for \$20, and B and C could share a cab for \$14. Finally, if all three were to share the cab, then it would cost \$25. The characteristic function then tells us the total cost that each coalition would need to pay to get its member(s) home, as shown in [figure 16.7](#).

S	$v(S)$
$\emptyset$	0
{A}	10
{B}	5
{C}	12
{A, B}	12
{A, C}	20
{B, C}	14
{A, B, C}	25

Figure 16.7

Characteristic function for the ride sharing game.

Distributing Payoffs: The Core

The characteristic function tells us the payoff that each coalition would receive if it were to form and defines the cooperative game. Now we wish to predict the outcome, by asking: (1) Which coalitions will form? and (2) How will they distribute those payoffs among their members?

The characteristic function can be used to describe any cooperative game. However, we typically restrict our attention to games that are *superadditive*, in which the value of a coalition is at least equal to the sum of its parts. A game is superadditive if, for any two disjoint coalitions S and T , $v(S \cup T) \geq v(S) + v(T)$. When a game is superadditive, the grand coalition is the most efficient, and thus we predict that all the players will form a single coalition, cooperate, and pool their payoffs.

How will the grand coalition distribute these payoffs? In an n -player game, we are looking for a payoff vector $x = (x_1, \dots, x_n)$, which assigns a payoff to each of the n players.

In non-cooperative game theory, the Nash equilibrium (and its associated versions for incomplete information and sequential games) is the standard solution concept. In cooperative games, there are several common solution concepts, which often predict different payoff distributions. We will consider two of the dominant solution concepts in cooperative game theory, called the *core* and the *Shapley value*. We will discuss each one, along with their applications, in the next two sections.

Imputation

As in the prisoner's dilemma game that we discussed earlier, it seems reasonable that the ultimate distribution of payoffs must be individually rational and collectively rational. These two conditions are so fundamental that we have a special term for allocations that satisfy them. We call an allocation satisfying both individual and collective rationality an *imputation* as described in definition 2.

Definition 2 (Imputation) Consider a cooperative game with a set of players $N = \{1, \dots, n\}$. A distribution $x = (x_1, \dots, x_n)$ is an imputation if:

(1) $x_i \geq v(\{i\})$ for all players i (it is individually rational)

(2) $\sum_{i=1}^n x_i = v(\{N\})$ (it is collectively rational)

Blocking Coalitions and the Core

Consider a game with the characteristic function described in [figure 16.8](#).

S	$v(S)$
$\emptyset$	0
{A}	6
{B}	3
{C}	3
{A, B}	20
{A, C}	24
{B, C}	12
{A, B, C}	30

Figure 16.8

Example 1.

Question: Which allocations are imputations?

A distribution (x_A, x_B, x_C) is an imputation if it satisfies the three inequalities required by individual rationality and the one equality required by collective rationality: $x_A \geq 6$, $x_B \geq 3$, $x_C \geq 3$, and $x_A + x_B + x_C = 30$.

Of course, there are infinitely many distributions that satisfy these four constraints. For instance, consider the following distributions:

Imputation 1: (10, 10, 10)

Imputation 2: (24, 3, 3)

Imputation 3: (16, 5, 9)

All of these are imputations, but do some seem more plausible than others? They all take into account the information in the first few lines of the table (what each individual could achieve) and the last line of the table (what the grand coalition could achieve). However, they do not reflect the information in between: what the *pairs* of players could achieve if they were to ditch the third wheel and form a two-member coalition on their own.

In the first imputation, the players simply share the grand coalition's surplus equally. While this may seem fair in that it gives the players equal shares, it assigns players A and C less than what they could collectively achieve if they were to form their own coalition. If A and C were to ditch B,

they could share 24 and each do better. We thus say that the coalition $\{A, C\}$ *blocks* this distribution, as described in definition 3.

Definition 3 (Blocking) Consider an imputation x . We say that a coalition S *blocks* x if there exists another imputation y such that

- (1) $\sum_{i \in S} y_i \leq v(\{S\})$ (the coalition S could feasibly achieve at least y if they chose to leave the grand coalition)
- (2) $y_i > x_i$ for all players i in S (y is better than x for all coalition members)

In the second imputation example, B and C are each given 3—the minimum they’re required to get under an imputation—while player A is given the rest. This satisfies the definition of an imputation, since B and C are given the amount that they can achieve independently. Thus, they would not be better off if they opted out of this agreement *and* failed to make an alternative agreement. However, they would be better off forming their own coalition $\{B, C\}$ and sharing its associated payoffs of 12. In this case, we say that the coalition $\{B, C\}$ *blocks* the distribution $(24, 3, 3)$.

What about the third imputation, $(16, 5, 9)$? It is individually rational, since each player gets more than they could achieve independently. It is collectively rational, since the payoffs sum to the maximum collective payoff of 30. Furthermore, there is no *set* of players who could do better ditching this agreement, forming their own coalition, and sharing the resulting payoffs: A and C collectively achieve 25 under this agreement and could only get 24 if they formed their own coalition; A and B collectively achieve 21 under this agreement and could only get 20 if they formed their own coalition; and B and C collectively achieve 14 under this agreement and could only get 12 if they formed their own coalition. Thus there is no player or set of players who could do better opting out of this agreement and going off on their own. We say that this distribution is *in the core*. The core (defined formally in definition 4) is the set of allocations that give each player and coalition of players at least the smallest payoff they could achieve on their own—that is, the set of imputations that cannot be blocked.

Definition 4 (The core) The core is the set of all imputations that cannot be blocked by any coalition. We say that an imputation is “in the core” if no set of players could do better by forming a coalition and redistributing that coalition’s value. Specifically, an imputation x is in the core if and only if:

$$\sum_{i \in S} x_i \geq v(\{S\}) \text{ for every coalition } S$$

In our example, an allocation is in the core if it satisfies the requirements for an imputation (i.e., $x_A \geq 6$, $x_B \geq 3$, $x_C \geq 3$, and $x_A + x_B + x_C = 30$), but also each coalition earns at least what it could get on its own: $x_A + x_B \geq 20$, $x_A + x_C \geq 24$, and $x_B + x_C \geq 12$. Thus, the requirements of the core take into account not only the first three lines of the table (the requirements for individual rationality) and the final line of the table (the requirement for collective rationality), but also the information in between: what each two-member coalition could achieve. In this example, this gives us a set of 7 expressions (1 equality and 6 inequalities) that must hold. Solving these expressions simultaneously, we find that there is a range of distributions (x_A, x_B, x_C) in the core, such that $x_A + x_B + x_C = 30$ with x_A between 14 and 18, x_B between 3 and 6, and x_C between 6 and 10.

We could also see this graphically. In the two-player bargaining games that we discussed in the previous two chapters, the boundary of the feasible set could be depicted as a line and each point represented a different allocation of the surplus between the two of them. In a three-player game like this one, we can represent the ways of sharing the grand coalition's value as a triangle, as shown in [figure 16.9](#). Let's focus on the left panel for now, which shows the core in our example. The dark triangle and its interior represent all the allocations of the surplus of 30 between players A, B, and C. The bottom left point of the triangle represents the allocation where player A gets all 30, while B and C get nothing. Likewise, the bottom right point is the allocation where B gets everything, and the point of the triangle indicates that C gets everything. The farther we get from the player's point, the less she is allocated. So, for instance, the bottom edge indicates distributions of the 30 between A and B, with C cut out completely, and points along the line that are closer to A means that A gets more. The allocation in which they equally split the 30 and get 10 each is located at the exact center of the triangle. So the triangle indicates all of the feasible allocations for the grand coalition—but which are in the core?

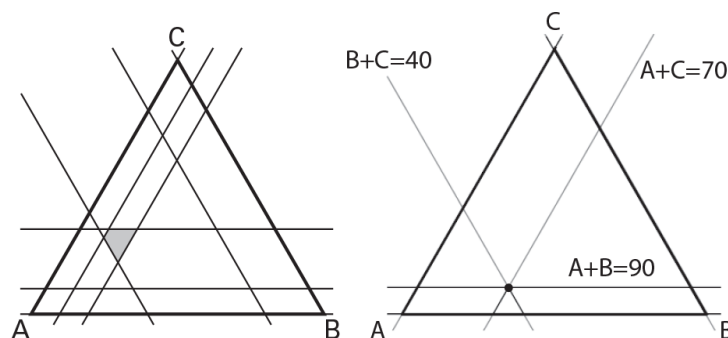


Figure 16.9

The core in Example 1 (left) and the merger game (right).

The other six light gray lines reflect the six inequalities that must also hold. To see what's going on with these, let's just focus on the two horizontal gray lines (but the interpretation of the lines that are parallel to the other two edges is analogous). The bottom edge of the triangle represents the allocations in which player C gets nothing, but those allocations cannot be in the core, since we know that $x_C \geq 3$. Thus, we are bounded away from the bottom edge, and any core allocation must be above that lower gray horizontal line, indicating that C gets at least 3. At the same time, we can't give C too much, because we also have $x_A + x_B \geq 20$. As a result, we could never have C getting more than 10, and so any core allocation needs to also be below the higher gray horizontal line, indicating that C gets less than 10. Likewise, the core allocations must fall between the two downward sloping diagonal lines and between the two upward sloping diagonal lines. Putting these six constraints together, the core allocations fall in the shaded region.

The Core in the Merger Game

In the merger game, the value of the grand coalition is 100, so we have $x_A + x_B + x_C = 100$. To satisfy individual rationality, we simply need each member to achieve nonzero payoffs. Finally, to be in the core, the allocation must satisfy three additional conditions: $x_A + x_B \geq 90$, $x_A + x_C \geq 70$, and $x_B + x_C \geq 40$. Solving these conditions simultaneously, we find that there is a unique core allocation, $(x_A, x_B, x_C) = (60, 30, 10)$.¹ This is also shown graphically in the right panel of [figure 16.9](#). Since the singleton coalitions can achieve 0, those constraints overlap perfectly with the edges of the triangle, and we only have the three gray lines representing the constraints for the two-player coalitions. Any core allocation must be on or below the horizontal line (indicating that A

and B collectively achieve at least 90), on or to the left of the diagonal gray line running from the bottom left to top right (indicating that A and C collectively achieve at least 70), and on or to the right of the diagonal gray line running from bottom right to top left (indicating that B and C achieve at least 40). There is a unique point at which these constraints hold simultaneously, indicating our unique core at (60, 30, 10).

Buyers and Sellers: Application to Markets

Consider again the three-person used car market, with the characteristic function we found previously. Let's find the core outcomes of this interaction. A distribution of payoffs (x_{Seller} , x_{Buyer1} , x_{Buyer2}) is in the core when the following expressions hold:

$$\begin{aligned} x_{Seller} &\geq 0, x_{Buyer1} \geq 0, x_{Buyer2} \geq 0 \\ x_{Seller} + x_{Buyer1} &\geq 400, x_{Seller} + x_{Buyer2} \geq 250, x_{Buyer1} + x_{Buyer2} \geq 0 \\ x_{Seller} + x_{Buyer1} + x_{Buyer2} &= 400 \end{aligned}$$

Solving simultaneously, we find that distributions in the core must give the seller between \$250 and \$400 of the surplus, while buyer 1 gets the rest and buyer 2 gets nothing. Why? The seller and buyer 1 could always ditch buyer 2, go off on their own, and share \$400 between them. So we can't have a distribution in the core that gives buyer 2 any amount of the surplus. If that were the case, the seller and buyer 1 would collectively receive less than $v(\{Buyer1, Seller\}) = 400$, and they could block that by going out on their own and sharing the \$400. Why does the seller need to get at least \$250? The seller doesn't need buyer 1 to achieve \$250, because this is what he could get by trading with buyer 2 instead. Thus, any distribution that gives the seller less than \$250 would be blocked by the coalition of the seller and buyer 2.

Another way to think about this situation is as a very small market. [Figure 16.10](#) graphs the supply and demand curves in this market. The supply curve is shown in black. The seller is willing to sell one car, so long as the price is at least \$300. If the price is less than \$300, the quantity supplied is 0. The demand curve is shown in gray. If the price is greater than \$700, neither buyer is willing to purchase, so the quantity demanded is zero. If the price is between \$550 and \$700, only buyer 1 is willing to purchase, so the quantity demanded is 1, and for any price below \$550, the quantity demanded is 2. To find our competitive market equilibrium, we look for the intersection of the supply and demand curves, and we find that our equilibrium price is between

\$550 and \$700. Under this outcome, the seller sells the car to buyer 1 (the only buyer who is willing to buy at a price above \$550). How do we interpret this price? If the price were \$700, this would be equivalent to the seller getting the entire \$400 surplus. A price of \$550 is equivalent to the seller getting \$250 of the surplus (i.e., receiving \$250 above her willingness to accept), while the buyer gets the remaining \$150. Thus, the equilibrium price range is perfectly equivalent to our core outcome. This is generally the case: the competitive equilibrium in a market is always in the core. This is just one application of the core solution concept. We will look at others later in the chapter.

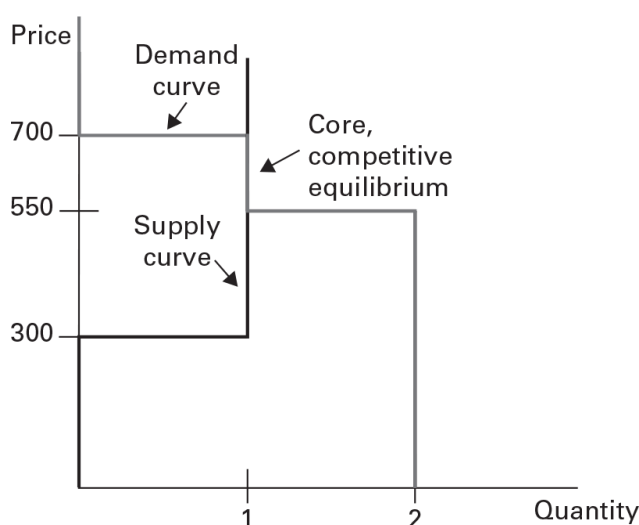


Figure 16.10

The supply and demand curves in the three-person used car game.

Dividing a Dollar

We have thus far seen examples in which there is either a continuum of allocations in the core or a unique allocation (as in the merger game). Let's look at an example in which the core is empty.

Consider three representatives who must decide how to divide a surplus between their three districts. Let's normalize the surplus to be 1. The outcome will be determined by majority rule, and so the votes of two out of the three players are needed to pass a proposed budget. In this situation, there is no distribution of payoffs across the three districts that is in the core.

Why? First, note that there is no core distribution in which all three members attain positive payoffs. Since any two players could always split the entire surplus between them, they do not need the vote of the third, so they

would be better off ditching the third person and splitting the surplus between them. Next, note that there is no core distribution in which one member attains positive payoffs while the other two receive nothing. The two players who received nothing could block this distribution and share the full surplus among themselves. Finally, note that there is no core distribution in which two members attain positive payoffs. Why? Imagine that you were the representative set to get nothing—what could you do about it? You could approach one of the other two representatives and propose an alternative allocation, in which she gets slightly more than under the current proposal while you get the rest. Both of you would thus be better off under this budget, and you don't need the third member's vote to pass it. Thus, any two members could always block a proposal in which two people are receiving positive shares.

This can be seen if we look at the characteristic function and the core requirements. Let's call the representatives A, B, and C. Then our characteristic function gives us $v(\{A\}) = 0$, $v(\{B\}) = 0$, $v(\{C\}) = 0$, $v(\{A, B\}) = 1$, $v(\{A, C\}) = 1$, $v(\{B, C\}) = 1$, and $v(\{A, B, C\}) = 1$. Any core allocation (x_A, x_B, x_C) must satisfy $x_A \geq 0$, $x_B \geq 0$, $x_C \geq 0$, $x_A + x_B \geq 1$, $x_A + x_C \geq 1$, $x_B + x_C \geq 1$, and $x_A + x_B + x_C \geq 1$. There is no allocation that simultaneously satisfies these conditions.

Distributing Payoffs: Shapley Value

The core solution concept provides the compelling and intuitive requirement that each player and set of players must do as well as they could if they opted out of the agreement and went off on their own. As we've seen, however, often the core is empty or contains a wide range of outcomes. In 1951, Lloyd Shapley devised a means of computing a value for each player, based on the average marginal contribution the player makes to any coalition she could join. Unlike the core, the *Shapley value* assigns a unique distribution of payoffs in any cooperative game. Since the Shapley value is not necessarily in the core, there may be sets of players who could make themselves better off by opting out of receiving their Shapley value and forming their own coalition. However, the Shapley value satisfies several other desirable properties that we'll discuss later.

To see how the Shapley value is calculated, let's first look at the example from earlier, with the characteristic function $v(\{A\}) = 6$, $v(\{B\}) = v(\{C\}) = 3$,

$v(\{A, B\}) = 20$, $v(\{A, C\}) = 24$, $v(\{B, C\}) = 12$, and $v(\{A, B, C\}) = 30$. Let's imagine building the grand coalition of $\{A, B, C\}$, starting with the empty set and adding one player at a time. As we add each player, we will consider how the surplus increases and refer to this increase as the *marginal contribution* of that player. We begin with the empty set, which has value $v(\emptyset) = 0$. Adding player A, the value increases from 0 to $v(\{A\}) = 6$. Thus, player A's marginal contribution is 6. Now we add player B, and the value increases from 6 to $v(\{A, B\}) = 20$. Thus, player B's marginal contribution is 14. Finally, we add player C and see that our value increases to $v(\{A, B, C\}) = 30$, so player C's marginal contribution was 10. This gives us one possible way of dividing the payoffs: (6, 14, 10).

But wait: Adding the players in this order—A then B then C—was arbitrary. We could have just as easily started with, say, player C, and then added B, then A, and our marginal contributions would have been very different (3 for C, 9 for B, and 18 for A).

To calculate the Shapley value, we thus consider *all possible* orders in which we could add players to get to the grand coalition, calculate each player's marginal contribution in each case, and then take the average over all these orders. In our example with three players, there are six possible orders in which we could build the grand coalition, shown in the first column of the table in [figure 16.11](#). For each of the six orders, we calculate the marginal value that each player adds as she joins the coalition. (Note that the first line replicates the original calculation we made, when we looked at the players joining in alphabetical order.) Note also that the marginal contributions must always sum to 30, the value of the grand coalition. Finally, we can take each player's average contribution over all six orders. In doing so, we find that player A's average marginal contribution is 14.33, while player B's is 6.83, and player C's is 8.83. (By necessity, these sum to 30.) This allocation is the game's Shapley value: it provides us with a unique means of distributing the grand coalition's payoff of 30 across the three members.

Order	Marginal Contribution			Total
	Player A	Player B	Player C	
A, B, C	6	14	10	30
A, C, B	6	6	18	30
B, A, C	17	3	10	30
B, C, A	18	3	9	30
C, B, A	18	9	3	30
C, A, B	21	6	3	30
Average	14.33	6.83	8.83	30

Figure 16.11

Computing the Shapley value in example 1.

Creating a table like this, which lists all the possible orders that the players could form the grand coalition, is a foolproof way to calculate the Shapley value of a game. The precise definition is given in definition 5. While this looks complicated, it is simply the formal expression of the procedure we followed when calculating the Shapley value of the previous example. Essentially, this says: Look at any coalition, S , which can be formed from the set of players, N , and that does not contain i (which is denoted in the equation as $S \subseteq N \setminus \{i\}$). We consider the marginal value of i if he were to be added to this coalition (i.e., $(v(S \cup \{i\}) - v(S))$). This is then weighted by the relative frequency of such coalitions ($|S|!(n-|S|-1)!$) as a fraction of all coalitions ($n!$), and summed over all of these coalitions.

Definition 5 (Shapley value) *Consider a cooperative game with a set N of n players and characteristic function v . The Shapley value assigns an amount to each player i in N equal to*

$$\sum_{S \subseteq N \setminus \{i\}} \frac{|S|!(n-|S|-1)!}{n!} (v(S \cup \{i\}) - v(S))$$

The Shapley value satisfies a number of beneficial properties. In fact, Shapley proved that it is the *only* distribution that simultaneously satisfies the following four axioms (Shapley 1951):

- *Efficiency.* The payoffs sum to the value of the grand coalition. That is, $\sum_{i=1}^n x_i = v(\{N\})$.
- *Symmetry.* If two players always contribute the same to any coalition, then they receive the same payment.

- *Additivity.* Consider two distinct cooperative games involving the same set of agents, with characteristic functions v_1 and v_2 . Now imagine that we model this as a combined game, with a characteristic function $v_{combined}$, such that $v_{combined}(S) = v_1(S) + v_2(S)$ for all coalitions S . The additivity axiom is satisfied if each agent's payment in the combined game is equal to the sum of his payments in the two separate games.
- *Dummy player.* Player i is a dummy player if player i always contributes exactly $v(\{i\})$ to any coalition he may join. The dummy player axiom is satisfied if dummy players are always allocated the amount that they could receive on their own.

Let's now see how to apply the Shapley value in various settings.

The Shapley Value When Dividing a Dollar

Let's return to the example of the three representatives choosing how to divide a dollar. We again have three players, and so there are six possible orders in which we can put the players together to build the grand coalition. Based on the characteristic function, we can consider the marginal contribution that each player makes as he is added for each of these orders. This is shown in [figure 16.12](#). Note that a player's marginal value is always 1 when she is the *second* player to join the coalition, at which point she is able to tip the vote. Her vote is valueless when she is the first person to join the coalition (since the coalition doesn't yet have the necessary number of votes to pass a budget) or the third person to join the coalition (since the coalition has already gained the necessary votes to pass the budget even before she joins). Note that each player is the second player to join the coalition in exactly two of the six orders. Thus, when we average over all six orders, we find that the Shapley value for each player is $1/3$. This makes intuitive sense. This is a symmetric game, and so it is only reasonable that the unique means of dividing the payoffs between the three players would give each an equal share. Thus, the Shapley value provides a reasonable solution, even though the core is empty.

Order	Marginal Contribution			Total
	Player A	Player B	Player C	
A, B, C	0	1	0	1
A, C, B	0	0	1	1
B, A, C	1	0	0	1
B, C, A	0	0	1	1
C, B, A	0	1	0	1
C, A, B	1	0	0	1
Average	1/3	1/3	1/3	1

Figure 16.12

Computing the Shapley value in the divide the dollar game.

The Shapley Value in the Merger Game

In the merger game, we found that there was a unique core outcome with $x_A = 60$ and $x_B = 30$, and $x_C = 10$. What is the Shapley value? The calculations are shown in the table seen in [figure 16.13](#):

Order	Marginal Contribution			Total
	Player A	Player B	Player C	
A, B, C	0	90	10	100
A, C, B	0	30	70	100
B, A, C	90	0	10	100
B, C, A	60	0	40	100
C, B, A	60	40	0	100
C, A, B	70	30	0	100
Average	46.67	31.67	21.67	100

Figure 16.13

Computing the Shapley value in the merger game.

The Shapley Value in Competitive Markets

Consider again the market game between the buyers and the seller. We found that any allocation in the core must give the seller between \$250 and \$400 of the \$400 surplus, while buyer 1 gets the remainder and buyer 2 gets zero. In contrast, the Shapley value gives buyer 2 a positive share, since in one of the

six scenarios, buyer 2 makes a positive contribution to the coalition. These calculations are shown in [figure 16.14](#).

Order	Marginal Contribution			Total
	Seller	Buyer 1	Buyer 2	
Seller, Buyer 1, Buyer 2	0	400	0	400
Seller, Buyer 2, Buyer 1	0	150	250	400
Buyer 1, Seller, Buyer 2	400	0	0	400
Buyer 1, Buyer 2, Seller	400	0	0	400
Buyer 2, Buyer 1, Seller	400	0	0	400
Buyer 2, Seller, Buyer 1	250	150	0	400
Average	241.67	116.67	41.67	400

Figure 16.14

Computing the Shapley value in the used car game.

Voting Power

One important application of the Shapley value is understanding the power of players in a voting game. Shapley and Shubik (1954) observed that a coalition's voting power is not simply proportional to its size, but rather is determined by the relative frequency with which its vote is pivotal. Consider a cooperative game in which the characteristic function assigns a value of 0 to any coalition that does not have enough votes to pass a proposal, and a value of 1 to any coalition that does. In this context, the Shapley value is known as the *Shapley-Shubik power index*.

To see this, let's go to Nassau County, New York, in 1964. Prior to the legislative system being declared unconstitutional by the US Supreme Court in 1993, the county governments in New York were run by boards of supervisors. Each town was typically given one seat on its county's board (with larger cities sometimes granted more seats). Since the towns often differed dramatically in size, each supervisor was given a different number of votes in an attempt to grant each town voting power commensurate with its population. For instance, the situation in Nassau County (birthplace of one of this book's authors) in 1964 is shown in [figure 16.15](#).

District	Population	% Population	Number of Votes	% Votes
Hempstead 1	728,625	57.1	31	27
Hempstead 2	–	–	31	27
Oyster Bay	285,545	22.4	28	24.3
North Hempstead	213,335	16.7	21	18.3
Long Beach	25,654	2.0	2	1.7
Glen Cove	22,752	1.8	2	1.7
Totals	1,275,801 (sic)		115	

Figure 16.15

Nassau County Board of Supervisors, 1964. Source: Shapley (1981).

There was a six-member legislature, with each member representing a municipality (with two members representing Hempstead). Each member was given a certain number of votes, shown in the fourth column of the table, to roughly represent the percentage of the population in their district. Looking at the total number of votes assigned (115), we see that a majority requires 58 votes. In practice, this meant that North Hempstead, Long Beach, and Glen Cove might as well have had no voting rights at all, since nothing that they could do, individually or collectively, would ever influence the vote. Even if all three of them joined up, that wouldn't be enough to tip the vote from one outcome to another. Instead, the two representatives from Hempstead and the Oyster Bay representative determine the outcome. Specifically, whichever way at least two of those three representatives vote will always determine the winning outcome. (If the two Hempstead representatives vote the same way, they achieve 62 votes, enough for a majority; if one Hempstead representative and the Oyster Bay representative vote the same way, they would collectively have 59 votes, also enough to decide the outcome.) The other three have no ability to affect the outcome and might as well stay home. Thus, the number of votes each representative gets does not really tell us the true story about their political power.

What if we look at the Shapley value? Just as in the divide the dollar game, we look at how often each player is able to tip the vote, on average. We find that the Shapley value assigns the first three representatives $1/3$, while the last three are assigned 0. This reflects what each member's true voting power actually is: the first three members are in a position similar to the three

players in the divide the dollar game, with two of the three of them always able to pass a proposal, and the last three might as well not be there at all.

Cost Sharing

Another major application of the Shapley value concerns how groups who could save money by joining together ultimately share their collective costs. For example, imagine that you're getting a cab home, and your cab fare is \$15. Your friend also needs a ride in a similar direction and would have to pay \$10 to get home. But if you split a cab, then the total ride comes to \$20. You and your friend are clearly better off sharing a cab and saving \$5—but then how do you decide how much each person should pay? In one scenario, imagine that you get into a cab intending to pay \$15. But then your friend pops in and says “Hey, I’m coming with you. I’ll pay the difference.” In this case, you pay \$15 and your friend pays \$5. Alternatively, maybe your friend got into the cab first, intending to pay \$10, and then you pop in and say, “I’m going to join you, and I’ll pay the extra \$10 fare.” In this case, you split the \$20 cost equally.

So which is fair? There is no need to agonize. The Shapley value tells us to simply average over the different ways in which you could form your ride-sharing coalition. In this case, there are only two orders in which we can build the grand coalition, so we average your marginal cost in each of them and find that you should pay \$12.50, while your friend pays \$7.50. What about in the three-player ride sharing game described earlier in the chapter? To find how much each of the three members should pay toward the \$25 ride, we consider the six orders in which the friends could form the grand coalition and calculate the marginal cost that each person adds to the coalition in each case. In doing so, we find that the Shapley value has person A paying \$9.50, person B paying \$4, and person C paying \$11.50, which you can confirm by making a table similar to those we used in the previous examples.

Behavior in Games

Can We Agree to Disagree?

The paradox of testing cooperative games in the lab is that the implementation must inherently provide some structure for the bargaining process. For instance, do people interact face to face? Can they send private messages and offers, or must everything be broadcast to the entire group? Is there a time limit on their negotiations? Experimental tests of cooperative game theory

concepts were predominantly conducted in the latter half of the twentieth century, starting with a paper coauthored by John Nash in the 1950s (Kalisch et al., 1954).

In the late 70s, Murnighan and Roth (1977, 1978, 1980) conducted a series of coalition formation experiments in games where one player has veto power. The veto player needs only to find a single partner in order to achieve the maximum payoff, and no other player can achieve a positive payoff without him. In the three-player version, for example, the experimenters told player A that he had a right shoe, while players B and C were told that they each had left shoes. Single shoes were worthless, but any coalition possessing a pair of one right shoe and one left shoe earned 100 points. Thus, the characteristic function is $v(\{A\}) = v(\{B\}) = v(\{C\}) = v(\{B, C\}) = 0$ and $v(\{A, B\}) = v(\{A, C\}) = v(\{A, B, C\}) = 100$. The core produces an extreme prediction: player A captures the entire surplus. In contrast, the Shapley value is 66.67 for player A and 16.67 for each of the others. Since the veto player requires only one of the two others to gain the surplus, this prediction can be interpreted as an expected value: player A gains 2/3 of the surplus, while one of the other two players gains 1/3, each with equal likelihood.

Rather than interacting face to face, subjects wrote proposals (and, in some treatments, messages) on slips of paper and the experimenters passed the slips to the relevant parties. In this game, Murnighan and Roth found that the vast majority of coalitions included two players, and on average, player A received a share of 65. That is, the players had no problem cutting the third player out of the deal and coming to an average arrangement in which player A gets a bit more than half of the surplus. They also looked at games with more than three players. As the number of additional players with left shoes increases, the Shapley value assigns a higher payoff to the veto player ($100 - 100/n$). Across a series of studies, Murnighan and Roth found that the Shapley value predicts average payoff distributions well across a wide range of group sizes, regardless of whether individuals could communicate outside their proposals, and that it also predicts the observed pattern that player A's payoff increases as the number of other players increase and there is more competition for his partnership. In contrast, outcomes approached the core prediction only in large groups that did not have access to communication channels outside their proposals, and only after several rounds had been played.

The merging companies game that we began the chapter with was developed and tested in the lab by Bolton, Chatterjee, and McGinn (2003). Subjects could send messages via email and the researchers varied whether communication was unconstrained, all messages were public, or two of the players didn't have each other's email addresses and could only communicate through the third. Unlike veto power games, the efficient outcome in this game requires the grand coalition, with the unique core specifying a payoff distribution of (60, 30, 10) and the Shapley value specifying (46.67, 31.67, 21.67). Notably, participants typically did not form the grand coalition. Instead, the very first two-way proposal was accepted about half of the time. The most common configuration was a two-player alliance between players A and B, who could achieve the greatest per-capita surplus, and they typically agreed on an equal or near-equal split of their coalition's value of 90.

Interestingly, it seems that participants were resistant to sharing the surplus unevenly within their coalition but had no problem cutting players out of the deal entirely. As one player A wrote to a player B, a bilateral agreement "can yield us the greatest profits each. If we split it, we get 45. Any other contract would not yield that much unless one of the parties took a smaller profit than the others" (Bolton et al., 2003, p. 592). How could player C attempt to compete with this outcome? Player C could make a direct competitive counterproposal by offering player A slightly more than 45, while she receives the remainder of their two-player value of 70, or by offering players A and B each slightly more than 45 while she receives the remainder of the grand coalition value of 100. The researchers find that the vast majority of the data can be explained by either members of the highest per-capita coalition (with direct lines of communication) teaming up and splitting the surplus equally, or the direct competitive counterproposal to this outcome.

Evidence-Based Adaptations of the Standard Model

Thus far, we have seen that none of the standard solution concepts in cooperative game theory can satisfactorily explain the outcomes in laboratory implementations. However, two things are clear: (1) equity concerns strongly influence which payoff allocations are deemed acceptable, and (2) it is unreasonable to expect participants to compute the Shapley value in their heads or anticipate the strategic pressures that eliminate all but the core outcomes. Reinhard Selten (1982, 1987)'s model of *equal division payoff bounds* instead assumes that players have limited cognitive capacities. Players

search for obvious clues regarding how much they should expect and use this to set a lower bound on the payoff that they aspire to achieve.

In games where the players are symmetric, the model assumes that each aspires to achieve an equal share. In games where players contribute different amounts to their coalitions, players assess their relative bargaining strength. They then expect at least an equal share in coalitions where they are the strongest member. Otherwise, they expect an equal share of any surplus they contribute by joining a coalition or replacing another member in a coalition.

The process is easiest seen through an example.² In the merger game, $v(\{A\}) = v(\{B\}) = v(\{C\}) = 0$, $v(\{A, B, C\}) = 100$, and $v(\{A, B\}) = 90 > v(\{A, C\}) = 70 > v(\{B, C\}) = 40$, indicating that A is stronger than B who is stronger than C.

We begin with the strongest overall player, A. Let's look for all the tentative lower bounds that A could consider demanding and assume that she aspires to the highest of them. Since A is the strongest member in coalitions $\{A, B, C\}$, $\{A, B\}$, and $\{A, C\}$, she should expect to receive at least an equal division in any of those coalitions (i.e., 33.3, 45, and 35, respectively). Furthermore, we could also consider what player A adds by substituting for someone else in a two-member coalition or by joining it to form the grand coalition, and assume that she expects at least an equal share of what she contributes. By necessity, the bounds that we already calculated are going to be greater than these benchmarks (since an equal share of a two- or three-member coalition's value will be at least as great as an equal share of what she adds to the value), but let's calculate these measures anyway for illustrative purposes. The two-member coalition without A has a value of 40, so player A would increase the value by 50 if she were to replace C and by 30 if she were to replace B. If she were to share these gains equally with the other member of the coalition, we have tentative lower bounds of 25 and 15, respectively. If instead player A were to *join* the other two members' coalition, the value would increase by 60, and an equal division of this gain would give her 20. We then define player A's highest tentative bound as the maximum across all of these benchmarks: $\max\{33.3, 45, 35, 25, 15, 20\} = 45$.

If we make a similar assessment for player B, we find that he is the strongest player only in the $\{B, C\}$ coalition and should expect at least half of that coalition's value, or 20. If he replaces another member in the two-member coalition, he would contribute at best 20, and an equal share of this

contribution gives him 10. If he joins the other two members to form the grand coalition, he contributes 30, again for an equal share of 10. We thus find that player B's highest tentative lower bound is 20. Finally, we consider player C, who is never the strongest member of a coalition and never contributes a positive amount by substituting for a member in any coalition. By joining the other two to form the grand coalition, player C adds 10, and so an equal share of this contribution would yield 3.33. Although player C is never the most profitable person to partner with in terms of the coalition's overall value, he could attempt to compete by offering one of the other players the best they could expect from the $\{A, B\}$ coalition. If A were to partner with B (whose highest tentative lower bound is 20), the highest share that A could walk away with is $90 - 20 = 70$. Thus, C could attempt to compete by offering A 70. But, since $v(\{A, C\}) = 70$, this would leave C with 0. Likewise, a competitive offer to B to join the $\{B, C\}$ coalition would yield a negative payoff for C. Thus, C's highest tentative bound is 3.33. Finally, the theory assumes that players reduce their aspiration levels in special cases where they are not mutually compatible or may round down to the next round number.

While the equal division payoff bounds model often predicts a wide range of payoffs, Selten found that it successfully predicted the vast majority of payoff divisions across four separate experiments, including 92% of the data in Murnighan and Roth's three-player games.

Exercises

(1) Consider a cooperative game between players A, B, and C. Player A has 2 left gloves and no right gloves. Player B has 3 left gloves and 10 right gloves. Player C has 12 left gloves and 8 right gloves. A set of gloves (with 1 right and 1 left) sell for \$100, but singletons cannot be sold.

- (a) Find the characteristic function.
- (b) Consider an allocation of payoffs across the three players, (x_A, x_B, x_C) . Find the conditions that must be satisfied for the allocation to be in the core.
- (c) Find the Shapley value. Is it in the core?

(2) Consider a market with two sellers (A and B) and two buyers (C and D). Player A is willing to accept any price of at least \$200 and B is willing to

accept \$300. Player C is willing to pay \$250 and D is willing to pay \$500.

- (a) Find the characteristic function.
 - (b) Find the core outcome.
 - (c) Graph the supply and demand curves. What range of prices is predicted by the competitive equilibrium? Confirm that this outcome corresponds with the core.
 - (d) Now imagine that player C's willingness to pay increases to \$400. Describe the core outcome and confirm that it corresponds to the intersection of the supply and demand curves.
- (3) Imagine that three companies are interested in interviewing you: A (located in Atlanta), B (located in Boston), and C (located in Chicago). Each one will pay the travel expenses associated with your trip.

If you fly out to each company individually, each one must pay the cost of getting you there from your home and back again. However, if you make more than one stop on the same trip, the companies can split the costs among them and save money. To be fair (and impress them), you decide to visit all three companies on a single trip, choose the itinerary that minimizes costs, and then divide the total costs according to the Shapley values.

Specifically, each *one-way* trip costs as follows:

Between Home and Atlanta: 700

Between Home and Boston: 600

Between Home and Chicago: 800

Between Atlanta and Chicago: 200

Between Atlanta and Boston: 400

Between Boston and Chicago: 400

(Note: Assume that costs are the same in both directions, so if you fly from Home to Atlanta and back Home, it costs $700 + 700 = 1,400$; if you fly from Home to Atlanta to Chicago to Home, it costs $700 + 200 + 800 = 1,700$, etc.)

- (a) The first step is to set up the problem as a three-player cooperative game by specifying the characteristic function. In this case, you can think of the characteristic function as giving the lowest possible airfare cost associated with each coalition.

Hint: For instance, the first coalition is A. If the costs are shared only by firm A, what is the lowest the costs could be? That gives you $v(\{A\})$. If the

coalition is A and B, what is the lowest that the collective costs can be? That gives you $v(\{A, B\})$. To find $v(\{A, B, C\})$, find the itinerary that minimizes the three firms' costs, given that you are visiting each one of them.

(b) How do you divide the total cost of the trip among the three companies? (Hint: To calculate the Shapley values, you must consider all the orders in which you can build the grand coalition up, one company at a time. These refer to the order in which the companies join the coalition, not the order in which you visit them. Remember that you're always choosing the itinerary that minimizes costs. Thus, for every order, you should find that the marginal costs sum to the answer you found for the grand coalition's value in part a.)

(5) Consider a committee with four members, each with different voting weights. Player A has 20 votes, B has 15, C has 8, and D has 7. Imagine that 30 or more votes are needed to pass any proposal.

(a) Does player A have the power to block a proposal?

(b) Find the Shapley-Shubik power index.

¹To see this, consider each of the conditions in turn. We know that the grand coalition achieves exactly 100, but A and B must collectively get at least 90. So C can get no more than 10. A and C collectively get 70, and since C is getting no more than 10, A must get between 60 and 70. B and C collectively get 40, and since C is getting no more than 10, B must get between 30 and 40. Finally, A and B collectively get 90, and we know A must get at least 60 and B must get at least 30, so we must have $x_A = 60$ and $x_B = 30$, and thus $x_C = 10$.

²The model must be applied to games in which $v(\{i\}) = 0$ for all i , and so games must first be zero-normalized by subtracting $v(\{i\})$ from all coalitions in which agent i is a member.

17

Matching Market Design

The design of matching markets relies on insights from both cooperative and non-cooperative game theory. Unlike the markets we typically think about in economics, in which the price mechanism works to allocate goods, matching markets assign agents to objects or to each other and no cash is exchanged. Building on the previous chapter on cooperative game theory, the notion of core stability is an important solution concept in matching markets because if there is a set of players who can do better by ignoring their assigned match and making their own arrangements, the matching mechanism will fail. In this chapter, we will examine several centralized mechanisms that have been developed to create stable matches. In these markets, agents submit a rank-ordered list of their preferences to a centralized clearinghouse and are assigned a match. This system, therefore, induces a non-cooperative game, in which agents can decide to strategically misrepresent their preferences to get a better match. In addition to producing a stable matching, another desirable property of a matching mechanism is that it is strategy-proof, such that each agent has a weakly dominant strategy to reveal their true preferences.

Imagine that your thirteen-year-old cousin is applying to public high school in New York City. There are three schools she can choose from, labeled 1, 2, and 3, and she is competing with four other students. Schools 2 and 3 each have two open seats, while school 1 has only one seat. (In reality, nearly 100,000 students use a similar process to vie for slots at over 400 New York City schools each year, but we'll focus on a simplified scenario where we can define the preferences of all the players involved.) Each student's preferences over schools are listed in [figure 17.1](#), where your cousin is labeled student C (e.g., your cousin's top choice is 2, 1 is her second choice, and she's not wild about 3).

A	B	C(ousin)	D	E
1	3	2	2	2
3	2	1	1	3
2	1	3	3	1

Figure 17.1

Each student's school preferences.

The students are asked to submit ranked lists of schools, and they are told that an algorithm will match them into a school, taking into account their stated preferences and each school's priority list, which is based on the students' test scores, proximity, and other factors. The schools' priorities over students are listed in [figure 17.2](#) (e.g., at school 1, student E has first priority, while your cousin has second priority).

1 (1 seat)	2 (2 seats)	3 (2 seats)
E	D	C
C	B	A
D	A	D
B	E	E
A	C	B

Figure 17.2

Each school's priority list.

First, imagine that the school system announces that the goal is to get as many students as possible into their first-choice school, so they will use the following procedure (used by the Boston Public Schools and elsewhere for many years): They first send each student's application only to their top-choice school. The school considers the applications of these students and assigns them seats in order of priority until they hit capacity. That is, school 1 could admit at most one student and, if they get more than one application, they accept the one with the highest priority score; schools 2 and 3 can admit at most two applicants, and if they get more than two applications they accept the two they received with the highest priority scores. The others are rejected. Then the school system considers students who did not receive a seat in this

first round and sends their applications to their second choices. Each school with room considers the students who listed it second (and were not already matched) and assigns them seats in order of priority until they hit capacity. Any rejected students are given their third-place choice. How do you advise your cousin to rank the three schools?

Now imagine that the school system announces a change in policy. As before, they begin by sending each student's application to their top-choice school. Rather than accepting students right off the bat, the schools simply *hold on* to as many applications as they have room for—again based on priority score—and outright reject the excess applications. The difference is that now the schools don't accept anyone just yet; the best answer a student can get at this point is “maybe—we're going to hang on to your application for now and see what happens.” Rejected students then have their applications sent to their second-choice school. Each school considers all the new applications *and* those that they are still holding onto, so the “maybe” crowd has to compete anew with the next wave of applicants. They again hold onto as many applications as they have seats, keeping those with the highest priority and rejecting the rest. Rejected applications are sent to the next school on the list that hasn't yet rejected them. This process continues until each student has received a “maybe” (or exhausted all their ranked options), at which point the matches become final and the “maybes” turn to “yeses.” Now how would you advise your cousin to rank the three schools?

Matching Markets

In 1962, David Gale and Lloyd Shapley published a short paper in *American Mathematical Monthly* that may at first seem like a clever bit of recreational mathematics without a clear practical application. They imagined a group of men and women, each of whom wishes to be married to a member of the opposite sex. Some ground rules: Each person could be married to, at most, one other person, and if A is married to B, then B had better be married to A as well. Each person has preferences over their possible spouses and typically it will not be possible for everyone to wind up with their first choice. Given this, Gale and Shapley asked: is there a way to marry everyone off that is *stable*, in the sense that there is no man and woman who each prefer each other to their spouses? Put differently, if the set of marriages were not stable, then a wild soap operatic drama would ensue, with various people running off

with each other's husbands and wives. Gale and Shapley proved that for all possible preferences, there always exists a set of stable marriages, which could be found through their *deferred acceptance (DA)* algorithm—which follows a procedure identical to the second process for allocating school seats described previously. While Gale and Shapley's paper applied this method to the problem of college admissions and tentatively suggested that “some of the ideas introduced here might usefully be applied to certain phases of the admissions problem,” the researchers acknowledged that it entered into “a world of mathematical make-believe” and it is unlikely they could have envisioned how widely their results would be applied over the next half century (Gale and Shapley, 1962, p. 14). Today, versions of Gale and Shapley's algorithm are used around the globe to match employers with employees and to assign students to schools.

The decade before Gale and Shapley's paper appeared, the labor market for medical residents unraveled. Medical residencies are the first job held by American doctors upon completing medical school. By 1945, the competition among hospitals for residents had become so intense that they were making offers to students early in their medical school career, often before the students had time to decide on their specialty and before there was much evidence about whether the students would even be successful doctors. The medical schools attempted to diffuse the situation by withholding information about their students until a specific date. In turn, the hospitals—afraid of losing out on their second choice if their top candidates declined—responded by issuing offers that expired within a matter of hours.

To resolve the chaos, representatives of the hospitals, students, and schools agreed to develop a centralized clearinghouse, or matching mechanism, now known as the National Resident Matching Program. After interviews, doctors submit their preference ranking of hospitals to the clearinghouse. Likewise, the hospitals submit their preference ranking of doctors, along with information about how many positions they have available. These rankings are entered into an algorithm and, at a pre-specified date and time, all the participants learn their match.

The matching process that they arrived at looked remarkably similar to the mechanism proposed by Gale and Shapley, as noticed by Al Roth in 1984, although several tweaks were made over the years to account for features such as spouses wishing to be matched in the same city. The system had to be

further redesigned in the 1990s as doctors lost confidence in the match and began opting out. To redesign the system, they approached Roth, a game theorist and experimental economist who has studied matching markets theoretically and in the lab and won the Nobel Prize in Economics in 2012 for founding the field of *market design*.

There are two broad issues that drive all matching market design problems. First, if we knew everyone's true preferences, what is the best way to assign matches? Second, how do we incentivize agents to truthfully reveal these preferences? To answer these questions, we need to blend insights from both cooperative and non-cooperative game theory. The matching markets discussed in this chapter rely on a matching mechanism, which collects the reported preferences of the agents and assigns a match. Each possible matching mechanism creates a non-cooperative preference revelation game, where the players are all the agents who submit a preference list, their available strategies are the set of possible rankings of the alternatives, and the outcome is determined by the mechanism. We therefore need to rely on non-cooperative game theory, because if it is not in the strategic interest of the agents to submit a list corresponding to their true preferences, then they will attempt to game the system to achieve a better outcome for themselves. Furthermore, if agents could do better by matching or trading with each other instead of accepting their assigned match, then the market will likely unravel. We therefore need to rely on cooperative game theory to determine which outcomes are stable (or, in the terminology of the previous chapter, in the core). That is, the equilibrium outcome of the preference revelation game also needs to be in the core of the cooperative game in which coalitions of agents can form to make their own arrangements outside the designed market.

In this chapter, we'll first consider one-sided matching markets, in which only one side of the market has preferences over their potential matches. For example, college students have preferences over their dorm room assignment, but the dorm rooms do not have preferences over occupants. Insights from this literature have been used to develop centralized systems to match kidney donors with recipients that have been credited with saving thousands of lives. Then we'll consider two-sided matching markets, in which both sides of the market have preferences over each other, such as doctors and hospitals or college students and universities. Finally, we'll consider the case of school choice, which is one of the most commonly studied applications of matching

market design. School choice is somewhat in between the one- and two-sided situations. The students have preferences over schools, but the public school seats don't have preferences over the students per se. Instead, the schools each have a *priority list* that ranks prospective students, which may be based on exam scores, distance from their home, necessary access to certain programs, or having siblings who already attend the school. In this sense, the situation is similar to the two-sided matching problem, except that the schools are not strategic actors and market designers are concerned only with the welfare of the students.

One-Sided Matching Markets

To examine a one-sided matching market, let's consider a situation in which agents have strict preferences over some indivisible object, such as houses. To begin, imagine that each agent initially has ownership over a house. However, this initial allocation isn't necessarily efficient: there may be alternative allocations that make some agents strictly better off without hurting anyone else. Say there are n agents, and each agent i currently owns home i . The agents can trade houses among themselves, but there is no money in the market. How should we orchestrate such a swap?

This model is due to Shapley and Scarf (1974). Other applications include workers who are initially assigned specific shifts or tasks and can potentially make trades among themselves. Specifically, a *house matching* assigns one house to each agent. In addition to looking for an efficient allocation of houses, there are other properties we may want the matching to satisfy. First, we may require that the matching is *individually rational*: each agent gets a house that they like as much as the one they originally owned. If the matching were not individually rational, then some agents would elect to not participate in the market and instead stick with the house that they already own. Second, we may want the matching to be *core stable*. We saw in the previous chapter that an outcome is in the core if no agent or coalition of agents could feasibly strike their own arrangement that makes the agents in the coalition better off. In the context of the housing market, a matching is core stable if no agent or coalition of agents could opt out of the assignment, trade among themselves, and get a (weakly) better house than what they were assigned.

A *mechanism* is a procedure for determining a matching. That is, the mechanism takes the set of agent preferences and the initial endowment and

produces a matching. Of course, the mechanism cannot look into the hearts and minds of the agents and determine what these preferences are. Instead, it can only rely on the agents' *reported* preferences. The mechanism therefore induces a revelation game, in which each player's strategy is the preferences she reveals. We say that a mechanism is *strategy-proof* if truthful revelation is a weakly dominant strategy in this preference revelation game.

Top Trading Cycle (TTC) Algorithm

A process called the *top trading cycle (TTC) algorithm* generates an assignment of houses to agents that is both core stable and efficient. The algorithm works as follows:

1. Initially, no agents are assigned to a house.
2. All agents not assigned to a house "point" at the owner of their most preferred house that is still in the market.
3. Any agents pointing to themselves are assigned to their own house and removed from the market.
4. Look for *cycles*: a closed loop of agents who are pointing to each other to form a full circle. We assign each of these agents to the house they are pointing at and remove them from the market. Then we return to step 2.

This process continues until all agents have been assigned a house. Note that participation is individually rational: step 3 ensures that the agents can do no worse than receiving the home that they originally owned, and they would never be forced to take a home that they like less than their current one. Furthermore, the way that this process is constructed ensures that there is no coalition of agents who could do better than their assigned homes by teaming up and trading among themselves. If such a coalition of people existed, they would necessarily have formed a cycle earlier in the algorithm and been assigned each other's homes instead. Thus, the TTC produces a matching that is core stable.

In addition, there is no benefit to lying to the mechanism by misreporting one's true preference ranking, as this could only cause someone to receive a less preferred house (Roth 1982). Furthermore, this is the *only* mechanism that is efficient, individually rational, and strategy-proof.

Let's look at an example with six homes and six homeowners. The agents' preference rankings are listed in [figure 17.3](#), from their most-preferred to least-preferred house. To begin, we have each homeowner point at the owner

of their most preferred home: agents 1 and 2 point at agent 3, agent 3 points at agent 5, etc. This is depicted in the top panel of [figure 17.4](#). Note that no one points at themselves, but we can identify a three-person cycle: agent 1 points at agent 3, agent 3 points at agent 5, and agent 5 points at agent 1. We therefore have them trade and assign them all to their most preferred houses. There could never be a stable assignment that gives them different homes, since this coalition would block that assignment. Next, we have the remaining agents (2, 4, and 6) point at the owner of their most preferred house that still remains available. Agent 6 points at herself, so we assign her to her own house and there are no any additional cycles. In the third iteration, only agents 2 and 4 remain, and they each prefer each other's house to their own, and so we have them trade. Thus the algorithm generates the following assignment: Agent 1 receives house 3, agent 2 receives house 4, agent 3 receives house 5, agent 4 receives house 2, agent 5 receives house 1, and agent 6 receives house 6.

		Agents					
		1	2	3	4	5	6
Houses	3	3	3	5	6	1	5
	2	4	4	4	2	2	6
	4	2	3	1	3	3	1
	1	1	2	4	4	4	4
	5	5	5	1	3	5	3
	6	6	6	6	5	6	2

Figure 17.3

Each homeowner's preference ranking of houses.

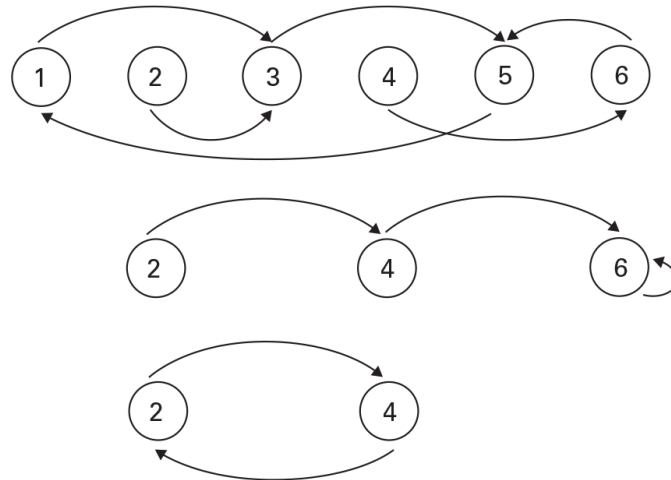


Figure 17.4

Finding the core allocation of houses using the TTC algorithm.

What if agents are not initially allocated to houses? In this case, it is less clear how to proceed. If there is an obvious hierarchy of agents, then we can use a mechanism called *simple serial dictatorship*: the agent with the highest priority chooses her preferred house, then the agent with the second-highest priority gets his preferred house, and so on. Another possibility is to simply assign houses to agents and then implement the TTC algorithm. Both of these mechanisms would yield efficient outcomes. In fact, if we take *any* efficient matching, there is both a hierarchy of agents that would produce that matching under simple serial dictatorship and an initial allocation that would produce that matching under the TTC algorithm.

House Allocation with Squatting Rights

Consider the following variation of the house allocation problem, which is both relevant to college housing and has far-reaching applications. As in the housing market described earlier, some students are already endowed with a room (from the previous term). However, others arriving on campus do not yet have a place to live, and some rooms have been vacated by students leaving campus. Furthermore, those who are already in their rooms often have squatting rights: they can always elect to stay put rather than entering the room draw and having to vie for a new spot. How should the rooms be allocated? The simplest method is *serial dictatorship*. A room draw list is generated randomly, while also taking into account seniority or other priorities. The housing office then moves down the list, in order, and assigns each person to their most preferred room that is still available. The problem

with this system, however, is that those who are already living on campus could end up with a worse room if they get a disadvantageous draw, and they may reason that it is safer to stay in their current space rather than enter the room draw and risk a downgrade. If those with squatting rights opt out of the match, the outcome may be inefficient, with coalitions of students each desiring each other's rooms.

An extension of the TTC algorithm, proposed by Abdulkadiroğlu and Sönmez (1999), ensures individual rationality (by guaranteeing that no one has to move to a worse space), while also taking into account that not all students are initially endowed with a spot. Under this system, the housing office moves down the priority list in order and, for each person, considers their most preferred room that has not yet been assigned. If that room is currently vacant, the person receives it. If it is currently occupied, then the *occupant* moves to the top of the priority list before proceeding any further, and the housing office first considers that person's most preferred unassigned room. For instance, maybe we have gotten to Aleks on the priority list, and she prefers a room that Ben currently lives in. At this point, there are three possibilities: Ben may prefer his own room to any other unclaimed space. In this case, Ben is assigned his own room, and we go back to Aleks and consider her next most preferred room. In this way, no students can ever wind up in a room they like less than their current spot. Alternatively, Ben may prefer a different vacant room, in which case we assign him there and Aleks is free to take his room. Finally, Ben may prefer a room that is occupied by someone else (say, Cham). In this case, Cham gets moved to the top of the priority list, and we consider her preferred choice. If at any point we identify a cycle (e.g., Aleks prefers Ben's room, Ben prefers Cham's room, Cham prefers Dylan's room, and Dylan prefers Aleks's room), then we assign each of them to their preferred choice and go to the next person on the priority list.

This system is called the *You Request My House—I Get Your Turn* algorithm. Notice that it generalizes the TTC. If all agents already occupy a room, it is identical to the TTC; if no one is endowed with a room, it is identical to serial dictatorship; and it can accommodate any hybrid while producing a core-stable outcome.

One of the most critical applications of the TTC algorithm and the You Request My House—I Get Your Turn variant has been kidney exchange. There are roughly 100, 000 people in the United States on the waiting list to

receive a kidney from a deceased donor, and many die each year before they are able to get the transplant they need. Receiving a kidney donation from a living friend or family member is also possible, and often a patient has someone who is willing to donate a kidney, but their donor is not necessarily a feasible match. However, it may be the case that, say, patient A's donor is a good match for patient B, while patient B's donor is a good match for A, and thus they can simply swap donors without having to join the waitlist. Or there may be larger coalitions of people who would each be made better off by engineering an exchange of kidneys between the donors and their beneficiaries. How can we orchestrate a swap of kidneys to benefit the most people and give them the best possible matches?

Roth, Sönmez, and Unver (2004) noted the similarities between the kidney exchange problem and the housing allocation problems we have discussed thus far. In response, they developed an algorithm known as the *Top Trading Cycles and Chains* algorithm, which is now applied to kidney exchange across New England and is credited with saving tens of thousands of lives. In this model, patients are analogous to students, while kidneys are analogous to dorm rooms. Patients who have a donor (even an incompatible one) are analogous to students who already occupy a room, while patients without a donor are analogous to newcomers without a room, and the cadaver kidney list is analogous to the unoccupied dorm rooms. Furthermore, a patient's chance of survival when receiving a kidney depends on blood type, size, age, and other factors, and so we can assume that patients have preference rankings over all possible kidneys that are based on their chances of survival with a given kidney. In addition to direct exchanges among a cycle of donors, Roth et al.'s algorithm allows for long chains of indirect exchanges, with the donor at the beginning of the chain giving a kidney to someone on the waitlist and the beneficiary at the end of the chain being placed at the top of the list. For example, we could have a three-person chain in which person A receives a kidney from B's donor, B receives a kidney from C's donor, C is moved to the top of the waitlist, and A's donor gives a kidney to someone on the waitlist. In practice, chains of up to 70 patients have been formed.

Two-Sided Matching

We've thus far looked at one-sided matching markets, where agents with strict preferences were matched with slots or objects that did not have their own

preferences for matches. In *two-sided matching markets*, agents on one side of the market are matched with agents on the other, and each have their own distinct preferences over their potential matches. Examples include labor markets, universities matching with students, and heterosexual marriage markets (since there must be two distinct sides of the market in this model). In some cases, the matching may be one-to-one (e.g., in a monogamous marriage market), while in other cases, multiple students or employees can be matched with a single university or firm. In all cases, however, the matching must be mutual (i.e., I cannot be your employee if you are not also my employer).

Let's begin by considering a one-to-one matching, by formally defining a *marriage problem* in definition 1 and the *matching function* that produces an outcome in definition 2.

Definition 1 (Marriage problem) *A marriage problem is a finite set of agents M , a finite set of agents W , a list of preferences for each agent in M over the set W (as well as the possibility of being unmatched), and a list of preferences for each agent in W over the set M (as well as the possibility of being unmatched).*

Definition 2 (Matching function) *The outcome of a marriage problem is a matching function, μ , that assigns each agent $m \in M$ to $\mu(m) = w \in W$ (they are matched to an agent in W) or $\mu(m) = m$ (they're matched with themselves, implying that they are unmatched) and assigns each agent $w \in W$ to $\mu(w) = m \in M$ (they are matched to an agent in M) or $\mu(w) = w$ (implying that they are unmatched), such $\mu(m) = w$ implies $\mu(w) = m$.*

Notice that there are two obvious ways in which a matching could fall apart: someone could decide she would rather be single than have to be with her current partner or a pair of individuals could decide they would rather ditch their matches and hook up with each other instead. Formally:

(1) A matching is *individually rational* if all agents prefer their assigned match to being unmatched. That is, for each i , $\mu(i) \succ_i i$ (where $\succ_i$ means "is preferred by agent i ").

(2) A matching is *blocked by a pair* if there is a pair of agents who each prefer the other to the assigned match. That is, for some pair (m, w) , $w \succ_m \mu(m)$ and $m \succ_w \mu(w)$.

A matching is *stable* if it is both individually rational and not blocked by any pair. The set of stable matchings is equivalent to the core—that is, the set of all matches for which there is no coalition of individuals who could do better by matching among themselves.

As a simple example, imagine four boys and four girls who are choosing their prom dates. Imagine it is the 1950s and they’re required to go with a member of the opposite sex. Each person’s preferences are shown in [figure 17.5](#). Note that Alia would rather stay home than go with anyone but Warren, while the others are willing to go with anyone.

Alia	Bette	Cheryl	Divya	Warren	Xander	Yi	Zev
W	W	W	Y	D	B	A	A
	X	X	X	A	D	D	B
	Y	Y	W	B	A	B	D
	Z	Z	Z	C	C	C	C

Figure 17.5

Each promgoer’s preferences over dates.

Consider the matching: Alia–Warren, Bette–Zev, Cheryl–Xander, Divya–Yi. Would everyone go along with this arrangement? Is there anyone who would rather stay home than go to prom with the assigned date, or any pair who would prefer to scandalously run off together instead of going with their match? Notice that Bette is matched with Zev, her least-favorite choice, and Xander is matched with Cheryl, his least-favorite choice. This matching isn’t stable because Bette and Xander would prefer to ditch their current dates and run off together. This assignment is not in the core.

Let’s consider this new matching, in which Bette and Xander are now together and the newly single Zev and Cheryl go to the prom together. Is this matching stable? Yes: everyone is dating someone he or she prefers to being unmatched and there is no pair that prefers each other to the assigned date. Zev and Cheryl are each other’s last choice; but, quite frankly, the two of them were everyone’s last choice, so they’re stuck with each other.

Deferred Acceptance Algorithm

Gale and Shapley’s *deferred acceptance (DA)* algorithm always produces a stable matching for both one-to-one and many-to-one matching problems. The

algorithm essentially acts out a process in which one side of the market (say, employers) makes offers to agents on the other side. These agents then assess the offers they received alongside the prospect of being unmatched, and the best among these options becomes their new provisional match. After these provisional matches have been made, employers who still have open slots make a new set of offers, and earlier provisional matches can be busted up if an agent with a provisional match prefers a new offer. The process continues until all employers fill all their slots or have exhausted their list of acceptable candidates. This is called the *deferred acceptance algorithm* because no offers are definitively accepted at the time they are made; the best answer a candidate can receive is a “maybe,” and final acceptances are deferred until the algorithm has run its course. But note that this is all happening behind the scenes. As far as the agents are concerned, they simply submit their preference ranking and then learn their final match, without witnessing any of the intermediate steps of the process.

DA algorithms are used not only for resident matching, but in various other labor markets, as well as for assigning students to public schools around the world and assigning advertisers to slots. Without a means of reliably finding an outcome in the core, the ultimate match wouldn’t be stable, and the system would unravel as members elect to negotiate their own agreements outside the matching program.

Let’s specifically examine the employer-proposing deferred acceptance algorithm (Gale and Shapley, 1962) for one-to-one matching problems, which works as follows:

1. Initially, all employers are active.
2. Each active employer makes an offer to its most preferred employee (who has not yet rejected them).
3. Each employee compares their current offers, being unmatched, and any provisional match. The highest-ranking outcome among these prospects becomes their new provisional match. The employee holds onto this offer and rejects all the others.
4. Employers whose current offers were rejected or whose previous provisional match rejected them now become active, and we return to step 2.

The process continues until all employers have filled their slots or have been turned down by all candidates they view as viable.

Consider the following example, in which employers A through E and job candidates 1 through 4 have the preferences listed in [figure 17.6](#). We'll assume that each employer has exactly one spot to fill. Note that some employers would rather leave their position unfilled than hire certain candidates, and that candidates 1 and 4 would rather be unemployed than work for employer E.

Employer:	A	B	C	D	E	Employee:	1	2	3	4
	2	3	2	4	1		C	B	E	A
	1	4	1	1	3		D	D	D	D
	3		3	2	4		A	A	A	C
	4				2		B	C	B	B
								E	C	

Figure 17.6

Employers' preferences over employees (left) and employees' preferences over employers (right).

To begin, all employers are active, with one open slot. Each makes an offer to their most preferred candidate: A makes an offer to 2, B to 3, C to 2, D to 4, and E to 1. Candidate 2 received two offers and prefers the one from employer A, so we provisionally match 2 with A. Candidates 3 and 4 received offers they view as better than being unemployed, so we provisionally match 3 with B and 4 with D. Candidate 1 received an offer from E, which is worse than being unemployed, so this offer is rejected. After this first round, C and E had their offers rejected and thus are active in the next round and make offers to the next candidates on their list: C to 1 and E to 3. 1 was previously unmatched and prefers C to being unemployed; 3 was provisionally matched with B but prefers the new offer from E, so this becomes his new provisional match. Thus, at the end of the second round, 1 is matched with C, 2 is matched with A, 3 is matched with E, and 4 is matched with D. Employer B is unmatched and in the next round makes an offer to the next candidate on its list, 4. Because 4 prefers the provisional match with D, she rejects this offer. The algorithm concludes here with each agent assigned their final provisional match. All firms have filled their slots except B, which exhausted its list of viable candidates and would rather the position be unfilled than hire 1 or 2.

In this version, the algorithm had the employers making proposals to the employees. However, we also could have run the algorithm with the

employees making proposals to the employers, and that might yield different results. The matching generated by the DA algorithm is always stable, but it isn't necessarily the *only* stable matching. Furthermore, some of these stable matchings may be more advantageous for the employers, while others are preferable for the employees. It has been shown that the DA algorithm in which the employers make the proposals is optimal for employers: each employer prefers its match as much as any match it could have been assigned under any other stable matching. The employer-proposing DA is thus also known as the *employer-optimal DA*. It also leads to the worst outcomes of any stable matching for the employees. Therefore, it is better to be a member of the side of the market that is making the offers in the DA algorithm. However, it doesn't change which people end up with *some match*: the set of agents who are unmatched is the same under all stable matchings.

Is the DA algorithm strategy-proof? Only partially: truth telling is a dominant strategy for agents on the side of the market that makes the proposals, but not those who receive the proposals. However, there isn't a better alternative: there is no mechanism that is both strategy-proof and core stable. Even if we were to relax the stability requirement, there is also no mechanism that is strategy-proof, efficient, and individually rational.

We can also extend the model and the DA algorithm to situations in which one employer can match with several candidates. In this case, the employer submits both its ranked list of acceptable candidates and how many people it has the capacity to hire. Then, in each stage, the algorithm has each employer make as many offers as it has open spots (i.e., its capacity minus its current number of provisional matches). Most of the theoretical properties of the one-to-one matching case extend to many-to-one matching, with a few exceptions. Notably, truth telling is no longer weakly dominant for the employer under the employer-optimal DA algorithm. Furthermore, in addition to manipulating the rank order of their preference lists, employers also may prefer to lie about their capacity.

School Choice

Alongside resident matching and kidney exchange, one of the most prominent and consequential applications of matching market design is the assignment of schoolchildren to public schools. In the school choice model, students are agents who have preferences over schools, but the public schools seats are

simply objects to be consumed and do not have preferences over students. In this sense, the situation resembles a one-sided matching problem. However, the students also have different *priorities* at each school. In some cases, the priority list may be identical across schools, such as when all students take a centralized exam and those with higher scores are given higher priority everywhere. In other cases, a school may grant higher priority to students who live within its district, have siblings at the school, or require access to special programs available at the school. In this case, the priority lists will differ across schools.

One trick that researchers have used is to treat the school choice problem as a two-sided matching problem, but with the school's priority list over candidates standing in for the school's preferences. However, there are a couple of ways in which the school choice problem diverges from the two-sided matching markets discussed in the previous section. First, since only the students actually have preferences over their match, we only consider their outcomes when determining whether a matching is efficient and disregard the schools' perspective. Second, because the schools do not have preferences, our definition of stability does not apply. Instead of considering whether a matching is blocked by some pair, the analogous condition in the school choice problem is called *elimination of justified envy*. A matching eliminates justified envy if, for each student i , there is no other student who is matched with a school that i prefers to her own match, yet had a worse priority at that school.

Putting this together, we'll say that a school choice matching is *stable* if:

- (1) It is individually rational: No student i prefers being unmatched to $\mu(i)$.
- (2) It is nonwasteful: No student prefers a school with an empty seat to $\mu(i)$.
- (3) It eliminates justified envy: Whenever i prefers j 's match to his own (i.e., whenever $\mu(j) \succ_i \mu(i)$), i has a lower priority at $\mu(j)$ than j does.

School Placement When All Schools Have the Same Priorities

When there is a single priority list that determines the students' rankings at all schools, then the matching problem is very straightforward. This is the situation, for instance, in the assignment of students to public high schools in Turkey on the basis of a centralized exam. With such an unambiguous priority list, the elimination of justified envy property is not just a nice condition to satisfy, but becomes critical for the perceived fairness of the mechanism: if a

mechanism placed students with lower exam scores into more preferred schools, then there would be an uproar. In this case, the obvious mechanism to use is serial dictatorship: the student with the highest priority (e.g., exam score) is matched with her most preferred school, then the student with the second-highest priority is matched with his most preferred school at which there is still an available seat, and so on. The simple serial dictatorship is the only mechanism that eliminates justified envy and is efficient.

Extending the Deferred Acceptance and Top Trading Cycle Algorithms to School Choice

The school choice matching problem becomes more complex when different schools have different priority lists. Simple serial dictatorship cannot be used and we need to consider other mechanisms, which differ in stability, efficiency, and strategy-proofness.

The so-called *Boston mechanism* was used to match public school students in Boston from 1999 to 2005. This is the first mechanism described in the example at the start of the chapter. Under the Boston mechanism, students submit their rankings of schools and each school submits its priority list and capacity. First, the mechanism considers the first-ranked choices of all students. It assigns all students to their first choice, in order of priority, until the schools reach capacity. Then, the mechanism moves on to the second choices of the students who are still unmatched and assigns as many unmatched students as possible to their second-ranked choice, and so on. While this system was created to match as many students as possible to their first-choice schools, the downside with this mechanism is clear: if a student does not rank a school first, then she essentially gives up her priority at that school to anyone who does rank it first. Consider a student who strongly prefers a popular school outside his district where he doesn't have a high priority and his second choice is a local school where he has high priority. It would be very risky to truthfully report his first choice. In the likely event that he doesn't get matched in the first round, then his second choice may have already been filled by students with lower priority there, whereas simply listing his local school as his first choice would essentially guarantee a spot. There is thus an obvious incentive to game the system by misrepresenting preferences, and this was not lost on the parents of Boston. Tips on how to strategically manipulate choice lists circulated at local parents' meetings, and even the Boston Public Schools guide advised: "For a better chance of your

‘first choice’ school ... consider choosing less popular schools. Ask Family Resource Center staff for information on ‘underchosen’ schools” (Abdulkadiroğlu and Sönmez, 2013, p. 31).

In response to these issues, Abdulkadiroğlu and Sönmez (2003) proposed two alternative strategy-proof mechanisms and consulted with the public school systems in Boston and New York to redesign their matching systems. The two mechanisms are extensions of the two strategy-proof mechanisms that we have already seen: the DA and the TTC algorithms.

The DA algorithm for school choice capitalizes on the analogy between two-sided matching markets and the school choice problem where agents have preferences over objects and objects have priorities over agents. We already know from the previous section that the student-proposing (or student-optimal) version is strategy-proof for students and that students prefer this matching to any other stable matching (i.e., any other matching that eliminates justified envy.) Of course, we could also run a school-optimal version of the DA algorithm, but since the goal is to maximize student welfare and incentivize truthful reporting by students, this would not make a lot of sense. The student-optimal DA starts the same as the Boston mechanism, by assigning students to their first choice up to the schools’ capacities. However, these matches are only provisional rather than final. Students who fail to get their first choice placement then move on to their second choice, and even if that school’s slots are already full, they can displace a student with a lower priority there.

The outcome of the student-optimal DA is preferred by students to any other match that eliminates justified envy. However, it is not necessarily efficient. If the primary concern is maximizing student welfare and priorities do not need to be strictly enforced (i.e., if we are willing to allow justified envy), then we may be able to find a more efficient match. To see this, consider the following simple example (originally due to Roth, 1982) with three schools (each with one seat) and three students. The students’ preferences and the schools’ priority lists are seen in [figure 17.7](#).

Student:	A	B	C	School:	1	2	3
	2	1	1		A	B	B
	1	2	2		C	A	A
	3	3	3		B	C	C

Figure 17.7

Students' preferences over schools (left) and schools' priority lists over students (right).

In this example, the only stable matching places student A at school 1, student B at school 2, and student C at school 3. Thus, this is the outcome of the student-optimal DA, which we can see by working through it. In the first step, A makes an offer to 2, B to 1, and C to 1. Since C has a higher priority than B at school 1, after this first step, A is provisionally matched with 2 and C with 1. In the second step, the unmatched B makes an offer to 2. Since B has a higher priority at school 2 than A does, B provisionally takes this spot, so we have B provisionally matched with 2 and C matched with 1. In step 3, the unmatched A makes an offer to 1. A has top priority at school 1, so he takes C's spot there, and we now have A matched with 1, B matched with 2, and C unmatched. C makes an offer to 2, which is not accepted. Finally, C makes an offer to school 3, which is accepted, and we arrive at the unique stable matching.

Although this matching is stable, it is not efficient when we only consider the preferences of the students. Consider the alternative matching in which student A is placed at school 2, B is placed at 1, and C is still placed at 3. Both students A and B are strictly better off with this matching, while student C is indifferent. The problem with this matching, and the reason that DA didn't implement it, is that it is not stable, because student C has justified envy of student B. Student B gets to attend school 1, which C likes better than her own match of 3, even though C had a higher priority at 1. (While stability implies efficiency in two-sided matching problems—where we take into account both sides' preferences—this is not the case in the school choice problem where we only take into account one side's welfare.)

If we're willing to accept justified envy to improve student welfare, how would we go about finding an efficient matching? In this example, it was possible for both students A and B to do better by essentially swapping their priorities at different schools. As we saw earlier in the chapter, the TTC algorithm determines how to exhaust all efficient swaps and thus can be used

to ascertain how students could efficiently swap priorities. Abdulkadiroğlu and Sönmez (2003) developed the following extension of the TTC to apply to public school choice:

Step 1. Each school is given a counter that keeps track of how many available seats it currently has. Initially, each school's counter is equal to its capacity. Each student points to her most preferred school (according to her stated preferences). Each school points to the student with the highest priority. There must exist at least one cycle, and each student and school can be part of at most one cycle. For each cycle, we remove the students, assign them to the schools they were pointing at, and reduce the schools' capacity counter by 1 for each student it was just assigned. If a school's capacity counter hits zero, we also remove it from the market.

Step k. Each remaining student points at her most preferred remaining school. Each remaining school points to the remaining student with the highest priority. There must exist at least one cycle, and each student and school can be part of at most one cycle. For each cycle, we remove the students, assign them to the schools they were pointing at, and reduce the schools' capacity counter by 1 for each student it was just assigned. If a school's capacity counter hits zero, we also remove it from the market.

Let's apply this algorithm to the previous example. In step 1, A points at 2, B and C point at 1, 1 points at A, and 2 and 3 point at B. This is shown in [figure 17.8](#). This produces a cycle with A pointing at 2, 2 pointing at B, B pointing at 1, and 1 pointing at A. We thus assign A to 2 and B to 1 and remove them from the market. The capacity at schools 1 and 2 are reduced to zero and removed as well. Since only C and 3 are left in the market, they also are matched, yielding the efficient outcome.

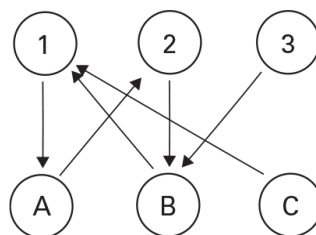


Figure 17.8

Finding the efficient matching using the TTC algorithm.

This procedure is strategy-proof and yields a match that is efficient, but, as we’ve seen, could produce justified envy. The choice between these two strategy-proof mechanisms, therefore, depends on how the market designers value the trade-off between the elimination of justified envy and efficiency. The student-optimal DA was implemented by New York City high schools in 2003 and the Boston Public Schools abandoned the Boston mechanism in 2005 in favor of student-optimal DA.

Example

Let’s now return to the example that we started with at the beginning of the chapter, where five students were vying for spots at three schools, and consider how these slots would be allocated under each of the mechanisms we’ve discussed. (With a slight modification, this example is based on an experiment by Pais and Pintér, 2008). School 1 has one available seat, while schools 2 and 3 each have two seats. The preferences of the students and the schools’ priority lists are shown in [figure 17.9](#).

						Schools’ Priority List:		
						1	2	3
						(1 seat)	(2 seats)	(2 seats)
Student:	A	B	C	D	E			
	1	3	2	2	2	E	D	C
	3	2	1	1	3	C	B	A
	2	1	3	3	1	D	A	D
						B	E	E
						A	C	B

Figure 17.9

Students’ preferences over schools (left) and schools’ priority lists over students (right).

Under the Boston mechanism, we would first consider each student’s first-choice school. A and B would each get their first choices (1 and 3, respectively). Students C, D, and E all want to attend 2, which has only two slots. Since D and E have higher priority, they are awarded the seats. Student C (your cousin in the motivating example) next prefers 1, but its slot has already been assigned, and so she is stuck with her last choice. Thus, it would have been reasonable to advise her to manipulate her preference list and place 1 at the top, since she would have received priority over A in the first round. Our final matching is as follows: A attends 1, B and C attend school 3, and D and E attend school 2.

Under the student-optimal DA, the first round is identical to the Boston mechanism, but now your cousin has the opportunity to bust up the tentative match made between school 1 and student A. She is the only student unmatched after the first round, so she makes a proposal to her second choice of 1. Since she has a higher priority than A there, she becomes 1's new tentative match, and A is now unmatched. He makes an offer to his second choice of 3, which still has a spot. Now all students are matched, and our final matching is as follows: A and B attend school 3, C attends 1, and D and E attend 2.

Finally, let's look at the TTC algorithm. The steps are shown in [figure 17.10](#). In the first round, each school points at the student with the highest priority and each student points to their first choice. Since school 2 and student D point at each other, they are matched. Student D is removed from the market, and school 2's capacity drops from 2 to 1. In the next round, we have a cycle in which B points at 3, 3 points at C, C points at 2, and 2 points at B. We thus assign B to attend school 3 and C to attend school 2 and remove them from the market. School 3's capacity drops to 1, and school 2 no longer has capacity. In the next round, only students A and E remain, and we have a cycle in which A points at 1, which points at E, who points at 3, which points at A. We thus assign A to 1 and E to 3. Our final matching is as follows: A attends school 1, B and E attend school 3, and C and D attend school 2. Notice that that E is justifiably envious of C's position at school 2, where E had higher priority.

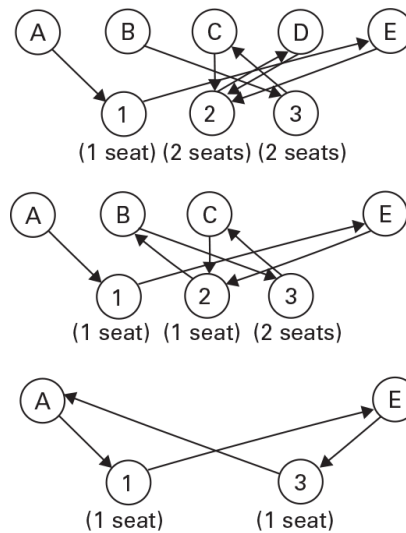


Figure 17.10

Finding the efficient matching using the TTC algorithm.

Behavior in Games

Lying to Strategy-Proof Mechanisms

One of the first experimental tests of school choice matching was conducted by Chen and Sönmez (2006). In each market, they recruited 36 subjects to play the part of students who were vying for 36 spots at 7 schools. The schools' priority lists were fixed in advance. Preferences were induced by explaining to subjects how their compensation would vary depending on which school they could match with, and these payments differed across subjects. They used two different sets of preferences and tested the Boston mechanism, the student-optimal DA, and the TTC algorithm. As predicted, they found far more misrepresentation under the Boston mechanism. While 70.8% of participants wound up with their *reported* top choice under this mechanism, only 28.5% received their *actual* top choice. Typically, this manipulation took the form of elevating one's district school, where they had a high priority rating. While less rampant than under the Boston mechanism, there was also still substantial misreporting under the strategy-proof DA and TTC, ranging between about a quarter and half of all reports across treatments. Pais and Pintér (2008) further found that manipulation increases in all three mechanisms when participants have more information about the schools' priorities. There is also evidence of misreporting under the DA algorithm outside the lab. Although it is less straightforward to determine agents' true preferences when they are not induced by experimenters,

researchers have gotten around this by noting that some college admissions clearinghouses required applicants to separately rank the same program with and without a fellowship. While the program with the funded fellowship was presumably preferred by all applicants, between 19% and 35% ranked it lower.

To the extent that participants play multiple rounds and have time to learn, truth telling rates typically increase overtime. Of course, most people don't participate in the same matching market repeatedly. Instead, families who are ranking schools typically learn by receiving advice from the school system authorities or from friends and neighbors. Most studies find that receiving correct advice typically increases truth telling, but people also seem more likely to follow incorrect advice (such as the suggestion to target more realistic schools), indicating the importance of authorities clearly communicating the strategy-proof nature of the mechanism to counteract misinformation.

When players do truthfully report in TTC and DA, is this because they recognize that it is a weakly dominant strategy, or is it because truth telling simply seems like an obvious default? To test this, Guillen and Veszteg (2021) also conducted reversed versions of each mechanism, in which participants' preferences are considered backward, such that players have a dominant strategy to report their worst outcome as the best, and so on. They find that substantially fewer people play the dominant strategy (of submitting their reverse-order preferences) under these backward mechanisms and people commonly submit their true preferences even in these cases, suggesting that truthful reporting is used as a default.

Among strategy-proof mechanisms, some are *obviously strategy-proof*, while others require the agents to engage in contingent reasoning (as advanced by Li 2017). For example, the second-price auction is strategy-proof—each person has a weakly dominant strategy of bidding their value—but it is not obviously so. Instead, it requires agents to think about the different possible contingencies of how the competing bids compare to one's own value, which we first broke down in chapter 3. In contrast, an auction where bidders drop out as the price increases until only one is left is strategically equivalent, but the dominant strategy is obvious: if the price is less than your value, then stay in, but once it hits your value, you should get out.

To test this, Li (2017) analyzed the failure of contingent reasoning in truthful reporting under the random serial dictatorship mechanism. In his study, there were four subjects in each market and four known monetary prizes to be allocated among them. Each player was assigned a priority score, which was a randomly drawn integer between 1 and 10 (with each number equally likely). The participants learned their own priority score (but not their ranking) and then simultaneously submitted their preference ranking over the monetary prizes. The player with the highest score got the first choice on her list (with ties broken randomly), then the player with the second-highest score got the highest-ranked remaining choice on his list, and so on. This mechanism is strategy-proof: players have a dominant strategy to submit a preference list that ranks the prizes by their monetary value. Li also considered a sequential version of the game, in which the players were called on in order of priority and could select from the remaining options when it was their turn. This mechanism is *obviously strategy-proof*, which implies that even a cognitively limited agent can recognize their dominant strategy. While strategically identical, the sequential version doesn't require the participant to engage in contingent reasoning to realize that she should truthfully report her preference regardless of which turn she is given. In the simultaneous game, 29% of the lists contained errors—typically reversing the best and second-best prizes—and over a third of games did not end with the dominant strategy outcome. This proportion dropped to 7% in the sequential version.

Evidence-Based Adaptations of the Standard Model

Why do people seemingly attempt to game the system at their own expense? As we saw in results from Li (2017), difficulty with contingent reasoning may play a role. Another explanation is that agents intentionally underrank more preferred outcomes to lower their own expectations and avoid being disappointed with the outcome. In this section, we consider how *expectations-based loss aversion* (Kőszegi and Rabin 2007, 2009) can incentivize agents to demote or omit attractive options in their preference list to minimize later disappointment.

Behavioral economics has long recognized that individuals often evaluate outcomes in terms of gains or losses from some reference point (Kahneman and Tversky 1979). Plausible reference points include the status quo (e.g., evaluating a salary offer relative to your current compensation) or expectations (e.g., evaluating a salary offer relative to what you believed you

would be offered). Furthermore, many people are *loss averse*, meaning that as we move away from one's reference point, losses hit harder than equal-sized gains. The coefficient of loss aversion, λ , indicates how losses are weighted relative to gains, with $\lambda > 1$ indicating loss aversion.

Expectations-based loss aversion was applied to the school choice problem as a potential explanation for preference misrepresentation independently by Dreyfuss, Heffetz, and Rabin (2019) and Meisner and Wangenheim (2019). Simply put, ranking an option highly sets this outcome as a reference point against which the final outcome will be compared. If an agent is highly loss averse, then she may prefer to rank a favored option lower—or to leave it off the list entirely—to avoid getting her hopes dashed. Under Kőszegi and Rabin (2009)'s model, agents gain utility from direct consumption, as well as from news that causes them to update their beliefs about current or future consumption. Applying this model to school choice, let's consider an agent who experiences three separate time periods. In the first period, she learns about the matching market procedure and submits her rank ordering. The list she submits causes her to rationally form expectations about her ultimate outcome and thus sets her reference point. In the second period, she learns the matching outcome. Relative to her expectations, this could be good news, in which case she gains positive utility in this period equal to the difference between the outcome and her prior expectation. Or it could be bad news, in which case she experiences a loss of λ times the difference between the outcome and her prior expectation. The more loss averse she is (the higher λ), the more bad news resonates relative to good news. In the third period, the agent gains the consumption utility associated with her match. The agent's only choice in all of this is what list to submit in period 1. Demoting or omitting a preferred option from her list in period 1 affects her utility in a few ways. It reduces the good news in the first period and the expected outcome in the third period, but it also reduces the potential for bad news in the second period. As a result, loss-averse agents with low chances of good outcomes would choose to rank these outcomes lower to begin with.

This pattern is consistent with Li (2017)'s data. Participants with the lowest priority score of 1 were three times more likely to misrepresent their preferences as those with the highest score of 10. Those with priority scores of 1 and 2 truthfully reported less than 60% of the time, and their next-most-common ranking listed the worst outcome as their top choice. To further

understand the intuition here, let's take an agent with loss aversion parameter λ and the lowest priority score of 1, and consider how she would evaluate her choice under several simplifying assumptions.

First, notice that the likelihood that an agent with a score of 1 would get first or second priority is exceedingly unlikely. Getting first priority would require all three other participants to also draw scores of 1 and then the agent to win the random tiebreaker. Thus, let's just assume that the only real possibilities are getting to choose third (with probability p) or fourth (with probability $1 - p$). Second, let's assume that the other agents are ranking their preferences truthfully. While we know this isn't the case, making this assumption allows us to ascertain whether truthful ranking is a dominant strategy even for loss-averse agents. These first two assumptions imply that the agent can only ever get the third- or fourth-best option. Finally, let's assume that the third-best option provides consumption utility x and the fourth-best one provides consumption utility 0. By ranking the fourth option best (or at least above the third best), the agent guarantees herself a utility of 0: her consumption is 0, she doesn't initially get any good news, and she guarantees that she never hears any bad news. In contrast, by honestly ranking the third best higher than the fourth best, the agent is choosing a lottery in which she has a p chance of getting x and thus gets the following utility according to the expectation-based loss aversion model:

$$\overbrace{xp}^{\text{period 1 news}} + \underbrace{p(x - xp)}_{\text{period 2 good news}} + \underbrace{(1 - p)(-xp)\lambda}_{\text{period 2 bad news}} + \overbrace{xp}^{\text{period 3 expected outcome}}$$

The first term indicates the good news that the agent gets in the first period: by ranking the third option better than the fourth, she has a p chance of getting x , and thus her expectation goes from 0 to the expected value of xp . The last term is the agent's expected consumption in the third period and thus also reflects the expected payoff from this lottery. The second and third terms reflect her second period expected utility from getting news about the outcome. There is a p chance that she gets good news: she learns that she got third priority, and thus her expectation goes from xp to x . This is reflected in the second term. Finally, there is a $1 - p$ chance that she gets bad news: she learns that she got fourth priority and her expectation drops from xp to 0, a loss that is magnified by the loss aversion parameter $\lambda \geq 1$. Combining terms, we get that this agent prefers to elevate her fourth choice and earn 0 when

$$2xp - xp(1 - p)(\lambda - 1) < 0$$

or

$$p < \frac{\lambda - 3}{\lambda - 1}$$

Consistent with the observation that more loss-averse agents will distort their preferences when they have low priority, this inequality is more likely to be satisfied when p is low and/or λ is high.

Exercises

(1) Consider the following house allocation problem with six homeowners and six houses shown in [figure 17.11](#). Each owner i currently owns house i . The owners' preferences over the houses are as follows:

1	2	3	4	5	6
1	4	4	6	3	2
2	2	1	5	1	5
3	3	3	4	2	3
4	1	5	3	5	4
5	6	6	2	4	6
6	5	2	1	6	1

Figure 17.11

Find the core allocation of houses to owners.

(2) Now let's consider a housing problem where only some people are initially endowed with a home. Specifically, assume that there are six students, labeled A through F, with preferences shown in [figure 17.12](#). The agents have priority in alphabetical order—that is, the housing office's room draw priority list is A, B, C, D, E, F. Agent B has squatting rights over house 2 and Agent E has squatting rights over house 5. The other agents and houses are unassigned. Use the You Request My House—I Get Your Turn algorithm to find the core stable matching.

A	B	C	D	E	F
1	4	5	5	1	2
2	2	1	2	2	5
3	3	3	3	3	3
4	1	4	6	5	4
5	6	6	4	4	6
6	5	2	1	6	1

Figure 17.12

(3) Consider the marriage market, shown in figure 17.13, described by Gale and Shapley (1962). There are three men, labeled α , β , and γ , and three women, labeled A, B, and C. Their preferences for agents on the other side of the market are

α	β	γ	A	B	C
A	B	C	β	γ	α
B	C	A	γ	α	β
C	A	B	α	β	γ

Figure 17.13

- Notice that there are six possible sets of marriages. Which of these are stable?
 - Use the men-proposing DA algorithm to find the optimal matching for the men. Verify that none of the other stable marriages you found in part (a) are weakly preferred by all the men.
 - Use the women-proposing DA algorithm to find the optimal matching for the women. Verify that none of the other stable marriages you found in part (a) are weakly preferred by all women.
- (4) Consider a two-sided matching market with five employers (A through E) and four job candidates (1 through 4). The preference rankings for the employers are shown in figure 17.14.

A	B	C	D	E
1	4	4	1	1
2	2	3	4	2
3	3	1	3	4
4	1	2	2	

Figure 17.14

(Note that employer E would rather leave the job unfilled than employ candidate 3.)

The job candidates' preferences over firms are given in figure 17.15.

1	2	3	4
B	C	E	A
C	A	D	D
A	B	A	E
D	D	B	B
E	E	C	C

Figure 17.15

Use the Gale-Shapley DA algorithm to find the stable matching that is optimal for the firms. (Remember that in each stage, each unmatched firm will make an offer to its most preferred candidate who hasn't yet turned them down. This can include candidates who are already provisionally matched. Then the candidates will evaluate their offers and their current match and decide which they like best. That will determine the new provisional match. This continues until all firms are matched or the unmatched firms have been rejected by all acceptable candidates.)

(5) Consider a school choice problem with six students (labeled A through F) and three schools (labeled 1 through 3). Each school has two available seats and the students' preferences are shown in figure 17.16:

A	B	C	D	E	F
1	2	1	1	2	1
2	3	3	2	1	2
3	1	2	3	3	3

Figure 17.16

Let's consider how various mechanisms will allocate the seats.

- (a) First, let's assume that all schools have the same priority list, which is reverse alphabetical order: F, E, D, C, B, A. What matching would be assigned under serial dictatorship?

For the remainder of the problem, we will assume that the schools have the priority lists shown in [figure 17.17](#).

1	2	3
A	C	E
B	D	F
C	E	A
E	A	B
D	B	C
F	F	D

Figure 17.17

- (b) What matching would be assigned under the Boston mechanism?
- (c) What matching would be assigned under the student-proposing DA mechanism?
- (d) What matching would be assigned under the TTC algorithm for school choice?

VI

Social Dilemmas

18

Social Dilemmas

A disproportionate share of behavioral and experimental research has concerned social dilemmas. Simply put, a social dilemma is a game in which rationally self-interested players end up at a suboptimal outcome, where the players as a group fail to take full advantage of the available surplus. With this chapter, we begin a new section of the book that develops this idea. In this chapter, we more formally define a social dilemma, discuss the history of the idea, provide several two-player examples, and summarize a few themes from the expansive behavioral literature on the topic.

It has been posited that one of the lesser appreciated but more important contributions of the development of markets was their civilizing effect on society. Before markets appeared, individuals produced but were hesitant to trade because the rules of exchange were unclear or not followed. As a result, welfare suffered. With the concentration of traders in a market, implicit rules of conduct arose through repeated interaction and institutions evolved that allowed producers to trade with confidence. As Montesquieu ([1748] 1961, p. 81) put it, “commerce ... polishes and softens barbaric ways as we can see every day.” To a great extent, the economy prior to the development of modern market institutions was seen as a *social dilemma* (Dawes, 1980). Although individuals had the knowhow to fabricate goods that others demanded, they feared that their output would be stolen by thieves or confiscated by the lord, so they produced little. If instead they could only trust other traders, exchange would make them all better off.

In other words, formal markets were thought to foster trust among traders. While this makes sense to some, does it ring true in the data? The latest wave (i.e., the seventh) of the World Values Survey continues to ask whether respondents across the world think that “most people can be trusted” or that they “need to be very careful” (World Values Survey n.d.). Indeed, there is considerable variation: 67% of Dutch respondents are willing to trust others, while only 3% of people in the Philippines are. More to the point, however,

the average frequency of trust in countries that are members of the Organisation for Economic Co-operation and Development (OECD) is 34%, but it is only 21% for nonmembers. Furthermore, the correlation between trusting and believing that “wealth can grow so there’s enough for everyone” versus “people can only get rich at the expense of others” is significant overall, but also much stronger among the OECD nations. In other words, it does appear that trust and the belief that the economy is a social dilemma that can be solved evolve more quickly among nations with strong market institutions, democracy, and the rule of law.

The Trust Game

If trust is so important for the economy, can we model its importance in a game? Let’s first consider a simple binary choice setting with two players. Say that both players are endowed with \$10 and the first-mover can invest all or none of it into a joint venture. If no investment is made, the game is over and players simply retain their endowments. Alternatively, an investment by player 1 is productive in the sense that it generates a surplus three times greater than the investment, which can be shared by the two players (i.e., \$10 is turned into \$30). It is the second-mover, however, who decides how the surplus will be divided as, for example, a boss would with the proceeds from the sale of the output produced by a worker. This second-mover can choose to either share the proceeds of the investment or keep it all for herself. [Figure 18.1](#) presents the extensive and normal forms of this interaction.

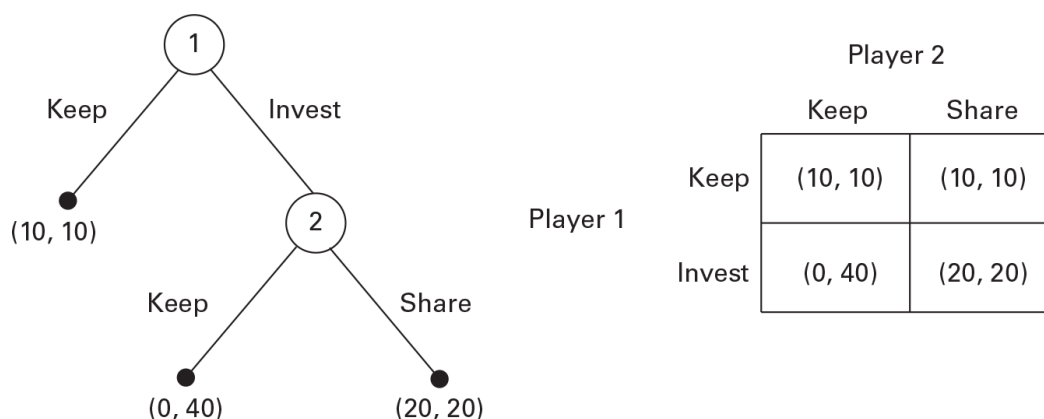


Figure 18.1
A binary trust game.

Considering the extensive form on the left of [figure 18.1](#), it should be clear that player 2 has an incentive to keep any surplus created by player 1's investment, and because of this incentive, player 1 has no reason to invest. That is, the subgame perfect Nash equilibrium (SPNE) prediction is for player 1 to keep her endowment and for player 2 to keep any surplus. You can also see, by switching to the normal form on the right of [figure 18.1](#), that this is the only equilibrium of the binary trust game.

Does this basic prediction change if we let the first-mover invest just a little instead of everything? In other words, what if we consider the continuous strategy version of this game, the one developed originally in Berg, Dickhaut, and McCabe (1995)? Here, player 1 can invest any amount, I , between \$0 and \$10 and player 2 can share any amount between \$0 and $3I$. If player 1 invests just a small amount, $3I$ won't be large, but what is key is that player 2 still has an incentive to keep it all, regardless of the magnitude. Hence, player 1, independent of the strategy space, should anticipate that player 2 will not be trustworthy. This belief implies that player 1 should continue to keep everything. The SPNE prediction remains the same.

From an efficiency point of view, what is important is that the players walk away with just \$20 collectively when they could have had \$40. As alluded to in the previous section, without some mechanism to force player 2 to at least repay player 1's investment, player 1 shouldn't trust or invest, and in equilibrium, the two players do not take full advantage of the surplus.

Social Dilemmas

Because there are so many important choices that can be characterized with similar incentives (e.g., public goods, team production, investment in human capital, producer cartels, common pool resources, voter turnout, and overpopulation), social dilemmas have become the purview of researchers in many fields, including political science, psychology, and sociology, in addition to economics. Given the prominence that these incentives play in various fields, it is important to provide as specific a definition as possible (see definition 1).

Definition 1 (Social dilemma) *A social dilemma is a situation in which individual incentives are at odds with group incentives. Individual rationality leads members of a group to an outcome that is suboptimal. More formally,*

the equilibrium of a social dilemma game is Pareto dominated by some other outcome.

In the binary trust game described earlier, the outcome in which the first-mover invests and the second-mover shares Pareto dominates the equilibrium outcome in which the players just end up with their endowments. In this trust-and-share outcome, both players earn \$20, but they only get \$10 in equilibrium. This, therefore, is a social dilemma. Notice, however, that not all alternative outcomes have to be better for the game to be a social dilemma. For instance, the outcome in which player 1 trusts and player 2 keeps does not Pareto dominate the equilibrium outcome because player 1 will be worse off there. The same rationale is true of the continuous-strategy trust game and more generally—whenever a sizable surplus is generated by interacting with another player, the resulting game has the potential to be a social dilemma if players can individually do better by acting selfishly. Returning to the examples listed before, because individuals have the incentive to free ride on the public goods provided by others, shirk when their teammates work, not invest in firm-specific human capital because their boss will “hold them up” (i.e., not compensate for the investment), produce more when the firm they are supposed to collude with shows restraint, extract more than their share of the carrying capacity in the common pool, not bother to vote, or even have too many kids, all these situations qualify as social dilemmas.

Not only does the volume of research indicate that social scientists have been interested in understanding social dilemmas for quite a while, this interest may extend back further than you might think. Indeed, it was Thomas Hobbes (introduced in chapter 2) ([1668] 1994) who modeled the “state of nature” as a social dilemma in the 17th century. In this case, his model is more consistent with a prisoner’s dilemma game, which, in some sense is a particularly tragic social dilemma because not only is the equilibrium outcome suboptimal, it is supported as a dominant strategy. In addition to recognizing the incentive problems embedded in social dilemmas, the social philosophers of the time were also interested in proposing solutions—mechanisms that allow players to take more advantage of the potential surplus. Hobbes, for example, in the 1651 edition of *Leviathan*, makes the case for a strong government to intervene, establish the rule of law, punish knaves, and enforce contracts. Surely, if the two players in the trust game can write a contract that can be enforced by a third party, like a scary leviathan,

then the dilemma is solved. The first-mover can invest everything knowing that the second-mover will be forced to share the surplus.

Other social philosophers followed Hobbes's contractarian impulses with various models of social dilemmas and alternative proposed solutions. In 1738, David Hume wrote in his *Treatise on Human Nature* about two neighbors trying to drain a meadow held in common and how it was in the interest of each to free ride on the efforts of the other, leading the work not to be done. Likewise, Jean-Jacques Rousseau's *Discourse on Inequality*, written in 1754, describes an alternative in the narrative description of the stag hunt (introduced in chapter 5). For Rousseau, a social dilemma game may have multiple equilibria and the dilemma is ending up at the Pareto-inferior one. Here, two hunters can work together to bring down a stag, resulting in a high payoff for each, or each can work alone to catch rabbits. Like Hobbes's prisoner's dilemma, being selfish and working alone result in an equilibrium with low payoffs but working as a group can also be supported as an equilibrium. As Rousseau put it, if each hunter can be persuaded to stay faithful to his post, then they can jointly achieve the Pareto-dominant equilibrium.

Finally, it is perhaps Immanuel Kant's *Groundwork of the Metaphysics of Morals* (1785) that most directly links to the modern notion of justice and a means of escape from the paradox of the social dilemma. Here, Kant understands the incentives of the social dilemma and offers advice that seems close to the maxim of the "golden rule." To paraphrase, Kant's *categorical imperative* requires us to act toward others in ways that we would want everyone else to act toward others. Put differently, he is proposing general laws for society to get out of the Pareto-inferior outcomes characterizing social dilemmas. If you would not like to be free ridden on (or to witness someone else being free ridden on), then don't do it to others. These ideas and those of the other social philosophers (including John Locke) formed the basis for John Rawls's (1971) more recent approach of using the *veil of ignorance* to determine a reasonable general rule to get from the *original position* (implicitly fraught like Hobbes's state of nature with incentive problems) to a society better able to take advantage of the full surplus. For Rawls, if you force people to formulate general rules without knowing which position they will occupy in the world (i.e., behind the veil of ignorance), they are likely to play it safe and make sure that if they do not end up in a privileged position,

they will still be taken care of. The use of such a *maximin* rule can be seen as a modern incarnation of the categorical imperative.

Continuous Social Dilemmas

So far, it may seem relatively straightforward to identify a game as a social dilemma. In addition to finding any Nash equilibria, we just have to check to see if there is an outcome that Pareto dominates at least one of the equilibria. Indeed, this is not hard in normal-form games or games with dominant strategies, but Pareto improvements can be difficult to identify in more complicated games.

Consider the following two-player game with continuous strategies and symmetric payoffs as an example. Player i chooses $x_i \in [0, 1]$ and earns a payoff of $u_i = x_i + x_j - 2x_i^2 + \frac{1}{12}x_j^2 + \frac{2}{3}x_ix_j$. Just by looking at this payoff, can you tell whether this is a social dilemma? Probably no, but look more closely at some of the elements of the function. The first and the third terms form a quadratic in one's own x , which indicates that, to some extent, early increases in x benefit the player a lot, but eventually the benefits diminish. Notice also that the other player's x always enters positively to your payoff, suggesting that she is generating a positive externality for you (like a public good). Finally, because the interacted last term is positive, your x and the other player's x are *strategic complements*—your action benefits you more when the other player is also choosing a higher x . In sum, greater x values seem to benefit you both, but you each also seem to have an incentive to free ride on the other player because the quadratic means that the marginal benefit of increasing your own x will fall. In other words, this game does seem like a social dilemma.

How do we more formally check to see if this game forms a social dilemma? We can always start by calculating the equilibrium. To do this, we need to find the best responses of the two players, in this case where $\frac{\partial u_i}{\partial x_i} = 0$. For the i th player, this derivative is $1 - 4x_i + \frac{2}{3}x_j$, which when set to zero at the maximum yields i 's best response function, $x_i = \frac{1}{4} + \frac{1}{6}x_j$. The intersection of i 's best response function and j 's, $x_j = \frac{1}{4} + \frac{1}{6}x_i$, leads us to the equilibrium: $x_i = x_j = \frac{3}{10}$. Plugging these x values back into the payoff function indicates that in equilibrium, the players earn a payoff of 0.4875. Can the players jointly do better if they choose something that is not the equilibrium? What if they increase their x values just a little (say, by 0.05)? At 0.35, the payoffs for both

rise to 0.5469, so indeed, this game seems to be a social dilemma; however, the 0.05 increment seems pretty arbitrary, doesn't it?

Instead of examining a series of small, arbitrary deviations in either direction, is there a way to get a sense of exactly how bad the dilemma is? That is, can we calculate some optimal outcome for the two players, a strategy profile that, in some sense, maximizes the joint payoffs for the players and see how this compares to what the players receive in equilibrium? The short answer is yes, but we need to be even more specific—what does maximizing the players' joint payoffs mean? Shall we maximize i 's payoff, j 's payoff, the product of their payoffs, or some other payoff weighting? In most cases, economists have decided to borrow a standard aggregation mechanism from the utilitarian social philosophers (e.g., Jeremy Bentham), calculating the profile of x values that maximizes the sum of the player payoffs. This notion is summarized in definition 2.

Definition 2 (Social optimum) *The social optimum of a game is the strategy profile that maximizes the total utility achieved by a group of players.*

To calculate the social optimum for the game described here, we act as a benevolent dictator and pick both x_i and x_j to maximize the sum of the player payoffs. In this case, $u_i + u_j = 2x_i + 2x_j - \frac{23}{12}x_i^2 - \frac{23}{12}x_j^2 + \frac{4}{3}x_i x_j$. Taking both derivatives, we find that $\frac{\partial(u_i+u_j)}{\partial x_i} = 2 - \frac{23}{6}x_i + \frac{4}{3}x_j$ and $\frac{\partial(u_i+u_j)}{\partial x_j} = 2 - \frac{23}{6}x_j + \frac{4}{3}x_i$. At the social optimum, both of these derivatives will equal zero, and with some algebra, we see that this occurs where $x_i = x_j = 0.80$. Hence, because the Nash equilibrium values of x are not equal to the socially optimal values of x , the game must be a social dilemma. As further evidence, if we plug the socially optimal strategy profile into the payoff function, we see that at the social optimum, each player achieves a payoff of 0.80, a substantial increase over what they achieve in equilibrium. You can see this graphically in the left panel of [figure 18.2](#).

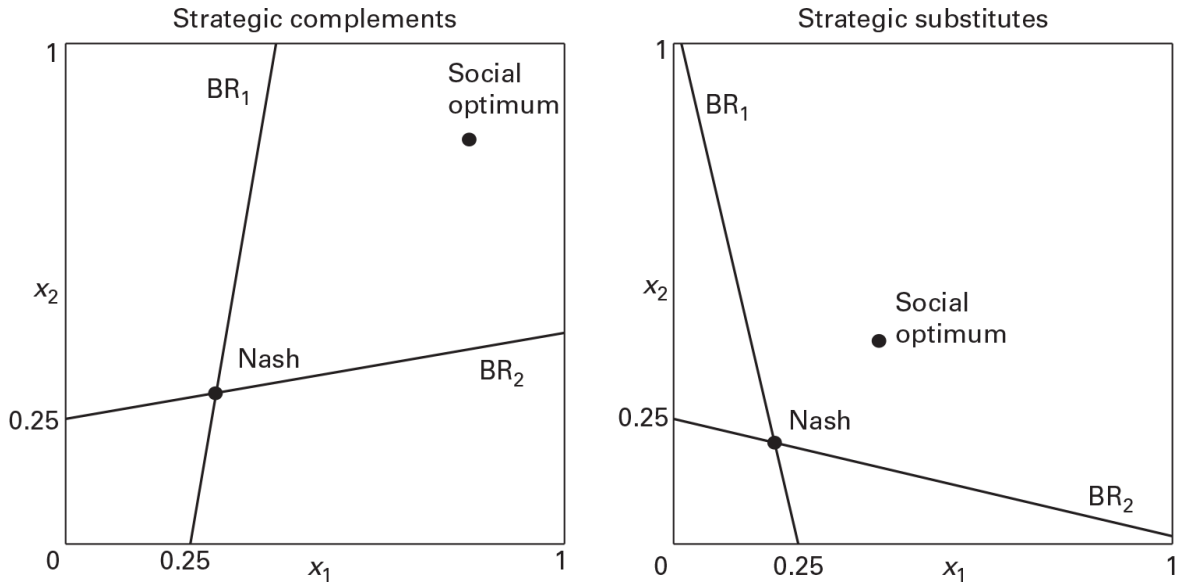


Figure 18.2

Equilibria versus social optima in two-player social dilemmas.

As another example, what if player strategies are *strategic substitutes* instead? To examine this situation and conduct our own experiment, let's maintain the positive externality and make the actions substitutable. This is easily done by flipping the sign of the last interacted term in the payoff function. We now have $u_i = x_i + x_j - 2x_i^2 + \frac{1}{12}x_j^2 - \frac{2}{3}x_ix_j$, and you can see that changing this one thing will reverse the sign of the slope of the best response functions for the two players. For player i , the derivative of the payoff function is $1 - 4x_i - \frac{2}{3}x_j$, which once set to zero can be rearranged to $x_i = \frac{1}{4} - \frac{1}{6}x_j$. Player j 's response function will look similar, and when the functions are solved simultaneously, we get the new Nash equilibrium of $x_i = x_j = 0.21$, with the associated payoff of 0.3061.

Using the same method as before, we first sum the payoffs for the players, getting $u_i + u_j = 2x_i + 2x_j - \frac{23}{12}x_i^2 - \frac{23}{12}x_j^2 - \frac{4}{3}x_ix_j$, and then let the benevolent dictator maximize this total payoff by choosing both x values. The derivative with respect to x_i is $2 - \frac{23}{6}x_i - \frac{4}{3}x_j$; it is $2 - \frac{23}{6}x_j - \frac{4}{3}x_i$ with respect to x_j , and after setting them both equal to zero at the optimum and using some algebra, we find $x_i = x_j = 0.39$ at the social optimum. Just to check to see that we have done it correctly, you can also verify that the payoff for each player at the social optimum is 0.3871, so they do, indeed, do better at the social optimum.

Strategic substitutes are depicted on the right side of [figure 18.2](#). Comparing the two graphs, it is interesting to see the implications of the switch from complements to substitutes. While the levels of x at both the equilibrium and social optimum are lower in the case of strategic substitutes, which makes sense because the other player's effort diminishes the benefit of your effort rather than amplify it, the spread between the socially optimal and the equilibrium x values is larger in the case of the strategic complements. Again, this makes sense because there is more room for cooperation when the players' choices positively complement each other.

We end this section by summarizing what we have learned about identifying social dilemmas. Remember that in social dilemmas, there will be a difference between the strategies chosen as part of a Nash equilibrium and the strategies that will lead to a Pareto-dominant outcome (except in coordination games like stag hunt). Hence, if you calculate an equilibrium and the resulting outcome is not Pareto dominant, then you have a social dilemma. This process is simple in normal-form games where there is dominance or not many strategies (like the trust game in [figure 18.1](#)). That said, the process is harder in games with continuous strategies and no dominance. In these instances, one can still identify the equilibria of the game but will have to work a bit harder to find the optimum benchmark against which to compare the equilibrium. The standard benchmark used in economics (and most of the social sciences) is the social optimum as defined in definition 2—the strategy profile that maximizes the sum of the individual payoffs. One thing to note, however, is that the concepts of Pareto dominance and social optimum are not identical unless we restrict attention to the symmetric outcomes of games, as we have done before. To be more specific, in games with asymmetries, the Pareto notion will only identify situations in which some players do better and none do any worse, while the utilitarian maximization of the total payoff could trade off the gains of some players against the losses of others.

Behavior in Games—Trust, Reciprocity, and Efficiency

In their original trust game experiment, Berg et al. (1995) had 32 pairs of participants from the University of Minnesota play the game. Contrary to the SPNE prediction, the majority of the first-movers (14 of 16) sent some money to a second-mover. In fact, five of them sent all of their \$10, and the average amount sent was \$5.16, far from the \$0 prediction. As for the second-movers,

they also defied the SPNE prediction by returning substantial amounts of money, \$4.66, on average.

So, to what extent is the trust game a social dilemma from an empirical or behavioral point of view? On the one hand, it clearly was not a dilemma for three of the groups in which the first-mover sent everything and the second-mover sent back enough to more than compensate the first-mover's investment. On the other hand, not everyone sent it all, and in two cases, first-movers who sent everything got screwed, getting back only \$1 or nothing. Overall, if everyone sent \$10, a total experimental surplus of \$1,280 (32×40) would be generated, but only \$815 was actually created in the experiment to share, just 64% of the total possible. In other words, there is considerable heterogeneity—some pairs can figure it out, while others are indeed trapped by the individual incentives. This variety of responses to the trust game is best appreciated visually on the left of [figure 18.3](#). Along the x -axis, you can see that some people sent a lot, some sent very little, but almost every amount was tried. In addition, there was a lot of heterogeneity in how much second-movers returned. On the right of [figure 18.3](#), you can also see that the original Berg et al. (1995) experiment was not a fluke because the same overall pattern of behavior was observed by Burks, Carpenter, and Verhoogen (2003) with 22 pairs of Middlebury College participants.

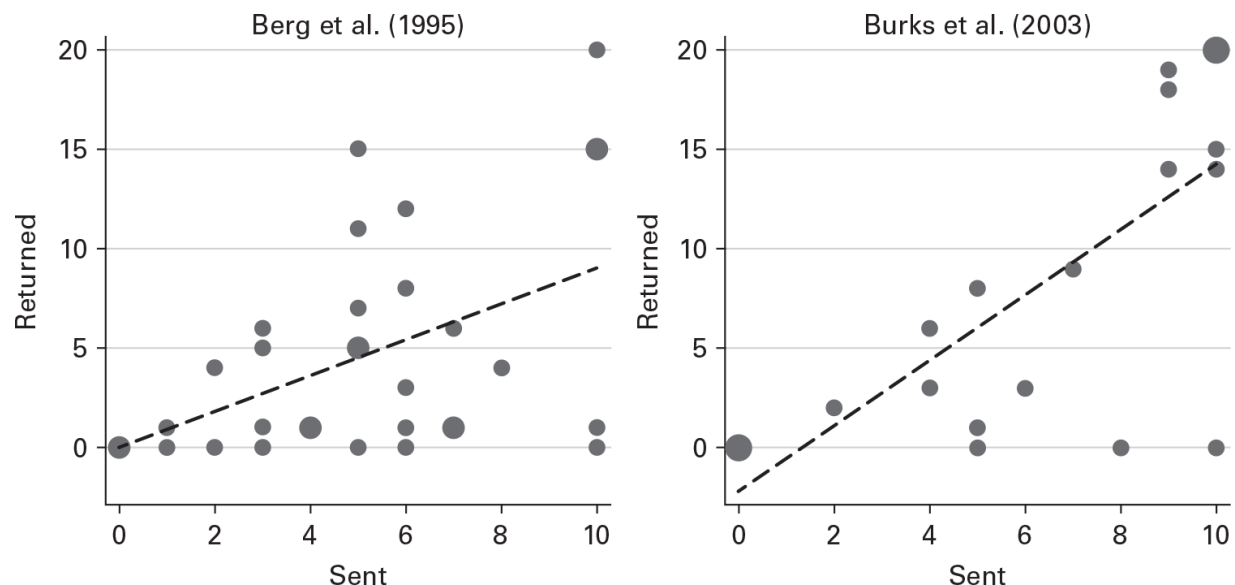


Figure 18.3

Trust game behavior (circle size indicates the number of identical observations).

For those pairs who figure it out by generating a substantial surplus to share, what is likely to be going on? One hypothesis has to do with trust and reciprocity. We already know from the beginning of this chapter that it is no surprise that many people are willing to trust in this game, but we haven't spoken much about why they might be willing to do so. Given anthropologists think of *reciprocity*—responding to another's action with an equivalent action—as a human universal, observed in almost every culture to one extent or another, it makes sense that first-movers, expecting even an anonymous second-mover to feel obliged to act reciprocally, would be willing to trust and send money.

Is there any evidence of this in the trust game data? One way to capture the relationship between trust and reciprocity in the data is to test to see if there is any connection between how much is sent and how much is returned. If the data in [figure 18.3](#) is just scattered randomly and there is no link between how much is sent and how much is returned (beyond the mechanical link that the surplus is three times the amount sent), then reciprocity is unlikely to be important. To the contrary, as one can see from the regression lines superimposed on the scatter plots in [figure 18.3](#), there is a strong positive relationship between how much first-movers send and the amount that second-movers return. In Berg et al. (1995), the estimate (i.e., the slope of the line) is that for each \$1 sent, 90 cents are returned ($p = 0.02$). Notice that the slope is positive, in support of the reciprocity hypothesis, but it is not large enough for the average first-mover to make money. The average first-mover actually loses 10% on her investment. The same estimate from the Burks et al. (2003) data is substantially larger, however; here, the average investor earns \$1.64 for each dollar invested ($p < 0.01$). Put differently, reciprocity seems to be present in both samples, but it is particularly strong among the participants from the relatively small, liberal arts community compared to the more anonymous large university setting.

We can also report more direct evidence of the importance of trust and reciprocity in solving two-player dilemmas. As a follow up to the original Berg et al. experiment, McCabe et al. (2003) compare a binary-move trust game like the one in [figure 18.1](#) to a game in which first-movers were forced to invest, and they had no choice but to send their entire endowment. The basic idea is that when the first-mover explicitly chooses to invest instead of being forced to, she signals to the second-mover the intention to form a

reciprocal-trust relationship. If this signal is interpreted correctly, theory suggests that reciprocity should be enhanced. In fact, when first-movers were forced to trust, second-movers shared the resulting surplus only a third of the time, but when the first-movers could credibly signal trust by explicitly sending their endowment, the second-movers responded reciprocally almost two-thirds of the time.

At the same time, one could also argue that the basic trust game paradigm is confounded by other potential reasons to send money. For example, what if players are just inequality averse or altruistic? As first-movers, players might send money to second-movers because they are motivated by altruism, not trust, and second-movers could send money back to first-movers just to reduce the inequality that the tripling creates. To subtract the social preference reasons for sending and returning money, Cox (2004) used a triadic design in which roughly a third of pairs play the standard trust game (game A); a third play game B, in which the second-mover can't send anything back; and the final third play game C, wherein the first-mover cannot send any money. Here, the rationale is that behavior in game A can be driven by both social preferences and trust/reciprocity, but the behavior of first-movers in game B can only be driven by altruism because the second-movers cannot reciprocate. Further, the choices of the second-movers in game C can only be motivated by inequality aversion because the first-movers could not trust. Comparing mean behavior in the three games, Cox found that first-movers send an average of \$5.97 in game A but only \$3.63 in game B, which means that trust accounts for 39% of first-mover behavior. Further, comparing second-mover behavior, in game A second-movers send back \$4.94, on average, while in game C they return only an average of \$2.06. Hence, reciprocity appears to account for 58% of second-mover behavior. While the trust game may be more complicated than first thought, overall it appears that the evidence is consistent with many players attempting to move toward a reciprocal-trust relationship when faced with a social dilemma.

We can develop the interaction of social/distributional preferences and reciprocity by borrowing the model presented and tested in Charness and Rabin (2002), which is also described in chapter 14. In this model, the authors develop a simple utility function to account for inequality aversion, altruism, and reciprocity. Suppose that x_i is the monetary payment received in a game

by player i and x_j is the payment to player j . Then the utility of player i is a weighted average of the player payments:

$$u_i(x_i, x_j) = [1 - (\rho r) - (\sigma s) - (\theta q)]x_i + [(\rho r) + (\sigma s) + (\theta q)]x_j$$

where ρ indicates how i cares about j 's payment when i is earning more, so that positive values reflect *superiority aversion* (not liking to have more than the other player), σ indicates how i cares about j 's payment when i is earning less, so that negative values reflect *inferiority aversion* (not liking to have less than the other player), and θ is a factor that indicates the strength of reciprocity. More specifically, when $x_i > x_j$, then $r = 1$ (it is 0 otherwise), and if ρ is positive then player i gets less satisfaction from her own payoff being large and more from the other player's payoff being large—she feels bad for having more. Likewise, when $x_i < x_j$, then $s = 1$ (it too is 0 otherwise), and player i may place lower (or even negative) weight on the other's payoff because she is doing worse than the other player. Finally, we assume that q is zero unless the other player misbehaves. In this case, if player j makes an unfriendly choice, then $q = -1$ and the weight player i puts on j 's payment shifts down to be less positive or more negative. Essentially, the parameters shift the weight the player assigns to the other player's outcome, depending on who is doing better or worse or who has behaved poorly.

Consider a few parameter configurations. When $\sigma \leq \rho < 0$, player i is competitive; she always likes to do better than player j , and her utility decreases as player j 's payment increases, regardless of whether j earns more or less than she does. By contrast, player i is inequality averse if $\sigma < 0 < \rho < 1$; her utility decreases both when she has more and less than player j . Considering reciprocity, if we assume that $\theta > 0$, then when $q = -1$, player i doesn't want j to do as well because j misbehaved.

We can use the Charness-Rabin utility specification to transform the payoffs of the basic binary trust game to envision the interplay between social preferences and reciprocity to see if the new predictions map nicely onto the empirical studies that were mentioned before. In [figure 18.4](#), we illustrate the transformed game along with the parameter values used to evaluate each set of payoffs. If player 1 decides not to play and just keeps her endowment, the material payoffs will be equal (\$10 for each player), so neither player will be superiority or inferiority averse (i.e., $r = s = 0$). In addition, there is no reason to interpret player 2's action as misbehaving because she didn't even get to make a choice. Hence, $q = 0$ for player 1. Because all the indicators (r, s, q)

are 0 for player 1, her utility is just equal to her own payment from this outcome (10). At the same time, player 2 could easily interpret player 1's choice as not trusting her, so $q = -1$ for player 2. Even with this feature, though, player 2's Charness-Rabin payoffs are just a weighted average of 10 and 10, so regardless of θ_2 , player 2's utility is likewise 10.

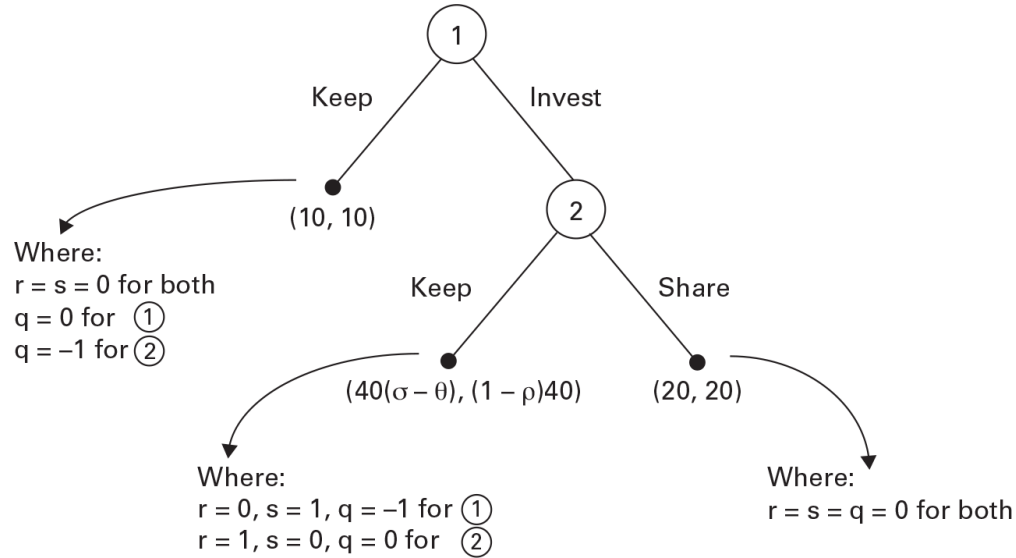


Figure 18.4

A transformed trust game.

If instead player 1 invests and player 2 shares, the utilities are easy to compute because both players did the right thing and again there is no inequality. This means that $r = s = q = 0$ for both players, and therefore $u_i = u_j = 20$. Finally, let's look at the utilities if player 2 keeps the surplus. For player 1, both inferiority aversion and reciprocity will be triggered, while only superiority aversion can reduce player 2's payoff. Player 2's payoff is given by $u_2 = 40(1 - \rho) + 0(\rho) = 40(1 - \rho)$, indicating that he will feel bad if he is superiority averse ($\rho > 0$). Player 1 will be even less happy with this outcome if she is inferiority averse (i.e., $\sigma < 0$) and if she is particularly reciprocal (i.e., θ is large). Her payoff is given by $u_1 = 0(1 - \rho + \theta) + 40(\sigma - \theta) = 40(\sigma - \theta)$.

Does any of this affect the SPNE prediction of the trust game? If she gets to make a choice, player 2 will choose to share the surplus instead of keeping it all if $20 > (1 - \rho)40$ or as long as $\rho > \frac{1}{2}$. Clearly, this precludes competitive players because ρ must be positive, but this condition could easily be satisfied by player 2s who value equity. Indeed, if player 2 is sufficiently superiority

averse, then player 1 will prefer to invest and earn \$20 rather than keep it and get \$10. As a result, although the reciprocity term in this model is mostly used to moderate or amplify one's distributional concerns, now there is a SPNE in which the social dilemma is solved—namely, player 1 invests and player 2 shares.

Exercises

(1) *Dilemmas?* Consider the normal-form games shown in [figure 18.5](#). Are any of these social dilemmas (why or why not)?

		Player 2	
		<i>A</i>	<i>B</i>
Player 1	<i>A</i>	3, 3	1, 4
	<i>B</i>	4, 1	2, 2

Game 1

		Player 2	
		<i>C</i>	<i>D</i>
Player 1	<i>A</i>	1, 0	3, 1
	<i>B</i>	3, 4	1, 5

Game 2

		Player 2	
		1	2
Player 1	<i>A</i>	1, 1	1, 0
	<i>B</i>	2, 0	0, 1

Game 3

		Player 2		
		<i>D</i>	<i>E</i>	<i>F</i>
Player 1	<i>A</i>	3, 15	9, 13	8, 9
	<i>B</i>	7, 12	7, 7	10, 12
	<i>C</i>	8, 3	6, 7	10, 5

Game 4

		Player 2		
		<i>D</i>	<i>E</i>	<i>F</i>
Player 1	<i>A</i>	3, 3	0, 4	1, 5
	<i>B</i>	4, 0	1, 1	1, 0
	<i>C</i>	5, 1	0, 1	2, 2

Game 5

Figure 18.5

(2) *Dilemma?* Consider the normal-form game shown in figure 18.6.

		Player 2	
		A	B
Player 1	A	1, 1	8, -1
	B	-1, 8	0, 0

Figure 18.6

- (a) If the game is played once, is it a social dilemma?
 (b) If it is played twice, is it a social dilemma?
- (3) *Give some, Take some.* Suppose that there are two players who each have exactly one coin to start with and face the decision to *contribute* the coin, do *nothing*, or *request* a coin. Any player who does not contribute keeps her coin and might be able to add to it because after the two players choose pure strategies, a benevolent dictator counts the contributions and requests and allocates additional rewards in the following way. If the net contributions are positive (i.e., the sum of the contributions is strictly larger than the sum of the requests), then each player also receives β times the net contribution. If the net contribution is less than or equal to zero, the players get no additional rewards.
- (a) Write the normal form of this game.
 (b) For what values of β is this a social dilemma?
- (4) *Continuous strategies.* Two players draw from a common fund where the benefit from what is withdrawn depends on how much has already been taken out. Specifically, where player 1 draws d_1 and player 2 draws d_2 , the benefit per unit withdrawn is $100 - d_1 - d_2$. Hence, the payoff to player 1, for example, is $\pi_1 = (100 - d_1 - d_2)d_1$.
- (a) Is this game an example of strategic complements or strategic substitutes?
 (b) What are the equilibrium draws for the two players?
 (c) Graph the best response functions for the players.
 (d) Is this a social dilemma?

(5) *Bosses and workers*. Consider the following employment situation. The boss offers his worker a fixed wage, w , and a piece rate, p , per unit of output produced, y . The worker then decides how much to produce. Producing output is costly for the worker, however. Suppose that the cost of production for the worker is $\frac{1}{2}y^2$. Combining all this, the worker's payoff is $\pi_W = w + py - \frac{1}{2}y^2$. The boss makes money selling the output produced by the worker. Let's say that the boss can sell each unit for v , but he has to pay p , so his net revenue is $(v - p)y$. Subtracting the fixed wage that he must also pay yields the boss's payoff as $\pi_B = (v - p)y - w$.

(a) Remember that this game is sequential (i.e., the boss can anticipate how the worker will react) and find the worker's subgame perfect equilibrium output level and the boss's equilibrium piece rate.

(b) Is this kind of employment contract a social dilemma?

(6) *Social preferences*. If either (or both) of two players *cooperates*, a surplus is generated of value V . If instead both players *shirk*, they each get a payoff of 1. If one player cooperates and the other shirks, the shirker gets the entire surplus and the cooperator gets nothing.

(a) Write the normal form of this game.

(b) What condition assures that this game is a social dilemma?

(c) Supposing that the two players have Charness and Rabin (2002) social/reciprocal preferences (as described at the end of the chapter), write the transformed normal-form game.

(d) For what parameter values is the social dilemma solved (i.e., which values make cooperating dominant)?

19

Public Goods

The provision of a public good is a classic social dilemma. Because individuals have an incentive to free ride on the contributions of others, they contribute too little and the public good is underprovided. Social scientists have spent a lot of resources trying to solve this particular dilemma and, interestingly, the success of the resulting mechanisms in the lab has not always been predicted by theory.

Americans gave \$427 billion to charity in 2018 (Giving USA, 2019). Although this includes donations from foundations, corporations, and bequests, the majority, \$292 billion, came from roughly half of individual taxpayers. This level of generosity may or may not surprise you, but the economics of philanthropy is a bit of a puzzle. For example, when you give to charity, you generate a positive externality for everyone, a *public good*, but the benefit to you is likely to be small, and is surely less than your donation. Hence, you give a dollar but get less than a dollar back in benefits—who would do that? Well, maybe an *altruist*, a person who values the benefits to others in addition to the benefits to themselves. After all, if there are a lot of beneficiaries, each receiving the small benefit you created, the sum of these benefits might soon outweigh the dollar you sacrificed.

At the same time, there are often other sources of funding for charities. If you learned, for instance, that your favorite charity secured a government grant would you still give? You might because this news could just reinforce your affinity for the charity, but an altruist might actually think twice. If the altruist wants to maintain some level of benefit for everyone (i.e., provide the public good), when the government kicks in a grant, the altruist no longer needs to supply those funds. In fact, government grants should crowd out altruistic donations, one for one! This is starting to seem complicated. Let's write a model of public good provision to examine the logic more carefully.

The Logic of Public Goods

Suppose that there are n potential donors in a community, indexed as $i = 1, \dots, n$. Each donor has wealth, $w_i > 0$, and contemplates making a contribution to the public good, c_i . Considering the benefits, contributions are aggregated over all n donors (giving us $\sum c_j$) and passed through the individual's public good production function, G_i . It seems reasonable to assume that this benefit function is weakly concave (i.e., that the second derivative of G_i is less than or equal to zero), such that the next dollar contributed by anyone carries a positive benefit, but that benefit decreases with scale. The resulting quasi-linear utility function for the i th community member is just

$$u_i = (w_i - c_i) + G_i(\sum c_j)$$

where the first term in parentheses is the donor's residual wealth after making a donation and the second term is the public good benefit that the donor receives from all the donations that are gathered.

We can take the derivative $\frac{\partial u_i}{\partial c_i}$ and set it equal to zero to figure out what the i th donor would do to maximize her utility. The derivative of the first bit of utility in the parentheses is just -1 , and the derivative of the public good function with respect to the choice variable, c_i , can be written as $\frac{dG_i}{dc_i}$. Hence, the optimizing donor contributes up to the point where $-1 + \frac{\partial G_i}{\partial c_i} = 0$ or

$$\frac{\partial G_i}{\partial c_i} = 1$$

Intuitively, the left side of the equation is the marginal benefit (MB) to donor i of contributing a dollar and the right side is the marginal (or opportunity) cost, the dollar that she could have kept for some other purpose.

Do rational donors contribute enough? As we know from chapter 18, we can test to see if contributing to a public good is a social dilemma by asking what a benevolent dictator, who seeks to maximize the sum of the donor utilities, would do. Summing across the n potential donors, we have

$$\sum u_i = \sum w_i - \sum c_i + \sum G_i(\sum c_j)$$

Notice that the sum of payoffs is simply a function of the total contributions. Letting C be the total contributions (i.e., $C = \sum c_j$), we can rewrite the sum of the payoffs as

$$\sum u_i = \sum w_i - C + \sum G_i(C)$$

In this case, the benevolent dictator chooses the total contributions needed from everyone to maximize total utility by calculating $\frac{\partial \sum u_i}{\partial C}$. Considering the three parts of the summed utility function in turn, C is absent from the first bit, so the derivative is just 0; and the derivative of the second part is $\frac{\partial C}{\partial C}$, which is just 1; but the derivative of the third part is a bit trickier. Notice that if we expand the third part, we will get

$$\sum G_i(C) = G_1(C) + G_2(C) + G_3(C) + \dots + G_n(C)$$

which means that the derivative with respect to C is the sum of a bunch of partial derivatives:

$$\frac{\partial \sum G_i(C)}{\partial C} = \frac{\partial G_1}{\partial C} + \frac{\partial G_2}{\partial C} + \frac{\partial G_3}{\partial C} + \dots + \frac{\partial G_n}{\partial C}$$

or, more simply, $\sum \frac{\partial G_i}{\partial C}$. Hence, at the social optimum, the first-order condition is

$$\sum \frac{\partial G_i}{\partial C} = 1$$

instead. Whereas the individual donor equated their own marginal costs and benefits of making a contribution, the social planner needs to equate the same marginal cost, \$1, to the sum of the marginal benefits, a calculus that will inevitably require larger donations because of the positive externalities each donation creates.

The rationale for why contributions will be too low at the Nash equilibrium is easier to see in [figure 19.1](#). As mentioned earlier, if we assume that the marginal benefit of another contribution is diminishing (i.e., downward sloping), the sum of the marginal benefits will become steeper when we add more potential donors. As a result, the optimal contribution, which is at the intersection of the marginal benefit and the marginal cost, in this case just \$1 for the forgone dollar donated, will increase. In other words, contributing to charity (aka the provision of a public good) is a social dilemma.

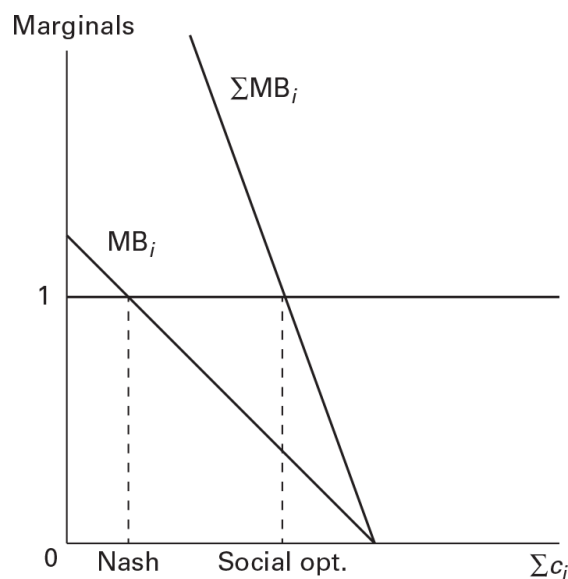


Figure 19.1

Public goods as a social dilemma.

Much of the experimental literature studying the provision of a public good utilizes a simple, linear version of the game just described. It is known as the *voluntary contribution mechanism (VCM)*, a concept we first introduced in chapter 9. Analyzing this game will both help you understand the experimental studies we will discuss later in the chapter and reinforce the basic public good incentives. The VCM is linear in the sense that the public good benefit function, G_i , is just a common benefit per dollar contributed (say α), multiplied by the total amount donated. Hence, the payoff for each potential donor becomes

$$u_i = (w_i - c_i) + \alpha \sum c_j$$

and the derivative, $\frac{\partial u_i}{\partial c_i}$, is just $-1 + \alpha$. Clearly, $\alpha - 1$ doesn't always equal zero, as it must at the Nash equilibrium, but when $\alpha < 1$, as is typically assumed, there is a dominant strategy—contribute nothing, regardless of what others are contributing. The intuition is straightforward. Every time I contribute a dollar, I generate a benefit of α for myself and everyone else, but regardless of the benefits I am receiving from the other donors (which I can't control), my benefits are never worth the sacrifice. On each dollar I contribute, I lose $1 - \alpha$.

Although the game has been vastly simplified, we can show that the linear VCM is still a social dilemma. If we sum across all n potential donors now,

we get

$$\sum u_i = \sum w_i - \sum c_i + n\alpha \sum c_j$$

and the last term simplified to $n\alpha \sum c_j$ because the public good benefit function is identical for everyone.

Taking the benevolent dictator's derivative, $\frac{\partial \sum u_i}{\partial C}$, yields $-1 + n\alpha$ and, as in [figure 19.1](#), we are now comparing the marginal cost of another dollar donated, 1, to the sum of the marginal benefits for all n potential donors, $n\alpha$. Further, so long as $n\alpha > 1$ or $\alpha > \frac{1}{n}$, the dictator insists that everyone donates everything.

Another way of looking at the VCM is to just compare the payoffs that donors receive at the two extremes. To focus on the logic, suppose everyone has the same endowment, w . If they play the Nash prediction and donate nothing, they just get w each because no public good is generated. If instead, they donate everything they will all earn αnw . The payoff when they all contribute everything is larger than when they contribute nothing, so long as $\alpha nw > w$, which again reduces to $\alpha > \frac{1}{n}$. In general, the VCM is a social dilemma when $\frac{1}{n} < \alpha < 1$. We now know the intuition for the first part of the inequality, but its purpose is really to ensure that the public good is productive enough to provide. When $\alpha < \frac{1}{n}$, the community is actually better off not producing the public good. It is like a lighthouse that no ships can actually see. The second inequality, $\alpha < 1$, if you recall, ensures that it is not a Nash equilibrium to provide the public good. Otherwise, the public good is so productive that it makes sense for each person to contribute everything to it, and thus there would be no social dilemma.

Exogenous Mechanisms to Increase Contributions

If providing a public good is a social dilemma, are there ways to get agents to contribute more? Let's first consider what we will call *exogenous mechanisms* to improve provision. By *exogenous*, we mean trying to change the incentives of the situation using interventions from people other than the decision makers themselves. For example, what if the benevolent dictator wanted to impose a system of taxes and subsidies to get the community to give closer to the socially optimal level instead of just explicitly choosing for them? What might such a system look like? Most important, the system will need to

penalize (i.e., tax) those agents who give too little, but it might also reward (i.e., subsidize) those who give more than their fair share.

As you can imagine, economists have thought a lot about how to structure the tax code to provide the right incentives, but so as not to lose the point of all this, we will consider a particularly uncomplicated scheme. What if the community members were taxed or subsidized proportional to how much they contributed below or above the average of everyone else? Suppose that $\beta > 0$ is both the tax and the subsidy (i.e., the incentives are symmetric on both sides of the community average contribution). In this case, the i th community member's tax (or subsidy) is just

$$t_i = \beta \left(c_i - \frac{\sum_{j \neq i} c_j}{n-1} \right)$$

where the $\sum_{j \neq i}$ indicates that donor i has been left out of the sum; it is the average contribution of everyone else (hence, the denominator is $n-1$ not n).

Imposing the new scheme on the community alters the payoffs, as seen in the augmented VCM payoff function:

$$u_i = (w_i - c_i) + \alpha \sum c_i + \beta \left(c_i - \frac{\sum_{j \neq i} c_j}{n-1} \right)$$

Notice that when i contributes more than the average, the difference at the end of the payoff function will be positive, and so β times the difference is the subsidy for those contributing more than average. When the difference is negative because donor i is contributing less than average, β times the difference is a tax.

Differentiating the augmented payoff function with respect to c_i for the individual yields $\frac{\partial u_i}{\partial c_i} = -1 + \alpha + \beta$. Now we can see that if the benevolent dictator sets $\beta > 1 - \alpha$, then the dominant strategy for each individual is to contribute everything, and the social optimum is achieved. In addition, notice that at the Nash equilibrium nobody pays a tax or receives a subsidy since everyone is contributing the same amount (so long as they have the same endowment).

While the tax scheme just described is nice because it implements the social optimum as a Nash equilibrium and it is revenue neutral (i.e., the taxes just balance the subsidies because they are both zero in equilibrium), the dictator still needs to monitor each community member, and this can be costly. Say that all this monitoring and bookkeeping costs $\$L + 1$; could the

dictator think of another way of spending less (such as $\$L$), to increase contributions?

What if the dictator held a contest in which the person who contributed the most was the most likely person to win a prize of $\$L$? Our guess is that you have already participated in this sort of contest to raise money for a public good because it is identical to the typical charity lottery, run at small stakes in church basements and at large stakes to fund public education in many states.

To account for the lottery, we augment the standard VCM payoff function with the expected value of the prize to the i th donor. In this case, we have

$$u_i = (w_i - c_i) + \alpha \sum c_j + \frac{c_i}{\sum c_j} L$$

Now every time you increase your contribution by a dollar, you increase your chances of winning the prize, $\$L$, and given that you contribute c_i in total, your chance of winning is $\frac{c_i}{\sum c_j}$. If it is helpful, think of the case that for every dollar donated, you get a lottery ticket, and your chance of winning is just the number of the total tickets sold that belong to you. This implies that your expected lottery prize is the product, $\frac{c_i}{\sum c_j} L$. Now let's investigate how potential donors will respond when the lottery is put in place.

This time, the individual's derivative is a bit harder because the quotient rule comes into play, but you can verify that the first-order condition that the i th community member should calculate is $\frac{\partial u_i}{\partial c_i} = -1 + \alpha + \frac{\sum c_j L - c_i L}{(\sum c_j)^2}$. Setting this equal to zero and solving for c_i are also a bit more complicated than normal, but to make things a lot easier, we can limit our attention to the symmetric Nash equilibrium, in which all the (identical) donors make the same contribution. At the symmetric equilibrium, we can assume that $\sum c_j = nc$, where c is the contribution they all make. This means that at the optimum, donor i is equating the derivative to zero. Rearranging and plugging in for the sum of the symmetric donations yield

$$1 - \alpha = \frac{ncL - cL}{(nc)^2}$$

To solve this expression for the Nash contribution, notice that a c cancels out of the right, leaving $(1 - \alpha)n^2c = nL - L$ after a little multiplication. Collapsing the right side and dividing gives us the Nash prediction, $c = \frac{L(n-1)}{(1-\alpha)n^2}$, which suggests that i 's contribution is now increasing in the size of the prize (L) and the marginal benefit from the public good (α) and decreasing in the

number of community members (n). Most important, this contribution is clearly positive, so it is an improvement over the baseline case, in which the dictator does not intervene (and donors contribute nothing). How big does the prize have to be to make sure that the donors contribute everything, as they still have to do at the social optimum? Well, to implement the socially optimal contribution level, we need the lottery contribution that we just solved for to be equal to w . Rearranging the resulting equality yields $L = \frac{(1-\alpha)n^2w}{n-1}$.

Can the dictator take this a step further by making the lottery revenue neutral like the tax scheme? This is what is known as a *pari-mutuel* lottery, an example of which is the common 50-50 lottery, in which half of the proceeds that are raised go to the charity and the other half form the prize given to the winner. If the share kept by the charity (i.e., the “handle”) is given by m , then the payoff function can be written as

$$u_i = (w_i - c_i) + \alpha (m \sum c_j) + \frac{c_i}{\sum c_j} ((1 - m) \sum c_j)$$

Notice that the amount raised for the public good is now only $m \sum c_j$ because the other $(1 - m) \sum c_j$ is used to finance the lottery. Let’s see how the donors should respond to these incentives. The derivative, $\frac{\partial u_i}{\partial c_i} = -1 + \alpha m + (1 - m)$, is much more straightforward now because the contribution sums cancel in the last term of the utility function. Remember that at the Nash equilibrium, this derivative is set to zero, yielding $m(\alpha - 1) = 0$. For perspective, recall that in the simple VCM without any intervention, the optimizing condition was that $\alpha - 1 = 0$. In that case, we saw that because α has to be less than 1 for the VCM to be a social dilemma, nobody contributed. But clearly if we divide both sides of the new condition by m , we just get the same result. This implies that the *pari-mutuel* lottery should lead to the same outcome wherein nobody contributes. In effect, when a donor contributes another dollar in this type of lottery, the positive externality to all the other donors from this contribution is simultaneously canceled by a negative externality resulting from the donor reducing their chances of winning the lottery (this is why the two contribution sums canceled in the *pari-mutuel* lottery utility specification).

Endogenous Mechanisms to Increase Contributions

Like the exogenous schemes discussed in the previous section, there have been a number of *endogenous* proposals to solve the public goods social

dilemma. In this case, the rationale for intervention comes from either the specifics of the situation or the proclivities of the donors. We will examine two such propositions.

First, suppose that the community realized that it only needs a finite amount of funds to produce the public good. This seems like a reasonable description of many (if not most) public goods. The community, for example, needs to raise $\$P$ to build a school or purchase a new fire truck that will serve the entire community. Let's also imagine that, as in these examples, unless $\$P$ is raised, the project cannot be completed, and whatever was contributed is in effect wasted—no benefit is produced. This is known as a *threshold* public good or a public good with a *provision point*.

Like the VCM, there is still a Nash equilibrium in which nobody contributes even though the amount needed is now finite. Instead of taking a derivative, think of it this way. Suppose that nobody is contributing—do you have an incentive to make a donation? Unless you can afford the entire fire truck (which will run you about \$750,000 these days), your modest contribution will be wasted. This is true for all n of the community members. Hence, if no one is currently contributing, nobody will take the initiative to contribute and the public good won't be provided.

There has got to be a point to imagining more details, so not contributing can't be the only equilibrium. Indeed, now imagine that the community is contributing right at this threshold. Given your current contribution, the community has collected exactly $\$P$. Should you change your contribution? Clearly, there is no point in contributing more, as the community has what it needs and the benefit you receive is fixed. What about contributing less—is there an incentive to free ride on the contributions provided by the other donors? In the standard VCM, there is a strong incentive to free ride, but can you see why adding the threshold removes this incentive? Now if you take back a dollar, the public good is no longer provided, and you don't get any benefit from it at all. Hence, so long as your share of the benefits from the fully provided public good are greater than your contribution, you no longer have an incentive to free ride. This discontinuity in the incentives provided by the threshold makes almost any allocation of contributions among the community members that is *just* sufficient to finance the public good a Nash equilibrium.

As an alternative endogenous mechanism, what if we let the community members police themselves? Specifically, suppose that p_{ij} is the punishment that community member i inflicts on member j when i thinks that j has not contributed enough. However, punishment is not free—the punisher needs to pay the cost of any sanctions. Likewise, suppose that p_{ji} is the punishment that community member i receives from member j for not contributing enough. If we account for the ability of the community to *mutually monitor* each other, i 's utility becomes

$$u_i = (w_i - c_i) + \alpha \sum c_j - \sum p_{ij} - \sum p_{ji}$$

As before, the underlying game is still a social dilemma and $\frac{\partial u_i}{\partial c_i} = -1 + \alpha$, but because monitoring is endogenous, the community members now have another lever to pull to maximize their utility (namely, p_{ij}). If i simply decides how much to punish the other group members in total, she needs to calculate $\frac{\partial u_i}{\partial (\sum p_{ij})}$ but we can see that this derivative is just -1 , indicating that every time i imposes another unit of punishment on some community member, it costs her a util. What this derivative indicates is that punishment is actually just another social dilemma.

There are two reasons for this. Notice that any benefit of punishing another community member will be diffuse. For example, if paying 1 to punish j results in j increasing her contribution by 1, i only gets α out of this, and we already know that this is not a good deal. Further, there is no reason why i shouldn't try to free ride on the punishment doled out by the other members of the community. If another player punishes those who don't contribute, you will still receive the benefits from this punishment even if you don't punish too. Hence, the only equilibrium of the mutual-monitoring version of the VCM is for nobody to punish, nobody to fear being punished, and everybody to contribute nothing. In the next section, we will see how this plays out in the lab.

Behavior in Games—Contributions to a Public Good

As the introduction to this chapter indicated, many people give to charity, but their reasons are likely to be varied (e.g., social pressure, religion, and tax breaks) and hard to sort out outside the lab. In a much simpler setting like the VCM, where the incentives could not be clearer (and more stacked against contributing), do people still give?

One of the first laboratory experiments using the VCM was conducted in the early 1980s by Isaac, Walker, and Thomas (1984). The authors instructed their participants to allocate an endowment of tokens between two accounts: an individual account that paid \$0.01 per token invested and a group account that paid an amount that depended on the number of people in the group. Specifically, i 's payoff was $u_i = 0.01(w - c_i) + a(\sum c_j)/n$, and normalizing by the benefit from the individual account, we have the public good $M \sum c_j$, where $M = \frac{a}{0.01n}$ is called the *marginal per capita return (MPCR)*. Using this setup, they examined experimental contributions as they varied the group size, MPCR, and experience.

Overall, Isaac et al. (1984) found that few participants free ride completely and across the 10 rounds, the average contribution is a bit more than 40% of the endowment. Further, even though it should not matter, given the dominant strategy in the game, contributions increased, on average, when the MPCR increased from 0.3 to 0.75, they decreased as the groups grew from 4 to 10 people, and more experienced people contributed less. In other words, it appeared as if participants were naturally inclined to contribute but reacted as they should to the incentives and may have been learning something as they gained experience.

So why is it that people contribute to public goods in the lab? One possibility, which is consistent with the results of Isaac et al. (1984), is that players are initially confused and learn to free ride as they gain experience in the game—in other words, they figure it out. Of course, this means that if we let the experiment go on for long enough, people will stop contributing because they realize that free riding is the dominant strategy in the VCM. This is precisely what Jim Andreoni (1988) attempted; however, his experiment added a clever innovation, as discussed in chapter 9. Imagine a group of five playing the VCM for 10 rounds. By the end of the 10th round, contributions have fallen to only 15% of the endowment after starting in the first round at closer to half. Instead of ending the session after the 10th round, Andreoni announced that because there was a bit of time left, the group would play another couple of rounds. If the participants are learning to free ride, it is pretty clear what should happen in the 11th round. Of course, the average contribution should continue to decline and be even less than 15%. Nope. Instead, the average in the 11th round rose back up to closer to 40%.

No standard learning model would predict that contributions will go back up after 10 rounds of decline, so what might be going on? One interpretation is that participants actually pretty quickly understand the incentives but are *conditionally cooperative*. What this means is that at the start of the experiment, these participants contribute a significant amount, despite the monetary incentive to free ride, and will continue to do so as long as the other players do so too. Further, if the other participants stop contributing, the only way to get back at them is by withholding your own contributions. So, while it appears they are learning to free ride, conditional cooperators are actually just doing the only thing they can to punish the free riders—free ride themselves—and, when the slate is wiped clean and the game is restarted, these players are willing to give cooperation a second chance.

As further evidence of this kind of *conditional cooperation*, Fischbacher, Gächter, and Fehr (2001) ran a version of the VCM using the strategy method, whereby each group member was asked for her full strategy of contributions, conditional on the contributions of the other members of the group. Specifically, each group member first made an unconditional contribution decision, identical to the previous VCM experiments. Then the member was presented with a contribution table and asked how much she wanted to contribute for each possible average contribution (rounded to the nearest integer) of the other members of her group. The heart of this study lies in the types of contribution functions that participants revealed.

Note that although the VCM was conducted using the strategy method, it is still a dominant strategy to contribute nothing, and consistent with this, roughly a third of the participants submit strategies in which they contribute nothing for all the possible average contributions of the others. Put differently, a third were free riders. However, the modal response of approximately half the participants was a rising contribution schedule, indicating that the more the others contributed, on average, the more they would be willing to contribute. In other words, half of the participants were indeed conditional cooperators.

The intuition seems straightforward, but can a simple model of conditional cooperation predict what we see in the lab? Let's continue with the specification of conditional cooperation introduced in Fischbacher et al. (2001). This means that community members are motivated by the average contribution of the other players. Suppose that we add just one parameter, $\gamma_i >$

0, which measures the conditional cooperativeness of player i . The larger the γ_i , the more the player responds to being different from the average contribution of the other players (so if $\gamma_i = 0$, the player doesn't care about the others' average and will be a standard free rider). Further, we can normalize any difference from the average of the other group members by dividing this difference by the endowment. This gives us something like the following:

$$u_i = (w_i - c_i) + \alpha \sum c_j + \gamma_i \left[1 - \frac{\left| c_i - \frac{\sum_{j \neq i} c_j}{n-1} \right|}{w_i} \right]$$

As you can see, when i contributes both less than and more than the average of the other players, the last term in the square brackets falls. As this term falls, the player receives less of her conditional cooperation benefit. Only when she contributes exactly the same amount as the average of the others does she get the full γ . Although it is not impossible to take the derivative of the absolute value, it can be less intuitive than thinking of this utility function as being piecewise. When i expects to be below the average contribution of the others, so $c_i < \frac{\sum_{j \neq i} c_j}{n-1}$, then the derivative of the utility function with respect to c_i will be $\alpha - 1 + \frac{\gamma_i}{w_i}$ which, depending on the size of γ_i , might be greater than zero, indicating that the player will contribute more. Similarly, when i expects to contribute more than the average, this derivative will just be $\alpha - 1 - \frac{\gamma_i}{w_i}$, which is surely less than zero, and the player will respond by contributing less. In sum, even a crude model of conditional cooperation can generate predictions in which contributions are positive.

There is another intuitive model that results in positive contributions in the VCM, and the rationale for it comes from surveying people who give to charity. Many of these people tell you that they feel good when they give, they feel a *warm glow* from contributing. Put differently, some people receive consumption benefits from contributing, in addition to whatever they get from the public good aspect of giving. Andreoni (1989) was among the first to consider a model of *impure* altruism. In our context, one could add yet a different term to the basic utility function, say $g(c_i)$, which is concave in one's contribution and independent from the public good benefits from contributing. We then have

$$u_i = (w_i - c_i) + \alpha \sum c_j + g(c_i)$$

and taking the derivative yields the first-order condition, $\alpha - 1 + \frac{\partial g}{\partial c_i}$, which, if the warm glow is strong enough, is sure to equal zero for some positive contribution. Here, if impure altruists receive enough of a warm glow from giving, they will contribute regardless of how much other people contribute or whether the government finances the public good too.

Evidence in favor of impure altruism comes both from the lab and empirical studies of naturally occurring tax records. Interestingly, the tax studies robustly find that government grants and subsidies to charities do not completely crowd out private contributions as they would if donors were pure altruists (see exercise 2). In the lab, Crumpler and Grossman (2008) devised a very clever experiment to estimate how much of altruism is likely to be due to the warm glow. In their study, like any standard dictator game, lab participants were told that they would have \$10 to divide between themselves and a charity that they chose. Unlike a standard dictator game, they were then told that the proctor of the experiment would also contribute to this same charity, and the charity would always receive exactly \$10. This was because the amount contributed by the proctor would be reduced by the amount that the participant gave. In other words, the charity would never receive more or less than \$10, regardless of how much the participant contributed. As you can see, in this setting there is no reason for a pure altruist to give because the total amount given cannot be increased. However, an impure altruist might still give because she can receive a warm glow from being the one to provide the public good.

While in similar dictator games without this contribution crowding, participants gave between 30% and 50% of their endowments on average, Crumpler and Grossman found that donations fall, as you might expect, but not by a lot. In their sample, they found that 57% of people give something when they could not change the amount going to the charity, and the average donation was 21% of the endowment. They conclude that warm glow giving exists and explains a significant portion of charitable contributions.

Experimenters have also asked whether they can coax participants into contributing more by implementing some of the mechanisms discussed here. Let's first consider the exogenous mechanisms: taxes, subsidies, and provision points. The experiment reported in Falkinger et al. (2000) examines the tax and subsidy scheme described previously, in which those who contribute less than the average are penalized and those who contribute more

are rewarded. The authors of this study compare standard VCMs to ones in which the tax scheme is implemented, while varying the MPCR, the group size, and the size of the tax-subsidy parameter, β . The data in the left panel of [figure 19.2](#) come from the treatments in which there are four people in a group and the MPCR is 0.4. Here, the authors found that, as in previous studies, without the mechanism, contributions in the control treatment start near half the endowment but decline over the course of the experiment, approaching the free-riding Nash prediction by the end. With the implementation of the tax and subsidy mechanism, however, donations increase immediately and approach the new equilibrium prediction of full contributions. This simple mechanism works well to enhance the provision of a public good.

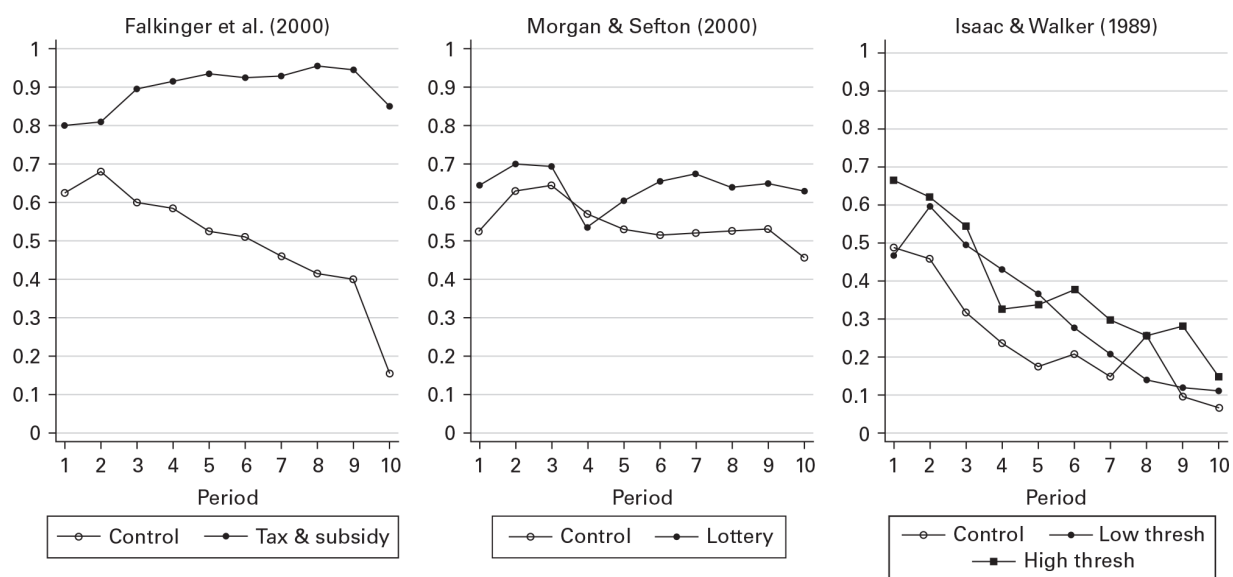


Figure 19.2

Contributions in the voluntary contribution mechanism.

In the middle panel of [figure 19.2](#), we reproduce the data from the first 10 periods of the lottery experiment conducted by Morgan and Sefton (2000). In their study, the groups also had four people, but the MPCR was 0.75. Contributions in the control treatment of this experiment tended to be higher than in the Falkinger et al. control, which is consistent with the fact, already mentioned, that contributions tend to increase with the MPCR. In this graph, we also see that adding a fixed-prize lottery can help increase contributions, as theory predicts. It takes a while, but after the fourth period, contributions in

the lottery treatment rise above those in the control and stay higher for the rest of the experiment. To a great extent, it appears that this intervention may be a good way to maintain, if not necessarily increase, contributions, especially if the prize is modest.

In the final panel of [figure 19.2](#), you will find the data from a threshold public good experiment. In Isaac et al. (1989), the authors compared a control VCM treatment with four-person groups and a MPCR of 0.3 to other conditions in which thresholds of differing sizes are imposed. Again, comparing control treatments across the graphs in [figure 19.2](#), we see that the higher the MPCR, the larger are contributions, on average. We also see in the right panel that thresholds affect contributions as predicted. In both cases shown, the threshold increases contributions on average, and there is weak evidence that the higher the threshold, the higher are contributions. However, what also comes with higher thresholds are lower rates of successfully providing the public good. In other words, raising the threshold does not tend to raise contributions at the same rate, so in many more instances, the threshold is not met and the public good is not provided.

The endogenous mechanism that has garnered the most interest from researchers, by far, is mutual monitoring and punishment. Given the opportunity to punish free riders directly rather than just by reducing one's contribution as in Andreoni's (1988) experiment, do lab participants take it? Or, do they realize that punishment is just a second-order social dilemma added on top of the contribution social dilemma? That is, do they realize that punishing free riders would be good, but what would be great is if someone else did it!

Fehr and Gächter (2000) is one of the first VCM studies that allowed participants to monitor (with anonymity) the contributions of the other members of their group and punish whomever they wanted to, at a cost. Specifically, by paying a fee, the monitor could impose punishment points on another player. The cost per point assigned by the punisher was increasing, and each point incurred by the target reduced her payoff by 10%. Again, remember that the Nash equilibrium here is where nobody punishes because it is costly to do so, and therefore everyone should try to free ride on the punishment done by others. Because of this, nobody will fear being punished, which means that they should continue to free ride. In sum, adding mutual monitoring and punishment should *not* increase contributions.

In [figure 19.3](#), we present the data from Fehr and Gächter's experiment in which players stayed in the same groups for all 10 rounds, as in many of the other studies we have already discussed. In one treatment, participants played the standard VCM (four person groups, $MPCR = 0.4$) without any monitoring or punishment for 10 rounds, and then they were given the ability to monitor and punish their groupmates. In this condition, contributions, as always, started near half the token endowment and fell to less than 20% of the endowment after 10 rounds. Once punishment was allowed, contributions jumped to more than 60% of the endowment in period 11 and continued to rise for the rest of the experiment. In a second treatment, participants were initially allowed to monitor and punish, and this caused contributions to start higher (60% of the endowment) and rise to nearly 90% by the end of 10 periods. However, when punishment was taken away, contributions tumbled in periods 11 through 20.

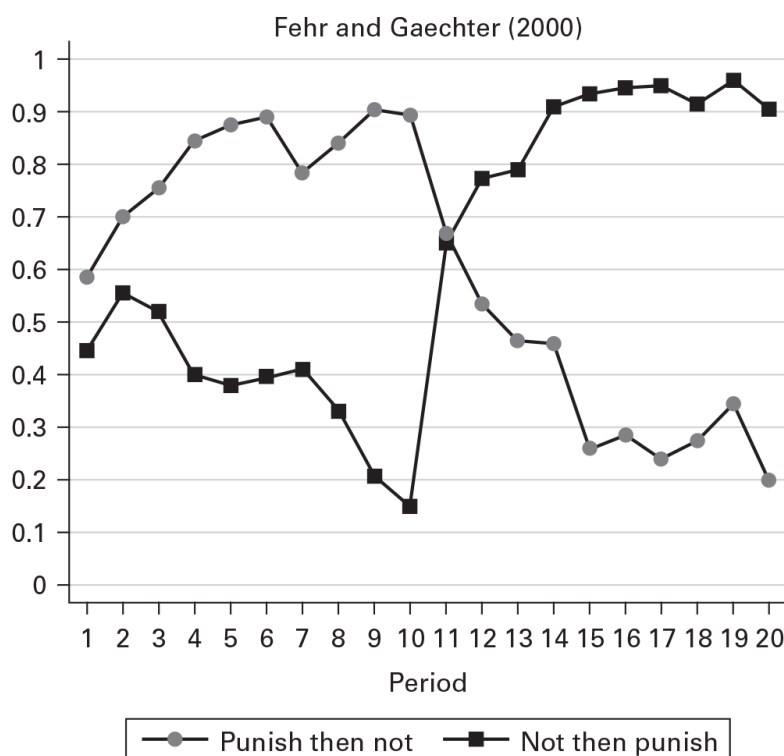


Figure 19.3

Contributions in the voluntary contribution mechanism with monitoring and punishment.

What is going on here? Clearly, punishment matters, and it yields levels of provision of the public good close to those seen with the Falkinger et al. tax

scheme (which, remember, was supported as an equilibrium). Is it the case that participants punish when they, in theory, shouldn't? Heck yes! It turns out that people love to punish each other in this game, and most of the punishment is prosocial—it is directed at free riders. In fact, Fehr and Gächter showed that the single most important factor for how much player i punished player j is how much less j contributed than the group average.

Subsequent to the Fehr and Gächter (2000) experiment, many other punishment experiments have been run in which researchers have varied the parameters of the game to better understand participant motivations (in fact, the density functions of contribution data plotted on this book's cover come from similar experiments conducted by one of the authors). For example, the ratio of the sanctions imposed to the cost of sanctioning has been altered to make punishment cheaper or more expensive, and we see that seemingly irrational participants behave according to the law of demand when it comes to punishment (i.e., the demand for punishment is downward sloping but very inelastic). Other studies have allowed *counterpunishment* by adding a third round to each period of the experiment, in which subjects can see how much they have been punished and then punish the people that punished them. As you can expect, this often turns into a counterproductive feud. Finally, experiments have also been run in which any punishment must be altruistic in the sense that punishers can never benefit from doing it. For example, imagine two groups playing the VCM side by side, but not benefiting from the public good created in the other group. It turns out that if you let people punish free-riders in the *other* group, they will do it, and it increases contributions in the other group.

All of the research has taught us that, indeed, people love to punish each other. In fact, neuroscience studies suggest that they have an innate sense of disgust when they witness free riding. This innate tendency to punish has been formalized as the notion of *strong reciprocity* in Bowles and Gintis (2004), and it is these agents who often ensure that public goods are provided.

Exercises

(1) *Public goods and bads*. There are n colony members indexed as $i \in \{1, \dots, n\}$ who all have an endowment, E , and can contribute any amount c_i (up to their endowment) to the community chest. The sum of the contributions to the

community chest is then multiplied by α , and this product is shared equally among all the colonists.

- (a) Write the payoff function for the i th community member and solve for the Nash equilibrium.
 - (b) Now suppose that in a different colony, instead of contributing, the colonists can take from the community chest. In this case, all the endowments are initially tossed into the community chest, and then colonists can take out any amount t_i up to their endowments. Further, whatever is left after all the colonists take what they want is then also multiplied by α and distributed equally. Write this new payoff function and solve for the Nash equilibrium.
 - (c) Are either of these two colonies facing a social dilemma?
- (2) *A nonlinear public good.* Suppose that n potential contributors to a public good indexed as $i \in \{1, \dots, n\}$ have Cobb-Douglas preferences, $u_i = (w_i - c_i)^\alpha (\sum c_j)^{1-\alpha}$, where w_i is donor i 's endowment, c_i is donor i 's contribution, and $(1 - \alpha)$ is the weight put on the public good by all donors.
- (a) Find the symmetric Nash equilibrium contribution (remembering that at the symmetric equilibrium, $\sum c_j = nc$).
 - (b) Now suppose that the government steps in and taxes each donor by an amount t and uses the sum of all the taxes collected to help finance the public good. How does this alter the payoff function for the i th donor, and what is the new symmetric Nash equilibrium contribution?
 - (c) Do taxes crowd out voluntary contributions at the new equilibrium? If yes, at what rate?
- (3) *Taxes and subsidies to promote contributions.* Reconsider the Falkinger et al. (2000) mechanism, for which the payoff of the i th player is $u_i = (w_i - c_i) + \alpha \sum c_j + \beta \left(c_i - \frac{\sum_{j \neq i} c_j}{n-1} \right)$.
- (a) Find the equilibrium contribution level.
 - (b) Now, suppose that $\beta = (1 - \frac{1}{n})$. What is the new equilibrium contribution (and what is special about this level of β)?
- (4) *Funding public goods with lotteries.* Suppose that an anonymous member of the community puts up a lottery prize, P , to try to get the other community members to contribute more to the public good.

- (a) If there are 10 total members of the community, how much will the representative member donate?
- (b) How large does P have to be to get the community members to contribute everything (i.e., fully fund the public good)?

(5) *A threshold public good.* A community of n individuals are each endowed with E . Each member decides how much of the endowment to contribute to a public good (c_i) and how much to keep ($E - c_i$). In this case, however, the public good has the same fixed value for everyone when it is provided, V , and it has a finite price, P . If the community raises at least P , the public good is funded and everyone gets V but if it raises less, the contributions are wasted and the public good is not provided.

- (a) Assuming that the public good is affordable (i.e., $nE > P$) but no one person can finance it alone ($E < P$), what are the symmetric Nash equilibrium levels of c_i for this threshold public good? Is it a social dilemma?
- (b) What if there is a money back guarantee? If the community does not raise enough funds to provide the public good, the contributions are returned. What are the symmetric Nash equilibria of this version of the game? Is this a social dilemma?
- (c) What if there are excess contributions such that $\sum c_j > P$ and the community implements the following rebate scheme? Players get a rebate that is proportional to their contribution. Hence, if the contributions exceed the price, one's payoff is $\pi_i = (E_i - c_i) + V + \frac{c_i}{\sum(c_j)}(\sum c_j - P)$, and if they fall short of the price, the payoffs are $\pi_i = E - c_i$. What symmetric outcome is Nash now?

(6) *How warm is the glow?* A recent field experiment confirms that the warm glow of giving is concave. In fact, the estimate is $g(c_i) = 2c_i^{\frac{1}{2}}$.

- (a) Predict the contribution of a warm glow giver with an endowment of \$10 and a marginal per capita return from the public good of 0.25.
- (b) Suppose that there are five identical donors. How happy is each? Are they happier than at the Nash prediction without warm glow?

(7) *Mutual monitoring with strong reciprocators.* Consider a simple setting in which there are just two potential donors to a public good. They each have an endowment of 10 and the MPCR is 0.75. Hence, the baseline payoff function

of each is just $u_i = (10 - c_i) + 0.75(c_i + c_j)$. Suppose that there are two possible strategies the donors can follow. They can *free ride* by contributing nothing or they can *reciprocate strongly* by contributing everything and punishing anyone who doesn't contribute everything. Suppose that the punishment inflicted by a strong reciprocator is p and the net psychic benefit they receive from catching and punishing a free rider is b .

- (a) Write a normal-form payoff table to represent the interaction between the two players.
- (b) Suppose that s is the probability that a donor adopts the strong reciprocity strategy. In equilibrium, how high must s be to ensure that it is the best response of the other player to contribute?

20

Common Pool Resources

Common pool resources (CPRs) are goods that can be readily accessed by many users and that experience congestion—and potential destruction—from heavy use. Examples include fisheries, forests, and irrigation systems. As the resource is tapped more heavily, each user's benefit or yield decreases, and it is less likely that the resource will be sustained for future use. Without regulations or agreements to govern the usage of common pool resources, users face a social dilemma. If each person follows their own individual self-interest, the resource will be overused in equilibrium, and each person will be worse off as a result. In this chapter, we'll look at games in which players make a one-shot decision regarding how much of the resource to extract, as well as dynamic games in which current extraction choices affect the future availability of the resource, and analyze how rules regulating the commons change the equilibrium predictions.

Imagine that you own a fishing boat. Each season, you can decide how much effort to put into harvesting fish from a particular fishing ground. The effort level that you put into extracting fish can be any number from 0 to 20, which you can think of as the number of full-time weeks during the season that you spend fishing. At the same time, seven other companies are making this same choice, and the amount of fish that your effort yields depends on how congested the fishing ground becomes.

Specifically, let's say that $\sum x_j$ is the total extraction by all eight boats. Then the total value of fish achieved by all eight individuals is given by the production function $11(\sum x_j) - \frac{1}{16}(\sum x_j)^2$. The graph of this function is depicted in [figure 20.1](#). Note that, initially, the more effort that is put into fishing, the greater the yield (i.e., the derivative is positive). However, the yield is increasing at a decreasing rate (i.e., the second derivative is negative). Eventually, the yield decreases with effort: if the total effort is greater than 88, spending more effort catching fish actually results in fewer fish being caught overall. The production functions of many natural resources have this concave shape.

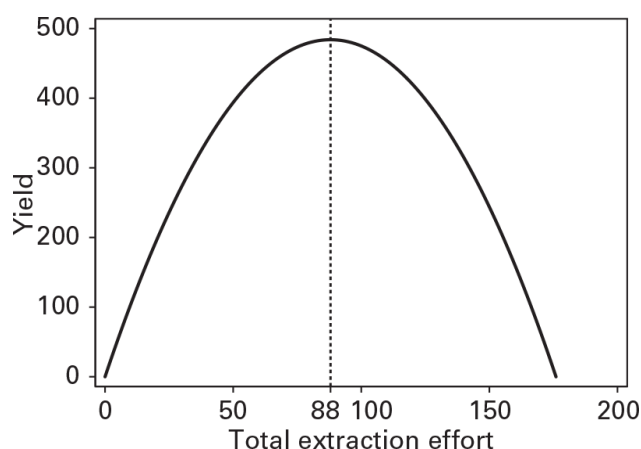


Figure 20.1

Yield of fish from extraction efforts.

The production function gives the total value of the extracted fish. But what portion of this yield does your company gain? Let's assume that each party's share is equal to their own effort as a fraction of the total effort. Each week that you *don't* spend fishing has a value to you of 2, which you can think of as the payment that you'd receive from instead putting this effort toward the next best use of your time, and this return does not depend on the choices of others.¹

How much effort do you put into fishing this season?

Common Pool Resources and the Tragedy of the Commons

In this scenario, no single individual owns the fishing ground, and thus it can be accessed by anyone. Furthermore, as more people use the good, it eventually becomes depleted, and the yields that each user can achieve diminish. We refer to this type of good as a *common pool resource*.

A common pool resource (CPR) is a good with two characteristics. First, it is difficult to exclude users from consuming the resource. Second, the yield that each user gains is subtractable—that is, one person's use reduces the benefits for the other users. Examples include both natural and human-made resources, such as fishing grounds, pastures, irrigation systems, road networks, forests, and the global atmosphere.

Figure 20.2 classifies goods along these two dimensions of excludability and subtractability. The first entry in the table is private goods, such as hamburgers or cars, which are the types of goods you've probably focused on if you've studied microeconomics. Private goods are readily excludable: once

you purchase a hamburger or a car, other people are legally excluded from using them without your permission. They are also highly subtractable: every bite of hamburger that you eat is a bite that someone else doesn't eat, and if you're driving a car, no one else can simultaneously drive the same car. The next entry in the table is club goods, which are excludable but can readily accommodate many users. Examples include private parks, beaches, golf courses, underused toll roads, and cable television subscriptions. While these goods may eventually experience some congestion (e.g., if you crammed millions of people onto a beach), the typical level of use implies that one user's presence won't significantly diminish another person's enjoyment. Having to share a beach with an extra person is not the same as sharing a hamburger with them.

		Subtractability	
		High	Low
Excludability	Easy	Private goods: Hamburgers, cars	Club goods: Parks, golf courses
	Difficult	Common pool resources: Fisheries, forests	Public goods: National defense, streetlights

Figure 20.2
Classification of goods by excludability and subtractability.

While private goods and club goods can be restricted to those who pay for the right to use them, common pool resources and public goods both have the feature that users cannot be easily excluded. The difference is that common pool resources are subtractable (or “rivalrous”), while pure public goods do not experience any congestion. Examples of public goods include national defense, public radio, and streetlights (as discussed in chapter 19). Once provided, a public good can be freely used by anyone, without one person's use impinging on another's enjoyment. In contrast, individual overuse of a common pool resource can destroy it for everyone, and for all time.

Both the provision of public goods and the use of common pool resources create social dilemmas among the potential users. As we saw in the previous chapter, individuals can access a public good regardless of whether they contribute to its funding, so they have little personal incentive to contribute.

As a result, the public good is underprovided in equilibrium relative to the optimal level. This “public goods problem” occurs because people consider only their own personal costs and benefits when they decide how much to contribute to the public good, and they ignore the positive benefits that their contribution would have for everyone.

Users of a common pool resource face the flip side of this problem. Consider a resource that has already been provided, such as a fishery or a congested road network, and individuals can freely access it. Each person can choose how much of the resource to use and, since it is subtractable, heavy use depletes the benefit for all users. If people make their usage choice considering only their own personal costs and benefits—and ignore the costs that their use imposes on others—then they will overuse the resource relative to the socially optimal usage.

The social dilemma that arises in the use of common pool resources is known as the *tragedy of the commons*. The name comes from a canonical example in which a group of herdsmen send cattle to graze in a common pasture. The herdsmen consider only the marginal payoff that they receive from adding an extra cow to the herd, while the negative consequences of their actions are spread across all the users. The effects of the tragedy are twofold. In the short run, the resource is overused relative to the social optimum, and thus the returns to each user are lower than what could be achieved if players were to consider the impact of their use on others’ yields and exercise some restraint. In the long run, the resource may not regenerate for future use and be ultimately lost forever.

A Static Common Pool Resource Game

The fishing game described at the beginning of this chapter provides a simple model of the tragedy of the commons. This setup is almost identical to the game studied in Ostrom, Gardner, and Walker (1994)’s classic book on common pool resource games, *Rules, Games, and Common Pool Resources*. Let’s solve for the equilibrium outcome and social optimum using the specific parameters from the example. Let x_i refer to the effort that individual i puts into extraction from the common pool resource. Then i ’s payoff is given by

$$u_i(x_i, x_{-i}) = 2(20 - x_i) + \frac{x_i}{\sum_{j=1}^8 x_j} \left(11 \sum_{j=1}^8 x_j - \frac{1}{16} \left(\sum_{j=1}^8 x_j \right)^2 \right) \quad (20.)$$

The first term tells us the payoff that i receives from the portion of his 20 units of effort that are not put into resource extraction. The second term tells us the payoff that i receives from the common pool resource. All members collectively extract $\sum_{j=1}^8 x_j$, so the total value they achieve is $(11 \sum_{j=1}^8 x_j - \frac{1}{16} (\sum_{j=1}^8 x_j)^2)$. Since the resource is subtractable, a unit consumed by one person cannot be consumed by anyone else, so this return must be divvied up between the users. In particular, each person receives a fraction of the good proportional to their effort, $\frac{x_i}{\sum_{j=1}^8 x_j}$. Notice that this factor thus captures the subtractability of the good.

Efficient Extraction

Let's first consider the social optimum. As defined in chapter 18, the social optimum is the outcome that maximizes the total payoffs of all the players. Imagine that a single corporation owns all eight fishing enterprises. What total extraction level should they choose in order to maximize the joint payoffs of all eight players? Let the total extraction level across all eight companies be given by $X = \sum_{j=1}^8 x_j$. Then the payoff summed across all the players is

$$\sum_{i=1}^8 u_i = 2(160 - X) + 11X - \frac{1}{16}X^2$$

Taking the derivative with respect to X , we get $-2 + 11 - \frac{1}{8}X$. Setting this equal to zero and solving for X , we find that total payoffs are maximized when $X = 72$. Thus, the group as a whole does best when the total extraction is 72. If each person equally contributes to (and benefits from) this extraction, then each chooses an effort level of 9. We will focus on this symmetric prediction, but note that any vector of efforts $(x_1, \dots, x_8)$ such that $\sum_{j=1}^8 x_j = 72$ is socially optimal.

We could also approach this problem in a slightly different way. If you have taken microeconomics, you know that people optimize by participating in an activity up to the point that their marginal benefit of doing so equals their marginal cost. In this example, the marginal cost of extraction is the opportunity cost of not putting that effort toward an alternative endeavor, which our model set at 2. The marginal benefit of extraction is given by the derivative of the concave production function, $11X - \frac{1}{16}X^2$, which we graphed at the beginning of the chapter and is shown in [figure 20.2](#). The derivative of this function equals the marginal cost of 2 when $11 - \frac{1}{8}X = 2$ or $X = 72$. The socially optimal extraction level of 72 is shown in the graph and occurs at the

point where the slope of the production function (i.e., the marginal benefit of extraction) equals the marginal cost of 2.

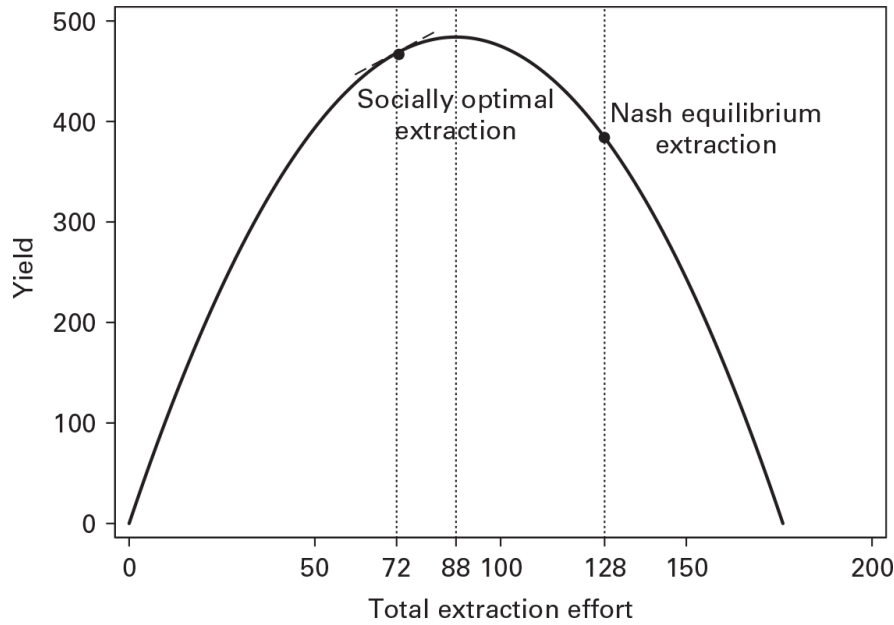


Figure 20.3

The production function with the socially optimal and Nash equilibrium extraction levels marked.

Nash Equilibrium Extraction

Now that we know what the extraction level will be if players are able to cooperate to maximize their joint payoffs, let's consider what it will be if players act independently in their own self-interest. To do so, we must find each individual's best response function, by taking the derivative of their payoff function and setting it equal to zero. But first, let's do ourselves a favor by simplifying the payoff function somewhat. The payoff function for player i given in equation (20.1) is rewritten here, but we've just distributed out the terms in parentheses:

$$u_i(x_i, x_{-i}) = 40 - 2x_i + 11 \frac{x_i}{\sum_{j=1}^8 x_j} \sum_{j=1}^8 x_j - \frac{1}{16} \frac{x_i}{\sum_{j=1}^8 x_j} \left(\sum_{j=1}^8 x_j \right)^2$$

Notice that the third and fourth terms each have $\sum_{j=1}^8 x_j$ in both the numerator and denominator, so we cancel and combine these terms to get

$$u_i(x_i, x_{-i}) = 40 + 9x_i - \frac{1}{16} x_i \sum_{j=1}^8 x_j$$

Finally, notice that $\sum_{j=1}^8 x_j$ sums up i 's effort along with the other seven players' efforts. Since we're going to need to take the derivative with respect to i 's effort, let's break this summation into the part that has our variable of interest, which player i can control (x_i), and the part that includes everyone else's effort and is treated as a constant ($\sum_{j \neq i} x_j$). Subbing in for $\sum_{j=1}^8 x_j = x_i + \sum_{j \neq i} x_j$, we get

$$u_i(x_i, x_{-i}) = 40 + 9x_i - \frac{1}{16}x_i^2 - \frac{1}{16}x_i \sum_{j \neq i} x_j \quad (20.)$$

Finally, taking the derivative with respect to x_i and setting it equal to zero, we get

$$9 - \frac{1}{8}x_i - \frac{1}{16} \sum_{j \neq i} x_j = 0$$

$$x_i = 72 - 0.5 \sum_{j \neq i} x_j$$

If the players' efforts are unconstrained, then the best response of each individual is to choose an effort level equal to the difference between the social optimum of 72 and one-half the sum of the other players' efforts. This equation is shown by the thin black line in [figure 20.4](#), where the horizontal axis is the total extraction of the other players ($\sum_{j \neq i} x_j$). (You can ignore the other lines for now; we'll return to them momentarily.) Note that if there are no other players with access to the resource or if the others are not putting any effort toward extraction, the player's best response would be the socially optimal total extraction level of 72. As the other players expend more effort, player i prefers to scale down her own effort, but only by 0.5 units for each extra unit of effort they choose. We could also rewrite the best response function slightly as $x_i = \frac{144 - \sum_{j \neq i} x_j}{2}$. This indicates that each player's best response is essentially to split the difference between 144 and what the other players are extracting. For instance, if the other players extract 100, player i 's best response is to extract 22, bringing the total extraction to 122. If the other players are collectively extracting at least 144, player i 's best response is to extract nothing. Finally, note that player i is constrained by our assumption that she has only 20 units of effort to expend. Given this constraint, player i 's best response is to extract the maximum of 20 whenever $72 - 0.5 \sum_{j \neq i} x_j \geq 20$. The best response function taking into account this constraint is given by the thick black line in [figure 20.4](#).

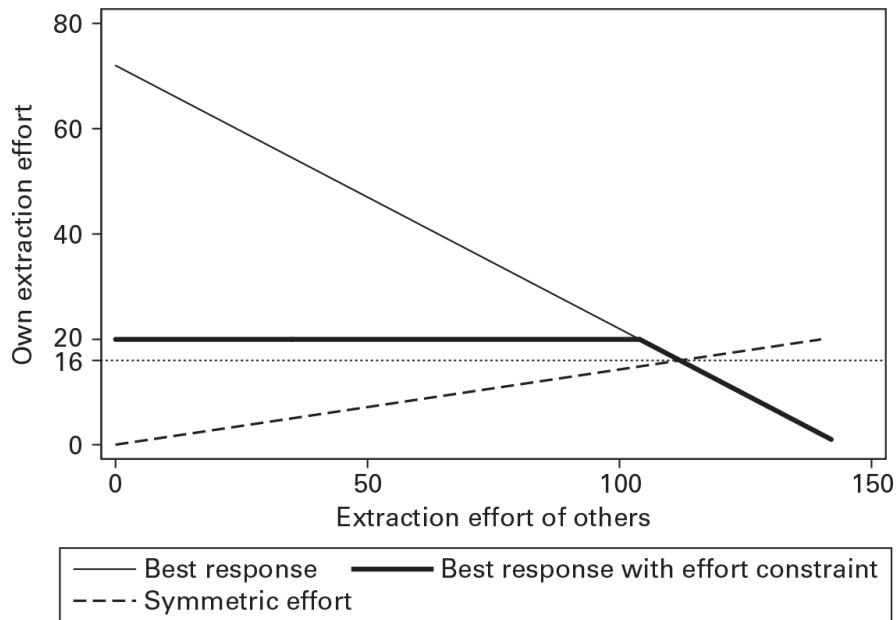


Figure 20.4

The best response function (thick black line) and the symmetric effort level (dashed line), with the symmetric Nash equilibrium at the intersection of the two lines (with each player extracting 16).

Recall the technique we learned in chapter 6 for finding the symmetric Nash equilibrium in a game with a large number of players. We consider whether there is an effort level x that is a best response to all seven other players choosing x . In this case, we can sub $7x$ for $\sum_{j \neq i} x_j$. That is, we want to find x such that

$$x = 72 - 0.5(7x)$$

Solving for x , we find that the symmetric Nash equilibrium effort level is 16. Thus, in equilibrium, the total extraction level is 128, vastly exceeding the socially optimal level of 72. We can also see this graphically. The dashed line in [figure 20.4](#) depicts the set of outcomes when everyone chooses the same effort level (i.e., when x_i is one-seventh of the total effort of the others). The intersection point at $x_i = 16$ demonstrates that this is the only effort level at which player i is best responding and all players are choosing the same effort level. The contrast between the social optimum and the Nash equilibrium extraction is shown in [figure 20.3](#).

Does Group Size Matter?

How does this result vary if we were to change the number of players in the game? Rather than eight people, let's imagine that we have n players. Note

that player i 's best response function in this game is essentially unchanged. She prefers to extract half of the difference between 144 and what the other n players collectively extract—the only difference is the number of other people whose efforts we're summing over. The symmetric Nash equilibrium is thus given by $x = 72 - 0.5(n-1)(x)$, or $x = \frac{144}{1+n}$, and the total equilibrium extraction level across all n players is $nx = \frac{144n}{1+n}$. If there were only a single player in this situation, she would choose to extract 72, or as close as she could get given constraints on her effort level. As the number of players grows, the total group extraction level increases and approaches 144 as the group size approaches infinity.

A General Model

We could also generalize this game further by considering any version with a quadratic production function. Let's allow the total endowment of effort to be given by a general parameter ω (rather than 20), the opportunity cost of extraction to be given by a general parameter c (rather than 2), and the production function to be given by a general quadratic function form $aX - bX^2$ (rather than $11X - \frac{1}{16}X^2$). Then the payoff function for player i is given by

$$u_i(x_i, x_{-i}) = c(\omega - x_i) + \frac{x_i}{\sum_{j=1}^n x_j} \left(a \sum_{j=1}^n x_j - b \left(\sum_{j=1}^n x_j \right)^2 \right)$$

We can solve for the social optimum and Nash equilibrium of this general model in the same way we did for the version with specific numbers. In this case, we find that the socially optimal total extraction is $\frac{a-c}{2b}$, while the symmetric Nash equilibrium is given by $\frac{1}{n+1} \frac{a-c}{b}$. You'll be asked to prove this in exercise 2 at the end of the chapter. The total extraction across all n people under the Nash equilibrium is thus $\frac{n}{n+1} \frac{a-c}{b}$. As n increases, $\frac{n}{n+1}$ approaches 1, and thus the total extraction approaches $\frac{a-c}{b}$: double the socially optimal level. Thus, our analysis of the fishing game holds more generally for any similar extraction model with a quadratic production function: so long as there is more than one person with access to the resource, it will be overused, and as the number of users grows large, the extraction will be close to twice the amount that is socially optimal.

Dynamic Common Pool Resource Games

In the previous section, we considered a one-shot common pool resource game. In this setting, we found that overuse in the current period reduced the amount of the resource that the users could harvest at the time, but we didn't consider the long-term consequences of these choices for future users. In reality, the future availability of resources such as groundwater systems, fisheries, and forests all depend on current extraction levels. Let's now consider multiperiod CPR games, in which the amount that is available to harvest in the current period depends on how much of the resource was left untapped in the previous period. Notice that this interaction is not that same as a repeated game, such as those we discussed in chapters 9 and 10. In a repeated game, players have an identical encounter in every period; in a dynamic CPR game, the available actions and payoffs in each period depend on the players' previous choices.

The Quadratic Production Function as the Steady State of a Dynamic System

The most common dynamic model of renewable natural resources assumes that the resource follows a logistic growth function that depends on the natural growth of the resource, its maximum capacity, and its existing size (see Schaefer, 1957). Let's say that the resource has a maximum carrying capacity of K , which we can think of, for instance, as the maximum number of fish that can thrive in the fishery. In each period t , the resource naturally grows by an amount $r \times w_{t-1} \times (1 - \frac{w_{t-1}}{K})$, where r is the resource's natural growth rate and w_{t-1} is the size of the resource after the previous period's extraction has occurred. Note that if the existing size of the resource equals the maximum K , then this term reduces to zero and no further growth takes place. Otherwise, the resource grows each period, with maximum growth occurring when $w_{t-1} = \frac{K}{2}$.

Now, let's add the humans to the model. The amount of the resource that can be extracted depends on how much effort the players put into extraction (X), as well as the current size of the resource (w_t) and a parameter α that captures what proportion of stock the players could harvest with each unit of effort. (In the context of the fishing example, this is sometimes referred to as the "catchability coefficient.") Specifically, the total harvested value in period t is given by $h(X) = \alpha X w_t$.

Let's consider the "steady-state" outcome, in which the amount that is harvested in each period is equal to the amount by which the resource grows, such that the resource neither grows nor shrinks over time and is perfectly sustained. Let X be the total effort put into extraction and w be the size of the resource. Then the total harvest amount is given by αXw . In the event that the resource is sustainably harvested at the exact rate that it grows, we have $\alpha Xw = rw(1 - \frac{w}{K})$. Solving for w , we find a steady-state resource size of $w = K - \frac{\alpha X}{r}K$. How much of the resource is harvested (e.g., how many fish are caught)? Plugging the steady-state value of w into αXw , we get that the total harvested value is $\alpha KX - \frac{\alpha^2 K}{r}X^2$. Notice that we wind up with a quadratic function of extraction effort (X), just as we had in the static model. Specifically, the harvest produced is of the form $aX + bX^2$, where $a = \alpha K$ and $b = -\frac{\alpha^2 K}{r}$. Thus, we can think of the production function that we used in the static model as the steady-state outcome of this dynamic system of resource regrowth.

However, depending on the growth process of the resource, players may have an incentive to not harvest sustainably and to deplete the resource past the point where it can regrow.

An Example of a Dynamic Game

Consider the following four-period dynamic CPR extraction game. This game is based on an experiment that Noussair, van Soest, and Stoop (2015) conducted with actual fishermen, who caught real fish in a pond that was restocked by the experimenters based on their earlier extraction choices. (The authors also paid them \$5 per fish, so they'd be properly incentivized even if they were unimpressed by the rainbow trout that they stocked the pond with.) To begin, the pond is stocked with eight fish, and the number of fish in the pond is adjusted according to the fishermen's harvesting choices. In each period, the fishermen independently choose how many fish to catch. If the sum of their choices is less than or equal to the existing number of fish, then each gets their chosen catch. What if the choices sum to more than the number of available fish? A common way to model the outcome is to assume that each person harvests simultaneously, at equal speeds, until either they have achieved their goal or the resource has been exhausted. Let's assume that the result of this process is that the resource is divided among the four fishermen as follows:

(1) Initially, we have 4 people who must be assigned a catch. If anyone chooses a number of fish less than or equal to $1/4$ of the stock, they are given this amount, and we no longer worry about them.

(2) Is the number of people who have not been awarded their catch equal to 3? If not, skip this step. If so, then anyone who has chosen a number of fish less than or equal to $1/3$ of the remaining stock is given this amount, and we no longer worry about them.

(3) Is the number of people who have not been awarded their catch equal to 2? If not, skip this step. If so, then anyone who has chosen a number of fish less than or equal to $1/2$ of the remaining stock is given this amount, and we no longer worry about them.

(4) Anyone who has not yet been assigned a catch receives an equal share of the remaining stock that has not yet been assigned.

The game lasts for four periods or until the stock has been depleted, whichever happens first. In the event that stock remains at the end of the period, it regenerates according to the growth function that we introduced previously, with $K = 8$ and $r = 2$. For the purposes of their experiment, Noussair et al. rounded the growth function to whole fish as shown in [figure 20.5](#).

Fish Remaining	Incremental Growth
0	0
1	1
2	3
3	3
4	4
5	3
6	3
7	2
8	0

Figure 20.5

Growth function used in the experiment by Noussair et al. (2015).

For example, if the fishermen collectively catch 6 fish in the first period, then 2 remain, and the stock grows by 3 fish, such that 5 fish are in the pond

for the second period. To find the subgame perfect Nash equilibrium (SPNE), we backward induct from the final period. Since there is no point in leaving any fish in the pond at the end of the game, all the remaining fish will be extracted in the final period, and the four fishermen will divide the catch equally. Next, let's consider the third period. Any fish that are left in the pond will regenerate according to the growth chart given here, and this additional stock will be shared equally among the fishermen in the final period.

Given this, is it ever a best response to allow fish to remain in the pond? Imagine, for instance, that you are considering letting 1 fish remain. If you caught the fish, you'd get the fish (and the \$5 prize). If you leave it alone, that fish will turn into 2 in the final period, which will be divided equally among the four of you, and your choice to leave it gets you 0.5 fish and \$2.50. Thus, you're better off catching the fish now rather than leaving it to grow and then sharing the larger stock. This same logic applies for any number of fish you could leave in the pond—for instance, if you leave 4 fish, you'll allow the stock to grow to 8 and wind up with a share of 2, but if you had grabbed those 4 fish now, you'd be better off. Thus, the players would also choose to maximally extract in period 3. By this same logic, we see that the players extract the maximal amount in every period. Thus, the SPNE prediction is that the resource will be fully depleted in the very first period, a total of 8 fish are consumed, and the players' joint payoff is \$40. In contrast, the socially optimal series of choices is for the fishermen to extract 4 fish in each of the first three periods, which allows the resource to regenerate to 8 fish for the next period, before extracting all 8 in the final period. This would allow them to jointly consume 20 fish, for a joint payoff of \$100.

Of course, this conclusion depends on the nature of the resource and its natural regrowth process. With a different growth process, agents may find it in their interest to decide not to fully deplete the resource at the first available opportunity.

Probabilistic Destruction of a Resource

An alternative dynamic model allows for the possibility that the resource may be depleted past the point of no return, at which point it will not regenerate and is lost forever. As current usage becomes heavier, the greater the likelihood that the resource will be destroyed and cannot be accessed in the future.

To analyze this type of interaction, let's continue with our same static game example, in which each individual has 20 units of available effort, the outside option pays 2, and the CPR production function is $11X - \frac{1}{16}X^2$ (where X is the total extraction by the group). Now let's assume that the game can last a maximum of 10 periods and each individual makes an extraction decision in each period. The probability that the resource survives the period depends on the group's total extraction that period and is equal to $1 - X/240$. If the resource is destroyed, then the game ends, and there are no further periods.

Let's first look for the socially optimal extraction level in each period, working backward from the 10th period. In the 10th and final period, the socially optimal extraction level is the same as in the one-shot game: 72, which yields the maximum group payoff of $2(160 - 72) + 11(72) - \frac{1}{16}(72)^2 = 644$. Now let's consider the group-level optimization problem in the 9th period. The group's *expected* payoff for the remainder of the game is equal to the immediate payoff from the current extraction choice, plus the probability that the resource survives into the next period, times the payoff of 644 that the group would receive at that time. Let X be the group's total extraction in period 9. Then its expected payoff function starting in the 9th period and going forward is equal to

$$2(160 - X) + 11(X) - \frac{1}{16}(X)^2 + \left(1 - \frac{X}{240}\right)(644)$$

Solving for the total extraction level that maximizes this payoff, we get $X = 50.5$. Thus, the group has an incentive to further restrict its usage in the 9th period to increase the chance that the resource will be available in the 10th period. At this level of extraction, there is a 79% chance that the resource will continue to exist. The group's expected payoff once they reach the 9th period is thus $2(160 - 50.5) + 11(50.5) - \frac{1}{16}(50.5)^2 + (0.79)(644) = 1124$. Now, we back up to the 8th period. The group's expected payoff for the remainder of the game is given by the immediate payoff from extraction, plus the probability that the resource survives the period, times the 1,124 that they expect to achieve if the game continues. That is, the objective in the 8th period is to choose X to maximize

$$2(160 - X) + 11(X) - \frac{1}{16}(X)^2 + \left(1 - \frac{X}{240}\right)(1124)$$

which occurs when $X = 34.5$. Continuing in this way, we find that the group's optimal extraction path is 21.4 in period 7 and 9.8 in period 6. Backing up to

period 5, we find that the optimal extraction level is negative, and so the group maximizes its joint payoff by not extracting at all. This is true in the earlier periods as well. Thus, the group is best off leaving the resource completely untouched in the first five periods in order to guarantee its survival, and then extract with increasing abandon as they near the end of the game. The socially optimal level of extraction in each period is shown in the second column of the table in [figure 20.6](#), alongside the likelihood that the resource survives the period under this strategy.

Period	Social Optimum		SPNE	
	Extraction	Chance of Survival	Extraction	Chance of Survival
10	72	0%	128	0%
9	50.5	79%	124.7	48%
8	34.5	86%	122.9	48.8%
7	21.4	91%	121.9	49.2%
6	9.8	96%	121.4	49.4%
5	0	100%	121.1	49.5%
4	0	100%	120.9	49.6%
3	0	100%	120.8	49.7%
2	0	100%	120.8	49.7%
1	0	100%	120.7	49.7%

Figure 20.6

Socially optimal and SPNE extraction levels in a 10-period dynamic common pool resource game.

Now let's consider the SPNE extraction path over the course of the game. We'll take essentially the same approach, except that we assume that each individual i chooses x_i in each period to maximize her own individual expected payoff for the remainder of the game moving forward. In the 10th period, the symmetric Nash equilibrium extraction effort is the same as in the one-shot game: 16, for a total extraction of 128 and payoff of 56 each.

Now let's calculate i 's best response in the 9th period. Recall that when we first calculated the Nash equilibrium in the one-period game, we first canceled terms in player i 's payoff function and split the total extraction effort ($\sum_{j=1}^8 x_j$) into the effort that i can control (x_i) and his group members' extraction ($\sum_{j \neq i} x_j$), arriving at equation 20.2. To get player i 's payoff starting in period 9, we must take this single-period payoff function and add the player's

expected payoff from the remainder of the game going forward. This is given by the payoff player i anticipates receiving in the 10th period (i.e., 56) times the probability that the game continues to the 10th period (i.e., $1 - \frac{\sum_{j=1}^8 x_j}{240} = 1 - \frac{x_i}{240} - \frac{\sum_{j \neq i} x_j}{240}$). We thus have

$$u_i(x_i, x_{-i}) = 40 + 9x_i - \frac{1}{16}x_i^2 - \frac{1}{16}x_i \sum_{j \neq i} x_j + 56 \left(1 - \frac{x_i}{240} - \frac{\sum_{j \neq i} x_j}{240} \right)$$

To solve for i 's best response, we take the derivative with respect to x_i and set it equal to zero, getting: $x_i = 70.13 - 0.5 \sum_{j \neq i} x_j$. The symmetric equilibrium extraction level x thus occurs when $x = 70.13 - 0.5(7)(x)$, or $x = 15.59$.

Given this, what is the expected payoff starting in period 9? The payoff from each player choosing $x = 15.59$ in period 9 is 58.8. The probability that the game will continue to the 10th period is $1 - \frac{(8)(15.59)}{240} = 0.48$, in which case the player will receive an additional 56. Thus, the player's expected payoff if he makes it to period 9 is $58.8 + 0.48(56) = 85.7$. We can now back up to the 8th period and calculate the best response of the player, knowing that if the resource survives the period, he can expect to get 85.7 in the remainder of the game. Rather than go through all the arithmetic for all 10 periods, let's calculate the symmetric Nash equilibrium extraction in any general period, assuming that we've already calculated the payoff that the player can expect if the resource survives the period. Let y be the player's expected payoff from the game continuing past the current period. Then player i 's expected payoff from the remainder of the game is

$$u_i(x_i, x_{-i}) = 40 + 9x_i - \frac{1}{16}x_i^2 - \frac{1}{16}x_i \sum_{j \neq i} x_j + y \left(1 - \frac{x_i}{240} - \frac{\sum_{j \neq i} x_j}{240} \right)$$

Solving for i 's best response, we get $x_i = 72 - \frac{8}{240}y - 0.5 \sum_{j \neq i} x_j$. The symmetric equilibrium extraction level x occurs when $x = 72 - \frac{1}{30}y - 0.5(7)(x)$, or $x = 16 - \frac{y}{135}$. Note that there is no possibility of the game continuing past the 10th period, so $y = 0$ in period 10, and we have $x = 16$. So our equation passes this initial sanity check. When we calculated the equilibrium extraction in period 9, we had $y = 56$ and we found $x = 16 - \frac{56}{135} = 15.59$. Our answer from period 9 thus checks out as well. In the 8th period, we found that each player can expect $y = 85.7$ if the game continues. Thus, the equilibrium extraction level in the 8th period is given by $x = 16 - \frac{85.7}{135} = 15.37$.

Following this process, we can compute the total group level extraction according to the SPNE, which is shown in the fourth column in [figure 20.6](#). Even in the early periods, the SPNE does not predict that the users will exhibit much restraint. If all users act in their narrow, individual self-interest, then the likelihood that the resource survives any given period is essentially a coin flip.

Governing the Commons

Up to this point, the conclusions of this chapter have been highly pessimistic: if users with free access to a common resource act independently and in their own self-interest, they will overexploit the resource, perhaps beyond its capacity to regenerate and survive into the future. This pessimistic prediction is at odds with the real, long-term success that many societies across the globe have had in sustaining their natural resources. Nobel laureate Elinor Ostrom devoted her career to researching the various formal and informal arrangements that communities have developed to govern and sustain their common resources. Around the world, users of shared resources have developed an incredible range of local rules that have proved successful for governing the communities' common resources. For example, Ostrom described visiting the Middle Hills of Nepal and encountering a single cow enclosed in a pen in the middle of the village. Her hosts explained that the animal was in "cow jail." According to the local arrangement, three other farmers could find a user in violation of the irrigation system policy, in which case they were authorized to bring the cow to the center of the village until the owner paid a fine. In the meantime, the cow's milk was distributed to the other villagers and the offending farmer suffered the shame of his cow being publicly jailed. Ostrom's work directly challenged the "tragedy of the commons" conclusion that the overextraction and destruction of common pool resources was inevitable.

Let's consider how we can incorporate into our model the various institutional arrangements that have been designed to cause individuals to internalize the negative externalities that their overuse imposes on others. Possibly the most straightforward regulation mechanism that we could consider is an authority who monitors users with a certain probability and fines those who are found to extract more than the socially optimal amount. Let's consider again the main example we've been using in this chapter,

where the payoff function for player i is given by $u_i(x_i, x_{-i}) = 2(20 - x_i) + \frac{x_i}{\sum_{j=1}^8 x_j} \left(11 \sum_{j=1}^8 x_j - \frac{1}{16} \left(\sum_{j=1}^8 x_j \right)^2 \right)$. Now let's imagine that each user is monitored with probability p and, if caught extracting more than the socially optimal level of 9, is hit with a fine of f dollars for each unit of effort beyond 9. In this case, we consider the player's expected payoff, since there is a risk of being monitored and fined. In the event that the player chooses $x_i > 9$, we must subtract an additional term from the payoff function equal to the expected fine: $pf(x_i - 9)$.

Solving for i 's best response, we get $x_i = 72 - 0.5 \sum_{j \neq i} x_j - 8pf$. In other words, the best response function for the player shifts downward, such that for each level of extraction completed by the other players, player i 's best response is to extract $8pf$ units less than she would have in the absence of the monitoring regime. Solving for the symmetric Nash equilibrium, we get $x = 16 - \frac{8pf}{4.5}$, indicating that the fine successfully reduces the equilibrium extraction downwards from 16. The higher the probability of being monitored, or the more harsh the punishment if caught, the less the individual will extract. To move the Nash equilibrium to the social optimum of 9, we can impose a combination of monitoring probability p and fine strength f such that $pf \geq 3.93$. For instance, if the probability that each person is monitored is $\frac{1}{2}$, then a fine of 8 for each unit extracted beyond 9 would incentivize an extraction rate of 9 units each.

In the general version of this game, the monitoring regime moves the symmetric Nash equilibrium from $\frac{1}{n+1} \frac{a-c}{b}$ to $\frac{1}{n+1} \frac{a-(c+fp)}{b}$ (or the social optimum, whichever is greater). Recall that c is the opportunity cost of extraction—what the user could earn by putting his effort elsewhere. The monitoring and fining regime essentially adds an additional expected cost for extraction to this existing opportunity cost, which is equal to the expected per-unit fine that one's extraction choice will incur.

While this monitoring and sanctioning regime can reduce overextraction, one downside is that an external authority must be employed, and whenever anyone is fined, wealth is drained from the community of users. One alternative is a system called “Carte di Regola,” which has been used in the Italian Alps since the 13th century to maintain commonly owned forests and pastures. Villagers agreed in advance to a certain set of rules, such as what constitutes overuse of the resource and what the fine would be for overusing. Villagers could monitor their fellow users at a cost to themselves and bring

violators to court. If the user was found to have violated the rules, then he paid a fine, at least part of which was awarded to the person who caught him. Thus, the villagers (who likely have more information about overuse by their neighbors than an external authority) are incentivized to monitor each other, and when someone is caught violating the rules, money simply changes hands among group members rather than this wealth being lost to the group.

Consider the same static CPR game that we have been looking at throughout. When we add the Carte di Regola system, the game now has two distinct steps. Users must first choose their extraction level, and then they observe the total extraction for the group and choose whether to monitor one or more of their group members. Let's say that the cost of monitoring is given by k , and any user caught extracting more than some level λ is slapped with a fine, f , for each unit in excess. As with the external monitoring regime, the best response function for each individual shifts down if she believes that there is a positive probability that she will be fined. Unlike the external monitoring regime, this probability is not fixed—it is based on the monitoring choices of the other group members. When is monitoring a best response? Only when group members believe that the sanctions they will collect are greater than the cost of monitoring. This occurs when $f(x_j - \lambda) > k$, where x_j is the effort of the target of their monitoring. In equilibrium, a player will choose to monitor another player j if and only if $x_j > \frac{k}{f} + \lambda$.

Thus, the symmetric Nash equilibrium occurs when each member chooses to extract x^* units and monitor with probability p^* , such that the following two equations hold simultaneously:

(1) $x^* = \frac{1}{n+1} \frac{a-(c+p^*f)}{b}$ (i.e., extraction rate x^* is a best response to each of the others choosing effort x^* and to probability p^* of being monitored)

(2) $p^* = \begin{cases} 0 & \text{if } x^* \leq \frac{k}{f} + \lambda \\ 1 & \text{if } x^* > \frac{k}{f} + \lambda \end{cases}$ (i.e., the monitoring choice p^* is a best response to the other group members' extraction levels, x^*)

This institution was first analyzed theoretically and in the lab by Casari and Plott (2003). The authors considered two different versions of the game, which differed in the strength of the sanctions. Let's look at these examples. First, let $k = 7$, $\lambda = 9$, and $f = 1$. Stated in words, it costs 7 to monitor a user, and if monitored, the user must pay 1 for each unit of effort he exerted in excess of the social optimum of 9. Is it in anyone's interest to monitor in this

situation? Monitoring is profitable only if the other user chooses x_j such that $7 < 1(x_j - 9)$, or $x_j > 16$. However, since the Nash extraction level even without monitoring is 16, no one has the incentive to ever choose an effort greater than 16, and so monitoring cannot be profitable in equilibrium.

In the second example, $k = 7$, $\lambda = 7$, and $f = 4$, and this is a system that moves the Nash equilibrium to the social optimum. In this case, it is in a user's interest to monitor if she expects the target of her search to extract at least 8.75 and, given that a player expects to be monitored, his best response is to extract $x = \frac{1}{n+1} \frac{a-(c+pf)}{b} = \frac{1}{8+1} \frac{11-(2+(1)(4))}{\frac{1}{16}} = 8.89$. Notice that the agreement in place among the users specifies that they can be fined if they exert effort greater than 7 (i.e., 2 units less than even the social optimum). However, this is necessary to incentivize agents to monitor each other for extraction levels that are even slightly in excess of the social optimum, which is in turn necessary to keep extraction at the socially optimal level.

Behavior in Games

Over-Overextraction

The earliest experimental work on common pool resources was conducted by Elinor Ostrom, Roy Gardner, and James Walker (1994). Participants were assigned to groups of eight, and each person in the group was given the same amount of tokens (either either 10 or 25). Each person was told that he could choose how to allocate his tokens across two different markets. Tokens invested in market 1 (private consumption) provided a guaranteed return of 5 per token. If they invested in market 2 (the CPR), the players were told that they'd receive a share of the output equal to the percentage of the total group investment that they personally invested. The CPR output was equal to $23 \sum_{j=1}^n x_j - 0.25(\sum_{j=1}^n x_j)^2$, which was presented to them in an abbreviated table that showed the returns for various levels of group investment. The Nash equilibrium predicts that total group investment in the CPR will be 64, while the social optimum is 36.

What happened? The results are graphed in [figure 20.7](#). The solid horizontal line at 36 shows the social optimum, while the dashed line at 64 shows the Nash equilibrium extraction. The solid black circles (connected with a black line) plot the group extraction level (averaged over all groups) in each period when participants had 10 tokens each. Notice that total extraction is actually remarkably close to the Nash equilibrium of 64. (However, at the

individual level, very few people actually contribute the equilibrium of 8.) This is in a marked contrast with other social dilemma games described in the previous two chapters, where behavior is generally between the self-interested Nash prediction and the cooperative outcome that is best for the group as a whole. In fact, extraction regularly *exceeds* the Nash equilibrium. Note that between the range of 36 and 64 (the two horizontal lines), there is tension between what is best for the group and what is best for the individual; but extracting above 64 is bad for both the group and the individual. Participants in this experiment are not only extracting more than what is optimal for the group—they're often extracting even more than what is optimal for them individually. This pattern is further exaggerated when participants had 25 tokens to use, as shown by the hollow squares connected by the dashed line in the figure. (Note that only 20 periods were conducted in this version, which is why the line stops abruptly.) Although the predicted extraction remains the same when people are given more tokens, they extract more, and exceed the Nash prediction in most periods.

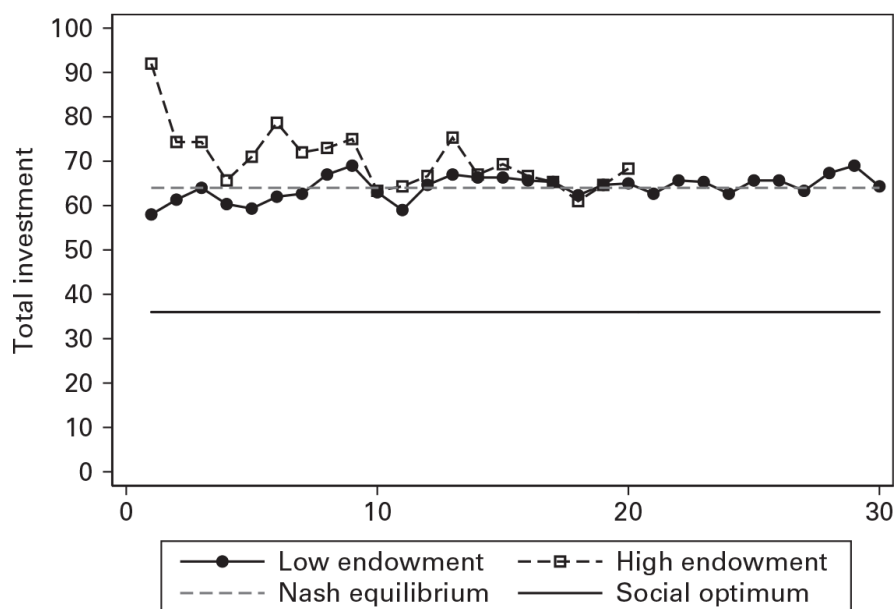


Figure 20.7

Group-level extraction in a CPR game (data from Ostrom et al. 1994).

So what is going on here? Why do we see such different behavior in the CPR game compared to the public goods game? One immediate difference in the two games is that the Nash equilibrium in the linear public goods game

(discussed in the previous chapter) is zero, on the boundary of the strategy set. Since players cannot contribute less than zero, any deviation from Nash play must necessarily be in the direction of cooperation—being less cooperative than the Nash prediction simply wasn't an option. In the CPR game, the Nash equilibrium is not in dominant strategies, its calculation is far less obvious, and participants are free to make mistakes in both directions.

Another difference between the public goods and CPR environments is that contributions in the former provide a positive benefit to group members, while in the latter, they impose a cost. Jim Andreoni (1995) asked whether this difference in framing could drive the lower cooperation rates. He assigned some participants to play a standard public goods game in which payoffs were defined in terms of the total amount contributed to the public good and investment in the group account was described as providing a return to the other group members. Other participants were assigned to play an identical game, with the payoffs reframed in terms of the number of tokens kept in each person's private accounts, and keeping tokens for oneself was described as reducing the earnings of the others. The differences in behavior were stark. Under the positive frame, contributions started at around half of the endowment and declined to 20% by the end of the 10 periods. Under the negative frame, contributions started closer to a quarter of the endowment and dwindled to only 1% in the final period.

A final difference that could drive the dissipating cooperation comes from the subtractable nature of the CPR returns, which imply that a single individual can extract nearly all the gains from cooperation provided by others' restraint and creates more temptation to defect on cooperative group members (see Cason and Gangadharan, 2015, for further discussion). For instance, in the public goods game, an individual who switches to fully free riding while his groupmates fully contribute could typically increase his payoff by a quarter or $3/8$; an individual in the CPR game who switches from the social optimum to best responding to his groupmates extracting the socially-optimal level could more than double his payoff.

While the finding that people regularly overextract a common pool resource—even beyond the self-interested levels—is certainly bad news, Ostrom et al.'s experiments also offered hope that users could endogenously regulate their collective extraction when given the opportunity. Simply allowing participants to communicate substantially lowered extraction in the

25-token game and brought the outcome to almost full efficiency in the 10-token version. Communication combined with a punishment mechanism, in which users could pay to reduce the earnings of their group members, further improved the situation, especially when the users implemented this mechanism by group vote. Unlike in the public goods game, punishment without communication doesn't tend to have a large impact on cooperation, likely because it is more difficult for participants to calculate the optimal benchmarks and develop a common understanding of how much extraction is appropriate without the ability to communicate and agree on which extraction levels should be met with punishment.

In fact, communication has been found to work better than external regulation, despite the fact that only the latter lowers the Nash prediction. Cardenas, Stranlund, and Willis (2000) had participants in rural Colombia play several rounds of a standard CPR game. They then introduced either communication or a relatively weak system of monitoring and punishment, which reduced the Nash extraction level, but not all the way to the social optimum. Because users under the monitoring regime could be punished if they extracted more than the optimal level, this system also had the advantage of communicating the social optimum to the participants. They found that both communication and the monitoring system initially lowered extraction, with the monitoring system having a much stronger effect when first implemented. However, it quickly backfired; after just a few rounds, people were extracting *more* than they did in the initial rounds without any regulation. This finding is consistent with a large body of literature in behavioral economics demonstrating that weak fines can lead to worse outcomes, by undermining people's intrinsic motivation to follow a rule or cooperate, and points to the value of societies governing their own common resources rather than relying on weakly enforced external regulation. Casari and Plott (2003)'s laboratory test of the Carte di Regola mechanism found that even when the sanctioning rule was weak—and users had no financial incentive to inspect others' extraction in equilibrium—about half of extractions were inspected and extraction decreased significantly; when the sanctioning rule was strong, nearly every decision was inspected and highly efficient extraction levels were maintained.

Evidence-Based Adaptations of the Standard Model

Proposed behavioral explanations for the patterns we have seen include spite, conformity, and inequality aversion. Let's focus on the inequality aversion model, which can explain the overextraction in Ostrom et al.'s baseline experiments, as well as the remarkable success of communication and sanctions, as demonstrated by Falk, Fehr, and Fischbacher (2002). Note that we discussed the two-player inequality aversion model in chapter 14. We'll now describe the n -player version, so it is worth skimming the next paragraph even if you recall the model from the earlier chapter.

Fehr and Schmidt (1999)'s *inequality aversion* model assumes that players experience psychological disutility when their own monetary payment differs from the other members of the group. There are two types of inequality that could affect a player's happiness with the outcome. *Advantageous inequality* occurs when her monetary payment is greater than the payments of others in the group, while *disadvantageous inequality* occurs when others receive more money than she does. Note that in the CPR game, the same player could simultaneously experience advantageous inequality (relative to those who extract less than she does) and disadvantageous inequality (relative to those who extract more), and both potentially cause her distress. Players may differ in how much they dislike each type of inequality, but it is generally assumed that disadvantageous inequality is more painful.

Formally, consider an n -player game, with a vector of earnings given by $(\pi_1, \dots, \pi_n)$. In the standard CPR game, π_i is simply the material payoff that player i receives according to the standard payoff function (such as equation (20.1)). Under the Fehr-Schmidt model, however, there will be additional terms in the player's final payoff function that capture the psychological disutility she experiences from advantageous and disadvantageous inequality. Specifically, player i 's payoff is given by the following expression, where α_i and β_i are individual-specific parameters reflecting player i 's distaste for disadvantageous and advantageous inequality, respectively:

$$u_i(\pi_1, \dots, \pi_n) = \pi_i - \frac{\alpha_i}{(n-1)} \sum_{j, \pi_j > \pi_i} (\pi_j - \pi_i) - \frac{\beta_i}{(n-1)} \sum_{j, \pi_i > \pi_j} (\pi_i - \pi_j)$$

where $\alpha_i \geq 0$, $\beta_i \geq 0$, $\beta_i \leq \alpha_i$.

The first term is simply the player's monetary earnings, while the second and third terms incorporate the player's disutility from disadvantageous and advantageous inequality, respectively. The second term comes into play only

when there are players who earn more than player i (i.e., when there are players j such that $\pi_j > \pi_i$, as denoted under the summation). In this case, we sum up all the extra dollars that are earned in excess of what player i is receiving and average them over the other $n - 1$ players. For each extra dollar that is being earned, on average, compared to player i , i 's payoff decreases by α_i . Similarly, the third term comes into play only when there are players who earn less than player i (i.e., when there are players j such that $\pi_i > \pi_j$), and this term captures the psychological cost of earning more than others. For each person earning less than player i , we sum up the amount by which i 's payment exceeds theirs and divide by $n - 1$ to get i 's average advantageous inequality. For each extra dollar that this average advantageous inequality increases, player i 's payoff decreases by β_i . If, as in the standard model, the players care only about their own personal earnings, not how their earnings compare to others in the group, then α_i and β_i are zero and the utility function collapses to $u_i(\pi_1, \dots, \pi_n) = \pi_i$.

How does inequality aversion change the Nash equilibrium predictions in the CPR game? Let's consider the equilibrium predictions for the setup that was used in Ostrom et al.'s experiments. To keep things simple, let's assume that all seven other players each choose the same extraction level x_j (as would be the case under any symmetric Nash equilibrium), and calculate player i 's best response to any x_j . The graph in [figure 20.8](#) shows the best response function for a self-interested player (dashed line) and for an inequality-averse player (solid line). The inequality-averse best response function is shown for parameters $\alpha_i = 4$ and $\beta_i = 0.6$, but the shape holds generally. The dotted line traces out the outcomes in which player i 's extraction matches the others.

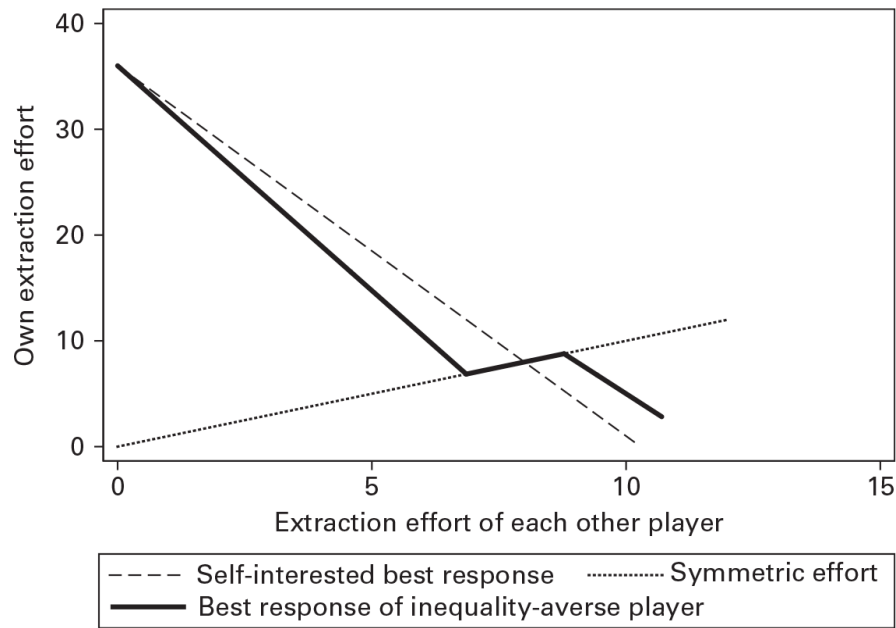


Figure 20.8

The best response function of a self-interested player (dashed line) and an inequality-averse player (solid line).

Notice that the inequality-averse player's best response function can be broken down into three distinct parts. For low levels of extraction by the others, she extracts less than a self-interested player, as can be seen by the solid line falling below the dashed line on the left side of the graph. Since the player dislikes earning more than others, she doesn't take full advantage of the other players' restraint and instead opts to extract less than the self-interested amount so as not to impose as much harm on them. The slope of this portion of the line depends on β : the more that the player dislikes advantageous inequality, the more restraint the player exercises and the more steeply the best response will decline over this region.² (If $\beta_i = 0$, then the best response will simply trace the self-interested best response function when $x_i > x_j$.) At the point where the black line hits the dotted diagonal line, the player's best response is to match the effort of the others, and the advantageous inequality disappears.

For relatively high levels of extraction by the other users, players generally find it in their self-interest to extract less than the others and allow other users to earn more. However, a player who is averse to disadvantageous inequality will not be as willing to yield and will devote more effort to extraction than would be in her own self-interest, for the sole purpose of depleting the

resource for the others. Hence, the inequality-averse player's best response function is above the self-interested player's best response in this region. While this excessive extraction lowers the player's own earnings, it imposes even greater costs on the heavier users. She is therefore willing to sacrifice her own earnings somewhat by overextracting in order to drag their earnings down closer to her own. The extent that she is willing to do this depends on her distaste for disadvantageous inequality: for higher levels of α , the player will extract more compared to the self-interested best response, and the slope of this portion of the dark line will be shallower.

Finally, between these two downward-sloping segments, the best response function traces the dotted line diagonal, as the player prefers to simply match the extraction of the others rather than face the disutility caused by earning more or less than they do. If all players have the same α and β parameters, then all will have the same best response function. In that case, any point along this upward-sloping portion of the best response function will be a symmetric Nash equilibrium. This is because this portion of the best response function overlaps with the diagonal dotted line, where all the players are choosing the same x . Thus, any extraction level on this portion of the curve is a mutual best response: if all seven other users choose x , then the eighth user prefers to choose x as well.

Therefore, the inequality-aversion model predicts a range of Nash equilibria, which surrounds the self-interested Nash equilibrium of 8. When users are averse to advantageous inequality ($\beta > 0$), there will be Nash equilibria in which each person extracts less than the standard prediction of 8, and so some level of cooperation can be sustained if they are able to coordinate on one of these equilibria. This provides at least a partial explanation for why communication often has such a large impact on cooperation: it enables players to coordinate on a more efficient equilibrium. In addition, when users are averse to disadvantageous inequality ($\alpha > 0$), there will be Nash equilibria in which each person extracts more than 8, consistent with some of the experimental results in groups without communication. Notice that the range of Nash equilibria (i.e., this upward-sloping portion of the best response function) widens as α and/or β increase. Specifically, when β increases, more cooperative outcomes are included in the set of Nash equilibria, and when α is larger, more spiteful outcomes can be equilibria.

What if the players differ in their preference parameters? Notice that an effort level x can be a symmetric Nash equilibrium only if x falls on the upward-sloping portion of the players' best response functions, such that all eight users best respond to x by choosing x themselves. Therefore, the lower limit of the range of Nash equilibria is determined by the smallest β among the players in the group, (which we'll write $\min(\beta_i)$) and the upper limit is determined by the smallest α (which we'll write $\min(\alpha_i)$). Specifically, the range of symmetric Nash equilibria is given by: $\left[\frac{(a-c)(1-\min(\beta_i))}{b(1+n(1-\min(\beta_i)))}, \frac{(a-c)(1+\min(\alpha_i))}{b(1+n(1+\min(\alpha_i)))} \right]$. In other words, if a single player in the group is self-interested ($\min(\alpha_i) = \min(\beta_i) = 0$), then the only symmetric equilibrium is the standard prediction of 8. In general, the group would be able to cooperate only as much as the person with the weakest aversion to advantageous inequality will allow. If they attempt to sustain an extraction level on the downward-sloping portion of this person's best response function, she would prefer to defect and take advantage of their cooperation for her own gain. This is the case even if communication is possible (unless communication itself changes the players' underlying preferences).

However, this model can also provide an explanation for why communication combined with sanctions is effective. Without sanctions, groups can only rely on their members' distaste for advantageous inequality to prevent them from selfishly extracting the gains from their group members' cooperation. With sanctions, groups may be able to leverage some group members' distaste for disadvantageous inequality to deter others from extracting more than them. In the typical sanctioning setup, users can pay some amount to reduce a group member's earnings by two to four times that amount. Although narrowly self-interested players would never be willing to sanction at a cost to themselves, those who are sufficiently averse to disadvantageous inequality would be willing to pay to bring down a heavy user's payoff so that it is closer to their own. If there are enough users with sufficiently high α parameters, their group members will anticipate that extracting more than these users will be met with an earnings reduction. As a result, people will be financially deterred from extracting more, even if they are not sensitive to advantageous inequality themselves.

Thus, inequality aversion or other related social preferences can provide an explanation for why communities are often able to sustain their common resources in spite of the monetary incentives to overextract.

Exercises

(1) Consider a CPR game between four players. Each player i has the following payoff function:

$$u_i(x_i, x_{-i}) = 5(50 - x_i) + \frac{x_i}{\sum_{j=1}^n x_j} \left(40 \sum_{j=1}^8 x_j - \frac{1}{4} \left(\sum_{j=1}^8 x_j \right)^2 \right)$$

- (a) What is the best response function for each player?
 - (b) Find the symmetric Nash equilibrium and the social optimum.
 - (c) Graph player i 's best response function, putting i 's extraction on the vertical axis and the total extraction by the other three players on the horizontal axis. Indicate the symmetric Nash equilibrium extraction level and the socially optimal extraction level on your graph, and label them with the appropriate numerical value.
 - (d) Now imagine that the players are averse to inequality. Specifically, the preference parameters for the four players are $\alpha_1 = 2$, $\alpha_2 = 4$, $\alpha_3 = 1.667$, $\alpha_4 = 2.5$, $\beta_1 = 0.667$, $\beta_2 = 3$, $\beta_3 = 0.75$, and $\beta_4 = 1$. Find the range of symmetric Nash equilibria given these preferences.
- (2) Consider a general CPR game between n players, with the general payoff function described in this chapter. Each player has an endowment of effort, ω , and can choose how much of this effort to spend on extraction or on a different activity, which pays c per unit of effort. The production function of the common pool resource is given by $aX - bX^2$, where X is the total extraction effort by all n users, and each user receives a payoff from the common pool resource equal to the effort that they devote to extraction as a proportion of the total extraction effort.
- (a) Prove that the socially optimal total extraction is $X = \frac{a-c}{2b}$.
 - (b) Prove that the symmetric Nash equilibrium has each person spending effort $\frac{1}{n+1} \frac{a-c}{b}$ on extraction.
- (3) Consider again the game between four players described in exercise 1. Now imagine that the game may last for up to five periods. After the fifth period, the game ends with certainty. Otherwise, it may end earlier if the resource is destroyed prior to the fifth period. In each period, the probability that the resource survives (and thus the game continues to the next period) is given by $1 - X/200$, where X is the total extraction.

- (a) Imagine that the group could agree to an extraction plan in order to maximize the total group earnings. How much would the group prefer to extract in each of the five periods? (Note that the extraction level may differ across periods.)
 - (b) Under the SPNE, how much will be extracted in each period?
 - (c) What is the probability that the resource survives past the first period if the players extract according to the SPNE prediction?
- (4) Consider the following alternative model of CPR extraction that has been used in the literature. There are five players and each player i chooses an extraction level x_i between 1 and 10. Each unit that has been extracted can be sold for a price of 110. However, extraction is increasingly costly. For each player i , the payoff is given by

$$u_i(x_i, x_{-i}) = 1000 + 110x_i - 2 \left(x_i + \sum_{j \neq i} x_j \right) - 3x_i \left(x_i + \sum_{j \neq i} x_j \right)$$

- (a) How much does each player extract according to the symmetric Nash equilibrium?
 - (b) Now imagine that each user is monitored with probability $\frac{1}{2}$. If any user is found to extract more than a single unit, then that person is fined 72. What is the Nash equilibrium extraction?
- (5) Consider the game used by Ostrom et al.: eight players simultaneously choose how much effort to exert, and each player i earns the following payoff:

$$u_i(x_i, x_{-i}) = 5(25 - x_i) + \frac{x_i}{\sum_{j=1}^8 x_j} \left(23 \sum_{j=1}^8 x_j - \frac{1}{4} \left(\sum_{j=1}^8 x_j \right)^2 \right)$$

Now let's assume that this community has agreed to the Carte di Regola system. The cost of monitoring is 9. Under their agreement, any user who has been found to extract more than 3 must pay a fine of 4.5 for each unit in excess of 3.

- (a) Imagine that you expect another user to choose extraction level x_j . For which levels of extraction would you find it profitable to monitor the user?
- (b) Find the symmetric Nash equilibrium extraction, x . In order for x to be a Nash equilibrium, it must be a best response to the other players

choosing x and, if x is greater than the value you found in part (a), to being monitored.

(6) Imagine that there are n lobstermen and each player i decides how many traps, t_i , to set. As more traps are set, the lower the catch from each trap. Specifically, the number of lobsters that player i catches is $t_i(1250 - t_1 - t_2 - \dots - t_n)$. Let's assume that the cost of setting and managing each trap is 100 and the per-lobster payoff is 2.

- (a) What is the payoff function for player i ?
- (b) Under the symmetric Nash equilibrium, how many traps does each lobsterman set?
- (c) As n increases, the total number of traps approaches what number?
- (d) What is the socially optimal number of traps? That is, if the lobstermen were to form a single company, how many traps would they choose to set to maximize their total payoff?

¹For example, imagine that your effort is 5, while the effort of each of the other 7 parties is 10. The total effort is thus 75, and the yield is 820. Your effort comprised 5/75, or 1/15, of the total extraction effort, so you personally receive a return of approximately 55 from your fishing endeavors. In addition, you earn a return of $2 \times (20 - 5) = 30$ from your other efforts, for a total profit of 85.

²How did we calculate the portion of the best response function when $x_i > x_j$? In this case, player i 's payoff is given by i 's earnings minus β_i times (i 's earnings minus j 's earnings): $50 + 18x_i - 0.25x_i^2 - 0.25x_i(7)x_j - \beta_i(50 + 18x_i - 0.25x_i^2 - 0.25x_i(7)x_j - (50 + 18x_j - 0.25x_j(7x_j + x_i)))$. Using $\beta = 0.6$, taking the derivative with respect to x_i , and solving, we find $x_i = 36 - 4.25x_j$, and this line intersects with the $x_i = x_j$ line when $x_j = 6.85$.

VII

Social Choice and Voting

21

Social Choice

How can a group or society whose members have different preferences make choices that affect the group as a whole? This chapter provides an introduction to the theory of social choice. We'll see that, when there are more than two alternatives, the collective preference of the group as a whole is often not self-evident. We will examine several direct voting rules, as well as model games in which political candidates determine the group's choice through the process of competing for votes. Along the way, we will learn three of the most fundamental results in the social sciences: Arrow's impossibility theorem, which says that there is no reliable way of incorporating all members' preferences into a coherent collective preference ranking that satisfies some reasonable properties; the Gibbard-Satterthwaite theorem, which says that dictatorship is the only voting rule that guarantees that no one has an incentive to strategically misrepresent their preferences; and the median voter theorem, which establishes the conditions under which there exists an alternative that will defeat all others in a direct competition.

Imagine that you are running for office. Your job is to choose where your policy platform falls along the left-right ideological spectrum. To do so, you must select a point in the $[0, 1]$ interval, ranging from most liberal (0, at the far left of the interval) to most conservative (1). You have a single competitor who will choose a platform simultaneously. Since you are both cynical politicians, you care only about winning and not whether your chosen platform aligns with your own values. The constituents, however, do care about what platform is enacted. Each voter i prefers policy platform p_i and the further the winning platform is from p_i , the less happy he is. Specifically, voter i 's payoff if the winning candidate's platform is x is $10 - 10|x - p_i|$. Voter i 's payoff is maximized when $x = p_i$, in which case it equals 10. Otherwise, the player's payoff decreases as the difference between x and p_i becomes greater. Imagine that there are seven types of voters in your district, each equally common, and everyone will turn out to vote. The ideal platforms for the various types are given by 0.1, 0.25, 0.35, 0.4, 0.5, 0.75, and 0.85, as shown in the interval in [figure 21.1](#). What platform do you choose?



Figure 21.1

Voters' preferred platforms.

Determining a Group's Preference Ranking

In the very first chapter of this book, we discussed the rational choice paradigm for analyzing individuals' decision making. We assumed that rational agents had preferences that are complete and transitive. As a result, they could rank or order the available options from most to least preferred, and there wouldn't be any weird cycles like $a > b > c > a$.

Now imagine that a group of people needs to make a *collective decision*. For example, perhaps a family of five needs to choose where to go to dinner, a hiring committee needs to choose which candidates they will give job offers, a community needs to decide how to allocate a budget surplus, the Supreme Court needs to rule on a case, or a nation's citizens need to elect a president. How can we take the preference rankings of each individual and aggregate them into a rational preference ranking for the group as a whole? This question is the domain of *social choice theory*, and the answer is surprisingly unsatisfying—so please prepare yourself now for disappointment.

When There Are Only Two Alternatives

To begin, let's imagine that there are only two alternatives, a and b , and each person in the group has strict preferences over these two options. How should we determine which alternative the group prefers?

The method that probably comes to mind most immediately is *majority rule*: whichever alternative is preferred by more group members is the group's preference. Other than being simple and intuitive, is there anything in particular to recommend this method? And what makes certain methods of aggregating a group's preference over two outcomes better than others, anyway?

For one, we may wish to treat all group members equally rather than give special priority to certain people's preferences. This condition is called *anonymity*: the votes aren't treated differently depending on who cast them. For instance, imagine that you and a friend each expressed a preference for a or b , such as by each casting a vote for your more preferred alternative. If the

two of you swapped ballots before submitting them, we wouldn't want the outcome to change. That would suggest that one person's vote carried more weight than the other's. Similarly, we may not wish to give priority to one *alternative* over the other. Imagine that everyone were to reverse their preference, such that everyone who preferred a now prefers b and vice versa. Then we would expect the group's preference to flip as well, a condition called *neutrality*. Finally, it probably makes sense that increasing the number of people who prefer a should never cause the group's preference to switch from a to b , a condition called *positive responsiveness*. If we agree that these three properties are essential, then that is equivalent to agreeing that majority rule is the only reasonable method for determining the group's preference. In 1952, Kenneth May proved that majority rule is the only means of generating a unique preference among two alternatives in a group with an odd number of members that simultaneously satisfies the anonymity, neutrality, and positive responsiveness properties.

So long as we are limited to two alternatives, determining a reasonable preference for the group as whole is thus straightforward and intuitive. The challenge arises as soon as we have three or more alternatives.

Condorcet Winners and Cycles

Consider a finite set of alternatives, $X = \{1, \dots, |X|\}$, where $|X|$ denotes the number of elements in the set. Let $N = \{1, \dots, n\}$ be a set of individuals, each with a complete and transitive preference ordering over X . A *preference profile* is an n -tuple of complete and transitive preference orderings $\rho = (\succ_1, \dots, \succ_n)$, which describes the preferences of all the individuals over the set X . In other words, for any two elements $a, b \in X$ and any agent $i \in N$, $a \succ_i b$ indicates that i strictly prefers a to b .

For example, consider a group of three friends who are trying to decide where to grab lunch before class, and there are four sufficiently speedy alternatives to choose among: the dining hall (D), a taco truck (T), the campus grill (G), and convenience store sushi (S). The preference profile is shown in [figure 21.2](#). For instance, player 1's preference ranking is $S \succ_1 T \succ_1 G \succ_1 D$.

Player 1	Player 2	Player 3
S	T	G
T	G	T
G	D	S
D	S	D

Figure 21.2

Preference profile 1.

How should the friends decide what to do for lunch? We would like to take this preference profile and use it to figure out a single preference ordering for the group as a whole. We will call this preference ranking the *social ordering*, and we will denote the group's preference by $\succ$. We will call the specific method for aggregating the individuals' preferences into a social ordering the *preference aggregation rule* as defined in definition 1.

Definition 1 (Preference aggregation rule) *A preference aggregation rule is a function that assigns a social ordering to any preference profile.*

In the case of our three friends, the simplest rule for determining the group's preference would be *plurality rule*: each person announces their top choice and the option with the most votes wins.¹ However, this essentially gets them nowhere. Player 1 votes for S, player 2 votes for T, and player 3 votes for G. Other than figuring out that no one likes the dining hall very much, they are no closer to figuring out the group's preferred lunch spot.

We know that majority rule works well for determining the group's preference over only two options, so maybe our friends can figure out the social ordering by comparing each pair of alternatives. Of course, the preference aggregation rule can't read each person's mind and ascertain that person's true preference over each pair. For now, we will assume that everyone votes *sincerely*—that is, players reveal their true preference and don't attempt to game the system. In this case, we can have them conduct a sort of round-robin tournament of pairwise majority-rule votes among each pair of alternatives. For instance, when sushi is pitted against tacos, sushi receives one vote (player 1), while tacos receives two votes. The group thus concludes that $T \succ S$. The outcomes of each of the pairwise votes are shown in [figure 21.3](#). Based on the series of votes, the group determines its preferences between each pair of options, as shown in the final column. This produces a

complete and transitive social ordering, such that $T \succ G \succ S \succ D$. The taco truck stands at the top of this list because it beats all the other alternatives in a head-to-head contest. We call an alternative that always beats all others in pairwise contests a *Condorcet winner*, named after the Marquis de Condorcet, a French mathematician studying voting processes in the 18th century. By the same token, a *Condorcet loser* is an alternative that loses to every other alternative in a pairwise majority rule vote (such as D in our example).

Contest	Outcome	Group Preference
S versus T	2 and 3 prefer T	$T \succ S$
S versus G	2 and 3 prefer G	$G \succ S$
S versus D	1 and 3 prefer S	$S \succ D$
T versus G	1 and 2 prefer T	$T \succ G$
T versus D	1, 2, and 3 prefer T	$T \succ D$
G versus D	1, 2, and 3 prefer G	$G \succ D$

Figure 21.3

Round-robin votes with preference profile 1.

Unfortunately, not all preference profiles have a Condorcet winner and, in these cases, pairwise majority-rule voting does not yield a rational social ordering for the group. Let's alter the preference rankings of our three friends ever so slightly, by flipping player 3's preference between T and S such that he now prefers sushi to tacos (see [figure 21.4](#)). We'll call this preference profile 2, which is shown in the table that follows. Does the group now simply have sushi as its top choice? Let's take a look at what happens when we conduct our pairwise tournament. We now have that $S \succ T$. However, it is still the case that $T \succ G$ and $G \succ S$. In other words, the social ordering contains a cycle: sushi is better than tacos, which are better than the grill, which is better than sushi. Thus, our social ordering does not satisfy transitivity. Even though we started with perfectly rational individual-level preferences, as soon as we attempted to aggregate the preferences to a group-wide ranking, we wound up with preferences that are irrational. This scenario is known as *Condorcet's paradox*: the majority's preferences may be cyclic, even if each individual's preferences are not.

Player 1	Player 2	Player 3
S	T	G
T	G	S
G	D	T
D	S	D

Figure 21.4

Preference profile 2.

Rather than requiring that each alternative face every other alternative, we could instead conduct a single-elimination tournament, such as that shown in [figure 21.5](#).

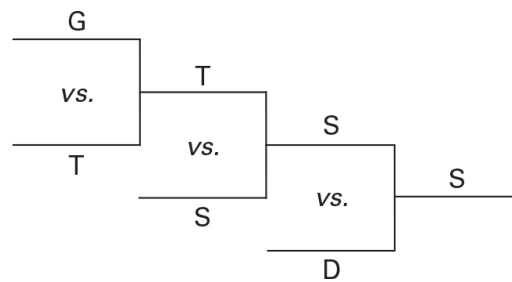


Figure 21.5

A single-elimination tournament.

In this example, option *G* first faces *T*, with the winner facing *S* and then the winner of that contest facing *D*. This process guarantees a unique winner, as you know if you've ever watched the World Cup or the National Collegiate Athletic Association (NCAA) basketball tournament. In addition, it rules out cycles: once an alternative is beaten by some other alternative *x*, then it is done; it cannot return to defeat an alternative that had bested *x* later in the tournament. If all the players cast their vote according to their true preferences shown in preference profile 2, then *T* defeats *G* in the first contest, then *S* defeats *T* in the second round and beats *D* in the final round. The good news is that we finally have figured out where the group is going to lunch. The bad news is that the outcome we found is completely dependent on the specific agenda of votes. If the initial vote featured a face-off between *T* and *S*, then *G* would ultimately win, and if the first vote pitted *S* versus *G*, then *T* would emerge victorious.

Furthermore, if our group members are game theory students, one of them may realize that, by backward inducting, it is possible to achieve a better outcome by misrepresenting their true preference in the initial vote. Let's consider the first agenda, in which G initially faces T . If player 2 casts his vote according to his true preference, he would vote for his top choice of T . But if he looked ahead, he would realize that T could never defeat S , so he will wind up with his least favorite lunch. However, his second choice, G , would go on to defeat S , and so by misrepresenting his true preferences and casting his initial vote for G , he winds up with a better outcome.

In fact, things can get even wackier than this. Consider preference profile 3 shown in [figure 21.6](#), which elevates D to player 3's second choice. Consider the same agenda of votes: T faces off against G , with the winner then facing S and the winner of that contest facing D . So long as everyone votes sincerely, T wins the first vote but is defeated by S in the second round, and finally D beats S . However, D is unanimously considered a worse choice than G . Thus, the choice of D is not Pareto efficient, since everyone is better off implementing G instead. Both players who voted for T in the first round would thus have an incentive to switch their votes to G in order to ensure that G goes through and eventually beats D in the final round.

Player 1	Player 2	Player 3
S	T	G
T	G	D
G	D	S
D	S	T

Figure 21.6
Preference profile 3.

This simple example of three friends trying to sort out their lunch plans underscores two of the main insights of social choice theory that we will explore in this chapter. First, voting rules matter: the exact way in which individual votes are aggregated can have a dramatic impact on the outcome. Second, strategy matters: often players can achieve more preferred outcomes by voting strategically, rather than in accordance with their true preferences.

Voting Rules Affect the Outcome

Consider a group of eight individuals, with the preference profile described in [figure 21.7](#). If the players were to vote sincerely under simple plurality, option D would win with three votes (compared to one vote for A and two each for B and C). However, D is not a Condorcet winner, and pairwise majority rule fails to generate a rational social ordering.

Player 1	Player 2	Player 3	Player 4	Player 5	Player 6	Player 7	Player 8
A	B	B	C	C	D	D	D
B	C	E	B	B	E	E	E
D	E	C	E	E	C	C	C
C	A	A	A	A	A	A	A
E	D	D	D	D	B	B	B

Figure 21.7
Preference profile 4.

Is there another preference aggregation rule that has more success than pairwise majority voting? Let's look at a few other common voting rules that are used when there are multiple alternatives.

Borda Count

Under the *Borda count* voting system, each voter ranks all the alternatives and submits this ranking as her ballot. An advantage of the Borda count, compared to plurality, is that it takes into account the full preference ranking of each individual rather than just each person's top choice. Each alternative is assigned points based on how highly it ranked on the ballot. The lowest-ranked alternative is assigned 1 point, while the next-highest alternative is given 2 points, and so on, with the top-ranked alternative assigned $|X|$ points. Finally, the points are summed up across all of the voters to get the Borda count for each alternative. The alternative with the highest Borda count is the group's first choice.

The Borda count is used for political elections in a small number of countries, as well as by many private companies and organizations. In sports, it is used to determine baseball Most Valuable Players (MVPs), Heisman Trophy winners, and NCAA team rankings.

Let's look for the Borda counts of each alternative in our example. Player 1 prefers $A > B > D > C > E$. If she votes sincerely, her top choice of A would receive 5 points, while B receives 4, D receives 3, C receives 2, and E receives 1. Summing up the points from each of the eight players' ballots, we have

$$A: 5 + 2 + 2 + 2 + 2 + 2 + 2 + 2 = 19$$

$$B: 4 + 5 + 5 + 4 + 4 + 1 + 1 + 1 = 25$$

$$C: 2 + 4 + 3 + 5 + 5 + 3 + 3 + 3 = 28$$

$$D: 3 + 1 + 1 + 1 + 1 + 5 + 5 + 5 = 22$$

$$E: 1 + 3 + 4 + 3 + 3 + 4 + 4 + 4 = 26$$

Thus, C wins, and our group's preference ranking is $C > E > B > D > A$.

Instant Runoff Voting

The *instant runoff voting* method also requires voters to submit a full ranking of alternatives. The procedure then works as follows: If any alternative receives more than half of the first-place votes, then this alternative wins. Otherwise, we identify the alternative that received the *fewest* first-place votes and eliminate that alternative from consideration. The voters who supported this eliminated alternative then transfer their vote to their next-most-preferred option that is still in the running. Thus, unlike plurality, people don't waste their vote by supporting an unpopular first choice—if their candidate isn't going anywhere, they simply toss their support to someone who still seems to have a shot. The process continues as long as there are more than two candidates in the running. In each step, we look for the option that is favored by the fewest people, eliminate it, and award its supporters' votes to their next best alternative. Finally, we reach two remaining alternatives, and whichever option is preferred by the majority wins. This voting method is also called *Hare voting*, after its development by Thomas Hare in 1861. Instant runoff voting is used in Australia, India, parts of the UK, and local governments around the US. It is also used to determine Academy Award winners.

Let's see how instant runoff voting works in our example. We begin by considering only the first-place votes. In the first step, our vote totals are $A(1)$, $B(2)$, $C(2)$, $D(3)$, and $E(0)$. Since E did not receive any votes, E is eliminated, but it had no supporters to redistribute. Since A received only one first-place ballot, next we eliminate A from consideration and award player 1's vote to her next-most-preferred option, B . Our vote totals among the remaining

options thus become $B(3)$, $C(2)$, and $D(3)$. Since C now has the fewest votes, we eliminate C . Players 4 and 5 can no longer cast their vote for C , so we award their votes to their next choice, B . Finally, in the two-horse race between B and D , B receives 5 votes and D receives 3 votes. B thus wins.

Approval Voting

The final method we'll consider in this section is called *approval voting*. Unlike under the Borda count and instant runoff voting, voters don't submit a ranked preference list. Instead, they essentially submit an up-down vote on each alternative, indicating whether they deem the alternative to be above the bar of acceptability. Each voter can choose which and how many alternatives he approves. Then the alternative that gets the most approval votes wins. A primary benefit of the approval voting system is its simplicity—voters simply choose which alternatives they're willing to support, and they don't need to worry about ranking them. The downside is that voters cannot communicate the intensity of their preferences. Approval voting has been used in local elections in the US and in Eastern Europe, as well as elections by various mathematical societies and by the United Nations to elect the secretary-general.

In our example, let's assume that most voters view their top four choices as reasonably acceptable, and only their last choice is untenable. However, let's say that player 1 considers only her top three choices to be above the bar. (Note that this is simply an assumption about the voter's preferences in our example—different voters may have different numbers of acceptable candidates.) If each voter in our example votes for each of the choices that she has determined are acceptable, then we have A receiving 8 votes (since everyone approves of A), B receiving 5 votes, C receiving 7 votes, D receiving 4 votes, and E receiving 7 votes. Thus, A wins. In this case, A benefits from the approval voting system not taking into account the intensity of preferences, since A was the second-to-last choice of essentially everyone, but the last choice for no one. In contrast, C was preferred to A by seven of the eight members.

We've now looked at five voting rules. Four produced a winner but yielded four different winners: D wins under plurality, C under the Borda count, B under instant runoff, and A under approval voting.

Arrow's Impossibility Theorem

The realization that voting rules can have such a dramatic impact on the group's ultimate choice naturally leads to the question: is there a reasonable, foolproof rule for aggregating individual-level preferences into a coherent group ranking? In addition to producing a complete and transitive group preference ranking, what other criteria would we expect a reasonable rule to satisfy? Consider the following four properties:

- **Universal domain.** A preference aggregation rule satisfies universal domain if it can be applied to any preference profile in which individuals have complete and transitive preference rankings. In other words, we don't put any additional restrictions on what preferences the individuals are allowed to hold.
- **Unanimity.** A preference aggregation rule satisfies unanimity if, whenever $a \succ_i b$ for all i in N , then $a \succ b$. That is, if the group members unanimously agree that a is better than b , then the group preference ranking must reflect this. While this seems eminently reasonable and straightforward, we have already seen an example in which unanimity was violated when choices were pitted against each other in a sequential, single-elimination tournament.
- **Independence of irrelevant alternatives (IIA).** A preference aggregation rule satisfies IIA if the relative positions of a and b in the social ordering depend only on the relative positions of a and b in the individuals' rankings. Essentially, this means that the presence, absence, or relative ranking of alternatives other than a and b should not alter how the social ordering ranks a and b . For example, imagine that our group of friends considered the lunch options of tacos, grill, and dining hall and decided that tacos ranked first. Then they remember that sushi is also available, and so they take a revote. What might we expect to happen? Maybe they would determine that tacos are still the best option, or maybe they decide that sushi is actually their preferred choice. But it would be odd if they suddenly decided that the grill was the best choice—that would imply that we added an alternative that should be irrelevant to the comparison between tacos and the grill and suddenly the group's preference between tacos and grill flipped. In fact, several of the examples we've already looked at violate IIA.

When we applied the Borda count to preference profile 4, we found that $C > E > A > B > D$. IIA says that, for example, the ranking of E above B should depend only on the relative rankings of E and B in each person's preference ranking, but not on where the other alternatives rank. But this is not the case. Imagine that we simply eliminated the presence of option A. Because each individual has well-defined preferences, this doesn't affect anyone's preference for E versus B, and we can simply strip A out of their rankings and reapply the Borda count rule. Yet now we find that B receives 21 points, C receives 21, D receives 19, and E receives 19 points. So our group's preference ranking is $B \sim C > D \sim E$. The presence of option A causes the group's relative preference for B versus E to flip.

- **Nondictatorial.** A preference aggregation rule is nondictatorial if there does not exist a player i such that $A \succ_i B$ implies that $A \succ B$. In other words, there is not a single agent, or dictator, whose preferences alone always determine the group's preferences.

This collection of criteria seem imminently reasonable as a minimal set of features we'd want our preference aggregation rule to satisfy. Unfortunately, we can't have them all. In the early 1950s, Kenneth Arrow proved that there is no method for aggregating the preferences of rational individuals over more than two options into a complete and transitive social ordering that satisfies all four criteria listed here (Arrow, 1950, 1951). *Arrow's impossibility theorem*, also known as the *general impossibility theorem*, is a core result in social science and is often viewed as necessitating a trade-off between coherence and fairness. The only way to guarantee that the group's preferences are always coherent is essentially to allow one individual to dictate the group's choice by imposing her own rational preference ranking on the group as a whole.

But perhaps we were too ambitious in insisting that the preference aggregation rule *always* work for all possible preference profiles. One of the criteria that Arrow found to be simultaneously incompatible with the others was universal domain—the preference aggregation rule needs to produce a coherent social ordering for any profile of individual preferences. What if we put some restrictions on the preferences our agents have? In the next section, we will think about arranging the alternatives spatially and consider a specific restriction on preferences called *single-peakedness*.

Spatial Voting

Spatial models of voting assume that all the alternatives can be arranged spatially according to certain issues or attributes. To begin, let's imagine that all the alternatives can be characterized by a single feature and lined up from left to right according to this feature. For example, if the alternatives are political candidates or platforms (as in the game at the beginning of the chapter), we can imagine arranging them ideologically along an interval ranging from most liberal to most conservative. If the alternatives are numerical policies, such as the tax rate or level of defense spending, we could arrange them from lowest to highest. Or if the alternatives are policies concerning gun control or abortion rights, we could imagine arranging them from least to most restrictive. In the example of the three friends considering their lunch options, we could think of the interval as capturing some single, continuous attribute that shapes their preferences, such as price, fanciness, how far they have to walk, or how salty the food is.

In this case, the group's collective decision problem is: pick a point along the interval that will be the choice for the entire group.

Single-Peaked Preferences

Let's say that the possible platforms of candidates for legislative office can be placed along the $[0, 1]$ interval shown in [figure 21.8](#), which ranges from most liberal (the far left of the interval) to most conservative. If you are familiar with American politics, we have included the ideological ratings of various prominent members of the 2017–2019 Senate for reference.²

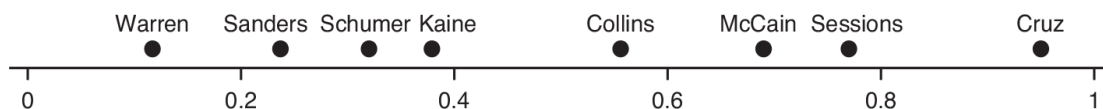


Figure 21.8

The ideological positions of various US senators.

Imagine that you had the power to choose any platform in this interval and enact it. Which point would you choose? This policy is known as your *ideal point*: the point in $[0, 1]$ that you most prefer. Now imagine that you graphed your payoff from each platform in this interval. What would that graph look like? By definition, your payoff must achieve its maximum at your ideal point. Furthermore, it probably strictly decreases as we get farther away from

that ideal platform in either direction. It would be somewhat odd if, for instance, your ideal point was, say, 0.5, and as we moved to the right, your payoff initially declined but then shot back up again. This would imply that you are a moderate who prefers extreme-right platforms to center-right platforms. In other words, the graph of your payoff over the interval is probably either strictly increasing over the whole range (if your ideal point is 1), strictly decreasing over the whole range (if your ideal point is 0), or strictly increasing up to your ideal point and then strictly decreasing (for any ideal point in the interior). As long as your payoff follows one of these shapes, we say that your preferences over this interval are *single-peaked* as described in definition 2.

Definition 2 (Single-peaked preferences) *A player i has single peaked preferences if the alternatives can be represented as points on a line and player i 's payoff is maximized at some point x^* , with i 's payoff strictly increasing to the left of x^* and strictly decreasing to the right of x^* . Formally, let $q: X \rightarrow \{1, 2, \dots, |X|\}$ be an ordering function that assigns a unique rank to each alternative in X . Player i has single-peaked preferences if there exists some $x^* \in X$ such that if $q(x_j) < q(x_k) < q(x^*)$, then $x^* > x_k > x_j$ and if $q(x_j) > q(x_k) > q(x^*)$, then $x^* > x_k > x_j$. A preference profile is single-peaked if there exists a common ordering q of alternatives for which all players have single-peaked preferences.*

Figure 21.9 depicts the scenario from the beginning of the chapter, in which seven voters each have ideal points along the $[0, 1]$ interval, given by 0.1, 0.25, 0.35, 0.4, 0.5, 0.75, and 0.85. The payoff function for each voter i with ideal point p_i is given by $10 - 10|x - p_i|$ and is graphed in this space. These voters have a specific type of single-peaked preferences called *Euclidean preferences*: they always prefer the alternative such that the Euclidean distance to their ideal point is the least.

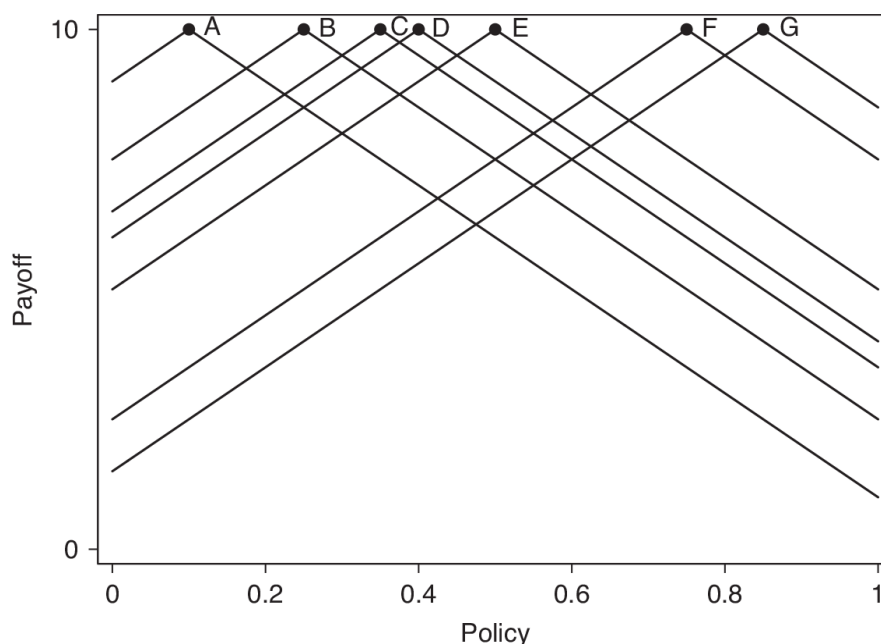


Figure 21.9

The payoffs from each possible platform for the seven votes in the opening example (labeled A through G). Each voter's payoff is maximized at their ideal point.

In this example, there is a Condorcet winner: that is, a single alternative that would beat all others in a head-to-head majority rule contest.

Question: Which alternative is the Condorcet winner?

Consider the point 0.5. Which policies, if any, could beat 0.5 in a head-to-head majority rule contest? Notice that any point to the right of 0.5 would definitely lose. However, a point just to the left of 0.5, such as 0.49, would gain the support of players A through D and win a majority. In fact, any point in the range (0.3, 0.5) would defeat 0.5 with the support of voters A through D. An alternative to the left of this range would lose the support of voter D, since it is farther from 0.4 than 0.5 is, and thus lose the majority. In other words, 0.5 can be beaten by any point closer to voter D's ideal point of 0.4. The election thus turns on the preferences of voter D, whose ideal point is the median among the group. This is true in general in this example; an alternative can defeat x only if it is more attractive to the median voter by being closer to 0.4 than x is. Of course, you can't get any closer to 0.4 than simply choosing 0.4—in which case this alternative is guaranteed to have voter D's support and thus win the majority vote.

The median voter's ideal point of 0.4 is thus the Condorcet winner. How do we know that there is no alternative x that could defeat 0.4? First, note that voter D would always support 0.4 over any $x \neq 0.4$. To win, x would thus need to receive the support of four of the other six voters. However, these voters are equally divided on whether they prefer points to the right or the left of 0.4. Any point that voters A , B , or C like better than 0.4 must fall to the left of 0.4 ($x < 0.4$), while any point that voters E , F , or G would support must fall to the right ($x > 0.4$). Thus, there is no point in this space that would be more appealing to at least four of these six people. The point 0.4 is thus unbeatable. Note that nothing about this logic requires Euclidean preferences specifically—only that preferences are single-peaked such that voters whose ideal points are to the left and to the right of the median voter are diametrically opposed in their view of which points would be better than the median.

We will more formally model elections between candidates later in this chapter, but note that this implies that choosing the median voter's ideal point in this game guarantees that you cannot lose no matter what platform your competitor adopts. Furthermore, the invincibility of the median voter's ideal point is not specific to this particular game. Duncan Black (1948) proved that the ideal point of the median voter is the Condorcet winner for *any* preference profile such that the agents have single-peaked preferences. This concept is laid out in theorem 1.

Theorem 1 (Median voter theorem [Black, 1948]) *Consider any preference profile satisfying single-peaked preferences. The ideal point of the median voter is the Condorcet winner—that is, it will defeat any alternative in a pairwise majority vote. Furthermore, pairwise majority rule produces a complete and transitive social ordering while satisfying unanimity, IIA, and nondictatorship.*

In other words, single-peakedness rules out cycles in the social ordering and eliminates Condorcet's paradox. So long as we are willing to relax the requirement of universal domain and impose the single-peakedness restriction on people's preferences, then all of Arrow's other conditions are satisfied. If preferences are not just single-peaked, but also Euclidean, then the median voter's preference ranking determines the group's ranking. In other words, we have a similar situation to the dictator lending his rational preference ordering to the group, except rather than one person being declared the dictator and

imposing his will regardless of the preference profile, the person whose rational preferences dictate the group's ranking is determined by the median.

Let's return to preference profile 1, for which we had a Condorcet winner, and the variation in preference profile 2, which caused such a headache for the decision makers. Let's somewhat arbitrarily line up the options from left to right in the order D, G, T, S . The preferences for the players in preference profile 1 are shown in the left panel of [figure 21.10](#). Notice that all three players' preferences are single-peaked given this order of alternatives, and the ideal point of the median voter is T . This aligns with our observation from earlier that T is the Condorcet winner. (Notice that if we had arbitrarily chosen to line up the alternatives in a different order, such as T, G, D, S , then the preferences of all three members would not be single-peaked. What is important is that we can find *some* common ordering, like that shown in the panel, such that everyone's preferences are single-peaked over this range.) In the second example, shown in the right panel, player 3's preference ranking over options T and S flip, such that her preferences are no longer single-peaked. Furthermore, there is no other way that we might line up these four options such that all three players would have single-peaked preferences, thus leading to our cycling problem.

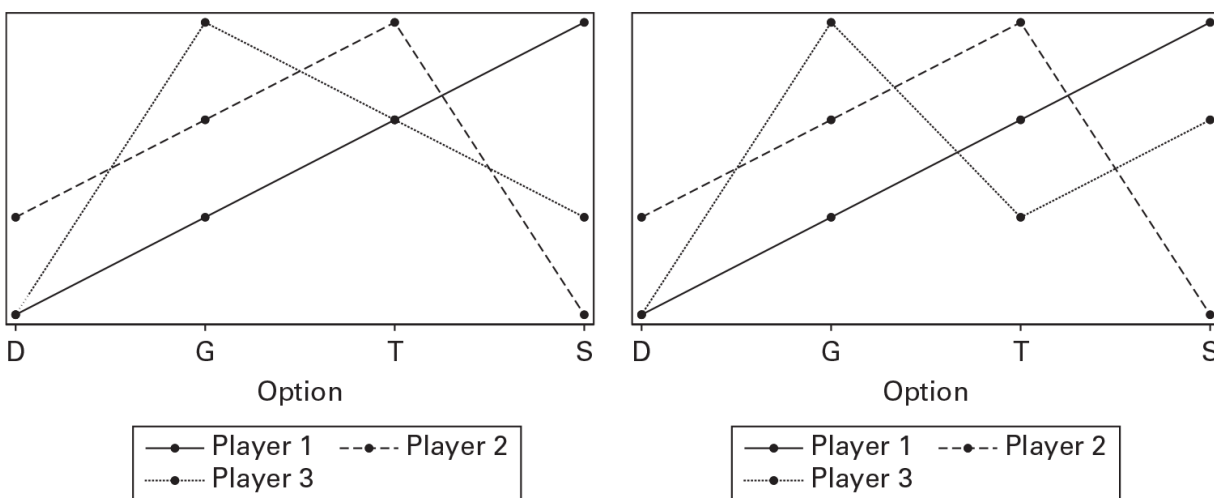


Figure 21.10

We can arrange the alternatives so that players' payoffs are single-peaked in preference profile 1 (left) but not preference profile 2 (right).

Multiple Dimensions and Radial Symmetry

While single-peakedness solves the challenge of preference aggregation uncovered by Condorcet and Arrow (among other researchers), in many cases it may be unrealistically restrictive to require that players have a shared understanding of a single attribute by which all alternatives can be aligned.

For instance, citizens who are evaluating political platforms may wish to separately consider fiscal policies and social policies. In this case, we may wish to represent the alternatives in two dimensions, where the horizontal axis describes how economically liberal or conservative the platform is and the vertical axis indicates how socially liberal or conservative it is. Each alternative is then associated with coordinates in this two-dimensional space. Now let's impose a two-dimensional version of the single-peakedness restriction on people's preferences. Just as in the unidimensional case, we assume that each voter has an ideal point in this space, and the farther an alternative is from her ideal point, in any direction, the less she likes it.

Consider an individual with ideal point a , shown in [figure 21.11](#). Imagine that you took a protractor and drew a circle of points that were equidistant from the player's ideal point, as shown by the concentric circles drawn around a in the figure. The player would be indifferent among all of the points that fall on the circle.

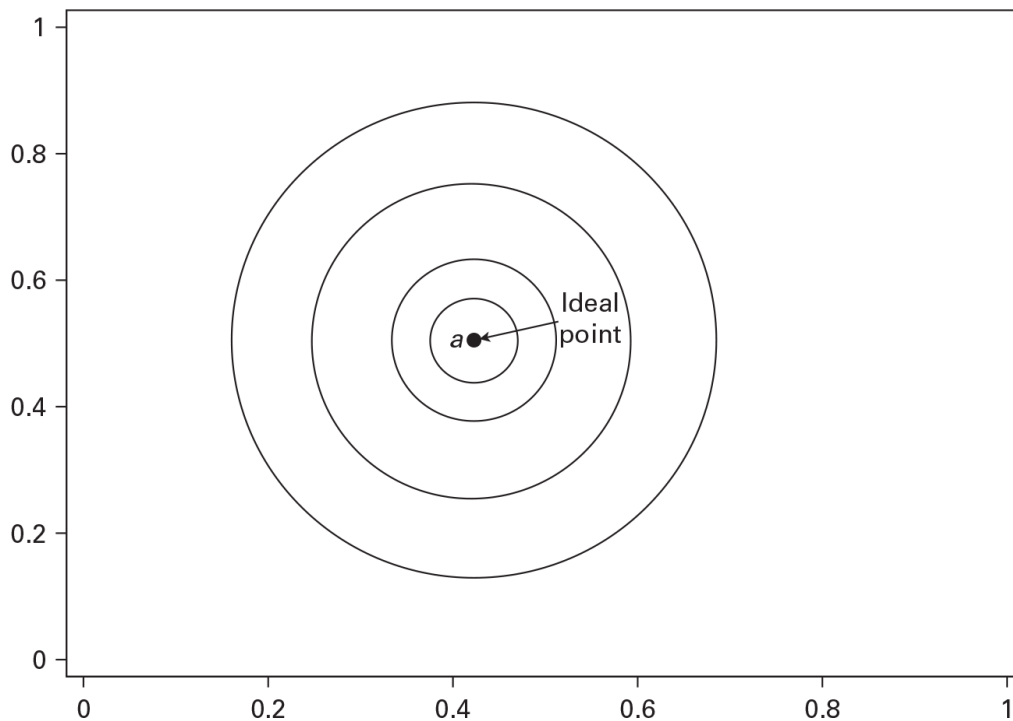


Figure 21.11

A voter's ideal point in two dimensions.

In addition, all points within the circle would be strictly preferred to those along the circle, which are strictly preferred to all points outside the circle. As we move to farther out circles, the player is strictly less happy. Thus, we could think of the space as a sort of target, with the player wishing to land on a policy as close to the bullseye (her own ideal point) as possible.

Consider a group of three players, A, B, and C, with ideal points a , b , and c shown in [figure 21.12](#), who are looking at alternative x .

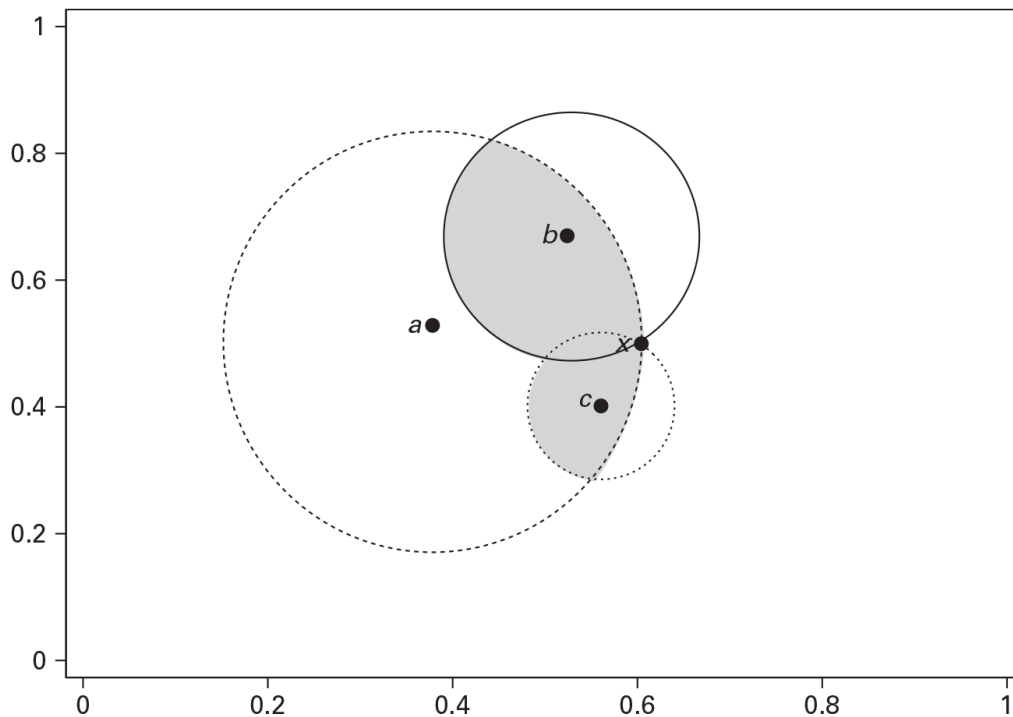


Figure 21.12

Three voters' ideal points and platforms that would defeat x .

What other alternatives could beat x in a head-to-head vote? The dashed circle traces all the points that A views as good as x , and so A would vote for any alternative on the interior of this circle over x . Likewise, B would vote for any challenger on the interior of the solid circle around b , and C would vote for any option on the interior of the dotted circle around c . The shaded region shows the alternatives that would receive the support of at least two of the three players and thus win a majority vote over x .

In a single-dimensional policy space, imposing the coherent structure of single-peaked preference profiles was sufficient to generate a Condorcet winner. Is there always a Condorcet winner in the analogous two-dimensional case?

First, let's consider a configuration of preferences such that all the voters' ideal points could be connected by a straight line, as in [figure 21.13](#).

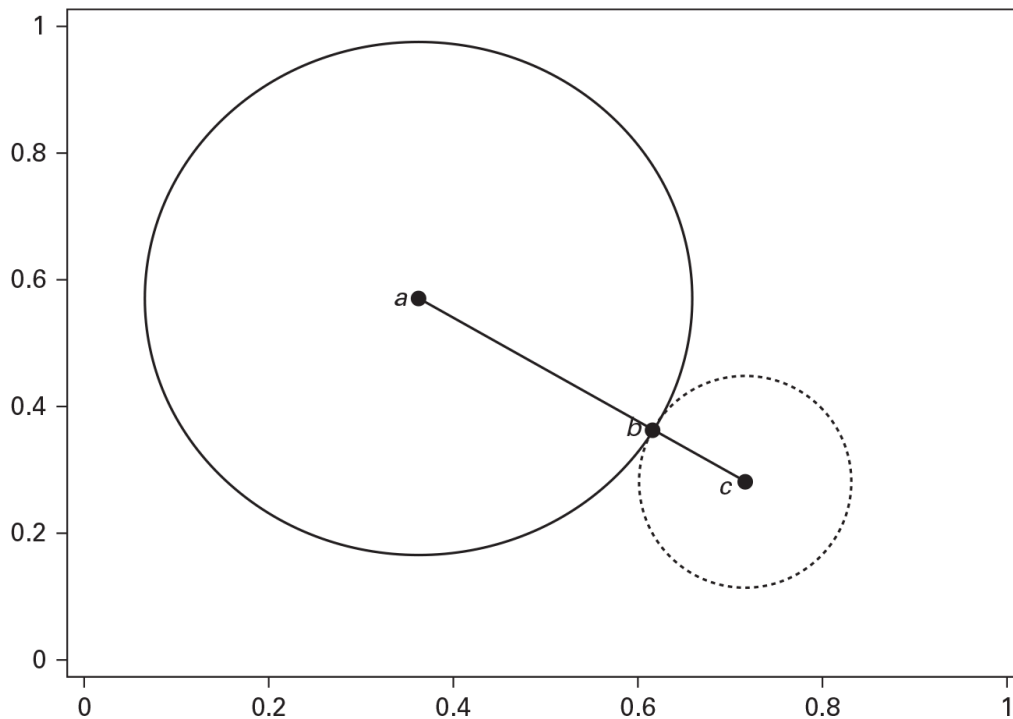


Figure 21.13

Three voters' ideal points and the platforms that would defeat b .

In this case, we're in much the same situation as the unidimensional case: b is the Condorcet winner, and the logic is similar to that of the median voter theorem. Since b is player B's ideal policy, he would never vote against it. So the only way to defeat b is to garner the votes of both players A and C. However, a and c fall on the exact opposite sides of b . As a result, the only alternatives that are closer to a than b is (and would thus win A's vote) are necessarily farther from c (and wouldn't gain C's vote), and vice versa. Therefore, there is no alternative that both A and C would prefer to b . We can also see this visually: all the points that A would prefer to b fall within the solid circle, while all the points that C would prefer to b fall within the dashed circle. Since there is no overlap, there is no point that both A and C would vote for over b .

Now let's add players to the group, but in a very specific way. For each player that we add to the group, we also add a counterpart, whose ideal point falls on the opposite side of b from her own. That is, if we were to draw a line between the ideal points of these two players, it would cross through b . This was already the case for players A and C, but now let's also add D and E, and F and G, whose ideal points are shown in [figure 21.14](#).

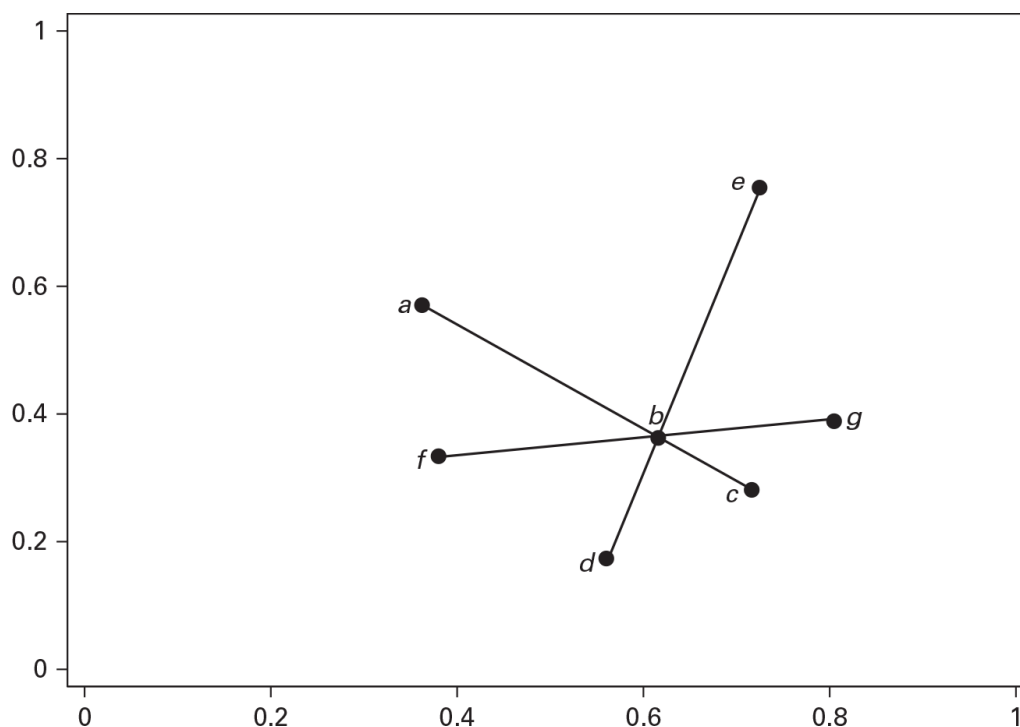


Figure 21.14
Platforms satisfying radial symmetry.

Note that the two ideal points in a pair don't need to be equidistant from b ; they just need to be situated on opposite sides, such that the players are diametrically opposed in their views of which alternatives would be an improvement. This preference profile satisfies *Plott's radial symmetry condition*, as described in definition 3.

Definition 3 (Plott's radial symmetry) *Let one player's ideal point be given by x^* . Further, assume that the ideal points of the other $n - 1$ players can be paired off, such that in each pair, the ideal points lie on a straight line with x^* and fall on opposite sides of x^* . Then the preference profile satisfies Plott's radial symmetry condition at x^* .*

In the carefully constructed preference profile shown in [figure 21.9](#), b is again the Condorcet winner. Why? To defeat b , an alternative would need to garner at least a four-vote majority. Again, B would never vote against b , so any challenger would need to receive the votes of at least four of the other six members. This means that it would need to win over the support of *both* members of at least one pair. But, by construction, members of a pair are diametrically opposed in their views of which options are better than b , so we

can never find an alternative that both members of the pair would choose over b . Thus, there can be no point in this space that would win a majority vote against b , so long as we have this carefully balanced preference profile. This insight is described in theorem 2.

Theorem 2 (Plott's radial symmetry theorem [1967]) *Consider a multidimensional spatial voting model, with an odd number of players, n , each with Euclidean preferences over alternatives satisfying Plott's radial symmetry condition at x^* . Then x^* is the Condorcet winner.*

What if we perturb this radial symmetry slightly, such as by nudging b a hair to the left as shown in [figure 21.15](#)? Then, quite frankly, all hell breaks loose. There is no longer a Condorcet winner. Worse, if we were to conduct a sequence of pairwise majority-rule votes, literally any outcome could arise depending on how we set the agenda. We can't even pare down the set of possible victors, as we did with the lunch choice example, when we found that G , T , or S could all win depending on the agenda of votes, but we could at least rule out D . Instead, any alternative could beat any other through some sequence of votes, a result known as *McKelvey's chaos theorem*, described in theorem 3.

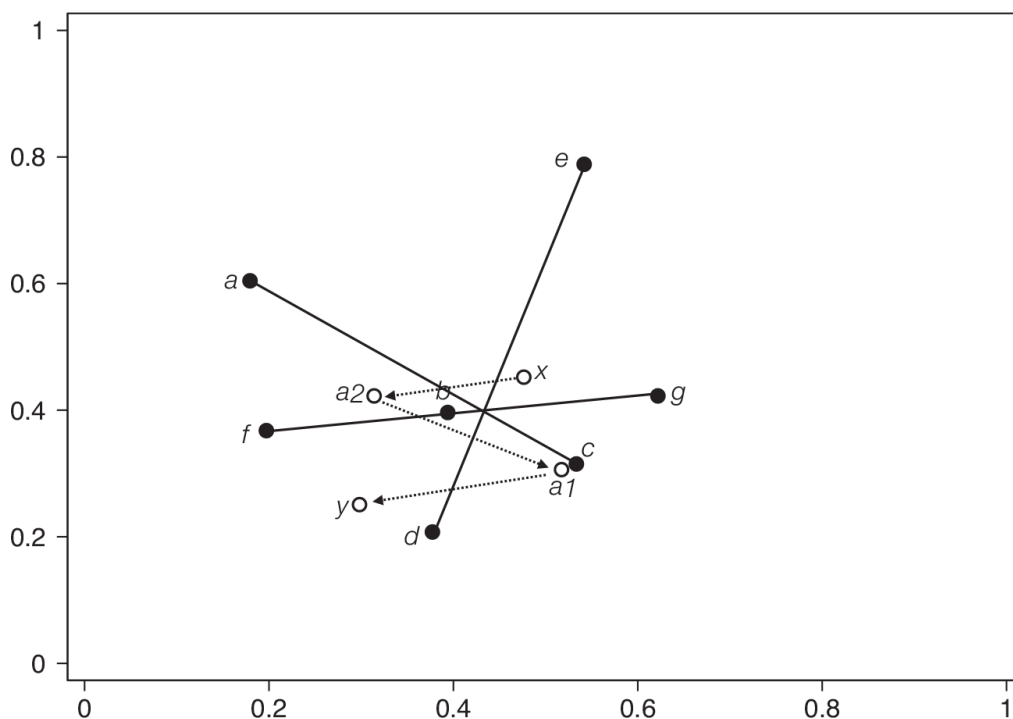


Figure 21.15

McKelvey's chaos theorem in action.

Theorem 3 (McKelvey's chaos theorem [1976]) *Consider a multidimensional spatial voting model, where players have Euclidean preferences over alternatives. One of the following is the case:*

- (a) Plott's radial symmetry condition holds, and there exists a Condorcet winner, or*
- (b) Consider any two alternatives, x and y . There exists a finite sequence of majority-rule votes by which y (ultimately) defeats x . That is, there exists a finite set of alternatives $\{a_1, \dots, a_t\}$ such that $y > a_1 > a_2 > \dots > a_t > x$.*

Let's look at McKelvey's theorem in the context of the example shown in [figure 21.15](#). Let's begin with some point, x , which we'll make the status quo: unless it is defeated by another point in a majority rule vote, x will be the policy. But perhaps the person who is setting the agenda of votes most prefers another point, y . Note that y could be any policy. If the agenda setter were to call a direct head-to-head vote between x and y , y would lose—players B, E, G, and C would all vote against it. But McKelvey's theorem tells us that there is some sequence of majority-rule votes that could lead to y 's ultimate victory. For instance, perhaps the agenda setter first calls a vote between x and a_2 . Players B, F, A, and D all vote for a_2 , which becomes the new status quo policy. Then the agenda setter calls a vote for a_1 , which defeats a_2 with support from E, D, G, and C. Finally, the agenda setter calls for a vote on y , which wins with support from D, F, A, and B. Thus, we were able to find a set of alternatives $\{a_1, a_2\}$ such that $y > a_1 > a_2 > x$.

Thus, as soon as we add a second dimension to the issue space, the reliable coherence of majority rule that we gained by imposing single-peaked preferences falls apart. If voters' ideal points don't follow Plott's radial symmetry requirements, then voters can be led (or misled) to adopt any outcome, provided that everyone votes sincerely.

Strategic Voting

Thus far, this chapter has focused on the conditions under which a preference aggregation rule can yield coherent a social ordering. In doing so, we've assumed that the input of this preference aggregation rule was the players' true preferences. Of course, a preference aggregation rule cannot look inside a player's head and ascertain their true feelings—it can only aggregate the preferences that players share through their votes. Up until this point, we've

assumed that players always voted sincerely, according to their true preference order. For instance, under plurality, players always vote for their top choice; under a sequence of pairwise majority-rule votes players always choose the alternative they prefer; and under the Borda count or instant runoff rules, players always submit their genuine preference ranking.

Given that this is a class on game theory, there is only so long that we can assume agents naively cast their votes according to their underlying preference ordering rather than strategically anticipating the votes of others and best responding. But could it be that voting sincerely actually is their best response?

Back in chapter 3, we learned that players in a second-price auction always have a dominant strategy to bid their own private value. Is there an analogous preference aggregation rule that similarly incentivizes the truthful revelation of one's preferences? Unfortunately, dictatorship is the only voting rule that guarantees that no one has an incentive to strategically misrepresent their preferences, a result proved independently by Gibbard (1973) and Satterthwaite (1975). We will begin by looking at several examples before formally stating the result in theorem 4.

The Problem of the Divided Majority

When citizens or pundits talk about strategic voting, they typically refer to voters' incentives to cast their ballots for one of the candidates with the best chance of winning, rather than "waste" their vote on a long-shot candidate who they prefer.

For example, consider the 2000 US presidential election, which all came down to the final vote tally in Florida, shown in [figure 21.16](#). The vote differential separating the Republican George W. Bush from the Democrat Al Gore was just 537—in percentage terms, the smallest margin of victory for any state in any presidential election.

Candidate	Vote Total
George W. Bush	2,912,790
Al Gore	2,912,253
Ralph Nader	97,488
Pat Buchanan	17,484
Others	23,095

Figure 21.16

US presidential election, Florida 2000.

In the wake of the election, Green Party candidate Ralph Nader was viewed as a spoiler candidate who cost Gore the presidency. The assumption was that Nader supporters' ideal points lay on the far left of the spectrum (at least, farther left than Gore's platform), so the preference ranking of the 97,488 Nader voters was $Nader \succ_i Gore \succ_i Bush \succ_i Buchanan$. Since they should have easily anticipated that their first choice would not win, the reasoning goes, these voters should have strategically cast their vote for their second choice, Gore, rather than wasting it on a sincere expression of their first choice. This story is a bit of an oversimplification for at least a couple of reasons. First, many Nader supporters actually did strategically vote for Gore, and those voters that "wasted" their vote on Nader were much more equally divided on their second choice than single-peaked preference assumptions would have us believe. One analysis of ballots concluded that at most 60% of Nader supporters would have voted for Gore over Bush in a two party-contest. While these voters would have been enough to tip the scales in this historically tight election, even the Socialist Workers' Party candidate James E. Harris's 562 votes would have been enough to swing the outcome. Second, despite the closeness of the election, the margin of victory still wasn't so narrow that any one voter had an incentive to unilaterally deviate from her choice. Unless a net total of 537 Nader supporters for whom $Gore \succ Bush$ switched their vote, Bush still would have won. (In chapter 22, we will take up the question of why people bother to vote at all, given that it is unlikely that their own individual vote will make a difference.)

This phenomenon is known as the *problem of the divided majority*—while a majority of voters may agree that, say, both Gore and Nader were better than Bush, if they vote sincerely, then they will divide their votes across two separate options and wind up with a worse outcome. If, however, they had

managed to agree on either Gore or Nader and coordinate their votes, then that choice would have won, but instead they ended up with their last choice. Under plurality voting, there is thus strategic pressure to abandon all but the top two expected vote-getters to give one's vote a chance to count in determining the outcome. For this reason, plurality voting naturally favors a two-party system (an observation known as *Duverger's law*).

Incentives for Strategic Voting

Let's return to the eight-person preference profile 4 in [figure 21.7](#) and see whether players have an incentive to vote strategically under the voting rules we examined previously. First, under simple plurality, we found that *D* wins with three votes. Players 1, 4, and 5 all have incentive to switch their vote to *B*, which would force a tie with *D* if any one of these voters were strategic, and would lead to *B*'s victory if at least two supported *B*. Similarly, players 2 and 3 have an incentive to switch their vote to *C*.

The Borda count is less susceptible to the problem of the divided majority, since it takes into account the full preference profile, not just each person's top choice. Does anyone have an incentive to submit a ranking that doesn't match their true preference ordering? When everyone votes sincerely, we found that *C* wins with 28 points, but *E* comes in a relatively close second with 26 points. Four of our voters prefer *E* to *C* and, keeping constant the others' votes, could engineer *E*'s victory by promoting it in their ranking while demoting *C*. For example, if player 6 were to rank her preferences as *EDABC*, then *E* would receive 27 points, while *B* and *C* would tie for second with 26 points each. It is perhaps not surprising that voters can manipulate the outcome of the Borda count, given that we know that this rule does not satisfy IIA and the group's preference over two alternatives can be reversed by individuals reordering other alternatives.

Under the instant runoff voting rule, we found that sincere voting led to a face-off between *B* and *D* in the final round, with *B* emerging victorious. Players 6, 7, and 8 would be particularly unhappy with this outcome, as *D* was their first choice and *B* was their last choice. However, if any one of them supported *C* from the beginning, then the final contest would come down to *C* versus *B*, and *C* would win.

Finally, we found that *A* wins under approval voting, provided that all eight voters sincerely express that *A* would be an acceptable option, while *C* and *E* come in second with seven votes each. Since *A* is the second-to-last choice for

most voters, any one of them could achieve a better outcome by retracting their support for A . For instance, player 4 could choose not to support either A or E (even though she views both as acceptable), which enables her first choice of C to tie.

Gibbard-Satterthwaite Theorem

Thus far, we've focused on preference aggregation rules, which take a preference profile and generate a social ordering. Strategic voting, however, is primarily concerned with whether one voter can affect the group's ultimate choice. We're therefore going to consider a *social choice function* (see definition 4), which takes a preference profile and produces a unique winner. In other words, instead of generating a full ranking of alternatives for the group, a social choice function simply spits out a choice. Furthermore, we will assume that the input of the function is the *reported* preference profile—that is, the expressed preferences of each group member—not necessarily the profile of true preferences.

Definition 4 (Social choice function) *A social choice function, f , is a mapping from the set of all preference profiles to X , such that f selects a single alternative in X for any reported preference profile.*

As with preference aggregation rules, there are certain reasonable criteria that we may wish our social choice function to satisfy, such as the group not being ruled by a dictator (nondictatorship) and the group not choosing an outcome that everyone agrees is worse than some other outcome (weak Pareto efficiency). These conditions are defined as follows:

- **Nondictatorship.** The social choice function is a dictatorship if there exists an agent i such that, for all reported profiles, f always outputs i 's top reported choice. The social choice function satisfies nondictatorship if such a dictator does not exist.
- **Weakly Pareto efficient.** The social choice function is weakly Pareto efficient if $x \succ_i y$ for all i implies that $f(\rho) \neq y$. In other words, when everyone prefers x to y , then y is not the social choice.

To these criteria, we may wish to add the requirement that players don't have any incentive to strategically misrepresent their own preferences to gain a better social choice. A social choice function is *strategy-proof* (see definition 5) if, like the second-price auction, players have a dominant strategy to report

their preferences truthfully. Otherwise, there are at least some situations in which a player could achieve a better social choice by submitting an alternative preference ordering instead of his own.

Definition 5 (Strategy-proof) *We say that a social choice function f is strategy-proof if, for any preference profile, each player i has a weakly dominant strategy to truthfully report his preferences. That is, consider any preference profile ρ and any reported preference profile $(R_1, \dots, R_n)$. The social choice function f is strategy-proof if, for any individual i , and any alternative preference for i , R_i' , $f(R_1, \dots, R_i, \dots, R_n) \succeq_i f(R_1, \dots, R_i', \dots, R_n)$ —that is, player i can't win a more preferred outcome by pretending that R_i' is her preference ordering instead of $\succ_i$. If f is not strategy-proof, then we say that it is manipulable.*

Putting these three criteria together, we arrive at the impossibility result proved by Gibbard and Satterthwaite, stated in theorem 4.

Theorem 4 (Gibbard-Satterthwaite theorem [Gibbard, 1973; Satterthwaite, 1975]) . *Suppose that there are at least three alternatives (i.e., $|X| \geq 3$). The only social choice function that is weakly Pareto efficient and strategy-proof is a dictatorship.*

There is also an alternative specification of theorem 4 that replaces the requirement of weak Pareto efficiency with the requirement that the social choice function be *onto*—that is, any alternative could be a winner under some profile of reported preferences.

Notice that the Gibbard-Satterthwaite theorem doesn't say that someone can *always* do better by being strategic; it says sincere voting is not a *dominant strategy* for any profile of preferences. (Furthermore, like Arrow's impossibility theorem, the Gibbard-Satterthwaite theorem assumes that any preference profile may hold. If we restrict the domain of preferences to those satisfying the single-peaked restriction, then there are social choice functions that are strategy-proof.) In other words, there may still be equilibria in which each player has a best response to vote sincerely.

In recent years, local governments throughout the US and UK have increasingly adopted instant runoff procedures, in large part to give voters the opportunity to voice their true preferences without worrying that they are wasting their votes. Yet the Gibbard-Satterthwaite theorem tells us that all nondictatorial social choice functions are susceptible to strategic voting under

certain conditions. So have these efforts been misguided? Or are some voting rules more readily manipulable than others, either by individuals or coalitions of likeminded voters who coordinate their ballots? Researchers have addressed this question by analyzing either how frequently an individual or coalition would regret voting sincerely, given the sincere votes of others (e.g., Chamberlin, 1985; Green-Armytage et al., 2016) or the extent to which players anticipate they can increase their expected payoff by voting insincerely, given their beliefs about the outcome and the strategic voting of others. Across both methods, instant runoff voting is regularly found to be one of the least manipulable, both by individuals or coalitions, which is good news for the municipalities have adopted this rule to curb strategic manipulation.

Candidate Competition Games

In a representative democracy, preferences for policies are generally aggregated through electoral competition among candidates, who adopt specific platforms with the understanding that the winning candidate will implement the platform that she campaigned on. In other words, the candidates' strategy profile determines the available alternatives, and the competition for votes ensures that the candidates' strategies are shaped by the profile of voters' preferences. In this section, we will analyze games in which candidates compete for votes by adopting a platform along the $[0, 1]$ interval.

The Hotelling Model

Models of spatial competition and the median voter result have their origins in a 1929 paper by Harold Hotelling, which analyzed competition between two producers who choose the extent to which they differentiate their products. For example, consider two ice cream vendors who are choosing where to set up their stands on a long boardwalk. Each wishes to choose a location to gain as many customers as possible. Vacationers are uniformly distributed along the beach and wish to minimize their walk, so they will patronize whichever ice cream stand is closest. (In the event of a tie, they choose each with equal likelihood.) We can represent the spots along the boardwalk using the $[0, 1]$ interval, and each vendor's strategy, s_i , is a location point in this strategy set. Since the customers are distributed uniformly along the beach, the median customer is located at 0.5 (smack in the middle of the boardwalk).

Let's say that the players adopt strategies (s_1, s_2) , as shown in [figure 21.17](#). All the customers who are located to the left of s_1 patronize stand 1, while vendor 2 gains the business of everyone to the right of s_2 . The midpoint between s_1 and s_2 is $\frac{s_1+s_2}{2}$, and this point divides the intervening space into those who are closer to s_1 and those closer to s_2 . Each vendor i gains a proportion of customers equal to the length of the interval containing all the points closer to s_i than s_j . That is, for $s_1 < s_2$, vendor 1 gains $\frac{s_1+s_2}{2}$ of the customers and vendor 2 gains the remaining $1 - \frac{s_1+s_2}{2}$.

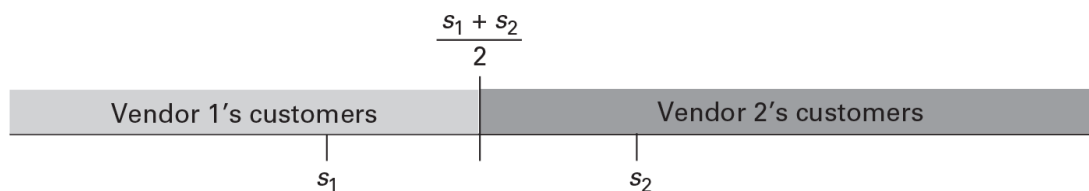


Figure 21.17

Proportion of customers gained by vendors located at s_1 and s_2 .

The unique Nash equilibrium of this game is $(0.5, 0.5)$: both vendors locate at the median. To see this, let's consider vendor i 's best response to any s_j . If $s_j = 0.5$, then player i 's best response is to locate at 0.5 as well, in which case the vendors split the market. If player i were to respond by choosing any point to the left or the right of 0.5, he would necessarily gain less than half of the market. Thus $(0.5, 0.5)$ is *one* Nash equilibrium. Is there any other? If s_j is any point other than 0.5, then player i 's best response is to locate *just next to* s_j , on the side closest to the center of the boardwalk. But this strategy profile cannot be a Nash equilibrium, because then player j would wish to wheel her cart to the other side of player i so that she is now slightly closer to the center. This process would continue until both carts were located at 0.5, at which point the vendors split the market, and neither could gain more customers by moving.

In equilibrium, the ice cream stands thus wind up right next to each other in the center of the boardwalk. This outcome isn't great for consumers, who always need to walk to the midpoint to get an ice cream. If the vendors were to instead locate at 0.25 and 0.75, then no one would need to walk more than a quarter of the length of the boardwalk. Likewise, Hotelling argued, consumers with diverse tastes would be better served if competing firms offered differentiated products, granting them more of a choice. But the competitive pressure pushes firms to mimic each other.

When Two Candidates Compete for Votes

Hotelling's spatial competition model was extended to the domain of political preferences and electoral competition by Duncan Black and advanced in Anthony Downs's classic 1957 book *An Economic Theory of Democracy*. Let's consider a game between two candidates. Each candidate i simultaneously chooses a platform $s_i \in [0, 1]$. There exists an odd number of voters with Euclidean preferences over $[0, 1]$. The voters observe the platforms (s_1, s_2) and simultaneously cast their votes. The candidate who receives the most votes wins and implements his platform. We'll assume that voter i always votes sincerely for the candidate whose platform is closest to her own ideal point p_i , since she has no incentive to be strategic in a single head-to-head vote between the alternatives. In the event that voter i is indifferent between the two candidates (e.g., if they choose the same platform or platforms that are equidistant from p_i), then the voter chooses a candidate by flipping a coin. Candidates care only about winning and earn a payoff of 1 if they receive the majority of the votes and -1 for a loss. If the strategy profile (s_1, s_2) implies that the election will be decided by a voter who was indifferent between the two candidates and tossed a coin, then $Eu_i(s_1, s_2) = 0$ for both players, and we'll call this a tie.

Let's consider the spatial voting game that we introduced at the beginning of the chapter, in which there are seven types of voters, each equally common, with ideal points given by 0.1, 0.25, 0.35, 0.4, 0.5, 0.75, and 0.85. The 0.4 platform is the median voter's ideal point, and by the median voter theorem, it is the Condorcet winner. The unique Nash equilibrium in this game is $(s_1, s_2) = (0.4, 0.4)$. The intuition of this result is straightforward, since we know that 0.4 cannot be beaten by any other platform. Let's first consider, logically, why this must be the unique Nash equilibrium, and then confirm this logic using best response analysis.

First, note that $BR_i(0.4) = 0.4$. If the other player chooses $s_j = 0.4$, then there is no s_i that allows i to win the election outright. By choosing $s_i = 0.4$, i forces a tie: Each voter randomizes over the candidates, and so i 's expected payoff is 0. Since this is higher than the payoff of -1 that i receives from choosing any other platform, and thus losing, 0.4 is the unique best response to 0.4. So $(0.4, 0.4)$ is a Nash equilibrium. How do we know that it is the only one? Consider any other strategy profile in which neither player is choosing $s_i = 0.4$. There are two possibilities: either one player earns more votes than the

other or they tie. Neither can be a mutual best response. In the first case, the losing candidate would prefer to switch to position 0.4 and win (for a payoff of 1 instead of -1); in the second case, each candidate would prefer to unilaterally deviate to position 0.4 and win outright (for a payoff of 1 instead of an expected payoff of 0).

Let's also analyze this game a bit more systematically by considering the best response of each of our candidates. We've already seen that the unique best response to $s_j = 0.4$ is $s_i = 0.4$. However, for any other platform that candidate j could adopt, i 's best response will not be unique. Rather, it will be an interval around 0.4 such that player i is at least *a bit* closer to the median than j is. So instead of a best response function, we are seeking player i 's best response correspondence.

To see this, imagine that player j chooses platform s_j on the left side of the spectrum, as shown in [figure 21.18](#). Which strategies are best responses? Strategies $s_i < s_j$ would lose, while $s_i = s_j$ results in a tie. We know that $s_i = 0.4$ would win, but there is a much broader range of winning platforms. If i chooses s_i just to the right of s_j , then i gains the support of everyone except voter A and wins B through G . However, i doesn't need the support of voters B or C to win the election, so she could keep venturing farther right. How far right could she push her platform and still emerge victorious? Only so far as to not lose the support of voter D , the median voter, whose ideal point is 0.4. Specifically, i could win with a position s_i to the right of 0.4 if and only if the distance between s_i and 0.4 does not exceed the distance between s_j and 0.4. That is, $s_i > 0.4$ beats $s_j < 0.4$ only if $s_i - 0.4 < 0.4 - s_j$, or $s_i < 0.8 - s_j$. This is the upper boundary of the interval of positions that are closer to 0.4 than s_j is, as shown in [figure 21.18](#).³ Thus, any $s_i \in (s_j, 0.8 - s_j)$ is a best response to $s_j < 0.4$. (Note that the end points of this interval are not included in the best response, as these platforms would yield a tie). By similar reasoning, if $s_j > 0.4$, then any platform s_i that is less than s_j and greater than $0.8 - s_j$ will win.

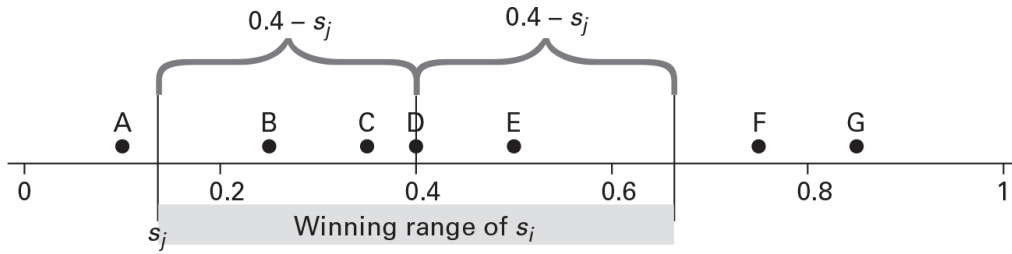


Figure 21.18

The best responses of player i to $s_j < 0.4$.

Thus i 's best response correspondence is given by

$$BR_i(s_j) = \begin{cases} (s_j, 0.8 - s_j) & \text{if } s_j < 0.4 \\ 0.4 & \text{if } s_j = 0.4 \\ (0.8 - s_j, s_j) & \text{if } s_j > 0.4 \end{cases}$$

This is graphed in [figure 21.19](#). When the other player chooses an extreme position, the candidate can win by choosing essentially any platform. However, as the other player's position approaches the median, the interval of platforms that would be more appealing to the median voter collapses. We thus have a race to the median by the two players. If we overlay both players' best response correspondences on the same set of axes, we find that the unique overlapping point is at $(0.4, 0.4)$. Thus, electoral competition drives the political candidates to adopt identical positions. Just as in the Hotelling model, voters are not afforded a choice and are indifferent between the two candidates.

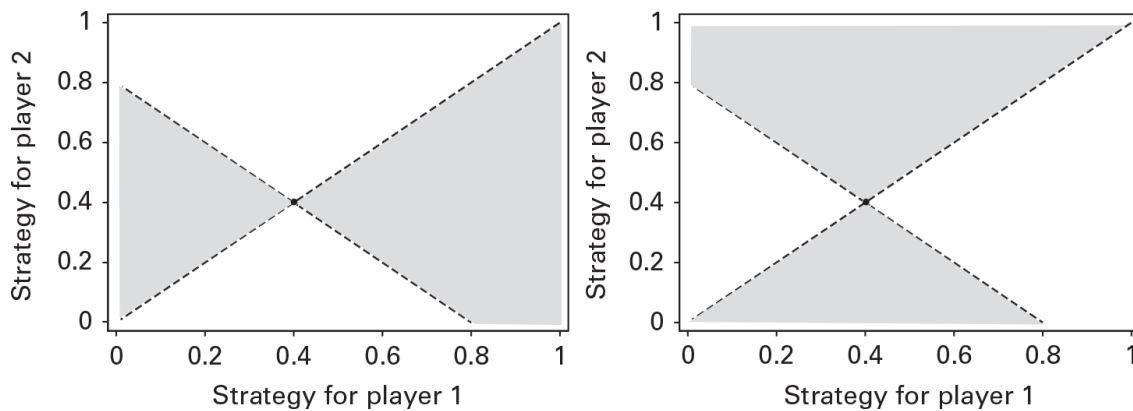


Figure 21.19

The best responses of player 2 to s_1 (left) and the best responses of player 1 to s_2 (right). When these graphs are overlaid $(0.4, 0.4)$ is the unique point of overlap.

Furthermore, we get the same results if voters are distributed continuously along this space, as in Hotelling's model. If the median voter in this distribution has ideal point m , then (m, m) is the unique Nash equilibrium. (You will be asked to find the best response functions and verify this result in exercise 3 at the end of this chapter.) This is likewise the case if the players care about maximizing vote totals rather than winning, similar to Hotelling's game.

So why do we not see the two major American political parties adopting the exact same platforms? While citizens on both ends of the political spectrum often gripe about the lack of distinction between the two parties (such as after the second debate of the 2000 presidential election, when Bush and Gore agreed on nearly every foreign policy question), few would argue that there is *no* daylight at all between the parties, and the parties' national committees typically have released official platforms every four years highlighting their differences.

There are several reasons why candidates may choose distinct positions that are not captured in our current model. Candidates may need to first win a primary with more ideologically extreme median voters, and worry about being branded as a flip-flopper and losing credibility if they then attempt to pivot to a much more centrist position for the general election. If voting requires effort, voters at the ends of the spectrum may not bother to turn out for candidates that they view as insufficiently different, so the candidates may need to choose distinct positions to shore up the support of their base. Or candidates may need to court the support of ideologically motivated donors who won't open their checkbooks unless they perceive one candidate to be a big improvement over the alternative.

The standard Hotelling-Downs model assumes only two candidates, but what if it is possible for a third-party candidate to enter after the major party candidates have chosen their positions? If the two dominant candidates have located at the median voter's ideal point, the third-party candidate could win a plurality by locating just to the left or the right. Thus, the two dominant candidates may wish to deter entry by choosing differentiated positions (as in Palfrey, 1984). In exercise 4, we'll look at a modified version in which candidates actually believe in something, rather than only caring about winning. In exercise 5, we'll look at a version of the game in which candidates differ in quality as well as position, and see that the higher-quality

candidate is more likely to take a centrist position, while the other takes an extreme position.

Finally, political science research over the last couple of decades indicates that party loyalty (or animosity toward the opposing party) drives voting behavior more than proximity between a candidate's platform and a voter's ideal point. The typical voter does not know the details of each candidate's positions on all the issues, but she does know whether she's a Democrat or a Republican, and so she may prefer to cast the vote that affirms her partisan identity. We refer interested readers to Achen and Bartels (2017) for a recent critique of what they call the "folk-theory" of democracy.

Behavior in Games

Voting Sincerely and Not Particularly Chaotically

The theoretical results presented in this chapter may have often read like a steady stream of bad news: dictatorships are the only preference aggregation rules that guarantee a coherent social ordering or prevent people from strategically misreporting their true preference rankings; plurality voting can lead to the election of a Condorcet loser when the majority is divided in their first choice; and, unless voters' preferences satisfy the knife-edge property of radial symmetry, chaos will erupt whenever the policy space is more than one dimension. The behavioral results, however, provide far more cause for optimism.

Experimental tests have generally found that when a Condorcet winner exists, candidates converge toward it, even when candidates have shockingly little information about voters' preferences and voters have very little information about the candidates' platforms. When a Condorcet winner doesn't exist, the outcomes would hardly be described as chaotic, with social choices tending to cluster in the interior of the voters' ideal points. Experiments also typically find that voters initially vote sincerely, but divided majorities can learn to coordinate using polls or other cues to prevent the victory of a Condorcet loser. Furthermore, work on *behavioral social choice* has found that the preference profiles that lead to many of the devastating results are fairly uncommon.

In some of the earliest experiments on social choice, Fiorina and Plott (1978) assembled five-person groups and told them that their job was to choose an alternative in two-dimensional space: literally, somewhere on the

blackboard that was set up in front of them. Each person was assigned an ideal point in this space and, to induce Euclidean preferences over the policy space, they were given payoff sheets such as that shown in [figure 21.20](#).

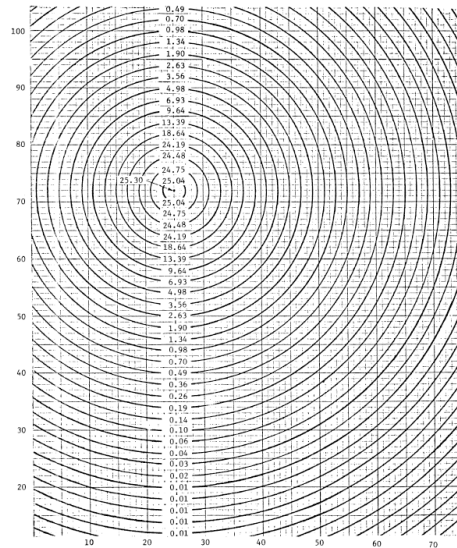


Figure 21.20

Subjects' payoffs over policy space (Fiorina and Plott, 1978).

If this were your payoff sheet, then your ideal point would be (25, 72), at the middle of the target. If the group agreed to this outcome, you would receive a payoff of \$25.30. Otherwise, your payoff would be determined by the numbers shown along the concentric circles overlaid on the policy space. For example, if the outcome were (20, 22), your payoff would be around 2 cents. Rather than having candidates determine platforms, the five group members could deliberate and vote via majority rule. There was an initial status quo policy, which gave a low payoff to everyone. At any point, a group member could raise his hand and propose a different coordinate on the blackboard. If the majority voted in favor, this point became the new status quo point. This process continued until someone made a motion to adjourn and, if the motion passed, the game ended and everyone was paid according to the current status quo point.

When a Condorcet winner existed, Fiorina and Plott found that outcomes tended to cluster nearby (with several committees hitting this point exactly), and the Condorcet winner was the best predictor of the group's outcome among 16 competing hypotheses. They also looked at configurations of ideal points in which the Plott conditions were not satisfied, but it was close, such

as the situation shown in [figure 21.15](#). While McKelvey's theorem predicts that anything could happen, Fiorina and Plott didn't find crazy sequences of votes, with the policy point being pushed all around the blackboard. The groups didn't take any longer to reach a decision, and the decisions that were reached tended to be clustered in a relatively small region of the policy space.

What happens when candidates compete for votes? McKelvey and Ordeshook conducted a series of experiments in the 1980s studying spatial voting games and the effects of information. In their initial experiments, pairs of subjects were assigned to compete as candidates and told the preferences or ideal points of a set of automated voters (who would always vote sincerely). In two-dimensional policy space with a Condorcet winner, platforms were initially spread over the entire policy space. However, by the final few periods in an 8- to 15-round game, platforms converged to the Condorcet winning point, with most hitting the point exactly (McKelvey and Ordeshook, 1982).

In subsequent experiments, the voters were also experimental subjects, who varied in their information about the candidates' platforms. For instance, McKelvey and Ordeshook (1985a) ran a series of elections with more than 40 voters each and two candidates who could choose a platform along the $[0, 100]$ interval. The candidates were not told the voters' ideal points. Voters knew their own payoff function and the number of other voters who were to the left and right of them; however, half did not learn the candidates' positions before they voted. While the overall median ideal point was 60, the median *informed* voter's ideal point was 75. Candidates first chose their positions, and then two polls were taken and the subjects voted. The results of two series of elections are shown in [figure 21.21](#) (the solid line shows one candidate's platform, while the dashed shows the other, and the circled point shows the winning platform in each election). As we can see, the candidates' platforms quickly converged—and they converged to the ideal point of the median voter—even though half of the voters are uninformed. Furthermore, 79% of uninformed voters (and 97.5% of informed voters) voted for their preferred candidate.

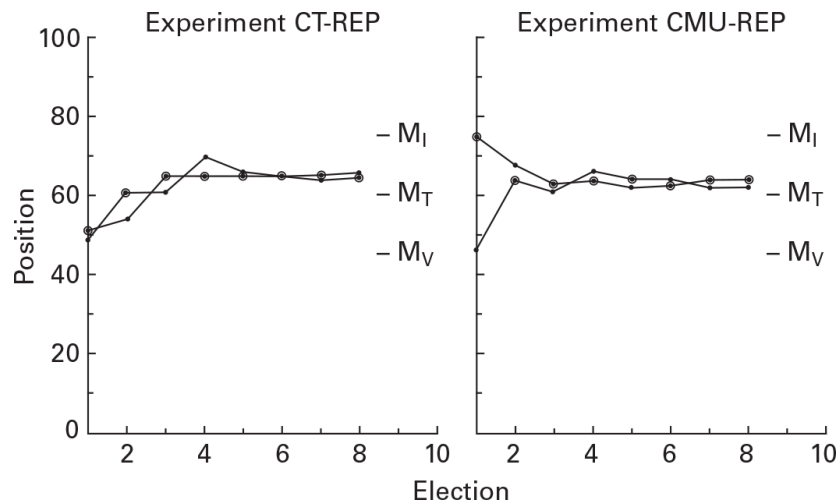


Figure 21.21

Candidates' platforms over time (McKelvey and Ordeshook, 1985a).

Since that worked well, why not push things even further and not inform *any* voters of the candidates' platforms? In another series of experiments by McKelvey and Ordeshook (1985b), candidates did not know the voters' ideal points. In each period, the candidates announced their positions, an interest group whose ideal point was at the origin announced its endorsement, and then a vote was held. Voters found out the winning position and their own payoff before the next round. Again, the authors found that the candidates' platforms converged toward the median voters' ideal points over time (although there was still considerable variation in the final rounds).

Finally, what if voters don't even know their *own* preferences? The theory of retrospective voting poses that voters cast their ballots primarily on the basis of how happy they were with the incumbent during the previous term. In this case, voters need not know anything about their own preferences over the policy space or the specifics of the candidates' platforms—they only need to broadly assess their overall happiness with the current officeholder's performance, as in the famous campaign question, "Are you better off today than you were four years ago?"

Collier et al. (1987) didn't inform the voters of their own utility functions or the candidates' policies (and candidates were also not informed of the voters' preferences). In the first period, one candidate was randomly selected to be the incumbent. After he choose his platform, each voter's payoff was calculated at that platform and communicated to the voter. The voter could then choose whether to keep the incumbent or vote him out of office (in

which case the other candidate became the incumbent in the next period). Again, there was convergence to the median voter's ideal point in many of the elections, even when voters and candidates weren't even told that preferences were single-peaked or there was an exogenous shock to preferences midway through the experiment.

Other experiments have examined whether experimental participants vote sincerely or strategically. Recall that voters may prefer to vote insincerely when they participate in a sequence of pairwise votes (in which case backward induction may lead some voters to vote down a preferred option early so as not to get a worse option later), or when voters face multiple alternatives. Furthermore, it may be the case that an individual can manipulate the outcome to her liking by being insincere, as in the examples we looked at, or that a group could coordinate to manipulate the outcome, as in the problem of the divided majority.

Eckel and Holt (1989) conducted an experiment in which participants held two pairwise votes over the set of alternatives $\{A, B, C\}$. In the first round, participants cast a vote between A and C . Then the winner faced B . There were three voters for whom $A > C > B$, three for whom $B > A > C$, and three for whom $C > B > A$. Thus, the preferences cycle. Furthermore, A would lose to B in the final round, while C would beat B . If everyone votes sincerely, A wins the initial vote but then loses to B in the second round. But if two of the three voters for whom $A > C > B$ initially cast their vote for C , then they would wind up with their second choice of C instead of their last choice of B . The authors varied both whether preference information was common knowledge and how frequently preferences shifted. They found that, initially, *no one* voted strategically. However, when preferences remained constant for at least a few rounds, the groups always converged to the strategic voting outcome (in which two of the three voters who can be strategic do so, and C wins). Notably, however, there were some voters who never learned to vote strategically. The authors concluded that sincere voting is generally a good assumption, except in cases where the same committee meets frequently and voters can learn each other's preferences.

Forsythe et al. (1993, 1996) examined strategic voting in divided majority problems. In their experiments there were 4 voters for whom $A > B > C$, 4 for whom $B > A > C$, and 6 for whom $C > A \sim B$. If everyone votes sincerely, C would win with 6 of the 14 votes. However, C is a Condorcet loser that would

be defeated in a majority-rule vote against either A or B. If the first two types of voters could coordinate their support on A or B, then their choice would win with 8 votes. Without a means of coordinating their behavior, they found that C won in 90% of the contests. However, when voters had access to polls that were conducted before the vote, then they were able to coordinate on A or B about two-thirds of the time. Furthermore, the researchers found that voting rules such as the Borda count and approval voting led to less strategic voting than did plurality.

Evidence-Based Adaptations of the Standard Model

Earlier in the chapter, we saw an example in which some voters would wish to strategically misreport their preferences, given that the others were voting sincerely. But what happens when voters strategically respond to the strategic votes of others? Under conditions of complete information, there will be an abundance of Nash equilibria whenever there are more than a few voters.⁴ This is because any vote is a best response except in the specific situation where the votes of others are divided so that your vote is pivotal. For instance, the distribution of Floridian votes across Bush, Gore, and Nader is a Nash equilibrium: no one person could change the outcome, and so no one had an incentive to unilaterally switch his own vote. In this section, we will apply the level- k reasoning model to a strategic voting game under various voting rules, and see how this can reduce the set of predicted outcomes.

For example, consider the preference profile in [figure 21.22](#). Notice that C is a Condorcet winner, and pairwise majority rule yields the rational preference ordering $C \geq B \geq D \geq A$. What is the Nash equilibrium under plurality voting, assuming that voters can be strategic by casting their vote for any option in the strategy set $\{A, B, C, D\}$? Even with just five voters and four options, there is a multitude of Nash equilibria, some of which do not seem particularly plausible. For instance, everyone voting for A is a Nash equilibrium, even though A is the last choice of 4/5 of the population. Nevertheless, if you believe that the other four voters will support A, then your vote won't make a difference anyway, so any strategy, including A, is a best response.

Player 1	Player 2	Player 3	Player 4	Player 5
D	D	A	B	C
C	C	B	C	B
B	B	C	D	D
A	A	D	A	A

Figure 21.22

Preference profile 5.

Let's instead analyze this plurality voting game using the level- k reasoning approach (e.g., Bassi, 2015). Recall that the level- k model is anchored on the behavior of a level-0 player who does not reason strategically. Researchers typically assume that a level-0 player randomizes uniformly over the available actions, or else she chooses a strategy that is particularly salient or obvious (see chapter 27 for a more detailed discussion of level- k reasoning). The obvious naive strategy in a voting game is to vote according to one's sincere preferences. Then, for any $k > 0$, the level- k strategy is the best response to all other players choosing the level- $(k - 1)$ strategy. (In our analysis, we will assume that players have complete information about the preference ranking of the others; an alternative approach would be to assume that players have beliefs about the distribution of types.)

Under the plurality voting rule, sincere voting implies that each person votes for their first choice, resulting in D's victory with two of the five votes. While the other three players all agree that B and C would both be better than D , if they vote sincerely, then they split their votes across options A , B , and C . The level-0 strategy for each player and the winning outcome if all players are level-0 are shown in the first row of the table in [figure 21.23](#).

Level	Player 1	Player 2	Player 3	Player 4	Player 5	Winner
Level-0	D	D	A	B	C	D
Level-1	D	D	B	C	B	D-B tie
Level-2	D	D	B	B	B	B
Level > 2	D	D	B	B	B	B

Figure 21.23

Level- k predictions under plurality.

Now let's assume that players vote strategically by anticipating the votes of the others and best responding. To do so, we will need to make some assumption about the intensity of players' preferences over options so that we can calculate their expected payoffs in the event of ties. For instance, do they prefer their second choice with certainty or a tie between their first and third choices? Let's assume that each player's payoff from their first, second, third, and fourth choices are 100, 70, 30, and 20, respectively.

Level-1 players assume that everyone else is a level-0 type that votes for the first choice option, and they thus best respond to the level-0 strategy. In this case, players 1 and 2 still prefer to vote for *D*, which will win if the others vote sincerely. However, all the other players would wish to switch to their second choice to force a tie with *D*. Specifically, player 3 would switch to *B*, player 4 would switch to *C*, and player 5 to *B*. If all players were level-1, we would thus have a tie between *D* and *B*. Finally, level-2 players best respond to the level-1 strategy profile. Player 4 would wish to switch back to *B*, allowing her first choice of *B* to win with three votes. The level-2 strategy of all the other players is the same as their level-1 strategy. In this example, each person's level-2 strategy is their best response to the others' level-2 strategies, and so the level-3 and above strategies are the same as the level-2 strategies, and this strategy profile is a Nash equilibrium. Thus, if all players are level-2 or higher, then we expect *B* to win.

Let's now consider this game under the Borda count rule in [figure 21.24](#). The level-0 sincere voting strategies are shown in the first row of the table that follows, along with the Borda counts and the winning option. If all players vote sincerely, then option *C* wins with 15 points. Option *B* comes in a close second, with 14 points. Given this, would level-1 players wish to vote strategically? Players 3 and 4 both prefer *B* to *C* and could reshuffle their votes to engineer *B*'s victory if the others vote sincerely. Specifically, player 3 could rank *B* first and downgrade *C* to last place, providing *B* with 15 points, while *C* and *D* tie for second place with 14 each. Player 4 already had *B* ranked first but had *C* ranked second. By moving *C* to last place on her list, and ranking her options *BADC*, she likewise could secure *B*'s victory by ensuring that *C* gets only 13 points to *B*'s 14. Note that the other players have no reason to manipulate their votes if they believe that the others will vote sincerely. Player 5 prefers *C* and so would not wish to manipulate the outcome. Players 1 and 2 prefer *D*, but they can't elevate *D* any further, nor

can they sufficiently downgrade both B and C so as to catapult D into the lead. Given that they can't have D , C is the next best option. The level-1 strategies are shown in the second row of the table, and if all players use level-1 reasoning then B wins.

Level	Player 1	Player 2	Player 3	Player 4	Player 5	Points	Winner
0	DCBA	DCBA	ABCD	BCDA	CBDA	(8, 14, 15, 13)	C
1	DCBA	DCBA	BADC	BADC	CBDA	(9, 15, 12, 14)	B
2	DCAB	DCAB	BADC	BADC	CBDA	(11, 13, 12, 14)	D
3	DCAB	DCAB	BACD	BACD	BCAD	(12, 14, 13, 11)	B
4	CDBA	CDBA	BACD	BACD	CBAD	(12, 13, 16, 9)	C
> 4	CDBA	CDBA	BACD	BACD	CBAD	(12, 13, 16, 9)	C

Figure 21.24

Level- k predictions under Borda count.

Likewise, we can calculate the level-2 strategies by finding the best response to level-1 play, and so forth. We see that D wins if all players use level-2 reasoning and B wins if all use level-3 reasoning. Each player's level-4 strategy is a best response to the others' level-4 strategies, so as long as everyone uses level-4 reasoning or higher, we reach a stable outcome in which the Condorcet winner of C wins. If most players use one or two levels of reasoning, as the experimental literature suggests, then we expect B or D to win under the Borda count. Thus, the level- k behavioral model allows us to narrow the set of possible outcomes under each voting rule if players can vote strategically.

Exercises

(1) Consider five voters with the preference profile listed in [figure 21.25](#). Find the Condorcet winner.

Player 1	Player 2	Player 3	Player 4	Player 5
A	E	A	H	J
C	J	B	G	D
F	G	I	C	F
I	H	F	D	B
D	F	C	B	H
B	C	D	F	A
E	I	H	J	I
G	A	E	A	C
H	B	J	I	E
J	D	G	E	G

Figure 21.25

(2) Imagine that there are three types of voters voting in the Iowa caucuses for the Democratic presidential nominee: 44% are Centrists/Party Establishment, 27% are Bernie Bro Progressives, and 29% are members of the Liberal Wing. There are five candidates: Kamala Harris (H), Joe Biden (B), Pete Buttigieg (P), Bernie Sanders (S), and Elizabeth Warren (W).

The preferences for each type of voter for these five candidates are listed in [figure 21.26](#).

Centrists (44%)	Bernie Bro Progressives (27%)	Liberals (29%)
B	S	W
P	W	H
H	P	S
S	H	P
W	B	B

Figure 21.26

For each preference aggregation rule listed here, first determine which candidate wins if everyone votes sincerely (nonstrategically). Then explain how one type of voter could achieve a better outcome if they were able to

coordinate their behavior as a group, such that every single member of that type misrepresented their preferences in a specific way (given that the other types vote sincerely).

- (a) *Simple plurality*: Each voter votes for her most preferred choice, and the candidate with the most votes wins.
- (b) *Plurality runoff*: Each voter votes for his most preferred choice. The top two vote getters then face each other in a runoff to determine the winner.
- (c) *Instant runoff*: Each voter submits their full preference ranking. The candidate with the fewest first place votes is eliminated, and those votes go to the next remaining candidate on the voter's list. The process continues until there is one remaining candidate.
- (d) *Borda count*: Each voter ranks the candidates in order of preference. Their top preferences receives 5 points, second best receives 4, and so on. The candidate with the most points wins.
- (e) *Approval voting*: Each voter announces which candidates are acceptable. The candidate with the most "OK's" wins. For the purpose of this example, assume that each voter views his top three choices as actually being acceptable. However, a voter being strategic could approve of as many of the candidates as she wants.

(3) Consider a game between two office-motivated candidates who care only about winning. Each simultaneously chooses a platform in the space $[0, 1]$. Voters have Euclidean preferences, and the median voter's ideal point is given by m . Find the best response functions for the candidates.

(4) Consider a game between two candidates for office, candidate R and candidate D. However, the candidates only care about the ultimate platform that is enacted, and not at all about who wins. The candidates have Euclidean preferences. Candidate R's ideal point is 1 and candidate D's ideal point is 0. Voters have Euclidean preferences, and the median voter's ideal point is given by m .

- (a) Is there a Nash equilibrium in which each candidate chooses m ? What is the best response of each candidate to the other choosing m ?
- (b) Is there a Nash equilibrium in which one candidate chooses m and the other chooses a platform other than m ? If yes, state what it is; otherwise, explain why this cannot be a mutual best response.

- (c) Is there a Nash equilibrium in which D chooses a platform less than m and R chooses a platform greater than m ? If yes, state what it is; otherwise, explain why this cannot be a mutual best response.
 - (d) Is there a Nash equilibrium in which D chooses a platform greater than m and R chooses a platform less than m ? If yes, state what it is; otherwise, explain why this cannot be a mutual best response.
 - (e) Is there a Nash equilibrium in which both R and D choose a platform less than m or both R and D choose a platform greater than m ? If yes, state what it is; otherwise, explain why this cannot be a mutual best response.
- (5) Consider a game between two office-motivated candidates who care only about winning, candidate A and candidate D. Each simultaneously chooses a platform from the set $\{0.3, 0.5, 0.7\}$. The median voter is equally likely to have an ideal point at each of these three positions. However, candidate A is viewed as a higher-quality candidate than D, and thus has an advantage. Each voter's payoff is given by $10 - 10|x - p_i|$, where x is the winning position and p_i is the voter's ideal point. In addition, each voter will gain a payoff bonus of 0.1 if candidate A wins.
- (a) Write this game in normal form, using the candidates' expected payoffs from each strategy profile.
 - (b) Find the mixed-strategy Nash equilibrium.
- (6) Consider a board of five local officials who must vote to approve the public school budget. The annual budget can be any point in the $[0, 100]$ interval. The officials have Euclidean preferences, with ideal points 25, 45, 60, 85, 90. However, the voters cannot choose any point in this space. Instead, the school superintendent proposes a budget, and the officials vote on whether to pass it. If the budget is not passed, then this year's budget defaults to the same as last year's budget, which was 40. The superintendent also has Euclidean preferences and prefers the maximal budget of 100.
- (a) What does the superintendent propose under the subgame perfect Nash equilibrium (SPNE)?
 - (b) Now let's see how your answer would change if we vary last year's budget. Let last year's budget be given by x . For each $x \leq 60$, find the superintendent's SPNE proposal. Explain the intuition behind your answer.

¹Unlike majority rule, the winning option under plurality rule with three or more alternatives doesn't necessarily have majority support.

²The ideology of each member of Congress is assessed according to the full voting record, using a methodology developed by political scientists Keith Poole and Howard Rosenthal in the late 1980s.

³Another way to see from [figure 21.18](#) that the upper boundary of winning positions is $0.8 - s_j$ is to note that the distance between s_j and the median position is $0.4 - s_j$, and so the farthest s_i could be is $0.4 - s_j$. Thus, the upper boundary of i 's winning platforms is $0.4 + (0.4 - s_j) = 0.8 - s_j$.

⁴The situation is different in games of incomplete information, in which voters assess the probability that various votes could be decisive, given the distribution of types playing different equilibrium strategies. For instance, under plurality, Palfrey (1989) showed that in any Bayesian Nash equilibrium, either only two candidates receive positive vote shares (in accordance with Duverger's law) or two or more candidates are essentially tied for second place, so voters can't figure out which one to abandon.

22

The Paradox of Voting

Why do people turn out to vote? Casting a ballot requires time and effort, and standard models assume that citizens weigh this cost against the expected effect of their vote on the outcome. The probability that one's vote will affect the outcome of an election depends on how many other people cast their ballots for each candidate, so this calculation requires voters to strategically anticipate the turnout decisions of others. In this chapter, we will analyze various equilibria in voter participation games and assess whether large-scale turnout can be consistent with voters rationally best responding to each other's voting strategies.

Imagine that you are a member of a team. Your team has 26 members and is competing against another team with 25 members. Each person must simultaneously make a choice of whether to participate or not. Whichever team has more participants wins, and each member of the winning team earns a payoff of 100, regardless of whether they personally participated. Independent of the outcome, each person also earns a bonus if he chooses *not* to participate. The bonus varies across people and ranges from 0 to 55 (with each bonus in the range equally likely). Before making your decision, you learn that your bonus for not participating is 10. What do you do? Would your behavior change if instead your team were twice as large as the other (i.e., 34 members versus 17)?

The Calculus of Voting

In the 2020 US presidential election between Donald Trump and Joe Biden, two-thirds of voting-age citizens cast a vote—the highest percentage turnout since 1900. Similarly, in the Brexit vote to exit the European Union in 2016, 72.2% of the voting-eligible population of the UK turned out to cast a vote. From the perspective of rational choice theory, however, the question is not why participation was not closer to 100% in these historic votes, but rather, why did anyone bother to turn out at all?

Let's consider the standard calculus of voting. For most people, voting is at least somewhat inconvenient. They usually need to either travel to a polling place or navigate the process of requesting a mail-in ballot. City-dwellers often must wait for hours outside their polling places for the chance to vote, sometimes in foul weather, and many voters must take time off from work to cast their ballot, resulting in lost income. Thus, there is some cost to voting, which reflects this inconvenience and varies across people.

Given that voting is at least somewhat costly, why does anyone bother to vote?

The Paradox That Ate Rational Choice

The *pivotal voter model* assumes that people vote in an election primarily to influence the outcome. We'll say that a player is *pivotal* if her vote would change the election result—that is, if the player had abstained, then either her preferred alternative would have lost by one vote (and her vote instead forces a tie) or would have tied (and her vote instead breaks the tie in favor of her choice).

Under the framework of rational choice theory, citizens choosing whether to vote must weigh the expected benefits against the costs. In the standard formulation, a voter will turn out only if the net value of voting (i.e., the benefits minus the costs) is positive, such that:

The net value of voting = (probability vote is pivotal)(value of tipping the vote) - c > 0

On the one hand, there is a fixed cost of voting, c , that is independent of the ultimate outcome. On the benefits side (the first term) is the expected effect of the individual's vote on her well-being. If the individual cares at all about the outcome, then she values tipping the vote toward her more preferred alternative. This gain from affecting the outcome could be substantial, especially if there is a dramatic difference between the candidates and/or the policies are poised to greatly affect the voter personally. However, this value needs to be weighted by the likelihood that one's vote is *pivotal* in deciding this outcome—that is, the voter's own vote was instrumental in bringing about an outcome that would not otherwise have been achieved if she had abstained.

In large elections, the likelihood that any one individual's vote will swing the outcome is vanishingly small. It has been estimated that around 1 in 100,

000 votes cast in US congressional elections are decisive, meaning that the candidate won by a single vote or tied. In advance of the 2008 US presidential election, Andrew Gelman, Nate Silver, and Aaron Edlin (2012) estimated that the average American voter had a 1 in 60 million chance of being decisive. As the probability of being pivotal decreases, the first term in this expression (the expected benefit of voting) goes to zero, while only the costs remain.

This conundrum was originally raised by Anthony Downs (1957) and has become known as the “paradox of (not) voting.” Political scientist Morris Fiorina (1990, p. 334) famously dubbed it “the paradox that ate rational choice.” There are two common resolutions to this paradox, which we will discuss in detail in the next two sections. The first is that voters may have reasons for voting that have nothing to do with influencing the outcome, such as feeling like they’ve done their civic duty or registered support for their party. The second is based on the observation that our previous calculation treated the calculus of voting as an individual decision problem with a fixed probability of being pivotal, rather than considering voters as players in a broader strategic interaction. For instance, we couldn’t have an equilibrium in which everyone reaches the conclusion that voting isn’t worth the cost and stays home, since then any one person could cast a solitary vote and tip the outcome to his side.

Noninstrumental Motives

The previous expression assumes that voters’ motives are purely *instrumental*: they vote because they aspire to influence the outcome. One way to rescue the rational choice theory approach is to assume that voters have *noninstrumental* motives for voting that are unrelated to their desire to actually influence the outcome. In this case, the act of voting itself brings them joy, even if they know that their vote alone will not affect the outcome. In other words, voting is not just a means to an end, but an activity that brings enjoyment in and of itself.

For instance, voters may derive direct utility from casting a vote if they believe that it is their civic duty to do so. Alternatively, voters may derive utility from expressing a particular preference or from affirming their partisan identity by registering support for their party. To illustrate this type of *expressive voting*, Brennan and Buchanan (1984) famously compared the act of voting to cheering for a favorite sports team. We go to games (or scream at the referees on our televisions) not because we believe that we’ll actually

influence the outcome, but because it brings us joy and helps us feel connected to a broader group. These types of direct consumption benefits are captured in the d term included in the following expression:

The net value of voting = (probability vote is pivotal)(value of tipping the vote) $- c + d$.

Because voters derive these benefits regardless of the outcome or their ability to influence it, as the probability of being pivotal goes to zero, the net value of voting is $- c + d$. Thus, the d term counterbalances the cost of voting, and citizens are incentivized to vote whenever $c < d$, even if they don't think their vote will matter at all. On the positive side, this provides a rational reason for voters to turn out. On the negative side, this motive has very little to do with the actual election at hand or citizens' desire to vote for the outcome they truly believe would be best, and instead involves the voters essentially working out their own issues in the voting booth.

While expressive voting could motivate voters who are unlikely to be pivotal to turn out, it can also distort the preferences they express once they are in the voting booth. This is because the alternative they actually prefer (and would choose if their vote were decisive) may diverge from the alternative that gives them the most joy to express. A vivid example can be seen in the aftermath of the Brexit election, when many of the British citizens who expressed shock and dismay with the outcome were Leave voters—the supposed winners. That morning on the BBC, for instance, “Adam from Manchester” explained that he didn't think his vote “was going to matter too much, because I thought we were just going to remain” and went on to say that he was now “quite worried.” Social media sites were filled with stories of Leave voters who regretted their choice as soon as they learned the outcome and would vote differently that very afternoon. Subsequent polls suggested that the Leave voters who regretted their choice was similar to the margin of victory, and that those with the greatest regrets had assumed that Brexit was unlikely, raising the possibility that if British voters had treated their vote as if it would be decisive, Britain's future might be quite different. Later in this chapter, we will also see evidence that people who believe they are unlikely to be pivotal often cast votes that are more partisan than what they truly believe (allowing voters to engage in “partisan cheerleading”) or more “ethical” than what is in their own self-interest (allowing voters to feel good about themselves without much consequence).

Voter Turnout as a Participation Game

Of course, if all voters rationally chose to abstain, then turnout would be zero—in which case any one individual could decide the election by casting a lone vote. Thus, it is important to consider the decision to vote not just as an individual-level choice, but as a strategic interaction between voters who are attempting to anticipate the behavior of others, and thus their likelihood of being pivotal.

To analyze this strategic interaction, Palfrey and Rosenthal (1983, 1985) modeled voter turnout as a participation game, in which each player makes a binary choice of whether to participate or not. There are n players, and each is a member of a team. Team A has n_A members, while team B has n_B members (and $n_A + n_B = n$). Each player simultaneously decides whether to participate at a cost c to themselves. Assume that c represents the balance of all the direct costs and benefits that the player incurs by voting that are independent of the outcome (e.g., bus fare, the opportunity cost of their time, sense of civic duty, or enjoyment from walking to the polling place). The team with more participants wins, and each member receives a benefit B , regardless of whether they personally participated. The losing team receives 0, and members who participated must still pay the cost. In the event of a tie, a coin flip determines the winner, so the expected benefit for each person is $B/2$. The game thus captures the situation in which players either prefer candidate A or B and make the choice between turning out for their preferred candidate or abstaining.

We will assume that $c \in (0, B/2)$. The lower bound implies that voting is at least somewhat costly, so a player's best response is to abstain when he is sure that he will not be pivotal. What if the player believes that he will definitely be pivotal? Recall that there are two ways to be pivotal: (1) his vote could flip a loss (an outcome associated with an expected benefit of 0) to a tie ($B/2$) or (2) his vote could flip a tie ($B/2$) to a win (B). In either case, a player who is pivotal increases his expected benefit by $B/2$ if he votes. The requirement that $c < B/2$ thus ensures that a player's best response is to vote if he believes that he will be pivotal. As we'll see, the number and type of equilibria depend on the number of players on each team and the relative costs, but note that zero turnout is never an equilibrium, so long as $B/2 > c$.

The strategic incentives of this game bear similarities with a couple of two-player games we're already familiar with. Notice that if $n_A = n_B = 1$, then we

have a prisoner's dilemma game. Both players have a dominant strategy to vote: if the other player doesn't vote, then you can guarantee a win rather than a tie by voting; and if the other player does vote, then voting gets you a tie rather than a loss. Thus, the Nash equilibrium has both players voting, resulting in a tie. Consequently, the players each earn an expected payoff of $B/2 - c$ in equilibrium, rather than the $B/2$ that they could achieve if both had stayed home. Alternatively, consider the situation in which one team has two players while the other has zero. In this case, we have a game of chicken, in which each player's best response is to mismatch the other player: if the other team member votes, then your own vote is unnecessary, but if she does not, then your vote is needed to win. There are thus two pure-strategy Nash equilibria, and each player prefers the one in which the other player votes and she gets to free ride on the winning outcome without incurring the cost.

Palfrey and Rosenthal (1983, p. 13) thus described the participation game as a "chicken's dilemma," where players want to free ride off of the votes of their teammates while simultaneously competing with the votes of the other team. In what follows, we will analyze this game for $n \geq 3$, in which case both motivations come into play simultaneously.

Pure-Strategy Nash Equilibria in the Participation Game

Let's first consider a game in which the teams are the same size: $n_A = n_B > 1$. In this case, there exists a unique pure-strategy Nash equilibrium in which everyone votes, resulting in a tie. There also exist symmetric mixed-strategy Nash equilibria (for certain ranges of costs), which we'll discuss in the next section.

Why? If everyone else is voting, then no one would wish to change their strategy to abstain, since this would result in a loss for their team instead of a tie. While they would gain c , they would lose $B/2 > c$. Thus, full participation is a Nash equilibrium. How do we know there is no other pure-strategy Nash equilibrium? Any other strategy profile must satisfy one of three possible cases:

- Case 1. The teams tie with less than full participation.
- Case 2. One team defeats the other by one vote.
- Case 3. One team defeats the other by more than one vote.

Case 1 cannot be a Nash equilibrium since any voter who abstains would prefer to participate and achieve a win for the team. Case 2 cannot be a Nash

equilibrium, since any voter on the losing team who abstained would prefer to participate and force a tie. And case 3 cannot be a Nash equilibrium, since any voter on the winning team would prefer to abstain (since her vote was unnecessary) and any voter on the losing team would also wish to abstain. Thus, the only pure-strategy Nash equilibrium involves full participation.

What about when the team sizes differ (say, $n_A < n_B$)? In this case, there does not exist a pure-strategy Nash equilibrium, and thus the only equilibria are in mixed strategies. Why? Unlike when the teams are equally balanced, there cannot be an equilibrium in which everyone participates. If all n_B members of team B were to vote, then team A could never win, and thus no member of team A would wish to vote. As before, let's consider the possible types of pure-strategy profiles and see why none involves a mutual best response. First, we cannot have a Nash equilibrium in which the teams tie, since this implies that at least one member of team B did not vote and would prefer to do so in order to win. Second, we cannot have a Nash equilibrium in which one team wins by more than one vote, since any participating member of the winning team would prefer to switch to abstain. Third, we cannot have a Nash equilibrium in which one team wins by exactly one vote and at least one member of the losing team abstained—in this case, an abstaining member would prefer to vote and force a tie. Finally, we cannot have a Nash equilibrium in which one team wins by exactly one vote and the losing team fully participated—in this case, there is no possibility that the losing team could tie or win, so each of its members would wish to abstain.

Thus, full turnout can be a rational best response when candidates have equal support. To predict the outcome when candidates are not exactly equally matched, we need to expand our search to mixed-strategy Nash equilibria, as discussed next.

Mixed-Strategy Nash Equilibria in the Participation Game

Let's begin by considering the conditions that must be satisfied in any equilibrium in which players choose mixed strategies. Then we will use this knowledge to search for equilibria of this type.

A player in the participation game will play a mixed strategy in equilibrium only if she is indifferent between voting and abstaining. This occurs when the cost of voting, c , is equal to the probability of being pivotal (i.e., the probability that one's vote flips the outcome from a loss to a tie or

from a tie to a win) times the expected benefit of being pivotal ($B/2$). In other words, the two terms in the calculus of voting formula must be equal, such that the net value of voting is zero. Otherwise, the player would strictly prefer either voting or abstaining, and a mixed strategy would not be a best response.

Let the probability that a team A member is pivotal be given by π_A and the probability that a team B member is pivotal be given by π_B . Then a mixed strategy is a best response for a team A member only if $c = \pi_A \times B/2$, and for a team B member only if $c = \pi_B \times B/2$.

Now that we know the condition that must generally hold for a player to play a mixed strategy in equilibrium, we will search for equilibria of this type. For most of what follows, we will normalize $B = 1$, but consider the equilibria for any general $c \in (0, 1/2)$.

Pure-Mixed-Strategy Nash Equilibria

We've seen that when $n_A < n_B$, there cannot be an equilibrium in which everyone chooses a pure strategy. But can there be an equilibrium in which members of one team play pure strategies while the other team plays mixed strategies?

Consider the strategy profile in which exactly n_A members of team B vote while the remaining $n_B - n_A$ members abstain, and the members of A each play the same mixed strategy in which they vote with probability p . Is this a Nash equilibrium?

To be a Nash equilibrium, three conditions must hold: (1) the members of A must be indifferent between voting and abstaining, (2) no abstaining member of B would do better by voting, and (3) no voting member of B would do better by abstaining. Let's examine each of these conditions.

First, a mixed strategy is a best response for the members of A only if they are indifferent between voting and abstaining, given that exactly n_A members of B vote and their fellow A members each vote with probability p . Given that n_A members of B are voting, team A can never win. However, it could tie in the event that every member of A votes. A member of A would be pivotal in forcing a tie only if all $n_A - 1$ other members of the team vote, which occurs with probability p^{n_A-1} (since each member of A votes with probability p , and we're interested in the event when all $n_A - 1$ of them vote). Thus, a member of A will be willing to play a mixed strategy if and only if $c = \frac{1}{2}p^{n_A-1}$ (remember that we normalized $B = 1$). Solving for p , we find that this implies that the

other members of A must be playing $p = (2c)^{\frac{1}{n_A-1}}$. Notice that the probability that a member of the minority party votes *increases* as voting becomes more costly. Why? We found this probability using the condition that a member of A would be willing to play a mixed strategy only if she is indifferent between voting and abstaining. As voting becomes more costly, she would wish to vote only if she has a higher guarantee that she could be pivotal, which requires the other members of A to vote with a higher probability.

Second, we must check that none of the abstaining members of B would prefer to vote, given that n_A members of B are voting and each member of A votes with probability p . Since n_A of their teammates are voting, they can never lose. However, if every member of A votes, then they would tie, and their vote would be pivotal in achieving a win. Thus, the probability that one of the abstaining members of B is pivotal is p^{n_A} . These players would thus wish to abstain only if the cost of voting is greater than the probability of being pivotal times the benefit of breaking a tie: $c > \frac{1}{2}p^{n_A}$. Recall that we found that members of A will mix only if $p = (2c)^{\frac{1}{n_A-1}}$. Plugging this in, we get $c > \frac{1}{2}(2c)^{\frac{n_A}{n_A-1}}$, or $(2c)^{n_A-1} > (2c)^{n_A}$, or $c < \frac{1}{2}$. Thus, abstaining can be a best response to n_A members of B voting and each member of A voting with probability $p = (2c)^{\frac{1}{n_A-1}}$ so long as the cost of voting is less than half the benefit of one's team winning.

Third, we must check that none of the voting members of B would prefer to abstain, given that $n_A - 1$ other members of B are voting and each member of A votes with probability p . Each of these voters would be pivotal in the event that B would lose without all their votes (i.e., n_A members of A vote) or there would be a tie without all their votes (i.e., $n_A - 1$ members of A vote). The probability of the former is p^{n_A} . The probability of the latter is $n_A p^{n_A-1}(1-p)$.¹

Thus, the probability that the player will be pivotal is $p^{n_A} + n_A p^{n_A-1}(1-p)$, and voting is a best response only if $c < \frac{1}{2}(p^{n_A} + n_A p^{n_A-1}(1-p))$. Again, we know that p must equal $(2c)^{\frac{1}{n_A-1}}$, so plugging in for p , we find $2c < (2c)^{\frac{n_A}{n_A-1}} - n_A(2c)^{\frac{n_A}{n_A-1}} + n_A(2c)$, or $2c(1 - n_A) < (1 - n_A)(2c)^{\frac{n_A}{n_A-1}}$, or $2c > (2c)^{\frac{n_A}{n_A-1}}$ (note that we must flip the inequality because $(1 - n_A) < 0$), or $c < \frac{1}{2}$.

Thus, there exists a Nash equilibrium in which all members of the smaller team play a mixed strategy and exactly enough members of the larger team turnout to guarantee a tie. Under this outcome, the probability that each

member of A votes is $p = (2c)^{\frac{1}{n_A-1}}$. As we already noted, the probability that a member of the minority party votes increases as the cost of voting increases. In addition, notice that as n_A becomes large, the probability that members of A vote goes to 1. Thus, voter turnout is substantial for large electorates in this equilibrium: close to twice the size of the minority party (since n_A members of B vote with certainty and close to n_A members of A vote).

A Three-Player “Chicken’s Dilemma”

In general, there may be multiple mixed-strategy Nash equilibria in the participation game. Recall that in symmetric games, we often look for equilibria in symmetric strategies: players who face the same situation play the same equilibrium strategy. In the participation game, members of the same team face the situation, but if team sizes differ, members of the other team face a somewhat different situation. We will thus focus on *quasi-symmetric* equilibria. In a quasi-symmetric Nash equilibrium, all members of the same team play the same strategy, but the strategies may differ across teams. In particular, let members of team A vote with probability p_A and members of team B vote with probability p_B . To understand the intuition, let’s start with a simple example in which $n_A = 1$ and $n_B = 2$. We’ll then see how to scale up this intuition for teams of any general size. Since the teams are unequally sized, there does not exist a pure-strategy Nash equilibrium. The member of team A will play a mixed strategy only if $c = \frac{B}{2} \times \pi_A$ or, since $B = 1$, $2c = \pi_A$.

What is the probability that the member of A is pivotal (π_A), given that the members of B each vote with probability p_B ? If neither member of B votes, then the player’s vote would flip the outcome from a tie (at zero votes each) to a win (1–0 in favor of A). This occurs with probability $(1 - p_B)^2$. The player’s vote would flip the outcome from a loss to a tie if exactly one member of B votes. This occurs with probability $2(p_B)(1 - p_B)$. Thus, the probability that the member of A is pivotal is given by $\pi_A = (1 - p_B)^2 + 2(p_B)(1 - p_B)$, which reduces to $1 - p_B^2$.² Thus, we have that the member of team A will play a mixed strategy only if $2c = 1 - p_B^2$, or $p_B = \sqrt{1 - 2c}$.

Next, what is the probability that a member of B is pivotal (π_B)? Note that this depends on both p_A (the probability that their opponent votes) and p_B (the probability that their teammate votes). The probability that a member of B would flip the outcome from a tie to a win is given by the probability that

both other players vote ($p_A p_B$) plus the probability that neither of the other two players vote ($(1 - p_A)(1 - p_B)$). The probability that a member of B would flip the outcome from a loss to a tie is given by the probability that the member of A votes and the other member of B does not ($p_A(1 - p_B)$). Thus, the probability that the member of B is pivotal is given by $\pi_B = p_A p_B + (1 - p_A)(1 - p_B) + p_A(1 - p_B)$, which reduces to $1 - p_B + p_A p_B$. Thus, we have that the member of team B will play a mixed strategy only if $2c = 1 - p_B + p_A p_B$. Combining this with our earlier condition that $p_B = \sqrt{1 - 2c}$, and solving simultaneously, we get that $p_A = 1 - \sqrt{1 - 2c}$ and $p_B = \sqrt{1 - 2c}$.

The Equilibrium Conditions for an n -Player Game

Now let's consider expanding the size of the electorate to any general n_A and n_B , such that $n_A \leq n_B$. The only thing that changes in our analysis is the pivotal probabilities, π_A and π_B . What are the ways in which a voter could be pivotal? First, the teams could tie without you, such that your vote would result in a win for your team. This occurs if the vote breakdown among the other players is 0 votes on each team, 1 vote on each team, etc., on up to $n_A - 1$ votes on each team. Second, your own team could be down 1 without you, in which case your vote would result in a tie. This occurs when the vote breakdown among the other players is 0 to 1, 1 to 2, etc., on up to $n_A - 1$ to n_A . We can calculate the probability of these pivotal events occurring using binomial formulas. Specifically, we have

$$\begin{aligned}\pi_A &= \sum_{k=0}^{n_A-1} \binom{n_A-1}{k} \binom{n_B}{k} (p_A)^k (1-p_A)^{n_A-1-k} (p_B)^k (1-p_B)^{n_B-k} \\ &\quad + \sum_{k=0}^{n_A-1} \binom{n_A-1}{k} \binom{n_B}{k+1} (p_A)^k (1-p_A)^{n_A-1-k} (p_B)^{k+1} (1-p_B)^{n_B-1-k} \\ \pi_B &= \sum_{k=0}^{n_A} \binom{n_A}{k} \binom{n_B-1}{k} (p_A)^k (1-p_A)^{n_A-k} (p_B)^k (1-p_B)^{n_B-1-k} \\ &\quad + \sum_{k=1}^{n_A} \binom{n_A}{k} \binom{n_B}{k-1} (p_A)^k (1-p_A)^{n_A-k} (p_B)^{k-1} (1-p_B)^{n_B-k}\end{aligned}$$

It may at first seem like there is a lot going on here, but these formulas capture the same process that we used in the example with three players. Let's focus on the first expression for π_A . The first term gives us the probability that a player on team A is pivotal because there would be a tie without her participation. The second term gives us the probability that the player is

pivotal because her team would lose by a single vote without her participation.

Let's examine the first term in more detail. We are summing over $k = 0$ to $k = n_A - 1$ to capture all the ways in which there could be a tie among the other players. When $k = 0$, we're looking at the probability that there is a $0 - 0$ tie without player i 's participation: that is, all other $n_A - 1$ teammates abstain and all n_B opponents abstain. Logically, we know that this occurs with probability $(1 - p_A)^{n_A - 1}(1 - p_B)^{n_B}$, and this is captured in the first term when $k = 0$. (The $\binom{n}{k}$ terms simply equal 1 when $k = 0$.) Likewise, for the other values of k up to $n_A - 1$, we're looking at the probability with which there is a $k - k$ tie without i 's participation; that is, k members of A vote, while the other $n_A - k - 1$ teammates abstain, and k members of B vote, while the other $n_B - k$ opponents abstain. The probability that k specific members of A vote while the other members abstain is given by $(p_A)^k(1 - p_A)^{n_A - k - 1}$. However, we need to consider the possibility that any combination of k members could be the ones to vote. The number of combinations of k voters that we could assemble from $n_A - 1$ members is given by the term $\binom{n_A - 1}{k}$. Likewise, the probability that k specific members of B vote while the remaining $n_B - k$ abstain is given by $(p_B)^k(1 - p_B)^{n_B - k}$, and we then need to multiply this by the number of combinations of k voters that we could assemble from the n_B members: $\binom{n_B}{k} = \frac{(n_B)!}{k!(n_B - k)!}$. The second term is calculated similarly, as is π_B , the probability that a member of team B will be pivotal.

Let's look at an example in which $n_A = n_B = 6$, which was used in an experiment by Schram and Sonnemans (1996) that we will discuss later in this chapter. In this game, the cost of voting is 100 and the benefit from one's team winning is 250. A mixed strategy is a best response in this game only if the cost of voting (100) equals the expected benefit $\binom{n_B}{k} = \frac{(n_B)!}{k!(n_B - k)!}$ times the probability of being pivotal): $100 = 125\pi_A$ and $100 = 125\pi_B$, where π_A and π_B are defined as before. Since both teams are the same size, all the players face a symmetric situation. Let's thus look for a fully symmetric Nash equilibrium, in which $p_A = p_B$. Setting p_A and p_B each equal to p in the equation for π_A and plugging into $100 = 125\pi_A$, we get

$$\begin{aligned}
100 = & 125 \times ((1-p)^{11} + 6(1-p)^{10}p + 30(1-p)^9p^2 + 75(1-p)^8p^3 \\
& + 150(1-p)^7p^4 + 200(1-p)^6p^5 + 200(1-p)^5p^6 \\
& + 150(1-p)^4p^7 + 75(1-p)^3p^8 + 30(1-p)^2p^9 + 6(1-p)p^{10} + p^{11})
\end{aligned}$$

Note that we also get an identical requirement for members of team B to play a mixed strategy. The left side of the equation gives the cost of voting, while the right side gives the expected benefit of voting, given that all other teammates and opponents vote with probability p . Both sides of this equation are graphed in [figure 22.1](#), and we see that there are two points where the cost of voting equals the expected benefit: when $p = 0.051$ and $p = 0.949$. Thus, there are two symmetric mixed-strategy Nash equilibria of this game—one in which everyone votes with near-certainty and one in which they abstain with near-certainty. Palfrey and Rosenthal (1983) found that when teams are equally sized, there are typically two symmetric mixed-strategy Nash equilibria, except when costs are sufficiently low. In this figure, we can see that when the cost of voting is around 56, there exists exactly one mixed-strategy Nash equilibrium in which each player votes with probability 0.5, and when the cost is less than this, there is no mixed-strategy Nash equilibrium. Furthermore, they showed that as the size of the electorate grows large, the two mixed-strategy Nash equilibria approach 0 and 1. Thus, there can be substantial turnout in equilibrium, so long as the teams are identically sized. In most situations, however, the equilibrium probability of voting approaches zero as the electorate grows.

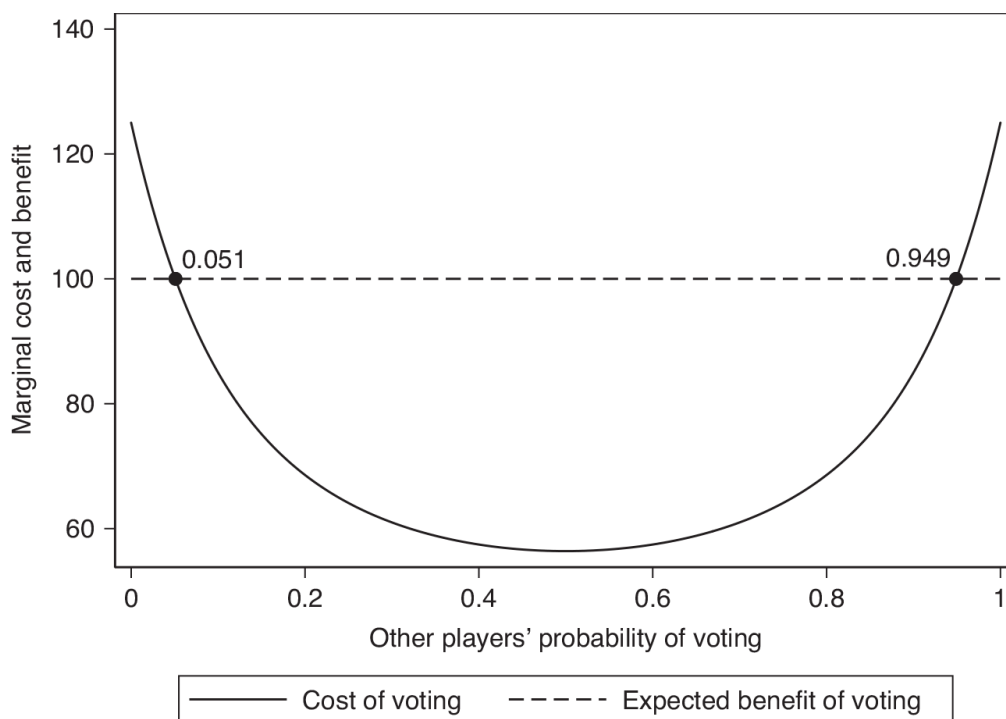


Figure 22.1

The cost of voting and the expected benefit of voting, over the probability that the other players vote.

The Participation Game with Incomplete Information

Thus far, we have assumed that all players face equal costs of voting. In reality, the cost of voting may vary dramatically across individuals. For instance, some people may be able to vote by mail at their convenience, while others must travel and wait in long lines. People also differ in their ability to travel and the opportunity cost of their time. Also remember that we think of c as the net balance of the direct costs and noninstrumental benefits that people derive from the act of voting, and so if people differ in their sense of civic duty or desire to express themselves, this would also be reflected in differences in costs.

Consider the following incomplete information game, studied by Levine and Palfrey (2007). The winning team members receive a payoff of 100, and each person's cost of voting is determined by a random draw from the uniform distribution over $[0, 55]$. Since players could gain only $\frac{B}{2} = 50$ by voting, even if their vote were definitely pivotal, those with costs greater than 50 would always choose to abstain. Likewise, those whose costs are almost 50 would prefer to abstain unless the probability of being pivotal is nearly 1, while those whose costs are close to 0 would prefer to vote unless there is

essentially no chance of being pivotal. In this game, there thus exists an equilibrium in *cutpoint strategies*: each player i 's strategy specifies a cost level, c_i^* , such that she always votes if her cost is below c_i^* and abstains if her cost is above it. We are thus looking for the critical value that cuts the range of costs into those for which voting is a best response and those for which abstaining is a best response. At this critical value, the cost of voting exactly equals the expected benefit and the player is indifferent between voting and abstaining.

Let's look at the situation in which $n_A = 1$ and $n_B = 2$, and find the quasi-symmetric Bayesian Nash equilibrium. That is, we are looking for cutpoints, c_A^* and c_B^* , that are used by all members of A and B, respectively. Thus, we must have $c_A^* = \frac{100}{2}\pi_A$ and $c_B^* = \frac{100}{2}\pi_B$. Recall that we have already found the pivotal probabilities for $n_A = 1$ and $n_B = 2$ to be $\pi_A = 1 - p_B^2$ and $\pi_B = 1 - p_B + p_A p_B$, where p_A and p_B were the probabilities that each other member of team A and team B voted. While we're assuming that players' strategies involve cutpoints rather than probabilities of voting, notice that these probabilities can be derived directly from the cutpoints that each player is using. The probability that a player on team A votes is given by the probability that her cost is less than c_A^* , which we can determine from the distribution of costs in the population. Recall that when a variable is uniformly distributed, the probability that it falls in a given interval is equal to the length of the interval as a fraction of the whole. In this case, the probability that a player's cost is below c_A is $\frac{c_A - 0}{55 - 0} = \frac{c_A}{55}$, as shown in [figure 22.2](#).

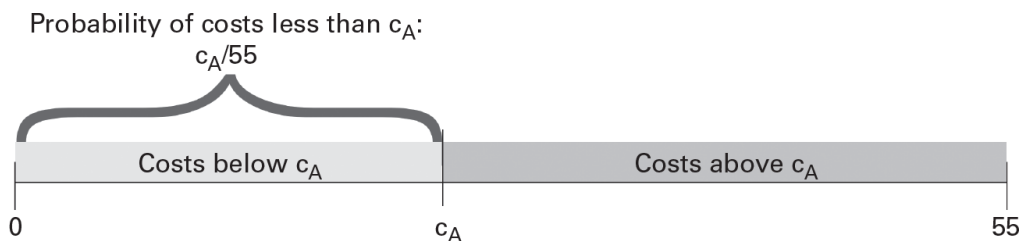


Figure 22.2

Costs are uniformly distributed from 0 to 55. The probability that costs fall in any given interval is equal to the length of the interval as a fraction of the whole (55). Thus, the probability that any given player's voting cost is less than c_A is given by the length of the interval from 0 to c_A (c_A) as a fraction of the whole: $\frac{c_A}{55}$.

Thus, c_A^* and c_B^* are best responses to members of team A using cutpoint c_A^* and members of team B using cutpoint c_B^* when

$$c_A^* = \frac{100}{2} \left(1 - \left(\frac{c_B^*}{55} \right)^2 \right) \quad \text{and} \quad c_B^* = \frac{100}{2} \left(1 - \frac{c_B^*}{55} + \frac{c_A^*}{55} \frac{c_B^*}{55} \right)$$

Solving simultaneously, we get $c_A^* = 29.54$ and $c_B^* = 35.2$, which implies that the member of team A votes with probability 0.537 and the members of team B each vote with probability 0.64. In this case, the team in the majority is slightly more likely to vote than the minority member. However, this is specific to the $n = 3$ case. When $n > 3$, Levine and Palfrey found that there is an *underdog effect*, in which members of the smaller team vote with a higher probability. In the game discussed at the beginning of the chapter, with $n_B = 26$ and $n_A = 25$, we have $c_B^* = 12.95$ and $c_A^* = 13.09$, and these become $c_B^* = 9.4$ and $c_A^* = 11.33$ when the game becomes more lopsided, with $n_B = 34$ and $n_A = 17$. In general, members of both teams are less likely to vote the greater the size disparity between the two parties and the larger the electorate. In fact, the equilibrium cutpoints approach zero as the electorate grows large. Thus, incorporating incomplete information eliminates the equilibria in which there exists substantial turnout in large electorates.

Behavior in Games

The first voter participation experiment was conducted by Schram and Sonnemans (1996). They used the $n_A = n_B = 6$ game we examined earlier, which has a pure-strategy equilibrium of full turnout as well as a mixed-strategy Nash equilibrium in which each person votes with probability 0.051 and one in which everyone votes with probability 0.949. Voter participation tended to start at around 50%, and declined toward 25% with repetition. They also found very similar results in games with $n_A = n_B = 14$ (for which the mixed strategies are more extreme) and $n_A = 8, n_B = 6$ (for which there exists only a mixed strategy in which team B votes with probability 0.05 and team A slightly less frequently). While the multiplicity of equilibria makes it difficult to interpret the results, the finding that voting declines over time in all three treatments suggests that voters may be converging toward the lower (more efficient) mixed-strategy equilibrium.

Levine and Palfrey (2007) studied a game with heterogeneous voting costs, in which the quasi-symmetric Bayesian Nash equilibrium specifies a unique cutpoint (and thus probability of voting), as previously discussed. They considered four group sizes (3, 9, 27, and 51) and varied whether team B had

one more person than team A, or team B was double the size of team A. The table that follows in [figure 22.3](#) shows the predicted voting probabilities and actual voting probabilities for each team and in each treatment. In many treatments, the turnout rates were remarkably close to the prediction, although there again seems to be a general trend in which turnout rates are closer to 0.5 than predicted. In particular, members of B vote less than predicted in the size 3 treatment, and everyone votes more than they should in the larger electorates (27 and 51). Comparing the turnout rates across treatments, essentially all the differences are in the predicted direction: members of B turn out more frequently than members of A when $n = 3$, but otherwise turn out less frequently; turnout is greater when the team sizes are closer, and, holding closeness constant, turnout is lower for larger electorates. Since all the participants played the game for multiple periods, with a different randomly drawn cost each time, the researchers were able to analyze the extent to which each player's voting was consistent with using some cutpoint rule, such that they vote if and only if their cost is less than some critical threshold. They found that most subjects' behavior was consistent with following a specific cutpoint rule in more than 90% of the periods, and nearly all subjects followed a cutpoint rule in at least 80% of periods.

n	n_A	n_B	Predicted p_A	Actual p_A	Predicted p_B	Actual p_B	QRE p_A	QRE p_B
3	1	2	0.537	0.539	0.640	0.573	0.549	0.616
9	3	6	0.413	0.436	0.374	0.398	0.421	0.395
9	4	5	0.460	0.479	0.452	0.451	0.468	0.463
27	9	18	0.270	0.377	0.228	0.282	0.297	0.275
27	13	14	0.302	0.385	0.297	0.356	0.348	0.345
51	17	34	0.206	0.333	0.171	0.266	0.245	0.230
51	25	26	0.238	0.390	0.235	0.362	0.301	0.300

Figure 22.3

Voter participation results from Levine and Palfrey (2007).

Thus, it seems that players respond to their costs of voting. To what extent does the participation decision correspond to the voter's belief that she will be pivotal? Duffy and Tavits (2008) conducted a participation game experiment, but they also elicited the participants' beliefs that their vote would be decisive and paid them according to the accuracy of their beliefs (in addition to their

payoff in the experiment). They found that the likelihood of voting increased with the subjects' beliefs that they would be decisive, but the correlation was far from perfect: many people whose stated beliefs were too low for the expected benefit to outweigh the costs voted anyway, while those who would find it in their interest to vote instead abstained. People also tended to overestimate the probability that they would be decisive.

Notice that if a player believes that she's unlikely to be pivotal, this could mean either that she believes that her team has no chance regardless of whether she votes, or that she believes that her own team will win in a landslide. According to the pivotal voter model, she should abstain in both cases. Is this the case, or do players behave differently depending on whether they think their team is ahead or behind? Several researchers incorporate uncertainty in team size and study the impacts of polls. They find that turnout generally increases the more closely balanced the teams are, but that people are more likely to turn out if they learn that they are in the majority than in the minority (Großer and Schram, 2010; Klor and Winter, 2007; and Agranov et al. 2018).

Agranov et al. (2018) also elicited beliefs about the ultimate breakdown of votes. Imagine that we were to plot the likelihood that a player votes over the predicted lead of her preferred alternative. The x -axis thus ranges from a total landslide loss (in which everyone else in the population votes against her preferred choice) to a landslide victory (in which everyone votes in favor of her preferred choice). The pivotal voter model would suggest that the probability that the player votes is hump-shaped over this range—low when either side is predicted to achieve a landslide victory, but high when the alternatives are in a close contest and there is more uncertainty about who will win. Instead, Agranov et al. (2018) found that the likelihood of voting increases over this range. As a result, landslides are more common when voters have access to polls, as those who anticipate defeat stay home and those who anticipate victory vote.

Evidence of Noninstrumental Voting: Voting Expressively and to Tell Others

Other experiments have found evidence that people vote for reasons other than influencing the outcome and voters who believe they are unlikely to be pivotal sometimes vote for a different option than what they would choose if they were decisive.

Feddersen, Gailmard, and Sandroni (2009) find that voters are more likely to vote against their own material self-interest (and instead vote for a more ethical option) when they believe that they are unlikely to be pivotal. In their experiment, the participants were divided into two groups, A and B. Some members of group B were active players who were invited to vote, while the other members of group B and all of group A were unable to participate. The voters have three choices: vote for option A (which would provide a payment for members of both groups), vote for option B (which would provide a higher payment for members of B, but nothing for members of A), or abstain and avoid the cost of voting. They know that one active player will be selected at random, and this person's choice implemented (with a coin flip used to determine the outcome in the event that this player abstains). Thus, the active players know with certainty the likelihood that they will be pivotal (i.e., 1 divided by the number of active decision makers). Note that option B always corresponds to the voters' material self-interest, but option A might be considered the morally superior alternative, since it provides a more fair payment to everyone. They find that people are significantly more likely to vote for option B when their vote is more likely to be pivotal.

Robbett and Matthews (2018) found evidence of partisan expressive voting: people who believe that they are unlikely to be pivotal are more likely to cast their vote in support of their own party rather than for what they truly believe. In the experiment, American political partisans were asked to vote on the answer to factual questions about politics (e.g., how much mean global temperature had increased or how the number of undocumented immigrants living in the US had changed over the course of the Obama administration). Each question was multiple choice and had an objectively true answer. Asking respondents to identify verifiable facts left no role for differences in preferences or values, and instead focused on their willingness to acknowledge facts that may conflict with their party's preferred views. All members of the group received a cash reward if the majority selected the right answer. They found a large gap in the responses provided by Democrats and Republicans, in the direction you would expect (e.g., Democrats are more likely to say that mean global temperature has gone up, while Republicans are more likely to say that the number of undocumented immigrants has increased). But does this gap arise because members of the different parties have genuine differences in beliefs? Or do they assume that they are not going

to be pivotal and thus elect to answer the question in a way that is more favorable to their own party? In a separate treatment, people answered the questions as decisive individuals, who received the cash reward if their own answer was correct. In this case, the gap between Democrats and Republican shrunk substantially, indicating that people give more partisan answers when they're voting than they would if they knew their choice were decisive.

DellaVigna et al. (2017) suggested that social image concerns influence people's choice to turn out: they vote because they anticipate that people will later ask them whether they voted, and they want to be able to answer honestly that they did. To test this, the authors left flyers hanging on the doors of homes in the Chicago area after the 2010 election, explaining that they would come back the next day to do a door-to-door survey. Some flyers explicitly explained that the survey would ask about their participation in the previous election. However, the authors already knew whether members of the household voted—they used the publicly available voter file to only target households where either everyone or no one in the home had voted in the previous cycle. They found that those who did not vote were substantially less likely to answer the door and respond to the survey when they knew that they would be asked about whether they had participated in the election—indicating that people do not wish to have to admit that they did not vote (or to lie about it). By varying the amount that participants were told they would receive for answering the survey, the researchers estimated the monetary value of voting to tell others to be between \$5 and \$15.

Evidence-Based Adaptations of the Standard Model

In Levine and Palfrey (2007)'s incomplete information game, we saw that the Bayesian Nash equilibrium tended to overpredict turnout in small elections and underpredict in large elections. Recall that the quantal response equilibrium model assumes that players don't always best respond by choosing the strategy associated with the highest payoff: instead, they are simply more likely to play a strategy the higher its payoffs compared to those of other strategies. As we've seen in previous applications, the quantal response equilibrium (QRE) generally predicts that participants play mixed strategies closer to 0.5 than the Nash equilibrium does, which suggests that the QRE prediction may organize the data better.

Under the Bayesian Nash equilibrium, each player used a cutpoint strategy. They abstain if and only if their costs are greater than some critical level,

equal to the point where the cost equaled the pivotal probability times the expected benefit. Under the QRE model, participants vote with some probability between 0 and 1 for each cost level, such that they are *more likely* to abstain the greater the costs compared to the expected benefit.

Let's consider the $n_A = 1$ and $n_B = 2$ example that we analyzed earlier. Costs were uniformly distributed between 0 and 55, and the expected benefit $B/2$ of being pivotal was 50. We found that under the Bayesian Nash equilibrium, the team A player voted whenever her costs were less than 29.5 and the team B players voted whenever their costs were less than 35. [Figure 22.4](#) depicts the team A player's probability of voting for each cost level. The different curves show the player's "better-response" functions for several values of λ , which is a parameter that describes how responsive the player's actions are to her payoffs (as explained in greater detail in chapter 26). The lighter-colored curves are associated with higher λ values, meaning that the player is more responsive to her payoffs (or, crudely, "more rational"). Note that when the cost is 29.5, the player's expected payoff from voting equals her expected payoff from abstaining and she votes with probability 0.5 for all values of λ . For costs less (greater) than 29.5, voting (abstaining) is her best response, and she votes (abstains) with probability greater than 0.5. For high λ , the player is highly responsive to differences in her expected payoffs for the two strategies, and we see that her strategy approaches the cutpoint strategy of voting with certainty for costs less than 29.5 and abstaining with certainty otherwise. For lower levels of λ , she is less sensitive to payoff differences and mixes with probability closer to 0.5 for most cost levels. Thus, the quantal response model predicts the individual-level deviations from sharp cutpoint strategies. For clarity, this is graphed assuming that the B players are playing the Nash cutoff strategy of 35. Note, though, that under a quantal response *equilibrium*, the team A player must better respond to the actual strategy employed by the team B players.

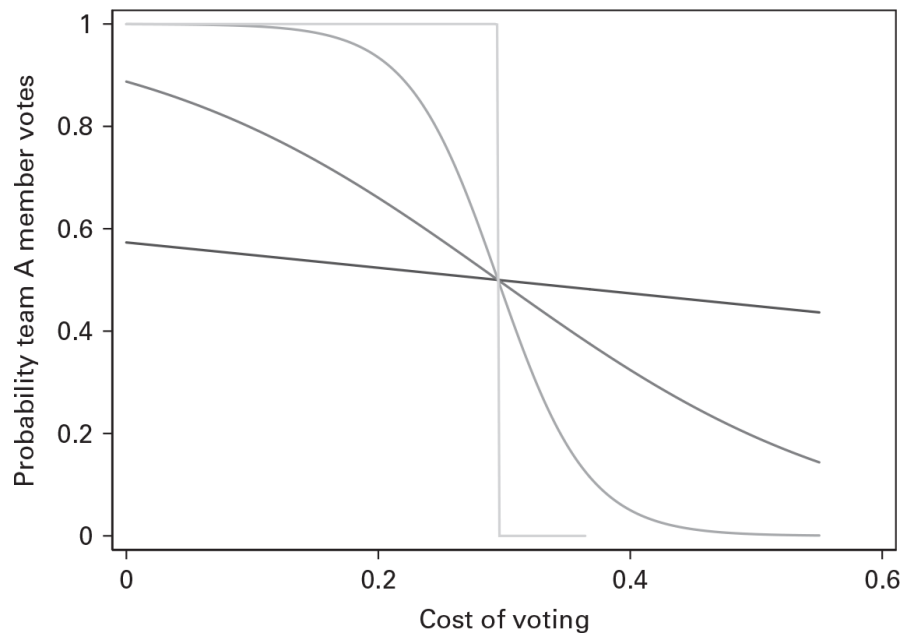


Figure 22.4

The probability that a member of team A votes for each cost under the quantal response model. The lighter the curve, the higher the λ value (i.e., the more responsive the player is to the difference in her payoff from voting and abstain). The figure is graphed for $\lambda = 1, 7, 28, 10000$.

The QRE predictions—in which players better respond to the other players better responding—are shown in [figure 22.3](#) for $\lambda = 7$. While the Bayesian Nash equilibria generally predicted the trends across treatments, the QRE also more closely predicts the turnout rates in larger and smaller elections.

The QRE model can also help explain the paradox of voting. We saw that the Bayesian Nash equilibrium cutpoint strategies approached zero when the electorate grew large (indicating that turnout should go to zero). This is not the case for the QRE, since players who make mistakes will still vote with some probability, even when they believe they are unlikely to be pivotal. For instance, imagine that the probability that a player is pivotal is actually 0. In this case, the payoff difference between voting and abstaining is $-c$. The quantal response function for each cost and $\lambda = 7$ is graphed in [figure 22.5](#). When $c = 0$, the player votes with probability 0.5, since voting and abstaining provide the same payoff. A citizen with positive costs abstains more often than not, but still votes with positive probability. Given that each person follows this strategy and that costs are distributed uniformly from 0 to 55, we can calculate that 17.5% of the population is expected to turn out. Thus, turnout does not go to zero even when the chance of being pivotal does. In addition, Levine and Palfrey (2007) point out that the costs of voting in this

game are relatively high (on average, 27.5) compared to the benefit of one's preferred candidate winning (100). If we were to instead assume that the costs for most of the population are on the lower end of this range, turnout would be higher (and approaches 50% as costs go to zero).

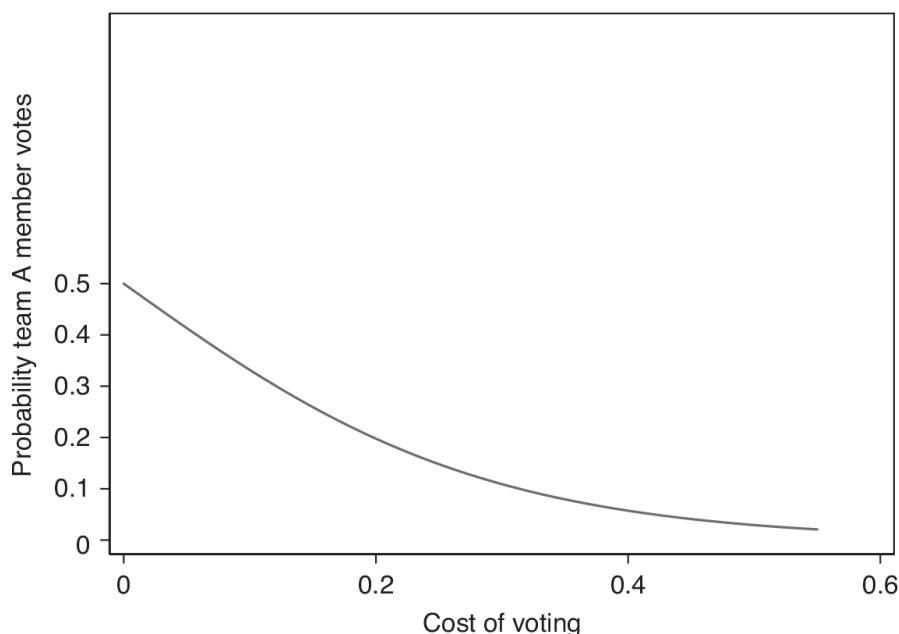


Figure 22.5

The probability with which a player votes under the quantal response model, given that they are pivotal with probability zero and there is no expected benefit of voting.

In other studies, we saw that people may be more likely to vote when they believe that their team will win. One reason why we may see voting probability increase when voters learn that they are in the majority is *bandwagon voting*: voters may derive utility from voting for the winner. Several researchers have also documented evidence of bandwagon voting outside the laboratory. For instance, Morton et al. (2015) found that voters in French territories in the Western Hemisphere followed the behavior of voters in mainland France who cast their ballots first. We can model the behavior of such bandwagon voters by adding another term to the expected benefits side of the calculation of whether to vote, such that they prefer to vote if

$$c < \pi \left(\frac{B}{2} \right) + a(\text{probability vote for winner})$$

where a is an individual-specific parameter capturing the value the voter receives if she votes for the winner (see Callander, 2007 and 2008, for models

of bandwagon voting in elections where everyone prefers the more competent candidate but receives different signals about who that is). In fact, for high-enough values of a , players may even prefer to vote for the candidate they like less if they believe that person will be more popular.

Agranov et al. (2018) used their subjects' choices to estimate a for each individual. They found substantial heterogeneity in the data, with between a third and a half of the subjects behaving consistently with $a = 0$, while others' choices seem strongly driven by a desire to vote for the winner. They found that approximately 78% of their data could be explained by a model in which players are one of two types. In addition, they found that there exists a Bayesian Nash equilibrium in which players who do not care about voting for the winner always abstain, voters who care about voting for the winner and learn that they are in the majority always vote, and voters who care about winning and learn that they are in the minority play a mixed strategy. Thus, this modification to the payoff function explains the patterns in the data, such that people vote with higher probability when they learn that they are in the majority.

Exercises

- (1) Consider a participation game with $n_A = 3$ and $n_B = 6$. What is the probability that a member of team A will be pivotal if the other A team members vote with probability $p_A = 0.8$ and B team members vote with probability $p_B = 0.25$?
- (2) Consider a participation game with $n_A = 0$ and $n_B = n > 0$. In this case, team B will win if at least one member participates, in which case each member will receive a payoff of 1. Otherwise, the teams will tie, and a coin flip will determine the outcome. Say that the cost of voting is 0.3. Under the symmetric Nash equilibrium, with what probability do members of B vote?
- (3) Consider a participation game with $n_A = 1$ and $n_B = 8$. Each member of the winning team earns 2, and the cost of voting is 0.25. There exists a quasi-symmetric Nash equilibrium in which all members of B vote with probability p and the member of A votes with probability $1-p$. Find p in this equilibrium. (Hint: Once you have derived the expression that must hold in equilibrium, you can plug both sides into a graphing calculator to find the intersection rather than solving algebraically.)

(4) Consider a participation game where the payoff to the winning team is 1 and the cost of voting is $c = 0.25$. Unlike the games we've looked at thus far, it is determined in advance that team A will be the winner in the event of a tie (and this is common knowledge). That is, team B will win only if they receive strictly more votes. Let $n_A = n_B$. Is there a pure-strategy Nash equilibrium of this game?

(5) Consider a participation game with $n_A = 1$ and $n_B = 2$, where the payoff to the winning team is 200 and the cost of voting is $c = 36$. It is determined in advance that team B will be the winner in the event of a tie (and this is common knowledge). That is, team A will win only if the one member of team A votes and neither member of B does. Find the quasi-symmetric Nash equilibrium in which each player chooses a mixed strategy.

(6) Consider a participation game with $n_A = n_B = 2$. The payoff to the winning team is 150 and the costs are uniformly distributed between 0 and 100. Find the symmetric Bayesian Nash equilibrium in which each player uses the same cutpoint strategy.

¹Why is this the probability that $n_A - 1$ members of A vote? The probability that one specific member of A does not vote while the remaining $n_A - 1$ do is given by $p^{n_A-1}(1 - p)$. However, any one of the n_A players could be the one not to vote, so we need to multiply this by n_A to indicate that there are n_A possible ways that this breakdown of votes could be generated.

²Note that it makes sense that the probability that the member of A is pivotal is $1 - p_B^2$ since the only situation in which the player isn't pivotal is when both members of B vote, which occurs with probability p_B^2 .

23

Voting with Private Information

We have thus far considered voting as a means of aggregating individuals' preferences. In this chapter, we will consider voting as a means of aggregating information, when people are not fully informed about the true state of the world. For instance, imagine that everyone's preferences are aligned, such as on a jury, where everyone prefers to convict the guilty and acquit the innocent. People may interpret the evidence differently, but on average each person is more likely to be right than wrong. If everyone votes sincerely, majority rule can allow the group to make a more accurate decision than any one individual could, and requiring unanimity can all but eliminate the likelihood of convicting an innocent defendant. However, we'll see that strategic voters may wish to vote contrary to their own information or free ride on the information acquisition of others. As a result, strategic voting can lead to perverse outcomes, such as unanimous voting rules increasing the likelihood of convicting an innocent defendant.

Imagine that you are serving on a 12-person jury. The jury's job is to decide whether to convict or acquit the defendant. Going into the trial, it is common knowledge that the defendant is equally likely to be guilty or innocent, and all jurors agree that it is best to convict the guilty and acquit the innocent. Everyone receives a payoff of -50 if an innocent defendant is convicted or a guilty defendant is acquitted, and receives a payoff of 0 if the jury gets it right.

Over the course of the trial, the jurors hear testimony and evidence that they may interpret differently. Let's say that each juror gets an independent signal of either "guilty" or "innocent." The probability that any one person interprets the evidence correctly is 0.7 . That is, if the defendant is actually guilty, then a juror gets the impression that he is guilty with probability 0.7 and innocent with probability 0.3 , and if he is actually innocent, then jurors get the impression that he is guilty with probability 0.3 and innocent with probability 0.7 . The signals are independently determined across people.

Each person simultaneously votes to acquit or convict the defendant. The defendant will be convicted only if all 12 jurors vote for conviction. Imagine that your signal is that the defendant is guilty. What do you vote for? What if your

signal is that the defendant is innocent? Would you behave differently if you knew that only a majority (at least 7 votes) was necessary for conviction?

Voting to Aggregate Information

In this chapter, we will consider a group of people who must make a choice between two alternatives, such as whether to convict or acquit a defendant, which candidate to vote for, or whether to approve a change in local tax policy. In the previous chapters, we assumed that people differed in their preferences over alternatives, but everyone had complete information over which alternative they personally preferred. Now let's consider the reverse situation. Imagine that everyone's preferences are aligned, but they have different interpretations or evidence about which alternative is "correct."

For instance, the defendant may be guilty or innocent, and everyone agrees that it is best to convict in the former case and acquit in the latter case. However, the jurors may interpret the evidence presented to them differently or draw different conclusions about whether a witness is credible. Similarly, people may all agree that it is best to elect the most competent candidate, but they have different indications of which candidate is actually better, or they agree that they should implement the tax policy that will stimulate the local economy but have different understandings of which policy would achieve that end.

Let's say that any one individual interprets the evidence correctly with probability p (and gets it wrong with probability $1 - p$). If a single person must make the decision for the group, they will thus make the right choice with probability p . However, there is greater wisdom in crowds. If there are n individuals, each with an independent chance of answering correctly, then the group as a whole has n independent signals about the correct answer. As the group grows larger, they thus have a substantial amount of evidence and the majority is almost certain to arrive at the correct answer. This logic was first described by the Marquis de Condorcet in 1785, in a result now known as *Condorcet's jury theorem*, as described in theorem 1.

Theorem 1 (Condorcet's jury theorem [Condorcet, 1785]) *Imagine that there are two possible outcomes, one of which is correct. If each individual votes for the correct outcome with probability $p > 1/2$, then adding more voters increases the probability that the majority votes for the correct*

outcome. As the number of voters goes to infinity, the probability that the majority votes correctly converges to 1.

Notice that there is an obvious downside to Condorcet's jury theorem: in cases where individuals' judgments are poor, such that they're more likely than not to get it wrong ($p < 1/2$), then voting makes the situation even worse. In that case, the probability that the group gets it right goes to 0 as the group size grows larger.

Consider our example from the beginning of the chapter, where 12 jurors each receive a signal that is correct with probability 0.7. If each person votes according to her own independent signal and the defendant is convicted only if the majority vote is in favor, then the group gets it wrong with probability 0.078.

Notice that this includes getting the decision wrong in either direction: convicting an innocent defendant or acquitting a guilty defendant. Society may view these outcomes to be equally bad, or they may prefer to minimize the likelihood of one at the expense of the other. The requirement that juries of 12 people must unanimously agree that the defendant is guilty in order to convict was designed specifically to minimize the likelihood of a wrongful conviction, albeit at the expense of a greater likelihood that the guilty would be acquitted. In our example, what is the probability that the jury would wrongly convict an innocent defendant if all jurors must unanimously agree on conviction? This would occur only if all 12 jurors got it wrong, which occurs with probability $(1 - p)^{12} = 0.3^{12}$, which is about 1 in 1.88 million.

This conclusion rests on a couple of assumptions. First, we're assuming that the jurors' signals are independent—in the case of a botched defense during trial, for instance, the signals may be correlated. In addition, this logic assumes that everyone always votes according to their signal. As we will see, when voters are strategic, then the requirement that a large jury unanimously agree to convict may have the exact opposite effect than intended: innocent defendants become *more likely* to be convicted than under majority rule, and this likelihood actually increases with jury size. To analyze the strategic interaction between players, let's begin by formally modeling the situation as a game.

The Jury Game

Consider a jury of n players, who must decide whether to *acquit* or *convict* a defendant. The model and its main results are due to Feddersen and Pesendorfer (1998). There are two possible states of the world: the defendant is either guilty (G) or innocent (I). Nature makes the defendant guilty with probability π . In most of what follows, we will assume $\pi = 0.5$, but we will also consider other values of π .

None of the jurors observes Nature's move directly. Instead, each person observes an independent signal that is correlated with the true state of the world: either g (she interprets the evidence as indicating the defendant is guilty) or i (she interprets the evidence as indicating innocence). Let's assume that each person has the same probability of getting it right and is more likely to be right than wrong. If the state of the world is G (the defendant is guilty), then each player observes the correct signal g with probability $p > 1/2$ and the incorrect i with probability $1 - p$. If the state of the world is I , then each player observes the incorrect signal g with probability $1 - p$ and the correct i with probability p . Thus, the player is able to update her beliefs about the likelihood that the defendant is guilty after observing her signal. Each player's signal is determined independently and is private information. The player's signal thus corresponds to her type, and a strategy must specify what she would do if she receives signal i and what she'd do if she receives signal g .

Assume that everyone prefers to acquit innocent defendants and convict guilty ones, and both of these outcomes are associated with payoffs of 0. If an innocent defendant is convicted, then each person receives a payoff of $-q$ and if a guilty defendant is acquitted, then each person receives a payoff of $-(1 - q)$, where $q \in (0, 1)$. Notice that q represents the juror's threshold for reasonable doubt: she will prefer a conviction if and only if the likelihood that she believes he's guilty is greater than q . When $q = 0.5$, she views convicting the innocent and acquitting the guilty as equally bad outcomes, and thus she would wish to convict whenever she believes the likelihood of guilt is greater than 0.5. When q is higher than 0.5, the juror views convicting an innocent person as the worst-case scenario and thus has a higher threshold for conviction—she'd need to be pretty sure that he was guilty before she'd prefer to convict.

The Optimal Decision of a Single Juror

Before we analyze the strategic interaction between the jurors, let's imagine that there is a single juror ($n = 1$) whose choice decides the fate of the defendant. Her strategy must specify whether she would acquit or convict if she receives signal g and if she receives signal l .

Let's first consider her expected payoff after receiving signal g . What is the juror's updated belief regarding the probability that the defendant is guilty? Specifically, we need to know $Pr(G|g)$: the probability that the defendant is guilty, given that the signal of g was observed. As we discussed in chapter 12, this can be calculated using Bayes' rule:

$$Pr(G|g) = \frac{Pr(g|G)Pr(G)}{Pr(g)} = \frac{Pr(g|G)Pr(G)}{Pr(g|G)Pr(G) + Pr(g|I)Pr(I)}$$

We know that $Pr(g|G) = p$ and $Pr(g|I) = 1 - p$, and that $Pr(G) = \pi$ and $Pr(I) = 1 - \pi$. Thus, we have

$$Pr(G|g) = \frac{p\pi}{p\pi + (1-p)(1-\pi)}$$

Likewise, we can compute the juror's updated belief that the player is innocent after receiving the guilty signal:

$$Pr(I|g) = \frac{(1-p)(1-\pi)}{(1-p)(1-\pi) + p\pi}$$

Now that we know the juror's updated belief that the juror is actually guilty or innocent, given her signal, we can calculate the juror's expected payoff from convicting and acquitting:

$$Eu(Convict|g) = p(G|g)(0) + p(I|g)(-q) = \frac{(1-p)(1-\pi)}{(1-p)(1-\pi) + p\pi}(-q)$$

$$Eu(Acquit|g) = p(G|g)(-1+q) + p(I|g)(0) = \frac{p\pi}{p\pi + (1-p)(1-\pi)}(-1+q)$$

The juror prefers to convict when $\frac{(1-p)(1-\pi)}{(1-p)(1-\pi) + p\pi}(-q) \geq \frac{p\pi}{p\pi + (1-p)(1-\pi)}(-1+q)$ or, rearranging, when

$$q \leq \frac{p\pi}{p\pi + (1-p)(1-\pi)}$$

Notice that the right side is simply equal to the juror's updated belief that the defendant is guilty after observing the guilty signal. That is, this corresponds to our earlier realization that the juror prefers to convict whenever her belief in his guilt exceeds her reasonable doubt threshold of q . We will continue to

use this shortcut to analyze the juror's optimal decisions throughout the chapter.

If $\pi = 0.5$, then this condition reduces to $p \geq q$: the player didn't have any prior indication one way or the other that the defendant was guilty before observing the signal, so she goes entirely on her signal and convicts if it is accurate enough to pass her reasonable doubt threshold.

Likewise, we can compute the player's updated beliefs after observing the signal of ι and calculate her best response in this case. We again find that she will follow her signal (in this case, choose acquit) so long as $q \geq \frac{(1-p)\pi}{(1-p)\pi + (p)(1-\pi)}$.

In the case that $\pi = 0.5$, then she will follow an innocent signal by voting to acquit so long as $q > 1 - p$. Putting this together with the condition for preferring conviction after viewing g , we get that the player will vote according to her signal whenever $1 - p < q < p$. For example, if $q = 0.5$ (she views convicting the innocent and acquitting the guilty as equally bad), then the player would always vote according to her signal.

When a player votes according to his own signal, we say that he is *voting informatively*. Based on the previous calculations, we conclude that an individual decision maker will typically follow his signal unless he has strong prior beliefs about the likelihood that defendants are guilty (π is far from $\frac{1}{2}$), his signal is only weakly correlated with the true state of the world (p is close to $\frac{1}{2}$), and/or he has a very high or low threshold for conviction (q differs from $\frac{1}{2}$). However, these calculations may change if the decision maker interacts strategically within a group of voters. Even if the juror prefers to follow his signal when he alone decides the fate of the defendant, he may prefer to strategically ignore his own information when voting in a group.

Voting Strategically against Your Own Evidence

Given that individuals would prefer to follow their own signals when they are the only decision maker, why would they have an incentive to strategically disregard their own information when they vote in a group? After all, the players share a common interest to make the best decision for the group, given their information. The difference is that individual decision makers know that their choice will always be decisive, and they only have their own signal to go by when updating their beliefs. In contrast, players who vote in a

group know that their own individual vote affects the outcome only if the other players' votes break down in a very specific way.

We say that a player's vote is *pivotal* if changing her vote would change the group's decision. For instance, under majority rule, a single player's vote would tip the outcome only if the other players split their votes between acquit and convict. Under unanimity, a player's vote is pivotal only in the event that every other person votes for conviction.

Strategic players have an incentive to assume that they will be pivotal and cast the vote that would be optimal in that specific scenario. If they are not pivotal, then there is no harm in following this strategy, since their single vote didn't influence the outcome anyway. If they are pivotal, then they voted optimally, given the breakdown of the other players' votes.

Furthermore, if the player believes that others are voting according to their private information, then she can infer the distribution of signals among the other players from the breakdown of their votes that would make her vote pivotal, and thus update her beliefs according to these inferred signals. In fact, if she believes that they voted informatively, then assuming a specific distribution of other players' votes means that she also assumes that she knows all n signals and can update her belief about the likelihood of guilt based on far more than her own private signal.

Thus, the incentive to vote contrary to one's own signal may arise when players strategically anticipate the breakdown of votes that would cause their own vote to matter, infer the signals that the other players must have received in that situation, and then base their decision on the full array of signals that they believe the group received.

Conditioning on Being Pivotal Under Unanimity

Imagine that you are serving on a jury that requires unanimous support to convict a defendant. If any other juror votes to acquit, then your own vote won't matter either way—the defendant will be acquitted. The only situation in which your vote would make a difference is if *every other juror* voted for conviction, such that your vote either seals the deal or lets him off the hook. But if every other juror votes for conviction—and you believe that they're all following their independent signals—then that is a pretty compelling indication that G is the true state of the world.

Let's consider the situation with $\pi = 0.5$. If all $n-1$ other jurors received signal g , what is your updated belief that the defendant is guilty? If you

received signal g as well, then we have: $\frac{p^n}{p^n + (1-p)^n}$. If instead you received signal i , your updated belief is $\frac{p^{n-1}(1-p)}{p^{n-1}(1-p) + p(1-p)^{n-1}}$.

In the example from the start of the chapter, with $n = 12$ and $p = 0.7$, your updated belief is 0.99996 if you receive a guilty signal and 0.9998 if you receive an innocent signal. Thus, jurors receiving an innocent signal have an incentive to ignore their own information and vote for a conviction, so long as they have a reasonable distaste for acquitting guilty defendants (i.e., q reasonably differs from 1). In this case, we cannot have an equilibrium in which everyone votes according to their own signal.

Note that this doesn't imply an equilibrium in which everyone fully ignores their information and votes to convict. If everyone convicted regardless of the signal, then the votes of the other jurors would be completely uninformative of their signals. Even if a juror conditioned on being pivotal, she would not be able to infer any additional information about the defendant's guilt from everyone else voting to convict, so she would go entirely off her own signal, as in the one-player case.

Conditioning on Being Pivotal Under Majority Rule

Conditioning on being pivotal leads to less extreme inferences under majority rule. When the number of jurors is odd, a player is pivotal only if the other jurors are evenly split. If everyone else votes according to their signals, this implies that exactly half received a signal of guilt and half received a signal of innocence. When $\pi = 0.5$, this means that the juror's own signal will determine whether she believes the defendant is more likely to be guilty or innocent. In situations when an individual decision maker would always follow her own signal if she alone were to decide the outcome, she would also follow her signal if she were voting under majority rule and believed others to be voting informatively.

In the example at the beginning of the chapter, where $q = 0.5$, $\pi = 0.5$, and $p = 0.7$, a single decision maker would prefer to convict if she saw a guilty signal and acquit if she saw an innocent signal. Now let's imagine that she is in a group of n jurors, each of whom votes according to his signal. She assumes that she is pivotal, and thus among the other $n - 1$ players, there were $\frac{n-1}{2}$ guilty signals and $\frac{n-1}{2}$ innocent signals. As a result, her own signal tips her choice one way or the other.

We can also see this mathematically using Bayes' rule. Let's say that the player sees a guilty signal. What is the player's updated belief of guilt after seeing $1 + \frac{n-1}{2}$ guilty signals and $\frac{n-1}{2}$ innocent signals?¹

$$Pr(G \mid \text{own signal of } g, \frac{n-1}{2} \text{ other } g \text{ signals, and } \frac{n-1}{2} \text{ other } i \text{ signals}) =$$

$$\frac{0.5(0.7)(0.7)^{\frac{n-1}{2}}(0.3)^{\frac{n-1}{2}} \binom{n-1}{\frac{n-1}{2}}}{0.5(0.7)(0.7)^{\frac{n-1}{2}}(0.3)^{\frac{n-1}{2}} \binom{n-1}{\frac{n-1}{2}} + 0.5(0.3)(0.7)^{\frac{n-1}{2}}(0.3)^{\frac{n-1}{2}} \binom{n-1}{\frac{n-1}{2}}}$$

This updated belief simply reduces to 0.7, indicating that the player goes by her own signal when assessing the defendant's guilt and behaves as if she were the lone decision maker.

However, this doesn't mean that players always prefer to vote informatively under majority rule. In particular, they will not vote informatively in a jury if they also would not vote informatively in the event that they were the only decision maker. Imagine a situation in which q is high (the jurors have a high bar to convict) and/or π is low (their prior belief is that the defendant is likely innocent). In this case, an individual decision maker may be unwilling to convict on the basis of a single guilty signal and would require multiple signals in the direction of guilt to update her belief that the defendant is guilty beyond q . When she votes in a jury where everyone else votes informatively, she assumes that she's pivotal only if there were an equal number of signals for guilt and innocence. If she receives a guilty signal, then she doesn't view a single extra piece of evidence in favor of guilt to be sufficient to convict, and thus she votes against her signal.

Thus, there is no "one size fits all" voting rule that guarantees informative voting. Instead, whether players wish to vote informatively or strategically depends on their prior beliefs, the strength of their signals, and their reasonable doubt threshold. In particular, there can exist an equilibrium in which players vote according to their signal only if the number of guilty signals that a lone decision maker would require before preferring conviction aligns with the number of votes that the voting rule requires for conviction. Imagine that a lone decision maker viewed all n signals before making her choice. Depending on her preferences and beliefs, there is a certain number of guilty signals that she would need to see before preferring conviction. If the voting rule also requires this number of votes to convict, then a player would be willing to vote informatively and we can have a Bayesian Nash equilibrium in which players vote according to their signals. In the next

section, we will search for Bayesian Nash equilibria in which players sometimes vote against their private information.

Bayesian Nash Equilibria: Convicting the Innocent under Unanimity

Under both types of voting rules, there exist Bayesian Nash equilibria in pure strategies, in which players completely ignore their signals and cast the same vote regardless of their evidence. In this case, almost any combination of voters choosing acquit no matter what and convict no matter what can be a Bayesian Nash equilibrium, so long as no one is pivotal in determining the outcome. For instance, under majority rule, all jurors voting to convict regardless of their signal and all jurors voting to acquit regardless of their signal are both equilibria, since no one person would be able to change the outcome, and thus no one has an incentive to cast more helpful votes. Likewise, all jurors voting to acquit is an equilibrium under unanimity, as is one person voting to convict while the rest vote to acquit, and so on.

In this section, we will search for symmetric Bayesian Nash equilibria in which players at least partially take into account their own signal, while also rationally best responding to the strategies of the other players. We saw in the previous section that there is an informative Bayesian Nash equilibrium under majority rule, so long as an individual decision maker would be willing to convict on the basis of a single signal, but that unanimity does not support an informative Bayesian Nash equilibrium except when the players have extreme preferences and wouldn't be willing to convict unless they received n guilty signals. We will now search for Bayesian Nash equilibria in the unanimity game.

Feddersen and Pesendorfer (1998) showed that so long as $q > 1 - p$, there is a symmetric Bayesian Nash equilibrium in which players vote to convict after seeing signal g , while players with signal ι play a mixed strategy, in which they vote to convict with probability c . Let's analyze the game with $p = 0.7$, $q = 0.5$, $n = 12$, and $\pi = 0.5$. To show that this is a Bayesian Nash equilibrium, we need to show that a player who sees the signal of ι is indifferent between conviction and acquittal, given that his vote is pivotal—that is, assuming that all $n - 1$ other players voted to convict. The player would be indifferent between conviction and acquittal only if his updated belief about the likelihood that the defendant is guilty is exactly equal to q , which is 0.5 in this game. That is, the player is willing to play a mixed

strategy only if his updated belief is that the defendant is equally likely to be guilty or innocent.

To calculate the player's updated belief, we use Bayes' rule to find the probability that the defendant is guilty, given that the player's own signal is i but all 11 other individuals choose to convict:

$$\frac{Pr(i|G)Pr(\text{another player votes to convict}|G)^{11}Pr(G)}{Pr(i|G)Pr(\text{another player votes to convict}|G)^{11}Pr(G) + Pr(i|I)Pr(\text{another player votes to convict}|I)^{11}Pr(I)}$$

We know that $p(i|G) = 1 - p = 0.3$, $p(i|I) = p = 0.7$, and $p(G) = p(I) = 0.5$. To calculate this expression, we just need to know the probability that another player chooses to convict given that the defendant is guilty and given that he is innocent. If the defendant is guilty, then the player correctly receives signal g with probability p and chooses to convict, and receives the incorrect signal of i with probability $1 - p$, but chooses to convict anyway with probability c . Thus, the probability that the player votes for conviction when the defendant is actually guilty is $p + (1 - p)c = 0.7 + 0.3c$. Similarly, the probability that the player votes for conviction when the defendant is actually innocent is $(1 - p) + (p)c = 0.3 + 0.7c$.

We therefore get that our player's updated belief of guilt is

$$\frac{(0.3)(0.7 + 0.3c)^{11}(0.5)}{(0.3)(0.7 + 0.3c)^{11}(0.5) + (0.7)(0.3 + 0.7c)^{11}(0.5)}$$

which equals 0.5 when $c = 0.824$. Thus, there is a symmetric Bayesian Nash equilibrium in which players whose own interpretation of the evidence points to innocence vote for conviction over 80% of the time.

Of course, just because jurors with innocent signals often vote to convict doesn't mean that the high bar of unanimity will be met and the defendant will actually be convicted. Given that people often vote against their own evidence in favor of conviction, what is the likelihood that an innocent defendant is convicted? The probability that any one player votes to convict, given that the defendant is truly innocent, is $0.3 + 0.7c = 0.3 + 0.7(0.824) = 0.877$. Therefore, the probability that all 12 jurors wrongly vote for conviction is $0.877^{12} = 0.207$.

So while we saw that players who vote according to their private information have essentially zero chance of convicting an innocent defendant under unanimity (approximately a 1 in 1.88 million chance), jurors who behave strategically inadvertently convict the innocent one-fifth of the time. [Figure 23.1](#) shows the probability of convicting an innocent defendant and the

probability of acquitting a guilty defendant under the symmetric mixed-strategy Bayesian Nash equilibrium for different-sized juries. As we can see, the likelihood of the jury convicting an innocent or acquitting a guilty defendant are both substantial. Furthermore, the probability of convicting an innocent defendant *increases* in larger juries: exactly the opposite of the intended high threshold of a 12-person jury. While the probability of acquitting a guilty defendant decreases with jury size, it never dips below 47%.

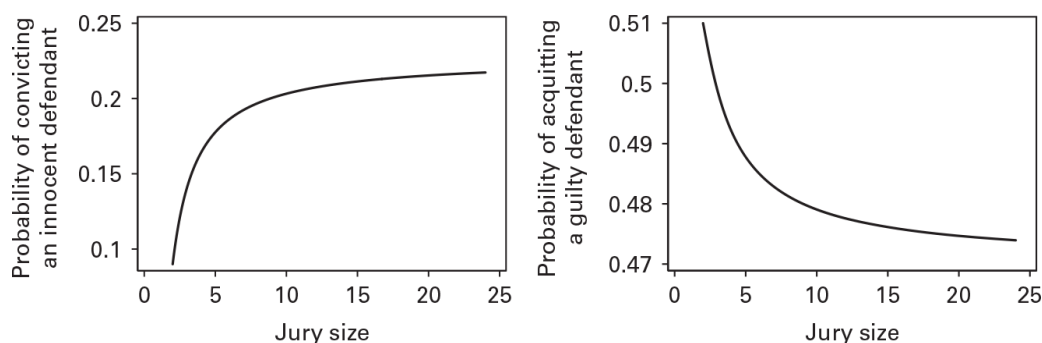


Figure 23.1

The equilibrium probability of convicting an innocent defendant (left) and acquitting a guilty defendant (right) over jury size for $q = 0.5$ and $p = 0.7$.

This is not just an artifact of this particular setup. Feddersen and Pesendorfer (1998) showed that, so long as an individual who observes $n-1$ guilty signals prefers conviction and $q > 1-p$, the only symmetric equilibrium in which players respond to their information requires a strategy in which players convict if they receive a guilty signal and play a mixed strategy if they receive an innocent signal. Furthermore, the probability c that players choose to convict after seeing i goes to 1 as n goes to infinity. If these assumptions are not satisfied, and players would instead prefer to acquit if they observe $n-1$ guilty signals but convict if they observe n guilty signals, then there is an informative voting equilibrium under unanimity, as we saw previously.

We have thus seen that the ability to support an equilibrium in which players reveal their private information depends on the extent to which the number of guilty votes required by the voting rule aligns with the number of guilty signals that they would require in order to be willing to convict if they were decisive. When the voting rule is misaligned with the players' preferences (as in the case of unanimity in our example), then they will vote

strategically, their true information will not be aggregated into the group's choice, and they may end up with a situation that everyone agrees is suboptimal. Yet this situation could be avoided if players shared their information with each other outside of a final vote in which they have incentive to condition on being pivotal. For instance, imagine that players first conducted a nonbinding straw poll. There exists an equilibrium in which players reveal their private information truthfully in the nonbinding poll and then everyone ignores their own information and votes according to the n signals that were revealed in the poll. In this case, we wind up with a situation in which players have access to all n signals and can cast the optimal vote given the actual profile of signals rather than the optimal vote for the specific situation in which their vote would be pivotal. In contrast, jurors who have different preferences or prior beliefs may have an incentive to misreport their information during the straw poll.

The Swing Voter's Curse

Often voters turn out for high-profile contests, such as presidential, gubernatorial, or senatorial elections, but fail to register a choice for items farther down the same ballot, such as local elections or referenda. In the previous chapter, we saw that abstention is rational when voting requires costly effort and the likelihood of being pivotal is low. But once someone is already in the voting booth to cast a ballot for some races, it is essentially costless to fill in a bubble or hit a button for the others. Feddersen and Pesendorfer (1996) proposed that abstention can be rational even when voting is costless if voters believe that others are better informed about the best choice. In fact, even players who have a strong prior belief that one candidate is better than another may achieve strictly higher payoffs from abstaining if they believe that it is highly likely that other players know the best choice with certainty.

In chapter 13, we saw that common value auctions are plagued by the winner's curse: learning that you won the auction suggests that you had a wildly optimistic assessment of the item's value and you just overpaid. Analogously, the phenomenon in which learning that you swung the election causes you to realize that someone had better information and you should have abstained is called the *swing voter's curse*. Imagine that players have a common prior belief that one option is better than another, but some players

have more accurate information. If a less-informed voter casts a vote according to her prior belief and then learns that her vote was pivotal, then she realizes that there must have been a close contest between the two options. The fact that the other option received support suggests that more-informed players voted for it—and her vote canceled out the vote of someone who had better information.

Consider a game with n voters who must choose between two options, A and B . There are two states of the world, which we'll also call A and B . Nature chooses state A with probability π . Each player simultaneously chooses between voting for A , voting for B , or abstention. Whichever option receives more votes is the group's choice. If the group's choice matches the true state of the world, then all players receive a payoff of 1, and if it mismatches, then all players receive a payoff of 0. In the event of a tie, a coin flip determines the outcome, and the players' expected payoffs are $\frac{1}{2}$.

While everyone has a common prior belief π that the true state is A , some players may receive a perfectly informative signal prior to casting their votes. Each player receives a signal with probability p , in which case the player learns the true state of the world with certainty, and thus is now perfectly informed. With probability $1 - p$, the player does not receive a signal and is unable to update her beliefs beyond π .

Rational Abstention

Let's consider a model in which some players are perfectly informed while others do not receive a signal. We'll look for Bayesian Nash equilibria in which players of the same type (informed or uninformed) play the same strategy.

Notice that when a player is informed, then she has a weakly dominant strategy to vote according to her signal. What about voters who are uninformed? When $\pi = 1/2$, voting is weakly dominated by abstaining. Since the player's vote would be a complete shot in the dark, even if everyone were uninformed, it wouldn't help the group in expectation, and if anyone in the group were informed, then it could hurt by canceling out the vote of someone who knew the true state.

When $\pi > 1/2$, then there can be an equilibrium in which the informed voters always vote for the true state and uninformed players play a mixed strategy in which they vote for A with positive probability and abstain the remainder of the time (and never vote for B). While players do not want to

cancel out the vote of informed voters, there is also a chance that *no one* will be informed, in which case they want to make sure that A wins, since it is more likely to be the right choice. Thus, such an equilibrium can exist when p is sufficiently low (such that it is unlikely that there are informed voters whose votes will be canceled out) and π is sufficiently high (such that the player is reasonably confident that A is the true state).

Let's look at an example with $n = 3$, $\pi = 0.9$, and $p = 0.2$. Informed voters will always choose to vote for the true state. Let's see whether there is a Bayesian Nash equilibrium in which uninformed voters play a mixed strategy, such that they vote for A with probability c and abstain with probability $1 - c$. The uninformed voter is willing to mix only if the expected payoff from voting equals the expected payoff from abstaining. Furthermore, these two expected payoffs differ only if the voter is pivotal: by voting for A , she either breaks a tie in favor of A or causes A and B to tie when A otherwise would have lost by one vote. Notice that being pivotal could be either a positive or a negative development, depending on whether the true state of the world is A or B . If the true state of the world is A , then the player would receive an expected payoff of 0 if B wins, 1 if A wins, and $\frac{1}{2}$ if they tie. If the player's vote allows A to tie rather than lose, then her vote increases her expected payoff by $\frac{1}{2}$ (from 0 to $\frac{1}{2}$, and if the player's vote breaks the tie in favor of A , then her vote also increases her expected payoff by $\frac{1}{2}$ (from $\frac{1}{2}$ to 1). Likewise, if the true state of the world is B and her vote for A is pivotal, then her vote caused her payoff to decrease by $\frac{1}{2}$. The difference in expected payoff between voting for A and abstaining is as follows:

$$Eu(\text{Vote for } A) - Eu(\text{Abstain}) = \frac{1}{2} Pr(A \text{ and } B \text{ otherwise receive equal votes and the state is } A) - \frac{1}{2} Pr(A \text{ and } B \text{ otherwise receive equal votes and the state is } B) + \frac{1}{2} Pr(A \text{ otherwise receives 1 fewer vote and the state is } A) - \frac{1}{2} Pr(A \text{ otherwise receives 1 fewer vote and the state is } B).$$

Let's examine each of these four terms or cases in detail. In the first case, the true state is A and options A and B would receive an equal number of votes without player i 's vote. By casting a vote, the player would thus allow the true state A to win instead of tie, and so her expected payoff increases by $\frac{1}{2}$. The probability of this combination of events is $Pr(A)Pr(A \text{ and } B \text{ otherwise receive equal votes} \mid A)$. We know that $Pr(A) = \pi = 0.9$. What is the likelihood that A and B receive equal votes if the true state is A ? Note that B

would never receive a vote if A is the true state of the world, since informed voters would vote for A and uninformed voters only vote for A or abstain. Thus, the only way there could be a tie, given that the true state is A , is if no one votes: that is, both other players are uninformed and do not cast a vote. Since we assume each player votes with probability c , this occurs with probability $((1 - 0.2)(1 - c))^2$. Thus, the probability of this case is $0.9((1 - 0.2)(1 - c))^2$.

In the second case, A and B would receive an equal number of votes without player i 's vote and the true state is B . By casting a vote, the player thus causes the incorrect option A to win instead of having a tie between A and B , and so her expected payoff decreases by $\frac{1}{2}$. The probability of this combination of events is $Pr(B)Pr(A \text{ and } B \text{ receive equal votes} \mid B)$. We know $Pr(B) = 1 - \pi = 0.1$. What is the likelihood that A and B receive equal votes if the true state is B ? There are two ways that A and B could tie. First, each option could receive zero votes, which occurs if both other players are uninformed and do not vote. This occurs with probability $((1 - 0.2)(1 - c))^2$. Second, each option could receive 1 vote, which occurs if one player is informed and votes for B while the other is uninformed and votes for A . This occurs with probability $2(0.2)(0.8)(c)$. Notice that the 2 at the beginning of this expression is necessary because there are two options for which player is informed versus uninformed. We can't have a tie in which each option receives more than 1 vote, since there are only two other players in the game. Thus, the probability of this case is $0.1(((1 - 0.2)(1 - c))^2 + 2(0.2)(0.8)(1 - c))$.

In the third case, the true state of the world is A and A would lose by 1 if the player doesn't vote. However, if the true state of the world is A , then no one would vote for B , and A could never be down by a vote. The likelihood of this event is thus 0.

In the fourth case, the true state of the world is B and A would lose by 1 if the player doesn't vote. By casting a vote, the player thus forces a tie (expected payoff of $\frac{1}{2}$) instead of allowing the correct option of B to win. Thus, her expected payoff decreases by $\frac{1}{2}$. The probability of this combination of events is $Pr(B)Pr(A \text{ otherwise receives 1 fewer vote} \mid B)$. $Pr(B) = \pi = 0.1$. What is the likelihood that A receives 1 fewer vote than B , given that the true state is B ? The only way in which A could receive 1 fewer vote is if one player is informed and votes for B while the other player is uninformed and

abstains, which occurs with probability $2(0.2)(0.8)(1 - c)$. Thus, the probability of this case is $0.1(2)(0.2)(0.8)(1 - c)$.

Summing up, we have that the payoff difference between voting for A and abstaining for an uninformed player is

$$Eu(\text{Vote for A}) - Eu(\text{Abstain}) = \frac{1}{2}(0.9((1 - 0.2)(1 - c))^2) - \frac{1}{2}(0.1(((1 - 0.2)(1 - c))^2 + 2(0.2)(0.8)(1 - c))) + \frac{1}{2}(0) - \frac{1}{2}(0.1(2(0.2)(0.8)(1 - c))).$$

An uninformed player will be willing to play a mixed strategy only if her payoff from voting for A is the same as her payoff from abstaining, such that this expression is equal to 0. This occurs when $c = 0.875$. Thus, we have a Bayesian Nash equilibrium in which players who are randomly bestowed with information always vote according to their information, while uninformed players vote for A with probability $c = 0.875$ and abstain with probability 0.125. When the likelihood that players are informed is higher or when π is lower, abstention becomes more common in equilibrium.

Voting to Counteract Partisans

Another reason why uninformed voters may have an incentive to cast a vote occurs when they know that others are biased in one direction or the other. Imagine that in addition to the n players, there are m partisans, who always favor A regardless of the true state of the world. These voters will always vote for A, even if they are uninformed or know the true state to be B.

Assume that $\pi = \frac{1}{2}$. Now what does the uninformed voter's decision look like? Again, she considers only the situations in which her vote is pivotal. If her vote matters to break or force a tie, then B must have received at least $m - 1$ votes from the other nonpartisan voters, indicating that there is a reasonable likelihood that informed players are supporting B. By casting her vote for B, the voter may offset the vote of one of the A-partisans and thus help the informed voters to determine the outcome. As a result, there exists a Bayesian Nash equilibrium in which partisans vote for A, informed voters vote for the true state, and uninformed voters play a mixed strategy in which they put positive weight on voting for B and abstention. The Bayesian Nash equilibria can be computed using the same process as in the previous example without partisans, except there are now m dummy players whose votes must be included in A's vote totals in each case. The greater the number of partisans, the more likely uninformed voters are to support B in equilibrium.

What if $\pi > 1/2$? The situation is less obvious, since the uninformed voters' prior beliefs work in the opposite direction from their incentive to

counteract A-partisans. However, so long as p is not too small, the Bayesian Nash equilibrium still requires the uninformed voters to play a mixed strategy in which they vote for B with positive probability and never vote for A. The greater the number of partisans is, the more likely uninformed voters are to support B in equilibrium, and the greater π is, the less likely uninformed voters are to support B.

Costly Information Acquisition and Rational Ignorance

We have thus far assumed that voters are randomly bestowed with information by Nature. More realistically, people choose whether to make the effort to become informed, such as by researching the candidates or issues prior to voting. However, seeking out information requires time and effort, and thus is associated with a cost. If a voter believes that she is unlikely to be pivotal, then she may prefer to remain “rationally ignorant” and free ride off the information acquisition of others rather than become informed herself.

Consider a game where states A and B are equally likely ($\pi = 0.5$), and each player receives a payoff of 1 if the group’s choice matches the true state (and zero otherwise). Abstention is not allowed. However, before casting one’s vote, each player can choose to view an informative signal at a cost to herself. Let’s analyze the game for $n = 3$. We’ll assume that the signal corresponds to the true state of the world with probability 1 and the cost of acquiring the signal is 0.08. The parameters of this game are due to Bhattacharya, Duffy, and Kim (2017).

Let’s backward induct by first considering the voting stage of the game, which occurs after players have chosen whether to acquire information. Since the alternatives are symmetric, we’ll assume that uninformed voters are equally likely to vote for A and B . Informed voters have a weakly dominant strategy to vote according to their own correct signal, since voting against one’s signal either doesn’t make a difference or swings the group’s choice to the incorrect outcome.

Now that we know how informed and uninformed voters will vote, let’s turn to the question of when information acquisition is a best response. A player will choose to acquire information only if the cost is less than or equal to the expected benefit. We’re assuming that the cost of becoming informed is $c = 0.08$. What is the expected benefit? By acquiring information, the player could flip the group’s decision from the incorrect to the correct state, in which

case her payoff would increase from 0 to 1. What is the probability that this happens? Choosing to be informed is beneficial only if (1) the other two voters split their votes across A and B and (2) the player would have voted incorrectly if she were not informed. Since we assume that uninformed players are equally likely to vote for either option, there is a 0.5 chance that the player would have voted correctly even if she were uninformed, so the probability of this latter event is $\frac{1}{2}$. Thus, a player will choose to acquire information only if:

$$0.08 < (1) \left(\frac{1}{2} \right) Pr(\text{Other players split their votes})$$

Note that if the other two players are informed, then there is no chance that they will split their votes and the third player would be pivotal. Thus, the player's best response to the two others becoming informed is to remain ignorant. If one or both of the other players are uninformed, then the probability that they will split their votes is 0.5, so the player's best response is to become informed. Thus, there are pure-strategy equilibria in which two of the three players acquire information.

Let's look for the symmetric equilibrium, in which each player chooses to acquire information with probability x . A mixed strategy is a best response only if the cost of becoming informed is equal to the expected benefit: $0.08 = (1) \left(\frac{1}{2} \right) Pr(\text{Other players split their votes})$. To find the equilibrium mixed strategy x , we must find the probability that the other players split their votes if they each become informed with probability x .

What is the probability that a player playing this mixed strategy correctly votes for the true state? There is an x chance that she becomes informed, in which case she votes correctly with probability 1, and a $1 - x$ chance that she remains uninformed, in which case she votes correctly with probability $\frac{1}{2}$. Therefore, the probability that each player accurately votes for the correct state is $x + \frac{1}{2}(1 - x) = \frac{1}{2} + \frac{1}{2}x$. Notice that when x is 1, the probability goes to 1, and when x is 0, the probability goes to $\frac{1}{2}$. The probability that she votes for the incorrect state is $\frac{1}{2} - \frac{1}{2}x$. To find the probability that the players split their votes, we multiply the probability that one votes correctly by the probability that one votes incorrectly, and we must also multiply by 2, since there are two ways that this situation can occur. Thus, we find that the players split their votes with probability $2 \left(\frac{1}{2} + \frac{1}{2}x \right) \left(\frac{1}{2} - \frac{1}{2}x \right) = \frac{1}{2} - \frac{1}{2}x^2$. Therefore, the symmetric mixed-strategy Nash equilibrium in which each player chooses to become

informed with probability x thus occurs when $0.08 = (1)(\frac{1}{2})(\frac{1}{2} - \frac{1}{2}x^2)$, or $x = 0.825$. In this example, the players thus seek information most of the time, and the likelihood that the group answers correctly is high. However, as n grows larger, it becomes less likely that players will be pivotal, and thus players become less likely to obtain information in equilibrium. Thus, contrary to the Condorcet jury theorem (theorem 1), increasing group size does not always increase the likelihood that the group will make the correct choice when information acquisition is costly.

In this example, voting was compulsory, such that players had to either become informed or risk leading the group astray with a random vote. What if voting is voluntary? In this case, players who forgo information choose to abstain, and thus only informed voters decide the election. Making voting mandatory can have contradictory effects on information acquisition depending on the size of the group and the accuracy of the signal. On the one hand, when voting is voluntary, people have less incentive to purchase information because they can simply abstain and not impose the negative externality of uninformed voting on the group. On the other hand, the expected number of voters is smaller, and there is no risk that one's informed vote would be canceled out by the noise of uninformed voters casting random votes, and so the expected benefit of purchasing information and voting is greater.

Behavior in Games

The jury voting game described at the beginning of this chapter was conducted in the lab by Guarnaschelli, McKelvey, and Palfrey (2000). They studied both three- and six-person juries, and they also varied whether the outcome was determined by majority rule or unanimity and whether participants could conduct a nonbinding straw poll before submitting their final votes. To avoid participants bringing their own preconceptions to the experiment, the authors did not frame the situation as jurors deciding the fate of a defendant. Instead, they told participants that there were two jars, each containing 10 colored balls. The “red jar” (corresponding to the guilty state of the world) contained 7 red balls and 3 blue, while the “blue jar” (corresponding to the innocent state of the world) contained the opposite configuration. One jar was randomly selected for the group, but participants did not know the color of the jar before making their choices. Instead, each

group member saw the jar on his computer screen, with all the balls grayed out. The participant could then click on one ball to view its color, corresponding to a signal that is accurate with probability 0.7. Finally, each participant voted for a color. Depending on the treatment, red won if it received the majority of votes or unanimous approval, and then group members each received a payment if the actual jar matched their group's decision.

As predicted, they found that players do vote strategically against their own signal under unanimity. While participants under majority rule tended to follow their signal, participants who received a blue (innocent) signal but knew that the decision of red (conviction) required unanimous support regularly voted against their own signal. In addition, this behavior increased in larger groups, as Feddersen and Pesendorfer (1998) predicted. Specifically, participants with a blue signal voted for red about 36% of the time in three-person juries (compared to the Nash prediction of 31%) and 48% of the time in six-person juries (compared to the prediction of 65%).

But although individuals voted strategically under unanimity, the group's final decision did not match the dire theoretical prediction that innocent defendants would be more likely to be convicted as the group size grew larger. Notice that members of six-member juries voted against their signal about half the time, but still significantly less than the Nash prediction of nearly two-thirds of the time. As a result, unanimous six-person juries actually had the *lowest* rates of convicting the innocent (less than 3%). When players had access to a straw poll, they tended to share their private signal initially and then cast their vote according to the straw poll results, although some players did still stick with their own signal.

Likewise, Goeree and Yariv (2011) found that players vote strategically under unanimity, but less than predicted, and as a result, the innocent were less likely to be convicted when unanimous support was required. They also allowed free-form communication prior to voting, which they found reduced the likelihood of incorrect decisions and reduced the impact of the specific voting rule. Participants used the chat opportunity to truthfully reveal their own information and then discuss the best approach for the group, and then participants tended to follow all the information revealed (while still sometimes overweighting their own signals).

Battaglini, Morton, and Palfrey (2010) tested the swing voter's curse in the lab. They varied both the prior probability of state A (although it was never high enough to incentivize uninformed players to vote for A) and the number of A-partisans. The predictions and behavior of uninformed voters are shown in [figure 23.2](#). As we can see, the results are remarkably close to the predictions when $\pi = 0.5$, but when π increases to $5/9$, players respond by overvoting for A. Without partisans, uninformed voters tend to rationally abstain when $\pi = 1/2$ but 20% voted in favor of their prior when $\pi = 5/9$. While voting for A is in line with the optimal decision in a one-person jury, it is contrary to the strategic decision that accounts for the swing voter's curse. In groups with A-partisans, players voted for B to counteract the effect of partisans, but again they were overly responsive to an increase in π .

Partisans	$\pi = 1/2$				$\pi = 5/9$			
	Voting for A		Voting for B		Voting for A		Voting for B	
	Nash	Data	Nash	Data	Nash	Data	Nash	Data
0	0	0.00	0	0.08	0	0.20	0	0.07
2	0	0.06	0.36	0.43	0	0.12	0.33	0.35
4	0	0.04	0.76	0.77	0	0.16	0.73	0.56

Figure 23.2

Votes from uninformed voters in Battaglini et al. (2010).

Evidence-Based Adaptations of the Standard Model

When we first learned about Bayesian games in chapter 11, we saw that people often act as if they don't fully appreciate that other players' actions are based on their private information and that they must best respond not only to the anticipated actions of others, but also to the information that they would infer from those actions. We looked at a couple of explanations for this: Eyster and Rabin (2005)'s cursed equilibrium model, which assumes that people underweight the extent to which other players' strategies are based on their information, and, more generally, that people may have difficulty engaging in hypothetical reasoning about future events. Let's look at how each of these explanations applies in the jury voting game.

According to the cursed equilibrium model, players correctly predict the relative frequency with which the other jurors vote to acquit or convict, but they underappreciate the extent to which their votes are based on private

information. Players who are cursed will therefore infer less information about other people's signals from the fact that they are pivotal, and not update their belief about the defendant's guilt as much as they would if they were fully rational. The parameter χ indicates the extent to which the player is cursed and fails to recognize the informational component of other players' strategies, with $\chi = 0$ indicating that the player is fully rational and $\chi = 1$ indicating that the player is fully cursed. If a player is fully cursed, then she assumes that players vote for conviction with the same probability regardless of their signal, and thus knowing that she is pivotal does not provide her with any information about the other players' signals. Therefore, she simply goes by her own signal and votes informatively.

In contrast, if players are only partially cursed, then they will vote against an innocent signal with positive probability under the cursed equilibrium, but less frequently than under the Bayesian Nash equilibrium. As a result, the partially cursed equilibrium model can explain the results of Guarnaschelli et al. (2000) (in six-person juries) and Goeree and Yariv (2011)—namely, that players often vote against their own innocent signals, but less so than the Bayesian Nash prediction, and as a result, actual false conviction rates are lower than predicted.

Specifically, a χ -cursed voter who receives an innocent signal has an updated belief that the voter is guilty, given by $(1 - \chi) \left(\frac{.3(.7 + .3x)^5}{.3(.7 + .3x)^5 + .7(.3 + .7x)^5} \right) + \chi(0.3)$. When $\chi = 1$, the player is fully cursed and thus goes off her own innocent signal, believing that the defendant is guilty with probability 0.3. When $\chi = 0$, the player's updated belief is the same as under the Bayesian Nash equilibrium, and thus the symmetric mixed-strategy Nash equilibrium is $x = 0.65$. For $\chi \in (0, 0.7)$, there is a unique symmetric cursed equilibrium in which each player chooses convict with probability $x < 0.65$ after receiving an innocent signal. Thus, the cursed equilibrium model provides an explanation for why players vote to convict when they have an innocent signal less frequently than predicted by the Bayesian Nash equilibrium.

Do voters fall victim to the swing voter's curse because they fail to appreciate the strategic reasoning of others? Or do they struggle more broadly with recognizing that a future event—learning that their vote is pivotal—can convey information that is relevant for their current choice? To address this, Esponda and Vespa (2014) transform the three-player jury game into an

individual decision problem, where the computer plays the role of the other two voters and always acts according to a known strategy. Specifically, participants are told that there is a jar of 10 marbles, p of which are blue. The computer will randomly draw a marble and cast two independent votes, while the subject must cast the third vote. If the chosen marble was red, the computer always votes red. If the chosen marble is blue, then each of the computer's votes are blue with probability q (with each of the votes determined independently). The subject must vote without seeing the marble and receives a cash prize if the majority of the votes match the marble that was drawn.

Participants played many rounds with varying p and q . However, in all cases, blue was the optimal choice, and this conclusion doesn't require the participant to engage in Bayesian updating. Why? If the chosen marble were red, then the computer would cast both of its votes for red and the participant's vote would not matter. Thus, the participant's vote can influence the outcome only when the true state is blue, and she is thus better off voting for blue in all cases. This experiment eliminates uncertainty regarding the strategies that others will play, as well as the need to calculate probabilities. To distinguish whether players have difficulty with extracting information from the knowledge that they are pivotal or have difficulty with hypothetical thinking, they also conducted a treatment in which participants learned the outcome of the computer's votes (i.e., two votes for blue, two votes for red, or one vote for each), and thus whether they are pivotal, before casting their own votes. Esponda and Vespa found that only about one-fifth of participants optimally vote for blue in the simultaneous game, while roughly four-fifths do so when the game is sequential. In other words, they don't have nearly as much trouble extracting the information that blue is optimal once they learn that they are pivotal. This indicates that players have difficulty engaging in hypothetical thinking and recognizing that a future event can be informative for their current decision.

Exercises

(1) Consider an individual juror who will decide, on her own, whether to convict or acquit a defendant. Let $\pi = 0.7$ and $p = 0.6$. For what range of q would the player vote informatively?

(2) Consider an individual juror who will decide, on her own, whether to convict or acquit a defendant. Let $\pi = 0.5$, $p = 0.6$, and $q = 0.7$.

- (a) Say that the juror views a single signal, which is accurate with probability $p = 0.6$. Will the juror vote to convict or acquit if the signal is t ? What if the signal is g ?
- (b) Now imagine that the juror views three independently drawn signals, each of which are accurate with probability $p = 0.6$. Thus, there are four situations: (1) she sees all guilty signals, (2) she sees one innocent and two guilty signals, (3) she sees one guilty and two innocent signals, and (4) she sees three innocent signals. Find the juror's optimal choice in each case.

(3) Consider a five-member jury that will convict a defendant only if all five players unanimously agree. Let $\pi = 0.5$, $p = 0.8$, and $q = 0.5$. Find the symmetric Bayesian Nash equilibrium, such that players vote for conviction if they receive a guilty signal and vote for conviction with probability x if they receive an innocent signal.

(4) Consider a three-player swing voter's curse game in which there are two states of the world, A and B. Players are perfectly informed of the true state of the world with probability $p = 0.5$, and completely uninformed otherwise. Players receive a payoff of 1 if the majority votes for the correct state, and a payoff of 0 otherwise.

- (a) Let the prior probability that A is the true state be $\pi = 0.85$. Is there a Bayesian equilibrium in which uninformed voters vote for A with positive probability? If so, say what it is. Otherwise, say why not.
- (b) Let the prior probability that A is the true state be $\pi = 0.8$. Is there a Bayesian equilibrium in which uninformed voters vote for A with positive probability? If so, say what it is. Otherwise, say why not.

(5) Consider a game where states A and B are equally likely, and each player receives a payoff of 100 if the group's choice matches the true state, and zero otherwise. Before casting one's vote, each player can choose to view a perfectly informative signal, which costs c .

- (a) Let $n = 3$. For what range of costs is there a symmetric mixed-strategy Nash equilibrium in which all players acquire information with positive probability?

- (b) Let $c = 12$. According to the symmetric mixed-strategy Nash equilibrium, what is the probability that each player acquires information?
- (c) Now imagine that the group size increases to 5, and c is still 12. According to the symmetric mixed-strategy Nash equilibrium, what is the probability that each player acquires information in the larger group?

¹How did we arrive at this equation? The numerator is equal to the probability that the defendant is guilty (0.5) times the probability that the player and exactly $(n - 1)/2$ of the other $(n - 1)$ members see guilty signals, given that the defendant is guilty. We can calculate this latter probability using the multiplication rule. First, the probability that the player sees a guilty signal, given that the defendant is guilty, is 0.7. Then the probability that $(n - 1)/2$ specific people see a correct guilty signal and $(n - 1)/2$ see an incorrect guilty signal is $(0.7)^{\frac{n-1}{2}} (0.3)^{\frac{n-1}{2}}$. However, we need to consider the possibility that any combination of $(n - 1)/2$ members could be the ones to see the correct signal. The number of combinations of $(n - 1)/2$ jurors that we could assemble from $n - 1$ members is given by the term $\binom{n-1}{\frac{n-1}{2}} = \frac{(n-1)!}{((n-1)/2)!(n-1/2)!}$. The denominator is calculated similarly and also includes the probability that the defendant is innocent times this combination of signals occurring conditional on innocence.

VIII

Behavioral Extensions of Standard Theory

24

Belief-Based Learning

With this chapter, we begin a section on models of strategic interactions when players' rationality is bounded. One way to summarize the motivation for this section is with a question: should we expect untrained players to achieve equilibrium, seeing as they have not taken a course in game theory? In other words, without knowing what a Nash equilibrium is, will naive players achieve it via some process of repeated play and learning? In this chapter, we begin with some of the earliest models of learning, those based on player beliefs. In the next two chapters, we consider models that expect even less from our players and assume that players respond to the payoff differences between strategies. In chapter 27, we allow for heterogeneity in the capabilities of our players.

Suppose, prior to reading the first couple of chapters of this book, that you and other equally naive players were asked to play game 1 (shown in [figure 24.1](#) and studied in Nyarko and Schotter, 2002) 10 times with the same partner. Let's assume that, even though you do not know the definition of a Nash equilibrium, you do intuitively understand the notion of a best response. Hence, you need to anticipate what the other player will do and then make the correct response.

		Other	
		Aay	Bee
You	One	(6, 2)	(3, 5)
	Two	(3, 5)	(5, 3)

Figure 24.1

Game 1, from Nyarko and Schotter (2002).

On the first round of play, you need to form a prior belief about what the other player will do. In this case, suppose that you are unsure, and therefore, you think that it is reasonable to assume that the other player will toss a coin. As a result, your prior is that $Pr(Aay) = Pr(Bee) = \frac{1}{2}$.

If you calculate the expected payoff of each of your pure strategies, you will find that $Eu(One) = 3Pr(Aay) + 3$ and $Eu(Two) = 5 - 2Pr(Aay)$. Further, inserting your prior, you calculate that $Eu(One) = 4.5 > 4 = Eu(Two)$, so you play *One*. The other player does the same and chooses *Bee* after concluding that $Eu(Aay) = 3.5 < 4 = Eu(Bee)$. In the next round, your best response to the other player choosing *Bee* is to switch to *Two*, and if you are now taking turns, the other player will then choose *Aay* on the third round. From there, you can see that if you each best respond to what the other player has done on the last round, the game cycles clockwise. What should the two of you conclude from this? One plausible hypothesis for both of you is that the person you are playing is using a mixed strategy in which both strategies receive equal weight. After all, if you just observe the history of play, each strategy is used half the time. The problem is that, although there is a mixed-strategy equilibrium, the weights in equilibrium are (0.4, 0.6) for both players, not (0.5, 0.5).

Let's see what happens when we allow the two of you to have slightly better memories. Suppose that you each can remember the entire history of play and may still suspect that the other player is mixing. On the first round of play, you make the same calculation as before and play *One*. The question is: how should the other player now respond? If she just counts the instances of each choice you make and starts off the count with 1 for each strategy (hence her $\frac{1}{2}$ prior), then after seeing your play, she updates her count to 2 *Ones* and 1 *Two*. Further, if she updates her prior according to these proportions, she now guesses that your mixture places $\frac{2}{3}$ weight on strategy *One*. Best responding to this belief using the expected utilities calculated earlier, we see that she chooses *Bee*. But, as you will see, play does not get stuck in the same cycles as before.

On the third round, you observe the other player's second-round choice and update your counts to 1 *Aay* and 2 *Bees*, yielding a belief that the other player is placing $\frac{1}{3}$ weight on strategy *Aay*. Your calculations indicate you should play *Two* again. In the next round, the other player's belief falls back

to $\frac{1}{2}$, so she plays *Bee* one more time. In the fifth round, your belief falls to $\frac{1}{4}$, causing you to play *Two* yet again. Now, however, in round six, the other player faces a dilemma—her belief falls to $\frac{2}{5}$ and she is indifferent between playing *Aay* and *Bee*. Suppose that she chooses *Aay* this time. This means that your prior rises to $\frac{2}{5}$ and now you are indifferent too. Interestingly, you can see that this process ends at the Nash equilibrium of the game in just six or seven rounds of play. In other words, we see that two players who are somewhat clever and have decent memories will quickly learn to play the Nash equilibrium of a relatively complicated game, all without knowing what it is that they have discovered.

In this chapter, we consider a class of models developed to describe the way that players learn over time in games that are repeated. According to the *belief-based* learning models that we cover, players form beliefs about the expected behavior of other players and choose actions based on how well they would perform against opponents playing as expected. After each round of the game, players then update their beliefs based on the feedback they receive. The important question is: to where do these processes converge?

Best Response Dynamics

One of the earliest notions of learning in games is the first one illustrated in this chapter—simply best responding to the strategy profile used most recently by the other players. This logic is often attributed to the model of quantity competition offered by Cournot (1838), which is described in more detail in chapter 6 and discussed by Fudenberg and Levine (1998). While the concept can be faulted for assuming, at the same time, both too much and too little cognition on the part of the players (i.e., they can calculate best responses but cannot remember more than one period of play), best response dynamics (see definition 1) is a powerful concept because in games with pure-strategy equilibria, the process *always* converges to a Nash equilibrium.

To understand how best response dynamics always converge to a pure-strategy Nash equilibrium (if one exists), recall our definition of a best response from chapter 4. Strategy s_i is a best response to strategy s_{-i} for the

other player if $u_i(s_i, s_{-i}) \geq u_i(s'_i, s_{-i})$ for all $s'_i \in S_i$. A shorthand for this is $BR_i(s_{-i}) = \operatorname{argmax}_{s_i} u_i(s_i, s_{-i})$. Using this notation, we have:

Definition 1 (Best response dynamics) *Suppose that players adjust their strategies in discrete time, $t = 1, 2, 3, \dots$, and in the initial round, the state of play is described by strategy profile θ_0 . On each round, players chose the pure strategy that is a best response to the previous period, $BR_i(s_{-i, t-1})$. Best response dynamics implies that $\theta_{i, t+1} = BR_i(\theta_{-i, t})$ and the dynamics achieves a steady-state $\theta_i^* = BR_i(\theta_{-i}^*)$ for all i .*

Put differently, best response dynamics converge to state θ^* whenever they get stuck in a state that they never leave. This happens whenever, given the previous round strategies, players choose to play the same thing during the next round and all future rounds. Further, by definition, because players are best responding to the strategies played in the previous round, any steady-state achieved must be a Nash equilibrium.

Let's consider an example in which best response dynamics converges. In game 2 (shown in [figure 24.2](#)), players need to coordinate off-diagonally to achieve equilibrium. What if they do not know this at the start of play and take turns using best response dynamics to learn how to play?

		Player 2	
		Up	Down
Player 1	Up	(1, 1)	(3, 2)
	Down	(2, 3)	(0, 0)

Figure 24.2

Game 2.

In addition, because their calculations have to start somewhere, let's say that their prior beliefs are that player 1 expects player 2 to use *Up* with probability $\frac{1}{5}$ and player 2 expects player 1 to place considerably more weight, $\frac{4}{5}$, on *Up*. That is, they expect to be near the (*Up*, *Down*) equilibrium to start. Calculating i 's expected payoffs based on the probability of j playing *Up*, $Pr_j(Up)$, we get $Eu_i(Up) = 3 - 2Pr_j(Up)$ and $Eu_i(Down) = 2Pr_j(Up)$.

Graphing the best responses for each of the players, we get [figure 24.3](#). As we can see, if player 1 expects player 2 to play *Up* with less than the mixed-strategy equilibrium weight, $\frac{3}{4}$ in this case, her best response is to play *Up* for certain. At the same time, if player 1 thinks that player 2 will place more than the equilibrium weight on *Up*, her best response is *Down* for sure. Because the game is symmetric, the best response function for player 2 looks the same after inverting the axes. At the three intersections of the best response functions are the Nash equilibria.

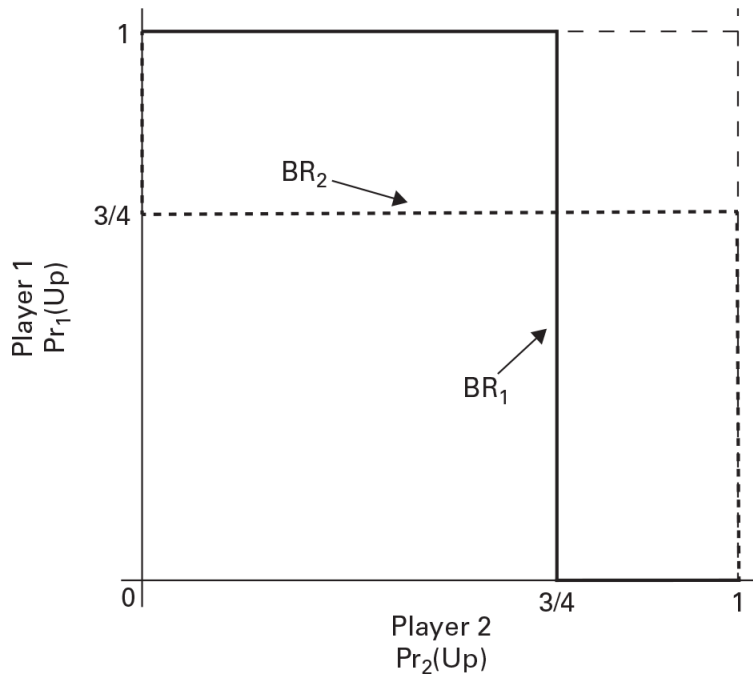


Figure 24.3

Best response functions for game 2.

Now let's consider the dynamics. To get the initial state of play (θ_0 from above), we let the players perform the expected utility calculations based on their priors. For player 1, the expected utility of playing *Up* ($\frac{13}{5}$) is larger than playing *Down* ($\frac{2}{5}$), and for player 2, *Down* narrowly beats *Up* in expected utils ($\frac{8}{5}$ versus $\frac{7}{5}$), so the initial state of play is (*Up*, *Down*) in the lower-right part of [figure 24.3](#). Because θ_0 constitutes a pure-strategy Nash equilibrium, regardless of which player goes next, it is clear that best responding will lead them to leave their strategies unchanged. In this case, the players start in equilibrium and learn to stay there.

What if we consider the other possible initial states of play? If we simply invert the player priors, they will start play at the $(Down, Up)$ equilibrium instead and stay there. If, however, both players expect that the other player is very likely to play down, so $Pr_i(Up)$ is low for each (i.e., they are in the lower-left part of [figure 24.3](#)), they will both best respond by playing Up . The result is initial play away from an equilibrium, but depending on which player goes next, they will end up at one of the pure-strategy equilibria after one round of best responding. Initial play can also be at $(Down, Down)$ if both players are convinced that the other player will play Up , but again, after one round of best responding, play is absorbed by one of the pure-strategy Nash equilibria.

One thing to note, however, is that, as we saw in the game described in the introduction to this chapter, play that starts near the mixed-strategy Nash equilibrium will not be absorbed by it. The reason is that best response dynamics requires players to pick pure strategies. If priors are right at the mixture, by construction, the expected payoffs for the two pure strategies will be the same, and yet players have to pick just one. If players randomize, they will either end up cycling, as in game 1, or drifting toward one of the pure-strategy equilibria, as would happen in game 2 and [figure 24.1](#).

Fictitious Play

As the examples we have worked through in this chapter should have illustrated, the initial priors of the players matter a lot for the resolution of best response dynamics. In other words, there is considerable path dependence in this type of belief-based learning. If player priors are closer to one pure-strategy Nash equilibrium than another, it is likely to be the absorbing state of the learning process. Noticing this means that the priors are not innocuous, a point first made by Brown (1951), who developed the notion of *fictitious play* as an alternative way to compute equilibria, even though the idea was also originally interpreted as a model of prior formation. To understand this interpretation, imagine a simple coordination game. Brown's idea was that players would go through a system of pre-play calculations to compute and harmonize their expectations on one of the equilibria (and this is why it is called *fictitious play*). Since Brown's

publication of the idea, however, fictitious play has been considered as an alternative belief-based model of learning—one that we will see is effectively a more sophisticated version of best response dynamics.

So what is fictitious play? Imagine that we assume players to be more symmetric in their cognitive abilities. This means that fictitious players will continue to be clever enough to best respond, but now they will also have the working memory capacity to match this cleverness, as in the very first example in this chapter. Specifically, let's assume that players can now remember the resolution of more than just the last round of play. In fact, suppose that they can remember all the prior moves of their opponent and they keep track of how many times each pure strategy is used. Finally, these tabulations are important partially because we will assume that fictitious players anticipate that their opponents might be using mixed strategies. These concepts are laid out mathematically in definition 2.

Regarding the details, in each round fictitious players best respond to their current assessment of their opponent's distribution of actions, and they update this assessment in a specific way. Like myopic best responders, fictitious players start with a prior belief about the weights that their opponent places on each pure strategy, but unlike best responders, they update these weights by counting the number of times each pure strategy is used. The weights assigned to each pure strategy in the fictitious player's expected utility calculations are just the proportions with which each pure strategy has been played by the opponent in the past.

Definition 2 (Fictitious play) *Say that $\kappa_{i,0}(s_{-i})$ is the initial assessment of a fictitious player about the weights that their opponent will use. After $t > 1$ rounds of play, the player updates this assessment according to the rule $\kappa_{i,t}(s_{-i}) = \kappa_{i,t-1}(s_{-i}) + 1$ if the opponent played strategy s_{-i} at time $t - 1$ and $\kappa_{i,t}(s_{-i}) = \kappa_{i,t-1}(s_{-i}) + 0$ if the opponent played another strategy. The probability with which the fictitious player expects the opponent to play strategy s_{-i} on round t is $\Pr_{i,t}(s_{-i}) = \frac{\kappa_{i,t}(s_{-i})}{\sum_{s'_{-i} \in S_{-i}} \kappa_{i,t}(s'_{-i})}$.*

Recall that this is just the procedure used by the players (with better memories) of game 1. Before we work through another example or two, it is important to point out that fictitious players will learn Nash equilibria like best responders; however, they also differ from best responders because

fictitious players can learn that their opponent is mixing too. Limiting our attention to two-person games, if a strict Nash equilibrium is played at any point of the fictitious play process, it will absorb play (i.e., it will be played forever after). The intuition is clear. If play lands at any combination of pure strategies for the two players and stays there, eventually the fictitious play tabulations will converge to predicting play at this strategy profile for sure. Further, if that strategy profile is a Nash equilibrium, best responding will guarantee the equilibrium will absorb the fictitious play dynamic. Considering mixed strategies, while fictitious players are required to respond with a pure strategy, as we saw with game 1, the expected weights that fictitious players calculate for their opponent will eventually coincide with the equilibrium mixed-strategy weights, in many cases.

As another example of fictitious play, consider game 3, shown in [figure 24.4](#). This is an asymmetric two-player game with three equilibria, two pure and one mixed. Let's examine the evolution of play of two fictitious players. Suppose that their tabulations start with one for each strategy so their priors are $Pr_1(Left) = Pr_2(Up) = \frac{1}{2}$. These priors are exactly in the middle of the state space; what do we think the fictitious players will learn, given this start?

		Player 2	
		Left	Right
Player 1	Up	(9, 10)	(0, 7)
	Down	(6, 0)	(9, 8)

Figure 24.4

Game 3.

The best response mappings for game 3 in [figure 24.5](#) can give us a sense of what to expect from fictitious play. You can see that the best response to the priors, $Pr_1(Left) = Pr_2(Up) = \frac{1}{2}$, is $(Down, Right)$, and as stated previously, given that $(Down, Right)$ is a strict Nash equilibrium, it quickly absorbs fictitious play. After the first round, the tabulations are 1 *Left*, 2 *Right* for player 1 and 1 *Up* and 2 *Down* for player 2, yielding the new expectations in round 2 of $Pr_1(Left) = Pr_2(Up) = \frac{1}{3}$. Of course, $(Down, Right)$

is an even better response to these beliefs, and so at the beginning of round 3, they fall to $Pr_1(Left) = Pr_2(Up) = \frac{1}{4}$, and this process continues until the expectations are approximately $Pr_1(Left) = Pr_2(Up) = 0$, indicative of the absorbing *(Down, Right)* equilibrium.

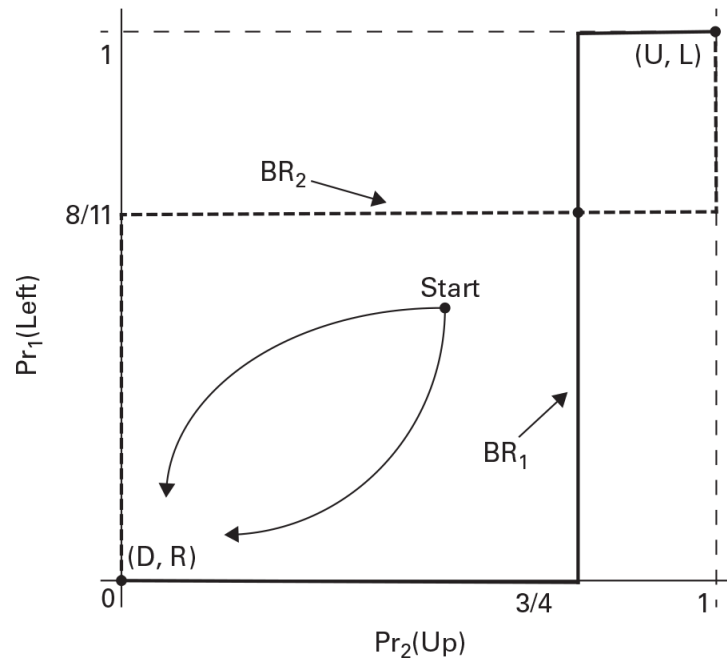


Figure 24.5

Best response functions for game 3.

What happens if the players start above the mixed-strategy equilibrium with priors of, say, $(\frac{7}{8}, \frac{7}{8})$? As you might expect, now the fictitious players converge on the other pure-strategy Nash equilibrium. As [figure 24.5](#) indicates, the best response to these priors is *(Up, Left)*, and again, fictitious play is quickly absorbed by this equilibrium. Finally, if the process starts arbitrarily close to the mixed strategy and outside the regions where a pure-strategy Nash equilibrium is the best response, we see that fictitious play can wander around a bit but will never be absorbed by the mixture. For instance, suppose that the priors are $(\frac{10}{12}, \frac{8}{12})$. Now the best response on round 1 is *(Up, Right)*, which is played for two rounds before the *(Down, Right)* equilibrium absorbs play.

In what sense, then, can fictitious play learn the mixed-strategy equilibrium? First, remember what we saw in game 1—the players' beliefs settled on the correct mixture after a few rounds. This would also happen in

the game of matching pennies, in which the mixed strategy calls for the players to toss the coin (i.e., $Pr(Heads) = Pr(Tails)$). In general, this is likely to happen in situations in which the mixed-strategy equilibrium is the *only* equilibrium. However, the outcome of fictitious play in games with multiple equilibria are harder to predict, as we have already seen.

In the game of chicken depicted here as game 4 in [figure 24.6](#), for example, we can show that two fictitious players can cycle back and forth across the mixed-strategy equilibrium if their choices are positively correlated whenever their beliefs coincide with the mixed-strategy equilibrium. In this formulation of chicken, the mixed-strategy equilibrium requires players to *Go* one-fifth of the time. If, for instance, the players start near this equilibrium with priors of, say, $(\frac{1}{4}, \frac{1}{4})$, they will both best respond by choosing *Stop* in the first round, but in the second round, they will be indifferent between *Go* and *Stop*.

		Player 2	
		Go	Stop
Player 1	Go	$(-10, -10)$	$(2, -2)$
	Stop	$(-2, 2)$	$(0, 0)$

Figure 24.6

Game 4: chicken.

Now, crucially, if when they are indifferent, they do the same thing (say *Go*), they stay out of the basins of attraction (see chapter 7) of either of the pure-strategy Nash equilibria. After a couple of rounds of *Stop*, the players' beliefs will return to the mixture, inducing indifference again, and so long as they continue to choose the same strategy when their beliefs match the mixed strategy, they will cycle back and forth across this state and their beliefs will converge on the mixed strategy in the long run. In some sense, this should be intuitive from the best response mappings of these games. Returning to [figure 24.5](#), so long as fictitious players cross the mixed strategy from northwest to southeast (or the reverse) in this coordination game, they stay out of the basins of attraction for the pure-strategy equilibria. However, if their play is not correlated in this sense, eventually,

when the players are indifferent, one player will make a choice that causes play to enter one of the basins of attraction, and it will quickly be absorbed.

As you can now tell, belief-based learning is a powerful but not infallible concept. The fact that in games with pure-strategy Nash equilibria, belief-based learning always leads to an equilibrium is powerful and yet, mixed strategies can sometimes throw a wrench in the works. Another shortcoming of this class of learning models is that, although players are assumed to have considerable memories and cognitive capabilities, they have little foresight. For example, recall that if both players in game 2 expect the other to be very likely to play *Down* initially, the two will best respond and end up at (Up, Up) . Further, the player making the next move (say, player 1) will clearly suffer because his best response is to play *Down* on the next round, and the game is absorbed into the $(Down, Up)$ equilibrium. Wouldn't a clever player 1 insist on player 2 going first? In general, belief-based learning does not account for players playing the long game, anticipating that current play can affect the equilibrium they will eventually reach. In the next section, we will see both whether belief-based learning predicts actual play and if this shortcoming matters.

Behavior in Games—Are Adaptive Beliefs Enough?

The problem with assessing models of belief-based learning experimentally is that, although we can directly observe the strategies chosen, it is hard to know for sure what players believe when making choices, or whether they even follow their beliefs. As we will see, this problem went unresolved until more recent experiments and in the beginning, much of the experimental research inferred beliefs by testing the extent to which play was consistent with a number of different learning algorithms.

One of the first experimental papers on best response dynamics and fictitious play was actually the assessment of experimental data gathered in nine other studies. As a result, the contribution of Boylan and El-Gamal (1993) was actually more econometric than experimental. The research design is clever, nonetheless. The authors programmed robots to play the same games that were played by humans in the other studies according to either best response or fictitious play and asked, essentially, which type of robot better corresponded to human play. What they find is interesting: in

games with a single pure-strategy Nash equilibrium, fictitious play seems to best describe what human players are doing, somewhat similar to the intuition described previously, but in games with multiple equilibria (and more complicated mixtures), best response dynamics matches human play better on average. However, aggregating across all the games and data, fictitious play seems like the more likely algorithm.

While an important contribution, Boylan and El-Gamal (1993) was not the final word on the topic. In fact, there were some obvious issues. For instance, the data came from experiments that were not designed to discriminate between learning models based on beliefs, only the two most prominent models were tested, and the econometric procedures assumed one model or the other—it couldn't assess the extent to which aggregate behavior is actually the product of various coexisting models interacting with each other.

To generate data to answer the belief-based learning question more directly, Cheung and Friedman (1997) conducted an experiment that examined five two-player games, (hawk-dove, coordination, prisoner's dilemma, buyer-seller, and battle of the sexes), exactly the types of interactions studied by Brown (1951) and those who followed him. Another innovation of Cheung and Friedman's study was that they created a more general learning model based on beliefs that nests best response and fictitious play as special cases. In addition, they had their participants play rather long games (160 rounds). As a result, when they econometrically assessed play, they could do so at the individual-player-level and, therefore, categorize individual players as either best responders (with what appear to have very short memories), fictitious players (with long memories), or somewhere in between. Indeed, the experimenters find considerable heterogeneity: among the three-quarters of participants that could be considered belief-based learners, approximately half seemed to be best responders and a third were fictitious players.

In another, related study, Duersch et al. (2010), conducted an experiment online in which participants were asked to play games against robots. The human players were randomly assigned to one of five robots, two of which are programmed to either best respond to the last round of play or gather and use the tabulations as a fictitious player. The purpose of the study is to explore one of the objections raised against belief-based learning models

listed previously—the assumption that belief-based learners do not have the foresight to try to manipulate their opponents into equilibria that favor them. Compared to participants who play other types of learning models (some of which are discussed in the next few chapters), participants randomly assigned to play against belief-based players earn higher profits, and those that participate in the experiment more than once do even better, indicating that they do try to manipulate their opponents.

Up to this point, all the experiments we have discussed inferred the beliefs of their participants based on their observed choices, but Nyarko and Schotter (2002) elicited them directly in their experiment. This experiment is interesting, not only because of the elicited beliefs but because it is also the source of the normal-form game analyzed in the introduction of this chapter (game 1). As you would if your class played this game, participants were asked to state their beliefs about what the other player would do at the beginning of each round. This is an important innovation because regardless of the model details (i.e., best responding or fictitious play), if belief-based learning explains behavior in games at all, it must be the case that actions are correlated with beliefs. It turns out that yes, human players clear this first hurdle: simply counting the instances in which play is consistent with any combination of the beliefs (stated, best response, fictitious play), the authors find that 75% of the time, a player's stated beliefs are consistent with the strategy chosen. The problem for our belief-based models, however, is that this correspondence only happens closer to 50% of the time for fictitious play and best response beliefs. Hence, while they don't necessarily form the most helpful beliefs, human players do respond to their beliefs and, as seen in the Duersch et al. (2010) study, they act more strategically than belief-based learning gives them credit for.

Exercises

(1) There are two best responding players, each of whom has a continuous utility function of the form $u_i = (1 - x_i - x_j)x_i$.

- (a) Calculate the best response function for each player and solve for the Nash equilibrium.
- (b) Graph the best response functions for the two players on the unit square.

(c) Suppose that both players expect the other to choose $x_i = \frac{1}{2}$ in the first round. Now assume that the players take turns best responding after the first round (say player 1 goes first on round 2, player 2 then goes on round 3, and so on). Build a table to illustrate seven rounds of play. Where do the players end up after these seven rounds? Does this make sense?

(2) Consider the asymmetric normal-form game shown in [figure 24.7](#):

		Player 2		
		C1	C2	C3
Player 1	R1	5, 1	8, 4.7	2, 3
	R2	2, 3	2, 1	4, 2

Figure 24.7

(a) Suppose that the initial weights placed on (C1, C2, C3) by player 1 are (1, 0, 4.32) and the initial weights placed on (R1, R2) by player 2 are (3, 5.7). Fictitious play converges faster if players take turns, so let player 1 go first. Calculate the results of fictitious play for 10 rounds. Does play converge? Why does your result make sense?

(3) Consider the normal-form version of matching pennies shown in [figure 24.8](#), in which the Matcher wins when the coins match and the Differencer wins when they differ:

		Differencer	
		H	T
Matcher	H	(1, -1)	(-1, 1)
	T	(-1, 1)	(1, -1)

Figure 24.8

(a) Calculate the mixed-strategy Nash equilibrium.

(b) Suppose that the initial weights placed on (H)eads and (T)ails by the two fictitious players, $w_i = (H_j, T_j)$, are $w_{\text{Matcher}} = (1.5, 2)$ and $w_{\text{Differencer}}$

$= (2, 1.5)$. What does the Matcher choose on the first round of fictitious play? Make a table (perhaps with a spreadsheet) with the tabulations, expectations, and best responses for the two players to assess how play progresses for 20 rounds. Importantly, let the players take turns best responding to their beliefs (so the Matcher goes on the odd rounds and the Differencer goes on the even rounds). What are the player beliefs at the end of these 20 rounds? To what steady state will the beliefs converge?

(4) The game shown in [figure 24.9](#) was discussed first by Lloyd Shapley in 1964:

		Column		
		<i>L</i>	<i>C</i>	<i>R</i>
Row	<i>T</i>	(0, 0)	(1, 0)	(0, 1)
	<i>M</i>	(0, 1)	(0, 0)	(1, 0)
	<i>D</i>	(1, 0)	(0, 1)	(0, 0)

Figure 24.9

- Calculate the mixed-strategy equilibrium.
 - Let the Row player go first and calculate the outcome of fictitious play for 10 rounds based on the following starting weights: for the Row player, $(L, C, R) = (1, 2.2, 1.1)$; for the Column player, $(T, M, D) = (1.1, 1.9, 2)$. Does anything seem odd?
- (5) Consider the coordination game shown here in [figure 24.10](#):

		Player 2	
		A	B
Player 1	A	(0, 0)	(1, 1)
	B	(1, 1)	(0, 0)

Figure 24.10

- Find *all* the Nash equilibria.

- (b) Suppose that the priors of the two players are that the other will play A with probability $\frac{1}{4}$. What would happen with two best responding players if player 1 best responds first?
- (c) Assume that the priors are still that the other will play A with probability $\frac{1}{4}$. What would happen if the two players use fictitious play instead (again, let player 1 go first)?
- (d) Now assume that the initial values with which the two fictitious players start their counts are the same $w_i = (A_i, B_i) = (1, \sqrt{2})$. How does play evolve for the two (and is there anything odd about the result)?

25

Evolutionary Game Theory

Compared to chapter 24, on learning, in this chapter we assume less about the agents playing games. We ask whether agents without much sophistication or memory will still achieve equilibrium when play is repeated. To this end, we will borrow from a branch of game theory pioneered by John Maynard Smith to explain the evolution of biological traits when they bestow differential fitness benefits on their hosts. Although the theory was developed to understand the evolution of phenotypes, the logic is isomorphic with a model of learning based on imitation. Put differently, we examine whether agents can achieve equilibria when all they are capable of doing is observing what others do and imitating the strategies that confer payoffs exceeding the population average.

Nearly all cats puff up when they come across another unknown feline. Their fur stands on end, making their tails look like pipe cleaners, and overall, they look bigger and maybe more “badass.” Some of them also howl, but despite these displays, they don’t always fight. Why? The obvious reason is that one can get hurt fighting, and an injured cat is less likely to survive. Because this makes sense, why would the cats consider fighting in the first place? The basic idea is that having a larger territory confers an advantage. For instance, if prey is scarce, the larger one’s territory, the greater is the likelihood of finding food, especially if you can keep other cats out. So this means that if you never fight, you might also not survive.

In a seminal paper published in *Nature* in 1973, John Maynard Smith and George Price examined the logic of animal conflict over resources from an evolutionary perspective. An abbreviated version of this logic is as follows: Suppose that our two cats are contesting a territory that provides a fitness advantage, V . What this means is that the cat that controls this territory and all its resources will have V more offspring than the cat that doesn’t. Extending this logic, if aggressiveness is hereditary, there will be relatively more aggressive cats in the next generation, and the question becomes: will the population evolve to an equilibrium in which all cats are aggressive?

To add a bit more structure while keeping our initial example simple, let's say that all cats are equally likely to win a fight and accrue this benefit. However, as mentioned previously, fighting for the territory is potentially costly. Suppose that half the time, any cat that fights with another for control of the territory suffers an injury that reduces its fitness by c . Putting this all together, if two cats meet and fight, the expected payoff (measured in fitness terms) is $\frac{1}{2}(V - c)$.

As we said, cats need not fight. If one aggressive cat is ready to fight and the other walks away, the aggressor gets the entire prize, V , and though the one that flees gets nothing, it also is unharmed. If you observe the neighborhood for long enough, you will also see two cats who initially puff up, but then compromise and share the territory. These cats each earn a payoff of $\frac{V}{2}$.

Using standard terminology, let's call the aggressive cats "Hawks" and the pacifist cats "Doves." Considering the normal form of the game implied by this scenario, depicted as the hawk-dove game in [figure 25.1](#), we can assess what should happen when Toughy and Bubba meet in your backyard.

		Bubba	
		Hawk	Dove
Toughy	Hawk	$(\frac{1}{2}(V - c), \frac{1}{2}(V - c))$	$(V, 0)$
	Dove	$(0, V)$	$(\frac{V}{2}, \frac{V}{2})$

Figure 25.1

The Hawk-Dove game.

Finally, notice that when $V > c$, in expectation, you'd like to have won a fight because $\frac{1}{2}(V - c) > 0$ so being a Hawk dominates. This suggests that, indeed, we should expect aggressive cats to take over the neighborhood. They are more likely to fight for the territory, win the benefits, and have more kittens.

If instead $V < c$ and the cost of fighting outweighs the benefits of gaining the territory, on average, the evolution of the neighborhood feline population is likely to be more complicated. Now there are two pure-strategy equilibria in which there is one cat of each type and a mixed-strategy equilibrium wherein cats play Hawk with probability $\frac{V}{c}$ and, as is intuitive, the higher is V ,

the more likely the cats are to play Hawk. In the next section, we develop a model of the evolution of these phenotypes (i.e., aggression and passivity) that predicts what will happen in the population.

Replicator Dynamics

Now imagine that there is an infinite population of cats whose baseline fitness is F_0 . We assume the population is infinite so that the choice of any individual cat will not affect the expected payoffs anticipated by any of the other cats. Further, the behavior of each cat is phenotypically determined to be either $H(awkish)$ or $D(oveish)$, and these cats are randomly paired to contest a small territory in the neighborhood.

If p is the frequency of $H(awkish)$ cats in the current population and $Eu[H, H]$, $Eu[H, D]$, $Eu[D, H]$, and $Eu[D, D]$ correspond to the fitnesses achieved in the contest (i.e., the normal form from before) for any possible matching of types (e.g., $Eu[H, D] = \frac{1}{2}(V - c)$ is the fitness payoff to a *Hawk* who meets a *Dove*), then we can write the expected fitnesses of the two types of cats as a function of the number of Hawks. Specifically,

$$F(D) = F_0 + pEu[D, H] + (1 - p)Eu[D, D]$$

Moreover, the mean fitness of the entire population is just a weighted average—the fraction of cats who are Hawks times the expected fitness of these Hawks in the current population plus the fraction who are Doves times the expected fitness of Doves. Put differently, the average fitness, $\bar{F}$, is

$$\bar{F} = pF(H) + (1 - p)F(D)$$

Evolution, based on asexual reproduction, dictates that the fraction of Hawks in the next generation (call it p_{t+1}) is determined by how many Hawks there are currently, p_t , and how well Hawks are doing in the current population when compared to the average (Maynard Smith and Price, 1973). We would, for example, expect evolution to be somewhat smooth, such that if there are a lot of Hawks around currently, although their numbers may increase or decrease in the next generation, there will still be quite a few in the next period. In other words, the population will not explode or crash within one or two generations. Further, if Hawks are currently doing better than the average, in fitness terms, they should expand in the population and if they are doing worse than average, they should shrink. To capture these dynamics, start with

$$p_{t+1} = p_t \frac{F(H)}{\bar{F}}$$

which, moving from stocks to flows to accentuate the difference, can be written as

$$p_{t+1} - p_t = p_t \frac{F(H)}{\bar{F}} - p_t$$

A little algebra gives us

$$p_{t+1} - p_t = p_t \frac{F(H)}{\bar{F}} - p_t \frac{\bar{F}}{\bar{F}}$$

or the simple difference equation:

$$p_{t+1} - p_t = p_t \frac{F(H) - \bar{F}}{\bar{F}}$$

The difference on the left side of this *replicator dynamic* (see definition 1) can be interpreted as the amount of growth or shrinkage in the number of Hawks from one period to the next and the right side indicates that, as mentioned before, if Hawks are currently doing better than average, the difference will be positive and Hawks will expand, but if they are doing worse than average, the difference will be negative and the number will shrink. If we assume that the differences per generation are not too big, we can approximate the evolutionary path of the population with the differential equation

$$\dot{p} = p_t \frac{F(H) - \bar{F}}{\bar{F}}$$

where $\dot{p}$ is just common shorthand for the time derivative, $\frac{dp}{dt}$.

Definition 1 (Replicator dynamics [based on Maynard Smith, 1982]) Say that an infinite population of agents are paired at random to play a symmetric game with strategies S in which p_t^i is the current fraction playing strategy $i \in$

S , $F(i)$ is the expected fitness of strategy i in the current population, and $\bar{F}$ is the average fitness of the entire population. The replicator dynamics are described by the system of equations $\dot{p}^i = p_t \frac{F(i) - \bar{F}}{\bar{F}}$ for $i \in S$.

Two things will allow us to simplify this replicator dynamic even further. First, to preview the purpose of all this, think about the cases in which $\dot{p} = \frac{dp}{dt} = 0$. When this condition is satisfied, the fraction of Hawks in the next generation will be exactly the same as in the current one. In other words, Hawks will neither expand nor contract and the population will have reached an equilibrium. If, for example, Hawks always do better than Doves, then $F(H) > F(D)$ regardless of p_t , and therefore, $F(H) > \bar{F}$ will also continue to be true until there are only Hawks left in the population (i.e., where $p_t = 1$). At this point, by definition, $F(H)$ will equal $\bar{F}$, the difference will be zero and $\dot{p}$ will equal zero too. Similarly, if Hawks always do worse than Doves, the opposite will be true. Hawks will continue to shrink in the population because $F(H) < F(D)$ and $F(H) < \bar{F}$ until there are no Hawks left. In this case, $\dot{p}$ will equal zero because p_t will eventually converge to zero.

Definition 2 (Evolutionary equilibrium [based on Maynard Smith, 1982]) A system of replicator equations $\dot{p}^i = p_t \frac{F(i) - \bar{F}}{\bar{F}}$ achieves an evolutionary equilibrium when $\dot{p}^i = 0$ for $i \in S$.

Notice, however, none of these results depend on the $\bar{F}$ in the denominator of the replicator dynamic equation. It is p_t and the numerator, $F(H) - \bar{F}$, that determine both the direction of change and the conditions under which an evolutionary equilibrium (see definition 2) is achieved. The $\bar{F}$ in the denominator only scales this evolution. If the average fitness in the population is high, evolution works more slowly than if it is low. While this makes sense, we will focus our attention on the direction and limiting states of evolution, not its speed. As a result, we can rewrite the replicator dynamic as follows, more or less ignoring the $\bar{F}$ in the denominator, and the population will continue to follow that same evolutionary trajectory:

$$\dot{p} = p_t [F(H) - \bar{F}]$$

Second, in situations in which there are only two types/strategies (and only in these situations), we can simplify the replicator dynamics by expanding $\bar{F}$. Recall that this weighted average is just $p_t F(H) + (1 - p_t)F(D)$, which, when substituted into the replicator equation, gives

$$\dot{p} = p_t[F(H) - p_t F(H) - (1 - p_t)F(D)]$$

This can be simplified to

$$\dot{p} = p_t[(1 - p_t)F(H) - (1 - p_t)F(D)]$$

and finally,

$$\dot{p} = p_t(1 - p_t)[F(H) - F(D)]$$

This simplification is the version of the replicator dynamics we will use for settings in which there is just one population and two phenotypes.

The Evolution of Hawks and Doves

As an example of how the replicator dynamic is used, we can calculate fitnesses and substitute them into the equation we just derived. Recall that the fitness of Hawks, considering the time increments, is

$$F(H) = F_0 + p_t E[H, H] + (1 - p_t) E[H, D]$$

Substituting the payoffs from the normal form given in [figure 25.1](#), we get

$$F(H) = F_0 + p_t \left[\frac{1}{2}(V - c) \right] + (1 - p_t)V$$

or, after some algebra,

$$F(H) = F_0 + V - \frac{1}{2}p_t(c + V)$$

The fitness of Doves, in general, is

$$F(D) = F_0 + p_t E[D, H] + (1 - p_t) E[D, D]$$

which, after substitution from the normal-form game, becomes

$$F(D) = F_0 + p_t 0 + (1 - p_t) \frac{V}{2}$$

Placing these explicit fitness payoffs into the replicator equation gives

$$\dot{p} = p_t(1 - p_t) \left[F_0 + V - \frac{1}{2}p_t(c + V) - F_0 - p_t 0 - (1 - p_t) \frac{V}{2} \right]$$

a differential equation that can be simplified to

$$\dot{p} = p_t(1 - p_t) \left[\frac{V}{2} - \frac{p_t c}{2} \right]$$

The next step is to figure out where $\dot{p}$ will be zero. These will be the *fixed points* of the dynamics—those phenotypical distributions from which the

population will no longer adjust. That is, these will be the equilibria of the evolutionary system.

In the version of the replicator dynamics we have settled on, it is easy to identify two population distributions that always have to be fixed points—namely, where p_t equals 0 or 1. These are places where evolution cannot get any traction. In the first case (where $p_t = 0$), there are no Hawks in the neighborhood to begin with and therefore, in the absence of mutations, Hawks can never evolve. In the second case, if $p(t) = 1$, every cat in the neighborhood is a Hawk to begin with, and now Doves can never evolve. In general, the replicator dynamics will always have fixed points that are characterized by any monomorphic initial state (i.e., where just one type is present at the outset).

We can also see, however, that if $\frac{V}{2} - \frac{p_t c}{2}$ equals zero, $\dot{p}$ will also be zero. This occurs when $p_t = \frac{V}{c}$, and it turns out to be no coincidence that this coincides with the mixed-strategy equilibrium of the underlying normal-form game. In fact, remember that if $V > c$, *Hawk* dominates, and there is a pure-strategy Nash equilibrium where $p_t = 1$, one of the fixed points identified previously. If instead $V < c$, there are off-diagonal pure-strategy equilibria and the mixed-strategy equilibrium just calculated. What is important is that the replicator dynamics will always identify Nash equilibria as fixed points. This fact is interesting for two reasons. First, it highlights the power of the Nash equilibrium concept because we see it has arisen in two very different disciplines. Second, and more important for the purposes of this chapter, these dynamics, with a bit of reframing, form the basis for another model of learning in games.

Instead of looking through the lens of genes, asexual reproduction, and fitness, consider the replicator dynamics written as $\dot{p} = p_t[Eu(H) - \bar{E}]$, substituting expected payoffs for fitnesses. Notice that we will get exactly the same fixed points. But how do we interpret this equation? Suppose that instead of cats (as smart as they are), we have a population of decision makers who can do just two things: discern their own payoffs and observe a random sample of payoffs accruing to other players. If these players use a simple decision rule, *use strategies in proportion to their payoff relative to the average*, we can expect them to learn Nash equilibria too. Further, this is a much simpler model of learning than best response dynamics (chapter 24)

because it is based on simple imitation, requiring less cognition on the part of our decision makers.

Evolution in the Stag Hunt

Let's now consider another example, this time explicitly adopting the imitation dynamics framing. In this example, we predict where imitation, as modeled by the replicator equation, will lead a population of agents who meet in pairs to play the stag hunt game.

In the stag hunt, agents consider two strategies, *Stag* and *Hare*, and the essential problem is one of social order—*Stag* is socially optimal but risky, as one can see in the normal form shown in [figure 25.2](#).

		Agent 2	
		Stag	Hare
Agent 1	Stag	(5, 5)	(0, 4)
	Hare	(4, 0)	(2, 2)

Figure 25.2

The stag hunt game.

We also see that there are three Nash equilibria in stag hunt. Conditional on the other agent playing *Stag*, also playing *Stag* is a best response and forms one equilibrium. Likewise, if the other agent is hunting *Hare*, then you will get nothing unless you do so as well. Finally, you can calculate that there is a mixed-strategy equilibrium in which *Stag* is played two-thirds of the time.

If players imitate others in proportion to how successful they are, we expect the population of hunters to evolve according to

$$\dot{s} = s(1 - s)[Eu(S) - Eu(H)]$$

where we drop the time subscripts for convenience, s is the current fraction of the population playing *Stag*, and $E(\cdot)$ denotes the expected payoffs as discussed before. The thing to remember when setting up this dynamic is that, by construction, whatever type/strategy you are following will end up being first in the difference. That is, if a is the fraction playing A and $1 - a$ is playing B , then the imitation dynamic will always take the form $\dot{a} = a(1 - a)[Eu(A) - Eu(B)]$, not $\dot{a} = a(1 - a)[Eu(B) - Eu(A)]$.

The expected payoff to playing *Stag* is $5s + 0(1 - s)$, or $5s$, and the expected payoff of playing *Hare* is $4s + 2(1 - s)$, which simplifies to $2 + 2s$. Hence, the imitation dynamic can be written as

$$\dot{s} = s(1 - s)[5s - (2 + 2s)]$$

or

$$\dot{s} = s(1 - s)[3s - 2]$$

The fixed points of this dynamic, where $\dot{s} = 0$, occur when $s = 0$, $s = 1$, and $s = \frac{2}{3}$. Put differently, imitation will stop (or not start) when all agents in the population are playing *Stag*, when they are all playing *Hare*, or when they are mixed, with two-thirds probability on *Stag* and one-third on *Hare*. Again, the replicator/imitation dynamic identifies the Nash equilibria as potential end points of the learning process.

There is one last thing to point out with regard to evolution and mixed-strategy equilibria. As this example and the one before it illustrate, the replicator dynamic will identify the mixed strategy of any game as a potential fixed point. But how should we interpret these mixtures? The standard way, as invoked earlier, is to posit that everyone is consciously mixing, rolling dice and placing exactly two-thirds probability on *Stag* in the last example and $\frac{V}{c}$ on *Hawk* in the previous one. However, part of the reason for adopting these sorts of dynamic models of learning is because we think that agents might have bounded rationality and are not able to make all these calculations. With this in mind, an alternative interpretation of the mixture might be more appropriate. Instead of everyone explicitly mixing, we can interpret the mixed strategy as the fraction of the population playing one strategy or the other. In other words, when exactly $\frac{V}{c}$ of cats are Hawks and the rest are Doves, the population will reach an equilibrium. Likewise, if two-thirds of hunters stalk Stag and the other one-third go for Hare, that population will be in equilibrium too.

The Stability of Evolutionary Equilibria

We have seen that the benefits of using evolutionary dynamics as a model of learning in games include that the interpretation (imitating strategies in proportion to how well they are doing in the current population) is intuitive, fewer cognitive requirements are placed on decision makers, and the dynamics will always select the Nash equilibria of the underlying game as

fixed points. One issue, however, is that there will often be more fixed points than Nash equilibria because any initial state composed of just one strategy/type is also a fixed point, by definition. Equally important is the fact that we have so far been unable to say anything about equilibrium selection. In this subsection, we will see that evolutionary dynamics are also useful because they provide sensible microfoundations for the selection of equilibria (as an alternative to the discussion in chapter 7).

To address this second issue, we exploit the dynamic nature of this learning model, examining the conditions under which evolutionary equilibria will be stable. The point is to posit that stable equilibria will be more likely to arise than unstable ones. To get a better sense of what we mean by stability, return to the hawk-dove game and recall that the replicator dynamic describing the motion of a single infinite population is $\dot{p} = p_t(1 - p_t)\left[\frac{V}{2} - \frac{p_t c}{2}\right]$, where p is the fraction of Hawks in the current population. Consider first the case in which $V > c$ and remember that this means that *Hawk* dominates *Dove*. This seems like a good first test of the dynamics because under any model of learning, we would expect all the players to learn to be a Hawk in this setting. But will they?

On the left of [figure 25.3](#), we set $V = 5$, $c = 3$, and plot $\dot{p} = p(1 - p)(2.5 - 1.5p)$ for every value of p between zero and 1. As we can see, there are two fixed points, $p = 0$ and $p = 1$, as expected; but what is more interesting in terms of stability is that $\dot{p} > 0$ for any other value of p . This means that the fraction of Hawks is always growing (hence the arrows of motion to the right) and implies that regardless of how many Hawks there are in the initial population (so long as there is at least one), the Hawks will take over. In other words, people will learn to play *Hawk* if they imitate others. Further, the all-*Hawk* equilibrium is *evolutionarily stable*—if there is a perturbation pushing the population to the left (perhaps because some people decide to experiment with *Dove*), the Hawks will quickly take over again. In this most straightforward of examples, you can now see how evolutionary stability can select among fixed points of the replicator dynamics—we’ve gone from two fixed points to one prediction that overlaps with the only Nash equilibrium of this version of the game.

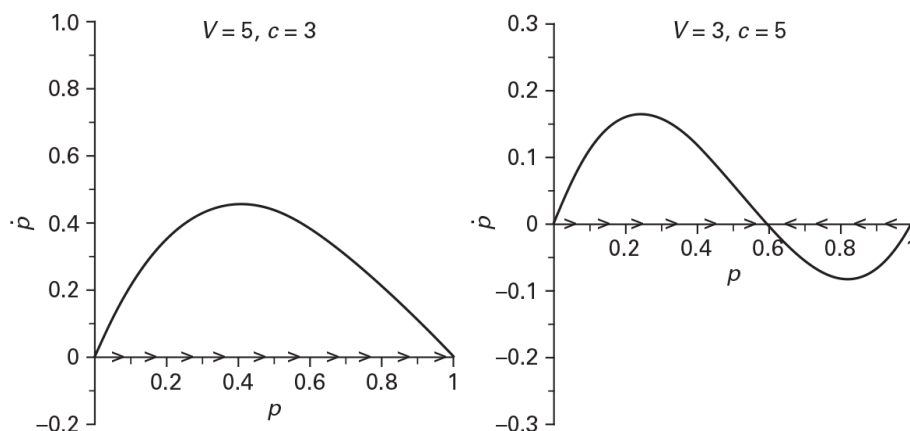


Figure 25.3

Replicator dynamics in the hawk-dove game.

Now let's complicate things and reverse our Hawk-Dove parameters. On the right side of [figure 25.3](#), we switched things so that $V = 3$, $c = 5$, and now $\dot{p} = p(1-p)(1.5 - 2.5p)$. One big difference is that there is now a viable mixed-strategy equilibrium, where $p = 0.6$. This means that this version of the game generates three fixed points of the replicator dynamics where $p \in \{0, 0.6, 1\}$. As always, there are fixed points where everyone is a Dove or everyone is a Hawk, but now there is also a fixed point at the mixed-strategy equilibrium. Are any of these fixed points stable?

In general, we place *arrows of motion* in the direction in which a population is evolving. If $\dot{p} > 0$, for instance, then the arrows should point to the right because the strategy associated with p is growing and p is getting larger. If the opposite is true and $\dot{p} < 0$, then the strategy associated with p is shrinking in the population and p is falling. Hence, the arrows point to the left. The quick way to determine stability is just to follow the arrows. In the graph on the left of [figure 25.3](#), this is easy because the arrows always point right, indicating that the all-Hawk equilibrium is *globally* stable and all initial conditions lead to this equilibrium.

The interpretation of the right side of [figure 25.3](#) is a little more complicated because arrows point in both directions; however, it should be clear that the monomorphic all-Hawk and all-Dove equilibria are not stable. The intuition for these not being stable is straightforward. If most of the population is playing *Dove*, near $p = 0$, then a mutant Hawk will do very well because it is guaranteed to meet a Dove and get all the resources. At the other end of the diagram, if almost everyone is playing *Hawk*, most players are

fighting and regretting it, and a lone Dove will do better because she stays out of the fray. Hence, the dynamics pull the population toward the center of the strategy space, and because the arrows are pointing toward the mixed-strategy equilibrium from both sides, it must be stable. Indeed, given our parameters, if a perturbation pulls the population above or below $p = 0.6$, it will be pushed back to this fixed point. In this case, the mixture in which $\frac{6}{10}$ of players are Hawks and $\frac{4}{10}$ are Doves is globally stable.

What about our other example? Are there any evolutionarily stable fixed points in the stag hunt? Given the parameters set for this game, we found that the fraction of *Stag* in the population should evolve according to $\dot{s} = s(1-s)[3s-2]$. Plotting this in [figure 25.4](#), we see that the dynamics are almost the opposite of those in the hawk-dove game with a mixed-strategy equilibrium. That is, when p is less than the mixed strategy ($p = \frac{2}{3}$), Stags should be decreasing in the population, but when $p > \frac{2}{3}$, they should increase. This makes sense. In the stag hunt, you should imitate whatever is relatively more common in the population. If there are not many people playing *Stag*, you are better off playing *Hare* because you are more likely to meet another person doing the same thing—coordination matters here. Likewise, when most people are playing *Stag*, you should learn to play *Stag* too. Hence, the two pure-strategy Nash equilibria are evolutionarily stable and the mixed-strategy equilibrium is unstable—any perturbation away from exactly $p = \frac{2}{3}$ will result in the population eventually converging to one pure-strategy equilibrium or the other.

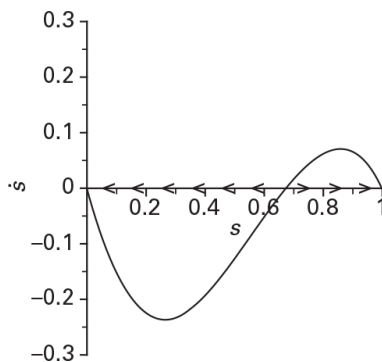


Figure 25.4

Replicator dynamics in the stag hunt.

What makes the stag hunt dynamic different is that none of the fixed points is globally stable. Here, the initial conditions determine which fixed point

obtains eventually. Given the parameters we have used, we can say that the all-*Hare* equilibrium is probably more likely because it has a larger *basin of attraction*. All the initial conditions between no Stags and $\frac{2}{3}$ Stag will converge to the all-*Hare* equilibrium. By contrast, the basin of attraction for the all-*Stag* equilibrium is comprised of just the remaining $\frac{1}{3}$ of the state space. The all-*Hare* equilibrium is more likely because it has the bigger basin of attraction, a concept isomorphic with the notion of risk dominance from chapter 7. This, of course, would change as expected if the payoffs changed. If, for instance, we increase the payoff to coordinating on *Stag*, the basin of attraction for the all-*Stag* fixed point would expand.

So what do all the evolutionarily stable fixed points we studied so far have in common? Just looking at the dynamic equations, it is not obvious, but if we look at the graphs, you should start to see a pattern: the dynamic appears to be downward sloping through all the stable fixed points. The reason for this may not be obvious, but there is a rationale to help explain it. Remember that the time derivative, $\dot{p}$ or $\frac{dp}{dt}$, is essentially a rate of change, and in these graphs, the larger the magnitude of these changes, the faster the population is evolving. By the same argument, when the absolute values of these time derivatives are small, the change slows. The idea is that the rate of change should slow as the system approaches a stable equilibrium and accelerate away from unstable equilibria.

In the first version of the hawk-dove game, at the beginning of this subsection, $\dot{p}$ is upward sloping when $p = 0$ because Hawks are almost guaranteed to meet Doves, and they do increasingly well. In fact, the dynamic speeds up until about $p = 0.4$, when Hawks start meeting each other too often, causing the growth in the Hawk population to start slowing. The dynamic continues to slow until it arrives at $p = 1$, the all-*Hawk* equilibrium. When we change the values of the resource and the cost of fighting, we again see the dynamic accelerate away from the all-*Dove* fixed point, where $p = 0$, but this time it starts slowing earlier because the cost of meeting another Hawk has risen. Coming from the right (i.e., the all-*Hawk* equilibrium), the growth of Doves accelerates and then slows near the mixed-strategy equilibrium for similar reasons. Finally, in the stag hunt, it is at the fixed points at the ends of the state space where the dynamic slows upon approach, in this case because as everyone starts playing the same strategy, that strategy can't be doing much

better than average, and hence, if you try something different, you won't do much better.

We can examine this notion of stability more closely by explicitly calculating the slope of the dynamic evaluated at the fixed points of, say, the second version of hawk-dove, in which $\dot{p} = p(1-p)(1.5 - 2.5p)$. Simplifying this expression a bit, we have $\dot{p} = 1.5p - 4p^2 + 2.5p^3$. If we take the derivative to calculate the slope with respect to p , $\frac{dp}{dp} = 1.5 - 8p + 7.5p^2$, and plug in the three fixed points, we find $\frac{dp}{dp}(0) = 1.5$, $\frac{dp}{dp}(1) = 1$, and $\frac{dp}{dp}(\frac{2}{3}) = -0.6$. Only the last of these, at the mixed-strategy equilibrium, is negative and we have evolutionary stability (see definition 3).

Definition 3 (Evolutionary stability [based on Hofbauer and Sigmund, 1998]) *Considering two-strategy games in a single infinite population, a fixed point of the replicator dynamic $\dot{p} = p_t \frac{F(i) - \bar{F}}{\bar{F}}$ (call it p^*) is evolutionarily stable if $\frac{dp}{dp}(p^*) < 0$.*

This definition formalizes the notion of evolutionary stability for the games we have considered so far: those played by a single infinite population of players, each of whom considers just two pure strategies. In the next section, we develop a set of criteria that allow us to identify evolutionarily stable equilibria without going to the trouble of writing the dynamics.

Evolutionarily Stable Strategies

Given the construction of the replicator equation, it should now be apparent that the key to evolutionary stability has to do with the difference in the expected payoffs of the two strategies. In this section, we examine this difference more closely and discover that we can write what turns out to be a very intuitive set of payoff criteria that assure a Nash equilibrium is evolutionarily stable under replicator dynamics. To see how to do this, we need to generalize our discussion a bit.

Suppose that we have the generic symmetric contest represented in [figure 25.5](#), in which agents play either X or Y . The expected payoff of anyone playing X against another agent playing X is denoted as $Eu[X, X]$. When playing X against another agent playing Y , one earns $Eu[X, Y]$, playing Y against X yields $Eu[Y, X]$, and when two Y players meet, they each earn $Eu[Y, Y]$, in expectation.

		Agent 2	
		X	Y
Agent 1	X	$Eu[X, X]$	$Eu[X, Y]$
	Y	$Eu[Y, X]$	$Eu[Y, Y]$

Figure 25.5
Determining an ESS.

Under what circumstances will the strategy X , for example, be evolutionarily stable? Referring back to our discussion in the previous subsection, we know that the arrows of motion have to point toward a fixed point where everyone is playing X , meaning that if a few players choose Y instead, they can't do better.

Returning to the replicator dynamics, if x now represents the fraction of players currently using X (and $1 - x$ is the fraction using Y), the growth of X in the population can be represented as $\dot{x} = x(1 - x)[Eu(X) - Eu(Y)]$. We now know, based on our discussion of the dynamics in the hawk-dove game, that one way to guarantee that strategy X will take over the population is if $Eu[X] > Eu[Y]$ for all x . To see this, notice that if $Eu[X] > Eu[Y]$, then $\dot{x}$ will always be positive or zero. This means that we have the graph shown in [figure 25.6](#), and X will always be growing in the interior of the state space.

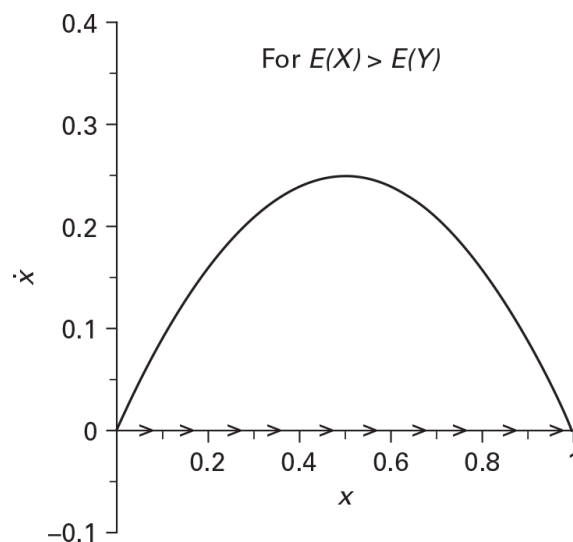


Figure 25.6
Assuring dynamic stability.

Let's examine this critical condition more closely. Based on the normal form discussed earlier, we can calculate that $Eu(X) = xEu[X, X] + (1 - x)Eu[X, Y]$ and $Eu(Y) = xEu[Y, X] + (1 - x)Eu[Y, Y]$. There are two ways that the first term can be greater than the second near the all- X fixed point. In the neighborhood of the all- X equilibrium, we know that x will be relatively large (and $1 - x$ will be small). In this case, $Eu[X, X] > Eu[Y, X]$ will be sufficient to ensure that $Eu[X] > Eu[Y]$. This will also guarantee that playing X is a Nash equilibrium. However, suppose that we weaken this condition a little and allow $Eu[X, X] = Eu[Y, X]$. If this is the situation, then you can see that we need for $Eu[X, Y] > Eu[Y, Y]$ to continue to have $Eu[X] > Eu[Y]$. These two conditions constitute a shortcut and guarantee that the all- X fixed point is dynamically stable. Together, they define an *evolutionarily stable strategy* (ESS). This is given more formally in definition 4.

Definition 4 (Evolutionarily stable strategy [based on Maynard Smith and Price, 1973]) Consider a single infinite population choosing between X and Y . X is an evolutionarily stable strategy (an ESS) if either (i) $Eu[X, X] > Eu[Y, X]$ or (ii) $Eu[X, X] = Eu[Y, X]$ and $Eu[X, Y] > Eu[Y, Y]$.

What is the intuition here? When will a population of agents continue to play X ? There are two cases to consider. First, reflect on the situation in which all the agents are already playing X . The only thing that would prevent the population from continuing to play X is if a deviant Y does better against X -players than staying with X does. To prevent this from happening, we can see in the normal-form game that we need for $Eu[X, X] > Eu[Y, X]$. In other words, if X -players do better against other X -players than switching to Y would, the population will remain all X . Second, what if nobody is currently playing X ? Can a mutant X -player successfully infiltrate a population of Y s? The X s can gain a foothold in the population and expand if playing X against Y does better than playing Y against itself. That is, if $Eu[X, Y] > Eu[Y, Y]$.

To see the utility of this ESS shortcut, we can return to the three games we already analyzed to check if the ESS criteria identify the same strategies that we found to be dynamically stable under the replicator dynamics. In the hawk-dove game when $V > c$, the first criterion requires that $Eu[H, H] > Eu[D, H]$ or, from the normal form at the beginning of the chapter, $\frac{V-c}{2} > 0$. This is clearly true, so we are done—*Hawk* is an ESS. As we saw in [figure 25.3](#), all *Hawk* is also a stable fixed point of the replicator dynamics.

What about the hawk-dove game when $V < c$? The first thing to do is to show that neither of the pure strategies is an ESS because they were not stable fixed points. For *Hawk*, we see now that $Eu[H, H] < Eu[D, H]$ because $\frac{V-c}{2}$ is negative. Hence, *Hawk* is no longer an ESS. For *Dove*, $Eu[D, D] = \frac{V}{2} < V = Eu[H, D]$, so wimpy Doves will never constitute an ESS. They will always be overrun by Hawks.

The mixed strategy, however, was dynamically stable. Is it an ESS? To check this, we start with the first criterion: when the mixture is played against itself, does it do better than playing *either* of the pure strategies against the mixture p^* ? That is, we have to check whether both $Eu[p^*, p^*] > Eu[H, p^*]$ and $Eu[p^*, p^*] > Eu[D, p^*]$ are true. If you calculate these expected payoffs, you should get $\frac{v(c-v)}{2c}$ for all three, which actually should not be a surprise. The exact payoff might not be obvious, but the fact that they are all equal should be. Remember from chapter 5, the purpose of playing a mixed strategy is to make the other player indifferent between her pure strategies so the payoffs have to be the same.

Hence, the first ESS criterion fails, but because the payoffs are all equal, satisfying the second one is still possible. To satisfy it, we have to make sure that both $Eu[p^*, H] > Eu[H, H]$ and $Eu[p^*, D] > Eu[D, D]$ are true. Put differently, does the mixture do better against the pure strategies than they do against themselves—can the mixture invade monomorphic populations of each type? We see that the mixture can invade a population of all Hawks because pure Hawks always fight with each other, while the mixture results in a fight only $\frac{V}{c}$ of the time. You can also see that the mixture does better against Doves than Doves do against themselves because $\frac{V}{c}$ of the time, the mixture extracts all V from the Doves and Doves always share with each other. Hence, we see that both $Eu[p^*, H] > Eu[H, H]$ and $Eu[p^*, D] > Eu[D, D]$ are satisfied and the mixture is an ESS. If you recall from [figure 25.3](#), the mixture was also dynamically stable when $V < c$.

Considering the stag hunt, we need to check to see if the opposite result obtains. Are the two pure strategies ESSs and the mixture not? Clearly, *Stag* is an ESS because $Eu[S, S] = 5 > 4 = Eu[H, S]$. The same is true for *Hare*, where we see that $Eu[H, H] = 2 > 0 = Eu[S, H]$. Recall that the mixture is $\frac{2}{3}$ *Stag*, so the expected payoffs of playing the mixture against itself, *Stag* against the mixture, and *Hare* against the mixture are all $\frac{10}{3}$. You can see the pattern—when a mixture is concerned, the first criterion will always fail because of this

built-in equality. What about $Eu[S, s^*]$ and $Eu[H, s^*]$ and the second criterion? If you play the mixture against *Stag*, you will get $\frac{2}{3}(5) + \frac{1}{3}(4)$ or $\frac{14}{3}$ which is slightly less than what you get from playing *Stag* against itself, 5. Finally, playing the mixture against *Hare* also does worse than playing *Hare* against *Hare*. Here, you get $\frac{2}{3}$ instead of 2. Again, we see that the ESS criteria predict which fixed points of the replicator dynamics will be stable.

Asymmetric Contests

So far, we have only considered symmetric games. In this section, we examine the use of evolutionary dynamics to predict the outcome in asymmetric contests. In general, what we will now assume is that instead of one infinite population, there are two, and pairs of players, one from each population, form at random to play the game. Once the game is finished, players return to their populations to compare payoffs. To get the hang of this, we will start with an example that is noteworthy both because the agents have different incentives and because the game has interesting dynamics.

The game, shown in [figure 25.7](#), is an asymmetric version of matching pennies in which two players compete for coins by choosing either *Heads* or *Tails* simultaneously. The Matcher wins the coins when both players choose either *Heads* or *Tails*, and the Differencer wins if the coins do not match. In the standard zero-sum case, such as the Princess Bride game that we analyzed in chapter 5, it is easy to see that the game has just one mixed-strategy equilibrium, in which both players flip the coins (i.e., they place $\frac{1}{2}$ the weight on either strategy). However, to accentuate the asymmetry in the game, not only do the players face opposite incentives when it comes to matching, we have changed the payoffs to make them more asymmetric.

		Differencer	
		Heads	Tails
Matcher	Heads	(2, 0)	(0, 1)
	Tails	(0, 3)	(1, 0)

Figure 25.7
Matching more pennies.

To figure out the mixed strategy of matching more pennies, let x be the probability that the Matcher plays *Heads* and y be the probability that the Differencer plays *Heads*. The expected payoffs for the two players are as follows:

$$\begin{aligned} Eu_M(H) &= 2y + 0(1 - y) = 2y & Eu_D(H) &= 0x + 3(1 - x) = 3 - 3x \\ Eu_M(T) &= 0y + 1(1 - y) = 1 - y & Eu_D(T) &= 1x + 0(1 - x) = x \end{aligned}$$

From these equations, we solve for $x^* = \frac{3}{4}$ and $y^* = \frac{1}{3}$.

Now let's think about the dynamics in the population of Matchers. The important thing to realize is that although the Matchers will ultimately compare their payoffs among themselves, those payoffs are actually determined by what is going on in the Differencer population. That is, they will be determined by y , not x . To see this, we form the replicator equation for the Matchers:

$$\dot{x} = x(1 - x)[Eu_M(H) - Eu_M(T)]$$

which, after substitution, becomes

$$\dot{x} = x(1 - x)[3y - 1]$$

The growth of playing *Heads* in the population of Matchers, $\dot{x}$, depends on how many *Heads* players are in the other (Differencer) population, y . Another thing that is important to point out is a simple matter of accounting. As the general form of the replicator dynamics indicates, the first term in the bracketed difference, $Eu_M(H)$, has to be the expected payoff to the strategy associated with $\dot{x}$, that is *Heads*.

Likewise, the first term in the brackets for the Differencer population dynamic, $\dot{y}$, has to be $Eu_D(H)$, the strategy defined by y , the fraction of Differencers playing *Heads*. Continuing this logic, the growth of *Heads* in the Differencer population is described by

$$\dot{y} = (1 - y)[Eu_D(H) - Eu_D(T)]$$

or

$$\dot{y} = y(1 - y)[3 - 4x]$$

and we see that the growth of *Heads* in the population of Differencers, $\dot{y}$, depends on the number of *Heads* players in the the Matcher population, x . The point is that because the two replicator equations are a function of both x

and y (i.e., $\dot{x}(y)$ and $\dot{y}(x)$), the two populations are connected and they will evolve together.

Given we now have two replicator equations, the fixed points of the resulting system of equations require that both $\dot{x}$ and $\dot{y}$ equal zero at the same time. As before, there are parameter combinations that are always fixed points—the situations in which there is only one type of player in each population. There are now four of these fixed points coinciding with (1) there are no *Heads* players in the two populations ($x = y = 0$), (2) all the players are using *Heads* ($x = y = 1$), (3) all the Matchers are using *Heads* and all the Differencers are using *Tails* ($x = 1, y = 0$), and (4) the opposite is true ($x = 0, y = 1$). In addition, you can see that the mixed strategy is also a fixed point, as it was when we examined symmetric games.

Hence, if the two populations start at the mixed strategy, they should stay there, but is this fixed point evolutionarily stable? What happens if there are a few jitters in the populations? To examine this possibility, we can graph the *phase diagram* of this system, which illustrates all the possible outcomes and evolutionary forces, and use a bit of analysis.

With two replicator equations, the state space becomes the unit square. At each vertex of this square, we have one of the four fixed points in which there is just one type of player in each of the two populations. [Figure 25.8](#) labels these points. In the lower left of the graph, both x and y are zero, so all the players in both populations are using *Tails*. Diagonally across from this position, all the players are using *Heads*, and the other two corners are where one population is all *Heads* and the other is all *Tails*.

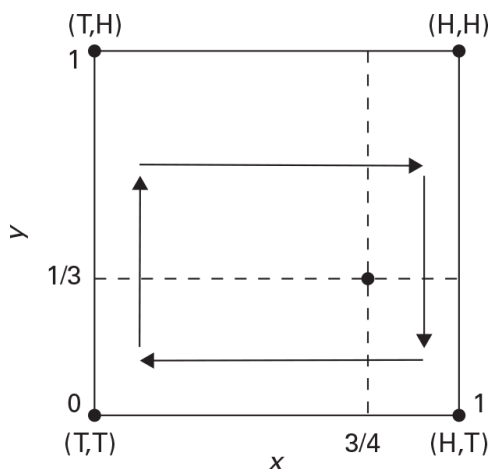


Figure 25.8

The dynamics of matching more pennies.

In [figure 25.8](#), you can also see the mixed-strategy Nash equilibrium. The question is: what happens to these two populations when they are pushed off the mixed-strategy distribution of types? First, consider the Matchers and notice that the sign of $\dot{x} = x(1-x)[3y-1]$ depends on the last bracketed term—the difference in the expected payoffs. When $y < \frac{1}{3}$, $\dot{x}$ will be negative because $x(1-x)$ will always be positive and $3y-1$ will be negative. This corresponds to the leftward-pointing arrow of motion in [figure 25.8](#). Put differently, whenever $y < \frac{1}{3}$, x , the fraction of *Heads* players, should be shrinking in the Matchers population because *Tails* is doing relatively better. Likewise, when $y > \frac{1}{3}$, $\dot{x}$ will be positive because *Heads* is now doing better because more of the Differencers are playing *Heads* too, as indicated by the rightward-facing arrow of motion.

Let's do the same thing for the Differencers. From $\dot{y}$, we see that *Heads* will increase in the Differencer population whenever $3-4x > 0$ or $x < \frac{3}{4}$. This is because enough of the Matchers are playing *Tails* for the *Heads*-playing Differencers to start doing well. When the opposite is true in the Matchers population (i.e., $x > \frac{3}{4}$, and a lot of them are playing *Heads*, it is the *Tails*-playing Differencers who start doing well and, therefore, *Heads* will shrink in the population. This is exactly what the arrows of motion indicate in [figure 25.8](#).

In [figure 25.8](#), you can also see that there is a tendency for the populations to cycle in matching more pennies. The same (in this case, clockwise) tendency can be seen when best responding in the original normal-form game. However, do we have any sense as to whether these cycles collapse inward (as in a whirlpool) or expand outward (as in the golden spiral)? Given the previous discussion of evolutionary stability, we might expect that the derivatives $\frac{d\dot{x}}{dx}$ and $\frac{d\dot{y}}{dy}$ can give us a clue. Although the details are more complicated (because we should consider $\frac{d\dot{x}}{dy}$ and $\frac{d\dot{y}}{dx}$ too), these derivatives, assessed at the mixed-strategy fixed point, do hint at what will happen if the population is jittered. Call the mixed-strategy equilibrium p^* . Doing the calculations, you should find that both $\frac{d\dot{x}}{dx}(p^*)$ and $\frac{d\dot{y}}{dy}(p^*)$ equal zero. What this suggests (but does not prove) is that while the equilibrium is not asymptotically stable, it is Lyapunov stable, meaning that the game will not collapse inward, but it also will not expand outward (Weibull, 1995). Simply put, if the populations are nudged a little off the mixed strategy, they will

enter a stable orbit very close to the mixed strategy. Therefore, for all intents and purposes, a small jitter will not be noticed.

We will now extend our discussion of asymmetric contests by considering an example of how to use evolutionary dynamics to select equilibria in extensive form games. Depicted here in [figure 25.9](#), is a market entry game in extensive form. In this game, there is a (C)hallenger firm that is deciding whether to enter a market with an established (I)ncumbent or stay out.

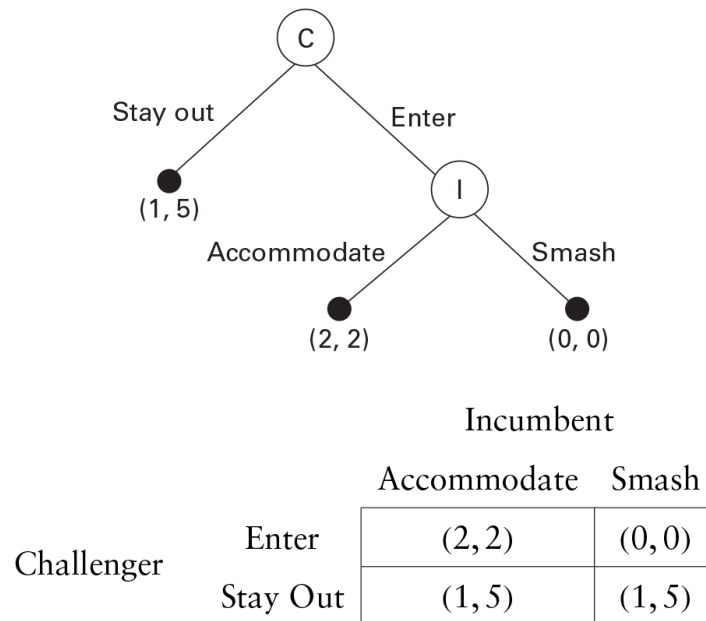


Figure 25.9

Market entry.

If the Challenger stays out, the firm continues to earn a nominal profit, 1, and the Incumbent, because it is a monopoly, does very well, earning 5. But, if the Challenger enters and the Incumbent accommodates the new firm in the market, the Challenger can double its profit. If instead, the Incumbent fights (i.e., *Smash*) the entrant, perhaps by lowering prices, both firms just break even.

The game in normal form is presented below the extensive form. Notice that because the Incumbent has a choice to make only if the Challenger enters, when the Challenger stays out, the payoffs are always (1, 5).

What are the Nash equilibria of market entry? There are two in pure strategies. First, if the Challenger chooses to *Enter*, the Incumbent's best response is to *Accommodate*, which forms an equilibrium. However, if the

Challenger is convinced that the Incumbent will *Smash* entrants, it is better off choosing to *Stay Out*. This, therefore, is also an equilibrium. But what is the problem with this second equilibrium? Smashing entrants is an empty threat, similar to NASA choosing Withhold in the Martian game from chapter 8. Once the Challenger has entered, Smashing is not a best response, since Accommodating the entrant increases the Incumbent's payoff from 0 to 2. What will be interesting in this example is whether evolutionary dynamics can cut through incredible threats.

Before we get to the dynamics, however, observe that there is another equilibrium in mixed strategies. It is not obvious, however. If you calculate the expected payoff to *Accommodate* for the Incumbent, you should get $5 - 3e$, where e is the probability that the Challenger Enters. If you calculate the expected payoff to *Smash*, you should get $5 - 5e$, and in expectation, it appears that *Accommodate* dominates. This is no surprise because you can see the weak dominance in the normal form of the game. Despite this, suppose that the Incumbent decides by tossing a coin. In this case, what are the expected payoffs for the Challenger? Staying Out always yields 1 for the Challenger, but if a is the probability of the Incumbent playing *Accommodate*, the expected payoff of entering is $2a$, which is also 1 when the Incumbent is mixing 50 – 50. As a result, there is a third, mixed-strategy equilibrium wherein the Incumbent mixes and the Challenger Stays Out.

Beginning with the Challenger, the growth of *Enter* in this population will be described by $\dot{e} = e(1 - e)[Eu(E) - Eu(SO)]$, where E and SO stand for *Enter* and *Stay Out*, respectively. Remember that the expected payoffs for the Challenger will actually depend on a , the fraction of accommodating Incumbents in the other population. Returning to the normal-form game, we calculate that $Eu(E) = 2a + 0(1 - a)$ and $Eu(SO) = 1$. If the Incumbent chooses to *Stay Out*, the game ends, so a plays no role. Putting this all together, we have

$$\dot{e} = e(1 - e)[2a - 1]$$

Switching focus to the Incumbent, $\dot{a} = a(1 - a)[Eu(A) - Eu(S)]$ describes the evolution of the strategy *Accommodate*, and we will see that the expected payoffs for the Incumbents will depend on what is happening in the Challenger population (i.e., e). From the normal form, we compute $Eu(A) = 2e + 5(1 - e)$ or $5 - 3e$ and $Eu(S) = 0e + 5(1 - e)$, which reduces to $5 - 5e$. Plugging these into the replicator equation, we get

$$\dot{a} = a(1 - a)[2e]$$

for the Incumbents.

Given $0 \leq e \leq 1$, you can already see that *Accommodate* should always be growing in the population of Incumbents. That is, we know that both of the first two terms in the replicator equation will be nonnegative, as will $2e$, so $\dot{a}$ will also always be positive when it is not at a fixed point. Again, this makes sense, given that *Accommodate* weakly dominates *Smash* in the normal form. Hence, in the phase diagram in [figure 25.10](#), the arrows of motion always point upward, indicating that *Accommodate* is always growing in the interior of the space.

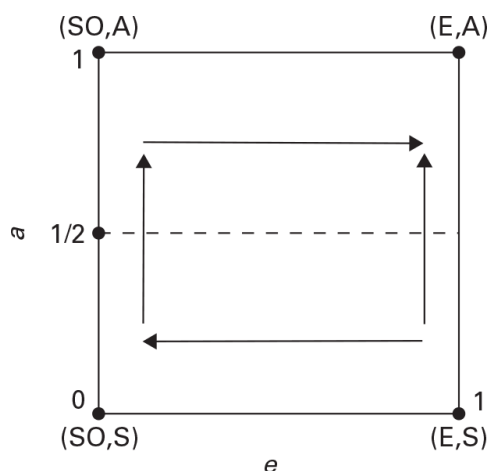


Figure 25.10

The dynamics of market entry.

For the Challengers, when $2a - 1 > 0$, that is $a > \frac{1}{2}$, $\dot{e}$ will be positive and *Enter* will be expanding. When the inequality goes the other direction and $a < \frac{1}{2}$, $\dot{e}$ will be negative—*Enter* will be contracting and *Stay Out* will be expanding in the population. In [figure 25.10](#), you can see that this means that the arrows of motion should point to the right (i.e., *Enter* is growing) when a is greater than $\frac{1}{2}$, and to the left (*Enter* is shrinking) when a is less than $\frac{1}{2}$.

What do these arrows of motion imply for the evolutionary stability of the fixed points in this asymmetric game? In [figure 25.10](#), we can see that the vertex of the phase diagram in the lower right, where every Challenger Enters and every Incumbent Smashes, is not stable because the arrows in both dimensions point away from it. Clearly, given that all the Incumbents are Smashing, a mutant Challenger who stays out will do much better, as will a

mutant Accommodating Incumbent facing a population of all entrants. You can also confirm that both $\frac{de}{dt}$ and $\frac{da}{dt}$ are positive at this fixed point, which corroborates the intuition.

Remaining in the lower half of [figure 25.10](#), if e should be shrinking toward the (SO, S) vertex/fixed point, is it stable? We already know that mutant entrants will not do well when the Incumbents are all mostly Smashing, so the dynamics should push the Challenger population back to the left, but remember that a is always positive, so if a player flips to *Accommodate*, this strategy should take off and continue to move away from the (SO, S) fixed point. Despite the forces continuing to reduce the number of entrants, because *Accommodate* should always be growing, the system will also roll right past the mixed-strategy equilibrium. Based on what we have already established, you can now see that the (SO, A) fixed point will suffer a similar fate as the (SO, S) one—while the growth of *Accommodate* pushes toward this vertex, entrants start to do better, and this pulls the system to the right. This leaves (E, A) , and here we see that the arrows from both directions point to this fixed point. This means that mutations will be repelled in both dimensions, and the only evolutionarily stable fixed point is the one in which all Challengers *Enter* and all Incumbents *Accommodate*.

From a theoretical point of view, an important corollary to the basic dynamic stability results we see in this example is that incredible threats do not survive evolutionary forces. Put differently, only subgame perfect Nash equilibria will be evolutionarily stable. Incumbents who stick to their guns and *Smash* entrants in this example do poorly and eventually fall out of the population as people start to imitate Incumbents who *Accommodate* entrants.

Behavior in Games—Can Evolutionary Dynamics and Imitation Predict Play?

While it is nice that evolutionary dynamics converge to Nash equilibria and select those that are subgame perfect, it would be even nicer if they predicted play in experiments as well as or better than the belief-based models described in chapter 24. The reason that it would be nicer is that a learning story based on imitation is much easier to square with what we know about the cognitive processes, which typically do not include the explicit calculation of best responses, used by real players. In this section, we will examine the experimental literature on using evolutionary dynamics to predict play.

As in the rest of this chapter, let's start with symmetric, one-population games. To keep this section as succinct as possible, we will present the underlying normal-form games played in the lab, the implied phase diagrams, and the experimental results. We leave as exercises at the end of the chapter the derivation of the replicator equations for each experiment we discuss.

In Embrey, Frechette, and Yuksel (2018), the authors experiment with the prisoner's dilemma presented in [figure 25.11](#). In this version of the game, *Defect* is not only dominant, but the “sucker's payoff” of 5 is particularly harsh. Knowing what we do about both behavior and evolutionary dynamics, it seems that people should quickly learn to *Defect* in this game. Our intuition about the dynamics is confirmed in the phase diagram shown in [figure 25.12](#) in which we see that not only should *Defect* evolve, but the magnitudes of $\dot{c}$, the growth rate of *Cooperate*, are large—players suffer a big penalty, on average, for Cooperating.

		Player 2	
		Cooperate	Defect
Player 1	Cooperate	(51, 51)	(5, 87)
	Defect	(87, 5)	(39, 39)

Figure 25.11

A prisoner's dilemma (Embrey et al., 2018).

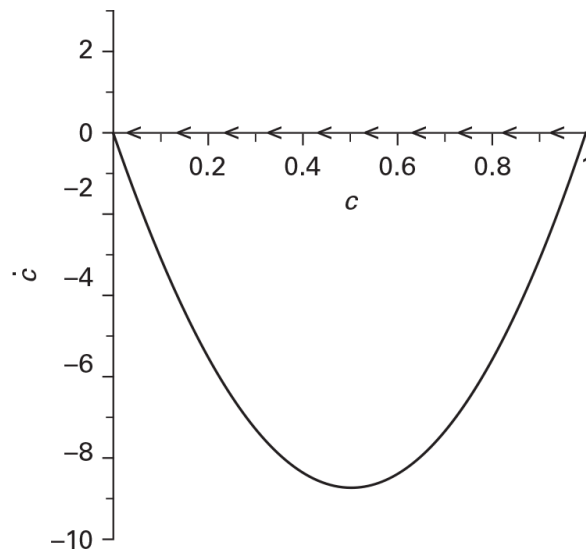


Figure 25.12

The Embrey et al. (2018) prisoner's dilemma normal-form and phase diagram.

To what extent do players cooperate in this environment? Do they learn to Defect, and if so, how long does it take? In the experiment, players were paired randomly to play a series of supergames in which this stage of the prisoner's dilemma was repeated eight times within each supergame. Between supergames, the players were rematched. In [figure 25.13](#), we present the average fraction of players choosing *Cooperate* in each of the eight stage repetitions. It is clear that players learn to Defect, but to get a sense of how well imitation might explain the decline of cooperation, in [figure 25.13](#) we also present the discrete time replicator dynamics (i.e., $c(t)$) starting in a population of half cooperators and half defectors.

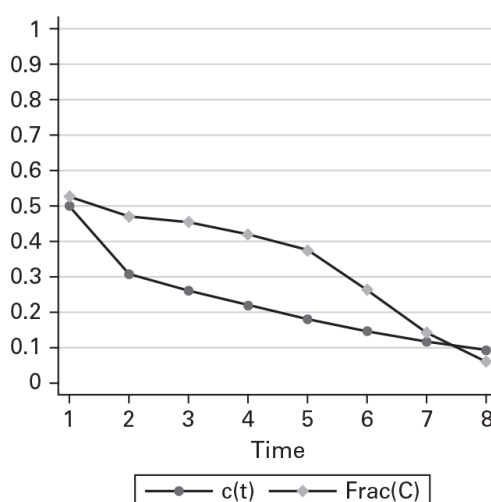


Figure 25.13

Cooperation in the Embrey et al. (2018) prisoner's dilemma.

To calibrate the graph, the replicator dynamics and data start in the same place; however, what is interesting is that they also end in the same place. This suggests that imitation might be a good model of learning in games. Of course, it is asking a lot for the two graphs to line up completely, but it is also interesting that the replicator dynamics predicts a sudden drop-off in the use of *Cooperate*, while the data suggests that most of the learning happens at the end of the supergame, not the beginning. It seems that real players are a bit more forgiving than theoretical ones, but when they've had enough, cooperation crumbles to levels predicted by the dynamics.

Is there any way to test the imitation story more directly? Remember the narrative we spoke of earlier. The idea is that players need to observe what other players are doing to calculate payoffs and compare them to the average.

Therefore, if players' information about what others are doing is limited, they should converge to a fixed point more slowly, if at all. Miller and Andreoni (1991) and Carpenter (2004) conducted related public good experiments with the same basic incentives as the prisoner's dilemma, but with a larger strategy space. Specifically, players had an endowment and could choose any amount to contribute to a public good, keeping the rest for themselves. Miller and Andreoni (1991) contributed by showing that, as in [figure 25.13](#), contributions in the public goods game fall at rates that also are consistent with evolutionary dynamics. Carpenter (2004) did this as well but also varied the amount of information that players have during the experiment, finding that when more imitation-relevant information is provided, players learn to free ride (aka Defect) quicker. This provides more direct evidence favoring imitation-based learning.

Now let's investigate how well evolutionary dynamics and imitation predict play in an asymmetric game. One of the most famous experimental games is Güth, Schmittberger, and Schwarze's (1982) ultimatum game, which is also one of the simplest formulations of bargaining (see chapter 14 too). In the general form of the game, a first-mover, the Proposer, makes an offer to a second-mover, the Responder, concerning how to share a monetary surplus provided by the experimenter. The Proposer can offer anything from the entire surplus to none of it, and all the Responder is asked to do is to accept or reject the offer. If the Responder accepts, the money is divided as proposed. If the Responder rejects the offer, however, the surplus vanishes and neither player gets anything.

Bolton and Zwick (1995) examined a discrete version of this game in the lab. The extensive form of their game appears in [figure 25.14](#). Here, the monetary surplus is 4, the Proposer can offer half to the other player or to keep 3 of the 4 for herself, and consistent with the lab data from the general form of the game, we assume that nobody rejects an equal split.

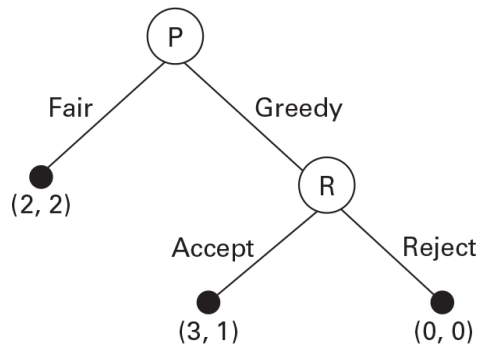


Figure 25.14

The mini-ultimatum game.

So, what should happen in the mini-ultimatum game? Working backward through the extensive form, you can see that the Responder's threat to reject a low offer is empty because Accepting yields 1 instead of 0. Knowing this, the Proposer maximizes her surplus by offering just 1. Being *Greedy* and choosing to *Accept* is, therefore, the subgame perfect Nash equilibrium. We now know that the replicator dynamic will select this fixed point, but will any other fixed point arise? Despite subgame perfection, what if enough Responders are willing to pay the dollar to punish greedy Proposers (as many are in ultimatum game experiments)? You can see that there is a critical value here. If fewer than $\frac{2}{3}$ of the Responders accept low offers, the Proposers will actually do better by being *Fair*.

We can see these dynamics more clearly in [figure 25.15](#), which plots the phase diagram for the mimi-ultimatum game. Here, f is the fraction of Proposers making *Fair* offers and a is the fraction of Responders who *Accept* unfair ones. Indeed, Accepting offers always pays more than Rejecting them, so a is increasing everywhere. Furthermore, when most Responders are playing their component of the subgame perfect Nash equilibrium (i.e., Accepting Greedy offers), there is no reason for Proposers to be Fair, so f falls. This means that *(Greedy, Accept)*, the subgame perfect Nash equilibrium, is evolutionarily stable according to the replicator dynamics.

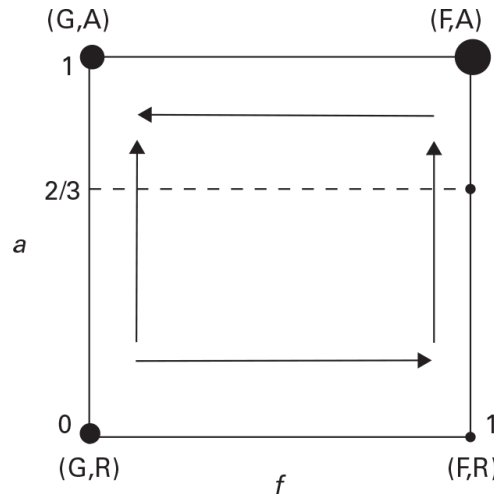


Figure 25.15

Evolutionary dynamics in the mini-ultimatum game.

You can also see in [figure 25.15](#), however, that in most of the state space, *Fair* offers should be increasing because it doesn't take that many Responders to start Rejecting to change the expected payoff of the two Proposer types. In fact, the entire right border of the state space from the (F, R) fixed point to the $\frac{2}{3}$ *Accept* threshold is neutrally stable (i.e., $\frac{da}{da} = 0$), indicating it will take a lot of perturbations to jostle the system into the basin of attraction of the subgame perfect Nash equilibrium. The reason to point this out is that the circles marking the fixed points in the phase diagram ([figure 25.15](#)) are of varying sizes, and these differences indicate the frequency with which play in Bolton and Zwick's experiment ended up at each outcome. While half of Proposers were Greedy, this didn't always work out for them, but offering half always did. Specifically, 50% of the games ended at (F, A) , 30% ended up at (G, A) , and the remaining 20% ended up at (G, R) . Taking the population as a whole, we see the rate at which real players *Accept* low offers is only $\frac{3}{5}$, slightly less than the $\frac{2}{3}$ critical value, indicating there is not sufficient evolutionary pressure to eliminate *Fair* offers, which persist throughout the experiment. In other words, evolutionary dynamics do a pretty good job of predicting play in this asymmetric game too.

As we often try to do, we end this chapter by asking whether there have been any theoretical advances developed to try to account for the differences between the existing theory of games and the behavior we find in the lab. It turns out that a simple and intuitive tweak to the replicator dynamics can

improve the prediction of evolutionary game theory in games like the mini-ultimatum game.

A stylized fact of ultimatum bargaining experiments is that you see a lot of play in the northeast area of [figure 25.15](#), where many offers are Fair and a fair number of Greedy offers are rejected. At the same time, the standard evolutionary dynamics (as a proxy model for imitation-based learning) do not identify any outcomes in this area of the strategy space as a stable fixed point. To address the gulf between the standard model and what we observe, Gale, Binmore, and Samuelson (1995) offered a modification of the replicator equation that coincides with another aspect of the related experiments—namely, that some small fraction of players (say, $0 < \delta < 1$) do not explicitly imitate; rather, they may just experiment with different strategies to get a sense of what works best.

To incorporate this sort of player experimentation into the model, let's assume that δ of the players randomly choose a pure strategy. What sort of dynamic would jibe with this behavior? Perhaps the simplest specification of experimentation (or what the authors call *evolutionary drift*) is just $\dot{p} = \frac{1}{2} - p$, where p is the fraction playing one of the strategies in any two-by-two game. You can see that the only fixed point of this dynamic is where $p = \frac{1}{2}$ and players randomize. Further, the derivative is $\frac{dp}{dt} = -1$ at this fixed point, so experimentation for this part of the population is dynamically stable.

To build experimentation into the learning model, suppose that the majority, $1 - \delta$, of players heed payoff differences and imitate according to the replicator equations, while the other δ experiment. This leads to the following replicator equations that account for experimentation/drift. At the extremes, when $\delta = 0$, we just have the standard replicator dynamics and when $\delta = 1$, everyone experiments:

$$\begin{aligned}\dot{f} &= (1 - \delta)f(1 - f)[E_P(F) - E_P(G)] + \delta \left(\frac{1}{2} - f \right) \\ \dot{a} &= (1 - \delta)a(1 - a)[E_R(A) - E_R(R)] + \delta \left(\frac{1}{2} - a \right)\end{aligned}$$

In [figure 25.16](#), we graph more detailed versions of the phase diagram for the mini-ultimatum game. On the left, we have an entire landscape of arrows of motion for the standard dynamics (i.e., $\delta = 0$) and a dark line that traces the predicted path of the two populations of players from an initial state in which a quarter of the Proposers make *Fair* offers and a quarter of Responders are

willing to *Accept* unfair offers. You can see that, as in [figure 25.15](#), only the subgame perfect Nash equilibrium is evolutionarily stable, even though a lot of players are rejecting unfair offers initially.

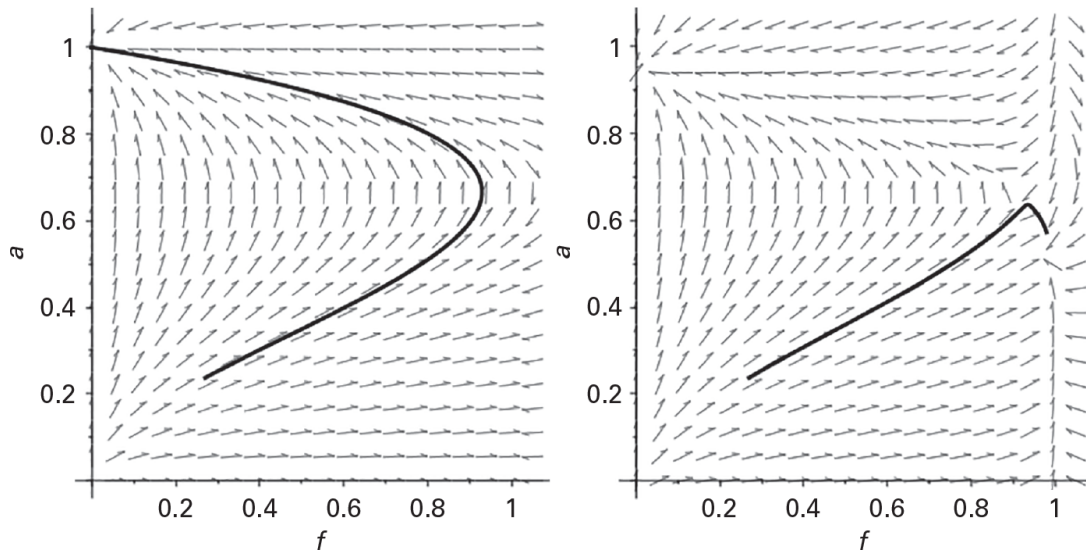


Figure 25.16

Mini ultimatum game phase diagrams without (left) and with (right) drift.

On the right side of [figure 25.16](#), we set $\delta = 0.01$ in the Proposer population, implying that 1 proposer in 100 experiments. In the Responder population, we increase delta to 0.10 so, effectively, 1 Responder in 10 randomly rejects a Greedy offer. Even with this small amount of evolutionary drift, we see that a second stable fixed point arises near the mixed-strategy equilibrium, precisely where a lot of experimental play occurs.

What is the intuition for the evolution of this sort of fairness norm? Without experimentation, the border where $f = 1$ below the mixed strategy where $a = \frac{2}{3}$ is neutrally stable because *Greedy* offers have been eliminated, and therefore *Accept* and *Reject* are both best responses. You can see this in the left panel of [figure 25.16](#) because the arrows are pointing at each other where $f = 1$. When experimentation is allowed, however, *Greedy* offers are constantly reintroduced to the game, and this puts evolutionary pressure on *Reject*; hence, the arrows now point upward where $f = 1$ in the right panel of [figure 25.16](#). At the same time, experimenting Responders are continually reintroducing *Reject* to the population too. The evolution of a stable fairness equilibrium in which *Greedy* offers are likely to be rejected, so Proposers act equitably, depends on the relative strength of these two forces. When the

experimental reintroduction of *Reject* outweighs the selection pressures against it, a fairness norm evolves, as it often does in the lab.

Exercises

(1) Consider the prisoner's dilemma used by Embrey et al. (2018) in [figure 25.11](#).

- Write the replicator equation for this single-population game.
- Does your model match the fixed points identified and the dynamics illustrated in the phase diagram in [figure 25.12](#)?
- What do you think will happen if the sucker's payoff of 5 increases to 20?

(2) Consider the following symmetric hawk-dove game:

		Player 2	
		<i>Hawk</i>	<i>Dove</i>
Player 1	<i>Hawk</i>	$(-2, -2)$	$(8, 0)$
	<i>Dove</i>	$(0, 8)$	$(4, 4)$

Figure 25.17

- Write the replicator equation for this game, wherein h is the fraction of players choosing *Hawk*.
 - Identify the fixed points. Are any of them evolutionarily stable?
 - Create the phase diagram for this game.
- (3) Consider the following asymmetric normal-form game, in which player 1 has a weakly dominant strategy and player 2 has a strictly dominant one:

		Player 2	
		<i>Left</i>	<i>Right</i>
Player 1	<i>Top</i>	$(1, 1)$	$(1, 0)$
	<i>Bottom</i>	$(1, 1)$	$(0, 0)$

Figure 25.18

- (a) Write the replicator equations for this game, wherein t is the fraction of players 1 choosing *Top* and l is the fraction of players 2 choosing *Left*.
 - (b) Identify the fixed points. Are any of them evolutionarily stable? Why are your results not surprising? Why are your results a little surprising?
 - (c) Create the phase diagram for this game.
- (4) Consider the following version of matching pennies in normal form:

		Differencer	
		<i>Heads</i>	<i>Tails</i>
Matcher	<i>Heads</i>	(1, 0)	(0, 1)
	<i>Tails</i>	(0, 1)	(1, 0)

Figure 25.19

- (a) Write the replicator equations for this game, wherein m is the fraction of Matchers choosing *Heads* and d is the fraction of Differencers choosing *Heads*.
 - (b) Identify the fixed points. Are any of them evolutionarily stable?
 - (c) Create the phase diagram for this game.
- (5) Consider the mini-ultimatum game studied by Bolton and Zwick (1995) and illustrated in [figure 25.14](#).
- (a) Write the replicator equations for this two-population game, where f is the fraction of Proposers making *Fair* offers and a is the rate at which Responders *Accept*.
 - (b) Does your model match the fixed points identified and the dynamics illustrated in the phase diagram in [figures 25.15](#)?
 - (c) Check the signs of $\frac{df}{dt}$ and $\frac{da}{da}$ to see if they are consistent with the subgame perfect Nash equilibrium being stable.
- (6) Now consider the following asymmetric game:

		Player 2	
		A	B
Player 1	C	(2, 2)	(0, -1)
	D	(3, 3)	(1, 4)

Figure 25.20

- (a) Write the replicator equations for this game, wherein a is the fraction of players 1 choosing A and c is the fraction of players 2 choosing C .
- (b) Identify the fixed points. Are any of them evolutionarily stable?
- (c) Create the phase diagram for this game.
- (7) Now consider the following hold-up game, where Workers *Invest* in human capital (or not), and then Bosses decide on a wage (*High* or *Low*) and Workers end the game by deciding whether to *Accept* the wage offer or *Reject* it. We assume that the High wage is always Accepted.

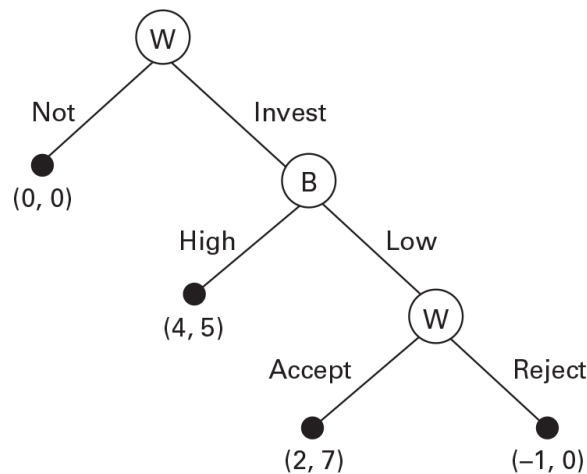


Figure 25.21

- (a) Write the normal form of the game, in which you consider two strategies for the Boss and four for the Worker: (*Not*, *Accept*), (*Not*, *Reject*), (*Invest*, *Accept*), and (*Invest*, *Reject*).
- (b) Write replicator equations for the two populations. (Hint: Because evolution will never select a dominated strategy, the normal form can be reduced to a two-by-two matrix.)
- (c) Create the phase diagram and identify any evolutionarily stable fixed points.

- (d) Assume now that a fraction, δ , of the players in the two populations mess up and choose randomly, while the remaining $1-\delta$ choose based on the replicator equations. What is the critical value of δ , such that the stable fixed points in part (c) are still stable? In other words, how much messing up will the dynamics tolerate?
- (8) Now consider the following variation on the stag hunt in which if both hunters go for *Hare*, they scare some of them away:

		Hunter 2	
		<i>S</i>	<i>H</i>
Hunter 1	<i>S</i>	(5, 5)	(0, 5)
	<i>H</i>	(5, 0)	(1, 1)

Figure 25.22

- (a) Identify any Nash equilibria.
- (b) Are either of the pure strategies an Evolutionarily Stable Strategy (ESS) according to the Maynard Smith and Price (1973) criteria in definition 4?
- (9) In the following version of the hawk dove game, there is a sense of territorial ownership, and to highlight this variant, we have added a third strategy, (*B*)*ourgeois*. The idea is that bourgeois players play *Hawk* if they currently own the territory, and they play *Dove* if they are invading the territory. If the two players are equally likely to own the territory in question, the expected payoffs to *Bourgeois*, in addition to the standard payoffs for *Hawks* and *Doves*, are listed here:

		Bird 2		
		<i>H</i>	<i>D</i>	<i>B</i>
Bird 1	<i>H</i>	(−1, −1)	(2, 0)	(0.5, −0.5)
	<i>D</i>	(0, 2)	(1, 1)	(0.5, 1.5)
	<i>B</i>	(−0.5, 0.5)	(1.5, 0.5)	(1, 1)

Figure 25.23

- (a) Is any strategy an Evolutionarily Stable Strategy (ESS) according to the Maynard Smith and Price (1973) criteria in definition 4 (check just the pure strategies and any mixed strategy that forms an equilibrium)?

26

Quantal Response Equilibrium

In this chapter, we consider the quantal response equilibrium (QRE) as an alternative (more general) solution concept to the Nash equilibrium. The Nash equilibrium requires players to correctly anticipate the behavior of others and then accurately calculate which action is best for themselves. In contrast, the QRE allows players to make mistakes by incorporating randomness into their choices. As in a Nash equilibrium, players' beliefs about the choices of others are accurate. However, the choices that players make in response to those beliefs are noisy. Rather than always perfectly best responding players better respond: they play all of their pure strategies with some positive probability but put more probability weight on those strategies with higher expected payoffs compared to those with lower expected payoffs. In equilibrium, the noisy choices of each player feeds back into the choices of the others, who correctly anticipate that the player will make mistakes and, in turn, better respond to the player's random choices.

Imagine that you are playing a matching pennies game, in which you and an opponent each have a penny, and you must simultaneously decide whether to place your penny with the Heads side up or the Tails side up. You then reveal your pennies to each other. If they match (with either both Heads up or Tails up), you gain a payoff of 1 while your opponent's payoff is 0. If they mismatch, then your opponent gains a payoff of 1, while your own payoff is 0. If you were to play a mixed strategy in this game, with what probability would you choose Heads?

Now imagine that you are playing an identical game with one small change. If your pennies match on Heads, you now will receive a payoff of 9 (while your opponent continues to earn 0). Now how often do you choose Heads? What if instead your payoff from a match on Heads was only $1/2$? How do you expect that your opponent's strategy would respond to each of these changes? The three versions of the game are shown in normal form in [figure 26.1](#), where you are the Row player.

Game 1				Game 2				Game 3			
Row		Column		Column		Column					
			Heads	Tails		Heads	Tails		Heads	Tails	
		Heads	(1, 0)	(0, 1)	Heads	(9, 0)	(0, 1)	Heads	(0.5, 0)	(0, 1)	
Tails	(0, 1)	(1, 0)	Tails	(0, 1)	(1, 0)	Tails	(0, 1)	(1, 0)			

Figure 26.1

Matching pennies game with Row player's payoff from (Heads, Heads) varying.

In all three versions of the matching pennies games, the only Nash equilibrium is in mixed strategies. In the first game, each player only cares whether they match or mismatch, but otherwise does not care whether they chose Heads or Tails. In this game, the unique Nash equilibrium is for each player to choose Heads with probability 0.5, a prediction that is fairly intuitive and is generally borne out in the data when real people play this game.

The second game is identical except that Row's payoff from matching on Heads increases from 1 to 9. But because the Nash equilibrium requires Row to choose his mixed strategy to make the other player indifferent—and Column's payoffs have not changed—Row should continue to play Heads with probability 0.5 in this second game.¹ Since Column's strategy must make Row indifferent, given his new payoffs, it is Column's mixed strategy that must change as we move from the first to the second game. Column must now play Heads less often (specifically, with probability 0.1) in order to deter Row from always opting for Heads. This unintuitive prediction that Row continues to play each of his strategies with equal probability, even as the payoff from the (Heads, Heads) outcome varies dramatically, is not consistent with people's choices in the lab. Instead, Row players tend to play Heads with a frequency far greater than 0.5 when the payoff from (Heads, Heads) is higher and far less than 0.5 when it is lower. In this chapter, we will explore an alternative solution concept to the Nash equilibrium, which produces predictions that are consistent with some of the systematic deviations from the Nash equilibrium that are observed in the matching pennies game and many other lab experiments.

Letting Players Make Mistakes

The Nash equilibrium solution concept has two features: (1) players form accurate beliefs about the strategies of others and (2) they always best respond

to those beliefs and implement the strategy associated with the highest expected payoff. As we have seen, behavior of real people often deviates from these sharp predictions. In cases of a unique mixed-strategy Nash equilibrium, people often use a very different probability mix when playing the game in the lab. Even in cases of a unique pure-strategy equilibrium, people often choose strategies that shouldn't ever be played according to the Nash equilibrium.

The *quantal response equilibrium* (QRE) is a generalization of the Nash equilibrium that allows probabilistic errors.² Specifically, it retains the first feature of the Nash equilibrium (that players form accurate beliefs about the strategies of others) but relaxes the second (that the players best respond to those beliefs with probability 1). Instead, players choose *all* their actions with some nonzero probability. Rather than perfectly best responding to their beliefs, they “better respond” by playing strategies associated with higher expected utilities more frequently than those with lower expected utilities. QRE recognizes that real people make mistakes in taking the action that is best for them. It incorporates the fact that people are unable to always perfectly maximize their payoffs and allows for the inherent randomness in people's behavior. The behavioral economist Colin Camerer has famously said that if John Nash had been a statistician rather than a mathematician, his equilibrium concept would look a lot like the QRE.

There are a couple of interpretations of this better response. The first is that people are always able to compute the expected payoff from each action, given the play of others, but they often fail to actually implement the strategy associated with the highest payoff. However, they are less likely to implement a strategy that is really bad for them than one that yields similar payoffs to their best response. The second is that players make errors in assessing the expected payoff that they would gain from each action. Since their assessments of the expected payoffs are noisy, they sometimes err in determining which action is their best response. Again, they are less likely to make these mistakes when there are big differences in their payoffs between actions than when the payoff differences are almost indistinguishable.

Better Responding: Quantal Response Functions

The quantal response equilibrium has its origins in psychological theories of probabilistic choice that allowed errors in decisions. In the 1950s, researchers

such as the mathematical psychologist Duncan Luce attempted to explain the observation that people's choices get noisier as the intensity of stimuli become more similar. For example, imagine that you are participating in an experiment in which you are asked to listen to two sounds. Your job is to indicate which is louder. Because you are human, you don't always perceive the sounds perfectly, and instead assess the loudness of each sound with some degree of error. Nevertheless, if the intensity of the two sounds differs dramatically, you are unlikely to make a mistake, so you correctly identify the louder noise with probability close to 1. However, as the two sounds become more similar, you are increasingly likely to get it wrong. And when they are essentially identical, the probability that you correctly select the one that is ever-so-slightly louder is roughly 0.5. Luce (1959) introduced a simple model of choice in this context, in which the relative likelihood that an option is chosen is equal to its relative intensity. Imagine that you evaluate n sounds, where sound 1 has loudness level l_1 , sound 2 has loudness l_2 , and so on. The likelihood that sound 1 is chosen is given by the *Luce ratio*: $\frac{l_1}{l_1 + l_2 + \dots + l_n}$.

Quantal response involves a similar idea for games. Rather than assessing the relative intensities of two sounds, players assess the relative magnitudes of the payoffs associated with their various actions. When the expected payoffs from two actions are the same, the player is equally likely to choose each of them. When the expected payoff from one action is higher than for the other possible actions, the player chooses this action more frequently, but not with perfect certainty as in the standard model.

Consider a generic two-player, simultaneous-move game, in which each player has two possible actions (also called a 2x2 game). The Row player has pure strategies A and B, while the Column player has pure strategies C and D. This generic 2x2 game is shown in [figure 26.2](#), with the payoffs x-ed out. The Row player forms some belief p about the likelihood that the Column player will play C and can then compute $Eu(A)$ and $Eu(B)$. In the standard model, the player would choose A with probability 1 if $Eu(A) > Eu(B)$, and with probability 0 if $Eu(A) < Eu(B)$. In this model, however, the Row player always chooses both strategies with some positive probability, with the probability of choosing A steadily increasing as $Eu(A) - Eu(B)$ increases. In other words, the player implements her best response with some error, and she is more likely to make errors when the payoff consequences are negligible (i.e., when $Eu(A) - Eu(B)$ is small) than she is when the differences in payoffs

are large. The exact probability with which Row selects A is given by a *quantal response function* (see definition 1).

		Column	
		p	$1 - p$
		C	D
Row	A	(xx, xx)	(xx, xx)
	B	(xx, xx)	(xx, xx)

Figure 26.2

A generic 2x2 game.

Definition 1 (Quantal response function) *A quantal response function is a mapping from a player's expected payoffs from each of her possible actions to a mixed strategy over those actions.*

There are many possible quantal response functions that could be used to model this situation. The simplest quantal response function is the Luce ratio, where we replace the loudness of sounds with the magnitude of payoffs. In this case, the probability with which Row plays A is given by $p_{row,A} = \frac{Eu(A)}{Eu(A) + Eu(B)}$. This implies that Row is equally likely to choose A and B in the event that $Eu(A) = Eu(B)$. Otherwise, Row chooses the action that yields the higher expected payoff more frequently. Holding $Eu(B)$ constant, Row is more likely to choose A as $Eu(A)$ increases.

In practice, the most commonly used quantal response function is the *logit quantal response function*, shown here for the 2x2 game example:

$$p_{row,A} = \frac{e^{\lambda Eu(A)}}{e^{\lambda Eu(A)} + e^{\lambda Eu(B)}} \text{ where } \lambda > 0 \text{ is a precision parameter} \quad (26.)$$

The logit quantal response function is similar to the Luce ratio, but with two major differences. First, each expected payoff, Eu , is replaced by the exponential function e^{Eu} . A major benefit of this specification compared to the Luce ratio is that the probability that it assigns to A is always positive, even if the expected payoffs are not (whereas the Luce ratio is defined only for positive payoffs). Just like the Luce ratio, it assigns a probability of 1/2 to each action when $Eu(A) = Eu(B)$. Second, included in the exponent is a so-called “precision parameter,” $\lambda > 0$. The value of λ specifies how responsive

the player is to differences in her payoffs. When λ is close to zero, the player is not responsive to the relative magnitudes of her payoffs, and her choice is essentially random. (Recall that $e^0 = 1$, so this fraction approaches $1/2$ as λ nears 0.) As λ approaches infinity, the player is increasingly sensitive to differences in her payoffs and the fraction approaches 1 if $Eu(A) > Eu(B)$ and 0 if $Eu(A) < Eu(B)$. In other words, the larger λ is, the more the player's behavior resembles perfect best responding. One can also think of λ as specifying the level of noise with which a player perceives her payoffs, where higher levels of λ indicate that her perception is more precise, and thus her choice is less noisy.

Notice that the equation (26.1) can also be rewritten as

$$p_{row,A} = \frac{e^{\lambda Eu(A)}}{e^{\lambda Eu(A)} + e^{\lambda Eu(B)}} = \frac{1}{1 + e^{-\lambda(Eu(A) - Eu(B))}} \quad (26.)$$

The expression on the right side of the equation can be somewhat less messy to deal with, and it makes it clear that the likelihood of choosing action A depends only on the superiority of A relative to B. To better understand the effect of λ , consider [figure 26.3](#). On the horizontal axis is the difference in expected payoffs from action A and B, ranging from -10 to 10 . Plotted over this range is the probability with which the player chooses A under different models. The standard best response function has the player choose A with probability 0 when the payoff difference is negative, with any probability when the payoff difference is 0, and with probability 1 when the payoff difference is positive. This is plotted as the dashed black line. In addition to the best response function, the logit quantal response functions are graphed for four values of λ : 0.003, 0.03, 0.3, and 3, where the darker gray curves are associated with the higher values of λ . When λ is very small, the probability that the player chooses A is close to 0.5, regardless of the payoff difference. As λ increases, the player's choice becomes more responsive to the relative payoffs from the two actions, and the quantal response functions approach the standard best response function. Notice that all the quantal response functions pass through the point $(0, 0.5)$, implying that players are always equally likely to choose their two actions when they are indifferent between them. In addition, note that the functions are all enclosed in the shaded regions (implying that players are always more likely to choose the action associated with the higher payoff), and they are all strictly increasing (implying that

players are strictly more likely to choose an action as its associated payoff increases relative to the other available action).

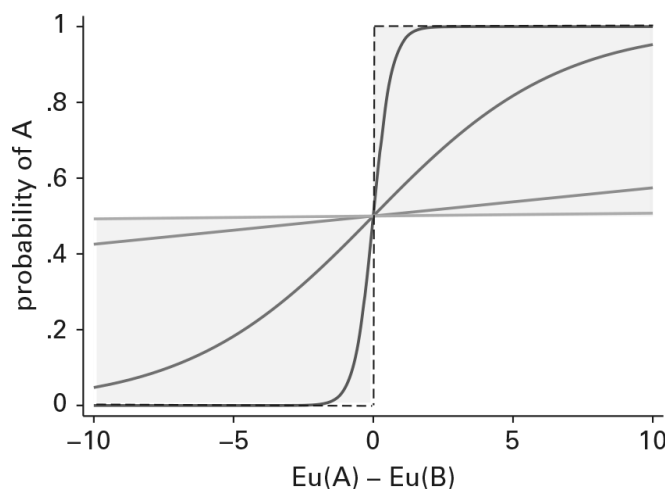


Figure 26.3

Logit quantal response functions over differences in payoffs (darker curves show higher λ values with the dashed black line showing the best response function).

The logit quantal response function can be easily extended to situations with more than two pure strategies. For a player with n pure strategies, labeled $s_1, \dots, s_n$, the denominator in equation (26.1) would be $e^{\lambda Eu(s_1)} + e^{\lambda Eu(s_2)} + \dots + e^{\lambda Eu(s_n)}$. For a player with a continuous strategy set, the denominator in equation (26.1) would be an integral. However, in this chapter, we will focus on games in which the players each face a binary choice.

Better Responding to Each Other: Quantal Response Equilibrium

The QRE incorporates these random response functions into a standard equilibrium analysis. In a quantal response equilibrium, the players are mutually better responding to the mixed strategies that are played by the others. In this way, QRE is analogous to the Nash equilibrium, while introducing the random choice described in the previous section. While the Nash equilibrium required each player's choice to be on their best response function, the QRE requires each player's choice to be on their quantal response function. Notice that this means that players do not simply implement Nash strategies with some randomness. Rather, there is feedback in the random play between the players. One player's mistakes are correctly

anticipated by the other player who, in turn, better responds to those mistakes, and so on.

As in a Nash equilibrium, players who are playing according to a QRE would not alter their strategies, even if they were to learn the strategies of others: each player is already on their quantal response function. Unlike in a Nash equilibrium, this doesn't mean that each player is fully maximizing their payoffs, given the strategies of others; rather, each player is better responding to the extent that they are able, given their precision parameter or cognitive capabilities. The QRE is formally defined in definition 2.

Definition 2 (Quantal response equilibrium [QRE]) *Consider a game with n players, where each player i has a quantal response function mapping her expected payoffs to a mixed strategy. A strategy profile, s^* , is a QRE if each player i chooses the mixed strategy s_i^* indicated by her quantal response function, given her expected payoffs associated with the mixed strategies played by the other players, s_{-i}^* .*

Better Responding in the Matching Pennies Game

Let's find the quantal response functions and QRE in the second matching pennies game, in which Row's payoff from a match on Heads is 9, as shown here:

First, let's think about the quantal response functions for each player. To simplify the notation, let's say that Column chooses Heads with probability p (and thus Tails with probability $1 - p$) and Row chooses heads with probability q (and Tails with probability $1 - q$), as shown in [figure 26.4](#). Regardless of which type of quantal response function we use, we're going to need to know the expected payoffs associated with each of the player's pure strategies (Heads and Tails), given the mixed strategy played by the other player. In this game, we compute as follows:

		Column	
		p	$1 - p$
Row	q	Heads	Tails
	$1 - q$	Tails	Heads
		(9, 0)	(0, 1)
		(0, 1)	(1, 0)

Figure 26.4

Matching pennies game with (Heads, Heads) payoff (9, 0).

$$Eu_{Row}(Heads) = 9(p) + 0(1 - p) = 9p$$

$$Eu_{Row}(Tails) = 1(1 - p) = 1 - p$$

$$Eu_{Column}(Heads) = 0(q) + 1(1 - q) = 1 - q$$

$$Eu_{Column}(Tails) = 1(q) + 0(1 - q) = q$$

Before we examine the logit quantal response functions, let's first look at the simpler Luce ratio. It tells us that the Row player should choose Heads with probability $\frac{Eu_{Row}(Heads)}{Eu_{Row}(Heads) + Eu_{Row}(Tails)}$. Using the expected payoffs we just computed before this implies that Row should choose q such that

$$q = \frac{9p}{9p + 1 - p}$$

Likewise, the Column player should choose Heads with a probability p that is based on the Luce ratio of her own expected payoffs from her two actions:

$$p = \frac{1 - q}{1 - q + q}$$

Under the QRE, both players must simultaneously play the mixed strategy given by their quantal response function. Solving these two equations simultaneously, we get $q = 3/4$ and $p = 1/4$. In other words, the Row player does switch to playing Heads more frequently under the QRE, as observed in the lab data, even using the simple Luce ratio as the quantal response function.

If we instead use the logit quantal response function, our equations for p and q are similar, except that they include the exponential function and the λ precision parameter:

$$q = \frac{e^{\lambda(9p)}}{e^{\lambda(9p)} + e^{\lambda(1-p)}}$$

and

$$p = \frac{e^{\lambda(1-q)}}{e^{\lambda(1-q)} + e^{\lambda(q)}}$$

To make these equations somewhat easier to manipulate, we can also rearrange terms, as described in equation (26.2), to rewrite them as

$$q = \frac{1}{1 + e^{-\lambda(10p-1)}}$$

and

$$p = \frac{1}{1 + e^{-\lambda(1-2q)}}$$

The exact quantal response functions depend on our choice of the parameter λ . The left panel of [figure 26.5](#) shows the Row player's quantal response function, given by the equation for q that we just found, for a precision value of $\lambda = 3$ (solid line), as well as the best response function (dashed line). The right panel shows the best response and quantal response functions for the Column player (given by the equation for p above, using $\lambda = 3$). Finally, the bottom panel graphs both quantal response functions (and best response functions) on the same axes.³

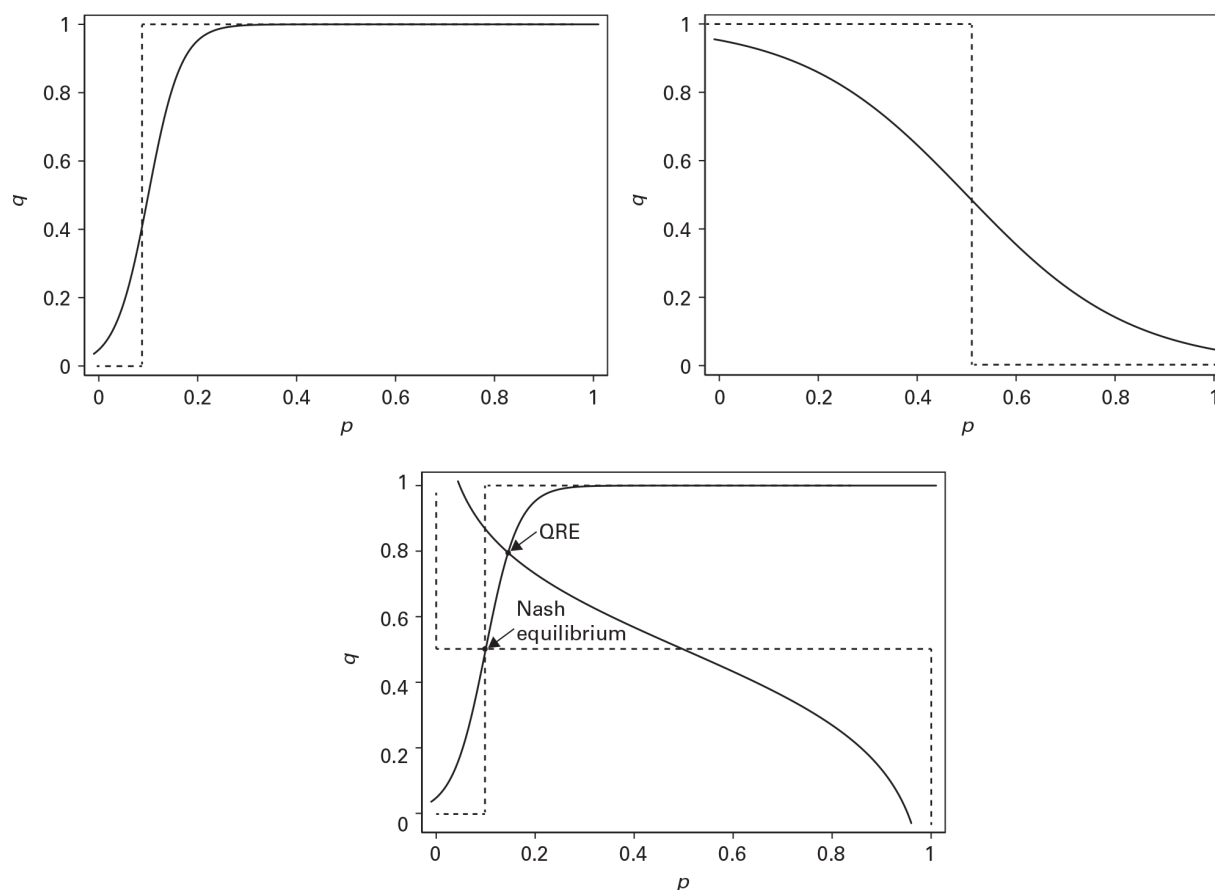


Figure 26.5

Best response functions and quantal response functions for Row (left), Column (right), and both (bottom) in matching pennies game 2 with (Heads, Heads) payoff for Row of 9.

As we can see from the bottom panel of [figure 26.5](#), the two quantal response functions intersect at $p = 0.145$ and $q = 0.795$. This is our QRE, using the logit quantal response function with $\lambda = 3$.⁴

We have thus seen that the QRE prediction has the Row player placing more weight on Heads, which is more consistent with participants' behavior in the lab. What about in the first version of the matching pennies game, where the payoff from (Heads, Heads) was 1 for the Row player? In this case, the average frequency with which experimental participants choose Heads tends to be close to the Nash equilibrium of 0.50 for both players. What does QRE predict? The left panel of [figure 26.6](#) plots the quantal response functions for the two players (again with $\lambda = 3$). Here, we see that the quantal response functions intersect at $(0.5, 0.5)$, which is also the Nash equilibrium. In fact, this is true for *any* λ that we could have chosen. Even more generally, whenever a Nash equilibrium involves all the players mixing uniformly over

their actions (i.e., playing each with equal probability), the QRE will always be the same as the Nash equilibrium, regardless of the game structure or the precision parameter. Why? Remember that in a mixed-strategy Nash equilibrium, each player chooses his mixture to make the other players indifferent. In the quantal response model, a player who is indifferent between her pure strategies always plays a mixed strategy, in which she places equal weight on each.⁵ This is true for any logit quantal response function (or any other reasonable quantal response function, as we'll discuss in the next section). In the context of matching pennies game 1, Row's choice of $q = 0.5$ makes Column indifferent between Heads or Tails, and thus Column's better response is always $p = 0.5$, and vice versa. Finally, the right panel of [figure 26.6](#) plots the quantal response functions for the two players when the payoff from (Heads, Heads) is $1/2$ for the Row player. Again, we see that the Row player responds to the changes in his own payoffs and underplays Heads relative to the Nash equilibrium of 0.5. While the Column player's best response to Row underplaying Heads is to play Heads all the time, her quantal response function causes her to underplay Heads relative to the Nash equilibrium as well.

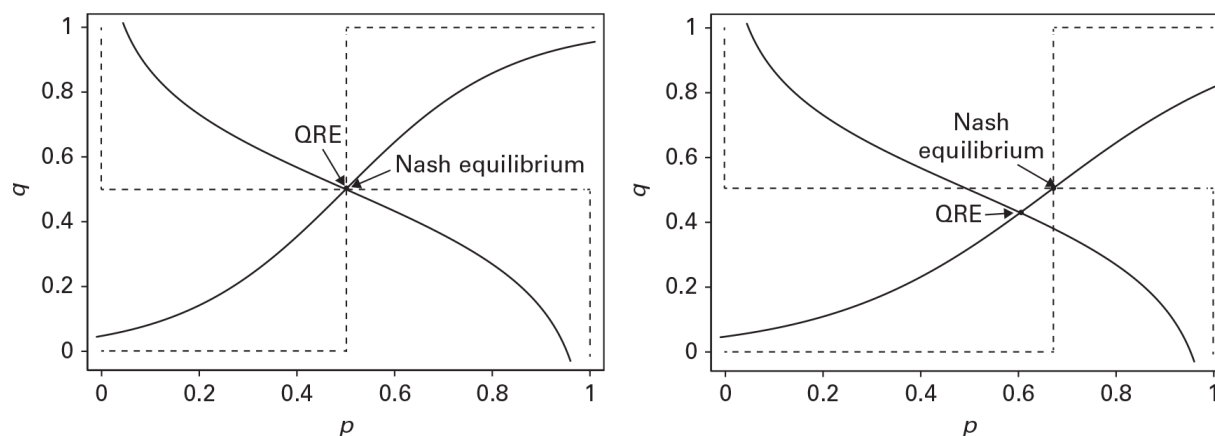


Figure 26.6

Nash equilibrium and QRE in matching pennies games 1 (left) and 3 (right).

Better Responding in Symmetric 2x2 Games

Calculating the QRE is simpler in symmetric games, in which each player faces an identical situation. In these games, there always exists a symmetric equilibrium, in which each player chooses the same strategy. We therefore need to compute only one mixed strategy (which will be the same for all

players in the symmetric equilibrium), rather than one for each player. For example, consider a game of chicken in which two players simultaneously decide whether to act Tough or to Concede. The game is presented in normal form as [figure 26.7](#). If both players are Tough, this is the worst outcome for both of them (payoff of 0). If one player is Tough while the other Concedes, that is the best outcome for the Tough player (payoff of 5), but not great for the player who Concedes (and therefore looks like a chicken, with a payoff of 2). If both players Concede, each gains a moderate payoff (payoff of 4).

		Column	
		Tough	Concede
Row	Tough	(0, 0)	(5, 2)
	Concede	(2, 5)	(4, 4)

Figure 26.7

Stay tough or concede game.

In this game, there are two pure-strategy Nash equilibria, (Tough, Concede) and (Concede, Tough). There is also a symmetric mixed-strategy Nash equilibrium in which each player is Tough with probability $1/3$. Let's find the symmetric QRE, in which each player chooses Tough with some probability p . Narrowing our search to outcomes in which players choose the same mixed strategy makes life easier, since we need only consider one variable p , rather than two as in the asymmetric matching pennies game.

We start by calculating the expected payoff for each player from each of her actions, assuming that the other player chooses Tough with probability p . For each player, $Eu(\text{Tough}) = 5(1 - p)$ and $Eu(\text{Concede}) = 2p + 4(1 - p)$. Thus, the difference in payoffs between the two actions is $Eu(\text{Tough}) - Eu(\text{Concede}) = 1 - 3p$. In the QRE, each player chooses p such that $p = \frac{1}{1 + e^{-\lambda(Eu(\text{Tough}) - Eu(\text{Concede}))}}$. Thus, we have

$$p = \frac{1}{1 + e^{-\lambda(1 - 3p)}}$$

Setting a specific value of λ , we can solve for the value of p that satisfies this equation. For example, if $\lambda = 2$, we find $p = 0.4$. As λ increases, p approaches the Nash equilibrium of $1/3$. As λ decreases to 0, p approaches $1/2$, or uniformly random play.

Structural Basis of Quantal Response Functions

You may wonder where the logit quantal response function came from. Is there any behavioral basis for using this type of specification? Are there any restrictions on the form that the quantal response function can take? And does the flexibility in the quantal response functions' specifications mean that any data can be rationalized as being consistent with the QRE model, so long as we choose a quantal response function and precision parameter that works with the data we're trying to explain?

Rather than starting with a specific form of quantal response function, the original work on QRE modeled random choice as the result of random error in how a player perceives his expected payoffs. Specifically, imagine that each player i perceives his payoff from action A as being the sum of the actual expected payoff of A and a random disturbance ϵ_{iA}/λ . Each of the player's actions has a different random error (drawn from the same distribution) added to it. The player would then choose A over B if and only if $Eu_i(A) + \epsilon_{iA}/\lambda > Eu_i(B) + \epsilon_{iB}/\lambda$, or $Eu_i(A) - Eu_i(B) > (\epsilon_{iB} - \epsilon_{iA})/\lambda$. As λ goes to infinity, the error terms become small and inconsequential, so the player always chooses the option with the higher expected payoff. As λ goes to 0, the error terms dwarf the true expected payoffs, and the player would simply choose the action whose random error term happened to be largest (which, since the error terms are drawn from the same distribution, means playing each action with equal probability).

So long as these errors are independently and identically distributed, the quantal response functions based on it will have the features of those shown in [figure 26.3](#)—that is, they will pass through the (0, 0.5) point and be nondecreasing. The exact distribution of the error terms determines the specific form of quantal response function that represents this choice. For example, if the errors are normally distributed, we would have a probit quantal response function; if they have an extreme value distribution, we would have the logit quantal response function that we've been studying.

One of the main criticisms leveled against the QRE model is that there is a free parameter, λ , which is typically estimated from the specific set of data that researchers are trying to explain. Rather than selecting λ first, and using that value to produce a single prediction, as we did earlier in this chapter, researchers use maximum likelihood techniques to search for the λ value that comes closest to fitting their data. They then typically report the QRE

predictions based on the estimated λ that fit the data best. It makes sense that this best-fitting model typically explains the data better than the Nash prediction, since the Nash equilibrium provides a specific point prediction, while the QRE has considerable leeway to match the average observed behavior.

Proponents of the QRE model have pushed back against this criticism in several ways. First, so long as the error disturbance terms are independently and identically distributed, or equivalently, some reasonable assumptions are made about the form of the quantal response function (known as *regular quantal response function*), it is not the case that *any* behavior can fall within its possible predictions by simply varying λ . For example, in the matching pennies game with Row's payoff of 9 from (Heads, Heads), all QRE fall in the range in which Row plays Heads with probability between 1/2 and 1, while Column plays Heads with a probability between 1/10 and 1/2. In the version in which Row's payoff is 0.5, all QRE fall in the range in which Row plays Heads with probability between 0 and 1/2, while Column plays Heads with probability between 1/2 and 2/3. In the game of chicken previously discussed, the QRE fall in the range of 1/3 to 1/2.

Of course, there is still a considerable range of behavior that could be consistent with the QRE model and how wide this range is depends on the structure of the game. Second, most experimental papers report several treatment conditions, and researchers estimate the λ that fits best across all of the treatments, which means that λ values are chosen to best explain the broad patterns across the treatments rather than as a means to fit the model to the specific averages in each distinct scenario separately. Researchers will also often use one set of data to estimate λ and then observe how closely the predictions match the findings in a different set of data from the same game.

In any case, it is clear that the QRE model's main power is its ability to broadly predict *how* behavior will deviate from the Nash equilibrium (often across all or most plausible λ values) in a variety of games, rather than in how well it precisely nails the relative frequencies of observed behavior by varying the free parameter.

Better Responding in the Volunteer's Dilemma

Consider again the volunteer's dilemma, first described in chapter 6. Each of n players faces a binary decision: to step up and volunteer to help someone in

need at a personal cost or to do nothing. In chapter 6, we solved for the symmetric mixed-strategy Nash equilibrium for general costs, benefits, and number of players. Here, however, we will consider specific parameters that were used in a lab experiment by Goeree, Holt, and Smith (2017), who also reported QRE predictions that were fitted to the data. Specifically, say that the benefit that each person in the group receives if someone volunteers is \$1, the cost to anyone who volunteers is \$0.20, and that the number of people in the group, n , is either 2, 3, 6, 9, or 12, which corresponds to the different treatments used in the experiment. In the model presented in chapter 6, the payoff to each person if *no one* volunteers was 0, but in this experiment, it was \$0.20.

In this game, there are always n pure-strategy Nash equilibria, each corresponding to one of the n players volunteering while the rest of the players do not. There is also a symmetric equilibrium in mixed strategies, in which each player chooses to volunteer with probability p and not volunteer with probability $1 - p$. Each player is willing to play a mixed strategy only if she is indifferent between volunteering or not, so under the Nash equilibrium, $Eu_i(\text{Volunteer}) = Eu_i(\text{Not Volunteer})$ for all players i . If player i volunteers, there will definitely be at least one volunteer, earning her \$1, but she must pay the cost of \$0.20, so we have $Eu_i(\text{Volunteer}) = 1 - 0.2 = 0.8$. If player i does not volunteer, then her payoff is either \$1 (if at least one of the other $n - 1$ players volunteers) or \$0.20 (if none of them volunteers). Since each of the other players is playing the same mixed strategy p , the probability that none of them volunteers is $(1 - p)^{n-1}$, and thus the probability that someone volunteers is $1 - (1 - p)^{n-1}$. We therefore have $Eu_i(\text{Not Volunteer}) = 1(1 - (1 - p)^{n-1}) + 0.2(1 - p)^{n-1} = 1 - 0.8(1 - p)^{n-1}$. Setting $Eu_i(\text{Volunteer}) = Eu_i(\text{Not Volunteer})$, we get

$$\begin{aligned} 0.8 &= 1 - 0.8(1 - p)^{n-1} \\ 0.8(1 - p)^{n-1} &= 0.2 \\ (1 - p)^{n-1} &= 0.25 \\ p &= 1 - 0.25^{\frac{1}{1-n}} \end{aligned}$$

Figure 26.8 plots this mixed-strategy Nash equilibrium p for each of the group sizes $n \in \{2, 3, 6, 9, 12\}$ (shown by the black line). As n increases, the likelihood of each individual personally volunteering decreases, as might be expected. The gray line plots the data from each of the group sizes in the

experiment. While there is a decline in volunteer rate with group size, it is not as dramatic as predicted by the Nash equilibrium: people undervolunteer in groups of size 2 and 3 while overvolunteering somewhat in larger groups.

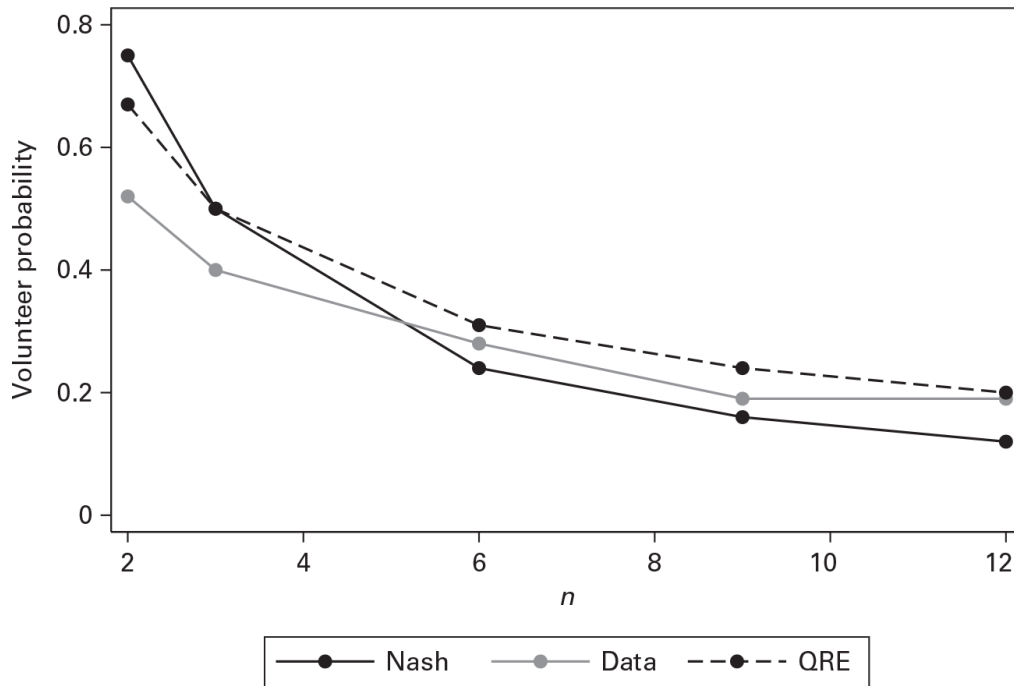


Figure 26.8

Nash equilibria, QRE predictions ($\lambda = 11$), and data in volunteer's dilemma (from Goeree et al., 2017).

The less intuitive, and perhaps more depressing, Nash prediction is that the likelihood that *no one* volunteers increases with group size: even though more potential helpers are available, the person is actually less likely to be saved. Specifically, the probability that no one volunteers, given that each of n people chooses to volunteer with probability p , is $(1 - p)^n$, or, plugging in for the equilibrium p that we computed earlier, $0.25^{\frac{n}{n-1}}$. The likelihood of this no-volunteer outcome is plotted in [figure 26.9](#) for each of the group sizes, where the likelihood of not receiving any volunteers is predicted to increase from 0.06 in the size 2 groups to 0.22 in size 12 groups. The data (gray line), however, indicate the exact opposite pattern: groups of size 2 neglect to have a volunteer in over 20% of observations, but this rate falls by half in groups of 12.

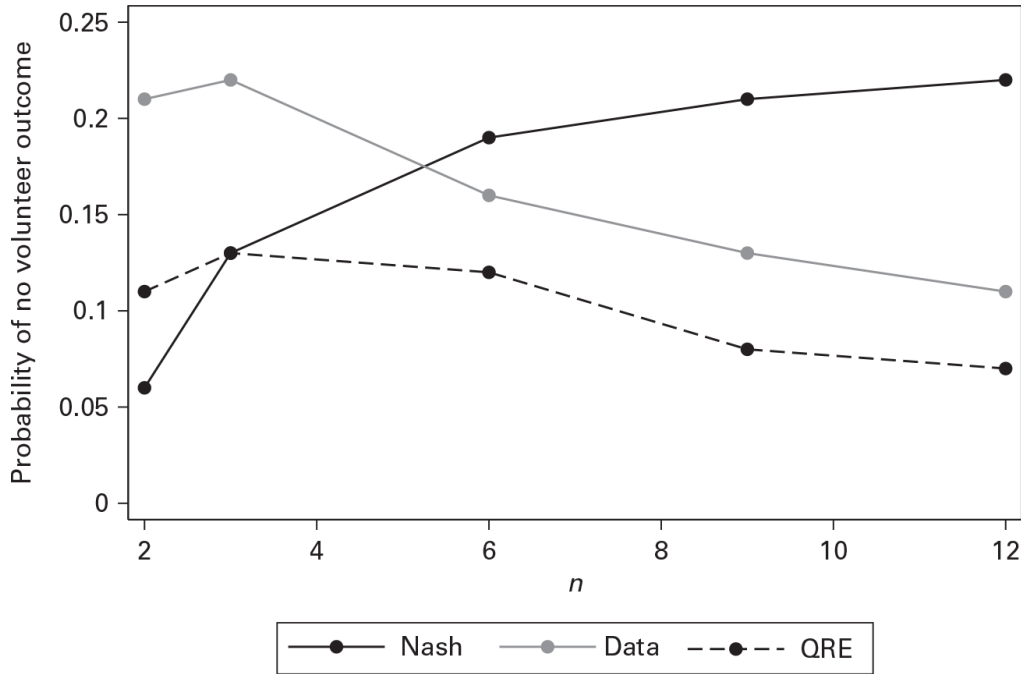


Figure 26.9

The probability that no one volunteers according to the Nash equilibrium predictions, QRE predictions ($\lambda = 11$), and data in volunteer's dilemma (from Goeree et al., 2017).

Now let's find the QRE predictions. In doing so, we will use the λ value of 11 that Goeree et al. (2017) found to best fit their data using the logit quantal response function. Under this QRE, each player is choosing volunteer with probability p such that

$$p = \frac{e^{11(Eu(\text{Volunteer}))}}{e^{11(Eu(\text{Volunteer}))} + e^{11(Eu(\text{NoVolunteer}))}}$$

Using the expected payoffs of volunteering and not volunteering that we already calculated, we have

$$p = \frac{e^{11(0.8)}}{e^{11(0.8)} + e^{11(1-0.8(1-p)^{n-1})}}$$

For each specific n , we can solve this expression numerically for p , for instance by plugging both sides into a graphing calculator and finding the intersection points of the two curves.⁶ This gives us the QRE prediction for each value of n , given the precision parameter $\lambda = 11$. These predictions are graphed as the dashed line in figure 26.8. While the QRE predictions also do not perfectly match the data under all the group size treatments, they are consistent with the finding that people undervolunteer in pairs but

overvolunteer in groups of size 6 and above. Thus, consistent with the data, QRE predicts that players will be less responsive to group size than predicted by the Nash equilibrium. Notice that the Nash prediction and the QRE perfectly overlap at 0.5 when $n = 3$. This is for the same reason that we discussed earlier in this chapter in the matching pennies game, when the Nash equilibrium involves all players playing each pure strategy with equal probability: when a player is indifferent between his two pure strategies (as he will be whenever the other players play the mixed-strategy Nash equilibrium), the QRE predicts that he will play both with equal probability.

Figure 26.9 also plots the QRE predictions for the likelihood that no one volunteers (dashed line), based on the predictions for p . As we saw, the Nash equilibrium predicts that the no-volunteer outcome will be more likely as the group size increases, while the data exhibited the exact opposite pattern. However, the QRE substantially underpredicts the likelihood of the no-volunteer outcome, especially for small groups. The QRE predictions better match the overall trend in the data, in that the no-volunteer outcome generally declines with group size.⁷

Extension to Sequential Games: Agent Quantal Response Equilibrium

The QRE can also be used to predict behavior in games in which players act sequentially. This framework is called the *agent quantal response equilibrium* (AQRE) and was developed by McKelvey and Palfrey (1998). The basic idea is that at each information set, a player is represented by a different agent who has the same payoffs as the player. The agent acts on the player's behalf by better responding at that information set, given how they anticipate the rest of the game will play out. We assume that a different agent implements a choice on behalf of the player at each of his nodes. Each player's agents act independently and cannot coordinate their actions, nor do they have access to private information held by other agents representing the same player. For instance, see the centipede game first discussed in chapter 8 and illustrated as a four-step game in figure 26.10, in which players 1 and 2 alternate making a decision of "Take" or "Pass." Each player has two decision nodes, so each is represented by two distinct agents. Agents 1 and 3 represent player 1 at the first and third nodes, while agents 2 and 4 represent player 2 at the second and fourth nodes.

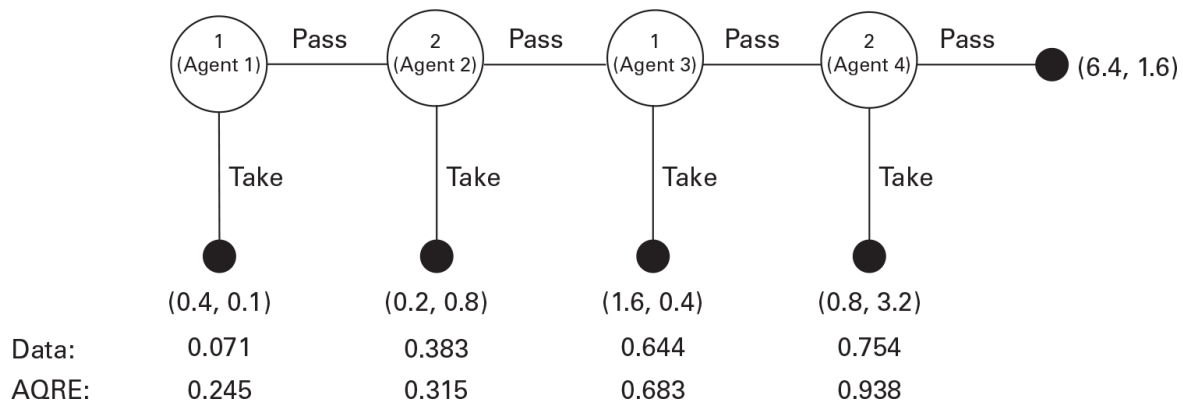


Figure 26.10

The centipede game with actual take rates in McKelvey and Palfrey (1992) and AQRE-predicted Take rates.

In games of perfect information, we can use backward induction to solve for the AQRE. At the final decision node of the game, the payoffs from each action can be computed by the agent. He then implements the mixed strategy given by his quantal response function. Then, we back up a step. The agent at the previous node has accurate beliefs about the strategy played at the final node, and she calculates the expected payoff from each action based on the mixed strategy played by the final agent. She then implements the mixed strategy given by her quantal response function. This process continues until we have calculated the AQRE mixed strategies of the agents at each decision node.

One benefit of the AQRE framework compared to the Nash equilibrium is that there are no information sets that are off the equilibrium path. Since every action is chosen with some positive probability, any information set can be reached in equilibrium. This means that we do not need additional requirements to rule out irrational play off the equilibrium path (as in chapter 8), and we also don't need to worry about Bayes' rule not applying in games of incomplete information (as we did in chapter 12).

Centipede Game

As an illustration of how to compute the AQRE, consider again the centipede game, first discussed in chapter 8. Imagine that there are two players and a tray with two piles of money, which they can pass back and forth between them. Whenever a player has the tray, he can either Take the larger pile of cash, which ends the game and leaves the other player with the smaller pile, or Pass the tray back to the other player, in which case the amount in each of

the piles is increased. In this perfect information game, players 1 and 2 alternate making decisions, and the piles of money initially on the tray are 40 cents and 10 cents. If player 2 passes in the final node, the game ends with player 1 receiving the larger payoff share.

The unique subgame perfect Nash equilibrium (SPNE) in this game is for each player to choose Take whenever it is his turn. [Figure 26.10](#) also shows the probability that each player Takes at each step in a lab experiment conducted by McKelvey and Palfrey (1992). While the SPNE predicts that player 1 always Takes in her first opportunity, only 7% of participants actually do. As the game goes on, players become increasingly likely to Take when given the opportunity.

Let's calculate the AQRE using the logit quantal response function with $\lambda = 1.7$, which McKelvey and Palfrey (1998) estimate fits the data best. Since there are four decision nodes, we can think about four agents, who each play a mixed strategy at their node. The agent's probability of Taking at a given node is computed using the logit quantal response function, based on the payoff from Taking and the expected payoff of allowing the game to continue, given the mixed strategies played by the subsequent agents.

To begin, we calculate the payoffs for agent 4 in the final node of the game. At this point, the agent can Take, for a payoff of 3.20, or Pass, for a payoff of 1.60. (Remember that the agent is representing player 2, so we use 2's payoffs.) So the AQRE probability of Take is given by

$$p_{Agent4,Take} = \frac{e^{1.7(3.20)}}{e^{1.7(3.20)} + e^{1.7(1.60)}} = 0.938$$

This AQRE prediction at the fourth node is listed in [figure 26.10](#). Next, we consider agent 3's AQRE strategy at the third node of the game. At this point, agent 3 can Take, for a payoff of 1.60, or Pass. The payoff from Pass depends on whether agent 4 Takes (payoff = 0.80) or Passes (payoff = 6.40). Given the probability of taking that we just computed for agent 4 (0.938), the expected payoff of agent 3 passing is thus $Eu_{Agent3}(Pass) = 0.938 \times 0.80 + (1 - 0.938) \times 6.40 = 1.147$. Plugging in the expected payoffs from each of agent 3's actions into the logit quantal response function, we find that the AQRE probability of Take at this node is

$$p_{Agent3,Take} = \frac{e^{1.7(1.60)}}{e^{1.7(1.60)} + e^{1.7(1.147)}} = 0.683$$

Continuing this process, we find that $Eu_{Agent2}(Take) = 0.80$ and $Eu_{Agent2}(Pass) = 0.683 \times 0.40 + (1 - 0.683) \times 0.938 \times 3.20 + (1 - 0.683) \times (1 - 0.938) \times 1.60 = 1.256$. Notice that agent 2 treats the mixed strategy of agent 4 as an independent probability (just as she treats the choice probability of agent 3), despite the fact that agents 2 and 4 are representing the same player. In other words, agent 2 cannot predict the actual action that agent 4 will take but only correctly anticipates her mixed strategy. We thus have $p_{Agent2, Take} = \frac{e^{1.7(0.80)}}{e^{1.7(0.80)} + e^{1.7(1.256)}} = 0.315$. Finally, $Eu_{Agent1}(Take) = 0.40$ and $Eu_{Agent1}(Pass) = 0.315 \times 0.2 + (1 - 0.315) \times 0.683 \times 1.6 + (1 - 0.315) \times (1 - 0.683) \times 0.938 \times 0.8 + (1 - 0.315) \times (1 - 0.683) \times (1 - 0.938) \times 6.4 = 1.061$. Thus, $p_{Agent1, Take} = \frac{e^{1.7(0.40)}}{e^{1.7(0.40)} + e^{1.7(1.061)}} = 0.245$.

The AQRE predictions that we just calculated are listed in [figure 26.10](#) below the actual Take frequencies observed in the data. While the AQRE model overpredicts the likelihood of Taking at the first and last nodes, it otherwise matches the general patterns in the data. First, it predicts that the likelihood of Taking in the early nodes is dramatically below the SPNE probability of 1—in fact, people are more likely to Pass than Take in these early periods. Notice that it makes complete sense for agents 1 and 2 not to want to Take very often, given how the rest of the game is anticipated to play out. At both of the first two nodes, we calculated that the expected payoff from Pass is actually higher than Take. A best responder would therefore always choose Pass at these nodes. Given the mechanics of the quantal response function (which puts more probability weight on better actions than worse ones), we simply predict players to better respond by choosing Pass at a higher rate than Take. In other words, it is not a mistake for agents 1 and 2 to Pass at these initial nodes, given the mistakes that agents 3 and 4 are subsequently expected to make; rather, the mistake that agents 1 and 2 make is ever choosing Take at all!

Second, the AQRE model predicts the increasing likelihood of Taking as the game proceeds. This is due to two factors. The stake size is increasing (in fact, doubling) on every turn, which makes the costs of mistakes higher and therefore reduces the likelihood that players will choose inferior options. Even keeping the precision parameter λ constant, people are better able to perceive these bigger payoff differences in their two actions and are thus more likely to play the action with the higher expected payoff. In addition, as the likelihood that subsequent players will mistakenly Pass diminishes, it is more

lucrative to choose to Take as well. As in the volunteer's dilemma and many other games, the fit of the QRE predictions can be improved if we also allow for heterogeneity in the players' λ values or preferences. While this is beyond the scope of our QRE discussion, chapter 27 will consider models in which players use different degrees of reasoning.

Exercises

(1) Consider the following prisoner's dilemma game:

		Column	
		Cooperate	Defect
Row	Cooperate	(3, 3)	(0, 5)
	Defect	(5, 0)	(1, 1)

Figure 26.11

Find the probability that the players choose Cooperate in the symmetric quantal response equilibrium, using the logit quantal response function, for each of the following λ values: 2, 1, and 0.5. Hint: Follow the example in the symmetric chicken game. That is, using the logit quantal response function, write the equation for the probability, p , that Row will choose Cooperate, assuming that the Column player is also playing Cooperate with probability p . Then solve for the symmetric p that each player chooses.

(2) Consider the game shown here between Joel and Midge, in which both players are simultaneously deciding what to do tonight. Each has a more preferred activity and a less preferred activity, and they disagree about which activity is better. However, what is most important is that they match each other's activity, and they each receive a payoff of 0 if they mismatch the other's activity (i.e., if both players pursue either their favored activity or their disfavored activity, rather than one compromising while the other gets his or her way).

		Midge	
		Favored	Disfavored
Joel	Favored	(0, 0)	(4, 1)
	Disfavored	(1, 4)	(0, 0)

Figure 26.12

- (a) Find the pure-strategy Nash equilibria of this game, as well as the symmetric mixed-strategy Nash equilibrium.
- (b) Find the symmetric QRE (using the logit quantal response function) for each of the following values of λ : 5, 1, and 0.1.
- (3) Consider the following game, in which two team members simultaneously decide how much effort to put into their work. Effort is costly, and it pays off only if both members work hard:

		Column	
		Work	Shirk
Row	Work	(12, 12)	(2, 6)
	Shirk	(6, 2)	(6, 6)

Figure 26.13

- (a) Find all the symmetric (pure- and mixed-strategy) Nash equilibria of this game.
- Since there are multiple symmetric Nash equilibria in this game, there may be multiple symmetric QREs. However, as we will see, the number of symmetric QREs will depend on the precision parameter.
- (b) Find all of the symmetric quantal response equilibria when $\lambda = 1$.
- (c) Find all of the symmetric quantal response equilibria when $\lambda = 0.1$.
- (4) The El Farol is a bar in Santa Fe, New Mexico, where everyone wants to hang out on Thursday nights. If it becomes too crowded, however, then it is no fun for anyone, so everyone would prefer to avoid the bar once it exceeds a certain capacity. This situation is known as the *El Farol bar problem*. Imagine a game with three players, each simultaneously deciding whether to head to the bar. If one or two of the three show up, then those players will have a

good time and earn a payoff of 1. Three, however, would be a crowd, and all three would earn a payoff of 0. Finally, any player who avoids the bar gets a payoff of 0.51, regardless of what the others choose to do.

- (a) Find the symmetric Nash equilibrium of this game.
- (b) Find the symmetric QRE when $\lambda = 5$.
- (c) Now imagine that the payoff from avoiding the bar were 0.91. In this case, would the QRE predict overattendance or underattendance relative to the Nash equilibrium?

(5) Consider again the credible threat game, discussed in chapter 8 and shown here. The unique SPNE is (Risky, Credible). However, if the first player suspects that the second might not play her best response, the first player may wish to take the safer payoff of 9.75. Find the AQRE for $\lambda = 1$.

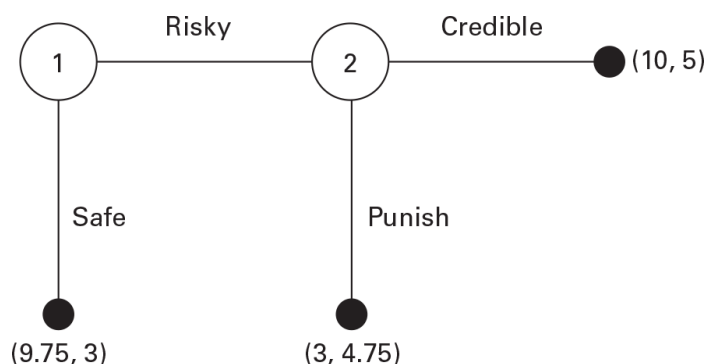


Figure 26.14

(6) Consider again the clean your room game, discussed in chapter 8 and shown here. The unique SPNE is (Dirty, (Live, Live)). Find the AQRE when $\lambda = 0.01$.

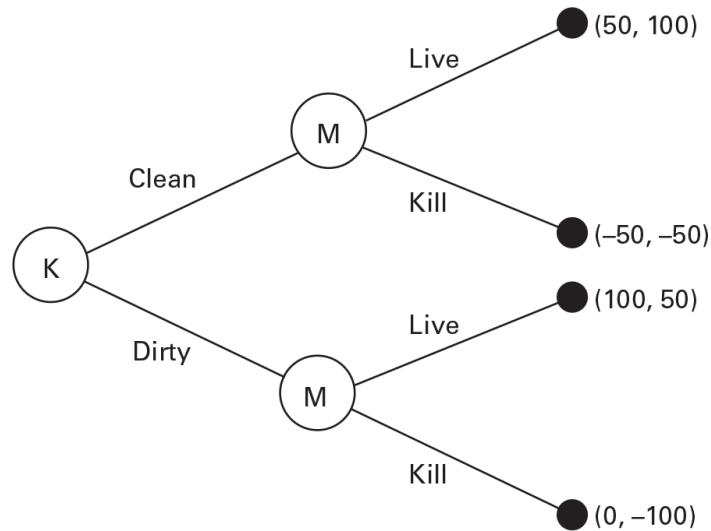


Figure 26.15

(7) Consider a variant of the centipede game used by Fey, McKelvey, and Palfrey (1996), in which the total payoffs do not change every time a player chooses Pass. Instead, the two payoff piles become more lopsided, such that the difference in payoffs for the two players increases as the game wears on. A four-step version is shown here:

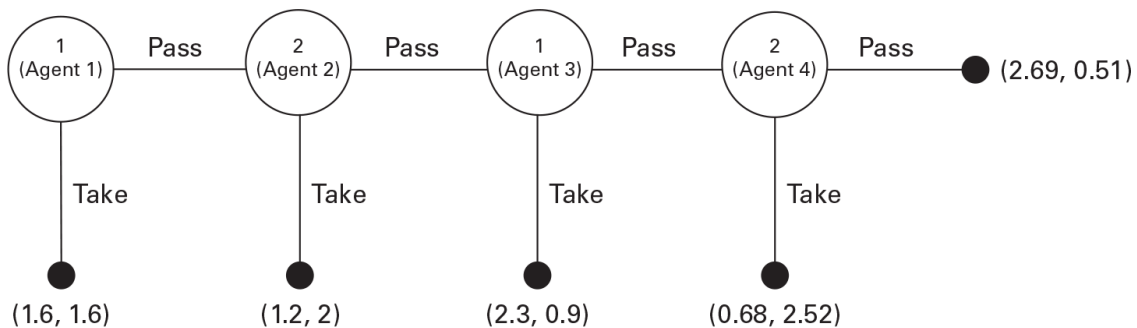


Figure 26.16

- Find the SPNE.
- Find the AQRE of this game, assuming that $\lambda = 1$.

¹Why? Intuitively, it seems that Row would be more attracted to Heads in this version. However, if Row were to play Heads with probability greater than 0.5, then Column would no longer be indifferent and would best respond by playing Tails all the time, in which case Row's best response would be playing Tails with probability 1, in which case Column's best response would be to play Heads, and so

on. Our precarious mixed-strategy equilibrium thus requires that Row continue to play Heads with probability 0.5 such that Column's best response remains a mixed strategy.

²The QRE model was originally developed in McKelvey and Palfrey (1995) and Goeree, Holt, and Palfrey (2016) provides an excellent in depth treatment.

³Note that, in order to graph Column's quantal response p on the same axes (where p is on the horizontal rather than the vertical axis), we need to rearrange the equation for p above such that q is isolated. Doing so, we have $p = \frac{1}{1+e^{-3(1-2q)}}$, or $p + pe^{-3+6q} = 1$, or $e^{-3+6q} = \frac{1-p}{p}$, or $-3 + 6q = \ln\left(\frac{1-p}{p}\right)$, or $q = \frac{\ln\left(\frac{1-p}{p}\right)+3}{6}$.

⁴The simplest way to arrive at this numerical solution is to choose a λ value, find the equations for p and q as we did above, and then use a graphing calculator (or mathematical software) to solve them simultaneously. This can be done by either plotting p and q on the same set of axes, as in the bottom panel of [figure 26.5](#), and finding the intersection, or by plugging the equation for p into the equation for q (or vice-versa). The latter approach would give you q on one side and a messy function of q (i.e., $\frac{1}{1+e^{\left(\frac{-30}{1+e^{6q-3}}+3\right)}}$) on the other, and you can then plug both into your graphing calculator to find the intersection.

⁵In other words, if other players are mixing uniformly, then player i must be indifferent between his actions (based purely on our initial assumption that all players uniformly mixing is a Nash equilibrium). And if a player is indifferent between his strategies, his quantal response function has him mix uniformly.

⁶For instance, to find the predicted p for $n = 3$, we would put in the equation $y = x$ and $y = \frac{e^{11(0.8)}}{e^{11(0.8)} + e^{11(1-0.8(1-x)^3-1)}}$. We find that these two curves intersect when $x = 0.5$, giving us our p value for group size $n = 3$.

⁷Goeree et al. (2017) showed that the fit between the data and the QRE model can be improved by allowing individual-level heterogeneity, although that is beyond the scope of this illustration.

Level- k Reasoning

In this chapter, we will consider a model of decision making in which players' strategic reasoning ability is limited. The players in the population differ in their level of reasoning about the actions of others. A level-0 player does not think about the incentives of the game or form beliefs about the other players' strategies. If there is an action that is particularly salient or focal, the level-0 player chooses that; otherwise, he randomizes uniformly over his actions. Higher-level players act as if they believe they are capable of using one more step of reasoning than everyone else and best respond to those beliefs: so a level-1 player's strategy is a best response to the level-0 strategy, a level-2 player's strategy is a best response to a level-1 player's strategy, and so on. Across a variety of games, many players act as if they use one or two (or sometimes three) steps of reasoning under this model.

Imagine that you are playing a game in which you can request money. You and one other player will simultaneously make a request for a dollar amount between \$11 and \$20. You can request only whole-dollar amounts. You will each get the amount that you request. In addition, if one player requests *exactly* \$1 less than the other, that person will get a bonus of \$20. How much money do you request?

Level- k Reasoning

In deciding how much money to request, perhaps you elected not to bother being strategic at all. Instead of predicting your opponent's request and attempting to undercut her, you ignored the other player entirely and simply asked for the maximum of \$20. In this chapter, we will call this type of nonstrategic play *level-0 reasoning*, since it is not based on beliefs about the other player's strategy. Or perhaps you did think about what the other player was likely to do, but you naively assumed that your opponent wouldn't and that she would simply ask for the maximum of \$20. Therefore, you concluded, your best response would be \$19. We'll call this best response to level-0 play *level-1 reasoning*. Or maybe you took it one

step further, and you assumed that your opponent would assume that you were naively choosing \$20, causing her to choose \$19, and conclude that your best response would be \$18. We'll call this *level-2 reasoning*. As we'll see in this chapter, behavior in a wide variety of games can be explained by assuming that a large proportion of players use level-1 and level-2 reasoning. Or you could take it a step further still. Perhaps you assumed that your opponent assumed that you were assuming that she would choose \$20; since you believe that she believes that you would choose \$19 to undercut her, you believe that she would choose \$18, and your best response thus would be \$17. While some players' behavior seems consistent with this *level-3 reasoning*, it is less common, and there is little evidence of higher-level reasoning than this.

This type of model of strategic thinking, called *level- k reasoning*, was introduced in experimental papers by Nagel (1995) and Stahl and Wilson (1994, 1995), which will be discussed in more detail in the subsequent sections. Specifically, each player is associated with a type, $k \in \{0, \dots, \infty\}$, that refers to how many steps of reasoning the player uses. Note that the type in level- k models is different from the type in Bayesian games: it refers to the reasoning ability of players rather than their incentives. Some players are willing and able to employ more steps of reasoning than others, and thus they are associated with higher- k types. The level- k model is anchored on the behavior of a strategically oblivious player who does not attempt to anticipate the actions or thought processes of others—that is, level-0 play. Although this type of player is rarely observed in lab experiments, the reasoning and choices of all other players are based upon it, and so a careful specification of level-0 behavior is essential to the model.

Most researchers assume that level-0 play simply involves the player choosing one of his pure strategies at random, each with equal probability (equivalent to a quantal response with zero precision discussed in chapter 26). In some games, however, there is a particularly obvious or focal strategy for a nonstrategic player to follow, such as making the maximum request in the 11–20 money request game, or always sending truthful messages and believing any messages received in cheap talk games. In these cases, researchers assume that level-0 play involves following these obvious strategies. For $k > 0$, the player's strategy is the best response to the strategy of a $k - 1$ player. That is, a level-1 type best responds to level-0

play, a level-2 type best responds to that level-1 strategy, and so on. The play of higher-level players is strategic, in the sense that they attempt to predict the other players' reasoning and strategies and then best respond to these beliefs.¹ However, it still may differ from equilibrium play, since their beliefs may not be correct. Instead, everyone assumes that everyone else's depth of reasoning is one level more simplistic than their own.

Level- k Reasoning and Nash Equilibrium

For some fairly simplistic games, low levels of reasoning are sufficient to reach equilibrium, while in other games, equilibrium will be reached only if all the players use high levels of reasoning, or it will never be reached no matter how great the k is of all the players. For instance, consider a game in which the Nash equilibrium is in dominant strategies, such as the prisoner's dilemma. A level-1 player would never play a dominated strategy such as Confess, since it is never a best response to anything (including level-0 play). Thus, if all players were level-1 or above, each would choose her dominant strategy, and the outcome would be the same as the Nash equilibrium. Likewise, equilibrium would be reached in a game that is dominance solvable in two steps if all players were level-2 or higher, since each believes the others to be level-1 and thus eliminates the possibility that the other players would choose a dominated strategy.

Consider the game in [figure 27.1](#). We saw in chapter 4 that the unique Nash equilibrium of this game is (X, B). Neither player has a dominant strategy. What level of reasoning is required of the players in order to reach the Nash equilibrium? First, let's start with level-0 play for each player: if they were level-0, both Sid and Nancy would play a mixed strategy, in which each pure strategy receives equal weight of $1/3$. Now consider the level-1 strategy for Sid. What is his best response if Nancy plays each of her actions equally? Sid's expected payoff from X is $Eu_{Sid}(X) = 1/3(1) + 1/3(3) + 1/3(5) = 3$. Likewise, $Eu_{Sid}(Y) = 3.67$ and $Eu_{Sid}(Z) = 4.67$. Thus, a level-1 Sid would best respond to level-0 Nancy by choosing Z . Doing the same calculations for Nancy, we find that level-1 Nancy would best respond to level-0 Sid by choosing B . Thus, if both players were level-1 types, Nancy would play according to her equilibrium strategy, but Sid would not, and the Nash equilibrium would not be reached. What about level-2 behavior? Level-2 Sid best responds to level-1 Nancy, who we just

calculated would play B. His best response is thus X. Level-2 Nancy best responds to level-1 Sid, who would be playing Z, and thus she would choose B. Thus, if Nancy is level-1 or higher and Sid is level-2 or higher, the Nash equilibrium outcome would be reached in this game.

		Nancy		
		A	B	C
Sid	X	(1, 1)	(<u>3</u> , <u>3</u>)	(5, 0)
	Y	(5, <u>4</u>)	(2, 0)	(4, 3)
	Z	(<u>7</u> , 0)	(1, <u>8</u>)	(<u>6</u> , 1)

Figure 27.1

Both players' best responses.

The 11–20 Money Request Game

In level- k models, assumptions about the level-0 strategy and the distribution of types are all that is needed to make predictions about the percentage of players choosing each strategy. The 11–20 money request game was designed by Arad and Rubinstein (2012) specifically to estimate the distribution of types in a situation in which level- k reasoning seems particularly natural. The game has a unique symmetric equilibrium in mixed strategies, which is shown in [figure 27.2](#). However, the mixed-strategy equilibrium is not intuitive, and laboratory participants deviate from the equilibrium distribution significantly.

Action	11	12	13	14	15	16	17	18	19	20
Nash equilibrium (%)	0	0	0	0	25	25	20	15	10	5
Results (%)	4	0	3	6	1	6	32	30	12	6

Figure 27.2

Nash equilibrium and experimental results of 11–20 request game (Arad and Rubinstein 2012).

As described earlier, the maximum request of \$20 is an obvious candidate for the level-0 strategy. However, one feature of this game is that \$19 is still the level-1 strategy (and thus \$18 the level-2 strategy, and so on) even if uniform random play is taken as the level-0 strategy. To see this,

notice that if your opponent is playing each of the 10 possible actions with probability $1/10$, then you have a $1/10$ chance of winning the \$20 bonus with requests 11–19—that is, the expected value of making any request $i \in \{11, \dots, 19\}$ is $i + 2$. Therefore, the best response to uniform random behavior is requesting \$19, associated with an expected payoff of \$21. Note also that the best response to a level-9 type choosing 11 is to choose 20 (you can't undercut them and win the bonus, so you might as well request the highest amount), and so once players reason their way down to 11, the cycle starts again at 20. Thus a request of \$20 is not only the choice of a level-0 player, but also of a level-10 player, level-20 player, level-30 player, and so on.

The experimental results in Arad and Rubinstein (2012)'s 11–20 money request game are listed in [figure 27.2](#) below the Nash equilibrium strategy, and they show that roughly three-quarters of participants choose 17, 18, or 19 (significantly different from the Nash equilibrium, which predicts that only 45% would make requests in this range). Furthermore, nearly every subject making one of these requests described their choice as the outcome of level-1, -2, or -3 reasoning in a post-experiment survey. For instance, a typical explanation for a request of 18 was “I request 18 ... since many people believe that the majority will request 20 and thus they will request 19” (Arad and Rubinstein 2012, p. 3566). In contrast, participants making lower requests did not explain their decision as resulting from level- k reasoning, but rather as guesses. Arad and Rubinstein conclude that the model that best fits their data involves participants of types 0 through 3, with noise. In fact, their estimated frequencies are 0.05, 0.13, 0.39, and 0.43 for these types, respectively. Other papers that we will see typically find that most participants' behavior is most consistent with level-1 or level-2 reasoning, and level-3 is fairly rare (and higher levels nonexistent). The higher frequency of level-3 play in this experiment may be due the subject pool or the 11–20 money request game, which was specifically designed to trigger level- k thinking.

Level- k Reasoning in Matrix Games

One of the first level- k studies was conducted by Stahl and Wilson (1995). They had participants play a series of 12 two-player symmetric normal-

form games, in which each player had three pure strategies. The games differed in whether they were dominance solvable, and whether there was a mixed-strategy Nash equilibrium or unique symmetric pure-strategy Nash equilibrium. The games were further designed to separate different decision rules, such that, for instance, level-1 behavior, level-2 behavior, and the symmetric Nash equilibrium each involved choosing different actions. They estimated that most players' behavior was well described as either level-0, level-1, level-2, Nash, or a worldly type who best responded to a mix of level- k and Nash behavior.

But just because a player's decisions are generally consistent with a particular level of reasoning, how do we know that his decision is actually a result of a level- k thought process? Costa-Gomes, Crawford, and Broseta (2001) designed an experiment in which they not only measured subjects' decisions, but also how they interacted with the payoff matrix before making those decisions. They had participants play 18 normal-form games with unique Nash equilibria, in which different types would play different sequences of actions across all the games. For instance, consider the two games in [figure 27.3](#), which depict their games 5B and 7B. A Row player playing according to the Nash equilibrium would choose M in the first game and Y in the second but would choose B and then X if he were a level-1 player (and assumed that level-0 played randomly), and B and then Y if he were a level-2 player. Likewise, Column would choose (R, C) if she were playing according to Nash, (L, A) if she were level-1, and (R, A) if she were level-2. By having participants play 18 such games, without feedback, they were able to determine whether their play was consistent with one of nine decision rules (which included level-1, level-2, and equilibrium play, along with several other plausible decision rules, such as minimax, maximizing joint earnings, and assuming that one's partner randomizes over his remaining actions after deleting dominated strategies).

		Column				Column		
		L	R	Row		A	B	C
Row	T	(76, 93)	(25, 12)		X	(67, 91)	(95, 64)	(31, 35)
	M	(43, 40)	(74, 62)		Y	(89, 49)	(23, 53)	(56, 78)
	B	(94, 16)	(59, 37)					

Figure 27.3

Games with unique Nash equilibria (Costa-Gomes et al., 2001).

To further assess whether a participant's thought process is consistent with a particular type, they used a software called MouseLab, which allows researchers to measure exactly what information a participant is looking at, when, and for how long. Specifically, the payoff information for both players is initially covered, like the cards at the start of a game of memory. The player can uncover any payoff at any time by clicking on the relevant cell. Before revealing another entry in the payoff matrix, the player had to cover up what he was looking at previously, so that he could see only one entry in the matrix at a time. Therefore, the researchers could determine exactly what payoffs the players based their decision on and the order in which they searched for information. For instance, a level-1 player need only look at her own payoffs, moving across the matrix to calculate her expected payoff from each of her own strategies. A level-2 player must first consider only the other player's payoffs, moving across the matrix to calculate his expected payoff from each of his strategies and determine his level-1 strategy, and then she must inspect her own payoffs only in the row or column associated with his level-1 strategy. For example, a level-2 Row player in the game on the right would search the matrix by first uncovering her opponent's payoffs from choosing A (91 and 49), then B (64 and 53), and then C (35 and 78). After determining that A is his level-1 strategy, she would look at her payoffs in the A column (67 and 89), and finally she would select Y. The researchers' analysis of both the decision and search data indicates that about 90% of the population is equally split between level-1 and level-2 types (with the next most common type following a decision rule that involves eliminating the other player's dominated

strategies and then best responding to the assumption that the other player mixes uniformly across his remaining strategies).

Level- k Reasoning in Guessing Games

The Beauty Contest

Consider again the beauty contest guessing game discussed in chapters 1 and 3: several players simultaneously submit guesses between 0 and 100, and the person whose guess is closest to the average guess times some target, p , wins a prize. Rosemarie Nagel (1995)'s beauty contest experiment is a foundational study in the level- k literature, as she showed that players' guesses tend to cluster around the level-1 and level-2 predictions. Nagel's study used groups of 15 to 18 subjects and p target values of $1/2$, $2/3$, and $3/4$. (Recall that the version of this game we discussed previously used $p = 2/3$ —that is, everyone is attempting to guess two-thirds of the average.) When $p < 1$, there is a unique Nash equilibrium of 0. When $p > 1$, everyone choosing 0 and everyone choosing 100 are both equilibria. (When $p = 1$, everyone simply wants to guess the average, and any outcome in which everyone makes the same guess is an equilibrium.)

Notice also that when $p < 1$, iterated elimination of dominated strategies leads to the Nash equilibrium of 0, albeit in an infinite number of steps. Consider the game when $p = 1/2$. The highest that half the average could be is 50, since guesses are bounded above at 100, and so guesses greater than 50 are weakly dominated by 50. But once guesses above 50 are eliminated, guesses above 25 are dominated, and once they are eliminated, guesses above 12.5 are dominated, and so on. When $p > 1$, however, there are no dominated strategies. We cannot use analogous reasoning to eliminate guesses at the low end of the interval, since every guess between 0 and 100 is the best response to some feasible average guess.

The level- k predictions in this game are straightforward. A level-0 player guesses randomly or simply chooses a number that he likes. A level-1 player therefore believes that her opponents randomize uniformly over the $[0, 100]$ interval and that the average guess will be 50, so their best response is $50p$. A level-2 player assumes that his opponents will all choose $50p$, and thus best responds with $50p^2$. In general, a level- k player in this game with target p will choose $50p^k$. One advantage of using the beauty

contest game to detect level- k behavior is that there is a wide strategy space that allows for distinguishing between types using different decision rules. In the matrix games described previously, often level-1 and level-2 behavior involved choosing the same action. There is no such overlap in the beauty contest game, where level-1 behavior (e.g., 33 when $p = 2/3$) differs substantially from level-2 behavior (22), which is very different from equilibrium behavior (0).

In Nagel's experiment, very few people played according to the equilibrium (in fact, no one chose the equilibrium guess of 0 in the $p = 2/3$ and $p = 1/2$ games). However, guesses tended to be in the neighborhood of $50p$ and $50p^2$, suggesting that many players were employing one or two levels of reasoning and either making minor errors or adjusting their guess slightly upward or downward. This pattern is even more prominent in a follow-up study in which the participants were newspaper readers who competed for prizes in the hundreds of dollars. The contests were announced in the *Financial Times* (UK), *Expansión* (Spain), and *Spektrum der Wissenschaft* (Germany), each with $p = 2/3$, and the number of respondents ranged from around 1,500 to 3,700. Since participants could submit their guess at their convenience, they were able to spend more time thinking about their strategy than the typical lab participant and could potentially consult with friends and family about how to approach the game. The results from each of the three newspapers were reported in Bosch-Domènech et al. (2002) and shown in [figure 27.4](#). In all three experiments, there are clear spikes at level-1 (33), level-2 (22), and equilibrium (0).

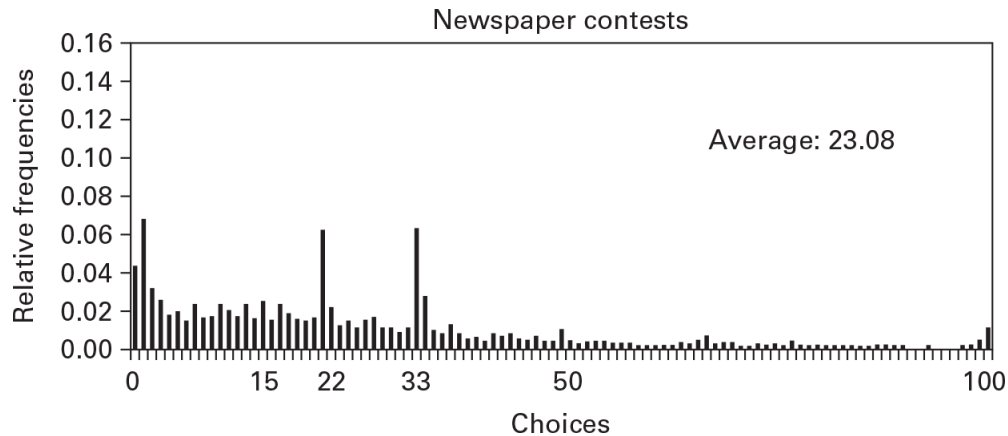


Figure 27.4

Beauty contest results from three newspaper experiments showing clear spikes at level-1 (33), level-2 (22), and equilibrium (0) (Bosch-Domènech et al. 2002).

The beauty contest game provides a strong indication that many people use level-1 and level-2 reasoning. However, the results could also be consistent with using a slightly different decision rule, in which participants perform an abbreviated process of iterated elimination of dominated strategies. In particular, consider a rule called D_k , in which a player performs k steps of iterated elimination of dominated strategies, and then best responds to the other players randomizing uniformly over their remaining strategies. In this game, a D_k player would make the same choice as a level- $k + 1$ player. For example, a player who performed one step of deletions in the $2/3$ game would eliminate all strategies greater than 67, assume that the average guess would thus be 33, and best respond by choosing 22, which would look the same as someone playing the level-2 strategy. Likewise, someone who performed two steps of deletions (eliminating all strategies greater than 44) and then best responded to the average remaining strategy would make the same choice as a level-3 player, and so on. Thus, these spikes could also be explained by players performing zero, one, or two steps of iterated elimination of dominated strategies and then best responding to the average of the remaining strategies.

Other Guessing Games

Costa-Gomes and Crawford (2006) designed two-player guessing games that distinguish between D_k and level- k decision rules. Instead of all players' guesses being bounded between 0 and 100, each player i has a

lower limit (either 100 or 300) and an upper limit (either 500 or 900), as well as a target $p_i \in \{0.5, 0.7, 1.3, 1.5\}$. The players simultaneously submit guesses within their own range, and each player earns payoffs that increase the closer she is to the opponent's guess times her own target.

Let's find the Nash equilibria in these guessing games. First, consider the situation in which both players' targets are less than 1. This situation resembles the beauty contest game with $p < 1$, in that both players would continually wish to undercut each other (albeit by different amounts), and—if not for the lower limits that restrict their guesses from falling all the way to zero—both players would guess 0 in equilibrium. Instead, one or both players hit the lower limit in equilibrium. The equilibrium guesses are thus determined by the more restrictive (higher) lower limit across the two players. In equilibrium, the player whose lower limit is greater guesses his lower limit, while the other player best responds by choosing her target times the other player's lower limit (or as close as she can get within her own allowable range). For example, consider a game in which player 1 has target $p_1 = 0.5$, lower limit 300, and upper limit 900, while player 2 has $p_2 = 0.7$, lower limit 100, and upper limit 500. Since player 1 has the more restrictive lower limit, he will choose his minimum possible strategy of 300. Player 2 could go all the way down to 100, but this wouldn't be her best response to player 1 choosing 300: she prefers to choose 300×0.7 , or 210. By similar reasoning, the equilibrium when both players' targets are greater than 1 involves the player with the smaller upper limit choosing his limit while the other player best responds.

Finally, let's consider the situation in which one player's target is greater than 1 and the other's is less than 1. Is the equilibrium at the maximum? Minimum? In between? Is there no pure-strategy equilibrium? The answer depends on the product of the two players' targets. If $p_1 p_2 < 1$, then the equilibrium is determined by the larger of the two lower bounds, just as when both targets are below 1. If $p_1 p_2 > 1$, then the equilibrium is determined by the smaller of the two upper bounds, just as when both targets are greater than 1. (When $p_1 p_2 = 1$, the equilibrium is not unique, so this scenario is not used in the experiment.)

One way to see this is to imagine players iteratively best responding to each other. Consider an example in which player 1 has $p_1 = 0.7$, lower limit

300, and upper limit 900, while player 2 has $p_2 = 1.3$, lower limit 100, and upper limit 900. Imagine that player 1 believes that player 2 guesses x . Player 1 best responds with $0.7x$, to which player 2 best responds with $1.3 \times 0.7x = 0.91x$. Each round of best responding drives guesses down to 91% of their previous value, until one or both players hit their lower limit—in this case, player 1 hits his lower limit of 300 and player 2 best responds by choosing $1.3 \times 300 = 390$. The best response functions and Nash equilibrium are shown in [figure 27.5](#), with player 1's guess plotted on the horizontal axis and player 2's guess on the vertical. Player 1's best response curve is shown in gray and player 2's in black. When possible, player 2 best responds by choosing 1.3 times player 1's guess, so the diagonal portion of player 2's best response has slope $p_2 = 1.3$. However, player 2 cannot guess below 100 or above 900, which is indicated by the horizontal portions of her best response function. Player 1's target is 0.7, and since player 1 is on the horizontal axis, the slope of his best response function in this graph is given by $1/0.7 = 1.43$ —slightly steeper than player 2's best response. Player 1's upper and lower limits are reflected in the vertical portions of his best response. There is a unique intersection at the equilibrium that we found earlier: Player 1 guesses his lower limit of 300 and player 2 guesses 390. Note that so long as $p_1 p_2 < 1$, the diagonal portion of player 1's best response will be above the diagonal portion of player 2's. This is because if $p_1 \times p_2 < 1$, then $p_2 < 1/p_1$ (i.e., the slope of player 1's plotted best response is steeper). Therefore, the only possible point of intersection occurs at one or both players' lower bounds: where player 1's curve shoots down vertically or player 2's curve shoots left horizontally. This confirms our earlier finding that when $p_1 p_2 < 1$, the equilibrium will occur on one or both lower bounds.

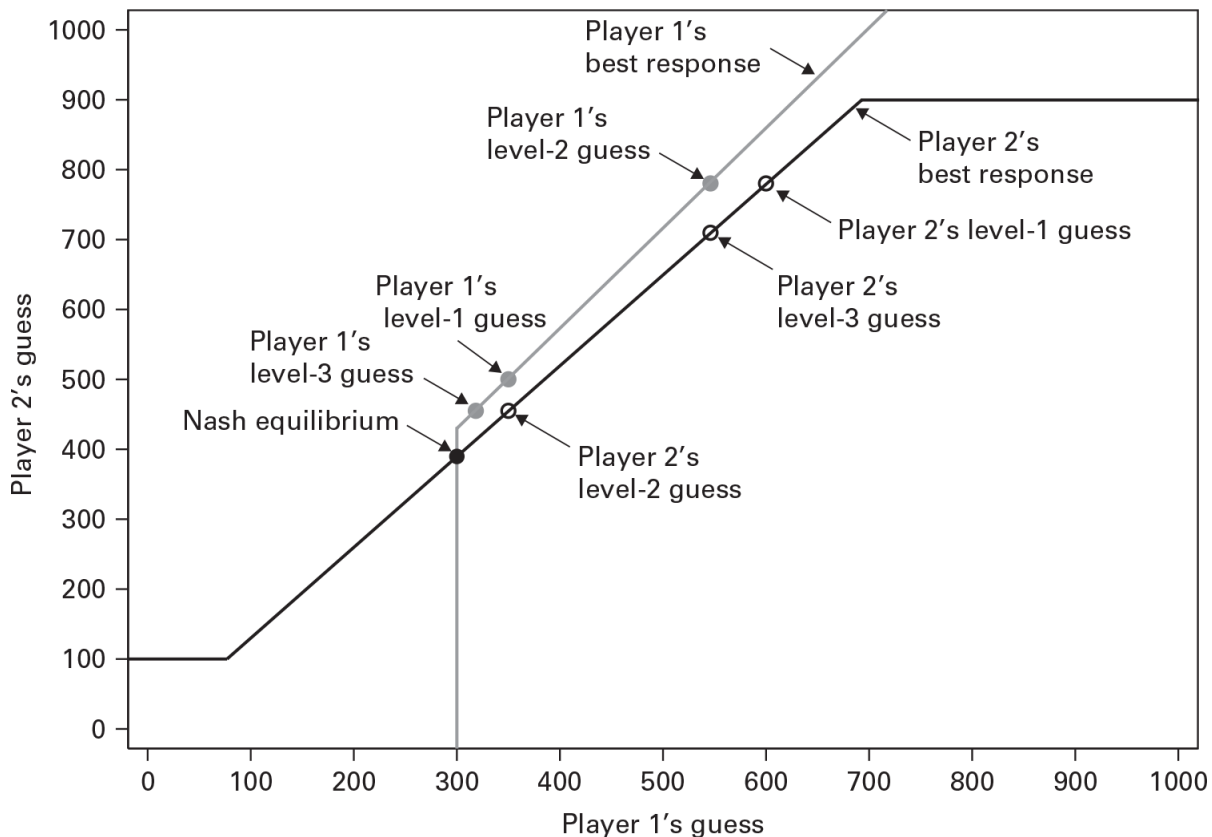


Figure 27.5

Best response functions for player 1 (gray) and player 2 (black) in the guessing game example. The Nash equilibrium is shown as the solid black dot, player 1's level-1 through level-3 predictions are shown as gray dots, and player 2's level-1 through level-3 predictions are shown as hollow black circles.

The level-1 and level-2 predictions are also shown on the graph. A level-1 player 1 best responds to the average guess of a player 2 who is randomizing uniformly over the $[100, 900]$ interval: $500 \times 0.7 = 350$. A level-1 player 2 best responds to the average guess of a player 1 who is randomizing uniformly over the $[300, 900]$ interval: $600 \times 1.3 = 780$. A level-2 player 1 best responds to a level-1 player 2 who chooses 780: $780 \times 0.7 = 546$; and a level-2 player 2 best responds to a level-1 player 1: $350 \times 1.3 = 455$, and so on.

We can also compute the Dk behavior. To compute the D1 strategy for player 2, we consider which guesses are undominated for player 1. Player 1 knows that player 2 can't guess above 900, so the highest p_1 times the guess of player 2 could be is $0.7 \times 900 = 630$. Thus, guesses above 630 are dominated for player 1. Player 2 thus assumes that player 1 randomizes

uniformly across the interval $[300, 630]$, with an average guess of 465, and best responds by choosing $1.3 \times 465 = 604.5$. Thus, this game allows separation in level- k and Dk predictions.

Looking only at the raw guesses, the researchers found that about half the participants' behavior conformed *exactly* to level-1, level-2, level-3, or equilibrium behavior in most of the games. That is, even though the strategy spaces had ranges that varied from 200 to 800 across the games (depending on the players' upper and lower bounds), players often hit the predicted level- k guess exactly (or within 0.5). In their econometric model based on both guesses and which information the players look at, they estimated that 44% of players act consistently with level-1 behavior, 16% with level-2 behavior, and 11% with equilibrium behavior. They do not find evidence of Dk types.

Level- k Reasoning in Sequential Games

Let's now consider how level- k reasoning models can also be applied to sequential games. To find a player i 's level- k strategy, we would first set the action at each node or information set for all players other than i to conform to those players' level- $k - 1$ strategies. Then, we backward induct over all of player i 's nodes or information sets to find her level- k strategy.

For example, consider the game depicted in [figure 27.6](#) between Kemba (who moves first) and Maya. Using backward induction, we can find the unique subgame perfect Nash equilibrium (SPNE), $((B, C, F, H, I), (P, W, X))$. To find the level-1 strategy for Kemba, we backward induct over each of Kemba's five nodes, assuming that Maya randomizes uniformly at each of her three nodes. At the three nodes where Kemba makes the last move of the game, he simply makes the choice that maximizes his payoff and does not need to consider Maya's behavior: thus, he plays C, F, and H at these nodes. At the node where he selects between I and J, he knows his payoff from J to be 250 and believes that his expected payoff from I is $0.5 \times 300 + 0.5 \times 100 = 200$, so he chooses J. Now, we back up to the first move of the game, at which point Kemba chooses between A and B. If he chooses A, he assumes that Maya will randomly choose P with probability 0.5 (which will give him a payoff of 270) and Q with probability 0.5 (which will give him a payoff of 60), for an expected payoff of 165. If he chooses B, he assumes

that Maya will randomly choose V (payoff of 10) or W (payoff of 250), for an expected payoff of 130. Thus, he chooses A over B. His level-1 strategy is thus (A, C, F, H, J). Employing the same procedure for Maya (fixing Kemba's strategy as uniform randomization over each of his nodes and backward inducting over Maya's nodes), we find her level-1 strategy to be (P, W, X).

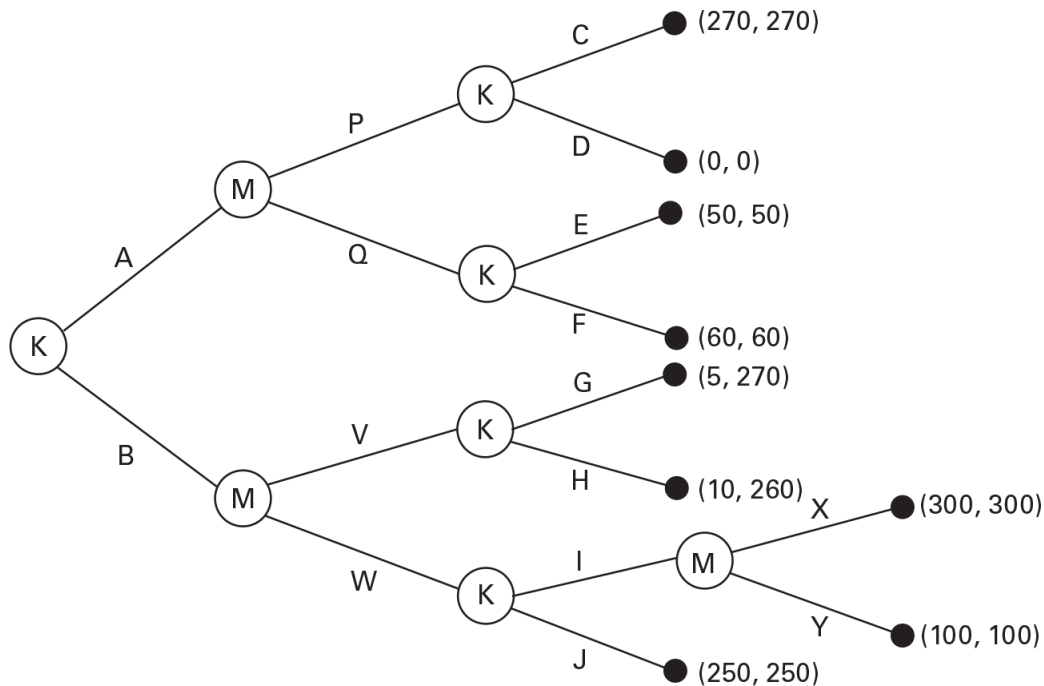


Figure 27.6

A sequential game between Kemba (K) and Maya (M).

To find Kemba's level-2 strategy, we fix Maya's strategy as (P, W, X) and then backward induct over Kemba's nodes: his choice at his final nodes remains C, F, and H, but his other choices change. Now that he believes that Maya will choose X, he is willing to choose I and let her make the final move of the game (giving them both a payoff of 300), rather than choosing J to get a payoff of 250. And now that he believes that she will choose W (and then X), he prefers to choose B rather than A.

The level- k predictions are shown in the table in [figure 27.7](#). Notice that so long as Kemba is type $k \geq 4$ and Maya is type $k \geq 3$, the SPNE will be played. Notice also that the players' strategies don't necessarily become increasingly consistent with SPNE play as their types increase. Maya plays

her SPNE strategy when she is a level-1 type (simply because it happens to be her best response to random play), but she would deviate from this strategy to (P, V, X) if she were level-2, and then again play it if she were level-3 or higher. Likewise, Kemba chooses his SPNE strategy when level-2 (in response to level-1 Maya coincidentally selecting her SPNE strategy), but he moves away from it when level-3 (because level-2 Maya moved away from hers), before returning to it when level-4 or above. In general, a player in a game of perfect information using level- k reasoning will play her SPNE strategy so long as she is within k moves from the end of the game. For example, a level-1 player does not play dominated strategy in his final move of the game; a level-2 player best responds to a player who would not play a dominated strategy in his final move, and thus she plays according to the SPNE when two moves away from the end of the game, and so on. This means that players will always play according to the SPNE so long as they are at least level- k , where k is the maximum number of moves from their first node to the end of the game (in our game, this is level-4 for Kemba and level-3 for Maya).

Level- k predictions		
Level	Kemba	Maya
Level-0	Randomize uniformly at each node	Randomize uniformly at each node
Level-1	(A, C, F, H, J)	(P, W, X)
Level-2	(B, C, F, H, I)	(P, V, X)
Level-3	(A, C, F, H, I)	(P, W, X)
Level-4	(B, C, F, H, I)	(P, W, X)
>Level 4	(B, C, F, H, I)	(P, W, X)

Figure 27.7
Level- k predictions.

In sequential games, level- k reasoning can thus be thought of as generating limited induction rather than full backward induction. There is, however, something odd about applying the level- k model to sequential games. Unlike the Nash equilibrium and quantal response equilibrium (QRE), level- k players often have incorrect beliefs about the strategies of

others. But as a sequential game unfolds, they may be able to correct these beliefs. For example, consider again the game shown in [figure 27.6](#) and assume that it is played between two level-2 players. Maya believes Kemba to be level-1 and thus playing (A, C, F, H, J). But in the very first move of the game, the level-2 Kemba instead plays B. Since this would indicate that he's not actually level-1, should Maya still play V (which is a best response only to the level-1 strategy)? Or should she call an audible and switch to W (a best response to all other types' strategies)? Updating her beliefs about the other player's type would also be odd, because the types in the level- k model refer to a player's *own* depth of reasoning, which in turn implies that they believe the other player to act according to a type who is one step lower. If Maya updated her beliefs after seeing A, such that she now thinks she's playing against a level-2 type (or a level-4 or greater type), under the level- k model, this means that she suddenly becomes capable of level-3 thinking (or level-5 or greater thinking). These issues are somewhat addressed in a dynamic level- k model for sequential games that was developed by Ho and Su (2013). In their model, all players are *cognitively capable* of any level of thinking, but their level- k type simply refers to their beliefs about the reasoning rule employed by others. Furthermore, they allow players in repeated games to update their beliefs, and thus their level- k type, after each repetition (although not between nodes within the same round of the game).

Level- k Reasoning in Games with Incomplete Information

In games with incomplete information, we have seen that players regularly fail to take into account the extent to which others' decisions are based on privately held information. For example, buyers in the lemons problem and in the acquiring a company game (discussed in chapter 11) often do not realize that a seller's willingness to sell them the car or company at a certain price is an indication that they know the car to be worth less to them. In chapter 11, we saw that behavior in many incomplete information games could be explained by the cursed equilibrium model, which assumes that players correctly anticipate the overall probability with which players choose various actions, but underestimate the extent to which actions are

based on private information about their type. In this section, we will apply the level- k model to games of incomplete information.

Consider the game presented in [figure 27.8](#), which was developed by Brocas et al. (2014) in an experiment designed to assess the limits of strategic thinking in incomplete information games. In this game, there are three possible states of the world, X, Y, and Z, each of which is equally likely. Each player simultaneously decides whether to Bet (B) or Not Bet (N). If one or both players choose not to bet, each gets a “sure thing” payoff of 10 that does not depend on the state of the world. If both players bet, then each gets a payoff that depends on the state of the world and differs across players (shown for player 1 in the top row and player 2 in the bottom row). Prior to choosing whether to bet, each player learns some information about the state of the world: player 1 learns whether the state is Z with certainty or not Z (in which case it may be either X or Y); player 2 learns whether the state is X with certainty or not X (i.e., Y or Z). The vertical lines in the table delineate the information sets for the two players. For example, player 1 cannot distinguish between X and Y (and so they are not separated in the table), but she can distinguish Not Z from Z (so there is a vertical line between these outcomes).

	State		
	X	Y	Z
Player 1	25	5	20
Player 2	0	30	5

Figure 27.8

Incomplete information game (Brocas et al., 2014).

Recall from chapter 11 that each player’s strategy in a Bayesian game specifies an action for each type, where the player’s type in this case refers to the information they learn: in this case, player 1 may be of type Not Z or Z, and player 2 may be of type X or Not X. In other words, players can condition their strategy on their private information. We will write player 1’s strategy as (*strategy for type Not Z*, *strategy for type Z*) and player 2’s strategy as (*strategy for type X*, *strategy for type Not X*), which is consistent with reading the information sets across the payoff table from left to right.

This game can be solved using iterated elimination of dominated strategies. When a player learns the state of the world with certainty, his optimal strategy is straightforward. Player 1 of type Z knows that not betting is associated with a sure payoff of 10, and if the players both bet, he will get a payoff of 20 (his payoff under the Z state). Thus, N is weakly dominated and player 1's best response is B if his type is Z. By the same reasoning, player 2 has a dominant strategy of N if she is type X.

Now, what about player 1 of type Not Z? Player 1 wants them to bet if the state is X, but not if it is Y. Player 1 doesn't know whether it is state X, but player 2 does—and we just found that player 2's optimal strategy is N if type X, since betting is associated with a payoff of 0 for her. Player 1 should thus reason that he will never be able to get the payoff of 25 that is associated with them betting under state X, since player 2 would never choose B under this scenario. Thus, player 1 of type Not Z would want to choose N. Notice that this reasoning requires player 1 to identify player 2's dominant strategy of N when type X. In other words, two steps of reasoning are necessary for player 1 to recognize that player 2's behavior is based on her private information. Thus, player 1's optimal strategy is (N, B), for types (Not Z, Z), respectively.

Now let's turn to player 2. We've seen that she has a dominant strategy of N when type X, but what about when Not X? Let's find her best response to player 1 choosing (N, B). She doesn't know whether the state of the world is Y or Z. If it is Z, then player 1's type is Z, he chooses B, and she will get 5 if she bets as well—worse than the sure payoff. If it is Y, then player 1's type is Not Z, he will choose N, and she will get the sure payoff of 10 no matter what she does. Therefore, she is better off playing N, since she can only do worse than 10 by betting. (Notice that this reasoning requires player 2 to identify player 1's dominant strategy when type Z, as well as player 1's optimal strategy when type Not Z, which was in turn based on player 2's own dominant strategy when type X.) Thus, player 2's optimal strategy is (N, N). We've thus found ((N, B), (N, N)) to be a Bayesian Nash equilibrium.

Another way to see this logic is to write the Bayesian game in normal form, as shown in [figure 27.9](#), and perform iterated elimination of dominated strategies. The first cell, for instance, gives the expected payoffs for the strategy profile in which both players bet regardless of type. In this

case, there is a $1/3$ chance of each state of the world and they bet in each state, so $Eu_1((B, B), (B, B)) = (1/3)(25) + (1/3)(5) + (1/3)(20) = 16.67$, and so on. In the next cell to the right, associated with $((B, B), (B, N))$, player 1 always bets, but player 2 bets only when the state is X; thus, the players get 10 when states Y or Z arise and, the $1/3$ of the time that the state is X, player 1 gets 25 and player 2 0, for expected payoffs of $(15, 6.67)$.

		Player 2			
		(B, B)	(B, N)	(N, B)	(N, N)
Player 1	(B, B)	(16.67, 11.67)	(15, 6.67)	(11.67, 15)	(10, 10)
	(B, N)	(13.33, 13.33)	(15, 6.67)	(8.33, 16.67)	(10, 10)
	(N, B)	(13.33, 8.33)	(10, 10)	(13.33, 8.33)	(10, 10)
	(N, N)	(10, 10)	(10, 10)	(10, 10)	(10, 10)

Bayesian Game where strategies are listed for types (Not Z, Z) and (X, Not X)

Figure 27.9

Incomplete information game (Brocas et al., 2014).

Once the Bayesian game is written in normal form, iterated elimination of dominated strategies allows us to delete (B, N) and (N, N) for player 1 and (B, B) and (B, N) for player 2, which is what we did previously when we found that player 1 would always want to bet in state Z and player 2 would never want to bet in state X. Then, we can delete (B, B) for player 1, which is what we did when we found that player 1 would not wish to bet in state Not X, given player 2's dominant strategy. Finally, we can delete (N, B) for player 2. We're thus left with $((N, B), (N, N))$, the Bayesian Nash equilibrium we found before. (Note that $((N, N), (N, N))$ is also a Bayesian Nash equilibrium in this game, since if a player believes with certainty that the other will never bet, then they will earn the sure payoff of 10 no matter what they do, and they are indifferent between all their strategies.)

Now let's consider the level- k predictions, assuming that level-0 behavior is characterized by players betting with probability 0.5, independent of their information about the state of the world. The predictions are summarized in the table [figure 27.10](#). Level-1 players believe that the other is equally likely to bet regardless of her information.

In this way, a level-1 player acts similarly to a fully cursed player in the cursed equilibrium model. A level-1 (and above) player would not choose a dominated strategy, and so notice that the rows associated with level-1 and higher have player 1 choosing B when type Z and player 2 choosing N when type X. Level-1 players, however, believe that the other player might choose a dominated strategy, so the other half of each of their strategies may differ from the Bayesian Nash equilibrium we found earlier.

Level- k predictions		
Level	Player 1	Player 2
Level-0	((0.5 B, 0.5 N), (0.5 B, 0.5 N))	((0.5 B, 0.5 N), (0.5 B, 0.5 N))
Level-1	(B, B)	(N, B)
Level-2	(N, B)	(N, B)
Level-3	(N, B)	(N, N)
Level >3	(N, B)	(N, N)

Strategies are listed for types (Not Z, Z) and (X, Not X)

Figure 27.10

Level- k predictions for the game of incomplete information.

Let's look for the remainder of the level-1 strategy for each player. A level-1 player 1 does not make the logical leap that betting under state Not Z will not be profitable due to player 2 never betting in state X. He simply compares his average payoff across states X and Y (15) with the sure payoff of 10, and chooses B. Thus, (B, B) is the level-1 strategy for player 1. This can also be seen from the game in normal form: assuming that player 2 chooses each column with probability 0.25, player 1's highest expected payoff occurs in row 1. By analogous reasoning, (N, B) is the level-1 strategy for player 2. The higher-level strategies are then computed by having each player best respond to an opponent using one lower level of reasoning. Notice that level-2 reasoning or higher is required for player 1 to reach the dominance-solvable equilibrium strategy (since he must believe that the other player will not choose a dominated strategy), and level-3 reasoning or higher is required for player 2 to reach the dominance-solvable

equilibrium strategy (since she must believe that the other player will believe that she will not play a dominated strategy).

Incomplete Unraveling of Information about Movie Quality

As we saw in chapter 11, when rational players with private information have the ability to verifiably communicate their type, information unravels, and in equilibrium, all information is fully revealed. We saw examples of prosocial players sharing their type and mediocre restaurants posting their health code ratings (those with the highest ratings post theirs, which causes those with moderate ratings to disclose, and so on). Here is an example from Hollywood studied by Brown, Camerer, and Lovo (2012): Movie studios generally know the quality of their films before releasing them, since they survey test audiences, but moviegoers do not necessarily know the quality of the film when deciding whether to see it. There is, however, an easy way to reveal this information: studios typically screen their films for critics in advance of their release. However, sometimes the studios choose to “cold-open” the movie without letting critics see and review it in advance. Why would they do this?

Under the standard model, consumers should assume that any movie that isn’t released to critics must be truly awful and steer clear, and so only the absolute worst films ever made would be cold-opened. For example, imagine that the quality of films, q , is uniformly distributed between 0 and 100. If studios screen above-average films and cold-open below-average ones, then consumers should assume any cold-opened film has $q < 50$ and, on average, is $q = 25$. Then this implies that studios who know their films to be better than 25 would actually prefer to screen them—consumers will learn that they are below average, but at least they wouldn’t think that they were as bad as $q = 25$. But if films are cold-opened only if $q < 25$, then consumers would assume that the average cold-opened film has quality 12.5, causing studios with films better than 12.5 to show them to critics, and so on, such that only films of quality 0 wind up being cold-opened in equilibrium.

In practice, however, over 10% of films are cold-opened, they earn higher box office revenue than films that are screened (controlling for quality), and moviegoers tend to be more disappointed in these films (controlling for quality), suggesting that consumers do not correctly update

their beliefs about the poor quality of the films being cold-opened and that studios thus find it in their interest to cold-open mediocre films. Brown et al. (2012) proposed that level- k reasoning can explain the failure of moviegoers to infer movie quality from the cold-opening decision. A level-1 moviegoer believes that the cold-open decision is random and communicates no information about the film's quality, and so assumes that a cold-opened film is of quality 50. Studios who best respond to level-1 moviegoers would thus cold-open below-average films and screen above-average ones. Studios who anticipate that moviegoers will be able to do an extra step of thinking would cold-open only those films of quality less than 25, and so forth. Thus, limited strategic thinking can explain the failure of information unraveling in this instance and similar situations.

Cognitive Hierarchy: Reasoning about the Reasoning of Others

In the model that we have considered thus far, players always acted as though they were exactly one step ahead of everyone else. A level- k player behaves as if he believes that everyone else is able to employ exactly $k - 1$ steps of reasoning, but he alone is able to complete the k th step. In the *cognitive hierarchy model* (Camerer, Ho, and Chong, 2004), players believe that there is variation in the strategic thinking of other players. However, they still believe that their own depth of reasoning is superior to everyone else's.

Specifically, assume that there is some distribution of types in the population, such that the proportion of players who use h levels of thinking is given by $f(h)$. A player of type k believes that the other players' types range from 0 through $k - 1$. Furthermore, the player has correct beliefs about the *relative frequency* of each of the lower types, even if she is oblivious to the fact that there are others in the population of type k and higher. Thus, a player of type k believes that the proportion of players of type h to be $\frac{f(h)}{\sum_{l=0}^{k-1} f(l)}$.

For example, consider the estimates of the proportion of each type in Arad and Rubinstein (2012), listed in the top row of [figure 27.11](#). A level-1 player believes 100% of the population of other players to be level-0. A level-2 player believes that some players are level-0 and some are level-1, but doesn't believe that anyone else is level-2 or above. Furthermore, she is

correct about the relative proportion of level-0 versus level-1 types—that is, she believes $\frac{0.05}{0.05+0.13} = 0.28$ to be the proportion who are level-0 and $\frac{0.13}{0.05+0.13} = 0.72$ to be the proportion who are level-1. Finally, a level-3 player believes $\frac{0.05}{0.05+0.13+0.39} = 0.09$ to be level-0, 0.23 to be level-1, and 0.68 to be level-2.

	Type			
	Level-0	Level-1	Level-2	Level-3
Actual population	0.05	0.13	0.39	0.43
Level-1 beliefs	1	0	0	0
Level-2 beliefs	0.28	0.72	0	0
Level-3 beliefs	0.09	0.23	0.68	0

The top row shows an estimate of how the population is distributed over level-0, level-1, level-2, and level-3 types, based on Arad and Rubinstein (2002). The other rows show how level-1, level-2, and level-3 players believe the population to be distributed.

Figure 27.11
Level- k estimates.

How would these beliefs change level- k behavior in the beauty contest game? Recall that the predictions in the $p = 2/3$ game are 33, 22, and 15 for levels 1 through 3, respectively. The prediction for level-1 is the same, since they still believe all the other players to be level-0 and choosing 50 on average. A level-2 player now believes that 28% of the population is guessing 50 on average and 72% are level-1 and choosing 33. Thus, they believe the average guess in the population to be $0.28 \times 50 + 0.72 \times 33 = 37.8$ and would guess 25, slightly higher than under the level- k model. A level-3 player now believes the average guess to be $0.09 \times 50 + 0.23 \times 33 + 0.68 \times 25 = 29$, and thus she guesses 19—again, a bit above the previous prediction of 15.

The cognitive hierarchy model typically uses the Poisson distribution to estimate the proportion of types. Under this distribution, the proportion of level- h types is given by $f(h) = \frac{\tau^h e^{-\tau}}{h!}$, where τ is a parameter, typically estimated from the data, that gives the mean and variance of the

distribution. Estimates of τ range from 3.7 (in a population of professional game theorists) to close to 0, and on average across several studies is around 1.3 – 1.6 (consistent with most people using one or two steps of reasoning).

Let's assume that $\tau = 1.5$. Then the proportion of level-0 players is given by $f(0) = \frac{\tau^0 e^{-\tau}}{0!} = 0.223$, the proportion of level-1 players is given by $f(1) = \frac{\tau^1 e^{-\tau}}{1!} = 0.335$, the proportion of level-2 players is given by $f(2) = \frac{\tau^2 e^{-\tau}}{2!} = 0.251$, and so on. Consider the normal-form game in [figure 27.12](#), which was one of the games used by Costa-Gomes et al. (2001). If this model is correct, how frequently would each action be played?

		Column	
		L	R
Row	T	(28, 37)	(57, 58)
	M	(22, 36)	(60, 84)
	B	(51, 69)	(82, 45)

Figure 27.12

Normal form game in Costa-Gomes et al. (2001).

The Row player has a dominant strategy of B, and therefore will choose B whenever he is level-1 or higher (since B is a best response to whatever strategy Column may choose, and thus to whatever level of reasoning Column may be using). Row will thus randomize uniformly over his three actions in the event that he is a level-0 type (which we just calculated occurs with probability 0.223) and play B with certainty when he is any other type (which occurs with probability $1 - 0.223 = 0.777$). T and M are therefore each played with probability $\frac{1}{3} \times 0.223 = 0.074$ (i.e., one-third of the time by the level-0 types) and B is played with probability $\frac{1}{3} \times 0.223 + 0.777 = 0.851$ (i.e., one-third of the time by the level-0 types and all the time by the other types).

The Column player plays L and R with equal likelihood when she is level-0. When she is level-1, she plays R, since it has higher average payoffs (assuming that Row is definitely level-0 and plays each of his actions with probability 1/3). When she is level-2, she assumes that Row is

either level-0 (to which her best response is R) or level-1 (to which her best response is L). In particular, she believes the likelihood that he is level-0 to be the relative frequency of level-0s compared to level-1s in the population: $\frac{0.223}{0.223+0.335} = 0.4$. Therefore, she thinks that there is a probability of 0.4 that he randomizes uniformly across his three strategies and a probability of 0.6 that he plays his dominant strategy of B. What, then, is her expected payoff from L and R? $Eu(L) = 0.4(1/3 \times 37 + 1/3 \times 36 + 1/3 \times 69) + 0.6(69) = 60.33$ and $Eu(R) = 0.4(1/3 \times 58 + 1/3 \times 84 + 1/3 \times 45) + 0.6(45) = 51.93$. Thus, the best response of a level-2 player (who assumes that 40% of the population is level-0 and 60% is level-1) is to play L. In other words, she assumes that there is a high-enough likelihood that she is dealing with a level-1 player playing his dominant strategy of B that she finds it in her interest to choose L. Likewise, the best response of Column players who are level-3 and greater is to play L, since higher-probability players assume that there is an even smaller likelihood that the other player is level-0. Therefore, R is played half the time by level-0 players and all the time by level-1 players, but by no one else. We thus expect to see R played with probability $\frac{1}{2} \times 0.223 + 0.335 = 0.446$, and L to be played with probability 0.554.

Compared to the other level- k model discussed previously, the cognitive hierarchy model has advantages and disadvantages. It incorporates the natural assumption that players believe that others vary in their depth of reasoning. Imagine that you are playing the beauty contest game with a large group of people: do you believe that everyone else in the room will engage in the same steps of thinking and make the same ultimate guess? The free parameter, τ , allows for flexibility in fitting the data. However, this requires researchers to estimate τ before they can calculate the strategies for players level-2 and higher. One main advantage of the standard level- k model is that researchers can generate predictions for each type without making any assumptions about the distribution, and then determine which individuals' behaviors are consistent with each type. As a result, the standard level- k model is used more frequently than the cognitive hierarchy model by researchers who wish to present alternative predictions to Nash that incorporate limited strategic thinking.

Exercises

(1) Consider the following normal-form game:

		Column		
		D	E	F
Row	T	(60, 45)	(30, 45)	(40, 50)
	M	(65, 10)	(20, 11)	(10, 12)
	B	(10, 40)	(5, 10)	(70, 50)
	X	(20, 200)	(15, 0)	(20, 0)

Figure 27.13

- (a) Assuming uniform randomization for the level-0 strategy, find the level-1, level-2, and level-3 strategies.
- (b) What level of reasoning is necessary for each player in order to play her Nash equilibrium strategy?
- (2) Consider a slightly different version of the incomplete information betting game discussed in this chapter. As in that game, player 1 observes whether the state is Z or Not Z, player 2 observes whether the state is X or Not X, and then they simultaneously choose to bet (B) or not bet (N). If both players choose B, their payoffs are as listed in the table here; if one or both choose N, each gets a payoff of 10 regardless of the state.

		State		
		X	Y	Z
Player 1	Player 2	15	0	5
	Player 2	15	5	25

Figure 27.14

- (a) Use iterated elimination of dominated strategies to find a Bayesian Nash equilibrium.
- (b) Assuming that level-0 behavior involves randomizing uniformly, find the level-1, level-2, and level-3 strategies for each player.

(3) Consider again the El Farol bar problem (introduced in exercise 4 in chapter 26). The El Farol is a bar in Santa Fe, New Mexico, where everyone wants to hang out on Thursday nights. If it becomes too crowded, however, then it is no fun for anyone, and so everyone would prefer to avoid the bar once it exceeds a certain capacity. Imagine a game with three players, each simultaneously deciding whether to head to the bar. If one or two of the three show up, then those players will have a good time and earn a payoff of 1. Three, however, would be a crowd, and all three would earn a payoff of 0. Finally, any player who avoids the bar gets a payoff of 0.51, regardless of what the others choose to do.

Assume that the level-0 strategy is uniform randomization and the population is distributed according to Arad and Rubinstein's estimates: 0.05 level-0, 0.13 level-1, 0.39 level-2, and 0.43 level-3. What percentage of the population do we expect to see at the bar on any given Thursday?

(4) Consider the following three-player sequential-move game. Find the level- k strategies for each of the three players for $k = 1, 2$, and 3:

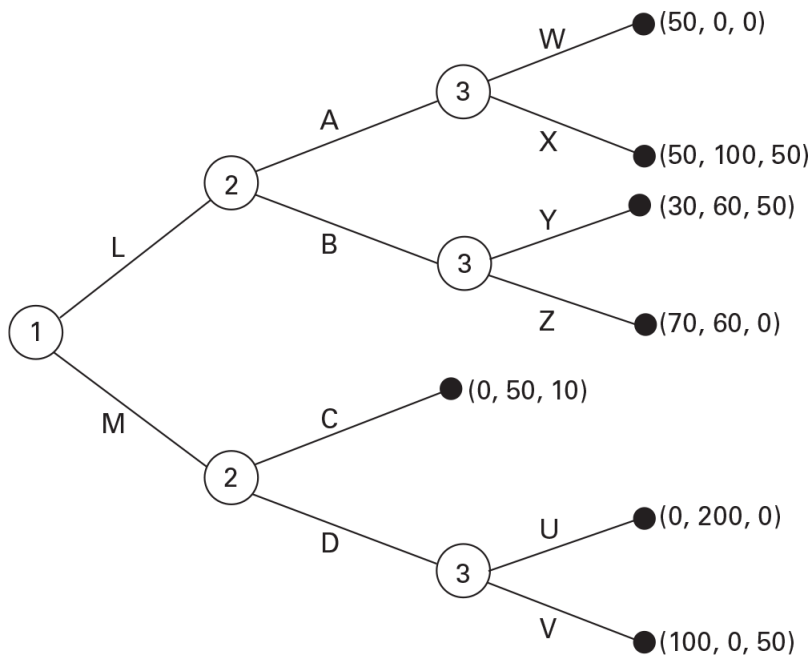


Figure 27.15

(5) Consider a guessing game like those discussed in this chapter. Two players simultaneously submit a guess within a certain range and want their

own guess to be as close as possible to the other player's guess times some target number, p . Each player has a different p and a different range within which their own guess must fall. Let's say that player 1's guess must be between 100 and 900 and he has target 0.5, while player 2's guess must be between 300 and 500 and she has target $p = 0.7$.

Find the Nash equilibrium guesses, as well as the guesses for level-1, level-2, and level-3 players.

(6) Consider the game show here between Row and Column:

		Column		
		L	C	R
Row	T	(40, 10)	(42, 11)	(35, 5)
	M	(50, 38)	(6, 0)	(30, 40)
	B	(25, 14)	(45, 15)	(40, 10)

Figure 27.16

(a) Find the level-1, level-2, level-3, and level-4 strategies for each player.

Now consider the Poisson cognitive hierarchy model and let $\tau = 1.2$.

(b) Find the beliefs for level-1, level-2, and level-3 players about how the population is distributed.

(c) Find the level-1, level-2, and level-3 strategies.

(7) There are many examples of level- k thinking in literature and film. Consider the following scene from the M. M. Kaye novel *The Far Pavilions* (discussed in Crawford, Costa-Gomes, and Iriberri, 2013): Ash is attempting to escape pursuers on a road that runs north–south. He is advised to go north, “since they will be sure you will be sure to go southward where the climate is kinder” (p. 25). Let's model the game in normal form as shown here: both parties prefer to go south in order to experience the nicer weather, and otherwise, Ash prefers to mismatch and the pursuers prefer to match.

		Pursuers	
		South	North
Ash	South	$(-5, 15)$	$(15, -10)$
	North	$(10, -5)$	$(-10, 10)$

Figure 27.17

Assume that the level-0 strategy is uniform randomization across north and south.

- Find the level- k strategy for each type for $k = 1, 2, 3, 4, 5, 6, 7$. Do you notice a pattern?
- What level strategy is the character using when he advises Ash to go north (“since they will be sure you will be sure to go southward where the climate is kinder.”)?

¹Many researchers do not consider level-1 players to play *strategically*, since they don’t attempt to predict the other player’s reasoning or consider their incentives and instead naively act as if the other player draws a strategy at random.

Psychological Game Theory

Standard game theory is outcome oriented. It assumes that players have preferences only over the ultimate outcomes of the interaction, which are uniquely determined by each player's action. In contrast, *psychological game theory* allows the process to matter too, such that preferences may also depend on one's own or others' beliefs. In this chapter, we will see how the framework of psychological game theory can be used to incorporate a wide variety of "psychological" or belief-based motives into strategic interactions, including self-esteem, reciprocity, and emotions like guilt.

Imagine that you own an iron foundry. One day, you are approached by a wealthy customer who wants to commission you to make a pair of custom gates for the entry to her estate. She promises she will pay top dollar if you construct "beautiful" gates that match the vision in her head. At first, you are thrilled. But then you start to think: what happens if you invest your time and materials into creating custom-made gates tailored specifically to her home and garish taste, only to have the customer claim that they are not sufficiently "beautiful"? Further, what if she offers you only a small fraction of your agreed-upon price?

This particular scenario appeared as a minor story in the book *Midnight in the Garden of Good and Evil*, and in that tale the customer feigned disgust with the gates and reneged on the deal, but then managed to ultimately acquire them very cheaply after the foundry slashed the price in a desperate attempt to recoup the cost of materials. This story depicts a general problem in economics known as the *hold-up* or *commitment* problem. This problem occurs when two parties could mutually benefit from transacting, but one party (A) must make a nonrecoverable, relationship-specific investment, such as creating a specific product, learning skills only needed at B's firm, or developing infrastructure that is useful only if transacting with B. Once A has committed to the relationship by making this investment, B can hold them up by renegotiating their agreement. Since the investment is valueless without

B's participation, B knows that A is in a tight spot and would have little choice but to agree. As a result, economic actors may avoid entering into agreements that require up-front investments, and efficient transactions can go unrealized.

Let's consider two different versions of this interaction between your company and the wealthy customer. In the first, once the gates are delivered and installed, you can't take them back. Imagine that the customer inspects the final product in her driveway and claims that they are worth little more than the price of the raw materials. She then makes you a take-it-or-leave-it lowball offer, and your only choices are to accept the smaller compensation or to storm off with nothing and let her keep the gates for free—a total loss for you. This scenario is depicted in the left panel of [figure 28.1](#) (the No Recourse version). In the first move, you choose whether to take on the project. If you choose In, the customer can choose whether to Honor your deal and pay you the agreed price or to Renege and offer a smaller price. In this case, you can choose whether to Accept or Reject. As we can see, there is a unique subgame perfect Nash equilibrium (SPNE) of ((Out, Accept), Renege).

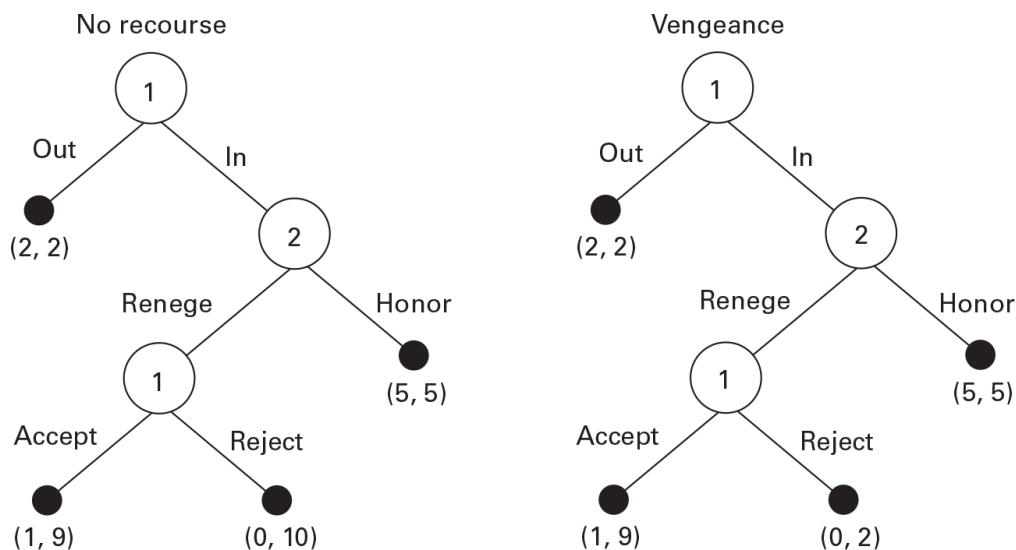


Figure 28.1
The hold-up problem.

Now imagine that the customer instead inspects the gates before they are installed. If she Reneges on the deal then you can either Accept the smaller price and allow her to keep the gates or you can storm off *with the gates* and deny her ownership of them. This scenario is depicted in the right panel of the

figure (the Vengeance version). It seems like this should change how the players act, but will it? Notice that Rejecting her lowball offer is still not a good option for you, because you are unable to resell the custom gates to a different customer. As a result, the unique SPNE remains ((Out, Accept), Renege). Yet Reject may have an added attraction in this version because it also harms the customer who wronged you. If you're sufficiently motivated to retaliate against players who you believe have intentionally harmed you, you may thus be willing to Reject, which would in turn keep the customer honest and make you willing to take on the job. Can the hold-up problem be solved by the motivation to reciprocate unkind behavior? In this chapter, we will consider a framework called *psychological game theory*, which allows players' payoffs to depend not just on outcomes, but also on their beliefs about others' choices and beliefs. This framework allows for the incorporation of a wide variety of belief-based, psychological motives into the analysis of strategic interactions, including players' desire to harm others who they believe were acting to intentionally harm them.

Psychological Games

Standard game theory assumes that players have preferences only over the ultimate outcomes of their interaction. As a result, the path of actions through the game pins down a terminal node that is uniquely associated with a payoff for each player, as we saw in chapter 2. Yet a bit of introspection and casual observation of other people's behavior reveal that our preferences don't just depend on actions, but also on our own beliefs about what others are doing (or should do) and the beliefs of others. That is, people don't just care about what everyone *does*, but also about what everyone *thinks*.

For example, we are often motivated to take certain actions purely because we want others to think that we are moral, smart, or cool. We typically view someone as more worthy of punishment if they *intentionally* rather than inadvertently harmed someone else, even if the ultimate outcome is the same, and our justice system often reflects this. We assess outcomes relative to what we had initially expected and lash out at others in anger and frustration when we are disappointed. We worry about letting others down and feel guilty when we do.

These belief-dependent motives can be incorporated into interactions using a general psychological game theory framework, which was developed by

Geanakoplos, Pearce, and Stacchetti (1989) and later extended by Battigalli and Dufwenberg (2009). The main characteristic of a psychological game is that ultimate payoffs can depend on the players' beliefs about the choices and beliefs of others. As Geanakoplos, Pearce, and Stacchetti write, "A player's emotional reactions cannot, in general, be independent of his expectations and of his interpretation of what he learns in a play of a game" (p. 61). To model these expectations, we typically focus on *first-order beliefs*, which are the probabilities that players assign to others' strategies (i.e., what one player thinks the others will do), and *second-order beliefs*, which are the beliefs that players have about others' first-order beliefs (i.e., what one player thinks that others think that he and the others will do). However, the theory can also incorporate preferences for higher-order beliefs.

Incorporating Guilt into Games

One of the most straightforward belief-dependent motives that may affect strategic interactions is simple guilt (Battigalli and Dufwenberg, 2007). A player suffers *guilt* whenever her actions cause another player to obtain a worse outcome than what she believes he had originally been expecting. In what follows, we will make the simplifying assumption (as in Battigalli and Dufwenberg, forthcoming) that a player feels a pang of guilt whenever another person receives a lower payoff than he had been expecting, regardless of the cause. In all of our examples, this outcome is due to the player's actions, and thus the definitions are overlapping for our purposes.

For example, imagine that Billie takes a cab ride in a foreign country and decides how much to tip her driver, Adam (an example due to Battigalli and Dufwenberg, 2009). Billie is unsure of the local tipping norm, and she doesn't want to let Adam down by giving him less than he expects. She has M dollars in her pocket and can tip any integer $t \in \{0, \dots, M\}$. Adam does not make a decision, and thus he wouldn't be considered a player in a traditional game-theoretic analysis of this encounter. In this case, Billie's decision is depicted in the tree in [figure 28.2](#), where the payoffs are the simple monetary outcomes for Billie and Adam. We could, alternatively, incorporate different distributional preferences for Billie (such as inequality aversion). Yet each end node would still be uniquely associated with a payoff for Billie and her best response simply involves choosing the t associated with the highest payoff.

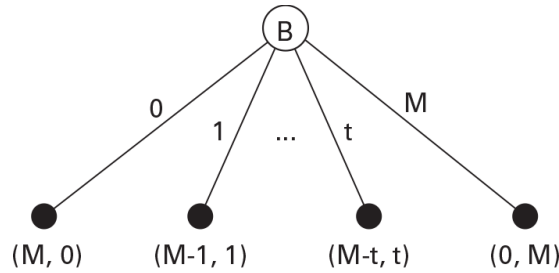


Figure 28.2
Billie's tipping dilemma.

However, if Billie cares about what Adam thinks, in addition to the ultimate monetary distribution, then we have a psychological game, in which Billie cannot determine her best response without first forming beliefs about what Adam expects. For instance, imagine that Adam's expectation of t is given by a and Billie's expectation of a is given by b . Billie feels guilty when she tips Adam less than what she thinks he is expecting (i.e., whenever $t < b$). Specifically, her ultimate payoff is given by $u_B = M - t - \theta \max\{0, b - t\}$, where θ captures Billie's guilt aversion. That is, Billie's payoff declines by θ for every dollar she shortchanges Adam relative to what she thinks he was expecting. Let's say that $\theta = 2$, and thus our game is depicted in extensive form in [figure 28.3](#). Notice that Billie's payoff is parameterized by b , which is based on her second-order belief. Thus, preferences are belief-dependent—the key feature of a psychological game—and Billie cannot determine her best response without first forming second-order beliefs.

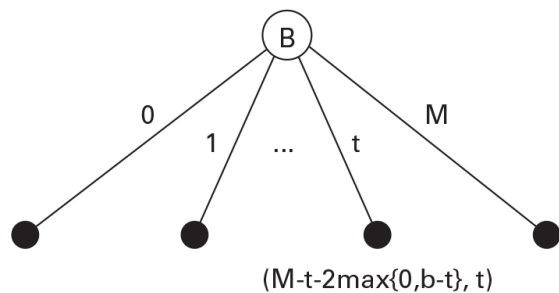


Figure 28.3
Billie's tipping dilemma with guilt.

We can further incorporate guilt aversion into games with multiple decision makers. In what follows, we will make a distinction between a player's *material payoff*—which will be the player's standard game-theoretic

payoff from the realized strategy profile and may be thought of in monetary terms—and a player's *psychological payoff*, which may depend on beliefs. In Billie's payoff function, her material payoff was given by $M - t$ (how much money she ends up with) and her psychological payoff was given by $-\theta \max\{0, b - t\}$ (the anguish she suffers for disappointing Adam). In general, we will say that a guilt-averse player has a payoff function that depends both on material and psychological motives, as described in definition 1.

Definition 1 (Guilt-averse preferences) *Consider a strategy profile s and let player j 's material payoff from s be given by $\pi_j(s)$. Let $E_{ij}(\pi_j)$ denote the payoff that player i believes player j expects to receive. A guilt-averse player i acts to maximize the expression: $u_i(s, E_{ij}(\pi_j)) = \pi_i(s) - \theta_i \max\{0, E_{ij}(\pi_j) - \pi_j\}$, where θ_i measures i 's guilt aversion.*

To consider the complexity of belief-dependent motives in a game with multiple decision makers, let's consider a trust game, which we studied in chapter 18. In the standard version of the game, one player first chooses whether to entrust the other with a sum of money, which the second-mover can either grab or return. Consider the following version portrayed in [figure 28.4](#), studied by Charness and Dufwenberg (2006). Player A first chooses between Out (i.e., not to trust), which gives both players a monetary payoff of 5, or In, in which case player B can make a choice. Player B can choose Grab, in which case she gets 14 and player A gets nothing, or Roll. If player B chooses Roll, then a die is rolled and A gains 0 if the number 1 comes up and 12 otherwise. Player B always gets 10 from rolling the die. Notice that, in expectation, this game reduces to a fairly simple binary trust game in which, if trusted, player B can choose between monetary distributions (0, 14) and (10, 10). The die roll component, however, gives player B plausible deniability, since player A won't know whether he got 0 because B screwed him over or because she tried to reward his trust and they got unlucky. Since player B maximizes her material payoff by choosing Grab, the unique SPNE is (Out, Grab), and both players earn only 5 rather than the 10 they could achieve with mutual cooperation.

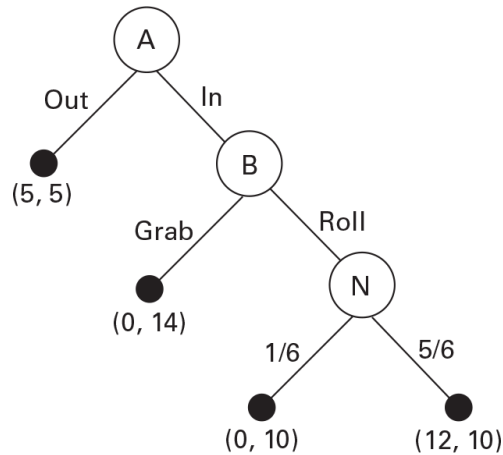


Figure 28.4

A trust game.

Now let's assume that, similar to Billie, player B is guilt averse. She earns psychological disutility whenever her actions cause A to earn less than A had been expecting. Let $\alpha \in [0, 1]$ be A's belief at the start of the game regarding the probability that B will choose Roll and let $\beta \in [0, 1]$ be B's second-order belief at the time that B moves. Furthermore, player B may hold some second-order belief at the beginning of the game, but once she observes that A has trusted her, she likely adjusts this belief upward. Given this, B's belief regarding how much A is expecting to get is $\beta(\frac{5}{6}(12) + \frac{1}{6}(0)) + (1 - \beta)(0) = 10\beta$. By choosing Grab, B believes she "lets down" A by an amount equal to the difference between what he gets (0) and what she thinks he was expecting to get, 10β . If B suffers disutility θ for every dollar that she lets A down, then her payoff from Grab is not 14 but rather $14 - 10\theta\beta$. While this game is simple, once again we see that we cannot simply backward-induct from B's final decision—her best response depends on her second-order beliefs and thus is endogenous to the broader strategic interaction.

Psychological Equilibria

Even though backward induction cannot be applied to psychological games, we can still solve for equilibrium outcomes. Geanakoplos et al. (1989) proved that an analog of the Nash equilibrium always exists in simultaneous-move psychological games and an analog of the SPNE (as well as its refinement of sequential equilibrium) exists in sequential psychological games.

While an extensive treatment of equilibria in psychological game theory is beyond the scope of this book, the basic idea of Geanakoplos et al. (1989)'s *psychological equilibrium* concept is straightforward. First, psychological equilibria require that players best respond to each other, which in this case means that players act to maximize their full payoffs in the psychological game (i.e., the sum of the psychological *and* material payoffs). That is, players' payoffs are belief dependent. As with standard game theory, subgame perfect psychological equilibria further require sequential rationality—players best respond at every information set, even those off the equilibrium path. Second, psychological equilibria impose the additional requirement that higher-order beliefs are consistent or accurate. That is, each player's first-order beliefs align with the actual strategy profile played by the others, each player's second-order beliefs match the other players' first-order beliefs, and so on.

In the tipping game with guilt aversion, this implies that, in the psychological equilibrium, Adam's expected tip $a = t$ (how much Billie actually tips him), and in turn, Billie's expectation, $b = a = t$. When $\theta < 1$, Billie's material concerns outweigh her guilt and she does not give a tip. However, when $\theta > 1$, Billie wants to match Adam's expectation, and we thus have $M + 1$ separate equilibria, each associated with one of the possible tip amounts (e.g., if Billie tips t , then in equilibrium, Adam anticipates receiving t , Billie accurately anticipates him expecting t , and thus she maximizes her psychological payoff by matching his expectation).

Before we return to the trust game with guilt, let's look at one other slightly simpler example of psychological equilibria.

Example: The Bravery Game

To solidify the concept of equilibrium in a psychological game, let's consider a classic example with a single decision maker from Geanakoplos et al. (1989). Peter can choose between two options: a dangerous choice of Bold or a safer choice of Timid. If Peter were by himself, he would prefer to be cautious and choose Timid ($u_1(\text{Timid}) = 3 > u_1(\text{Bold}) = 2$). However, Peter's friend will take note of his actions. She prefers that he be Bold and, if she expects him to choose Bold, then he doesn't want to disappoint her.

Let's say that Peter chooses Bold with probability p , his friend expects that he will choose Bold with probability q , and he believes that she expects him to choose Bold with probability r . The game is shown in [figure 28.5](#), where

the first payoff is Peter's and the second is his friend's. In the psychological equilibrium, the following conditions must hold:

- Peter is best responding: he chooses Bold if and only if $2 - r > 3 - 3r$.
- The friend has correct beliefs: $q = p$,
- Peter has correct beliefs: $r = q$.

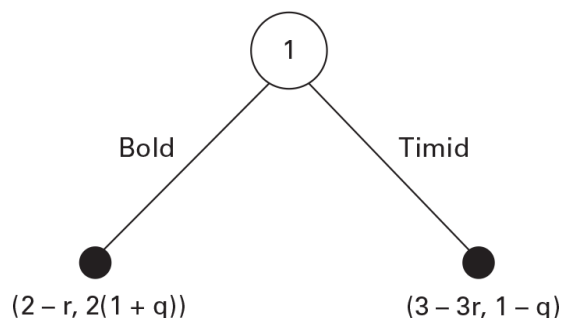


Figure 28.5

The bravery game.

Putting these conditions together, we can see that there are three equilibria in this psychological game. In the first, Peter chooses Timid ($p = 0$), his friend correctly anticipates this ($q = 0$), and Peter also correctly anticipates her low expectations ($r = 0$), such that Timid is indeed a best response ($2 - 0 < 3 - 3(0)$). In the second, Peter chooses Bold ($p = 1$), his friend correctly anticipates this ($q = 1$), and Peter correctly anticipates her high expectations ($r = 1$), such that he feels compelled to choose Bold ($2 - 1 > 3 - 3(1)$). In the third, Peter plays a mixed strategy of $p = 1/2$. Consistency of beliefs implies that $q = r = 1/2$, so Peter is indeed indifferent between Bold and Timid and willing to mix ($2 - 1/2 = 3 - 3(1/2)$).

Psychological Equilibria in the Trust Game with Guilt Aversion

Let's return to the trust game with a guilt-averse player B. Assume that player B chooses Roll with probability p and Grab with probability $1 - p$. Player A's belief that player B will choose Roll is given by α , and player B's expectation of α is β . Given the equilibrium requirement that beliefs match behavior and lower-order beliefs, we must have $\beta = \alpha = p$. A best responds by choosing In only if $5 \leq 10\alpha$ or $\alpha \geq \frac{1}{2}$. Since $\alpha = p$, this implies that we can have an equilibrium in which A chooses In only if $p \geq 1/2$. Finally, B best responds by choosing Roll if $14 - 10\theta\beta \leq 10$. Since $\beta = p$, this implies that we can have an equilibrium in which B chooses Roll only if $\theta p \geq 0.4$.

For low θ , guilt aversion is not a strong motivator of behavior, and our unique SPNE is the same as in the standard game: (Out, Grab). However, for sufficiently high θ (≥ 0.4), we can also have a pure-strategy equilibrium of (In, Roll). Since A anticipates that $\alpha = p = 1$, he prefers to choose In (because $5 \leq 10(1)$), and since B anticipates that $\beta = \alpha = 1$, she prefers not to let him down (because $\theta(1) \geq 0.4$). Player B's guilt thus generates an additional, more efficient equilibrium. Notably, when this outcome occurs, player B never actually experiences the emotion of guilt. Nevertheless, her aversion to this emotion is what allows cooperation to transpire in equilibrium.

Even if player B is guilt averse, we can still have a pure-strategy equilibrium of (Grab, Out). Under this equilibrium, A rationally anticipates $\alpha = p = 0$. Because A has low expectations, B wouldn't feel guilty about choosing Out, no matter how high θ were. However, as pointed out by Battigalli and Dufwenberg (2009), this outcome is not broadly compatible with a *psychological forward induction* argument, in which players deductively reason about each other's beliefs based on their observed behavior. Specifically, if B were to see A choose In, then she would logically conclude that $\alpha \geq 1/2$. If $\theta > 0.8$, player B would then prefer to choose Roll after seeing In and updating her beliefs accordingly. Thus, this equilibrium can survive the forward induction argument only if θ is not too high. Finally, we can also have mixed-strategy equilibria, in which player B chooses $p = 0.4/\theta$.

As in the tipping game, guilt aversion thus leads to sets of strategies and beliefs that are self-fulfilling: when B is trustworthy, A has high expectations of B, and these high expectations in turn motivate B to be trustworthy so as not to let A down. While guilt aversion may seem intuitive and plausibly consistent with our real-world experience, is there any evidence that this type of simple guilt motivates player B's trustworthiness in this type of game? This game was conducted in the lab by Charness and Dufwenberg (2006). They included a treatment in which player B could send a message to player A prior to the start of the game, and they also elicited first- and second-order beliefs by asking players A to guess the proportion of Bs choosing Roll and players B to guess the average guess of As, with rewards provided for correct guesses. They find that communication substantially increases the likelihood that player B chooses Roll, and that messages in which player B explicitly promised to play Roll were especially likely to be followed by (In, Roll).

There is evidence that this effect works through beliefs, consistent with the guilt-aversion argument: the ability to send messages had a significantly positive effect on both first- and second-order beliefs and Bs with higher second-order beliefs were more likely to choose Roll.

Intention-Based Reciprocity

Consider again the hold-up game described at the start of the chapter. In both versions of the game, the unique SPNE is ((Out, Accept), Renege). But do you think that this outcome may be more likely to occur in one version of the game than the other? Consider the final move of the game, in which player 1 is deciding to Accept the unfair offer for the product after player 2 has Reneged. Even though Accept is associated with a higher monetary payoff, in the case of the Vengeance version, player 1 may be willing to sacrifice the payoff of 1 to take revenge on player 2 for betraying their agreement. In the No Recourse version on the left of [figure 28.1](#), in contrast, Rejecting both gives player 1 a lower payoff and gives player 2 a higher payoff, and thus would not be a particularly productive way of extracting revenge.

In fact, this intuition is borne out in laboratory results. Dufwenberg, Smith, and van Essen (2013) find that about a third of players 1 Reject in the Vengeance version (and none do in the No Recourse version). Those in the role of player 2 seemed to anticipate this reaction: while very few chose Honor in the No Recourse version, over a third did in the Vengeance version. Finally, this all seems to translate into greater willingness to enter in the Vengeance version—over three-quarters of players 1 chose In in the Vengeance version, compared to 50% in the No Recourse version. In other words, the threat of rejection was real in the Vengeance version, and it could be used to enforce cooperation and the more efficient outcome of 5 each (rather than the material payoff of 2 that each players would receive from the SPNE).

One potential explanation for the rejection rates in the Vengeance version is inequality aversion, which is purely based on preferences over outcomes, like in standard games. For example, the Fehr-Schmidt (1999) inequality aversion model (described in more detail in chapters 14 and 20) assumes that player 1 bears psychological costs of α_1 for every extra dollar that player 2 earns more than him. In this case, player 1's payoff from choosing Accept is $1 - 8\alpha_1$ and from Reject is $0 - 2\alpha_1$, and rejection would be preferable whenever

$\alpha_1 > 1/6$ —that is, when player 1 is sufficiently averse to earning less than player 2.

Yet the outcome-oriented, inequality-averse explanation misses a crucial aspect of the interaction—that people may be especially motivated to reduce someone’s earnings when that person has done them harm. In addition to being intuitively appealing, there is experimental evidence indicating that intentions matter, as well as outcomes. We saw one such example in chapter 14, when we observed that responders in the ultimatum game were more willing to reject unfair offers when the proposer *could have* made them a fair offer and elected not to. Another example comes from the moonlighting game, which was developed by Abbink, Irlenbusch, and Renner (2000) and further analyzed with respect to the role of intentions by Falk, Fehr, and Fischbacher (2008). In the game, two players each begin with 6 tokens. First, player A can send any number between -6 (i.e., take tokens) to 6 to player B. Negative tokens are subtracted from player B’s total and added to player A’s, while positive tokens are subtracted from player A’s total and multiplied by 3 before being added to player B’s (similar to sending money in the trust game). Then, player B can choose to return any number of tokens ranging from -6 to 18. Positive tokens are added to player A’s total and subtracted from player B’s (e.g., consistent with B sacrificing to reward A), while negative tokens are also subtracted from B’s total and then multiplied by 3 before being subtracted from A’s total (e.g., consistent with B sacrificing to hurt A). Under the unique SPNE, player B neither punishes nor rewards in the final move (i.e., returns no tokens) and player A thus takes everything in the first move (i.e., sends -6 tokens).

In Falk et al.’s experiment, player B had to list her full strategy—that is, how many tokens to return for each of the 13 different token amounts that player A could send in the first move. Furthermore, there were two different treatments. In the *Intentions* treatment, player A’s choice actually determined how many tokens he sent. In the *No Intentions* treatment, the number of tokens sent was determined by a die roll, and even though this choice affected A’s payoff, he had no control over his actions. The tokens chosen by B in response are shown in [figure 28.6](#):

A's tokens sent:	−6	−5	−4	−3	−2	−1	0	1	2	3	4	5	6
Intentions	−8.09	−6.91	−5.97	−4.70	−2.97	−2.73	−0.88	1.24	1.73	2.64	4.58	3.64	6.55
No Intentions	−1.43	−2.35	−1.52	−2.26	−0.30	−0.57	−0.78	0.57	−0.39	−0.30	1.65	1.09	1.39

Figure 28.6

Average tokens returned by player B for each number of tokens sent by player A (Falk et al., 2008).

These results clearly indicate that intentions matter. In the No Intentions treatment, there is a weak positive relationship between tokens sent by A and tokens returned by B, indicating that some people do care about ultimate outcomes, even when they were unintentional. However, the median response for all levels of tokens sent was to return zero tokens. In contrast, the relationship between tokens sent and returned is much stronger when player A could act with intent, indicating that the participants cared about reciprocating their partners' intended kind and unkind acts. The difference in tokens returned between the treatments is significant for all numbers of tokens sent by A except 0.

The Charness and Rabin (2002) model, which was discussed in chapters 14 and 18, considers preferences over final allocations while also taking into account whether the decision maker feels wronged by their partner. This model assumes that a player's payoff is a weighted sum of their own earnings and the earnings of their partner and that these weights depend on whether the other person is earning more or less than they are. Furthermore, the weight that the player places on the other person's payoff *shifts down* if that person has previously been unkind toward them. Specifically, player i 's payoff from allocation (x_i, x_j) is given by

$$u_i(x_i, x_j) = (\rho r + \sigma s + \theta q)x_j + (1 - \rho r - \sigma s - \theta q)x_i$$

where $r = 1$ if $x_i > x_j$ and 0 otherwise, $s = 1$ if $x_i < x_j$ and 0 otherwise, $q = -1$ if player j has misbehaved and 0 otherwise, and ρ , σ , and θ are preference parameters. ρ and σ indicate how much weight i places on j 's payoff, where positive values indicate that i is willing to sacrifice to increase j 's payoff and negative values indicate that i is willing to sacrifice to decrease j 's payoff. Inequality-aversion preferences are captured by parameter values $\sigma < 0 < \rho$, such that the player wants to help someone who is earning less and hurt someone who is earning more than them. It is generally assumed that $\theta > 0$,

so the weight placed on the other player's payoff is shifted down when the person has been unkind.

How does this model capture the final choice in the Vengeance version of the hold-up game, where player 1 is choosing between (1, 9) and (0, 2)? First, assume that player 1 is simply choosing between these two allocations and has no prior history with the other player. In this case, $u_1(1, 9) = \sigma(9) + (1 - \sigma)(1) = 8\sigma + 1$ and $u_1(0, 2) = \sigma(2) + (1 - \sigma)(0) = 2\sigma$. The player thus prefers rejection when $2\sigma > 8\sigma + 1$ or $\sigma < -1/6$. If, however, player 1 believes that player 2 has been unkind, then the choice is between $u_1(1, 9) = (\sigma - \theta)(9) + (1 - \sigma + \theta)(1) = 8(\sigma - \theta) + 1$ and $u_1(0, 2) = (\sigma - \theta)(2) + (1 - \sigma + \theta)(0) = 2(\sigma - \theta)$, and rejection is more appealing when $\sigma - \theta < -1/6$. Since $\theta > 0$, this condition is more likely to hold than the condition without the θ term, and willingness to reject thus increases when player 2 has been unkind.

In this way, the Charness-Rabin model provides a straightforward extension that incorporates intentions into distributional preferences. *But what does it really mean for one player to believe that the other has misbehaved or treated him unkindly?* In many interactions, this assessment depends on one or both players' expectations and is thus endogenous to the strategic analysis of the game. For example, consider the game in [figure 28.7](#): player 1 can either choose Stay and the players equally split \$10 or choose Reach, in which case player 2 can choose between two lopsided allocations of the \$10. Is choosing Reach a kind or unkind action? This necessarily depends on what player 1 expects player 2 to do. If she expects player 2 to selfishly take the \$9 for himself, then it was very kind of her to let him do so—in which case he might want to think hard about his selfishness. But if player 1 chooses Reach, expecting that player 2 will feel compelled to give her \$9, then she was unkind—in which case he might feel no qualms in being selfish and choosing Take.

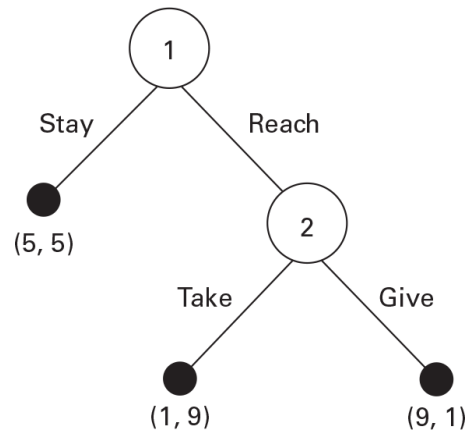


Figure 28.7
Reach!

Intuitively, reciprocating player 1's intentions then requires player 2 to surprise player 1: Give when player 1 had been expecting him to Take (and thus was being kind by letting him decide), and Take when player 2 had been expecting him to Give (and thus was being unkind by letting him decide). Since psychological equilibria require players' hierarchy of beliefs to be accurate, we thus cannot have a psychological subgame perfect equilibrium in pure strategies in a game like this, in which one player is motivated to upset the other's expectations. This game illustrates that, without being more specific about beliefs, the assessment that one player has misbehaved may be too arbitrary. We will return to its solution in exercise 3 at the end of the chapter.

Reciprocity in Simultaneous-Move Games

The first paper to use psychological game theory to model players' desire to reciprocate each other's fair and unfair behavior is Rabin (1993). In fact, this was also the first work to develop a means of deriving a psychological game (with specific psychological payoffs) from a standard game's material payoffs. Rabin's model applied to simultaneous, two-player games and took into account players' first- and second-order beliefs. Throughout, we will continue to use the notation α_j to refer to player i 's first-order belief (i.e., i 's belief about j 's strategy) and β_i to refer to i 's second-order belief (i.e., i 's belief about j 's belief about i 's strategy). Notice that the subscripts refer to whose strategy we are considering, not whose belief.

Just as the first step of incorporating guilt aversion in to the examples was to determine how much one player “let down” another, the first step to incorporating reciprocity involves defining a measure of *how kind* one player was to the other. Note that our specification regarding the players’ kindness and their payoff functions will diverge slightly from Rabin (1993)’s specification, to be consistent with later work presented in the next section. However, this divergence doesn’t affect the broad conclusions in the examples we present here.

Let i ’s kindness toward j be given by $\kappa_{ij}(s_i, \alpha_j)$ —that is, the assessment of i ’s kindness depends on both what i does (s_i) and what i believes j does (α_j). Intuitively, i ’s kindness is given by the material payoff that i believes j will actually get (as a result of their joint actions), and how this compares to other material payoffs that j *could have gotten* if i had played a different strategy (holding fixed what i thinks j is doing, α_j). The material payoff that i believes j will get as a result of i playing s_i is $\pi_j(s_i, \alpha_j)$. To assess how kind i is, we need to compare this payoff to some reference point. What should that reference point be? Rabin (1993) and its successors use a fairly simple rule of thumb: player j ’s *comparison payoff* is simply the average of the lowest and highest material payoffs that i believes j could get (given that j is playing according to α_j).

There is one twist, however, which is that if i has an inefficient strategy (that always makes everyone weakly worse off and at least one person strictly worse off), then presumably avoiding a strategy that would hurt everyone shouldn’t necessarily be assessed as a kind act. We therefore do not include these inefficient strategies when considering the minimum payoff that i could impose on j . Specifically, let E_i be the set of efficient strategies in S_i . The comparison payoff is then given by $\pi_j^c(\alpha_j) = \frac{1}{2}(\max\{\pi_j(s_i, \alpha_j) | s_i \in S_i\} + \min\{\pi_j(s_i, \alpha_j) | s_i \in E_i\})$. Finally, i ’s kindness is simply given by the difference between the payoff that i believes j gets as a result of his actions and the comparison payoff based on the other alternative material payoffs that j could have gotten if i had chosen a different (but efficient) strategy:

$$\kappa_{ij}(s_i, \alpha_j) = \pi_j(s_i, \alpha_j) - \pi_j^c(\alpha_j)$$

Let’s look at the specific example in [figure 28.8](#) to make these kindness calculations more concrete:

		Player 2	
		X	Y
Player 1	A	(5, 5)	(2, 10)
	B	(0, 15)	(10, 0)
	C	(−100, −100)	(−50, −50)

Figure 28.8
Kindness game.

Let's say that player 1 believes that player 2 will choose X . How kind are each of player 1's pure strategies? First, we compute player 2's comparison payoff. Given that player 1 believes that player 2 plays X , 1 could choose to give 2 a material payoff of 5, 15, or -100 . However, C is an inefficient strategy for player 1, which always gives both players lower payoffs, and so we don't include this when considering the minimum payoff that player 1 could give player 2. That is, avoiding the bonkers strategy of C shouldn't, in itself, be considered a kind act. We thus have that player 2's comparison payoff is $\frac{1}{2}(5 + 15) = 10$. We then compute the kindness of each of player 1's strategies as the payoff that player 2 gets relative to this comparison: $\kappa_{12}(A, X) = 5 - 10 = -5$, $\kappa_{12}(B, X) = 15 - 10 = 5$, $\kappa_{12}(C, X) = -100 - 10 = -110$.

While player i knows how kind he intended to be toward j (based on his own actions and beliefs about j 's actions), j can only guess at how kind i intended to be. This is computed in the same way as κ_{ij} , except that it is based on j 's belief about i 's strategy (i.e., α_i instead of s_i) and j 's second-order belief about her strategy (i.e., β_j instead of α_j). Let j 's assessment of i 's kindness toward her be given by

$$\tilde{\kappa}_{ij}(\alpha_i, \beta_j) = \pi_j(\alpha_i, \beta_j) - \pi_j^c(\beta_j)$$

Now we need to incorporate this kindness into the players' payoff functions. Specifically, we'll assume that the player's payoff is given by the sum of his material payoff and a psychological component based on the interaction between his own kindness to j and his belief about j 's kindness to him:

$$u_i(s_i, \alpha_j, \beta_i) = \pi_i(s_i, \alpha_j) + \theta_i \times \kappa_{ij}(s_i, \alpha_j) \times \tilde{\kappa}_{ji}(\beta_i, \alpha_j) \quad (28.)$$

Notice that the second term is positive whenever the signs of κ_{ij} and $\tilde{\kappa}_{ji}$ match (i.e., i was kind to a player who was kind to him or unkind to a player

who was unkind to him). If the signs mismatch, then i suffers disutility. We can now define a *reciprocity equilibrium* (Rabin 1993; also described in definition 2), which is a psychological equilibrium in a game where players care about reciprocity as described previously (i.e., based on their first- and second-order beliefs).

Definition 2 (Reciprocity equilibrium) *Consider a two-player, simultaneous-move game. A strategy profile (s_1, s_2) is a reciprocity equilibrium if both of the following are true:*

- (1) *The players' beliefs are correct: $s_1 = \alpha_1 = \beta_1$ and $s_2 = \alpha_2 = \beta_2$*
- (2) *Each player i 's strategy s_i is a best response given α_j and β_i . That is, $u_i(s_i, \alpha_j, \beta_i) \geq u_i(s_i', \alpha_j, \beta_i)$ for all $s_i' \in S_i$, where the player's payoff is given by equation (28.1).*

We'll now look at examples of reciprocity equilibria in two classic simultaneous-move games.

Prisoner's Dilemma

Consider the following prisoner's dilemma game in [figure 28.9](#). Is there a reciprocity equilibrium in which the players both choose Cooperate? To see this, we must calculate player i 's payoff from choosing Cooperate, given that he believes that player j will choose Cooperate ($\alpha_j = \text{Cooperate}$) and he believes that j believes that he will choose Cooperate ($\beta_i = \text{Cooperate}$), and compare this to i 's payoff if he were to instead unilaterally deviate to Defect (keeping $\alpha_j = \beta_i = \text{Cooperate}$). Notice that a player who considers deviating can change only his own strategy—he cannot also change his first- or second-order beliefs. Within the context of a player's payoff function, this means that switching to a different strategy can change the player's material payoff and how kind he intends to be to his partner, but *not* how kind he believes his partner intends to be toward him.

		Player 2	
		Cooperate	Defect
Player 1	Cooperate	(4, 4)	(0, 6)
	Defect	(6, 0)	(2, 2)

Figure 28.9
A prisoner's dilemma.

To see whether player i would want to deviate from Cooperate, let's first consider how kind player i is to player j by choosing Cooperate when $\alpha_j = \beta_i = \text{Cooperate}$. In this case, we have $\kappa_{ij}(\text{Cooperate}, \text{Cooperate}) = \pi_j(\text{Cooperate}, \text{Cooperate}) - \pi_j^c(\text{Cooperate}) = 4 - \frac{1}{2}(4 - 0) = 2$. Likewise, $\tilde{\kappa}_{ji}(\text{Cooperate}, \text{Cooperate}) = 2$. Thus, player i 's psychological payoff from Cooperate when believing that j will Cooperate and that j believes that i will Cooperate is $u_i(\text{Cooperate}, \text{Cooperate}, \text{Cooperate}) = 4 + \theta_i(2)(2) = 4 + 4\theta_i$.

To see if this is a best response, we now need to calculate how kind it is for player i to instead choose Defect when $\alpha_j = \beta_i = \text{Cooperate}$. In this case, player i 's material payoff increases to 6, but his kindness toward j changes as well, to

$$\kappa_{ij}(\text{Defect}, \text{Cooperate}) = \pi_j(\text{Defect}, \text{Cooperate}) - \pi_j^c(\text{Cooperate}) = 0 - \frac{1}{2}(4 - 0) = -2.$$

Thus, player i 's payoff from choosing Defect when believing that player j will Cooperate and that player j believes that player i will Cooperate is $u_i(\text{Defect}, \text{Cooperate}, \text{Cooperate}) = 6 + \theta_i(-2)(2) = 6 - 4\theta_i$.

Thus, Cooperate is a best response for player i when $4 + 4\theta_i \geq 6 - 4\theta_i$ or $\theta \geq 1/4$. That is, when i cares enough about reciprocity, the motive to “sign-match” j 's kindness outweighs the motive to gain a higher material payoff. Thus, we can have a reciprocity equilibrium in which both players choose Cooperate so long as $\theta \geq 1/4$ for each player. Notice that reciprocity preferences don't eliminate the (Defect, Defect) equilibrium. If a player anticipates that the other will Defect, then she thinks that he is unkind, so choosing Defect in response both maximizes her material payoff *and* reciprocates his unkindness.

Battle of the Sexes

Consider the Battle of the Sexes game in [figure 28.10](#), in which the two players are simultaneously choosing whether to go to a baseball game or a soccer game. Andrea prefers baseball, while Jeff prefers soccer, but they both prefer to be together and thus get payoffs of 0 if they mismatch on their activities. There are two pure-strategy Nash equilibria in the game, in which the players always match (i.e., (Baseball, Baseball) and (Soccer, Soccer)).

		Jeff	
		Baseball	Soccer
Andrea	Baseball	(2, 1)	(0, 0)
	Soccer	(0, 0)	(1, 2)

Figure 28.10
Baseball or soccer?

These are also both reciprocity equilibria in the psychological game. To see that (Baseball, Baseball) is a reciprocity equilibrium, let's consider whether Baseball is a best response for Andrea believing that Jeff is choosing Baseball and that Jeff believes she's playing Baseball, and vice versa. The relevant kindness functions for calculating player 1's best response are as follows:

$$\begin{aligned}\kappa_{AJ}(\text{Baseball}, \text{Baseball}) &= 1 - \frac{1}{2}(1 + 0) = \frac{1}{2} \\ \tilde{\kappa}_{JA}(\text{Baseball}, \text{Baseball}) &= 2 - \frac{1}{2}(2 + 0) = 1 \\ \kappa_{AJ}(\text{Soccer}, \text{Baseball}) &= 0 - \frac{1}{2}(1 + 0) = -\frac{1}{2}\end{aligned}$$

Andrea's payoff from Baseball is thus $2 + \theta_A(1/2)(1) = 2 + \theta_A/2$, while if she deviated to Soccer, her payoff would be $0 + \theta_A(-\frac{1}{2})(1) = -\theta_A/2$. Choosing Baseball is thus a best response for all θ_A , since switching to Soccer both yields a lower material payoff and makes Andrea feel bad for being unkind to a player who was kind to her. Likewise, Jeff's payoff from Baseball is $1 + \theta_J(1)(1/2) = 1 + \theta_J/2$, whereas if he switched to Soccer, it would be $0 + \theta_J(-1)(\frac{1}{2}) = -(1/2)\theta_J$, so Baseball is a best response. Thus, (Baseball, Baseball), and likewise (Soccer, Soccer), are both equilibria even when incorporating fairness considerations.

However, Rabin (1993) also showed that if people care enough about reciprocating each other's unkind behavior, then we could also wind up with an unintuitive reciprocity equilibrium of (Baseball, Soccer)—that is, each person stubbornly goes to their own preferred activity, and each ends up with a material payoff of 0. While this could not be a Nash equilibrium in the standard version of the game—since neither player is best responding—this can be a reciprocity equilibrium in the psychological game if people are sufficiently motivated to reciprocate unkind behavior. Why? Notice that going to your own preferred game while believing that your partner is also going to their preferred game both involves a material sacrifice (bringing your own payoff down to 0 instead of 1) and is unkind to the other person (giving them a payoff of 0 instead of 2). But if you believe that the other person is being equivalently unkind to you, then you may be willing to make this sacrifice in order to reciprocate their unkindness and hurt them in return.

To show that this type of spiteful play can be a reciprocity equilibrium, we first calculate the kindness associated with this outcome. If Andrea chooses Baseball while believing that Jeff is choosing Soccer and that Jeff believes that she is choosing Baseball, how kind is Andrea? In this case, Andrea imposes a payoff of 0 on Jeff, while the comparison payoff is $\frac{1}{2}(2 - 0) = 1$. Thus, Andrea's kindness is -1 . Likewise, Jeff's kindness from choosing Soccer while believing that Andrea chooses Baseball is also -1 . Thus, the psychological payoff for each player i from this spiteful outcome is $0 + \theta_i(-1) = -\theta_i$. Would, say, Andrea wish to deviate by switching to Soccer? In doing so, she would earn a material payoff of 1 instead of 0. What about her psychological payoff? First, she still believes that Jeff is being unkind to her (he is choosing Soccer believing that she is choosing Baseball), so she still perceives his kindness to be -1 . However, by switching to Baseball, she is being kind to him: she gives him a payoff of 2, while the comparison payoff is 1, for a kindness term of 1. Thus, Andrea's payoff from deviating to Soccer is given by $1 + \theta_A(1)(-1) = 1 - \theta_A$. Thus, choosing Baseball is Andrea's best response when $\theta_A \geq 1 - \theta_A$ or $\theta_A \geq 1/2$, and so long as each player's θ parameter is at least $1/2$, the spiteful equilibrium of (Baseball, Soccer) can arise in the psychological game.

Notice how the existence of this spiteful equilibrium alongside the others in which the players match is purely driven by the feature that the players' payoffs depend on their beliefs, not just their actions. If you were to learn that

(Baseball, Soccer) and (Soccer, Soccer) were both Nash equilibria in a standard game, you would conclude that Andrea must be indifferent between Baseball and Soccer when Jeff chooses Soccer. But that's not what's going on here. For $\theta_A \geq 1/2$, Andrea *strictly prefers* Baseball under the first equilibrium (in which she ascribes bad intentions to Jeff), and yet she *strictly prefers* Soccer under the second (in which she ascribes good intentions to Jeff). This would not be possible if Andrea's payoffs depended purely on the actions being played, not also on Jeff's intentions.

Reciprocity in Sequential Games

In standard game theory, applying the Nash equilibrium concept to sequential games can sometimes lead to implausible predictions, as we saw in chapter 8. This problem arises because the Nash equilibrium requires players to best respond only on the equilibrium path, and so in sequential games, we focus on subgame perfect Nash equilibria, which require players to best respond at all of their information sets, even those that wouldn't be reached in equilibrium.

However, there is an added wrinkle when it comes to sequential play in psychological games. At the start of the game, players have some initial beliefs about the strategies (and, therefore, the kindness) of others. But as play unfolds and players learn what the others actually do, they may need to revise their beliefs about the strategies of others and, with it, their perception of how kind they are. To address these issues, Dufwenberg and Kirchsteiger (2004) provided a model for extending reciprocity motives to sequential games.

To make the problem more concrete, consider their sequential prisoner's dilemma game in [figure 28.11](#). If we were to apply the reciprocity equilibrium concept from the previous section to this game, we would find that (Cooperate, (Cooperate, Cooperate)) is an equilibrium for sufficiently high θ . Under this equilibrium, both players believe that the other is being kind by Cooperating with them and, as such, they are willing to reciprocate Cooperation. In player 2's case, this Cooperation is unconditional: he Cooperates even at his rightmost node, after player 1 has chosen Defect, and this can be consistent with our existing definition of a reciprocity equilibrium, because player 2 would never find himself in this position anyway. But is it plausible that player 2 would really Cooperate if he were to find himself at his rightmost node? Notice that there are two issues at hand: player 2's best response off the equilibrium path and his beliefs off the equilibrium path. To

address the first, Dufwenberg and Kirchsteiger's model requires sequential rationality, such that the players best respond at all points of decisions. But if player 2 clings to his initial assessment of player 1's kindness, then Cooperate may actually be his optimal response at this node. Yet this clearly makes little sense, since at this point player 2 *knows* that player 1 has been unkind by Defecting, and this type of unconditional Cooperation is both at odds with how people actually play this game in the lab and the notion of reciprocity. Thus, the model also requires players to reevaluate their kindness assessments as the game unfolds.

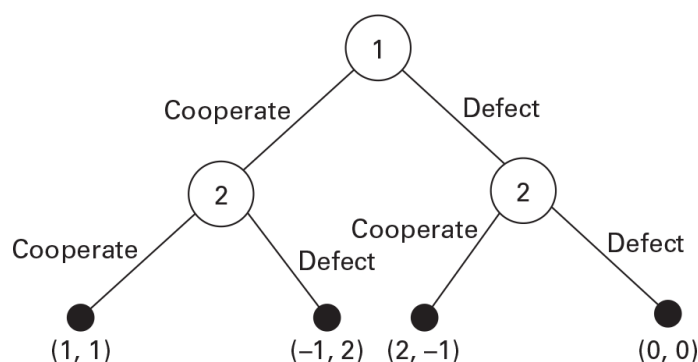


Figure 28.11

The sequential prisoner's dilemma.

In particular, Dufwenberg and Kirchsteiger's model keeps a running tally of each players' beliefs as the game is played. The model assumes that each time a new subgame is reached, the players update their beliefs about each others' strategies, so as to put probability weight of 1 on actions that have already been chosen. At each point of decision, the player updates her assessment of the other player's kindness based on her current beliefs, and chooses the optimal strategy based on this assessment. Their solution concept of *sequential reciprocity equilibrium* captures these features. We will consider the two-player version and work through a couple of examples illustrating it.

Definition 3 (Sequential reciprocity equilibrium [Dufwenberg and Kirchsteiger, 2004]) A sequential reciprocity equilibrium in a two-player, sequential-move game is a strategy profile (s_1, s_2) such that:

(1) Players have accurate first- and second-order beliefs ($s_1 = \alpha_1 = \beta_1$ and $s_2 = \alpha_2 = \beta_2$); and

(2) *Players best respond in each subgame, by maximizing the reciprocity-enhanced payoff given by equation (28.1), where first- and second-order beliefs are updated based on the history leading up to the subgame.*

Dufwenberg and Kirchsteiger (2004) proved the existence of a sequential reciprocity equilibrium in every psychological game with reciprocity motives. When $\theta_i = 0$ for all i , the sequential reciprocity equilibrium is equivalent to an SPNE in the standard game.

Sequential Reciprocity in the Prisoner's Dilemma

Let's now consider whether mutual cooperation in the sequential prisoner's dilemma can be a sequential reciprocity equilibrium. There are actually a multiplicity of sequential reciprocity equilibria in this game, which depend on the θ values of the two players and are exhaustively cataloged by Dufwenberg and Kirchsteiger (2004). However, *all of them* require that player 2 Defect after observing player 1 Defect, consistent with the intuition we just discussed. Because player 1's choice of Defect necessarily lowers player 2's payoff, this is an unambiguously unkind action, and thus player 2 has no incentive to sacrifice his own material payoff to help player 1 after observing this unkindness.

Next, let's consider whether Cooperate can be a best response for player 2 after player 1 has chosen to Cooperate. At this node, how kind are each of player 2's actions? Here, player 2 can either give player 1 a payoff of 1 or -1 , so the comparison payoff $\pi_1^c = \frac{1}{2}(1 - 1) = 0$. We therefore have that $\kappa_{21}(\text{Cooperate}) = 1 - 0 = 1$ and $\kappa_{21}(\text{Defect}) = -1 - 0 = -1$. The final piece that we need to compute player 2's payoff is to assess how kind player 2 believes player 1 was in choosing Cooperate. This depends on player 2's second-order belief, β_2 —that is what player 2 believes player 1 thought he would do. Let p represent the probability that player 2 believes player 1 assigns to player 2 choosing to Cooperate after player 1 has Cooperated. That is, we need to think about player 2's belief about player 1's belief about the probability that player 2 chooses Cooperate if they arrive at the leftmost decision node. Let this probability be given by p .

By choosing to Cooperate, player 2 believes that player 1 intends to give him the expected material payoff: $p(1) + (1 - p)(2) = 2 - p$. In contrast, if player 1 had Defected, then she would give player 2 a material payoff of 0 (since we already know that player 2 would Defect after a Defection). Thus,

player 2's comparison payoff is $\frac{1}{2}(2 - p + 0) = 1 - \frac{p}{2}$. Thus, we have that player 2's assessment of player 1's kindness is given by $\tilde{\kappa}_{12}(\text{Cooperate}, p) = \pi_2(\text{Cooperate}, p) - \pi_2^c = 2 - p - (1 - \frac{p}{2}) = 1 - \frac{p}{2}$. Putting this all together, we have player 2's full payoff from Cooperate is $1 + \theta_2 \times 1 \times (1 - \frac{p}{2})$ and from Defect is $2 + \theta_2 \times -1 \times (1 - \frac{p}{2})$. Again, notice that the player 2's material payoff and kindness change if he switches to Defect, but not his assessment of 1's kindness. Player 2 thus has a best response to Cooperate whenever $\theta_2(2 - p) \geq 1$. Of course, in equilibrium, beliefs must be correct, and so player 2 choosing Cooperate must imply that $p = 1$. Our inequality thus becomes $\theta_2 \geq 1$. We conclude that we can have a sequential reciprocity equilibrium in which player 2 chooses Cooperate if and only if $\theta_2 > 1$. By this same reasoning, we can have an equilibrium in which player 2 chooses Defect, which would require $\theta_2(2 - p) \leq 1$ and $p = 0$, or $\theta_2 \leq 1/2$. Finally, for intermediate levels of θ_2 between $1/2$ and 1 , player 2 must play a mixed strategy, which would require him to be indifferent between Cooperate and Defect (such that $\theta_2(2 - p) = 1$), implying that he chooses Cooperate with $p = \frac{2\theta_2 - 1}{\theta_2}$.

Let's focus the situation in which θ_2 is at least 1 , such that player 2's best response is (Cooperate, Defect). In this case, there is a sequential reciprocity equilibrium in which player 1 Cooperates and, furthermore, it does not require player 1 to care about reciprocity.

To see why, we first compute player 1's payoff from choosing Cooperate, while believing that player 2 chooses (Cooperate, Defect) and that player 2 believes she will choose Cooperate. In this case, player 1's action gives player 2 a material payoff of 1 rather than the 0 she could have given him, and so her kindness is $\kappa_{12} = \frac{1}{2}$. In return, player 2's kindness from Cooperating after player 1 Cooperated is 1 (calculated as stated previously). Thus, player 1's full payoff from Cooperate is $1 + \theta_1(\frac{1}{2})(1) = 1 + \frac{\theta_1}{2}$. What if player 1 were to unilaterally deviate to Defect, while still believing that player 2 is playing (Cooperate, Defect) and that player 2 believes that she is choosing Cooperate? In this case, player 1's material payoff becomes 0 and her kindness flips to $-\frac{1}{2}$. However, her assessment of player 2's kindness, $\tilde{\kappa}_{21}(\beta_1, \alpha_2) = 1$, is unaffected by 1's deviation to Defect.

This is a somewhat subtle result that is worth expanding on. You might wonder: If player 1 correctly anticipates that switching to Defect will cause

player 2 to Defect, and if player 1 assesses mutual Defection as unkind, why does player 1's payoff from switching to Defect include a positive assessment of player 2's kindness? The reason is that player 1's choice is driven by her belief of how kind player 2 intends to be *at the time player 1 moves*. This depends on what player 1 believes about player 2's strategy (which has not changed) and what 1 believes 2 believes about 1's strategy (which also has not changed). While player 1 can change her action by unilaterally deviating to Defect, she cannot unilaterally change player 2's beliefs. Given what player 1 believes player 2 believes at the time that 1 moves, she thus doesn't think that 2 is intending to harm her by playing (Cooperate, Defect).) Thus, player 1's payoff from Defect is $0 + \theta_1(\frac{1}{2})(-1) = -\frac{\theta_1}{2}$. We can then conclude that player 1's payoff from Cooperate is higher than Defect for all θ_1 . If player 2 always reciprocates, then player 1 earns a higher material payoff by Cooperating. Even if player 1 doesn't care about reciprocity ($\theta_1 = 0$), she prefers to Cooperate, and this motive only becomes stronger if she cares about reciprocity.

In addition, notice that there is a sequential reciprocity equilibrium in which player 1 *does* care about reciprocity and Defects even though she accurately anticipates that player 2 is playing (Cooperate, Defect). This may sound counterintuitive at first, but the basic idea is that player 1 believes that player 2 believes she'll Defect, and thus player 1 assesses player 2's strategy of (Cooperate, Defect) as being unkind in this situation. This is because, after player 1 Defects, player 2 could give her a payoff of 2 but gives her 0 instead. Even though player 1 would earn a higher material payoff from Cooperating, she chooses Defect to punish player 2 *in advance* for this presumed transgression. Specifically, if player 1 believes that player 2 is playing (Cooperate, Defect), and that player 2 believes she is choosing Defect, then her payoff from Defect is $0 + \theta_1(\frac{-1}{2})(-1) = \frac{\theta_1}{2}$, while her payoff from Cooperate is $1 + \theta_1(\frac{1}{2})(-1) = 1 - \frac{\theta_1}{2}$. Player 1 thus prefers to Defect when $\theta_1 > 1$.

Sequential Reciprocity in the Mini-Ultimatum Game

As another example of sequential reciprocity equilibria, consider the mini-ultimatum game shown in [figure 28.12](#). Player 1 can offer a Fair split of \$10, in which case each player's material payoff is \$5. Alternatively, player 1 can be Greedy and offer only \$1. Player 2 can Accept or Reject, in which case they both get nothing. In the standard analysis of this game, there is a single

SPNE of (Greedy, Accept). However, let's see if there are sequential reciprocity equilibria in which the second-mover Rejects instead.

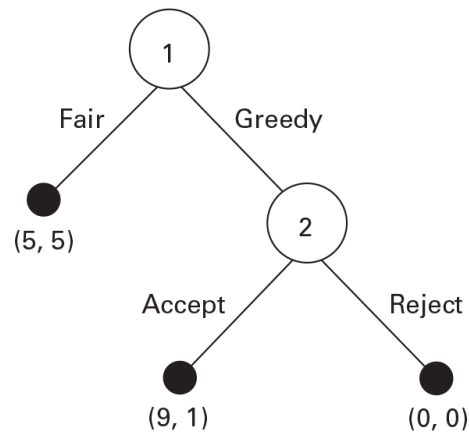


Figure 28.12

The mini-ultimatum game.

Let's start with the second-mover's choice. She can elect to Accept and get a material payoff of \$1, or Reject and get a material payoff of \$0. Thus, she would choose Reject only if she is sufficiently motivated by reciprocity. How kind would each of these choices be? Notice that Reject is an inefficient choice that hurts both players. Since Accept is the only efficient choice, the comparison payoff for player 1 is simply equal to the \$9 he would get if she accepts (since \$9 is both the maximum payoff that she can give him and the minimum payoff associated with an efficient choice). Choosing Accept is thus neither kind nor unkind—it is simply the only efficient option (i.e., $\kappa_{21}(\text{Accept}) = \pi_1(\text{Accept}) - \pi_1^c = 9 - 9 = 0$). On the other hand, Rejecting is clearly unkind: $\kappa_{21}(\text{Reject}) = \pi_1(\text{Reject}) - \pi_1^c = 0 - 9 = -9$).

Whether player 2 wants to sacrifice her own material payoff to hurt player 1 depends on how kind she believes him to be at the moment she makes her choice. Whatever she thought of player 1's kindness at the start of the game, once he chooses Greedy, she knows that he elected to give her a material payoff of \$1 or \$0 instead of \$5, and now she believes him to be unkind. Exactly *how kind* she believes him to be depends on her second-order beliefs—whether she thinks he made his choice expecting her to Accept (and wind up with \$1) or Reject (and wind up with \$0). Let p be player 2's second-order belief about the probability she'll Accept. In this case, player 1's choice of Greedy gives player 2 an expected material payoff of $p(1) + (1 - p)(0) = p$.

Alternatively, he could have given her a payoff of \$5 by choosing Fair, and so her comparison payoff is $\frac{1}{2}(5 + p) = 2.5 + \frac{p}{2}$. Thus, player 2 assesses player 1's kindness after choosing Greedy to be $\tilde{\kappa}_{12}(\text{Greedy}, p) = p - (2.5 + \frac{p}{2}) = \frac{p}{2} - 2.5$.

Putting this all together, player 2's choice involves a comparison of $u_2(\text{Accept}) = 1 + \theta_2(0)(\frac{p}{2} - 2.5) = 1$ and $u_2(\text{Reject}) = 0 + \theta_2(-9)(\frac{p}{2} - 2.5) = \frac{45\theta_2 - 9p}{2}$. She will therefore choose Reject whenever $\frac{45\theta_2 - 9p}{2} > 1$. In equilibrium, her second-order belief must be correct—that is, if she chooses Reject in equilibrium, then we must have $p = 0$, and our condition collapses to $\frac{45\theta_2}{2} > 1$. Thus, we can have player 2 choose Reject as part of a sequential reciprocity equilibrium if $\theta_2 > \frac{2}{45}$.

Let's now return to player 1's choice while assuming that $\theta_2 > \frac{2}{45}$, and so player 2 is playing Reject. Player 1 thus chooses between giving himself a material payoff of \$5 (Fair) or \$0 (Greedy, anticipating rejection). If player 1 is not motivated by fairness intentions ($\theta_1 = 0$), then he will, somewhat ironically, choose Fair, simply out of self-interest.

What if he does care about fairness? In making his choice, he is choosing between a material payoff of \$5 or \$0 for player 2. Player 2's comparison payoff is thus \$2.50 and the choice of Fair is kind ($5 - 2.5 = 2.5$), while Greedy is unkind ($0 - 2.5 = -2.5$). Finally, we need to assess how kind he believes player 2 is toward him by choosing Reject. Note that even though player 1 believes player 2 is choosing Reject, he doesn't view this choice as unambiguously unkind at the time he makes his choice—instead, his assessment of her kindness depends on what he believes she believes he will do. Let q be player 1's second-order belief attached to his choice of Fair. If $q = 1$, then he doesn't think that player 2 believes that she is influencing his payoff at all by choosing Reject—thus, he thinks that she is neither kind nor unkind. It is only when $q < 1$ that he thinks of her as unkind, and her kindness is decreasing as q goes to 0. Specifically, by choosing Reject, he believes that she believes that she is giving him an expected material payoff of $q(5) + (1 - q)(0) = 5q$. If she instead chose Accept, then this expected material payoff would be $q(5) + (1 - q)(9) = 9 - 4q$. His comparison payoff is thus $\frac{1}{2}(9 - 4q + 5q) = 4.5 + \frac{q}{2}$. Player 1's perception of player 2's kindness is thus given by $\tilde{\kappa}_{21}(\text{Reject}) = 5q - 4.5 - \frac{q}{2} = 4.5q - 4.5$. Consistent with our intuition, notice that this goes to 0 when $q = 1$, such that player 1 didn't think player 2 would think she was making a choice. If 1 thinks that 2 didn't anticipate acting, 1 doesn't assess her strategy as being either kind or unkind. In

addition, notice that this term goes to -4.5 when $q = 0$. In that case, 1 believes that 2 intentionally is trying to give him \$0 instead of \$9 with certainty. Putting together player 1's material payoff, kindness toward player 2, and belief about player 2's kindness toward him, we have $u_1(\text{Fair}) = 5 + \theta_1(2.5)(4.5q - 4.5) = 5 + 11.25\theta_1(q - 1)$ and $u_1(\text{Greedy}) = 0 + \theta_1(-2.5)(4.5q - 4.5) = 11.25\theta_1(1 - q)$. The choice of Fair is a best response when $5 > 22.5\theta_1(1 - q)$.

Is there a sequential reciprocity equilibrium in which player 1 chooses Fair? Since beliefs are correct in equilibrium, we must have $q = 1$, and our condition for choosing Fair becomes $5 > 0$. We can thus have an equilibrium of (Fair, Reject) for all values of θ_1 . This makes sense because player 1 believes that player 2 believes that she isn't influencing his payoff in equilibrium, and therefore he doesn't view her as being kind or unkind at the time he moves. As a result, he acts to maximize his material payoff by choosing Fair to avoid Rejection. Is there a sequential reciprocity equilibrium in which player 1 chooses Greedy? For equilibrium beliefs to be correct, we must have $q = 0$ in this case, and our condition for player 1 to prefer Greedy is $5 < 22.5\theta_1(1 - 0)$ or $\theta_1 > 5/22.5$. Similar to the (Baseball, Soccer) reciprocity equilibrium in the Battle of the Sexes, this sequential reciprocity equilibrium is associated with a mutually destructive outcome, in which both players accurately anticipate each other's bad intentions and sacrifice both their payoffs in order to retaliate. From the start of the game, player 2 believes that player 1 is going to be Greedy, and she plans to hurt him in response. Therefore, at the time player 1 moves, he surmises that player 2 has bad intentions and *preemptively* hurts her in retaliation.

The Hold-up Problem with Vengeful Players

There is some debate in the literature regarding whether positive reciprocity is as strong a motive as negative reciprocity (e.g., Charness and Rabin, 2002). In the hold-up game experiment, in particular, it seems that negative reciprocity is the principal driver. Both the No Recourse and Vengeance versions offer similar opportunities for player 2 to reciprocate player 1's trust by making good on the offer, and yet it is only in the Vengeance version—where player 1 has the opportunity to take revenge—that player 2 Honors with any real frequency and most players 1 are willing to choose In.

Dufwenberg, Smith, and van Essen (2013) thus proposed that behavior in their game may be consistent with a *vengeance equilibrium*, which is the same solution concept as the sequential reciprocity equilibrium except that players are only motivated to reciprocate unkind behavior. That is, instead of caring equally about their perception of the other player's kindness or unkindness, the players only care about their perception of their partner's unkindness. To capture this, let $\tilde{\kappa}_{ij}^- = \min\{0, \tilde{\kappa}_{ij}\}$. Now let's assume that the players' psychological payoff is given by the same expression we've been using (equation 28.1) except that we replace $\tilde{\kappa}_{ij}$ with $\tilde{\kappa}_{ij}^-$:

$$u_i(s_i, \alpha_j, \beta_i) = \pi_i(s_i, \alpha_j) + \theta_i \times \kappa_{ij}(s_i, \alpha_j) \times \tilde{\kappa}_{ji}^-(\beta_i, \alpha_j)$$

The second term thus comes into play only when i perceives j as unkind; if he thinks that she has been kind, then the second term goes to zero, and his only concern is maximizing his material payoff.

What are the vengeance equilibria in the hold-up game? In the No Recourse version, the only vengeance equilibrium is the same as the standard SPME: ((Out, Accept), Renege). Why? First notice that player 1 has no incentive to choose anything other than Accept in the final move of the game, since this action is consistent with both maximizing his material payoff *and* punishing player 2's unkindness. What about player 1's choice at the start of the game? Given that equilibrium beliefs must be correct, he assumes that player 2 correctly anticipates that he is choosing Out, and thus he doesn't think that she believes she is affecting his payoff by choosing Renege (since she won't move anyway). As a result, he does not assess her as being either kind or unkind, and so he simply acts to maximize his material payoff by choosing Out (to get a payoff of 2 rather than 0). Finally, what about player 2's choice at the second node? Conditional on reaching this node, she believes he chose In with certainty, which could only be considered a kind act, and his subsequent choice of Accept is neither kind nor unkind. Since player 2 doesn't attribute any bad behavior to player 1, she simply acts to maximize her material payoff, by choosing Renege. Even if player 1 were to surprise her with his kindness by choosing In, she still wouldn't Honor because she only reciprocates unkind behavior.

In the Vengeance version, in contrast, player 1 will choose Reject in equilibrium if he is sufficiently vengeance-minded and the more efficient strategy profile ((In, Reject), Honor) can be a vengeance equilibrium.

Let's first consider when player 1 would choose to Reject. With respect to his material payoff, Accept gives him a payoff of 1 and Reject gives 0. With respect to his kindness toward player 2, he only has one efficient strategy, such that the comparison payoff for player 2 is 9. As in the mini-ultimatum game, Accept is neither kind nor unkind, while Reject is clearly unkind ($\kappa_{21} = -7$). Finally, to assess how kind he believes player 2 is, we need to consider his second-order belief that he will accept—let's call it q . By choosing Renege, he thus believes that she expected to give him a material payoff of $q(1) + (1 - q)(0) = q$. In other words, if she expects him to Reject with certainty, then she expects to give him a material payoff of 0 and if she expects him to Accept with certainty, her choice is associated with giving him a material payoff of 1. However, she could have given him a material payoff of 5 by choosing Honor, so his comparison payoff is $\frac{1}{2}(5 + q)$ and his assessment of her kindness is $q - \frac{5+q}{2} = \frac{q-5}{2}$. Putting this together, we get that player 1's payoffs are $u_1(\text{Accept}) = 1 + \theta_1(0)(\frac{q-5}{2}) = 1$ and $u_1(\text{Reject}) = 0 + \theta_1(-7)(\frac{q-5}{2}) = \frac{(5-q)(7\theta_1)}{2}$. If player 1 chooses Reject in equilibrium, then we must have $\frac{(5-q)(7\theta_1)}{2} > 1$ and, since beliefs must be correct, $q = 0$, yielding the condition $\theta_1 \geq 2/35$.

Let's thus assume that $\theta_1 \geq 2/35$ and see why ((In, Reject), Honor) is a vengeance equilibrium in this case. First, consider player 1's initial choice of In or Out. In equilibrium, he believes that player 2 is choosing Honor and thus is kind. Since he doesn't attribute any bad intentions to her, he simply acts to maximize his own material payoff, which he correctly anticipates is higher from choosing In (5) than Out (2). At player 2's decision node, she likewise does not attribute any bad behavior to player 1. Her equilibrium second-order beliefs are that she believes that player 1 believes that she will play Honor, and thus his only actual effect on her payoff was to give her 5 (by choosing In) rather than 2 (by choosing Out). Since, in equilibrium, player 1 doesn't actually have to follow through and Reject, this part of his strategy doesn't harm player 2, and she doesn't attribute unkindness to him. She thus acts to maximize her own material payoff by choosing Honor (5) instead of Renege (2). We therefore find that player 1's vengeance motives—along with the opportunity to take revenge that exists in the Vengeance version of the game—can prevent player 2 from Reneging on the deal, and thus make player 1 willing to choose In and make the deal in the first place.

There is also a more unfortunate vengeance equilibrium of ((Out, Reject), Renege). While we leave details for exercise 6 at the end of this chapter, the intuition is similar to the perverse equilibrium of (Greedy, Reject) that we just found in the mini-ultimatum game: both players attribute unkind motives to each other, which in turn incentivizes them to behave unkindly themselves. The only thing that prevents the existence of a vengeance equilibrium in which player 1 plays In and the scorched-earth outcome plays out on the equilibrium path, is that player 1 recognizes that player 2 gets her lowest payoff of 2 regardless of whether he chooses Out initially or ultimately Rejects, so he can punish her just as effectively—and at lower cost to himself—by playing Out.

We have thus seen that incorporating negative reciprocity motives into the hold-up game explains the laboratory evidence that the ability to take revenge can lead to more efficient outcomes than predicted by the standard theory. More broadly, we have seen that psychological game theory provides a flexible framework for incorporating a vast array of psychological motives that are based on players' beliefs. We anticipate that this framework will be a particularly productive avenue for behavioral game theory research in the years ahead, which will allow researchers to develop more standardized methods for analyzing strategic interactions that include reciprocal motives, emotions, image concerns, and the desire to adhere to social norms.

Exercises

(1) Consider a rebellious high school student who is deciding whether to study for her first math exam. If she studies, she will ace the test. Otherwise, she will get a C. However, she is also concerned about upending the teacher's expectations. Whatever he expects of her studiousness, she wants to prove him wrong. Let p be the probability that the student chooses to study, q be the teacher's belief that the student studies, and r be the student's second-order belief. Then the student's payoffs are given by $u_1(\text{Study}) = 5 + \theta_1(1 - r)$ and $u_1(\text{Don't Study}) = 3 + \theta_1 r$.

- (a) For what range of θ can there exist a psychological equilibrium in which the student chooses to Study with probability 1?
- (b) For what range of θ is there only a psychological equilibrium in which the student plays a mixed strategy? Find her equilibrium mix (as a function of θ).

(2) Consider the game of chicken shown in [figure 28.13](#), in which two players simultaneously choose between Hang In and Bail. Both (Hang In, Bail) and (Bail, Hang in) are Nash equilibria. We will now see that if the players are sufficiently concerned with fairness, these cannot be reciprocity equilibria.

		Player 2	
		Hang In	Bail
Player 1	Hang In	$(-2, -2)$	$(2, 0)$
	Bail	$(0, 2)$	$(1, 1)$

Game of chicken

Figure 28.13

- (a) Calculate the kindness of player 1 toward player 2 for each of the four outcomes. That is, find $\kappa_{12}(\text{HangIn}, \text{HangIn})$, $\kappa_{12}(\text{Bail}, \text{HangIn})$, $\kappa_{12}(\text{HangIn}, \text{Bail})$, and $\kappa_{12}(\text{Bail}, \text{Bail})$.
- (b) Show that (Hang In, Bail) cannot be a reciprocity equilibrium if either player is sufficiently concerned with fairness. How high does θ_1 need to be for player 1 to prefer to unilaterally deviate to “Bail”? How high does θ_2 need to be for player 2 to prefer to unilaterally deviate to “Hang In”?
- (3) Consider again the Stay or Reach game depicted in [figure 28.7](#). Assume that player 2 has the reciprocal preference parameter θ_2 . Show that for low levels of θ_2 , the only behavior for player 2 that could hold in a sequential reciprocity equilibrium is Take, while for higher levels of θ_2 , player 2 must play a mixed strategy.
- (4) Consider the sequential prisoner’s dilemma described in [figure 28.11](#). We saw that, so long as $\theta_2 > 1$, player 2 prefers to match player 1’s cooperation. In this case, there is a pure-strategy sequential reciprocity equilibrium in which player 1 always Cooperates (regardless of θ_1) and a pure-strategy sequential reciprocity equilibrium in which player 1 always Defects (if $\theta_1 > 1$). There is also an equilibrium in which player 1 plays a mixed strategy. What is the range of θ_1 for which this mixed-strategy sequential reciprocity equilibrium exists, and with what probability does player 1 cooperate?

(5) Consider the following trust game shown in figure 28.14 (which also resembles the hold-up problem if we take as a given that player 1 Accepts the offer). For what range of θ_1 , θ_2 is (In, B) a sequential reciprocity equilibrium?

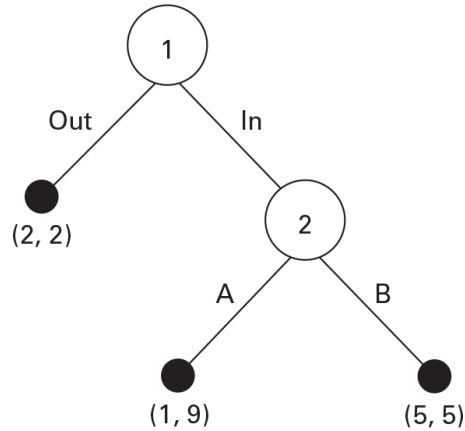


Figure 28.14

(6) In the hold-up game, we saw that player 1 prefers to reject whenever $\theta_1 > 2/35$ and there is a vengeance equilibrium of ((In, Reject), Honor). Let's now show that there is also a vengeance equilibrium of ((Out, Reject), Renege).

- Show that Out is a best response for player 1 at the initial node, given correct first- and second-order beliefs regarding the remainder of the strategy profile (i.e., player 2 will Renege and player 1 will Reject).
- Show that Renege is a best response for player 2 at the second node, given correct, updated first- and second-order beliefs at this node.
- Finally, let's see why ((In, Renege), Reject) is *not* a vengeance equilibrium. Show that In is not a best response for player 1 at the initial node, given correct first- and second-order beliefs regarding the remainder of the strategy profile (i.e., player 2 will Renege and player 1 will Reject).

Appendix

The main text of this book assumes that students have had some experience with set notation and have taken a first course in calculus, and thus know what the intersection of two sets is, how to optimize a function of a single variable, and how to take a simple integral; and that they have some familiarity with probability. For students who do not feel confident in their mastery or recollection of this material, this appendix provides all the mathematical background needed to understand the concepts in the book, and also expands on some of the material within the text regarding optimizing a function of multiple variables using partial derivatives, the rules of probability, choice under risk, and intertemporal choice, for those who would find a more in-depth treatment beneficial.

Set Notation

A *set* is a collection of any number of distinct objects (i.e., each specific object need be listed only once in a set). Sets are very flexible constructs and can contain numbers, physical objects like cars, or soccer clubs like Liverpool FC and Leeds United. For our purposes, sets will typically contain players like Buyer and Seller, actions like opt in or go left, or full strategies like tit-for-tat and trigger. Each of these examples are *elements* of a set. In terms of notation, set elements are separated by commas and encompassed by brackets. For example, here is the set of coffee shops in town, denoted as C :

$$C = \{\text{Royal Oak}, \text{Carol's Hungry Mind}, \text{Vermont Coffee Company}, \text{Haymaker}\}$$

In some cases, sets can include a large number of items like the 50 best small coffee roasters in the Northeast, which can be abbreviated as $R = \{1, 2, 3, \dots, 50\}$, and in other cases, the set can be large and *finite* but with a flexible maximum, like the number of players in an *n-person game*, the set $N = \{1, 2, 3, \dots, n\}$. Finally, sets can also have an infinite number of elements, like the set of positive integers (aka natural numbers), which can be expressed as $\mathbb{N} = \{1, 2, 3, \dots\}$ or $\mathbb{N} = \{n: n > 0 \text{ and is an integer}\}$.

In cases in which strategies are continuous, such as being asked to pick any number strictly between zero and 1, we have an *interval* set, which can be represented as $\{x \mid 0 < x < 1\}$, and if we want to include the end points, we

would have $\{x \mid 0 \leq x \leq 1\}$. As a shorthand for these sorts of intervals, we can also just use the beginning and ending points of the interval and parentheses or square brackets. The first set given here can be abbreviated further as $(0, 1)$, and the second one as $[0, 1]$.

There are also various basic *set operations*. For instance, the *union* of two sets, A and B , is a new set that contains all the distinct elements from the two sets and is denoted as $A \cup B$. Hence, if $A = \{a, b, c\}$ and $B = \{3, 6, 9\}$, then $A \cup B = \{a, b, c, 3, 6, 9\}$. If we looked at the union of A and $B' = \{b, 3, 6, 9\}$ instead, it would be the same. The *intersection* of two sets is the set of all the common elements. Say that $C = \{a, 9, x, 3, q\}$, then the intersection of sets A and C , denoted as $A \cap C$, is $\{a\}$, and the intersection of sets B and C , $B \cap C$, is $\{3, 9\}$. Notice, however, that the set $A \cap B$ is empty because the two sets have no common elements. In this case, we can use the notation for the empty set such that $A \cap B = \emptyset$. Finally, when one larger set contains all the elements of a smaller (or equal-sized) set, then the second set is known as a *subset* of the first. For example, say that set $W = \{1, 2, 3, 4, t, y, u, i\}$ and set $V = \{2, t, y, u\}$, then V is a subset of W , which we write as $V \subset W$.

The *Cartesian product* of two sets, written as $A \times B$, is a new set that contains all the ordered pairs (a, b) , where a is an element of set A and b is an element of set B . The shorthand for being an element is $a \in A$ and $b \in B$. Considering the sets A and B in our example,

$$A \times B = \{(a, 3), (a, 6), (a, 9), (b, 3), \dots, (c, 9)\}$$

and this Cartesian product contains 3 A elements $\times$ 3 B elements, or 9 pairs.

Optimizing a Function of a Single Variable

Standard game theory is built on rational choice theory. Rational decision makers are always able to rank their possible outcomes from most to least preferred (with no cycles), such that we can assign numerical values called *utilities* or *payoffs* to each outcome, with higher utilities associated with more preferred outcomes. A rational decision maker then optimizes by acting in a way that maximizes the payoff.

In some cases, the decision maker's payoff may be a continuous function of one or more variables. For example, consider a firm whose profits depend on its production quantity, a student whose payoff depends on how much effort she puts into each of her courses, or a politician whose payoff depends on how much money he spends on social media advertising.

Let's imagine that your payoff, u , is a function of how many ounces of espresso you drink today, x . Specifically, let $u = u(x) = 100x - 5x^2$. The graph in [figure A.1](#) shows how your payoff changes as we vary x . As a rational decision maker, you want to drink the amount of espresso that maximizes your payoff. Without knowing this function, you'd likely approach the problem through trial and error. Perhaps on the first day you drink $x_1 = 5$, and you observe that your happiness is 375. Not bad, but maybe you can do better. So the next day, you expand your consumption to $x_2 = 7$ and observe that you're more alert and productive—your payoff goes up to 455. In other words, your change in x (given by $\Delta x = x_2 - x_1 = 2$) was positive and resulted in a positive change in payoff ($\Delta u = 455 - 375 = 80$). As a result, $\Delta u / \Delta x > 0$. So long as this is the case, you can improve your payoff by increasing your consumption.

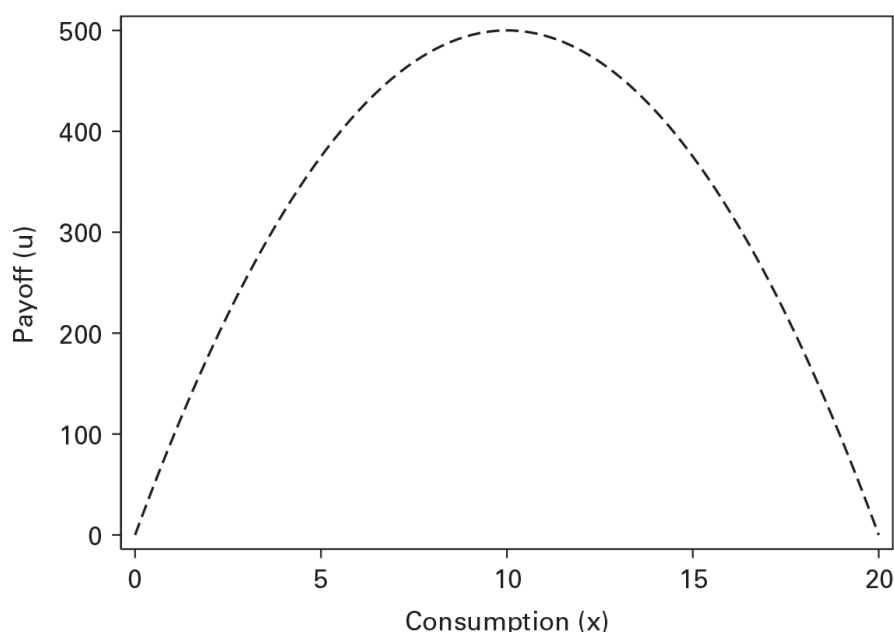


Figure A.1

Payoff as a function of espresso consumption.

Since increasing your espresso input worked so well yesterday, perhaps you go crazy and drink 18 ounces the next day. Now you find that you're jittery and nauseous, and your payoff drops to 180. That is, your change in espresso intake was positive, but the change in payoff was negative. In other words, $\Delta u / \Delta x < 0$. This indicates that you overshoot (so to speak), and you can improve your payoff by scaling back on your espresso consumption.

So now what do you do? You'll want to dial in on the maximum using smaller and smaller values of Δx . And the spot you want to dial in on is the point at which, for infinitesimally small changes in x , $\Delta u/\Delta x$ is equal to zero. It is only at that point that, if you increase espresso intake slightly, you won't be happier, and if you decrease your consumption slightly, you also won't be happier. So what we're looking for is x^* , such that this slope, the derivative of the function at x^* is zero. The concept of a derivative is explained in definition 1.

Definition 1 (Derivative) *The derivative of a function is the instantaneous rate of change. Let y be a function of x . The derivative of $y = f(x)$ with respect to x measures how y changes with changes in x and is given by*

$$\frac{dy}{dx} = f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

The derivative of a function may be a function itself, and the derivative defined at point x_0 is

$$\left. \frac{dy}{dx} \right|_{x=x_0} = f'(x_0) = \lim_{h \rightarrow 0} \frac{f(x_0+h) - f(x_0)}{h}$$

Now we're equipped to say, formally, what we realized about your approach to espresso consumption. If x^* is a maximum, then it had better be the case that the derivative of $f(x)$ at x^* is zero. If it is positive, we know that we could increase your payoff by increasing consumption slightly, and if it is negative, we know that we could increase your payoff by decreasing consumption slightly. This requirement is called the *first-order condition for a maximum* (see definition 2).

Definition 2 (First-order condition for a maximum) *If a function f attains its maximum at x^* , then*

$$\left. \frac{df}{dx} \right|_{x=x^*} = 0$$

To find the maximum, we must thus take the derivative of our function and find the x such that the derivative is zero. The rules for finding derivatives are listed here for your convenience:

Constant rule. Let c be a constant. Then the derivative of c is given by

$$\frac{d}{dx}[c] = 0.$$

Constant multiple rule. Let c be a constant. Then the derivative of cx is given by $\frac{d}{dx}[cx] = c$.

Power rule. Let c be a constant such that $c \neq 0$. Then the derivative of x^c is given by $\frac{d}{dx}[x^c] = cx^{c-1}$.

Sum rule. Let $f(x)$ and $g(x)$ be functions of x . Then the derivative of $f(x) + g(x)$ is given by $\frac{d}{dx}[f(x) + g(x)] = f'(x) + g'(x)$.

Difference rule. Let $f(x)$ and $g(x)$ be functions of x . Then the derivative of $f(x) - g(x)$ is given by $\frac{d}{dx}[f(x) - g(x)] = f'(x) - g'(x)$.

Product rule. Let $f(x)$ and $g(x)$ be functions of x . Then the derivative of $f(x) \cdot g(x)$ is given by $\frac{d}{dx}[f(x) \cdot g(x)] = f(x)g'(x) + f'(x)g(x)$.

Quotient rule. Let $f(x)$ and $g(x)$ be functions of x . Then the derivative of $\frac{f(x)}{g(x)}$ is given by $\frac{d}{dx}\left[\frac{f(x)}{g(x)}\right] = \frac{f'(x)g(x) - f(x)g'(x)}{[g(x)]^2}$.

Logarithmic function rule. The derivative of $\ln(x)$ is given by $\frac{d}{dx}[\ln(x)] = \frac{1}{x}$.

Exponential function rule. The derivative of a^x is given by $\frac{d}{dx}[a^x] = a^x \ln(a)$.

Chain rule. The chain rule allows us to see how one variable affects another through an intermediary variable. Let f be a function of g , which is, in turn, a function of x . The derivative of $f(g(x))$ is given by $\frac{d}{dx}[f(g(x))] = f'(g(x)) \cdot g'(x)$.

To find the optimal x in our example, we first take the derivative of $u = 100x - 5x^2$ with respect to x . In doing so, we have $du/dx = 100 - 10x$. We then set the derivative equal to 0 and solve for the x^* that satisfies our first-order condition: $100 - 10x^* = 0$, or $x^* = 10$.

Does this tell us that your payoff is maximized when espresso consumption equals 10? Not quite: It tells us that if there is, in fact, a level of espresso that maximizes your payoff, then 10 is the only candidate for that maximum. It is possible, though, that our function might not have a maximum, and we could in fact be at a minimum, which would be a bummer for you as the espresso drinker. This is because the first-order condition for a minimum is exactly the same—the derivative of the function is also zero at its minimum. This concept is explained further in definition 3.

Definition 3 (First-order condition for a minimum) *If a function f attains its minimum at x^* , then*

$$\left. \frac{df}{dx} \right|_{x=x^*} = 0$$

To know that we're truly at a maximum, we need to check the second-order condition (see definitions 4 and 5), which we do by calculating the second derivative. The second derivative is the derivative of the derivative. In other words, the derivative tells you the slope as you move along the graph; the second derivative tells you how the slope changes. To calculate the second derivative, you take the derivative, and then you do it again. The second derivative of f with respect to x is written as $\frac{d^2f}{dx^2}$ or $f''(x)$.

Definition 4 (Second-order condition for a maximum) *If a function f attains its maximum at x^* , then*

$$\left. \frac{d^2f}{dx^2} \right|_{x=x^*} < 0$$

Definition 5 (Second-order condition for a minimum) *If a function f attains its minimum at x^* , then*

$$\left. \frac{d^2f}{dx^2} \right|_{x=x^*} > 0$$

If the second derivative is negative (and our first derivative was zero), then we know we're at a maximum. If it is positive, we know we're at a minimum. So why does this make sense? Maybe it seems sort of backward that we're looking for a negative derivative when we're at a maximum. But to see why this is, let's continue our coffee example in [figure A.2](#), with the maximum at $x^* = 10$. If we're to the left of x^* , then payoff is increasing in x , so the derivative over this region is positive. Exactly at x^* , we know that the derivative is zero. To the right of x^* , the derivative is negative. Remember that the second derivative is telling us how the first derivative changes. Well, we've gone from a positive derivative, to zero, to negative. So the derivative is decreasing, which means that the derivative of the derivative—the second derivative—is negative if we're at a maximum.

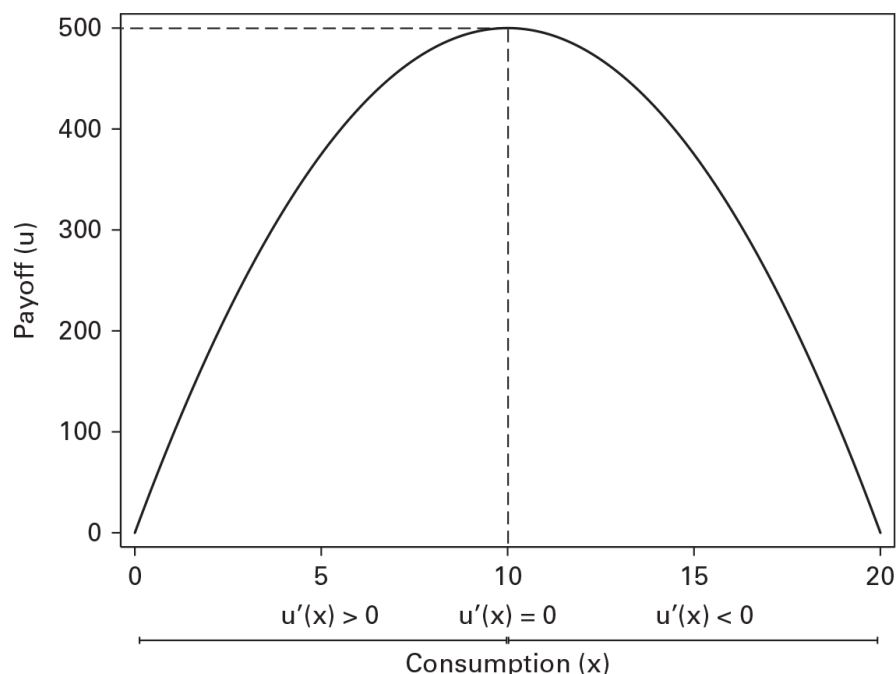


Figure A.2

The maximum occurs at the point where the first derivative is zero and the second derivative is negative.

In our example, the second derivative is $d^2u/dx^2 = -10 < 0$, so we know that our function is maximized at $x^* = 10$.

Optimizing a Function with Variables out of the Decision Maker's Control

In the previous section, we saw how to optimize a payoff function of a single variable, such as espresso consumption. More generally, the decision maker's payoff may be a function of multiple variables. However, not all these variables are necessarily under the decision maker's control. This is particularly relevant in game theory, where players' outcomes depend on the decisions of others.

Sticking with our espresso example, let's imagine that your payoff depends both on the amount of espresso that you drink, x , and the percentage of cloud cover that day, c . Specifically, let $u = 100x - 5x^2 - 10c + cx$.

When considering how to maximize this function, there are two important things to note. First, the amount of cloud cover is outside the decision maker's control. You thus still have only one lever for influencing your happiness, which is your espresso consumption, and there is nothing that you can do

about the weather (without moving somewhere else). Second, although you can't change c , it still is an important part of your own decision, since the optimal level of espresso varies depending on the cloudiness. We can see this in [figure A.3](#), which depicts graphs of your payoffs for $c = 0$, $c = 50$, and $c = 100$. As we can see, the level of espresso that maximizes your payoff is greater the cloudier it is outside: on a cloudless day, the optimal level of espresso is 10, but it rises to 15 on a partly cloudy day, and to 20 on a day that's completely soaked in. To figure out your optimal consumption more generally, for any level of cloud cover c , we can take the derivative of your payoff function with respect to x , holding c constant, which is called the *partial derivative* of f with respect to x (see definition 6).

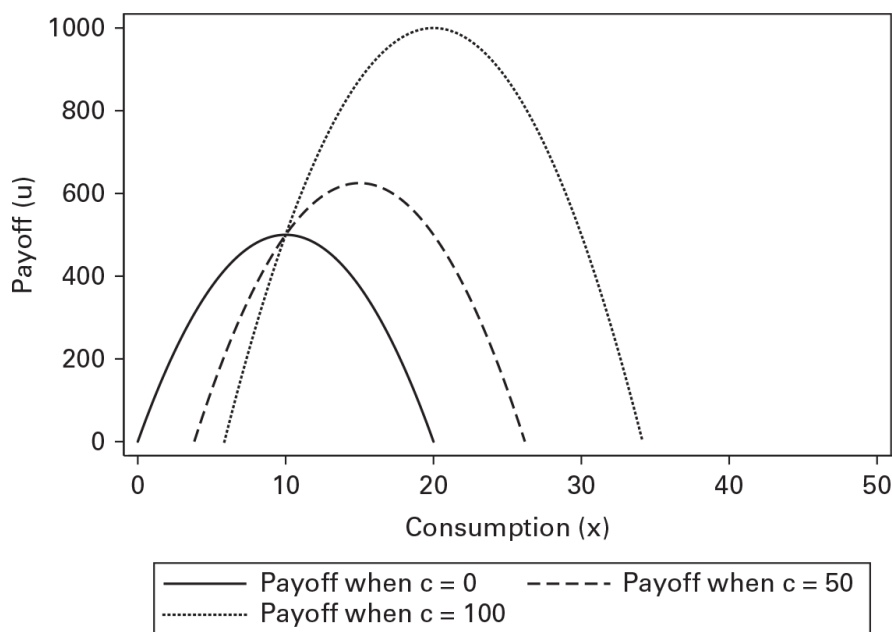


Figure A.3

Payoff as a function of espresso consumption at different levels of cloud cover.

Definition 6 (Partial derivative) Let f be a function of n variables $(x_1, x_2, \dots, x_n)$. The partial derivative of f with respect to x_1 is the instantaneous rate of change moving in the direction of x_1 . That is, it measures how f changes with changes in x_1 while holding constant x_2 through x_n and is written as $\frac{\partial f}{\partial x_1}$.

Perhaps you've heard of the expression *ceteris paribus*, meaning that all other conditions remain the same. The partial derivative is the mathematical representation of this. It tells us how a change in one variable affects our

outcome, holding our other variables constant. In our example, the partial derivative of u with respect to x , $\partial u / \partial x$ tells us the rate at which our payoff increases when we scale up our espresso consumption, holding constant the cloud cover for the day. To take the partial derivative with respect to x , we simply treat x as our variable and c as we would any other constant, such as 4 or π . We thus have

$$\frac{\partial u}{\partial x} = 100 - 10x + c$$

Now if we set the partial derivative equal to zero and solve for x , we find the optimal level of espresso consumption for any type of day: $100 - 10x^* + c = 0$, or $x^* = 10 + c/10$. As we saw in [figure A.3](#), when $c = 0$, $x^* = 10$, when $c = 50$, $x^* = 15$, and when $c = 100$, $x^* = 20$. But instead of starting with some level of c and optimizing each of these functions separately, the expression tells us the optimal x^* for any weather conditions we could encounter. In other words, once you check the weather forecast or otherwise form beliefs about c , then you know your optimal x .

The last thing to consider when it comes to differentiation is that sometimes we may want to make broader claims about how incentives may affect behavior, so we try to generalize our theories by considering an entire class of functional forms. For example, suppose that we do not know the exact relationship between the number of shots of espresso the customer orders from the barista, the cloud cover, and her utility, but we know that there is some *implicit* function $F(u, x, c) = 0$, which can be solved for any of the arguments, including the one in which we have been interested, $u = f(x, c)$. Furthermore, we can still let the customer try to maximize her utility by taking the derivative with respect to the number of espresso shots, x . What's important here is the notation. Given that we don't know the *explicit* relationship between x and u , as we did before, our notation has to reflect this fact. Here, taking the *total* derivative of the function $f(x, c)$, we get

$$du = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial c} dc$$

and this catalogs all the ways that u can change. The customer's utility can change when she gets some more espresso, and the amount that her utility will change depends on the effect of another shot of espresso on utility (captured by the partial derivative $\frac{\partial f}{\partial x}$) times the number of shots consumed, dx . The

customer's utility can also change when the cloud cover changes, and this effect is determined by the effect that a unit shift in cloud cover has, $\frac{\partial f}{\partial c}$, and the number of units that the cloud cover changes, dc . As good experimenters, we can't change everything at the same time, so let's examine what happens when the cloudiness of the day is unchanged (i.e., $dc = 0$). In this case, we just have

$$du = \frac{\partial f}{\partial x} dx$$

which can be rearranged as

$$\frac{du}{dx} = \frac{\partial f}{\partial x}$$

and this expression $\frac{\partial f}{\partial x}$ is the *implicit derivative* of the customer's utility when the cloud cover is constant. Further, even without the explicit function we had before, we can still be sure that utility finds a critical value when $\frac{\partial f}{\partial x} = 0$. However, just to check, recall that before, we assumed an explicit function for utility, $u = 100x - 5x^2 - 10c + cx$, and you can see that if we find $\frac{\partial f}{\partial x}$, we get $100 - 10x$. Setting this equal to zero, we find $x^* = 10$, the same critical value. The point is that when we generalize our thinking and use $\frac{\partial f}{\partial x}$ instead of the explicit derivative, the logic is unchanged—this derivative, regardless of explicit form, must be zero at a critical value.

Integration

Let's consider your coffee problem from a different perspective. Suppose, reasonably, that while experimenting with your espresso intake, all you know is the increment or decrement to your happiness when you changed your consumption. You start caffeine-free on the first day, and after your first shot, your utility meter jumps up by 90. The next morning, you go for a double shot and find the increment remains positive, but now the difference from the day before is only 80. After a few more experiments, you estimate that the marginal utility of consuming x shots of espresso is $100 - 10x$.

What if you are tired of experimenting and want to predict your utility from espresso more generally? Is there a way to go from this marginal function up to the total? Well, yes; we know that we took a derivative to get to $100 - 10x$ not long ago, and we also know that the function that we took the

derivative of was just $100x - 5x^2$. It turns out that we can use definition 7 to *integrate* this derivative, $u'(x)$, to get back to the primitive function, $u(x)$.

Definition 7 (Indefinite integral) *Let f be a function of x . The integral of $f(x)$ with respect to x is $\int f(x) dx = F(x) + F(0)$, where $F(0)$ is a constant of integration.*

In our example, we want to integrate the marginal utility of espresso consumption, $u'(x) = 100 - 10x$, with respect to the consumption of espresso, x , to get the total utility provided by any espresso consumption level, $u(x) = 100x - 5x^2$. Well, how do you do that? In the cases we will consider here, this is done by simply finding the *antiderivative* of $u'(x)$. Put differently, you just do the opposite of taking the derivative—ask yourself what function you had to take the derivative of in order to get this marginal function.

To get from $u(x)$ to $u'(x)$, you used the simple power rule, reducing the power and multiplying by the original power. For the antiderivative, you do the opposite: increase the power by 1 and divide by this power. In our example, we can increase the power from 0 to 1 on the first term and divide by the power to get $100x$. Doing the same for the second term gives us $\frac{10x^2}{2} = 5x^2$. Putting these terms back together and adding our constant of integration, we get

$$\int (100 - 10x) dx = 100x - 5x^2 + u(0)$$

The only loose end here is the constant of integration, $u(0)$. Intuition suggests that we need this because starting with the marginal, we can never know for sure if there was a constant added to the original utility function, perhaps to account for your baseline happiness in the absence of any coffee.

Probability

The concept of probability provides a means of analyzing random events. Let X be the set of all possible outcomes or states that could be realized. For example, if you are flipping a coin, then $X = \{Heads, Tails\}$, if you are rolling a die, then $X = \{1, 2, 3, 4, 5, 6\}$, or if you're considering how many minutes it will take you to complete an assignment that's due in 24 hours, then $X = [0, 1140]$. In each case, the outcomes are mutually exclusive (only one will occur) and exhaustive (no outcome outside of X could occur). For instance, if you flip a coin, you will either get Heads or Tails. The coin will not land on its

side, roll across the room, and get wedged vertically in the floorboards. If this were a possibility, then it would need to be included in set X .

The relative likelihood of each state in X is given by a *probability distribution* over X . Let the set of outcomes in X be indexed as $x_1, \dots, x_n$. Then, for each element $x_i \in X$ (i.e., x_i in the set X), the probability distribution assigns a number, or *probability*, $p(x_i) > 0$, such that $\sum_{x_i \in X} p(x_i) = 1$.¹ In the coin flip example, the probability distribution assigns a number $p(\text{Heads})$ to Heads and a number $p(\text{Tails})$ to Tails, such that each is nonnegative and sum to 1. The probability associated with an outcome tells us the relative frequency with which that outcome occurs. In other words, if we were to flip our coin many, many times under exactly identical conditions, what is the proportion of times that it would come up Heads versus Tails? Or perhaps we are considering an Olympic speed skating race between four teams and the outcomes are the teams that could win: namely, $X = \{\text{Russia, USA, South Korea, Canada}\}$. Then our probability distribution assigns a number between 0 and 1 to each of these possible outcomes, such that the four numbers sum to exactly 1, and indicates the relative frequency with which each outcome will occur.

When the outcome space is continuous, such as the real interval $[a, b]$, a probability distribution that we will often use is the *uniform distribution*. A random variable is uniformly distributed on $[a, b]$ if the probability that the realized value falls in some smaller interval within $[a, b]$ is simply equal to its relative length. That is, all intervals of the same length are equally probable. More specifically, consider a smaller interval $[c, d]$ within $[a, b]$, such that $c \geq a$ and $d \leq b$. The probability that the realized outcome is within $[c, d]$ is given by the length of this smaller interval $[c, d]$ as a fraction of the length of the entire interval $[a, b]$: $\frac{d-c}{b-a}$.

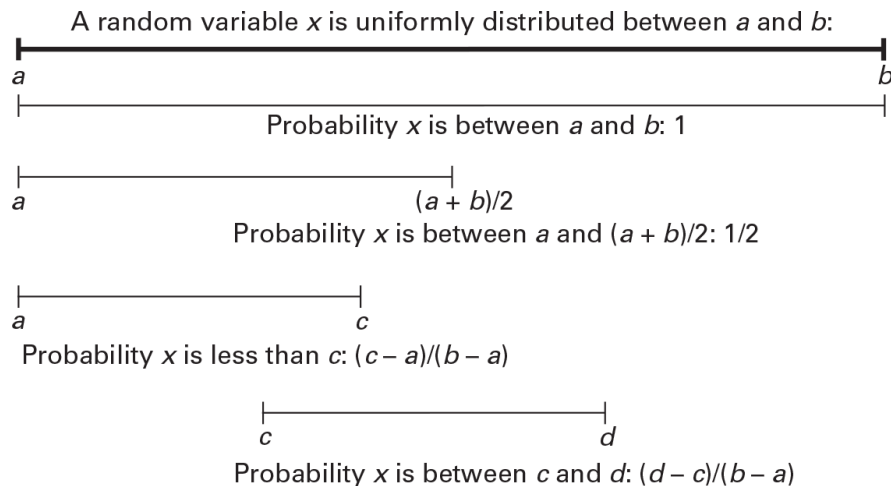


Figure A.4

The uniform distribution over $[a, b]$. All intervals of the same length are equally probable, and the probability that the random variable falls in the interval is equal to its length as a fraction of $(b - a)$.

Once we know the probability distribution, then we can consider the probability of *sets of outcomes*, also called *events*. For example, perhaps you roll a die and you're interested in the event in which a number greater than or equal to 5 appears. What is the probability of this event? So long as it is impossible that *both* outcomes can occur simultaneously (i.e., a single roll of the die can't produce both a 5 and a 6), then we can find the probability of this event by adding up the probabilities of the states in the event.

Probability addition rule. Consider two outcomes A and B , which are mutually exclusive. The probability that either A or B occurs is given by $p(A) + p(B)$.

Now let's consider the probability of two events *both* occurring. For instance, perhaps you are interested in the probability that the American speed skating team wins *and* it snows. Our computation of this probability depends on whether the two events are independent. We say that two events are *independent* if the outcome of one does not affect the probability of the other. Since the speed skating event is indoors, it seems likely that the probability that the Americans win will not be influenced by whether it snows, and presumably the probability that it snows would also not be influenced by who wins the race. Since the two events are independent, the probability that they both occur is simply given by the product of the probabilities of each occurring.

Probability product rule. Consider two events A and B, which are independent. The probability that both A and B occur is given by $p(A)p(B)$.

For example, let's imagine that you flip a coin three times in a row. What is the probability that you get a Heads on all three flips? Let the probability that the coin comes up Heads on any given flip be $p(\text{Heads})$. Then the probability that it comes up on all three flips is given by $p(\text{Heads}) \cdot p(\text{Heads}) \cdot p(\text{Heads}) = p(\text{Heads})^3$. It follows that the probability that the coin comes up Tails *at least once* is given by just $1 - p(\text{Heads})^3$.

Expected Value

Let a *gamble* (or a lottery) be a probability distribution over a known set of outcomes. Often the different outcomes are associated with different numerical values, such as money earned or points scored. In these cases, we can calculate the *expected value* of the gamble (see definition 8): if the gamble were to be conducted many times under identical conditions, what do we expect to get each time, on average? The expected value is the average outcome, weighted by the probability distribution over the outcomes. That is, for each possible outcome, we take the value of the outcome multiplied by the probability of the outcome occurring, and then we sum up over all the possible outcomes.

Definition 8 (Expected value) Consider a gamble X with possible outcomes $\{x_1, x_2, \dots, x_n\}$, each with respective probabilities $(p_1, p_2, \dots, p_n)$. The expected value of X is given by $EV(X) = \sum_{k=1}^n p_k(x_k) = p_1x_1 + p_2x_2 + \dots + p_nx_n$.

As an example, consider the choice faced by American football teams after scoring a touchdown. It can attempt a 33-yard extra point kick, which has a fairly high probability of success (by one estimate, the probability of a football team making this field goal is 0.94) and earns them 1 point. Or it can attempt a 2-point conversion: it gets the ball on the 2-yard line and has one chance to get the ball into the end zone. The 2-point conversion has a lower probability of success (0.48), but, if successful, the team earns 2 points. What are the expected values associated with these decisions? Going for the single extra point is like participating in a gamble where the team has a 0.94 chance of getting 1 point and a 0.06 chance of getting 0 points. Thus, the expected value is $0.94 \times 1 + 0.06 \times 0 = 0.94$. Going for the 2-point conversion means participating in a gamble with a 0.48 chance of getting 2 points and a 0.52

chance of getting 0 points, so the expected value of the 2-point conversion is $0.48 \times 2 + 0.52 \times 0 = 0.96$.

Given that the expected value is greater when a team goes for 2 points, why would anyone ever go for a single point? The football team's objective is not to maximize its points, but to win the game. Depending on the game situation, 1 versus 2 points might differ in importance. So in determining how people make decisions over lotteries, we're not just going to look at the expected value that they get from different gambles. We have to think about their utility or payoff that they get from the different outcomes in the gamble. The next section describes a model of decision making under risk that involves the decision maker maximizing his *expected payoff* rather than the expected value.

Expected Payoff: A Model of Choice under Risk

Now imagine that a decision maker must choose between two gambles. Which does the decision maker prefer? The standard model of choice under risk is called *expected utility theory*. Expected utility theory was developed by John von Neumann and Oskar Morgenstern in the first chapter of the very first game theory book (von Neumann and Morgenstern, 1944), as a preliminary step before developing the theory of games, and it is still the dominant theory of choice under risk in economics.

To determine which lottery a decision maker prefers, we must consider both the probability of each outcome *and* how happy the decision maker would be with each outcome. We will thus look at the expected value of the utilities or payoffs that the individual gets. This is commonly called the *expected utility* of the gamble. However, since we're using the terminology of *payoff* to describe how happy a player in a strategic game is with each outcome, we'll use the term *expected payoff*. That is, the *expected payoff* of a gamble is equal to the average payoff that the individual gets weighted by the probability distribution. The concept is explained further in definition 9.

Definition 9 (Expected payoff) *Imagine that individual i faces a risky decision or gamble X with possible outcomes $\{x_1, x_2, \dots, x_n\}$, each with respective probabilities $(p_1, p_2, \dots, p_n)$. Individual i 's expected payoff from X is given by $Eu_i(X) = \sum_{k=1}^n p_k u_i(x_k) = p_1 u_i(x_1) + p_2 u_i(x_2) + \dots + p_n u_i(x_n)$.*

According to expected utility theory, a decision maker prefers gamble X to gamble Y if and only if $Eu_i(X) \geq Eu_i(Y)$. In situations without risk, we assume that rational decision makers act to maximize their payoffs. When there is risk, we assume that rational decision makers act to maximize their expected payoffs.

As an example, imagine a decision maker who is leaving the house and deciding whether or not to grab an umbrella. She doesn't want to be caught in the rain without it, but she also doesn't want to schlep around the umbrella if it isn't raining. Let's say that her payoffs for the different possibilities are as follows:

	Rain	Sun
Umbrella	40	60
No Umbrella	10	100

Figure A.5
Payoffs from different outcomes.

Should she take the umbrella or not? If it is raining, then she prefers to take it, and if it is sunny, she prefers to leave it. So her decision depends on the likelihood that she thinks it is going to rain. Let's say that the weather forecast gives a 40% chance of rain. Her choice is whether to take the umbrella or not. We therefore compute the expected payoff of each action and compare them. Taking the umbrella involves participating in a gamble where there is a 0.4 chance of winding up in the rain with an umbrella (payoff of 40) and a 0.6 chance of winding up in the sun with an umbrella (payoff of 60). Thus, her expected payoff from taking the umbrella is $Eu(\text{Take Umbrella}) = 0.4 \times 40 + 0.6 \times 60 = 52$. Leaving the umbrella involves participating in a gamble in which there is a 0.4 chance of winding up caught in the rain without an umbrella (payoff of 10) and a 0.6 chance of winding up in the sun and not having to carry her umbrella (payoff of 100). Thus, her expected payoff from taking the umbrella is $Eu(\text{Leave Umbrella}) = 0.4 \times 10 + 0.6 \times 100 = 64$. Since her expected payoff from the Leave Umbrella lottery is higher than the expected payoff from the Take Umbrella lottery, she leaves it at home. Of course, this depends on the payoffs from each of the four different options and the probability with which it will rain. As the likelihood of rain increases, eventually it would be in the decision maker's interest to take the umbrella.

Choice over Time

How do individuals make choices that involve trade-offs across time periods? Many choices that we face involve costs and benefits that occur at different points in time, and individuals may prefer to take an action that is associated with a lower immediate utility, knowing that it will pay off somewhere down the line. For example, should you spend money on consumption today or save money and delay your consumption? Do you want to work on a big assignment now, or do you prefer to push that cost to the future? Do you want to quit smoking or begin a new workout routine today, knowing that it will not be fun right now but will ultimately benefit your future self? Each of these cases involves a calculation in which current costs and benefits are weighed against future costs and benefits, and the ultimate decision depends on *how patient* the decision maker is.

Most people are at least somewhat impatient: they give more weight to the costs and benefits that they will experience immediately than they do to outcomes that won't occur until later. In this case, we say that people *discount the future* by underweighting—or discounting—the costs and benefits that they don't experience until later. We represent this impatience mathematically by assuming that each individual i has some discount factor δ_i , which is a number between 0 and 1 that indicates the weight that they put on outcomes that occur one period in the future. In what follows, we'll drop the subscript i and simply remember that the discount factor δ is an individual-specific parameter that captures how patient a particular individual is. In our analysis of games in the main text, different players may have different discount factors.

For example, imagine that a decision maker can choose between option A, receiving a payoff of 120 today, or option B, receiving a payoff of 120 a year from now. (We can think of the 120 as a monetary payment, with the individual's utility or payoff simply equal to $u_i(120) = 120$, or we could think of the individual receiving a payoff of 120 from some other outcome or event, such as eating a favorite meal.) Most people would say that they strictly prefer the payoff today. This means that when assessing such a person's choice, we can't simply compare options A and B by comparing their payoffs of 120 and 120. Getting 120 in a year is not the same as having 120 today—it's worse. To reflect that it is worse, we need to discount the payoff that the decision maker won't receive for another year. We thus multiply it by the decision maker's

discount factor δ to determine its present value (PV) to the decision maker. Remember that the discount factor δ is less than 1 so, by multiplying the payoff by δ , we are discounting (reducing) it.

In our example, the PV of getting a payoff of 120 today is just 120, but the PV of having 120 in a year is less than 120: it is $\delta \times 120$. Discount factors that are closer to 1 represent greater patience, with the decision maker not discounting the future much. In contrast, a discount factor close to 0 implies that the decision maker puts almost no weight on the future—if she has to wait to get something, she might as well not get it at all.

We said that our decision maker preferred to have 120 today, which implies that the PV of waiting a year ($\delta \times 120$) must be less than having it today (120), and so it must be the case that $\delta < 1$ for our decision maker. In other words, this person is not perfectly patient. In fact, if we asked whether she'd prefer a payoff of 120 today or 130 a year from now, perhaps she would also say that she wants the lower amount immediately. In this case, the PV of having 130 in a year is $\delta \times 130$, and if this person likes that outcome less than just having 120 now, then it must be the case $\delta \times 130 < 120$, or $\delta < 0.923$. Foregoing the extra 10 is a perfectly reasonable choice, so long as this person is a bit impatient. We could imagine continuing to scale up the payoff that she gets in a year, until she is finally willing to wait for the larger payoff. Perhaps when offered the choice between 120 today and 150 in a year, the person states that she is indifferent, but if we increase the amount that she will get from waiting any more, then she definitely prefers to wait. In this case, this person views the PV of receiving 150 in a year as being equal to getting 120 today. That is, for this person, $120 = \delta \times 150$, indicating that she has a discount factor of $\delta = 0.8$.

Now we've seen how players evaluate payoffs across two different time periods. How do they assess a stream of payoffs over multiple time periods? So for instance, perhaps we offered the decision maker the choice of option A, 1, 000 today and in each of the next three time periods, and option B, 100 today and in the next two time periods, but then 5, 000 in the third time period, as shown in the following table.

	Today	Period 1	Period 2	Period 3
Option A payoff stream	1,000	1,000	1,000	1,000
Value today:	1,000	$\delta(1,000)$	$\delta^2(1,000)$	$\delta^3(1,000)$
Option B payoff stream	100	100	100	5,000
Value today:	100	$\delta(100)$	$\delta^2(100)$	$\delta^3(5,000)$

Figure A.6
Present value of streams of payoffs.

Since the decision maker is making this choice today, we have to think about how he values each of these payoff streams today (i.e., their PVs). We know that the value of 1, 000 today is just 1, 000 and the value of 100 today is just 100. We also now know that the value of waiting one period to get 1, 000 is $\delta(1, 000)$ and the PV of waiting a period to get 100 is $\delta(100)$. What about when the decision maker needs to wait more than one time period?

The standard model of choice over time is called the *discounted utility model*, which was introduced by Paul Samuelson in 1937. This model assumes that all intertemporal choice considerations can always be boiled down to the single variable δ . Specifically, the discount factor reflects not just how people discount having to wait from now to one period from now, but also how people always discount waiting an extra period, regardless of when that time period occurs. In other words, δ reflects not just how much the decision maker dislikes having to wait a period from now instead of getting something immediately, but also how much he dislikes having to wait for 32 time periods instead of 31 time periods. According to the discounted utility model, we discount the ultimate payoff by δ for each period that the decision maker needs to wait.

So to find the present value of a payoff that the decision maker gets two periods from now, we multiply by δ and then multiply by δ again, or by δ^2 . Another way to think about this is to imagine that you're going to get \$100 in two months, where a month is a time period. Now imagine that you're already one month into this wait, so you only have to wait one more month. At that point, the PV at that time of getting 100 in one month is $\delta \times 100$. Now, what happens if we step back to today? In one month, the PV is going to be worth $\delta \times 100$, but we're still one month out from that, and so we need to discount this $\delta \times 100$ by δ , giving us $100 \times \delta \times \delta$ or $100 \times \delta^2$. Likewise, if we need to find the PV of a payoff that occurs T periods from now, we multiply it by δ^T . Thus,

in our payment plans listed in the table, we can find the present values of each of the future payoffs. Then, to get the present value of this full stream of payoffs, we simply add up the present values of all the payoffs. For option A, this is $1,000 + \delta(1,000) + \delta^2(1,000) + \delta^3(1,000)$, and for option B, this is $100 + \delta(100) + \delta^2(100) + \delta^3(5,000)$. Which of these plans the decision maker prefers depends on his δ . If the decision maker is relatively patient (specifically, if $\delta > 0.827$), then he'll prefer option B. Otherwise, he'll prefer option A.

Now what if there were an infinite number of periods? For instance, let's imagine that the decision maker gets some payoff a in every period for all eternity. Then her payoff stream is the following infinite sum:

$$\text{payoff stream} = a + \delta a + \delta^2 a + \delta^3 a + \dots$$

Since δ is less than 1, this sum converges. To be able to compare it with various payoff streams, we need to know what it converges to. To see this, first look at all the terms after the first a . Since they all have a δ in them, we can factor out δ , which gives us $a + \delta(a + \delta a + \delta^2 a + \dots)$. Now notice that what is in the parentheses is just the payoff stream itself that we started with. That is, we have $\text{payoff stream} = a + \delta(\text{payoff stream})$. If we rearrange our terms, we get $\text{payoff stream} = \frac{a}{1-\delta}$, and thus we have found what our infinite sum converges to. The present value of this infinite stream of a 's is $\frac{a}{1-\delta}$.

As a final example, let's imagine that our decision maker receives a payoff of 5 today, and then 20 for all periods for all eternity. Thus, the payoff stream is given by $5 + \delta 20 + \delta^2 20 + \delta^3 20 + \dots$. What does this converge to? We have our 5 that we get initially, which simply has a PV of 5, and then we have an infinite stream of 20s, which we know has a PV of $\frac{20}{1-\delta}$. However, we have to wait a period before our stream of 20s begins, and so the value $\frac{20}{1-\delta}$ must be discounted by δ . Thus, the present value of this payoff stream is: $5 + \frac{20\delta}{1-\delta}$

¹The notation $a \in X$ denotes that a is an element of the set X . $\sum_{x_i \in X} p(x_i)$ is the sum of the probabilities associated with each element in X .

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